

ABSTRACTS

SIOP ABSTRACTS

IPSO LIVE Q&A: WELCOME AND IPSO SESSION 1: LIVER
TUMORS 21-10-2021 8:00 AM - 9:00 AM

O0001 / #1661 | Preliminary Experience in Treating Pediatric
Hepatoblastoma with Indocyanine Green Fluorescence

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Background and Aims: Locally advanced hepatoblastomas pose unique clinical challenges. Despite chemotherapy, surgical resection remains the mainstay of cure. These tumors can be managed either by radical liver resections (RLR) or liver transplantation. Despite prior experiences with Indocyanine-green (ICG) in liver tumors, its potentials in children remains unclear. The aim of this study is to report the role of ICG in RLR.

Methods: Review of demographics, surgical technique, complications and survival of 3 patients with Hepatoblastomas, resected under ICG-guidance.

Results: Case-1, female, 12 months-old, presented with abdominal mass, Alpha-fetoprotein (AFP) 273,000. Images reported a multifocal PreTEXT-III, P+V-. Histology: epithelial/fetal Hepatoblastoma. Good response after 3 cycles of intermediate risk (IR) chemotherapy. ICG was administered 72 hours before surgery. A non-anatomical left-extended hepatectomy was performed (segments 2, 3, 4 and 5). Intraoperative margins were extended guided by ICG. Case-2, male with prenatal liver mass. Imaging confirmed a mass, originated in the caudate lobe, PreTEXT-III, P+V+C+. AFP 380,000. Histology: embryonal/epithelial hepatoblastoma. After 3 cycles of IR chemotherapy, there was a good response. ICG was administered 72 hours before surgery. Resection of the caudate lobe was performed. Case-3, male, presented with liver mass and precocious puberty. AFP 199,000, b-HCG: 567. Images reported a multifocal PreTEXT-IV, P+ V-. Histology: epithelial/mesenchymal Hepatoblastoma. After 6 cycles of IR chemotherapy, there was a good response. ICG was administered 70 hours before surgery, a right-extended hepatectomy plus wedge-resection of a segment 2 ICG-avid lesion, was performed. Non intra-or-postoperative complications were reported. We achieved negative microscopic margins in all patients. No recurrence at 7, 4 and 2-months follow-up.

Conclusions: ICG-fluorescein guidance was safe and feasible for tumor demarcation, enhancing the accuracy of resection, allowing negative margins in tumors adjacent to vascular mayor vessels, also providing surgical visualization of multifocal disease after treatment, increasing the indications of radical resections over liver transplantation. Long-term follow up is needed.

IPSO LIVE Q&A: WELCOME AND IPSO SESSION 1: LIVER
TUMORS 21-10-2021 8:00 AM - 9:00 AM

O0002 / #1038 | CTNNB1 Circulating Tumour DNA as a
Putative Biomarker for Hepatoblastoma

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Background and Aims: Driver mutations in CTNNB1 are a hallmark of hepatoblastoma and offer a common biomarker for a liquid biopsy approach that is based on the presence of CTNNB1 circulating tumour DNA (ctDNA). Objectives: (A) To investigate the utility of a universal next-generation sequencing (NGS) assay to detect CTNNB1 ctDNA in patients with hepatoblastoma. (B) To compare the levels of: (1) NGS ctDNA, (2) ddPCR ctDNA, (3) serum Alpha-fetoprotein (AFP, current hepatoblastoma biomarker), and verify if they correlate with tumour burden and treatment response.

Methods: We developed and tested a custom digital NGS assay covering exons 2 to 4 of CTNNB1 for detection of somatic mutations in plasma. Our cohort included 19 patients, 16 with Sanger sequencing confirmed CTNNB1 somatic mutations in the matched tumours.

Results: NGS limit of detection was evaluated at 2% variant allele frequency (VAF) and accurately detected mutations in 11/16 (69%) of the Sanger confirmed cases in samples: taken at initial diagnosis in 8/9 (89%, including 2 patients with no diagnostic biopsy), following neoadjuvant chemotherapy in 4/9 (44%), post-operatively for fully resected localized disease in 0/4 (0%), and metastatic recurrence in 1/5 (20%), suggesting a positive correlation between CTNNB1 variants and tumour burden. Available matched ddPCR data (7 patients, 21 samples) showed similar ctDNA levels for VAF>2%. ddPCR demonstrated higher sensitivity and was able to detect ctDNA in additional 7 NGS-negative samples, with VAF detected as low as 0.085%. NGS ctDNA levels did not correlate with AFP levels in treatment naïve samples taken at initial diagnosis. Nevertheless, longitudinal samples (5 cases, 12 samples) showed similar dynamics of both ctDNA and AFP, reflecting response to treatment.

Conclusions: ctDNA is a good surrogate marker of tumour burden and shows correlation with treatment response. CTNNB1 NGS poses an opportunity for a universal somatic genotyping at disease presentation of hepatoblastoma.

IPSO LIVE Q&A: WELCOME AND IPSO SESSION 1: LIVER TUMORS 21-10-2021 8:00 AM - 9:00 AM

O0003 / #1659 | Novel Technique for Suprahepatic Inferior Vena Cava Tumoral Thrombus in Hepatoblastoma

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Background and Aims: In unresectable, Post-TEXT III and IV hepatic neoplasms, liver transplantation (LT) is the only curative option. The presence of tumoral thrombus (TT) add a challenge. There is scarce literature for suprahepatic inferior vena cava TT (SHVC-TT). The aim of this study is to report a surgical alternative technique.

Methods: Review of demographics, surgical technique, postoperative complications and survival, of 2 cases of Hepatoblastoma with SHVC-TT, treated with LT, cava vein resection and suprahepatic-atrial shunt (SSHA).

Results: Case-1, female, 20 months-old, presented with abdominal mass, alpha-fetoprotein (AFP) 146,000. Images showed a multi focal, Pre-TEXT-IV tumor, pulmonary metastases (PM) and SHVC-TT. Histology reports fetal epithelial Hepatoblastoma. Post high-risk (HR) chemotherapy, there was resolution of PM and partial regression of TT. The patient was listed for deceased-LT. A left-lateral segment from an adult was implanted. An In-block hepatectomy with retro and SHVC resection was performed. An iliac vein graft was used to perform the SSHA without IVC replacement.

Case-2, female, 24 months-old, presented with abdominal mass, AFP >600,000. Images reported a multifocal PreTEXT-III, PM+, P+ and SHVC-TT. Histology: Predominantly fetal pattern Hepatoblastoma. Treated as per HR protocol. Post-TEXT-III, total PM and partial TT response. Living-donor (LD) transplantation was performed, with SHVC ligation without replacement of IVC and a SSHA using the left internal jugular vein from the same LD.

Intra and post-operative results: Warm ischemia time: 43 and 46 minutes, An-hepatic phase: 58 and 65 minutes. Histology showed negative margins. Hospital stay 60 and 47 days. Both patients are disease-free without complications related to their SSHA at 11 and 8 months follow-up.

Conclusions: We present a novel surgical technique, allowing a successful LT. Since these patients had total long-term thrombus of the IVC, and developed collateral drainage veins, we thought, replacement was not necessary. Longer follow-up is needed to determine long-term surgical complications and event-free survival.

IPSO LIVE Q&A: WELCOME AND IPSO SESSION 1: LIVER TUMORS 21-10-2021 8:00 AM - 9:00 AM

O0004 / #690 | Mesenchymal Hamartoma of Liver: Outcomes of Surgical Management

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Background and Aims: Mesenchymal hamartoma (MH) is a benign liver tumor with good long term outcome after surgical management. The aim was to evaluate the presentation, diagnosis, and outcome of surgical management of MH at a tertiary care center, over 15 years.

Methods: Retrospective analysis of children with a diagnosis and treated for MH managed at our center from January 2007 to January 2021

Results: Sixteen patients with a diagnosis of mesenchymal hamartoma of liver were included in the study. There were 9 (56%) males and 7 (44%) females. Mean age was 21 months (range 15 days to 96 months). While 2/16 (12.5%) were antenatally diagnosed, 2/16 (12.5%) were diagnosed incidentally, remaining 12/16 (75%) were diagnosed on investigations for abdominal distention. Ultrasonography and computed tomography were suggestive of a multiloculated, multiseptated thick walled solid lesion with cystic component in all children. 3/16 (19%) had preoperative severe respiratory distress and underwent preoperative ultrasound guided drainage of the largest cystic space. Right lobe was involved in 93% children. All underwent surgical resection: Non-anatomical in 5/16 (31%) and anatomical resection in 11/16 (69%; right hepatectomy in 9/11 and extended right hepatectomy in 2/11). Post-operative complications were observed in 2/16 (12.5%; bile leak and bleeding in one each). Two children had gross residual lesion after non anatomical resection and one of them had a recurrence 1 year after surgery which needed a re-resection. Histopathology was confirmatory of mesenchymal hamartoma in all patients with feature of chronic cholecystitis in 31%.

Conclusions: Most children with MH present with abdominal distention and liver resection was curative in majority of patients, with recurrence noted only in one (6%). The long term outcome is good, with 100% survival.

IPSO LIVE Q&A: WELCOME AND IPSO SESSION 1: LIVER TUMORS 21-10-2021 8:00 AM - 9:00 AM

O0005 / #1662 | Surgical and Oncologic Outcomes of a Large Series of Malignant Liver Tumors Treated at a Comprehensive Pediatric Cancer Center

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Background and Aims: At our institution, liver tumors represent 5% of all patients with cancer. The aim of our study is to compare our series with the international reports, in terms of surgical and oncologic outcomes.

Methods: A retrospective review of medical records was conducted for patients with primary malignant liver tumors treated at our institution from 2008 to 2018. The data collection included the demographic data, clinical presentation, imaging and staging, pathology reports, laboratory results, operative reports, presence of metastases, metastasectomy, type of chemotherapy and doses, and follow up and survival. The statistical analysis was done with the aid of the SPSS package, with descriptive statistics, square chi, and survival functions.

Results: We identified 149 patients with malignant liver tumors during this period. The diagnosis was hepatoblastoma in 82.6%, hepatocarcinoma in 6.7%, NOS malignant liver tumor in 3.4%, and embryonal sarcoma in 7.4%. The mean age at diagnosis was of 50.7 months. The PRETEXT classification was I for 12.1%, II for 43.6%, III for 34.9% and IV for 8.7%. Distant metastases were present in 22.8%. Upfront surgical resection was performed in 36.7%, and the rest received neoadjuvant chemotherapy; this down-staged the POSTEXT status in 77.2% of the patients. Overall, right resections were performed in 61.1%, left resections in 16.8%, mesohepatectomy in 6%, non-anatomical resections in 8.7%. This series includes one patient with orthotopic liver transplant, and one case without resection. The mean operative time was of 58% of those with lung metastases required resection. The survival analysis showed an overall survival of 57.7%, with 20.8% lost to follow up, and 12.1% dead with progressive disease. Stratified by PRETEXT stage, I and II had the best outcomes, with 80 and 71%.

Conclusions: Our series has acceptable outcomes, considering the setting of a Country with limited resources. It could be improved with earlier referral.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (1) 21-10-2021 8:15 AM - 9:45 AM

O0006 / #207 | The Pandemic Pivot - Lessons Learned in Program Delivery

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Background and Aims: Our organization has survived the initial ravages of the pandemic in 2020. Organizational sustainability in 2021 and beyond continues to be a challenge many of us face as the Covid-19 pandemic rages on globally. How did we address program delivery going forward?

Methods: PAUSE - Adjusting programs and services to ensure compliance with the ever-changing environment and evolving public health and government regulations is not feasible. Temporarily suspending programs was often the prudent course of action. This allowed time to understand the bigger picture and to take into account unfolding scenarios, properly assess program administration challenges and ensure adequate resources moving forward. PRIORITIZE - Fundraising in a pandemic isn't easy and many organizations were not able to secure the necessary funds and/or retain staff to maintain program delivery. There are things to consider when prioritizing programs for ongoing delivery: Impact - How does this program make a difference? How many patients/families are impacted? Cost - How much does it cost? Delivery - How does this program work? Who is involved the administration of the program? Keeping a program is not always just about cost. You may have the necessary dollars but not the hands to make it possible. The impact of a program may be life changing for a smaller number of individuals benefitting then a lot of people benefitting a little bit.

Results: PROCEED - Do we continue the program or not? The pandemic itself sometimes dictates whether or not the program continues. Other times, program changes and revisions - "the pandemic pivot" we implemented and the program continued.

Conclusions: Surviving 2020 and the global Covid-19 crisis resulted in our organizations' having to modify program delivery while retaining some continuity with its' previous versions for ongoing sustainability. Ongoing evaluation of each program continues to ensure impact and viability.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (1) 21-10-2021 8:15 AM - 9:45 AM

O0007 / #1170 | Response of a Civil Society Organization for Ensuring Adequate Supplies of Personal Protective and Infection Control Equipments During Covid-19 Pandemic

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Background and Aims: COVID-19 epidemic led to a sharp surge and shortages in the requirement of personal protection equipment (PPE) and sanitization supplies at pediatric oncology units centers across. This affected morale and safety of health workers and in some centers compromised delivery of optimal healthcare. Cankids, a national society for change in childhood cancer that strives to provide holistic care and support to in children and families coping with childhood cancer in India, stepped in to overcome this challenge. The aim of the study was to assess the impact of this initiative over a 5-month period.

Methods: In the time period from April-August 2020, Cankids received requests for PPE, masks and sanitization supplies from 35 units treating childhood cancer across India. The organization reached out to the

manufacturers and stockists to meet the demand at discounted tariffs. The delivery to the centers was achieved by road, rail and air transport with the assistance of Airport Authority of India, Indian railways, and various courier services. Members of parents support group were mobilized to produce over 10,000 handmade cloth masks for distribution to parents and patients at various centers. Funds for this initiative were raised through crowd funding.

Results: With the efforts described above we were able to meet the demand of all 35 centers in entirety This comprised of 2080 PPE, 8138 3 layered Masks, 4988 N95 masks, 1450 Surgical Gowns, 606 Head Caps, 28300 Shoe Covers, 280 Face Shields, and 13250 pairs of gloves in addition to hand sanitizers, and disinfectants. The total expenditure incurred was INR 25,05,176.0 (\$ 33,684.85).

Conclusions: With concerted efforts and support from various stakeholders and agencies involved, timely delivery of these articles could be ensured to all centers. As a support group this exercise has empowered us to meet unforeseen challenges in the future.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (1) 21-10-2021 8:15 AM - 9:45 AM

00008 / #788 | Let Me Cover U: Rainbow's New Actions Under Pandemic

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Background and Aims: Since the early 2020, the COVID-19 broke up and spread the whole world without early warning. The live paces were disordered and nearly everyone was under the panic with limited information and materials. The families with children cancers suffered more: 1. Hard to go to the hospital 2. Lack of protection (masks and sterilizing equipment, see hand washing etc.) 3. Extra Expenses (Nucleic Acid Test (NAT), CT etc.) 4. Panic and worries under uncertainty While charities faced new challenges: 1. All activities in wards had to stop 2. Difficulties in fund raising At different period, Rainbow Foundation tried various efforts to continue its services.

Methods: Rainbow designed the program "Let me cover U" to deal with different situation of the pandemic. 1. Physical Covering: Children Masks for kids around 0-3 years, 3-5 years, 5-8 years. Same and even above the standards of surgical masks. Colourful with designing signs. Small bottle with special designed. Protection-set. Zero-distance distribution (by mail or other ways) 2. Psychological Cover: New Online Activities 2.1 Online Traveling 2.2 Online funding 3. Offline activities under pandemic.

Results: 1. Around 7,000 person times got protection materials. (March 2020 - March, 2021) 2. 135 Online activities covered 40,000 person-times, more than 200,000 person times participated in this program. (March 2020 - March, 2021) 3. USD150,000 DONATIONS at one live show.

Conclusions: 1. No Geographically Boundary, Serve More Kids: From around 10 children in one offline activity to about 2000+ kids with their families online. 2. Attract More Volunteers: A smart phone is enough Low cost. More participation. 3. More ways of Fund Raising: Donators can see the patients and moved deeply 4. Attract more brands and companies: More companies can directly join the activities.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (1) 21-10-2021 8:15 AM - 9:45 AM

00009 / #1465 | Oncovid: Information about Cancer Care During the Pandemic

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Background and Aims: The covid-19 pandemic challenges health professionals to reorganize care for pediatric cancer patients and their families - such as clinical care as consultations, tests and procedures and also psychological care on social and emotional scope. In this context information is essential, being used to educate and clarify about covid-19, avoiding departure or pause of the treatment by patients and their families and being used by health professionals as a support and coping tool.

Methods: Beaba, a non-profit organization for demystifying cancer and providing information about the disease and treatment, developed information materials on covid-19 using the precepts of Information and Communication Technology. The creation and production of the information materials followed the Agile Methodology, using the World Health Organization (WHO) as a scientific source of information and Information Architecture and Design for adequacy, accessibility and usability of the material, respecting the characteristics and particularities of the pediatric oncology environment. Social media were used to broadcast them, enabling quick and accurate updates, considering the frequent changes resulting from new discoveries about the disease.

Results: Information materials covered general topics such as symptoms, transmission and risk groups of covid-19 and specific topics such as care for immunocompromised patients and protective measures for health professionals in the oncology field. Educational materials were also created as communication support tools: "How to talk to children about covid-19" and "We are together, but apart".

Conclusions: More than 50 thousand people were impacted by the materials through dissemination on the Instituto Beaba's digital channels and by the shares made by patients, health professionals and institutions. The information could be used as a prophylactic and protective way, especially at a time when one of the best "medicines" available is education.

IPSO LIVE Q&A: IPSO SESSION 2: NEUROBLASTOMA I

21-10-2021 9:30 AM - 10:15 AM

O0010 / #1682 | Tumor Resection of High-Risk Abdominal Neuroblastoma after Abdominal Irradiation

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Background and Aims: Background: In some conditions the sequence of local treatment in patients with abdominal high-risk neuroblastoma differs for various reasons. As consequence, some patients have to undergo surgery after abdominal irradiation. Aim of this study was to analyse the course of such patients from a surgical standpoint.

Methods: We retrospectively analyzed data of patients with high-risk abdominal neuroblastoma who underwent surgery at our institution between 08/2006 and 10/2019 after having been irradiated beforehand. Data evaluation included patients' and tumor characteristics, surgical results and oncological outcome.

Results: We identified 12 patients meeting the inclusion criteria. All patients had received irradiation at other institutions before being referred to us. Irradiation had been applied because i) tumors had been judged as unresectable (n=3), ii) in cases of incomplete resections (n=8), or iii) in case of relapses. Median age at surgery was 5.9 years (1.4- 14.7). Gross Total Resection was accomplished in 11/12 patients. Major surgical complications occurred in 9 patients: vascular reconstructions including patching or vascular replacement were necessary in 9 patients, one patient had to undergo autotransplantation of a kidney, in that same patient the other kidney lost function because of arterial stenosis and had to be removed. After a median follow-up of 2.5 years (0.4- 14.1), 10 children are alive without evidence of disease, one patient is under treatment for tumor recurrence and 1 patient died of disease. Histological workup revealed vital tumor compounds in 9/12 patients with a median tumor vitality rate of 70% (0-90).

Conclusions: Children with abdominal high-risk tumors judged to be unresectable should receive a central surgical review before radiotherapy is applied as the only planned local therapy. Whenever possible, surgery should precede irradiation within the concept of local therapy in affected patients.

IPSO LIVE Q&A: IPSO SESSION 2: NEUROBLASTOMA I

21-10-2021 9:30 AM - 10:15 AM

O0011 / #861 | Two-Cavity Approach for Resection of Pediatric Abdominal Neuroblastic Tumors

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Background and Aims: An abdominal approach alone is often not sufficient for resection of neuroblastic tumors, because in complex cases these tumors are often extending into the thoracic compartment. Literature addressing the thoraco-abdominal approach for resection of abdominal neuroblastoma in children is scarce. The aim of this study was to evaluate the efficacy of this approach based on our experience in a reference pediatric onco-surgical center.

Methods: Between March 2003 and December 2020 eighteen patients (12 female, 6 male) with neuroblastic tumors (12 neuroblastomas, 5 ganglioneuroblastomas, 1 ganglioneuroma) underwent tumor resection through a two-cavity approach. A retrospective review of patient's records was carried out.

Results: The median age at operation was 48 months (5-132). The surgical access was realized via thoraco-abdominal approach (n=6), transverse laparotomy with incision of the diaphragm (n=9), transverse laparotomy with lateral thoracotomy (n=2), and transverse laparotomy with thoracoscopy (n=1). Gross total resection (GTR) was achieved in 13 patients, a near-GTR (>95%) in 4 cases and an incomplete resection (>90%) in 1 case. The median duration of surgery was 307.5 minutes (99-990), the median intraoperative blood loss was 62 ml/kg (12.5-180). Median length of stay on intensive care was 4.6 days (1-14). Median length of hospital stay was 16.5 days (6-60). Postoperative complications occurred in 8 patients, 2 patients required surgical reintervention (1 reconstruction of the aorta, 1 surgical hemostasis).

Conclusions: The two-cavity approach for resection of neuroblastic tumors in children is a feasible technique with no added morbidity, which allows complete tumor resection in complex cases and avoids two separate operations with equal results in selected cases.

IPSO LIVE Q&A: IPSO SESSION 2: NEUROBLASTOMA I

21-10-2021 9:30 AM - 10:15 AM

O0012 / #1321 | Delayed Surgery in High Risk Neuroblastoma to Prevent Intra-Operative Renal Injury

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Background and Aims: High-risk neuroblastoma (HRN) presents a therapeutic challenge despite multimodal treatment. Management protocols recommend surgery soon after completion of induction chemotherapy (surgery according to protocol; PS). In our centre, surgery is postponed if there is significant risk of damage to the kidney (delayed surgery; DS). The aim of our study was to review the surgical management of HRN in a UK regional centre and assess whether DS impacts long-term outcomes.

Methods: We conducted a retrospective review of consecutive HRN patients that underwent surgery at a single quaternary paediatric oncology unit between January 2008 and December 2019. We collected demographic and clinical data, including details of surgical management. Analysis was performed to determine whether DS affected peri-operative outcomes, as well event-free survival (EFS) and overall survival (OS).

Results: A total of 84 HRN children (35 males; 42%) underwent surgery in our centre. Median age (range) at diagnosis was 2.5 (1.5-4.3) years. Tumour resection was performed within 5 weeks following completion of induction chemotherapy in 49 patients (58%; PS group) and was delayed in 35 patients (42%); 13 due to poor response (15%; excluded) and 22 due to risk of renal injury (27%; DS group). Median time (range) from completion of chemotherapy to surgery in the DS group was 19 (8-79) weeks ($p < 0.0001$ vs. PS). Complete tumour resection was achieved in 20 DS patients (91%; $p = 0.7$ vs. PS) with no requirement for unplanned nephrectomy, and no differences in intra/post-operative complications between groups. EFS in DS patients was 59% and OS was 64% and were similar to those in the PS group [EFS: 61%, OR (95%CI): 0.9 (0.3-2.5), $p = 0.9$ vs. DS; OS: 71%, OR (95%CI): 0.6 (0.2-1.9), $p = 0.4$].

Conclusions: Surgery for HRN can be delayed to mitigate the risk of intra-operative renal injury with no effect on resectability, peri-operative and long-term outcomes.

IPSO LIVE Q&A: IPSO SESSION 2: NEUROBLASTOMA I
21-10-2021 9:30 AM - 10:15 AM

O0013 / #1391 | A 10 Year Single Centre Retrospective Review of Surgical Outcomes in Low and Intermediate Risk Neuroblastoma

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Background and Aims: To evaluate the impact of initial management in outcomes for low- and intermediate-risk neuroblastoma at our centre.

Methods: Retrospective data over a 10-year period (2009-2019). We analysed patients managed through three different groups - observation, primary surgery or surgery following chemotherapy. Primary outcomes included overall survival (OS) and event free survival (EFS) over a 5-year follow up period. Secondary outcomes measured were surgical complications.

Results: A total of 43 patients were identified as low (32 patients) and intermediate (11 patients) risk neuroblastoma (24 males, 19 females). The median age at diagnosis was 5 months (range 0 to 98 months). Overall survival at follow up was 95% and event-free survival was 72%. A total of 28 (65%) operations were performed as both primary surgery and surgery following chemotherapy, with 12 (43%) post-operative complications observed in 11 patients. The observation group were all patients with low risk neuroblastoma. There were 10 (31%) patients in this group. Three went on to require further intervention and 1 neonate died. Primary surgery was performed in a total of 16 patients, 12 patients (38%) with low risk and 4 patients (36%) with intermediate risk neuroblastoma. There was an overall 50% surgical complication rate. All were disease free at follow up. Chemotherapy was commenced in 17 patients, 10 (31%) low risk and 7 (63%) intermediate risk neuroblastoma. Eleven patients had surgery, all but one patient had ongoing Image Defined Risk Factors (IDRF). Three (17%) patients had surgical complications. Sixteen patients were in remission and one died during follow up.

IPSO LIVE Q&A: IPSO SESSION 2: NEUROBLASTOMA I
21-10-2021 9:30 AM - 10:15 AM

O0014 / #541 | Roadtoidrfscore - A Delphi Study on the Definition of the Surgical Complexity Index in Neuroblastoma

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Background and Aims: Following the retrospective work presented at the 52nd SIOP Congress on the possibility of assigning a severity score to IDRFs in Neuroblastoma (NB) surgery, a task force was set up with the aim of defining a preliminary Surgical Complexity Index (SCI) for risk stratification in NB surgery.

Methods: The task force composed of 15 world-renowned surgeons for the experience acquired in NB surgery was involved in a Delphi consensus study. Twenty-seven shared items classified in three domains (pre-operative, intra-operative, and post-operative) and predictive and/or indicative of surgical complexity, were scored according to the following classification: standard risk, somewhat increased risk, quite increased risk, very high risk.

An agreement was defined as the achievement of at least 75% consensus focused on a single answer or, alternatively, the simultaneous fulfillment of the following conditions: 1) at least 60% consensus focused on the prevailing risk category; 2) at least 75% consensus in the sum between the prevailing category and the immediately closest category.

Results: It took 3 Delphi rounds to reach an agreement on 25/27 items (92.6%). All experts completed the first 2 rounds, 86.7% of them completed the third round.

Based on this agreement, a severity score was established for each item ranging from 0 to 3, with an overall SCI range varying from a minimum of 0 to a maximum of 29 points for a given patient.

Conclusions: The definition of the SCI scoring system speeds us to the subsequent retrospective data collection in order to calculate the SCI for each patient operated on for NB and finally validate the SCI itself. Following steps will consider the individual analysis of the IDRFs and the attribution of a severity score for each of them

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (2) 21-10-2021 10:15 AM - 11:45 AM

O0015 / #208 | To Zoom or Not to Zoom - The Switch to Virtual - Does It Work?

J. Lamont

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Background and Aims: In March 2020, the global pandemic and subsequent Public Health regulations saw the cancellation of "in person" support groups and educational seminars. Other programs and services were modified to ensure the safety and well-being of both staff, patients and families. The message "you are not alone" was the focus of communications.

Methods: A needs assessment was conducted in June 2020 to determine the ongoing impact of Covid-19 on mental and financial health, members' level of comfort with virtual platforms, and which platforms were being used.

Results: This resulted in: A virtual "Kids, Cancer & Covid-19" education series via the Zoom platform with Dr. Donna Johnston MD, FRCPC, FAAP Chief, Division of Hematology/Oncology CHEO Session 1 - June 25 with 38 participants, Session 2 (Back to School version) - August 22 with 65 participants. A marked increase in member participation from the typical In-person attendance of up to 12 parents. A "Speaker Series" has been implemented in 2021, with virtual sessions being offered in Feb, March, April, May, Sept, Oct, Nov. A professional facilitated virtual paint-night social gathering via Zoom for moms on December 1, 2020 attended by 18 parents. Virtual social gatherings for parents planned for April and November 2021. A pivot to virtual counselling sessions with our community-based psychologist, who at any given time is working with 3 - 5 oncology families, which otherwise was sporadic. This is ongoing on a private, secure platform utilized by professional health care providers

Conclusions: Engaging parents virtually decreases the need to travel to a venue where additional expenses may be incurred for child-care and are generally more convenient to access. A silver lining of this pandemic has been an increase in individuals using virtual platforms for support groups, counselling and social gatherings.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (2) 21-10-2021 10:15 AM - 11:45 AM

O0016 / #658 | Effectiveness of a Smartphone App to Provide Psychological Support and Empowering Children and Parents during Covid 19 Pandemic

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Background and Aims: Due to COVID-19 pandemic and additional restrictions at the Department of Hemato-oncology of the Mother and Child Health Care Institute of Serbia, children and their parents did not have access to psychological support from the association. In order to provide online psychological support, with a special focus on empowerment of children and parents, the smartphone application "Feelings" was created, which enables the creation of a personalized video with specific and individual content and messages for each patient.

Methods: The pilot study lasted two months, with participation of twenty-five children (aged 2-12) and their mothers. A personalized two minutes video was created for each child and contains favorite photos of the child, the child's favorite tune and with encouraging mother's voice messages. The personalized video was shown to the children as a passive activity while the child performed other routine actions. Our assumption was that parents would notice reduced psychological

negative reactions to medical interventions, as well as the importance of parent's more frequent verbal encouragement of a child.

Results: Evaluation of the survey showed: 87% of mothers estimate that the use of video is a good method for providing support to children; 69% of mothers reported that their child seemed calmer in general; 42% of mothers believe that the psychological reaction of the child decreased prior to medical intervention; 91% of mothers reported that they pay much more attention to the importance of more frequent verbal encouragement of the child.

Conclusions: Parents reported that video had a positive effect on reducing the child's reaction to stress. Also, this type of empowerment positively influenced the parents to pay more attention to the verbal encouragement as the mother's voice has a positive effect on the child's emotional development and increases child's capacity to cope with stress. This pilot project requires to be customized for teens.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (2) 21-10-2021 10:15 AM - 11:45 AM

O0017 / #1068 | Shared Care for Children with Cancer in India Through Social and Healthcare Partnerships during the Covid-19 Pandemic

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Background and Aims: As part of the multi-fold strategies to cope with the unprecedented crisis of COVID-19, a number of hospitals in India were designated as COVID-only-hospitals by government of India. This posed unique challenges for pediatric-oncology units at these hospitals forcing them to discharge admitted patients at short notice. Cankids, a national society for change in childhood cancer in India stepped in to forge partnerships with other centers who were in a position to take over the ongoing treatments in a twinning-model.

Methods: As soon as a center was designated as Covid-19-only-facility, a virtual-meeting was organized with a partner center willing to take over the ongoing care of these patients. Each case was discussed at-length and care transferred to the other center. The transport of the patients was coordinated by Cankids. Where required arrangements were made for families at "Home-away-from-Home" managed by Cankids. The partner centers were requested to extend maximum discounts and support to these patients. The entire financial burden was borne by the organization.

Results: A total of 58 patients from six-hospitals were supported from April-November 2020. No patient abandoned treatment. The median age was 8 years and their hospital of origin are in Bihar, Uttar Pradesh, West Bengal, and Delhi. The expenditure incurred for this exercise was INR 6,183,249 (\$ 84129.04), at an average of INR 41,465 (\$) per patient. 291 trips for the transport were arranged a cumulative stay of 174 days was facilitated at "Home away from Home". The patients were transferred back to the primary center as and when it was feasible for them. One patient was succumbed during the study period due to refractory disease.

Conclusions: The twinning model allowed continuing delivery of optimal care at no additional financial burden to the family. It has also paved the way for future partnership between the involved centers.

CCI LIVE Q&A-ADAPTING TO COVID: BEST PRACTICES AND EFFECTIVE INNOVATIONS (2) 21-10-2021 10:15 AM - 11:45 AM

O0018 / #1426 | The Chilean Experience in the Provision of Holistic Rehabilitation for Children with Cancer

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Background and Aims: In Chile, 500 children are diagnosed with cancer every year. Treatment is granted for all of them through public health policies and NGO collaboration. To address the acute and long-term consequences of treatment and the disease, the Oncological Rehabilitation Center Fundación Nuestros Hijos (CROFNH) provides a multidisciplinary approach to the rehabilitation needs for children and adolescents with cancer. Here, we report the experience of the CROFNH, its multidisciplinary approach, and the successful telerehabilitation strategy implemented in response to the emergency situation of the Covid-19 pandemic during 2020.

Methods: Direct data of the clinical specialties and coverage indicators of the CROFNH are used to offer a view of the scope of services available.

Results: The services offered in the CROFNH include kinesiology, occupational therapy, neuropsychology, speech therapy, physiatry, special education, nutrition, psycho-oncology, and dentistry. From September 2014 to December 2020, 53,481 sessions were given to more than 480 patients on active treatment (79%), follow-up (17%), and palliative care (4%). Specialties of kinesiology, occupational therapy, neuropsychology, and speech therapy count for 75% of the sessions. As lockdown and restricted mobility were in place during the Covid-19 pandemic, a program of synchronous telerehabilitation was implemented on March 2020. The clinical specialists provided individual and/or group online therapy sessions of 45-60 minutes to the patients

and their parents. As of December 2020, a total of 8,549 synchronous telerehabilitation sessions have been granted to 194 children and adolescents including: special education, physiatry, speech therapy, kinesiology, neuropsychology, nutrition, psychology, and occupational therapy.

Conclusions: With its integrated services in the medical treatment of children and adolescents with cancer, the CROFNH helps to reduce the impact of treatment-related side effects in children's daily lives, improves their quality of life, and contributes to maximizing their possibilities of becoming independent and functional adults up to the maximum of their capacities.

IPSO LIVE Q&A: IPSO SESSION 3: SARCOMA 21-10-2021

12:45 PM - 1:25 PM

O0019 / #983 | Pediatric Chest Wall Neoplasms – A 20 Year UK CCLG Single Center Experience Defining Management and Outcomes

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Background and Aims: Chest wall tumours are rare in children and there is varied literature concerning the management and outcomes of these heterogenous lesions. This current study report highlights the accumulating experience of surgical management with particular focus on those requiring chest wall resection and body wall reconstruction.

Methods: All index patients (<18yrs) with chest wall tumours attending a single UK CCLG oncology center during 2001 - 2021 were identified using electronic patient records (EPR) and a pathology database registry. Data were analysed and reported with regard demographics, radiology investigative studies, operative surgery and multidisciplinary oncology management.

Results: Over a 20 year period, 28 pediatric patients with chest wall tumours were identified, aged 4 days - 18 years (13M : 15F). A variety of pathological lesions were confirmed histologically with primitive neuro-ectodermal tumour (PNET) being the most frequent neoplasm (25%). Other chest wall diagnoses included lipoblastomas, lymphangiomas, sarcomas, hamartomas amongst other diagnoses which were rare. Presenting clinical features included a palpable lump (74%) and chest pain (13%). Ten patients (36%) required rib resection(s), four (14%) of these cases subsequently requiring artificial prosthetic patch insertion for chest wall reconstruction. A single patient had a latissimus dorsi muscle flap with 12 (43%) cases also receiving adjuvant multimodal chemotherapy / radiotherapy. Surgical approaches were tailored to each individual patient depending on the size and

anatomical location of the lesion. The most common morbidity(s) documented were wound seromas (11%) in the short term and scoliosis (7%) on longer term follow up.

Conclusions: Whilst the pathology of chest wall tumours are varied, a significant number of these lesions are malignant and require complete surgical resection (R0) where possible. Chest wall reconstruction following tumour excision remains a challenge particularly in those patients where rib resection(s) are required and an individualised approach is thus highly recommended.

IPSO LIVE Q&A: IPSO SESSION 3: SARCOMA 21-10-2021

12:45 PM - 1:25 PM

O0020 / #1764 | Ewing Sarcoma / Primitive Neuroectodermal Tumor of the Thorax in Children and Adolescents. Identification of Prognostic Factors and Treatment Results

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Background and Aims: To describe the clinical characteristics of children and adolescents with ES / PNET in the chest wall, and to evaluate the result of multidisciplinary management, and its relationship with the t (11,22) (q24q12) translocation.

Methods: Patients with ES / PNET treated at the Federico Gómez Children's Hospital (HIMFG in Mexico City, Mexico) and Children's Hospital Dr Renan Esquivel (in Panama City, Panama) from 01/01/2001 to 04/30/2017. Review of medical records, molecular biology studies (RT-PCR determination of chromosomal translocation EWS-FLI and EWS -ERG) on tissue samples previously obtained for pathological diagnosis.

Results: Thirteen patients were analyzed. There were 8 girls and 5 boys; the mean age was 13 ± 3.19 years. Location of the tumor: rib cage 5, scapula 5, and thoracic vertebrae 3. Pulmonary metastases were present at diagnosis in 8 cases. The tumor size ranged from 3 cm to 20 cm. The operations included 4 scapulectomies, 5 thoracotomies with chest wall resections, 1 clamshell thoracotomy, 2 paravertebral resections, and 1 laminectomy were performed. EWS-FLI 1 was identified in 10 patients, and EWS -ERG in 1. Autologous stem hematopoietic cell rescue was used in 2 cases. The overall survival rate was 60% after 2 years of diagnosis.

Conclusions: The absence of metastasis at diagnosis gave 80% survival at 2 years of diagnosis. The presence of EWS-FLI or EWS -ERG was possible to determine by RT-PCR from two different centers. Further multicenter research is necessary for the identification of prognostic factors and to treat accordingly.

IPSO LIVE Q&A: IPSO SESSION 3: SARCOMA 21-10-2021

12:45 PM - 1:25 PM

O0021 / #1065 | Utility of Circulating Tumour DNA in Ewing Sarcoma

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Background and Aims: New risk-stratification markers are needed to identify patients with Ewing sarcoma (ES) unlikely to benefit from standard therapy as well as to monitor treatment response to new agents. Objective: To investigate the utility of circulating tumour DNA (ctDNA) in patients with ES for molecular risk-stratification and post-treatment monitoring.

Methods: We developed and assessed custom digital next generation sequencing (NGS) assay that captures intronic regions of the gene EWSR1 as well as the coding regions of the genes STAG2 and TP53, to detect mutations in the circulation of patients diagnosed with ES.

Results: ctDNA mutations were detected in samples taken from patients at: initial diagnosis in 6/8 (75%, 4/5 localised and 2/3 metastatic), post-neoadjuvant chemotherapy in 2/4 (50%), post primary resection in 0/2 (0%), and recurrent disease in 0/3 (0%), suggesting a positive correlation between ctDNA and tumour burden. EWSR1 fusions were detected in 6, STAG2 loss-of-function mutations detected in 4, and TP53 mutations detected in 2 patients, with overall variant allele frequencies (VAF) of 5-57%. Serial blood samples collected from diagnosis throughout the course of treatment of three patients showed a good correlation between ctDNA VAF dynamics (disease progression/regression) and diagnostic imaging studies. In addition, samples taken prior to targeted treatment (PARP inhibitor Olaparib) and 4 cycles post-treatment for one patient having STAG2 somatic mutation, showed no change to ctDNA VAF, in accordance with lack of treatment response.

Conclusions: We show that ctDNA detectable with our NGS assay is a good surrogate marker of tumour burden and shows correlation with treatment response.

IPSO LIVE Q&A: IPSO SESSION 3: SARCOMA 21-10-2021

12:45 PM - 1:25 PM

O0022 / #1313 | Non Rhabdomyosarcomas in a Lower and Middle Income Country: A Report from South Africa

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Background and Aims: Non-rhabdomyosarcoma soft tissue sarcomas (NRSTS) are a diverse type of sarcoma that account for less than 5% of paediatric cancers. Given the rarity of data on NRSTS in paediatric patients in Africa and South Africa, the aim of the study was to describe the management and outcomes of children with NRSTS at Chris Hani Baragwanath Academic Hospital (CHBAH). In doing so, this study may provide insight into potential areas for improvement.

Methods: This is a retrospective qualitative study on NRSTS in patients below 18 years old managed by paediatric oncologists and surgeons at CHBAH, from 1st January 2009 to 31st December 2018.

Results: A total of 45 patients with NRSTS were identified, all black African, 51% male, 49% female, and 62% were below the age of 10 years. Sixteen histological subtypes were identified, the most common subtype were fibroblastic and myofibroblastic tumours, with malignant peripheral nerve sheath tumours and synovial sarcomas being the next most frequent. Thirty six percent received neo-adjuvant chemotherapy and or radiotherapy, and 49% adjuvant therapy. Surgery consisted of microscopic negative margin resection (R0) in 33% who had an 80% survival, microscopic positive margin resection (R1) in 27% with a 42% survival, gross residual or non-resection (R2) in 20% with a 44% survival, and an amputation in 20% with a 78% survival. Metastases were present in 31% with a survival of only 21%, as opposed to a 74% survival in those without. There is an overall survival of 64%, and a 36% mortality mainly from metastases followed by disease progression. Inferior survivals were reported with the higher grade and stage of the tumours.

Conclusions: Overall survival is higher in lower grade and stage disease that have undergone complete resection, and this study is important to low and middle income countries (LMIC) to show that good survival rates are achievable.

IPSO LIVE Q&A: IPSO SESSION 3: SARCOMA 21-10-2021

12:45 PM - 1:25 PM

O0023 / #1323 | The Role of HIPEC (Hyperthermic Intraperitoneal Chemotherapy) in Pediatric Oncology - Preliminary Report from a Single Center

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ogy And Intensive Therapy, Gdansk, Poland, ⁸Medical University of Gdansk, Department Of Pathology And Neuropathology, Gdańsk, Poland

Background and Aims: INTRODUCTION: Peritoneal metastases in children are mainly seen in sarcomas (like desmoplastic small round cell tumor – DSRCT), rhabdomyosarcomas, Gastrointestinal Stromal Tumors or primary disseminated ovarian tumors. The main purpose of HIPEC is to remove the visible macroscopic disease with complete cytoreduction (CRS). Several papers presented in the literature deal with the CRS and HIPEC in children and adolescents, however the experience in pediatrics is still limited. Thus far, the HIPEC procedure has been carried out on patients over 2 years of age. The most common indication for the surgery and the best outcome was experienced by patients with desmoplastic small round cell tumor (DSRCT). Most patients received intraperitoneal cisplatin.

Methods: OUR EXPERIENCE: Between 2018 and 2020 eight patients were treated with cytoreduction and HIPEC at the Department of Surgery and Urology for Children and Adolescents of the Medical University of Gdansk (Poland). Those were the first pediatric HIPEC procedures performed in Poland.

Results: One of the patients was 6-months-old during the procedure and was the youngest patient and the only infant in the world reported in the literature. In 7/8 cases a complete resection (R0) was achieved. Four of the patients remains in a complete remission (follow-up 7months - 27 months) and three of them died 5-9 months after the procedure due to the disease progression. One patient died shortly after the procedure due to severe toxicity associated with heavy chemotherapy pretreatment and kidney damage before HIPEC in the setting of the second relapse of DSRCT.

Conclusions: CONCLUSIONS: CRS and HIPEC is technically possible also in infants. For its safe course patients selection and technique modification are necessary. In the world literature the best HIPEC outcome was experienced in the treatment of DSRCT and a complete cytoreductive surgery preceding HIPEC directly seems to be the key factor in survival.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID
21-10-2021 2:00 PM - 2:45 PM

O0024 / #1071 | Treatment of Sacrococcygeal Teratoma in Children by Single Incision of Sacrococcygeal Region and Evaluation of Anorectal Function after Operation

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Background and Aims: Tumors of Altman type I and part of type II should be removed by sacrococcygeal incision according to traditional views, while abdominal sacrococcygeal incision should be considered for some cases of type II, III and IV with high presacral extension. Since 2015, sacrococcygeal single incision has been taken as the preferred route for all Altman types of sacrococcygeal teratomas in our hospi-

tal. The purpose of this study is to evaluate the feasibility and safety of this route in the treatment of all Altman types of sacrococcygeal teratomas.

Methods: From January 2015 to December 2018, children with sacrococcygeal teratoma treated by surgery in our hospital were analyzed, and the anorectal function of children in this cohort was followed up and evaluated. The recorded data included: age at admission, surgical approach, Altman classification and histopathological classification. The anorectal function was evaluated according to Li Zheng's score: defecation, purgation and incontinence.

Results: A total of 56 children were included in the study. Altman type was classified, and type I in 19 cases, type II in 26 cases, type III in 10 cases, and type IV in 1 case. All the 55 patients were operated by sacrococcygeal approach, and the tumor and tailbone were completely removed. Only one patient received abdominal sacrococcygeal operation because of the serious Inflammatory adhesion between the tumor and the pelvic. According to Li Zheng's anorectal function score method, 50 cases were good, 6 cases were moderate, 0 case was poor. The results showed that there was no significant difference in anorectal function scores among children of different ages, genders and pathological types, and different Altman types.

Conclusions: Sacrococcygeal single incision is suitable for most children with sacrococcygeal teratoma, the anorectal function is satisfactory after operation, thus it can be considered as the first choice for children with sacrococcygeal teratoma.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID
21-10-2021 2:00 PM - 2:45 PM

O0025 / #1083 | The Effect of Surgical Resection and First-Line Chemotherapy on Treatment Outcome of Pediatric Malignant Mediastinal Germ Cell Tumors: Taiwan Pediatric Oncology Group Report

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Background and Aims: Primary mediastinal germ cell tumors (GCTs) are rare pediatric neoplasm and their outcome are worse than GCTs of other sites. We reviewed the outcomes of pediatric patients registered in Taiwan Pediatric Oncology Group (TPOG) and investigated the prognostic factors, focusing on surgical resection and first-line chemotherapy regimens.

Methods: Retrospective collected data of children with malignant extracranial GCT registered to TPOG from January 1, 2005 to December 31, 2019. were investigated to identify the relationship between disease outcome and clinical characters and treatment modality.

Results: There were 54 pediatric patients with median follow-up of 26.3 months. The 5-year overall survival (OS) and event free survival (EFS) were 56% and 42%. Strong unfavorable clinical characters include higher stage (5y OS 100%, 100%, 73%, 20% for stage I, II, III, IV, $p < 0.001$), and histology of choriocarcinoma and yolk sac tumor (5y OS 100%, 35%, 0%, 100%, 71% for immature teratoma, yolk sac tumor, ghoriocarcinoma, germinoma/seminoma, mixed GCT, $p < 0.001$). Radiotherapy had no influence ($p = 0.95$) to outcome. Surgical resection yielded better outcome for non-metastatic disease (5y EFS 0% vs 67%, $p < 0.001$, OS 93% vs 0%, $p < 0.001$) but only improved EFS for metastatic disease (5y EFS 38% vs 9%, $p = 0.034$; OS 38% vs 9%, $p = 0.093$). Upfront or delayed resection showed no difference in non-metastatic group (5y OS 100% vs 87%, $p = 0.121$) while upfront resection yielded inferior outcome in metastatic group (5y OS 0% vs 50%, $p = 0.032$). Comparing firstline cisplatin-containing regimen and non-cisplatin containing regimen (most carboplatin-containing) showed no difference either in metastatic group (5y EFS 12% vs 75%, $p = 0.08$) or non-metastatic group (5y EFS 47% vs 67%, $p = 0.37$).

Conclusions: Outcome of choriocarcinoma is dismal and requiring better treatment. Surgical resection, either upfront or after neoadjuvant chemotherapy, was recommended for resectable diseases. Cisplatin-containing regimen are not superior to carboplatin-containing regimen as 1st line therapy and may be avoided for toxicity concern.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID 21-10-2021 2:00 PM - 2:45 PM

O0026 / #575 | Outcomes of Benign Extracranial Germ Cell Tumors – Experiences from A Tertiary Referral Centre in India

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Background and Aims: To report the epidemiological aspects and outcomes of children with extracranial germ cell tumours (GCT) presenting to our centre.

Methods: A retrospective study was conducted and all patients with extracranial GCTs managed from January 1998 through December 2018. Patients with increased alpha-fetoprotein (AFP) or with malignant foci on histopathology were excluded. Prospectively maintained data sets of the included patients were evaluated for patient demographics, presenting symptoms, their duration, radiologic and histologic findings, surgical procedures, postoperative complications, and recurrence.

Results: The study group included 100 patients with 41 males and 59 females. The median age at presentation was 9.5 months (10 days – 16 years). The largest subgroup in our cohort was of sacrococcygeal teratoma (SCT) ($n = 47$) followed by Gonadal ($n = 16$), Mediastinal ($n = 13$) and Retroperitoneal GCT ($n = 10$). The most common presentation was of a palpable mass reported in 76/100 of the patients. Conclusive preoperative histopathology was available in only 7/100 of the complete cohort. All the patients underwent operative intervention with 97/100 of them undergoing complete excision. Immature teratoma was present in 14 of the patients with the highest incidence in gastric teratoma ($n = 4/9$, $p = 0.0004$), the most common grade being Grade 3 immature teratoma. The median follow-up was 27 months and the 5-year-event-free survival of 90.6+/-9.8%. The most significant complication was bladder-bowel dysfunction documented in 6/47 (12.7%) patients of SCT. There was a total of 4 (4%) recurrences in the cohort noted only in SCT ($n = 3/47$, 6%) and gonadal tumours ($n = 1/16$, 6%). All the recurrences were benign. There was no significant difference in recurrence in immature and mature teratomas.

Conclusions: The current study is a large series of extracranial teratomas which highlights the need for complete excision. The study emphasizes regular follow up especially in children with SCT for the development of bladder-bowel dysfunction or recurrence.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID
 21-10-2021 2:00 PM - 2:45 PM

O0027 / #555 | Intraoperative Fine Needle Aspiration in Pediatric Thyroid Disease: A Novel Technique

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Background and Aims: Fine needle aspiration (FNA) for thyroid nodules is not well-tolerated in children without sedation or anesthesia. Postoperative pathologic analysis may require a return to the operating room for completion thyroidectomy or nodal clearance. We present a novel technique in which FNA is completed in the operating room and hypothesize that it provides a sensitive, efficient, and safe approach to guide surgical resection that could avoid preoperative FNA or the need for reoperation.

Methods: Intraoperative FNA was completed by the surgeon at the time of thyroid lobectomy. The specimen was passed to the pathologist who completed microscopic evaluation with a mobile pathology cart in the operating room, providing results within minutes as the surgeon continued the dissection. This novel technique is reviewed retrospectively to determine the primary outcome of thyroid reoperation associated with the use of intraoperative FNA. All patients who underwent thyroid resection from 2016-2019 at our tertiary children's hospital were considered. Secondary outcomes were the concordance rates between preoperative cytology, intraoperative cytology, and final surgical pathology and the proportion of patients whose preoperative FNA could have been replaced with an intraoperative FNA.

Results: Over three years, 47 patients underwent thyroid surgery. Seven patients underwent intraoperative FNA. Nineteen patients underwent preoperative FNA. Intraoperative cytology was concordant with final surgical pathology in six cases (85.7%). One patient (14.3%) required reoperation for intraoperative FNA under-diagnosis. Five patients underwent preoperative and intraoperative FNA biopsy of the same target; cytology was concordant in four of these cases (80.0%). Based on clinical presentation and preoperative imaging, 11 preoperative FNAs (57.9%) could have been avoided in favor of intraoperative FNA only.

Conclusions: Intraoperative FNA is a previously unreported strategy in the surgical management of pediatric thyroid disease. This sensitive, safe, and novel technique could save children from traumatic awake FNAs and prevent repeat exposure to anesthesia and subsequent operations.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID
 21-10-2021 2:00 PM - 2:45 PM

O0028 / #997 | Long Term Complications Associated with Thyroidectomy in Children with Thyroid Cancer: Experience in an Oncologic Referral Center in a Developing Country

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Background and Aims: Thyroid cancer accounts for less than 1.5% of all cancers in childhood. Patients have advanced locoregional disease at the time of diagnosis. Surgery is crucial in terms of overall and disease-free survival. The morbidity related to thyroid cancer surgery in pediatrics is underestimated and poorly described in the literature.

Methods: This retrospective study evaluates long-term postoperative complications in children who underwent thyroid cancer surgery at a Colombian referral cancer center. We focused the analysis in two main complications: permanent hypoparathyroidism and permanent damage to RLN (recurrent laryngeal nerve).

Results: Fifty-seven patients had confirmed DTC and were operated at our institution (thyroidectomy and/or MRND). 45% presented permanent hypoparathyroidism and 14% RLN permanent injury. Multifocality (OR 4.2, 95% CI, 1.19-14.83), neck dissection (OR 3, 95%, 0.24-37.67), and reoperation (OR 2.3, 95% CI, 0.69-7.61) relate to a higher risk of permanent hypoparathyroidism. Bilateral tumor compromise (OR 5.6, 95% CI, 0.95-33.2), extension to adjacent soft tissue (OR 3.5, 95% CI, 0.4-30.95), lymphovascular invasion (OR 5.3, 95% CI, 0.96-29.19), and metastasis (OR 15.9, 95% CI, 1.79-140.59) had a higher risk of RLN injury.

Conclusions: Thyroid cancer surgery in pediatric patients is a surgical challenge. Several factors are associated with permanent hypoparathyroidism and permanent injury to the RLN, which are more prevalent with an oncologic diagnosis. In our cohort, we had a high presence of these complications, especially related to extensive local disease and reintervention. We found multifocality and metastases as significant risk factors for permanent hypoparathyroidism and permanent injury of RLN. More than a third of the operated patients were referred after initial surgery in another institution. We believe that centralizing patients care with DTC is essential in developing countries; this would reduce the number of reoperations and improve TT and cervical lymph node dissection morbidity.

IPSO LIVE Q&A: IPSO SESSION 4: GERM CELL & THYROID
 21-10-2021 2:00 PM - 2:45 PM

O0029 / #1787 | Well-Differentiated Thyroid Carcinoma in Children: Experience at A Single LMIC Institution

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Background and Aims: Differentiated thyroid carcinoma (DTC) is rare in children, with distinctive clinical, pathologic and molecular characteristics. Disease tends to be biologically aggressive, with advanced stages of disease at presentation. In spite of this, response to treatment and long-term survival are excellent.

Methods: With IRB approval, we performed a retrospective chart-review of 57 pediatric patients, treated at our Institution for DTC, between 2000-2021.

Results: The final cohort (N=36) included 27 females and 9 males (mean age 13.4+/-2 years). Diagnosis was confirmed by FNA 58.3% (21/36), cervical lymph-node biopsy 30.5%, during hemithyroidectomy 11.2%. Thyroglobulin was elevated in 25 and anti-TG antibodies in 11 patients. History of neck radiation (6/36) for ALL (3) and Hodgkin disease (3). Five had Hashimoto's thyroiditis. Cervical lymph-node involvement was present in 98% of patients.

All underwent total thyroidectomy, 31 had modified bilateral neck-dissection, and five received (level VI). Average operating time was 250+/-80 minutes, and mean blood-loss was 60+/-50 ml. Length-of-stay was 3 (2-4) days. No nerve injuries in this cohort. Multi-focality was 51%. Histologic variant was papillary in 28 and follicular in eight patients. Temporary hypocalcemia developed in 50%, and proved permanent in 7%. Local recurrence rate was 30% (9/36), all underwent surgical re-exploration. Three proved non-malignant. All received radioiodine. All are alive without radiological evidence of disease, after a median follow-up of 13.5 years (0.3-20 years).

Conclusions: Survival of pediatric DTC is excellent. Thus, reducing treatment-related morbidity is a priority. We found a significant rate of postoperative complications. Large tumors, reoperations and previous radiation exposure or chronic inflammatory conditions might explain the high incidence of perioperative morbidity. These may be reduced by centralization of care, and use of surgical navigation devices. Indocyanine-green (ICG) is a promising tool to decrease the incidence of long-term hypocalcemia. Life-long surveillance is mandatory.

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0030 / #567 | Indocyanine Green-Guided Pediatric Tumor Resection: Approach, Utility, and Challenges

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Background and Aims: Incomplete tumor resection increases the risk of local recurrence. However, the standard of care approach to distinguishing tumor tissue is less than optimal, as it depends on a conglomeration of preoperative imaging and visual and tactile indicators in real time. This approach is associated with a significant risk of inadequate resection; therefore, a novel approach that delineates the accurate intraoperative definition of pediatric tumors is urgently needed. ICG is a fluorescent dye that when administered intravenously accumulates passively in the tumor because of enhanced permeability and retention (EPR) effect, thereby providing tumor contrast for intraoperative real-time tumor recognition. This retrospective study examines the feasibility of ICG-guided tumor resection in common childhood solid tumors such as neuroblastoma, sarcomas, hepatic tumors, pulmonary metastases, and other rare tumors. Pediatric dosing and challenges related to the optimization of tumor-to-background ratio are also examined.

Methods: This study was approved by the St. Jude institutional review board and waiver of informed consent was approved. We retrospectively reviewed the charts of all patients who underwent fluorescence-guided tumor resection at our institution from 2019 through 2020. Data on preoperative diagnosis and imaging findings, ICG dose, timing of ICG administration, operative notes, Iridium system video recording, intraoperative fluorescence, background noise, histopathology report, and complications were collected.

Results: Fifty-five patients (28 males and 27 females; median age 10 years [range <1-21 years]) underwent fluorescence-guided tumor resection. Of them, eight underwent two procedures and one patient underwent three procedures. The total number of procedures done was 65, including 37 thoracic, 19 abdominal (other than nephron-sparing resections) and 9 trunk and extremity operations. The sensitivity of NIR to identify tumor was 88% and specificity was 77%.

Conclusions: ICG-guided tumor localization is a feasible adjunct for most pediatric solid tumors and is highly sensitive for tumor tissue but its specificity is low

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0031 / #576 | Increasing the Efficiency of Hyperthermic Intraperitoneal Chemotherapy (HIPEC) by a Combination with a Photosensitive Drug in Paediatric Rhabdomyosarcoma

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Background and Aims: Cytoreductive surgery (CRS) in combination with hyperthermic intraperitoneal chemotherapy (HIPEC) is an option in advanced peritoneal sarcomatosis. Nevertheless, CRS and HIPEC are not successful in all patients. Therefore, an enhancement of HIPEC using photodynamic therapy might be beneficial. For this approach, we tested a combination of the photosensitizer Hypericin with HIPEC in an animal model.

Methods: Alveolar rhabdomyosarcoma cells were xenotransplanted into NOD/LtSzscid IL2R γ null mice (n=80) inducing intraperitoneal sarcomatosis. All groups received Hypericin (100 μ g/200 μ l) intraperitoneally with and without cisplatin-based (30 or 60mg/m²) HIPEC (37 or 42°C, for 60 min) (five groups, each n=16). Hypericin-based photodynamic therapy (PDT) of a representative tumor bulk was performed in all groups for ten minutes. Tumor dissemination was documented visually and by using Hypericin-based fluorescence guidance. Tumor and tissue samples, harvested at the end of the perfusion, were evaluated regarding morphology (H&E staining), proliferation (Ki-67 staining) and apoptosis (TUNEL-assay).

Results: A time dependent Hypericin uptake even in smallest tumor nodes (< 1 mm) was found. Hypericin-based fluorescence guidance detected a higher tumor dissemination compared to the visible tumor spread. Immunohistochemistry revealed a Hypericin penetration across the tumor surface. Hypericin-based PDT without HIPEC induced marginal apoptotic effects only at the tumor surface. Combining Hypericin with HIPEC revealed cisplatin-concentration dependent decrease in proliferation capacity and induction of apoptosis across determined cell layers of the tumor surfaces.

Conclusions: Hypericin as fluorescing photosensitizer offers an intraoperative diagnostic advantage detecting the exact intraperitoneal tumor dissemination. The combination of Hypericin and cisplatin-based HIPEC was feasible in vivo. Effects on tumor proliferation and apoptosis induction across the tumor surface were observed. Further studies combining Hypericin and HIPEC as new therapy approach will follow to establish a clinical application.

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0032 / #775 | Artery of Adamkiewicz: Preoperative Visualization and Impact on Surgery in Posteromedial Paediatric Tumors

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Background and Aims: Surgery for posteromedial paediatric tumors carries the risk of injury to the Artery of Adamkiewicz (AKA). Cases of severe neurological complications in children following tumor resection in this area have been reported. Spinal diagnostic subtraction angiography (DSA) is the method of choice for visualization of AKA. Data on the preoperative use of spinal DSA in children for this purpose are rare.

Methods: Between 2002 und 2021 36 children with posterior mediastinal tumors were evaluated for surgery at our department. DSA was used to assess AKA during the preoperative workup in 10/36 children with left sided or bilateral tumor localized at the vertebral level T8 to L1. We analyzed patient's data, tumor characteristics as well as diagnostic and therapeutic procedures.

Results: Tumor entities were neuroblastoma (n = 3), ganglioneuroblastoma (n = 2), ganglioneuroma (n = 2), local relapse of a hepatocellular carcinoma (n = 1), and neurofibroma (n = 2). Median age of the children at examination was 79 months (16 – 217), 3 of them were younger than two years. The AKA could be visualized in all children. In 5 children, the tumor and AKA were on the same side and at the same level; consequently, 2 of these children underwent irradiation instead of tumor resection. In 1 child with thoracoabdominal neuroblastoma resection of the abdominal and irradiation of the thoracic tumor part was performed. Two children with neurofibroma underwent surgery in the form of intracapsular resection. Spinal DSA and subsequent operations were carried out without complications.

Conclusions: Spinal DSA is a safe and reliable method for preoperative identification of AKA for surgery in posteromedial paediatric tumors. With its possible impact on the decision regarding local treatment it can help to avoid critical situations during or after surgery and to determine the most appropriate local therapy.

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0033 / #918 | Sacrococcygeal Teratoma – What are the Long Term Outcomes? A UK CCLG Surgeons Group Nationwide Study

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Background and Aims: Sacrococcygeal teratoma (SCT) is a rare paediatric germ cell tumour (1:40 000). Long-term data regarding urinary tract and bowel function after SCT resection is limited. This surgeon-led UK multicentre study aimed to critically analyse long-term functional sequelae in patients following resection of SCT.

Methods: A multicentre study registry amongst UK paediatric surgical oncology centres was established. All index cases of newborns/ infants and children <16 years with SCT diagnosis during 2005-2015 were included.

Results: 110 patients treated at 10 surgical oncology centres were included. Mean age at surgery was 1 day [IQR 0-7 days]. Seventy (64%) patients were female, 40 male. Diagnosis was established antenatally in 52% (57) cases. 61% of patients were Altman Stage I or II, 39% were Altman Stage III or IV. Follow-up data was available on 105 out of 110 (95%) patients. Median length of follow-up was 60 months [IQR 43-87 months]. Eight (7%) patients had recurrent disease at a median time of 9.5 months [IQR 7-50.5 months] following operation. Thirty-seven out of 110 patients (34%) developed bladder and/or bowel dysfunction. Fifteen patients required urinary clean intermittent catheterisation (CIC). Bowel dysfunction was managed purely medically in 19 patients; 7 patients required surgery for management of bowel dysfunction (ACE [n=2], colostomy [n=4], anorectal reconstruction [n=1]).

Conclusions: In this UK nationwide study, 34% of patients developed bladder or bowel dysfunction after primary SCT resection. Full assessment of bladder and bowel function is mandatory during follow-up of all patients having SCT resection(s). A multi-disciplinary surgical care pathway is crucial to facilitate best practice, monitor long term health and improve patient quality of life (QoL).

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0034 / #891 | Tumor Characteristics, Treatment Approach and Outcomes in Children with WAGR Syndrome and Wilms Tumor – An Analysis of The International WAGR Syndrome Association Registry (IWSA)

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Background and Aims: WAGR syndrome (Wilms Tumor Aniridia Genitourinary Malformations and range of development delays) is a rare disease. Children with WAGR have a predisposition to Wilms Tumor (WT). We present an epidemiological analysis of tumor characteristics, treatment, and associated congenital risk factors in WAGR patients with WT.

Methods: The IWSA conducted a survey between 2015-2021. Demographic, surgical, chemotherapeutic, radiotherapy and congenital anomalies that predisposed to WT were analyzed. Categorical comparisons were made with chi-squared test and continuous comparisons were made using t-test and ANOVA.

Results: Parents of 145 children with WAGR completed the questionnaire, of whom 64(44.1%) had WT and 12(8.2%) had nephrogenic rests. Median length of follow-up from the diagnosis of WT was 6.7 years (IQR 2.5-13.4 years), median age at diagnosis was 26 months (IQR 18-26 months). One child was diagnosed after 5 years of age. Fifty-four(84.3%) had unilateral stage 1-2 disease and 59(92.1%) had favorable histology WT(FHWT). One child developed bilateral WT (BWT); only 2 had stage IV disease. Treatment included preoperative chemotherapy for 47 children and primary nephrectomy for 15. At least one kidney was removed in 38. Nephron sparing surgery (NSS) was associated with preoperative chemotherapy 19/24(79.1%) compared to upfront surgery 5/24(20.8%). The duration of chemotherapy was between 1-12 months in 50 children and was more prolonged in 8 (duration was unknown for 6 children). Chemotherapy included a two-drug regimen for 21 and a three-drug or more intensive therapy in 34. Relapse occurred in 3(4.6%) children. There was one death. No congenital genitourinary anomalies in children with WAGR (e.g. hypospadias, cryptorchidism) were associated with the development of WT.

Conclusions: In this analysis of children with WAGR syndrome and WT their disease was primarily unilateral and FHWT with infrequent relapse. Regrettably, NSS was performed in only 1/3 of children.

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0035 / #932 | The First National Delphi Consensus Guideline on The Surgical Management of Benign Ovarian Tumours in Children and Adolescents

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Background and Aims: No widely agreed protocols exist for the management of benign ovarian tumors (BOT) in children. Patients present in a multitude of settings, and to various specialties including paediatric oncology surgeons, paediatric general surgeons and gynaecologists. Recent studies demonstrate significant risk of metachronous disease. The lack of unified guidance and heterogeneity of approaches present a substantial risk for suboptimal management. This study aimed to generate a multi-specialty consensus statement to clarify current perioperative controversies and establish clear follow-up guidance for children with BOTs.

Methods: A prospective consensus study with a two-round confidential e-Delphi consensus survey was distributed amongst a multi-specialty expert panel; concluded by two semi-structured videoconferences.

Results: Consensus guidance was developed for 4 domains: Pre-operative, intra-operative management, follow-up and referral to adolescent gynaecology. Summary of salient points: 1) Children with BOTs should receive the same management as other children with potentially neoplastic lesions. This includes pre-operative discussion at the oncology multi-disciplinary meeting to risk stratify tumours, and management by specialists with expertise in ovarian-sparing surgery and laparoscopy. 2) Ovarian-sparing surgery for BOTs should be performed wherever possible to maximise fertility preservation. 3) Ovarian masses detected during emergency diagnostic laparoscopy/laparotomy should be left in situ and investigated appropriately (imaging/tumour markers) prior to planned resection. 4) Follow-up: The first appointment should be scheduled with ultrasound within 3 months following resection, then within 2 years after resection, then every 2 years until 16 years. Following resection of an ovarian tumour all patients should be referred to adolescent gynaecology to discuss implications on future fertility/ fertility preservation. The referral should take place once the patient is post-pubertal (16 years).

Conclusions: This first multi-specialty consensus guideline emphasizes the importance of treating children with BOTs according to oncological principles, highlights the need for maximal fertility preservation by ovarian-sparing surgery for BOTs, and outlines a clear follow-up protocol which facilitates detection of metachronous disease early.

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0036 / #1216 | Thyroid Cancer in Children: A Multicenter International Study Highlighting Clinical Features and Surgical Outcomes of Primary and Secondary Tumors

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Background and Aims: Thyroid gland malignancy is rare in pediatrics (0.7% of tumors); only 1.8% are observed in patients < 20 yrs with a higher prevalence recorded in females and adolescents. Risk factors include genetic syndromes - MEN disorders, autoimmune disease and ionizing radiation exposure. Radiotherapy is also linked with increased risk of secondary thyroid cancers. The present study describes the clinical features and surgical outcomes of primary and secondary thyroid tumors.

Methods: Institutional data was collected on pediatric patients with thyroid cancer during 2000 - 2020 from 8 International Surgical Oncology centers. Statistical analysis was performed using GraphPad Prism.

Results: Of 298 cases of thyroid cancer, only 19 (6.4%) were secondary tumors. Primary thyroid malignancies were more likely to be multifocal in origin (odds ratio [OR] 1.993, 95% confidence interval [CI] 0.7466-5.132, p 0.2323), had bilateral glandular location (OR 2.847, 95% CI 0.6835-12.68, p 0.2648) and proved metastatic at 1st diagnosis (OR 1.259, 95% CI 0.3267-5.696 p>0.999). Secondary tumors showed a higher incidence of disease relapse (OR 1.556, 95% CI 0.4579-5.57, p 0.4525) and surgical morbidity (OR 2.042, 95% CI 0.7917-5.221, p 0.1614) including hypoparathyroidism and recurrent laryngeal nerve injury. Overall survival (OS) was 99% at 1 year and 97% after 10 years. No EFS differences were evident with primary vs. secondary tumors (Chi square 0.7307, p 0.39026).

Conclusions: This multicenter study demonstrates excellent survival for pediatric thyroid malignancy. Secondary tumors exhibit greater disease relapse (15.8% vs 10.5%) and a higher incidence of surgical related complications (36.8% vs 22.2%).

IPSO LIVE Q&A: IPSO SESSION 5: THE ROBERT ARCECI BEST OF IPSO 22-10-2021 8:00 AM - 9:30 AM

O0037 / #1100 | Prognostic Significance and Tumor Biology of Regional Lymph Node Disease in Patients with Embryonal Rhabdomyosarcoma: Report from The European Soft Tissue Sarcoma Study Group (EPSSG)

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Background and Aims: Introduction: Regional lymph node disease (RLND) is a component of the risk-based treatment stratification in rhabdomyosarcoma (RMS). The purpose of this study was to determine the contribution of RLND to prognosis for patients with nonmetastatic embryonal RMS (ERMS).

Methods: Between 2005 and 2016, of the 1760 children enrolled onto EpSSG RMS 2005 protocol we analyzed 143 patients (pts) diagnosed with ERMS and RLND. Local therapy with radiotherapy (primary tumor and involved nodes) began at week 12.

Results: Results: most pts had tumors with unfavorable characteristics: tumor size >5 cm 70%, T2 invasiveness 70%, IRS III 84%, and parameningeal (PM) (40%). The 5-year event-free survival (EFS) was 65.2%, Overall Survival (OS) 70%. N1 ERMS pts had a significantly better outcome than those with N1 alveolar RMS (EFS 65% vs 49%; OS 70% vs 51.2%; p=0.03). Outcomes were significantly better for pts with favorable vs unfavorable tumor site (EFS 73% vs 50%, OS 77% vs 53% p=0.002) and those with tumor size < 5 cm vs >5 cm (EFS 72.5% vs 49.4%, OS 76% vs 53.9%, p=0.0009). IRS II pts have also a significantly better outcome vs IRS III pts (EFS 84.5% vs 52.2%, OS 89% vs 56.5%, p=0.0006). EFS and OS was not statistically different comparing 94 pts receiving and 32 not receiving nodal irradiation (65.5% vs 75%, 69.6% vs 77.7% p=0.3) but in the first group unfavorable features (PM site, size, age, IRS III and invasiveness) were more frequent. On univariate analysis, larger tumor, unfavorable site, invasiveness, IRS group III, correlated with worse prognosis. On multivariate analysis,

IRS Group remained as the only significant prognostic variable but it does not seem to be a consistent estimator.

Conclusions: Conclusion: Regional lymph node disease does not alter prognosis for ERMS. The role of radiotherapy to nodes has to be reconsidered in a larger prospective study.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021 10:00 AM - 11:00 AM

O0038 / #322 | Extended Lymph Node Sampling Does not Lead to Increased Complications in Children with Renal Tumors

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Background and Aims: Children's Oncology Group protocols mandate lymph node sampling (LNS) during extirpative surgery for pediatric renal tumors. While data suggests a minimum of 6-10 LNs should be sampled, LNS is often underperformed. Concerns over morbidity associated with extended LNS have led to hesitancy in adopting a formal LN dissection template. We hypothesized that complications in children undergoing LNS for renal tumors would be rare, and not associated with the number of LNs sampled.

Methods: A single-institution, retrospective review of patients aged 0 – 18yrs undergoing extirpative renal surgery with LNS for a suspected malignancy between 2005-2019 was performed. Patients with an unknown number of LNs sampled were excluded. Perioperative complications were categorized based on the Clavien-Dindo (CD) scale. A “clinically significant” (CS) complication was defined as any CD complication ≥ III, small bowel obstruction (SBO), ileus, chylous ascites, organ injury, or wound infection/abscess. Patients were separated into 2 cohorts for analysis using a cut-point of ≤ v >6 LNs sampled. Univariable and multivariable analysis (MVA) was performed.

Results: 150 patients met inclusion criteria. Median patient age was 37.8 months (IQR 20.3-62.4). 24 patients (16%) had a CS complication, the most common of which was ileus/SBO (n=18). No CD grade IV or V complications were noted. There was no difference in complications, OR time, or length of stay between the two cohorts. On MVA, sampling >6 LNs was not predictive of CS complications (p=0.958). Left-sided tumors were protective against complications (p=0.019).

Conclusions: More extensive LNS during surgery for pediatric renal tumors does not appear to lead to an increase in complications. A standardized LNS template for pediatric renal tumors may lead to increased protocol adherence and more accurate staging without placing the patient at increased risk of morbidity.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021
10:00 AM - 11:00 AM

O0039 / #814 | Defining The Role of T-Regulatory Cells (T-Regs) and Cytokines in The Tumour Microenvironment in Wilms Tumour (WT)

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Background and Aims: Interaction between developing tumor and the immunologic microenvironment is active. A tiny subset of CD4+ T cells that co-express IL2 are called T-regs most actively interact with the tumor. **Aims:** To evaluate T-regs in patients with WT before and after tumour resection and evaluate the cytokine profile(IL-2, IL-10 and TGF- β).

Methods: Blood mononuclear cells were analysed for total T-helper(Th) and T-regs(CD4+CD25+Foxp3+) at presentation and three months following resection. Serum IL-2, IL-10 and TGF- β levels were estimated. Immunohistochemistry for Foxp3+ CD4+T-cells, IL-2 and IL-10 in a subset of patients.

Results: Thirteen children with mean age 34.8 months were enrolled. Median baseline T-reg was 1.61(0.4-8.32, mean 2.71 $\pm$ 0.67) as opposed to 3.05(1.65-4.86, mean 3.15 $\pm$ 1.16) in controls. Significant negative correlation ($p=0.01$) between total T helper cells (Th) before and after resection was observed. Before resection baseline mean 15.64 $\pm$ 5.39 (9.95-24.29), post-resection mean 11.66 $\pm$ 5.40 (4.50-20.83) was noted. T-regs showed a negative trend with tumour resection ($p=0.064$): Pre-resection mean 2.71 $\pm$ 2.44, (range 0.40-8.32), and post-resection mean 1.71 $\pm$ 1.11(range 0.39-3.86). IL-2 levels were significantly lower ($p=0.029$) in patients than controls (34.56 $\pm$ 8.43 pg/ml vs. 63.43 $\pm$ 2.08 pg/ml). Patients demonstrated higher than normal levels of TGF- β ($p=0.052$), mean 7963.31 $\pm$ 5.73 pg/ml (range: 2312.27-24,663.52 pg/ml) vs. control 3351.27 $\pm$ 2.63 (range: 1219.89-8408.20 pg/ml). IL2, IL10 and Foxp3 positive monocytes were significantly greater in the tumour microenvironment compared with normal renal tissue. IL-2, IL-10 and TGF- β showed a negative correlation with the T-regs (Correlation coefficients -0.267,-0.528 and -0.069 respectively; $p=0.069$).

Conclusions: Decrease in Th and T-regs suggests that resection removes the immune stimulus. Immunosuppressive role of TGF- β is suggested by the negative correlation with number of Th. Larger percentage of IL-10 positive monocytes with lower than normal levels of T-reg support theories of T-regulatory type 1 (Tr1) genesis from the Foxp3+ T-cell population in the tumour microenvironment.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021
10:00 AM - 11:00 AM

O0040 / #822 | Loss of Heterozygosity at 11P13, 11P15 Locus and its Correlation with Histopathology and Outcome in Wilms Tumor

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Background and Aims: To evaluate the incidence of loss of heterozygosity (LOH) at 11p13 and 11p15 in Wilms tumor (WT) in the Indian population and study its correlation with histopathology and outcome.

Methods: Prospective evaluation of 75 nephrectomy samples was done. LOH of 11p13 and 11p15 were correlated with 2 year overall survival (OS) and event free survival (EFS) by Kaplan Meir analysis.

Results: LOH at 11p13 was observed in 28/75 (37.3%) and LOH11p15 in 17/75 (22.5%) samples. LOH11p13 was not observed in any of the epithelial predominant favorable histology (FH) tumors while it was observed in all the anaplastic tumors (UHF). LOH 11p15 was not observed in any of the anaplastic tumors. Among these 75 cases there were 11 deaths and 9 recurrences. There were 4 deaths among 28 with LOH11p13 and 7 among the 47 without LOH11p13 giving a 2-year OS of 84% (95CI 63-93) and 83% (95CI 67-91) respectively ($p=0.98$, hazard ratio 0.98). Two year EFS was 73 % (95CI 57-84) and 68 % (95CI 46-82) for those with and without LOH11p13 respectively ($p=0.66$; HR 1.22). There were 2 deaths among 17 with LOH11p15 and 9 among 49 without LOH11p15 giving a 2-year O.S. of 87 % (95CI 58-96) 82% (95CI 68-90) respectively ($p=0.68$; HR 0.72). Two year EFS was 75% (95CI 46-89) and 70 % (95CI 55-81) respectively ($p=0.70$; HR 0.81).

Conclusions: The incidence of LOH at 11p13 and 11p15 in WT is in the Indian population is similar to western data. LOH at 11p13 and 11p15 is not correlated with OS or EFS in our patients.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021
10:00 AM - 11:00 AM

O0041 / #1271 | Long Term Outcomes for Patients with Anaplastic Bilateral Wilms Tumors on Aren0534

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Background and Aims: Most relapses in patients with unilateral WT occur by two years off therapy. However, for BWT there is concern that late events (ie after 3 years) are more common. This is particularly concerning for patients with anaplastic histology who outcomes are the poorest. The purpose of this study is to examine the long term outcomes in patients with anaplastic BWT from ARENO534

Methods: ARENO534 study patients with anaplastic BWT. (focal anaplasia – FA and diffuse anaplasia DA) were included in the study. All alive/dead anaplastic BWT are now 5 years or more from end of therapy. Location, timing and outcomes of events are reported from the end of therapy.

Results: There were 25 patients with anaplastic BWT. Median age of enrollment was 43.1 months (24.1- 113.7). Median follow up for those patients who are alive is 6.5 years. There were 8 patients with FA and 17 with DA. Of the 8 patients with FA only one had FA in both kidneys. 7/8 are alive with one patient dying from local relapse due to FHWT at 5 years off therapy. A second patient with stage II FA had progressive of disease while on therapy. This patient did not have their nodes sampled. Pathology at relapse was DA. For DA patients, only 4/17 had DA in both kidneys. 12/17 are currently alive. Three died from progression, two from relapse (one local and one both local and lung) within a year after end of therapy. Both patients had positive margins. Of the 12 who are alive, 2 relapsed within a year and both had positive margins. Of the 10 who are alive without an event 5 had positive margins.

Conclusions: Patient with BWT and anaplasia often have discordant pathology. Late events are rare. Margin status at the time of definitive surgery although worrisome does not seem to impact outcomes

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021 10:00 AM - 11:00 AM

O0042 / #1724 | A 5-Year Comparison Between Open and Laparoscopic Nephrectomy in Pediatric Patients with Wilms' Tumor Treated at A Developing Country

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Background and Aims: Videolaparoscopic nephrectomy in Wilms' Tumor was introduced with promising results. Studies are needed to assess surgical and oncological results. This study aims to compare the surgical and oncological results of nephroureterectomy laparoscopic versus open for the treatment of children with Wilms' tumor.

Methods: Retrospective analysis of pediatric oncology patients diagnosed with Wilms' Tumor, submitted to surgical treatment by open or laparoscopic approach treated with SIOP protocol. Pre-chemotherapy and pre-surgery tomographic images were reviewed and the tumor measurements were collected, the relationship between tumor and adjacent structures were evaluated (central vessels, spine). Information related to surgery, pos-operative period and patient evolution were collected.

Results: RESULTS: Fifty patients were included in the study, of which 38 underwent open nephroureterectomy and 12 by videolaparoscopic approach. All patients in the videolaparoscopy group received neoadjuvant chemotherapy, and 4 patients in the open group underwent primary surgery. The mean tumor volume was higher in the open surgery group both at diagnosis as in pre-surgery moment ($p < 0.001$). The distance between hilar vessel and central vessel was obtained at two time-points: at diagnosis and after neoadjuvant chemotherapy. Pre-operatively the average distance in the open group was 5.42 mm and in the laparoscopy group, 10.83 mm ($p = 0.005$). The average number of lymph nodes sampled was 8.2 in the open group and 2.1 in the laparoscopy group ($p < 0.001$). In patients undergoing laparoscopy, there was no local recurrence. The overall survival rate for the open surgery and laparoscopy groups were 86.8% and 100%, respectively. The event-free survival for the open group was 84.2% and for the laparoscopy group it was 91.7%.

Conclusions: CONCLUSIONS: Videolaparoscopic nephrectomy is a viable alternative to open surgery in selected cases, reducing long-term morbidity and without increasing rupture rates. Despite the smaller lymph node sampling in the laparoscopy group, the oncological results were not affected.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021 10:00 AM - 11:00 AM

O0043 / #1306 | Bilateral Renal Tumors in Children: The First 5 Years' Experience of National Centralization in the Netherlands

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Background and Aims: Renal tumors account for 6% of childhood cancers and mostly represent Wilms tumors (WT). Survival rates of unilateral WT and nephroblastomatosis are promising, however, management of bilateral renal tumors often induces dilemmas, weighing treatment efficacy against preservation of kidney function.

Methods: We retrospectively describe characteristics and outcome of all patients with bilateral kidney tumors treated in the first 5 years of the national Princess Máxima Center for Pediatric Oncology.

Results: Twenty-five patients presented with bilateral kidney tumors (2015-2020)(15% of all kidney tumors). One family refused informed consent. Diagnosis was WT/nephroblastomatosis in 23 patients (median age 25 months, range 7-59) (19% of all WT/nephroblastomatosis patients) and renal cell carcinoma with contralateral cysts (RCC) in one 15-years old child. Four WT patients presented with metastases. All WT/nephroblastomatosis patients received preoperative chemotherapy, the RCC patient had surgery only. Tumor volume reduction (>10%) to preoperative chemotherapy was observed in 43/59 tumors with information on response, stable disease in 10/59 tumors and local tumor progression (>10%) was found in 6/59 tumors. Nephron-sparing surgery (NSS) was performed in 11/22 patients (bilateral NSS 4/11, total nephrectomy with NSS in 6/11, 1/11 unilateral NSS), unilateral nephrectomy in 11/22 and no surgery in 2. Highest local stage and histological risk group after surgery revealed stage I (n=7), II (n=4) and III (n=9), intermediate risk (n=15) and high risk (n=5) in 20 patients with at least one WT. Predisposition was clinically and/or genetically confirmed in 18/24 patients. At a median follow-up of 30.5 months, 22/24 patients are alive, of which experienced local relapse. Two patients died of toxicity, which were the only ones who had compromised kidney function and were on dialysis starting 3 and 7 months post-surgery.

Conclusions: Our results emphasize that bilateral renal tumors represent a challenging group, underscoring the need for multidisciplinary expertise centers for these patients.

IPSO LIVE Q&A: IPSO SESSION 6: RENAL TUMORS 22-10-2021 10:00 AM - 11:00 AM

O0044 / #817 | Tumor Characteristics of Urgent Pediatric Nephrectomy Patients: 15 Years' Experience in Siop Protocol Center

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Background and Aims: Following the UKW3 trial UK surgical practice has followed SIOP protocols with neoadjuvant chemotherapy followed by nephrectomy for children with Wilms tumour (WT). However, a number of patients may require an urgent nephrectomy as a deviation from the intention-to-treat protocol. Aim: To look at indications for urgent WT nephrectomy and associated complications in children.

Methods: We performed a retrospective analysis of all patients with renal tumours in a single institution, over a 15 years period between 2005 and 2020. Patients who required urgent nephrectomy either before or during neoadjuvant chemotherapy were included. We excluded patients under six months of age, predominantly cystic tumours and non-Wilms pathology.

Results: Out of a total of 84 patients with a renal tumour, 65 (77%) patients fulfilled the inclusion criteria. Fifty nine out of 65 (90%) children completed their neoadjuvant chemotherapy and proceeded to a planned nephrectomy. The remaining six (10%) required urgent nephrectomy. Four of them were males. Four cases were on the left side. No patient had syndromic WT. Median age was 27 months. Nephrectomy was required in two patients before commencing chemotherapy, two patients completed only one cycle and in two more patients four cycles were given, but still required an unplanned urgent nephrectomy. The median volume of tumour was 1392ml. Two patients required additional resections. The indications for early surgery were: mass effect in two patients, bleeding in two patients, bowel obstruction in one patient, diagnostic dilemma in one patient and marked enlargement of the tumour in one patient. Histopathological staging was Stage I in three, Stage III in three patients. Three patients were stromal predominant. All six cases were intermediate risk pathology. There were no deaths or intraoperative complications. None needed reoperation.

Conclusions: Despite the aim to follow the protocol, some patients require unplanned nephrectomy. Large tumour volume and pre-operative rupture may be contributing factors.

Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE
ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021
10:15 AM - 11:45 AM

O0045 / #127 | Efficacy and Safety of Olanzapine as an Add on
Therapy in Children Receiving Highly Emetogenic Chemotherapy:
A Randomized Double Blind Placebo Controlled Trial

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lar Pharmacology And Pharmacy, New Delhi, India

Background and Aims: Olanzapine, a second-generation antipsy-
chotic, is recommended in adults for chemotherapy induced vomit-
ing (CIV) prophylaxis, but not in children due to limited prospective
data. We did this phase 3 trial to evaluate the safety and efficacy
of olanzapine in children receiving highly emetogenic chemotherapy
(HEC).

Methods: Patients aged 3-18 years scheduled to receive HEC were
randomly assigned to either the olanzapine or placebo group. All
patients received intravenous ondansetron (0.15 mg/kg) and dex-
amethasone (0.15 mg/kg) 30 minutes before HEC, followed by oral
ondansetron (0.15 mg/kg/dose) for 48 hours. Participants in the olan-
zapine group received olanzapine syrup once daily (0.15 mg/kg/dose,
max. 5 mg) on day one and day two, while those in the control group
received a placebo syrup in the same dosage and same schedule. The
primary objective was to compare the complete control rates between
the groups in the acute, delayed, and overall phases. The secondary
objectives were to evaluate the safety of olanzapine and the need for
rescue medications.

Results: One hundred and twenty eight patients were randomly
assigned either to the olanzapine group (n=63) or the control group
(n=65). We performed an intention to treat analysis. Complete con-
trol of vomiting between the olanzapine and the placebo group was
(60% vs 54%; p=0.46) in the acute phase, (73% vs 48%; p=0.005) in
the delayed phase, and (48% vs 34%; p=0.117) in the overall phase
respectively. Grade 1 and 2 sedation was greater in the olanzap-
ine group (46% vs 14%; p<0.001). There was no grade 3/4 seda-
tion or other major adverse effects reported. A significantly higher
proportion of patients (29%) in the placebo group required rescue
medications for vomiting compared to in the olanzapine group (14%)
(p=0.025).

Conclusions: Olanzapine significantly improved complete control of
vomiting in the delayed phase with no major adverse effects in the pedi-
atric age group.

Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE
ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021
10:15 AM - 11:45 AM

O0046 / #277 | The Effect of Ready-To-Use Therapeutic Food
(RUTF) On Body Composition in Undernourished Children with
Cancer – Results of a Randomized, Controlled Phase 3 Trial (RCT)

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Background and Aims: Children being treated for cancer often
develop deleterious changes in body composition (BC). We assessed
the impact of RUTF on changes in BC in undernourished children with
cancer.

Methods: In this RCT, severely and moderately undernourished
children with cancer were randomized 1:1 to receive either
standard nutritional therapy (SNT) or SNT+RUTF for 6 weeks
(CTRI/2015/06/005896). The primary outcome was weight gain >10%
from baseline and secondary outcome improvement in BC, measured
by dual-energy x-ray absorptiometry. Total body mass (TBM), fat
mass (FM), fat-free mass (FFM), lean mass (LM), and appendicular
lean mass (ALM) were obtained and analyzed according to the 2019
International Society for Clinical Densitometry guidelines. Fat mass
index (FMI), fat-free mass index (FFMI), lean mass index (LMI) and
appendicular lean mass index (ALMI) were obtained by dividing the
individual components (in grams) by height in metre². Changes in BC
were evaluated at 6 weeks.

Results: Between July 2015 and March 2018, 260 subjects were
enrolled and 126 were analyzable in both arms at 6 weeks. More
children on SNT+RUTF had weight gain [98 (77.8%) vs 81(64.2%)
p=0.025] than on SNT. The median increases in FMI and FFMI were
416 and 4130 in the SNT+RUTF arm compared to 144 (p=0.013) and
3524 (p=0.006) in the SNT arm. There were no significant differences
in median LMI and ALMI between RUTF (28 and 211) and SNT arms (26
and 156, p=0.793 and p=0.227 respectively).

Conclusions: Children who consumed RUTF showed an absolute
increase in TBM reflecting the increase in weight, with a disproportion-
ate increase in fat mass possibly due to the high concentration of lipids
and simple sugars in RUTF. Changes in BC are reflected better in indices
because of changes in TBM. These findings must be interpreted with
caution in children who are undernourished with severely depleted fat
stores at baseline.

Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021 10:15 AM - 11:45 AM

O0047 / #279 | Effective Nutritional Intervention Improves The Health-Related Quality of Life (HRQL) of Undernourished Children with Cancer – Results of A Randomized, Controlled Phase 3 Trial (RCT)

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Background and Aims: Undernourished children experience low HRQL. We studied the effect of ready-to-use-therapeutic food (RUTF) and weight gain on HRQL in undernourished children with cancer.

Methods: In this RCT, 260 severely and moderately undernourished children were randomized 1:1 to receive standard nutritional therapy (SNT) or RUTF+SNT for 6 weeks (CTRI/2015/06/005896). While the primary objective was comparison of weight gain>10%, a secondary objective was assessing the impact of the interventions on HRQL. This was assessed using the proxy-respondent version of Health Utilities Index (HUI) interviewer-administered questionnaire. Each of the 6 attributes/domains of HUI2 (sensation, mobility, emotion, cognition, self-care and pain), and 8 attributes of HUI3 (vision, hearing, speech, ambulation, dexterity, emotion, cognition and pain) received a single-attribute score, on a scale of 0.00 to 1.00, which was used to obtain overall HRQL. The defining scores for disability are HUI2: none(1.00), mild(0.91 to 0.99), moderate(0.80 to 0.90), and severe(<0.80), and HUI3 : none(1.00), mild(0.89 to 0.99), moderate(0.70 to 0.88), and severe(<0.70).

Results: More children in the RUTF+ SNT arm - 98 (77.8%)- than in the SNT arm - 81 (64.2%)- achieved a weight gain of >10%(p=0.025). Lowest mean values at baseline for HUI2 were mobility 0.94, emotion 0.94, self-care 0.95 and pain 0.95; for HUI-3 were ambulation 0.92, emotion 0.92, dexterity 0.95 and pain 0.95. Children with weight gain(n=176) had a higher proportion of no/mild disability compared to those with less/no weight gain (n=69): overall HUI2 scores: 106 (60.2%) vs 30 (43.4%, p=0.03), overall HUI3 scores : 83 (47%) vs 20 (29%, p=0.01), HUI2 mobility: 152(86.4%) vs 50(72.4%, p=0.02), emotion: 159 (90.3%) vs 54(78.2%, p=0.03), pain: 142(80.7%) vs 52(75.3%, p=0.029), HUI3 emotion: 139 (78.9%) vs 46(66%, p=0.049), dexterity: 153 (87%) vs 53 (76.8%, p=0.079).

Conclusions: In severely and moderately malnourished children with cancer, administration of RUTF and weight gain were associated with

improved HRQL. Effective nutritional interventions can improve overall HRQL, and in numerous domains.

Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021 10:15 AM - 11:45 AM

O0048 / #120 | Neutropenic Diet Versus Standard Indian Diet in Reducing The Incidence of Febrile Neutropenia Following Chemotherapy in Pediatric Oncology Patients: A Randomized Controlled Trial

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Background and Aims: Neutropenic diet (ND) has been hypothesized to decrease the rate of febrile neutropenia (FN) occurring post-chemotherapy for pediatric cancers. Despite widespread use, it has not shown to be of benefit by randomized controlled trials (RCT) in western countries. No RCT has been conducted in India to evaluate its efficacy against the standard Indian diet. This pilot RCT of Neutropenic versus standard Indian diet aimed to evaluate its efficacy in decreasing FN rates.

Methods: Children, aged 3-14 years, with cancer, scheduled to receive strongly myelosuppressive chemotherapy were randomized to receive either neutropenic diet or standard Indian diet for one chemotherapy cycle. They were followed up till the end of the chemotherapy cycle. FN rate was recorded as the primary outcome while the focus of infection, antibiotic length, the requirement for hospital admission, adherence to diet were recorded as secondary outcomes.

Results: Forty-Two (42) patients were enrolled and randomized to receive either the neutropenic diet(n=21)or standard diet (n=21). The groups were similar in baseline characteristics. Twelve patients (57%) in the neutropenic diet and nine patients(43%) in the standard diet arm developed FN, the difference was not statistically significant (RR 1.33 95% CI 0.71-2.47, p = 0.35). Patients in ND arm had significantly higher chance of developing Neutropenic Enterocolitis (NEC) (33.33% vs 4.76% p=0.044, OR =10 95% CI 1.36 -117.3, RR=7, 95% CI 1.28 - 41.96) Patients in ND arm had higher mortality rate (14.29% vs 0%, p=0.23) and requirement for admission (47.6% vs 19.05%, p=0.06), however it did not reach statistical significance. There was no significant difference between the two arms in antibiotic length (13.5 vs 10 days, p= 0.63) Adherence was similar in both groups (95% vs 98%, p=0.6)

Conclusions: Neutropenic diet is not effective in reducing FN rate and is associated with a significantly higher rate of NEC when compared to the standard diet.

**Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE
ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021
10:15 AM - 11:45 AM**

**O0049 / #590 | Prevalence of Malnutrition in Patients with
Solid Tumour at Cancer Diagnosis in Pediatric Oncology Units in
South Africa**

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Background and Aims: In Africa, more than 45% of children diagnosed with cancer are classified as malnourished based on body mass index (BMI), or 59% when mid-upper arm circumference (MUAC) is included. This study aims to determine the prevalence of malnutrition in children diagnosed with a solid tumour in South Africa

Methods: Children from five South African pediatric oncology units (POUs), diagnosed with a solid tumor had their nutritional assessment done by measuring height/length, weight, BMI and MUAC within 72h after diagnosis between October 2018 and December 2020. Z-scores for height-for-age (H/A), weight-for-age (W/A), BMI-for-age (BMI/A) were determined with WHO AnthroPlus 2007; MUAC with MUAC growth charts (<5 years WHO 2007; >5 years Mramba 2017). Two standard deviations ($\leq -2SD$) below normal defined malnutrition ($\leq -2SD$ MUAC/A), stunting ($\leq -2SD$ H/A), underweight (≤ -2 W/A) and wasting ($\leq -2SD$ BMI/A). Frequencies and correlations were calculated using SPSS v27 statistical software.

Results: A hundred and seventy-nine (179) patients were included with a M:F ratio of 1:0.9. The median age was 3.8 years (range 0.3 to 15.3 years; mean 5.1 years). The majority (68%) had advanced disease (29% stage 3, 39% stage 4). Malnutrition was present in 24% (n=42) of children measured by MUAC, while 17% (n=31) were stunted, 12% (n=22), wasted and 9% (n=14) underweight. There was a significant correlation between MUAC and both underweight ($p < 0.000$) and wasting ($p < 0.008$). Wasting correlated with the UNICEF MUAC colour band ($p = 0.003$) and had a trend towards significant association with the province of residence ($p < 0.008$) and gestational age at birth ($p < 0.087$). Underweight patients had a trend towards significant asso-

ciation with advanced disease ($p \leq 0.091$) and gestational age at birth ($p < 0.098$).

Conclusions: Children with solid tumours in South Africa were less malnourished (24%) than reported by other African countries. MUAC is a sensitive measurement as it is independent of the large tumour burden.

**Free Paper Session (FPS) FPS 01-LIVE Q&A-SUPPORTIVE CARE
ANTIEMETICS, NUTRITION AND CONSTIPATION 22-10-2021
10:15 AM - 11:45 AM**

**O0050 / #1378 | Secondary Impacts of Constipation in Acute
Lymphoblastic Leukemia in U.S. Children's Hospitals**

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Background and Aims: Children with acute lymphoblastic leukemia (ALL) suffer a litany of chemotherapy-induced side effects. Constipation secondary to vinca alkaloids, environmental changes, and opioid use are common issues for children newly diagnosed with leukemia. This study investigated the morbidity associated with constipation including infections, mucositis, and healthcare burden in hospitalized children with ALL receiving induction chemotherapy.

Methods: We analyzed data from 48 children's hospitals in the Pediatric Health Information System, extracting patients 1-21 years of age with ALL, hospitalized for induction from October 2015 through December 2019. Admission was identified by the presence of vincristine, L-asparaginase, steroids, and intrathecal cytarabine. Constipation was defined as the presence of a constipation diagnosis code or having received ≥ 2 unique anti-constipation medications during hospitalization. Data were analyzed using nonparametric statistics, comparisons of outcomes between those with and without constipation were presented as odds ratios (OR).

Results: We identified 2,587 (56%) patients with constipation out of a total of 4,623 unique ALL patients in induction. Compared to patients without constipation during induction and adjusting for length of stay, patients with constipation were significantly more likely to have mucositis (aOR=2.23; $p = 0.0014$), perirectal issues (aOR=3.21; $p = 0.0091$), abdominal x-ray (aOR=2.40; $p < .0001$), physical therapy consult (aOR=1.57; $p < .0001$), or occupational therapy consult (aOR=1.30; $p = 0.0294$). The median length of induction hospitalization was significantly greater in those with constipation compared to those without constipation (10 days vs. 8 days; $p < .0001$).

Conclusions: Children with ALL suffering from constipation during induction therapy have more infections, mucositis, and healthcare burden compared to children with ALL and no evidence of constipation during induction therapy. Further research should explore the

causative relationship between constipation and infections. Increased attention should be given to constipation prophylaxis, management, and treatment in patients with ALL at the start of induction therapy, particularly in patients with complications or prolonged hospitalizations.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

O0051 / #487 | Significant Improvement in Survival of Advanced Stage Childhood and Young Adolescent Cancer in The Netherlands Since The 1990S: A Population-Based Study

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Background and Aims: This is a nationwide study aiming to describe trends in cancer survival and mortality for children and young adolescents in the Netherlands including the unique information on stage at diagnosis.

Methods: All neoplasms in patients younger than 18 years, diagnosed between 1990-2015 (N=14,100), were derived from the Netherlands Cancer Registry. Traditional cohort-based and period-based analysis were used to estimate 5-year and 10-year observed survival (OS). OS was estimated overall and for the time periods 1990-99, 2000-09, and 2010-15. Changes over time were evaluated by parametric survival models adjusted for follow-up time. Time trends in mortality rates were evaluated by average annual percentage change.

Results: Since the 1990s, 5-year and 10-year OS of childhood and young adolescent cancer have improved significantly by 9 and 8 percent-points, reaching survival rates of 81% and 77%, respectively. Favourable trends in survival were observed for both genders, all age groups, and most diagnostic (sub)groups, and were particularly pronounced for advanced disease. Compared to 1990-99, the risk of dying within five years of diagnosis was significantly reduced in the most recent periods (2000-09: HR=0.8; 2010-15: HR=0.6) after adjustment for age, gender, and follow-up time. Nonetheless, the prognosis of children and young adolescents suffering from central nervous system tumours, neuroblastoma, and osteosarcoma remained poor, with 5-year OS <70% and/or 10-year OS <65%. Childhood and young adolescent cancer mortality in the Netherlands decreased by an average of 3% annually between 1990 and 2018.

Conclusions: Significant progress has been realized in the prognosis of childhood and young adolescent cancer in the Netherlands since the 1990s. Survival improvements were especially evident for patients with advanced stages and were also reflected by declining mortality rates.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

O0052 / #519 | Cancer Risk in Children with Congenital Heart Disease: Results from a Swedish Register-Based Study

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Background and Aims: Birth defects have been consistently associated with elevated childhood cancer risk; however, the relationship between congenital heart disease (CHD) and childhood cancer remains conflicting and poorly understood. Considering the increasing patient population with CHD after improvements in their life expectancies, insights into this relationship are particularly compelling. Thus, we aimed to elucidate the relationship between CHD and cancer in Swedish children and adolescents.

Methods: All individuals registered in the Swedish Medical Birth Register (MBR) between 1973 and 2014 were included in this study (n=4,170,980). Individuals with CHD (n=68,776) were identified from the MBR and National Patient Register, whereas cancer diagnoses were retrieved from the Swedish Cancer Register. The relationship between CHD and childhood cancer (<20 years at diagnosis) was evaluated using Cox proportional hazards regression models.

Results: Among children with CHD, 221 had a cancer diagnosis during childhood. We observed increased risks of childhood cancer overall, leukemia, lymphoma, and hepatoblastoma in children with CHD, but after adjustment for Down syndrome, only the increased lymphoma (HR=1.55, 95% CI 1.05-2.31) and hepatoblastoma (HR=3.77, 95% CI 1.76-8.11) risk remained. However, when restricting to individuals with CHD diagnosed from the MBR only, i.e. those diagnosed near the time of birth, the risk for childhood cancer overall (HR=1.45, 95% CI 1.23-1.71) and leukemia (HR=1.41, 95% CI 1.08-1.84) was more pronounced, even after controlling for Down syndrome. Finally, a substantially elevated lymphoma risk (HR=8.05, 95% CI 4.02-16.14) was observed in children with complex CHD.

Conclusions: We found associations between CHD and childhood lymphomas and hepatoblastomas, not explained by a diagnosis of Down syndrome. Stronger associations were observed in complex CHD. The findings can help inform both the medical monitoring of children with CHD and future research aiming to elucidate the causal mechanisms underlying the relationships between CHD and childhood carcinogenesis.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

O0053 / #1047 | From Research to Policy: Developing A Pediatric Palliative Care Advocacy Tool in 11 Eurasian Countries

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Background and Aims: The "Assessing Doctors' Attitudes on Palliative Treatment" (ADAPT) study was conducted in 11 Eurasian countries as an evidence-based, comprehensive, and culturally-relevant assessment of physician knowledge regarding the early integration of palliative care into pediatric oncology practice and perceptions of structural barriers affecting access to consultation. While study findings were published in 2020, regional stakeholders identified a demand for an advocacy tool to deliver country-specific results and policy recommendations to inform local stakeholders and implement policy change.

Methods: The ADAPT advocacy tool was developed based on study findings and delivers country-level data, reviewed by Eurasian and American pediatric palliative care and oncology experts to appeal to local health and educational stakeholders. In parallel, a survey was developed to understand the clinical, educational, and policy goals of those requesting reports and how they will be used to advocate for the indicated goals. The country report and survey were translated to English, Russian, and Mongolian.

Results: After iterative rounds of review, we created country-specific 2-page reports which display data and figures describing data on pediatric palliative care education, access, barriers to and timing of integration with childhood cancer care, alongside clinical and policy recommendations. These reports were distributed to regional collaborators in 11 countries. Survey results (n=31) demonstrated that the primary goals of advocacy included increasing access to pediatric palliative care services (77%), establishing a community-based pallia-

tive care and hospice network (70%), and increasing opportunities for palliative care specialization (70%).

Conclusions: We describe the development of an evidence-based advocacy tool utilizing published data to create local health, education, and policy change on-the-ground in Eurasia. By increasing accessibility of published findings through visual summary, translation to the local language, and adaptation to the knowledge of a broader audience, this work can improve the access to and quality of palliative care delivered to children with cancer.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

O0054 / #1286 | Family Functioning in Siblings of Childhood Leukemia Survivors, Long-Term After Diagnosis: A French Childhood Cancer Survivor Study for Leukemia (Lea Cohort)

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Background and Aims: The experience of pediatric cancer is a life-altering event for all the family. Siblings must adapt to additional responsibilities, reduced availability of their parents and various conflicting emotions. Although psychological problems have been investigated and may persist over time, literature concerning long-term impact on siblings, especially on family functioning (FF), is scarce. We aimed (1) to explore the link between the FF perceived by siblings of childhood leukemia survivors and the time since diagnosis and (2) to specify its determinants.

Methods: Recruited through the LEA cohort, a French multicenter long-term follow-up program, siblings over 11 years old of childhood leukemia survivors were asked to complete the Family

Assessment Device (FAD). In accordance with authors' threshold values, a score equal to or above 2.00 defined a problematic FF. Univariate and multivariate analyses were used to identify sibling's and survivor's sociodemographic, psycho-behavioral, environmental, health-related and cancer-related characteristics likely to be associated with siblings' FF.

Results: We included 605 siblings (mean age 22.2 ± 8.0 years, mean follow-up time from diagnosis 14.1 ± 6.8 years). A first logistic regression showed FF did not differ according to the time since diagnosis but siblings of survivors with ≥ 2 sequela were significantly more likely to perceive impaired family functioning ($OR=1.49$, $p=0.04$). A second model, including familial economic status, showed siblings with household financial difficulties reported familial dysfunction ($OR=1.49$, $p=0.04$). Depending on siblings' age, univariate analysis on psycho-behavioral characteristics showed a weak social support and emotional state seemed to be linked with a poorer perceived FF.

Conclusions: Our study contributed to identify profiles of siblings who may encounter difficulties within the family. To avoid late consequences, more consideration and support should benefit to siblings, also frequently considered as the "forgotten children". Further research using Structural Equation Modeling is needed to explore links between FF and psycho-behavioral factors, such as coping strategies.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

00055 / #560 | Retrospective Cohort Study of Outcomes for Transgender and Gender Diverse Pediatric and Young Adult Patients with Cancer

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Background and Aims: Approximately 0.5-2% of the population identify as transgender; however, few studies have examined outcomes for transgender patients with cancer. Moreover, little is known about the outcomes for transgender and gender diverse pediatric patients with cancer.

Methods: A retrospective cohort study of 13 patients who were evaluated at Dana-Farber Cancer Institute/Boston Children's Hospital between 1978-2020 with a cancer diagnosis before the age of 21 years and self-identified as transgender or gender diverse. Clinical and treatment

characteristics, gender-affirming care (GAC), and long-term outcomes were recorded.

Results: Seven patients (54%) identified as transmasculine, three (23%) as transfeminine, two (15%) as non-binary, and one (8%) as gender fluid. Five patients (38%) were designated male at birth. Median follow-up time was 11.4 years (range: 1.2-41.3). Six (46%) patients had hematologic malignancies and seven (54%) had solid tumors. Ten (77%), nine (69%), and eight (62%) received chemotherapy, surgery, and radiotherapy, respectively. Five (38%) patients' long-term survivorship care notes had mixed gender-specific pronoun usage. Eight (62%) patients received gender-affirming hormone therapy (GAHT) and five (38%) patients received gender-affirming surgeries after cancer diagnosis. One patient (8%) received GAHT before diagnosis of an androgen receptor-positive serous borderline ovarian tumor. After cancer diagnosis, the patient had a brief break from GAHT and restarted after multidisciplinary discussion regarding risks and benefits with the patient and family. Eleven (85%) patients' oncology notes reported discussion with other specialty providers, including endocrinology, regarding the intersection of GAC, particularly safety of GAHT, and cancer care or associated toxicities.

Conclusions: Oncology providers will benefit from more education regarding caring for transgender patients, including correct pronoun usage. Transgender pediatric patients with cancer undergo treatments or suffer from toxicities that may affect GAC, highlighting the need for multidisciplinary care. Future studies need to incorporate experiences of transgender pediatric patients with cancer to further improve care.

Free Paper Session (FPS) FPS 02-LIVE Q&A-EPIDEMIOLOGY AND POLICY 22-10-2021 10:15 AM - 11:45 AM

00056 / #697 | Role of Non-Chromosomal Birth Defects on the Risk of Developing Pediatric Hodgkin Lymphoma: A Children's Oncology Group Study

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Background and Aims: Non-chromosomal birth defects are an important risk factor for several childhood cancers. However, these associations are less clear for Hodgkin lymphoma (HL). Therefore, we sought

to more fully elucidate the association between non-chromosomal birth defects and HL risk by leveraging data from a large study of childhood and adolescent HL.

Methods: Information on cases (n=517) diagnosed with HL between the ages of 0-14 at Children's Oncology Group Institutions for the period of 1989-2003 was obtained. Healthy control children (n=784) were identified and individually matched to cases on sex, race/ethnicity, age and geographic location. Parents completed comprehensive interviews. To test the association between birth defects and HL risk, conditional logistic regression was applied to generate adjusted odds ratios (aORs) and 95% confidence intervals (CIs), adjusted for maternal age at index child's birth, level of maternal education and annual household income level. We conducted these adjusted conditional logistic regression analyses overall and stratified by cofactors of interest (tumor Epstein-Barr viral status, histologic subtype and age group).

Results: Children born with non-chromosomal birth defects were not more likely to be diagnosed with HL at 0-14 years of age (aOR: 0.95; 95% CI: 0.73-1.23). A nonsignificant inverse association was detected for those children born with any birth defect and EBV-positive HL (aOR: 0.52; 95% CI: 0.25-1.08). Those born with two or more birth defects had a suggested increased risk of developing HL between the ages of 10-14 years (aOR: 2.09; 95% CI: 0.97-4.50).

Conclusions: Previous assessments of being born with a non-chromosomal birth defect and developing childhood HL have been limited. Using data from the largest case-control study of HL in those <15 years of age, we did not observe strong associations between being born with a birth defect and HL risk.

CCI LIVE Q&A-SESSION 1: PALLIATIVE & PSYCHOSOCIAL SUPPORT 22-10-2021 10:15 AM - 11:45 AM

O0057 / #681 | Psychological Support of Bereaved Families in Ukraine: Project by Tabletochki

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Background and Aims: In Ukraine, there is no integrated palliative care for children with cancer, not to say about the psychological support for bereaved families. Seeking to address this deficit, Tabletochki Charity Foundation launched the palliative care program in 2015 and the psychological support of bereaved families project in 2018.

Methods: Psychological support for families in grief is provided in groups supervised by experienced psychologists. Each group consists of 18-20 participants from all over Ukraine. The participants meet bimonthly in Kyiv for one day during 1.5 year (the life cycle of a group). Guided by a psychologist, the parents spend this day together sharing their experience, supporting each other, reflecting on their emotional status and seeking for the meaning of their loss and their lives. The families are encouraged to come in full (both parents and siblings).

Siblings have their separate activities with a pediatric/adolescent psychologist and volunteers, combining therapeutic work and entertainment. By the end of the 1.5-year cycle, the group has two memorable events – a 3-day outdoor camp in the Carpathian mountains and a graduation day.

Results: Since 2018, 46 bereaved parents have graduated from the groups and 44 more are under way. This project started with supporting mothers but currently up to 60-70% of participants come in couples. In 2020 we involved a pediatric psychologist to work with siblings which boosted the full family format. The parents who attend or graduated from the groups provide very positive feedback, they advise other bereaved families to join the project, and establish long-lasting friendly relationship with other families.

Conclusions: The bereaved families need psychological support to live through their loss and grief. The group format where the families receive both professional and peer-to-peer support turned out to be very successful. The family format when both parents and siblings come in the meetings is to be encouraged.

CCI LIVE Q&A-SESSION 1: PALLIATIVE & PSYCHOSOCIAL SUPPORT 22-10-2021 10:15 AM - 11:45 AM

O0058 / #1097 | Psychometric Evaluation of The Chinese Version of The Resilience Scale for Children with Cancer

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Background and Aims: A valid and reliable assessment tool that accurately measures the resilience of children with cancer is vital for appropriately planning interventions that can enhance resilience in such children and foster the development of their coping mechanisms and positive mental well-being. However, no tool with effective psychometric properties is available for assessing resilience in Chinese children with cancer. This study aimed to test the psychometric properties of a traditional Chinese version of the Resilience Scale for Children (RS-10) and examine its factorial structure via a confirmatory factor analysis (CFA). **Methods:** One hundred and eighty-six Hong Kong Chinese children with cancer were invited to participate in the study. The psychometric properties of the traditional Chinese version of the RS-10 were assessed, namely its content validity, convergent and discriminant validity, construct validity, internal consistency and test-retest reliability.

Results: The newly translated traditional Chinese version of the RS-10 demonstrated adequate internal consistency, excellent test-retest reliability, good content validity and appropriate convergent and discriminant validity. The CFA results demonstrated that there was a good fit between the factor structure of the Chinese version of the RS-10 and the observed data, thereby confirming the construct validity of this instrument.

Conclusions: The traditional Chinese version of the RS-10 was found to be a reliable and valid tool for assessing the resilience of Hong Kong Chinese children with cancer. The newly developed traditional Chinese version of the RS-10 is an appropriate clinical research tool for evaluating the effectiveness of nursing interventions in enhancing the resilience of and promoting mental well-being in children with cancer.

CCI LIVE Q&A-SESSION 1: PALLIATIVE & PSYCHOSOCIAL SUPPORT 22-10-2021 10:15 AM - 11:45 AM

00059 / #1328 | An Exploration of Impact of Peer Relationships on Resilience in Children with Cancer

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Background and Aims: Children receiving cancer related treatment often go through a multitude of changes during treatment. Socialization, self-image, along with coping is effected during these months of treatment. This research focused on the exploration of peer relationships that were formed during active treatment and how those relationships effected resilience in children because such researches are lacking in a non-profit hospital settings, which observe peer relationships of patients in LMIC.

Methods: Qualitative interviews were conducted in the Pediatric Oncology and Hematology Unit of the Indus Hospital and Health Network (IH&HN) of patients undergoing active treatment.

Results: It was found that the interaction between pediatric patients who were undergoing treatment for cancer with other children undergoing the same treatment and experiencing similar conditions was seen to impact their resilience in terms of connection, confidence and control in a positive manner. It was found that their interactions with each other helped promote a feeling of collective experience and provided them with the support they required to undergo daily treatment processes.

Conclusions: Peer relationships have been seen to play a vital role in the functioning and building of resilience of children with pediatric cancer.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

00060 / #211 | Inter-Clinician Delineation Variation for a New Highly-Conformal Flank Target Volume in Children with Renal Tumors: A Siop-Renal Tumor Study Group International Multicenter Exercise

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Background and Aims: Recently, the SIOP-RTSG developed a highly-conformal flank target volume definition for children with renal tumors. The aims of this study were to evaluate the inter-clinician delineation variation of this new target volume definition in an international multicenter setting and to explore the necessity of quality assurance.

Methods: Six pediatric renal cancer cases were transferred to ten radiation oncologists from seven European countries ('participants'). These participants delineated the pre- and postoperative Gross Tumor Volume (GTV_{pre/post}), and Clinical Target Volume (CTV) during two test phases (case 1-2 and 3-4), followed by guideline refinement and a quality assurance phase (case 5-6). Reference target volumes (TV_{ref}) were established by three experienced radiation oncologists. The Dice Similarity Coefficient between the reference and participants (DSC_{ref/part}) was calculated per case. Delineations of case 5-6 were graded by four independent reviewers as 'per protocol' (0-4mm), 'minor deviation' (5-9mm) or 'major deviation' (≥10mm) from the delineation guideline using 18 standardized criteria. Also, a major deviation resulting in underestimation of the CTV_{ref} was regarded as an unacceptable variation.

Results: A total of 57/60 delineation sets were completed. The median DSC_{ref/part} for the CTV was 0.55 without improvement after sequential cases (case 3-4 vs. case 5-6: p=0.15). For case 5-6, a major deviation was found for 5/18, 12/17, 18/18 and 4/9 collected delineations of the GTV_{pre}, GTV_{post}, CTV-T and CTV-N, respectively. An unacceptable variation from the CTV_{ref} was found for 7/9 participants for case 5 and 6/9 participants for case 6.

Conclusions: This international multicenter delineation exercise demonstrates that the new consensus for highly-conformal postoperative flank target volume delineation leads to geometrical variation among participants. Moreover, standardized review showed an unacceptable delineation variation in the majority of the participants. These findings strongly suggest the need for additional training and centralized pre-treatment review when this target volume delineation approach is implemented on a larger scale.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0061 / #329 | Image Guided Brachytherapy for Paediatric Pelvic Tumours: Optimising Efficacy and Minimising Toxicity

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Background and Aims: Image guided brachytherapy (IGBT) is an appealing option in the treatment of paediatric solid tumours, with the capability to deliver a high radiation dose to a localised target with rapid dose drop off. IGBT offers an alternative local strategy directed towards the reduction of late side effects and preservation of function. This is a prospective case series of sequentially treated children with pelvic tumours.

Methods: Radiological assessment of tumour response was done after initial chemotherapy and pre-plans created. IGBT implants were performed under general anaesthesia with radiation oncology and surgical team present. Post-operative imaging was acquired and individual adaptive treatment plans created. Dose objectives were adjusted from paediatric protocols and adult BT literature. Quality assurance, safety protocols and workflow were adapted depending on the implant. Children remained sedated and ventilated throughout treatment (twice daily). A subset of six patients underwent plan comparisons between IGBT and proton therapy to investigate optimal patient selection.

Results: Between 2012 and 2020, 12 children from Australia and New Zealand were treated with IGBT with a median age of 18 months (range 11m–16 yrs). Tumour types comprised 7 rhabdomyosarcoma, 4 undifferentiated sarcoma and 1 recurrence of Wilms tumour. IGBT techniques included transperineal interstitial implants, transabdominal implants and customised 3-dimensional applicators. IGBT as the sole local treatment modality was used in 5 children. Doses of 5–6.5Gy over 5 fractions were delivered to 95% of the target volume. Ten of 12 children are alive without disease at median follow up 31 months (range 3–104 months). Five of six comparative plans had lower growth plate dose using IGBT.

Conclusions: IGBT is an ideal primary local treatment modality option in paediatric patients with pelvic tumours that may preserve function, minimise side effects and is delivered over a shorter time period. It requires a collaborative approach between surgical, paediatric and radiation therapy colleagues.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0062 / #395 | Identifying Predictors of Inter-Fractional Anatomical Variations in Paediatric Abdominal Radiotherapy

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Background and Aims: Proton therapy is an advantageous radiation modality for children due to its normal tissue-sparing capabilities. However, proton beams are sensitive to density changes, which may lead to beam under/over-ranging. This study used longitudinal imaging to identify patient variables predictive of inter-fractional anatomical variations during abdominal radiotherapy.

Methods: Body contour and gastrointestinal gas were semi-automatically segmented on 21 planning computed-tomography (CT) and 77 weekly cone-beam CT (CBCT) scans for 21 patients treated with intensity modulated arc therapy for abdominal high-risk neuroblastoma (median age 4, range 2–19 years). Sex, age, feeding tube and general anaesthesia (GA) were explored as predictive variables for inter-patient variability in gas volume (ΔV_{gas}) and body contour shape (d_{avg}) throughout radiotherapy. ΔV_{gas} was defined as the standard deviation of the gas volume over all scans available per subject; d_{avg} was defined as the mean value (over all CBCTs per subject) of the average three-dimensional separation between CT and CBCT body contours. All analyses considered only common field-of-views between rigidly-aligned CT/CBCTs.

Results: We found a ΔV_{gas} and d_{avg} of $71.0 \pm 51.6 \text{ ml}$ and $1.10 \pm 0.40 \text{ mm}$, respectively. Most patients exhibited in-room reduction in gastrointestinal gas and body contour compared to planning (81% and 86%, respectively). ΔV_{gas} was greater for subjects $\leq 4 \text{ y}$ ($P < 0.01$) and under GA ($P < 0.01$); multivariate analysis identified GA as the strongest predictor for ΔV_{gas} ($P = 0.01$). d_{avg} was greater in patients without feeding tubes ($P = 0.04$). No other variables predicted for ΔV_{gas} and d_{avg} . A moderate correlation between reduction (from planning) in gastrointestinal gas and body contour ($R = 0.50$) suggests that, whilst body contour

changes are affected by abdominal distension, weight fluctuations and setup variations may contribute to some degree.

Conclusions: Age, GA and feeding tubes were identified as potential predictors of inter-fractional anatomical variations. Our findings may help identify patients who benefit from adaptive/robust planning pathways and more frequent surface and/or image-guided protocols to monitor changes.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0063 / #800 | The Use of Low Dose Cone Beam Computed Tomography Protocols for Image Guided Paediatric Radiotherapy on Abdominal And/Or Pelvic Target Volumes

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Background and Aims: This study explored the use of low dose Cone-Beam Computed Tomography (CBCT_{LD}) in Image-Guided Radiotherapy (IGRT) for children with an abdominal and/or pelvic planning target volume (PTV). The aim was to compare CBCT_{LD} with standard dose CBCT (CBCT_{SD}) using qualitative assessment of relevant IGRT aspects. The study also aimed to determine optimal patient separation in the left-right direction (L-Rsep) where CBCT_{LD} could be implemented without compromise on image usability.

Methods: Thirty-Five CBCT's were acquired on 10 children. (Abdomen PTV n=7, Abdomen + Pelvis PTV n=3) CBCT_{SD} was acquired on the first fraction of each course; remaining acquired images were CBCT_{LD}. CBCT dose was measured using clinical phantoms. Five clinicians (2x therapeutic radiographers, 2x clinical oncologists, 1x dosimetrist) assessed each CBCT for seven clinically relevant aspects. These included kidney, bladder, bowel/rectum, tumour-insitu, external contour and bone visibility. Comparison between clinician feedback and lateral patient separation (L-Rsep) was used to determine optimal indications for CBCT_{LD} in IGRT analysis.

Results: All clinicians reported that both external contour and bone could be clearly visualised in all CBCT_{LD}. Kidneys were reported to be clearly seen in 40.7% of CBCT_{LD} (61.3% if L-Rsep≤18.5cm), compared to 77.7% on CBCT_{SD}. Bowel gas was reported to be clearly visible in 73.3% of CBCT_{LD} (85.7% if L-Rsep≤18.5cm), and 87.7% for CBCT_{SD}. Bladder and rectum were visualised clearly on CBCT_{LD} in 48.4% and 55.8% cases respectively (83.3% and 61.1% if L-Rsep≤18.5cm), compared to 28.5% and 57.2% for CBCT_{SD}. Tumour-insitu was not seen on included CBCT's due to small residual volume.

Conclusions: CBCT_{LD} is an excellent IGRT tool for use in assessing external contour and bone position, whilst kidney, bladder and

bowel/rectum assessment was most suitable for patients with a L-Rsep≤18.5cm at level of PTV. CBCT_{LD} can account for up to 4.3mSv dose reduction per image when compared to CBCT_{SD}.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0064 / #327 | Experiences and Lessons Learnt from Twenty Years of Implementation of National Telemedicine-Based Shared Competence for Pediatric Radiotherapy

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Background and Aims: Pediatric radiotherapy comprises a plethora of diagnoses and challenges for radiotherapy departments. The rare incidence of pediatric malignancies infers difficulties in securing and maintaining state-of-the-art competence among radiation oncologists and for providing adequate training for young physicians. In Sweden, only six centers give pediatric radiotherapy, each center treating 10-30 children annually. With the intention to secure competence, the Swedish Pediatric Radiotherapy Group was formed in the year 2000. We report on the lessons learnt and the utility and benefits from these 20 years of work.

Methods: The group has become an established basis for pediatric radiotherapy with bi-monthly telemedicine conferences. In 2008, we started a registry (dosimetric and follow-up data) for Swedish children receiving radiotherapy. A national monitor secures a 100% enrollment.

Results: During these first 20 years almost all children due for radiotherapy have been reviewed in the teleconferences. Treatment recommendations for challenging clinical situations have been formulated, and selection of optimal treatment techniques, as proton treatment was introduced as an alternative to photon techniques, have been facilitated. Dummy-runs were conducted on new treatment protocols for rhabdomyosarcoma, Hodgkin's lymphoma and cranial tumors in order to discern ambiguities or possible risks of misinterpretation for radiotherapy guidelines before implementation. Treatment guidelines for re-irradiation of intracranial tumors have been developed, and the experience of introducing proton treatment in 2015 continuously evaluated. Also, guidelines for delivering CSI, ependymomas and craniopharyngiomas have been formulated. On a case-by-case basis, also children from neighboring countries lacking similar groups have been discussed. A core-curriculum for pediatric radiotherapy is being developed.

Conclusions: The experiences strongly support the benefits of regular telemedicine collaboration between radiotherapy centers providing pediatric radiotherapy. It may enhance treatment quality in centers with few pediatric cases and provides an excellent tool for education and further training in radiotherapy.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0065 / #356 | Pulmonary Sequelae After Reduced Intensity Conditioning Regimen with Low-Dose Total Body Irradiation

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Background and Aims: Idiopathic pneumonitis is a treatment-related morbidity that emerges after total body irradiation (TBI). A reduced intensity conditioning (RIC) regimen with low-dose irradiation has been recently applied for the treatment for both malignant and non-malignant diseases, to support engraftment with minimized toxicity. This study analysed the risk of pulmonary complications, including idiopathic pneumonitis after low dose TBI (LDTBI), in RIC.

Methods: We performed a retrospective analysis of patients who underwent LDTBI and 3 Gy single-fraction as a conditioning regimen in a children's hospital between 2010 and 2020. Any respiratory symptoms that emerged 120 days after transplantation and having been subjected to radiological examination were listed as pulmonary complications. These pulmonary complications were classified into three categories according to American Thoracic Society Research Statement: group 1, idiopathic pneumonia syndrome group; group 2, infection group; and group 3, respiratory disorder caused by other specific aetiologies. The factors causing idiopathic pneumonia, including disease status, graft versus host disease (GVHD), and radiotherapeutic parameters were explored.

Results: LDTBI was introduced in a total of 57 patients, with median age of 4.4 (0.3-20.9) years. The primary diseases for stem cell transplantation were: malignant disease (n=8), non-malignant disease (n=44). Pulmonary complications were observed in nine patients. Among those with the idiopathic pneumonitis, group 1 accounted for 2 patients (3.5%). Acute GVHD was observed in 26 patients. All two idiopathic pneumonia patients suffered from acute GVHD.

Conclusions: The incidence of pulmonary complications after LDTBI was lower than 15% that reported for TBI in myeloablative conditioning, and most events were caused by infection or other organ failure. The relationship between idiopathic pneumonia and acute GVHD pre-

dicted that idiopathic pneumonia after LDTBI is attributed not only to pulmonary damage by irradiation, but also to immunological hyperreaction after stem cell transplantation.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0066 / #507 | Cine MR Based Margin for Respiratory Movement for Radiotherapy of Pediatric Abdominal Tumors

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Background and Aims: 4D CT is often used for the evaluating the extent of respiratory motion in the thoracic and abdominal region for use in radiation treatment. However, improved image contrast and lack of ionizing radiation makes MRI an attractive alternative. The aim of this study was to evaluate the use of movie-like 2D cine MRI sequences to measure the extent of respiratory motion for radiotherapy planning in pediatric patients treated with proton therapy for abdominal tumors.

Methods: Sagittal and coronal cine MRI sequences were added to an MRI acquired in general anaesthesia for target delineation for five pediatric patients with abdominal tumors. The median age was 3 years (7 months - 4 years). Cine MRI sequences were acquired at 4Hz for 16 seconds with 2-3 coronal and 2-7 sagittal images. The displacement length (d) between expiration and inspiration was visually assessed in three directions (AP, CC, LR) as maximal (d_v) in CTV near structures that were easy to visualize. This was compared to deformable registration between the inspiration and expiration respiratory phase and d was measured at 90% ($d_{90\%}$) and at maximum (d_{max}).

Results: Visual evaluation resulted across all patient in median(range) of $d_{v,AP}$ =1mm (0mm 1mm), $d_{v,LR}$ =1mm (0mm 2mm), and $d_{v,CC}$ =2mm (2mm 5mm). Similarly, deformable registration resulted in a median(range) 90% displacement of $d_{90\%,AP}$ =1.6mm (1.3mm 2.3mm), $d_{90\%,LR}$ =1.7mm (1.3mm 3.7mm), and $d_{90\%,CC}$ =2.8mm (2mm 3.4mm). The maximal deformation length was $d_{max,AP}$ =3.3mm (2.7mm 12.0mm), $d_{max,LR}$ =3.9mm (3.7mm 6.4mm), and $d_{max,CC}$ =5.3mm (3.7mm 6.0mm).

Conclusions: Visual evaluation was closer to the 90% central range than to the maximal deformation vector length. However, further validation using a 4D MRI phantom to establish a ground truth is needed as well as direct comparison to 4D CT. Cine MR may be used as an alternative to 4DCT for evaluation on abdominal motion with the largest motion found in the CC direction.

PROS LIVE Q&A-FREE PAPER SESSION 1 22-10-2021 10:15 AM - 11:45 AM

O0067 / #1411 | Is The Spleen An Organ at Risk in Pediatric Radiotherapy?

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Background and Aims: Background : Splenectomy in young children has long been associated with a risk of bacterial infections. Radiation may also cause long-term splenic dysfunction, risking potentially fatal late sepsis. Purpose: To assess the doses received to the spleen in pediatric patient who receive abdominal radiation.

Methods: A retrospective dosimetric analysis of 3D radiotherapy plans of 29 pediatric patients treated from 2014 to 2020, in our department. Those patients has received abdominal irradiation. Radiotherapy plans were reviewed to assess spleen dosimetry. Mean doses to the spleen were considered.

Results: The mean patient age was 5,4 years (range from 2 to 14 years old). Patients were treated for medulloblastoma (20.7%) with a total dose of 54 Gy, wilm's tumor (37.9%) with a dose of 14.4 Gy, neuroblastoma (13.8%) with a dose of 36 Gy, synovial sarcoma(3.4%) with a dose of 45 Gy, Rhabdomyosarcoma (3.4%) with a dose of 54 Gy, ewing sarcoma of the spinal epidural (3.4%) with a dose of 45 Gy, pinealoblastoma (3.4%) with a dose of 54 Gy, primary neuroendocrine tumor (10.3%) with a dose of 54 Gy and embryonal tumor (3.4%) with a dose of 54 Gy. The median mean dose to the spleen was 1.328 Gy (range: 0.138 - 45.897 Gy). 6 patients (20.7%) received a mean spleen dose >10 Gy, but only one patient received a dose >40 Gy. This patient had a primary left renal synovial sarcoma receiving postoperative radiotherapy with a dose of 45 Gy in 25 fractions. No cases of septicemia or bacterial meningitis were noted in this study;

Conclusions: The spleen may receive some incidental dose when the target volume is in the abdomen. SIOP-Europe Radiation Oncology Working Group recommend routine delineation of the spleen and advocate antibiotic prophylaxis and/or (re)vaccination, to prevent « overwhelming post-splenectomy infection » if the mean spleen dose is >10 Gy.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP - HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY 22-10-2021 10:15 AM - 11:45 AM

O0068 / #147 | Hearing Loss after Radiation and Chemotherapy for Central Nervous System and Head and Neck Tumors in Children

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Background and Aims: Hearing loss (HL) is a serious secondary effect of pediatric head and neck and central nervous system tumor treatment. Radiation (RT) and platinum chemotherapy independently increase HL risk; however, combined modality treatment is routinely used and its effect on HL risk is not well-studied. We created a model for HL, including a nomogram.

Methods: In this single-institution retrospective study, 171 patients were treated with RT, with or without chemotherapy, and had longitudinal (≥ 2) audiological evaluation. SIOP grades were assigned to 2,420 assessments of 342 ears. For the nomogram, grade ≥ 3 was considered clinically significant. Multivariable weighted ordinal logistic regression was fitted to evaluate the effect of clinicopathologic features on HL.

Results: Patients underwent a median of 6 (range 2-23) assessments over a median of 3.1 (range 0.1-15.2) years. Cisplatin was given to 63% of patients; 34% received carboplatin. Mean cochlear doses on the right and left were 36.8 Gy (standard deviation [SD] 16.5) and 37.0 Gy (SD 16.2), respectively. Multivariable regression revealed that mean cochlear dose (odds ratio [OR] 1.04 per Gy, 95% confidence interval [CI] 1.02-1.05, $p < 0.001$), time since RT (OR 1.2 per year, 95%CI 1.2-1.3, $p < 0.001$), cisplatin (OR 5.33, 95%CI 2.9-9.9, $p < 0.001$), and carboplatin (OR 2.3, 95%CI 1.27-4.17, $p = 0.006$) were associated with worse HL. There was no synergistic effect of RT and cisplatin (interaction term, $p = 0.4$) or RT and carboplatin (interaction term, $p = 0.9$). Cumulative incidence of high-frequency HL (>4 kHz) was >50% at 5 years post-RT if mean cochlear dose was >20 Gy, while incidence of HL across all frequencies continued to increase beyond 5 years post-RT.

Conclusions: RT and chemotherapy have an additive, not synergistic, effect on HL risk. Mean cochlear radiation dose, time since RT, and platinum chemotherapy use were associated with HL. This modelling can guide follow-up to ensure that educational accommodations and assistive devices are in place.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP - HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY 22-10-2021 10:15 AM - 11:45 AM

O0069 / #577 | Cisplatin Ototoxicity: Examination of the Impact of Dosing, Infusion Times, and Schedules

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Background and Aims: Sensorineural hearing loss is a well-known side effect of cisplatin (CDDP). There is limited research on the effect of dosing, infusion times, and schedules of CDDP administration and their impact on hearing loss.

Methods: A retrospective chart review of 993 pediatric patients from August 1990 to March 2015 was conducted using stringent inclusion criteria to characterize patients with hearing loss. Of the 993 patients, 248 received CDDP and 216 had sufficient infusion data to assess for sensorineural hearing loss attributable to CDDP. Demographic and clinical characteristics were summarized by descriptive statistics; univariate and multivariate logistic regressions were performed to examine the relationship between hearing loss and specific parameters of cisplatin administration. Stepwise variable selection procedure was performed in the multivariate model building to identify independent outcome measures of hearing loss and the corresponding ototoxicity grade using the grading scale from the International Society of Pediatric Oncology (SIOP).

Results: Of the 216 patients reviewed, 153 patients were evaluable for analysis. Sensorineural hearing loss was identified in 72.6% of the patients. Multivariate analysis revealed age [OR=0.90 (0.84-0.97), $P=0.0086$], radiation to any part of the body, [OR=3.20 (1.29-7.93), $P=0.012$], amount infused per dose (mg/m^2) [OR=1.018 (1.002-1.033), $P=0.029$], and cumulative cisplatin exposure (mg/m^2) [OR=1.004 (1-1.008), $P=0.027$] were associated with hearing loss. These same variables were also found to be associated with worsening SIOP grade. The number of doses per cycle, prescribed infusion times, number of cycles, and total number of doses of CDDP failed to be significantly associated with hearing loss.

Conclusions: The amount of CDDP infused per dose is a significant risk factor for sensorineural hearing loss. Considerations in designing regimens that reduce the amount of CDDP infused per dose may reduce the risk of hearing loss. Randomized prospective trials are needed. Acknowledgements: Drs Camet and Spence share first authorship.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP - HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY 22-10-2021 10:15 AM - 11:45 AM

00070 / #587 | Screening and Interventions for Childhood Cancer Survivors at Risk for Cardiomyopathy: Consensus and Controversy

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Background and Aims: Treatment with anthracyclines and chest radiation (CR) are risk factors for cardiomyopathy in childhood cancer survivors (CCS). Evidence to guide screening/prevention are lacking. We used systematic consensus-building methodology to define: 1) expert consensus recommendations and 2) controversial areas in managing asymptomatic CCS at risk for cardiomyopathy.

Methods: A Delphi panel of 40 physicians from US/Canada with expertise in CCS (pediatric and radiation oncology, onco-cardiology, pediatric cardiology, internal medicine) participated in 3 iterative, open-ended, anonymous rounds of questionnaires for consensus development. In Round 1, panelists described their clinical practice related to screening and management of asymptomatic findings on echocardiogram including: technically inadequate, ejection-fraction (EF) 45%, EF 50% on serial-echocardiograms, grade 1 diastolic-dysfunction, and EF normal/global longitudinal strain (GLS) >-16%. In Rounds 2 and 3, panelists re-appraised their responses in view of the group modal responses and rationale. "Consensus" defined as > 90% agreement.

Results: Response rate was 100% for all rounds. Panelist reached consensus on laboratory testing for modifiable cardiovascular risk-factors and referral to cardiology for EF < =50%, for diastolic-dysfunction, and for cardiac MR if repeat echocardiograms were technically inadequate. In the scenario of abnormal global strain, consensus was for close follow-up with repeat echocardiogram, and high level of agreement for referral to cardiology (88%). High level of agreement was reached on frequency of echocardiogram screening based on anthracycline dose (85%) and during pregnancy (88%). Areas of disagreement (<70%) included CR dose threshold for echocardiogram screening; use of serum bio-markers (natriuretic peptides, BNP/NT-proBNP) to monitor asymptomatic left ventricular dysfunction (ALVD); use of both ACE-inhibitor and Beta-blocker vs ACE-inhibitor only for ALVD and ACE-inhibitor vs no medication for borderline ALVD.

Conclusions: Despite a paucity of evidence, expert physicians reached consensus on most management decisions for cardiovascular monitoring of CCS. Controversial areas included screening after chest radiation and medical management of ALVD.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP -
HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY
22-10-2021 10:15 AM - 11:45 AM

O0071 / #1204 | Medical Follow-Up after Childhood Cancer:
Are Survivors with An Increased Risk for Cardiomyopathy
Regularly Followed-Up?

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Background and Aims: Many late effects related to cancer treatment, especially cardiovascular diseases, can be prevented or reduced by prevention or early detection. Thus, adherence to screening for cardiomyopathy in childhood cancer survivors (CCS) at high-risk of cardiovascular outcome is crucial. We aimed to analyse the medical follow-up of CCS at increased risk for developing a cardiomyopathy using data from the French administrative healthcare database.

Methods: This study involved 3,249 5-year CCS from the French childhood cancer survivor study (FCCSS) cohort. Survivors who received ≥ 250 mg/m² of an anthracycline dose and/or ≥ 30 Gy on $\geq 10\%$ of the volume of the left ventricular were defined as subjects at high-risk for developing a cardiomyopathy. Medical follow-up corresponded to the completion of an echocardiogram within the five previous years and was identified through data from the French administrative healthcare database. The study of demographic and clinical determinants associated with non-adherence to medical follow-up was performed using multivariable logistic regression models. In addition, we matched survivors at high-risk of cardiomyopathy who attended a long-term follow-up (LTFU) consultation to those who did not using propensity scores, and we analysed the impact of the attendance to a LTFU consultation on the completion of echocardiography using Cox regression models.

Results: Only 31% of the 642 CCS at high-risk of cardiomyopathy had an echocardiogram within the five previous years. Males (OR 1.58, 95% CI 1.06 to 2.33) and CCS <50 years old (OR 2.81, 95% CI 1.50 to 5.25) were more likely to not adhere to this medical follow-up. Going to a LTFU consultation was strongly associated with the completion of echography (HR 6.66, 95% CI 4.42 to 10.00).

Conclusions: Despite the risk of treatment-related cardiovascular outcomes, less than one third of childhood cancer survivors at high-risk of cardiomyopathy adhered to the medical follow-up. LTFU consultations highly improve adherence to surveillance.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP -
HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY
22-10-2021 10:15 AM - 11:45 AM

O0072 / #1269 | Cognitive and Academic Functioning in
School-Age Survivors of Retinoblastoma

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Background and Aims: Retinoblastoma is the most common intraocular childhood cancer and is typically diagnosed in young children. Excellent survival rates in developed countries have led to increased focus on ocular salvage and vision preservation while reducing toxicity of medical therapy. However, little is known about long-term functional outcomes. The current study examines cognitive and academic functioning in school-aged survivors.

Methods: Sixty-nine survivors of retinoblastoma underwent an assessment of cognitive and academic functioning. Survivors (age=10.89 $\pm$ 1.07 years; 49.3% male; 56.5% unilateral disease, 68.1% treated with enucleation, years since diagnosis=9.37 $\pm$ 1.16) completed the WASI-II and select subscales from the WISC-IV to assess cognitive functioning, and the WJ-III Tests of Achievement to assess academic functioning.

Results: IQ scores were within the average range (95.46 $\pm$ 15.36), though were significantly below the normative mean ($t(62)=-2.35$, $p=.02$). Further, a larger proportion of survivors demonstrated below average scores (27%) than expected in the normative sample (16%, $\chi^2=5.67$, $p<.02$). IQ was significantly lower among survivors who underwent enucleation (92.60 $\pm$ 15.02) versus those who did not (101.60 $\pm$ 14.61; $t(61)=2.32$, $p=0.03$). Processing speed (94.77 $\pm$ 16.33, $t(61)=-2.52$, $p=.02$), working memory (95.64 $\pm$ 15.05, $t(65)=-2.36$, $p=.02$), and math calculation scores (92.80 $\pm$ 19.85, $t(62)=-2.75$, $p<.01$) were also significantly below the normative mean; however word reading was not (97.70 $\pm$ 16.86, $t(62)=-0.62$). The proportion of survivors with below average scores was consistent with normative expectations (16%) for processing speed (22.6%), working memory (19.7%) and word reading (12.7%), but was significantly larger for math (25.4%, $\chi^2=4.14$, $p<.05$).

Conclusions: On average, school-aged survivors of retinoblastoma demonstrated cognitive and brief academic functioning that is below the normative mean, but still well within the average range overall. Survivors treated with enucleation demonstrate lower IQ scores than

those who were not. A notable minority of survivors demonstrate below average difficulties with IQ and math, suggesting a need for intervention. Further research is needed to understand the impact of enucleation on cognitive functioning.

Free Paper Session (FPS) FPS 03-LIVE Q&A-SURVIVORSHIP - HEARING LOSS, NEUROCOGNITION AND CARDIOMYOPATHY 22-10-2021 10:15 AM - 11:45 AM

O0073 / #1421 | Late Toxicity Comparison of Alkylating-Based Maintenance with Cyclophosphamide (VAC) Versus Ifosfamide (VAI) In Ewing Sarcoma Survivors Treated In The Randomized Clinical Trial Euro-Ewing99-R1 in France

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Background and Aims: In efficacy analysis of Euro-EWING99-R1 randomized trial, after a common induction of 6 VIDE (vincristine-ifosfamide-doxorubicin-etoposide), cyclophosphamide (VAC) was concluded as non-inferior to ifosfamide (VAI) in consolidation treatment of standard-risk Ewing sarcoma (SR-EWS). We present the results of the comparative late effects analysis of VAC vs. VAI conducted in the French cohort of 5-year free-of-relapse survivors.

Methods: Of 267 French randomized patients, 172 SR-EWS 5-year free-of-relapse survivors were included in analysis (sex-ratio M/F=1.3, median age at diagnosis=14years, range=1-43): 84 patients in VAC (median cumulative doses: cyclophosphamide=9.7g/m²,

ifosfamide=59.4g/m²) and 88 in VAI (ifosfamide=97.1g/m²). Long-term cardiac, renal and gonadal toxicities were prospectively collected. Cumulative probabilities of each toxicity were estimated using Kaplan-Meier method and compared between both arms by log-rank test.

Results: With a median follow-up of 10 years (range=5-17), 5 late relapses (range=5-13years), 5 second malignancies (range=6-9years) and 1 death as first-event were recorded resulting in 10-year event-free survival of 94% (95%CI=90-98), similar between VAC and VAI (93% vs. 95%, p=0.63). Overall, we estimated the 10-year cumulative probabilities of cardiac and renal toxicities at 4.5% (95%CI=1.1-7.7%) and 34.9% (95%CI=26.8-42.1%), respectively. Toxicity was moderate (cardiac: 73% grade-1, 27% grade-2; renal: 85% grade-1, 15% grade-2). Cardiac toxicity was similar in both arms, whereas probability of renal toxicity was higher in VAI (at 10 years, 43.0% vs. 25.7%, p=0.02), resulting from a significant difference in glomerular toxicity (31.1% vs. 13.1%, p<0.01) without significant difference in persistent tubular toxicity (p=0.55). Gonadal toxicity was observed in 19/66 explored post-pubertal men (29%) and in 19/67 women (28%), without significant difference between VAC and VAI.

Conclusions: Renal and gonadal toxicities represent major issues in SR-EWS survivors, with significantly higher nephrotoxicity in VAI arm compared to VAC, without significant gonadal toxicity reduction. This supports the need to explore late toxicity of VDC/IE regimen, with similar cumulative doses of cyclophosphamide and ifosfamide than VIDE/VAC.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0074 / #632 | Direct Correlation of MRI with Histopathology in Pediatric Renal Tumors Through the Use of A Patient-Specific 3D-Printed Cutting Guide: A Pilot Study

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Background and Aims: In pediatric renal tumors there is no histopathological assessment until after surgery. Imaging could play a crucial role in the pre-operative characterization of renal tumor types. Direct correlation of radiology and pathology could reveal imaging features in these frequently highly heterogeneous lesions. With this pilot study the feasibility of a patient-specific 3D-printed cutting guide is investigated to ensure corresponding orientation between magnetic resonance imaging (MRI) and histopathology.

Methods: Based on the preoperative MRI-scan prior to total nephrectomy, a 3D-model of the kidney and tumor was manually developed in

3D Slicer. This allowed us to create a cutting guide enabling an MRI-orientated transversal slicing of the specimen. The pathologist used the cutting guide for gross dissection of the specimen, ensuring macroscopic slices of 5 millimeters each. The feasibility of this technique was determined using structured measurements including overlap calculation with the dice similarity coefficient and qualitatively through questionnaires among involved experts.

Results: In this pilot, the cutting guide has been applied to seven patients with Wilms tumor. Their median age at diagnosis was 50 months (range 4-100 months). Digital correlation in a total of 28 slices resulted in a median dice similarity coefficient of 0.91 (range 0.53-0.97). The positioning and slicing of the specimen was overall rated as 'easy' and the median macroscopic slice thickness of each specimen ranged from 5-6 millimeters. Tumor consistency strongly influenced the practical application of the cutting guide.

Conclusions: A patient-specific 3D-printed cutting guide seems to provide a feasible tool for the direct correlation of MRI and histopathology of pediatric renal tumors. The clinical implementation and design of the cutting guide are currently optimized, to ensure a methodological workflow for future prospective studies.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0075 / #1207 | Usefulness of Navigation Surgery for Pediatric Neoplastic Diseases

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Background and Aims: During surgery for pediatric neoplastic diseases, an identification of small tumor lesions or vital neurovascular bundles is often difficult. In this study, we reviewed our experience of navigation surgery for these diseases.

Methods: From 2002 to 2020, 8 patients underwent navigation surgery. They included 1 with intrahepatic lesion and 2 with lung metastases of hepatoblastoma, 1 with recurrent thyroid medullary carcinoma (MTC), 1 with ectopic parathyroid in the mediastinum, and 3 with cervical lymphangioma.

Results: Fluoroscopy-aided resection with preoperative lipiodol-marking was performed for lung metastasis of hepatoblastoma. Although the pulmonary nodule was identified easily, it required CT-guided injection just before surgery under general anesthesia. From 2017, fluorescence-guided surgery using indocyanine green (ICG) fluorescence imaging system (PDE®) was introduced for hepatoblastoma and its lung metastasis. PDE clearly visualized the lesions in the liver or on the surface of the lung, except the lesion at a depth of over 1cm. Radio-guided surgery using gamma radiation detector (Navigator system GPS®) with preoperative radioisotope injection was performed for MTC (^{123}I -MIBG) and ectopic parathyroid ($^{99\text{m}}\text{Tc}$ -sestamibi), and

high scintillation counting spot was well noted at the lesion site. Nerve monitoring system (NIM®) was useful to preserve facial nerve during debulking surgery for cervical lymphangioma.

Conclusions: Navigation surgery for pediatric oncologic disorders are still developing, but promising technique. Intraoperative navigation using various modalities based on the physiological specificity of diseases makes surgery safer and curative.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0076 / #745 | Long-Term Results of Transpupillary Thermotherapy in Children with Retinoblastoma

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Background and Aims: Background: Retinoblastoma is the most frequent intraocular malignancy of childhood. Transpupillary thermotherapy is one of the main methods of small retinoblastoma treatment. However, only a few studies have been published on the use of transpupillary thermotherapy in children with retinoblastoma and these publications are devoted to the certain aspects of the application of this method. Aim: To evaluate the efficacy of transpupillary thermotherapy for retinoblastoma.

Methods: In a retrospective study, we reviewed case series of 177 children with retinoblastoma (1156 tumors in 224 eyes) treated with transpupillary thermotherapy at the S. Fyodorov Eye Microsurgery Federal State Institution in Moscow between October 2011 and December 2020. The median patient age at the time of treatment was 16.8 months. The mean tumor thickness was 1.1 mm (from 0.2 to 4.5), the mean basal diameter was 2.2 mm (from 0.3 to 13.4). Transpupillary thermotherapy was performed using a diode laser with the following parameters: wave length-810 nm, spot diameter-1200 microns, power from 200 to 800 mW (mean - 350 mW), exposure-from 3 to 15 s in the application mode, and continuous in the scanning mode.

Results: Overall tumor control was achieved in 92.7% (1072 tumors). Tumor recurrence was observed in 7% (82 tumors) and these tumors were treated with other local methods. 209 eyes (93%) were preserved. 15 eyes (7%) were enucleated due to tumor progression, total retinal detachment, vitreous hemorrhage or subatrophy of the eyeball. Complications after transpupillary thermotherapy included local vitreous hemorrhage (2 eyes; 0.9%), pretumoral hemorrhage (3 eyes, 1.3%), local vitreous opacities (2 eyes; 0.9%), iris atrophy and anterior synechiae (1 eye, 0.45%). The median follow-up was 35.5 months.

Conclusions: Transpupillary thermotherapy is an effective treatment for retinoblastoma with excellent local tumor control and few complications.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0077 / #782 | Gastrointestinal and Oncologic Outcomes of Pediatric Gastrointestinal Lymphoma Following Upfront Resection or Biopsy

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Background and Aims: Complete upfront resection of pediatric gastrointestinal lymphomas is recommended over biopsy whenever feasible, but either approach may have adverse sequelae. We sought to compare gastrointestinal and oncological outcomes of pediatric gastrointestinal lymphomas who underwent attempted upfront resection or biopsy of the presenting bowel mass.

Methods: We retrospectively reviewed charts of children with gastrointestinal non-Hodgkin lymphomas treated on LMB89 and LMB96 protocols at KK Women's and Children's Hospital from 2000–2019 who underwent upfront gastrointestinal surgery, and compared resection and biopsy groups.

Results: Of 33 children with abdominal lymphomas, 20 had upfront gastrointestinal surgery – 10 each had resection or biopsy. Most had Burkitt lymphoma (n=18 (90%)). The resection group included 6 hemicolectomies for irreducible intussusception, 2 ileal resections and 2 appendectomies. The biopsy group included 6 laparoscopic, 3 open and 1 percutaneous procedures. Patients with resections had fewer post-operative gastrointestinal complications compared to biopsies (10% vs. 60%, $p=0.057$), but significantly longer time to chemotherapy initiation (median 11.5 vs. 4.5 days, $p<0.001$), with only 3 resection patients successfully down-staged surgically. Despite this, relapse and survival outcomes did not differ significantly between biopsy and resection groups, though was better with LMB96 protocol with stage II/III disease. Second surgeries were required in 30% and 40% of resected and biopsied patients, respectively, at median 4.6 months, with all 4 patients in the biopsy group requiring resection and anastomosis at their previously diseased bowel segments for strictures or suspected residual disease.

Conclusions: Children with upfront resection had low rates of surgical downstaging and greater delay in chemotherapy initiation than those

with biopsies, but fewer gastrointestinal complications and subsequent surgeries. This suggests that while surgeons overestimate the likelihood of complete upfront disease clearance, the similar survival rates regardless of upfront surgery reflect the beneficial effects of newer protocols. Nevertheless, second surgeries are still frequently required, even after completion of therapy.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0078 / #480 | Adrenocortical Tumors in Children: Treatment and Outcomes

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Background and Aims: Adrenocortical tumors (ACT) are rare tumors in children accounting for 0.2% of all pediatric malignancies, with paucity of outcome data especially in resource challenged settings. The aim was to evaluate the presentation, management and outcome over 28 years.

Methods: A retrospective study was conducted to evaluate the presentation, management, and outcomes of children with adrenocortical tumors who presented from January 1992 to December 2019.

Results: During the study period, 28 children were included. Adenoma was present in 9 children and the remaining 19 children had adrenocortical carcinoma. The median age of presentation was 48 (3–156 months) months. Nearly half of the children with carcinomas presented with a mass compared to only 10% adenomas ($p=0.049$). Hypertension was the most common presenting sign. Virilization was present in 60% of the patients and in most cases was associated with Cushingoid features. Three children (15.7%) had metastasis at presentation. All children underwent surgical excision with a spill in 6 and gross residue in 2 patients. Recurrence was observed in 5 patients after a mean duration of 11.8 months. The 4-year overall survival and event-free survival were 100% in patients with adenomas compared to 75% and 44%, respectively, in carcinoma. On comparing the effect of spill on stage 1 and 2 tumors, only one of the four patients (25%) with a spill was disease-free whereas 9/11 (81%) without spill had disease-free survival ($p=0.07$). No patient with stage 3 or stage 4 was disease-free.

Conclusions: ACT are uncommonly encountered, with a better outcome noted in adenomas compared to carcinomas. There was no

disease-free survival in incompletely resected or metastatic tumors. Tumor spill may also have an adverse outcome in completely resected tumors.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0079 / #1105 | Early Postoperative Complications after Abdominal Solid Tumor Surgery in Childhood

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Background and Aims: Postoperative period after abdominal solid tumor surgery is critical regarding complications. This study aimed to detect incidence and treatment of complications.

Methods: Files were reviewed retrospectively for age, diagnosis, post-operative complications and management between the years 2010-2019 in our center. Complications before 30th postoperative day was assumed early and included. Intraoperative or late complications were excluded.

Results: Median age of 393 cases was 48 months (21-108). The overall complication rate was 10% and they were chylous leak (n=8, 2%), collection in surgical field (n=8, 2%), acute renal failure (n=7, 1.7%), hemorrhage (n=5, 1.3%), intestinal obstruction (n=4, 1%), surgical site infection (n=3, 0.7%), thromboembolism (n=3, 0.7%) and peripheric neuronal event (n=2, 0.5%). Complications were managed conservatively in 26 cases, with minimally invasive methods (stenting, percutaneous drainage) in 11 and by surgery in 3. Median hospital stay was 14 (12-25) days among patients with complications and 6 (5-7) days in others ($p<0.01$). Five patients received chemotherapy in surgical ward due to prolonged stay. Only postoperative mortality happened in a patient who had acute hepatic failure due to portal venous thrombosis. Median age and percentage of benign or malignant disease were not different in groups with or without complications ($p=0.11$, $p=0.24$). Males had more complications compared to females ($p=0.02$). Having more than one surgery aiming tumor resection was associated with increased postoperative complications ($p=0.002$). Incidence of complications were highest after extragonadal germ cell tumor (35%), hepatoblastoma (31%) and pancreatic pseudopapillary tumor (27%) resection. Hepatic (29%), pancreatic (25%) and renal (15.5%) tumors were associated with highest complication rate.

Conclusions: Postoperative complications of abdominal tumor surgery are diverse in children. They are related with prolonged hospital stay and need for medical or surgical interventions. Number of tumor surgery, gender, organ of origin and tumor type have an influence on risk of complications.

IPSO LIVE Q&A: IPSO SESSION 7: NEW TECHNOLOGIES & MISCELLANEOUS 22-10-2021 11:30 AM - 12:30 PM

O0080 / #989 | The Use of Pre-Operative Sympathetic Blockade in Patients with Adrenal Neuroendocrine Tumors and Paraganglioma: 15 Year Single Institution Experience

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Background and Aims: The World Health Organisation states that pheochromocytomas and paragangliomas are indistinguishable at a cellular level and defines pheochromocytomas as intra-adrenal paragangliomas. These are rare diagnoses and experience is limited for anaesthesiologists in the perioperative management of life threatening emergency surgeries from events such as hypertensive crises, tachyarrhythmias, cardiomyopathy and multiorgan failure. Our aim is to assess the need for pre-intervention initiation of sympathetic blockade based on tumour site, histology and catecholamine properties.

Methods: Records of 123 children who underwent surgical intervention for adrenal neuroendocrine tumors or paraganglioma from 2006-2021 were retrospectively reviewed. Data was accessed through electronic records, examining tumour histology and urinary catecholamine results, and medical chart review, examining pre-operative blood pressure, use of sympathetic blockade, and intraoperative haemodynamic stability. To date all electronic medical records and 60 medical charts have been reviewed.

Results: 117 patients had Neuroblastic tumors, 3 had paragangliomas, and 3 had pheochromocytomas. Of patients with Neuroblastic tumors, 2 patients were hypertensive pre-operatively. One received a sympathetic blockade and none developed intraoperative haemodynamic instability. Two patients received sympathetic blockade, despite being normotensive pre-operatively with one developing intraoperative hypotension. Only 1 patient with pheochromocytoma had pre-operative hypertension, and all were given preoperative sympathetic blockade. None had episodes of intraoperative haemodynamic instability. No patients with paraganglioma had preoperative hypertension and none received sympathetic blockade. All patients with pheochromocytoma and paraganglioma had elevated preoperative urinary catecholamines.

Conclusions: Current analysis has demonstrated low risk of hemodynamic instability relating to resection of neuroblastoma or paraganglioma. We see no reason to advise for routine sympathetic blockade for surgical intervention in cases of neuroblastoma. There is much evidence of the risk associated with intra-adrenal paraganglioma. Based on this we would advise pre-interventional sympathetic blockade in line with local guidelines. For all other paragangliomas we recommend no routine use of sympathetic blockade prior to surgery.

Nursing FREE PAPER SESSION 1-PSYCHOSOCIAL CARE AND ETHICS 22-10-2021 2:30 PM - 4:00 PM

O0082 / #612 | Efficacy of a Mobile-Based Resilience Training Programme on Resilience, Depressive Symptoms and Quality of Life Among Parents of Children with Cancer: A Randomised Controlled Trial

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Background and Aims: Caring for children with cancer can be a stressful experience for parents and may have negative impacts on their physical and psychological well-being. Although previous studies have shown that resilience was associated with positive psychological well-being, there is a paucity of interventions designed to enhance the resilience of parents of children with cancer. This study aimed to examine the effectiveness of a mobile-based resilience training programme in reducing depressive symptoms and enhancing resilience and quality of life among parents of children with cancer.

Methods: A two-arm parallel-group randomised controlled trial was conducted in three tertiary hospitals in China. Parents of children who received a diagnosis of cancer within the past year were invited to participate in the study. The participants were randomly assigned either to the experimental group (n=52) to undergo an 8-week mobile-based resilience training programme, or to the control group (n=51) to receive an 8-week placebo intervention. Participants' resilience, depressive symptoms and quality of life were assessed at baseline and 2 and 6 months after starting the intervention.

Results: The participants in the experimental group showed statistically significant higher levels of resilience (67.96 ± 15.8 vs 58.27 ± 19.0 ; $P < 0.001$) and lower levels of depressive symptoms (40.17 ± 9.9 vs 46.04 ± 10.9 ; $P < 0.001$) than those in the control group at 6 months of follow-up. The intervention showed statistically significant effects in improving resilience ($\beta = 6.082$; $P = 0.010$) and decreasing depressive symptoms ($\beta = -2.772$; $P = 0.041$) relative to the control group. While no statistically significant intervention effect was detected for quality of life ($\beta = 0.020$; $P = 0.383$).

Conclusions: The mobile-based resilience training programme effectively enhanced resilience and alleviated depressive symptoms in parents of children with cancer. It is highly recommended that healthcare professionals incorporate this resilience training programme when providing psychological care for parents of children with cancer.

Nursing FREE PAPER SESSION 1-PSYCHOSOCIAL CARE AND ETHICS 22-10-2021 2:30 PM - 4:00 PM

O0083 / #1410 | The Hematology/Oncology/Stem Cell Transplant Advancing Resiliency Team (HART): Promoting Resiliency While Supporting a Healthy Work Environment

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Background and Aims: Caring for pediatric hematology, oncology, and HSCT patients is emotionally and physically taxing on the interdisciplinary team. Peer support reduces isolation, increases connection, builds cohesion, and instills hope in staff (Yalom & Leszcz, 2005). The Hematology/Oncology/Stem Cell Transplant Advancing Resiliency Team or (HART) program at Boston Children's Hospital (BCH) was developed in 2019 to enhance staff resilience, create a healthier work environment, and improve team morale by offering an easily accessible on-site program led by five nurse coaches available twenty-four hours a week through a real time nurse-driven, peer-to-peer education and support system.

Methods: Since inception, electronic surveys were sent via email to staff on the Hematology/Oncology/Stem Cell Transplant units prior to the start of the HART program to explore individuals' perceptions of resiliency strategies and again at six-months and twelve-months to evaluate the programs' effectiveness.

Results: As of March, 2021, there have been 162 HART shifts worked and over 2000 encounters recorded. A HART coach encounter is categorized into work-related, patient-care based, integrative therapy, and or personal concern. Registered nurses make up the largest discipline utilizing HART services, followed by clinical assistants and patient experience representatives with the majority of encounters being work-related 42% (n=621).

Conclusions: Staff reported "feeling more connected to co-workers", "making time for breaks during a shift", and "feeling more supported by leadership" in the one-year survey. Staff feedback indicates that peer-to-peer support connects our team members through facilitation of resources, conversations and integrative therapy offered by HART.

Nursing FREE PAPER SESSION 1-PSYCHOSOCIAL CARE AND ETHICS 22-10-2021 2:30 PM - 4:00 PM

O0084 / #309 | Visualize The Invisible-Needs and Wishes for Emotional Support in Survivors of Childhood Cancer

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Background and Aims: This study aims to identify the needs and wishes for emotional support among survivors of childhood cancer.

Methods: Survivors of childhood cancer (seven men; seven women), aged 19–33 years, participated in semi-structured interviews

(November–December 2020). The ages when they were diagnosed were 1–12 years. The interviews lasted between 45–85 minutes. A thematic analysis was used to identify themes.

Results: Needs Processing late effects of cancer treatment The participants described visible and invisible as well as physical and psychological late effects and reported that they found these to be difficult to process with people who had not had similar experiences. Processing psychosocial health The participants cited psychosocial issues, such as anxiety and social phobia, and the lack of appropriate forums for addressing these issues. Meeting others who share similar experiences The participants conveyed a need to meet others who had had similar experiences and who would be able to provide genuine understanding and support through the experiences that they had lived through. Wishes Digital tools for connecting with people who had had similar experiences The participants expressed desires to connect with people who had had similar experiences in terms of issues such as diagnosis and late effects, with the help of digital tools. Different forms of communication Certain formats of communication facilitate trust in others. Communication via text and video conferencing meet different needs, such as those based on personality and topics for discussion. A safe place with varying degrees of anonymity The need for feelings of security was expressed as a prerequisite for communication, including the option to remain anonymous.

Conclusions: This study emphasized the needs and wishes of childhood cancer survivors to meet others who had had similar experiences in digital spaces that offer a secure place, with options for a variety of methods of communication and levels of anonymity.

Nursing FREE PAPER SESSION 1-PSYCHOSOCIAL CARE AND ETHICS 22-10-2021 2:30 PM - 4:00 PM

O0085 / #578 | Morally Difficult Situations and Questions Raised During Moral Case Deliberations - A Swedish Nationwide Overview from Paediatric Oncology

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Background and Aims: Healthcare professionals are daily facing various moral challenges. Moral case deliberation (MCD) is a method of ethics support with the goal to assist healthcare professionals to deal with morally challenging situations. In Sweden, MCDs are performed in the care team at all paediatric oncology centres, and are led by healthcare professionals trained to facilitate MCDs. The aim of this study was to describe different kinds of morally difficult situations and the following questions raised during MCDs in Swedish childhood cancer care.

Methods: Data was collected by using facilitator-questionnaires from MCDs (n=66), performed at all six paediatric oncology centres between 2017 and 2020. The questionnaires were completed after each MCD by the facilitator (n=15) and included a) information about

the difficult situation, and b) the moral question raised. Data was analysed in accordance with qualitative systematic text condensation.

Results: The different kinds of difficult situations, preceding the moral questions, were; medical-, social -and cultural situations. Themes of moral questions were: What is the right care action? How much and when is it right to let the child decide? and How much and when is it right to let parents influence care? One example of a moral question in the last theme was: 'Should we force parents to accept treatment or not?'

Conclusions: This study contributes to an increased understanding of what kinds of difficult situations and moral questions that were raised during MCDs in Swedish paediatric oncology. All moral questions relate to shared decision-making. The results will be useful in paediatric oncology training and as a foundation for future studies of moral questions and ethics support in general.

Nursing FREE PAPER SESSION 1-PSYCHOSOCIAL CARE AND ETHICS 22-10-2021 2:30 PM - 4:00 PM

O0086 / #585 | The Role of Psychology in Reducing Distress at the Transition off Directed Cancer Treatment

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Background and Aims: The Standards for Psychosocial Care for Children with Cancer and Their Families emphasize the importance of assessment/consultation throughout the cancer trajectory, yet little is known about the adjustment of families at the transition off therapy and the impact of psychosocial and cognitive assessment/consultation in promoting positive adjustment. Thus, the objectives of this research were to: (1) characterize distress related to transition, (2) examine differences in reports of receiving information and feeling prepared for transition between families who did and did not receive psychological and cognitive assessment/consultation, (3) explore links between feelings of preparedness and distress ratings.

Methods: Caregivers (n=65; 90% white; 94% female) of children who completed active cancer treatment responded to a questionnaire assessing distress, feelings of preparedness, and feeling informed surrounding the transition. Caregivers also reported on social, emotional, and cognitive/academic concerns for their child and emotional concerns for family members. Frequency analyses were used for objective 1. Objective 2 was explored using ANOVAs, with receipt of psychological consultation and cognitive assessment examined separately. An ANOVA and correlation analyses were used for objective 3.

Results: Nearly 23% completed a transition-related psychology consult and 44% completed a cognitive assessment. Overwhelmingly, caregivers "agreed" or "strongly agreed" with feeling an array of difficult emotions (65–85%) and family struggles (52%) related to the

transition. Patient emotional concerns (51%) and school problems (41%) were also endorsed. Caregivers of patients who completed a consultation/cognitive assessment were significantly more likely to report being informed, able to navigate support, and prepared to manage difficulties. Feelings of preparedness were associated with decreased caregiver, patient, and family distress.

Conclusions: Though ending active cancer treatment is a point of celebration for families, it is also marked with distress. Providing psychoeducation, emotional support, and cognitive/academic guidance may increase caregiver efficacy to manage the transition and decrease distress.

IPSO LIVE Q&A: IPSO SESSION 8: MISCELLANEOUS 22-10-2021
3:00 PM - 3:45 PM

O0087 / #1611 | Randomized Comparative Study of Fixation of Tunneled Catheter with Biological Glue in Pediatric Oncology Patients

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Background and Aims: A peripherally inserted central venous catheter (PICC) and centrally inserted central venous catheter (CICC) are used in several situations in clinical practice, including for the infusion of chemotherapeutic antineoplastic drugs. In pediatric oncology, these catheters have been increasingly used as they are easy to handle, allow insertion even in clotting or platelet disorder, and can have more than one route. These also allow blood draws for testing, which brings comfort to the patient. The benefit of sutureless devices and catheter tunneling has been demonstrated in recent years. Additionally, studies have been published on the use of biological glue for catheter fixation, which reduces extraluminal infection by sealing the catheter entry point in the skin. However, this benefit has not been explored in comparative studies on tunneled catheters fixed with sutureless devices. Therefore, we compared the use of PICC and CICC fixed with or without cyanoacrylate glue in an oncologic pediatric population.

Methods: It is a prospective, longitudinal, randomized study on patients undergoing tunneled PICC and CICC implantation at a Pediatric Oncology Hospital using the ultrasonography-guided Seldinger micropuncture technique with posterior tunneling and fixation with a Statlock® stabilization device. The study included 50 patients who were equally divided into two groups.

Results: The mean age was 83.68 months in the glueless group and 84.09 months in the glue group. No differences were observed between groups regarding patient age, the type of hospitalization, implanted vein, implantation side, catheter survival, and need for platelet transfusion. The mean duration of catheter implantation was 90 and 96.52 days in the glueless and glue groups, respectively.

Conclusions: PICC and CICC catheters fixation with glue showed no difference regarding the duration of use, catheter survival, or infection and traction rates.

IPSO LIVE Q&A: IPSO SESSION 8: MISCELLANEOUS 22-10-2021
3:00 PM - 3:45 PM

O0088 / #634 | Fluoroscopic Assessment of Central Venous Catheter Tip Position in Children

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Background and Aims: Central venous catheters (CVCs) are commonly required in oncological practice, and are a major source of morbidity associated with treatment. Obtaining optimal CVC tip position ensures adequate catheter function whilst minimising the risk of complications. In paediatric practice, tip position is typically assessed fluoroscopically at the time of insertion. This study aimed to determine the reliability of this method.

Methods: The intra-operative fluoroscopic images of one hundred and two consecutive CVC insertions in a single paediatric surgical centre were independently reported by the clinician performing the procedure and by two radiologists. Each reporter described CVC tip position using the following categories: superior vena cava (SVC), cavoatrial junction (CAJ), or right atrium (RA). Inter-observer reliability was assessed using Cohen's kappa test, and fluoroscopic reports were compared to subsequent computed tomography (CT) where available.

Results: Inter-observer reliability between clinicians and radiologists was poor ($\kappa = 0.13$). Inter-observer reliability between radiologists was significantly better ($p < 0.01$), but was still only moderate ($\kappa = 0.53$). Radiologists were significantly more likely to report a proximal tip location (SVC, CAJ) than clinicians, who were more likely to report a RA position ($p < 0.01$). Tip position on CT was significantly more likely to correlate with the radiologists' interpretation of the fluoroscopic images ($p < 0.05$).

Conclusions: The inter-observer reliability of fluoroscopic assessment of CVC tip position in children is, at best, moderate. In this study, clinicians inserting CVCs were significantly more likely to over-estimate tip insertion length than radiologists when compared to subsequent CT – and were therefore likely to insert CVCs more proximal than intended. Clinicians inserting CVCs should be aware that the CAJ is typically distal to commonly used radiological landmarks, such as the lower border of the right main bronchus.

IPSO LIVE Q&A: IPSO SESSION 8: MISCELLANEOUS 22-10-2021
3:00 PM - 3:45 PM

O0089 / #1541 | Surgical Treatment of Invasive Fungal Infection in Pediatric Oncology

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Background and Aims: Invasive fungal disease (IFD) affects primarily patients with a compromised immune system, mainly children with a hematological malignancy. This type of infection is often aggressive and related with high mortality, despite the improvement of medical treatment. New evidences report that the association of medical treatment and surgery may ameliorate the outcome. The aim of this study was to analyze the surgical experience of a single-center in the treatment of fungal infections in the pediatric oncological populations.

Methods: Retrospective study (2000 to 2020) of a single-center experience. We reviewed the medical record of all patients with hematologic malignancies. We analyzed presentation and diagnosis of IFD, medical treatment, indication for surgery, type of surgical treatment and outcomes including complications (rated according to Clavien Dindo Classification), eventual delay in the resume of the oncological treatment, resolution of infection, relapse and survival.

Results: Forty patients (19M, 21F), median age 8 years (0-17) at the diagnosis of IFD. The main indications for surgery were no benefits from medical treatment and advance of fungal infection (67%), the need to confirm the diagnosis (12.5%), and only two patients needed an emergency operation for rapid worsening of symptoms. The most common site of infection was the lung (80%), followed by maxillofacial region (17.5%) and the most frequently identified pathogen was *Aspergillus* (25%). The most common procedures were lobectomy (performed in 16 patients) and lung resection (8). Complications were mostly treated by medical approach. Only in 7 cases mycoses were persistent after surgery. The median time of resumption oncological treatment was 40 days (range 0-150). Reason of death was predominantly related to refractory malignancy (18).

Conclusions: Surgery is a feasible tool in the multimodal treatment of invasive fungal infection in pediatric oncological patients, with excellent resolution rate. Overall mortality depends on the underlying malignancy.

IPSO LIVE Q&A: IPSO SESSION 8: MISCELLANEOUS 22-10-2021
3:00 PM - 3:45 PM

00090 / #992 | Implementation of a Protocol for the Management of Perianal Soft Tissue Infections in Pediatric Patients with Oncologic Diagnosis

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Background and Aims: Perianal soft tissue infection (PSTI) in pediatric patients with cancer is a common infectious complication with high associated morbidity and mortality. After different therapeutic approaches across the years, in 2017, we designed a management protocol for PSTI based on a severity classification we created and the primary treatment with rectal povidone-iodine irrigations.

Methods: We conducted a case series study on patients 0-15 years, who had an oncologic diagnosis and associated PSTI, managed by the Pediatric Oncology Surgery Service at Central Military Hospital. This new protocol aims to avoid the morbidity associated with debridement and colostomy in oncologic children. Included patients from June 2017 to December 2019 underwent evaluation, severity classification and management with povidone-iodine rectal irrigations. Duration of symptoms, chemotherapy regimen, need for surgical management and related deaths were considered.

Results: Data were collected from 30 patients with cancer and associated PSTI. Primary diagnosis was hematopoietic malignancies (73.3%, n 22). We found a higher frequency of PSTI in the presence of neutropenia (63.3%, n 19). A more remarkable presentation of PSTI was observed with the use of Vincristine, Doxorubicin and Asparaginase. Time to resolution of the infection was between 6-10 days according to the severity classification. Debridement and colostomy were performed on just one patient (3.3%). During the studied period there were no related deaths.

Conclusions: PSTI in pediatric oncologic patients represents a severe entity that can quickly progress, leading to death. Debridement and colostomy represent standard management, with considerably associated morbidity. We propose a clinical severity classification for PSTI, which can guide treatment (medical vs. surgical). We also expose our experience using povidone-iodine rectal irrigations and strict clinical surveillance, avoiding surgery in most cases and resolving symptoms without increase in morbimortality. We consider this is a novel and effective treatment option in the management of oncologic children with PSTI.

IPSO LIVE Q&A: IPSO SESSION 8: MISCELLANEOUS 22-10-2021
3:00 PM - 3:45 PM

00091 / #1771 | Pediatric Surgical Oncology as Dedicated Specialty to Consolidate Multidisciplinary Childhood Cancer Care in Panama

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Background and Aims: From 2018, upon the return of the first fellowship-trained pediatric surgical oncologist, the multidisciplinary care of children and adolescents with cancer was reorganized at the major referral public hospital in Panama City. Although our case series is small, this report aims to show the impact of such reorganization as a model of institutional effectiveness

Methods: Review of the implementation of standardized surgical protocols at the José Renán Esquivel Hospital (HDNDJRE) from March 2018 to March 2020 as a new routine, reinforced with the development of two surgical sessions in July 2018 and August 2019 with the participation of international expert surgeons from Mexico and Poland.

Results: In 2018, Five renal tumors were operated after neoadjuvant chemotherapy, and they received postoperative radiation therapy. Four survived, and one died with disease progression, atrial thrombus, and extension to the pulmonary artery. During the 2018 session, an adolescent with primary rhabdomyosarcoma of the breast who received neoadjuvant therapy underwent radical mastectomy with axillary lymph node dissection. Two other patients were operated in that session after neoadjuvant chemotherapy, one with thoracic Ewing's sarcoma, and one with a massive retroperitoneal rhabdomyosarcoma, and two complex benign tumors, a cervicothoracic lipoblastomatosis and a patient with a massive pelvic desmoid tumor. During the 2019 session, a subtotal scapulectomy for a patient with congenital fibrosarcoma, that failed neoadjuvant chemotherapy and responded to immunotherapy, one massive abdominal neurofibroma with the involvement of the hepatic hilum, and recurrent mediastinal lymphatic malformation that received sclerotherapy with direct vision through thoracoscopy.

Conclusions: The introduction of oncological surgery emphasizing molecular biology in individualized treatment allowed for more organized and knowledgeable personnel. The application of new procedures and treatments grow the arsenal that institutions can use to battle pediatric cancer. Also, the cooperation of international entities dramatically aids in the progress of less developed countries.

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0092 / #363 | Brainstem Toxicity After Posterior Fossa Proton Radiation Therapy in the Prospective Multicenter Pediatric Proton/Photon Consortium Registry (PPCR)

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Background and Aims: Controversy exists as to whether proton radiotherapy (PRT) has been associated with higher rates of symptomatic brainstem toxicity than photons in pediatric brain tumors. We aim to report the incidence, severity, risk factors, and management of PRT-related symptomatic brainstem toxicity in prospectively enrolled patients on the multicenter Pediatric Proton/Photon Consortium Registry (PPCR).

Methods: We evaluated the clinical data and PRT plans from 11 PPCR institutions on <22 years-old patients treated with >50Gy_{RBE} for posterior fossa (PF) primary brain tumors, censoring/excluding patients requiring PF re-irradiation. Brainstem contours were standardized and plans scored as meeting or not the suggested brainstem goal/maximal constraints outline in Indelicato et al. Acta Onc, 2014 paper: D_{max/0.1cc} 56.6/58 Gy_{RBE}, D_{10%} 55.4/56Gy_{RBE}, D_{50%} 52.4/54 Gy_{RBE}.

Results: 502 patients treated with PRT from 2012-2020 were reviewed. Median follow-up was 3.3 years (0.11-9.3), age at PRT 6.6 (0.47-21.9), and dose prescribed 54 Gy_{RBE} (non-toxicity: 51-59.4; toxicity: 54.0-59.4). Cumulative incidence of brainstem toxicity at 3/5 years was 1.3%. Of the five (1.0%) with brainstem toxicity (two grade 2, three grade 3), 3 had ependymoma, 2 medulloblastoma, 3 were male, 3 received standard-dose chemotherapy, and 2 had PF syndrome. In the affected patients, age at PRT was 4.1 years (1.9-8.9) and 4.5 years (2.5-10.0) at the time of brainstem toxicity, which occurred 4.8 months (3.9-12.3) after PRT. All required medical intervention: steroids (n=4), bevacizumab (n=3), hyperbaric oxygen (n=1), and blood transfusion (n=1, for sickle cell anemia); four improved, two of them resolving with treatment. Median Karnofsky-score was 80% (50-100) before PRT and at the latest follow-up. Only 40% (n=2) met the suggested proton brainstem constraints, while 75.7% of the non-toxicity patients did (p=0.10).

Conclusions: Symptomatic brainstem toxicity is rare in PPCR-enrolled patients treated with proton radiotherapy and is comparable to other proton or photon cohorts. Most children with brainstem toxicity improve with standard medical intervention.

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0093 / #1368 | Development and Validation of a Delineation Atlas of Neurovascular Structures

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Background and Aims: Radiation-induced neurovascular late-effects in pediatric patients with brain tumor are currently being investigated in the HARMONIC prospective cohort study (<https://harmonicproject.eu/>). In this project, dose/volume effects to relevant neurovascular structures will be studied. We therefore developed an atlas for these structures, and explored if doses to the suprasellar cistern (SC) could be used as a surrogate for the Circle of Willis (CW) doses.

Methods: Neuroradiologists developed an initial atlas of the CW, its 10 individual arteries and the SC. A single observer applied this atlas to thirty proton therapy plans from pediatric patients with brain tumors to delineate the neurovascular structures, and retrieved their associated dose metrics. The agreement between doses to the CW vs. SC was specifically investigated. The neurovascular structures were also contoured in three test patients by three independent observers, and the differences were assessed visually as well as dose-wise (Dmean/Dmax). This final step was used for refining the contouring guidelines.

Results: For the thirty validation patients, there was an excellent agreement between the Dmax/Dmean to the CW and the SC with a linear coefficient of determination R^2 value of 0.99 for both metrics. Examining inter-observer variation, visual assessment showed good spatial correlation between all individual arteries for all three patients, especially in the cranio-caudal extent. This translated into similar doses to the CW, with a Dmean (in percent prescription dose) of 69.4%/68.1%/68.3% (patient 1), 19.8%/18.9%/17.8% (patient 2) and 4.7%/5%/3.7% (patient 3).

Conclusions: Our data indicate that the doses to the SC can be used as a surrogate for the dose to the CW. Following the atlas, all individual arteries could be located with high precision by the three independent observers. This knowledge could guide the treatment planning process in an effort to spare neurovascular structures.

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0094 / #1052 | Patterns of Failure in Children, Adolescents and Young Adults with Primary Spinal High-Grade Gliomas

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Background and Aims: Primary spinal high-grade gliomas(S-HGG) are rare aggressive tumors with dismal prognosis. They are managed similarly to intracranial glioblastomas, but outcomes in young patients are not well-described. We conducted a retrospective review to study the clinical features of S-HGGs in children, adolescents and young adults.

Methods: S-HGG patients < 29 years of age who received radiation therapy (RT) at our center from 2007-2019 were analyzed for patient, tumor, and treatment characteristics. Kaplan-Meier estimate and Cox proportional hazard regression method were used for survival analysis.

Results: A total of 18 patients were identified with a median age of 19.4 years (range 0.7-28.4y). All had a pathological diagnosis with grade 3 tumor in 4 patients and grade 4 in the remainder. The thoracic cord was the most common site of involvement. Three patients had a gross total resection and 15 underwent sub-total resection/biopsy. Thirteen patients received photon-based radiation (IMRT = 5, 3-D CRT= 8) and five received proton therapy. Median RT Dose was 50.4 Gy (range 43-54 Gy) with 88% receiving > 45 Gy and two patients receiving craniospinal irradiation. 88% patients received concurrent and/or adjuvant chemotherapy, most commonly temozolomide. 89% of patients had recurrence of which eleven (61%) relapsed locally, six (33%) relapsed in brain and five (28%) developed leptomeningeal disease. Bevacizumab was the most common systemic treatment at recurrence. Median OS from diagnosis was 18.2 months (range 9-105 m) and median PFS after RT was 8.4 months (range 3-73 m).

Conclusions: Patients with S-HGGs have a poor prognosis with over a third of the patients developing distant recurrence despite chemoradiation. Although majority of patients had gross disease at RT and received a lower RT dose than cerebral high-grade gliomas, only 61% patients failed locally. A better understanding of the molecular drivers of S-HGGs is needed to develop more effective treatment options and is the subject of ongoing analysis.

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0095 / #1087 | Proton Therapy in the Treatment of Non Metastatic Intracranial Germinoma: Early Result

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Background and Aims: Proton beam therapy (PBT) has the potential to achieve long term local control with reduced radiation-related toxicity in patients with non metastatic intracranial germinoma (GCT). However, the available data is scarce

Methods: Data from the Prospectively registry study "KiProReg" on patients with localized CNS GCT treated with PBT between 2015 and 2020 was analyzed (DRKS00005363). Side effects were documented according to CTCAE v4.0. Overall survival, local control and toxicity were analyzed

Results: 22 patients were included in this analysis (15 male/7 female, median age 12 years (range, 6-16 years). Tumor site was suprasellar, pineal or bifocal in 54%, 23%, and 23% of patients, respectively. All patients had received previous chemotherapy according to the trial SIOP CNS GCT II, except one patient. 23% of patients had underwent ventriculostomy, or 9% partial resection. In five patients whole ventricle therapy (WV1, 24 Gy) only was applied due to complete response after chemotherapy, whereas in 17 patients an additional boost to the tumor bed up to 40 Gy. was performed. The median dose to selected critical organs was: hippocampus 27.2 Gy (range, 24.6-34.7), thalamus 27.8 Gy (range 23.7-39.4), temporal lobe 13.9 Gy (range, 14.3-20.8), hypothalamus 31.7 Gy (range, 23.7-40.2), pituitary gland: 32.2 Gy (range, 23.9-40.1), cerebellum 13 Gy (range,10.3-18.6).The median follow up time (FU) from diagnosis to last contact was 24.5 months (range, 7-40 months). The treatment was well tolerated. Until now, no acute or late toxicity $\geq$ grade 3 was observed. All patients were alive at last FU without local relapse. Metastases was reported in only 1 patient (in left occipital dura) 18 months after the end of PBT.

Conclusions: Early results display good feasibility of PBT and excellent tumor control in patients with non metastatic CNS GCT. Still, longer FU and larger cohorts are necessary to verify efficacy, feasibility and particularly long-term outcome

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0096 / #1464 | Craniospinal Irradiation using Robust Multi-Isocentric Volumetric Modulated ARC Radiotherapy: Initial Clinical Experience

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Background and Aims: To evaluate the clinical experience of the first patient cohort treated with a robust multi-isocentric volumetric modulated arc radiotherapy (RMI-VMAT) technique for craniospinal irradiation (CSI).

Methods: Twenty patients were treated with RMI-VMAT for the first phase of craniospinal irradiation. Two or three isocenters were used depending on the length of the combined CSI planning target volume (PTV). Robust min-max optimization was used to control the impact of patient set-up errors in the superior-inferior direction on the overlap from adjacent isocenters. Partial arcs were used at each isocentre to avoid treating through highly attenuating support structures and the patient's arms and shoulders. The second phase was delivered using standard single arc VMAT. Post-treatment follow-up was performed until 8th Jan 2021, with progression free survival defined as the interval from end of treatment to the most recent MRI available.

Results: Patients had a median age at diagnosis of 8 years (range, 3 - 32), and 19/20 were pediatric or young adults. Sixteen patients were diagnosed with medulloblastoma, six of whom were metastatic. Eleven patients required feeding assistance, either prophylactically (2/11) or due to nausea and vomiting (4/11), posterior fossa syndrome (3/11), or mucositis (2/11). Reported acute toxicities at Grade 3+ were: decreased WBC (16/20), neutropenia (13/20), reduced platelet count (8/20), nausea (5/20), vomiting (2/20), oral mucositis (1/20), and pharyngeal mucositis (1/20). Three patients reported Grade 4 neutropenia or decreased white blood cell count (WBC), one of whom reported both. One patient was lost to post-treatment follow-up overseas. Median overall survival was 399 days (range 175 - 756 days), with one patient deceased after 508 days. Median progression free survival was 131 days (range 34 - 490).

Conclusions: RMI-VMAT for CSI can be implemented safely with an acute side effect profile similar to experience with conventional field-based photon techniques.

PROS LIVE Q&A: FREE PAPER SESSION 2 22-10-2021 3:15 PM - 4:15 PM

O0097 / #768 | Treatment Outcome of Paediatric Medulloblastoma - A Single Institution Experience in Sabah Women and Children's Hospital, East Malaysia

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Background and Aims: Treatment outcome of paediatric medulloblastoma has improved with advancement of treatment modalities, especially in low and middle income countries. This study aimed to present the demographics and treatment outcome of paediatric medulloblastoma at Sabah Women and Children's Hospital in East Malaysia.

Methods: This is a 5-year retrospective study. Twelve cases of paediatric medulloblastoma from 2015 to 2019 were identified. However, two were excluded because their parents refused treatment after surgery and one was excluded for missing records. The data was analysed using SPSS version 22.

Results: The median age at diagnosis was 7.5 years (range: 4.3-13.2 years). The male-to-female ratio is 1.25:1. Headache (77.8%), vomiting (77.8%), and ataxia (66.7%) were the most common presenting symptoms. The median interval between onset of symptom and diagnosis is 7 weeks (range: 4-24 weeks). Four children had M0 disease (44.4%), three had M2 disease (33.3%), and two had M3 disease (22.2%). All had post-operative residual tumour volume of $>1.5\text{cm}^3$ and were categorized into high-risk group. Histopathological examination report showed classic histology subtype in all the children. Five children received radiotherapy within ≤ 5 weeks of surgical resection (55.6%); the remaining four received radiotherapy within longer interval (range: 7-10 weeks). They received craniospinal irradiation plus tumour bed boost, with additional spinal boost for those with spinal involvement. This was followed by chemotherapy with CH 455 Packer Protocol. As of April 2021, all children are still living and none has relapsed. The mean duration of survival is 3 years (range: 1.3-5.5 years). Hearing impairment and hypothyroidism were each reported in 22.2% ($n=2$) of the cases respectively.

Conclusions: Our study showed good treatment outcome in children treated for medulloblastoma. However further follow-up is needed to assess the progression-free survival and overall survival of these children as well as the rate of long term complications.

Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM

O0098 / #152 | Clinical Study of MAP2K1-Mutated Langerhans Cell Histiocytosis in Children

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Background and Aims: To analyze the genetic and clinical features of children with MAP2K1-mutated Langerhans cell histiocytosis (LCH).

Methods: The clinical data of patients with MAP2K1-mutated LCH from July 2017 to October 2020 were collected and

compared with all patients with either the BRAF^{V600E} mutation or no known mutation of the MAPK pathway at the same period.

Results: Thirteen mutations of the MAP2K1 gene were detected in 37 patients with LCH. The median age of the patients at disease onset was 4.56 (1.54, 9.42) years, and the median follow-up time was 17.67 (4.82, 24.92) months. The mutations of MAP2K1 were mainly concentrated at p.53-62 and p.98-103, and the most common mutation site was c.172_186del (32.4%). The BRAF^{V600E} mutation group and the no known mutation of the MAPK pathway group included 133 and 59 patients, respectively. Compared with the BRAF^{V600E} mutation group, the MAP2K1 mutation group had an older age of disease onset, a lower incidence of multisystem involvement, a higher incidence of single system multiple bone involvement ($P<0.05$). There was no significant difference in clinical features compared with the no known mutation group. The 2-year progression-free survival (PFS) rate of first-line treatment in MAP2K1-mutated patients was $65.6\% \pm 9.5\%$, without a significant difference among the three groups. The 2-year PFS of second-line chemotherapy in the MAP2K1 mutation group was $35.7\% \pm 26.7\%$, which was close to that of the BRAF^{V600E} group ($P>0.05$) but significantly lower than that of the no known mutation group ($P=0.011$). Lung involvement was an independent risk factor in patients with MAP2K1 mutations [HR (95% CI) = 6.312 (1.769-22.526), $P=0.005$].

Conclusions: Compared with the BRAF^{V600E} mutation group, the patients with MAP2K1 mutations were mainly characterized by single system multiple bone involvement, with later disease onset and less involvement of risk organs. The prognosis of MAP2K1-mutated patients with lung involvement may be poor.

Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM

O0099 / #281 | Childhood Cancer Survival Gap by Health Insurance Type in Colombia: A Report from Vigicancer Surveillance System

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Background and Aims: Since 1993, Colombia, a middle-income country, shifted to a universal healthcare system with current coverage of more than 95% of its inhabitants. Herein, we describe pediatric cancer survival gaps by health insurance type in nine large Colombian cities.

Methods: We included prospectively collected data of children (<15 years) from VIGICANCER (Childhood Cancer Outcomes Surveillance System) from nine cities with a mean population of 1.8 million inhabitants (range: 0.35 to 7.7 million). We grouped cities with >100 cases/year (3 cities) or <100 cases/year (6 cities). We used Kaplan-Meier for survival analyses and Cox regression to adjust the hazard ratios (aHR) by covariates (age, sex, ethnicity, residence, city group, and tumoral group).

Results: From 2013 to 2019, 2431 children were registered in the 9 cities. Patients' median age was 6 years, 56% were male, 36% resided in a province capital, and 8% were afro-Colombians. Ninety-six percent of patients had health insurance coverage. Insurance type was semi-private in 40%, public in 51%, and other (military/police/teachers) in 5%. Cohort 5-year overall survival (OS) was 58% (95% CI: 56, 61); 61% for hematological malignancies, 46% for brain tumors, and 58% for extra-cranial solid tumors. Semi-private vs. public OS gap at 2, 3, and 5 years was 6%, 8%, and 11%, respectively; $p < 0.001$. The aHR of mortality for public compared to semi-private insurance was 1.4 (95% CI: 1.2, 1.7).

Conclusions: We show that the pediatric cancer OS gap by health insurance type in Colombia is a widespread phenomenon despite city size and region. Determinants of pediatric cancer survival are complex and insurance coverage plays a pivotal role. Although achieving universal health coverage is an immense public health advancement, it seems that is not sufficient to improve outcomes and additional measures must be taken to address pediatric cancer survival gaps.

Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM

O0100 / #257 | Urinary 3-Methoxytyramine is a Biomarker for MYC Activity in Patients with Neuroblastoma

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Background and Aims: Elevated urinary 3-methoxytyramine (3MT) level at diagnosis was recently put forward as an independent risk factor for poor prognosis in neuroblastoma. Here, we investigated the biological basis underlying the putative association between elevated 3MT levels and poor prognosis.

Methods: Urinary 3MT levels and prognosis were investigated in both retrospective Italian (n=90) and prospective Dutch (n=95) cohorts. From the previously published retrospective Dutch cohort (n=301), patients with matched urinary 3MT measurements and gene expression data were used to generate a 3MT gene signature. The 3MT gene signature score was then used to predict survival outcome in the Children's Oncology Group (n=247) and the German Pediatric Oncology Group (n=498) and compared to other known gene signatures. Immunohistochemistry of MYCN and dopamine β -hydroxylase (DBH) proteins was performed on primary tumors.

Results: Elevated urinary 3MT levels were associated with poor prognosis in both retrospective Italian and prospective Dutch cohorts. Moreover, elevated urinary 3MT levels were associated with eight differentially expressed genes, providing a 3MT gene signature that successfully predicted poor clinical outcome. Even among low-risk patients, high 3MT signature score was associated with poor 5-year overall survival (72% vs. 99% among low-risk patients with a low 3MT signature score), and the 3MT signature score was correlated with MYC activity in the tumor ($R=82\%$, $P<0.0001$). Finally, a strong MYCN and weak DBH staining of tumors derived from patients with elevated urinary 3MT levels was observed, linking MYC activity in the tumor to both catecholamine biosynthesis and elevated urinary 3MT levels.

Conclusions: Elevated urinary 3MT is a promising biomarker for poor prognosis and reflects increased MYC activity in the tumor. Therefore, urinary 3MT levels should be measured at diagnosis and may assist in assessing risk.

Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM

O0101 / #195 | The Association of Neighborhood Effects and Frailty in Childhood Cancer Survivors: A Report from The St. Jude Lifetime Cohort Study

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Background and Aims: Childhood cancer survivors experience reduced physiologic reserve, known as frailty, earlier and at higher rates than their peers. A survivor's residence may affect their individual health behaviors and overall health status. The association between these "neighborhood effects" and the development of frail states in survivors has not yet been investigated.

Methods: Participants in the St. Jude Lifetime Cohort Study with geocoded residential addresses were included for analysis (N=3,911, 47.2% female, 81.2% white, mean age at assessment 33.5±9.9 years). Pre-frail/Frail states were defined as having 2/≥3 of sarcopenia, muscle weakness, poor endurance, slow walking speed, and exhaustion obtained from direct assessments. Survivors completed questionnaires regarding individual health behaviors. Neighborhood effects were characterized at the census tract level using publicly-available geospatial health data and methodology (e.g., rurality/urbanicity, access to exercise opportunities and healthy food, and neighborhood socioeconomic status). A series of nested multivariable logistic regression models identified associations between neighborhood effects and pre-frailty/frailty, adjusting for physical activity, diet, smoking, treatment exposures, and chronic medical conditions.

Results: Compared to non-frail survivors (n=2,941, 75.7%), pre-frail (n=720, 18.5%) and frail survivors (n=226, 5.8%) were more likely to live in neighborhoods with reduced access to exercise opportunities (45.2% vs 89.2% vs 97.0%, P<0.0001), lower neighborhood socioeconomic status (46.6% vs 53.8% vs 59.7%, P<0.0001), and that were considered rural (13.1% vs 15.4% vs 20.8%, P=0.0026). Survivors had a 13.3% (95% CI 6.9-20.1%) increase in the odds of being frail/pre-frail if they lived in a resource-poor neighborhood as opposed to a resource-rich neighborhood when accounting for other variables such as demographics, individual health behaviors, treatment exposures, and chronic medical conditions.

Conclusions: The neighborhood in which a childhood cancer survivor resides as an adult is associated with pre-frailty/frailty. This study provides valuable information for creating and implementing interventions that use community factors to mitigate frailty and improve health outcomes in survivors.

**Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM**

00102 / #1139 | Beta-Catenin Mutation Detecting System in Serum cfDNA in Children with Hepatoblastoma

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Background and Aims: In pediatric live tumors (PLTs), especially, hepatoblastoma, activation of Wnt/ β -catenin signaling is mainly attributed to the somatic mutations in the exon 3 of β -catenin gene. Accord-

ingly, the alterations in β -catenin may be used as markers in the diagnosis and prognosis of PLTs. Circulating cell-free DNA (cfDNA) has been described in various malignancies as a diagnostic and prognostic biomarker. We tried to establish the screening system for detection of these mutation using real time PCR or digital PCR.

Methods: CfDNA and tumor DNA were isolated from 30 patients with HB. CfDNA was also isolated from the serum of 5 healthy individuals. CfDNA and tumor DNA samples were analyzed by the following two methods: 1) Detection of point mutations: digital PCR system using two different fluorescent probes to CTNNB1 exon 3. 2) Detection of deletion of exon3: bridged nucleic acid-clamp real-time PCR (BNA-clamp PCR) using PNA for exon 3. The CTNNB1 gene of tumor DNA samples were analyzed by Sanger sequencing.

Results: In a total of 30 cfDNA samples of HB patients were analyzed, all 9 of whom had a point mutation in CTNNB1 exon 3 in tumor DNA were detectable by digital PCR system. In the remaining 21 cases, CTNNB1 deletion was detected in 16 tumor DNA samples by Sanger sequencing. In these 16 cases, CTNNB1 deletion was detectable in 13 cfDNA samples by BNA-clamp PCR. Three cfDNA samples whose deletion was undetectable by BNA-clamp PCR harbored the large deletion beyond the PCR lesion. The 5 cfDNA of healthy volunteers showed no mutation and deletion signals.

Conclusions: Our results indicate that digital PCR and BNA-clamp PCR using cfDNAs of HB patients detects somatic mutations and deletions in CTNNB1 with high accuracy. This cfDNA analysis might become one of the useful liquid biopsy tools and also a useful monitoring marker for relapse/recurrence of HB.

**Free Paper Session (FPS) FPS -LIVE Q&A SIOP AWARD SESSION
22-10-2021 5:00 PM - 6:30 PM**

00103 / #1448 | Highly Sensitive Detection Method of Retinoblastoma Genetic Predisposition and Biomarkers

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Background and Aims: Retinoblastoma is a malignant tumor of the retina affecting infants and young children. Nearly half of patients are predisposed to retinoblastoma by a germline RB1 pathogenic variant. Non-hereditary retinoblastoma is mainly caused by inactivation of both RB1 alleles at a somatic level. Several polymorphisms have been reported as biomarkers of retinoblastoma risk, aggressiveness or

invasion. The most informative genetic testing is obtained from tumor DNA. For long, access to tumor DNA has been warranted by the frequent indication of enucleation, which has decreased due to advances in conservative approaches. Recent studies showed that tumor cell-free DNA can be analyzed in aqueous humor from retinoblastoma patients. Here we describe a next generation sequencing method relying on unique molecular identifiers for a highly sensitive detection of retinoblastoma genetic predisposition and biomarkers in a single analysis.

Methods: A custom gene panel analysis was performed with SureSelect XT-HS2 enrichment (Agilent) and sequencing on NextSeq 500 (Illumina). It is adapted for genomic DNA extracted from peripheral blood, or tumor DNA extracted from tumor fragment, aqueous humor or plasma.

Results: The gene panel was evaluated on 30 samples from 23 retinoblastoma patients. It enabled the detection of RB1 point variants, including low-level mosaic variants, large genome rearrangements and loss of heterozygosity, and biomarker analysis such as MYCN amplification.

Conclusions: The access to tumor cell-free DNA improves the diagnosis of genetic predisposition in case of conservative ocular therapy and provides access to biomarkers guiding the treatment strategy. The analysis of a gene panel is cost-effective and can be easily implemented in diagnostic laboratories.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

00104 / #1259 | Implementing The Painsquad Multidimensional Pain Assessment app for Youth with Cancer: Nurse Perspectives on Barriers and Facilitators

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Background and Aims: Most youths with cancer experience pain, negatively affecting health-related quality of life. To improve outcomes, barriers to pain assessment and treatment must be mitigated using effective, but also well-implemented, assessment tools. PainSquad is a valid and reliable, multidimensional app designed to support youths (8-18 years) with cancer to self-report cancer pain in real-time and in any environment. Partnering with nurse practitioners (NPs) and registered nurses (RNs), we used an evidence-based implementation toolkit and knowledge translation strategies to implement the PainSquad app into routine pediatric oncology practice. Following a 6-month implementation period, we aimed to evaluate the implementation process and effectiveness from the perspectives of our partner nurses.

Methods: Audio-recorded, semi-structured, qualitative interviews were conducted with all participating NPs and RNs. Interviews were transcribed and coded in duplicate according to the Consolidated Framework for Implementation of Research (CFIR) constructs to

uncover intervention, setting, individual, and process themes impacting implementation success. Coded data were then analyzed using a template analysis approach.

Results: Two NPs and three RNs were interviewed. Four main themes were derived that each described patterns of barriers and facilitators to implementation: (1) characteristics of PainSquad (including clinical advantages of the intervention); (2) internal context and setting of the practice (including a need to improve compatibility between PainSquad and existing workflows and systems); (3) implementation partner characteristics and attitudes (including a need to optimize familiarity with PainSquad); and (4) PainSquad implementation process within the practice (including a need to engage champions and formally appointed implementation leaders to support implementation).

Conclusions: Several nurse-specified barriers and facilitators to implementing a pain assessment app for youth into routine pediatric oncology practice were identified. Future PainSquad implementation efforts will require continued partnerships with stakeholders to ensure process fit in current workflows and the engagement of a central team lead to support the project.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

00105 / #1250 | "Just Ease The Stress of It": Perspectives and Intervention Suggestions from Parents Managing Cancer Pain in Young Children at Home

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Background and Aims: The majority of children with cancer experience pain despite the availability of evidence-based treatment guidelines. Further, a significant proportion of pain episodes occur in environments where management options are limited (i.e., at home). Young children receiving outpatient cancer care may be particularly vulnerable to undermanaged pain because of their limited ability for pain self-report and their reliance on parents to implement pain management interventions or seek medical treatment. We aimed to understand parents' experiences of assessing and managing their young children's cancer pain at home, as well as the impact of children's pain on the child, parent, and family.

Methods: We recruited parents of children (2-11 years) with cancer spending >25% of their cancer care time outside of the hospital and who experienced pain in the previous week. Parents were recruited

from two pediatric tertiary care centers to participate in audio-recorded, semi-structured qualitative interviews. We conducted qualitative interviews until data saturation was reached. Audio-recordings were then transcribed, coded, and analyzed using thematic analysis.

Results: Twenty parents participated. Four themes, each comprising several subthemes, were identified: (1) the child's pain experience at home (pain severity, frequency, type, cause); (2) currently used home-based methods to assess and treat young children's cancer pain; (3) the emotional, care coordination, and practical impact of caring for a child with pain at home; and (4) solutions needed to better support parents in home-based pain management.

Conclusions: Parents recognized pain in their younger children and its negative impact on the child and family. Parents reported using a number of management solutions at home, however, co-developing interventions with parents to further assess and treat their child's pain may result in better managed pain and, ultimately, improved quality of life for young children.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

00106 / #1496 | Trends in Gram Negative Infections in Children Undergoing High Dose Chemotherapy over the Last 5 Years in a Tertiary Pediatric Hemato-Oncology Unit in India

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Background and Aims: Gram negative sepsis is the main cause of mortality in children undergoing high dose chemotherapy in India. Nursing care is paramount to decreasing mortality in this cohort. We analyzed the trends in the rates of infection over a five-year period and aim to use this data to introduce nursing interventions to combat infections

Methods: We performed a retrospective analysis of the blood culture positive infection in the recipients of high-dose chemotherapy for hematolymphoid malignancies and hematopoietic stem cell transplantation (HSCT) up to 18 years of age, from January 2016 to December 2020. Gram negative bacilli (GNB) were further classified as Extended Spectrum Beta Lactamase (ESBL) if there was resistance to cephalosporins but sensitive to carbapenem, Carbapenem Resistant Enterobacteriaceae (CRE) if there was resistance to carbapenem and sensitive to colistin and polymyxin and Pan Drug Resistance (PDR) if there was resistance to colistin and polymyxin

Results: A total of 558 children underwent conditioning chemotherapy for HSCT, and 22 for malignancies (15 acute myeloid leukemia; seven Burkitts lymphoma). Culture positive sepsis was documented in 82 HSCT recipients (14.7%) and 14 children with malignancies (63%). Across the spectrum, GNB was predominant (75.6%). *Klebsiella pneumoniae* was the most common isolate (46.7%), followed by *E. coli* (20.9%) and *Pseudomonas* (12.9%). The rates of CRE were 45.1%, ESBL

9.6%. There were no isolates of PDR in our cohort. In 12.9% children, bacteria such as *Acinetobacter*, *Ochrobacterium*, *Bacteroides* were isolated.

Conclusions: The rates of GNB infections are alarmingly high. The trends have been stable over the last 5 years with no reduction or increase in drug resistant infections. Stringent infection control measures and adherence to central line handling protocols as per CLABSI bundle will help in prevention. We plan to present the data for nurse training sessions in our unit to make meaningful interventions and reduce morbidity and mortality in children.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

00107 / #597 | Single-Institution Practice-Based Evidence Project Using Targeted Interventions to Reduce Central Line-Associated Bloodstream Infections in Children with Cancer and Those Undergoing Hematopoietic Stem Cell Transplant

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Background and Aims: Children with cancer are at disproportionate risk for central line-associated bloodstream infections (CLABSI) relative to other hospitalized children. This ongoing single-institution project evaluated changes in CLABSI rates following targeted CLABSI-reduction strategies using a practice-based evidence approach.

Methods: The setting was a 32-bed inpatient pediatric hematology/oncology/stem cell transplant unit in the United States. CLABSI identified using National Health Safety Network (NHSN) surveillance criteria from January 1, 2006 – December 31, 2019 were included. Data were organized into four time periods: • Pre-intervention baseline (2006-2010). • Children's Hospital Association (CHA) best practice bundles (2011-2012). • Formalized supportive cares (2013-2017 3rd quarter). • Antibacterial prophylaxis for patients with high-risk hematologic malignancies (2017 4th quarter-2019). Conditional Poisson regression models were used to evaluate the effect of time (intervention period) on CLABSI rates with post-hoc Tukey pairwise comparisons between each of the four time periods. Clinical characteristics of patients who developed CLABSI were compared using chi-square tests.

Results: From 2006 through 2019, 227 patients experienced 310 CLABSI. Most (69%) occurred in patients with leukemia; however, events involving patients with leukemia decreased following implementation of antibacterial prophylaxis ($\chi^2=32.51$; $p<0.01$). Clinically important decreases in CLABSI rates from baseline (4.84/1,000 line days) occurred with implementation of CHA bundles (3.29/1,000 line days); however, this difference was not significant ($p=0.16$).

CLABSI rates decreased from baseline with the addition of formalized supportive cares [2.66/1,000 line days; incidence rate ratio (IRR)=0.60; $p < 0.01$], and with the use of antibacterial prophylaxis (1.66/1,000 line days; IRR=0.35; $p < 0.01$). Post-hoc comparisons indicated decreased CLABSI rates with the use of antibacterial prophylaxis compared with CHA bundles alone (IRR=0.49; $p=0.011$) and CHA bundles plus formalized supportive cares (IRR=0.58; $p=0.046$).

Conclusions: Results demonstrate sustained success using a practice-based evidence approach to guide CLABSI-reduction interventions. Follow-up research, applying machine learning algorithms, may identify additional risk factors and inform future interventions.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

O0108 / #1723 | Peripherally Inserted Central Catheter in Pediatric Oncology: A 3-Year Experience of a Tertiary Oncology Center in a Developing Nation

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Background and Aims: The presence of a venous access device is important in the anticancer treatment for patients requiring chemotherapy or parenteral nutrition. The aim of this study is to determine the indications and the complications associated with the use of peripherally inserted central catheter (PICC) in paediatric oncology patients of a tertiary oncology care hospital of a developing nation with a PICC dedicated nursing staff.

Methods: All paediatric oncology patients, less than 18 years of age, who underwent PICC line insertion at a tertiary oncology centre from January 2018 to December 2020 were enrolled in the study. A retrospective analysis of demographic features, primary diagnosis, length of catheter stay, complications and indication for removal was done.

Results: A total of 114 PICC were inserted in 110 patients. The total catheter days were 14927 with 116.5 median catheter dwell time. Seventy-eight patients with haematolymphoid and 32 patients with solid tumor malignancies underwent PICC insertion. The overall complication rate was 44.73% with 18.4% complications warranting removal of PICC. The most common complications were catheter related infections (19%, 1.47 per 1000 catheter days), mechanical complications (fracture and displacement) (19%, 1.47 per 1000 catheter days) and thrombosis (6%, 0.46 per 1000 catheter days). No patient had pneumo/hemothorax with PICC insertion. The most common complication causing removal of PICC was infection, accounting for 44% cases of PICC removal. It was found that in the hands of dedicated and trained nursing staff the PICC line infection rate went down from 28% in 2018 to 16.8% in 2019 to 2020.

Conclusions: Our study suggests that the PICC is an effective and easy to handle CVAD with low rates of complications if dealt by a trained

and a PICC dedicated nursing staff. PICC can be successfully utilized in paediatric oncology patients as a means of safe and long-lasting venous access.

Nursing FREE PAPER SESSION 2-SUPPORTIVE CARE: QUALITY IMPROVEMENTS 22-10-2021 5:00 PM - 6:30 PM

O0109 / #1242 | Reduce Mortality in Children with Cancer after Implementation of a Pediatric Early Warning System (PEWS): A Multicenter Study in Peru

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Background and Aims: Pediatric cancer patients suffer frequent deterioration requiring care in an intensive care unit (ICU). Unfortunately, in many hospitals, ICU access is limited due to lack of pediatric intensivists and shortage of ICU beds. To address these issues, three centers in Peru (INEN–Ministry of Health; Rebagliati and Almenara Hospital – Social Security ESSALUD) joined an international network for the implement of a Pediatric Early Warning System (PEWS) in their Pediatric Oncology Units (POUs). The aim of this study is to evaluate the impact of PEWS implementation on mortality during clinical deterioration in these units.

Methods: This is a prospective cohort study of all pediatric patients admitted to the POU at the 3 collaborating hospitals. Pre-implementation data were compared to 12 months post-implementation at each center (INEN: 4/2017-4/2018 vs 2/2019-1/2020, Rebagliati: 8/2018-6/2019 vs 12/2019-11/2020, and Almenara: 6/2018-4/2019 vs 2/2020-1/2021). Clinical deterioration events, defined as unplanned ICU transfer or floor ICU-level interventions, were identified and clinical characteristics extracted from patient charts.

Results: During the study period, 253 deterioration events were recorded among 101,716 inpatient days across the 3 centers. Comparing the period before and after PEWS implementation, the rate of

clinical deterioration increased from 2.7 to 3.8/1000-inpatient-days at INEN, decrease at Rebagliati Hospital from 2.15 to 1.04 and Almenara Hospital from 2.85 to 1.38. There was a decrease in mortality from 47% to 34% of events (27% decrease) at INEN, from 47% to 42% of events (11% decrease) at Rebagliati Hospital and 82% to 60% of events (26.8% decrease) at Almenara Hospital.

Conclusions: PEWS is an effective intervention to improve quality of inpatient care in limited setting allowing for early identification of deterioration with timely ICU transfer, resulting in reduced mortality. Our experience in Peru supports the global adoption of this intervention to improve childhood cancer survival.

**Free Paper Session (FPS) FPS 04-LIVE Q&A-NEW
DRUGS/EXPERIMENTAL THERAPEUTICS 23-10-2021 8:40 AM - 9:40 AM**

O0110 / #842 | Precision Medicine for High-Risk Paediatric Cancers – Molecular Tumour Board Recommendations and Treatment Responses (Zero Childhood Cancer Program)

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Background and Aims: Advances in our knowledge of the molecular landscape of cancer has led to the development of targeted therapy and personalised anti-cancer treatment. The PRISM trial (ZERO Childhood Cancer Program) evaluates the benefits of a comprehensive precision medicine approach for high-risk paediatric cancer (survival <30%).

Methods: WGS/RNA-seq/methylation results were discussed in a national Molecular Tumour Board (MTB) with 5 tiers of therapeutic recommendations: Tier 1 and 2, clinical evidence in same and different cancer type, respectively; Tier 3 and 4, preclinical evidence in same and different cancer type, respectively; Tier 5, consensus opinion.

Results: 386 patients (43% diagnosis and 57% relapse/refractory) were enrolled over 3 years, with median time to MTB of 6 weeks. A total of 500 recommendations (21% Tier 1, 38% Tier 2, 32% Tier 3, 14% Tier 4 and 5% Tier 5) were made in 261 (67%) patients (median: 1 recommendation/patient; range 1 - 6). 85% were single-agent recommendations and 15% were combination treatments. The PI3K/mTOR, MAPK and cell cycle/cyclin pathway were the 3 most frequently targeted pathways. 87/261 (33%) patients received MTB recommended treatment (median start time 5.3 weeks). Of the first 43 patients who received

MTB recommended treatment, 35 could be evaluated for response: 11% complete response (CR), 20% partial response (PR), 40% stable disease (SD) and 29% progressive disease (PD). The duration of SD was ≥ 24 weeks in 9/14 patients. The clinical benefit rate (CR, PR and SD > 24 weeks) was 57%. The proportion of tier 1 and 2 recommendations was similar across all response groups. The CR+PR, SD and PD group consisted of 45%, 50% and 50% of tier 1 and 2 recommendations, respectively.

Conclusions: The ZERO platform provided personalised therapeutic options for 67% of patients and clinical benefit was observed in majority who received the recommended treatment.

**Free Paper Session (FPS) FPS 04-LIVE Q&A-NEW
DRUGS/EXPERIMENTAL THERAPEUTICS 23-10-2021 8:40 AM - 9:40 AM**

O0111 / #1235 | Academic, Phase I/II Trial on T Cells Expressing a Third-Generation GD2 Chimeric Antigen Receptor and Inducible Caspase-9 Safety Switch for Treatment of Relapsed/Refractory High-Risk Neuroblastoma

F. Del Bufalo¹, G. Del Baldo¹, A. Mastronuzzi¹, A. Serra¹, M.A. De Ioris¹, A. Castellano¹, I. Caruana¹, V. Bertaina¹, M. Sinibaldi¹, S. Di Cecca¹, B. De Angelis¹, M.C. Garganese², P. Merli¹, G. Li Pira¹, D. Pagliara¹, G. Leone³, C. Quintarelli¹, F. Locatelli⁴

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Background and Aims: Treatment of high-risk neuroblastoma (HR-NB) remains a major challenge, with an unsatisfactory long-term survival of <50%. Immunotherapy using chimeric antigen receptor (CAR)-expressing T-cells targeting disialoganglioside-2 (GD2) represents an attractive option.

Methods: At Ospedale Pediatrico Bambino Gesù in Rome, we opened an academic, phase I/II clinical trial on the use of autologous, 3rd-generation (CD28-4.1bb) GD2-CAR T-cells expressing the suicide gene inducible caspase-9 (iC9-GD2-CAR), for treatment of patients with relapsed/refractory (r/r) HR-NB. After obtaining informed consent, the study consecutively enrolled children and young adults (aged 1 - 25 years), using a dose escalation/de-escalation design for the phase I. All patients received a fludarabine/cyclophosphamide-based lymphodepletion.

Results: Twenty-six heavily pre-treated children were enrolled between February 2018 and March 2021. We did not experience any production failure. Three dose levels were tested (namely: 3, 6 and 10 $\times 10^6$ CAR T-cells/kg) and no dose limiting toxicities were recorded, defining the recommended dose as 10 $\times 10^6$ CAR T-cells/kg. Cytokine

release syndrome (CRS) occurred in 19/26 patients (73%) and reached grade 4 in 1 patient only, requiring the activation of the suicide gene, with a rapid clinical resolution. iC9-GD2-CAR T-cells expanded in vivo and were detectable in peripheral blood by flow-cytometry in 25/26 patients, up to 18 months after infusion (mean+SEM peak value: 81.76+34.81 CAR T-cells/ul). Transduced cells were also detected in bone marrow and cerebrospinal fluid. Twenty-one children responded to the treatment (overall response rate: 80.8%), with 7 patients obtaining complete response (CR) and 1 treated with non-evidence of disease maintaining this state; 2-year event-free and overall survivals are 33.74% and 44.5%, respectively.

Conclusions: Use of iC9-GD2-CAR T-cell in an academic setting is feasible and safe in treating highly resistant/relapsed HR-NB. Peculiar, sometimes severe, toxicities can develop and the activation of the suicide gene iC9 is able to control side effects. Importantly, iC9-GD2-CAR T-cells are able to determine a sustained anti-tumor effect.

**Free Paper Session (FPS) FPS 04-LIVE Q&A-NEW
DRUGS/EXPERIMENTAL THERAPEUTICS 23-10-2021 8:40 AM -
9:40 AM**

O0112 / #485 | Efficacy and Safety of Larotrectinib in Pediatric Patients with Non-Central Nervous System (CNS) Tropomyosin Receptor Kinase (TRK) Fusion-Positive Cancer: An Expanded Dataset

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Background and Aims: Neurotrophic tyrosine receptor kinase (NTRK) gene fusions are oncogenic drivers in various tumor types across all ages. Larotrectinib is a first-in-class, CNS-active, highly selective TRK inhibitor approved for patients with TRK fusion-positive cancer, demonstrating a 78% objective response rate (ORR) across 175 pediatric and adult patients with non-CNS cancers (McDermott et al, ESMO 2020). We report data in an expanded dataset of pediatric patients with TRK fusion-positive cancer.

Methods: Pediatric patients (<18 years [y]) with non-CNS TRK fusion-positive cancer in larotrectinib clinical trials (NCT02637687, NCT02576431) were included. Response was investigator-assessed (RECIST v1.1).

Results: As of July 20, 2020, 78 patients (median age 2.1y, range 0.1-17.8y; 60% male) were included. Tumor types included infantile fibrosarcoma (56%), other soft tissue sarcoma (37%), thyroid cancer (3%), congenital mesoblastic nephroma (3%), and melanoma (1%). Overall, 69% of patients received ≥ 1 prior treatment. ORR with larotrectinib was 88% (95% CI 79-94): 30 (39%) complete response (CR; 4 pending confirmation, 9 pathological CR), 37 (47%) partial response (4 pending confirmation), 7 (9%) stable disease, 1 (1%) progressive disease, 1 (1%) not determined, 2 (3%) not evaluable. Median time to response was 1.8 months. The 36-month rates for duration of response, progression-free survival and overall survival were 63% (95% CI 47-79), 60% (95% CI 45-75), and 97% (95% CI 91-100), respectively. Treatment duration ranged from 0.26+–50.6+ months. At data cut-off, 19 patients had progressed; 10 continued treatment post-progression for ≥ 4 weeks. Treatment-related adverse events (TRAEs) were mostly Grade 1–2. Three patients (4%) discontinued treatment due to TRAEs.

Conclusions: In this expanded dataset of pediatric patients with TRK fusion-positive cancer, larotrectinib demonstrated rapid, durable efficacy and a favorable safety profile. These results illustrate the importance of offering NTRK gene fusion testing to pediatric patients with solid tumors, especially those lacking the usual driver alterations.

**Free Paper Session (FPS) FPS 04-LIVE Q&A-NEW
DRUGS/EXPERIMENTAL THERAPEUTICS 23-10-2021 8:40 AM -
9:40 AM**

O0113 / #574 | A Phase 1/2, Open-Label Study of Repotrectinib in Pediatric and Young Adult Patients with Advanced or Metastatic Malignancies Harboring ALK, ROS1, or NTRK1-3 Alterations

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Background and Aims: ALK, NTRK1-3, and ROS1 are oncogenic drivers in a subset of pediatric tumors. Repotrectinib is a potent next-generation ROS1/TRK/ALK tyrosine kinase inhibitor (TKI) designed to overcome mutations that may render other conventional TKIs ineffective.

Methods: The ongoing CARE study (NCT04094610) is a Phase 1/2 trial of repotrectinib in pediatric and young adult patients up to age 25. Phase 1 objective is to determine the recommended Phase 2 dose (RP2D) in patients <12 years. Phase 2 objective is overall response rate (ORR) by RECIST v.1.1 or RANO in the following cohorts: 1: NTRK fusion (TKI-naïve); 2: NTRK fusion (TKI pre-treated); and 3: other ALK, ROS1 or NTRK alterations. Repotrectinib is administered orally in 28-day continuous cycles at the established adult RP2D (≥ 12 years) or a weight-based dose (<12 years). This is the first report of preliminary safety and efficacy.

Results: As of 2 April 2021, 7 patients (ages 1-16 years) have been treated (4 Phase 1, 3 Phase 2). Tumor types included anaplastic ependymoma, glioma, inflammatory myofibroblastic tumor (IMT), glioblastoma multiforme (GBM), sarcoma, and congenital mesoblastic nephroma; 6 were NTRK+ and 1 ROS1+. Prior therapies ranged from 1-5 lines. Three (43%) patients received a prior TKI. In 6 efficacy evaluable patients ORR was 50% (2 confirmed PR in IMT and sarcoma, 1 unconfirmed CR in GBM). Repotrectinib was generally well tolerated with a safety profile similar to adults. Most TEAEs were Grade 1 or 2. The most frequent TEAEs (in ≥ 3 patients) were fatigue, anemia, dizziness, headache, hypermagnesemia, and nausea. No TEAEs led to dose modifications. No DLTs were observed. PK exposure was similar to that observed in adults.

Conclusions: Repotrectinib demonstrated encouraging clinical activity in pediatric patients with ROS1+ and NTRK+ tumors including patients with CNS disease, and a favorable safety profile similar to that seen in adults.

Free Paper Session (FPS) FPS 05-LIVE Q&A-SOFT TISSUE SARCOMAS - NON-RHABDOMYOSARCOMA SOFT TISSUE SARCOMAS 23-10-2021 8:40 AM - 9:40 AM

00114 / #253 | Pediatric Non-Rhabdomyosarcoma Soft Tissue Sarcomas: The Prospective NRSTS 2005 Study by the European Pediatric Soft Tissue Sarcoma Study Group (EPSSG)

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Background and Aims: A standardized approach to pediatric non-rhabdomyosarcoma soft tissue sarcomas (NRSTS) is still lacking. We report the results of the NRSTS 2005 protocol developed specifically by the European pediatric Soft tissue sarcoma Study Group (EpSSG).

Methods: The study included two prospective non-randomized trials on patients, one on localized "adult-type" NRSTS, the other on localized synovial sarcoma. The present analysis is on the two trials combined. Patients were stratified by tumor grade, surgical stage, tumor size, nodal involvement and – for synovial sarcoma – tumor site. Patients were then divided into four treatment groups: 1) surgery alone; 2) adjuvant radiotherapy; 3) adjuvant chemotherapy ($\pm$ radiotherapy); or 4) neo-adjuvant chemotherapy ($\pm$ radiotherapy). The main chemotherapy regimen was ifosfamide plus doxorubicin; the number of cycles ranged from 3 to 7 depending on the stage of the disease.

Results: The cohort examined here included 569 patients <21 years (median age 12.6 years) (206 with synovial sarcomas, 363 with "adult-type" NRSTS), treated between August 2005 and December 2016 at 100 centers in 13 countries. With a median follow-up of 80.0 months (IQR 54.3-111.3), five-year event-free survival (EFS) and overall survival (OS) rates for the whole cohort were 73.7% (95% CI 69.7-77.2) and 83.8% (95% CI 80.3-86.7), respectively. EFS and OS were: 91.4% and 98.1% in the surgery alone group (n=250); 75.5% and 88.2% in the adjuvant radiotherapy group (n=17); 65.6% and 75.8% in the adjuvant chemotherapy group (n=93); and 56.4% and 70.4% in the neoadjuvant chemotherapy group (n=209).

Conclusions: Together with the Children's Oncology Group (COG) ARST0332 trial, the EpSSG NRSTS 2005 study represents a benchmark for this tumor group, and defines the risk-adapted standard of care. Adjuvant treatment may be safely omitted in the low-risk population. Improving the outcome for patients with high-risk, initially-resected "adult-type" NRSTS, and those with initially unresectable disease remains a major clinical challenge.

**Free Paper Session (FPS) FPS 05-LIVE Q&A-SOFT TISSUE
SARCOMAS - NON-RHABDOMYOSARCOMA SOFT TISSUE
SARCOMAS 23-10-2021 8:40 AM - 9:40 AM**

**O0115 / #272 | Low Grade Soft Tissue Sarcomas in Children
and Adolescent, Young Adults: The Benefit from a National
Network**

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S. Bonvalot⁹, V. Minard-Colin¹⁰, C. Fayard¹¹, S. Pannier¹², J.-Y. Blay¹³,
L.-R. Le Nail¹⁴, F. Le Loarer¹⁵, H.J. Brisse¹⁶, F. Chotel¹⁷, G. Pierron¹⁸,
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Background and Aims: Complete surgical resection constitutes the mainstay of treatment for low grade soft tissues sarcomas (LG-STS). Local relapse is the most frequent tumor event, especially if positive margins (R1 margins). Our aims were 1/ to assess the impact of the national network on patients' care and 2/ to evaluate the role of the immediate re-excision in children, adolescents-young adults with LG-STS incompletely resected.

Methods: National retrospective multicentric study of young patients (≤ 25 yo) treated for LG-STS from 2005 to 2017 in SFCE and NETSARC+ centers who benefited from a pathology/biology review through the NETSARC+ national network and included in the Sarcoma "ConticaBase".

Results: A total of 104 patients were identified (median age 17 years, range 0-25; sex ratio 1) mainly affecting limbs (53%) or trunk (40%) sites. Histology included low-grade fibromyxoid sarcoma (47%), angiomatoid fibrous histiocytoma (14%), plexiform fibrohistiocytic

tumor (11%) and other histotypes (28%). Tumors were localized in 98% (1N+ and 1M+ tumors). Histological diagnosis was corrected by expert review in 9%. Molecular analysis (RT-PCR/FISH) contributed to diagnosis in 45% sarcomas with known transcript. After a median follow-up of 4.5 years (range: 0-12.3), 9 local relapses, 2 distant metastases occurred and 1 patient died. Overall, 5-year EFS is 88.4% [95%CI: 81.8-95.9]. The 5-year EFS for R1-patients (n=51) with (n=24) and without (n=27) immediate re-excision are 90.9% [95%CI: 79.7-100.0] and 80.9% [95%CI: 65.5-99.9], respectively (p=0.33). The 42 patients directly treated within reference centers had more frequent diagnostic biopsy (75% vs. 25%; p<0.001), more complete surgery (R0: 79% vs. 21%; p<0.001) and need less re-excision (17% vs. 54%; p<0.001).

Conclusions: This large series indicates that LG-STS in youths are rare and share an overall favorable prognosis. Immediate management in reference centers benefits patients. Main tumor events are local. After incomplete resection, risk of tumor relapse in LG-STS is 14%. Acknowledgements to Conticabase and NetSarc+ national networks.

**Free Paper Session (FPS) FPS 05-LIVE Q&A-SOFT TISSUE
SARCOMAS - NON-RHABDOMYOSARCOMA SOFT TISSUE
SARCOMAS 23-10-2021 8:40 AM - 9:40 AM**

**O0116 / #1360 | Extracranial Rhabdoid Tumours: Results of a
French Series of Patients Treated with a Dose Compression
Strategy According to EPSSG Recommendations**

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Background and Aims: Extracranial malignant rhabdoid tumours (ECRTs) are tumours that mainly affect young children and have a poor prognosis. In 2014, EpSSG developed treatment recommenda-

tions consisting in intensive dose chemotherapy every two weeks by VDCy (vincristine-doxorubicin-cyclophosphamide) and IE (ifosfamide-etoposide) associated with early surgery and irradiation of tumor sites.

Methods: A retrospective study was conducted on children treated in France by these new recommendations up to January 2019.

Results: Thirty-five patients were identified. The primary tumour was in the kidney for 11, in the liver for 6 and in miscellaneous soft parts for 18 patients. The median age at diagnosis was 17.5 months (range 1.2 – 198.2). Distant locations (M+ or synchronous tumor) were present in 37.1% (13/35) at diagnosis. SMARCB1 germline pathogenic variant was detected in 18.2% of patients (6/33 evaluable). Overall tolerance was good with a relative dose intensity from 87.4 to 97.4%. The mean interval between two courses was 18.5 days. There was no permanent chemotherapy-related toxicity but one patient died secondary to respiratory failure occurred 2 months after bi-pulmonary radiotherapy. Surgical resection was performed in 82.9% (19 R0, 7 R1, 3 R2) and local radiotherapy in 48.6% of patients. After a median follow-up of 50.4 months (range 16.5 – 134.1), the 2-year overall survival (OS) and event free survival (EFS) were 47.6% (95% CI 30.2-63.1) and 42.9% (95% CI 26.5-58.3) respectively. On univariate analyses, localized disease (M0) and gross total resection were significantly associated with favorable outcomes.

Conclusions: Intensive dose chemotherapy with VDCy/IE can be administrated with no increase in short-term toxicity, including infants. However, the outcome remains poor for patients without gross total resection or with metastatic or multifocal disease. These patients could be stratified into a high-risk group that requires a new immediate therapeutic approach such as targeted agents combined with multimodal therapy.

Nursing LIVE Q&A: MCM NURSING SESSION 23-10-2021
8:40 AM - 9:40 AM

O0117 / #640 | A Qualitative Evaluation Methodology for Four Sanofi Espoir Foundation “My Child Matters” Nurse-Led Project/Interventions in Low- or Middle-Income Countries

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Background and Aims: Four of 37 successful Sanofi Espoir Foundation (SEF) supported “My Child Matters” (MCM) Nursing projects (in collaboration with International Society of Paediatric Oncology [SIOP]) were selected for evaluation to determine impact. We describe the methodology of the evaluation process during a global pandemic.

Methods: A SIOP MCM Nursing Task Force coordinator oversaw the evaluations with SEF leadership. El Salvador (infection control), Tanzania (awareness course), and Indonesia and the Central African Republic (CAR) (basic pediatric oncology nursing course) projects reflected

content and geographic diversity. Local (or native) professional qualitative researchers with fluency in local languages and knowledge of national/local healthcare systems were identified. English and French terms of reference, consent forms, and interview guides were written. Evaluators conducted short literature reviews on the project/intervention topic to contextualize findings.

Results: SEF leadership managed communication with CAR evaluator. COVID-19 precluded observational and in-person evaluation activities (except in CAR). Nurse Project Leaders, nursing associations and/or relevant NGOs, and stakeholders were enthusiastic and shared their experiences and insights by phone or VOIP. Evaluations were completed in 2 months (except CAR), despite COVID-19's social and economic impact on healthcare practices and staff and family's experiences. Evaluator funding transfers were challenging. Respondents varied considerably due to project diversity. Final reports analyzed by the SIOP coordinator and a second SIOP Task Force member determined commonalities and a final summary report submitted to SEF and SIOP.

Conclusions: Evaluations conducted in local languages by local or native qualitative professionals are possible during pandemic lockdowns and crises in resource-limited countries. Nurse Project Leaders and stakeholders had a shared understanding of culture, health system, governmental hierarchy, or patient/family economic situations. Communication in local languages allowed for detailed respondent answers and avoided the pitfalls of evaluator translation. The projects' impact and lessons learnt for the future were identified despite evaluation methodological complexities.

Nursing LIVE Q&A: MCM NURSING SESSION 23-10-2021
8:40 AM - 9:40 AM

O0118 / #270 | My Child Matters Nursing Projects in Four Low- and Middle-Income Countries: A Qualitative Evaluation

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Background and Aims: The Sanofi Espoir Foundation began a My Child Matters (MCM) Nursing award program in low- and middle-income countries with the International Society of Paediatric Oncology, in 2015. Thirty-seven education, practice or research projects have been funded. An evaluation of award projects aimed to determine barriers, facilitators, impact, and recommendations.

Methods: MCM Nursing Projects from four countries were selected: Tanzania, Indonesia, Central African Republic (CAR) and El Salvador. Local or native professional qualitative evaluators were identified in each country. Local permissions to conduct evaluations were obtained from relevant authorities. Participants were identified by the evaluator in consultation with the nurse awardee and project partners. All participants gave written or audio consent; in-depth interviews were performed and recorded. Transcribed interviews were analysed by the

evaluator for content and themes and a final report in English (CAR: French) was written. The reports were compiled and analysed for common patterns.

Results: All nurse project leaders had outstanding characteristics and/or skills and their leadership was acknowledged locally. The role of local Nursing Associations and/or NGO collaboration was key. Project goals were met and exceeded: 1) Nurse Learners increased their paediatric oncology competence and confidence in nursing practice in 3 countries. 2) Parent empowerment/engagement in the treatment process increased (El Salvador); communication with nurses improved (CAR). 3) Health facilities improved: hospital-based nursing care (Indonesia), referral process to paediatric oncology care (Tanzania), new specialty unit (CAR), and permanent infection control nurse (El Salvador). Bureaucratic and operational issues challenged project initiation. Recommendations for future award projects were made in all settings.

Conclusions: This evaluation demonstrated that the MCM Nursing Awardees and their collaborators achieved their project goals and more. Collaboration with professional organizations, local authorities, and other stakeholders, for example NGOs, seemed to be a success factor and is likely to be essential for the sustainability of projects.

Free Paper Session (FPS) FPS 06-LIVE Q&A-RETINOBLASTOMA AND BRAIN TUMOR BIOMARKERS 23-10-2021 8:40 AM - 9:40 AM

00119 / #1781 | Ophthalmic Artery Chemotherapy (OAC) as Conservative Treatment for Retinoblastoma Patients: Risks of Metastatic Relapse or Recurrence in Patients with Prolonged Treatments

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Background and Aims: Treatment is challenging for patients with advanced intraocular tumors or a single preserved eye, as prolonged attempts of eye-preserving treatments increase the risk of metastatic spread.

Methods: We retrospectively evaluated a cohort of patients treated with more than 6 cycles of OAC between January 2014 and June 2019 to assess an association with metastatic spread.

Results: Thirty-nine of 72 patients received more than 6 OAC cycles (n:31 with bilateral retinoblastoma). Median age at diagnosis was 6 months (range: 0-29) and median age at first OAC was 12 months (range: 6-52). A total of 459 procedures were conducted in 49 eyes (n:15 group D, n:34 group B/C). OAC was used as monotherapy in 14 eyes and after systemic chemotherapy in 35 eyes (24 eyes belonged to

patients younger than 6 months at diagnosis). Intravitreal chemotherapy was used in 21 eyes. Chemotherapy agents included carboplatin, topotecan and melphalan. Thirty-three eyes received 6-10 cycles while 16 eyes received 11 or more cycles (8 were single preserved eyes). With a median follow-up of 67 months (range: 35-128), the eye survival probability was 0.9 (CI95%: 0.84-0.96). Five enucleated eyes presented high risks pathology factors and received adjuvant chemotherapy. The overall survival was 97.4% and no patient received external-beam radiotherapy. Three patients had metastatic relapse at 49, 31 and 52 months of retinoblastoma diagnosis. All of them had bilateral disease and only one eye with preserved vision. They received 8, 11 and 16 cycles of OAC respectively and brachytherapy. Only one patient survived after high-dose chemotherapy treatment.

Conclusions: The ocular preservation rate is 90% in patients who received 6 or more cycles of OAC combined with other modalities. However, 12.5% of patients who received 11 or more cycles developed systemic metastatic relapse. Prolonged attempts to salvage the eye should consider this risk

Free Paper Session (FPS) FPS 06-LIVE Q&A-RETINOBLASTOMA AND BRAIN TUMOR BIOMARKERS 23-10-2021 8:40 AM - 9:40 AM

00120 / #428 | Molecular Diagnosis of Retinoblastoma by Circulating Tumor DNA Analysis

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Background and Aims: The analysis of circulating tumor DNA (ctDNA), a fraction of total cell-free DNA (cfDNA), might be of special interest in retinoblastoma patients. Since the accessibility to tumor tissue is very limited in these patients, either for histopathological diagnosis of suspicious intraocular masses (biopsies are proscribed) or for somatic RB1 studies and genetic counseling (due to current successful conservative approaches), we aim to validate the detection of ctDNA in plasma of non-hereditary retinoblastoma patients by molecular analysis of RB1 gene.

Methods: In a cohort of 19 intraocular unilateral non-hereditary retinoblastoma patients for whom a plasma sample was available at

diagnosis, we performed high-deep next-generation sequencing (NGS) of RB1 in cfDNA. Two different bioinformatics/statistics approaches were applied depending on whether the somatic RB1 status was available or not.

Results: Median plasma sample volume was 600 μ L [100-1000]; median cfDNA plasma concentration was 119 [38-1980] and 27 [11-653] ng/mL at diagnosis and after complete remission, respectively. In the subgroup of patients with known somatic RB1 alterations (n=11), seven of 9 somatic mutations were detected (median allele fraction: 6.7%). In patients without identified somatic RB1 alterations (n=8), 6 candidate variants were identified for 7 patients.

Conclusions: Despite small tumor size, blood-ocular barrier, poor ctDNA blood release and limited plasma sample volumes, we confirm that it is possible to detect ctDNA with high-deep NGS in plasma from patients with intraocular nonhereditary retinoblastoma. This may aid in diagnosis of suspicious cases, family genetic counseling or follow-up of residual intraocular disease.

Free Paper Session (FPS) FPS 06-LIVE Q&A-RETINOBLASTOMA AND BRAIN TUMOR BIOMARKERS 23-10-2021 8:40 AM - 9:40 AM

00121 / #1399 | A DNA Damage Repair Signature in Pediatric High-Grade Gliomas: XRCC1 and PARP1, Key Biomarkers

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Background and Aims: Despite their low incidence, pediatric high-grade gliomas (pHGGs), including diffuse intrinsic pontine gliomas (DIPGs), remain the leading cause of mortality in pediatric neuro-oncology. Resistance to standard therapies like irradiation is a frequent occurrence in pHGGs, especially in those harboring genomic alterations such as histone mutations. Recent insights have underlined the predominant role of DNA damage repair (DDR) as a way to classify adult HGG tumors. Furthermore, the evidence obtained in histone-mutated gliomas indicates that transcription-replication conflicts can induce replication stress associated to DDR abnormalities. Therefore, this study aimed to establish a DDR signature able to classify specifically pHGGs and identify good responders to radiation.

Methods: We performed transcriptomic analyses to discriminate seven pHGGs from ten pediatric pilocytic astrocytomas based on the expression of specific DDR genes. The resulting transcriptomic signature was refined through comparison with a signature previously

established in adult HGGs. Specifically, we challenged both signatures against datasets from published transcriptomic analyses of DIPG and sus-tentorial pHGGs. To correlate DDR protein expression levels to histone and TP53 mutations, we performed an immunohistochemical assessment of several DDR factors in a collection of 21 pHGG diagnostic samples. Functional inhibition of selected DDR factors exposed by our signature was carried out in 5 pHGG patient-derived cell lines (PDCL).

Results: We identified a pediatric-specific gene signature, that cluster two groups of pHGGs, including the DIPGs, based on DNA repair and cell cycle machineries. Among those genes, we validated significantly XRCC1 and PARP1 as two major theranostic DDR biomarkers. Their variations are able to discriminate responders to radiotherapy and, when using PDCL, we showed that the PARP-1/XRCC1 expression balance might be correlated with resistance to PARP1 inhibition.

Conclusions: We provide evidence that DDR gene-based signature is able to stratify pHGG patients and PARP1 overexpression, associated to XRCC1 expression and TP53 mutations, is a new theranostic target.

Free Paper Session (FPS) FPS 06-LIVE Q&A-RETINOBLASTOMA AND BRAIN TUMOR BIOMARKERS 23-10-2021 8:40 AM - 9:40 AM

00122 / #193 | Genome-Wide Association Study Across Pediatric Central Nervous System Tumors Implicates Shared Predisposition and Points to 1Q25.2 (PAPPA2) and 11P12 (LRRC4C) as Novel Candidate Susceptibility Loci

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Background and Aims: Central nervous system (CNS) tumors constitute the most common form of solid neoplasms in children, but knowledge on genetic predisposition is sparse. In particular, whether susceptibility attributable to common variants is shared across CNS tumor types in children has not been investigated. The purpose of this study was to explore potential common genetic risk variants exhibiting

pleiotropic effects across pediatric CNS tumors. We also investigated whether such susceptibility differs between early and late onset of disease.

Methods: A Danish nationwide genome-wide association study (GWAS) of 1,097 consecutive patients (< 15 years of age) with CNS tumors and a cohort of 4,745 population-based controls.

Results: For both the overall cohort and patients diagnosed after the age of four, the strongest association was rs12064625 which maps to PAPP2A at 1q25.2 ($p = 3.400 \times 10^{-7}$ and 9.668×10^{-8} , respectively). PAPP2A regulates local bioavailability of insulin-like growth factor I (IGF-I). IGF-I is fundamental to CNS development and is involved in tumorigenesis across a wide range of different cancers. For the younger children, the strongest association was provided by rs11036373 mapping to LRRC4C at 11p12 ($p = 7.620 \times 10^{-7}$), which encoded protein acts as an axon guidance molecule during CNS development and has not formerly been associated with brain tumors.

Conclusions: This GWAS indicates shared susceptibility attributable to common variants across pediatric CNS tumor types. Variations in genetic loci with roles in CNS development appear to be involved, possibly via altered IGF-I related pathways.

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS
23-10-2021 10:30 AM - 12:00 PM

00123 / #810 | Pancreatoblastoma in Children: The Expert/Partner Diagnostic and Therapeutic Recommendations

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Background and Aims: Pancreatoblastoma (PBL) is a rare malignant epithelial pancreatic neoplasm affecting typically young children. Complete surgical resection is the mainstay of treatment but optimal therapeutic protocol has to be established.

Methods: We present international consensus recommendations for the management of pediatric PBL, established by the European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT) within EU-funded PARTNER (Paediatric Rare Tumors Network-European Registry) project. The Consensus Conference Standard Operating Procedure methodology was adopted to grade the levels of evidence and strength of recommendations.

Results: Pancreatoblastoma can involve any part of pancreas and is usually diagnosed in advanced stages. Serum alpha-fetoprotein is often elevated reflecting tumor burden [Level IV; Grade A]. Histopathological examination is mandatory for diagnosis. Preoperative radiologic work-up should include abdominal MRI and chest CT [IV; A]. Complete surgical resection with negative margins (R0) is the cornerstone of treatment [IV; A] and should be performed in experienced centers. We recommend 2 courses of PLADO (cisplatin, doxorubicine) after R0 surgery and 4 PLADO in case of microscopic residues (R1) or resected involved lymph nodes. In unresectable and metastatic PBL, neo-adjuvant PLADO (2-6 courses) is advised [IV; B]. Feasibility of complete resection should be evaluated every 2 courses. In case of inadequate response, ICE (ifosfamide, carboplatin, etoposide) [IV; C] or VAC (vincristine, dactinomycine, cyclophosphamide) regimens are alternatives [V; C]. The impact of radiotherapy in incompletely resected PBL remains unproven, and potential side effects in young children have to be considered. Long-term monitoring of pancreatic, auditory, cardiac and renal functions is mandatory [IV; B].

Conclusions: The prognosis is usually favorable in completely resected pediatric PBL. Unresectable and/or metastatic PBL may become amenable to complete delayed surgery after neo-adjuvant PLADO chemotherapy. Discussion in multidisciplinary teams with oncologists and surgeons and enrolment of patients in the national or European database are mandatory. EU Funding: PARTNER GA n.777336 - 3rdHP (2014-2020).

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS
23-10-2021 10:30 AM - 12:00 PM

00124 / #251 | Cutaneous Melanoma in Children and Adolescents: The Expert/Partner Diagnostic and Therapeutic Recommendations

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Background and Aims: Cutaneous melanoma is rare in children and suffers from a shortage of knowledge and clinical expertise. Its clinical management is often challenging. Young patients are treated following the same principle as for adults, but concern remains about their access to clinical trials and new drugs, which dramatically changed the natural history of advanced melanoma.

Methods: We present the internationally-recognized recommendations for the diagnosis and treatment of pediatric melanoma, established by the European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT) within the EU-funded project called PARTNER (Paediatric Rare Tumours Network-European Registry).

Results: Surgery is the cornerstone of treatment; complete excision should ensure 2–5 mm margins according to tumor thickness. The indication for sentinel lymph node biopsy (SLNB) should be the same of adults, in relation to disease thickness ($\geq 0.8\text{mm}$) and other features such ulceration, lymphovascular invasion, mitotic rates and others. When nodal metastases are found, whether proceeding with complete lymph node dissection (CLND) or with careful ultrasound monitoring is under debate (CLND is generally indicated in case of extracapsular spread, ≥ 3 positive sentinel nodes, multifocal/extensive disease). Recommendations for advanced melanoma are extrapolated from large, randomized, phase III clinical trials on adult patients. BRAF inhibitors/MEK inhibitors and immune checkpoint inhibitors are the current standard treatment in metastatic melanoma and have also been explored in the adjuvant setting in patients with resected stage II–III disease.

Conclusions: Main recommendations are: to discuss pediatric patients in multidisciplinary teams that include both pediatric oncologists and specialists in adult melanoma; to enroll patients in prospective trials, if available (e.g. to include adolescents in adult phase I/II trials); to col-

lect data in national-international databases; and to develop an effective international collaboration between pediatric and adult melanoma groups to facilitate the transfer of potentially-effective new agents from the adult to the pediatric setting. EU Funding: PARTNER GA n.777336 - 3rdHP (2014-2020).

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS 23-10-2021 10:30 AM - 12:00 PM

O0125 / #333 | Thymoma and Thymic Carcinoma in Children and Adolescents: The Expert/Partner Diagnostic and Therapeutic Recommendations

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Background and Aims: Thymic tumors are epithelial tumors arising from the anterior mediastinum and constitute 0.2 to 1.5% of all adult malignancies but are exceptional in pediatric population. Thymic epithelial tumors (TETs) encompass a variety of histologic subtypes associated with different clinical outcomes.

Methods: Authors present the consensus recommendations established by the European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT) within the EU-funded PARTNER project (Paediatric Rare Tumours Network-European Registry) based on the evidence mainly collected from published retrospective series (Level IV), case reports (Level V) and personal expertise (Level V). The “strengths” of recommendations are categorized by additional grading (Grade A to C).

Results: Initial tumor assessment should include chest CT scan or thoracic MRI with contrast (Level IV; Grade A). PET-CT scan in case of thymic carcinoma suspicion is recommended (Level IV; Grade B).

Normal AFP and beta-HCG serum level should be verified (Level IV; Grade A). Fine-needle aspiration is not considered sufficient for TET diagnosis (Level IV, Grade C). Histopathological evaluation is mandatory [Level IV; Grade A]. Complete resection without adjuvant therapy remains the optimal treatment option in early stages of TETs (Level IV, Grade A). There is no optimal chemotherapy evidence-based established for children (Level IV; Grade C). Neoadjuvant treatment with cyclophosphamide-doxorubicin-cisplatin (CAP), cisplatin- etoposide (PE) or carboplatin-paclitaxel is recommended for unresectable stage III-A to IV-A TETs (TNM classification; Level III, Grade B). Adjuvant chemotherapy is recommended for thymic carcinoma regardless of quality of resection (R0, R1 or R2) or for thymoma stages III-IV-A Masaoka with R2 resection (Level III, Grade B). For thymic carcinoma with R2 resection, adjuvant chemotherapy $\pm$ delayed surgery, and radiotherapy are recommended (Level IV, Grade A).

Conclusions: Multidisciplinary team approach is mandatory at diagnosis and during treatment. Centralization of data is necessary to improve knowledge pediatric on TETs. Funding: CHAFAEU Grant Nr: 777336.

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS 23-10-2021 10:30 AM - 12:00 PM

00126 / #347 | Salivary Gland Carcinoma in Children and Adolescents: The Expert/Partner Diagnosis and Treatment Recommendations

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Background and Aims: Salivary gland carcinomas (SGC) are rare in children. No standardized diagnosis and treatment guidelines are available. Complete surgical resection constitutes the mainstay of treatment. However, the indications of adjuvant therapy remain controversial and may be challenging in view of the potential long-term toxicities.

Methods: We present the consensus recommendations established by the European Cooperative Study Group for Pediatric Rare Tumors (EXPERT) according to the ESMO scale, based on evidence collected from published series, case reports and personal expertise.

Results: Surgery designed to achieve microscopically complete resection constitutes the cornerstone of the treatment. Upfront systematic neck dissection should be reserved to cases with clinically or radiologically suspected lymph node involvement. Second-look lymphadenectomy may be considered for high-grade and/or advanced tumors. Adjuvant radiotherapy of the tumor bed should be discussed in the case of high-grade and advanced tumors with or without lymph node involvement. Radiotherapy for low- or intermediate-grade tumors is generally not recommended but could be considered in the presence of poor prognostic factors. There is no consensus on the optimal dose and volume of radiotherapy; a high dose (60Gy) is generally recommended. Adjuvant cervical radiotherapy should be considered for high-grade tumors when complete neck dissection is deemed infeasible or in the presence of several positive nodes or extracapsular spread. It may be considered for non-high-grade with more than 3 involved nodes or with extracapsular spread. Chemotherapy is generally reserved for palliative treatment of recurrent and/or metastatic disease not amenable to further surgery or radiotherapy, in the setting of a prospective clinical trial whenever possible as well as targeted therapy.

Conclusions: Young patients with SGC should be managed according to standardized recommendations, including first-line surgery, and neck dissection $\pm$ adjuvant radiotherapy for high-risk tumors. Whenever possible, patients should be enrolled into prospective registries. EU Funding: PARTNER GA n.777336 - 3rdHP (2014-2020).

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS 23-10-2021 10:30 AM - 12:00 PM

00127 / #348 | Consensus Recommendations from the Expert/Partner Groups for the Diagnosis and Therapy of Sex Cord Stromal Tumors in Children and Adolescents

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Background and Aims: The European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT) is partner in the European Union funded project designated PARTNER. As part of this and previous projects, EXPeRT is continuously developing consensus recommendations in order to harmonize standard care for very rare tumors (VRT) of children and adolescents in Europe.

Methods: For this work, a literature review as well as experience from different national registries has been evaluated. Levels of Evidence were graded according to the Consensus Conference Standard Operating Procedure methodology, and the strength of recommendation. Due to the rarity of these tumors, most recommendations have been developed based on evidence collected from published series, case reports and personal expertise, finally evaluated through a consensus process within the PARTNER group. This paper presents the recommendations for the diagnosis and treatment of children and adolescents with sex cord stromal tumors.

Results: The clinical approach to sex cord stromal tumors of the testis (TSCST) and ovary (OSCST) depends on histological differentiation and tumor stage. Tumor resection constitutes the cornerstone of treatment. Virtually all TSCSTs present as localized non-metastatic tumors, with excellent prognosis after complete resection. In contrast, prognosis of OSCSTs may be adversely affected by tumor spillage or presence of metastases, which may predominantly occur within the peritoneal cavity and abdominal lymph nodes. In these cases, cisplatin-based chemotherapy is recommended. Of note, some SCSTs may develop in the context of genetic predisposition syndromes e.g. DICER-1, so that specific follow-up is indicated, such as screening for thyroid disease.

Conclusions: SCSTs should be diagnosed and treated according to standardized recommendations that include reference pathology, genetic testing for tumor predisposition syndromes in selected cases, and stratified adjuvant chemotherapy in patients with unfavorable risk profile. To ensure high quality of diagnosis and therapy, patients should be enrolled into prospective registries. EU Funding: PARTNER GA n.777336 - 3rdHP (2014-2020).

Free Paper Session (FPS) FPS 10-LIVE Q&A-RARE TUMORS
23-10-2021 10:30 AM - 12:00 PM

00128 / #1630 | Morphologic Spectrum of INI1 [Integrase Interactor 1] Deficient Tumours in Children

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Background and Aims: Structural or functional deletion of INI1/hSNF5 gene located on chromosome 22q11.2 is a predisposing factor to the development of variety of tumours. Although INI1 is a unifying factor, tumours with INI1 loss are diverse. We undertook this retrospective study with the aim of studying the spectrum of all INI1 deficient tumours in paediatric patients as compared to adults.

Methods: We reviewed histopathology and complete immunohistochemistry profile of tumours from 137 patients including adults which showed loss of INI1.

Results: 52% of patients showing loss of INI1 in their tumours were children. A strong association of loss of INI1 expression and 'rhabdoid' phenotype across different tumour types was demonstrated. Commonest tumours seen in children were extra-renal rhabdoid tumour (64%), rhabdoid tumours of kidney (24%) which constituted pure 'rhabdoid' morphology. Poorly differentiated chordomas (5.84%) and epithelioid malignant peripheral nerve sheath tumours (5.10%) were other tumours showing INI1 loss. Non paediatric patient population had wider morphologic and diagnostic spectrum associated with loss of INI1 and included myoepithelial carcinomas (5.84%), renal medullary carcinomas (1.46%) and INI-1 deficient Carcinoma of the Sinonasal Tract. Only 6% epithelioid sarcoma were seen in children, the rest being in young adults.

Conclusions: Tumours with INI1 loss are diverse with respect to their clinical presentation, prognosis, morphology and also immunohistochemistry profile. Rhabdoid phenotype associated with loss of INI1 is a strong indicator of specific entities in children such as renal or extra renal rhabdoid tumours; while in adults, the morphologic and diagnostic spectrum associated with INI1 loss was found to be much wider. Presence of pure rhabdoid phenotype was less often seen in adults as compared to children. All tumours with rhabdoid morphology and loss of INI1 loss were uniformly aggressive tumours in children, dominantly exemplified by rhabdoid tumours. Epithelioid sarcoma showed the widest morphologic spectrum of all tumours.

Free Paper Session (FPS) FPS 07-LIVE Q&A-NEUROBLASTOMA - BIOLOGY 23-10-2021 10:30 AM - 12:00 PM

00129 / #1227 | Multiparametric Flow Cytometry Highlights B7-H3 as a Possible Novel Diagnostic/Therapeutic Target In GD2NEG/Low Neuroblastoma Variants

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Background and Aims: High-Risk neuroblastoma (HR-NB) relapses in more than 30% of cases, despite using aggressive therapies including targeting of GD2. The presence of GD2^{neg/low} NB variants and/or the surface expression of different immune checkpoint ligands including B7-H3 might contribute to therapy resistance. The main aim of this study was to set a fast, unbiased method to unequivocally identify, quantify and characterize Bone Marrow (BM) infiltrating tumor cells.

Methods: We used a Multiparametric Flow Cytometry (MFC) panel to analyze 41 BM aspirates from 25 HR-NB patients, comparing results with Cytomorphological Analysis (CA) and/or immunohistochemical analysis (IHC). Spike-in experiments assessed the sensitivity of MFC. To find novel prognostic markers possibly integrating the MFC panel, Kaplan-Meier analysis on 498 primary NBs has been performed.

Results: No false-positive were detected, and MFC showed high sensitivity (0,0005%). Optimized MFC identified CD45^{neg}CD56^{pos} NB cells in 11 out of 12 (91.6%) of BM indicated as infiltrated by CA, 7 of which co-expressed high levels of GD2 and B7-H3. MFC detected CD45^{neg}CD56^{pos}GD2^{neg/low} NB variants expressing high surface levels of B7-H3 in two patients diagnosed at 53 and 139 months of age. The first patient had a non-MYC amplified tumor with unusual TH^{pos}PHOX2B^{neg} phenotype, which relapsed 141 months post-diagnosis with BM infiltration and a humerus lesion. This patient did not receive anti-GD2 immunotherapy during the therapeutic protocol. Kaplan-Meier analysis highlighted an interesting dichotomous prognostic value of different ligands involved in NB recognition by the immune system.

Conclusions: Our study describes a specific, sensitive and fast MFC analysis allowing a precise quantification of i) BM tumor burden; ii) surface expression of GD2; iii) surface expression of different immune checkpoint ligands. MFC might usefully support other routinely used diagnostic and prognostic tools, improving diagnosis, prognosis, and orienting novel personalized treatments in patients with GD2^{neg/low} NB, who might benefit from innovative therapies combining B7-H3 targeting.

Free Paper Session (FPS) FPS 07-LIVE Q&A-NEUROBLASTOMA - BIOLOGY 23-10-2021 10:30 AM - 12:00 PM

O0130 / #1241 | A Risk Signature of Four Autophagy-Related Genes for Predicting Neuroblastoma Survival is Associated with Tumor Immune Microenvironment

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Background and Aims: Treatment for neuroblastoma (NB) has been challenging. Though autophagy plays an important role in NB, however there's no signature model based on the characteristics of autophagy-related (ATG) gene pairs. We aimed to construct and validate an ATG-gene-based prognostic signature model for NB.

Methods: After identifying differentially expressed ATG genes between INSS stage 4 and INSS stage No-4, we constructed an ATG genes risk model by Kaplan-Meier, univariate Cox regression, least absolute shrinkage and selection operator regression and multivariate Cox regression, followed by internal and external validation using K-M and ROC analysis. Then, we studied the correlation between risk score and immune microenvironment.

Results: This signature model based on four hub ATG (BIRC5, GRID2, RNASEL and HK2) can accurately predict the long term overall survival of NB patients (10 years) and is an independent prognostic factor for NB patients. For this we constructed a nomogram based on ATG-signature risk score and three other independent clinical factors (INSS_Stage, Age, MYCN-status). The results of the nomogram show that our signature can bring clinical net benefits. Since gene set enrichment analysis (GSEA) suggested that the signature was strongly associated with immune cell functions, CIBERSORT, LM22 matrix and Pearson correlation were further performed, showing that the risk signature was significantly correlated with T regulatory cells (Cor = 0.115), T follicular helper cells (Cor = 0.312), CD8+ T cell (Cor = 0.137), CD4+ T cell (Cor = 0.123), CD4 memory resting cells (Cor = 0.279), plasma cells (Cor = 0.282), macrophage M2 cells (Cor = 0.159), B memory cells (Cor = 0.207) and immune checkpoint genes consisting of PD-L1 and CTLA4.

Conclusions: We identified and independently validated an ATG gene risk signature for NB, as well as discovering its significant association with NB immune microenvironment. Low risk patients in our model may benefit more from immune checkpoint therapy.

Free Paper Session (FPS) FPS 07-LIVE Q&A-NEUROBLASTOMA - BIOLOGY 23-10-2021 10:30 AM - 12:00 PM

O0131 / #1683 | Near-Infrared Photoimmunotherapy Targeting GD2—Shedding New Light on Neuroblastoma Treatment

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Background and Aims: In neuroblastoma (NB) treatment, interventions targeting tumour-specific surface markers may enhance the efficacy of surgery generating anti-tumour immunological response from the ablated tumour residues. Here, we investigated the near-infrared photoimmunotherapy (NIR-PIT), a novel therapeutic approach combining the cytotoxicity of photodynamic therapy with the specificity of immunotherapy to enhance the antigenicity and adjuvanticity of NB tumour residues following surgery.

Methods: The anti-GD2 antibody Dinutuximab-beta was conjugated to the phthalocyanine dye, IR700DX, which, when excited with the light at 690 nm, produces cytotoxic and immunological responses. The anti-GD2-IR700DX binding specificity was confirmed by flow cytometry and confocal microscopy in NB cell lines expressing the GD2 antigen at different levels. Cell viability was measured with the CellTiter-Glo assay while the main hallmarks of immunogenic cell death (ICD) were investigated following the administration of NIR-PIT (8 and 16 J/cm²) to NB cancer cells. Proof-of-concept anti-GD2-IR700DX-PIT (100 µg, 100 J/cm²) in vivo studies were performed using mice bearing subcutaneous NB xenografts.

Results: The conjugate showed effective targeting to GD2 positive NB cells and blocking with an excess of non-labelled anti-GD2 antibody significantly reduced the median fluorescence in all tested cell lines, confirming probe specificity. Phototoxicity studies in vitro showed a significant decrease in cell viability in a dose dependent manner. Furthermore, the anti-GD2-IR700DX-PIT triggered the ICD leading to release of danger-associated molecular patterns (DAMPs). In vivo studies confirmed that the conjugate specifically targets GD2 positive tumours and significantly inhibits tumour growth post-irradiation.

Conclusions: Our preclinical feasibility study showed the therapeutic potential of anti-GD2-IR700DX-PIT for NB treatment, indicating this novel tool could become part of the armoury to improve loco-regional control and reduce the risk of microscopic residual disease after surgery.

Free Paper Session (FPS) FPS 07-LIVE Q&A-NEUROBLASTOMA - BIOLOGY 23-10-2021 10:30 AM - 12:00 PM

00132 / #910 | Development of an Immune-Related Gene Expression Signature for Long Term Prognosis Prediction in Neuroblastoma Patients

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Background and Aims: Neuroblastoma is one of the most malignant tumors in children. immune-related genes (IRGs) play an essential role in neuroblastoma. But the expression signature has rarely been investigated in prognosis. We aimed to identify and assess prognostic IRGs signature to predict overall survival (OS) in the patients.

Methods: IRGs were obtained from Immunology Database and Analysis Portal. Differentially expressed IRGs were identified based on Therapeutically Applicable Research To Generate Effective Treatments database. Bioinformatics analysis using the Gene Ontology and the Kyoto Encyclopedia of Genes and Genomes platforms was performed. The Lasso cox regression analysis was performed to screen hub IRGs for overall survival, which were used to calculate the risk score of each patient. The prognostic nomogram based on risk scores and clinical factors was constructed. Receiver operating characteristic curve and calibration plot were used to evaluate the efficiency of the model. Finally, we prepared a Shiny R application to automatically calculate the predicted OS rate.

Results: The OS-related prognostic model was constructed based on 15 IRGs (AGT, LBP, PROC, FGHR3, PRLH, AR, ULBP2, KLRK1, PDF, IL1B, PDGFRA, S100A14, GH1, NRTN, HNF4G) and significantly stratified patients into high- and low-risk groups in terms of OS (P < 0.001). The area under the curves of the model were 0.87 in 3- year survival, 0.876 in 5- year survival, 0.919 in 10- year survival. Risk score based on the expression of 15 hub genes and 2 clinical factors (INSS stage and MKI index) were identified as significant prognostic factors. A nomogram was constructed by the above 3 poor factors, the calibration curves showed optimal agreement between the probability as predicted by the nomogram and the actual probability. A interactive online shiny-website was constructed finally.

Conclusions: Our IRGs based prediction model is a reliable prognostic and predictive tool for long term overall survival in neuroblastoma.

Free Paper Session (FPS) FPS 07-LIVE Q&A-NEUROBLASTOMA - BIOLOGY 23-10-2021 10:30 AM - 12:00 PM

00133 / #905 | Development of an Autophagy-Related Gene Expression Signature for Long Term Prognosis Prediction in Neuroblastoma

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Background and Aims: Neuroblastoma is one of the most malignant tumors in children. Autophagy-related genes (ARGs) play an essential role in neuroblastoma, but the expression signature has rarely been investigated in prognosis. We aimed to identify and assess prognostic ARGs signature to predict overall survival (OS) in the patients.

Methods: 234 ARGs were obtained from The Human Autophagy Database. Differentially expressed ARGs were identified based on Therapeutically Applicable Research To Generate Effective Treatments database. Bioinformatics analysis using the Gene Ontology and the Kyoto Encyclopedia of Genes and Genomes platforms was performed. The Lasso cox regression analysis was performed to screen hub prognostic ARGs for overall survival, hub genes were used to calculate the risk score of each patient. The prognostic nomogram based on risk score and clinical factors was constructed. its performance was evaluated by receiver operating characteristic (ROC), and calibration curves. Finally, we prepared a Shiny R application to automatically calculate the predicted OS rate.

Results: The OS-related prognostic model was constructed based on 4 ARGs (VTN, EXOC4, BIRC5, CFTR) and significantly stratified patients into high- and low-risk groups in terms of OS ($P < 0.001$). The area under the curve of the model were 0.7 in 3- year survival, 0.672 in 5- year survival, 0.819 in 10- year survival. Risk score based on the expression of 4 hub genes and 3 clinical factors (DNA ploidy, INSS stage and MKI index) were identified as significant prognostic factors. A nomogram was constructed by the above 4 poor factors, the calibration curves showed optimal agreement between the probability as predicted by the nomogram and the actual probability. A interactive online shiny website was constructed finally.

Conclusions: Our ARGs based prediction model is a reliable prognostic and predictive tool for long term overall survival in neuroblastoma.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

O0134 / #1341 | The Power of Video Storytelling: "20 Stories for 2020: Pediatric Oncology Nurses in the Year of the Nurse and the Midwife"

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Background and Aims: "20 Stories for 2020: Pediatric Oncology Nurses in the Year of the Nurse and the Midwife" is a five-part video series released in September 2020 to raise awareness of pediatric oncology nursing globally. Video storytelling is used to engage, educate and influence audiences using personal narratives, but less is known

about the effect on the storytellers. We sought to assess the impact of storytelling on participating nurses.

Methods: Six months following the release of the videos, an anonymous survey was sent to the participants. Using a 5-point Likert scale, nurses were asked to rate their agreement with 9 statements regarding their experience. An open-ended question regarding whether they would change their submission was also included.

Results: Twenty-one pediatric oncology nurses from 19 countries participated in the project and were sent the survey. Responses were received from 17 (81% response rate). All nurses agreed that seeing their stories woven together created a sense of connection to the global pediatric oncology nursing community (94% strongly agree, 6% somewhat agree) and to the World Health Organization Year of the Nurse and the Midwife. Respondents also agreed that telling their story allowed for meaningful reflection on their practice (88% strongly agree, 12% somewhat agree). Nurses felt that they were able to share their stories with confidence and that they received adequate preparation (100% somewhat or strongly agree); only 2 (12%) would make changes to their video. Most nurses (88%) would participate in a video storytelling format again; the other 12% were neutral towards future participation.

Conclusions: Video storytelling was a successful tool for advocacy and disseminating powerful narratives of pediatric oncology nurses. Our survey reveals that storytelling also provided a mechanism for nurses to connect with each other and share their experience. Organizations should consider using video storytelling to engage and empower nurses in advocacy campaigns.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

O0135 / #1394 | Nurse Specialists of the Who Global Initiative for Childhood Cancer Offer Webinars to Promote International Nursing Engagement

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Background and Aims: Expert paediatric oncology nursing is essential to the success of the WHO Global Initiative for Childhood Cancer (GICC), which aims to increase global survival to 60% by 2030. The Nurse Specialists of the GICC is an informal coalition of expert paediatric oncology nurses from high- to low-income countries who advocate for nurses' engagement and visibility at all levels of decision making and activity within the Initiative. Two webinars were conducted with the purpose of updating paediatric oncology nurses from across the globe on the progress of nursing within the Initiative and to showcase nurses' engagement in WHO focus countries.

Methods: Two webinars were held following the launch of the WHO 'CureAll' framework. Times were aligned for Eastern and Western hemispheres. Both webinars were in English; Western hemisphere provided Spanish translation. Each webinar gave an overview of the Initiative, an outline of work of the Nurse Specialists Group, and the perspective and priorities of focus country nurse leaders (Peru, the Philippines and Zambia).

Results: Participants included 147 attendees from over 33 countries. Work-to-date by the Nurse Specialists was highlighted including engagement in GICC work streams; publications; priority-setting and establishing connections with focus country nurses. Nurses from focus countries discussed their role in local implementation of the Initiative and described nursing priorities and challenges including increased nurse specialised education and training; safer working environments, and the development of local evidence-based guidelines.

Conclusions: The webinars provide an opportunity for paediatric oncology nurses to learn about the nursing activities within WHO GICC. The number of registrants demonstrated international interest and engagement. The Nurse Specialists plan to continue strengthening collaboration with focus country nurse leaders and WHO non-state members, and to offer ongoing forums for nurses from across the globe who are interested in learning about and advocating for the nursing goals and priorities within the Initiative.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

00136 / #504 | International Dissemination of the Baseline Standards for Pediatric Oncology Nursing

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Background and Aims: Pediatric oncology requires a nursing workforce with specialized education and competencies to attain optimum patient outcomes. Education alone is not enough if nurses lack the resources and support to practice quality nursing care. The Baseline Standards for Pediatric Oncology Nursing, published in 2014, outline six elements required to promote the delivery of quality pediatric oncology nursing care and create effective practice environments. The SIOP Nursing Baseline Standards Task Force is devoted to advocating and disseminating the standards and sought to appraise and assess the current status of awareness and implementation globally.

Methods: Literature searches were used to document publications and citations, and global endorsements and website access/downloads were measured. A repository was created to serve as a database of known standard initiatives globally.

Results: Ongoing efforts to promote and implement the standards, now published in 8 languages, have resulted in 7 publications with over 81 citations, 14 endorsements by global organizations, a pre-SIOP "Leadership and Advocacy Workshop" and numerous presentations at local, national and international conferences. Since April of 2019, baseline standards' information has been accessed 524 times from the SIOP website. Exemplar repository initiatives span 49 sites in 29 countries and include: situational assessments (17 hospitals in Africa, Eurasia, South America); advocacy and health policy (mandated staffing ratios in Peru); a tiered orientation program (25 Latin American hospitals); chemotherapy safety (Philippines, Vietnam); staff non-rotation (Vietnam); and implementation of all Standards (Pakistan).

Conclusions: Advocacy efforts have increased awareness and implementation of the baseline standards. The standards provide structure and objective goals for pediatric oncology programs striving to improve patient outcomes. To achieve the WHO Global Initiative for Childhood Cancer target of doubling the global childhood cancer survival rate to 60%, achievement of baseline nursing standards for pediatric oncology must be prioritized and appropriately resourced by hospital administrators, governments, and other stakeholders.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

00137 / #893 | Assessing the Impact of Baseline Nursing Standards for Pediatric Oncology Nurses Advocacy by the Siop Nursing Group

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Background and Aims: Majority (80%) of childhood cancer cases are in low- and middle-income countries (LMIC) with limited specialized medical/human resources and access to appropriate treatment. Many children arrive at a facility with progressive disease resulting in poor health outcomes. A most effective method to improve the outcomes of children with cancer is to have and retain specialized pediatric oncology nurses. The International Society of Paediatric Oncology (SIOP) Pediatric Oncology for Developing Countries (PODC) committee established baseline nursing standards for pediatric oncology nurses in LMIC. To measure the awareness and implementation of SIOP nursing baseline standards, a survey was conducted across PODC membership.

Methods: An 8-question survey on Google forms requested responses on profession, country, standards awareness and implementation in respondents' organizations and interest in further contact. The survey was distributed through emails and social media platforms through the PODC Nursing Working Group and Education and Training Group.

Results: Forty-six responses (87% nurses, 4.3% doctors, 2.2% nurse practitioners and 6.6% lecturers) were received from Bolivia, Cameroon, Chile, India, Pakistan, Indonesia, Italy, Philippines, South Africa and US. Almost all (89.1%) stated having heard about the standards; 10.9% were unaware. About half (54.3%) had heard about the standards through SIOP or a conference; 47.8% were informed by colleagues. Almost half (47.8%) were implementing ≥ 1 standard; 23.9% had partially addressed ≥ 1 and 17.4% had successfully implemented ≥ 1 standard and 10.9% had not yet addressed the standards. Most commonly cited standards included continuing nursing education (75.6%), multidisciplinary team work (61%) and evidence-based practices (56.1%). Almost all (91.3%) respondents wished to be contacted again.

Conclusions: SIOP PODC baseline nursing standards have been well received to improve quality of care provided to patient and their families. An action plan is being shaped to create focus group discussion forums to share information about the baseline nursing standards implementation progress and challenges across LMIC.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

00138 / #1168 | Peruvian Pediatric Oncology Nursing Partnership to Achieve SIOP Pediatric Oncology in Developing Countries' Nursing Baseline Standards and Improve Nursing Outcomes

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Background and Aims: In Peru, the WHO Global Initiative for Childhood Cancer (GICC), Ministry of Health and GICC Nursing Committee's strategy includes a pilot research study for applying the six International Society of Paediatric Oncology (SIOP), Pediatric Oncology in Developing Countries (PODC) Baseline Standards for Pediatric Oncology Nursing in Low- and Middle-Income Countries at the Instituto Nacional de Enfermedades Neoplásicas (INEN) in Lima. This is the first study to establish an external assessment of the Standards, and in the future aims to identify an association between achievement of the Standards and outcomes. Despite the pandemic, progress has continued and will be reported.

Methods: This multiphase improvement study includes: 1) an onsite assessment of the Standards (complete), 2) developing metrics for each standard (on-going), 3) implementing all six standards (pending resolution of pandemic), 4) associating achievement of the Standards with patient outcomes (planned). Two initial site assessment visits were conducted January-February 2020. COVID-19 required pivoting to online follow-up communication in monthly meetings with INEN nurse leadership and key stakeholders for ongoing project implementation-strategy planning.

Results: The quantitative assessment revealed opportunities for improving adherence to and measurement of each Standard (e.g., improving staffing ratios). The qualitative review found staff had minimal knowledge of the WHO GICC and role of US partners, only initial integration of the newly implemented local nurse educator role due to staffing, and room for additional nursing practice and human resource support. Post-assessment, hospital leaders immediately hired 11 new nurses to improve staffing ratios. In monthly meetings, INEN nurse leadership identified internally monitored quality indicators relevant to each Baseline Standard to inform process and outcome measures for the proposed study.

Conclusions: Evidence from this study may inform other Ministries of Health in WHO selected countries and those who wish to improve nursing practice as a strategy for increasing overall survival even in resource-limited settings.

Nursing FREE PAPER SESSION 3-DISSEMINATION OF KNOWLEDGE AND THE USE OF THE BASELINE STANDARDS 23-10-2021 10:30 AM - 12:00 PM

00139 / #1143 | Establishing a Virtual Multidisciplinary Team Meeting Between Cameroon, Central Africa, and Leeds, UK, Childhood Cancer Services

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Background and Aims: Multidisciplinary team (MDT) meetings provide a regular, structured forum of a core group of professionals to provide expert decision-making in individual patient care. We describe our experience in establishing a virtual paediatric oncology MDT between twinning partners in Cameroon (4 hospitals) and the UK.

Methods: Monthly video conferencing MDT meetings were established in March 2019, initially using Google Hangouts and later Zoom platforms. Core team members included oncologists, nurse practitioners, nurses, surgeons, pathologist, and radiologist. A Standard Operating Procedure was developed with a plan to discuss all new patients, and patients needing wider discussion. Case details, including images were circulated to team members in advance, using formal proformas.

Results: Between March 2019 and March 2021, there were 269 discussions of 201 patients in 21 meetings. Patients have been varied in age (up to 24 years). Eighty seven (43.3%) patients had NHL, 16 (8.0%) had Wilms tumour, 20 (10.0%) had Acute Lymphoblastic Leukaemia, 19 (9.5%) had Retinoblastoma and 59 (29.4%) had other malignancies. Four of the patients did not have a malignancy. The majority of patients had stage 3 or 4 disease. Nine patients had relapsed disease. The therapy was changed in 10 (5.0%) and 15 (7.5%) of children received MDT initiated therapy. Core team members reported positive benefits from the MDT including defining goals of care, shared decision making, professional education and team-building. Internet connectivity was the major challenge experienced.

Conclusions: The virtual MDT enabled regular support to the local care team remotely and improved quality of clinical decisions. Virtual MDT meetings between geographically dispersed teams are possible, and have proved particularly valuable during the COVID-19 pandemic when in-person visits are not permitted. This model requires little additional resources and can be replicated in twinning programmes across LMIC settings.

Free Paper Session (FPS) FPS 08-LIVE Q&A-RENAL TUMORS
23-10-2021 10:30 AM - 12:00 PM

00140 / #267 | Circulating Tumor DNA as A Biomarker in Favorable Histology (FH) Wilms Tumor (WT) Patients: Analysis from Children's Oncology Group (COG) Trial AREN0533

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Background and Aims: The utility and clinical relevance of circulating tumor DNA (ctDNA) analyses has not been established in the context of children with Wilms Tumor (WT). We evaluated the detection of ctDNA and selected risk markers in the serum and urine of patients with WT and compared the findings to matched diagnostic tumor samples.

Methods: Fifty of 395 children with Stage III or IV WT enrolled on COG AREN0533 had banked pre-treatment blood, urine and tumor. Ultra-low passage whole genome sequencing was employed to detect ctDNA in serum and urine samples. Copy number changes in 1q, 16q and 1p in serum and urine were compared with clinical sequencing data from matched tumor samples. Event-free survival (EFS) was compared between patients with and without ctDNA detection.

Results: ctDNA was detected in the serum of 41/50 (82%) and in the urine in 13/50 (26%) patients. 1q gain was identified in 25/41 positive serum samples, 16q deletion in 14/41 and 1p deletion in 14/41. Agreement between serum ctDNA detection and tumor sequencing results was as follows: 77% for 1q gain, 88% for 16q deletions and 70% for 1p deletions, with κ -coefficients of 0.56, 0.74 and 0.29, respectively. ctDNA analysis revealed 1q gain (n=13) or 1p deletion (n=3) in serum samples from children in whom these alterations were not detected in tumor samples. There was a trend toward worse EFS in patients with vs. without serum ctDNA detection (4-year EFS 80% vs 100%).

Conclusions: ctDNA demonstrates promise as an easily accessible prognostic biomarker with potential to overcome limitations of tumor heterogeneity and sampling. The observed trend toward more favorable outcome in patients in whom ctDNA was not detected requires validation. ctDNA profiling should be further explored as a non-invasive diagnostic and prognostic tool in the risk-adapted treatment for patients with WT.

Free Paper Session (FPS) FPS 08-LIVE Q&A-RENAL TUMORS
23-10-2021 10:30 AM - 12:00 PM

00141 / #369 | Impact of Age on Overall Survival Among Children with Wilms Tumor: A Population-Based Registry Analysis

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Background and Aims: For pediatric Wilms tumor (WT), international trials have reported conflicting findings on whether the association between age and worse overall survival (OS) is due to age as an

independent risk factor, or the more advanced presentation of older children at diagnosis. We sought to address this question using a large population-based registry.

Methods: The Surveillance, Epidemiology and End Results (SEER) database was queried for all patients diagnosed with WT under the age of 20. The association between age and OS was assessed using univariate and multivariate Cox proportional hazards regression, adjusted for clinicopathologic and treatment covariates.

Results: In this study 1,868 females and 1,595 males were diagnosed with WT between 1975 and 2016. More advanced stage, larger primary tumor size, lymph node involvement, disease requiring radiotherapy, and omission of surgery were associated with worse OS ($P < 0.05$). More advanced stage, larger primary tumor size, and disease requiring radiotherapy were also associated with older age, while bilateral disease was associated with younger age ($P < 0.001$). Histologic grade was excluded due to sparse availability ($< 9\%$ of patients). Each year of age conferred an incremental hazard ratio (HR) of 1.08 ($P < 0.001$). This association persisted in multivariate analysis, independent of the aforementioned covariates (HR = 1.06, $P = 0.040$). The rise in OS HR was most pronounced around ages 2 and 15 (HR_{age 3–15 vs. 0–2} = 1.6, $P = 0.039$; HR_{age 16–19 vs. 3–15} = 2.7, $P = 0.022$), independent of covariates. Thirty-year survival rates were 88%, 84%, and 58% among patients diagnosed at 0–2, 3–15, and 16–19 years of age, respectively.

Conclusions: Diagnosis of pediatric WT at an older age was found to be independently associated with worse OS. While exploration of additional confounders (such as heterogeneity of treatment) and prospective studies are warranted, more aggressive treatment of older children based on age alone, especially as they approach early adulthood, may be considered in the multi-disciplinary management of WTs.

Free Paper Session (FPS) FPS 08-LIVE Q&A-RENAL TUMORS 23-10-2021 10:30 AM - 12:00 PM

00142 / #457 | Effect of Positive Abdominal Lymph Nodes (+LN) Alone or in Combination with LOH of 1P or 16Q in Stage III Favorable Histology Wilms Tumor (FHWT)

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dren's Hospital, Pediatric Surgery, United States of America, ¹²Children's National Medical Center, Oncology, United States of America, ¹³Cook Children's Medical Center, Hematology/oncology, United States of America, ¹⁴IWK Health Centre, Hematology/oncology, Canada, ¹⁵Children's National Medical Center, Hematology/oncology, United States of America, ¹⁶Dana Farber Cancer Institute, Oncology, United States of America

Background and Aims: COG AREN0532 demonstrated decreased event-free survival (EFS) for Standard Risk (SR) Stage III Favorable Histology Wilms Tumor (FHWT) patients with +LN in combination with LOH of 1p or LOH 16q. To better establish these effects, AREN0532 patients were pooled with like-treated patients from COG biology study AREN03B2.

Methods: A total of 519 SR Stage III FHWT DD4A-treated patients were analyzed (442 AREN0532, 77 AREN03B2). EFS was compared by study, LN, and LN +/- singular LOH using log-rank tests and Cox regression. Patients with anaplasia, without nodal sampling at upfront or delayed nephrectomy, combined LOH of 1p and 16q, or post-chemotherapy blastemal predominant histology were excluded.

Results: EFS did not differ by study enrolled, supporting pooling. Overall, 186 (36%) had +LN, 45 (9%) LOH 1p, and 94 (18%) LOH 16q. Patients with +LN vs -LN had significantly worse EFS regardless of LOH status (+LN 4-year EFS = 84%, -LN 4-year EFS = 94%, HR = 2.81, $p = 0.0001$). Among 380 patients with neither LOH 1p nor LOH 16q, +LN remained significant (+LN 4-year EFS = 86%, -LN 4-year EFS = 97%, HR = 3.6, $p = 0.0005$). Among patients with LN+ and LOH 1p or LOH 16q, 4-year EFS estimates were 73% and 79% respectively. Examining LN and singular LOH together, patients with +LN and either LOH 1p or LOH 16q had significantly worse EFS (4-year = 76%) vs. those with -LN or no LOH (4-year = 92%), with HR = 3.01 ($p = 0.0004$).

Conclusions: Findings in this expanded cohort confirm +LN as an adverse prognostic factor independently and in combination with LOH of 1p or 16q. Therapeutic augmentation for those with +LN and LOH 1p or 16q is proposed on the next COG study.

Free Paper Session (FPS) FPS 08-LIVE Q&A-RENAL TUMORS 23-10-2021 10:30 AM - 12:00 PM

00143 / #841 | Subtotal Necrosis in Unilateral Non-Anaplastic Wilms' Tumours is an Indicator of Excellent Prognosis

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Background and Aims: In the SIOP Wilms tumour (WT) trials preoperative chemotherapy has been used as the first line of treatment. Completely necrotic WTs (CN-WTs) are classified and treated as low-risk tumours. The aim of the study was to evaluate whether regressive type WTs (RT-WTs, 67-99% chemotherapy-induced changes - CIC) showing a good response to preoperative chemotherapy had a comparably excellent survivals as seen in CN-WTs. We also tried to prove what percentage of CIC could be established as a safe cut-off point for RT-WTs.

Methods: A retrospective study of 2,117 unilateral, non-anaplastic WTs from the UK-CCLG and GPOH WT studies (2001-2020) treated according to the SIOP-WT-2001 protocol.

Results: There were 126 (6.0%) patients with CN-WTs and 773 (36.5%) with RT-WTs, stages I-IV. The 5-year RFS and OS for patients with CN-WTs were 95.3% ($\pm 2.1\%$ SE) and 97.3% ($\pm 2\%$ SE), and for RT-WTs 86.4% ($\pm 1.4\%$ SE, $p < 0.01$) and 95.2 ($\pm 0.9\%$ SE, $p = 0.59$), respectively. RT-WTs were sub-divided into sub-totally necrotic WTs ($> 95\%$ CIC) (STN-WT96-99) (124 patients) and the remaining of RT-WT (RR-WT67-95) (649 patients). CN-WT and STN-WT96-99 groups showed significantly better RFS than RR-WT67-95 ($P = 0.003$ and $P = 0.02$, respectively), whereas OS were superimposable ($97.3 \pm 1.5\%$ SE for CN-WT; $97.8 \pm 1.5\%$ SE for STN-WT96-99; $94.7 \pm 1.0\%$ SE for RR-WT67-95).

Conclusions: WTs with subtotal necrosis following pre-operative chemotherapy ($> 95\%$ CIC) shared the same excellent RFS and OS as CN-WTs, and although it was achieved by more treatment, treatment reduction in postoperative treatment of patients with STN-WT96-99 may be justified. If introduced, 16% of patients with RT-WT would benefit from reduction in total duration of treatment, exclusion of radiotherapy for stage III patients, improvement of quality of life of patients and less access to hospital.

Free Paper Session (FPS) FPS 08-LIVE Q&A-RENAL TUMORS
23-10-2021 10:30 AM - 12:00 PM

00144 / #509 | A Trainee Led Audit of the Use of Paediatric Renal Tumour Biopsy in the United Kingdom

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Background and Aims: In 2018, UK national guidance was revised to recommend biopsying paediatric renal tumours only when defined clinical, biochemical or imaging features suggestive of a diagnosis other than Wilms' tumour (WT) are present. This, the first national audit led by the Paediatric Oncology Trainees Group, aimed to assess compliance with these guidelines.

Methods: Data were collected retrospectively by trainees at 12/20 national principal treatment centres. Patients registered on to the Improving Population Outcomes for Renal Tumours of childhood (IMPORT) study from June 2018 to May 2019 with a unilateral tumour were included. Data collated included demographics, type of biopsy, imaging findings, serum calcium and lactate dehydrogenase and urinary catecholamines.

Results: Data were available for 73 patients (80% of all cases enrolled on IMPORT), 36% underwent a tumour biopsy; all were percutaneous cutting needle biopsies. Only 40% patients received all the pre-treatment investigations to determine whether biopsy was recommended. Based on results of the investigations performed, a biopsy was recommended in 28 cases (38%). The most common indications for biopsy were atypical imaging appearances or abnormal biochemistry (54% and 36% respectively). Seven (10%) patients had a biopsy which was not recommended according to the new guidance and all these patients had WT. Nine patients (12%) did not have a biopsy despite it being indicated and 3 of these had a non-WT diagnosis. Of the 62 children with WT, 25 (40%) underwent a biopsy.

Conclusions: Trainee led audits are an effective tool to assess national clinical practice. Rates of biopsy in children with WT remain higher in the UK than the 18% reported in the international SIOP-WT-2001 study. Increasing compliance in performing the recommended biochemical investigations and following indications for biopsy would improve the pre-operative detection of non-WT and reduce the proportion of WT patients undergoing a biopsy.

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE
INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM -
12:00 PM

00145 / #280 | Service Evaluation of A New UK National Febrile Neutropenia Protocol Using The "Aus" Rule for Risk Stratification

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Background and Aims: New UK guidelines advised using the 3-item Australia-UK-Swiss ("AUS") rule to identify patients with febrile neutropenia (FN) who could be discharged home after a shorter duration of intravenous (IV) antibiotic treatment. Reduced hospital stay was particularly attractive during the COVID-19 pandemic. This one-year prospective service evaluation aimed to assess the effectiveness and safety of this new protocol.

Methods: Recommended minimum duration of inpatient stay for IV antibiotics (ranging from 4-48 hours) was determined using the AUS score. Patients had to meet eligibility criteria to be considered for homecare management with oral antibiotics. Anonymised episode data relating to predefined outcomes was contemporaneously collected via Qualtrics software. Regular pre-specified analyses were shared with participating centres and national professional networks.

Results: 13 centres contributed 735 presentations of 458 patients (range 3-178 presentations/centre). 20% episodes were eligible for homecare (range 0% to 60%/centre). Median admission duration stratified by AUS score was 72, 96, 120 and 168 hours for AUS 0-3, respectively. Equivalent duration for those eligible for homecare was 26, 30, 53 and 69 hours.

There were 60 cases of re-presentations within 7 days of the original FN episode. 18% homecare patients were readmitted ($p = 0.501$ for difference with previous pooled estimate from ambulatory regimens). Reasons for re-presentation in this group were new symptoms (40%), persistent symptoms (35%) and positive blood cultures (25%). The reason for representation was not significantly different between those eligible for homecare and those not ($p=0.110$). Overall, 13 children were admitted to ICU and 7 children died (2 of infection, 5 of other causes) - none were eligible for, or treated on, the homecare pathway.

Conclusions: This large multi-centre cohort found 20% patients with FN can have shorter inpatient stays and safely complete oral antibiotics at home without significant readmissions or adverse outcomes

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE
 INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM -
 12:00 PM

00146 / #165 | Fever and Neutropenia Outcomes and Areas
 for Intervention: A Report from Succour – Supportive Care for
 Children with Cancer in Africa

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Background and Aims: Death during paediatric cancer treatment is common in sub-Saharan Africa. Using the infrastructure of Supportive Care for Children with Cancer in Africa (SUCCOUR), our objective was to describe fever and neutropenia (FN) characteristics and outcomes in Sub-Saharan Africa in order to identify potential areas for future intervention.

Methods: A multi-country, multi-centre prospective, observational cohort study was conducted in sub-Saharan Africa in Malawi, Kenya, Ghana and Zimbabwe. Data were collected from September 2019 to March 2020. Children below 16 years with newly diagnosed cancer treated with curative intent were included. Data were abstracted in real time using standardised case report forms by trained personnel. Characteristics and outcomes of FN during the first three months of treatment were documented.

Results: 252 patients were included (median age 6.0, range 0.2–15.0 years, 54% male). The most common cancer was Burkitt lymphoma (63/252, 25%). Among 104 FN episodes, 21 (21%) were associated with prolonged neutropenia (> 1 week) and 32 (31%) were associated with profound neutropenia (absolute neutrophil count < $0.1 \times 10^9/L$). In 10/104 (10%) episodes, empiric antibiotics were started within one hour following fever onset and in 16/104 (15%) episodes, a blood culture was obtained before starting antibiotics. Malaria parasitaemia was detected in 4/104 (4%). A total of 11/104 (11%) patients died in the FN episodes.

Conclusions: Although most FN was not associated with prolonged or profound neutropenia, 11% resulted in death. Future efforts to target for reducing mortality include doing blood cultures prior to giving antibiotics and earlier initiation of empiric antibiotics. We aim to develop and implement a local care pathway on the management of FN and to evaluate the impact of this intervention.

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE
 INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM -
 12:00 PM

00147 / #916 | Do Children with High-Risk Febrile
 Neutropenia Require Intravenous Antibiotic Therapy Till
 Neutrophil Recovery? Results of a Randomized Controlled Trial
 from India

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 G. Narula, M. Prasad, N. Roy Moulik, C. Dhamne, V. Bhat, B. Parambil,
 S. Srinivasan, S. D Banavali
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Background and Aims: The optimal duration of antibiotic therapy in high-risk febrile neutropenia(HRFN) is debatable. Although predominant practice is continuation of therapy until neutrophil recovery, there is evolving evidence for early discontinuation. We evaluated safety, feasibility, and cost effectiveness of short course antibiotic therapy in children with HRFN.

Methods: Open label randomized controlled trial was conducted at Tata Memorial Hospital, India. Eligible children(afebrile for 48hours, sterile blood culture, no focus of infection and clinically stable) aged 2 to 15 years, with HRFN were randomized on day-5 to either Arm-A[continuation of antibiotics till absolute neutrophil count(ANC) > 100/cumm and rising on two consecutive days] or Arm-B(discontinuation of antibiotic therapy on day-5). Enrolled patients were monitored for total duration of antibiotic therapy(including treatment for recurrent fevers), rate of recurrent febrile episodes and re-hospitalizations, complications/ICU admissions until ANC recovery(>500/cumm). Statistical analysis was done using SPSS Version 24.0.

Results: A total of 93 patients were randomized (February,2016-February,2020): 44 to Arm-A(standard duration) and 49 to Arm-B(early discontinuation). Baseline characteristics were comparable between the two arms, most common malignancy was acute myeloid leukemia followed by acute lymphoid leukemia. The mean duration of antibiotic therapy in Arm A and B was 12.3 days(SD=4.9 days) and 8.4 days(SD=4.1 days) respectively($p<0.01$). Arm-A had less recurrences in fever(38%($n=17$) vs 63%($n=31$); $p=0.02$) and re-hospitalization rates(11%($n=5$) vs 61%($n=30$); $p<0.01$). There was no significant difference in complication rate, ICU admission or mortality($p=0.43$) between the two arms. The incidence of culture positive sepsis was also similar between the two groups($p=0.47$). Mean cost of treatment was higher for Arm-A (\$656USD, SD=\$356USD) in comparison to Arm-B (\$502USD, SD=\$171USD) ($p=0.01$).

Conclusions: Short course of empirical antibiotic therapy in stable high-risk febrile neutropenia is a feasible and safe strategy without increase in mortality/ICU admissions. The reduction in duration of hospitalisation and reduced cost of treatment will have a significant impact in many resource limited settings.

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM - 12:00 PM

00148 / #1300 | Early Discontinuation of Empiric Antibiotic Therapy in Children Treated for Cancer with Febrile Neutropenia

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Background and Aims: Febrile neutropenia (FN) is a life-threatening complication for children with cancer. Early administration of broad-spectrum antibiotics has significantly contributed to improved outcome. In many countries, children with culture negative FN remain hospitalized on intravenous antibiotic treatment until neutrophil count recovery, resulting in prolonged antibiotic courses and increased length of hospital stay. The aim of this study was to explore the safety of short course antibiotic regimens in children with culture negative FN.

Methods: A prospective study of children treated for hematologic malignancies and solid tumors admitted with FN at the Paediatric Oncology department between February 2017 and December 2020. All children received empiric treatment with Cefepime or Piperacillin/Tazobactam combined with an aminoglycoside. Children with negative blood cultures regardless of their absolute neutrophil count, had their antibiotics discontinued after 48 hours of defervescence. On discharge, children and carers were instructed to return if fever reappears. Daily follow-up visits were scheduled for clinical examination and blood count.

Results: A total of 36 cases with FN met the inclusion criteria for early discontinuation of empiric antimicrobial treatment. Median age at admission was 4.7 years(IQR=3.5-9.4) with male predominance(69.4%). All children were on intensive chemotherapy. Median neutrophil count at discontinuation was 160/mm³(IQR=0-250/mm³). Early discontinuation reduced the length of hospitalization by 180 days (median 3 days/case-IQR=2-6). Median length of hospitalization was 2 days(IQR=2-3), compared with a median value of 6 days(IQR=5-11) until neutrophil recovery($p<0.0001$). Median length of hospitalization for the historical control group was 7 days(IQR=5-10)-($p<0.0001$). There were no readmissions due to fever or infection during the stage of neutropenia.

Conclusions: Results from this study demonstrate that early discontinuation of antibiotics in oncology patients with FN is safe and advantageous, regardless of neutrophil count. Reduced exposure to antibiotics, shorter hospitalization and minimized risk of hospital acquired infections are among the beneficial secondary outcomes observed in this study.

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM - 12:00 PM

00149 / #656 | Can Ganciclovir be Replaced with Valganciclovir as a Preemptive Therapy for Cytomegalovirus Infection in Hematopoietic Stem Cell Transplantu (HSCT) Recipients?

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Background and Aims: Cytomegalovirus (CMV) infection remains an important cause of morbidity and mortality after HSCT. Oral valganciclovir (VGC) has been shown to be effective in pre-emptive treatment of CMV in patients undergoing solid organ transplantation, but data is lacking in patients undergoing HSCT.

Methods: Patients <18 years of age who underwent HSCT between January 2017 to December 2020 were included. CMV reactivation was defined as plasma CMV DNA load >1000 copies/ml quantified by polymerase chain reaction assay performed weekly for the first 100 days and bi-weekly thereafter. VGC was given orally and doses were calculated based on body surface area and creatinine clearance. Admitted patients received intravenous Ganciclovir (GCV) and were switched to VGC after discharge.

Results: 152 patients underwent HSCT during the study period. Donors were HLA-matched - 88, haploidentical - 43, and matched unrelated - 21. Sixty-five (42.7%) patients had CMV reactivation. All the patients received VGC while in 6 patients GCV followed by VGC was used. The median time from HSCT to CMV reactivation was 51 days (range: 18 to 137). The median value of CMV copies at the start of CMV treatment was 3852 copies (range: 1621 to 28522). The median time to the first negative CMV DNA result was 12 days. There was no significant difference in CMV clearance and treatment duration between the patients who received either VGC or VGC followed by GCV. None of the patients developed CMV disease. Majority of the patients who were started on VGC tolerated it well except one who had drug-induced aplastic anemia. Three patients in whom VGC was used had to be switched to Cidofovir for some other reason.

Conclusions: VGC as preemptive therapy was found to be safe and effective in our patients. In a resource limited country where cost is always an issue, VGC offers excellent alternative to intravenous GCV.

Free Paper Session (FPS) FPS 09-LIVE Q&A-SUPPORTIVE CARE INFECTIOUS COMPLICATIONS 23-10-2021 10:30 AM - 12:00 PM

00150 / #1422 | Barriers and Enablers of Implementation of Pediatric Early Warning Systems (PEWS) in Resource-Limited Pediatric Oncology Centers

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Background and Aims: Pediatric Early Warning Systems (PEWS) are tools for early identification of deterioration in hospitalized patients. Implementation of PEWS improves patient outcomes, particularly in pediatric hematology-oncology (PHO) patients in resource-limited settings. This study evaluates barriers and enablers of PEWS implementation in these settings.

Methods: Semi-structured qualitative interviews were conducted with 71 staff involved in implementation of PEWS at 5 centers in 4 countries in Latin America. Centers were purposefully sampled to include those that had implemented quickly (3-4 months) and those that took longer (10-11 months). The interview guide was developed using the Consolidated Framework for Implementation Research (CFIR). Interviews were conducted in Spanish by bilingual members of the study team using a web-based platform, audio recorded, and professionally transcribed and translated into English. A codebook was developed using a priori codes from the CFIR and iteratively revised based on review of 9 transcripts. Two coders independently analyzed all transcripts using MAXQDA, achieving a kappa of 0.8-0.9. Analysis was based on CFIR domains.

Results: Participants included physicians (45%), nurses (45%), and administrators and data managers (10%). All 5 CFIR domains (characteristics of individuals, inner setting, outer setting, intervention characteristics, and implementation process) were mentioned by all participants as important in implementation of PEWS. Particular emphasis was placed on early communication with hospital leadership (engaging leaders), convincing stakeholders of the importance of PEWS (stage of change), having adequate human and material resources for implementation, and the need for adaptation, both of PEWS to the center (PEWS adaptation) and the center to PEWS (site adaptation) for successful implementation.

Conclusions: This is the first study describing barriers and enablers to PEWS implementation in resource-limited settings. Findings demonstrate that many identified barriers can be converted to facilitators through targeted effort by the implementation team, and provides guidance on strategies for successful improvement interventions like PEWS globally.

CCI LIVE Q&A-SESSION 2: ENHANCING COLLABORATION, JOINING FORCES ... 23-10-2021 12:30 PM - 2:00 PM

O0151 / #240 | Overview of a Successful National Childhood Cancer Advocacy Strategy in Canada

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Background and Aims: Childhood cancer is the number one cause of death by disease for Canadian children. In Canada, funding for childhood cancers has not been a priority for our federal government. There were also many existing federal systems that were barriers to ensuring Canadian children got fast, fair and safe access to trials and treatments. To address this unmet need, childhood cancer survivor Helena Kirk formed a national advocacy campaign and became the leading voice in a formal national advocacy organization.

Methods: A multi faceted strategy was developed and deployed to advance a proposal that had consensus built with patient advocates, organizations, pediatric oncologists/researchers and federal politicians from across Canada.

Results: The immediate results included a proposal that was endorsed by 40 cancer/childhood organizations, 30 leading pediatric oncologists and 86 MPs/Ministers. A constitutional legal opinion was secured, front page national newspaper and significant national media coverage achieved. Mid-term results included 2 roundtables (Dec 2018 and Nov 2019) with leaders from Health Canada, Canadian Institutes for Health Research Institute for Cancer Research, leading pediatric oncologists and researchers and patient advocates. Significant systemic changes resulted from these round-table discussions including Health Canada opening an Office of Pediatrics and streamlining the approval process for pediatric clinical trials and treatments. Longer term results included an announcement by the Liberal government of a \$30 million dollar investment in childhood cancer in their 2019 election platform and Prime Minister Justin Trudeau's inclusion of childhood cancer as a priority in the new Minister of Health's mandate letter (Dec 2019).

Conclusions: When a collaborative national advocacy campaign is developed and led by survivors and patient advocates with relevant professional backgrounds, it can create the political pressure for policy and funding improvements. With a multi-faceted approach, we drove the political pressure for health systems change and increased funding for childhood cancer in Canada

CCI LIVE Q&A-SESSION 2: ENHANCING COLLABORATION, JOINING FORCES ... 23-10-2021 12:30 PM - 2:00 PM

O0152 / #488 | Lebanon Cancer Cooperation: Cancer NGOS in Lebanon Unite for Public Awareness, Capacity Building, & Advocacy

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Background and Aims: Lebanon has a population of 6.8 million, yearly average of around 11,500 cancer diagnoses, and an increasing number of NGOs supporting cancer patients; with no sustainable coordination for optimal impact. The Children's Cancer Center of Lebanon (CCCL), initiated (July 2019) the creation of "Lebanon Cancer Cooperation" a network for cancer NGOs with a vision to reduce the burden of cancer on patients & families in Lebanon through: Advocacy; NGO capacity building programs; Public awareness.

Methods: We, at the CCCL, identified lack of sustainable cooperation between cancer NGOs and find opportunities for it. With the patronage & presence of HRH Princess Dina Mired, cancer control international advocate, then Union for International Cancer Control President, we invited cancer NGOs to a gathering and proposed the idea of the Cooperation. A presentation and discussion was done between CCCL Chair Dr. Cesar Bassim and NGO representatives. Forms were filled by representatives to confirm their interest in the Cooperation and outline priorities. Following the positive reply, meetings were done with Ministers of Public Health and Social Affairs to get their patronage too; which had positive replies. Subsequently, a meeting was organized for interested NGOs and we agreed on communication channels and meeting frequencies: giving all members a voice and equal rights. Monthly meetings were hosted by CCCL and discussions were made to come up -by end 2019- with the Cooperation's mission, vision, and framework.

Results: Since its initiation, the LCC members are meeting on a monthly basis coordinating national efforts. LCC has 50% engagement rate; and in April 2021 launched its first social media campaign in partnership with AstraZeneca promoting public awareness about the importance of regular cancer screening to ensure early detection, despite all the challenges posed by COVID.

Conclusions: National coalitions like LCC can have a great impact on the lives of cancer patients and communities.

CCI LIVE Q&A-SESSION 2: ENHANCING COLLABORATION, JOINING FORCES ... 23-10-2021 12:30 PM - 2:00 PM

O0153 / #338 | Developing a Working Paper to Cultivate Ethically Sustainable Relationships Among Stakeholders When Nonprofits Fund Pediatric Cancer Clinical Trials

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Background and Aims: Nonprofit organizations provide significant funding support for drug development research in pediatric cancer. The regulatory requirements that govern clinical trial sponsors, investigators, and institutions do not apply to nonprofits, however, potentially raising questions about the ethical, legal, and social responsibilities of nonprofits to the communities they serve. These include transparency in communications with donors and patient communities, especially when trial recruitment is involved. In this presentation, we will explore ethical questions surrounding nonprofit scientific review, conflicts of interest, and research priorities in patient communities.

Methods: We conducted a systematic review of the literature to answer the following question: What are the ethical, legal and social commitments of nonprofits to i) patients/families, ii) researchers, iii) sponsors, and iv) donors when making funding decisions to support new drug research and development in pediatric cancer. We applied a deductive analytical approach to responsibilities relevant to each stakeholder group identified in the literature using the “framework for patient involvement in regulatory processes” by Haerry and colleagues (2017).

Results: Our search produced 7397 articles indexed in PubMed and Web of Science. A collaborative working group of the Coalition Against Childhood Cancer (CAC2) screened 214 full text articles published in English. A total of 54 studies met the inclusion/exclusion criteria. Key themes that emerged from the review included strategies for minimizing conflicts of interest and ensuring transparent reporting and communication. In contrast, ethical standards for identifying and negotiating research partnerships as well as methods for setting research priorities were underrepresented.

Conclusions: Nonprofit organizations are taking on central roles in funding pediatric cancer clinical trials, but guidance on how to foster ethically sustainable relationships with all stakeholders in the pediatric clinical trial enterprise is markedly scant. Results from this review will inform a working paper that highlights relevant issues for nonprofits to consider when funding pediatric cancer clinical trials.

CCI LIVE Q&A-SESSION 2: ENHANCING COLLABORATION, JOINING FORCES ... 23-10-2021 12:30 PM - 2:00 PM

00154 / #1461 | Empowering Foundations, Strengthening Public-Private Collaboration for Childhood Cancer in Latin America

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Background and Aims: Social inequalities hinder access to treatment for children with cancer who otherwise have good possibilities of cure.

Civil society, public, and private sectors need to team up to build the necessary governance and resources that offer better opportunities to overcome this disease. We present a project that aims to empower and strengthen Non-Governmental Organizations in Latin America in order to improve collaboration between civil society and public and private health institutions towards the provision of comprehensive services for childhood cancer.

Methods: This project will work through three phases: (1) A self-evaluation and peer to peer evaluation system to improve organizational capacities (2) A playbook to share knowledge on good integral care practices (3) Training on successful programs among the NGOs.

Results: The first phase of the project started in August 2020. Due to the pandemic, it was developed in an online format, with monthly meetings with training for and discussions of organizational criteria to meet. Representatives of 14 non-profit organizations working for childhood cancer in Peru, Bolivia, Colombia, Costa Rica, Honduras, Ecuador, Paraguay, Chile, Brazil, and Mexico are participating. Auto-evaluation and peer-to-peer evaluations have started and are expected to conclude by September 2021. Main achievements of this phase are the participants' engagement in sharing knowledge and experiences, discussing administrative and organizational strategies to foster valuable strengths for short and long-term growth, and better self-positioning in the childhood cancer reality in their own countries.

Conclusions: Foundations and NGOs are of strategic value in reducing the inequalities that limit the possibilities of curing children and adolescents with cancer. If institutions adopt strategic vision, develop institutional credibility, and rely on technical and scientific knowledge, they can strengthen their partnerships with public services and substantially improve childhood cancer outcomes in Latin America.

CCI LIVE Q&A-SESSION 2: ENHANCING COLLABORATION, JOINING FORCES ... 23-10-2021 12:30 PM - 2:00 PM

00155 / #1124 | Partnerships in Time of Crisis

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Background and Aims: The political and military conflict that started in the Middle East, specifically Syria in 2011 had major consequences affecting CCCL. A number of refugee children were diagnosed with cancer. As the leading Cancer Center, CCCL actively responded to the critical medical needs of those patients.

Methods: In 2012, when the crisis aggravated, the number of refugee children approaching CCCL increased dramatically. In 2013, CCCL collaborated with SJCRH to support refugee children with cancer. However, due to the budget and the space constraints, a structured action plan was required to respond to the medical needs of those Children.

Results: In 2019, we collaborated with 7 hospitals across Lebanon and started supporting refugee patients there. Since treatment of cancer requires patients to stay near the hospital, we collaborated with other NGOs to ensure accommodation in hotels near the hospital. Due to

these partnerships, we were able to support more than 850 refugees by March 2021. On August 4th 2020, a devastating explosion hit Beirut. Hundreds of people died and thousands were injured, many hospitals were damaged. This all came where the country was facing COVID-19 and a deep economic crisis. CCCL as a result, was facing a major problem in fundraising. Despite all this, CCCL offered support in covering the treatment of children with cancer all over Lebanon. Moreover, CCCL coordinated with damaged hospitals to move patients to hospitals outside Beirut to make sure their treatment is not affected. Therefore, we had to expand more, thus our partners reached 14 hospitals. We also worked with all pediatric oncologists in Lebanon to make sure all children are taking the needed treatment without any delay.

Conclusions: Due to these efforts in 2020, CCCL allowed more than 377 new patients to have access to the right treatment and get the needed support in 14 hospitals in Lebanon.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS
23-10-2021 2:30 PM - 4:00 PM

O0156 / #1444 | Cell-Specific Gene Expression and Oncogene-Induced Cellular Senescence in Pediatric Hodgkin Lymphoma

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Background and Aims: The rarity of Hodgkin Reed-Sternberg (HRS) cells within Hodgkin lymphoma (HL) tumors (<1%) has limited the characterization of HRS cell gene expression, thus pathogenic mechanisms, in pediatric HL. Our objective was to define transcriptomes of HRS cells and tumor-infiltrating lymphocytes in order to inform pathogenic mechanisms and identify potential therapeutic targets in pediatric HL.

Methods: Multi-parameter flow cytometry was used to sort HRS cells, CD4+/CD8+ T-cells, and CD20+/30+ B-cells from pediatric subjects' HL lesions and control tonsils. Gene expression profiles (GEPs) were assessed using Affymetrix GeneChip HTA 2.0. GEPs of HL cells were compared to respective controls using a univariate t-test. Significance was determined using a multivariate permutation test to estimate false discovery. Differentially expressed genes (DEGs) were analyzed through gene set enrichment analysis (GSEA) and ingenuity pathway analysis (IPA). Immunohistochemistry (IHC) was performed on HL lesions to validate results of GSEA/IPA.

Results: GEPs were compared for HL samples: HRS vs. control CD20+/CD30+ (1934/3846 DEGs), HL CD4+ vs. control CD4+ (635 DEGs), HL CD8+ vs. control CD8+ (2 DEGs). Transcriptomic analysis (HRS vs. control CD30+) revealed that genes were significantly up/down-regulated in pathways related to cellular senescence: Cyclins and Cell Cycle Regulation (z-score=-3.0), G1/S Checkpoint Regulation (z-score=2.4), and G2/M DNA Damage Checkpoint Regulation (z-score=2.2). Upstream analysis of the G2/M pathway demonstrated

that senescence markers (P21, GADD45) were significantly upregulated in HRS cells (fold-change=2.27, 2.9), while proliferation markers (Cyclin B1, 2), were significantly downregulated in HRS cells (fold-change=0.09, 0.06). IHC demonstrated positive staining of senescence markers p21, p53, and senescence-associated beta-galactosidase within HRS cells.

Conclusions: Flow-sorting can feasibly characterize specific cell populations within HL tumors. HRS cells in this cohort expressed gene signatures consistent with oncogene-induced senescence – a novel finding in pediatric HL. These findings support further pre-clinical and clinical testing of novel therapeutic strategies that target pathologic senescence in pediatric HL.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS
23-10-2021 2:30 PM - 4:00 PM

O0157 / #627 | Whole-Body MRI Versus an FDG-PET/CT-Based Reference Standard for Staging of Paediatric Hodgkin Lymphoma: A Prospective Multicentre Study

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Background and Aims: To assess the concordance of whole-body MRI (WB-MRI) and an FDG-PET/CT-based reference standard for the initial staging in children with Hodgkin lymphoma (HL)

Methods: Children with newly diagnosed HL were included in this prospective, multicentre, international study and underwent WB-MRI and FDG-PET/CT at staging. Two radiologists and a nuclear medicine physician independently evaluated all images. Discrepancies between

WB-MRI and FDG-PET/CT were assessed by an expert panel. All FDG-PET/CT errors were corrected to derive the FDG-PET/CT-based reference standard. The expert panel corrected all reader errors in de WB-MRI DWI dataset to form the intrinsic MRI data. Inter-observer agreement for WB-MRI DWI was calculated using overall agreement, specific agreements and kappa statistics. Concordance for correct classification of all disease sites and disease stage between WB-MRI (without DWI, with DWI and intrinsic WB-MRI DWI) and the reference standard was calculated as primary outcome. Secondary outcomes included: positive predictive value, negative predictive value and kappa statistics. Clustering within patients was accounted for using a mixed effect logistic regression model with random intercepts and a multilevel kappa analysis.

Results: Sixty eight children were included. Inter-observer agreement between WB-MRI DWI readers was good for disease stage ($\kappa=0.74$). WB-MRI DWI agreed with the FDG-PET/CT-based reference standard for determining disease stage in 96% of the patients versus 88% for WB-MRI without DWI. Agreement between WB-MRI DWI and the reference standard was excellent for both nodal (98%) and extra-nodal (100%) staging.

Conclusions: WB-MRI DWI showed excellent agreement with the FDG-PET/CT-based reference standard. The addition of DWI to the WB-MRI protocol improved the staging agreement.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS 23-10-2021 2:30 PM - 4:00 PM

00158 / #565 | Dose-Intensive Chemotherapy Without Radiation for Pediatric Hodgkin Lymphoma in Low Resource Setting – Single Center Experience

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Background and Aims: Excellent survival for paediatric Hodgkin Lymphoma (HL) in high-income countries has been achieved with a combination of chemotherapy and radiation. The adoption of dose-intensive regimens in paediatric HL has reduced the need for radiation. Outcomes for paediatric HL in sub-Saharan Africa (SSA) remain poor, with only 50-60% of children surviving. More than 50% of countries in SSA lack access to radiation, and concerns for treatment-related toxicity limit the use of dose-intensive regimens. We investigated the impact of dose-intensive chemotherapy on survival for paediatric HL in Malawi – a country without access to radiation.

Methods: This was an IRB-approved retrospective cohort of children with biopsy-proven HL at Kamuzu Central Hospital in Lilongwe, Malawi, from 2015 to 2019. Patients (age ≤ 16 years) treated with ABVE-PC (adriamycin, bleomycin, vincristine, etoposide, prednisone, and cyclophosphamide) – a dose-intensive regimen used commonly in North America, were included in this analysis. Chemotherapy was administered every 21 days. The 2-year Progression-Free-Survival

(PFS) and Overall Survival (OS) were estimated using Kaplan-Meier analysis.

Results: Twenty-nine patients were included in the analysis. The median age was 11 years (IQR 4); 66% (n=19) had B symptoms, and 62% (n=18) presented with high-risk disease (stage IIB with bulk, IIIB or IV). The median number of cycles was 6 (range 4 - 8). At the last treatment follow-up, 86% (n=24) were alive and in clinical remission, and 14% (n=4) had died. No treatment-related deaths were observed. Three out of the four relapses occurred in patients who had treatment interruptions. The 2-year PFS and OS were 81% and 90% respectively.

Conclusions: Dose-intensive chemotherapy resulted in excellent survival without increased treatment-related mortality. With appropriate supportive care, this approach should be considered in low-resource settings, particularly when radiation is unavailable. A reduction in dose intensity arising from treatment interruptions was associated with an increased risk for relapse.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS 23-10-2021 2:30 PM - 4:00 PM

00159 / #482 | Results from an Open-Label, Phase 1/2 Study of Frontline Brentuximab Vedotin Plus Adriamycin, Vinblastine, and Dacarbazine in Pediatric Patients with Advanced Stage Hodgkin Lymphoma

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Background and Aims: In pediatric Hodgkin lymphoma (HL) settings, combined radiation and alkylating chemotherapy regimens are associated with serious sequelae. C25004 (NCT02979522) is a Phase (Ph) 1/2 open-label study assessing safety and efficacy of frontline brentuximab vedotin + adriamycin, vinblastine, and dacarbazine (A+AVD) in pediatric patients with advanced-stage classical HL (cHL).

Methods: Eligible patients aged 5 to <18 years, had untreated stage III/IV cHL, Lansky/Karnofsky Performance Status (LKPS) ≥ 50 , and bidimensional measurable disease by radiography. All patients received up to 6 28-day cycles of A+AVD (days 1 and 15 of each cycle). Ph1 primary objectives were assessment of safety, tolerability, and recommended Ph2 dose (RP2D) of brentuximab vedotin dosed at 48 mg/m² or 36 mg/m². Ph2 primary objectives included assessment of objective response rate (ORR), the proportion of patients achieving PET-negativity (Deauville score 1–3) after 2 cycles (PET2-), and the proportion of patients completing 6 cycles at the RP2D.

Results: In total, 59 patients were enrolled (Ph1 n=8, Ph2 n=51), median age was 14 years (range 6–17), 53% were male, and 51% had a LKPS of 50–90. RP2D was 48 mg/m²; all patients completed 6 cycles of A+AVD. ORR at end of treatment was 88% (95% CI: 77, 95), 76% of patients (95% CI: 63, 86) achieved CR. PET2- rate was 90% (95% CI: 79, 96). After 17.3 months (median) follow-up 13 patients (22%) had progressed. The most common any-grade treatment-emergent adverse events were vomiting (85%) nausea (75%), and neutropenia (58%); 41% of patients experienced serious AEs. The most common serious AE was febrile neutropenia (17%). There were no on-study deaths.

Conclusions: Frontline brentuximab vedotin (48 mg/m²) +AVD provided high rates of ORR, CR, and PET2- in pediatric patients with advanced cHL, aligning with previously reported adult data (ECHELON-1 study, NCT01712490). Patients continue in follow-up.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS
23-10-2021 2:30 PM - 4:00 PM

O0160 / #1177 | The Profiling of Circulating Tumor DNA in Pediatric Mature B-Cell Non-Hodgkin Lymphoma(B-NHL): A Multicenter, Prospective Clinical Study

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Background and Aims: The next-generation sequencing (NGS) based on liquid biopsy has been an emerging technology to identify tumor-specific genetic aberrations in adult lymphoma. However, there were few studies on the genomic profiling of plasma circulating tumor-derived DNA (ctDNA) in pediatric mature B-NHL.

Methods: The tumor tissue (FFPE) and plasma samples from the newly diagnosed patients were collected and sequenced by 475 genes panel before, during and post treatment. Clinical stage system, risk stratification and treatment of pediatric mature B-NHL were according to the modified BFM-95 protocol.

Results: A total of 35 pediatric mature B-NHLs were enrolled. The somatic mutations and the TOP 5 genes detected in the 25 tissues and 24 baseline plasma samples, were 200 vs 212, and MYC(68%), ID3 (44%), DDX3X(36%), TP53(28%), ARID1A(28%)vs MYC(52.2%), ID3(43.5%), DDX3X(39.1%), TP53(34.8%), GNA13(26.1%), respectively. The sensitivity of plasma ctDNA to testing tissue mutations was 63.8% in the 15 matched samples (10 samples from BL and 5 samples from DLBCL/HGBCL), and the sensitivity in BL and DLBCL/HGBCL were 59.5% and 75.0%, respectively. Among the 24 mature B-NHL patients, the abundance of ctDNA in patients with stage III-IV (18/19) was significantly higher than that of stage I-II patients (5/5) (P=0.005). Until last follow-up, 21 patients have completed anti-tumor treatment and post-treatment PET-CT, and 19 patients have done ctDNA testing synchronously. We noticed that PET-CT reflected residual tumors were observed in 3 patients, but both of the biopsy pathology and ctDNA testing showed no tumor residual in 2 of these 3 patients.

Conclusions: Plasma ctDNA testing by NGS was practicable in pediatric mature B-NHL. The abundance of ctDNA is significantly related to tumor burden. CtDNA testing was more sensitive than PET-CT for residual disease assessment. Nevertheless, sample size expansion is required to verify such conclusions.

Free Paper Session (FPS) FPS 14 -LIVE Q&A-LYMPHOMAS
23-10-2021 2:30 PM - 4:00 PM

O0161 / #1494 | Burkitt Lymphoma Outcomes Following Implementation of Tanzania National Treatment Guidelines

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Background and Aims: Evidence-based treatment guidelines elicit best-practice performance from healthcare practitioners and improve patients' outcomes. National treatment guidelines for the treatment of Burkitt lymphoma were developed and implemented in Tanzania in 2016. The following study evaluated guideline compliance and patient outcomes following guideline implementation.

Methods: All pediatric patients (<18yo) diagnosed with Burkitt lymphoma from 1/2016-12-2019 were eligible for evaluation. Demographic information, presentation, and treatment compliance were extracted from hospital records. Compliance was measured based on completion of recommended laboratory testing, imaging, and chemotherapy based on the published National treatment guideline. Outcome measures included treatment abandonment and 1-yr overall survival.

Results: A total of 66 patients were reviewed, 59.1% female (27/66), average age of 6.7 years. The average symptom duration was 2.6 months (SD 3.1), and the most common reported symptoms were jaw swelling (46.9%) and abdominal swelling (39.1%). Most (98.2%, n=55) were high risk staging. A total of 44.4% (n=28) were diagnosed using cytology or histopathology. 19% had a clinical diagnosis. There was a 39% compliance with national Burkitt treatment guideline compliance, with 31% of patients abandoning treatment, and a 45% 1-yr overall survival rate.

Conclusions: Overall guideline compliance was low, resulting in a lower than expected 1-yr overall survival as compared guideline use during clinical trial (45% vs 67%). Complex care coordination and limited provider training are critical challenges in the implementation of treatment guidelines. The mitigation of these health system barriers are needed to improve guideline compliance and impact treatment outcomes.

Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR CLINICAL 23-10-2021 2:30 PM - 4:00 PM

00162 / #1270 | Identifying Therapeutic Options for Pediatric Patients with Central Nervous System (CNS) Tumors Through Precision-Guided Deep Sequencing of DNA and RNA

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Background and Aims: The Pediatric Precision Genomics Program (PPGP) at Indiana University School of Medicine provides both CLIA approved tumor and germline DNA/RNA sequencing for children with high-risk, relapsed, or refractory cancers. Over time, the PPGP began sequencing more upfront diagnoses, particularly for our neuro-oncology patients in whom DNA variations are common, to elucidate prognostic features, open avail-

ability of clinical trials, and guide selection of novel molecular therapeutics.

Methods: DNA and RNA whole exome sequencing is performed on tumor and/or germline samples via CLIA approved labs (Paradigm™, NANTHealth®, Tempus®, Foundation Medicine®, or Ashion®). Via data agreements between these companies, VCF files are downloaded for institutional DNA, RNA, and germline analyses. RNAseq and germline data are generated for research analysis only by these companies and the PPGP communicates to families RNAseq and germline variations based on pathogenicity from our internal analysis.

Results: A total of 70 out of 71 (99%) patients with CNS tumors referred to the PPGP since April 2016 had successful sequencing. Actionable variants were identified in 61 of the 70 patients (87%) with 28 (40%) that received PPGP-directed targeted therapy. Of these 28 patients, 7 had partial response (25%) to therapy with 6 enrolling on clinical trials based on sequencing. The most abundant actionable finding was a BRAF gene fusion or missense mutation in 35% of patients. Germline analysis found 13 of 71 patients (18%) had variants in 7 known cancer predisposition genes, with biallelic loss confirmed in 6 patients. These patients all had additional genetic counseling and confirmatory testing.

Conclusions: Despite lack of in-house sequencing efforts like other large pediatric institutions, the PPGP provides sequencing and analysis of tumor and germline DNA/RNA in pediatric CNS tumors by partnering with CLIA approved sequencing companies. This analysis uncovers the benefit to expanded sequencing in this population and emphasizes its importance for informed decision-making of treatment plans.

Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR CLINICAL 23-10-2021 2:30 PM - 4:00 PM

00163 / #646 | The Therapeutic Potential of Smart Polymeric Nanocontainers in Pediatric Central Nervous System Malignancies

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Background and Aims: Pediatric pilocytic astrocytomas represent >20% of all primary brain tumours and medulloblastomas account for 20% of all malignant cases. Little progress has been made regarding their curative therapeutics compared to the good progress made revealing the several molecular/genetic factors that drive their development. Current chemotherapeutic agents lack in specificity and selectivity, resulting in unpleasant side effects for the child. To improve drugs' efficacy and subsequently the therapeutic outcome, one approach is to encapsulate them within nanocontainers (NCs).

Biocompatible and biodegradable polymeric NCs can cross the blood-brain-barrier without causing any damage, due to their nano-size and suitable modified surface. The aim was to synthesize polymeric NCs capable of delivering first-line chemotherapy drugs at the tumour site to increase their efficacy and reduce their side effects.

Methods: Hollow NCs were synthesised following three steps: 1) PMAA core synthesis, 2) Coating it with MAA and EGDMA copolymers, and 3) Removal of the PMAA core to obtain hollow NCs P(MAA-co-MBA-co-EGDMA). Throughout the synthesis steps, polymeric NCs were characterized structurally by Fourier-Transform-Infrared-Spectroscopy and Dynamic-Light-Scattering, and morphologically by Scanning-Electron-Microscopy. Daunorubicin -as gold-standard drug- and cisplatin were loaded in the nanocontainers. The NCs were used for the following biological evaluations: haemolysis assay, wound healing assay, MTT assay, and fluorescence microscopy.

Results: The NCs' compatibility and loading capacity were investigated. The haemolysis assay showed <8% haemolysed RBCs for the concentrations of nanocontainers tested. Daunorubicin and cisplatin were loaded in NCs by 55% (± 10). Free NCs were shown to be non-cytotoxic in MTT assay and loaded NCs showed good cellular uptake in fluorescence microscopy.

Conclusions: Re-synthesis of the polymeric NCs will take place to further optimise their loading capacity for a sustained drug release and prolonged drug biodistribution, whilst additional experiments will be performed to evaluate their efficacy in pilocytic astrocytoma and medulloblastoma cell cultures and mouse models using different drugs.

Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR CLINICAL 23-10-2021 2:30 PM - 4:00 PM

00164 / #143 | Quantifying Intrathecal Drug Delivery Utilizing Programmable Ventriculoperitoneal Shunts

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Background and Aims: Programmable ventriculoperitoneal shunts (pVP shunts) are increasingly utilized for intraventricular chemotherapy, radioimmunotherapy, and/or cellular therapy. Shunt adjustments allow optimization of drug concentrations in the thecal space with minimization in the peritoneum. Drug delivery quantification using several types of pVP shunts has not been reported.

Methods: We performed a retrospective analysis on patients with CNS tumors and pVP shunts at Memorial Sloan Kettering Cancer Center from 2003-2020, noting shunt model. CSF flow through the pVP

shunt was evaluated using In-111-DTPA scintigraphy at approximately 4 hours and 24 hours after injection. pVP shunts were calibrated pre-injection to minimize peritoneal flow and re-calibrated to baseline setting 4-5 hours following injection. Scintigraphy studies quantified ventricular-theal and peritoneal drug activity at these 2 time points.

Results: Twenty-one CSF flow studies were administered to 15 patients, ages 1-27 years. Diagnoses included medulloblastoma (N=10), metastatic neuroblastoma (N=3), pineoblastoma (N=1), and choroid plexus carcinoma (N=1). Models of pVP shunts included Aesculap Miethke proGAV (N=3), Aesculap Miethke proGAV2.0 (N=3), Codman HAKIM (N=2), Codman Certas Plus (N=1), Medtronic STRATA (N= 5), and Sophysa Polaris (N= 1). All 21 studies (100%) demonstrated ventriculo-theal drug activity. 29% (6 of 21) of the studies had no peritoneal uptake visible by imaging. 73% (16 of 21) of the studies had minimal peritoneal uptake (<12%), and 24% (5 of 21) demonstrated moderate peritoneal uptake (12-37%). Models of pVP shunts measuring minimal to no peritoneal uptake included: Aesculap Miethke proGAV (N=2), Aesculap Miethke proGAV2.0 (N=3), Codman HAKIM (N=2), Codman Certas Plus (N=1), Medtronic STRATA (N= 3), and Sophysa Polaris (N= 1).

Conclusions: Successful drug delivery to the ventriculo-theal space can be accomplished using pVP shunts: 80% of studies have minimal (<12%) peritoneal drug activity. Though efficacy varies by shunt model, low numbers preclude conclusions regarding model superiority. CSF flow scintigraphy studies reliably assess drug distribution.

Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR CLINICAL 23-10-2021 2:30 PM - 4:00 PM

00165 / #1727 | Radiomics and Dosiomic Analysis in Medulloblastoma Patients Treated with Craniospinal Irradiation for Predicting Radio-Induced Neurotoxicity

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Background and Aims: The purpose of this study is to apply a retrospective exploratory MR-CT-based radiomics and dosiomic analysis based on machine-learning technologies, to investigate imaging-based biomarkers of clinical outcomes in paediatric patients affected by medulloblastoma.

Methods: Data of 50 medulloblastoma patients treated with radiotherapy were retrospectively collected. For all patients planning CT and MR images at baseline to the last follow-up are available. In a wider and deeper research, multiparametric data were considered, including: MRI before radiotherapy and dose distribution of the radiotherapy plan. Features from those images were extracted with PyRadiomics and analysed with two different programs, a homemade script and RadAR (Radiomics Analysis with R). Ten features were associated with radio-induced toxicity occurrence. Random Forest classifier was trained on different combinations of the available data. RadAR performs analysis of radiomic features, implementing multiple statistical methods. To compare multiple features together, scaling and normalization were applied. An unsupervised analysis based on hierarchical clustering was used together with the Fisher's exact test to estimate statistical significance.

Results: The machine learning performance was trained and evaluated within a 5-fold cross validation method. At each iteration, a different test set consisting of an equivalent number of subjects with and without toxicity was used. Individual dose, T1w, T2w and FLAIR data were not predictive, leading to average accuracies of 0.44 ± 0.21 , 0.56 ± 0.16 , 0.59 ± 0.16 and 0.41 ± 0.07 , respectively. While combining all multiparametric information a better prediction performance was obtained: 0.62 ± 0.09 . All the features available were predictive with a specificity of 0.79 and sensitivity of 0.65.

Conclusions: This multimodal and multiparametric approach could have a large impact for precision medicine, as radiomic biomarkers are non-invasive and can be applied to imaging data that are already acquired in clinical settings.

**Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR
CLINICAL 23-10-2021 2:30 PM - 4:00 PM**

00166 / #1335 | Oral Vinorelbine Exposure Variability Among Pediatric Patients: Potential Role of ABCB1, ABCG2, and UGT2B7 Genetic Variants

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Background and Aims: Oral Vinorelbine (VNR) is a promising drug in pediatric cancers requiring prolonged treatment or metronomic chemotherapy. Better understanding of pharmacokinetics (PK) of oral VNR in children would help predicting drug exposure. Here we characterized the population PK (Pop-PK) of VNR in children and studied several factors likely to explain the variability in VNR exposure and subsequent clinical outcome.

Methods: We collected VNR PK samples from 36 patients of the OVIMA multicentric phase II study in children with recurrent / progressive Low-Grade Glioma. Pop-PK analysis was performed using a non-linear mixed-effects modeling approach on Monolix®. Parameters were estimated by the SAEM algorithm. Fifty SNPs on PK-related genes were genotyped. The Influence of demographic, biological and pharmacogenetic covariates on VNR PK parameters was investigated using a stepwise multivariate procedure.

Results: A three-compartments model, with a delayed double zero-order absorption and a linear elimination, best described VNR pharmacokinetics. Typical population estimates (between-subject variability, between-occasion variability) for the apparent central volume of distribution (Vc/F) and elimination rate constant (k_{10}) were 803 L (47%, 41%) and 0.60 h^{-1} (18%, 16%), respectively. Model diagnosis showed good descriptive and predictive performances. Following covariate analysis, BSA, white blood cells count, and ABCB1-rs2032582 SNP on the Vc/F were clinically relevant. Additionally, ABCG2-rs2725264 A/G or G/G carriers and UGT2B7-rs7668258 T/T carriers showed a higher k_{10} (i.e., 30% and 42% compared to reference alleles, respectively) and included in the final model. Conversely, age and sex had no significant effect on VNR PK.

Conclusions: ABCB1, ABCG2, and UGT2B7 variants showed a significant impact on VNR systemic exposure. Univariate and multivariate simulations showed that the identified covariates on VNR plasma exposure could have an impact on both efficacy and toxicity outcomes. Thus, a personalized dosing strategy considering those covariates would help reducing PK variability to optimize the efficacy/toxicity balance of VNR.

**Free Paper Session (FPS) FPS 15 -LIVE Q&A-BRAIN TUMOR
CLINICAL 23-10-2021 2:30 PM - 4:00 PM**

00167 / #432 | Phase II Trial of Oral Vinorelbine in Children with Progressive or Recurrent Unresectable Low-Grade Glioma

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Background and Aims: Many patients with progressive unresectable low-grade glioma (PULGG) require several lines of treatment, leading to long-term side effects and/or impacting quality of life. Previous studies showed that intravenous vinorelbine, a semi-synthetic vinca-alkaloid, has some activity in recurrent LGG.

Methods: Main objective of this multicenter phase-II trial was to assess efficacy of oral vinorelbine (OV) in children with recurrent/progressive PULGG, in terms of Progression-Free Survival (PFS) using a Two-stage Minimax Simon design. Patients aged 6-18 years with histologically confirmed PULGG or typical optic pathway glioma were enrolled. OV was given at the dose of 60 mg/m² at days 1, 8 and 15 during cycle-1, then 80 mg/m² for subsequent cycles depending on safety, for 12 cycles.

Results: Thirty-eight patients were enrolled and treated between May-2014 and December-2018. Median age was 11.5 years (range, 6-17). Median time between diagnosis and study entry was 6.8 years (1.5-16). Two children have type-1 neurofibromatosis. Molecular analysis revealed a KIAA1549-BRAF fusion in 15/26 patients (58%), and a V600E-BRAF mutation in 1/27 patient. Prior treatment included a median of 2.5 (1-9) lines of systemic treatment, with vinca-alkaloid for all patients, surgery (60.5%) and/or radiotherapy (18.4%). Median number of treatment cycles was 12 (3-18). Treatment was prematurely stopped for 15 patients (39%), none due to toxicity. PFS and Overall Survival at 9 months, 2 and 5 years were 75.9%, 66.1%, 33.7% and 97.4%, 94.5%, 71.4%, respectively. Median relative dose-intensity (RDI) was 94%, with only 9 patients (23.7%) with RDI<80%. Most frequent toxicity was grade 3+ neutropenia (15 patients, 39.5%). Only 2 patients experienced grade 3 gastro-intestinal adverse event classified as treatment-related.

Conclusions: Oral vinorelbine appears a well tolerated outpatient treatment, leading to significant disease control in heavily pre-treated children with PULGG. These findings suggest that it should be considered earlier in the management of children with PULGG.

Free Paper Session (FPS) FPS 11-LIVE Q&A-NEUROBLASTOMA CLINICAL 23-10-2021 2:30 PM - 4:00 PM

00168 / #406 | Determining the True Incidence of Neuroblastoma in South Africa

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Background and Aims: According to the Surveillance, Epidemiology, and End Results (SEER) Program, the incidence for neuroblastoma (NB) in the USA is 10.5/ million children. The reported South African incidence (2015) is 2.7/ million children, based on the South African Children's Tumour Registry (SACTR), a clinical data-based registry. We determined the true incidence of NB in South Africa using multiple data sources.

Methods: We conducted a probabilistic record linkage using data from three sources: NB cases from nine paediatric oncology units in SA, the South African National Cancer Registry (a national pathology-based cancer registry) and the SACTR. For the purpose of record linkage, the cases from the nine oncology units were merged with the SACTR. We included patients diagnosed between 2000 and 2016.

Results: There were 463 cases from hospital-based records, 312 and 603 cases from the South African National Cancer Registry (SA-NCR) and SACTR respectively. After excluding duplicates (n=148) and patients with insufficient data (n=14), 824 cases were included for linkage. Only 329 records (39.9%) matched between these respective registries. The SA-NCR and SACTR reported between 23 - 51 cases per annum and 18 - 57 cases per annum respectively. The SA-NCR incidence for children under 15-years varied between 1.5-2.8/ million children and between 1.74-2.6 /million children in the SACTR. A probabilistic record linkage from all the sourced data resulted in a combined incidence of 2.9 cases/ million children.

Conclusions: South Africa still reported a lower incidence of NB than the SEER Program. The reasons for a lower incidence are not clear, but might be due to a true lower incidence, underdiagnoses or under-reporting of NB cases. Underdiagnoses may be due to spontaneous regression of tumours, death before the diagnosis of NB was made or misdiagnosis as other diseases.

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00169 / #1526 | Minimal Residual Disease Detection in the Bone Marrow of Neuroblastoma Patients by Automatic Immunofluorescence Plus Fish (AIPF) and QRT-PCR - A Multicentre Feasibility Study

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Background and Aims: More than half of neuroblastoma (NB) patients' present with bone marrow (BM) infiltration at diagnosis. Cytomorphology (CM) is the gold standard to detect BM disease, however qRT-PCR using an mRNA marker panel containing adrenergic (ADRN) and mesenchymal (MES) neuroblastoma markers, as well as automatic immunofluorescence plus FISH (AIPF), have been demonstrated to be feasible, more sensitive, and clinically relevant. In this study we compare BM disease detection by cytomorphological examination of BM smears, AIPF, and qPCR.

Methods: 312 BM samples from 124 Austrian and Dutch non-high-risk- and high-risk NB patients, taken at diagnosis and during treatment (HR-NBL-1/SIOPEN and NBL2009), were prospectively obtained. AIPF for GD2/CD56/DAPI was performed at the CCRI (A) and qRT-PCR at Sanquin (NL) using the ADRN mRNA panel: β -glucuronidase (GUSB), paired-like homeobox 2b (PHOX2B), tyrosine hydroxylase (TH), cholinergic receptor alpha 3 (CHRNA3), and growth-associated protein 43 (GAP43). CM was performed independently at local sites.

Results: In total 308 samples were analysed by all techniques, 172 (= 55.8 %) were positive for one or more techniques. Both AIPF and qRT-PCR detected BM disease more frequent than CM, with qRT-PCR to a greater extent (168 qRT-PCR vs. 94 AIPF vs. 46 CM positive samples). AIPF and qRT-PCR double positive samples (n=90) show a high correlation of calculated infiltration levels. All three techniques were able to detect high level tumour cell infiltration. At very low levels, n=4 samples were positive by AIPF only (range 2.1e-5 – 1.0e-4%) and n=37 by PHOX2B qRT-PCR only (range 6.9e-6 – 0.5%). Interestingly, in the course of treatment a fraction of patients with negative samples showed positivity in subsequent samples.

Conclusions: Our findings demonstrate feasibility of cross-border prospective analysis and the sensitivity for the detection of BM disease in NB patients through the validation of AIPF and qPCR.

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00170 / #1512 | Practice Patterns for Primary Site

Radiotherapy in High-Risk Neuroblastoma at US Proton Centers: A Report from the Pediatric Proton/Photon Consortium Registry (PPCR)

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Background and Aims: Radiotherapy to primary site is part of high-risk neuroblastoma (HRNB) treatment paradigms in US and Europe. Dose of 21.6 Gray (Gy) has been utilized within the Children's Oncology Group (COG) since benefit of 22Gy vs 10Gy shown in CCG3891 (Haas-Kogan, 2003). Dose escalation to 36Gy for residual disease was introduced on ANBL0532 and accepted as standard until 4/2019; historical comparison then demonstrated no benefit. The study below was carried out to evaluate impact of this change on practice patterns within the US proton therapy community.

Methods: Patients treated for HRNB with proton therapy to the primary site between 2014-2019 were enrolled on the PPCR. Kaplan-Meier method was used to evaluate DFS and OS. Relationship between maximal surgical resection and radiation dose was compared using Chi-square.

Results: Proton therapy delivered to 147 patients with HRNB at 13 US centers; 86 (59%) male and 101 (70%) white. All were treated according to standard chemotherapy protocols, 22% enrolled on clinical trial. OS at 5y was 68% (95% CI 52%-79%) and DFS 52% (95% CI 36%-67%). Median radiation dose was 27.3Gy (range 15-39.6Gy, mean 30.9Gy); 48% of pts received <24Gy and 44% $\geq$ 36Gy. Maximum surgical resection was reported as biopsy (Bx)/subtotal (STR) (n=60, 51.7%), near total (NTR) (n=9; 7.8%), or gross total resection (n=47, 40.5%). Most patients (85%) were treated before 4/2019. In this group, 49% of patients received $\geq$ 36Gy after Bx/STR/NTR compared with 28% after GTR (p = 0.063). For patients treated after 4/2019, 20% received $\geq$ 36Gy after Bx/STR/NTR compared with 0 after GTR (p = NS).

Conclusions: Our data suggest that findings from COG resulted in major shift in practice patterns, resulting in many fewer patients receiving dose escalation (including those not enrolled on trial). Future analysis will allow further understanding of impact of radiation dose on late-effects/ disease-related outcomes in the setting of HRNB.

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00171 / #679 | Naxitamab Treatment of Refractory/Relapsed High-Risk Neuroblastoma (R/R HR NB); Subgroup Analysis of Updated Interim Efficacy and Safety Data for The Registrational Phase II Trial

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Background and Aims: High-risk neuroblastoma represents a treatment challenge; frequently metastatic at presentation and residual disease in bone or bone marrow (BM) increases the risk of relapse. Naxitamab was recently approved in the US in combination with GM-CSF for the treatment of R/R HR NB in the bone or BM in patients ≥ 1 year of age who demonstrated a partial response, minor response, or stable disease to prior therapy. Here we present updated interim subgroup analyses from the pivotal study supporting this accelerated approval.

Methods: Study 201 (NCT03363373) is an international multicenter phase II study of naxitamab in patients with R/R HB NB with disease in bone/BM. Naxitamab was administered over ≥ 30 min in the outpatient setting on Days 1, 3 and 5 at 3 mg/kg/infusion in combination with GM-CSF at 250 $\mu\text{g}/\text{m}^2/\text{day}$ on Days -4 to 0 and at 500 $\mu\text{g}/\text{m}^2/\text{day}$ on days 1 to 5. Subgroups assessed include refractory vs relapsed disease and disease by compartment (bone/BM).

Results: The safety population comprised 48 patients, the efficacy population 36 patients. The overall response rate (ORR) was 58% (21/36) with a complete response rate (CR) of 44% (16/36) and a partial response rate of 14% (5/36) as per independent review. The ORR was 47% (CR 29%) for relapsed disease and 68% (CR 58%) for refractory disease. Compartmental responses were 68% (CR 47%) in bone and 75% (CR 75%) in BM. The safety profile was manageable, similar to previously reported interim results.

Conclusions: Naxitamab provided a clinically meaningful response in patients with R/R HR NB with bone/BM disease, regardless of disease status and compartment. The safety profile was manageable in the outpatient setting. Considering current treatment options and unmet medical need, naxitamab presents a unique treatment option for these patients.

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00172 / #343 | First in Children Phase I Clinical Study of Oral Olaparib in Pediatric Patients with Refractory Solid Tumors

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Background and Aims: There is no established standard chemotherapy for recurrent pediatric solid tumors. Since some of these tumor cells show dysfunctions in homologous recombination repair, the goal is to conduct a phase I study of olaparib, a poly(ADP-ribose) polymerase (PARP) inhibitor. In this clinical trial, the aims are to evaluate the safety, tolerability, and efficacy of olaparib in pediatric patients with refractory solid tumors and to recommend a dose for phase II trials.

Methods: This phase I trial assessed the PARP inhibitor olaparib (AZD-2281) as monotherapy. In this open-label, multicenter study, olaparib tablets (62.5, 125, and 187.5 mg/m² b.i.d.) will be administered orally every day (1 cycle 28 days) in a standard 3 + 3 dose-escalation design. Patients aged 3 to 18 years with recurrent pediatric solid tumors are eligible. Pharmacokinetic and pharmacodynamic analyses will also be performed.

Results: Fifteen patients were enrolled; 15 received olaparib monotherapy, which was safe and well-tolerated. The recommended phase II dose (RP2D) for everyday administration was 187.5 mg/m² b.i.d. Among 13 evaluable cases, two patients achieved RECISTv1.1 partial response. A patient with metastatic Wilms harboring ATM mutation and loss of heterozygosity and A patient with neuroblastoma. Pharmacokinetics were dose-proportional and by comparison with previously published adult data, the value of C_{max} in 187.5 mg/m² bid in the child was similar to the data of the 200 mg

bid cohort in adults. In terms of AUC₀₋₁₂, the value of 187.5 mg/m² bid in the child was similar to the 200 mg bid cohort but lower than the 300 mg bid cohort in adults. Pharmacodynamic studies demonstrated substantial inhibition of PARP activity.

Conclusions: This is the first report of a PARP inhibitor as monotherapy in children. Olaparib was well tolerated, with preliminary antitumor responses with DNA damage response defective pediatric tumor observed.

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O0173 / #857 | Long Term Follow-Up of Treatment with CH14.18/SP2/0 in Children with Metastatic High-Risk Neuroblastoma: Long-Term Toxicity and Quality-Of-Life

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Background and Aims: ch14.18/SP2/0 treatment significantly improves prognosis in high-risk neuroblastoma (HR-NB) patients. Studies assessing long-term outcome, impact on long-term sequelae and quality-of-life are lacking.

Methods: Between 1992 and 2003 a cohort of 63 children (aged 0.4-17.6 years, median 3.1 years) received ch14.18 for primary HR-NB (INSS-stage 4 in 59 patients (93.6%), 4 (6.4%) stage 3 with MYCN-amplification). All patients received induction-chemotherapy according to German NB90/NB97-protocol, all but 7 received HD-chemotherapy+autologous stem cell transplantation (ASCT). Four to eleven cycles of ch14.18 were administered, 100mg/m²/cycle. We evaluated overall- and event-free-survival, risk-factors and conducted a questionnaire-based survey in long-term survivors to determine the proportion with severe toxicity and impact on quality-of-life (EORTC-QLQ-C30).

Results: Overall-survival and event-free-survival 25 years after initial diagnosis were 54.2% and 46.0%, 27 patients died, causes were progression/relapse (n=22), secondary neoplasm (MDS=4, Adenocarcinoma=1); Lost to follow-up: 5 patients. Thirty-one long-term survivors were identified with median follow-up of 24.6 years

(15.5-32.2 years). Long-term survivors underwent ASCT (n=24), tandem-transplant (n=1), allogeneic SCT (n=1). Additionally, radiotherapy (n=15) and/or mIBG-therapy (n=17). Twenty patients were evaluated for QoL and long-term effects. In all survivors long-term sequelae occurred: Ototoxicity in 15/20 (75%), cardiac toxicity in 3/20 (15%), pulmonary toxicity in 1/20 (5%), nonreproductive endocrine toxicity in 17/20 (85%), renal toxicity 3/20 (15%). Height/weight z-scores were less than -1.7 in 2/20 (10%). 3/31 (9.4%) developed secondary malignancies: MPNST(1), colon carcinoma(1), basaloma (1), all within radiation-field. All QoL-scores were comparable to those of general population, with women showing slightly lower functioning and higher symptom-levels. All patients attained secondary education, 8 graduated from university. Two female patients have children of their own.

Conclusions: Immunotherapy with ch14.18 can induce long-term survival in HR-NB patients. The high incidence of long-term sequelae underlines the importance of careful follow-up. EORTC-QLQ-C30 scales were comparable to general population. With the limitation of non-randomization, no ch14.18-specific long-term sequelae could be identified.

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O0174 / #458 | Predictors of Severe Toxicity and Poor Efficacy of Reinduction Chemotherapy in First Relapse of B-Acute Lymphoblastic Leukemia: Report from the Children's Oncology Group (COG) AALL1331

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Background and Aims: Standard treatment for B-ALL first relapse begins with reinduction chemotherapy. Predictors of severe toxicity and poor efficacy would identify patients for whom alternative approaches could be prioritized.

Methods: Patients aged 1-30 years received reinduction with vincristine, dexamethasone, pegaspargase, mitoxantrone and intrathecal chemotherapy (UKALLR3 mitoxantrone arm), followed by risk stratification and randomization to chemotherapy vs. blinatumomab. We reviewed reinduction response [morphologic and flow minimal residual disease (MRD)] and adverse events (AEs) by age (<12 vs 12-17 vs 18+ years), sex, site of relapse [marrow vs extramedullary (EM)] and duration of first remission (<18 months vs. 18-36 months vs >36 months). Pearson's chi-squared test or Fisher's exact test was used to evaluate associations. P values are two-sided and a p value <0.05 was considered statistically significant.

Results: Of 662 patients starting reinduction, 24 (4%) died during reinduction (22 AEs; 2 disease progression). Death was associated only with shorter first remission duration ($p=0.006$). Of those surviving, 44 (7%) had treatment failure (M3 marrow and/or persistent CNS disease) and 67 (11%) had non-fatal severe AEs (grade 4 febrile neutropenia, infection, sepsis or mucositis). Treatment failure was associated with site of relapse (marrow greater than EM, $p=0.029$) and shorter first remission duration ($p<0.001$). Severe AEs were associated only with older age ($p=0.014$). Of patients without death or treatment failure, 272 (46%) were MRD-positive ($\geq 0.01\%$); MRD-positivity was associated with older age ($p=0.038$) and marrow relapse ($p<0.001$).

Conclusions: Patients with early relapse are at higher risk of toxic death and early treatment failure. Older patients are at higher risk of non-fatal severe AEs and MRD positivity. Patients with bone marrow relapse are at higher risk of early treatment failure and MRD positivity. Males and females have similar risks of poor reinduction outcomes.

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00175 / #549 | Real World Treatment of Pediatric Patients with Down Syndrome and Relapsed/Refractory (R/R) B-Cell Acute Lymphoblastic Leukemia (B-ALL) Treated with Tisagenlecleucel

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Background and Aims: Individuals with Down syndrome and B-ALL (DS-ALL) have increased toxicity with chemotherapy and those who relapse have poor outcomes. Treatment with tisagenlecleucel, a CD19-targeted Chimeric Antigen Receptor T-cell (CART), demonstrated favorable outcomes of r/r B-ALL in pediatric patients. The aim of this study was to determine the safety and efficacy of tisagenlecleucel in patients with r/r DS-ALL treated in a real world setting.

Methods: Retrospective data from 185 patients treated with tisagenlecleucel at 15 centers was extracted from the Pediatric Real World CART Consortium database and data from patients with DS-ALL was analyzed.

Results: Fifteen patients with DS-ALL received tisagenlecleucel. The median age at diagnosis was 7 years (range 3-17), the median age at infusion was 10 years (range 6-17), and 10/15 patients were male. Disease status at the time of CART included primary refractory (n=4), 1st relapse (n=10), and ≥ 2 relapses (n=1). No patients received a stem cell transplant (SCT) or CD19 directed therapy prior to infusion.

Cytokine release syndrome $\geq$ grade 3 occurred in 3/15 and 2 patients had neurotoxicity. One patient experienced an infection $\geq$ grade 3. Complete remission (CR) by morphology occurred in 14/15 (93%) patients at day 28, and 13 also became minimal residual disease negative. Two patients died, 1 was refractory and died of disease after 303 days, and 1 died of infection while in remission after 123 days. Relapse occurred in 3/15 patients (CD19+ n=2, CD19- n=1). Ongoing remission in 10 patients at the time of last follow up ranged from 4 to 25 months (median 13). Of these 10, 4 received no additional treatment, 5 had loss of B-cell aplasia and received additional treatment (SCT n=2, 2nd CART n=2, both n=1). One patient received a 2nd CART infusion with continued B-cell aplasia.

Conclusions: Tisagenlecleucel was well tolerated and effective in patients with r/r DS-ALL.

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00176 / #1458 | Bridging Chemotherapy Prior to CD19-Specific CAR T-Cell Therapy in Children/Young Adults with Relapsed/Refractory Acute Lymphoblastic Leukemia

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Background and Aims: CD19-specific chimeric antigen receptor (CAR) T-cells achieve response and durable remission in children/young adults with relapsed/refractory (R/R) B-cell acute lymphoblastic leukemia (B-ALL). Chemotherapy is utilized in the bridging period (during CAR T-cell manufacturing) with a goal of halting or slowing disease progression while minimizing complications in this heavily pretreated population. We report clinical outcomes of the bridging treatment strategies in a cohort of pediatric/young adult R/R B-ALL patients since 2013.

Methods: This study included patients enrolled on clinical trial NCT01860937 or referred to MSKCC for commercial CAR T-cell therapy (tisagenlecleucel). Bridging chemotherapy (given after T-cell collection and prior to CAR T-cell infusion) was defined as high intensity if myelosuppression was expected for > 7 days. Outcome comparison analyses were performed in high versus low intensity bridging therapy groups and 0-1 versus 2+ lines of bridging therapy groups. Statistical analysis was performed using Fisher's exact and Chi-square tests.

Results: 48 patients had T-cell collection, 35 of whom proceeded to CAR T-cell infusion and are the subjects of this analysis. The day 28 complete response (CR) rate was 82%, of which 93% were MRD negative as measured by flow cytometry. There was no significant difference in CR rate between patients who received 19-28z or 19-BBz (p=0.30), high or low intensity bridging chemotherapy (p>0.99), or 0-1 or 2+ lines of bridging chemotherapy (p=0.07). There were no differences in CRS and ICANS between high or low intensity and 0-1 or 2+ lines, but patients who received 2+ lines of bridging chemotherapy had more infections during the bridging period (p=0.001).

Conclusions: In this analysis of children/young adults with R/R B-ALL, receiving 2+ lines of bridging chemotherapy was associated with higher infection, but no difference in CRS, ICANS, or CR. Selection of a bridging regimen thus requires consideration of limiting the number of cycles to minimize infectious complications.

Free Paper Session (FPS) FPS 12-LIVE Q&A-ALL AND HIGH RISK POPULATIONS INCLUDING IMMUNOTHERAPY 23-10-2021
2:30 PM - 4:00 PM

00177 / #544 | Outcome and Prognostic Significance of Minimal Residual Disease in Infants with KMT2A-Germline Acute Lymphoblastic Leukemia Treated on The Interfant-06 Protocol

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Background and Aims: The outcome of infants with KMT2A-germline ALL is much better than of infants with KMT2A-rearranged ALL, but still worse than of non-infant ALL patients. Here, we describe the outcome and prognostic factors for infants with KMT2A-germline ALL treated on Interfant-06 protocol.

Methods: 167 infants with KMT2A-germline ALL were enrolled in Interfant-06. Univariate analysis on prognostic factors (age, WBC at diagnosis, prednisolone response and CD10 expression) was performed on KMT2A-germline infants in complete remission at the end of induction (EOI) (n=163). Bone marrow minimal residual disease (MRD) was measured in 73 patients, by real-time quantitative PCR of immunoglobulin genes and/or T-cell receptor genes, at various time points (EOI, n= 68, end of consolidation (EOC) n= 56, and before OCTADAD (n=57)). MRD results were classified as negative, intermediate ($< 5 \times 10^{-4}$), and high ($\geq 5 \times 10^{-4}$).

Results: The 6-year event free survival [SE] and overall survival [SE] was 73.9% (3.6) and 87.2% (2.7). 28 of 31 (90%) relapses occurred early, within 36 months of initial diagnosis. Treatment related mortality was 3.6%. Surprisingly, age < 6 months was a favorable prognostic factor with a 6-years disease free survival (DFS) [SE] of 91% (9.0) compared to 71.7% (4.2) in infants > 6 months, $p=0.04$. Of the MRD timepoints, MRD at end of induction (EOI) was most prognostic for outcome. Patients with high EOI MRD, had worse outcome with 6y DFS (SE) 61.4% (12.4), compared to patients with negative EOI MRD (6y DFS (SE) 87.9 % (6.6)) or intermediate EOI MRD (6y DFS (SE) 76.4 % (11.3) ($p=0.02$).

Conclusions: Young age at diagnosis and low EOI MRD are favorable prognostic factors in infants with KMT2A-germline ALL. However, the prognostic value of MRD is not as strong as in older children with ALL. This can partly be explained by the different genetic makeup of infants with KMT2A-germline ALL.

Free Paper Session (FPS) FPS 12-LIVE Q&A-ALL AND HIGH RISK POPULATIONS INCLUDING IMMUNOTHERAPY 23-10-2021
2:30 PM - 4:00 PM

O0178 / #141 | Rapid Minimal Residual Disease Clearance with More Stringent Age-Related Dose Criteria Improves Prognosis in Infants with KMT2A-Rearranged Acute Lymphoblastic Leukemia

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Background and Aims: KMT2A-rearranged acute lymphoblastic leukemia (KMT2A-rALL) is a rare and dismal disease in infants. The JPLSG MLL-10 trial has shown that its prognosis could be improved (3-year-EFS: 66.2%) by introduction of high-dose cytarabine in early consolidation and by increasing the treatment intensity at the induction phase based on Interfant-06 protocol by adopting stricter age-related chemotherapeutic dose reduction criteria. MRD at end of induction (EOI) and at end of early consolidation (EOC) was analyzed to investigate the favorable outcome observed in the MLL-10 trial.

Methods: MRD analysis through polymerase chain reaction (PCR) that targets leukemia-specific Ig/TcR genes (PCR-MRD) and/or 4-color flow cytometry (FCM-MRD) was measured in 70 infants with KMT2A-r ALL. The PCR-MRD results were classified as negative, intermediate (positive under quantitative range or $< 5 \times 10^{-4}$), and high ($\geq 5 \times 10^{-4}$); while FCM-MRD results were classified as negative ($< 0.01\%$) and positive ($\geq 0.01\%$).

Results: PCR-MRD was evaluated in 37 (53%) patients: 20/27 (54%/73%), 10/5 (27%/14%), and 6/4 (16%/11%) cases were negative, intermediate, and positive at EOI/EOC, respectively. One case each had no data at EOI or EOC. In particular, of 28 patients ≥ 4 months of age (M) at diagnosis receiving 100% dosing, the rate of negativity was high (16/21; 57%/75%) at EOI/EOC, while the rate in 9 cases receiving 67% ($< 2M$) and 75% ($2- < 4M$) dosing, respectively, was 4/6 (44%/67%). Of the remaining 33 cases that could not be evaluated for PCR-MRD, the number of FCM-MRD positive cases was 11 (33%) at EOI, which decreased to 3 (9%) at EOC.

Conclusions: More patients achieved a PCR-MRD negative state in the MLL-10 trial than those in Interfant-06 (54%/73% vs 19%/40% at EOI/EOC). These results indicate that rapid clearance of leukemic blasts in early phase is responsible for better outcome of the MLL-10 trial. Intensification, employing less reduced dose, is likely to have contributed to the given results.

Free Paper Session (FPS) FPS 12-LIVE Q&A-ALL AND HIGH RISK POPULATIONS INCLUDING IMMUNOTHERAPY 23-10-2021
2:30 PM - 4:00 PM

O0179 / #1605 | Blinatumomab Modulates the B-All Bone Marrow Immune Microenvironment and Induces T-Cell Infiltrates that Correlate with Therapeutic Efficacy

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Background and Aims: Blinatumomab is a CD3 T-cell engager targeting CD19 approved for relapsed/refractory (R/R) and measurable residual disease (MRD)-positive B-ALL. Blinatumomab induces activation of multiple circulating T-cell subsets, however bone marrow immune effector dynamics during blinatumomab therapy in B-ALL have not been investigated.

Methods: Patients treated with blinatumomab for R/R B-ALL from 2011-2020 were retrospectively identified. Marrow core biopsies were collected immediately prior to blinatumomab administration (pre-treatment) and during or immediately following 1-4 treatment cycles (post-treatment). Biopsies were analyzed histologically and immunohistochemically (CD3, CD4, CD8) for T-cell infiltrates. B-ALL involvement was determined by aspirate differentials and MRD flow cytometry. RNA sequencing-based gene expression profiling (EdgeSeq Immuno-Oncology Panel, HTG Diagnostics) was performed on paired pre- and post-treatment biopsies from 4 patients, with differential expression analyzed using DESeq2 within HTG Reveal software.

Results: We identified 18 patients with R/R B-ALL treated with blinatumomab (median age: 13 years, range 1-23). Fourteen pre-treatment and 24 post-treatment marrow core biopsies were examined. Pre-treatment marrow B-ALL involvement ranged from 0.5-95% (median: 58%). Post-treatment biopsies with MRD<0.1% response to blinatumomab showed greater CD3+ T-cell percentages than post-treatment biopsies with ≥0.1% B-ALL (16.2% vs. 4%, $p=0.002$). Analysis of paired pre- and post-treatment marrows revealed patients with MRD<0.1% response showed increased post-treatment T-cell infiltration ($p=0.04$), which was not observed in patients with ≥0.1% disease post-therapy. CD8+ T-cell predominant lymphoid and lymphohistiocytic aggregates increased in MRD<0.1% responders following multiple blinatumomab cycles (100% vs. 60% pre-treatment), but not in patients with ≥0.1% post-treatment disease. Differential RNA expression analysis showed post-treatment upregulation of activated T-cell and macrophage markers alongside T-cell chemoattractant and leukocyte adhesion transcripts.

Conclusions: Increased marrow infiltration of activated T-cells and macrophages correlates with response to blinatumomab in R/R B-ALL. Marrow T-cell aggregates can serve as a novel indicator of sustained response to blinatumomab, and further investigations may suggest immunomodulatory approaches to improving therapeutic efficacy.

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0180 / #883 | Epigenetic Silencing of CD9 Drives Immune Evasion and Disease Progression in Pediatric Acute Myeloid Leukemia

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Background and Aims: We recently reported the importance of CD9 in pediatric ALL but its role in AML remains unknown. In this disease context, we pursued to characterize its prognostic significance, elucidate its regulation and function, and identify its role in leukemia immunosurveillance.

Methods: Patients were stratified based on CD9 status for comparison of long-term survival. Epigenetic control of CD9 was investigated by ChIP-sequencing and confirmed by HDACi treatment. Impact of CD9 on leukemia aggressiveness was measured by competition and colony formation assays. Influence of CD9 on leukemia progression was evaluated in xenograft models, coupled with single-cell transcriptomics to dissect the underlying mechanisms. Immune-related CD9 interactors were identified by IP-MS, followed by proof-of-function experiments in an immune-reconstituted mouse model.

Results: CD9 expression of AML patients was significantly lower than normal BM donors (13.2% vs. 48.4%). Among 81 AML cases, blasts of 32 patients (39.5%) were CD9+. The 5-year RFS rate of CD9+ patients was significantly lower than CD9- patients (34.1% vs. 61.2%). A marked decrease of H3K9/27Ac occupancy of the CD9 locus was observed in AML than in ALL cells (4.8-14.2-fold), and strongly correlated with CD9 repression ($r=0.585-0.719$). Panobinostat exposure elevated CD9 expression (3.1-32.2-fold) and restored activating histone marks in AML samples. Overexpression of CD9 reduced proliferation and clonogenicity of AML cell lines. NOD/SCID mice receiving CD9+ AML exhibited a drastic reduction of leukemic load by 70.7-91.8%. Mechanistically, CD9 promoted basal and cytokine-induced MHC I/II expression through the JAK-STAT axis, and regulated their intracellular trafficking by physical binding. Preliminary data showed that CD9 could enhance BM infiltration of cytotoxic T cells and thereby mount an effective immunity against AML in humanized NSG mice.

Conclusions: Our data established CD9 as a novel tumor suppressor and immune regulator in pediatric AML, and inspired a new combinatorial epigenetic/immunotherapy for this rare but aggressive malignancy.

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0181 / #323 | Kit Mutations at Exon-17 Correlate with Adverse Survival in Children with Core-Binding Factor Acute Myeloid Leukemia (CBF-AML)

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Background and Aims: KIT mutations (KIT+) are common in CBF-AML and have been associated with varying prognostic significance. We sought to define the clinical significance of distinct KIT mutations in this subgroup.

Methods: Children less than ≤15-years of age with CBF-AML diagnosed between January-2017 and December-2019 were included. KIT mutation was detected by Sanger-Sequencing (till December-2017) and subsequently by panel based next generation sequencing. For the later, computational analysis was done using open-source tools according to the GATK best practices. Treatment consisted of four cycles of intensive chemotherapy followed by maintenance therapy for 1 year. Clinical outcomes were analysed according to mutation status [KIT+ vs. wild-type KIT(KIT-)] and mutation location.

Results: KIT data was available for 90(87%) of 103 patients with CBF-AML. KIT+ were detected in 37(41%) of 90 patients; 26(70%) involved only exon-17, 6(16%) only exon-8, 3(8%) both exons, and 2(6%) alternative exons. The incidence of KIT+ was 38%(n=28) and 56%(n=9) amongst patients with t(8,21) and Inv16 respectively. Baseline demographics was comparable between the 2 groups. The incidence of induction failure [1(3.4%) vs 2(4%), p=0.9], post induction-I MRD >0.01% [10(34%) vs 19(58%), p=0.7] and toxic deaths [6(18%) vs 7(14%), p=0.5] were comparable between KIT+ and KIT- patients respectively. After a median follow-up of 21(95%CI:18.1-23.9) months, KIT+ CBF-AML had a poorer 2-year overall survival (52.3±9.1% vs 76.3±6.0%; P = 0.042) and a higher 2-year cumulative incidence of relapse (CIR) (40±10% vs 15.8±5.5%; P = 0.022) to those with KIT-. Amongst KIT+, relapse was higher for exon-17 mutation alone as compared to other exon mutations (2-year CIR 47.2±12% vs 16.7%±15.2%; p=0.180). On multivariate analysis, KIT+ was associated with inferior survival [HR:3.4(95%CI:1.3-9.1; p=0.014)].

Conclusions: Our findings indicate that KIT+, especially at position exon-17 is a poor prognostic factor in pediatric CBF-AML. The role of tyrosine kinase inhibitors and bone-marrow transplantation needs exploration.

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0182 / #1539 | Improved Outcome in Pediatric De Novo AML – Results of The AML-BFM 2012 Trial

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Background and Aims: The Study AML-BFM 2012 aimed to improve the outcome by further intensification of the induction and an improved risk group stratification. The newly defined high-risk group includes allogeneic stem cell transplantation (alloSCT) in 1st CR. Clofarabine (C) was combined with liposomal daunorubicin (Dx) and cytarabine (A; CDxA). The standard arm was ADxE, developed in the AML-BFM 2004 study. Patients were stratified to standard (SR), intermediate (IR) and high risk (HR) based on genetics and treatment response.

Methods: AML-BFM 2012 was an multi-center, prospective, randomized study for childhood AML. Since 11/2017 the auxiliary drug Dx, was not available anymore there had to be replaced with idarubicin (I; AIE). As consequence, the trial had to be early terminated. Trial data were compared to the AML-BFM registries 2012/ 2017, recommending the similar standard treatment and risk groups stratification.

Results: The 5-year OS of 82±3% and EFS of 69±4% improved significantly compared to AML-BFM 2004 (EFS 55±3%, OS 72±2%, p<0.02). CDxA (n=55) and ADxE/AIE (n=107) did not differ: intent-to-treat (all 5-yrs) EFS 72±6 vs 69±5% and OS 86±5% vs. 82±4%, p>0.5, as-treated: CDxA (n=50) EFS 73±6%, OS 86±5%; ADxE (n=70), EFS 67±6%, OS 81±5%; AIE (n=38) EFS 76±7%, OS 87±6%; all p>0.5. Reasons of death were early death (n=9), TRM (n=2) and alloHSCT (n=6). The AML-BFM registries 2012/2017 (n=136) showed similar EFS 66±4% and OS 82±6%. The alloSCT in 1st CR significantly improved the EFS of children with HR-characteristics from 30±6% (AML-BFM 2004) to 60±7%. The significant differences between the risk groups in study AML-BFM 2004 were abolished.

Conclusions: The EFS and OS of children with AML significantly improved. Treatment with Clofarabine is feasible but not superior. A better risk group stratification defined a more appropriate HR-patient group. The progress in alloSCT, reduction of TRM and the rescue of refractory AML contributed an overall survival above 80%.

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID
LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0183 / #621 | The Origin and Clonal Evolution of
Therapy-Related Paediatric Acute Myeloid Leukaemia

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Background and Aims: Together with a rising number of paediatric cancer survivors emerges an increased incidence of late effects, such as therapy-related acute myeloid leukaemia (t-AML). Here, we aim to increase understanding of the origin and clonal evolution of paediatric t-AML.

Methods: We studied mutation accumulation and mutational patterns in 17 paediatric patients with t-AML by whole genome sequencing (WGS) of both t-AML blasts and normal haematopoietic stem and progenitor cells (HSPCs) at different timepoints between the first cancer diagnosis and the t-AML.

Results: Our data confirm that paediatric t-AML is mainly driven by gene fusions, in specific MLL-rearrangements (67% of our patients), while few other driving mutations are identified. In contrast to HSPCs from de novo paediatric AML, normal HSPCs in t-AML patients show an increased clonal mutational load compared to our previously established healthy baseline. Mutational signatures SBS5 and HSPC, which act in a clock-like manner in healthy blood, had expected contributions to the spectra of these HSPCs. However, the large contribution of treatment-related signatures in most patients indicates that different chemotherapies can leave a distinct mutational footprint in DNA of both normal HSPCs and t-AML blasts. Specifically, thiopurine and cisplatin treatment caused a clear mutational pattern, with thiopurine-caused mutations more frequently present in t-AML blasts than normal HSPCs. Moreover, we identified four novel signatures, with a thus far unknown cause. Finally, in two t-AML cases, we identified a rare type of MLL-rearranged cells with stem-like properties which were closely related to the t-AML blasts but phenotypically similar to normal HSPCs.

Conclusions: The variance in mutational effects of different chemotherapies could indicate that, in children, not all cancer treatments have large mutagenic effects. These differences might affect a patient's risk to develop t-AML and could, in the future, influence the development of treatment regimens in paediatric oncology.

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID
LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0184 / #1789 | How to Diagnose and Manage
SAMD9/SAMD9L MDS Predisposition Syndromes: Development
of a Multidisciplinary Expert Opinion Statement

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Background and Aims: Recently discovered activating germline heterozygous SAMD9 and SAMD9L mutations cause a severe blood disorder with high potential for the development of myelodysplastic syndrome (MDS) in need of better understanding and guidelines for diagnosis and therapy. In November 2019, 40 international clinical and basic science experts discussed current knowledge and management strategies. A major aim for this collaboration was to define the disease from its biological roots to its clinical presentation, improving accuracy of diagnosis and timely detection of complications.

Methods: A steering committee initially formulated questions to be addressed by meeting attendees. Identified subspecialty fields provided multidisciplinary expert opinion including Hematology, Oncology, Bone Marrow Transplant, Endocrinology, Neurology, and Genetics. The meeting consisted of literature review, review of current models to study disease, and international experience with SAMD9/9L families. Moderators then summarized each question followed by debate with attendees. Following the meeting, surveys were distributed to all attendees to ensure a 75% consensus for finalized consensus statements. This transparent methodology to develop expert opinion was utilized given the novelty of disease and lack any clinical trial data.

Results: Statements were drafted on the following subjects: (1) Suspicion of diagnosis, (2) Diagnostic evaluation (3) Hematologic and multidisciplinary evaluation, and (4) Management of patients with SAMD9/9L mutations. Statements underwent surveys leading to finalized consensus statements. A recommended comprehensive initial evaluation and ongoing disease surveillance was decided upon. Standardized treatment approaches for most clinical manifestations of SAMD9/9L syndromes have been recommended. No clear recommendation could be made yet for patients with MDS and monosomy 7 with stable counts.

Conclusions: Recommendations are derived from expert opinion gathered from the SAMD9/9L symposium. This evolving collaboration represents an attempt to close the critical knowledge gap that exists in identifying, diagnosing and treating patients with germline SAMD9/9L mutations

Free Paper Session (FPS) FPS 13-LIVE Q&A-ACUTE MYELOID LEUKEMIA 23-10-2021 2:30 PM - 4:00 PM

O0185 / #292 | Comparison of Haploidentical Hematopoietic Stem Cell Transplantation Utilizing Post-Transplant Cyclophosphamide in Children with Hematological Malignancies to Matched Sibling and Matched Unrelated Donor Bone Marrow Allografts

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Background and Aims: Introduction of Post-Transplant Cyclophosphamide (PTCy) as graft-versus-host disease (GVHD) prophylaxis has made haploidentical hematopoietic stem cell transplantation (HSCT) a standard approach in adults, but pediatric experience is limited. Based on adult data, decreased donor availability and need to cryopreserve cells during the COVID19 pandemic, our center has increasingly used peripheral blood stem cells (PBSCs) from haploidentical donors (HaD) utilizing PTCy. Here we compare outcomes of traditional donor choices including matched sibling donor (MSD) and HLA matched unrelated donor (MUD) with those receiving HaD PBSC.

Methods: Retrospective single-center study. GVHD-free relapse-free survival (GRFS, defined as absence of acute GVHD grade III-IV, relapse, death or chronic GVHD requiring systemic therapy), overall survival (OS), relapse free survival (RFS), incidence of acute and chronic GVHD (cGVHD) were analyzed.

Results: Of 104 allogeneic transplants for leukemia/MDS from January 2014-December 2020 26 patients underwent HaD transplants with PBSC, 31 patients MSD, and 47 patients MUD bone marrow transplants. Demographic/HSCT characteristics were not significantly different, apart from remission status, with HaD cohort having more patients in $\geq$ CR3 prior to HSCT ($p < 0.01$). Median follow-up time of entire cohort-573 days. GRFS at 18 months for HaD was 61% (95CI 43.3%-85.9%), for MUD 44.6% (95CI 31.8%-62.5%) and for MSD 62.1% (95CI 45.7%-84.3%), $p = 0.26$. OS and RFS at 18 months for HaD were 80.7% (95CI 61.7%-100%) and 73.8% (95CI 55.5%-98.1%), for MUD 83.4% (95CI 72.8%-95.5%) and 70.3% (95CI 57.9%-85.3%) and for MSD 80.9% (95CI 66.9%-97.7%) and 66.5% (95CI 50.5%-87.5%), respectively. Incidence of chronic GVHD was higher with MUD 31.7% versus 10% and 9.2% in the MSD and HaD cohorts ($p = 0.02$).

Conclusions: Haploidentical PBSC transplant with PTCy had comparable outcomes to MSD and MUD BM and less cGVHD compared to MUD BM. Logistical advantages and lower resource burden make Haploidentical HSCT with PBSC an alternative to MUD donors in children. Further prospective studies are needed.

IPSO LIVE Q&A: IPSO SESSION 11: THORAX 23-10-2021 4:30 PM - 5:10 PM

O0186 / #1547 | Evaluation of Pulmonary Nodules in Metastatic Lung Disease

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Background and Aims: Minimally-Invasive Surgery in Oncological Surgery is slowly gaining acceptance: the use of thoracoscopy (TS) for pulmonary nodules (PN), in particular, is often debated, mainly due to concerns about the possibility to identify all PN with this technique. This study aims to compare the outcome of PN removal, TS versus thoracotomic manual lung palpation (TT).

Methods: We evaluated surgical approach and history of the nodule (form targeting to removal), to histological result. The primary outcome was the identification of variation in the ratio between neoplastic (positive) and non-neoplastic (negative) nodules.

Results: Between January 2007 and December 2018, 41 patients (median age 14 years), underwent 65 surgical procedures for pulmonary metastases: 35 TT and 30 TS (6 converted to TT for difficult localization of the PN). An average of 2 PN were targeted at pre-operative imaging overall (range 1-6, total 125) and same average removed during surgery (range 1-15, total 173). In 20 procedures (4 TS, 16 TT), more nodules than targeted were identified and removed during surgery (1 to 13/patient, total 47; 3 during TS, 44 during TT). Twenty-nine nodules (62%) were histologically confirmed to be metastases not targeted pre-operatively; all were in 5 patients with further multiple imaging-targeted nodules. The positive/negative ratio is similar in imaging-identified and surgically removed nodules among TS and TT approach, 1,9 and 1,8 respectively ($p = ns$).

Conclusions: TS may useful to remove targeted PN in selected groups of patients and following multidisciplinary discussion. Manual lung palpation in this series does not seem to increase the percentage of positive nodules overall. The presence of multiple nodules seems to benefit of manual palpation, while few nodules are efficiently addressed by target excision. Further prospective studies are needed to confirm the present finding.

IPSO LIVE Q&A: IPSO SESSION 11: THORAX 23-10-2021

4:30 PM - 5:10 PM

O0187 / #1783 | Fluoroscopy-Assisted Thoroscopic Resection of Lung Nodules Marked with Lipiodol

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Background and Aims: Pulmonary nodules (PN) in patients treated for malignancies are a common finding. Given the advances in high-resolution CT scanning, small PN can be detected, in this setting, histological study is essential to distinguish between metastases and other non-neoplastic lesions. Minimally invasive surgery (MIS) is, at the same time, a diagnostic and therapeutic tool in some of these patients. Nonetheless, there are nodules that may not be visualized by thoracoscopy. Thus, different preoperative marking methods have been developed to increase the effectiveness of MIS. The aim of this review is to describe and analyze lipiodol-marking method for thoroscopic biopsy in pediatric oncology patients with PN.

Methods: Description of the lipiodol-marking technique, and retrospective review of patients managed with this technique, description of demographics, complications and effectiveness of the procedure.

Results: Thirteen PN in 8 patients with osteosarcoma, 4 PN in 2 Ewing sarcoma, 1 PN in 1 hepatoblastoma and 1 PN in 1 Wilms tumor, were marked with lipiodol (1 to 4 PN per patient, 2 bilateral-synchronous procedures). Marking-Procedure: Under general anesthesia, a CT guided 22 Gauge spinal-trocar was inserted adjacent to the PN and 0,05 to 0,2 ml of lipiodol was infiltrated, the needle was removed and anesthesia was reversed. Immediate to 24 hours after this procedure, the patient was taken to MIS biopsy, using radiology-guidance to navigate intraoperatively. One out of 19 marking-procedures presented post-marking pleuritic pain and 3 patients developed immediate pneumothorax, drained in the same marking-procedure with needle aspiration, without the need of a chest tube; 2 presented extravasations of lipiodol to the blood vessels adjacent to the PN, although this not affect the MIS biopsy. All PN under study, were successfully removed.

Conclusions: The lipiodol-marking technique is safe, with low rate of complications, and provide adequate guidance for thoroscopic PN biopsy in pediatric oncology patients.

IPSO LIVE Q&A: IPSO SESSION 11: THORAX 23-10-2021

4:30 PM - 5:10 PM

O0188 / #1487 | Intraoperative Electromagnetic Navigation Bronchoscopy (IENB): A New Technique to Guide Video-Assisted Thoroscopic Surgical Biopsies in the Pediatric Oncology Population

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Background and Aims: The usefulness, diagnostic yield and accuracy of lung biopsies in pediatric oncology patients are variable. Here we describe our preliminary results using intraoperative electromagnetic navigation bronchoscopy (IENB) to increase the surgical yield and accuracy of lung biopsies in these patients.

Methods: From May 2018 until October of 2020 all lung biopsy requests from the pediatric oncology service were performed using IENB technology according to an IRB approved protocol. IENB tattooing using methylene blue, indocyanine green (ICG) or both, followed by Video-Assisted Thoroscopic Surgery (VATS) were performed in the same operating room and during the same anesthetic. Data was collected retrospectively. Data points included diagnosis, technical success, pathologic diagnosis, alteration in treatment management and complications.

Results: A total of 10 biopsy procedures were performed on 8 patients during the time period of the study. Age at time of surgery ranged from 88 to 244 months. All had successful IENB with tattooing. All biopsies were diagnostic, none of the pathology returned as normal lung. No procedures had to be converted to open thoracotomy. There were no technical failures or procedure complications from either IENB tattooing or the surgical biopsy. One patient had a total of 11 biopsies, 6 from the right lung and 5 from the left, performed at 2 separate procedures. Another had 2 biopsies, one from the right lung and one from the left performed at the same operation. In 7 of the 8 patients treatment changes were made based on either pathologic or microbiologic results of their biopsy.

Conclusions: Here we present the first described experience of IENB tattooing of lung lesions in pediatrics. We conclude that IENB is a safe and effective technique for guiding lung biopsies and may improve the diagnostic accuracy of lung biopsies in these patients. Our preliminary findings warrant further investigation into this technique.

IPSO LIVE Q&A: IPSO SESSION 11: THORAX 23-10-2021

4:30 PM - 5:10 PM

O0189 / #1538 | Children Presenting with a Malignant Chest Mass in a Lower to Middle Income Country: Experience From a Single South African Centre

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Background and Aims: Chest wall and mediastinal tumours are rare in the paediatric population. A paucity of literature exists relating to these tumours in Sub-Saharan Africa. Understanding the incidence, management and outcomes can lead to improved management and survivals in a Lower to Middle Income Country (LMIC), hence the aim of the study

was to determine the numbers, types, management and outcomes at Chris Hani Baragwanath Academic Hospital (CHBAH), South Africa (SA).

Methods: A retrospective review of patients below 18 years of age at CHBAH presenting with a malignant chest mass managed by paediatric surgery and oncology from 1st January 2009 to 31st December 2018.

Results: Twenty patients were identified over the ten-year period with mediastinal tumours (15) and chest wall tumours (5). Mediastinal tumours were 5 Non-Hodgkin lymphomas (NHL), 4 Hodgkin lymphomas (HL), 4 neuroblastomas (NBL), 1 thymoma and 1 teratoma. The average age of presentation of patients with lymphoma was 11 years, thymoma 14 years, teratoma 13 years, and neuroblastoma 2 years. All 4 NBL's were surgically resected after chemotherapy with a survival of 75%, the thymoma and teratoma underwent primary resection with a 100% survival. NHL had a lower survival rate of 40% compared to HL of 100%. The chest wall tumours comprised of 2 rhabdomyosarcoma's (RMS) - 8 and 11 years old, 1 chondrosarcoma - 8 years, one malignant peripheral nerve sheath tumour (MPNST) - 15 years, and 1 myofibroblastic tumour - 2 years, all were in females. Patients with RMS and chondrosarcoma presented with advanced stage disease, did not undergo surgical resection and demised. Patients with MPNST and myofibroblastic tumour underwent surgical resection and remain alive.

Conclusions: The majority of patients presenting with mediastinal masses had lymphomas followed by NBL, whereas thoracic wall tumours comprised of a heterogenous group of sarcomas. NHL, RMS and chondrosarcoma had the worst outcomes.

Opening/Closing LIVE: AWARD SESSION SCHWEISGUTH PRIZE LECTURE 23-10-2021 4:50 PM - 5:20 PM

00190 / #700 | Mutation and Microsatellite Burden Predict Response to PD-1 Inhibition in Children With Germline DNA Replication Repair Deficiency

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Background and Aims: Cancers arising from germline DNA mismatch-repair or polymerase-proofreading deficiencies (MMRD and PPD) in children harbour the highest mutational and microsatellite insertion/deletion (MS-indel) burden in humans and are lethal due to inherent resistance to chemo-irradiation. Although immune checkpoint inhibitors (ICI) have failed to benefit children in previous studies, we hypothesized that hypermutation caused by MMRD and PPD will improve outcomes following ICI in these patients.

Methods: Clinical data was collected as a part of the International RRD Consortium Registry Study. Multi-platform analyses, including exome sequencing, bioinformatics analyses for mutation burden, COSMIC mutational signatures, microsatellite indel and neoantigen calling, protein expression using immunohistochemistry, immune inference analysis, T-cell receptor (TCR) rearrangement profiling and flow cytometry, were performed with prospectively collected tissue and blood samples. Predictors for response and survival were assessed using uni and multivariate analyses.

Results: ICI treatment of 45 progressive/recurrent tumours including metastatic cancers from 38 patients across 7 cancer-types revealed durable objective responses in majority, culminating in 3-year survival of 41.4%. High mutation burden predicted response for ultra-hypermutant cancers (>100 mutations/Mb) enriched for combined MMRD+PPD. MS-indels predicted response in MMRD tumours with lower mutation burden (10-100 mutations/Mb). Both mechanisms were associated with increased immune infiltration (CD8+T-cell infiltration, PD-L1 expression) even in "immunologically-cold" tumours such as gliomas, contributing to the favorable response. Pseudo-progression (flare) was common (27%), accompanied by a dramatic clonal increase in the T-cell repertoire, and increased immune-infiltration in the microenvironment. Serial flow cytometry analysis of peripheral blood samples showed that flare was associated with an expansion of activated CD8+ T-cells expressing Ki67 and 4-1BB. Critically, patients with flare continuing ICI treatment achieved durable responses.

Conclusions: Our study demonstrates improved survival for patients with tumours not previously known to respond to ICI, including CNS and synchronous cancers, and identifies the dual roles of mutation burden and MS-indels in predicting sustained responses to immunotherapy.

Nursing POSTER SESSION 1 23-10-2021 5:25 PM - 6:25 PM

00191 / #221 | Heparin Versus Normal Saline: Flushing Effectiveness in Managing Central Venous Catheters in Pediatric Oncology Patients

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Background and Aims: To maintain a catheter's patency, heparin has traditionally been utilized as the standard of practice. However, multiple recent studies in adults have demonstrated normal saline to be just as safe and effective. There is insufficient evidence on the effects of intermittent flushing with normal saline versus heparin to prevent occlusion and/or increased alteplase usage in pediatric patients with central venous catheters. Additionally, heparin is a costly medication. Flushing once daily at home with a lower concentration of heparin may provide a meaningful cost difference. Due to the lack of evidence, a project to evaluate the effectiveness between the current standard of heparin flushing to saline flushing in the management of central line catheters within the pediatric oncology population would likely have a significant impact. Our project aim is to evaluate the effectiveness of the standard flushing practice in the management of central venous access devices in pediatric oncology patients. We expected to see no change in alteplase usage as well as a decrease in cost of the heparin flushing supplies

Methods: Pre-intervention, the pediatric flushing guidelines both inpatient and outpatient included heparin for the tunneled lines as noted here in the project schema. After project implementation the flushing guidelines now included all tunneled lines be flushed with saline in the new push pause method. Heparin changes included decrease in dose from 3ml (100u/ml) to 2ml (10u/ml) for tunneled CVL, ports remained unchanged. Data was evaluated for five months in all pediatric oncology patients with tunneled central lines.

Results: Sixty-two patients were included in the study. There was no difference in alteplase usage rate between Pre and Post implementation (2.17 vs. 2.42 per 1000 line days, p=0.89). The cost per single lumen decreased \$154 after the implementation.

Conclusions: We recommended a formal flushing guideline using saline and lower heparin concentrations for pediatric oncology patients.

Nursing POSTER SESSION 1 23-10-2021 5:25 PM - 6:25 PM

00192 / #463 | "Effect of the Quality Improvement Program Evat Tool (Early Warning Assessment Scale) on the Empowerment and Job Satisfaction in Nursing Staff. Pilot Study"

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Background and Aims: Pediatric cancer patients are at high risk of clinical impairment. The EVAT (ESCALA DE VALORACIÓN DE ALERTA TEMPRANA : EARLY WARNING ASSESSMENT SCALE) tool includes active nursing participation as a cornerstone in the detection and timely treatment of clinical deterioration events, promoting active

exercise of the profession, empowerment and job satisfaction. **OBJECTIVE:** To assess whether the implementation of the EVAT program in the oncology service promotes the empowerment and job satisfaction of nursing personnel

Methods: A transverse cohort was designed. To measure the level of nursing empowerment, the Spreitzer psychological empowerment in Spanish scale was applied and to measure the level of satisfaction at work, THE LABORAL S10/ scale was used. We also analyzed the gender, age, schooling, years of seniority and marital status. Descriptive statistics, one-way ANOVA testing and X² was used according to variables. We also evaluated Alpha Cronbach to confirm the reliability of both questionnaires

Results: We included 15 staff questionnaires that used the tool and 25 without it. A statistically significant difference was observed in the overall score obtained: the labor satisfaction score was 68.2 (64-72) vs 57(56-61) $p = 0.008$ as well as in empowerment 82(81-83) vs 66(63-68) with and without use of the tool, respectively. Working seniority negatively influences the perception of empowerment in a clinical but not significant way. The reliability score was of .77 and .74 in both questionnaires.

Conclusions: The Evat tool is a program to improve the quality of care, aimed at the timely detection of events of clinical deterioration, but also promotes autonomy, decision making, sense of belonging, work in team and leadership in nursing staff in a country with limited resources. Our sample is very small but significant, and necessary to reproduce this study in other paediatric oncology units.

Nursing POSTER SESSION 1 23-10-2021 5:25 PM - 6:25 PM

00193 / #1788 | Evaluation of Nurses Competencies on Vital Signs and Physical Exploration Needed for PEWS Implementation

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Background and Aims: Background: Pediatric Early Warning Scores (PEWS) have improved the prognosis for pediatric oncologic inpatients. The major component of the scale is related to vital signs and physical exploration performed by nurses. One preliminary step of the implementation path should focus on analyzing knowledge gaps to create educational programs to unify techniques and knowledge to enhance the predictive value of the tool. Aim: The study aimed to develop an instrument to evaluate nurses' knowledge and identify the educational needs to work on them.

Methods: A questionnaire (38 multiple option questions) developed from literature and expert input and validated by subject experts. One pilot study was used to evaluate nurses' knowledge of vital signs and physical exploration related to PEWS needs. Descriptive statistics were applied for analysis.

Results: A total of 79 nurses participated. The overall correct answer rate was 83.5%. Only 18.5% of nurses were considered to have insufficient knowledge and should be favored to gain more training about these themes. Overall, the questions related to cardiovascular exploration like pediatric pulses and pathological respiratory patterns with correct responses rate from 52%-58%. The highest scores were found related to neurological exploration, cardiovascular findings (capillary refill, rhythm), and regular respiratory patterns with rates above 95%.

Conclusions: Conclusion: These results strongly suggest that even highly experimented centers on PEWS have an educational opportunity to improve the accuracy of their clinical findings. The findings from this kind of evaluation allow focusing the educational interventions on the founded gaps. Further research to validate the instrument is needed, and a post educational reevaluation is recommended to confirm improvements.

Nursing POSTER SESSION 1 23-10-2021 5:25 PM - 6:25 PM

00194 / #1229 | Building Mhealth Capacity to Manage Pain in Young Children with Cancer: A Pain Management App Needs Assessment with Parents

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Background and Aims: Pediatric cancer pain is common and has wide-ranging negative consequences for children and parents. Young children with cancer are particularly vulnerable to undermanaged pain in the home setting due to limited pain self-report ability and reliance on parents for pain treatment. App-based digital interventions can help to equip parents with evidence-based pain management strategies, but no such interventions have been developed for parents of young children with cancer. We aimed to explore the perceptions of parents regarding the value of an app to assist in managing young children's cancer pain and key aspects needed to make the app successful.

Methods: Parents of children aged 2-11 years who experienced cancer-related pain in the previous week were recruited from two pediatric tertiary care centers. Parents participated in audio-recorded, semi-structured interviews. Interviews were conducted until data saturation was achieved and then transcribed, coded independently in duplicate, and analyzed using thematic analysis.

Results: Twenty parents participated. The concept of a mobile app was endorsed as a valuable means to support pain management and enhance child quality of life. General app characteristics that

parents desired included simple navigation, accessibility in multiple languages, and a family-centered nature whereby young children could be engaged in the pain management process. Specific app functions identified as priorities were: (1) multidimensional assessment of pain, (2) real-time and evidence-based physical, psychological and pharmacological pain management support, (3) capacity to receive virtual pain management support from clinicians, and (4) a means to connect with other families of children with cancer.

Conclusions: Parents of young children with cancer expressed the need for an app-based solution to support pediatric cancer pain management. The insights of parent stakeholders related to digital pain management should be reflected in the principles that become the basis of app design.

Nursing POSTER SESSION 1 23-10-2021 5:25 PM - 6:25 PM

00195 / #1660 | The Effects of Using A Board Game on the Distress of Children with Cancer

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Background and Aims: Most kids with cancer experience some level of distress associated with their diagnosis and treatment. Distress is defined as any unpleasant feeling, emotion, or experience that affects your ability to cope with your disease. Aim: to evaluate the effects of using a board game on the distress of children with cancer.

Methods: A before-and-after study was performed at an institute specialized in the treatment of cancer in childhood in the city of São Paulo. Thirty-five children with cancer aged eight to 12 years undergoing chemotherapy participated in the study. The intervention consisted on using a board game, which had been developed and validated by the researcher. As a pre-intervention, a sociodemographic questionnaire and the kidcope instrument were administered. The instrument has a question for the child describes a difficult situation. Then there are three additional questions to assess the level of distress, self-assessed by the child in the situation. After the intervention, the kidcope was administered again. The quantitative data were submitted to descriptive and inferential analysis ($p \leq 0.05$).

Results: The child reported the main difficult situations experienced. Most were categorized as a personal problem, being submission to venipuncture (31.43%); discomfort caused by signs and symptoms related to the disease and treatment (20%); and complications from clinical or surgical conditions (14.29%). There were also reports categorized as problems of an interpersonal nature, including the interaction of the child during the communication of difficult news about the diagnosis (22.86%); and the distance from other family members, in this case due to the search for treatment outside the home (11.43%). There was a reduction in the total distress score ($p < 0.001$).

Conclusions: For the children, the game helped them change the focus from the problem by interacting with a healthcare professional. The board game is a useful tool to decrease the level of distress in children with cancer.

IPSO IPSO SESSION 12: NEUROBLASTOMA II 23-10-2021 5:40 PM - 6:20 PM

00196 / #1057 | Neuroblastoma Molecular Risk-Stratification Via Circulating-Free DNA Copy Number Profiling

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Background and Aims: Segmental chromosomal aberrations (SCA) and MYCN amplification (MNA) are biomarkers for risk-group stratification in neuroblastoma, currently assessed on tissue biopsy. Increase in demand for tumour genetic testing for neuroblastoma diagnosis is posing a challenge to current practice, as the small size of the core needle biopsies obtained are required for multiple molecular tests. Objective: To investigate the utility of circulating free DNA (cfDNA) in patients with neuroblastoma for molecular risk-stratification.

Methods: Various pre-analytical conditions were tested to optimize cfDNA for SNP microarray currently used for neuroblastoma diagnosis. Plasma samples from 12 patients diagnosed with neuroblastoma were assessed for SCA and MNA using the Infinium CytoSNP-850K Beadchip array, with MNA status assessed independently using digital-droplet PCR (ddPCR).

Results: cfDNA samples generally yield highly degraded DNA and thus perform poorly in array-based applications such as SNP whole-genome genotyping. This could be overcome using pre-analytical DNA repair procedures, reaching a limit of detection of 20% variant allele frequency (VAF) for SCA, and 10% for MNA (our lowest tested VAF). MNA detection is highly affected by physical perturbations of the blood sample (mimicking sample shipment), which could be overcome using specialized collection tubes. 11/12 (92%) patients tested positive for SCA using SNP array: 8/8 (100%) and 1/2 (50%) at initial diagnosis with metastatic and localised disease, respectively, and 2/2 (100%) at metastatic relapse; of which 7 also presented MNA. cfDNA and matched tumour DNA profiles were generally identical. 7/7 (100%) of MNA and 0/5 (0%) non-MNA cases were tested positive using ddPCR.

Conclusions: These results reflect the feasibility of ctDNA SCA profiling in molecular risk-stratification, with MNA status orthogonal conformation via ddPCR in a clinically relevant time frame, and suggest that ctDNA analysis could potentially reduce tissue requirements currently embedded in the management of NB.

IPSO IPSO SESSION 12: NEUROBLASTOMA II 23-10-2021

5:40 PM - 6:20 PM

00197 / #1392 | Abdominal Anatomical Zones: A Novel Surgical Classification for Neuroblastoma

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Background and Aims: Surgery is integral to the local management of high-risk neuroblastoma (HRN). Imaging-defined risk factors (IDRF) are used to classify tumours into surgical risk groups but do not define tumour extent based on location. We propose a novel classification based on abdominal zones (AZ) to better characterise these tumours.

Methods: We conducted a retrospective review of consecutive HRN patients that underwent surgery at a single paediatric oncology unit between January 2008 and December 2019. We collected clinical data including details of tumour location/extent on imaging prior to surgery. The abdomen was divided into eight AZ based on anatomical landmarks. Analysis was performed to determine whether tumour distribution within these AZ affected peri-operative outcomes, event-free survival (EFS) and overall survival (OS).

Results: A total of 84 HRN children underwent surgery. Median age (range) at diagnosis was 2.5 (1.5-4.3) years. Preliminary analysis indicated that the total number of "tumour-positive" AZ impacted outcomes but did not correlate with IDRF number ($r=0.3$, $p=0.1$). Based on ROC analysis, patients were divided into 2 groups; G1: tumour in ≤ 2 AZ ($n=36$), and G2: tumour in ≥ 3 AZ ($n=48$). There were no differences between groups on baseline parameters (including number of IDRF; $p=0.08$), but only 5 G1 patients (14%) had involvement of midline AZ (96% in G2, $p<0.0001$). Complete resection was achieved in 71% of G2 children (100% in G1, $p=0.008$), who also had higher incidence of intra-operative complications (including bleeding; $p=0.002$). G2 patients had worse EFS [48% vs. 72% in G1; OR (95%CI): 0.4 (0.1-0.9), $p=0.02$] and OS [46% vs. 94% in G1; OR (95%CI): 0.05 (0.01-0.2), $p<0.0001$].

Conclusions: Tumour location/extent defined by AZ correlated to surgical complications and outcomes despite a comparable number of IDRF. This AZ classification may allow clearer comparison of tumours and better prediction of surgical complexity than IDRF alone.

IPSO IPSO SESSION 12: NEUROBLASTOMA II 23-10-2021

5:40 PM - 6:20 PM

00198 / #1616 | Structured Reporting of Surgery for Neuroblastoma and Peripheral Neuroblastic Tumors is Essential & Easy : How to Use The "International Neuroblastoma Surgical Report Form" (I.N.S.R.F.)

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Background and Aims: The role of surgery in neuroblastoma (NBL) and peripheral neuroblastic tumors (pNBT) such as ganglioneuroblastoma, is diagnostic (biopsy) or therapeutic (resection). Due to various tumor locations and frequent adherence or encasement of vascular or neural structures and vital organs, surgery in NBL & pNBT may be challenging, with important operative morbidity and mortality. The hazards of surgery and its complications in NBL & pNBT are not well described and its role in the treatment of high-risk disease remains debated. To optimize the recording of important perioperative data and intra- and postoperative outcomes, a standardized surgical case report form was created with international consensus. This presentation aims to make pediatric surgical oncologists acquainted with the standardized surgical report form and will demonstrate its correct use. It also aims to

stimulate its use in routine clinical practice, to engage all colleagues for structured reporting of NBL & pNBT surgery.

Methods: The "International Neuroblastoma Surgical Report Form" (I.N.S.R.F.) was published in 2020 as a joint initiative by the surgical committees of the pediatric oncological cooperative groups SIOPEN, COG and GPOH. We created a short instructive video to demonstrate its correct use in clinical practice, in a step-by-step approach.

Results: The brief instructive video will demonstrate clearly the correct and straightforward use of the I.N.S.R.F. in clinical practice. The different parts of the I.N.S.R.F. offer structured and standardized recording of important perioperative data and of intra- and postoperative complications.

Conclusions: The I.N.S.R.F. provides standardized recording of important perioperative data in NBL & pNBT surgery in a structured and straightforward manner. Its correct use is demonstrated in a short instructive video. Routine use of the I.N.S.R.F. in clinical practice, to document every surgical procedure for NBL or pNBT, is highly recommended, as it will ensure surgeon connectivity, facilitate surgical outcome analysis and improve clinical care.

IPSO IPSO SESSION 12: NEUROBLASTOMA II 23-10-2021

5:40 PM - 6:20 PM

00199 / #1780 | The Surgery of Mesenteric or Portal Vein Obstruction in Children with Periportal Malignant Tumors

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Background and Aims: We hypothesize that meso-Rex bypass (MRB) or portosystemic shunt (PSS) are effective at relieving portal venous (PV) obstruction secondary to malignant disease in children.

Methods: Ten patients with PV obstruction and periportal malignancy underwent a MRB or PSS (1997-2019, IRB 2006-12729). Paired and independent t-tests were used for comparison ($p < 0.05$ considered statistically significant).

Results: Patients presented the following tumors: five hepatoblastomas, one undifferentiated embryonal sarcoma of the liver (UESL), one bile duct rhabdomyosarcoma, one pancreatic neuroendocrine tumor, one yolk sac tumor, and one ganglioneuroblastoma. Four children had tumor excision and PV reconstruction simultaneously while six were done sequentially. Mean age at tumor excision was significantly younger in the sequential group ($4.2 \text{ years} \pm 2.7$ vs. simultaneous $9.8 \text{ years} \pm 3.3$, $p = 0.024$). Mean age at diagnosis of venous obstruction was similar in both groups (sequential $6.0 \text{ years} \pm 4.6$ vs. simultaneous $9.2 \text{ years} \pm 3.2$, $p = 0.26$). Mean age at PV surgery was similar (sequential $10.2 \text{ years} \pm 6.1$ vs. simultaneous $9.8 \text{ years} \pm 3.3$, $p = 0.91$). 4 patients had symptomatic variceal bleeding and 5 advanced hypersplenism prior to PV surgery. 8 patients underwent a primary MRB; one thrombosed and was converted to a PSS. Two patients underwent a primary PSS because no suitable intra-hepatic PV was present. There was no differ-

ence in mean age at PV surgery between types of PV procedure (MRB $8.9 \text{ years} \pm 4.0$ vs. PSS $12.9 \text{ years} \pm 6.6$, $p = 0.26$). Overall, median length of follow up was 1.5 year (21 days-8.5 years). There was no recurrence of variceal bleeding. Palpable spleen size diminished ($8.6 \text{ cm} \pm 4.5$ to $1.0 \text{ cm} \pm 1.7$, $p = 0.011$) and platelet count increased ($75.4 \text{ thou/uL} \pm 43.3$ to $135.2 \text{ thou/uL} \pm 65.5$, $p = 0.01$) significantly after PV surgery. One patient died (UESL recurrence). All others survived with patent vascular bypasses and no tumor recurrence at last follow-up.

Conclusions: We conclude the meso-Rex bypass can successfully restore portal circulation after venous obstruction from malignant disease without adversely affecting oncological outcomes. Portosystemic shunts can offer similar palliation if the meso-Rex bypass option is not available.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY 24-10-2021 8:00 AM - 9:30 AM

00200 / #1295 | Health-Related Quality of Life in Childhood Acute Lymphoblastic Leukemia: A Cohort Single Center Study

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Background and Aims: Children with acute lymphoblastic leukemia (ALL) and their families are at risk for poor Health-Related Quality of Life (HRQOL) due to physical, emotional, social and school problems. Only few studies assessing HRQOL during multiple treatment phases have been conducted, particularly in resource poor countries. This study presents a serial measurement of HRQOL of children with ALL and their parents during cancer treatment and investigates socio-demographic and clinical factors related to HRQOL.

Methods: A prospective HRQOL cohort study of children with ALL and their parents was conducted at an academic hospital in Indonesia. HRQOL was measured using PedsQL™ 4.0 Generic Core Scale and PedsQL™ 3.0 Cancer Module at three different treatment phases: induction phase (T1), consolidation phase (T2), and mid of maintenance phase (T3). Parents of all eligible children and patients aged 5-18 years were requested to complete both questionnaires.

Results: In total, 113 children aged 2-18 years and 221 guardians participated in this study. Of all children, 51% were diagnosed with standard risk ALL and 49% with high risk ALL. HRQOL improved during the course of treatment according to both proxy- and self-reports. Children (2-9 years) had a lower total HRQOL score than adolescents (10-18 years) throughout all three treatment phases of cancer treatment. Subscales of physical health, social functioning, school functioning, treatment anxiety and communication were significantly lower in children compared to adolescents based on proxy-reports. High risk

ALL patients had a significantly lower score for physical health compared to standard risk patients during T2 and T3 measurements.

Conclusions: HRQOL improved during the course of cancer treatment. For young children interventions to reduce treatment anxiety and to guide physical, social, school and communication problems are required. More attention is needed for the physical health of high risk ALL patients during consolidation and maintenance.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY
24-10-2021 8:00 AM - 9:30 AM

O0201 / #483 | An Honest Look in the Mirror: Patient-Reported Outcome Measurements are Rarely Used in Paediatric Cancer Clinical Trials - A Systematic Review of Clinical Trial Registries

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Background and Aims: Patient-reported outcomes (PROs) are the gold standard to assess patients' subjective health-status. Both the Food and Drug Administration (FDA) and the European Medicines Agency (EMA) recommend their use as endpoints in paediatric clinical trials to support claims for medical product labelling. The aim of this study was to assess the actual use of PRO instruments in childhood cancer clinical trials.

Methods: In June 2020 ClinicalTrials and EudraCT were systematically searched for all trials registered since 2007 investigating anti-cancer treatments in children and adolescents (≤ 21 years). To investigate which trial characteristics are associated with the use of PROs, Chi-Square or Fisher exact tests and a multivariate binary logistic regression were calculated.

Results: Of 4789 identified trials, 711 were included. The most frequent reason for exclusion was an upper age limit >21 years. Of all included trials, only 8.2% used PROs as endpoints and only 0.6% as primary endpoint. The most commonly used questionnaire was the PedsQLTM (32.8%), followed by PROMIS scales (12.1%). Another 24.1% reported measuring quality of life without specifying the tool. In 19.0% only pain or acceptability was assessed. No association was observed between the use of PROs and trial region, number of centres, trial phase, or intervention type (all $p > .05$). PRO implementation did not substantially increase over time. Only 20.3% of closed studies had published results.

Conclusions: Despite recommendations of regulatory agencies, PRO assessment is extremely rare in paediatric oncology clinical trials. The results shed light on the delay of progress in paediatric oncology due to its dependency from adult oncology. As research

in adult cancer patients has already shown, PROMs are easily applicable and useful tools both for research and clinical practice. More efforts should be undertaken to facilitate implementation of PRO in paediatric trials to guarantee patient-centred research and treatments.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY
24-10-2021 8:00 AM - 9:30 AM

O0202 / #647 | Impaired Health-Related Quality of Life of Male and Female Survivors of Childhood Cancer: Results from The DCCSS-Later 2 Psycho-Oncology Study

M. Van Gorp¹, L. Van Erp¹, A. Maas¹, L. Kremer², E. Van Dulmen-Den Broeder³, W. Tissing^{2,4}, J. Loonen⁵, H. Van Der Pal², A. De Vries^{2,6}, M. Van Den Heuvel-Eibrink², C. Ronckers², D. Bresters⁷, M. Louwerens⁸, M. Van Der Heiden-Van Der Loo², G. Huizinga⁴, H. Maurice-Stam¹, M. Grootenhuis¹

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Background and Aims: With increased survival in pediatric oncology, long-term health-related quality of life (HRQoL) of childhood cancer survivors (CCS) is becoming even more relevant. Previous studies often disregarded differences in the impact of childhood cancer on HRQoL between sexes. Therefore, we aimed to compare HRQoL of Dutch male and female childhood cancer survivors (CCS) to the general population using the TNO-AZL Questionnaire for Adult's HRQoL (TAAQoL).

Methods: As part of the Dutch Childhood Cancer Survivor Study LATER cohort (1963-2001) part 2, CCS (age ≥ 18 years, diagnosed ≤ 18 , time since diagnosis ≥ 5 years) filled out the TAAQoL, including 12 domains of HRQoL. Dutch general population reference data were available ($n=2476$). Proportions of CCS with impaired HRQoL (score $< 25^{\text{th}}$ percentile of general population scores) were compared to references using logistic regression analyses corrected for age and sex ($p < 0.004$). Interactions of group with sex were evaluated and included to model separate odds ratios (ORs) if significant ($p < 0.05$).

Results: Both male and female CCS ($n=1766$, mean age: 35.9 years [SD: 9.4], 51% male) had higher odds of impairment compared to references in the domains gross (OR male: 1.7, female: 2.3) and fine motor function (OR: 2.1), cognitive function (OR male: 2.7, female: 3.8), sleep (OR: 1.6), vitality (OR: 2.1) and depressive emotions (OR: 1.3). In addition, female CCS had higher odds of impairment in the domains pain (OR:

1.9), daily activities (OR: 1.9) and sexuality (OR: 1.4) and lower odds of high aggressive emotions (OR: 0.6). No differences compared to references were found for social function and positive emotions.

Conclusions: Dutch adult CCS more often have impaired HRQoL than the general population in several domains, most pronounced in cognitive function. Compared to male CCS, female CCS experience impairment in HRQoL more often and in more domains and accordingly may need more attention.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY
24-10-2021 8:00 AM - 9:30 AM

O0203 / #1213 | Measuring Fear of Cancer Recurrence in Survivors of Childhood Cancer: The Fear of Cancer Recurrence Inventory – Child (FCRI-C)

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Background and Aims: Fear of cancer recurrence (FCR) is among the most common and distressing concerns after cancer. However, no validated measures exist to evaluate FCR in survivors of childhood cancer. The aim of this study was to adapt the Fear of Cancer Recurrence Inventory – Short Form (FCRI-SF), a widely used adult measure, for use with children ages 8-18 years.

Methods: The existing 9-item FCRI-SF was reviewed for relevance and language by a panel of eight experts (disciplines included pediatric oncology, psychology, nursing, child life, and patients). Cognitive interviews with survivors of childhood cancer (N=3) assessed the clarity and face validity of the adapted measure. 103 childhood cancer survivors ($M_{\text{age}}=14.1$, $SD=3.0$, 42.7% female) then completed the FCRI – Child version (FCRI-C). We examined the reliability, factor structure, and construct validity of the FCRI-C including associations with child current age, child sex, intolerance of uncertainty, body vigilance, and pain catastrophizing.

Results: All FCRI-SF items were retained for the FCRI-C but the language was simplified. The internal consistency of the FCRI-C was very good ($\alpha=.89$). A one-factor structure was a good fit to the data, explaining 54.8% of the variance. Greater FCR was strongly associated with higher intolerance of uncertainty ($r=.52$, $p<.001$) and moderately associated with greater body vigilance ($r=0.49$, $p<.001$) and pain catastrophizing ($r=.40$, $p<.001$). Child sex and FCR were signifi-

cantly correlated with girls reporting higher levels than boys ($r=.20$, $p<.05$). There was no relationship between FCR and child current age ($p=.27$).

Conclusions: This research describes the development and preliminary validation of a measure to assess FCR in survivors of childhood cancer. The FCRI-C demonstrated strong reliability and construct validity. Future studies will further examine the psychometric profile of the FCRI-C in a separate sample and its relationships with other health outcomes such as health-related quality of life and healthcare utilization after childhood cancer.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY
24-10-2021 8:00 AM - 9:30 AM

O0204 / #194 | Parents' Experiences with Whole-Exome Sequencing in Pediatric Renal Cancer

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Background and Aims: In pediatric renal cancer, germline whole-exome sequencing (WES) contributes to the identification of predisposing factors, facilitating surveillance for second tumors and family counseling. Little is known about experiences and needs of parents of childhood cancer patients regarding WES. Our qualitative interview study explores these experiences and needs in order to improve counseling and support.

Methods: 33 interviews were conducted with parents after they had been approached for a nationwide germline WES study in children with renal tumors. This comprised a cancer panel analysis and subsequent optional exome-wide trio-analysis in case of normal panel analysis. The interviews were analyzed using an inductive thematic approach.

Results: Parents reported both individual and altruistic motives for participating. Altruistic motives such as contributing to science and helping future patients appeared more important after the child's treatment had been finished. Several families approached shortly after diagnosis reported feeling overwhelmed by the information. Parents often chose exome-wide (trio-)analysis over cancer panel analysis, although they had significant difficulties distinguishing between these approaches during the interviews. Families who received negative results felt relieved, but some worried about yet undiscovered factors. In families in which a predisposition was identified, parents did not experience significant distress although in some families distress from the predisposition seemed to add up to already existing emotional burden.

Conclusions: Families are generally positive about WES, however we identified several challenges pertaining to timing, consent and follow-up. We recommend approaching families during a relatively stable phase in their child's treatment trajectory. Separating consent and

counseling for panel analysis and exome-wide analysis might help parents to make a deliberate decision. Attention should be paid to families who already have a high emotional burden.

Free Paper Session (FPS) FPS 18-LIVE Q&A-PSYCHOONCOLOGY
24-10-2021 8:00 AM - 9:30 AM

**00205 / #539 | Support Needs of Dutch Young Adult
Childhood Cancer Survivors**

L. Van Erp¹, H. Maurice-Stam¹, L. Kremer², W. Tissing^{2,3}, H. Van Der Pal², L. Beek¹, A. De Vries², M. Van Den Heuvel-Eibrink^{2,4}, B. Versluys⁵, M. Van Der Heiden-Van Der Loo², M. Van Gorp¹, G. Huizinga³, M. Grootenhuys⁶

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Background and Aims: Studies about support needs of young adult childhood cancer survivors (YACCS) have previously focused mainly on information needs. This study assessed support needs and its determinants (socio-demographic, medical and psychosocial functioning) in Dutch YACCS.

Methods: YACCS (aged 18-30, diagnosed ≤ 18 years, time since diagnosis ≥ 5 years) cross-sectionally filled out a questionnaire regarding their need for various types of support (concrete information, personal counselling, and peer contact) in eight domains (physical consequences of childhood cancer, social-emotional consequences, relationships and sexuality, fertility, lifestyle, school and work, future perspective, insurance and mortgage), and the 'Pediatric Quality of Life Inventory Young Adults', 'Hospital Anxiety and Depression Scale', 'Checklist Individual Strengths' (fatigue). Descriptive statistics were used to describe support needs. Linear regression analyses were used to identify characteristics associated with support needs.

Results: Most YACCS reported a need for support in one or more domains (88.0%). Over 50% of participants reported a need for concrete information in the domains lifestyle, fertility, and physical consequences of childhood cancer and 25-50% in the domains insurance and mortgages, future perspective, social-emotional consequences of childhood cancer. In the domains lifestyle, physical as well as emotional consequences of childhood cancer 25-50% reported a need for counselling. Overall need for support was positively associated with middle ($\beta=.26$, $p=.024$) and high ($\beta=.35$, $p=.014$) compared to low educational attainment and subclinical anxiety ($\beta=.22$, $p=.017$), and negatively associated with social functioning ($\beta=-.37$, $p=.002$).

Conclusions: YACCS report the strongest need for support, in particular for concrete information, in the domains lifestyle, fertility, and physical consequences of childhood cancer. Associated factors were mostly socioeconomic and psychosocial in nature. Psychosocial care should be an integral part of LTFU for YACCS, with screening for psychosocial problems, the provision of psycho-education and tailored interventions, and adequate referrals to more specialized care if necessary.

Nursing FREE PAPER SESSION 4-EDUCATION OF NURSES AND PARENTS 24-10-2021 8:00 AM - 9:30 AM

**00206 / #594 | Teaching Pediatric Oncology Nursing at
Kamuzu College of Nursing**

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Background and Aims: In the Malawian health system, nurses are of paramount importance, but most never receive specialized pediatric oncology nursing (PON) training. While there are many nursing schools in Malawi, none have PON-related courses. Nurses are often the frontline health workers that see pediatric patients, but don't know how to recognize cancer. Kamuzu College of Nursing (KCN), within Malawi University, is recognized as an excellent public nursing and midwifery college with special commitment to academic rigor. The aim of this project was to develop a partnership between KCN faculty and Global HOPE Nursing to offer specialized training to nursing students.

Methods: This project uses the foundation courses developed to provide a formal structure to Global HOPE Nursing education. The two Global HOPE foundation courses were used to teach KCN nurses, each course includes modules that contain readings, short video lectures, case studies, group discussion questions and exam questions. Thirteen KCN graduate students specializing in paediatrics enrolled in the two courses. Students met for class with the Global HOPE Nurse Educator who taught through face-to-face lectures and group discussions. Due to COVID 19, schools were closed for 2 months and this disrupted the clinical allocations and final examinations for students. However, after schools re-opened, students finished their clinical work and took the final examinations.

Results: All 13 students successfully completed the courses that involved 32.5 hours of classroom work including face-to-face presentations and ZOOM. Students also completed 80 hours of clinical experience on the pediatric cancer ward.

Conclusions: This project serves as a model platform for other sub-Saharan Africa locations where nursing education programs would enhance capacity. It also provides an ongoing opportunity for continuing professional development in the specialty and provides accessible oncology information, hence impacting on early recognition and referral of oncology patients.

Nursing FREE PAPER SESSION 4-EDUCATION OF NURSES AND PARENTS 24-10-2021 8:00 AM - 9:30 AM

O0207 / #592 | Using Project ECHO to Promote Pediatric Oncology Nursing Education in Sub-Saharan Africa

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Background and Aims: To maximize cures for children with cancer in low- and middle-income countries (LMIC), excellence in nursing is at the center of all that is needed. While the burden of childhood cancer care is the highest in LMIC, opportunities for continuing nursing education and specialization are extremely limited. The aim of this project was to implement an education network for nursing caring for children with cancer across sub-Saharan Africa.

Methods: The Extension for Community Health Outcomes (ECHO) model is used by Global HOPE Nursing to implement a learning platform that provides nurses in sub-Saharan Africa access to nursing expertise in the care of children with cancer. Case-based learning strategies are used at monthly Project ECHO nursing seminars to provide access to nursing expertise in the care of children with cancer. Early career nurses specializing in pediatric oncology are mentored to serve as Global HOPE Nursing Project ECHO presenters.

Results: During the first 17 months of implementation 598 participants from 18 countries attended an ECHO seminar with an average of 35 participants per seminar. To date, 187 unique participants attended the first 17 Global HOPE Nursing Project ECHO Seminars with 42% of these individuals (n= 78) attending more than one. Nurses were from Global HOPE Centers of Excellence (44%) and community hospitals or schools of nursing (56%). Participants felt more confident in providing childhood cancer nursing care after the Project ECHO seminar and evaluations demonstrated a case-based approach to learning was extremely successful.

Conclusions: A key focus of the Global HOPE Nursing Project ECHO is to build a knowledge network to increase capacity of pediatric oncology nursing experts in sub-Saharan Africa. Formal education programs that build the capacity of specialists in pediatric oncology nursing are essential to improve global cure rates for children with cancer.

Nursing FREE PAPER SESSION 4-EDUCATION OF NURSES AND PARENTS 24-10-2021 8:00 AM - 9:30 AM

O0208 / #321 | Structured Evaluation of Pediatric Oncology Websites to Create a Resource Database for Low- and Middle-Income Country Nursing Education

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Background and Aims: Websites offer a plethora of information for creating pediatric oncology nursing curricula or training materials in low- and middle-income countries (LMIC). Unfortunately, website construction is not regulated, so identifying accurate information can be challenging. Three expert pediatric oncology nurses (1 LMIC and 2 with experience working in LMIC), identified relevant websites for a structured evaluation and to create a database resource.

Methods: A search of PubMed, CINAHL and Google Scholar from 2011-2021 was conducted for reviews of pediatric oncology websites useful for nursing education. Search terms included pediatric, oncology, nurs*, education, websites or internet. English-language websites known to the authors were collected and additional sites were identified through a targeted Google search. Website currency, relevance, authority, accuracy and purpose were evaluated with an author-adapted CRAAP tool (maximum 30 points). Pediatric content, reading level and language were appraised.

Results: No earlier pediatric oncology website review was located. Eighty-six websites were identified as relevant for pediatric oncology nursing curricula or training content; 25 were excluded as non-pediatric, paywall, non-functional links, fundraising websites, or patient support. The majority (70%) had explicit pediatric oncology content; 39% of websites offered translations. The vast majority (92%) were high school reading level or higher. The range of CRAAP scores was 9-30; 36/61 websites scored ≥ 25 were considered excellent and highly recommended, 16/61 were recommended (20-24), 9/61 (≤ 19) were less useful.

Conclusions: Guidance on website utility for LMIC pediatric oncology nursing curricula and training material has been unavailable. Our findings benefit nursing educators seeking appropriate, accurate and timely information for teaching materials for nurses, patients and families. The large number of unreferenced, outdated, author unidentified, or not fit-to-purpose websites makes it critical for expert evaluation and a continually updated resource available to nurses working in pediatric oncology, especially in LMICs.

Nursing FREE PAPER SESSION 4-EDUCATION OF NURSES AND PARENTS 24-10-2021 8:00 AM - 9:30 AM

O0209 / #1000 | Development and Implementation of Nursing Educational Material for Parents/Caregivers of Children with Cancer: A Peruvian Pilot Study

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Background and Aims: Education to the parents/caregivers of children with cancer is a fundamental role of nursing to avoid complications, provide quality care, promote adherence to treatment and maintain basic standards of care. Aim: design educational material for parents/caregivers of children with cancer on critical information about cancer and childcare.

Methods: Within the framework of the WHO Global Initiative for Childhood Cancer in Peru, a multicenter working group including key nursing leaders, education coordinators, nurse educators and other multidisciplinary team members was established in May 2020. A comprehensive needs assessment survey for parents/caregivers of children with cancer was distributed in June-July 2020. The survey was conducted online and in person in 9 institutions (Lima and Arequipa) to examine the preferred models and content for online and onsite educational talks, and audiovisual and printed material. A self-administered questionnaire was given to selected 13 parents/caregivers to determine their comprehension of the proposed resources.

Results: A total of 365 parents/caregivers participated (37.9 years median age). Most respondents (63.3%) were parents of children in active treatment. Almost half had only primary or secondary education (45%). Respondents ranked as most important: cancer treatment and side effects (25%), nutrition (15%), warning signs (12.5%) and home care (8.2%). Most respondents (70.9%) preferred to receive information through educational talks and videos (64.7%). Pilot educational material (online lectures and seven educational videos) was developed August-October 2020 through online educational collaboration in each institution, with on-going content and structural refinement. No parent/caregiver had difficulty understanding the newly created educational material.

Conclusions: Assessing educational needs of parents/caregivers of children with cancer provides a starting point in the design of educational strategies. Continued consultation with parents/caregivers to evaluate new educational material is essential to ensure acceptance and relevance.

Nursing FREE PAPER SESSION 4-EDUCATION OF NURSES AND PARENTS 24-10-2021 8:00 AM - 9:30 AM

O0210 / #227 | Interprofessional Spiritual Care Education in Pediatric Hematology-Oncology

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Background and Aims: In the care of adolescents and young adults (AYA) with cancer and chronic blood disorders, emerging evidence calls providers to give greater attention to spirituality. Spiritual perspective and hope-derived meaning are among the most significant protective factors in the Resilience in Illness Model (RIM) for AYA. Evidence-based oncology guidelines recommend that interdisciplinary team members screen all patients for spiritual distress and attend to their spiritual suffering and wellbeing. A lack of spiritual care training has been a key barrier to adequate delivery of this aspect of care.

Methods: In order to provide interprofessional spiritual care education, clinicians treating AYA were recruited from the hematology-oncology department of a pediatric hospital. The Interprofessional Spiritual Care Education Curriculum (ISPEC[®]) online curriculum was provided alongside six in-person group discussion sessions. Discussions were co-led by a chaplain and an oncologist. Pre- and post-education surveys measured participants' ability, frequency, confidence, and comfort to recognize and address spiritual concerns at baseline and 1, 3, and 6 months post-training. Responses were analyzed using ANOVA and multiple pairwise comparisons.

Results: Participants (n=21) included physicians, advanced practice providers, nurse coordinators, and psychosocial clinicians. Significant improvements were observed in interprofessional spiritual care ability (36%, p<.0001), frequency (56%, p=.0135), confidence (32%, p<.0001), and comfort level (31%, p=.0181) baseline vs. 1-month, with several gains sustained through six months.

Conclusions: Benefits of interprofessional spiritual care education were observed in clinicians' skills, self-efficacy, and frequency of

engaging in generalist spiritual care. Positive effects were maintained, which is vital for the continuous integration of patients' spiritual needs and strengths in their care. Interprofessional spiritual care education effectively prepares hematology-oncology clinicians from diverse disciplines to provide generalist spiritual care within their scopes of practice, enhancing the whole-person care for AYA with cancer and chronic blood disorders.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0211 / #680 | Clinical Applications of RNA-Seq in Childhood Acute Lymphoblastic Leukemia (ALL): Results from the Dana-Farber Cancer Institute (DFCI) All Consortium Protocol 16-001

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Background and Aims: The molecular hallmark of ALL is characterized by recurrent, prognostic genetic alterations, many of which are cryptic by conventional cytogenetics. RNA-seq is a powerful next-generation sequencing technology with the ability to simultaneously identify cryptic gene rearrangements, sequence mutations and gene expression profiles in a single assay. We examined the feasibility and utility of incorporating RNA-seq into a prospective multicenter phase 3 clinical trial for children with newly-diagnosed ALL.

Methods: Patients enrolled on the DFCI ALL Consortium Protocol 16-001, who consented to optional studies and had available material for testing underwent RNA-seq.

Results: RNA-seq was performed in 173 patients with ALL (B-ALL: n=136; T-ALL: n=37). Median time for library preparation, sequencing, bioinformatic analysis and data interpretation was 17 [7-51] days. Median time from sample receipt to final report delivery was 35 [15-63] days. RNA-seq identified at least one alteration in 157 patients (91%) and fusion detection was 98% concordant with results obtained from conventional cytogenetic analyses for 46/47 (98%) gene fusions. An additional 49 gene fusions were identified by RNA-seq, many of

which confer prognostic or therapeutic significance. Gene expression profiling and fusion detection enabled further molecular classification into the following B-ALL subgroups: ETV6-RUNX1/ETV6-RUNX1-like (n=31; 23%), high hyperdiploid (n=34; 25%), TCF3-PBX1 (n=7; 5%), KMT2A-rearranged (n=5; 4%), Philadelphia (Ph) chromosome-positive/Ph-like (n=19; 14%), DUX4-rearranged (n=11; 8%), PAX5-altered (n=7, 5%), ZNF384-rearranged (n=4, 3%), MEF2D-rearranged (n=1; 1%) and NUTM1-rearranged (n=1; 1%). RNA-seq identified 141 nonsynonymous mutations in 93 patients (54%); the most frequent were RAS-MAPK pathway mutations (36%). Among 68 patients with both low-density array and RNA-seq data for the Ph-like gene signature prediction, results were concordant in 63 patients (93%).

Conclusions: RNA-seq is feasible in the context of a multicenter prospective clinical trial. We identified several clinically-relevant genetic alterations not detected by conventional methods, supporting the integration of this technology in frontline trials for pediatric ALL.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0212 / #151 | Precision Co-Targeting of the Thymic Stromal Lymphopoietin Receptor in Childhood Crlf2-Rearranged Acute Lymphoblastic Leukemia

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Background and Aims: Philadelphia chromosome-like acute lymphoblastic leukemia (Ph-like ALL) is associated with high rates of chemoresistance and relapse. CRLF2 (cytokine receptor-like factor 2) rearrangements occur in 50% of Ph-like and 60% of Down Syndrome (DS)-ALL and induce constitutive kinase signaling. Current clinical trials are studying chemotherapy with the JAK inhibitor ruxolitinib in patients with CRLF2-rearranged Ph-like ALL, but results are not yet known. While chimeric antigen receptor T-cell (CAR) immunotherapies have induced remarkable remissions in children with relapsed/refractory B-ALL, approximately 50% of CD19CAR-treated patients will relapse again, many with CD19 antigen loss. New therapies are needed to prevent relapse and overcome immunotherapeutic resistance.

Methods: We previously developed thymic stromal lymphopoietin receptor (TSLPR; encoded by CRLF2) CAR T-cells and demonstrated potent preclinical activity in Ph-like ALL models (Qin Blood 2015), which led to a soon-to-open phase 1 clinical trial. In the current studies, we hypothesized that combinatorial targeting with bispecific TSLPRxCD19CAR or TSLPRxCD22CAR or with TSLPRCAR+ruxolitinib will have superior activity against CRLF2-rearranged Ph-like and DS-ALL.

Results: TSLPRCART treatment of CRLF2-rearranged ALL cell line and patient-derived xenograft (PDX) models potently inhibited leukemia proliferation in vitro and in vivo and induced long-term 'cure' of xenograft mice. However, co-administration of TSLPRCART+ruxolitinib markedly diminished in vivo T cell proliferation, blunted cytokine production, and facilitated leukemia relapse, which could be abrogated by delaying ruxolitinib. Importantly, ruxolitinib co-treatment prevented severe TSLPRCART-induced cytokine release syndrome (CRS) and animal death. Interestingly, ruxolitinib withdrawal led to return of T-cell functionality with redetection of TSLPRCART in peripheral blood, IFN- γ production, and leukemia clearance upon CRLF2+ ALL rechallenge.

Conclusions: TSLPRCART has potent preclinical activity against both CRLF2-rearranged Ph-like and DS-ALL. Ruxolitinib co-administration impaired in vivo TSLPRCART functionality, but was protective against life-threatening CRS. Ongoing studies are defining optimal TSLPRCART+ruxolitinib sequence(s) and assessing in vivo efficacy of bispecific TSLPRCARTs in CRLF2-R Ph-like and DS-ALL PDX models for future clinical translation.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0213 / #1332 | Nanopore RNA Sequencing as a Diagnostic Tool for Acute Leukemia in Low Resource Settings

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Background and Aims: With no opportunities for prevention but available highly cost-effective treatment opportunities, pediatric cancer survival fundamentally depends on accurate diagnosis. Comprehensive modern diagnostics are unavailable for many low- and middle-income countries (LMIC). We sought to develop a low-cost leukemia lineage and genomic classification approach using Oxford Nanopore Technology RNA sequencing.

Methods: We performed Nanopore RNA sequencing on 124 acute leukemia specimens across well-characterized lineage and genomic subtypes. Using only gene expression profiling, we trained a set of pairwise and one-vs-all partial least squares discriminant analysis classifiers among lineage and genomic subtypes. The output vectors from these were combined into a feature matrix for a support vector machine classifier. The composite model provides a set of predicted probabilities for each lineage and genomic subtype.

Results: Samples averaged 756,119 reads (465 Mbp) over an average of 11,755 unique genes per sample with a median read length of 840 bp. Overall, ninety seven percent (121/124) of cases were classified correctly into leukemia lineage (AML, B-ALL, T-ALL). The 117 cases with predicted probability above 80% were classified with 100% accuracy. Within the B-ALL subgroups of ETV6-RUNX1, hyperdiploid, TCF3-PBX1, KMT2A rearranged, and Ph+/Ph-like, 47/50 (94%) were correctly classified. Examining only the 40 cases of these core B-ALL subgroups with a predicted probability greater than 50%, all were classified correctly. The 18/23 (78%) of AML cases with high predicted probability for genomic subtypes were all correctly classified into core subgroups of KMT2A rearranged, core-binding factor, or "other".

Conclusions: Our research lays the groundwork for an innovative diagnostic approach for acute leukemia in LMIC through unbiased RNA sequencing using Nanopore at material costs of \$100-300 USD per sample and sequencer costs of \$1000 USD. We expect genomic classification to improve with increasing samples for training the classifier. Next steps include expanding the genomic classification and developing the assay within the LMIC context.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0214 / #740 | Next-Generation Sequencing Revealed that the Biology of Childhood Acute Leukemia in Iraq is Probably Different

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Background and Aims: Molecular analyses in hematological malignancies may provide information about the etiological factors in leukemogenesis. Since molecular analyses are not yet available in Iraq; the country of repeated wars, we conceived of performing next-generation sequencing (NGS), to disclose the genomic landscape of Iraqi childhood

acute lymphoblastic leukemia (ALL) and acute myelogenous leukemia (AML).

Methods: From June 2016 to February 2020, bone marrow paired samples (at diagnosis and remission) were collected from 66 Iraqi children aged < 16 years, including 55 ALL, and 11 AML. Flinders Technology Associates (FTA) cards were used to transfer the samples from Iraq to Japan. We performed whole-exome sequencing, targeted sequencing-based structural variation detection, and whole-genome sequencing.

Results: Among the 55 evaluated Iraqi children with ALL, there were (49, 89.1%) precursor B-ALL (B-ALL), and (6, 10.9%) precursor T-ALL (T-ALL) cases. TCF3-PBX1 was the most recurrent fusion gene (11, 22.4%). Moreover, RAS pathway mutations including NRAS, KRAS, FLT3, PTPN11, and BRAF, were abundant (22, 44.9%) in B-ALL, and they were significantly associated with high hyperdiploidy (10/12, 83.3%). Whereas no RAS mutation was identified in the 6 cases with T-ALL. Additionally, concerned with the AML subset, 5 out of 11 cases with AML had acute promyelocytic leukemia. Similarly, RAS mutations were frequent in patients with AML (4, 36.4%).

Conclusions: Apart from disclosing the unprecedented high frequency of TCF3-PBX1, NGS results confirmed our previous finding of frequent oncogenic RAS in Iraqi childhood ALL and AML compared to other regions. Our results suggest that the biology of childhood acute leukemia in Iraq is probably different, and the environment related to war and war-aftermath in Iraq may play a role along with the geographic or ethnic background. **Acknowledgments:** We appreciated the 53rd-Annual SIOP acceptance for our abstract to be published in the official congress publications.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0215 / #223 | Protein Kinase D-Dependent Downregulation of Immediate Early Genes Through Class Iia Histone Deacetylases in Acute Lymphoblastic Leukemia

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Background and Aims: Acute lymphoblastic leukemia (ALL) is one leading cause of cancer-related death in children, and cure rates for relapsed/refractory ALL remain suboptimal, highlighting the need for novel targeted therapies. We previously uncovered that ALL blasts are vulnerable to energy/ER-stress via AMP-activated protein kinase (AMPK) activation.

Methods: In order to identify genome-wide metabolic stress regulated genes, we used RNA-Seq and compared mRNA profiles in ALL cells treated with AICAR, an AMPK activator. RT-qPCR and ChIP assays were employed to validate the RNA-Seq data in Bp-ALL, T-ALL, and other mammalian cell models. Pharmacological inhibitors/activators,

siRNA/shRNA, kinase assays, and genetic models were used to investigate AICAR-induced IEGs downregulation mechanism. We confirmed the in vivo relevance of our data, using an NSG mouse model.

Results: RNA-Seq identified several immediate-early genes (IEGs) as a subset of genes downregulated by AICAR. AICAR-induced IEGs downregulation was blocked by an adenosine uptake inhibitor but not with agents blocking its conversion to ZMP and AMPK activation, indicating AICAR was responsible for IEGs reprogramming. Further investigation established this mechanism was AMPK-independent. We demonstrated that AICAR directly inhibited PKD kinase activity and identified PKD as responsible for IEGs downregulation. PKD inhibition induced DUSPs downregulation, and decreased phosphorylation and nuclear export of HDAC4/5, which lowered histone H3 acetylation and decreased NFκB(p65) recruitment to IEGs promoters. Treatment with PKD inhibitors (CRT, NaPP1) led to growth inhibition and apoptosis in ALL cells and primary patient cells. Finally, we showed that treatment with CRT, regorafenib (multikinase inhibitor, including the NFκB pathway) and the combination CRT+regorafenib led to statistically significant survival of NSG mice engrafted with ALL-Luc cells.

Conclusions: We identified a previously unreported PKD-dependent adaptive survival mechanism in response to energy/metabolic stress in ALL through regulation of HDAC-IEGs and DUSPs expression. Further investigations to confirm a potential therapeutic role for targeting PKD in ALL are warranted.

Free Paper Session (FPS) FPS 16-LIVE Q&A-ALL - BIOLOGY AND BIOMARKERS 24-10-2021 8:00 AM - 9:30 AM

O0216 / #525 | Bioinformatic Analysis Identifies Specific Co-Expression of LNCRNA/MRNA in Pediatric B-Cell Acute Lymphoblastic Leukemia

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Background and Aims: Acute lymphoblastic leukemia (ALL), as the most common cancer in children, is a biologically and clinically heterogeneous disease. Pediatric B-cell ALL (B-ALL) constitutes an aggressive neoplasia in which new biomarkers and therapies are required to obtain a better prognosis for the patients. Many long ncRNAs (lncRNAs) are involved in key biological processes. Dysregulation of lncRNAs has been described in numerous human diseases, including cancer, leading to oncogenic or tumor-suppressive activities. However, few lncRNAs have been directly associated with pediatric B-ALL. Here, we established lncRNA/mRNA expression profile associated with B-ALL in children.

Methods: We performed a comparative study of the lncRNAs and associated protein-coding genes profiles in 42 pediatric B-ALL samples and 4 bone marrows from healthy donors by microarray analysis. To validate our results, we analysed multiple external pediatric B-ALL datasets. We studied the possible gene expression regulation mechanisms of the lncRNA/mRNA candidates. Lastly, we compared the expression profiles and survival data of patients with B-ALL against different subtypes from these cohorts.

Results: We identified a subset of lncRNAs that were differentially expressed between different pediatric B-ALL subtypes and healthy bone marrow samples. The co-expression of TCL6/TCL1B and AL133346.1/CCN2 were consistently correlated across all datasets. B-ALL samples showed a higher expression of AL133346.1/CCN2 than T-ALL samples in external datasets. Gene Ontology Analysis showed overrepresented cancer-related pathways among the differentially expressed lncRNA/mRNAs. Survival analysis demonstrated that patients with B-ALL with "high" expression levels of lncRNA TCL6 and mRNA CCN2 showed higher overall survival (OS) than those with "low" levels.

Conclusions: The classification of patients based on differentially lncRNA/mRNA, analysed in this study, could improve the diagnosis of pediatric B-ALL. "High" TCL6 and CCN2 levels could be associated with better clinical outcome of pediatric patients with B-ALL. These potential biomarkers could allow a better stratification and prognosis of children with B-ALL.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMIOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

O0217 / #1476 | Trends and Variations in Global Chemotherapy Pricing: Data from 95 Countries over Eight Years

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Background and Aims: Lack of transparency in chemotherapy pricing hinders efforts by governments and stakeholders to develop coordinated procurement policies, negotiate drug prices, and perform cost effectiveness analyses in pediatric cancer care. This study aimed to assess patterns and variations in global chemotherapy pricing with implications for the treatment of childhood cancer in low- and middle-income countries.

Methods: Data sources included Management Sciences for Health (MSH) data and IQVIA MIDAS data, which reflects estimates of marketplace activity obtained under license from IQVIA. MIDAS data on chemotherapy prices, manufacturers, and purchase volumes were retrieved on February 5, 2020 for the time period 2012-2019. Chemotherapy agents were selected based on their inclusion in front-line regimens for acute lymphoblastic leukemia, Burkitt lymphoma, Hodgkin lymphoma, low grade glioma, retinoblastoma, and Wilms tumor. Data were aggregated by region and income classification, and descriptive statistics were used to determine central tendency and variability of chemotherapy prices. Cumulative prices for treatment regimens were compared for each cancer subtype across income classification.

Results: A total of 212,429 data points were obtained from 95 countries: 41 high-income countries (HICs), 26 upper-middle income countries (UMICs), 19 lower-middle income countries (LMICs) and nine low-income countries (LICs). Forty-three percent of the data were from HICs, 27.4% from UMICs, 20% from LMICs, and 9.5% from LICs. While median drug prices for all chemotherapy agents were greater in HICs, prices varied substantially within income groups and did not correlate with gross domestic product (GDP). Prices of treatment regimens were higher for non-adapted protocols, hematologic malignancies, and higher risk stratification or stage. An inverse correlation was observed between volume of drug purchased and variance in drug prices.

Conclusions: As governments and stakeholders in LMICs scale health-care systems for childhood cancer treatment, the findings of this study should inform efforts to develop evidence-based pricing policies, negotiate drug prices, and perform cost-effectiveness analyses in childhood cancer care.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMIOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

O0218 / #1442 | A Cloud-Based Dynamic Visualization Tool to Estimate Global Costs Associated with Childhood Cancer Medications

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Background and Aims: Reliable access to quality chemotherapy is a barrier to effective global childhood cancer control efforts. To estimate global childhood cancer drug needs and costs, we developed a mathematical model to help policy makers and administrators plan procurement and formulate budget impact analyses. As data needs are context specific, this study aimed to develop a publicly-available dynamic visualization tool.

Methods: A mathematical model was used to estimate the global cost of cytotoxic medicines for children with cancer by location, drug type, and cancer type from 2015-2030. Relevant drugs from the World Health Organization (WHO) Essential Medicines List for Children were included. Total cost of chemotherapy was estimated by multiplication of disease burden, cumulative drug amount and drug costs under proper model assumptions. Burden was estimated using publicly available incidence and survival estimates, and stage/risk distribution from registry and literature estimates. Cumulative drug doses were calculated from standard regimens and age-based body surface area. Cost inputs were obtained from global data sources. R (<http://www.R-project.org>, version 4.0.4) and R Shiny package (<https://CRAN.R-project.org/package=shiny>, version 1.5.0) were used to develop the mathematical model and a graphical user interface (UI).

Results: The UI is available at <https://chemoglobe.stjude.org>. Results are based on 84,672,000 simulations. Interactive stacked bar plots present chemotherapy costs associated with 14 childhood cancer categories by WHO regions, World Bank income groups and United Nations subregions. Users may also select among six parameters and 32 options to dynamically appraise uncertainty in estimates. In performance testing, the tool takes eight seconds to update after parameter adjustments.

Conclusions: We successfully deployed version 1.0 of a cloud-based dynamic visualization tool to help stakeholders plan drug procurement and explore potential policy interventions relevant to their setting. Additional visualizations to allow point-and-click analyses of drug-specific costs, country estimates and temporal trends in case-finding are in development.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMIOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

00219 / #1775 | Comparative Costs in Diagnostic Evaluation of Acute Leukemia in the United States and Two Lower-Middle Income Countries

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Background and Aims: The current paradigm of diagnostic development in low- and middle-income countries (LMICs) is based on incremental implementation of expensive and complex high-income country testing approaches. Oxford Nanopore Technology (ONT) is a sequencing platform with very low capital costs and low material costs with emerging potential to replace traditional cancer diagnosis practices. This study aims to estimate the total and materials-only pathology costs for acute leukemia in the United States (US) and two LMICs compared with ONT.

Methods: Diagnostic costs (USD, 2021) for acute leukemia were calculated based on standard evaluations in hospitals in the US, Pakistan, and El Salvador. US costs were estimated using Centers for Medicare-Medicaid Services Current Procedural Terminology code charges for karyotyping, flow cytometry, fluorescent in situ hybridization (FISH), and limited molecular testing. Flow cytometry, karyotyping and FISH are used at Indus Hospital, Pakistan, while flow cytometry alone is used at Hospital Nacional de Ninos Benjamin Bloom, El Salvador. Costs at both institutions were identified using hospital administrative data. A secondary micro-costing analysis of ONT mRNA sequencing material costs across a range of sequencing depths (5x10⁵ – 5x10⁶ reads/sample) was completed for comparison.

Results: The estimated costs of routine diagnostic testing for acute leukemia in the US were \$3,475 for acute lymphoblastic leukemia (ALL) and \$5,493 for acute myeloid leukemia (AML). The per sample ALL/AML cost of diagnostic testing was \$820 (\$280 materials-only) in Pakistan (flow cytometry, karyotyping, FISH) and \$612 (\$261 materials-only) in El Salvador (flow cytometry only). The material cost for unbiased ONT mRNA sequencing ranged from \$102/sample for shallow to \$291/sample for deeper sequencing.

Conclusions: Access to acute leukemia diagnostic tests varies globally and is expensive. ONT sequencing is an emerging diagnostic technology that may be able to deliver cost savings while also allowing for comprehensive diagnosis of leukemia in LMICs using a singular testing platform and supply chain.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

O0220 / #1492 | Development and Validation of a Childhood Acute Lymphoblastic Leukemia Policy Model Constructed Using Real World Population-Based Data

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Background and Aims: While new effective but expensive immunotherapies are being approved for the treatment of childhood acute lymphoblastic leukemia (ALL), their cost-effectiveness remains unknown. We therefore aimed to use real-world population-based healthcare utilization and cost data to build and validate a microsimulation policy model for childhood ALL that allows flexible and rapid economic analyses of new agents

Methods: Children aged 0-17 diagnosed with primary ALL 2002-2012 in Ontario, Canada were identified through a provincial registry. Detailed disease, treatment, and outcome data were chart abstracted. Costs (2018 Canadian dollars) were obtained through linkage to population-based healthcare databases. Multistate survival analysis and linear regression modeled outcomes and costs associated with various disease states [e.g. relapse, bone marrow transplant (BMT)] respectively. A microsimulation model was developed using the fitted models and a simulated ALL population that was validated by comparing outcomes and costs across various patient subgroups.

Results: The overall simulated population had a 5-year survival of 94.0% (95th confidence interval 92.6-95.4), and a cumulative incidence of relapse of 10.4% (9.0-12.3). Average 5-year cost was \$252,700 (246,400-259,000). Comparing simulated subgroups yielded expected results. For example, superior outcomes and higher 5-year costs were seen among children with B-lineage vs. T-lineage disease [94.2% (92.4-95.7) vs. 88.5% (83.9-92.6); \$253,900 (246,400 - 261,400) vs. \$246,100 (221,100-271,034)]. Similar results were seen compar-

ing end of induction minimal residual disease negative vs. positive patients [96.6% (95.5-97.6) vs. 87.8% (82.3-91.3), \$246,100 (238,100-254,100) vs. \$286,200 (261,800-310,600)].

Conclusions: Projections from the microsimulation model are consistent with observed trends in Ontario and with published clinical data. Analyses are currently underway examining the cost-effectiveness of nelarabine in the treatment of upfront T-lineage ALL and of blinatumomab in the treatment of high-risk relapsed B-lineage ALL.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

O0221 / #1176 | Mapping of Resources and Priorities for Pediatric Hematology/Oncology in 11 Eurasian Countries: Report from Eurasian Alliance in Pediatric Oncology

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Background and Aims: Mapping of healthcare resources is important to optimize use of referral pathways, aid in prioritization of resource requests, and streamline communication with the ministry of health (MoH). This study describes resource mapping of pediatric hematology-oncology (PHO) services in 11 countries of the Eurasian Region, represented by countries in Eastern Europe and Central Asia.

Methods: Eleven countries part of the Eurasian Alliance in Pediatric Oncology (EurADO) provided data for mapping including: demographics and personnel, pediatric population, pediatric cancer cases, total workforce, palliative care and rehabilitation services availability, cancer registry presence, and three priority asks for the MoH to improve PHO care in the country. The generated country map includes: heat map (incidence/100,000 children), institutions providing PHO care (numbers and types of beds, equipment for diagnostics and treatment, workforce), blood bank supply, and educational resources. Additionally, a unified regional map of all 11 countries was created with proposed referral pathways.

Results: In total, 97 maps were prepared (85 regional maps of one of the countries, 11 individual country maps, and 1 regional map) representing 131 institutions. Analysis of PHO services identified limitations in the PHO workforce (3 countries reporting 0.3-0.5 pediatric oncologist per 100,000 children), diagnosis and treatment, with only 5 countries reporting access to all basic diagnostics (both laboratory and instrumental) and treatment (including surgery and HSCT). Palliative and rehabilitation care was available in only 7 and 5 countries, respectively. Population cancer registries are organized in 3 countries; hospital PHO registries in 8. Map review with local PHO leaders identified common asks for MoH and governmental authorities to improve PHO care, including improvement of diagnostics, cancer registration, and education (unification of PHO specialty).

Conclusions: Resource mapping allows for understanding of PHO services in Eurasia, informs development of referral pathways for diagnostics and treatment, and helps identify priority asks to improve PHO care.

Free Paper Session (FPS) FPS 17-LIVE Q&A-EPIDEMIOLOGY - COSTS AND RESOURCES 24-10-2021 8:00 AM - 9:30 AM

O0222 / #599 | Disparities in the Cost of Care: Work Flexibility, Social Support, and the Financial Impact of Cancer

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Background and Aims: Prior studies have documented disparities in financial burden after a cancer diagnosis. This study aims to characterize such disparities experienced by diverse families affected by pediatric cancer. We hypothesize that differing caregiver work flex-

ibility and social support may account for disparities in financial burden.

Methods: Cross-sectional survey of 118 caregivers of children treated at University of California San Francisco Benioff Children's Hospitals for any cancer requiring chemotherapy and/or radiation from March 2015 to March 2019. Financial burden was characterized by financial toxicity (COST score), income loss, and household material hardship (HMH; food, housing, or energy insecurity). The survey is composed of adaptations of 5 validated surveys assessing a family's finances, workplace flexibility, financial stress, HMH, and social support. Caregivers completed the survey in-person or via phone, email or mail. Multivariable logistic regression models were used to estimate odds ratios and corresponding 95% confidence intervals.

Results: The population sampled was 66% White, 20% Hispanic, 59% high income (>400% federal poverty level), and 64% highly educated (minimum bachelor's degree). Mean percent income loss was 9%. The prevalence of HMH increased from 23% of families at baseline to 39% at 12 months after diagnosis. Financial toxicity was significantly increased in low- and middle-income families compared to high-income ($p < 0.01$). Work flexibility was significantly higher in high-income families compared to low- ($p = 0.02$) and middle-income ($p < 0.01$), but was not associated with HMH. Social support was strongly associated with reduced HMH ($p = 0.01$).

Conclusions: HMH increases in the first year after pediatric cancer diagnosis. While higher income families reported greater work flexibility, this did not impact their likelihood of HMH. Social support, however, is associated with lower risk of HMH, suggesting that interventions to strengthen social support may mitigate some of the financial hardship of pediatric cancer.

CCI LIVE Q&A- SESSION 3 : SURVIVORSHIP 24-10-2021

9:45 AM - 10:15 AM

O0223 / #580 | Meaning in Childhood Cancer Survival: Latin American Young Adults Thriving with Support or Adrift Without it

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Background and Aims: Childhood cancer survivors transition to adult healthcare, and sometimes receive support for wellbeing. We explored treatment engagement in Peru, El Salvador and Mexico in a SIOP PODC Task Force study across 10 resource-limited countries. Survivors shared positive and negative experiences during and following treatment.

Methods: An ethnographic investigation of recent survivors ($n=30$) was facilitated by local Childhood Cancer International (CCI) organizations. A pediatric oncopsychologist and medical anthropologist (native

Latin American [LATAM]) conducted focus group discussions and in-depth interviews, recorded and transcribed data and analyzed content. **Results:** Survivors mentioned local CCI as main facilitators of social and therapeutic opportunities. A small group of El Salvadorean survivors meet regularly to share feelings and plan activities. In south-west Mexico, a nascent survivor group wrote a play about their cancer experience. In both countries, survivors visit hospitalized children to bring hope and relief. They also consider being with peer survivors important. Peruvian survivors are not yet supported and mentioned poor emotional and friendship bonds with peers, fellow survivors, and others. They described continued interrupted schooling (for family and economic reasons). Transition to adult healthcare made them feel alone/abandoned without a trusted pediatric oncologist to share fears/frustrations. Peruvian survivors wonder about their battle to survive if life does not resemble their wishes or ideals.

Conclusions: Although small and nascent positive support in Mexico and El Salvador emphasizes the role of CCI members in survivorship, in Peru, consequences of missing support was highlighted by life disappointments. These survivors are in an impossible situation of self-managing emotions and health while failing to fulfill their imagined life after cancer. Survivors without support beyond Peru and throughout LATAM may also be at risk. Childhood cancer survivors deserve the option of formal support, including psychological counseling, to share experiences while giving meaning to life after cancer and realizing their imagined healthy adult lives.

CCI LIVE Q&A- SESSION 3 : SURVIVORSHIP 24-10-2021 9:45 AM - 10:15 AM

O0224 / #1116 | How Survivors Community Affects Identity Formation of Childhood Cancer Survivors

S. Zakaria

YOAI (Indonesian Childhood Cancer Foundation), Cancer Buster Community (survivors Community), West Jakarta, Indonesia

Background and Aims: One of the essential parts of every individual to become mature is by developing a sense of self or identity. There are many factors in life that can hinder the development of identity, especially when someone has had a chronic disease at a young age such as childhood cancer. Having good social support is very helpful for the survivors of children with cancer to overcome the difficult situations. After finishing the treatment, identity issues have become one of the problems that occur in childhood cancer survivors. Due to different personalities, environments, and family backgrounds, some childhood cancer survivors struggle with the issues. Such as low self-esteem, intense emotion of sadness, and low resilience which leads to anxiety and insecurity that are sometimes experienced by survivors. To help them get through the situation, the presence of the childhood cancer survivors' community could help the survivors to develop an identity and shape their personalities so they can overcome the problems.

Methods: Cancer Buster Community's activities and programs (such as cancer survivors camp, roadshow, hospital visit and many more) could help the survivors to be more exist and active in society, by interacting with other survivors and engaged in positive activities.

Results: By joining the community and getting involved in the programs, survivors were able to interact with various sub-identities of an individual that could help to shape the learning, thinking, perceptual environment, and knowledge. It can also boost their positive self-esteem which will allow the survivors to shape a positive identity.

Conclusions: Social support like community could be one of the sources to help survivors of children with cancer finding and shaping positive identity. It also enables them to realize how important and inspiring they are to other people.

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS 24-10-2021 10:30 AM - 12:00 PM

O0225 / #561 | Clinical and Molecular Analysis of Smoothened Inhibitors in SHH Medulloblastoma

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Background and Aims: Background. Smoothened inhibitors (SMOi) have shown activity in Sonic Hedgehog (SHH) medulloblastoma, however this therapeutic class was not developed in children due to severe effects reported on growth. We hereby report long-term follow up of young patients treated with SMOi for recurrent medulloblastoma.

Methods: Clinical data on response and toxicity from patients treated with vismodegib or sonidegib from 2011 to 2019 for a SHH medulloblastoma were retrospectively reviewed. Methylation analysis and whole exome sequencing were performed whenever possible.

Results: All patients with a somatic PTCH1 mutation responded to SMOi (6/8), including two prolonged complete responses. One patient was free of disease 8.2 years after treatment. SMOi was challenged again for three patients. Two of them had a response, one with SMOi

alone, the other one in combination with temozolomide despite previous progression under monotherapy. SMO resistance mutations were found in patients from biopsy at relapse. Combination with temozolomide or surgery plus radiotherapy was associated with very long disease control in two patients. The most severe adverse events were myalgia and growth plate fusion with metaphyseal sclerosis. Normal growth velocity was recovered for one patient although her final height was below estimated target height.

Conclusions: Targeting SMO in mutated PTCH1 is an interesting strategy for long-term responses. Combination of SMOi with chemotherapy or surgery and local radiotherapy is an appealing strategy to prevent early resistance and diminish SMOi exposure, especially in young patients. Inhibition of SHH pathway causes growth and development impairment but partial recovery of the growth velocity is possible.

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS
24-10-2021 10:30 AM - 12:00 PM

O0226 / #607 | What Does A Non-Response to Induction Chemotherapy Imply in High-Risk Medulloblastomas?

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Background and Aims: High-risk medulloblastomas (HR-MB) may not respond to induction chemotherapy, with either post-induction stable (SD) or progressive disease (PD). There is no consensus regarding their optimal management.

Methods: A retrospective, multicentre study investigated non-responder HR-MB patients treated according to the PNET HR+5 protocol (NCT00936156) between 01/01/2009 and 31/12/2018. After two courses of etoposide and carboplatin induction chemotherapy, patients with SD or PD were analyzed. Upon clinician's decision, the protocol was either pursued with tandem high-dose chemotherapy (HDCT) and craniospinal irradiation (CSI) (continuation group) or it was modified (switched group).

Results: Forty-nine patients were identified. After induction, 37 patients had SD and 12 had PD. The outcomes were significantly better for the SD group: the 5-y PFS and OS were 52% (95% confidence interval [95% CI] 35-67) and 70% (95% CI 51-83), respectively in the SD group, while the 2-y PFS and OS were 17% (95% CI 3-41) and 25% (95% CI 6-50), respectively, in the PD group ($p < 0.0001$). The PNET HR+5 strategy was pursued for 3 patients in the PD group, of whom only one survived. In the SD group, it was pursued for 24/37 patients whereas 13 patients received miscellaneous treatments including a 36 Gy CSI with a 18 Gy boost on tumor bed in 12 cases. The 5-y PFS were 78% (95% CI 54-90) versus 0% and the 5-y OS were 78% (95% CI 54-90) versus 56% (95% CI 23-79) in the continuation group and in the switched group, respectively. In the SD group, multivariate analysis revealed that MYC amplification, molecular group 3, and a switched strategy were independent prognostic factors for progression.

Conclusions: Patients with post-induction SD may benefit from HDCT and CSI, whereas improvement of the way patients with early PD are treated will require new therapeutic approaches.

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS
24-10-2021 10:30 AM - 12:00 PM

O0227 / #1664 | Optical Genome Mapping Reveals Novel Structural Variants in Pediatric Brain Tumors

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Background and Aims: Brain tumors are a leading cause of cancer-related mortality in children. Research shows that structural variants (SVs) play an important role in tumorigenesis and growth. However, most genetic studies have relied on short-read based exome/genome sequencing (SRS) that have limited ability to detect SVs. Optical genome mapping (OGM) utilizes ultra-long DNA molecules to construct and evaluate much longer genomic fragments, making it effective in identifying large SVs. In this study, we utilized OGM for the detection of novel clinically relevant SVs in pediatric brain tumors.

Methods: Ultra high molecular weight DNA was extracted from a cohort of pediatric brain tumors and matched normal samples ($n=50$). OGM was performed using nanochannel chip arrays on a Saphyr

instrument and SVs (large insertions/deletions, inversions, translocations) were identified. We then used nanotatoR to subclassify SVs into functional pathways and determined which genes impact specific cell developmental functions.

Results: Tumor-associated, low population frequency SVs were identified in all samples. The most common SV events were deletions and translocations. Nearly all samples had a SV that overlapped with a gene known to have clinical significance. CDKN2A/B deletions were most common in high grade tumors. OGM also detected clinically relevant, previously unknown fusions (i.e. NTRK2-KANK1, C11orf95-NCOA1 fusion) and deletions (NF1, ATRX) that may have affected therapeutic management. We also discovered novel SVs overlapping genes and miRNAs that could impact tumorigenesis.

Conclusions: OGM effectively identifies novel clinically relevant SVs in pediatric brain tumor samples, including SVs that were not discovered other methods of testing. This is a promising new method to map the full SV spectrum of pediatric brain tumors. Future studies will expand testing to a larger cohort of samples to better understand the full spectrum of SVs and begin clinical analysis to determine the effect of SVs on response to treatment/prognosis.

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS
24-10-2021 10:30 AM - 12:00 PM

O0228 / #639 | Targeting H3K27 Mutant Diffuse Midline Glioma Via Glucosylceramide Synthase Inhibitors

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Background and Aims: H3K27M mutant diffuse midline glioma (H3K27M mut) is an aggressive brain tumor affecting primarily children. Radiotherapy is the standard of care but provides even in combination with various systemic antineoplastic agents unsatisfactory treatment results with a median overall survival of around 12 months. Glycosphingolipids (GSL) are amphipathic lipids particularly abundant in the brain where their amount and expression patterns change drastically during the embryonic to postnatal stages. The biosynthesis of GSL begins with the formation of glucosylceramide from ceramide. This step is catalyzed by the glucosylceramide synthase (UGCG), which can be inhibited by the small molecular compounds eliglustat and miglustat. Here we hypothesized that GSLs are deregulated in H3K27M mut and represent a new therapeutic target.

Methods: The expression of GSLs and ceramide was assessed by mass spectrometry in an in vitro model of H3K27M mut, before and after

treatment with eliglustat. The effect of eliglustat alone or in combination with ionizing radiation was analyzed in proliferation assays.

Results: The GSL profile indicates a shift toward GSL typical of proliferative embryonic cells. Treatment with eliglustat resulted in a decrease in GSLs and increase in ceramide. Eliglustat inhibited the proliferation of H3K27M mut primary cells and increased the efficacy of radiation in an additive manner.

Conclusions: The shift in GSL composition suggests that H3K27M mut derives from proliferative precursor cells. Modulation of GSLs via UGCG inhibitors sensitizes the cells to radiation. This is potentially due to the increase in ceramide concentration, which is considered a mediator of apoptosis. Because UGCG inhibitors are already used for treating children with Gaucher's disease, a rapid access of patients to new innovative therapies is possible.

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS
24-10-2021 10:30 AM - 12:00 PM

O0229 / #930 | Zfta Fusion-Positive Ependymomas Show a Dismal Outcome at Relapse

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Background and Aims: ZFTA fusion-positive ependymomas (ST-EP-ZFTA) are the most common subtype of supratentorial ependymoma in children, featuring a 75% 5-year-OS at primary diagnosis. Data on survival and treatment of these patients at recurrence is scarce.

Methods: In this analysis of survival and treatment of 19 patients with recurrent ST-EP-ZFTA from the German multi-centre HIT-REZ-studies. Survival was estimated by Kaplan-Meier-method, using time from 1st recurrence to death (OS) or next recurrence/progression or death (PFS), 95% confidence-intervals (95%-CI) are given.

Results: Median time to first recurrence was 19.0 months (range: 2.6 – 57.0). Median PFS and OS were 10.3 months (95%-CI: 4.4 – 17.2) and 33.4 months (95%-CI: 18.9 – 50.1), respectively. 3-year-OS was 47.6%, while 5-year-OS was 0%. Metastases at first recurrence were present in 57.9%, of which 72.7% were located intracranially.

Gross-total/near-total resections (GTR/NTR) were achieved in 73.7%. Interestingly, no differences in median PFS and OS were found between patients with GTR/NTR (PFS: 7.3 months (95%-CI: 3.5 – 14.2), OS: 33.4 months (95%-CI: 17.2 – 50.1)) and patients with subtotal / no resection (PFS: 20.5 months (95%-CI: 2.5 – NA), OS: 57.4 months (95%-CI: 9.7 – NA)). Median PFS was improved if radiotherapy was applied (17.2 months (95%-CI: 7.3 – 23.0)) compared to when it was not used (4.4 months (95%-CI: 2 – 13.2)). However, no difference in median OS was found between patients who received radiotherapy vs. those who did not (33.4 (95%-CI: 9.7 – NA) vs. 29.6 months (95%-CI: 16.6 – NA)). Chemotherapy slightly improved PFS (12 (95%-CI: 2.5 – 17.2) vs. 5.9 (95%-CI: 2 – 20.5) months)

Conclusions: Our cohort of patients with recurrent ST-EP-ZFTA shows worse outcome, without survivors five years after recurrence. Surprisingly, we found no survival advantage for complete resection. However, case-numbers are still small and larger series are needed to verify results. Supported by German Children's Cancer Foundation

Free Paper Session (FPS) FPS 21 -LIVE Q&A- BRAIN TUMOR BIOLOGY AND POOR PROGNOSIS/RECURRENT TUMORS
24-10-2021 10:30 AM - 12:00 PM

O0230 / #1782 | Long-Term Outcomes for Children with Atrt Treated with Intensive Multimodal Therapy: An Update on DFCI #02-294

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Background and Aims: Atypical teratoid rhabdoid tumor (ATRT) is a highly malignant neoplasm primarily affecting young children. Understanding the cancer genomics of this disease has not yet been incorporated into standard treatment approaches and outcomes remain poor. We now provide an update on the long-term survival outcomes of a multi-institutional phase II clinical trial for children with newly diagnosed ATRT.

Methods: Between 2002-2006, 20 eligible patients were treated on this prospective trial, DFCI #02-294 (NCT:00084838). Patients underwent surgery, induction, chemoradiotherapy and continuation chemotherapy, according to a modified IRS-III-based regimen, includ-

ing intrathecal chemotherapy. Survival outcome data was collected for all previously treated patients.

Results: Twenty eligible patients were enrolled on the initial study, (median follow-up, 5.2 years; range:0.14-15.0). Median age at diagnosis was 2.2 years (range:0.2-19.5). Twelve patients completed the intended treatment course: eight discontinued study early due to disease progression (N=4), toxic death (N=2), treatment toxicity (N=1) and non-compliance (N=1). Five- and 10-year overall survival were 49.6% (95%CI: 26.6-68.8) and 44.7% (95%CI: 22.7-64.4), respectively (median survival, 5.2 years). Eight patients are long-term survivors, with median follow-up, 14.7 years (range:10.2-15.0). Median age of the survivor group was 1.9 years at diagnosis (range:0.2-19.5). All survivors had localized disease at presentation and received radiation therapy as part of their treatment regimen. Of these, six had complete/near-complete resection upfront. Of the 10 patients who died of disease, 5 had metastatic disease & 7 underwent biopsy/partial resection only. Long-term neuro-psychological outcomes for a subset of this cohort will be discussed.

Conclusions: These long-term survival outcomes remain the highest reported for this malignant cohort despite their young age. A subset of patients with ATRT can be cured with the use of multimodal therapy but for many, despite aggressive standard regimens currently employed, novel strategies based on recent genetic and epigenetic understanding of this disease are needed.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0231 / #693 | Developing a Preliminary Core Set of Nursing-Sensitive Indicators for International Pediatric Oncology Nursing Practice: A Mixed Methods Delphi Study

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Background and Aims: Nursing sensitive indicators (NSIs) measure factors influencing nursing care quality and patient outcomes. However, a core set of NSIs for international pediatric oncology nursing practice does not currently exist. The purpose of this research was to develop a preliminary core set of NSIs for international pediatric oncology nursing that would be important, actionable, and feasible to measure across varied resource settings and countries.

Methods: A multiphase mixed methods research design intersected with a classical Delphi method was utilized. Through purposive snowball sampling, 122 expert pediatric oncology nurses from 43 countries participated. Round One: Panelists identified 5 potential NSIs/Constructs. Open-ended responses were coded and categorized through descriptive content analysis and integrated into Round

Two. Round Two: Panelists selected their top 10 NSIs/Constructs and ranked them by importance. Mean-importance scores were calculated through reverse-scoring; the top 10 NSIs/Constructs were integrated into Round Three. Round Three: Panelists ranked the top 10 NSIs/Constructs by order of importance, then rated each NSI/Construct for relevance and feasibility of measurement by Likert-scale. Rounds Two and Three were analyzed using descriptive statistics.

Results: Eighty-five (70%) panelists from 38 countries completed all Delphi survey rounds. The preliminary core set of NSIs/Constructs identified by the expert panel, and ranked in order of importance, were: safe chemotherapy administration and handling, infection prevention/control, pediatric oncology nursing orientation program, early warning score system/recognition of patient deterioration, chemotherapy/biotherapy education/course, pain assessment/management, symptom assessment/management, patient and family education, palliative/end of life care, and continuing nursing education/competency. All NSIs/Constructs were rated actionable; all but palliative/end of life care were rated as feasible to measure.

Conclusions: Findings provide insight into common attributes of international pediatric oncology nursing practice that are important, actionable, and feasible for quality measurement and benchmarking. These findings will inform future studies to finalize a core set of NSIs for pediatric oncology nursing internationally.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0232 / #919 | Global National Cancer Control Plans: Nurse Participation and Child and Nursing Content

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Background and Aims: The 2020 WHO and International Council of Nursing Report called for increased nursing leadership in health policy development and decision-making to ultimately influence effective health systems and social care infrastructure. National Cancer Control Plans (NCCP) set a country's course over an extended period. Nurse participation in NCCP development helps consider factors beyond general public health or biomedical treatment that may negatively affect cancer control. Aim: Locate NCCP across WHO regions to determine nurse participation, whether nursing was addressed, and/or whether childhood cancer action/outcomes were included.

Methods: A search was conducted on the International Cancer Control Partnership (ICCP) portal for NCCP. Internet searches revealed no additional NCCP.

Results: A total of 112 NCCP were located for 194 countries (57%). While 50% were expired, many other countries' NCCP had no end

date or only had NCD plans. The ratio of countries per region with NCCP ranged from 74% Europe to 36% Southeast Asia. Nurses were identified in NCCP development committees (or authors knew contributors as nurses) in 12% of NCCP. AFRO and European Regions each had 5 (24% and 13% respectively) and Americas had 4 (27%) plans with nurse contributors. Nursing was mentioned in 60% of plans and actions/outcomes for childhood cancer were included in 71%. Nursing was addressed in the domains of education/training and workforce/staffing while childhood cancer topics included prevention, awareness/detection, guidelines/protocols, and survivorship.

Conclusions: There is a paucity of nurses identified in NCCP development worldwide. This diminishes nurse leaders' professional involvement at decision-making levels. The limited NCCP with nursing action/outcomes demonstrates insufficient recognition of nursing's vital role in cancer control. There is a lack of prioritization of childhood cancers. Health authorities would be well advised to create/maintain free-standing NCCP, place on the ICCP portal for accessibility, and include nursing and childhood cancer action/outcomes to improve cancer control.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0233 / #229 | The Ghanaian Pediatric Oncology Nursing Program Promotes Nursing Clinical Expertise and Leadership

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Background and Aims: Children with cancer in low- and middle-income countries are four times more likely to die of the disease than those in high income countries. Expert nurses are critical to a successful pediatric oncology program, yet few opportunities exist for nurses across Ghana to receive specialized nursing education. Aligned with priorities of the 2019 WHO Global Initiative for Childhood Cancer (GICC) workshop in Ghana, a collaborative was formed to build an accredited Pediatric Oncology Nursing Program. The COVID19 pandemic hit during planning, forcing a rapid transition to a blended curriculum of remote and in-person teaching.

Methods: The one-year program is a combination of didactic theory and clinical practicums, modeled on a curriculum previously created for the Ghana College of Nurses and Midwives by the Hospital for SickKids in Toronto. The program kicked off with a virtual three-day preceptor

workshop, followed by theory blocks delivered by international faculty which incorporated breakout rooms, interactive lectures, and group work. On-site teaching and clinical practicums are delivered by local clinical experts. Learning is assessed through written assignments, presentations, examinations, and competence assessment.

Results: Of eighteen enrolled students, sixteen completed on-line evaluation forms, and reported satisfaction with content, delivery and assignments for Block One modules. Student ratings of each module ranged between an Overall Mean Score of 4.21 to 4.38 (of 5). All students scored competent or above for Competency Skills Assessment during the first clinical practicum. All enrolled students successfully passed the end of module examinations.

Conclusions: An accredited training program prepares students to lead nursing practice, education and training within their own institutions, and to serve as leaders for future cohorts of the program, aligning with the WHO GICC goal to expand capacity to deliver best practice in childhood cancer care.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0234 / #596 | Building Pediatric Oncology Nursing Leadership in Sub-Saharan Africa

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Background and Aims: Pediatric oncology nursing training is a challenge to effective clinical oncology management in sub-Saharan Africa due to lack of expertise in the field. Global Hematology Oncology Pediatric Excellence (HOPE) set out to provide nurse leaders with training opportunities in leadership and management and to promote individual professional development by building and enhancing core competencies and strategic workplace management skills.

Methods: Global HOPE nurse leaders from Botswana, Malawi and Uganda participated in a year-long program focused on developing leadership skills. Participants completed five modules: essential nursing leadership skills, high-performance teams, diplomatic communication, strategic management and practice models, and nursing quality care. Each module consisted of lectures, discussion, readings, group activities conducted at their practice site, and forum presentations. Participants met monthly via ZOOM during the program.

Results: All nurse leaders completed the program requirements. Program results are presented by discussing completed activities designed to promote leadership skills. Learning activities included personal assessments of individual strengths and weaknesses, and interviews with nurse leaders at their institutions followed by oral presentations on key elements of leadership. Participants learned about high perfor-

mance teams and interviewed their own site teams to explore barriers to team performance; recommendations for change to improve team performance were developed and shared with their teams. Effective communication strategies were explored using several change and practice management models to gain knowledge of strategic management. Participants chose a practice model and presented the framework during group discussion. Quality improvement processes were learned, and all nurse leaders completed the open course on Quality Improvement that is part of the Institute for Healthcare Improvement.

Conclusions: Global HOPE is making strides to enhance vital leadership skills among oncology nurse leaders. This translates to better patient care while shaping future approaches to oncology nursing in sub-Saharan Africa.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0235 / #417 | Tuck-In Rounds: A Standardized Evening Rounding Process to Improve Communication and Care Coordination

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Background and Aims: The goal for overnight patient care is different than during the day and involves timely recognition and interventions for patients with acute deterioration. To achieve these goals, clear communication between the medical team and nurses is crucial. At our institution, the cross covering medical teams and nurses routinely participate in night rounds. Prior to this effort, the content and structure of these rounds were not standardized and has led to lack of awareness of contingency plans, multiple follow-up conversations, and preventable errors. Our team sought to improve the quality of overnight patient care by targeting nurse to provider communication processes. The specific aim was to increase contingency plan discussion during night rounds from 20% to 80% in 1 year without increasing average time spent.

Methods: The multi-disciplinary quality improvement (QI) team used QI approaches per the Institute for Healthcare Model for Improvement. The team identified key-drivers necessary to increase contingency plan discussion. Interventions included implementing a rounding checklist in several phases and enforcing compliance through observation. Average frequency of contingency plan discussion and time spent per patient were plotted on control charts. Centerlines were shifted based on pre-established criteria for special cause detection in the context of interventions.

Results: From August 2019 to June 2020, average contingency plan discussion increased from 20% to 97%, without increasing time spent per patient. In addition, both nurses and providers reported increased perceptions in the quality of night shift care plan development.

Conclusions: Using QI methodologies, we successfully increased contingency plan discussion during night rounds without increasing time spent per patient. The revised process sets an expectation for the night shift: developing a clear plan for detecting and responding to patient deterioration. Further efforts are on-going to incorporate this standardized rounding process to new employee orientation.

Nursing FREE PAPER SESSION 5-NURSING RESEARCH AND LEADERSHIP 24-10-2021 10:30 AM - 12:00 PM

O0236 / #520 | What Matters to Children with Cancer a Systematic Review of Qualitative Studies and Patient-Reported Outcome Measures (PROMS)

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Background and Aims: Several patient-reported outcome measures (PROMs) for health-related quality of life (HRQOL) in children with cancer exist. However, these have been criticized for not accurately reflecting issues that matter to children and for seldom involving children in PROM development. We aimed to collect and compare the items assessed by existing PROMs and issues mentioned by children or parents in qualitative studies.

Methods: A systematic literature search on PubMed, PsycInfo, COS-MIN, and Cochrane was conducted. Studies were included if (a) children (8-14 years) with cancer undergoing treatment completed PROMs or (b) children or their parents provided qualitative information on HRQOL issues of the child. Issues identified in qualitative studies and PROM items were mapped onto the conceptual framework of HRQOL in children with cancer proposed by Anthony et al. (2014).

Results: A total of 2551 records were identified of which 316 papers were included. Among the included studies, 78 collected qualitative data and 277 used PROMs (39 combined both methods). 142 different PROMs and more than 4000 individual items were identified. The major domains of physical, psychological, social, and general health proposed in the conceptual framework by Anthony et al. (2014) were confirmed and the majority of items mapped well onto the model. Most identified issues assessed physical function or symptoms while

aspects of psychosocial functioning were often under-assessed in existing PROMs even though these emerged as key constructs in the qualitative studies.

Conclusions: This study offers a comprehensive overview of HRQOL issues for children with cancer. While physical aspects of the disease are covered, most PROMs lack a comprehensive assessment of key psychosocial issues and thus overlook fundamental features of the cancer experience of children. This calls for the development of a more comprehensive PROM for children with cancer including items which children themselves have identified as relevant and important to them.

Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP GENERAL 24-10-2021 10:30 AM - 12:00 PM

O0237 / #1709 | Pain and Frailty in Childhood Cancer Survivors: A Report from The St. Jude Lifetime Cohort Study

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Background and Aims: Childhood cancer survivors experience reduced physiologic reserve, known as frailty, earlier and at higher rates than their peers. Survivors also are at-risk for pain that may impact daily activities and precipitate or exacerbate frail states. Associations between pain and the development or acceleration of frailty in survivors has not been investigated.

Methods: 1,299 survivors (mean[SD] age 30.7[7.0] years; time since diagnosis 22.7[7.4] years) recruited from the St. Jude Lifetime Cohort completed clinical assessments at baseline and a median follow-up of 5.3 years. Frailty states were defined as having 2 (pre-frail) or ≥ 3 (frail) of the following: sarcopenia, muscle weakness, poor endurance, slow walking speed, and exhaustion. Survivors completed self-report measures of pain and pain interference. Multinomial logistic regression, using odds ratios (OR) and 95% confidence intervals (95% CI), examined associations between pain and frailty with adjustment for age, sex, race/ethnicity, and chronic health conditions.

Results: 5.5% of survivors were frail at baseline and 5.9% progressed from not frail/pre-frail to frail at follow-up; 15.5% reported pain with interference at baseline and 16.9% reported increased pain at follow-up. Compared to survivors without pain, survivors with pain and daily interference were more likely to be pre-frail (OR: 2.55; 95% CI, 1.72-3.78) or frail (OR: 4.58; 95% CI, 2.45-8.55) at baseline. Pain at baseline was not significantly associated with frailty progression; however, survivors who had persistent (OR: 2.23; 95% CI, 1.20-4.13) or increased pain (OR: 3.59; 95% CI, 1.94-6.63) from baseline to follow-up were more likely to progress from not frail/pre-frail to frail at follow-up.

Conclusions: Pain that interferes with daily functioning is associated with pre-frail and frail states in survivors. These results may inform interventions that target pain and functional daily activities to mitigate the risk of frailty and improve health outcomes in survivors.

**Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP
GENERAL 24-10-2021 10:30 AM - 12:00 PM**

O0238 / #1712 | CEP72 Genotype is Associated with Prevalent Motor Neuropathy in Survivors of Childhood Acute Lymphoblastic Leukemia

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Background and Aims: Acute peripheral neuropathy affects 20-60% of children during treatment for acute lymphoblastic leukemia (ALL) with symptoms persisting in up to 40% of survivors, impacting mobility and quality of life. CEP72 genotype (rs924607) is associated with acute vincristine-induced neuropathy. We aimed to evaluate the association between CEP72 genotype and peripheral neuropathy in childhood ALL survivors.

Methods: Participants included 990 (5+ years) childhood ALL survivors (mean age 27.0±8.6 years) treated at St. Jude Children's Research Hospital (mean time from diagnosis 21.1±8.4 years). Peripheral neuropathy was assessed using the modified Total Neuropathy Score (mTNS) and functional tasks. Sensory or motor neuropathy was classified present if signs and symptoms from the mTNS and functional tasks met National Cancer Institute Common Terminology for Adverse Events (v5) criteria (Grade≥2). CEP72 genotype was determined from DNA in peripheral blood by whole genome sequencing or imputation from the Affymetrix SNP array 6.0.

Results: Neuropathy was present in 16.0% (motor or sensory), 12.5% (sensory) and 7.2% (motor) of survivors. The prevalence of motor neuropathy was higher in individuals with rs924607-T allele (TT genotype, 10.9%, p=0.01; CT genotype, 8.46%, p=0.02) than in those without (CC genotype, 4.5%). In models adjusted for clinical predictors including cumulative vincristine dose (median 54.2 mg/m², IQR 11.7-61.6), survivors with TT (OR 2.47; 95% CI 1.15-5.31) and CT (OR 1.93; 95% CI 1.06-3.51) genotypes had significantly increased motor neuropathy risk compared to those with CC genotype. The CT genotype was associated with higher total (β =0.48; 95% CI 0.15--0.82) and motor (β =0.33; 95% CI 0.10-0.57) scores on the mTNS than the CC genotype.

Conclusions: Screening for rs924607-T allele may identify children treated for ALL at greatest risk for experiencing motor neuropathy decades after treatment. Work supported by the National Institutes of Health (R01CA036401 (W.E.E.), U01CA195547 (M.M.H, L.L.R.)), Cancer Center Support Grant (CA 21765 (C.W.M.R.)), and the American Lebanese-Syrian Associated Charities.

**Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP
GENERAL 24-10-2021 10:30 AM - 12:00 PM**

O0239 / #311 | Bone Mineral Density Surveillance for Childhood, Adolescent, and Young Adult Cancer Survivors: Recommendations from The International Late Effects of Childhood Cancer Guideline Harmonization Group (IGHG)

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Background and Aims: Childhood, adolescent and young adult (CAYA) cancer survivors are at increased risk of (very) low bone mineral density (BMD). Clinical practice surveillance guidelines are important for timely diagnosis and treatment of these survivors, which could improve BMD parameters and prevent fragility fractures. Discordances across current late-effects guidelines necessitated international harmonization of BMD surveillance recommendations.

Methods: The IGHG established a panel of 36 experts from 10 countries, representing a range of relevant medical specialties. The evidence of risk factors for low (Z-score ≤ -1) and very low (Z-score ≤ -2) BMD and fractures, surveillance modality, timing of BMD surveillance, and treatment of (very) low BMD were evaluated and critically appraised, and harmonized recommendations for CAYA cancer survivors were formulated. We graded the recommendations based on the quality of underlying evidence and the balance between potential benefits and harms.

Results: BMD surveillance is recommended for survivors treated with cranial or craniospinal radiotherapy, and is reasonable for survivors treated with total body irradiation. Due to insufficient evidence, no recommendation can be formulated for or against BMD surveillance for survivors treated with corticosteroids. This surveillance decision should be made by the survivor and healthcare provider together, after careful consideration of the potential harms and benefits and additional risk factors. We recommend to perform BMD surveillance using dual-energy X-ray absorptiometry at entry into long-term follow-up (between 2-5 years following treatment cessation), and if normal (Z-score > -1), repeat at 25 years of age when peak bone mass should

be achieved. Between these two measurements and thereafter, BMD surveillance should be performed as clinically indicated based on BMD and ongoing risk assessment.

Conclusions: This guideline provides BMD surveillance recommendations that may improve health outcomes by facilitating more consistent care for CAYA cancer survivors internationally. In addition, it promotes strategically planned ongoing research that will inform future guideline updates.

**Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP
GENERAL 24-10-2021 10:30 AM - 12:00 PM**

O0240 / #1206 | Effect of Genetic Variation in CYP450 on Gonadal Impairment in European Female Childhood Cancer Survivors, a Candidate Gene Approach: Results from the Pancarelife Study

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Background and Aims: Female childhood cancer survivors (CCSs) carry a risk of therapy-related gonadal dysfunction. Alkylating agents (AA) are well-established risk factors, yet inter-individual variability in ovarian function is observed. Polymorphisms in CYP450 enzymes may explain this variability in AA-induced ovarian damage. We aimed to evaluate associations between previously identified genetic polymorphisms in CYP450 enzymes and AA-related ovarian function among adult CCSs.

Methods: Anti-Müllerian hormone (AMH) levels served as a proxy for ovarian function in a discovery cohort of adult female CCS, from the pan-European PanCareLIFE cohort (N=743) (age (years): median 25.8, IQR 22.1-30.6). Using two additive genetic models in linear and logistic regression, nine genetic variants in three CYP450 enzymes were analysed in relation to cyclophosphamide equivalent dose (CED) score and their impact on AMH levels. The main model evaluated the effect of the variant on AMH and the interaction model evaluated the modifying effect of the variant on the impact of CED score on AMH levels. Results were validated, and meta-analysis performed, using the USA-based St. Jude Lifetime Cohort (N=391) (age (years): median 31.3, IQR 26.6 - 37.4).

Results: CYP3A4*3 was significantly associated with AMH levels in the discovery and replication cohort. Meta-analysis revealed a significant main deleterious effect (Beta(95% CI): -0.706 (-1.11 - 0.298) p-value = 7×10^{-4}) of CYP3A4*3 (rs4986910) on AMH levels and the estimated relative AMH level is five-fold lower with CYP3A4*3 compared to someone with similar CED and no alternative alleles. CYP2B6*2 (rs8192709) showed a significant protective interaction effect (Beta(95% CI): 0.527 (0.126 - 0.928) p-value = 0.01) on AMH levels in CCSs receiving more than 8000mg/m² CED.

Conclusions: Female CCSs CYP3A4*3 carriers had five-fold lower AMH levels, and CYP2B6*2 has a protective effect on AMH levels. Identification of risk-contributing variants may improve individualised counselling regarding treatment-related risk of infertility and fertility preservation options.

**Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP
GENERAL 24-10-2021 10:30 AM - 12:00 PM**

O0241 / #556 | Genetic Susceptibility to Longitudinal Cognitive Decline Following Craniospinal Irradiation (CSI) in Pediatric Patients with Central Nervous System (CNS) Tumors

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Background and Aims: Survivors of pediatric CNS tumors exhibit considerable variability in cognitive performance following CSI, which may be partly explained by inherited genetic susceptibility. Goals of this study were to: 1) replicate candidate genetic variants previously associated with cognitive performance in both a) pediatric CNS tumor survivors and b) the general population, and 2) identify novel variants associated with cognitive decline using a genome-wide approach.

Methods: Intelligence (IQ), working memory (WMI), and processing speed (PSI) were longitudinally collected from patients treated with CSI (n=249) enrolled on St. Jude medulloblastoma clinical trials or an observational cohort at Texas Children's Hospital. Genotype-by-time interactions were evaluated using mixed-effects linear regression to identify common (>1%) variants associated with cognitive performance change. Novel variants associated with cognitive decline (p<5.0E-5) in individuals of European ancestry (n=167) were considered replicated if they demonstrated consistent genotype-by-time interactions (p<0.05) in individuals of other ancestries (n=82) and achieved genome-wide statistical significance (p<5.0E-8) in a meta-analysis across ancestry groups.

Results: Participants were primarily males (65%) diagnosed with medulloblastoma (98%) at a mean age of 8.9 years. Overall, 1,080 neurocognitive evaluations were collected (median = 4, range: 2-11 per participant). One of the five loci previously associated with cognitive outcomes in pediatric CNS tumors survivors demonstrated significant time-dependent IQ declines in our population (PPARA rs6008197, p=0.02). Two of 371 general population loci were associated with declines in IQ after Bonferroni correction (rs9348721 near BTN1A1,

$p=1.1E-5$; rs31771 near RPL7P20, $p=3.8E-5$). In genome-wide analyses, we identified and replicated novel loci associated with accelerated declines in IQ (rs116595313 near FHDC1, meta-analysis $p=3.8E-8$) and WMI (rs17774009 near BVES, meta-analysis $p=6.3E-11$).

Conclusions: We present evidence that inherited genetic variants involved in baseline intellectual functioning and novel susceptibility loci jointly influence the degree of treatment-associated cognitive decline in pediatric CNS tumors. Genetic variants may inform risk stratification and efforts to preserve cognitive function.

**Free Paper Session (FPS) FPS 19-LIVE Q&A-SURVIVORSHIP
GENERAL 24-10-2021 10:30 AM - 12:00 PM**

O0242 / #898 | Educational Achievements Among Survivors of Childhood Cancer— Findings from the Saliccs Research Program

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Background and Aims: Attaining upper secondary education is important for future career opportunities, and some survivors of childhood cancer may have difficulties achieving this. Survivors of CNS tumours seem to be at particular risk, but for survivors of other cancer types the evidence is less consistent. In a population- and register-based study including Denmark, Finland, and Sweden, we investigated educational achievements among survivors of childhood cancer and identified particular risk groups.

Methods: Within the SALiCCS (Socioeconomic Consequences in Adult Life after Childhood Cancer in Scandinavia) research program, we included all children, diagnosed with cancer at ages 0-14 during 1971-2005, alive by age 25 ($n=7,629$), their age-, sex- and country-matched population comparisons ($n=35,411$) and siblings ($n=6,114$). Annual individual information on education was retrieved from national registers. We assessed the relative risk (RR) of educational delay, and of failing to achieve upper secondary education by age 25, using log-binomial regression.

Results: Overall, 80% of survivors and 84% of population comparisons and siblings, had achieved upper secondary education by age 25. The RR of failing to achieve upper secondary education was 1.24 (95% CI 1.18-1.31) and differed by cancer type (e.g., survivors of lymphoid leukaemia 1.11 (95% CI 1.00-1.25); survivors of CNS tumours 1.73

(95% CI 1.60-1.87)). Children of parents with high educational level were more likely to achieve a higher education themselves, however, in this group the risk difference of failing (comparing survivors and comparisons) was somewhat larger. A lower proportion of survivors than comparisons achieved upper secondary education without delay.

Conclusions: In this study we demonstrate that some survivors of childhood cancer experience difficulties in educational achievements. This differed by cancer type; while only small differences in achievement of upper secondary education by age 25 were observed for survivors of some cancer types, differences were pronounced among survivors of CNS tumours.

**Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM**

O0243 / #409 | Retrospective Analysis of Outcomes for Pediatric Acute Lymphoblastic Leukemia in South American Centers

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Background and Aims: Acute lymphoblastic Leukemia (ALL) is the most common pediatric malignancy worldwide. While the overall survival for ALL in high-income countries is over 90%, the estimated survival in low-and middle-income countries ranges from 22-79%. This study retrospectively analyzed the outcomes of children with ALL in South America, at 5 pediatric centers in Bolivia, Ecuador, Peru, and Paraguay.

Methods: A retrospective review examined data for children less than 18 years old diagnosed and treated for ALL these centers between

2013-2017. Descriptive indices of demographic, clinical, and biological parameters were analyzed, as well as reasons for treatment failure.

Results: Interim analysis of 595 patients included patients with B-cell (92.1%) and T-cell (7.4%) ALL, with a mean age of 7 years. At the time of diagnosis, 49.2% of patients were categorized as standard risk and 50.8% high risk. Most patients (58.2%) were CNS1 status, however CNS status was not evaluated in 38.7% of patients and not available in 1%. MRD evaluation was obtained in 66.7% of total patients. Of 588 patients, 485 experienced chemotherapy delays with a median of 17 days (1-100). Delays were attributed to infection (63.8%, 204/320), multiple drug related toxicities (43.8%, 140/320), resource constraints including bed availability (21.9%, 70/320), and abandonment (5%, 16/320). The 5-year event-free and overall survival (EFS, OS) were 62.0±3.5% and 74.9±3.2%, respectively. EFS events included: 4 induction failures, 41 induction deaths, 124 relapses (70 marrow, 32 CNS, 17 combined, 5 other), and 27 remission deaths.

Conclusions: EFS and OS estimated in this study support recently published estimates for the region. This work provides important insight into causes for treatment delay and treatment-related mortality such as infections and toxicities, which may be related to treatment intensity and contribute to mortality and relapse. The details of this analysis provide opportunities for interventions to further improve outcomes for ALL patients in this region.

Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM

O0244 / #873 | Primary Outcome of Pediatric B All Patients Treated on ICIcle Pilot Protocol : Retrospective Analysis from a Tertiary Cancer Center in India

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Background and Aims: Risk-stratified and response-adapted therapy is considered standard for Childhood Acute Lymphoblastic Leukemia(cALL). Response-assessment by Minimal Residual Disease(MRD) on Flowcytometry(FCM) has become fundamental to therapeutic decisions. We piloted ICIcle-2014, a multicenter-trial for cALL at our center for 5-years before commencing on-trial enrollment. Outcomes of B-ALL patients on this strategy were analyzed.

Methods: B-ALL patients <15-years age treated on ICIcle-pilot protocol from Feb-2013 – Feb-2018 were included. Risk-stratification was by NCI criteria to standard and intermediate risks. Children with high-risk cytogenetics, CNS involved and poor Day-8 prednisolone-response(PR) were further classified as high-risk. MRD by 10-color multiparametric-FCM at the end-of-induction(EoI) ≥0.01% were subsequently treated as high-risk. 3-year Event-free and Overall Sur-

vival(EFS & OS) were estimated by Kaplan-Meier method, and risk-factors by Odds-ratio using Jamovi (Version-1.6).

Results: There were 1406 patients with median follow-up 32-months(range:24-84), median age 5-years(range:1-15), M:F-2:1, median baseline total leucocyte-count(BTLC)-32000/μL, CNS-involved-57(4.2%), and Testis-involved-21(2%). Cytogenetics showed No translocations:1029(73.19%), ETV6-RUNX1:176(12.5%), TCF3-PBX1:114(8.1%), BCR-ABL:53(3.7%), MLLr:24(1.7%), HLF-TCF3:2(0.14%), iAMP21:2(0.14%) (6 patients(0.43%)-no records). Day-8PR was Good in 1124(85%) (79 not-evaluated:5-early deaths, 74-no records). On initial risk-stratification, Standard Risk(SR) were 431(30.76%), Intermediate Risk(IR):492(35.12%), and High Risk(HR):478(34.12%). Of these, 1322 patients evaluated for EoI-MRD (84 not evaluated:79 induction-deaths, 5-no records); 967(73%) were Negative, 355(27%)-Positive. Three-year EFS and OS 69.1±1.4% and 80.4±1.1% respectively. By final-risk, 3-year EFS for SR, IR and HR were 85.5±4.3%, 76.5±4.9%, 65.8±4.1%, and OS, 93.68±3.0%, 88.39±3.6%, 78.39±3.5% respectively. Univariate analyses showed significant hepato-splenomegaly, BTLC>33.4/μL, MLLr, Poor D-8PR, Initial and Final Risk-stratification, and EoI-MRD as significant, of which only BTLC>33.4/μL(p<0.001%), MLLr(p=0.004%)and EoI-MRD(p=0.001%) retained significance on multi-variate analyses.

Conclusions: A risk-stratified, response-adapted strategy for B-ALL was piloted successfully in a single-center, yielding improved outcomes and the risk/response parameters held true, with poor biology and EoI-MRD emerging as the most important risk-factors for outcome. Acknowledgements: ICIcle-2014 Study Group, National Cancer Grid, Indian Council of Medical Research

Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM

O0245 / #175 | Novel Insights into Pediatric Acute Lymphoblastic Leukemia Ophthalmic Relapses from a Nationwide Cohort Study

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Background and Aims: Ten to fifteen percent of children with acute lymphoblastic leukemia (ALL) relapse following treatment. Of these, only few display ophthalmic relapses, which owing to their scarcity are largely undocumented leaving clinicians with few diagnostic and therapeutic recommendations, despite serious functional sequelae.

Methods: We conducted a national French multicenter spanning over 20 years to collect all clinical, radiological, biological, and therapeutic data and outcome for children with ALL ophthalmic relapses.

Results: From 2000 to 2020, 31 ophthalmic relapses were diagnosed, with an estimated rate of ophthalmic relapse of 2.8%. Among them, 19 B-ALL and 12 T-ALL: 25 patients (81%) had concomitant involvement of Central Nervous System (CNS), 18 (58%) a combined bone marrow relapse, and only 1 had an isolated ophthalmic relapse. The most frequent symptoms were reduced visual acuity (n=22, 71%) and ocular pain (n=7, 23%). Ten (32.6%) had a bilateral ophthalmic involvement and 20 patients (65%) had papilledema. Combined medullary relapse was not a significant worse prognosis factor (OR 3.39 p:0.15) contrary to hypopyon (p:0.04). Total body irradiation (p = 0.001) was associated with an improvement in overall survival. More patients alive in complete remission (CR) underwent an allogenic hematopoietic stem cell transplantation (OR 0.13; p:0.01). All the patients who received chimeric antigen receptor T cells (n=4) relapsed afterwards. Ten children (31%) died, 9 from a refractory disease and 1 from toxic death. Five patients have recently relapsed. Among the 16 survivors in CR, 8 display severe ophthalmic sequelae including complete blindness with a median follow-up of 19.9 months.

Conclusions: Although rare, ophthalmic relapses have a significant impact on the functional prognosis of survivors. These observations confirm the important role of CNS prophylaxis in 1st line and indicate the need to define optimal management approaches for ophthalmic relapse, including cerebral irradiation. Their management must be multidisciplinary, with a fundamental role for ophthalmologists.

Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM

00246 / #351 | Chemotherapy-Induced Peripheral Neuropathy Persists Despite Reduction of Vincristine Frequency in Children with B-Acute Lymphoblastic Leukemia: A Report from Children's Oncology Group AALL0932

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Background and Aims: Children with B-acute lymphoblastic leukemia (B-ALL) are at risk for chemotherapy-induced peripheral neuropathy (CIPN) from vincristine. AALL0932 randomized reduction in vincristine/dexamethasone (every 4 week [VCR/DEX4] vs every 12 week [VCR/DEX12]) during maintenance in the average-risk subset of NCI standard-B-ALL (SR AR B-ALL) and did not find a difference in disease-free survival. We longitudinally measured CIPN, overall, and in VCR/DEX4 vs VCR/DEX12.

Methods: SR AR B-ALL patients from AALL0932 ≥3 years old, without prior neuromuscular disease, and with an English/Spanish speaking parent were evaluated at four time-points (end-consolidation [n=150], start of maintenance [n=108], month 18 of maintenance [n=102], 12 months post-therapy [n=73]. CIPN was assessed by motor (hand/ankle strength, ankle dorsiflexion/plantarflexion range of motion [ROM]), sensory (finger/toe vibration), and functional (dexterity; Purdue Pegboard, gait; 6 Minute Walk) impairment. Age/sex matched z-scores and proportion impaired (z-score < -1.3) were measured longitudinally and compared between groups.

Results: Among 150 participants (mean age 5.1 years [SD=1.7], 48.7% female) at end consolidation, there was frequent impairment in motor/sensory function (59.3% handgrip strength, 31.8% ankle strength, 32.5% plantarflexion ROM, 27.9% dorsiflexion ROM, 27.4% toe vibration, 13.8% finger vibration) and daily function (83.5% gait, 55.0% dexterity). By 12 months post-therapy, only handgrip strength (p<0.001) and gait (p=0.02) had improved; there was no improvement in sensation, ankle strength/range of motion, or dexterity (vs end-consolidation). In VCR/DEX4 (n=50) vs VCR/DEX12 (n=63) there was no difference in frequency of impairments at month 18 of maintenance or 12 months post-therapy except impaired ankle dorsiflexion ROM (VCR/DEX4 46.7% vs. VCR/DEX12 14.7%, p=0.007). Group comparisons did not change after adjusting for baseline functioning.

Conclusions: SR AR B-ALL patients frequently experience CIPN early in therapy that can persist at least 12 months post-therapy. Except dorsiflexion ROM, outcomes were similar in VCR/DEX4 vs VCR/DEX12. Children with SR AR B-ALL should be monitored for CIPN, even with reduced vincristine frequency.

Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM

O0247 / #1006 | Functional Brain Activity Associated with Working Memory in Long-Term Survivors of Pediatric Acute Lymphoblastic Leukemia: Sex-Based Differences

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Background and Aims: Background: Long-term survivors of pediatric acute lymphoblastic leukemia (ALL) are at increased risk for developing neurocognitive deficits and corresponding brain dysfunction. The aim of this study was to investigate functional brain activity associated with working memory in long-term pediatric ALL survivors, with a particular focus on sex-based differences.

Methods: Functional magnetic resonance imaging (fMRI) and neurocognitive testing were obtained in 123 survivors treated on the St. Jude Total 15 treatment protocol (46% male; median [min-max] age = 14.2 [8.3-26.5] years; time since diagnosis = 7.7 [5.1-12.5] years). fMRI was obtained on a 3T scanner during a n-back working memory task. Functional neuroimaging data were preprocessed (realigned, slice time corrected, normalized and smoothed) and analyzed using Statistical Parametric Mapping with contrasts developed for the 1-back, 2-back, and 2-back vs 1-back conditions, which reflect varying degrees of working memory load. Group level fMRI contrasts were stratified by sex and adjusted for age at evaluation and methotrexate dosage (i.e., methotrexate area-under-the-curve).

Results: Female survivors displayed less activation (i.e., reduced blood-oxygen-dependent-level signaling) in the right parietal operculum, supramarginal gyrus and inferior occipital gyrus, and reductions in the bilateral superior frontal medial segment and anterior cingulate gyrus during increased working memory task load (FWE corrected $p < 0.01$, adjusting for age and methotrexate dose) in comparison to males. Although there were no differences in mean performance accuracy by sex, female survivors displayed slower reaction times during the 2-back condition compared to males ($p < 0.05$). Male survivors demonstrated greater deficits in cognitive flexibility as measured using a standardized neurocognitive battery ($p < 0.05$).

Conclusions: These results suggest that female survivors display a reduced pattern of activation in the working memory network, which may reflect an unidentified sex-based chemosensitivity.

Free Paper Session (FPS) FPS 20 -LIVE Q&A-ALL CLINICAL
24-10-2021 10:30 AM - 12:00 PM

O0248 / #1265 | Quality of Life in Mothers and Fathers of Children with Acute Lymphoblastic Leukemia in Sweden, Finland and Denmark

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Background and Aims: Introduction: Children with acute lymphoblastic leukemia (ALL) have excellent survival with current therapy. This shifts the focus to the burden of therapy. Serious toxicity is common and the knowledge how the child's leukemia and its treatment impacts the quality of life of parents is incomplete. Aims: The aim was to assess health-related quality of life (HRQoL) in mothers and fathers of patients from Sweden, Finland and Denmark, who had been treated on the NOPHO ALL-2008 protocol 2008-2019.

Methods: The 36-Item Short Form Survey Instrument (SF-36) was used to determine parent HRQoL, approximately 6 months after completed treatment. 526 parents of 310 children treated for ALL were recruited 2010-2019. Normative data for SF-36 from Norway was used as reference. The influence of parental background factors on their HRQoL was analysed in linear regression models.

Results: Participating parents scored lower than the reference population on both physical and mental summary indexes (-1.16, 95% CI [-1.88, -0.45] and -7.72, 95% CI [-8.84, -6.61], respectively). Mothers scored lower than fathers in both indexes. Mothers were more likely than fathers to stop working and take care of the affected child. Higher income was associated with higher physical HRQoL. Higher mental HRQoL was associated with male gender and living in Finland or Denmark (compared to Sweden). Correlation in scores between spouses was weak (0.278, $p < 0.001$ and 0.375, $p < 0.001$, physical and mental scores respectively).

Conclusions: Parents of children treated for ALL report reduced HRQoL after cessation of therapy. The effect on HRQoL was more pronounced for mothers. These findings suggest that the need for support is different for mothers and fathers.

CCI LIVE Q&A-SESSION 4 : BUILDING CAPACITY AND DEVELOPING ENGAGEMENT 24-10-2021 10:45 AM - 12:15 PM

O0249 / #674 | Enhancing Pediatric Cancer Education in Guatemala and Possibly Latin America: Small Steps to Improve Early Diagnosis of Pediatric Cancer

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Background and Aims: Pediatric cancer can be highly curable disease with timely and correct diagnosis. Formal training to healthcare providers (HCP) in early warning signs of cancer avoids diagnostic error and, along instructions for referral, could reduce lag time and improve outcomes. Aim: To train HCP in early signs, initial approach, and instructions for referral a pediatric patient with suspected cancer.

Methods: Through the financing of My Child Matters Program and collaboration of Francisco Marroquín University we developed an online open-course named "Introduction to Pediatric Oncology". Registry and access to learning materials is free on website opencourses.ufm.edu. The course is given in Spanish, asynchronous and structured in 4 calendar weeks. It addresses diagnosis of pediatric cancers and management of common emergencies. The activities include 16 videos and podcasts, 9 readings, 11 evaluations, 4 live sessions and 4 discussion forums. After approval of evaluation activities, participants can apply for a digital certificate for the course completion.

Results: By April 05th 2021, 969 participants from 27 different countries were registered in the first launch of the course. Forty seven percent (n=457) were from Guatemala. Thirty one percent (n=303) completed one or more evaluations and 25% (n=243) started week 4. Of participants who started fourth week, 79% were female (n=193); 47% (n=115) were 21-30 and 38% (n=93) 31-40 years old. Fifty six percent (n=137) were physicians, 27% (n=65) medicine students and 10% (n=24) nurses. Fifty seven percent (n=137) invested 1-3 hours/week to complete course activities and 70% (n=169) had completed the learning activities and approved the evaluations. Live sessions have reached 2446 views, and the discussion forums generated 286 comments.

Conclusions: There is a high interest in learning about pediatric cancer signs and diagnosis in Guatemala and Latin America. The impact of this learning experience on early diagnosis of pediatric cancer is still to be determined.

CCI LIVE Q&A-SESSION 4 : BUILDING CAPACITY AND DEVELOPING ENGAGEMENT 24-10-2021 10:45 AM - 12:15 PM

O0250 / #262 | Abandonment of Treatment; A Report from The Collaborative African Network for Childhood Cancer Care and Research – Cancare Africa

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Background and Aims: 'Treatment abandonment' (or discontinued treatment), is a common and preventable cause of childhood cancer treatment failure in LMIC. In sub-Saharan Africa cost is an overriding reason. In Malawi, a comprehensive package of strategies to enable parents to complete treatment of their child, including no costs for families, was implemented. This reduced 'treatment abandonment' from 19% to 5%. Using the infrastructure of CANCaRe Africa, we studied 'treatment abandonment' in Sub-Saharan Africa, to assess potential areas for future intervention.

Methods: A multi-country, multi-centre, prospective, observational cohort study was conducted in five hospitals in Malawi, Cameroon, Kenya, Ghana and Zimbabwe. Children below 16 years of age, with newly diagnosed cancer, treated with curative intent were included. Data were abstracted in real time using standardised case report forms by trained personnel. Frequency and details of factors associated with abandonment of treatment were documented during the first three months of treatment.

Results: We included 252 patients (median age 6.0, range 0.2–15.0 years, 54% male). The most common cancer was Burkitt lymphoma (63/252, 25%). Seven percent of patients (18 of 252) abandoned treatment within three months of starting. Median travel time to the hospital was four hours (range 0.25 - 24 hours). 65% (163 of 252) of patients had to borrow money to reach the hospital. 79% (200 of 252) of patients were counselled on the need to complete treatment. Of the five centres; three are providing free accommodation to families, one provides free medical treatment and full cover of transport costs, three provide meals to the patients and their families.

Conclusions: Funding is needed to implement known successful interventions to prevent abandonment of treatment. We aim to implement a comprehensive package to enable parents to complete treatment, including no costs for families and improved counseling, in all centres participating in CANCaRe Africa.

CCI LIVE Q&A-SESSION 4 : BUILDING CAPACITY AND DEVELOPING ENGAGEMENT 24-10-2021 10:45 AM - 12:15 PM

O0251 / #1555 | Exploration of Patient and Parent/Caregiver Engagement in Ten Limited-Resource Countries by A SIOP PODC Task Force

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Background and Aims: Literature on patient and caregiver engagement in pediatric oncology is mostly from high-income countries. The SIOP PODC Engagement Taskforce conducted a literature review in 2018, which produced a total of 32 articles. Six publications were from low- and middle-income countries (LMICs): South Africa, Malawi, Ghana, India, Lebanon and Romania. In 2017, the Taskforce initiated a study to give voice to families on childhood cancer treatment engagement in resource-limited settings.

Methods: Taskforce co-Chairs created terms of reference and recruited countries through CCI/SIOP meetings and personal contacts. Professional qualitative researchers performed individual interviews and focus group discussions in local languages with recent patients: 13-18 years, £2 years off treatment and survivors, 5-8 years off treatment; and parents/caregivers with children £5 years off treatment. CCI-associate NGOs supported local researchers to reach respondents and offered housing/food if needed for researchers. Ten countries participated and a 5-day workshop to share data was held before SIOP/CCI in Lyon, 2019.

Results: During the Lyon workshop, researchers from Ethiopia, Morocco, South Africa, Mexico, El Salvador, Peru, India, Iran, Georgia and Greece (Albanian families seeking treatment in Greece) shared data commonalities/differences and early thematic analysis within the group including 2 physicians and 2 qualitative nurse researchers. The most common theme was 'communication'. Two publications on core components of communication (information needs, decision making, healing relationships, patient self-management, managing uncertainty, engendering solidarity [validation], and responding to emotions), guided the group discussion. Further analysis of communication and other themes along with planning for sharing all findings with stakeholders completed workshop activities.

Conclusions: Patient and caregiver engagement during childhood cancer treatment in LMICs is not well documented or understood by healthcare stakeholders. Analyses of communication and other thematic findings continue. Our intention is to describe respondents' engagement challenges and recommendations and create potential interventions for improved engagement in these settings according to respondent preferences and priorities.

CCI LIVE Q&A-SESSION 4 : BUILDING CAPACITY AND DEVELOPING ENGAGEMENT 24-10-2021 10:45 AM - 12:15 PM

O0252 / #1741 | Rehabilitation as A Way of Addressing Cognitive Function Alteration Experienced by Children after Radiation Therapy: Parents' Engagement The Key Determinant Factor

S. Ghods

MAHAK - Society to Support Children Suffering from Cancer, Board of Directors, Tehran, Iran

Background and Aims: Children with brain tumors and leukemia often experience adverse effects on their cognitive functions after radiation therapy. To assess this, MAHAK utilizes the Wechsler Intelligence Test to measure children's IQ scores.

Methods: A pre-test score of 80 or more qualifies the participating child for annual post-tests. If results indicate a declining score after the first year, rehabilitation will be initiated. Simultaneously external/environmental factors are taken into consideration and following the changes in the subsequent post-test results, a risk-laden decision needs to be taken whether to continue or terminate the effort.

Results: This idea was formulated in 2009 at MAHAK, but the actual practice was launched in 2017. Till March 2021, more than 459 tests have been taken in total. Of the participants, 60 have deceased since, many* refused to partake in the rehabilitation plan thereafter, with only 7 cases remaining actively engaged. Resistance is especially prevalent among rural families due to various factors including socio-cultural taboos associated with children's cancer. In addition, children's physical problems, language differences, and low IQ scores are the factors that prevent us from taking the test. *This variable cannot be measured accurately.

Conclusions: The resistance exhibited by families to participate in such programs needs to be better understood and tackled. Many causes having universal roots, the effort to overcome resistance needs to be approached universally as well. Effective multi-disciplinary approach to both treatment and post-treatment interventions is pretty well established as best practices. Without active engagement by families, this effort has no chance for success. Discovering and implementing mitigating factors signals a long and arduous road ahead. To succeed, we suggest an information-sharing methodology inclusive of all conference attendees sharing challenges and solutions on an ongoing basis. MAHAK is prepared to do the same and take the lead on this crucial effort.

Nursing POSTER DISCUSSION SESSION 2 24-10-2021 12:30 PM - 2:00 PM

O0253 / #314 | Addressing the Emotional Wellbeing of Nurses Working in Paediatric Oncology in Sub-Saharan Africa

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Background and Aims: Nurses working within paediatric oncology in low- and middle-income countries have heightened exposure to death and dying, pain and suffering, treatment abandonment, along with staff shortages and high workloads. A Nurses Wellbeing Group was formed as a sub-group of the SIOP Sub-Saharan African Nursing Network to explore ways to increase and improve the emotional wellbeing support available to nurses working within paediatric oncology in the region.

Methods: A survey was used to collect the views of a convenience sample of nurses working in paediatric oncology in the region. The survey focused on what strategies nurses currently use to manage wellbeing and resilience and how provision of emotional wellbeing support could be improved. Survey Monkey was used to circulate the survey which was available in English and French.

Results: A total of 64 surveys were completed by nurses from 11 countries. Nurses reported three main aspect of their work that have an emotional impact: Witnessing children in pain and suffering (94%), death of a child (91%) and caring for children at end of life (88%). The most common methods for managing emotional wellbeing were: discuss with a colleague (88%), think positively (81%), faith/religion/spirituality (81%). The majority of nurses (83%) stated they were not provided with sufficient support with emotional wellbeing at work. Four interventions were highlighted as areas where nurses need greater support: Access to a psychologist, case study meetings to discuss emotional impact of work, greater support and recognition from management and more training in emotional wellbeing and resilience.

Conclusions: The survey results demonstrate unmet need in the emotional wellbeing support available to nurses working in paediatric oncology. Strategies are needed to implement professional and structured support. The Nurses Wellbeing Group is developing a support plan and exploring ways to embed this support across hospitals in Sub-Saharan Africa.

Nursing POSTER DISCUSSION SESSION 2 24-10-2021 12:30 PM - 2:00 PM

O0254 / #201 | Broadening Collaboration in Pediatric Oncology Nursing in the Netherlands: Working Together for a High Standard Shared Care

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Background and Aims: In the Netherlands, complex low volume pediatric cancer care, is concentrated in one national pediatric oncology center. The Princess Máxima Center works together closely with 16 hospitals, the so-called Shared Care centers, based on the principle: central treatment if necessary, close to home if possible. This means that children can have less complex aspects of their treatment done in their own region. However, shared care involves new challenges for nurses. One of the important issues is the alignment of nursing practices between the Princess Máxima Center and shared care centers. For children and parents, it is important that shifting between our center and the shared care center is perceived as a fluent process.

Methods: Therefore, we have established a nursing committee, consisting of nurses working at the Princess Máxima Center and the shared care centers. Within the committee, different subjects are discussed, such as inter-hospital handoff exchange, protocols and education. The committee participates twice a year in shared care meetings. Working groups within the committee meet separately during the year.

Results: The first results of the work of the committee look promising. Knowledge exchange has improved, through providing access to the online educational and guideline portals of our center. Furthermore, patient handoffs have been adapted to the needs of the shared care centers. Patient diaries contribute to the alignment of care, ensuring that children are approached in the same manner. Shared care centers have access to our patient safety system, to be able to report (potential) adverse events. This contributes to the continuous two-way monitoring and improvement of care. The committee is currently examining the educational needs with the aim of developing a custom-fit educational program.

Conclusions: The establishment of a nursing committee shows that through inter-hospital nursing collaboration a higher standard of care can be achieved for children with an oncological disorder.

Nursing POSTER DISCUSSION SESSION 2 24-10-2021 12:30 PM - 2:00 PM

O0255 / #239 | Attitude and Empathic Communication Skill Development after Training Program in Nursing Students and Pediatric Nurses

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Background and Aims: Empathic communication is an effective strategy to improve quality of care in pediatric oncology. It helps nurses to access the needs of children, however, this skill requires good attitude and practicing. By receiving fund from Sanofi-Esprit Foundation (My Child Matters), the training program for nursing students and pediatric nurses in King Chulalongkorn Memorial Hospital was initiated to develop attitude and empathic communication skill. This study aims to demonstrate the outcomes of the training program.

Methods: The training program for nursing students consisted of 3 activities: self-awareness by wrote 3 letters, active listening skill by role play and the difference of sympathy and empathy by video discussion questions. After being trained, they were assigned to practice communication with children in oncology ward and report self-assessment. For the pediatric nurses training, there were 2 activities: active listening and empathic communication by role play. After training, the nurses had practiced in pediatric ward for 3 months and wrote the difficult case report. Finally, they shared positive and negative experience together, as well as effective ways to overcome communication barriers. The outcome were evaluated by 5-point Likert scale and qualitative method.

Results: Total 202 nursing students and 32 pediatric nurses participated in the training program. The participants had positive attitude in communication skill (mean 4.80/5 for nursing students and mean 4.65/5 for nurses). They also perceived self-efficacy to communicate with children (mean 4.65/5 for nursing students and 4.73/5 for nurses). The most communication skill they used is active listening, nonverbal and empathic communication. The nurses' reflection about barriers were workload, limitation of time and the child's fear.

Conclusions: The empathic communication is an important tool to connect with children and improve quality of care. The training program should be started early when they are nursing students to develop awareness and practicing should be continued among pediatric nurses.

Nursing POSTER DISCUSSION SESSION 2 24-10-2021 12:30 PM - 2:00 PM

O0256 / #301 | Does an End of Treatment Telephone Clinic Benefit Patients and Parents During the Transition Following the Completion of Treatment?

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Background and Aims: Increasing survival rates for children and young people with cancer is a recognised achievement for both medical science and individuals themselves. With more patients having successful outcomes and reaching end of treatment, there is a need to ensure adequate support through the unique transition period from patient to survivor. Literature highlights challenges experienced during this adaptation with additional support being required. This audit aimed to evaluate how beneficial an end of treatment telephone clinic was for patients – or their parents – who had a solid tumour diag-

nosis, to provide support at the end of treatment. The clinic was run by Clinical Nurse Specialists. Objectives included whether the Clinical Nurse Specialists sign-posted patients/parents to other services and how the clinic could be used to best serve their needs during transition.

Methods: Patients/parents who had received a telephone call as part of the clinic were invited to complete a mixed methods questionnaire. The questionnaire focused on their experiences at end of treatment and support provided. Qualitative themes and descriptive statistics are reported.

Results: Fifty patients received the questionnaire with a 64% response rate. Eighty-four percent of respondents felt they benefitted, all reported they felt prepared for the end of treatment. Patients/parents were signposted to services such as psychology in 91% of cases. Telephone calls were received 4-6 weeks following end of treatment. Some responses suggested two phone calls at differing times. A clearer explanation for the purpose of the clinic is required, as not all respondents understood. Participants benefitted from reassurance they were not alone, the healthcare team was available for advice regarding ongoing recovery, surveillance and concerns.

Conclusions: An end of treatment telephone clinic is an effective way to provide support following end of treatment. A clearer explanation of the clinics purpose will be provided and giving a choice in timings will be considered.

Nursing POSTER DISCUSSION SESSION 2 24-10-2021 12:30 PM - 2:00 PM

O0257 / #138 | Caring for a Young Child Receiving Brachytherapy in Singapore – A Nursing Experience

Y.-J. Chen, Y.S.L. Lim, Y.Y. Lim

KK Women's and Children's Hospital, Ward 76, Singapore

Background and Aims: An 11 months old boy diagnosed with embryonal rhabdomyosarcoma of the bladder and prostate was referred to KK Women's and Children's Hospital. Surgery was deemed challenging in view of high surgical morbidity and the consensus was to proceed for brachytherapy. As the institution does not have radiation facilities, a cross-institution, multidisciplinary team in collaboration with National Cancer Centre Singapore was formed to deliver the treatment plan.

Methods: Documenting the novel experience of nursing a young child receiving brachytherapy in Singapore.

Results: Urinary symptoms and pain are the commonest anticipated complications for brachytherapy over the bladder and prostate region. Apart from pre-emptive interventions such as bladder irrigation and anticipatory analgesia administration for comfort measures, timely assessment has allowed for escalation and individualisation of care. The application of hip spica to immobilise patient and prevent catheter dislodgement, while critical, increased the challenges faced by the nursing team. Special precaution for limb circulation and skin assessment should be considered and managed accordingly. Young children

may be at risk of inadequate oral intake due to the limited positioning for safe feeding during treatment, nutritional consideration should also be prioritized to ensure holistic management of the child.

Conclusions: Frequent symptoms assessment and the provision of prompt interventions for comfort measures are crucial during treatment. A multidisciplinary approach is pivotal to the success of the brachytherapy treatment and providing the best care to patient and their family.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0258 / #598 | Developing an Online Paediatric Palliative Course for Healthcare Professionals in Sub-Saharan Africa

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Background and Aims: Worldwide, there is tremendous need for paediatric palliative care (PPC) services. However, in resource-poor settings, where services are highly needed, programs and training are scarce. To ensure feasible, acceptable, accessible, affordable and effective PPC services, healthcare providers need to be trained with an affordable and effective program. The aim of this project was to develop an online PPC course for healthcare providers in sub-Saharan Africa.

Methods: The online course was developed by multidisciplinary PPC experts from Botswana, Malawi and Uganda. Five modules with 15 lectures provide the course foundation: introduction to palliative care, communicating with children and families, cultural, spiritual and bereavement considerations, symptom assessment and management, and care at the time of death. Each module contains videotaped lectures and case studies completed by the student. At the end of each module, a quiz is completed before moving on to the next module. Course certificates are awarded when all modules and coursework are finished.

Results: During the first months of the course launch, 105 participants enrolled and 52 completed the course. Students who completed the course were from Malawi (54%), Botswana (34%), Uganda (10%) and outside sub-Saharan Africa (2%). The majority of course participants

were nursing/medical students (44%) followed by practicing nurses (33%) and medical officers (23%). Course evaluation evaluated knowledge gained and self-efficacy in end-of-life care: 96% agreed that the course met their educational needs and they felt more confident in providing care for a child receiving palliative care.

Conclusions: There is an urgent need for PPC training of health providers in resource-poor settings as the majority of children needing palliative care live in these countries. This online PPC course confirms the effectiveness of online distance-based learning and provides essential education that increased the number of providers comfortable in this important area of specialized care.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0259 / #721 | Assessment of Physician Perceptions of Pediatric Palliative Care for Children with Cancer In Latin America

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Background and Aims: The World Health Organization (WHO) designates early integration of palliative care as an ethical responsibility in the management of children with serious illness. While structural barriers may influence the capacity to provide pediatric palliative care (PPC) for children with cancer in resource-limited settings, underlying physician perceptions also impede early integration of PPC within cancer care.

Methods: The Assessing Doctor's Attitudes on Palliative Treatment (ADAPT) survey was developed for physicians who care for children with cancer and was initially distributed in Eurasia. Survey questions evaluated provider perceptions on the scope of palliative treatment, physician responsibility, timing of palliative care integration, and ethical issues. We incorporated recommendations from the WHO 2018

guidelines into survey items to assess provider alignment with WHO standards for PPC provision. The survey was modified for use in Latin America, including translation to Spanish. Descriptive statistics were used to summarize the data.

Results: A total of 874 physicians from 17 countries completed the survey. The majority of physicians (56%) had no formal PPC training and 35% had no access to PPC experts for consultation. On average, providers' perspectives on PPC were aligned with the WHO guidelines (83% alignment, range 57% - 96%). However, only 50% of respondents felt comfortable addressing physical symptoms, 34% felt comfortable addressing emotional symptoms, and 25% felt comfortable addressing grief and bereavement for the patient's family. Forty-two percent felt the integration of pediatric palliative care occurred too late in the course of treatment. Ninety-six percent of survey participants desired further education and training in PPC.

Conclusions: While physician's perspectives aligned for the most part with WHO guidelines, we were able to identify opportunities for improving training around symptom management and supportive care for patients and families. These findings will inform the development of targeted interventions to enhance PPC education and resources for physicians in Latin America.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0260 / #1511 | Perspectives of Pediatric Oncologists and Palliative Care Physicians on the Therapeutic use of Cannabis in Children with Cancer

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Background and Aims: Children with cancer are increasingly using cannabis therapeutically. The purpose of this study was to determine the perspectives and practices of pediatric oncologists and palliative care physicians regarding the use of cannabis for medical purposes among children with cancer.

Methods: A self-administered, voluntary, cross-sectional, de-identified online survey was sent to all pediatric oncologists and palliative care physicians in Canada between June to August 2020. Survey domains included education, knowledge and concerns about cannabis, views on its effectiveness, and the importance of

cannabis-related research. Data were analyzed using descriptive statistics.

Results: In total, 122/259 (47.1%) physicians completed the survey. Although 62.2% of the physicians completed some form of training about medical cannabis, nearly all desired to know more about the dosing, side-effects, and safety of cannabis (114/119; 95.8%). Physicians identified a possible role of cannabis in the management of nausea and vomiting (102/119; 85.7%), chronic pain (86/119; 72.3%), cachexia/poor appetite (80/126; 67.2%), and anxiety or depression (51/126; 42.9%). Only four (0.3%) physicians considered cannabis to be potentially useful as an anticancer agent. Nearly all physicians reported that cannabis-related research for symptom relief is essential (108/118, 91.5%) in pediatric oncology, whereas 51.7% (61/118) expressed that future studies are necessary to determine anticancer effects of cannabis.

Conclusions: Our findings indicate that most pediatric oncologists and palliative care physicians identify a potential role for cannabis in symptom control in children with cancer. Well-conducted studies are needed to generate evidence for cannabis use and promote shared decision-making with pediatric oncology patients and their caregivers.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0261 / #563 | The Effect of Family-Centered Pediatric Advance Care Planning for Teens with Cancer (FACE®-TC) on Congruence in Treatment Preferences Between Teens and Their Families

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Background and Aims: Pediatric advance care planning (pACP) is not yet standard of care. The efficacy of pACP for teens with cancer beyond a single site is unknown. We evaluated the efficacy of FACE®-TC in a multi-site RCT on patient-family congruence in end-of-life treatment preferences.

Methods: Single blinded, intent-to-treat randomized clinical trial. Adolescent/family dyads were randomized at a 2:1 ratio to receive either FACE®-TC or Treatment as Usual (TAU) at four pediatric hospitals. Dyads received 3 weekly 60-minute FACE®-TC sessions: pACP Survey; Next Steps: Respecting Choices interview; and Five Wishes. Control dyads received TAU, plus pACP information. The Statement of Treatment Preferences measured decisional congruence immediately post-intervention.

Results: One hundred twenty-six dyads underwent randomization (83 dyads to FACE-TC and 43 dyads to TAU). Categorizing overall

agreement as poor, good, or perfect, 63/73 dyads (86%) in FACE-TC had good-to-perfect agreement vs. 19/42 dyads (46%) in TAU ($p < 0.0001$). The largest difference was in situation "treatment would extend my life by no more than 3 months and side effects are serious:" 56/73 (77%) agreement among intervention adolescents/families, compared to 15/42 (36%) of TAU ($p < 0.0001$). The interaction of female patient and intervention for good versus poor agreement was significant ($\beta = 3.01$, $SE = 1.35$, $p = 0.0262$).

Conclusions: FACE-TC significantly increased Statement of Treatment Preference Congruence between adolescent/family dyads compared to controls. Females benefited most.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0262 / #1245 | Development of Pediatric Bereavement Care Intervention in LMIC

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Background and Aims: Background and aims: Bereaved families have expressed bereavement care services as imperative and effective aspect of care in grieving process. Objective of this paper is to describe the development of bereavement care intervention at a tertiary care, non-profit hospital and to analyze parental responses following child's death.

Methods: Method: In this qualitative study 118 calls were made from January 2016 till December 2016 to bereaved parents whose children passed away with chronic illnesses and in critical care unit of The Indus Hospital & Health Network, Pakistan. A standardized transcript was prepared and the calls were made by Spiritual care counselor. The purpose of the calls was to provide condolences on behalf of the hospital, to explore parents/ caregiver experiences following child's death and to identify bereaved parent's need for further intervention.

Results: Overall 118 calls were made in 2016, out of 118 total 108 (92%) calls were received, 22 (20%) of the phone calls were responded by caregivers other than parents and 10 (8%) calls were not received. Maximum two attempts were made. Generally received feedback ranged from families expressing gratitude for calls. Few of them expressed anger regarding EOL experiences of the bereaved. In terms of satisfaction with treatment and hospital staff, general feedback was satisfactory few of them expressed skeptical concerns regarding treatment. Few bereaved talked about the deceased and the details surrounding their death. In terms of coping with their child's death, most parents reported coping through patience and connecting with a higher being to manage their feelings of distress.

Conclusions: Conclusion: Overall experience reflects Spiritual Counselor and condolence calls can play integral role in addressing the needs of bereaved families and is considered as a positive supportive intervention for them. Key Words: Condolence calls, bereaved Families, grief, LMIC.

Poster Discussion POSTER DISCUSSION 01-LIVE-PALLIATIVE CARE 24-10-2021 12:30 PM - 1:30 PM

O0263 / #515 | Relationship Between the Choice of End-Of-Life Location by Parents of Children with Cancer and the End-of-Life Care Desired by the Parents

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Background and Aims: Japan has embarked on the establishment of freestanding "children's hospices", as places for children with cancer and their families to live their lives fully. The purpose of this study was to determine the relationship between parents' evaluations of their choice of end-of-life location and the care they desire following a diagnosis of pediatric cancer.

Methods: The data for this cross-sectional study was collected through 117 bereaved family members of children who had died of cancer. The Decision Regret Scale (DRS) was used to evaluate the parents' choice of end-of-life location and to examine the relationship between the evaluations and the 46 survey items on "the care that the parents desired" following a diagnosis of cancer and when their children had 1 month to live.

Results: The median value (first quartile - third quartile) of the total DRS score in this study was 67.5 (range, 46.3-85). When the relationship between the total DRS score and the care desired by the parents following a diagnosis of cancer was examined, a positive correlation of 0.3 or higher was seen for "professional mental care for the families" ($r = 0.378$, $p < .01$), "children talking about themselves to someone other than family members" ($r = 0.323$, $p < .05$), and "continuing care at home through home medical care" ($r = 0.307$, $p < .05$). At one month to live, a negative correlation was found for "participating in events at school/daycare center" ($r = -0.335$, $p < .05$), "the child participating in various events" ($r = -0.358$, $p < .01$), and "the parent participating in events with the child" ($r = -0.361$, $p < .01$).

Conclusions: These findings suggest the need for a comprehensive system that provides pediatric palliative care, including care at the hospital and the home, to achieve the end-of-life care desired by the parents, as well as the need for a system that provides consultations for the children and their families and an environment.

Poster Discussion POSTER DISCUSSION 02-LIVE-CLINICAL
PRACTICE GUIDELINES 24-10-2021 12:30 PM - 1:30 PM

00264 / #1158 | An Evidence-Based Guideline for Social
Restrictions in Children with Cancer

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Background and Aims: In current clinical practice, recommendations regarding social restrictions for children with cancer are often not evidence-based. Critically reviewing the evidence and recommendations is therefore of great importance as these social restrictions (e.g. swimming, school attendance, sports) can impair the quality of life of these children severely. Therefore, a clinical practice guideline (CPG) was developed to establish an overview of the available evidence and provide recommendations for both clinicians, children and their parents.

Methods: A systematic literature review was performed, including dual appraisal of all citations. The GRADE methodology was used to assess, extract and summarize the evidence. When evidence in children with cancer was limited, additional evidence was extracted from adult cancer guidelines. A comprehensive multidisciplinary panel was assembled, comprising 24 professionals and patient representatives. Multiple in-person meetings were held to rank outcomes, discuss evidence, complete evidence-to-decision frameworks and formulate recommendations. Final recommendations were unanimously supported by all panel members.

Results: Nine studies including more than 1400 children with cancer formed the evidence base for the recommendations. The expert panel assessed all evidence and translated it, transparently, into recommendations. In total, more than 20 recommendations were formulated for social restrictions in children with cancer.

Conclusions: In this clinical practice guideline, we provide both evidence-based recommendations and best practice statements regarding social restrictions in children with cancer. With these recommendations we provide guidance for clinicians, children and their parents, and contribute to improving quality of life for children with cancer.

Poster Discussion POSTER DISCUSSION 02-LIVE-CLINICAL
PRACTICE GUIDELINES 24-10-2021 12:30 PM - 1:30 PM

00265 / #955 | Recommendations for Age-Appropriate Testing,
Timing and Frequency of Audiological Monitoring During
Childhood Cancer Treatment: An International SIOP Supportive
Care Consensus Report

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Background and Aims: Ototoxicity is an irreversible direct as well as late effect of certain childhood cancer treatments. Audiological surveillance during therapy as part of the supportive care pathway enables early detection of hearing loss, decision making about ongoing cancer treatment, and where applicable the timely use of audiological interventions. Pediatric oncological clinical practice and treatment trials have tended to be driven by tumor type and tumor specific working groups. Internationally accepted standardized recommendations for monitoring hearing during treatment have not previously been agreed on.

Methods: An Ototoxicity Task Force was formed under the umbrella of the International Society of Pediatric Oncology Supportive Care Committee, consisting of international audiologists, otolaryngologists and leaders in the field of relevant pediatric oncology tumor groups. Expert driven consensus meetings were organized, aimed at providing standardized recommendations on age-directed testing, timing and frequency of monitoring during cancer treatment based on literature and consensus. Consensus statements were prepared by the core group, adapted following several videoconferences and finally agreed upon by the expert panel.

Results: The consensus reached was that children who receive ototoxic cancer treatment (platinum agents, cranial irradiation, and/or brain surgery) require a baseline case history, monitoring of their middle ear and inner ear function, and assessment of tinnitus at each audiological follow-up. As a minimum, age-appropriate testing should be performed before and at the end of treatment. Ideally, audiometry with counselling before each cisplatin cycle should be considered in the context of the individual patient, specific disease, feasibility and available resources.

Conclusions: This is an international multidisciplinary consensus report providing standardized supportive care recommendations on hearing monitoring in children undergoing potentially ototoxic cancer treatment. The recommendations are intended to improve the care of children with cancer and facilitate comparative research on the timing and development of hearing loss caused by different cancer treatment regimens.

Poster Discussion POSTER DISCUSSION 02-LIVE-CLINICAL PRACTICE GUIDELINES 24-10-2021 12:30 PM - 1:30 PM

00266 / #1169 | An Evidence-Based Guideline for Platelet Transfusions in Children with Cancer

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Background and Aims: Children with cancer often suffer from thrombocytopenia. Platelet transfusions are important for both prevention

and treatment of bleeding in thrombocytopenic children with cancer. In current clinical practice, recommendations regarding thresholds for administering platelet transfusions are often not evidence-based. This is problematic, given the important balance between the overuse of platelet transfusions versus risk of bleeding. Therefore, a clinical practice guideline (CPG) was developed to establish an overview of the available evidence and provide recommendations for clinicians.

Methods: A systematic literature review was performed, including dual appraisal of all citations. The GRADE methodology was used to assess, extract and summarize the evidence. When evidence in children with cancer was limited, additional evidence was extracted from adult cancer guidelines. A comprehensive multidisciplinary panel was assembled, comprising 20 professionals and patients representatives. Multiple in-person meetings were held to discuss evidence, complete evidence-to-decision frameworks and formulate recommendations. Final recommendations were unanimously supported by all panel members.

Results: Five studies including more than 1600 children with cancer formed the evidence base for the recommendations. The expert panel assessed all evidence and translated it, transparently, into recommendations. In total, more than 15 recommendations were made regarding platelet transfusions in children with cancer.

Conclusions: In this clinical practice guideline, we provide both evidence-based recommendations and best practice statements regarding platelet transfusions in children with cancer. With these recommendations we aim to provide guidance for clinicians and contribute to improving outcomes for children with cancer.

Poster Discussion POSTER DISCUSSION 02-LIVE-CLINICAL PRACTICE GUIDELINES 24-10-2021 12:30 PM - 1:30 PM

00267 / #1445 | Recommendations for the Surveillance of Education and Employment Outcomes in Survivors of Childhood Cancer: A Report from the International Late Effects Guideline Harmonization Group

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Background and Aims: Educational achievement and employment outcomes are critical indicators of quality of life in survivors of childhood, adolescent, and young adult (CAYA) cancer. This study aimed to provide an evidence-based clinical practice guideline (CPG) with internationally harmonized recommendations for surveillance of education and employment outcomes in survivors of CAYA cancer diagnosed before 30 years of age.

Methods: This CPG was developed by a multidisciplinary panel under the umbrella of the International Late Effects of Childhood Cancer Guideline Harmonization Group. After evaluating concordances and discordances of four existing CPGs, we performed a systematic literature search in PubMed (MEDLINE). We screened articles for eligibility, assessed quality based on evidence-based methods provided by Cochrane Childhood Cancer, and extracted and summarized the data from included articles. The quality of evidence was graded using an adapted version of the Grading of Recommendations Assessment Development and Evaluation (GRADE) criteria (i.e., Level A, high level of evidence; Level B, moderate-low level of evidence; Level C, very low level of evidence). We formulated recommendations based on the evidence and clinical judgement.

Results: Of 2655 articles identified, 64 articles originating from 17 countries were included. Survivors were more likely to have lower educational achievement (Level C), complete their education with a delay, not attend mainstream education (both Level B), and to be unemployed than comparisons (Level C). Key risk factors for poor outcomes included primary diagnosis of a central nervous system tumor and experiencing late effects. We recommend that healthcare providers be aware of the risk of educational and employment problems, implement regular surveillance, and refer survivors to specialists if problems are identified.

Conclusions: We present a harmonized CPG that aims to facilitate evidence-based care, positively influence education and employment outcomes, and ultimately minimize the burden of disease and treatment-related late adverse effects for survivors of CAYA cancers.

Poster Discussion POSTER DISCUSSION 02-LIVE-CLINICAL PRACTICE GUIDELINES 24-10-2021 12:30 PM - 1:30 PM

O0268 / #48 | Evidence-Based Recommendations for Nursing Monitoring and Management of Immunotherapy-Induced Cytokine Release Syndrome: A Systematic Review Children's Oncology Group

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Background and Aims: Children with leukemia and lymphoma are increasingly receiving immunotherapies. The most common adverse reaction of immunotherapies is cytokine release syndrome (CRS). Accurate nursing assessment, early identification, and management of clinical CRS, mitigates the risk of CRS related morbidity and mortality. However, nurses lack standardized guidelines for monitoring and managing CRS. This systematic review examined the evidence of CRS following administration of immunotherapies to support the development of nursing practice recommendations.

Methods: A systematic search of the literature was performed in March 2018. Inclusion criteria were at least one pediatric participant, a diagnosis of cancer and immunotherapy-induced CRS, published since 2000, English language, available in full-text, and relevant evidence to the topic. Articles were appraised using the GRADE framework and the AGREE II instrument. A level of evidence quality (high, moderate, low, or very low) was determined for the body of evidence and recommendation statements (strong or weak) were developed.

Results: Six studies and one clinical practice guideline were included. Six nursing practice recommendations (five strong, one weak) were developed based on low or very low-quality evidence. Of the six recommendations, two reflect pre-infusion monitoring, one focuses on monitoring during and post-infusion, and three pertain to the nurse's role in CRS management. Biomarker utility in predicting CRS risk and severity was also examined.

Conclusions: Nursing practice recommendations will support nurses to recognize CRS and intervene quickly to aid in minimizing CRS related morbidity and mortality. Although this evidence is of low quality, it serves to augment the growing body of literature needed to develop institutional algorithms and practice guidelines needed for nursing care of patients at high-risk for CRS.

Poster Discussion POSTER DISCUSSION 03-LIVE-DESCRIPTION AND IMPACT OF THE COVID-19 PANDEMIC 24-10-2021 12:30 PM - 1:30 PM

O0269 / #1643 | The First Wave of The Covid-19 Pandemic Did Not Affect Childhood Cancer Incidence In Canada: A Report from Cyp-C

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Background and Aims: The COVID-19 pandemic has a major impact on access to healthcare resources. Our objectives were to describe the impact of the COVID-19 pandemic on Canadian childhood cancer incidence, clinical trial enrollment and proportion of patients presenting with metastatic disease.

Methods: We performed a retrospective population-based cohort study of all new cancers diagnosed between December 2017 and February 2020 (baseline period) and between March and August 2020 (pandemic) in children aged <15 years old diagnosed at one of 17 Canadian pediatric oncology centers. The autoregressive integrated moving average model was used to determine the projected monthly age-standardized incidence rates (ASIR). Observed and projected ASIRs (based on cases from 2001 to February 2020) were compared using mean absolute percentage error and coincidence test.

Results: ASIRs during COVID-19 from March-May 2020 and June-August 2020 were 155.7 and 163.2 per million, respectively, while quarterly baseline ASIRs ranged between 149.6 and 175.8 per million. No statistically significant difference between the projected and observed ASIRs was found for all new cancer diagnoses (p-value of coincidence test=0.83), or when stratified by cancer type (leukemias and lymphomas: p=0.18; central nervous system tumors: p=0.49; extracranial solid tumors: p=0.33) or by geographical area (p-values: 0.20 to 0.98). When comparing pandemic periods of March-May 2020 and June-August 2020 with baseline quarterly rates, there was no difference in the proportion enrolled in clinical trials (pandemic: 10.8% and 14.1% vs. baseline median [interquartile range, IQR]: 9.1% [7.0-10.0%]) or presenting with metastatic disease (pandemic: 18.9% and 27.3% vs. baseline median [IQR]: 22.6% [20.7-23.8%]).

Conclusions: No statistically significant change was observed in the incidence of childhood cancer, enrollment on clinical trials, or in the proportion of children with metastatic disease during the first wave of the COVID-19 pandemic in Canada. Continued surveillance is necessary to ascertain potential long-term negative effects of the COVID-19 pandemic for children with cancer.

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O0270 / #248 | Impact of Covid-19 Pandemic on New Diagnoses and Abandonment Rates of Children with Cancer in Indonesia

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Background and Aims: More than a year into the COVID-19 pandemic, its impact on children with cancer is not fully described. In low- and middle-income countries (LMIC), under-diagnosis and abandonment of treatment were major issues prior to the pandemic; whether these have worsened in COVID-19 pandemic is not known. We assessed the impact of the COVID-19 pandemic in one Indonesian tertiary hospital upon the new diagnoses and abandonment rates. We also explored whether parental characteristic and attitudes regarding COVID-19 was associated with the risk of abandonment

Methods: A retrospective cohort design compared two cohorts. The first comprised of patients with childhood cancer admitted before and during the COVID-19 pandemic (November 2019-May 2020) and the second was comparison cohort a year before the pandemic (November 2018-May 2019). Both newly and previously diagnosed patients were included. Two outcomes were compared between cohorts: the rate of new diagnoses (Poisson regression) and proportion of abandonment (Fishers exact test). An additional analysis used questionnaires to determine parental reasons for abandonment.

Results: We found decrease in the new diagnoses rate during and during pandemic (-0.555 vs -0.403; p <0.001) respectively compared with the same period the year prior. A trend towards higher abandonment was seen in April 2020 as compared to April 2019 (12.8% vs. 8.6%; p=0.06); a substantial and statistically significant increase was seen in May 2020 (14.8% vs. 6.7%; p<0.001). In addition to lower maternal education, increased parental perceptions of risk related to COVID-19 was also significantly associated with abandonment.

Conclusions: The COVID-19 pandemic has clearly impacted the ability to diagnosis Indonesian children with cancer as well as treatment abandonment rates. Factors related to parental decision on abandon treatment include maternal education level and parental perception of COVID-19 risk. Interventions to mitigate these risks are urgently needed to protect the outcomes of LMIC children with cancer during COVID-19 pandemic

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**O0271 / #554 | Impact of the Covid-19 Pandemic on Pediatric
Oncology Providers Globally: A Mixed-Methods Study**

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Background and Aims: COVID-19 disrupted pediatric oncology (PO)
care globally, including increasing demands on healthcare providers
(HCPs) as they adapted to continue caring for children with cancer.
This sequential mixed-methods study sought to characterize the pan-
demic's impact on pediatric oncology HCPs globally.

Methods: A 60-item cross-sectional survey focused on changes to clin-
ical care, resource availability, and effects on clinicians due to the
pandemic. A geographically and socioeconomically diverse subgroup
of institutions was purposefully selected from respondents for focus
groups that explored resilience strategies employed during the pan-
demic.

Results: The survey included 311 responses from 213 institutions
representing 79 countries. Sixteen institutions participated in 19
multi-disciplinary focus groups in 8 languages. Decreased clinical staff
availability was the most cited impact on PO care (51% of institu-
tions). Staffing modifications, found in 66% of institutions, included
decreased provider availability, changes in roles or responsibilities,
and transfer outside the specialty area. Physical effects included
frequent COVID-19 illness; 8% of respondents reported PO provider
deaths at their institution. Fifty percent of providers did not always
have the minimally necessary personal protective equipment. HCPs
also experienced emotional and psychological distress and financial
concerns. Impacts were described across all resource levels, and most
acutely during local peaks of infection. Focus groups revealed that such
impacts were balanced by increased teamwork and communication,
contributions outside usual roles, policies aimed at optimizing safety,
and feelings of making a difference.

Conclusions: Our study highlighted the effects the COVID-19 pan-
demic had on HCPs across all resource settings. Findings suggest that
despite multiple difficulties, balancing elements protect and motivate
providers during times of undue stress, allowing continuity of care for
pediatric oncology patients. To protect providers and their patients,
healthcare systems should consider implementing measures such as
frequent communication, structured debriefing, safety policies, proce-

dural algorithms, psychological support and safety equipment during
future catastrophic events.

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**O0272 / #1287 | Transitioning to Virtual Pediatric Hematology
and Oncology (PHO) Prioritization Workshops Due to Covid-19
Pandemic: The Profile Tool Experience**

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Background and Aims: The Abbreviated Version of the Pediatric
Oncology Facility Integrated Local Evaluation (ProFILE) tool helps
stakeholders set priorities for collaborative work and PHO care
improvement. In-country, in-person, prioritization workshops were
an essential part of the Abbreviated ProFILE implementation pro-
cess. The COVID-19 pandemic forced a rapid transition to a vir-
tual workshop format. We describe the transition process and its
outcomes.

Methods: A brainstorming session was conducted with St. Jude Global
regional coordinators with experience conducting ProFILE. The ses-
sion allowed forecasting challenges of transitioning to a virtual format
for the three ProFILE implementation phases. Potential technology
tools and solutions were identified. Bi-weekly meetings were estab-
lished with regional operations teams to plan the design-thinking exer-
cises and ensure good fit between exercises and workshop objectives.
The initial workshop prototype underwent rapid-cycle iteration based
on facilitator and participants' feedback for each cohort.

Results: Five prioritization workshops have been conducted virtually
during the COVID-19 pandemic (Palestine, Eurasia, Chile, Cameroon &
Ethiopia). Three workshops were fully virtual, two used a hybrid format
(whereby local teams met in person, St. Jude regional team assisted vir-
tually). A total of 241 (range 33-55) stakeholders participated. Fully vir-
tual workshops were delivered as 2-hour sessions during a 1-2-week
period. Hybrid format workshops were delivered as 6-8 hours ses-
sions during consecutive days. Eight design-thinking/quality improve-
ment exercises were converted to a virtual format and delivered using
the MURAL platform and Zoom breakout sessions. Each cohort set at
least three country or regional priorities after completing the exercises.
Internet and technology barriers were common and addressed on an
individual basis.

Conclusions: Rapid adaptation to the unprecedented challenges posed
by the COVID-19 pandemic has been possible. The Abbreviated Pro-
FILE in-person workshops were successfully transitioned to a virtual
format. Five additional virtual workshops are scheduled for this year.

This novel approach can be adapted to other collaborative activities to improve PHO care globally.

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**O0273 / #710 | Covid-19 Reinfection vs Reactivation in
Pediatric Cancer. A Report of 27 Cases from Unidad Nacional De
Oncología Pediátrica (UNOP), Guatemala**

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Background and Aims: SARS-CoV-2 infection has a wide range of illness severity and immunologic response. After the first COVID-19 episode reinfection can occur and for immunocompromised children it's uncertain the time of occurrence and clinical presentation. We aim to describe the clinical characteristics of pediatric patients with cancer and documented COVID-19 reinfection.

Methods: A retrospective, observational report of patients under 18 years of age with cancer and SARS-CoV-2 reinfection at UNOP. Patient was considered recovered of COVID-19 when asymptomatic for 2 weeks + 2 negative PCR nasopharyngeal swabs or more than 3 months after diagnosis. Reinfection was considered in patients recovered from first COVID-19 episode and new documented positive PCR swab. Time for reinfection, cancer diagnosis, severity of symptoms, absolute neutrophils count (ANC) and outcome are described.

Results: From May 15th 2020 to April 14th 2021, 220 patients tested positive for SARS-CoV-2 with 27 (12%) reinfections documented. Fourteen (52%) were males, median age 6 years (0-17), median time for reinfection 125 days (50-230). Eighteen (67%) patients were leukemias and lymphomas, 12 (69%) were asymptomatic and detected in a routine PCR swab before the consolidation therapy, 1 of them developed neurological symptoms with documented methotrexate leukoencephalopathy. Six (31%) had constitutional symptoms, 1 had ANC <500, 3 required intensive care (IC) and 2 oxygen therapy. Of 9 (33%) patients with solid tumors, all receiving chemotherapy except one patient, 6 (67%) had constitutional symptoms, 4 (44%) had ANC <500/mL, 1 patient required IC and oxygen in nasal canula. None of the patients died during acute phase of COVID-19 reinfection.

Conclusions: COVID-19 reinfection in our sample was more frequent than previously reported in other populations. A complete viral sequencing is needed to confirm true reinfection and exclude COVID reactivation in these patients.

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**O0274 / #196 | Covid-19 Infection in children with Cancer
after the First Wave in Turkey: A Study of the Turkish Pediatric
Oncology (TPOG) and Hematology (TPHD) Societies**

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Background and Aims: Children with cancer are at increased risk for COVID-19 than their healthy peers. We had evaluated nationwide 51 cases with cancer and COVID-19 before June, 2020. This study evaluates 180 additional children with cancer and COVID-19 in Turkey after the first wave.

Methods: Data of 180 children with cancer and COVID-19, of whom six underwent SCT, diagnosed in 25 centers in Turkey were assessed. Results were compared with results in the first wave.

Results: Median age was 92 (3-240) months, 58% were male. Cancer types were hematological malignancies in 48%, solid tumors in 52%. Cases were confirmed with PCR in 169 (93%), 11 were probable cases with clinical-radiological findings and contact history. 105 had contact in the family. 131 (72.7%) had asymptomatic/mild, 24 (13.3%) moderate, 25 (13.8%) severe/critical disease. Twenty-nine were in ICU. Ten had multisystem inflammatory syndrome (MIS-C). Most common presenting signs were fever and cough. Most frequent system involved was the respiratory system. Radiologically confirmed pneumonia was observed in 30.7% of patients. Central nervous, hematopoietic, gastrointestinal, musculoskeletal systems and multisystem involvement were found in 5.6%, 3.9%, 3.3%, 2.8% and 3.6%, respectively. Chemotherapy was interrupted in 50.5%. Treatment was given as recommended by Ministry of Health. Mortality due to COVID-19 was 3.9%. Having SCT ($p=0.018$), mixed infection ($p=0.015$), radiological findings in the chest CT ($p<0.001$) were associated with increased incidence of critical disease.

Conclusions: Number of cases and mortality due to COVID-19 (3.9% vs 1.9%) has increased after the first wave, while severe/critical cases decreased (14% vs 20.7%). Having stem cell transplantation, mixed infection, lung involvement significantly increased the severity of the disease. Evaluation and sharing of national data of children with cancer/SCT and COVID-19 within a time frame, and accumulation of international data leads to a better understanding of the disease, treatment and risk factors to guide healthcare professionals.

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00275 / #1524 | Impact of Covid-19 Pandemic on Adolescents and Young Adults with Cancer in Canada: A National Survey

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Background and Aims: Adolescents and Young Adults (AYAs) diagnosed with cancer encounter unique challenges and suboptimal health outcomes. The COVID-19 pandemic has triggered many societal changes that have further impacted these domains and require exploration to provide comprehensive care to them. Our study sought to determine the impact of the COVID-19 pandemic on the mental health and cancer care of AYAs with cancer living in Canada.

Methods: We conducted an online, voluntary, self-administered, cross-sectional, 49-item bilingual survey of AYAs with cancer living in Canada from January to February 2021. The survey included Kessler Psychological Distress Scale to assess psychological distress (k-10), Insomnia Severity Index to evaluate sleep and the UCLA loneliness scale to measure loneliness. Univariable and multivariable analyses were done to find the correlates of high psychological distress and negative impact on cancer care. Self-reported free text items underwent qualitative thematic analysis.

Results: The survey was completed by 1063 participants, of which 805 responses (mean age $\pm$ SD: 30.27 $\pm$ 5.27 years) were included in the analysis. The high psychological distress (K-10 $\geq$ 25) was present in 68% of AYAs. Independent associations of high psychological distress included the presence of clinical insomnia, loneliness and pre-existing chronic health conditions. Nearly one-fourth had their cancer affected impacted by pandemic: delayed or cancellations of appointments (16%), diagnostic tests (12%), or treatments (11%). AYAs with a pre-existing mental or chronic health condition were at increased risk for having a negative impact of the pandemic on their cancer care. Common themes for optimizing cancer care during and after pandemic included improving healthcare visits, enhancing personalized care, and strategies for preventing COVID-19.

Conclusions: AYAs with chronic physical or mental health conditions are at higher risk of experiencing inferior outcomes during this pandemic. Enhanced mental and physical health supports, personalized care, and patient engagement may help optimize the care of AYAs with cancer.

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00276 / #660 | SHARE-ing Experiences, Needs, and Decision Making of Parents of Children with Cancer in the United Kingdom During Covid-19

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Background and Aims: The COVID-19 pandemic has had far-reaching impacts on the lives of parents who have a child with cancer; including work, social connections, healthcare delivery, and access to support. The SHARE study aimed to explore changing experiences, information and support needs, and decision-making of parents with a child with cancer during the COVID-19 pandemic in the United Kingdom.

Methods: An online survey using closed and open questions was developed. Parents of a child with cancer (aged 0-18 years) were asked to complete the survey in March 2020- first lockdown, June 2020- when lockdown eased and February 2021- third lockdown. Descriptive quantitative analyses and qualitative thematic content analysis were undertaken.

Results: Responses from 310 parents/caregivers were included. Over time, there was a decrease in worries about the virus (T1-85.4%, T2-58.9%, T3- 58.3%) and the view that the hospital is no longer considered a safe place (69.6%, 37.0%, 31.4%). Vigilance about child's cancer symptoms remained high (93.4%, 93.2%, and 88.6%). Qualitative analysis revealed that initial worries of parents related to perceived risk that their child had greater susceptibility to the virus. Parental information needs at T1 were about the child's risk of catching the virus, and receiving tailored information to their child. Updated guidance from charities and clinical staff indicating children with cancer were not at greatly increased risk alleviated parents' worries. T2 needs focused on advice on children returning to school. Parents also felt more relaxed going outside, however they remained cautious. Parents described difficulties in access to healthcare support due to more restrictive hospital visiting policies. Worries highlighted at T3 centered on new variants of the virus.

Conclusions: The COVID-19 pandemic impacted on the lives, routines, and access to care for families affected by cancer. Efficient, clear, and timely communication may help support parents during this challenging time.

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00277 / #1396 | Covid-19 Vaccine Hesitancy among Parents of Children with Cancer

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Background and Aims: Despite United States COVID-19 vaccination efforts, vaccine hesitancy remains a public health concern. Little

research exists on COVID-19 vaccine hesitancy among caregivers of children with cancer. We aimed to examine: 1) the mediating role of parent-reported COVID distress on the association between COVID exposure and vaccine hesitancy, and 2) whether treatment status (i.e., on/off-treatment) moderated associations.

Methods: Parents of children with cancer (n=356; 97% mothers; 94% White) were recruited February–April 2021 via Facebook and Momcology, a pediatric cancer community-based organization. Parents completed the COVID-19 Exposure and Family Impact Scale (CEFIS) and a modified Vaccine Hesitancy Survey (scale 1-4) online. A series of mediation models assessed the role of COVID-19 distress (pandemic-related caregiver/child distress) on associations between COVID-19 exposure (i.e., direct exposure/disruption due to pandemic) and vaccine hesitancy. Moderation models examined the role of treatment status (i.e., on/off-treatment), on associations between COVID-19 exposure, distress, and vaccine hesitancy. Covariates included parent age, child age at diagnosis, and family income.

Results: Parents reported moderate vaccine hesitancy (M=2.03, SD=0.79). Mediation analysis revealed a significant indirect effect of distress (95% CI [0.00, 0.03]) on the association between higher exposure and higher vaccine hesitancy ($c' = 0.04$, $p = 0.02$); higher exposure was associated with higher distress ($a = 0.36$), which was then associated with lower vaccine hesitancy ($b = -0.03$), but there was no moderating effect of treatment status. When compared to those on-treatment, simple moderation models revealed a stronger association between higher exposure and higher vaccine hesitancy ($b_1 = 0.04$, $p = 0.04$), as well as lower distress and higher vaccine hesitancy ($b_1 = -0.05$, $p = 0.01$), for those off-treatment.

Conclusions: Higher parent-reported COVID exposure and lower distress may be important factors related to vaccine hesitancy among parents of children off cancer treatment. Providers of childhood cancer survivorship care should address vaccine hesitancy and potential health risks. Education may encourage vaccine uptake when they become available for children.

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00278 / #671 | No Increase in Psychosocial Stress of Dutch Children with Cancer and Their Caregivers During the Covid-19 Pandemic

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Background and Aims: Anxiety caused by the COVID-19 pandemic may increase psychosocial stress, especially in vulnerable populations

like children with cancer and their caregivers. Therefore, we aim to compare their psychosocial functioning in the first months of the COVID-19 era to the pre-COVID-19 era.

Methods: Children with cancer and their caregivers in outpatient care participate in regular psychosocial monitoring and screening with the KLIK patient reported outcome measure (PROM) portal. Health-related Quality of Life (HRQoL) and fatigue of children and distress of caregivers were assessed every 3-6 months using the PedsQL generic and multidimensional fatigue scales and the distress thermometer for parents (DT-P). To correct for dependency of data within families or participants, mixed effect linear regression was used to compare pre-COVID-19 era (January 2020 – March 13th) to COVID-19 era (March 13th – May 2020) scores. Clinical distress was analyzed using a logistic generalized estimating equation model. Models were corrected for personal and medical characteristics and interaction effects for children in or after treatment were evaluated.

Results: One or more questionnaires were available from (caregivers of) 799 children with cancer (75% response rate, pre-COVID-19/COVID-19 era: n=494/438, 17% in both era). A majority of children was male, after treatment and had hematological cancer. A smaller percentage of caregivers had clinical distress in the COVID-19 era than the pre-COVID-19 era (OR [CI]: 0.59 [0.42; 0.83, p=0.002]). No other statistically significant differences (p<0.01) were found.

Conclusions: Apparently, experienced stress during the COVID-19 era did not affect HRQoL and fatigue of children with cancer in outpatient care. Results may be explained by experience with medical traumatic stress, appropriate support from (psychosocial) healthcare professionals and changes in daily life (home schooling and less travelling for medical appointments and employment).

Poster Discussion POSTER DISCUSSION 04-LIVE-IMPACT OF COVID-19 ON PSYCHOONCOLOGY 24-10-2021 12:30 PM - 1:30 PM

00279 / #1500 | The Impact of the Covid-19 Pandemic on the Quality of Life of Children with Cancer

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Background and Aims: Little is known about the impact of the COVID-19 pandemic on the quality of life (QOL) of children with cancer. Factors like social distancing and fear of infection may compound psychosocial and health challenges. We aimed to: 1) examine associations between COVID-19 exposure, distress, and QOL in children with cancer, and 2) identify potential moderation based on child's cancer status (i.e., time since diagnosis, on/off treatment).

Methods: Parents of children with cancer (N=356; 97% mothers; 94% White) were recruited February-April 2021 via Facebook and Momcology. Parents completed the COVID Exposure and Family Impact Scale (CEFIS) to assess COVID exposure (i.e., 25-item subscale about direct exposure/disruption due to pandemic), impact (10-item subscale) and distress (i.e., two-item subscale gauging caregiver/child distress), and the PedsQL to assess child QOL. Controlling for parent age, income, child age, and child sex, we examined the indirect effect of COVID distress on the association between COVID exposure and child QOL, as well as the moderating role of cancer status.

Results: Children had lower QOL scores (M=61.99) than the clinical cutoff (M=65.42). Mediation analyses revealed a significant indirect effect of COVID distress (95%CI[-1.12,-0.40]) in the association between COVID exposure and QOL, such that higher exposure was associated with higher distress (a=0.39, p<.001), which was then related to lower QOL (b=-1.84, p<.001). Treatment status did not affect this indirect path, however the association between distress and QOL was stronger as time since diagnosis increased (95%CI[-0.11,-0.004]).

Conclusions: Parents who report greater COVID distress may also report lower QOL in their children with cancer, especially as time since diagnosis increases. Clinicians should be aware of the negative impact of the pandemic on parents and screen for COVID-related distress. Additionally, results highlight the importance of long-term, family-centered care, regardless of children being on or off treatment.

Poster Discussion POSTER DISCUSSION 04-LIVE-IMPACT OF COVID-19 ON PSYCHOONCOLOGY 24-10-2021 12:30 PM - 1:30 PM

00280 / #982 | The Psychosocial Impact of Covid-19 Pandemic on Families of Children with Cancer in Syria

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Background and Aims: COVID-19 pandemic has affected all aspects of health care around the world. Childhood cancer care in Syria was at the beginning of the capacity-building phase after end of war when the pandemic emerged. The aim of this abstract is to describe the psychosocial impact of the covid-19 pandemic on families of children with cancer treated at an NGO-based pediatric oncology unit in Damascus, Syria.

Methods: A 21-item survey was distributed to families of children treated at BASMA Pediatric Oncology Unit in Damascus in May 2020. The unit provides free treatment for all enrolled patients. The patient's caregiver filled the survey. The survey had 17 questions focusing on the epidemiological information and the direct social, financial and psychological impact of the pandemic on families. We used the Coronavirus Anxiety Scale (Lee et al, 2020) to measure families' anxiety and associated symptoms at the onset of pandemic.

Results: Fifty families (n=50) had filled the survey. Most families (n=42, 84%) reported excessive indirect financial costs, including travel (n=23, 46%) and living costs during the curfew. Major anxiety (CAS>9) was found in 11 participants (22%). Thirty-one participants (62%) had dizziness and lightheadedness when hearing or watching news about COVID-19. Twenty-five participants had sleep disorders (50%). Appetite disorders were seen in sixteen parents (28%). Seven of the participating caregivers (14%) suffered from stomach disorders when thinking about the pandemic.

Conclusions: COVID-19 pandemic had a major psychosocial impact on families of children with cancer in Syria. Interventions are needed to decrease this effect.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

O0281 / #993 | Epidemiology of Pediatric Oncology in the State of Sao Paulo, Brazil

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Background and Aims: Cancer is an important cause of morbidity and mortality in children and adolescents worldwide. Despite improvements in cancer registries in Brazil, there are few studies about epidemiology of pediatric cancer. The objective of the study is to evaluate incidence, mortality and survival of pediatric cancer in the state of Sao Paulo.

Methods: Observational study based on public data from National Cancer Institute (INCA). The estimate of incidence on cancer in the population up to 19 years old was calculated from the incidence estimates of the INCA (2018) and the historical series of Population-Based Cancer Registries of the state of Sao Paulo. Mortality data and potential years of life lost were obtained from INCA Mortality Atlas. The survival estimate was calculated using the equation $[1 - (\text{crude mortality rate}/\text{crude incidence rate})] \times 100$.

Results: The estimated incidence of pediatric cancer in the state of Sao Paulo in 2018 was 1,850 cases and a rate of 150.2 cases/million children and adolescents. The rate in the female population was 150.1 cases/million and 150.3 cases/million in the male population. The incidence rate in the 15-19 age group (169.2 cases/million) was higher than in 0-14 age group (142.8 cases/million). In 2018, there were 470 pediatric cancer deaths in the state, which corresponded to 0.8% of the total cancer deaths and 7% of the total years of life lost due to cancer. The crude mortality rate from pediatric cancer in the state was 38.2 deaths/million. The estimated survival of childhood cancer was 74%.

Conclusions: Data on cancer are essential for monitoring and controlling the disease. The findings of the study reveal that the mortality rate of childhood cancer in the state has a significantly impact on the total potential years of life lost, and can serve as a subsidy for the formulation of public policies.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

O0282 / #355 | Neurocognitive Outcome in Survivors of Childhood Acute Lymphoblastic Leukaemia: Experience from a Tertiary Care Centre in India

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Background and Aims: Neurocognitive deficits are worrisome late effects among childhood cancer survivors affecting quality of life. Data from low middle-income countries is scarce. Our aim was to determine the prevalence and spectrum of neurocognitive deficits in childhood acute lymphoblastic leukemia(ALL) survivors and evaluate associated factors.

Methods: We conducted a cross-sectional study at an academic research hospital in India on survivors of childhood ALL between July 2016 to November 2017. Prevalence of neurocognitive deficits was assessed based on the Full Scale Intelligence Quotient(FSIQ) calculated using Weschler's Intelligence Scale for children-4^{INDIA}. Neurocognitive function in discrete domains like verbal comprehension, perceptual reasoning, working memory and processing speed were assessed and correlated to demographic characteristics and treatment received.

Results: 70 survivors of childhood ALL(mean age 10.5(± 3.2) years, 77% male and 5.0(±2.8) years from diagnosis) underwent detailed neurocognitive evaluation. The mean FSIQ was 86.1±20.5, with significant neurocognitive deficit(FSIQ <90) being prevalent in 50%(95% CI 38-62%) of the cohort. The proportion of survivors with deficits in individual domains of verbal comprehension, perceptual reasoning, working memory and processing speed were 49%, 50%, 47% and 44% respectively. The odds of having deficits in neurocognitive function was higher when a child belonged to lower socioeconomic strata(OR 5.7, 95% CI 1.8-18.6, p=0.004), had parents with less than primary school education(OR 95% CI for maternal education 4.6(1.5-13.7), p=0.007 and paternal education 4.3(1.1-17.2), p=0.041), and whose birth order was higher(OR 95% CI 20.1(2.5-164.6), p=0.005). Age at diagnosis, age at assessment, dose of radiation, receiving intravenous methotrexate or cytarabine, did not correlate with significant neurocognitive deficit.

Conclusions: Our data shows a high prevalence of neurocognitive deficits in childhood cancer survivors. Socio demographic factors, rather than disease related factors appeared to play a major role. This may imply the need to establish socio culturally appropriate assessment tools to initiate early screening and develop preventive and rehabilitative interventions.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

O0283 / #595 | The Evaluation of Bone Health in Childhood Solid Tumor and Lymphoma Survivors

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Background and Aims: Long-term survival in childhood cancer survivors is increasing and bone health's one of the areas that requires follow-up. Vitamin D insufficiency, low bone mineral density (BMD), and increased fractures in survivors have been reported. However, research on solely solid tumor and lymphoma cases involving both vitamin D and BMD are few. This study aims to investigate bone health of solid tumor and lymphoma survivors involved in Istanbul Faculty of Medicine, Pediatric Hematology and Oncology, by evaluating vitamin D and BMD measurements.

Methods: Seventy cases diagnosed with solid tumors and lymphoma before age 18 and whose treatment was completed were included in the study, between May 2019- April 2020. Forms about sun exposure, bone fracture, and physical activity were filled. Information about the diagnosis and treatments were obtained from medical files. Anthropometric measurements, biochemical data, 25(OH)D and BMD values were evaluated prospectively. Also, anthropometric measurements, biochemical and 25(OH)D values of 70 control cases under age 18, without a known disease were recorded.

Results: This study consisted of 40% lymphoma and 60% solid tumor cases. Average age at diagnosis and study time were 6.6 and 11.4 respectively and time from diagnosis was 4.9 years. 45.7% of cases and similarly 47.1% of control group had vitamin D insufficiency ($p=0.927$). The average Z-score, corrected volumetrically, was normal (-0.3). 30% of cases had low and 2.9% had very low Z-score. Girls were more prevalent in vitamin D deficient group ($p=0.011$). Z-scores improved as time from diagnosis increased ($p=0.020$).

Conclusions: Vitamin D insufficiency was found similar in childhood lymphoma, solid tumor survivors and healthy population, but the fact that 70% of cases had a history of vitamin D use was remarkable. Longitudinal studies characterizing the natural history of, and risk factors for, BMD deficits among childhood cancer survivors are warranted.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

O0284 / #498 | Implementation of Home-Based End-of-Life Care to Facilitate Decision-Making Regarding Place of Death in Children with Cancer in a Resource-Limited Setting

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Background and Aims: While home is often the preferred place for end-of-life care (EOLC) in high-income settings, resource-limited settings, such as Mexico, often lack comprehensive programs that facilitate EOL based on family's preferences. Despite a pediatric cancer palliative care program was established at Hospital General-Tijuana in 2011, only 37% of patients died at home. We implemented an enhanced home-based EOLC to facilitate parental decision-making regarding place of death of their children with cancer in Mexico.

Methods: In 2016, a multidisciplinary EOLC team was formed which included pediatricians (1), nurses (1), psychologists (2); and social workers (1). The team received training in EOLC, symptom management, particularly pain, bereavement, and techniques to facilitate decision-making, and family education on EOL. Interventions included home-based visits and phone calls to provide EOLC and psychosocial interventions, and education and support to patients and their families. In 2019, an anesthesiologist and two pediatric oncologists joined the team; one-to-one educational sessions and psychosocial support interventions during hospital visits were added; and frequency of in-home interventions was increased.

Results: From 2016 to 2018, we conducted 25 homevisits/year and 53 phone interactions/year; 53% of patients died at home and 47% died at the hospital annually. From 2019 to 2020, after the enhanced strategies, we conducted 38 home visits/year, 334 phone/video calls/year and 44 one-to-one interventions/year during hospital visits, resulting in, 83% of children dying at home and 17% at the hospital annually.

Conclusions: In a resource-limited setting, we demonstrated the feasibility of providing home-based EOLC and facilitating parental decision-making regarding place of death for their children with cancer. Importantly, there was a significant shift in place of death from hospital to home. Although the frequency of home visits were reduced during the 2020 COVID-19 pandemic, in-home EOLC services and subsequently place of death remained unchanged given the increase in support provided by phone/video calls.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

O0285 / #1128 | Haploidentical Stem Cell Transplantation for Pediatric Patients with High Donor-Specific Antibody (DSA) Titers

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Background and Aims: The application of haploidentical stem cell transplantation (haplo-SCT) greatly expands the donor pools for recipients of allogeneic hematopoietic stem cell transplantation (HSCT). Patients with high donor-specific antibody (DSA) titers have higher risks of graft failure particular for patients that require repeated trans-

fusions of blood products before transplant. Different approaches have been developed to reduce DSA. Current there are no consensus or standard guidelines for treatment of pediatric haplo-SCT recipients with high DSA titers due to limited evidences in published data.

Methods: Fourteen haplo-SCT patients aged 2 to 17 years were tested to have high DSA titers (HLA type I and/or type II, MFI $\geq$ 4000) between January 2020 and September 2020. Primary diseases included thalassemia major (13 patients) and severe aplastic anemia (1 patient). Patients were selectively given one or more approaches to reduce DSA titers, including high-dose intravenous immunoglobulin, rituximab, bortezomib, and plasma exchange in addition to a cyclophosphamide-busulfan based conditioning chemotherapy. The prophylaxis against graft-versus-host disease (GVHD) consisted of post-transplant cyclophosphamide, mycophenolate mofetil, and tacrolimus.

Results: Clinical data of 2 girls and 12 boys were analyzed. Haplo grafts were collected from the siblings (5 patients), fathers (8 patients), and mothers (1 patient), respectively. The mean CD34⁺ cell dose was 12.52 $\times$ 10⁶/kg. Neutrophil and platelet engraftments were achieved by a median of 20 days (range 8-33 days) and 20 days (range 8-40 days), respectively. No patients had graft failure nor had developed poor graft function. Only two patients developed grade I-II acute GVHD. No chronic GVHD was observed. Reactivation of cytomegalovirus or Epstein-Barr virus was not observed. All patients were alive without complications with a median follow-up period of 293 days (range 214 to 384 days).

Conclusions: Our data suggest that proper approaches of DSA desensitization are effective to limit graft failure and to reduce transplant-related complications in pediatric haplo-SCT recipients.

Poster Discussion POSTER DISCUSSION 05-LIVE-BEST OF GLOBAL ONCOLOGY (PODC) 24-10-2021 12:30 PM - 1:30 PM

00286 / #1178 | Outcomes of Haploidentical Hematopoietic Cell Transplantation in Children with Non-Malignant Disorders

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Background and Aims: Haploidentical allogeneic hematopoietic cell transplantation (HCT) is a valuable curative option for children with non-malignant disorders. Haploidentical HCT using post-transplant cyclophosphamide (PTCy) offers an easy and affordable option in resource-limited settings. This approach has been mainly used in children with hemoglobinopathy and severe aplastic anemia (SAA), but data on children with primary immunodeficiency (PID) are scarce.

Methods: We conducted a retrospective single-center study on the outcome of haploidentical HCT in children with non-malignant disorders. The primary outcome was overall survival (OS). Secondary out-

comes were graft-vs-host-disease (GvHD)-free relapse-free survival (GRFS) and cumulative incidence of acute and chronic GvHD.

Results: We gathered data from 44 patients underwent HCT from 2015-2020. The indications for HCT were PID (n=16), SAA (n=14), inherited bone marrow failure (n=8), metabolic disorders (n=4), and hemoglobinopathy (n=2). Median age at HCT was 6.8 years (range 0.7-20). The conditioning regimens were myeloablative (n=17), or reduced intensity (n=27). Bone marrow was used in 18 patients and peripheral blood in 26 patients. Serotherapy was used in 37 patients. After a median follow-up of 20 months (range 4-70), 2-year OS was 89% and 2 year GRFS was 64%. We did not identify factors associated with OS or GRFS. This must be interpreted cautiously given the modest size of the study population. Cumulative incidence of acute GvHD grade 2-4 was 20% and of moderate to severe chronic GvHD was 5%. Eight patients (18%) experienced graft failure (6 primary and 2 secondary). CMV reactivation developed in 55%, and BK hemorrhagic cystitis in 20%. All surviving patients had full donor chimerism at last follow-up except for 5 patients with stable mixed chimerism.

Conclusions: Haploidentical HCT using PTCy is feasible and effective for patients with non-malignant disorders. PTCy is a valid option in resource limited settings. Reduction of conditioning intensity to decrease conditioning related toxicity need to be prospectively assessed.

Free Paper Session (FPS) FPS 22-LIVE Q&A-BONE TUMORS-NÖLLENBURG PRIZE 24-10-2021 2:30 PM - 3:30 PM

00287 / #1387 | Telomere Size is Associated with Relapse in Osteosarcoma

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Background and Aims: Osteosarcoma is a malignant neoplasm that originates in the progenitor cells of the bone, mostly affecting long bones. It presents features of chromosomal instability, high levels of mutations and epigenetic changes. For the last 30 years there is no significant changes in the survival rates. Understanding the underlying mechanisms of osteosarcoma tumorigenesis may help to develop more effective treatments. The telomere is essential for chromosome-end protection and genomic stability.

Methods: We evaluated telomere size in osteosarcoma samples and searched for associations with clinical and pathological characteristics in 41 children and adolescents. DNA was extracted from the osteosarcoma samples and was submitted to a real time polymerase chain reaction (qPCR) with SybrGreen. Cawthon method was applied for relative measurement of telomere length and the t-test was used to compare

groups. There were 21 males and 17 females aged from 6 to 28 years at diagnosis (mean 13, median 13).

Results: Telomere length was associated with relapse but not with gender, body mass index, bone locations, metastasis at diagnosis, histology or death. Patients that relapsed following end of treatment presented increased telomere length compared to those patients that did not relapse.

Conclusions: Characterization of expression of genes that controls telomere size should help to understand the mechanism associated with osteosarcoma progression. Telomere length could be used as a prognostic marker in patients with osteosarcoma.

Free Paper Session (FPS) FPS 22-LIVE Q&A-BONE TUMORS-NÖLLENBURG PRIZE 24-10-2021 2:30 PM - 3:30 PM

O0288 / #1544 | A Three-Gene Signature to Predict Survival in Pediatric Osteosarcomas (OS) and to Follow Patients Towards Liquid Biopsy: A Collaborative Work in OS2006 SFCE Protocol

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Background and Aims: The overall survival of pediatric OS is stable worldwide since two decades. The management of OS is lacking new approaches to classify closely the patients and adapt thereafter the chemo-treatments since diagnosis. The objectives of our molecular study, using allelotyping, quantitative PCR (qPCR) and droplet digital PCR (ddPCR), were to determine the survival impact of specific genes involved in bone de-differentiation and study them in the initial tumor biopsy, in the tumor resection after neoadjuvant chemotherapy and on circulating DNA at different treatment steps. This panel was originally identified in the tumor cohort of the French protocol OS94 and demonstrated a significant prognostic impact of 3 genes MET, TWIST and APC. To go further and understand signature significance, the same study was performed in the recent OS2006 protocol.

Methods: We performed retrospectively in the cohort of OS2006 protocol (144 patients at diagnosis, 32 patients with resected tumors) the molecular targeting of those 3 genes. Allelotyping, copy number variations by qPCR and ddPCR were used. A preliminary cohort of 28 patients with liquid biopsies at diagnosis, before tumor surgery and at the end of chemotherapy was screened by ddPCR with this gene panel.

Results: We were able to determine frequent rearrangements in the 3 genes and a statistically significant correlation was made between the presence of gene rearrangements and a worst outcome. The 3-gene signature was significantly predicting the outcome of more than 70% of OS patients. Furthermore, on the tumor resection, the persistence or appearance of new clones was significantly correlated to a higher risk of relapse. This 3-gene signature was also usable for residual disease assessment on plasmatic DNA and we were able to follow closely patients' evolution.

Conclusions: This molecular signature was able to provide evidence that this approach is a powerful tool to be used in pediatric OS follow up.

Free Paper Session (FPS) FPS 22-LIVE Q&A-BONE TUMORS-NÖLLENBURG PRIZE 24-10-2021 2:30 PM - 3:30 PM

O0289 / #546 | EWS/FLI1 Hijacks UHRF1 to Drive the Malignancy of Ewing Sarcoma

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Background and Aims: Ewing sarcoma (EwS) is genetically characterized by a balanced chromosome translocation forming the EWS/ETS chimeric oncoprotein, most frequently EWS/FLI1. Low mutation burden raises the possibility that epigenetic alterations play a role in the malignancy of EwS. It has been suggested that the chimeric protein participates in aberrant DNA methylation of EwS while its mechanism remains elusive. UHRF1 maintains the faithful inheritance of DNA methylation by taking advantage of multiple domains. We studied the interaction of EWS/FLI1 and UHRF1 in DNA methylation in EwS, as well as the mechanisms of the specific DNA methylation pattern in EwS.

Methods: Immunohistochemistry was performed to check the expression of UHRF1 in different donors including EwS. GEO datasets were used to check and correlate their expression with different statuses and prognosis. Immunofluorescence, immunoprecipitation, and GST-pulldown were performed to detect their direct interaction. In vitro and in vivo experiments were performed to study the therapeutic efficiency of targeting UHRF1 in EwS.

Results: UHRF1 is highly expressed in EwS, and high expression of UHRF1 correlates with recurrence ($P < 0.001$) and metastasis ($P < 0.001$). Knockdown of UHRF1 limits tumor growth in vitro ($P < 0.0001$) and in vivo ($P < 0.01$). Our work demonstrates EWS/FLI1 binding to the SRA domain of UHRF1 to maintain the stability of each other. DNA methylation array indicates 2/3 of the hypomethylated sites mediated by UHRF1 are overlapping with sites hypomethylated by EWS/FLI1, further confirming that EWS/FLI1 interacts with UHRF1 to drive the specific DNA methylation pattern of EwS.

Conclusions: High expression of UHRF1 correlates with a dismal prognosis in EwS; EWS/FLI1 binds to SRA domain of UHRF1 to maintain each other's stability and drive the aberrant DNA methylation pattern of EwS.

Free Paper Session (FPS) FPS 22-LIVE Q&A-BONE TUMORS-NÖLLENBURG PRIZE 24-10-2021 2:30 PM - 3:30 PM

O0290 / #1299 | Modern Therapy for Spinal and Paraspinal Ewing Sarcoma: An Update of the University of Florida Experience

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Background and Aims: In 2009, we published a comprehensive review of our institutional outcomes treating children with spinal and paraspinal Ewing sarcoma. Multimodality therapy was associated with a fair disease control but significant toxicity, including a 37% rate of late Grade ≥ 3 toxicity. The objective of this study is to review children treated with modern technology and treatment regimens available since the last report.

Methods: Between 2010-2021, 32 consecutive pediatric patients with non-metastatic spinal or paraspinal Ewing sarcoma were treated at the University of Florida. The median age at the time of diagnosis was 9.8 years old (range, 2.1-21.8 years). The median maximum tumor diameter was 5 cm (range, 3-19 cm) and 30/32 tumors possessed an EWSR1 translocation. At diagnosis, 28/32 patients had significant motor, bowel, or bladder deficits. Patients were treated with chemotherapy regimens according to contemporary North American and European protocols. Prior to radiotherapy, 14 patients had a gross

total resection while the other 18 underwent a biopsy or subtotal resection with cord decompression. All patients were treated with proton therapy, and 6 with hardware stabilization also received a component of intensity modulated photon therapy (IMRT/VMAT).

Results: With a median follow-up of 4.1 years (range, 0.7 – 9.4 years), the 5 year local control, progression free survival, and overall survival is 92%, 79%, and 85%, respectively. Twenty of 30 living patients recovered from their original motor, bowel, or bladder deficits. Overall, 22% of patients experienced new-onset CTCAE Grade 3 late toxicity related to multimodality treatment: kyphosis (n=4), esophagitis (n=2) and chronic kidney disease (n=1). No patients developed grade ≥ 4 toxicity.

Conclusions: Compared to our prior institutional experience, advances in multimodality treatment offer an improved therapeutic ratio for children with spinal and paraspinal Ewing sarcoma. With appropriate management, most patients can be cured with recovery of long-term neurologic function and modest side-effects.

Free Paper Session (FPS) FPS 23-LIVE Q&A-GERM CELL AND LIVER TUMORS 24-10-2021 2:30 PM - 3:30 PM

O0291 / #456 | Histologic Subtype Predicts Disparate Outcomes in Pediatric Hepatocellular Carcinoma – A Pediatric Surgical Oncology Research Collaborative Study

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Background and Aims: Hepatocellular carcinoma (HCC) is a rare cancer in children with variant histologic subtypes. There are a paucity of reliable data to guide clinical expectations and predict prognosis. To allow for characterization of clinical outcomes of variant histologic HCC subtypes, a multi-institutional collaborative dataset was developed.

Methods: Patients (0-20 years) with HCC (1990-2018) were identified at 15 hospitals participating in the Pediatric Surgical Oncology Research Collaborative (PSORC). Clinical data collected included demographics, staging, treatment and outcomes. Patients were subdivided into the following histologic subtypes: HCC with or without underlying liver disease, fibrolamellar carcinoma (FLC), and hepatoblastoma with HCC features (HB-HCC). Univariate and multivariate analyses were used to identify predictors of mortality and clinical outcomes.

Results: 262 children were identified with hepatocellular neoplasms: 110 HCC (69 normal liver, 34 inflammatory/cirrhotic, 7 unknown); 119 FLC; and 32 HB-HCC. We found notable differences in clinical presentation and behavior amongst tumor subtypes, including an increase in lymph node involvement in FLC and higher PRETEXT stage in HCC. Factors associated with mortality ($p < 0.05$) included HCC, elevated AFP, multifocality, and PRETEXT stage. Multivariate analysis identified increased risk of mortality for HCC vs FLC (HR 2.2, $p = 0.004$) and in patients deemed unresectable (HR 3.45, $p < 0.001$). Achievement of disease-free status at any point predicted survival (HR 0.15, $p < 0.001$).

Conclusions: This large pediatric HCC dataset allows comprehensive analysis of outcomes for patients with these rare cancers. Our data demonstrate that pediatric HCC subtypes should not be considered equivalent entities, as FLC and HCC present with different anatomic patterns and differential outcomes. This dataset will be used as a foundation to elucidate the impact of histologic characteristics on specific outcomes, with the goal of designing optimal risk-stratified treatment algorithms for these patients.

Free Paper Session (FPS) FPS 23-LIVE Q&A-GERM CELL AND LIVER TUMORS 24-10-2021 2:30 PM - 3:30 PM

O0292 / #375 | The Outcomes of Living Donor Liver Transplantation for Hepatoblastoma in Japan: An Analysis of Cases Listed in the Registry of the Japanese Liver Transplantation Society

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Background and Aims: Liver transplantation (LT) has been increasingly performed for unresectable hepatoblastoma (HB) with acceptable results. We conducted a national survey of cases undergoing living donor liver transplantation (LDLT) for HB to evaluate their outcomes.

Methods: We conducted a national survey of patients undergoing LDLT for HB between October 1990 and April 2018 and the cases were followed until December 2019. We mainly analyzed the impact of tumor recurrence and lung metastasis before LT on outcomes.

Results: Ninety-eight patients (61 males and 37 females with a median age at LDLT of 3.0 years) underwent LDLT during the study period. A total of 30 of 98 (30.6%) patients received salvage liver transplantation, including 25 patients for recurrent tumor and 5 patients for deteriorating liver function after liver resection. There was no significant difference in the 1- and 5-year patient survival rates between patients undergoing primary LT and those who received salvage LT for tumor recurrence (89.7%, 81.6% vs. 88.0%, 76%) ($P = 0.526$). Of the 68 patients who received primary LT, 21 had lung metastases. Among these 21 patients, complete radiographic clearance of lung metastases was achieved in 10 at the time of LT, while the remaining 11 required metastasectomy before LT. New lesions appeared in the lungs of two patients during neoadjuvant chemotherapy. On comparing the prognosis by the presence of lung metastases, the recurrence rate was significantly worse in the patients with lung metastases at the time of the diagnosis than in those without lung metastases ($P = 0.043$), whereas there was no marked difference in the mortality rate ($P = 0.742$). The mortality rate tended to be higher in patients with the appearance of lung metastasis during NAC than in others ($P = 0.086$).

Conclusions: LT can be curative for unresectable HB, including cases with lung metastases and after recurrence.

Free Paper Session (FPS) FPS 23-LIVE Q&A-GERM CELL AND LIVER TUMORS 24-10-2021 2:30 PM - 3:30 PM

O0293 / #516 | Targeting Oncogenic Micrornas in Malignant Germ Cell Tumours with Locked Nucleic Acid (LNA)-Based Inhibitors

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Background and Aims: MicroRNAs (miRNAs) are short, non-protein-coding RNAs that regulate gene expression, primarily through mRNA

degradation. We have shown that all malignant germ-cell-tumours (mGCTs) overexpress two miRNA clusters, miR-371~373 and miR-302/367, resulting in downregulation of functionally significant mRNAs/pathways. Here, we inhibited these overexpressed miRNAs in mGCT cells in vitro to determine their functional significance and explore their potential role as therapeutic targets.

Methods: RT-qPCR confirmed that miR-371~373 and miR-302/367 levels in mGCT cell lines [seminoma (SEM), yolk sac tumour (YST) and embryonal carcinoma (EC)] corresponded to those in tissue samples of the relevant tumour subtype. MiRNA inhibitors composed primarily of locked nucleic acids (LNAs) were designed to target key mature miRNAs. Gene expression profiling of cells treated with inhibitor or control was performed using mRNA microarray. Sylamer analysis was used to identify direct mRNA targets of these miRNAs, and key pathways analysed using KEGG and Reactome for functional validation.

Results: Targeting mature miRNAs of the miR-302 family (miR-302a-d) using LNA-based inhibitors resulted in reproducible growth inhibition in SEM and YST cells, although EC cells were more resistant to treatment. In the mGCT cells with reduced growth, Sylamer revealed enrichment of upregulated mRNAs that contained the seed-complementary-region for miR-302a-d in their 3' untranslated regions. Pathway analysis of these upregulated mRNA targets primarily demonstrated involvement in cell cycle regulation. This was functionally validated through cell cycle analysis using flow cytometry.

Conclusions: LNA-based miR-302 family inhibition results in cell cycle arrest in two main subtypes of mGCT and may represent a new therapeutic approach.

Free Paper Session (FPS) FPS 23-LIVE Q&A-GERM CELL AND LIVER TUMORS 24-10-2021 2:30 PM - 3:30 PM

00294 / #460 | Risk of Germ Cell Tumors is Increased in Children with Birth Defects

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Background and Aims: Some studies indicate that birth defects are associated with germ cell tumors (GCTs). However, few investigations have evaluated associations by histologic subtype, location,

age at diagnosis, sex, and type of birth defect. Therefore, we leveraged a large cohort of children diagnosed with GCTs to perform a detailed analysis of the associations between birth defects and these malignancies.

Methods: Cases (N=638) were participants in the Children's Oncology Group GaMETES Study, diagnosed with GCTs between July 2008 and December 2015 and aged <20 years. Controls matched on sex and maternal race/ethnicity were randomly selected from births in three U.S. states (Oklahoma, North Carolina, and Texas) at a 10:1 ratio. Information on birth defects diagnoses were obtained from parental interview (cases) and statewide birth defects registries (controls). Logistic regression was used to calculate odds ratios and 95% confidence intervals (ORs; 95% CIs) to estimate the association between birth defects and GCTs, overall and by subtype.

Results: Overall, the prevalence of birth defects and chromosomal anomalies in GCT cases was 8% compared to 4% in controls (2.0, 1.4-2.7). Non-chromosomal defects were associated with mixed/other GCTs (1.9, 1.0-3.3) and yolk sac tumors (1.9, 0.8-3.9). Genitourinary defects were associated with yolk sac tumors (3.4, 1.0-8.6), gonadal tumors (2.2, 0.9-4.5), and extracranial/extragenital tumors (2.2, 0.5-6.0). In stratified analyses, associations with non-chromosomal defects were stronger among males, but increased odds of yolk sac (2.1; 0.6-5.4) and extracranial/extragenital tumors (1.7; 0.4-4.8) were also observed among females.

Conclusions: Our results suggest that non-chromosomal birth defects, particularly genitourinary defects, are associated with GCTs in children and adolescents. Associations were strongest for yolk sac or mixed/other tumors, and may be specific to males. These findings could be leveraged to assist in the discovery of risk factors for these tumors and to understand sex ratio disparities.

Free Paper Session (FPS) FPS 24-LIVE Q&A-SOFT TISSUE SARCOMAS - RHABDOMYOSARCOMA 24-10-2021 2:30 PM - 3:30 PM

00295 / #538 | Congenital Rhabdomyosarcoma: Not as Bad as We Thought a Report from the European Paediatric Soft Tissue Sarcoma Study Group

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Background and Aims: Congenital rhabdomyosarcoma (RMS) represents a challenging disease due its characteristics and the difficulties in delivering treatment in this immature population.

Methods: We analyzed patients with congenital RMS enrolled in the European paediatric Soft tissue sarcoma Study Group protocols. We defined as congenital, RMS diagnosed in the first two months of life and, as infants, those diagnosed >2 to 12 months of age.

Results: 115 children aged <1 year were registered, including 24 congenital RMS. In the latter group, all, except one patient (PAX3-FOXO1-positive metastatic RMS), had favorable histology and localized disease. Three patients had VGLL2-CITED2/NCOA2 fusion. No radiotherapy was given, and complete tumor resection was achieved in 10 patients. Chemotherapy doses were adjusted to age and weight. Only two patients required further dose reduction for toxicity. The 5-years EFS and OS were 75.0% (95% C.I. 52.6-87.9) and 87.3% (65.6-95.7) in the congenital and 68.4% (57.6-77.1) and 86.3% (77.1-92.0) in the infant group, respectively. Progressive disease was the main cause of treatment failure in the congenital group.

Conclusions: Patients with congenital RMS presented with favorable disease, allowing weight and age-adjusted doses of chemotherapy and avoidance of irradiation, without compromising outcome.

Free Paper Session (FPS) FPS 24-LIVE Q&A-SOFT TISSUE SARCOMAS - RHABDOMYOSARCOMA 24-10-2021 2:30 PM - 3:30 PM

00296 / #1133 | Role of 18F-FDG-PET/CT in the Staging of Metastatic Rhabdomyosarcoma: A Report from The European Paediatric Soft Tissue Sarcoma Study Group

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Background and Aims: Initial staging of rhabdomyosarcoma is crucial for prognosis and to tailor the treatment. The standard radiology workup (SRW) includes MRI, chest CT and bone scintigraphy, but ¹⁸F-FDG-PET/CT (PET-CT) use is increasing. The aim of our study is to evaluate the impact of PET-CT in the initial staging of patients with metastatic rhabdomyosarcoma enrolled in the European protocol MTS2008.

Methods: Two authors retrospectively reviewed the SRW and PET-CT reports comparing the number and sites of metastases detected. For bone marrow involvement, PET-CT and bone marrow aspirates/biopsies were compared.

Results: Among 263 metastatic patients enrolled from October 2008 to December 2016, 121 had PET-CT performed at diagnosis, and for 118/121, both PET-CT and all radiological reports were available for review. PET-CT showed higher sensitivity than SRW in the ability to detect locoregional (96.2% vs. 78.5%, p-value=0.0013) and distant lymph node involvement (94.8% vs. 79.3%, p-value=0.0242), but sensitivity was lower for intrathoracic sites (lung 79.6% vs. 100%, p-value=0.0025). For bone metastasis, PET-CT was more sensitive than bone scintigraphy (96.4% vs. 67.9%, p-value=0.0116). The PET-CT sensitivity and specificity to detect marrow involvement were 91.8% and 93.8%, respectively. The mean number of metastatic sites was 1.94 (range 0-5) with PET-CT and 1.72 (range 0-5) with SRW. In 4 (3.4%) patients, PET-CT changed the stage from localized to metastatic disease.

Conclusions: PET can help to identify all involved sites in metastatic RMS patients and can change the stage from localized to metastatic in a limited number of children. This is due to its higher ability to recognize lymph nodes and bone involvement. SRW remains essential to detect lesions in intrathoracic sites. PET-CT could replace bone scintigraphy to study bone involvement.

Free Paper Session (FPS) FPS 24-LIVE Q&A-SOFT TISSUE SARCOMAS - RHABDOMYOSARCOMA 24-10-2021 2:30 PM - 3:30 PM

00297 / #1450 | Prospective Evaluation of Reduced Dose Alkylator Chemotherapy in Combination with Radiotherapy for Standard Risk Rhabdomyosarcoma Subgroup C: A Report from The EPSSG RMS2005 Study

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Background and Aims: EpSSG standard risk non-alveolar rhabdomyosarcoma patients: IRS stage II/III disease at favourable sites, including orbit, head and neck non- parameningeal (HNnonPM), genitourinary non-bladder- prostate (GUnonBP), were eligible for RMS2005 study Subgroup C.

Methods: Subgroup C patients were recommended to receive reduced dose alkylator chemotherapy with five cycles of Ifosfamide, Vincristine and Actinomycin-D (IVA) chemotherapy (30g/m² Ifosfamide) plus radiotherapy, or nine cycles of IVA (54g/m² Ifosfamide) without radiotherapy. Surgical/ delayed excision of the primary tumour (DPE) outcomes were recorded. Univariate analyses were performed.

Results: 359 patients including orbit 164 (45.7%), HNnonPM 77 (21.5%) and GUnonBP 118 (32.8%). Five-year event free (EFS) & overall survival (OS) were 77.4% (CI 72.5- 81.6%) & 93.5% (CI 90.1- 95.8%) respectively; no significant differences based on IRS group or site. 232 patients (65%) received radiotherapy. Reduced dose alkylator chemotherapy and radiotherapy significantly improved 5-year EFS: 86.3% (17/ 132 relapsed) versus 65.7% (38/ 113 relapsed) ($p=0.0002$) for 9 cycles IVA without RT, but not 5-year OS: 93.7% and 93.9% ($p=0.8003$) respectively. 80/ 248 (32.3%) IRSIII cases had DPE; only 4 had macroscopic residual. For the 60 patients with complete microscopic (R0) resection, there was no significant difference in EFS or OS with adjuvant radiotherapy. 16 patients with favourable disease based on size and age achieved a complete remission with IVA. Local failure occurred in 8 (50%); 7 receiving 9 IVA alone, and 1 receiving 5 IVA, 4 VA and radiotherapy. For orbital tumours 5-year EFS was 22.3% (19/ 26 relapsed) without radiotherapy.

Conclusions: Excellent outcomes were observed for EpSSG Standard Risk Subgroup C rhabdomyosarcoma. For those receiving radiotherapy, reduced dose alkylator chemotherapy should be the standard. Reduced EFS but not OS is observed with the omission of radiotherapy. Adjuvant radiotherapy may be safely omitted for RMS2005 Standard Risk Subgroup C rhabdomyosarcoma where there has been a delayed R0 resection.

Free Paper Session (FPS) FPS 24-LIVE Q&A-SOFT TISSUE
SARCOMAS - RHABDOMYOSARCOMA 24-10-2021 2:30 PM -
3:30 PM

O0298 / #403 | Long-Term Adverse Effects Following Treatment
for Paediatric Head and Neck Rhabdomyosarcoma, Results of a
Trans-Atlantic Multi-Center Cross-Sectional Cohort Study

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Background and Aims: Local therapy for head-neck rhabdomyosarcoma (HNRMS) patients includes radiation therapy potentially combined with surgery. Adverse effects (AEs) of local treatment are prevalent in survivors. Efforts are being made to limit AEs through more conformal radiation techniques and the development of new combined modality strategies. Theoretical differences between strategies exist but the actual clinical benefit in survivors remains unclear. The primary purpose of this study was to describe the prevalence and severity of AEs in HNRMS survivors. Also, the data allowed a comparison of AEs between four local treatment strategies: definitive external beam radiotherapy with photons, definitive external beam radiation with protons, surgery combined with brachytherapy (AMORE), and surgery combined with external beam radiotherapy.

Methods: A trans-Atlantic, international, multicenter cross-sectional cohort study was performed. Survivors with ≥ 2 years follow-up after treatment for a primary pediatric HNRMS were systematically assessed by a multidisciplinary team systematically following a predefined list of AEs according to the Common Terminology Criteria for Adverse Events v 4.0 system during outpatient clinic visits.

Results: Ninety-nine survivors with a median follow-up of 9 years were included. Eighty-three percent of survivors experienced at least one grade ≥ 2 AE, survivors had a median of 8 different AEs. Musculoskeletal deformity, cataract, hearing impairment, speech abnormalities and eyelid disfunction were the most common AEs. Tumor size ≥ 5 cm and younger age (0-5y) at diagnosis were independent risk factors for a grade ≥ 3 AE. Our results suggest variation in type and severity of AEs between different local treatment strategies, but these differences were not statistically significant.

Conclusions: HNRMS survivors are at high risk for a diverse range of toxicity. The nature of AEs varies between local treatment strategies and data from this study may be used to optimize side-effect surveillance. Both dose-effect analyses as well as long-term clinical outcomes are necessary to properly tailor local therapy for children with HNRMS.

Nursing FREE PAPER SESSION 6-COMPLEMENTARY METHODS
AND CLOSING REMARKS 24-10-2021 3:00 PM - 4:00 PM

O0299 / #1618 | Aromatherapy in the Pediatric Oncology
Population

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Background and Aims: Aromatherapy using essential oils is a popular integrative medicine practice. Despite protocols with a list of approved hospital-provided oils for inhalation, many families on the Cancer Transplant Unit at Primary Children's Hospital (PCH) continue to use personal oils. Staff also identified a need for updated education and clarity regarding aromatherapy practices. This project included a review of the literature and survey of other pediatric oncology programs in the United States to update hospital practices and staff/family education regarding essential oil aromatherapy.

Methods: Sources published between 2010-20 with attention to aromatherapy to reduce nausea, pain and anxiety among pediatric and oncology populations were identified. A survey regarding hospitals' essential oil policies, mode of distribution, approved oils, documentation, and staff education was posted on the Association of Pediatric Hematology/Oncology Nurses discussion board. An industrial hygiene specialist was consulted to discuss the safety of inpatient diffusers.

Results: The literature review yielded 10 articles and a textbook chapter. Studies suggested aromatherapy is effective to reduce nausea, anxiety and pain among hospitalized children. Two authors noted the need for studies with larger sample sizes.

Nurse educators, managers and aromatherapists from 20 inpatient pediatric oncology units completed the survey. All reported allowing at least one essential oil, most frequently lavender. Five reported using QueaseEASE personal nasal inhalers (PNI) with favorable responses. Five hospitals allowed diffusers; hospitals that prohibited diffusers cited concerns maintaining low-microbial, scent-free, and safe environments. Eight hospitals train nurses in the use of essential oils and require documentation of any oil administration. Eleven respondents indicated direct inhalation as the preferred method of administration, with 7 prohibiting all other methods.

Conclusions: The PCH aromatherapy policy has been updated to prohibit diffusers hospital-wide. A PNI blend promoting digestive ease was added to PCH supply. Staff and family educational materials are being updated using FDA-approved language. PCH is collaborating with the hospital aromatherapist and unit educators to train staff on essential oil administration, including PNIs and nontransdermal patches.

Nursing FREE PAPER SESSION 6-COMPLEMENTARY METHODS AND CLOSING REMARKS 24-10-2021 3:00 PM - 4:00 PM

O0300 / #1221 | Validating a Questionnaire to Assess the Use of Complementary Health Approaches in Children with Cancer: Phase I

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agement And Evaluation, Canada, ⁶Hospital for Sick Children, Child Health Evaluative Sciences, Canada

Background and Aims: Complementary health approaches (CHA), such as herbs and yoga, are commonly used by children with cancer (CWC) globally. Recently, our team published a systematic review of CHA questionnaires in pediatrics, and 35 questionnaires were identified. None of these questionnaires were thoroughly validated, but the "Which Health Approaches and Treatments are you using?" (WHAT) questionnaire had evidence of initial validation and could be adapted for CWC. A sequential 4-phased approach is now being used to validate the WHAT questionnaire. The current study is the first phase and aims to reach consensus on the content of the WHAT questionnaire for CWC.

Methods: An electronic Delphi survey was conducted to reach consensus on the domains and items of the WHAT questionnaire for CWC. We invited CWC, their parents, health care providers (HCP), CHA providers and researchers. Parents and CWC were recruited from SickKids and the Ontario Parents Advocating for Children with Cancer group. HCP and CHA providers and researchers were recruited from the International Society of Pediatric Oncology working groups, and Pediatric Complementary and Alternative Medicine Research and Education Network. Based on previous similar research, the target response rate was 17%.

Results: Sixty participants completed the survey (14 CWC, 13 parents, 21 HCP, and 12 CHA providers/researchers). The response rates for CWC/parents, and HCP/CHA providers and researchers were 61% and 31%, respectively. Consensus was reached on all domains and items of the WHAT questionnaire after one Delphi cycle. The domains found to be the most relevant were CWC use of CHA, factors associated with CHA use, perceived impact of CHA use, and communication about CHA use.

Conclusions: Consensus was reached by CWC, parents, HCP, CHA providers and researchers on the content of the WHAT questionnaire. Next steps are further development and then psychometric testing of the adapted WHAT questionnaire in pediatric oncology.

Nursing FREE PAPER SESSION 6-COMPLEMENTARY METHODS AND CLOSING REMARKS 24-10-2021 3:00 PM - 4:00 PM

O0301 / #1074 | Social Robots in Pediatric Oncology: Opinions of Health Care Providers

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Background and Aims: The use of social robots is increasing in pediatric oncology. Research suggests potential positive effects of social

robots for children, but it is unknown how health care providers think about this innovation. We aimed to describe the opinions of health care providers in pediatric oncology regarding social robots and to examine differences in their opinions based on their background characteristics and experience with social robots.

Methods: Health care providers working in pediatric oncology care were approached globally to participate in an online survey. We reported frequencies of answers to multiple choice questions and a thematic approach was used to determine themes in open-end questions. Differences in opinions were studied using logistic regression analyses.

Results: Health care providers' ($n = 286$) first thoughts about social robots were mostly positive (81.0%). They saw value in social robots (83.6%), mostly during hospitalization or procedures and when used for distraction. They were willing to use one (67.8%), but also expected difficulties (67.1%), mainly concerning technical errors. Psychosocial staff members less often saw value in social robots, more often expected difficulties and were less willing to use a social robot than other health care providers ($ps < 0.05$). No other differences were found.

Conclusions: Social robots may be a promising addition to standard care, since health care providers in pediatric oncology were generally positive about this innovation. However, psychosocial staff members were more reluctant. Important potential barriers were defined that should be addressed when wanting to implement social robots in pediatric oncology care.

PPO ORAL

O0302 / #192 | Objective and Subjective Sleep in Children with a Newly Diagnosed Primary Brain Tumor

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Background and Aims: Children with brain tumors are at risk to develop sleep problems. The prevalence of sleep problems and whether they arise early or at a later stage remains unclear, although this information could be used to provide timely interventions. The aim is to examine sleep and contributing factors shortly after brain tumor diagnosis.

Methods: Patients with a primary brain tumor aged 6-16 years old were recruited for a longitudinal study. Inclusion started January 2020 and is expected to be completed September 2021. Sleep was measured for the first time within three months after diagnosis by actigraphy and questionnaires. Actigraphic outcomes were compared to 47 age-matched healthy Dutch children with t-tests. Self-reports and parent-reports (both 6-16 years) of PROMIS questionnaires (Sleep Disturbance (SD) and Sleep Related Impairment (SRI)) were compared to healthy Dutch children using t-tests. T-scores $\geq 1SD$ were considered in

the clinical range. The influence of age and sex on disrupted sleep were explored with linear regression models.

Results: On average, patients ($N=41$, 70% male, 73% response rate, 24% supratentorial tumors, 39% midline tumors, 37% infratentorial tumors) were assessed 1.8 months after diagnosis ($SD=1.1$) at the age of 12.1 years ($SD=2.9$). In patients compared to controls, actigraphy showed shorter sleep onset latency ((SOL), 8 minutes earlier, $p=0.03$). No differences in sleep duration or efficiency were found. Parents reported higher child SD (mean 53.7 vs. 50.0, $p=0.03$) and SRI (mean 54.8 vs. 50.0, $p=0.01$), with 29% and 37% in the clinical range, respectively. No differences in self reports or for age and sex were found.

Conclusions: These preliminary results showed that shorter SOL and higher parent reported child SD and SRI may reflect fatigue, indicating sleep problems may already occur close to diagnosis and should be monitored systematically. The follow-up of this cohort will provide more insight into the development of sleep problems.

PPO ORAL

O0303 / #449 | Self-Reported Emotional Distress in Children with Advanced Cancer: Results of the Pediquet Study

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Background and Aims: Children with cancer experience symptoms of anxiety and depression at much higher rates than healthy children. While emotional distress is associated with lower health-related quality of life in children with cancer, we do not understand the patterns, correlates, and potential causes of these symptoms. This information is critical to develop and implement appropriate treatments. This study sought to describe patient-reported emotional distress in children with advanced cancer and associated factors.

Methods: Children (age at enrollment 2-18; $N=104$) or their parents reported on symptoms weekly to monthly using the PediQUEST Memorial Symptom Assessment Scale (PQ-MSAS) and the Pediatric Quality of Life Inventory 4.0 as part of a randomized controlled trial. The primary outcome was emotional distress (yes/no response to any of the PQ-MSAS emotional distress items (nervousness, sadness, worry)). High distress was defined as at least one reported symptom above the established cut point. Univariate generalized linear mixed models were used to examine associations between any emotional

distress and demographic factors, disease and treatment characteristics, physical symptoms, and quality of life. All models were adjusted for intervention arm.

Results: 920 surveys were reported by the patients or their parents and 913 had complete information and were included in analyses. Emotional distress was reported in 371 (41%) surveys of which 281 (76%) reported high distress. Reporting of any emotional symptom was significantly associated with a change in disease status in the past 10 days ($p=.006$), inpatient admission in the past 10 days ($p=.007$), less time since last disease progression ($p=.002$), reports of pain ($p<.0001$), drowsiness ($p<.0001$), nausea ($p=.03$), and anorexia ($p=.01$), and more difficulty functioning in school ($p=.01$).

Conclusions: Emotional distress is prevalent among children with advanced cancer and is commonly highly distressing. While treating physical symptoms may mitigate some emotional distress, interventions directly targeting psychological symptoms are also warranted.

PPO ORAL

O0304 / #514 | Insomnia, a Sleeping Problem in Adolescents and Young Adults within Pediatric Oncology: Who is at Risk?

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Background and Aims: Insomnia entails difficulty with initiating or maintaining sleep which considerably impacts daily functioning. It is associated with negative health outcomes and impaired quality of life, which poses a health risk for adolescents and young adults who had cancer. Knowing who is at risk enables targeted treatment. This study's aim is to determine the prevalence of insomnia in adolescents and young adults across all cancer diagnoses and associated risk factors.

Methods: Patients (12-30 years old) who were at least 6 months after treatment for diverse childhood cancer types and within 10 years after diagnosis were invited to fill out questionnaires including: the Insomnia Severity Index (ISI-score 0-28) and sociodemographic/medical ques-

tions. Logistic regression was used to evaluate risk factors. Aimed inclusion rate is 500 participants, which will be reached summer 2021.

Results: 382 patients participated thus far (response 52.5%): 49.7% females, mean age 17.2 years, 48.2% hemato-oncology, 30.4% solid tumors, 21.5% neuro-oncology. Overall, 136 (35.6%) reported insomnia symptoms, which varied in severity: 22.3% subthreshold (ISI-score 8-14), 11.8% moderate-severe (ISI score 15-21) and 1.6% severe (ISI-score 22-28). Of those with insomnia, 55.0% would like to receive treatment. Risk factors for insomnia entailed: females (OR 2.49, 95% CI 1.61-3.84), young adults 19-26y (OR 1.67; 95% CI 1.08-2.59) and older age at diagnosis (OR 1.08, 95% CI 1.0-1.16). No association was found with diagnosis or treatment type. In the total group 15.2% used sleep medication, mostly melatonin (12.0%). However, the majority of the sleep medication users (66.7%) still suffered from insomnia.

Conclusions: Insomnia is common after treatment for childhood cancer, independent from cancer diagnosis or treatment type. Females, young adults, and those with an older age at diagnosis seem at risk. Considering the patients' wish for treatment of their sleep problems and the frequent use of sleep medication with seemingly insufficient effects, more efficient treatment options are warranted.

PPO ORAL

O0305 / #1518 | Predictors of Parental Mental Health and Child Quality of Life Among Infants and Toddlers With Cancer

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Background and Aims: Infancy and early childhood is a time of rapid brain development which is foundational for physical and mental well-being throughout the lifespan. However, treatment and parental psychosocial effects on infants and toddlers with cancer are poorly understood. The current study aims to examine how treatment factors (e.g., time since diagnosis) and parental psychosocial factors (e.g., parental stress) impacts quality of life (QOL) longitudinally in young children (0-4 years) diagnosed with cancer.

Methods: Children ($n=19$ male, 54.3%) 0-48 months of age (median 31.06 months) on active cancer treatment (mean time since diagnosis [days] 356.46) and their parents ($n=33$ female, 94.3%) were recruited. Parents completed questionnaires at three time points (baseline[T1], 6 months[T2], 12 months[T3]) related to parent mental health (Brief Symptom Inventory) and stress (Parenting Stress Index), and child QOL (Pediatric Quality of Life Inventory). Data on treatment related factors were collected through medical charts. Demographic information was collected at T1.

Results: A multiple regression model was used to predict child QOL and parental mental health at 12 months using the following predictors: child sex, child age[FSMS1] at T1, parental post-traumatic stress[T1], parental education, and child QOL[T1]. The multiple regression model was significant for predicting child QOL, $F(6, 14) = 232.58$, $p=0.037$,

adj. $R^2=0.390$. QOL at T1 emerged as a significant predictor of QOL at T3 ($p=0.027$) where higher QOL at baseline predicted higher QOL 12 months later.

With respect to parent mental health, the multiple regression model was significant $F(6, 14)=279.38$, $p=0.032$, adj. $R^2=0.406$. Parental post-traumatic stress[T1] ($p = 0.003$) significantly predicted parent mental health where greater post-traumatic stress at T1 predicted worse mental health [FSMS1] at T3.

Conclusions: It is important to identify modifiable targets for intervention to improve the QOL and mental health for families of infants with cancer within the first 12 months of diagnosis.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0001 / #1007 | Cytogenetics in Children with Acute Lymphoblastic Leukemia

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Background and Aims: The karyotype is recognized as a prognostic factor of acute lymphoblastic leukemia in the world. In Morocco, mortality of acute leukemia remains higher than in developed countries. This is due to several factors including toxic mortality and frequent relapse. The aim of our study is to define and classify the chromosomal abnormalities among newly diagnosed children with ALL.

Methods: We reviewed all patients newly diagnosed of childhood ALL from 2006 to 2019 hospitalized in our oncology pediatric unit in Casablanca children's hospital. All the patients were treated by Marall O6 protocol.

Results: Among 189 Moroccan children examined with ALL in this period, 143 (76%) patients yielded results, with 78 (41,2%) showing a normal karyotype and 66 (35%) abnormal karyotype. Number abnormalities were observed in 34 cases (53%) [7 (10,6%) pseudohyperploid, 15 (22.7%) hyperdiploidy and 2 (3%) had diploidy, 1 nearploid (1,5%) 6 trisomy, 1 monosomy, 1supplementary X] Structural abnormalities were observed in 27 (41%), of which 9 (13.6%) cases were translocations and other structural aberrations 4 cases with addition, 9 cases with deletion, 2 dicentric chromosome, 1 duplication and 1 isochromosome were also observed. A complex karyotype was observed in 6 cases (9%). Theses patients were stratified in two groups 19 cases (13.2%) with non-favorable karyotype and 124 case (86.7%) with favorable karyotype.

Conclusions: Our data are in agreement with those published in the literature. This confirms the frequency of cytogenetic abnormalities and their prognostic pertinence.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0002 / #1025 | Clinical and Laboratory Features Associated with Thrombosis in Children with Cancer During Intensive Care Stay

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Background and Aims: Children with cancer develop risk for thrombosis during their treatment. The principal aim of this study is to describe clinical and laboratory test features in children with cancer that develop thrombosis during stay in a pediatric-oncology intensive care unit.

Methods: A retrospective design for the study was established. Patients with diagnosis of thrombosis during their stay in the PICU were identified from October 2014 to January 2021 in a single pediatric oncology center in Mexico. Clinical features, known risks factor and laboratory test were found on the medical record. Descriptive analysis of clinical findings was performed.

Results: Thrombosis prevalence of 1.87% was found in 9 patients of 481 admissions to PICU in a 6 year period. Median age was 10.9 years. Acute lymphoblastic leukemia (ALL) was present in 6 patients (66.6%). Medical diagnosis was found in 7 (77.7%) vs 2 (22.2%) surgical patients. Main site of thrombosis was in central nervous system in 4 patients (44.4%). Thrombosis was developed in less than 7 days in 77.7% of the cases. Main risk factors were presence of central venous line (100%), decrease mobility (88.8%), steroid use (77.7%), age >8 years (66.6%) and diagnosis of infection (66.6%). For patients with ALL high risk diagnosis and L-asparaginase use. D- Dimer, procalcitonin and C- reactive protein levels were elevated in all patients. Currently 4 of 9 patients survived and continue with treatment.

Conclusions: Children with cancer admitted to PICU have increased risk for thrombosis. Prevalence of thrombosis is low, but when is present it could lead to a lethal outcome. Candidate patients for thrombosis risk are those with ALL, central line, infection diagnosis, steroid and L- aspar use, decrease motility and high risk diagnosis with increased D Dimer, CRP and procalcitonin. Multicenter studies are necessary to confirm this risk factors and determine which patients could be benefited from thromboprophylaxis.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0003 / #1037 | Childhood Acute Lymphoblastic Leukemia: Definitive Risk Classification by Early Minimal Residual Disease in the Induction Therapy and Survival in a Hospital in a LMIC

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Background and Aims: Background: Minimal residual disease (DRM) is the best marker to assess the therapeutic response in children with acute lymphoid leukemia (ALL). It has been important to reduce the intensity of treatment in low-risk groups, reducing hospitalizations and deaths due to infectious complications. Aims: to evaluate the hospitalizations and death due to infections related to assistance of children with ALL, according to definitive risk classification on 19th day of treatment. **Methods:** Retrospective cohort including ALL under 18 with ALL monitored at the Pediatric Oncology Service of Instituto Medicina Integral Prof. Fernando Figueira (IMIP) in the period from 2011 to 2018 classified according to MRD 19th day. Patients with standard provisional risk were classified into three groups: very low risk (MRD <0.01%), standard risk (MRD 0.01 to 4.99%) and high risk (MRD 5%) according to MRD. The studied variables were sociodemographic and clinical data. The studied outcomes were hospitalization due infection in the induction of treatment, recurrences and death according to the leukemia subtype and risk classification. The analyzes were performed on STATA 13.1 and involved Student's t-test and association chi-square when relevant.

Results: Three hundred and twenty-four children with ALL were included, of these, 273 (84.3%) had ALL-B and 51 (15.7%) ALL-T. There were 152 hospitalizations due to care-related infections during treatment induction in 87 (69%) of the 126 evaluated patients. The statistical analysis performed showed a lower number of hospitalizations for care-related infections, during treatment induction, relapses and deaths for children classified as very low risk compared to standard and high risk classifications ($p=0.045$).

Conclusions: It is crucial to frequently evaluate chemotherapy protocols to reduce complications resulting from treatment and increase survival.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0004 / #1040 | Epidemiology of the Death by Pediatric Lymphoblastic Leukemia in Brazil

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Background and Aims: Lymphoblastic leukemia is the most common type of cancer in childhood, representing about one third of neoplasms. Although the majority of children with acute lymphoblastic leukemia have a good response to chemotherapy, there are still cases where the disease persists.

Methods: Data were collected using the Mortality Information System and related to information from the Live Born Information System. Data were collected about death by lymphoblastic leukemia in children between 0-14 years old during the period between 2000-2019 in Brazil. The parameters analyzed were: prevalence, region of the country, sex, race and age. The assessment was made by the number of deaths by 100,000 live births.

Results: Lymphoblastic leukemia represented 56.55% of deaths by leukemias, with 7,670 cases. The average number of deaths per year was 383 and a trend has not been observed over the years. The distribution of deaths was higher in the North (16/100,000) and proportional in the other regions, with 13/100,000 in the South, 11/100,000 in the Southeast and 12/100,000 in the Northeast and in the Midwest. The incidence concerning sex was 14/100,000 for boys and 11/100,000 for girls. Yellows and browns had a lower proportion (9/100,000 and 10/100,000, respectively), while the other races had a higher proportion (white and indigenous with 15/100,000 and black with 16/100,000). The number of deaths in children <1 year old (302) was significantly lower compared to 1-4 years old (2,024), 5-9 years old (2,801) and 10-14 years old (2,543).

Conclusions: A higher incidence of deaths by lymphoblastic leukemia was observed in males, over one year of age, and in the North region. The incidence was lower in yellow and brown children. The observed patterns can be a guide for future studies to identify the factors that interfere in the incidence of deaths and possible ways of preventing.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0005 / #1079 | Procalcitonin vs C-Reactive Protein, Biomarkers for Risk Stratification of Febrile Neutropenia among the Children with Acute Leukemia

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Background and Aims: Febrile neutropenia(FN) is a frequently occurring treatment related complication with significant morbidity and mortality for childhood acute leukemia. Early diagnosis and assessment of severity are essential steps for early comprehensive treatment to reduce FN related morbidity and mortality. Biomarkers like C-reactive

protein(CRP) and procalcitonin(PCT) can be used to assess and predict the adverse outcomes in FN. This study is aimed to evaluate the role of procalcitonin and CRP as a biomarker for risk stratification of FN.

Methods: This prospective observational study was conducted in the Department of Paediatric Haematology and Oncology, BSMMU in 30 Children's with acute leukemia aged 1 to 17.9 years with FN. Procalcitonin, CRP, Complete blood count, Blood C/S, Urine C/S and chest xray were evaluated within 24 hours of onset of FN.

Results: Of the 30 patients, 6 patients (20%) and 2 patients (6.7%) showed positive blood and urine culture, respectively. *Escherichia coli* and *klebsiella* were the most frequent organism isolated. The median PCT levels were significantly higher in bacteremic patients than patients without-bacteremia (26.10 μ g/l vs 0.6 μ g/l, P=0.011). The median CRP levels in the bacteremia and without-bacteremia patients were 70.18mg/L and 60.15mg/L, respectively(P = 0.437). The median PCT level in patients who died was 9.71 μ g/l compared to 0.6 μ g/l in patients who improved(P=0.033). There was no differences in median CRP in patients who died and those who improved (P=0.059). The PCT threshold, 0.59 μ g/l, was taken as a cutoff value, with area under the curve (AUC) of 0.840 with sensitivity and specificity of 100% and 95% for predicting bacteremia. AUC for CRP was 0.604 with sensitivity and specificity of 83.3% and 95%.

Conclusions: PCT is a useful biomarker for the prediction of bacteremia with better prognostic efficacy than CRP in patients with FN and can be used as an adjunct for risk-stratification of FN patients.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0006 / #1081 | Risk Factors of Neutropenic Enteropathy During Induction Phase of Treatment of Childhood Acute Lymphoblastic Leukemia

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Background and Aims: Although there has been dramatic progress in the treatment of acute lymphoblastic leukemia but the major challenges remain in control of infection, coagulopathy & relapse. Apart from blood stream & lung infections, gastrointestinal tract is a frequent site of infection especially during profound & prolonged neutropenia and neutropenic enteropathy is a life threatening one causing mortality up to 30%. Objective of this study was to recognize the risk factors of neutropenic enteropathy during induction phase of treatment of childhood acute lymphoblastic leukemia.

Methods: This prospective case control study was conducted in the Department of Pediatric Hematology & Oncology, Bangabandhu Sheikh Mujib Medical University from November 2018 to January 2021. Newly diagnosed ALL children of both sexes with age ranging from 1 to 18 years without any co-morbid condition were included in this study. Total ninety-eight children who developed neutropenia

were analyzed among which 36 patients were in case group who developed enteropathy and 62 patients were in control group who did not developed enteropathy. Proper history & thorough physical examination were carried and complete blood count, serum electrolytes, serum albumin, blood C/S, stool C/S, abdominal X ray, abdominal USG were performed in all cases.

Results: Mucositis (Odd ratio – 252.7, p value <0.001), profound (Odd ratio – 183.0, p value <0.001)& prolonged (Odd ratio – 507.5, p value <0.001) neutropenia (duration >7 days), septicemia (Odd ratio – 488.0, p value <0.001), use of daunorubicin (Odd ratio – 5.37, p value 0.004) were found to be risk factors for developing neutropenic enteropathy in univariate analysis. But in multivariate analysis, prolonged neutropenia and septicemia found to be significant.

Conclusions: Prolonged & profound neutropenia, septicemia, mucositis, use of Daunorubicin were the risk factors for the development of neutropenic enteropathy in univariate analysis. However, prolonged neutropenia and septicemia found to be significant in multivariate analysis

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0007 / #1123 | Translocation T (1:19) in Acute Precursor B-Cell Lymphoblastic Leukemia in Pediatric Patients- Pakistani Population

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Background and Aims: Acute lymphoblastic leukemia (ALL) is one of the commonest pediatric malignancies. The translocation t (1:19) is commonly seen in acute Precursor -B cell (Pre-B) lymphoblastic leukemia, which results in E2A-PBX1 fusion gene. It was once considered as high risk cytogenetic feature associated with poor prognosis but with modified therapies it is no longer considered as one. We analyzed the impact of t (1:19) translocation on the outcomes of Pre B ALL. **Methods:** From January 2012 till January 2018, 450 patients were diagnosed with precursor B-cell acute lymphoblastic leukemia among which 25 patients had translocation t (1:19). The outcomes were compared with patients without any translocation or cytogenetic abnormalities. Overall survival (OS) and event free survival (EFS) were calculated using Kaplan-Meier curve method.

Results: When compared with the control group, no statistical significance was seen in the age of presentation and white blood cell count. The 2-year of over-all survival was 70.8% in the t (1:19) group and 69.5% in the comparison group, whereas event-year survival was 70.8% and 60.0% respectively. Relapse disease was seen in 4 four patients with t (1:19) translocation.

Conclusions: Limited studies are available on the biology and cytogenetics of acute leukemia in Pakistani population, translocation t (1:19) positive patients do not need treatment intensification.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0008 / #113 | Association of MTHFR Gene Polymorphisms and its Influence on Methotrexate Toxicities in Paediatric ALL Patients

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Background and Aims: Methotrexate (MTX) has been an important drug which is widely used in many ALL treatment protocols. MTHFR polymorphisms are widely studied in relation to toxicity of methotrexate. We report study of incidence of MTHFR polymorphisms at our institute and its influence on Methotrexate toxicities.

Methods: Our study included 42 Paediatric ALL patients. MTHFR gene polymorphisms (A1298C & C677T) were detected using Real time PCR done on peripheral blood. Incidence of febrile neutropenia, mucositis, AST, ALT levels, Serum bilirubin & serum creatinine levels were evaluated after 48 hrs of methotrexate infusion.

Results: MTHFR gene polymorphisms were detected in 18 out of 42 patients (42.8%). 28 out of 42 patients had received high dose methotrexate therapy (Methotrexate > 500mg /m²) out of which 10 patients were having MTHFR gene polymorphisms and 18 patients were having no polymorphism. Febrile neutropenia (Grade 3) was seen in 4 patients (40%) with MTHFR gene polymorphism compared to none in patients with no polymorphism (p=0.01). Grade 2 mucositis was seen in 3 patients (30%) of MTHFR gene polymorphisms compared to 1 patient (5%) with no polymorphism (p=0.11), none had grade 3 or more mucositis. AST & ALT enzyme elevation (Grade 1) was seen in 6 patients (60%) with MTHFR gene polymorphism compared to 1 patient (5%) with no polymorphism (p=0.003), none had grade 3 or higher enzyme elevation. Serum bilirubin was elevated (Grade 1) in 1 patient (5%) with MTHFR gene polymorphisms compared to none with no polymorphism (p=0.35). Serum creatinine levels were not elevated in any patient in any patient with or without MTHFR gene polymorphisms.

Conclusions: Our study has showed that in patients with MTHFR gene polymorphisms who are receiving high dose methotrexate therapy, there was statistically significant increased incidence of febrile neutropenia and AST/ALT enzyme elevation compared to patients with no polymorphism.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0009 / #1149 | Different Disorders with The Same Therapy – Outcomes in Children with T Acute Lymphoblastic Leukaemia and Lymphom

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Background and Aims: T-cell lymphoblastic lymphoma (T-LBL) and lymphoblastic leukemia (T-ALL) are malignancies that arise from the transformation of immature precursor T cells. The clinical features of T-LBL and T-ALL will help modifying therapies and improve outcomes in this unique group of children

Methods: We conducted a retrospective study in our department at Apollo Cancer Centre in Chennai, India from 2004 to 2019 over a 15 year period. All children from 1 to 18 years with a diagnosis of T-ALL or T-LBL were included. Data collected from medical records included the clinical profile at presentation, response to induction therapy, regimen related toxicity and disease free survival using a uniform chemotherapy protocol. All children received a UK MRC ALL 2003 regimen B protocol with four doses of high dose methotrexate

Results: A total of 73/478 (15%) children were diagnosed to have T lymphoblastic leukaemia / lymphoma. Mediastinal mass was the predominant presentation in 35/73 (47.9%) children. T LBL was documented in 17/73 (23.2%) children. 49/56 (87.5%) children with T-ALL and all 17 children with T-LBL demonstrated optimal response to induction. Regimen related events included cerebral thrombosis (3), grade 4 pancreatitis (2) and fungal sepsis (2) and avascular necrosis of the femoral head (2). The relapse rates were high in the T-ALL group with 19/56 (33%) lost to progressive disease and 3/17 (17.6%) in the T-LBL group. Mortality was related to refractory seizures in 1 child, graft versus host disease in 2 children and glioblastoma multiforme in 1 child. The overall survival is 33/56 (58.9%) in T-ALL and 14/17 (82.5%) in T-LBL group after a minimum follow up of 24 months

Conclusions: Malignancies of immature T cells carry a poor prognosis compared to precursor B lymphocytes. T-LBL tends to present in a dramatic manner but have a superior survival compared to children with T-ALL

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0010 / #115 | Mixed-Phenotype Acute Leukemia in Children: An Observational Study for Clinical Characteristics and Outcome

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Background and Aims: Background: Mixed phenotypic acute leukemia (MPAL) in pediatrics is a rare subtype of acute leukemia that fails to be classified by assigning to certain lineage and its treatment

remains largely without consensus. Aim: To assess the clinicopathological characteristics and outcomes of MPAL patients diagnosed based on the European Group for Immunological Characterization of Acute Leukemias (EGIL) or the 2008/2016 World Health Organization (WHO) classification.

Methods: Patients and methods: A retrospective descriptive study of 42 patients, less than 18 years old, newly diagnosed with mixed phenotypic leukemia during the period from July 2007 till December 2017.

Results: Eighty-three percent of the patients (35/42) fulfilled both EGIL and WHO criteria while 6 patients were fulfilling EGIL score only. The immunophenotyping was T/Myeloid in 24 patients (57.1%) and B/Myeloid in 16 (38.1%). Three patients had MLL gene rearrangement, 3 patients had monosomy 7 (2 associated with Philadelphia chromosome) and 8 patients with FLT3-ITD >0.4. Two patients died before start of chemotherapy, ten patients (25%) started ALL induction chemotherapy and all achieved complete remission (CR) with no induction deaths, while out of 30 patients (75%) who started AML induction therapy, five (16.6%) died during induction and 17 (56.7%) achieved CR. Four patients out of the eight with induction failure were shifted to ALL therapy and 10 patients received maintenance ALL therapy after ending AML treatment. The 5-year event-free survival, relapse-free survival (RFS) and overall survival were $56.7\% \pm 8\%$, $76\% \pm 8\%$, and $61\% \pm 8\%$ respectively; with a median follow-up duration of 31.6 months. FLT3-ITD showed a significant higher relapse rate having a 5-year RFS of $41.7\% \pm 22\%$ compared with $88.9\% \pm 7\%$ for the wild-type, $P = .036$.

Conclusions: Conclusion: This study showed the superiority of ALL induction therapy having higher remission and lower induction mortality rates. Showing high incidence of FLT3-ITD in our patients and their poor outcome, FLT3 inhibitor should be considered.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0011 / #1193 | Treating Acute Lymphoblastic Leukemia in a Developing Country: Triumphs and Ongoing Challenges

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Background and Aims: Current outcomes for Acute Lymphoblastic Leukemia (ALL) exceed 80%. Optimal management in India is fraught with numerous challenges. We report outcomes from a tertiary referral center in North India over an 18 year period (2003-2020).

Methods: Data of all consecutive patients 1-18 years of age who presented to the center was analyzed retrospectively. Data pertaining to demographic profile, laboratory findings, immunophenotype, cytogenetics, presence of CNS disease was documented. Patients were classified as Standard Risk (SR) or High Risk (HR) based on age, WCC at diagnosis, immunophenotype and cytogenetics. Treatment was given in risk-stratified, response adapted manner. Cranial RT was given only

to patients who had CNS involvement at diagnosis. Response to treatment in the form of steroid responsiveness and End of induction bone marrow examination morphologically and MRD (when available) was documented. Primary events of interest were refractory disease, toxic deaths and relapse.

Results: A total of 372 patients were diagnosed as ALL during this period. Of these 15 patients abandoned/ opted to take treatment at another center. Data for 357 patients was analyzed. The median age at diagnosis was 5.5 years, with a male to female ratio of 2:1. Out of these, 300 (84%), 53 (15%) and 4 (1%) patients had B-Lineage, T-Lineage and biphenotypic ALL respectively. Eleven (3%) patients had CNS disease at presentation. There were 17 (4.8%) toxic deaths. A total of 33 (9.2%) patients relapsed. The 5 year EFS and OS of the entire cohort is 79.3% and 81.8%. The EFS and OS of patients classified as SR and HR were 86.3%, 90%, 73.1% and 74.4% respectively.

Conclusions: There continues to be gender disparity in children with ALL presenting for treatment. With current strategies, optimal outcomes could be achieved for patients with SR disease. However, outcomes need to be optimized for patients with HR disease with incorporation of novel strategies.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0012 / #1203 | Circrna Involvement in Childhood Acute Lymphoblastic Leukemia

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Background and Aims: Childhood acute lymphoblastic leukemia (ALL) represents one of the most frequent causes of death by disease in children. Deeper insights in the genetic basis of the disease and their influence on the progress and treatment response are critical to discovering new prognostic markers and novel therapeutic targets that can open new doors in precision medicine. In recent years, a new type of non-coding RNAs, circRNA, has emerged as a promising biomarker for cancer diagnosis, however little is known about their roles in ALL development and treatment response. Thus, we investigate and characterize circRNAs in ALL.

Methods: We have performed a transcriptome analysis (RNA-Seq) of 129 ALL patients (104 B-ALL and 25 T-ALL) and 10 controls from two independent cohorts. CircExplorer was used to quantify, annotate, and characterize circRNAs transcription. Differential expression analyses between subtypes and their controls were performed using DESeq2.

Results: A total of 79,349 circRNAs were identified in the discovery cohort. Their size was on average 820 bp, with a mean of 5 exons per circRNA. The overall number of circRNA variants per gene was 14,6 circRNAs. Unsupervised clustering analysis showed that circRNAs could properly stratify cALL subtypes. Differential circRNA expression between ALL and controls, showed 390 circRNAs upregulated and 336 downregulated. 50% of circRNAs were unique for specific subtypes. A total of 254 circRNAs from the discovery cohort were validated in an independent cohort, which characterize subtype-specific expression profiles. Furthermore, functional predictions identified circRNA-microRNA as well as circRNA-protein interactions, suggesting different mechanism of action in molecular subtypes of childhood ALL.

Conclusions: We have characterized and identified subtype-specific circRNAs in ALL. In addition to protein-coding genes and other non-coding RNAs, we have shown that circRNA expression profiles can accurately classify ALL subtypes.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0013 / #1208 | Assessment of Copy Number Alterations in T- Cell Acute Lymphoblastic Leukemia in Indian Population

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Background and Aims: T-cell acute lymphoblastic leukemia (T-ALL) is a molecularly heterogeneous disease and comprises distinct subtypes defined by the chromosomal rearrangements that usually involve one of the T cell receptor (TCR), but non-TCR driven translocation, including whole chromosome and subchromosomal gain and losses are the common contributors of disease progression and have demonstrated its utility for patient prognosis in T-ALL.

Methods: Multiplex ligation-dependent probe amplification (MLPA) reactions were performed on 128 newly diagnosed pediatric T-ALL patients using the SALSA P383-A2 MLPA probemix, containing 56 probes, for detection of deletion and duplication in 13 different chromosomes regions affected by recurrent CNAs in T-ALL: STIL-TAL1, LEF1, CASP8AP2, MYB, EZH2, MLLT3+MTAP+CDKN2A/B, NUP214-ABL1, PTEN, LMO1, LMO2, NF1+SUZ12, PTPN2, and PHF6. For this 100 ng input genomic DNA was denatured and hybridized. The reactions, including 12 negative control samples, were performed according to the instructions. The amplified probes were analyzed using Coffalyser.net software. After intrasample and intersample normalizations, copy number status at each locus was estimated. Relative peak ratios between 0.8 and 1.2 were considered normal, while values below or above indicated losses or gains of genetic material, respectively.

Results: Highest percentage of deletion events were observed in PHF6 (69.53%), ARHGEF6 (69.28%), SH21A (60.15%) and CDKN2A (46.87%), while PTPN2 (10.15%), NUP214 (5.46%), EZH2 (6.25%)

exhibited highest duplication events. Out of 128 T-ALL patients, 88.28% of patients exhibited at least one deletion event, while 73.43% of patients exhibited at least one duplication. A total of 92.96% of patients exhibited CNAs (either deletion or duplication). STIL-TAL1, NUP214-ABL1, and LMO2-RAG2 fusion were observed in 3.90%, 2.34%, and 0.78% of T-ALL patients respectively.

Conclusions: The observed molecular heterogeneity in T-ALL may provide molecular basis for variations observed in clinical response in T-ALL.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0014 / #1214 | Prognostic Factors and Outcomes of Infant Acute Lymphoblastic Leukemia (ALL), Hypodiploid ALL and Mixed-Phenotype Acute Leukemia (MPAL) In Canada: A Population-Based Study

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Background and Aims: The prevalence, prognostic factors and outcomes of children diagnosed with infant acute lymphoblastic leukemia (ALL), hypodiploid ALL and mixed-phenotype acute leukemia (MPAL) in Canada are unknown. The main objective of this study was to describe the prevalence, prognostic factors and outcomes of three rare ALL subtypes in Canada.

Methods: This was a retrospective study using the Cancer in Young People-Canada (CYP-C) database, a nationwide population-based childhood cancer surveillance program. Patient characteristics were presented for all patients and by subtype. Event-free survival (EFS) and overall survival (OS) were described by the Kaplan-Meier method and compared using the log-rank test for all patients and stratified by subtype. Univariate and multivariable Cox proportional hazard

regression models of EFS and OS were performed, overall and stratified by subtype.

Results: Among 2626 children aged 0-14 years diagnosed with B-ALL between 2001 to 2018, 227 (8.6%) patients were identified to be infant ALL (n=139; 5.3%), hypodiploid ALL (n=43; 1.6%) or MPAL (n=45; 1.7%). The median age and white blood cell (WBC) at diagnosis were 7 (4-10) months and $105.5 (23.8-358.1) \times 10^9/L$ for infant ALL; 7 (3-14) years and $12.8 (5.0-26.1) \times 10^9/L$ for hypodiploid ALL; 7 (4-12) years and $12.9 (3.8-54.4) \times 10^9/L$ for MPAL. The 5-year EFS and OS were significantly worse in the infant ALL subgroup ($38.9 \pm 4.6\%$ and $58.9 \pm 4.6\%$) compared to hypodiploid ALL ($63.0 \pm 8.6\%$; $p=0.0006$ and $75.2 \pm 7.2\%$; $p=0.0273$) and MPAL ($58.8 \pm 8.1\%$; $p=0.0025$ and $74.9 \pm 7.0\%$; $p=0.0353$) respectively. For the entire cohort, presenting $WBC \geq 50 \times 10^9/L$ retained significance in multivariable analyses of EFS (HR: 2.4, 95% CI 1.6-3.6) and OS (HR: 2.5, 95% CI 1.3-4.8).

Conclusions: Infant ALL, hypodiploid ALL and MPAL represent 8.6% of childhood ALL in Canada and are associated with poor prognosis. $WBC \geq 50 \times 10^9/L$ at diagnosis represents an independent adverse factor of EFS/OS, irrespective of ALL subtype.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0015 / #1238 | Synergistic Activity of Combined Targeting of BCL-2 with MCL-1 or BCL-XL in B-Cell Precursor All

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Background and Aims: Targeting cell death pathways by BH3-mimetics has become a promising treatment approach in different types of cancer. However, given the abundance of different anti-apoptotic proteins in B-cell precursor acute lymphoblastic leukemia (BCP-ALL) cells, targeting single proteins may not be sufficient. In this study, we evaluated the activity of BH3-mimetics selectively antagonizing BCL-XL (A-1331852) and MCL-1 (S63845) alone and in combination with the BCL-2-specific antagonist venetoclax in BCP-ALL.

Methods: Half-maximal effective concentrations (EC_{50}) of BH3-mimetics were analyzed by cell death assays in cell lines (N=7) and patient-derived xenograft (pdx) samples (N=27). BH3 profiling and immunoprecipitation were used to analyze BCL-2 family proteins. Combination effects of BH3-mimetics were evaluated in a BCP-ALL pdx mouse model.

Results: We found heterogeneous activity of venetoclax, A-1331852 and S63845 in BCP-ALL. Interestingly, several venetoclax-resistant leukemias were sensitive to MCL-1- and/or BCL-XL-antagonism. Functional analysis revealed that venetoclax-treated ALL cells switched dependence from BCL-2 to MCL-1 and BCL-XL. Our mechanistic studies showed alternated binding of the apoptosis activator BIM

between BCL-2 and MCL-1 upon exposure of BCP-ALL cells to venetoclax or S63845. Importantly, antagonizing both anti-apoptotic proteins by venetoclax and S63845 led to disruption of both BIM/BCL-2 and BIM/MCL-1 complexes leading to BIM-release and mitochondrial apoptosis signaling. Accordingly, synergistic cell death induction was identified upon combined antagonism of BCL-2 and MCL-1 in BCP-ALL cell lines and pdx samples. Moreover, we found clearly enhanced anti-leukemia activity in mice transplanted with a high-risk KMT2A-MLLT1 positive infant pro-B ALL, indicating synergistic anti-leukemia activity of combined BCL-2 and MCL-1 inhibition in vivo.

Conclusions: Taken together, our data show switched dependence of BCP-ALL cells on different BCL-2 family proteins upon exposure to selective BH3-mimetics. Attacking this mutual interplay by co-inhibition of BCL-2 with MCL-1 or BCL-XL showed synergistic activity in primary patient-derived xenograft samples ex vivo and in vivo.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0016 / #1247 | Mondo-A Maintains Aggressiveness in Common Acute Lymphoblastic Leukemia (B-ALL) by Modulating the Metabolic Stress Response and can be Targeted by a Designer Drug

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Background and Aims: Cancer cells are characterized by rapid proliferation. Therefore, they must adapt to proliferation-induced metabolic stress. One mechanism to accomplish this is to reprogram gene expression in a metabolism dependent manner. MondoA (also known as MLXIP), a member of the MYC interactome, is a metabolic sensor. The role of MondoA in malignancy is not fully understood.

Methods: We found MondoA overexpression associated with lower survival in pediatric B-ALL. Using CRISPR/Cas9 and RNA interference, we observed that MondoA mediates the transformational capacity of B-ALL cells in vitro and the malignant potential in vivo. MondoA expression in patient data sets correlates with a decrease of metabolic pathways. Experimentally, MondoA repressed activity of the Krebs cycle by compensating metabolic stress in via limitation of oxidative phosphorylation through decreased PDH activity. That mechanism was successfully targeted by a designed low-molecular weight competitive inhibitor (Vega & Kelly) interfering with binding of MondoA to the promoter of its canonical downstream target.

Results: Glutamine withdrawal greatly increased apoptosis in the absence of MondoA and highlights the inability to alleviate the metabolic stress associated with the loss of MondoA. Surprisingly, MondoA also induced glutamine anaplerosis. Of further interest, metabolomic analysis also revealed MondoA induction of serine biosynthesis from the 3-phosphoglycerate intermediate. In addition, MondoA suppressed MYC-targets and enhanced pharmacological MYC inhibition. ChIP-Seq analysis revealed redistribution of MYC DNA binding towards MondoA binding sites upon loss of MondoA.

Conclusions: In summary, MondoA provides leukemia cells with cellular plasticity, allowing cells to switch rapidly to glycolysis, depending on the microenvironmental context. In the absence of MondoA, cells become highly glutamine dependent, warranting further research into combining MondoA inactivation with glutamine restriction as a therapeutic strategy. Our results provide new insight into the role of MondoA in pediatric B-ALL and support the hypothesis that inhibition of MondoA offers a therapeutic option which warrants further investigation.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0017 / #1288 | Cerebrovascular Disease (CVD) Preceding B-Acute Lymphoblastic Leukemia Diagnosis, Case Report

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Background and Aims: CVD precedes or follows cancer diagnosis in adults. In children, malignancies increase risk of thrombosis/CVD, CVD is associated ALL's complications or treatment but not preceding diagnosis. We aim to document the first case of CVD preceding ALL diagnosis.

Methods: Case report

Results: We describe a 4-year-old girl, previously healthy, presenting with bilateral basal ganglia's stroke, MRI/CT confirmed narrowing distal left internal carotid artery (ICA) and middle cerebral artery (MCA) M1 segment. Diagnostic work-up showed: EKG first degree AV-block,

weak ANA, elevated ESR, anti-cardiolipin and ANCA, dsDNA, RF, Infectious and Thrombosis panel negative. She received prednisone for 8 weeks for possible vasculitis. Six months later, she was diagnosed with standard risk B-ALL, CNS negative. MRI showed new areas of stenosis M1 segment both MCA. She received induction as per COG-AALL1731 (including PEGaspargase). No neurological/thrombotic complications during induction. Post-induction, she was randomized to SR-Avg-B-ALL (neutral cytogenetics and MRD negative) on AALL1731 (included blinatumomab #2 cycles). She presented with TIA 5-months later. MRA/CTA showed white matter changes in the left frontal area/focal stenosis bilateral terminal ICAs, proximal M1-A1 segments. Angiogram confirmed less pronounced/very mild stenosis of the supra-clinoid ICA bilaterally with extension into the proximal M1-A1 segments. Genetic testing negative for Moya-Moya. She developed a new left frontal stroke 7-months later (during delayed intensification, 3 weeks after Pegaspargase), coagulopathy was resolving at this point. MRA (vessel wall) showed wall enhancement within the supraclinoid ICA, left and right A1 and both proximal MCAs. Stable arterial stenoses. She was treated with prednisone, low dose aspirin and Cyclophosphamide with the presumptive diagnosis of vasculitis. Infectious and rheumatological work-up, EKG, Echocardiogram, MRA chest/abdomen normal. No further strokes during maintenance.

Conclusions: We described a child with CVD preceding and overlapping with ALL diagnosis and treatment, etiology is still unclear. Multiple center collaboration is required to further understand these disorders in children.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0018 / #1304 | Influence of Genetic Factors on Long-Term Treatment-Related Cardiovascular Complications in Survivors of Childhood Acute Lymphoblastic Leukemia: The Petale Study

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Background and Aims: A significant number of survivors of childhood acute lymphoblastic leukemia (ALL) suffer from treatment-related late adverse effects, with cardiovascular complications being one of the leading causes of treatment-related morbidity and mortality. To further address this issue, using two different approaches - a candidate genes and an exome-wide association study, we analyzed cardiovascular complications in relation to common and rare genetic variants derived through whole-exome sequencing data from childhood ALL survivors (PETALE cohort). Additionally, top-ranking associations were assessed through stratified and multivariate analyses.

Methods: Analysis of common variants was performed using allelic ratio, chi-square, Fisher's test, and logistic regression in PLINK and SPSS software. The effect of rare variants was evaluated using the SKAT-O test.

Results: Significant associations were detected between left ventricle ejection fraction (LVEF) and fractional shortening (FS), and common variants in the TTN gene ($p \leq 0.0007$). An association between LVEF and rare variants enrichment in the NOS1 and ABCG2 of doxorubicin pathway ($p = 0.002$ and $p = 0.003$) was also noted, whereas exome-wide approach detected contribution of rare variants in the NOD2 gene in relation to LVEF ($p = 2 \times 10^{-6}$) and NOD2 and ZNF267 in relation to FS ($p = 1 \times 10^{-6}$ and $p = 2 \times 10^{-6}$, respectively). Following stratification according to risk groups, we also detected the modulatory effect of the CBR1, ABCC5, ABCG2, and AKR1C3 genes. The association with common TTN variants in the independent St. Jude Lifetime Cohort was not significant. For the 46 rare variants requested for replication, none of the 8 non-monomorphic rare variants passing SJLIFE quality control were replicated.

Conclusions: Current findings may help to understand the influence of genetic factors on long-term treatment-related cardiovascular complications. Further studies are needed to confirm whether the described genetic markers may be useful in identifying patients at increased risk of these complications.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0019 / #1324 | A Pre-Clinical Platform to Assess Efficacy for Precision Molecular Targeted Therapies in Pediatric Acute Lymphoblastic Leukemia

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Background and Aims: Precision medicine in pediatric cancers is predicated on targeting druggable pathways identified via genomic sequencing of patient-derived tumours. However, significant challenges remain in predicting efficacy that includes choice between several targets, and between multiple agents for each target. Here, we outline an integrative approach to assess the comparative efficacy for

targeting genetic lesions identified in pediatric acute lymphoblastic leukemia (ALL).

Methods: Patient bone marrow (BM) leukemia (24 B-ALL and 10 T-ALL) were sourced from BC Children's Hospital Biobank. Tumour gene variants were characterized with the OncoPrint™ Childhood Cancer Research Assay. BM-ALL were expanded as xenografts (PDX) in NSG mice and viably cryo-preserved. Ex vivo drug assays were conducted with inhibitors targeting MEK, CDK4/6 and PARP. Relative chemosensitivity was assessed by comparing IC₅₀ or EC₅₀ for each agent and PDX-ALL pairing.

Results: Of 15 B-ALL with MEK targetable variants, 5 exhibited enhanced sensitivity (~100 fold) to the MEK inhibitors, trametinib and selumetinib. In contrast, all 15 B-ALL used as non-MEK targeted controls did not exhibit enhanced sensitivity. Sensitivity to CDK4/6 inhibitors of B-ALL with loss of CDKN2A ($n = 17$) was not significantly different when compared to non-targeted controls ($n = 14$). Similarly, differential sensitivity to PARP inhibitors (6 targeted, 14 controls) was not observed. Other variants assessed included targets of RTK, JAK, mTOR and AZH2 ($n \leq 2$ for each class), with results remaining inconclusive at this time.

Conclusions: The efficacy of precision molecular targeted therapies may be improved with pre-clinical functional testing for confirmed sensitivity of patient derived tumours. Such preparative evaluation can be accommodated prior to tumour relapse, in particular for gene variants that are maintained through tumour progression.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0020 / #1330 | Association Between NUDT15 Genetic Polymorphism and Mercaptopurine Myelotoxicity in Syrian Children with Acute Lymphoblastic Leukemia

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Background and Aims: NUDT15 genetic polymorphism is known to be associated with frequent hematopoietic toxicities during childhood acute lymphoblastic leukemia (ALL) mercaptopurine therapy. The aim of this study is to test this association in a group of Syrian ALL children, in addition to quantify the prevalence of the corresponding SNP in patient population.

Methods: Between January 2018 and May 2020, ALL children treated at BASMA Pediatric Oncology Unit and at Al Bairouni Hospital in Damascus were enrolled. Low- and intermediate-risk patients were treated as per the EORTC 58951 protocol, while high-risk patients

were treated as per the St Jude TOTXV protocol. DNA was extracted from patient EDTA-anticoagulated peripheral blood samples. NUDT15 specific primers were used to amplify a 450 bp long amplicon containing the rs116855232 site, then amplicons were sequenced to define patient genotypes and allele frequency. Mercaptopurine dose intensity was calculated during maintenance therapy as the percentage of the tolerated dose to the planned dose to maintain $WBC > 2000/\mu l$. Early-onset leukopenia was defined as $WBC < 2000$ during the first 8 weeks of maintenance therapy.

Results: A total of 92 patients were enrolled (63% low-risk, 25% intermediate-risk and 12% high-risk). The TC genotype was found in 5 patients (5.4%) and the TT genotype was found in 1 patient (1.1%), with allele frequency ($C=0.962$ and $T=0.038$). The mercaptopurine dose intensity was 82%, 57% and 5% for the genotypes CC, TC and TT respectively ($P: 0.01$). A significant association was found between early onset neutropenia and NUDT15 polymorphism (TC or TT), OR: 6.8 (95% CI: 1.22-38.11, $P: 0.04$). None of the patients with NUDT15 polymorphism had significant liver transaminases elevation (SGPT/SGOT $> 10\times$ normal level)

Conclusions: Our study confirms that NUDT15 rs116855232 polymorphism is associated with mercaptopurine hematopoietic toxicity but not with hepatotoxicity in a population of Syrian children with ALL. The incidence seems to be slightly higher than neighboring countries.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0021 / #1331 | **Acute Lymphoblastic Leukaemia (ALL)**

Induction Mortality: An Indicator of Quality Care in a Paediatric Oncology Unit

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Background and Aims: The induction phase of chemotherapy has a mortality rate of around 1-2%. 1/10th of all ALL-associated deaths and 50% of all treatment related deaths are accounted to induction mortality. Induction mortality is the highest in the first week of induction treatment and the induction mortality rate is also higher in children who require mechanical ventilation. The aim is to assess induction mortality and to identify factors affecting induction mortality in Paediatric oncology units in developing countries

Methods: This is a retrospective study, patients who have expired during the induction phase were included in this study. Their data was collected from 2019 Jan -2020 Dec from Pediatric Oncology Units (POU) of Indus hospital & health care network.

Results: In total, 50 cases of induction mortality males were 28 (56%) and the mean age was 9.6 ± 5.6 . B-cell were 39 (78%) in which HR were 30 (77%) while SR were 9 (23%) and T cell were 11 (22%) in which HR were 10 (91%) and SR were (9%). Positive Blood culture were 25 (50%). The mean CRP was 144.4 ± 58.1 and 50% of patient stays in hospital were > 14 days in which 23 (46%) patients were ventilated. Infections were found in 35 (70%), bleeding in 8 (16%) and drug toxicity in 7 (14%). In 2019 induction mortality was 26 (52%) while in 2020 it was 24 (48%).

Conclusions: The death rate during induction therapy is high in our setup. The major cause of death in the present study was infection, followed by bleeding. Increased rate of infections was seen particularly in low income families. A good antibiotic cover should be given to treat infections and most importantly, patient's caretaker should be educated regarding complete patient care.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0022 / #1346 | **Blast Size Index: A Flow Cytometry Derived Novel Parameter to Determine Aneuploidy in Pediatric Precursor B-Lineage Acute Lymphoblastic Leukemia Patients**

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Background and Aims: Nearly 40% of pediatric precursor B-lineage acute lymphoblastic leukemia (B-ALL) patients have aneuploidies of prognostic relevance. Aneuploidies are commonly identified through either laborious conventional cytogenetics (CG) or resource-consuming flow cytometric (FCM) DNA index (DI) techniques. Both these techniques are not widely available across low and middle-income countries (LMIC). In a flow cytometer, the size of cells is expressed as forward scatter (FSC). The size of a lymphoblast is largely determined by its nuclear content. In the current study, we have introduced and evaluated the utility of a no-cost and novel FCM parameter called blast size index (BSI) which can determine the ploidy of lymphoblasts without using any nucleic acid binding dyes.

Methods: 139 pediatric B-ALL patients were included in the study. CG-ploidy and FCM-ploidy were assessed using conventional cytogenetics and FxCycle™ Violet dye-based DI, respectively. BSI was calculated as the ratio between the FSC-geometric mean (FSC-GM) of blasts and lymphocytes within the sample. The correlation between DI and BSI was analyzed, and BSI across all common aneuploidies was determined.

Results: FCM-ploidy was 80% concordant with CG-ploidy. There was a significant correlation between DNA content, i.e., DI and size (BSI) of the lymphoblasts ($r = .833$, $p < .001$). The mean ($\pm$ S.D) BSI of diploid, low-hyperdiploid, and high-hyperdiploid blasts were $1.08 (\pm 0.08)$, $1.16 (\pm 0.12)$, and $1.32 (\pm 0.10)$, respectively. There were significant

differences in BSI between diploid vs low-hyperdiploid ($p = .011$), diploid vs high-hyperdiploid ($p < .001$) and low- vs high-hyperdiploid ($p < .001$) blasts.

Conclusions: BSI is a simple, novel, and no-cost parameter that can be calculated even with a single laser flow cytometer. BSI can be effectively used as a surrogate for aneuploidy assessment in LMIC where sophisticated flow cytometry and cytogenetics laboratories are not widely available.

E-Poster Topic: *Disease Orientated / ASO1 Haematology / ASO1.a Acute Lymphoblastic Leukaemia*

PV0023 / #1413 | T-Lymphocyte Activation and Pd-1 Expression Profile in Childhood B-Cell Precursor Acute Lymphoblastic Leukemia at Diagnosis and After Induction Remission Treatment

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Background and Aims: Suppressive effects of the immune system were demonstrated in hematological neoplasms, by mechanisms like the induction of immunosuppressive molecules. This process induces exhaustion in T lymphocytes, characterized by the expression of inhibitory molecules and the loss of effector function, which may be correlated with the response to treatment. Our aim is to evaluate the association between T lymphocyte activation and PD-1 expression and their correlation with BCP-ALL treatment.

Methods: We have analyzed bone marrow samples from patients diagnosed with BCP-ALL, 28 samples at diagnosis, and 16 samples on day 78 of treatment (D78). A panel of monoclonal antibodies was used: CD4-FITC, CD8-PE, CD45RA PE-Cy7, CD279 PerCPCy5.5, CD3-APC, and CD28 APC-H7. The acquisition of 1,000,000 events was performed in FACSCanto II and analyzed using Infinicyt software. Statistical analyzes were performed using the GraphPad Prism software and results with p -value < 0.05 were considered statistically significant.

Results: We observed at D78 an inversion on the proportion of CD4+/CD8+ T cells ($p = 0.01$) and a decrease in CD4+ Naive T cells ($p = 0.04$) in contrast to an increase in Early Memory ($p = 0.01$). In addition, an increase in CD8+ Naive T cells at D78 ($p = 0.01$). Regarding PD-1 expression, we observed that CD4+ and CD8+ T cells from patients evaluated at diagnosis expressed high levels of PD-1, which remained high at D78 ($p = 0.001$). Simultaneously, we saw that patients with MRD+ results presented at diagnosis higher percentages of cells expressing PD-1 at intermediate levels than patients with MRD- results ($p = 0.01$) and that patients with high-risk NCI presented a higher percentage of T cells expressing high levels of PD-1 compared to standard risk ($p = 0.04$).

Conclusions: We have found evidence that changes in the immune system, especially at T-lymphocytes PD-1 expression, are associated with the therapeutic response of pediatric patients with BCP-ALL.

E-Poster Topic: *Disease Orientated / ASO1 Haematology / ASO1.a Acute Lymphoblastic Leukaemia*

PV0024 / #1440 | Aberrant Expression of Nuclear Enriched Abundant Transcript 1 Predicts Poor Prognosis in Pediatric B-Cell Precursor Acute Lymphoblastic Leukemia

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Background and Aims: B-cell precursor Acute Lymphoblastic Leukemia (BCP-ALL) represent more than 80% of B-ALL in childhood. Although, the BCP-ALL is associated with a good outcome, and the cure rates approaching 90%, it remains a leading cause of cancer mortality. Therefore, identifying novel prognostic and therapeutic targets become an urgent goal nowadays. Recent innovative approaches have identified non-coding RNAs (lnc-RNA) as novel molecular biomarkers that regulate various biological and pathological processes. Nuclear Enriched Abundant Transcript 1 (NEAT1) is a lnc-RNA which has recently identified as crucial component of nuclear paraspeckles. Notably, aberrant expression of NEAT1 has been reported in malignancy with distant pathogenic roles. Meanwhile, many solid cancers exhibit up-regulation of NEAT1, downregulation was reported in leukemias. The lnc-NEAT1 is direct target of TP53, by which it contributes to its tumor-suppressor function, via sponge of miR-34a. We evaluated the impact of miR-34a, lnc-NEAT1 and TP53 apoptotic pathway in pediatric BCP-ALL, and to evaluate its prognostic value. Additionally, to find out the correlation of lnc-NEAT1 and miR-34a as well as the TP53 gene.

Methods: The gene expression level of lnc-NEAT1, miR-34a and TP53 was measured in 70 peripheral blood (PB) samples of children with BCP-ALL at diagnosis and normalized to 50 healthy controls.

Results: A significant difference between BCP-ALL patients and healthy control for the expression of TP53, lnc-NEAT1 and miR-34a was detected ($p < 0.01$). It was found that TP53 is up regulated by 1.4 folds in BCP-ALL; whereas, lnc-NEAT1 expression is decreased by 3.1 folds, and miR-34a was downregulated by 6.25 folds. Additionally, the expression profile of the three biomarkers was significantly correlated to each other.

Conclusions: Lower expression level of lnc-NEAT1 predicts poor prognosis in pediatric BCP-ALL and it could serve as decision guide for selection of proper therapy, and therapeutic target in order improve disease outcome.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0025 / #1443 | Longitudinal Assessment of Phenotypic Heterogeneity and Subpopulation Trajectories in B-ALL Disease Progression Samples

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Background and Aims: With the continued poor outcome of relapsed pediatric acute lymphoblastic leukemia (ALL), patient-specific approaches for early relapse prediction and novel therapeutic interventions are urgently needed. A leukemia-associated immunophenotypic pattern is often defined for each patient at diagnosis, but the poorly understood phenotypic shifts that occur during chemotherapy and with ALL progression have limited the evaluation of subpopulation trajectories. Here, we used patient-derived xenografts (PDX) to define stable phenotypic subpopulations within the bulk leukemia.

Methods: Primary ALL samples collected at diagnosis and/or relapse were obtained from the BC Children's Hospital Biobank and engrafted into non-conditioned NSG mice. Organs harvested from PDX were evaluated using cross-standardized clinical flow cytometry protocols and instrument settings to permit direct comparison of marker expression on patient- and PDX-derived blasts.

Results: Combining the clinical immunophenotyping depth and analytical sensitivity of MRD diagnostics with high-dimensional computational analysis algorithms, we demonstrate the phenotypic stability of ALL subpopulations in patients and PDX. The phenotypic clarity and minor population expansion achieved by xenografting enabled identification of rare subpopulations that were undetected in the clinical flow evaluation. Furthermore, comparison of subpopulation hierarchies in matched diagnosis and relapse samples revealed an outgrowth of previously minor populations that is consistent with the clonal fluctuations described in genomics studies.

Conclusions: These results support that the phenotypic shifts associated with clinical disease progression may reflect outgrowth of distinct minor subpopulations rather than a general plasticity of marker expressions on the bulk population. By incorporating the immunophenotypic complexity of leukemic populations, our novel analytical platform could greatly expand the utility of PDX for investigating the biology of ALL progression, enhancing MRD monitoring, and assessing therapies directed at elimination of the relapse-driving blast subpopulations.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0026 / #1451 | Deregulation of Long Non-Coding RNAs in Childhood Acute Lymphoblastic Leukemia

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Background and Aims: Identification of genomic biomarkers predicting response to anticancer agents, as well as recognition of novel therapeutic targets and strategies are the central research problems of cancer precision medicine, especially, in childhood acute lymphoblastic leukemia (ALL), where refractory and relapsing patients still represent one of the most frequent cause of death from cancer in pediatrics. In this context, we aimed to establish the potential of long non-coding RNAs (lncRNAs) as new therapeutic targets in ALL through expression profiling.

Methods: An exhaustive systematic search to identify studies examining lncRNA expression in ALL was done on PubMed and Scopus. Additionally, a transcriptome analysis (RNA-Seq) in a cohort of 64 Spanish patients (54 B-cell precursor (BCP) ALL, 10 T-ALL) and 6 controls (3 pre-B CD19+ and 3 pre-T CD3+) was performed. Expression levels by gene was determined with the HTseq-count software, using gene models from Ensembl75 combined with (non-overlapping) lncRNA transcripts provided in Casero et al. Differential expression analyses between subtypes and their controls were performed using DESeq2.

Results: The systematic review included a total of 23 articles showing 4 signatures defined by 32 lncRNAs, related to diagnosis, subtype classification, prognosis, and treatment response in ALL. RNAseq results displayed a total of 199 lncRNAs differentially expressed in BCP-ALL and 166 in T-ALL versus controls. Out of the 20 most significant deregulated lncRNAs, 4 were common for both BCP-ALL and T-ALL and 8 showed the same dysregulation as that shown in the literature. No association was found between the variants detected in those lncRNAs and the expression changes.

Conclusions: The analysis of the expression of lncRNAs could allow to improve the knowledge of the mechanisms involved in the pathogenesis of ALL.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0027 / #1466 | Identification of Molecular Disorders in Pediatric Patients with Acute Leukemia (AL)

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Background and Aims: Acute leukemias are the most common childhood cancer and comprise an heterogeneous variety of hematologic malignancies characterized by accelerated clonal proliferation of immature hematopoietic cells that shows different levels of differentiation, therefore genomic tools are very useful to identify associated cytogenetic alterations. However, in Mexico, molecular studies and cytogenetic classification are limited. The purpose of this paper is to identify cytogenetic alterations in pediatric population with AL at Pediatric Hospital of Sinaloa

Methods: This is a descriptive, cross-sectional and prospective study, that included fifty-seven bone marrow or peripheral blood samples from patients less than 18 years old with newly diagnosis of AL, admitted to the Pediatric Hospital of Sinaloa between April 2018 and March 2021, those samples were processed and analyzed by FISH and karyotype. The FISH study was performed according to the lineage reported in the Immunophenotype. Probe panels were used for the translocations with highest incidence in pediatric population.

Results: Fifty-seven patients with AL were included, of which 47 patients (82%) had diagnosis of Acute Lymphoblastic Leukemia (ALL) and 10 patients (18%) Acute Myeloid Leukemia (AML). In Fifty-eight percent (n=33) of patients were identified cytogenetic alterations by FISH, the most frequent alterations identified in ALL were: t(12;21) (p13;q22), iAMP21 and t(9;22) (q34;q11) and in AML were: CBFβ, t(15;17) (q24;q21) and t(8;21) (q22;q22). Based on karyotype, 21 patients (36%) had chromosomal alterations, the most frequent was hyperdiploidy.

Conclusions: The results of the identification of molecular alterations represent an important resource for risk stratification and targeted treatment in patients with AL. It is important to expand the studies for detection of molecular alterations in leukemias and to know the epidemiology in the Mexican pediatric population.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0028 / #1467 | NUDT15: Incidence, Clinical Significance of NUDT15 Enzyme Polymorphisms in a London Population of Paediatric All Patients Treated at Great Ormond Street Hospital, London

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Background and Aims: NUDT15 polymorphisms have shown reduced metabolism activity of mercaptopurine (6MP) by 75-100% causing poor tolerance of 6MP. In patients with NUDT15 polymorphism the aim is to confirm if it caused the expected level of haematological toxicity when checked against patients who are wildtype. London has an extremely diverse population with the majority being people of a mixed and Asian ethnicity. NUDT15 polymorphisms are prevalent in Asian ancestry. Is there a prevalence and correlation of NUDT15 polymorphisms in paediatric patients treated for ALL in London (at Great Ormond Street Hospital) and ethnicity?

Methods: 83 subjects who completed treatment were chosen to retrospectively test correlation between ethnicity and NUDT15 polymorphism. In addition to prevalence, clinical effect on 6MP tolerance during maintenance was also explored in relation to NUDT15 variants detected. In 52 patients, weekly neutrophil results and maximum tolerated 6MP dose as a percentage of the maximum total dose was obtained for the duration of MC period.

Results: Found no significant trend between ethnicity and NUDT15 polymorphisms. Prevalence for NUDT15 polymorphism is between 13-26%, with a notable occurrence in Asian ethnicity subjects (P=0.009). Two clinically relevant variants were detected showing subjects only tolerated a mean 6MP dose of approximately 40% and 60% throughout maintenance. Whilst the groups tested were too small to elicit statistical significance, clinically such a reduced tolerance to 6MP is certainly worth exploring. On average these 2 groups had great fluctuations in neutrophils over the maintenance phase suggestive of poor understanding of the affect of inherited polymorphisms.

Conclusions: In conclusion NUDT15 variant and ethnicity proved no trend, suggesting the need to test all Paediatric ALL patients in London, as prevalence is similar to that of TPMT variants, which are screened as standard of care. Those with a recognised variant tolerated clinically lower 6MP doses than expected.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0029 / #1474 | Very Early Remission and Increased Apoptosis with the use of Pentoxifylline in Children with Acute Lymphoblastic Leukemia

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Background and Aims: Among patients with acute lymphoblastic leukemia, there are still cases of relapse and death secondary to

resistance to chemotherapy or apoptosis. One of the mechanisms is the activation of the IKK/NF- κ B signaling pathway that leads to the expression of genes that can interfere with apoptosis. Pentoxifylline (PTX) can block the phosphorylation of serine 32 of I κ B, thus preventing the NF- κ B activity. To evaluate the minimal residual disease, survival, and modification in the percentage of apoptosis and senescence of mononuclear cells of the bone marrow with the use of pentoxifylline administered during the induction therapy in patients with ALL.

Methods: Phase II, controlled versus placebo, randomized, double-blind clinical trial. Pediatric patients with ALL during induction therapy were assigned to the pentoxifylline or placebo group, with 22 patients each. Bone marrow aspirates were performed on day-1, day-8, day-15, and day-22. We performed flow cytometry minimal residual disease, determination of apoptosis (annexin-V/propidium iodide test), and senescence (β -galactosidase activity) in each sample.

Results: On day-8, patients in the pentoxifylline group showed a higher percentage of apoptotic cells of $41.3\% \pm 6.8$ vs 19.4 ± 4.3 in the placebo group ($p=0.029$); also on day-15 with $35.0\% \pm 4.1$ vs $14.2\% \pm 1.6$ showed in the placebo group ($p < 0.01$). No difference was observed in the senescence. On day-8, patients in the PTX group showed MRD of $0.25\% \pm 0.18$ vs $18.2\% \pm 6.6$ ($p < 0.01$) in the placebo group; on day-15 PTX group showed MRD of $0.09\% \pm 0.07$ vs $1.4\% \pm 1.1$ ($p = 0.02$). In the entire cohort, on day-8 patients with MRD $< 0.01\%$ had 3-year OS of 81.6% vs 58.3% ($p=0.03$); on day-15 patients with MRD $< 0.01\%$ had 3-year OS of 77.9% vs 54.5% ($p=0.03$).

Conclusions: Patients in the PTX group showed MRD $< 0.01\%$ earlier, on day-8 and 15. Furthermore, they presented a higher percentage of apoptotic cells without showing an increase in senescence. In the entire cohort, patients with MRD $< 0.01\%$ earlier showed better survival.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0030 / #148 | A Prospective Multicentric Randomized Double-Blind Trial on the Safety and Efficacy of Ketamine Mouthwash in Control of Pediatric Chemotherapy-Induced Mucositis

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Background and Aims: Oral mucositis is a debilitating adverse effect of chemotherapy treatment. Ulcerations define oral mucositis in oral and gastrointestinal tract mucosa. The occurrence is multifactorial,

including age, malignancy type, chemotherapeutic agents, and intensity of administered chemotherapy regime. This prospective randomized, double-blind placebo-controlled trial was targeted to assess the efficacy and safety of ketamine mouthwash in decreasing chemotherapy-induced severe OM pain in children.

Methods: This prospective, double-blind, randomized placebo-controlled study (NCT01566448) was conducted on 40 Patients aged between 7 to 14 years old with grade 3 or 4 oral mucositis. Patients were randomized to receive ketamine mouthwash (4 mg/mL) or placebo three times daily as needed. Pain score was assessed at baseline and one hour after and every 4 hours till 72 hours after administration. The outcome variables of efficacy are defined as a decrease in pain score; Safety outcome is defined as any adverse events related to ketamine mouth wash.

Results: Result: A statistically significant reduction in pain scores in comparison to baseline and one hour after and second and third days detected ($p < 0.0001$, $p = 0.0003$). Pain scores significantly reduced at two and third days of the assessment, respectively ($p = 0.0006$, $p = 0.0001$). No patients showed adverse effects related to ketamine administration.

Conclusions: Differences were detected between the two groups' reduction of Mucositis pain intensity. Oral tolerance improved significantly on the second and third day among the Ketamine group compared with the placebo group. Noticeable statistical differences were seen for the onset of effect and duration of effect in the Ketamine group in all three study days. Ketamine mouthwashes caused a clinically significant decrease in the pain scores and revealed an acceptable safety profile. Hence it could be an initiator of a new era in the treatment of severe mucositis.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0031 / #1489 | Clinical Presentation and Outcomes of Pediatric Acute Lymphoblastic Leukemia: The Experience from a Tertiary Care Center in Mexico

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Background and Aims: Acute lymphoblastic leukemia is the most common cancer in childhood in México; the national overall survival for children treated between 2008 and 2014 was 49.6%. This study aimed to document survival at a tertiary care center and identify the factors associated with lower survival.

Methods: This retrospective cohort study included all patients under 18 years of age treated with adapted Total XV therapy between 2011 to 2015 at the tertiary care center. Survival analysis was performed using SPSS.

Results: Eligibility criteria identified 211 patients; the mean follow-up was 4.7 years. Abandonment of therapy occurred in 17 patients (8%). The median age at diagnosis was 7 years; 31% were >10 years old and 2.3% were <1-year-old. At diagnosis, 20.3% of the cases had hyperleukocytosis, 89% were CNS1, 1.8% CNS2, and 9% CNS3. The OS was 68.5% and EFS was 65.9%. For standard-risk patients by NCI criteria, the OS was 84.4% and an EFS was 81.1%. Of the 211 patients, 61 died (31.4%); 17 (8.7%) died during induction, 12 (5.7%) died of sepsis, 18 (9.2%) died in remission, and 23 (10.9%) died due to relapse. Day 19 MRD was <0.01% in 78.6%. Among the 40 relapses (20%), most were isolated CNS relapse (9%). The factors associated with lower EFS were age, karyotype, hyperleukocytosis, sepsis during induction, and day 19 MRD; in the multivariate analysis, the last three variables remained statistically significant.

Conclusions: The OS and EFS were 68.5% and 65.9% respectively, lower than those reported in developed countries but higher than the national average. The OS and EFS for standard-risk patients were 84.4% and 81.1% respectively. Sepsis during induction, treatment abandonment, and relapse were important contributors to lower survival. Day 19 MRD serves as a valuable prognostic factor in our population.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0032 / #1508 | Cytogenetics and Molecular Alterations in Pediatric T-ALL

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Background and Aims: T-cell Acute Lymphoid Leukemia is an aggressive hematological malignancy, representing 15% of childhood ALL. Currently, specific genetic abnormalities with importance in the diagnosis and prognosis of are described. Cytogenetic and molecular tools have identified oncogenes involved in the pathogenesis of disease. This study aimed to evaluate the clinical-genetic characteristics of T-ALL, between 0-18 years old, admitted at Center of Pediatric Oncohematology, from 2004 to 2021.

Methods: Thirty nine patients with T-ALL were studied by cytogenetics: G-Banding, FISH with HOX11L2(TLX3), HOX11(TLX1), TCR-AD break-apart probes and BCR-ABL single fusion (to ABL1 gene amplification), MCB (chromosomes 6,11,17,19) and BACs (TCRB, TAL2) and by PCR techniques to KMT2A-MLLT3, KMT2A-MLLT10, KMT2A-MLLT1, SIL-TAL1, HOX11(TLX1) and HOX11L2(TLX3) rearrangements.

Results: Average age was 11 years, 46% were older than 12 years, the majority male (2.25:1). Classical cytogenetics showed normal karyotypes in 29.6%; numerical alterations in 37% (tetraploidy - three, triploidy - one, trisomies - five cases); structural changes were found in

40.7%, being five translocations (three non-recurring- t(3;13), t(17;19); t(6;11;17), and two t(11;14)). FISH showed-TLX3 rearranged in 25% and two deleted. Three patients had TCR- $\alpha\beta$ rearranged (two due to t(11;14)), five- SIL-TAL1 and one HOX11. ABL1 amplification was negative in all the 20 patients studied. Rearrangement by PCR was performed on 25 patients (three had HOX11L2, one HOX11, five SIL-TAL1). The overall survival was 69.2%, high-risk patients were: 3-12 year old males, WBC count above $50 \times 10^3/\text{mm}^3$ and thrombocytopenia $<20,000/\text{mm}^3$. Among 12 patients dead, five had normal karyotype and one trisomy 9. Six patients had structural alterations: two-t(11;14), and t(17;19), t(3;13),7q-, Xq isochromosome one of each. **Conclusions:** This study demonstrated the importance of molecular studies in T-ALL. Patients with t(11;14), t(17;19) and t(3;13) had early recurrence and died. As in the literature, the prognosis of TLX3 rearrangement is still doubtful (four -alive and one death).

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0033 / #1528 | Acquired Venetoclax-Insensitivity in B-Cell Precursor Acute Lymphoblastic Leukemia is Characterized by High MCL-1 and Can Be Overcome by Co-Targeting

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Background and Aims: Deregulation of cell death is a hallmark of many cancers and contributes to leukemogenesis and treatment failure in B-cell precursor acute lymphoblastic leukemia (BCP-ALL). While pro-apoptotic BCL-2 homology domain 3 (BH3) proteins induce cellular death, apoptosis induction is counter-regulated by anti-apoptotic molecules like BCL-2. Selective inhibition of BCL-2 by Venetoclax (VEN) shows high activity in relapsed/refractory CLL and promising results in preclinical and clinical studies in ALL. But despite high activity, resistance can be acquired over time. In this study, we modeled VEN resistance in BCP-ALL and investigated underlying mechanisms in order to identify potential strategies to overcome VEN insensitivity. **Methods:** Starting from the BCP-ALL cell line RS4;11, five parallel VEN insensitive lines were generated. After continuous exposure to increasing concentrations of VEN over time, all lines displayed increasing EC₅₀ values from 4 nM to 26.2 μM indicating VEN insensitivity.

Results: Further characterization showed that VEN insensitive lines showed no growth advantage and equal cell death response to conventional chemotherapeutic agents when compared to control lines. Neither mutations nor changes in expression of the target molecule BCL-2 were found. Interestingly MCL-1, another anti-apoptotic regulator was significantly upregulated. In line, investigating the functional role of different BCL-2 family members by BH3 profiling, we identified clearly reduced BCL-2 - but increased MCL-1 dependency upon

VEN insensitivity. Co-targeting BCL-2 and MCL-1 with VEN and the MCL-1 inhibitor S63845 synergistically induced cell death. Mechanistically, precipitation analyses showed reciprocal binding/release of pro-apoptotic BIM to/from BCL-2 and MCL-1.

Conclusions: Taken together, we show that acquired VEN resistance in BCP-ALL is characterized by upregulated expression of counter-regulatory MCL-1 sequestering BCL-2 released BIM which can be overcome by simultaneous inhibition of BCL-2 and MCL-1.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0034 / #155 | Management of Lymphoblastic Chloromas of The Central Nervous System as a Relapse of Acute Lymphoblastic Leukemia in Pediatric Patients Treated with Stem Cell Transplantation

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Background and Aims: Acute Lymphoblastic Leukemia (ALL) is a common cancer in pediatric patients. Stem Cell Transplantation (SCT) is considered a curative treatment; however, a small subset of patients may experience a bone marrow relapse or CNS recurrence as a solid tumor known as a lymphoblastic chloroma. These lesions may occur in isolation or associated with overlying but non-communicating soft tissue lesions, which requires its own biopsy and evaluation. We review the literature and discuss the optimal management for patients with these unique recurrences.

Methods: Clinical presentation: A nine-year-old female with multiple CNS and systemic relapses of ALL who underwent SCT complicated by graft versus host disease (GVHD), presented to our Children's Hospital complaining of headaches, myalgias and frontal scalp swelling. MRI demonstrated a 4.8cm enhancing right frontal lobe hemorrhagic mass, as well as an overlying, non-contiguous soft tissue mass. The intervening bone was intact without evidence for invasive disease. Given the wide differential for these lesions, a craniotomy was performed for tissue diagnosis and debulking, as well as an excision of extracranial mass via the same scalp incision. Pathology confirmed that the tumors were two separate chloromas. Postoperatively, a novel treatment plan was implemented that included systemic and intrathecal chemotherapy, focal proton beam radiotherapy (PBRT), as well as systemic CAR T-cell therapy.

Results: Not Applicable

Conclusions: ALL can present as a CNS lymphoblastic chloroma, even in patients previously treated with a SCT. Of note, GVHD may also influence chloroma occurrence in these cases. Patients with CNS relapse should be screened for chloromas in other parts of the body and vice versa. CNS chloromas are treated with subtotal surgical resection and adjuvant therapy, and when possible, the en-bloc resection of the soft tissue mass within the same operative corridor. Novel treatments

for this patient additionally included PBRT and CAR T-cell therapy, as described.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0035 / #1570 | Real World Indications and use of Chimeric Antigen Receptor T Cell Therapy in Pediatric Patients with Relapsed or High-Risk Leukemia

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Background and Aims: The Pediatric Real World CAR Consortium (PRWCC) consists of 30 oncology centers across the United States that provide CAR T cell therapy to pediatric patients with relapsed and refractory leukemia. We aimed to study the referral patterns for patients treated with tisagenlecleucel (Kymriah) to understand utilization in the pediatric real-world setting and characterize patients referred with upfront refractory disease and treated in first relapse.

Methods: We reviewed 185 patients treated across 15 centers to analyze treatment histories and clinical outcomes. Our objective was to characterize pathways for tisagenlecleucel use in the pediatric B-ALL setting and examine overall-survival (OS), event-free-survival (EFS), duration of remission (DOR) and duration of B-cell aplasia (DBA).

Results: Of the 30 patients treated due to upfront refractory disease, 7 patients (23%) met standard definition of primary refractory disease with $\geq 5\%$ bone marrow lymphoblasts at end of induction, 23 patients (77%) had $\leq 5\%$ lymphoblasts detectable by flow cytometry. Of the 67 patients infused upon first relapse, indications for CAR referral spanned: chemo-refractory disease (N=54, 81%), contraindication to hematopoietic stem cell transplantation (HSCT) (N=15, 22%), relapse after HSCT (N=9, 13%) and contraindication to chemotherapy (N=1, 1.5%). Patients in 2nd or greater relapse had a range of relapses prior to receiving CAR T cell therapy (Figure). Outcomes of patients treated in first relapse were comparable to those in 2nd or greater relapse. Duration of B cell aplasia was longer in patients treated with upfront refractory disease (p=0.03).

Conclusions: Definitions of primary refractory disease in the real-world setting expand beyond morphologic detection. The majority of upfront refractory tisagenlecleucel referrals (77%) had EO1 lymphoblast levels of $\leq 5\%$ and disease detectable by flow cytometry alone. We identify that patients treated in first relapse achieve comparable survival outcomes to patients treated due to upfront refractory disease or disease in 2nd or greater relapse.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0036 / #1572 | Successful One-Bag Rapid Desensitization to Erwinia Asparaginase

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Background and Aims: Asparaginase is a critical component of therapy for acute lymphoblastic leukemia (ALL). Adverse effects, including hypersensitivity reactions (HSR) and immune-mediated drug clearance, can limit its use, negatively impacting patient outcomes. Some

patients, who are intolerant of both pegaspargase (Escherichia coli-derived) and Erwinia asparaginase (Erwinia chrysanthemi-derived) can successfully complete desensitization to pegaspargase. However, others with ongoing HSR or silent asparaginase inactivation have previously required discontinuation of asparaginase, leading to inferior outcomes. In developing an Erwinia asparaginase desensitization protocol, the aim is to overcome this challenge for a patient following HSR to both forms of asparaginase and silent inactivation following pegaspargase desensitization.

Methods: To avoid wasting a drug that is chronically on drug shortage and to combat a 4-hour drug expiration, we designed a 10-step, 185-minute desensitization regimen. With premedications of ondansetron, methylprednisolone, diphenhydramine, montelukast, and famotidine, our patient received Erwinia asparaginase 60,000 units (24,490 unit/m²) prepared in 500 mL in a single-bag, rapid-titration protocol. The infusion initiated at 0.5 mL/hr with rate titrations every 15 minutes to a maximum rate of 400 mL/hr, which was replicated for each Erwinia asparaginase dose.

Results: This patient tolerated Erwinia asparaginase desensitization without HSR or inactivation. At 48 hours post-Erwinia asparaginase administration, asparaginase enzyme activity remained therapeutic with levels of 0.18, 0.15, 0.104, 0.105, and 0.111, meeting the goal of a trough > 0.1 IU/mL. After the sixth dose, a 72-hour level resulted at 0.023 IU/mL, confirming that this patient required every 48-hour dosing versus standard Monday, Wednesday, Friday dosing.

Conclusions: We successfully developed a one-bag, rapid desensitization protocol for Erwinia asparaginase based on modifications of other published chemotherapy desensitization regimens. Desensitization to Erwinia asparaginase using this protocol was accomplished without HSR and more importantly, with maintenance of appropriate asparaginase enzyme activity, despite prior HSR to both formulations and inactivation following pegaspargase desensitization.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0037 / #1621 | First Wave of Covid-19 Pandemic and Health Care Delivery in Greece: Results from the Nationwide Registry of Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST)

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Background and Aims: The epidemiology of COVID-19 among children with cancer is not well known. We aimed at assessing the impact on timely diagnosis-registration and health care delivery of children with cancer given the severe restrictive measures implemented during the first wave.

Methods: We compared data from active registration by the Nationwide Registry for Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST) for six types of cancer (leukemias, lymphomas, brain tumors, neuroblastomas, Wilms tumors and liver tumors) during January-August 2020 vs. the respective average of the 5-year period 2015-19. Alongside, interviews with health personnel of collaborating pediatric cancer departments were conducted.

Results: The total number of cases registered during the first 8-month period of 2020 (131) showed no deficit vs. the 130 expected (average 2015-19); specifically, 54 leukemia cases were diagnosed vs. 49.4 expected. Thereafter, a peak was shown in July followed by a deficit during the August vacation. Treating physicians reported high burden of extra administrative work load and voluntary training of families in tele-medical support. Considerable routine pediatric visits were postponed or cancelled; indeed, diagnosis was delayed in a boy (unilateral stage III Wilms tumor at diagnosis) born with unilateral polycystic kidney, whose suggested ultrasound was postponed until after lockdown. Likewise, parental fear of exposure to COVID-19 may have delayed diagnosis (death the day after admission from a second malignancy, Acute Myeloid Leukemia) in a 14-year-old girl with sarcoma.

Conclusions: Fluctuations in childhood cancer registration possibly reflect diverse practices of parents and physicians in health care delivery in Greece, a country considered as a "success story" for maintaining relatively low infection rates by imposing lockdown at the beginning of the pandemic. The changing health care delivery remains a challenge for the protection of this fragile population exquisitely exposed to the indirect effects of the COVID-19 pandemic.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0038 / #1623 | Covid-19 Among Children with Cancer in Greece (2020): Results from the Nationwide Registry of Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST)

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Background and Aims: The impact of coronavirus disease 2019 (COVID-19) varies by age and chronic disease group. We aimed to assess the magnitude and severity of COVID-19 among Children with Cancer (CwC) in Greece.

Methods: An electronic survey entailing demographic characteristics, cancer type, COVID-19 infection, treatment, and outcome information among CwC diagnosed at ages 0-14 years (1/1/2010-31/12/2020) and alive until 31/12/2020 was electronically addressed (November 2020) to members of treating groups and complementary sources collaborating with the Nationwide Registry of Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST) and to referral centers of Pediatric Clinics for severe COVID-19 cases representing all seven Medical Schools across the country.

Results: No COVID-19 cases were reported during the first eight months of 2020. During the last four months, however, among the prevalent CwC cases, fifteen were reported with SARS-CoV-2 (median age: 9.8, range 4.9-18.6 years; 7/15 boys), 12 living in Northern Greece. Although the total estimated COVID-19 infection rate (6.5/thousand CwC) was low, the infection rate among CwC in Northern Greece was over four times higher (28.5/thousand CwC), as the region experienced the heavier disease burden during the second wave of the pandemic. Four children with ALL were receiving maintenance treatment, which was interrupted for 2-weeks. Interruption/short delay of treatment was also reported in one patient with low grade glioma and another one with osteosarcoma. Six were childhood cancer survivors (Hodgkin's disease N=2, ALL N=2 and neuroblastoma N=2). Household contact was the source of COVID-19 infection in 14 cases. Fever was the main symptom in 8/10 patients presenting with symptoms, and the course of COVID-19 infection was mild and self-limited. One child was hospitalized because of fever with neutropenia. No adverse outcome was detected in this case series.

Conclusions: Further surveillance of COVID-19 in CwC is required to assess the direct and indirect effects of the pandemic in this vulnerable population

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0039 / #1626 | Effect of Virtual Reality-Based Exercise Approaches on Quality of Sleep in Children with Acute Lymphoblastic Leukemia: A Randomized Controlled Trial

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Background and Aims: Sleep plays a pivotal role both in the normal growth and development and also for psychological well-being of healthy children. Sleep therefore is more important for a child suffering from a chronic illness such as cancer. Sleep disturbances are common in patients with cancer. The effect of exercise interventions to improve sleep quality is controversial. Our aim was to determine the changes in sleep quality and to investigate the effect of virtual reality-based exercise (VRBE) approaches on sleep in patients with acute lymphoblastic leukemia (ALL) after treatment.

Methods: This randomized controlled trial was carried out on 38 children who were subdivided into two groups; group I-(exercise group) included 17 children (11 ALL and 6 healthy siblings) who applied VRBE in 2 days/ a week, 45 minutes of each sessions for 12 weeks and group II-(control group) included 21 children (13 ALL and 8 healthy siblings) who were managed with supportive measures. Patients were evaluated for sleep disorders with polysomnography (PSG) and "Children's Sleep Habit Questionnaire(CSHQ)" before and after intervention. The VRBE comprised of aerobic exercise in different four games which called free jogging, hula hoop, step, and step plus by the Nintendo Wii Fit Plus.

Results: Before exercise at initial evaluation, ALL patients and healthy siblings had similar quality of sleep in terms of PSG and CSHQ findings. After exercise intervention, when ALL patients and healthy siblings are combined as an "exercise group", total time asleep improved statistically significant and they had more time spent in deep sleep (stage 3) after exercise. Respiratory disturbance index (RDI) of hypopnea, apnea/hypopnea index and number of apnea improved significantly after exercise.

Conclusions: Cancer can have an impact on sleep quality and the exercise interventions are useful for children with cancer. Novel types of exercises (VRBE) have positive effects on sleep quality in children with ALL.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0040 / #1684 | NKG2D CAR T Cells as Therapeutic Tool for Acute Pediatric Leukemia

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Background and Aims: Acute leukemia is the most frequent cancer in children. Interactions between Natural Killer Group 2D (NKG2D) receptor, expressed in cytotoxic immune cells, and its ligands

(NKG2DL), that are upregulated in leukemic blasts, are important for anti-leukemia immune surveillance. Nevertheless, ligand shedding by tumor cells may occur, which reduces NKG2DL expression and may cause NKG2D downregulation. In this study, we analyzed the efficacy of NKG2D-CAR memory T cells against pediatric acute leukemia and explored the negative impact of soluble NKG2DL (sNKG2DL) on NKG2D-CAR expression and cytotoxicity.

Methods: The expression of NKG2DL was analyzed in leukemia cell lines and in blasts from leukemia patients at different disease's status by flow cytometry. Cytotoxicity of NKG2D-CAR T cells against leukemia cells was evaluated in vitro by performing conventional 4-hours Europium-TDA assays. NOD.Cg-Prkdc^{scid}Il2rg^{tm1Wjl}/SzJ (NSG) mice were engrafted with 1x10⁶ T-ALL Jurkat-luc cells and treated with single or multiple infusion of NKG2D-CAR T cells. Leukemia progression was monitored weekly. sNKG2DL concentration was measured in the sera of leukemia patients by ELISA. To evaluate the effect on cytotoxicity and NKG2D-CAR expression, NKG2D-CAR T cells were cultured with different concentrations of sNKG2DL.

Results: NKG2DL were heterogeneously expressed in leukemia cell lines and leukemic blasts. NKG2D-CAR T cells were cytotoxic against leukemia cell lines, being myeloid and T-ALL cells more susceptible than B-ALL, and primary leukemic blasts, with specific lysis over 25%. In a murine model of human T-ALL, leukemia progression was temporarily controlled in treated mice, especially when multiple infusions were administered. The sera of patients showed higher levels of sNKG2DL than healthy donors. Only supraphysiological levels of sNKG2DL impaired NKG2D-CAR expression and NKG2D-CAR T cells' cytotoxicity.

Conclusions: NKG2D-CAR memory T cells are cytotoxic against pediatric leukemia cells in vitro and in vivo, and thus could be a novel therapeutic approach for children suffering from acute leukemia.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0041 / #1711 | Methotrexate-Induced Complications in Childhood Cancer: A Report from Armenia

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Background and Aims: Methotrexate (MTX) belongs to antimetabolites and is frequently used in a wide range of treatment regimens of childhood malignancies. MTX-induced toxicity (MTXIT) is characterized by mild to severe neurologic, renal, dermatologic toxicities, mucositis, nausea and vomiting. The endpoint of the study is to identify the incidence and severity of MTXIT among children with ALL and NHL

receiving treatment at the Clinic of Chemotherapy of Muratsan Hospital Complex of Yerevan State Medical University between 2008-2018, and at the Hematology Center after R. Yeolyan between 2018-2020.

Methods: Retrospectively the data of 98 patients (≤18 y/o) with NHL and ALL were retrieved from medical records including demographic features, clinical presentation, dose of MTX. The severity of the toxicity is evaluated according to the National Cancer Institute Common Toxicity Criteria.

Results: From the cohort 72.7% were male, median age was 6.84 and 58.1% were patients with ALL. All patients received MTX of whom 78 (79.6%) received 1g/m², 11 (11.22%) 2g/m² and 9 (9.18%) 5g/m². The 86 (87.8%) patients developed oral mucositis (Grade 1-2) and only 9 (9.1%) had Grade 3-4, 30 (30.6%) patients had also cutaneous complications, 25 (25.5%) had nausea and vomiting, 2 of them (2.04%) had renal complications. The plasma MTX level monitoring was not available till 2018, so MTX administered at a dose of 1.0g/m² and only one, who was diagnosed with primary CNS lymphoma received a high dose MTX (5g/m²). From 2018, the MTX plasma level monitoring is available. Supportive treatment including folate rescue therapy, aggressive intravenous hydration, urine alkalization were performed to all patients.

Conclusions: As the study was retrospective there were definitely unreported cases of MTXIT. However, our data showed that the majority of patients developed MTXIT during their treatment. Non-availability of monitoring of MTX plasma level was a significant problem and fortunately, it has been available in Armenia since 2018.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0042 / #1729 | Evaluation of The Effect of Bisphosphonate Treatment on Symptomatic Vertebral Fractures in Children Treated for Acute Lymphoblastic Leukemia

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Background and Aims: Vertebral fractures (VF) in children with acute lymphoblastic leukemia (ALL) are often symptomatic and life-impairing. Prevalence of children with VF at ALL diagnosis has been reported as 16% with peak incidence occurring during the first year after diagnosis. Moreover, only 15.8% of these patients will have vertebral reshaping 24 months after ALL diagnosis. Intravenous bisphosphonates (BP) can increase bone density and decrease bone pain. Our aim was to describe the bone health trajectory of a cohort of children with symptomatic VF treated with BP during their ALL treatment.

Methods: This retrospective cohort study included children treated with DFC111-001 and DFC116-001 protocols. As per institution

protocol, lateral spine X-rays were done within the first month of diagnosis, annually, and if new back pain occurred during the treatment period. Patients with symptomatic VF after induction treatment were treated with IV BP. VF were graded according to the Genant classification.

Results: 139 children (52% male, 5.2 years -IQR 3.3-10.5-) were included. Prevalence of patients with VF at ALL diagnosis was 10%. Sixteen patients with 24 symptomatic VF were treated with BP for a median duration of 18 months (IQR 13.8-21). Among BP-treated patients, six (37.5%) presented VF at diagnosis, while the first VF for the remainder patients occurred within the first year after diagnosis. After BP initiation, patient-reported pain subsided in all children. One incidental VF 6 months after BP initiation occurred in one patient. At the end of ALL treatment, 46.7% of BP-treated patients had completely reshaped VF. Among those with sustained VF at 24 months of diagnosis, 50% had improved Genant grading.

Conclusions: We report preliminary results from the first study exploring VF trajectory in children with ALL treated with BP for symptomatic VF. Prospective trials are needed to confirm if BP influence VF reshaping in this population.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0043 / #1736 | Integrated Dissection of BCP-ALL Regulatory Landscape Reveals Enhancer Implication in Disease Progression

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Background and Aims: The rapidly expanding characterization of genetic events occurring in cancer is generally not accompanied by a systematic assessment of patients' epigenome. Acute Lymphoblastic Leukemia (ALL) is the primary cause of death from hematological disease in children and new approaches particularly effective in relapsed patients constitute a significant emergency in clinical practice. Here we show that chromatin accessibility dynamics play a major regulatory role contributing to ALL heterogeneity and sustain disease evolution.

Methods: Tissues from Healthy donors (N=6) and BCP-ALL patients currently treated at Bambino Gesù Children's Hospital at time of Primary onset (N=11), Remission (N=7) or Relapse (N=9) were CD19+ magnetically sorted and we generated genome-wide accessibility pro-

files using the Assay for Transposase-Accessible Chromatin (ATAC-seq) at the coverage of 50-60 million reads of each sample.

Results: Differential analysis of Healthy profiles and Primary tissues gives evidence of dynamic shifts in the 2 states' chromatin layout. We identified 688 regulatory regions being downregulated and 758 upregulated during cancer early stages which, recapitulate the characteristics of ALL interrogating the Disease Ontology database and gene ontology (Pval< 10⁻⁵). In addition, we demonstrate that regulatory activity dynamically changes during cancer progression. Relapsed patients are dominated by regulatory elements (N≈6000) originally represented at diagnosis that shrink under endocrine treatments and subsequently re-expand, driving the arising of relapse. H3K27ac and H3K4me1 ChIP-seq performed on a subset of Primary, Remission, and Relapse matched samples confirm that a large percentage of these elements are active enhancers increasing the risk of relapse significantly. Subsequent analysis of the cistrome landscape suggests the activity of ERG transcription factor in binding the selected subset of active enhancers further confirmed by ERG ChIP-seq.

Conclusions: This work suggests that chromatin accessibility shifts can be used to trace phenotypic changes throughout the journey of ALL-patients and provide novel insights in targeting cancer phenotypical modulators such as ERG.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0044 / #1737 | Novel Liquid Biopsy Markers of Central Nervous System Involvement in Pediatric Acute Lymphoblastic Leukemia

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Background and Aims: Refractory central nervous system (CNS) involvement is among the major causes of therapy failure in childhood acute leukemia. Applying contemporary diagnostic methods, CNS disease is often underdiagnosed. To explore more sensitive and less invasive CNS status indicators, we examined microRNA (miR) expressions in body fluids of patients with pediatric leukemia. The main intercellular transporters of miRs, small extracellular vesicles (EV) were also characterized in cerebrospinal fluid (CSF) samples.

Methods: In an acute lymphoblastic leukemia (ALL) discovery cohort, 47 miRs were screened using Custom TaqMan Advanced Low-Density Array gene expression cards. As a validation step, a candidate miR family was further scrutinized with TaqMan Advanced miRNA Assays on serial CSF, bone marrow (BM) and peripheral blood samples (n=132). Next, small EV-rich fractions were isolated with ultracentrifugation and the samples were processed for immunoelectron microscopy with anti-CD63 and anti-CD81 antibodies, simultaneously.

Results: Regarding the discovery study, principal component analysis identified the role of miR-181-family (miR-181a, -181b, -181c) in clustering CNS-positive (CNS⁺) and CNS-negative (CNS⁻) CSF samples. We were able to validate miR-181a expression differences: it was about 52 times higher in CSF samples of patients with CNS⁺ ALL compared to CNS⁻ cases (n=8 vs. n=10, $\Delta FC=52.30$, $p=1.5E-4$). The sensitivity of CSF miR-181a measurement highly exceeded those of conventional cytospin in the initial diagnosis of CNS⁺ ALL (90.0% vs. 54.5%). Decrease in miR-181a level from the diagnosis of ALL to the 33rd day of therapy was also detectable in BM samples among CNS⁺ patients ($\Delta FC=-187.96$, $p=0.006$). Pellet resulting from ultracentrifugation of CNS⁺ CSF samples from ALL patients showed atypical CD63⁺/CD81⁺ small EVs in high density by immunoelectron microscopy.

Conclusions: After validating in international cohorts, quantification of miR-181a or a specific EV subtype might provide novel tools to monitor CNS disease course and further adjust CNS-directed therapy in pediatric ALL.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0045 / #1745 | Analysis of Febrile Neutropenia Episodes During Intensive Chemotherapy in Pediatric Acute Lymphoblastic Leukemia

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Background and Aims: Introduction: Infections are an important cause of morbidity and mortality in children receiving chemotherapy for acute lymphoblastic leukemia(ALL). Though gram negative organisms are the most common etiology, causes of infection are heterogenous.

Methods: A retrospective analysis was performed on febrile neutropenia(FN) episodes in children with acute lymphoblastic leukemia who were treated on the BFM 2002 protocol at our centre from January 2018 to March 2021. FN episodes were analysed in all intensive phases of chemotherapy.

Results: There were 90 FN episodes in 38 children.5/38(13%) patients had T ALL, 32/38(84%) patients had B ALL while 1/38(2.6%) had mixed phenotypic acute leukemia(B/Myeloid). Male to female ratio was 1.7:1.The mean age of the cohort was 5.4 years. 23/90(25%) episodes were noted in induction phase1A, 24/90(26%) episodes in induction1B, 10/90(11%) episodes during protocol M while 25(27%) episodes occurred during protocol II. 8 FN episodes were observed in 2 patients who received high risk blocks. A focus was found in

only 30/90(33%) of the episodes. In these, foci of infection were as follows: Pneumonia in 7/30(23%), perianal infection in 10/30(33%), bone marrow aspiration site infection in 2(6%), enterocolitis in 9(10%) while 1(3%) had necrotising fasciitis. CRBSI was proven in 5/90(5.5%).1 patient had otomycosis in which aspergillus was isolated. Causative organism was identified in 28/90(31%)episodes, these included: Pseudomonas in 3/28(10.7%), E.coli in 2/28(7.1%), Klebsiella in 3/28(10.7%), Staphylococcus aureus in 1/28(3.5%), enterobacter in 1/28(3.5%), stenotrophomonas in 1/28(3.5%), candida species in 3/28(10.7%), mucor in 1/28(3.5%), parvovirus in 11/28(12.2%), dengue in 1/28(3.5%) and Sars Cov-2 in 2/28(7%) patients. 12/90(13%) episodes required ICU care. 4/38(4%) patients succumbed to septic shock,3 in induction1A and 1 in induction1B.

Conclusions: Conclusion: Managing FN in children with ALL is challenging but good microbiology laboratory support and appropriate antimicrobial therapy improves outcomes in these children. Periodic review of infections helps to improve antimicrobial policies of the hospital

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0046 / #176 | A Pharmacokinetic Study of Native E.Coli Asparaginase for Acute Lymphoblastic Leukemia Treated with Thaipog Protocol

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Background and Aims: Asparaginase is one of the essential chemotherapies used to treat acute lymphoblastic leukemia (ALL). Given its protein structure, asparaginase antibody production may cause a subtherapeutic level of asparaginase activity and result in an inferior outcome.

Methods: Prospective study of asparaginase activity among patients who received native E.coli asparaginase 10,000 IU/m² according to ThaiPOG ALL protocol. The blood samples were drawn after the first dose with a total of 9 samples during the next 48 hours. The asparaginase activity is measured by the coupled enzymatic reaction with activity greater than 0.1 IU/ml considered at the therapeutic level. Pharmacokinetic data provides the peak activity level (C_{max}), time to reach the maximum concentration (T_{max}) and area under a curve (AUC_{0-48h}) being elucidated.

Results: Ten patients (five males and five females), median age 6.5 years, were enrolled. The median asparaginase activity of seven cases who are eligible for calculation reached T_{max} within 24 hours (range 6-48 hours) with mean C_{max} 3.60±0.34 (range 3.02-4.11) IU/ml. Mean AUC_{0-48h} is 143.23±36.94 IU.h/mL (range 71.07 - 180.12 IU.h/mL). The post-48-hour activity showed a mean of 3.19±0.24 IU/ml

(range 2.77-3.51 IU/ml) which implied an adequacy of activity over 48 hours and proper for the 12 day period. One relapsed ALL patient who was subsequently treated with high risk ALL induction therapy showed an extremely low AUC of asparaginase activity which coincided with urticaria after asparaginase injection. Subsequently, the asparaginase antibody was demonstrated in this patient.

Conclusions: Native E.coli asparaginase-based ThaiPOG protocol provides a compelling pharmacokinetic effect compared to previous studies. Plasma asparaginase activity and antibody testing are recommended for a relapsed patient or an suspected allergic reaction to the asparaginase. Patients in this pharmacokinetic study showed that low asparaginase activity may benefit from switching to an alternative form of asparaginase to maintain leukemia treatment efficacy.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0047 / #219 | Providing Care for Pediatric Acute Lymphoblastic Leukemia Patients During the Covid-19 Pandemic in Mexico

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Background and Aims: Mexico has some of the highest COVID-19 death rates globally and is facing an humanitarian crisis. Continuing the care of patients with acute lymphoblastic leukemia (ALL) has represented a significant challenge. We aim to document the compliance to the treatment in pediatric patients with ALL during the pandemic, the incidence and outcomes of patients infected with SARS-Cov2 in our institution.

Methods: We included patients with ALL, <16 years old on active treatment in "Hospital Universitario José Eleuterio Gonzalez" from March to December 2020. We documented the element of the non-compliance treatment (reason, duration, impact), and the outcomes of patients infected with COVID-19.

Results: We analyzed 223 visits from 67 patients. The chemotherapy application was timely, and according to the local scheme in 83%(n=183) of the records reviewed. Modifications of treatment occurred in 38 visits (intrinsic in 12.1%, n=27 and extrinsic in 4.9%, n=11) secondary to absence of inpatient beds (2.2%), lack of financial support (1.3%), and shortage of chemotherapy (0.9%). According to the treatment phases compliance was lower than 80% in consolidation (66%) and intermediate maintenance (60%); the rest of the phases, we reached the adherence goal. Fourteen patients (20%) were diagnosed with COVID-19; two patients (14%) at diagnosis, 5 (35%) during intensive chemotherapy phases, and 2 (14%) during HSCT. Seven patients (50%) had pneumonia, 5 (35%) required oxygen, and 2 (14%) had

multisystem inflammatory syndrome. The median time of hospital stay was 9.5 days. The 30-day mortality rate was 7%.

Conclusions: The compliance of treatment was >80% in our hospital, however ours is a tertiary center that don't reflect the global scenario in our contry. It is essential to each center define alternatives for minimizing the risk of infection, providing a safe environment and overcome the lack of hospital rooms and shortage of chemotherapy to continue the care of patients with ALL.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0048 / #246 | Improving Risk Based Treatment Stratification by Minimal Residual Disease (MRD) in Childhood Acute Lymphoblastic Leukemia. A Single Center Experience in Paraguay

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Background and Aims: To develop methodologies to improve the risk stratification of pediatric patients in Paraguay, the Sanofi Espoir Foundation with the My Child Mathers program are developing the quality care in pediatric oncology project. The goal of this project is to improve the risk-based treatment stratification of patients with ALL, by training human resources and developing guidelines for standardizing properly the procedures to diagnosis and stratification into risk groups. The guidelines collaborated to develop a risk-directed therapy based on the study of Minimal Residual Disease (MRD). This study aims to evaluate the feasibility of the MRD as a strategy to ensure risk-based treatment stratification in pediatric ALL.

Methods: From 6th June 2019 to 2nd Feb 2021, 53 patients with ALL were enrolled in a treatment guide, 50 with ALL-B and 3 with LLA-T. The median age was 5 years (range 7 months to 18 years). The diagnosis of ALL was based on WHO 2016 criteria. The MRD level was assessed by bone marrow flow cytometry on day 15th and day 29th. The cutoff level for considering MRD was equal to or greater than 1% and 0,01% for day 15th and day 29th.

Results: The MRD was positive in 7/53 (13%) of the patients, 3/7 (43%) on day 15th and 4/7 (54%) on day 29th. At the diagnosis 6/7 (86%) of the patients were classified as low risk and one patient as intermediate risk, due to the MRD they were stratified as intermediate and high risk respectively.

Conclusions: The MRD measurement allows to re-assign the patients into risk groups for the treatment of ALL and contribute to evaluating the quality in the diagnosis and treatment as a part of a pediatric quality care program.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0049 / #317 | Gut Bacterial Gene Changes Following Pegaspargase Treatment in Pediatric Patients with Acute Lymphoblastic Leukemia

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Background and Aims: Treatment of pediatric acute lymphoblastic leukemia (ALL) with pegaspargase exploits the dependency of ALL cells on exogenous asparagine. Pegaspargase depletes asparagine, consequentially effecting aspartate, glutamine and glutamate levels. The gut as a confounding source of amino acids (AAs) and the gut microbiome's role in AA metabolism has not been examined. This study investigated asparagine, aspartate, glutamine and glutamate in stool samples along with microbial genes, whose products interact with these AAs, to understand how the gut microbiome may impact or be impacted by pegaspargase treatment.

Methods: We examined shotgun metagenomic sequence data from 88 stools samples in 26 ALL patients and identified microbial genes whose products interact with the AAs of interest. In addition asparagine, aspartate, glutamine and glutamate were examined in a subset of 39 stool samples from 15 patients over the course of pegaspargase treatment. Changes over the course of treatment in AAs and genes were investigated (Kruskal-Wallis test, $\alpha < 0.05$). The impact of antibiotic use on taxa and gene changes was also examined.

Results: Stool asparagine declined significantly after treatment, and 31 microbial genes changed, both increasing and decreasing. Changes were complex, and included genes involved in AA metabolism, nutrient sensing, and pathways found increased in other cancers. While antibiotic use did impact taxa our observations of gene changes in the stool suggest that the taxa affected by antibiotic use differ from those that changed over treatment.

Conclusions: Intravenous pegaspargase impacts systemic levels of asparagine, but is not thought to reach the gastrointestinal tract. Thus, the decline in stool asparagine is either due to bacterial asparaginase or movement of asparagine from the gut to the blood. We identified

one gene (iaaA) with only limited asparaginase activity, leaving open other mechanisms for asparagine decline. Understanding pathways that change AA availability, including by microbes in the gut, could be useful in optimizing pegaspargase therapy.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0050 / #410 | Facilitating Implementation of Blinatumomab in Low- & Middle-Income Countries (LMIC)

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Background and Aims: In LMICs, there is a significant delay from time of novel therapy development to local implementation. Blinatumomab is an effective treatment for pediatric patients with B-acute lymphoblastic leukemia. To improve access to blinatumomab in LMICs, we developed a product-only donation program paired with a multi-level, multifaceted implementation effort to characterize the determinants and strategies needed to administer and manage blinatumomab treatment in LMICs.

Methods: Key stakeholders from industry, multidisciplinary pediatric oncology clinicians in high and LMICs, and implementation scientists collaborated to identify barriers and facilitators to blinatumomab delivery in a resource-limited setting. Based on these determinants, we created learning objectives and a robust training package including adapted eligibility criteria and guidelines for administration and management, multidisciplinary training sessions, and workflow tools to facilitate implementation. We conducted synchronous multidisciplinary training sessions at 3 pilot sites in India and Pakistan and continue to provide virtual consultation and technical assistance to institutions while patients receive blinatumomab.

Results: A training curriculum consisting of 12 role-specific provider materials was developed including information about blinatumomab background and mechanism of action, compounding, administration requirements, adverse events and management, and patient monitoring. Seventy-eight providers completed three 90-minute online training sessions (25 physicians, 17 pharmacists, 31 nurses, 4 representatives from the clinical trial unit). To date, 4 patients across the 3 centers met eligibility criteria and have begun enrollment, 2 patients have successfully received blinatumomab.

Conclusions: Through diverse inclusion of key stakeholders, we developed a multilevel implementation strategy to improve access to blinatumomab and facilitate safe administration for patients in LMICs. Ongoing next steps include formative evaluation of the training package, assessment of implementation outcomes including feasibility, acceptability, and appropriateness, and development of an Implementation Roadmap for Novel Therapies to facilitate both translation of blinatumomab to subsequent sites and a rubric for expeditious and life-saving translation of future novel therapeutics to LMICs.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0051 / #414 | Venous Thromboembolic Events in Adolescent and Young Adult Patients with Acute Lymphoblastic Leukemia

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Background and Aims: Acute venous thromboembolism (VTE) is a serious treatment-related complication of acute lymphoblastic leukemia (ALL) chemotherapy. Risk factors associated with VTEs in adolescent and young adult (AYA) patients being treated on intensified pediatric ALL protocols are unknown. Obesity and hypoalbuminemia are known risk factors of VTE in adults. In this study, we investigated if hypoalbuminemia or obesity is associated with an increased risk for VTE in this population.

Methods: Sixty-six AYA patients, 15-26 years old, with newly diagnosed ALL from 2009-2020 were identified. Patients were grouped into two cohorts based on the presence or absence of VTE (controls). Demographic data, laboratory results, treatment-related morbidities, type of central venous catheter (CVC) used, site and timing of VTE, hypoalbuminemia across treatment phases, and survival data were analyzed using SAS.

Results: Fourteen patients (21%) experienced 17 VTEs. The majority of VTEs occurred during induction chemotherapy (47%), after asparaginase administration (88%), in the upper extremities (59%), were associated with a CVC (53%), and were treated with anticoagulation (94%). Median time to first VTE was 27 days. Hypoalbuminemia during induction did not increase VTE risk ($p=0.9$) and was not predictive of VTE

severity ($p=0.2$). Obesity was not predictive of VTE risk ($p=0.9$). There were no statistical differences in age at diagnosis, sex distribution, race/ethnicity distribution, treatment-related morbidities, relapse free survival (79% vs 81% at 5 years), and overall survival (93% vs. 81% at 5 years) in patients who experienced a VTE compared to controls, respectively.

Conclusions: AYA patients with ALL experienced similar rates of VTE to their pediatric counterparts. Unlike adults with cancer, hypoalbuminemia or obesity were not associated with an elevated risk of VTE. Collaborative prospective studies are needed to help identify risk factors predictive of VTE in this population.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0052 / #418 | Allergic Reactions to E. Coli Asparaginase are Associated with Decreased Asparaginase Activity in an Indonesian Pediatric Population with ALL

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Background and Aims: Asparaginase is essential for acute lymphoblastic leukemia (ALL) treatment. However, its use is limited by neutralizing antibodies, which is especially problematic in countries with limited access to alternative asparaginase preparations. ASP antibody levels and activity was measured during remission induction and correlated with allergy manifestations.

Methods: E.coli ASP was dosed at 7500 IU/m². ASP IgG antibody levels were quantified at the beginning and end of induction. ASP activity was measured 24 hours after administering the 1st and 5th dose (standard-risk) or 7th dose (high-risk patients), and within 24 hours in case of allergic reactions. All subjects were monitored for allergy, according to CTCAE version 3. Parametric statistical analysis was performed using unpaired t-tests, ANOVA for repeated measurements and the non-parametric Chi-square test for patient characteristics.

Results: Of 63 consecutive patients, ASP antibody levels and activity were available in 41 cases. Allergic manifestations occurred in 13/41 subjects (19 events), with urticaria being the most frequent (9/19). Severity was grade 2 and 3 in 9/19 and 10/19 events, respectively. There were no significant differences in subject characteristics based on allergic reactions. The 5th dose was the most frequent time of onset. Antibody levels at the end of induction did not differ from those at

baseline ($p > 0.05$). There was no association between the occurrence of allergic reactions and the level of antibodies ($p > 0.05$). ASP activity in the allergic group (180.7 ± 195.9 mIU/mL) was lower than in the non-allergy (311.9 ± 160.5 mIU/mL; $p < 0.05$). Using a 24-hour level of 100 mIU/mL as a threshold for adequate ASP activity, 6/13 patients with allergy had adequate levels, compared to 26/28 patients without ($p < 0.05$).

Conclusions: E.coli ASP IgG antibodies was not associated with allergy manifestation while ASP activity was. Other types of antibodies might have contributed to the lower ASP activity.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0053 / #442 | Transcriptome Study in Patients with Acute Lymphoblastic Leukemia Reveals a Higher Frequency of Single Nucleotide Variants and Insertions/Deletions in Relapsed Patients

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Background and Aims: Patients diagnosed with acute lymphoblastic leukemia (ALL) are stratified into risk groups based on biochemical, cytogenetic and molecular parameters, and early response to therapy. Although considerable progress on survival rates has been achieved, some patients still relapse and/or die in all risk groups. The identification of new biomarkers, such as single nucleotide variants (SNVs) and small Insertion/Deletions (InDels) with biological and clinical relevance might contribute to improve the initial stratification based on molecular profiles.

Methods: We collected bone marrow samples at diagnosis from 39 patients with ALL from three hospitals in Argentina. We performed paired-end transcriptome sequencing (RNAseq). Clinico-pathological characteristics were recorded (mean follow-up: 32 months; range: 5-73). We used the RNAmut software to evaluate SNVs/InDels in 114 selected genes and 79 selected fusion genes. All found variants were analyzed and filtered based on: 1) mutant reads ≥ 15 and variant allele fraction $\geq 20\%$; 2) variant frequency in 1000 genomes database < 0.01 ; 3) annotation in ClinVar; 4) association with leukemia phenotype; 5) prediction of pathogenicity (Polyphen); 6) oncogenicity predictions (OncoKB); and 7) previously reports in patients with ALL (COSMIC).

Results: We found a total of 9,594 variants in the 39 transcriptomes analyzed. After annotation and filtering, we identified 17 SNVs/InDels in 12 patients and 4 fusion genes in 5 patients. We observed a significant association between the presence of SNVs/InDels with disease relapse (OR=33; 95%CI=1.63-669.99; $p=0.023$) compared to patients without SNVs/InDels. No associations were found with MRD at day 15 nor Risk Group.

Conclusions: The associations observed warrant further analysis after a longer follow-up, and in a validation cohort. Our results show that it is possible to detect oncogenic mutations at diagnosis using RNA samples from bone marrow aspirates. Their assessment might contribute to the detection of patients with an increased risk of relapse.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0054 / #453 | Identification of Pharmacogenetic Predictors of Methotrexate-Induced Mucositis in Children with Cancer

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Background and Aims: Methotrexate is effectively used to treat several cancers, yet some patients develop mucositis, a painful inflammation and ulceration of mucosal linings found at various locations throughout the body, including the eyes, ears, mouth, rectum, esophagus, intestine, vagina and urethra. In severe cases, mucositis is life-threatening due to increased susceptibility to infection or the inability to proceed with anti-cancer drug treatment. Although 28 genetic variants have been significantly associated with methotrexate-induced mucositis, their predictive value remains uncertain due to limited replication attempts or inconsistency between studies. In this work, we aimed to discover and replicate genetic variants predictive of methotrexate-induced mucositis.

Methods: We recruited children with acute lymphoblastic leukemia (ALL) from six Canadian pediatric oncology centres and collected genomic DNA along with detailed clinical/demographic information. Rigorous characterization of mucositis was also conducted to determine the probability it was methotrexate-induced. An observational case-control study design using logistic regression will be used to identify genetic predictors of mucositis. Cases suffered serious mucositis defined as CTCAE/WHO grade ≥ 2 toxicity, controls did not experience mucositis (grade 0), and children with mild mucositis (grade 1) were excluded to improve case/control separation. To determine the robustness/consistency of findings, sensitivity analyses will assess the impact

of oncology centre, treatment regimen, clinical factors (e.g., radiation) and case definition.

Results: 585 methotrexate-treated children were enrolled in this study, with 130 (22.2%) cases, 367 controls, and 88 children with mild mucositis. We found that age, methotrexate regimen and radiation therapy impact mucositis risk. Genome-wide genotyping data has been generated, and genomic/sensitivity analyses are underway to determine the impact of genetic variants.

Conclusions: Ultimately, identified genetic variants predictive of mucositis will form the basis of a mucositis-predictive test to help select the safest and most effective treatment strategy.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0055 / #469 | Childhood Acute Lymphoblastic Leukemia: Four Years Evaluation of Protocol 2013 and 2016 in Yogyakarta, Indonesia Single Center Study in A Low-Middle Income Country

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Background and Aims: Prognosis of childhood ALL is excellent in high-income countries but far lower in low- middle-income countries. The national Indonesian ALL2013 protocol resulted in more toxicity and drop-out cases than expected. Therefore, we marginally modified it into a pilot ALL2016 protocol in Sardjito General Hospital Yogyakarta, Indonesia. Our catchment area has many low-income families.

Methods: Changes to the ALL2013 protocol: no anthracyclines in SR, dexamethasone replaced prednisone in reinduction for HR and some drugs were rescheduled, but total intensity remained similar. Results were analysed by Kaplan-Meier method at 4 years after diagnosis.

Results: From February 2013 to November 2018, 383 children with ALL were diagnosed, 21 were excluded, mainly because of previous treatment. ALL2013 included 174 patients (106 SR and 68 HR) and ALL2016 188 (91 SR and 97 HR). There was improvement in OS in ALL2016 (67.0% vs 60.3%; $p=0.087$) and EFS (50.0% vs 37.9%; $p=0.012$). In the HR group, both OS and EFS were significantly better in ALL2016 (EFS 43.3% vs 23.5%; $p=0.010$ and OS 59.8% vs 45.6%; $p=0.036$). In the SR group, the EFS and OS for ALL2016 only showed a trend to improvement (EFS 56.7% vs 47.2%; $p=0.114$ and OS 74.4%

vs 69.8%; $p=0.298$), due to late relapses. In ALL2016 the number of toxic deaths decreased (10.4% to 5.5%). Both SR and HR group showed a smaller number of dropouts in ALL2016. Note that in ALL2016 the percentage of HR cases was higher (51.6% vs 39.1%).

Conclusions: Small changes in protocol led to major improvements in outcome: the aims of less initial toxicity and less drop-out were achieved. Other factors might have influenced better outcome: general health insurance improved in 2014, with better access to medication. But non-adherence and non-availability of medicines in some periods led to many interruptions of treatment with higher relapse risk.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0056 / #508 | Practice of L-Asparaginase Usage: A Survey Among Healthcare Providers Treating Children with Cancer in India

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Background and Aims: L-Asparaginase is the most essential chemotherapeutic agent in the therapy of Acute Lymphoblastic Leukemia (ALL). Intensification of L-Asparaginase has led to significant improvement in survival in children with ALL in the West. In LMIC like ours, the outcomes in ALL are inferior compared to the published literature hence the following survey was conducted to understand the practice of using this agent among oncologists treating children with cancer in our country.

Methods: A questionnaire comprising of 25 aspects of L- Asparaginase usage was designed and a survey link was created and directed to healthcare providers involved in treating children with cancer in India. The responses were later analyzed using descriptive statistical methods.

Results: Of the total 75 responses recorded, 50 (65%) responders had more than 5 years of experience in Pediatric oncology and were treating at least 5-10 newly diagnosed ALL patients monthly. Fifty-three (70%) responders utilized both native and PEGylated L- Asparaginase and 22 (30%) oncologists used exclusively native E.coli L-Asparaginase. Most common route of administration was intramuscular route (70%). Seventy percent of respondents utilized native form at a dose of 10,000IU/m² and 20% at 6000IU/m². Doses used for PEGylated L-Asparaginase were 1000IU/m², 2500IU/m², and variable doses in 50%, 38% and 12% of responses respectively. Though serum Asparaginase assay (SAA) was not measured routinely in most of centers, 35 (45%) health care providers perceived performing SAA helps to make clinical decisions and 39 (50%) were not sure about the usefulness of SAA.

Conclusions: This survey shows that there is a wide variation in L-Asparaginase usage among health care providers caring for children with cancer in our country. As L-Asparaginase is a pivotal component of ALL therapy, uniformity in its usage and dosing with the possibility of monitoring SAA may be one of the critical steps towards improving outcomes in ALL in our country.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0057 / #572 | Persistence of Proteomes, Targetable Lesions, and Therapy Sensitivity Through Disease Evolution in Pediatric Acute Lymphoblastic Leukemia

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Background and Aims: Timing is a serious challenge to precision medicine in pediatric cancer: how to identify and test personalized treatments for each child with relapsed or refractory disease in a timeframe that enables effective clinical intervention. Here, we evaluate a change in personalized pediatric oncology from an approach that is currently reactive at cancer relapse to one that is proactive at diagnosis through retrospective analysis of proteomes, genetic lesions and responses to molecular-targeted drugs for progression samples from children with acute lymphoblastic leukemia (ALL).

Methods: ALL progression samples from 11 patients or control marrow samples from age-matched donors (n=3) were retrospectively sourced from BC Children's Hospital. Proteomes were performed and analyzed for equivalence across the whole proteome and within 269 pediatric cancer driving gene products, which were also sequenced (NGS). Public NGS data for paired ALL progression samples (n=69, St. Jude Hospital) was analysed. Samples were treated with drugs informed from published target-agent data.

Results: The highest equivalence of protein expression occurred between paired patient samples rather than similar disease states or random pairings. Versus normal samples, 33 proteins (of defined cancer drivers) were significantly changed in disease samples; for these proteins, expression (log2FC over normal) was similar in paired Dx-R samples, particularly for overexpressed proteins (r=0.63, p<2.2e-16). NGS indicated retention of variants in 10 of 11 BCCH patients or 56 of 80 (70%) patients when including SJH data. Only 7 (of 80) patients lack shared variants through disease progression. Finally, viably-frozen matched Dx-R/R1-R2 samples from four patients showed similar responses to four molecular-targeted drugs.

Conclusions: Changes in variant frequencies mark disease evolution. But, paired ALL progression samples show relatively stable proteomes, targetable lesions and therapy sensitivity. Thus, initiation of a precision medicine workflow when pediatric ALL is first diagnosed has the poten-

tial to identify disease sensitivities that are likely to persist at disease relapse.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0058 / #602 | Switching to Native L-Asparaginase in Childhood Patients with Acute Lymphoblastic Leukaemia and Hypersensitivity Reaction During First Exposure to PEG-Asparaginase. A Pilot Study

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Background and Aims: Asparaginase is one of the crucial elements in treatment of paediatric acute lymphoblastic leukaemia (ALL). With lower antigenicity, PEG-asparaginase has become the first line treatment in many protocols. When hypersensitivity develops, Erwinase is used as second line treatment but has the drawback of high-cost and supply is not stable. Allergic reactions to PEG are increasingly recognized. We postulate that subjects with hypersensitivity reactions during first exposure to PEG-Asparaginase may be unrelated to Asparaginase component and switching to native L-Asparaginase is a safe and effective second line treatment.

Methods: We performed a retrospective review of paediatric patients with ALL receiving treatment in Hong Kong Children's Hospital. We identified subjects with allergic reactions to PEG-Asparaginase during first exposure and subsequently switch to native L-Asparaginase. Severity grading of PEG-Asparaginase hypersensitivity is according to NCI CTCAE criteria. Information on L-Asparaginase post-dose activity and adverse reactions towards L-Asparaginase were collected.

Results: From Jul 2020 to Dec 2020, 5 patients developed hypersensitivity to intravenous PEG-Asparaginase at the first exposure and had switched to native L-Asparaginase. Median age was 9.1 years (range 1.2–14.1 years). All but 1 patient received diphenhydramine and H2 blocker as pre-medication. Four had CTCAE grade 2 reaction and 1 had grade 3 reaction with anaphylactic shock. Hypersensitivity manifested at a median infused dose of 451unit/m² of PEG-Asparaginase (range 73–1158units/m²). Regular asparaginase activity monitoring in 4 subjects revealed adequate activity (range 0.27–1.1IU/ml, ref range >0.1IU/ml) at 48–166 hours after L-Asparaginase infusion. They received a median of 18 doses (range 11–30 doses) of L-Asparaginase without allergic reactions. The remaining subject received 15 doses of L-Asparaginase and developed anaphylaxis at the 15th dose. Erwinase was subsequently used without further reactions.

Conclusions: In subjects with hypersensitivity reaction towards PEG-Asparaginase during first exposure, most of them did not develop allergic reactions after switching to native L-Asparaginase. Asparaginase activity was maintained.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0059 / #626 | The Prevalence of Metabolic Syndrome and its Components in Survivors of Standard-Risk Acute Lymphoblastic Leukemia

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Background and Aims: The 5-year survival rate in standard-risk acute lymphoblastic leukemia (ALL) has approached 95%. Long-term survivors of ALL face many adverse health consequences. The purpose of this study was to determine the prevalence of metabolic syndrome (MS) in children and adolescents survivors of standard-risk ALL.

Methods: The study recruited 54 children with standard-risk ALL, who were off therapy for one year or more. Age, gender, birth weight, body mass index (BMI) of the patient and the parents, waist circumference, and blood pressure, lipid profile, and fasting glucose were assessed. For nutritional status, BMI percentile curves standardized for Turkish children were used. International Diabetes Foundation criteria were used for the diagnosis of MS.

Results: The median age of patients (24 male, 30 female) was 16,3 (9,8-22,5) years. Median time off-therapy was 9,47 (range 1,13-14,56) years. Twenty-six patients had waist circumference >90th percentile according to the standards of Turkish children. Fourteen patients (out of 26) did not have any other criteria for MS whereas nine patients (out of 26) had only one criterion. The prevalence of MS was 5,5%. The prevalence of obesity and abdominal obesity were 11% and 48,1%, respectively. The number of patients with overweight and obesity significantly increased in comparison to pretreatment assessment. Age, gender, birth weight, and parental obesity did not have an impact on MS development.

Conclusions: Up to 33.6% of adult survivors of childhood ALL were reported to have MS, depending on the criteria used for diagnosis. Metabolic syndrome prevalence was found to be low in patients with standard-risk ALL in our study, but MS components have been observed with increasing frequency. In conclusion, blood pressure, weight, and waist circumference should be carefully monitored as well as necessary preventive measures and lifestyle modifications should be recommended in the early and late courses of survivorship.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0060 / #638 | Toxicity Profile and Tolerance of High Dose Methotrexate Therapy in Childhood Acute Lymphoblastic Leukemia - A Prospective Study from a Tertiary Cancer Centre in India

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Background and Aims: Limitations in the availability of trained staff and adequate laboratory support pose problems in administration of HDMTX in childhood ALL in the developing world. In this study we assessed the toxicity profile and clearance of MTX in a Low-Middle Income countries (LMIC) setting with detailed monitoring of clinical and lab parameters along with limited serum MTX estimation.

Methods: This is a prospective observational study done from 1st January 2019 to 31st December 2019. Children with age <14 years diagnosed with High Risk ALL who received a total of 4 HDMTX courses were included. Administration of HDMTX was done by trained nursing staff at a dose of 5gram/metre² in an inpatient setting with strict hydration and alkalinization with urine pH monitoring. Data regarding toxicity profile, leucovorin rescue and serum MTX levels starting at 54-hour from start of HDMTX until subsequent normalization (<0.2 micromole/liter) were analyzed.

Results: During the study period, 488 HDMTX infusions were administered. In 59.3%(n-289/488) and 35.7%(n-174/488) infusions, serum methotrexate level normalization occurred at 54-hour and 78-hour from start of HDMTX infusion respectively. In the remaining 5% infusions, serum methotrexate level normalized at beyond 78-hour time point. The incidence of severe oral mucositis, leukopenia, thrombocytopenia and hepatotoxicity were 2.7%, 16%, 11.5% and 13.9% respectively. The incidence of skin toxicity, nephrotoxicity, neurotoxicity and infection were 13.1%, 0.6%, 1.2% and 24.4% respectively. There was no treatment related mortality. Majority (60%) of infusions which had toxicities had normal MTX clearance.

Conclusions: Even in developing/LMIC countries like India, HDMTX at 5 gram/metre² can be safely administered. This can be achieved by meticulous hydration, urine alkalization, administration by trained staff nurse, serum MTX level monitoring at 54-hour time point and adjusted dosing of leucovorin rescue in an inpatient care setting with serious toxicities like severe oral mucositis, nephrotoxicity and neurotoxicity limited to <3% of infusions.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0061 / #709 | Expression of Infiltrating Bone Marrow Natural Killer Cell Receptors in Pediatric Acute Leukemia

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Background and Aims: Leukemia is the most common childhood cancer. Despite ~70-90% children with acute lymphoblastic leukemia (ALL) or acute myeloid leukemia (AML) recover, the patients with relapsed disease present a poor prognosis. Immunotherapy with natural killer (NK) cells has emerged as a safe and promising alternative for the treatment of leukemia in children. We aimed to analyse the infiltrating bone marrow (BM) expression of NK surface receptors in pediatric acute leukemia.

Methods: From March 2019 to March 2021 a total of 20 BM samples were collected from pediatric patients with B-cell ALL (n=16) and AML (n=4) at La Paz University Hospital, Madrid, Spain. Infiltrating BM NK cells phenotype were monitored and analysed by multiparametric flow cytometry. Also, K562 cell line was co-cultured with BM mononuclear cells to examine NK cell degranulation.

Results: In patients with B-cell ALL, expression of NK activating receptors Nkp46, NKG2D was high while inhibiting receptors as TIM-3, LAG-3, and PD-1 was low. However, their ligands were increased in blasts at diagnosis. In B-cell ALL relapse, Nkp44, NKG2D activating receptors and inhibitor receptor ligands expression were reduced whereas TIM-3, LAG-3, and PD-1 expression did not change respect to diagnosis. In patients with AML, NK activating receptors NKG2D, CD57, CD69 and Nkp46 were the most activating receptors expressed at diagnosis while the inhibitory receptors expression were low. The only patient analysed at relapse exhibited increased CD69 and decreased NKG2D and Nkp44 receptors. In this patient, blasts showed

an increased expression of ligands Galectin-9 and PD-L1 whereas NK cells exclusively overexpressed Galectin-9 receptor, TIM-3. NK cell degranulation capacity was reduced at induction and relapse respect to diagnosis.

Conclusions: NK cells from patients with acute leukemia show more expression of activating than inhibitory receptors at diagnosis. In addition blasts express high ligands for inhibitory receptors. At induction and relapse, the expression of receptors is highly heterogeneous.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0062 / #711 | Asparaginase Activity Monitoring and Management of Asparaginase Hypersensitivity Reactions: Results of a Canadian Survey

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Background and Aims: The primary formulation of asparaginase (ASP), pegaspargase (PEG), is immunogenic and can result in anti-ASP antibody formation, which can decrease its effectiveness without causing any clinically-apparent reaction (CAR) (a process called silent inactivation [SI]). When a patient has SI or severe reaction/anaphylaxis, switching to Erwinia ASP is warranted. As obtaining Erwinia is problematic, it is important to switch only when necessary. The only reliable way to distinguish between a CAR and SI is by measuring an ASP level, but levels are not routinely measured at all Canadian centers. The objective of this study is to identify variations in practice regarding ASP monitoring and Erwinia use.

Methods: A 21-item survey was developed using OPINIO software and distributed to all Pediatric Hematology-Oncology physicians in Canada from Feb-Oct 2020 after institutional ethics approval.

Results: Respondents represented 15 hospitals across each region of Canada (response rate=41.5%). Only 39.2% of respondents reported routinely measuring ASP levels, yet 53% of respondents have modified therapy from PEG to Erwinia in up to half of their patients. The most common reason for not measuring ASP levels was not knowing how to use levels clinically (25.5%) or if measuring levels was required (21.6%). There was variation in the timing of levels and their target, but all respondents target an ASP level of at least 0.1 IU/mL. The most common reason to modify therapy from PEG to Erwinia was for a Grade 3-4 infusion reaction or for ASP levels that are subtherapeutic. Switching occurred up to 25% of the time.

Conclusions: It is imperative to use ASP levels to determine when a switch from PEG to Erwinia is justified in order to conserve supply. Given the variation in practice identified by this survey, the authors intend to develop a practice guideline on ASP activity monitoring that can be implemented at each center.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0063 / #731 | Low Bone Mineral Density in Survivors of Childhood Acute Lymphoblastic Leukaemia

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Background and Aims: Improvement in overall survival of childhood acute lymphoblastic leukaemia (ALL) has shifted the focus of care towards late effects of treatment. One of the chronic health concerns among the survivors is reduced bone mass acquisition due to chemotherapy, reduced physical activity and nutritional deficiency.

Methods: This single-centre, cross-sectional study aimed to determine the prevalence of low bone mineral density (BMD) among our cohort of childhood ALL survivors and examine possible factors which may contribute to low BMD. Over a period of 7 months, seventy-eight subjects (36 males, 42 females) aged 5 years and above, were enrolled into the study. These subjects had completed treatment for more than one year and remained in remission. Dual-energy x-ray absorptiometry (DXA) scan was used to measure lumbar spine (LS) Z-score and total hip (TH) Z-score.

Results: Median age at diagnosis was 4.7 years (IQR 25th 3.1; 75th 6.6) while median age at study entry was 13.2 years (IQR 25th 10.0; 75th 18.2). Majority (71.8%) of them was diagnosed with standard risk ALL. Twenty (25.6%) out of 78 survivors have low BMD for age, with mean LS Z-score of -2.14 (± 0.58) and mean TH Z-score of -2.30 (± 0.76). Five of them had sustained a fracture during or immediately post-treatment. Survivors with normal body mass index were found to have significantly lower BMD compared to those who were overweight/obese. Otherwise, no disease or treatment-related factors were found to be a significant risk factor to develop low BMD.

Conclusions: Our study demonstrated that 25% of our childhood ALL survivors had low BMD. Monitoring of bone health should be incorporated in the follow-up plan of survivors of childhood ALL to allow early identification and treatment of bone mineral deficits.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0064 / #762 | Pediatric Acute Lymphoblastic Leukemia Outcomes in Mexico: A Report of 578 Patients from 5 Institutions

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Background and Aims: Cure rates for acute lymphoblastic leukemia (ALL) exceed 90% in high-income countries, where access to timely diagnosis, molecularly risk-stratified therapy, and comprehensive supportive care exists. Reports from middle-income countries describe lower survival rates due to deficiencies in these factors. This retrospective study aimed to analyze outcomes for ALL at 5 centers in Mexico.

Methods: Patients <18 years-of-age with newly diagnosed B- and T-cell ALL treated between January 2011 and December 2015 were included. Clinical and biological characteristics and their association with outcomes were examined.

Results: Overall, 578 patients with a median age of 6.3 years were included. B-cell immunophenotype was identified in 498 (86.2%) patients. Mean white blood cells (WBC) at diagnosis was $61.2 \times 10^3/\text{mm}^3$. CNS status was 1 in 496 (85.8%), 2 in 14 (2.4%), and 3 in 30 (5.2%). 470 patients (81.3%) were classified as high risk at diagnosis. Based on NCI criteria, 192 (38.6%) patients were inappropriately assigned risk group. FISH and PCR were performed for 160 (27.7%) and 137 (23.7%) patients, respectively. MRD during induction was performed in 352 (60.9%) patients, with 73.9% having a result <0.01%. Median follow up was 4.5 years. Sixteen patients (2.8%) had induction failure. Induction mortality rate was 10.7%, while death due to toxicity in remission was 4.2%. Treatment abandonment rate was 1.9%. Five-year cumulative incidence of CNS relapse was 10.6%. Five-year EFS was 57.3% for B-cell ALL and 43.7% for T-cell ALL. Five-year OS was 69.3% for B-cell ALL and 53.5% for T-cell ALL. Five-year OS are 79.9% and 64.1% for standard-risk and high-risk groups. On multivariable logistic regression, high WBC, age, and positive MRD were associated with EFS.

Conclusions: Outcomes in this multi-institutional cohort describe poor outcomes, influenced by early mortality, high CNS relapse rate, and incomplete and inconsistent risk-stratification. Treatment with appropriate molecular risk-stratification and de-intensification could lead to improved outcomes.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0065 / #807 | Patient-Derived Xenograft Models Recapitulate The Proteome and Functional Capacity of Primary Pediatric Acute Lymphoblastic Leukemia but Divert in Select Pathways

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Background and Aims: Patient-derived xenografts (PDX) of pediatric leukemia recapitulate genomic and epigenomic aberrations in patients. The proteome is remodelled by genomic and host factors and it remains unclear how similar the proteomes of PDX and primary patient blasts are. Since the proteome state determines the functional capacity of cells and their response to treatment it remains unknown which aspects of blast behaviour can be accurately assessed in PDX models. To address this, we studied global protein abundance, phosphorylation, and protein maturation by proteolytic processing in 11 pediatric B- and T- cell ALL patients and 19 corresponding xenografts.

Methods: We measured protein abundance, phosphorylation, and limited proteolysis in paired patient and xenografted pediatric acute B- and T- cell lymphocytic leukemia, and in pediatric leukemia cell lines by mass spectrometry. Targeted next-generation sequencing was utilized to examine genetic abnormalities in patients and corresponding xenografts.

Results: The proteomes of pediatric ALL xenografts most closely matched primary cell proteomes from their patient of origin with >50% of proteins being equivalently abundant. Leukemia-specific changes in protein levels were largely recapitulated in xenografts. As expected, PDX blasts lacked proteins constituting the interface to the host's immune system. We found increased expression of cell cycle proteins indicating a high proliferative capacity of xenografted cells. Structural genomic changes and mutations were reflected at the protein level in patients. Comprehensive N-terminomics revealed deregulated protein processing in leukemic cells. The post-translational modification landscape was primarily shaped by leukemia type. Reflecting the more highly dynamic nature of post-translational modifications they showed clear host-specific differences and only limited patient-specific differences.

Conclusions: This study highlights key processes that differ between patient and PDX derived cells and confirms that murine patient-derived xenografts, in most aspects, constitute an accurate model system supporting the analysis of most biological processes and phenotypes of pediatric acute lymphoblastic leukemia.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0066 / #826 | Avascular Necrosis of Bone in Chinese Children with Acute Lymphoblastic Leukaemia

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Background and Aims: Avascular necrosis of bone (AVN) is a debilitating complication of acute lymphoblastic leukaemia (ALL). While it is extensively studied in Western study groups and therapeutic trials, very few studies focus on Chinese. This study aims to evaluate the incidence, risk factors and clinical outcomes of AVN among Chinese ALL patients.

Methods: This study is a retrospective, territory-wide cohort study. In total, 509 ALL patients were identified from 3 Hong Kong ALL study protocol databases (ALL-IC-BFM 2002, CCLG-ALL 2008, CCCG-ALL 2015). The cumulative incidence of AVN in these 3 studies is calculated. By comparison between the AVN and non-AVN groups, risk factors for development of AVN are identified. The clinical and radiological outcomes of AVN patients are also reported.

Results: A total of 24 patients were identified to have AVN, the overall incidence of AVN for the cohort was 4.7%. The timing of AVN diagnosis did not differ among the three study protocols. The most important risk factor for development of AVN was age, with children >10 years old having cumulative incidence of 16.7% while children <10 years old with only 0.5%. Another risk factor was T-cell ALL which had higher incidence than B-cell ALL (32% vs 24%, p-value 0.043). Other patient-related or therapy-related risk factors were not significant. Among the 24 patients with AVN, 6 of them required orthopaedic interventions in view of their progressive and severe disease. On follow up, 12 of 22 hip joints presented with progressive disease radiologically, however all AVN patients had no or mild joint pain and minimal functional disturbance no matter received operations or not.

Conclusions: The incidence of AVN in Chinese ALL patients was comparable to Western studies. Age >10 years old was recognized to be the most important risk factor for development of AVN. Overall functional outcomes were satisfactory in all AVN patients.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0067 / #828 | Healthcare Utilization and Costs Associated with Acute Lymphoblastic Leukemia (ALL) in Children with and without Down Syndrome (DS): A Population-Based Study

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Background and Aims: Children with DS and ALL are at increased risk of treatment-related morbidity and mortality compared to non-DS children, requiring increased levels of supportive care. Despite this, little is known about the healthcare utilization and cost associated with DS-ALL. We examined the healthcare utilization of this important sub-type of ALL patients in order better inform future evaluations of high cost new therapies.

Methods: We identified all children (1-17 years) diagnosed with B-lineage ALL in Ontario, Canada between 2002-2012 through a comprehensive provincial registry. Detailed demographics, disease, treatment, and outcome data were chart abstracted. Linkage to population-based health services databases identified all outpatient and emergency department (ED) visits, hospitalizations, and physician billings. Healthcare utilization costs (2018 Canadian dollars) were determined through validated algorithms. Event-free and overall survival (EFS, OS), healthcare utilization rates, and costs were compared between DS and non-DS patients using regression models, adjusting for demographic, disease, and treatment covariates.

Results: Of 711 patients, 28 (3.9%) had DS. No EFS or OS differences were observed between DS and non-DS patients. Adjusting for all covariates, DS children experienced substantially higher rates of ED visits [rate ratio (RR) 1.5, 95% confidence interval (95CI) 1.2-2.0; $p=0.001$] and inpatient days (RR 2.5, 95CI 1.4-4.5; $p=0.002$) compared to non-DS children. No difference in outpatient visit rates was found (RR 1.1, 95CI 0.9-1.3; $p=0.41$). The median 5-year healthcare utilization cost was \$247,700 among DS patients [interquartile range (IQR) 200,900-354,500] and \$196,200 among non-DS patients (IQR 148,900-280,300; $p=0.02$). In adjusted analyses, DS-associated costs were 50% higher (RR 1.5, 95CI 1.2-1.9; $p<0.002$).

Conclusions: Healthcare utilization and treatment costs of DS-ALL patients are substantially higher than those of non-DS ALL but result in comparable outcomes. Previous cost-effectiveness analyses of new agents may therefore underestimate cost-effectiveness in DS populations. Our data provide a baseline for future DS-specific cost-effectiveness studies.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0068 / #831 | **Dynamics of Anti HBS Titre in the Children During Maintenance Phase of Chemotherapy for Acute Lymphoblastic Leukemia-Experience from a Tertiary Cancer Centre of India**

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Background and Aims: Children with Acute lymphoblastic leukemia (ALL) are at high risk for developing hepatitis B due to immunosuppression secondary to chemotherapy, multiple invasive procedures, injections and multiple blood transfusions. This prospective study aimed to

evaluate the dynamics of Anti-Hbs titres and seroconversion to vaccination in children with ALL during maintenance chemotherapy.

Methods: Anti-HBs titre and HbsAg were measured for all children of age < 14 years with ALL at the time of diagnosis and twice during the maintenance phase of chemotherapy 6 months apart. Children were categorized into three groups- non protective(<10 mIU/ml), low protective(10-100 mIU/ml) and high protective(>100 mIU/ml). Hepatitis B Immunoglobulin/vaccine was given to children as per their titres status at diagnosis. A booster dose was again given to children with non protective first maintenance titre. The two Anti-Hbs titres during maintenance were compared and rate of seroconversion and HbsAg positivity were determined.

Results: Total 124 children with ALL (13=nonprotective titre, 111=protective titre) were recruited. At initial maintenance estimation, 15 patients had non protective titres and received booster dose of hepatitis-B vaccination while 109 children retained protective titre. At second estimation during maintenance, 7 patients remained non protective while 117 children had protective titres. Seroconversion occurred in Thirteen children in non protective group after booster dose of vaccination. Five children in low protective group lost their protective titres during maintenance phase. Loss of protective antibody titre was not seen in any child with high protective titre. No one developed HBsAg positivity during chemotherapy.

Conclusions: During maintenance phase all children with high protective titre retained their titre while 7.3% children of low protective group lost their titres. Seroconversion to booster dose of hepatitis B vaccination during maintenance chemotherapy in non protective group is very high (86.6%). Anti Hbs titre may be estimated again at the start of maintenance chemotherapy of ALL and a booster dose of hepatitis B may be administered if titres are low protective.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0069 / #846 | **Clinical Characteristics and Outcomes of Pediatric Relapsed Acute Lymphoblastic Leukemia in a Resource-Limited Setting**

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Background and Aims: Acute lymphoblastic leukemia (ALL) is the most common malignancy of childhood. About 15-20% of cases experience relapse. In developed countries, patients with relapsed ALL have a survival rate of 60 to 70%. There is limited data on survival in developing countries and many patients are treated in palliative intent. We aim to determine clinical features and outcomes of paediatric relapsed ALL in a resource-limited setting without the availability of bone-marrow transplant.

Methods: A retrospective chart review of all paediatrics patients with relapsed ALL treated with curative intent at a tertiary care cancer

centre from January 2010 to December 2018. Percentages were used for descriptive data and Kaplan-Meier analysis for survival.

Results: Of the total 750 patients treated for ALL during the study period 74 (9.8%) relapsed. Of these eight (11%) received palliation so the final study population was 66. The mean age of the relapsed population was 7.1 years (+/-2.8) with Male: Female ratio of 1.9:1. Sixty-two (96%) patients had B-lineage and 4(6%) T-lineage ALL. Timing of relapse was very early, early and late in 18%,42% and 39% respectively. Location was extra-medullary, bone-marrow and combined in 36%, 50% and 14% respectively. Forty-eight (65%) were treated as per UKALL R2/R3 and 17(26%) as per COG AALLO2P2. Five-year overall survival(OS) was 39% (median survival 23 months (95%CI 2.9-43). OS was better in patients with late relapse (50%), combined or isolated extramedullary relapse (45%) and those treated as per AALLO2P2 (69%). Very early relapse had poor survival (20%). Twenty-four of the total 36 deaths (66.6%) were due to sepsis.

Conclusions: Overall survival is low for patients in resource-limited countries but selected populations like those with late relapse and isolated extra-medullary relapse have a better outcome and should be offered curative treatment. Sepsis is the most common cause of death and improvement in supportive care may improve survival.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0070 / #858 | Acute Leukemia Patients in the PICU: Prognostic Factors and Mortality

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Background and Aims: Pediatric patients with acute leukemia (AL) are a high-risk population for infection and life-threatening conditions that require PICU admission, which carries an increase of the mortality rate. Studies predicting mortality among patients with AL without an history of HSCT are scarce. We investigated their clinical and epidemiological characteristics to determine prognostic factors for mortality.

Methods: Retrospective single-center analytic-observational study conducted from January 2011-December 2018 in a tertiary-care center. AL patients from 28 days-18 years admitted to the PICU were included, excluding those with history of HSCT or CAR T-cell therapy. We collected epidemiological and clinical characteristics, laboratory and microbiology results and outcome of the patients.

Results: Forty-three patients required urgent PICU admission (21.2%, 35 ALL and 8 AML) in 63 episodes. Main reasons were sepsis (21, 33.3%), hyperleukocytosis (12, 19%), respiratory failure (11, 17.5%) and seizures (8, 12.7%). Nineteen patients (30.2%) required inotropic

support while mechanical ventilation (MV) was used in 15 patients (23.8%). Three patients died during the PICU admission (3/63, 4.7%). Univariate analysis showed that MV and cardiopulmonary resuscitation increased mortality at 60th days (OR 13.7, 95%CI 1.2-152 and OR 38 (OR 38, 95%CI 2.3-618). Gram-negative sepsis was independently associated with one-year mortality (OR 13.5, 95%CI 1.4-130.2, p=0.024) in the multivariate analysis. Readmission to the PICU was not a risk factor for mortality at day 60.

Conclusions: In our population, the main cause of PICU admission in patients with AL was infection, which was associated with more severity and longer length of stay. Gram-negative sepsis was predictor of mortality.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0071 / #876 | Critical Care Needs and Outcome of Paediatric Patients with T-Cell Lymphoblastic Lymphoma/Leukemia Presenting with Superior Vena Cava Syndrome

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Background and Aims: Superior Vena Cava Syndrome (SVCS) is a life-threatening oncological emergency affecting children with T-cell Acute lymphoblastic leukemia (ALL)/Lymphoblastic Lymphoma (LL). We describe critical care needs and outcome of these patients.

Methods: Medical records of all children diagnosed with T-Cell ALL/LL between May2016 and March2021 were retrospectively analyzed.

Results: Of 120 paediatric LL/ALL patients, 23 patients (19%) had T-cell ALL/LL. Eleven (48%) amongst them presented with SVCS. All were males, aged 4-16years (ALL-6/11, LL-5/11). All presented with cough, dyspnea and mediastinal-mass. Examination findings included pleural-effusion (5/11), muffled heart sounds (5/11), cardiac-tamponade (4/11) and distended neck veins (4/11). Hyperleukocytosis was seen in 2/11. Prompt diagnosis could be reached by flowcytometry in 90% patients (peripheral smear-3, pleural-fluid-4, pericardial-fluid-2, bone marrow-1). Mediastinal-mass biopsy was performed only in 1 patient after starting treatment. Eight patients required critical care unit stay (median-4.5days, range:2-13days). Pleurocentesis (3/11) and pericardiocentesis (5/11) were performed when indicated. Oxygen was needed in 9/11 and endotracheal-intubation in 2/11. All were started on corticosteroids soon after diagnosis of SVCS. The median time to symptom relief was 4 days (1-13days) and that for mediastinal -mass resolution to >50% was 7 days (4-20 days). As per BMF95 risk stratification 64% had standard-risk, 18% each had high-risk and very high-risk disease and were treated accordingly. During Induction one patient died of refractory disease and one abandoned therapy. Of the remaining 9 patients; 7 achieved CR1 on day33 and 1 achieved CR1 by day79. One patient with

refractory disease on day 79 died later on. One patient relapsed on treatment and underwent successful allogeneic hematopoietic stem cell transplant after achieving CR2. At a median follow up of 8-months (4-54 months) overall survival is 78% and event-free survival is 63%.

Conclusions: T-cell ALL/LL patients with SVCS with good critical care can have improved outcomes. Flowcytometry can be handy tool for quick diagnosis.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0072 / #881 | Comparative Study of Two 5-Year Periods of Pediatric all Patients in a Third Level Hospital in Mexico

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Background and Aims: Background and Objectives: Acute Lymphoblastic Leukemia (ALL) is the most common pediatric cancer worldwide. Its incidence in Mexico is approximately 6.9/100000 children per year. In middle and low-income countries, the survival rate of all ALL in children varies between 15-50%. Among the main indicators of disease-free survival for ALL are: the Minimal Residual Disease (MRD) and the presence of infections. In order to improve these aspects, our hospital has collaborated since 2018 in some projects of St. Jude Children's Hospital, such as the ALL Retrospective Study 2011-2015.

Methods: Based on the retrospective ALL study with St. Jude Children's Hospital, a cross-sectional study of pediatric ALL patients was conducted in the periods of 2011-2015 (first period) and 2016-2020 (second period). Data on ALL patients was obtained using a standardized format by St. Jude Children's Hospital. Descriptive statistics were used.

Results: Of the 1249 analyzed files, 242 were ALL cases. Over the entire period (2011 - 2020), the average MRD assessment was 67.85%. An increase of high-risk cases was reported, from 4.17% in the first period to 9.47% in the second period. The percentage of patients with infection reduced in 26.32% from the first period to the second one, with no increases in mortality.

Conclusions: According to the data obtained from the 2011-2015 ALL collaborative retrospective project with St. Jude Children's, no changes were reported in terms of ALL prevalence or MRD uptake. Despite the increase in high-risk cases in the second period, we did not report an increase in mortality, and we observed a decrease in the incidence of infections. This may be due to the implementation of St. Jude's collaborative programs such as EVAT and Golden Hour. Moreover we register a decreased of 56.67% in the mortality rate for each 1000 patients in second period.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0073 / #890 | A Novelty Nudge

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Background and Aims: Aims: To improve our management of neutropaenic sepsis, through the use of novelty. Neutropaenic fever remains the commonest haematological and oncological emergency. The intensive cytotoxic chemotherapy regimens we inflict on children with malignancy in addition to their disease, often causes profound neutropaenia and places the child at risk of neutropaenic sepsis.

Fever and neutropaenia in children with malignancy carries a mortality rate of 1% and infection related mortality in children with ALL sits at 30% of deaths. Fever is often the only sign of potentially fatal infection. 1 hour is the national guideline for delivering antibiotics. We were keen to know our local performance and apply a 'novelty nudge' to improve the delivery of antibiotics.

Methods: We audited our current performance over a 1 month period. Only 40% of patients received antibiotics within 1 hour with a maximum of 3 hours between the child being seen and antibiotics given. This was most often due to a lag between blood results being known and acted upon. In an attempt to rectify this, we introduced a novelty 'duck'. When a child presented with fever their bloods were taken and sent to the labs. The duck timer was set for 30 minutes. At 30 minutes the alarm would sound prompting to check the bloods and contact the doctor to agree or not to the antibiotics. We then re-audited our performance following duck implementation.

Results: Our novelty duck improved the time the doctor assesses the child and the appropriate delivery of antibiotics. There was a 55% improvement in the time patients received antibiotics (40% to 95%). The visible presence of our duck encouraged interest and the duck was an advantageous prompt.

Conclusions: We respond well to 'a novelty nudge'. We feel such an approach improves our performance and ultimately the care, safety and outcomes for our patients.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0074 / #909 | Bone Health in Long-Term Survivors of Acute Lymphoblastic Leukemia (ALL) Addressed by Peripheral Quantitative Computed Tomography (PQCT)

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Background and Aims: Loss of bone mineral is a common concomitant of the treatment of ALL and may persist long into survivorship. Measurement of bone mineral density (BMD) by dual energy X-ray absorptiometry is limited to two dimensionality and cannot distinguish trabecular from cortical bone. These limitations are overcome by pQCT which was used to assess bone health in long-term survivors.

Methods: Subjects (N=74, males=40) more than 10 years from diagnosis (median 15 years), at a median age of 21 years, underwent pQCT at metaphyseal (trabecular bone) and diaphyseal (cortical bone) sites in the non-dominant radius and tibia for measures of geometry, density and architecture using conventional and custom software. These metrics combine to provide composite measures of bone strength.

Results: Bone mineralisation was lower than normal in both trabecular and cortical bone, as measured by Z scores of vBMD ($p=0.0265$ and $p<0.0001$ respectively). The Connectivity Index, a surrogate measure of the strength of trabecular bone, was lower in females than males at the 4% radius site (0.20 ± 0.04 vs 0.24 ± 0.05 , $p<0.0001$). Likewise, the Polar Strength Strain Index, a measure of the strength of cortical bone, was lower in females at the 38% tibia site (1245.3 ± 137.9 vs 1743.1 ± 410.1 , $p<0.0001$). Survivors of standard risk ALL had greater connectivity than survivors of high risk disease (0.24 ± 0.05 vs 0.21 ± 0.05 , $p=0.023$) and more compact trabecular bone, reflected in a lower average hole size (0.85 ± 0.55 vs 1.21 ± 0.82 mm², $p=0.029$). There were no significant differences in any of the metrics at any of the sites between subjects who had and had not a history of fractures, cranial irradiation or use of a bisphosphonate.

Conclusions: These long-term survivors of ALL had somewhat compromised bone health, but data in comparable healthy populations are limited. Longitudinal studies in larger and more ethnically diverse cohorts will provide further insights into bone health in this vulnerable population.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0075 / #921 | Chemotherapy Tolerance in a Child with Xeroderma Pigmentosum with B- Acute Lymphoblastic Leukemia

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Background and Aims: Xeroderma pigmentosum (XP) is an autosomal recessive disorder caused by defect in nucleotide excision repair and is associated with increased incidence of skin cancer. Lymphoblastic leukaemia in Xeroderma pigmentosum is extremely rare and there

are very few case reports of the same. We report a case of a 11-year-old boy with xeroderma pigmentosum, who developed B - Acute lymphoblastic leukaemia and treated with Modified BFM (Berlin-Frankfurt-Münster) Protocol for High Risk ALL.

Methods: An 11-year-old boy, who is diagnosed to have xeroderma pigmentosum at the age of 6 months, was referred with history of fever for 10 days, vomiting and decreased appetite for 5 days. There was no history of bone pains or bleeding manifestations. On physical examination, there was no pallor, icterus, cyanosis, clubbing, lymphadenopathy or bleeding manifestations. He had multiple diffuse hyperpigmented macules all over the body with sparing of palms and soles, conjunctival congestion and pterygium. Bone marrow flow cytometry was suggestive of B- Lymphoblastic Leukaemia.

Results: He is treated with Modified BFM (Berlin-Frankfurt-Münster) Protocol for High Risk ALL (Acute Lymphoblastic Leukemia). He completed induction, phase 1b, interim maintenance, protocol M with high dose methotrexate (3 grams/m² for 2 cycles and 5 gram/m² for 2 cycles) so far. Tolerance to ALL chemotherapy was good in our XP patient except for the transient derangement of liver function tests during induction treatment, grade 2 mucositis and P. aeruginosa blood stream infection during protocol M.

Conclusions: Though rare, a child with XP requires to be investigated for acute leukaemia on development of bicytopenia. Tolerance to ALL chemotherapy was good in our XP patient except for the transient derangement of liver function tests during induction treatment.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0076 / #924 | Relevance of Flow Cytometric Immunophenotypic Subcategorization and End of Induction Measurable Residual Disease Assessment in Pediatric and Adult Precursor T-Lineage Acute Lymphoblastic Leukemia Patients

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Background and Aims: Precursor T-lineage acute lymphoblastic leukemia (T-ALL) patients expressing myeloid/stem cell antigens are classified into early-T-cell precursor ALL (ETP-ALL) or near-ETP-ALL based on CD1a, CD8, and CD5 expression. The clinical and laboratory characteristics of ETP-ALL patients have been frequently documented, however, the literature on 'near-ETP-ALL' patients is sparse.

Methods: The clinical and laboratory profile, flow cytometric end-of-induction minimal residual disease (EOI-MRD) and survival outcome of our treatment-naïve T-ALL patients were analyzed. For diagnosis, a 6-tube 10-color FCM panel was used. EOI-MRD was assessed in bulk-lysed first-pull BMA samples using a single-tube 11-antigen 10-color panel and a median of 2.3 million events were analyzed.

Results: Among 81 consecutive T-ALL patients diagnosed, 21% (n=17) and 19% (n=15) were ETP-ALL and near-ETP-ALL, respectively. EOI-MRD was detectable in 39% (31.6% of pediatric and 52.4% of the adult) of samples tested (n=59). The frequency of EOI-MRD positivity was higher among ETP-ALL (75% positive, $p=0.001$) and near-ETP-ALL (71% positive, $p=0.009$) patients as compared to conventional-T-ALL (con-T-ALL) patients (22.5% positive). There was upregulation of CD8 ($p=0.046$) and CD38 ($p=0.046$) expression in the EOI-blasts of con-T-ALL and ETP-ALL samples, respectively. 2-year OS, RFS and EFS among our T-ALL patients (pediatric vs adult) was 79.5% vs 39.8% with $p<0.001$, 84.3% vs 60.4% with $p=0.026$ and 80.3% vs 38% with $p<0.001$, respectively. By univariate analysis, the 2-year EFS & RFS of pediatric T-ALL patients was independent of the immunophenotypic T-ALL subtype and was influenced only by EOI-MRD status. However, the 2-year survival characteristics among adult T-ALL patients were EOI-MRD independent and were influenced only by the presence of near-ETP-ALL phenotype.

Conclusions: Both ETP-ALL and near-ETPALL showed adult-age predilection and had high EOI-MRD positivity. The EOI-MRD positive status (irrespective of T-ALL sub-type) and the mere presence of near-ETP ALL immunophenotype (irrespective of EOI-MRD status) influenced the 2-year survival among pediatric and adult T-ALL patients, respectively.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0077 / #931 | Patterns of Treatment Failure and Toxicities in B-ALL and Opportunities for Future Interventions: Lessons Learnt from a Pilot Study at a Single Center in India

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Background and Aims: We piloted ICiCLE-2014, a multicentre trial for childhood B-Acute Lymphoblastic Leukemia (B-ALL) for 5 years before commencing on-trial enrolment and outcomes were analysed for patterns of treatment failure to identify areas for improvement.

Methods: B-ALL patients ≤ 15 years of age treated on ICiCLE-pilot protocol from February-2013 to February-2018 were included. End-of-Consolidation (EoC) multiparametric-flow cytometry defined Minimal Residual Disease $\geq 0.01\%$ (MRD) was considered refractory-disease. Patients with Relapsed/Refractory (R/R) B-ALL were offered salvage chemotherapy followed by Allogenic-Stem Cell Transplant (Allo-SCT), or palliated when treatment was not feasible. 3-year Overall Sur-

vival(OS) by Kaplan-Meier method and Odds-ratio for risk-factors were estimated using Jamovi (Version-1.6).

Results: Among 1406 eligible patients, with 3-year OS of $80.4 \pm 1.1\%$, there were 79(5.6%) induction deaths[5(6.3%)-early treatment, 42(53.2%)-sepsis and 32(40.5%)-other causes]. A further 86(6.1%) died before a relapse event (45%-sepsis, 55%-other causes). Ninety-four (6.6%) patients had refractory disease, while 245(17.4%) experienced first relapse [94(38%)-very early(≤ 18 -months); 124(50%)-early(18-36 months), 27(11%)-late relapse(> 36 months)]. Of these patients, 157(64.1%) and 58 (23.7%) had isolated medullary and extramedullary relapse whereas 30(12.2%) had combined relapses. Sixty(85%) of refractory-disease and 61(24.5%) of relapsed patients were offered salvage therapy. Only 6 patients underwent subsequent allo-HSCT, while 5 had 2nd-relapse. Median and 3-year OS for refractory disease was 45-months and 63.8%(53.8%-75.6%, 95%CI), while for Relapse it was 27-months and 26.6%(19.7%-35.8%, 95%CI), respectively. Multivariate analysis for refractory disease showed End-of-Induction(EoI) MRD as highly significant [HR:50.93(16.09-207.85, $p<0.001$)], while for relapsed disease EoI-MRD [HR:1.99(1.12-3.52, $p=0.019$)], Male[HR:1.52(1.09-2.14, $p=0.014$)] and hyperdiploidy [HR:0.45(0.26-0.79, $p=0.005$)] also had an impact. For induction mortality, only Initial-Risk treatment retained significance on multivariate analysis ($p=0.021$ for IR and HR opposed to SR).

Conclusions: The ICiCLE-Pilot protocol gave good overall outcomes but had increased sepsis-related mortality on high-risk treatment. Simultaneously, EoI-MRD remained the most important predictor of relapse/refractory disease, signifying that future strategies should focus concurrently on addressing Treatment-toxicities while achieving early MRD-negative status. Acknowledgements: ICiCLE-2014 Study Group, National Cancer Grid, Indian Council of Medical Research.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PV0078 / #939 | Characterizing the Complications Associated with Central Venous Access Devices in Childhood Acute Lymphoblastic Leukaemia Patients During Maintenance Therapy

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Background and Aims: Current Canadian guidelines indicate that a central venous access device (CVAD) be sustained during maintenance therapy of childhood Acute Lymphoblastic Leukemia (ALL) for venous access. While CVADs replace the need for repeated venipuncture, they are prone to complications. We aim to characterize the incidence of CVAD-associated complications, episodes of febrile neutropenia (FN), and blood products received during maintenance therapy of pediatric ALL patients in order to improve CVAD guidelines.

Methods: In this retrospective population-based study, patients aged 0-18 years diagnosed with ALL at the I.W.K. Health Centre in Nova Scotia, Canada, requiring a CVAD during maintenance therapy between January 2008 to December 2018 were included. Patient charts were reviewed for episodes of FN and blood product administration. Characteristics of CVADs including number of lines and reasons for removal were collected from a pre-existing database. The data were analyzed with both descriptive and inferential statistics.

Results: A total of 192 patients were included. The subtypes of ALL included B cell (91.7%), T cell (6.8%), Mixed (0.5%), and Philadelphia (1.0%). Of these patients, majority were standard-risk ALL (60.4%), and the remaining were high-risk ALL (39.6%). High-risk ALL patients had a greater mean number of CVADs and a significantly higher proportion of CVADs lost compared to standard-risk ALL. However, the reason for loss of CVAD was not differentiated between groups. The number of FN episodes and blood products between groups were not statistically different.

Conclusions: Guidelines outside Canada support the removal of CVADs and administration of intravenous chemotherapy peripherally for pediatric ALL patients every 3 months during maintenance therapy. While high-risk ALL patients had a significantly higher number of CVADs and CVAD loss, there were no differences in the number of FN episodes or need for blood products between the risk groups. Further research is required to support the need of CVADs during maintenance therapy.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0079 / #966 | Prophylactic Use of Levofloxacin Leads to Reduction in the Febrile Neutropenia Episodes During Induction Chemotherapy in Children of Acute Lymphoblastic Leukemia and Lymphoblastic Lymphoma

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Background and Aims: Over the past years the outcomes of acute lymphoblastic leukemia /lymphoma (ALL/LBL) have dramatically improved due to intensive chemotherapy regimen and good supportive care in developed countries. But in low-middle countries, infections are still the leading cause of morbidity and mortality. The induction mortality at our institute from 2015 till 2016 was reported to be 10 percent and infection was major contributing factor. Antibiotic prophylaxis is routinely used in acute myeloid leukemia and in patients undergoing transplant, but no definite clinical guideline is available in pediatric lymphoblastic leukemia/lymphoma.

Methods: A single- center, observational cohort study was carried out at Shaukat Khanum Memorial Cancer Hospital and Research Centre. Data was collected prospectively of patients diagnosed with ALL who received levofloxacin prophylaxis during the induction phase and was compared with historical controls who did not receive antibiotics prophylaxis.

Results: Total 100 patients were enrolled, among which 61 patients received levofloxacin prophylaxis, whereas 39 patients did not receive any antibiotic prophylaxis. The mean age of presentation of both groups was 3.3 years, no significance difference was seen in the clinical characteristics. The patients started on levofloxacin prophylaxis had lower incidence of febrile neutropenia episodes than patients who did not receive any prophylaxis (p- value 0.01)(odd ratio [OR]: 0.28, 95% Confidence Interval (CI): 0.1 – 0.78). No significant resistance against antibiotics was seen. No significant difference was seen in the induction mortality between the two groups(p value- 0.37).

Conclusions: Levofloxacin prophylaxis reduced the rate of febrile neutropenia episodes among patients, but it did not affect the infection related mortality. Further study regarding antibiotic prophylaxis needs to be carried out and long-term resistance against antibiotics to be considered.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PV0080 / #975 | Hepatic Sinusoidal Obstruction Syndrome Post Chemotherapy in Pediatric and Adolescent Age: Case Series of Six Patients in Qatar

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Background and Aims: Hepatic sinusoidal obstruction syndrome (HSOS) is a life-threatening complication of hematopoietic stem cell transplant and chemotherapy treatments for a variety of tumors. HSOS is associated with high incident of morbidity and mortality. Case presentation We report here the cases of 6 patients who developed HSOS after chemotherapy. All patients were diagnosed applying the new the European Society for Blood and Marrow Transplantation (EBMT) criteria specific for children and adolescence, whose application gave the advantage of proper and timely recognition of the HSOS. 2 of the patients had ataxia telangiectasia as underlying potentially predisposing syndrome.

Methods: All patients received supportive treatment and ursodeoxycholic acid therapy, 3 out of 6 received additionally defibrotide. The median duration of the SOS episodes was 9 days. 5 out of 6 HSOS episodes resolved, 1 patient died due to multi-organ failure, being probably the underlying condition of ataxia-telangiectasia as an aggravating factor.

Results: We report 6 patients who developed HSOS after chemotherapy for hemophagocytic lymphohistiocytosis, Ewing sarcoma, leukemia

and chemotherapy plus radiation (Wilms tumor) in the last 2 years at Sidra Medicine. All patients were diagnosed applying the new the European Society for Blood and Marrow Transplantation (EBMT) criteria specific for children and adolescence, whose application gave the advantage of proper and timely recognition of the HSOS. 2 of the patients had ataxia telangiectasia as underlying potentially predisposing syndrome. All patients received supportive treatment and ursodeoxycholic acid therapy, 3 out of 6 received additionally defibrotide. The median duration of the SOS episodes was 9 days. 5 out of 6 HSOS episodes resolved, 1 patient died due to multi-organ failure, being probably the underlying condition of ataxia-telangiectasia as an aggravating factor.

Conclusions: We report on the course of HSOS and outcome in six cases of postchemotherapy/radiotherapy. The diagnosis was based on the new EBMT criteria specific for children and adolescence. Timely diagnoses and management of HSOS is essential to avoid multi-organ failure, especially that prevention and treatment options are available nowadays.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0081 / #1073 | Comparison of Outcomes after Hematopoietic Cell Transplantation from Haploidentical and Unrelated Donors in Pediatric Patients with Acute Myeloid Leukemia

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Background and Aims: We investigated the outcomes of allogeneic hematopoietic cell transplantation (HCT) in children and adolescents with acute myeloid leukemia (AML) and compared the outcomes between haploidentical family donor (HFD) and unrelated donor (URD).

Methods: Fifty-nine pediatric patients with AML underwent allogeneic HSCT between January 2011 and December 2019. Donors were haploidentical family (HFD) for 27 patients and unrelated volunteer for 32. Forty-three (HFD 15, URD 28) received HSCT in CR1, 9 (HFD 7, URD 2) in CR2, 3 (HFD 2, URD 1) in CR3, and 4 (HFD 3, URD 1) in PR. HCT from HFD received ex vivo $\alpha\beta$ T-cell depleted graft.

Results: Of a total of 59 patients, one died at D+6 before engraftment and the remaining all 58 patients achieved sustained neutrophil engraftment at a median of 11 days (range, 9-18). Neutrophil engraftment was significantly faster in HFD compared to URD (10 days for HFD and 12 days for URD, $P=0.008$). Cumulative incidence (CI) of acute GVHD grade 3-4 was not different between HFD and URD. But CI of moderate/severe chronic GVHD was significantly lower in HFD than URD (12% for HFD and 43.2% for URD, $P=0.02$). The TRM and relapse rate (RR) at 3 years post-transplant were not different

according to donors (TRM, 7.4% for HFD and 9.8% for URD; RR, 11.1% for HFD and 19.3% for URD, $P>0.05$). At a median follow up of 70 months (range, 14-123), 3-y EFS were 82% for HFD and 70% for URD, respectively. In a multivariate analysis, the incidence of moderate/severe chronic GVHD was significantly less in HFD-HCT compared to URD-HCT (HR 0.261, $P=0.036$).

Conclusions: In this study, the outcome of haploidentical HCT was comparable with HCT from URD in pediatric patients with AML, demonstrating that HCT from HFD is a realistic option for patients who lack a matched related or unrelated donor.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0082 / #1092 | An Outcome Analysis of Childhood Acute Promyelocytic Leukemia Treated with ATRA and Arsenic Trioxide, and Limited Dose Anthracycline

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Background and Aims: The overall survival of Acute Promyelocytic Leukemia (APL), reported in recent studies, is approaching to 90% wherein, arsenic trioxide (ATO) and all-trans retinoic acid (ATRA) are used as the mainstay of treatment with either limited or no use of anthracycline and cytarabine. This study is aimed to ascertain the outcome of children with APL using similar approach.

Methods: A total of 30 patients with APL, registered from January 2015 to December 2018, were reviewed. Diagnosis was established on bone marrow aspirate and confirmed by the presence of PML-RARA translocation. Treatment protocol was based on Australian APML 4 study performed by Australian Leukemia Lymphoma Group (ALLG). Lumbar puncture was not performed as it was not part of the protocol due to the risk of bleeding.

Results: The mean age in current cohort was 9 years with 53% males. Seven (23.3%) patients died and three (10%) abandoned treatment during induction. Twenty patients completed the intensive phase of chemotherapy and all (100%) of them attained molecular remission (MR). One patient dropped out after MR whereas, 19 remain on follow up with no evidence of disease, reflecting disease free survival (DFS) of 95%. With a median follow up of 2.5 years (range 2.1-4.8 years) the 5 years Kaplan-Meier estimate of OS was 63% and 73%, with and without abandonment, respectively. Analysis of outcome according to risk groups revealed inferior outcome of high risk (HR) group (38% and 50% with and without abandonment, respectively) in contrast to standard risk (SR) group which showed better outcome (82% and 88% with and without abandonment, respectively).

Conclusions: The attainment of 100% molecular remission and absence of relapse supports the effectiveness of this regimen. Moreover, it is found to be less toxic and therefore, can be conveniently managed in day-care settings.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0083 / #1198 | Ambulatory and Hospital Care for Patients Diagnosed with Leukemia in Brazil: A Quantitative Analysis

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Background and Aims: Leukemia is a group of hematological cancers. Acute lymphocytic leukemia (ALL) and acute myeloid leukemia (AML) are the two most common types of leukemia in childhood. In Brazil, there are few population-based studies on the treatment of leukemia and this information is essential for planning public policies. The objective is to evaluate the panorama of hospital and ambulatory system for patients with leukemia up to 19 years of age in the Unified Health System, in 2009-2018.

Methods: Descriptive study based on secondary data from the Ambulatory Information System (AIS) for authorizations for highly complex procedures (AHCP) for chemotherapy and Hospital Information System (HIS) by authorizations from hospital admissions (AHI) of patients diagnosed with ALL or AML from 0 to 19 years in 2009-2018.

Results: 21,331 patients with leukemia up to 19 years old were treated in the ambulatory system in 2009-2018 being 76.2% ALL patients and 17.0% AML patients. The main procedure performed in these patients was cancer chemotherapy in childhood and adolescence - first line. In the hospital information system, 29,597 patients were treated, 69.8% ALL patients and 15.0% AML patients. Of these, there were 146,374 admissions events, being 76.5% ALL patients. The main procedure performed was hospitalization for acute/chronic leukemia chemotherapy. The male gender and the group up to 9 years old were more frequent in ambulatory and hospital system.

Conclusions: The results indicate a wide panorama of leukemia treatment in Brazil in 2009-2018. However, the available records do not allow assess patient's journey, identify places of care, treatment outcome and quality of care. Therefore, it is necessary to create a national cancer registry that makes it possible to monitor the care with the objective of continuous improvements.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0084 / #1510 | Two New Kindreds with Germline CEBPA Mutations and Acute Myeloid Leukemia (AML)

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Background and Aims: We present two unrelated patients with AML and family histories of leukemias. Patient #1, a 12-year-old male,

had numerous maternal family members with leukemia including an aunt with leukemia at 5 years and second leukemia at 20 years. Patient #2, a 6-year-old female, had a brother with childhood AML. Here we report the genetic evaluation and disease course of these patients.

Methods: We performed FoundationOne genomic profiling on diagnostic marrows and germline genetic testing on skin fibroblasts. We sequenced blood samples from the patients' parents for patient-specific mutations.

Results: FoundationOne results revealed bi-allelic heterozygous pathogenic variants in CEBPA: S28Rfs*132 and Q311_Q312insF (Patient #1) and E309_T310insQRNVE and H24fs*84 (Patient #2). Skin fibroblast sequencing confirmed germline heterozygous CEBPA mutations. The healthy 36-year-old mother of Patient #1 and the healthy 50-year-old father of Patient #2 carry germline CEBPA mutations. CEBPA mutations occur in 10-15% of pediatric AML with 70-80% having biallelic mutations, and are associated with favorable outcomes. Germline CEBPA mutations occur in 5-10% of all bi-allelic CEBPA-mutated AML, have autosomal dominant inheritance, and high AML penetrance. Of AML patients with germline CEBPA mutations, 91% achieve complete remission, with 10-year OS of 67%. There is a 56% incidence of a secondary leukemia, which carries a 10-year 60% mortality rate, thus consideration of hematopoietic stem cell transplant is warranted. Neither patient had a suitable donor, so both were treated with chemotherapy alone, and are doing well off therapy.

Conclusions: Treatment decisions for individuals with germline CEBPA-mutant AML are complex. Such patients have high risk for subsequent leukemias, but transplantation carries risks and data support that second leukemias are chemosensitive. One must incorporate age of diagnosis, risks for therapy-associated toxicities, donor availability, and patient/family preferences. These cases highlight the importance of evaluating children with myeloid malignancies for inherited germline predispositions.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0085 / #1591 | Outcome Prediction of Chronic Myeloid Leukaemia (CML) in Children

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Background and Aims: We evaluated the existing risk assessment tools for CML in children.

Methods: A total of 55 patients from 1.4 to 18.0 years with newly diagnosed CML between 1996 and 2019 were included. Forty-nine patients presented in the chronic phase, thirty-six of whom were treated with upfront tyrosine kinase inhibitor (CP-TKI group); one presented in the accelerated phase and 4 in the blastic phase. Treatment, survival, responses, and tolerance were evaluated.

Results: The median follow-up time was 8.7 years (range, 2 months to 24.3 years). All patients in the CP-TKI group received imatinib as their first TKI treatment. Allogenic stem cell transplantation was performed in one patient after complete cytogenetic response was achieved with imatinib and in one patient with imatinib failure. Dasatinib and nilotinib were prescribed as second-line TKI in 5 patients and 4 patients respectively. The 10-year overall survival (OS), progression-free survival (PFS) and event-free survival (EFS) of TKI treated group was 97%, 91.4% and 72.3% respectively. The rates of major molecular response and deep molecular response of TKIs were 81.2% and 67.5% at 60 months. The EUTOS long-term survival (ELTS) risk grouping did not predict OS, PFS or EFS. The IMAFAIL risk groups are correlated with the risk of imatinib failure.

Conclusions: TKIs resulted in excellent long-term overall and progression-free survival in children and adolescents with newly diagnosed CML in the chronic phase. Further studies are required to modify the existing prognostic scoring system or develop new ones for children.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0086 / #1642 | Clinical Profile and Outcomes of Childhood Acute Myeloid Leukemia with Myeloid Sarcoma

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Background and Aims: Myeloid sarcoma(MS) is defined as a solid tumour consisting of abnormal myeloblasts that occurs outside the bone marrow. The impact of MS on the prognosis of AML have been unclear. We present our experience of treating paediatric AML patients with MS using chemotherapy with or without radiotherapy.

Methods: Children less than ≤15-years of age with AML diagnosed between January,2014 and December,2019 were included. Baseline evaluation included bone marrow aspiration flow cytometry, cyto-

netics by FISH and molecular techniques. Chemotherapy consisted of standard 3+7 induction, 3 courses of High-dose cytarabine followed by maintenance-therapy. Emergent radiotherapy was given in patients with impending or existing organ impairment due to MS whereas adjuvant radiotherapy to the patients with significant residual disease or extensive disease at presentation.

Results: Out of 345 registered AML patients 44(12.7%) had MS. The M:F ratio was 1.9 and median age was 95(32-178)months. Median baseline total leukocyte count was 17,910 (3260-74700)/cumm. Cytogenetic/Molecular abnormalities seen were RUNX1-RUNXT1(n=35,79.54 %), CBFB-MYH11 (n=2,4.54 %), KMT2A-r (n=3,6.8%), FLT3-ITD (n=2,4.54 %). MS was unifocal in 35(79.54%) patients and multifocal in 9(20.5%) patients. The most common location was orbital(n=27,61.4%) with 10(22.7%) patients presenting with threatened vision followed by para-spinal(n=6,13.6 %) of which 3(6.5%) presented with features of cord compression. Twenty-five(56.8%) of the 44 patients received radiotherapy(RT): 13(29.54%) patients received emergent RT and 10(22.72%) patients received adjuvant RT. Patients who received RT and who did not receive RT had comparable 3-year RFS(54.7±12.5 vs 61.9 ±12.3 p=0.824) and 3-year OS(44.4±12.6 vs 50.7±12, p=0.851).

Conclusions: Myeloid sarcoma did not confer an adverse outcome in children with AML. Patients who did not have any residual disease and did not receive radiation had comparable outcome to those who did not have MS.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0087 / #1719 | Genomic Characterization in a Cohort of Pediatric Patients with Acute Myeloid Leukemia in Colombia

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Background and Aims: Characterize the genomic alterations (chromosomal and molecular) in pediatric patients diagnosed with acute myeloid leukemia (AML) and their co-mutation patterns.

Methods: Samples with AML diagnosis in the ages of 0-18 years were included between 2016-2021 from Fundación HOMI and Clínica Infantil Colsubsidio (Bogotá, Colombia). Chromosomal study and Next Generation Sequencing panel (NGS) of 30 genes for myeloid disorders were carried out in Servicios Médicos Yunis Turbay- Genetics Institute (Bogotá, Colombia).

Results: Cohort of 47 patients, 30 male and 17 females. Thirty-one patients presented cytogenetic alterations (66%), of which 12 (26%) had Core-Binding Factor alterations (7, t (8; 21); 4, inv (16)), 8 patients (17%) with KMT2A rearrangements and 11 (23%) with other chromosomal alterations. Forty patients (85%) had at least one driver mutation. Mutations were identified in 16 genes, among which are: FLT3 gene in 16 (34%) patients (8, ITD; 8, TKD), NRAS 14 (30%), KIT 9 (19%), WT1 6 (13%), KRAS 6 (13%), CEBPA 4 (9%), PTPN11 4 (9%), RUNX1 4 (9%), ASXL1 2 (4%), NPM1 2 (4%), ETV6 2 (4%), TET2 1 (2%), IDH1 1 (2%), EZH2 1 (2%). All patients with inv (16) had FLT3 and KIT gene mutations, 3 had NRAS and 1 had WT1 mutation. Four patients with t(8;21) had concomitant involvement of KIT, 2 of ASXL1, 1 of NRAS, 1 of ETV6, 1 of FLT3. In the group with KMT2A rearrangements, 2 had mutations in PTPN11, 1 in NRAS, 1 in FLT3, 1 in KRAS and 1 in CEBPA.

Conclusions: We were able to identify that 89% of patients had either a chromosomal or a molecular alteration as a driver mutation. These results provide the possibility to correlate the genotypic findings with clinical outcomes, allowing to refine the prognostic stratification and diagnostic algorithm in Colombia, and to incorporate personalized treatments in the near future.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0088 / #191 | Prevalence and Prognosis of Anemia at Diagnosis in Chronic Myeloid Leukemia in Children

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Background and Aims: The clinical presentation of the pediatric chronic myeloid leukemia (CML) is commonly severe at diagnosis. Anemia appears to be frequent at diagnosis but its prevalence and its prognosis impact are not well-known.

Methods: We selected from the international pediatric registry I-CML-Ped-Study all patients under 18 years at CML diagnosis, in chronic phase and treated upfront with imatinib. We examined the hemoglobin level at diagnosis to assess the prevalence of anemia and its impact on response to tyrosine kinase inhibitors (TKI).

Results: 430 patients were included. The prevalence of anemia at diagnosis was 81%, with a mean hemoglobin level at 96.4 g/l. 122 patients presented with moderate anemia, and 110 with severe anemia according to the WHO guidelines. Moderate and severe forms were significantly associated with younger age at diagnosis, presence of splenomegaly, asthenia and increased leukocytosis and basophils. Times to achieve the major and complete molecular responses were increased for patients with moderate and severe anemia. Resistance to imatinib was significantly more frequent for patients with moderate and severe anemia, with more TKI switches. However, hemoglobin level had no significant impact on survival.

Conclusions: Anemia at diagnosis of pediatric CML is frequent, and may be considered as an independent prognosis factor.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PV0089 / #271 | Outcome of Pediatric Acute Myeloid Leukemia in Low- and Middle-Income Countries: A Systematic Literature Review

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Background and Aims: Survival of pediatric acute myeloid leukemia (AML) in low- and middle-income countries (LMICs) seems extremely poor, but literature is scarce. We systematically reviewed the literature on treatment outcomes of pediatric AML in LMICs.

Methods: The electronic databases PubMed, EMBASE, the Cochrane Library, and Scopus were searched from inception up to January 2020. Inclusion criteria were: 1) carried out in a LMIC, as classified by the World Bank; 2) a study population of ≥ 5 children (0-19 years) with de novo AML; 3) reporting of at least one of the following outcomes: complete remission (CR), early death (ED), relapse, probability of event-free survival (pEFS) or overall survival (pOS). Relevant data from sole abstracts and abstracts in the International Society of Pediatric

Oncology (SIOP) abstract books (2010-2019) were studied separately, but not included in our main analyses and conclusions.

Results: Twenty-seven articles, including 2,945 children with AML were eligible. CR rates varied from 32%-100% (n=19 studies). ED rates varied from 3%-59% (n=18 studies). Treatment-related mortality (TRM) rates were mainly above 10%. Relapse rates varied from 10%-83% (n=15 studies), but were mostly between 30%-45%. The 5-year pEFS and pOS varied from 24%-63%, and 10%-72%, respectively (n=7 studies). Abandonment rates varied from 0%-78% (n=16 studies). Data studied from 23 sole abstracts and 30 SIOP abstracts supported these findings.

Conclusions: This first systematic review on treatment outcomes of pediatric AML in LMICs yields poor outcomes compared with high-income countries. Treatment outcomes varied markedly between LMICs. There was a wide variation in the reporting and analysis of treatment outcomes across studies, which should become more uniform. The substantially inferior treatment outcomes are mostly attributed to high abandonment, ED, and TRM rates, and low salvage rates after relapse. High ED and TRM rates are most likely caused by suboptimal supportive care and limited capacity.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0090 / #412 | Evaluation of Chest Radiographs in Children with Newly Diagnosed Acute Myeloid Leukemia

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Background and Aims: Acute myeloid leukemia (AML) is the second most common leukemia in children. At diagnosis, chest x-rays (CXRs) are evaluated to guide initial management, but their findings and utilities are not well known.

Methods: Findings of CXRs obtained at diagnosis and within 7 days of treatment initiation and their associations with clinical characteristics, initial management, and adverse events were evaluated in children with newly diagnosed AML.

Results: CXRs of 302 patients who were treated on 3 clinical trials (AML97, AML02, and AML08) at St. Jude Children's Research Hospital revealed abnormal findings in 164 (54.3%) patients. Pulmonary opacity (n=99, 32.8%), perihilar thickening (n=40, 13.2%), pleural effusion (n=22, 7.3%), and lymphadenopathy (n=19, 6.3%) were common. Pul-

monary opacity was more commonly seen in patients receiving diagnoses at a younger age (median age [years]; 7.0 vs. 10.5, P<0.01) and with French-American-British classification (FAB) M4/M5 (P<0.01); perihilar thickening was associated with younger age (median age [years]; 2.6 vs. 10.4, P<0.01); pleural effusion was associated with FAB M4/M5 (P=0.04); and lymphadenopathy was associated with central nervous system involvement (P=0.01). Pulmonary opacity and pleural effusion at diagnosis were associated with less-invasive central venous access (P=0.01 for both), increased use of cytoreductive chemotherapy or leukapheresis (P=0.01 and P=0.04, respectively), intensive care unit admissions (P<0.01 and P=0.02, respectively), and respiratory support (P=0.01 for both). Lymphadenopathy was associated with respiratory support (P<0.01). Patients with pulmonary opacity were more likely to experience infection (hazard ratio [HR]=1.45, 95% confidence interval [CI]=1.09-1.93) and hemorrhage (HR=2.34, 95% CI=1.18-4.67) during the first course of induction. Patients with bronchial/perihilar thickening were more likely to have febrile neutropenia (HR=1.77, CI=1.15-2.73) and gastrointestinal side effects (HR=1.61, CI=1.01-2.56).

Conclusions: Abnormal CXR findings at diagnosis and within 7 days of treatment initiation can help guide initial management and predict patient complications.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0091 / #552 | A Novel Classification of Hematologic Conditions in Patients with Fanconi Anemia - A Report from The German Fanconi Anemia Registry

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Background and Aims: Fanconi anemia (FA) is an inherited bone marrow failure syndrome associated with an increased risk of myelodysplastic syndrome (MDS) and acute myeloid leukemia (AML). The bone marrow in patients with FA is typically hypocellular and can mimic other conditions such as acquired severe aplastic anemia. The presence of cytopenia and myelodysplastic features alone are not sufficient for the diagnosis of MDS in FA - with the exception of distinct subtypes such as refractory anemia with ring sideroblasts or in cases with rapid rise in cellularity in whom MDS needs to be considered. In patients with FA, the diagnosis of MDS requires an elevated blast count and/or other

definite signs of transformation, such as duplication of chromosome 3q, deletion 7q, or monosomy 7. Until now, however, an internationally accepted classification of myeloid neoplasia in patients with FA is lacking.

Methods: We studied 86 patients enrolled in the German FA registry and performed an algorithm for the diagnosis of benign and malignant hematologic conditions in FA that is based on blast percentage and cytogenetics.

Results: In addition to FA-BMF, FA-MDS-without excess blasts (FA-MDS-non EB), FA-MDS with excess blasts (FA-MDS-EB), and FA-AML, the classification includes the new category FA with cytogenetic aberration of indeterminate potential (FA-AIP). The new classification FA-AIP is characterized by cytogenetic aberrations such as duplication of chromosome 1q, deletion 6p, or deletion 7p that remain stable over a prolonged period without transformation.

Conclusions: The proposed diagnostic algorithm avoids the misdiagnosis of myeloid neoplasia and averts the immediate indication for HSCT. In addition, the new algorithm serves as a model for the classification of myeloid neoplasms in patients with a germline predisposition. The concept of AIP may also apply to other inherited bone marrow failure syndromes such as Shwachman-Diamond syndrome.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0092 / #611 | Genetic Characteristics and Survival of Pediatric Patients with Diagnosis of Acute Myeloid Leukemia in the National Unit of Pediatric Oncology (UNOP), Guatemala

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Background and Aims: Las características genéticas de la leucemia mieloide aguda (LMA) guían el tratamiento e identifican a los niños con mal pronóstico. La UNOP ha realizado un panel citogenético básico que incluye t(8; 21), t(15; 17), Inv16, FLT3/ITD y FLT3/D835 desde 2012. Nuestro objetivo es identificar la correlación de estas características con la supervivencia general (SG).

Methods: A descriptive, retrospective analysis of patients <18-year-old newly diagnosed with AML from January 2013 to December 2018 at UNOP. The treatment consisted in two induction cycles with Cytarabine 100 mg/m²/q12h 8-day and Doxorubicin 30 mg/m²/d 3-day (AD) followed by 3 cycles of consolidation based on high-dose cytarabine. In 2018, the cycles AD were substituted by 2 cycles MAG (mitoxantrone 10 mg/m²/d 3-day + cytarabine 10 mg/m²/q12h 10-day + granulocyte-colony stimulating factor 5 mcg/kg/d 10-day). The cytogenetic test was performed at the beginning of therapy. Morphological complete remission (CR) was evaluated after the first induction cycle. Were excluded: Down syndrome, myelodysplastic syndromes and treatment refusal. Death or abandonment were considered events.

Results: Of 151 patients diagnosed, 136 were eligible. The median age was 8.5 years (1-18), 78 males (57%). The test was performed in 86 (63%) and was positive in 35 (40%), t(8; 21) was positive in 24 (28%), FLT3/ITD in 6 (7%) and others 5 (6%). There were 29 (22%) induction deaths, 10 (8%) resistant disease and 94 (70%) CR. OS was 47 (35%), patients with t(8;21) positive had significantly higher OS (54%) compared with FLT3/ITD (17%), or without genetic alteration (29%) (p=004).

Conclusions: The OS of AML in Guatemala treated only with intensive chemotherapy is poor; t(8;21) could be a predictor of better prognosis. Efforts to complete the genetic testing in all patients should be made.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0093 / #651 | Genomic Eligibility Algorithm at Relapse for Better Outcomes (GEARBOX): A Decision Support Tool for Matching Children with Relapsed Acute Myeloid Leukemia to Clinical Trials

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Background and Aims: Therapy for children with relapsed or refractory leukemia remains challenging. Delays in testing and clinical trial enrollment means most children do not receive targeted therapies during reinduction. The Leukemia & Lymphoma Society (LLS) and pharmaceutical have partnered the Pediatric Acute Leukemia (PedAL) study, which includes rapid central testing and development of treatment regimens. This will be an integrated multi-site, multi-therapy trial matching patients to precision treatment based on their unique tumor biology. We created a novel platform for trial matching, identifying, and facilitating patient enrollment within 48-72 hours of relapse or refractory disease identification. Developed at UChicago, GEARBOX (Genomic Eligibility Algorithm at Relapse for Better Outcomes) helps physicians rapidly match children with relapsed/refractory AML to trials based on leukemia biologic. GEARBOX facilitates trials identification by synthesizing tumor biology and trial eligibility criteria, improving access to precision therapies through the PedAL initiative and improving outcomes for children with hard-to-treat leukemia.

Methods: GEARBOX's backend parses clinical trial criteria from clinicaltrials.gov through manually curated and natural language processing-derived information. Clinicians enter clinical, genomic, and biomarker data into the frontend to identify potential trials. For each trial listed, additional important information is available.

Results: In the pilot, open trials are listed with eligibility parameters and links to additional information with new trials added as they become available. Users enter clinical, genomic, and biomarker data; the list of potentially-relevant trials is refined in real time. The cloud-based system will be available to any clinician.

Conclusions: This is a breakthrough matching tool geared towards children with relapsed and refractory leukemia. The platform streamlines access to critical available trial information, significantly reducing time to enrollment. Future innovations include direct consumption of data from source systems and automatic ingestion of clinical data, reducing data entry time and increasing accuracy of predictions. GEARBox will be available for general use by January 2022.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0094 / #708 | Pediatric Langerhans Cell Histiocytosis: A Single Center Experience

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Background and Aims: Langerhans cell histiocytosis(LCH) is a rare disease that usually affects the bones. The aim of this study is to evaluate the demographic, clinical characteristics, treatment options and survival rates in LCH in a single center.

Methods: From 1988 to 2021, the medical files of 53 patients with LCH were evaluated retrospectively.

Results: The median age at diagnosis was 4.5 years (2 months-12 years), Male/Female: 0.9. Median time to diagnosis was 8 weeks. Initial presenting symptoms were swelling and pain for 77% of patients. 11% of the patients presented with skin rash, 5% with polyuria-polydipsia. Single system disease rate was 68% (36/53) and 25/36 with unifocal bone involvement. 32% of patients had multisystem involvement and 10% had risky organ involvement. Seventeen (32%) patients were followed without any treatment after excision-curettage and incisional biopsies. Systemic chemotherapy (CT) was given to 60% of the patients; fourteen of these patients treated with LCH-III protocol, four of them with LCH-IV protocol, fourteen of them according to DALHX-83/90 protocol. Local surgery for 62% of patients, radiotherapy (RT) given to 22% of patients. The disease relapsed in 12 patients; once in nine, twice in two patients and four relapses in one. All relapses were treated with chemotherapy. For relapsed/refractory disease, etoposide-cytarabine based chemotherapy regimens were given. One patient is still alive and in remission after 4 relapses. Only 4/53 patients died due to disease progression. Seven patients have endocrine and one patient orthopedic problems as late effects of the disease and treatment. The median follow-up period was 8 years (1 month-30 years), only 3 patients were lost to follow-up. Overall survival rate was (OS) 92% in 1 year; 5 and 15 years. Event-free survival rate (EFS) was 82% at 1 year, 64% at 5 and 15 years.

Conclusions: Sixty-eight percent of the patients with LCH had single system involvement. The higher recurrence rate in multisystem disease was statistically significant. The survival and overall survival rates of all patients were consistent with the literature. Vinblastine-steroid combination as chemotherapy and low dose RT were effective in treatment for most of the cases.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0095 / #730 | 3Q21 Deletion Affects GATA2 and is Associated with Myelodysplastic Syndrome in Children

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Background and Aims: 3q21 deletion is a rare genetic insult that presents as a clinically variable, devastating condition. Hemato-oncologic manifestations are possible in this condition given the deletion of several hematologic candidate genes. The goal of this project was to report the hematologic manifestations of a patient with 3q21 deletion at our institution and to conduct a systematic review of the hematopathologic consequences of 3q21 deletion.

Methods: We conducted a systematic literature review for studies describing 3q21 deletion. Our systematic search strategy included Ovid MEDLINE, EBM Reviews, Embase, Scopus, and Web of Science. Two independent researchers reviewed all abstracts.

Results: Systematic search revealed 327 abstracts. After review and exclusion, we included 13 manuscripts that described 15 unique cases of 3q21 region deletion. Four hemato-oncologic candidate genes are included in band 3q21: RPN1 (acute myeloid leukemia, chronic granulomatous disease), GATA-2 (immunodeficiency, lymphedema, and myelodysplasia), and ZNF148 (colorectal cancer), and GP9 (Bernard-Soulier Syndrome Type C). Including our patient, there were two patients who developed MDS (13%, n=2/15). Both patients had evidence of chromosome 7 monosomy. The previously described female patient presented with pancytopenia complicated by gingivitis and staphylococcal infection at the age of 11 years old. The patient at our institution was a 9-year-old female who presented with intermittent neutropenia and lymphocytopenia and new onset mild thrombocytopenia with no recent increase in infection. Both patients required hematopoietic cell transplantation.

Conclusions: Patients with 3q21 deletion have a predilection for myelodysplastic syndrome, which is likely due to GATA-2 deletion. Akin to other patients with germline GATA-2 mutations, patients with 3q21 deletion may require regular complete blood count monitoring and annual bone marrow evaluation to monitor for development of MDS.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0096 / #852 | Hand Surgeons are Important in the Identification of Fanconi Anaemia

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Background and Aims: Fanconi Anaemia (FA) is associated with physical anomalies of which hand anomalies are frequent. We describe the clinical epidemiological features of FA as assessed by Hand Surgeons and relate it to age of diagnosis and other features of Fanconi Anaemia.

Methods: Photographs and radiographs of FA cases from prospectively collected data set at the National Cancer Institute were analysed and classified by 2 consultant hand surgeons using Bayne and Klug (radius anomalies), Blauth (thumb hypoplasia) and Wassel (thumb duplication) classification systems. The data were analysed focusing on age at diagnosis, type of limb anomaly and any potential association with other known clinical anomalies of VACTERL-H and PHENOS phenotypes.

Results: 48 cases were identified. The median age at diagnosis was 5 years (IRQ 0.33-8.75). 29/48(60%) cases had a limb anomaly; thumb hypoplasia was the commonest abnormality. Having a severe or mild abnormality was equally likely. 15/29(52%) children had surgery for their limb anomaly. Age at diagnosis was younger in the cases with a severe limb abnormality; only 50% with a milder limb abnormality were diagnosed by 2 years of age. 25/29(86%) with a limb anomaly had VACTERL-H or PHENOS but 4/29 (14%) did not.

Conclusions: The commonest limb anomaly in Fanconi Anaemia is thumb hypoplasia. Limb anomalies can vary involving the thumb and forearm. Patients with more severe anomalies are diagnosed early whilst those with milder limb anomalies are older at diagnosis. Our findings add to the evidence that all children with a radial limb anomaly identified by a hand surgeon should be referred for FA screening.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0097 / #859 | Improving Survival by Reduced Toxicity Regimens for Pediatric Acute Myeloid Leukemia

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Background and Aims: Outcomes of pediatric acute myeloid leukemia (AML) in the developing world are inferior due to high treatment

related mortality (TRM). We describe here outcome of pediatric AML treated as per reduced toxicity protocols.

Methods: Outcomes of AML patients aged <18 years were retrospectively studied. All except those with Acute promyelocytic leukemia (APML) were treated with 2 cycles of 3+7 (Daunorubicin 45 mg/m² x 3 days and Cytarabine 200 mg/m² x 7 days) chemotherapy followed by 4 cycles of high dose cytarabine (HIDC) (12 g/m²). Those eligible for allogeneic stem cell transplant (SCT) were transplanted after 2 cycles of HIDC. All APML cases were treated with Arsenic & all-trans-retinoic acid (ATRA). Children with high risk APML received additionally a course 3+7 as induction chemotherapy.

Results: Out of 26 AML cases managed at our centre from January 2016 to January 2021, 11 were excluded from the study (Nine referred with relapsed/refractory AML and two abandoned therapy). Remaining 15 cases, were classified as favourable-13, intermediate-1 & high-risk-1 according to European leukemia Net risk-stratification. Among favourable group; 5 had APML and all are in complete remission (CR)-1 with median follow up of 40 months (16-60 months). Of the remaining 8 favourable cases, 6 are in CR-1 with median follow up for 14 months (6-30 months) and 2 relapsed. Intermediate-risk patient refused matched sibling donor and later relapsed and died. High-risk patient underwent haploidentical SCT and remains disease-free at 5 years. TRM was 0%. Of the 3 relapsed cases, 1 died of refractory disease and 2 underwent allogeneic SCT in CR2 and both died (one relapsed and other died of acute graft vs. host disease). Overall, 80% are alive and disease free at median follow-up of 20 months (range 6-60 months).

Conclusions: Less toxic chemotherapy protocol both for AML and APML can help improve survival in the developing world.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.b
Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PV0098 / #971 | Twenty-Year Experience with Paediatric Acute Myeloid Leukaemia in a Resource Constrained Oncological Service

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Background and Aims: Survival gains in paediatric acute myeloid leukaemia (AML) in high-income countries have not been mirrored in lower- and middle-income countries (LMIC). This study aimed to determine the disease spectrum, outcome and complications of paediatric AML in a resource constrained oncological service.

Methods: From 1998 to 2017, the clinical details, treatment, outcome and complications of all paediatric patients with AML were analysed. Treatment was according to the Berlin-Frankfurt-Munster (BFM) AML 1998 protocol.

Results: Over the 20-year study period 119 children, with a mean age of 7 years, were diagnosed with AML. Fifty (41%) were classified as

standard risk and 54 (44%) as high risk. Included in the cohort, were 23 patients diagnosed with Acute Promyelocytic Leukaemia (APL). Overall, 81 (68%) patients died; 52 (64%) were due to disease-related causes and 29 (36%) due to sepsis. Thirty-six (44%) of the deaths had occurred by the end of induction. Infections occurred in 93 (78%) patients and sepsis-related deaths increased from 19% to 29% over the 2 decades. Overall survival in our cohort was 32% with an event free survival of 23%. The 5-year survival rate for children with APL was 30%.

Conclusions: Paediatric AML survival was found to be poor, with no improvement over the 2 decades. Induction and sepsis related mortality rates are unacceptably high. This data will be used to guide strategies to improve survival of paediatric AML in our setting. Targeting early mortality and children with APL have the potential for significant benefit.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0099 / #1094 | Spatial Distribution of Incident Pediatric Burkitt Lymphoma in Central and Northern Malawi and Association with Malaria Prevalence

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Background and Aims: Burkitt lymphoma (BL) accounts for about 90% of pediatric lymphomas in sub-saharan Africa. Plasmodium falciparum (Pf) malaria is considered an etiological factor of BL. We describe the distribution of incident pediatric BL in Malawi and association with malaria prevalence (PfPR).

Methods: We prospectively enrolled patients with incident pathologically confirmed BL at the only facility for cancer treatment in Northern and Central Malawi (2013-2018). District-level 6-year average BL incidence rate (BLIR) was calculated using pediatric population estimates from the Malawi Census. District-level PfPR (2010-2014) was extracted from Ministry of Health/KEMRI Wellcome Trust malaria report. Geographic visualization of BLIR and PfPR was conducted in QGIS. The relationship between PfPR and BL was assessed using simple regression.

Results: Of 218 cases, 143 (67%) were male and median age 9 years (IQR 6-12). Two-hundred (92%) were from Central Malawi and 69 (32%) from the capital Lilongwe. Four (2%) were HIV-positive. Central districts had higher BLIR than Northern, with the highest observed in Mchinji (1.3 cases per 100,000) and lake districts Nkhosha (BLIR=1.0) and Salima (BLIR=1.0). Northern districts Rumbi and Chitipa had the lowest BLIR (0 and 0.1). Visual comparison of BLIR and

PfPR maps clearly showed a positive relationship, excluding outliers Ntchisi and Likoma. Ntchisi (Central; high relative PfPR) had low BLIR (0.3) compared to neighboring districts. Likoma (Northern; low relative PfPR) had the highest BLIR (3.0) (although Likoma only had one crude case, its population is extremely small). Statistical analysis supported these findings. A 1% increase in PfPR predicted an increase of 0.03 BL cases per 100,000 (excluding Likoma) ($p=0.001$).

Conclusions: Our study supports evidence for an etiologic role of Pf in BL. Limitations include lack of data from the South and potential confounders such as proximity to treatment facilities. Future studies include evaluating spatial clustering and developing models to compare observed versus expected district-level BLIR.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0100 / #1412 | Hodgkin Lymphoma Treatment Using Abve-PC in Children and Adolescents in Botswana

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Background and Aims: Hodgkin Lymphoma (HL) is among the most curable pediatric cancers with overall survival of over 98% in high income countries (HICs). ABVE-PC (Adriamycin, Bleomycin, Vincristine, Etoposide, Prednisone, Cyclophosphamide) has become a standard protocol in HICs with limited published data from Africa.

Methods: A retrospective review was conducted of all children and adolescents diagnosed with HL registered in the Botswana Pediatric Oncology Database (BPOD) between January 2012 and March 2021 that were treated with ABVE-PC. Demographic, diagnostic and treatment data were collected and analysed. Overall survival (OS) was estimated using Kaplan-Meier analysis.

Results: Overall, 19 children were diagnosed with HL during the study period with 78.9% treated with ABVE-PC ($n=15$). Among the ABVE-PC cohort, 93.3% were male, median age was 9.1 years (range 6.0-17.5 years), 46.6% were underweight ($< 5\%$ BMI for age) and 2 were human immunodeficiency virus (HIV) positive. Cervical lymphadenopathy was the commonest presenting sign at 73.3%. HL was complicated by hemophagocytic lymphohistiocytosis (HLH) ($n=2$), pleural effusion ($n=2$) and autoimmune hemolytic anemia (AIHA) ($n=1$). Most had nodular sclerosing histology 66.6% and were at least stage 3 (73.3%). Treatment was initiated as high risk ($n=9$) and intermediate risk ($n=6$) with individualized radiotherapy ($n=9$). Two patients died with both deaths associated with HLH and occurring within the first six months of diagnosis (2 weeks and 6 months) while three patients required salvage therapy. With a median of 1.9 years of follow-up (InterQuartile Range (IQR), 1.1-5.2 years), the two-year OS was 86%

(95% Confidence Interval (CI) 55-96%). Treatment toxicity included cytopenia-associated treatment delayed (n=1, 6.6%) and radiation-induced hypothyroidism (n=2, 13.3%).

Conclusions: ABVE-PC was well tolerated with positive outcomes in a cohort of children and adolescents diagnosed with HL in Botswana. Radiation based on response individualized treatment is suitable in low resource settings where HL presents late with multiple unfavorable risk factors.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0101 / #1434 | Brentuximab Vedotin in Children with Hodgkin's Lymphoma

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Background and Aims: Brentuximab vedotin (BV) is an anti-CD30 monoclonal antibody combined with antineoplastic agent (monomethyl auristatin E). Although it is widely used in adult patients with lymphoproliferative disorders, its role in children is yet to be established. The present study assesses the effectiveness of BV-based therapy in children with relapsed or refractory Hodgkin's lymphoma (R-R HL).

Methods: BV was administered to 37 children (median age of 16 years) with R-R HL in RM Gorbacheva Research Institute, Pavlov University. Median follow up was 759 (87-1780) days. Nodular sclerosis was the most frequent histological subtype (n=29, 78%). At the moment of BV start advanced disease was observed in 22 patients (68%). Refractory disease was registered in 21 (57%) and relapse in 16 (43%) cases. Median number of prior therapy lines was 4 (1-6). BV 1.8 mg/kg triweekly monotherapy was administered in outpatient clinic (n=20, 54%). In 17 (46%) cases BV was combined with bendamustine (n=10, 59%) 120 mg/m² on Days 1-2 of each 3-week cycle, DHAP (n=3, 18%), nivolumab (n=3, 18%), nivolumab+bendamustine (n=1, 3%). Median number of BV infusions was 4 (2-12). Response to therapy was assessed according to Lugano criteria.

Results: Twenty five patients (67%) responded with complete response in 10 (27%) and partial response in 15 (40%) cases. Overall survival (OS) at 4 years according to Kaplan-Meier method was 70% and progression-free survival (PFS) – 30%. Following complications were registered: polyneuropathy (n=3, 8%), herpes zoster (n=2, 5%), anaphylactic shock (n=1, 3%), Quincke edema and urticaria (n=1, 3%). All side effects were cured without lethality or long-term disability.

Conclusions: BV-based therapy is effective in children with R-R HL and results in remission in the majority of patients. PFS remains sub-optimal but OS may be regarded as satisfactory for this heavily pre-treated group. BV-based therapies are characterized by low toxicity that enables their implementation in outpatient clinic.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0102 / #1546 | The Challenge of Local Relapse in Patients with T-Cell Lymphoblastic Lymphoma: Single Center Experience

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Background and Aims: The outcome of T-cell lymphoblastic lymphoma (T-LL) has improved dramatically over the past few decades. However, patients with relapsed or refractory lymphoblastic lymphomas still have poor prognosis despite aggressive therapy. Understanding the patterns of relapse among these patients may help in modifying their upfront approach

Methods: This is a retrospective study of pediatric patients with T-LL treated with modified BFM95 regimen at a single center during 9-year period (Jan2008 to Jan2017). Data were collected from our electronic medical records after obtaining an IRB approval. We used descriptive analysis; with Kaplan-Meier method to estimate EFS and OS, and logrank-test to perform specific univariate comparisons.

Results: We identified 44 patients (34 males, 76%; 10 females, 24%) with a median age of 9.9 years (range, 0.85 to 16.9). The 4-year event free survival (EFS) and overall survival (OS) were 84.6%±5.8% and 87.8%±5.0%, respectively. Seven patients (16% of total, 6 males, 1 female) relapsed at a median of 1.1 years after diagnosis; they all had achieved very good partial response post induction. Stages at diagnosis were II (N=1), III (N=2) and IV (N=4) and all had had bulky mediastinal masses at presentation. Six relapses were local with recurrence of a mediastinal mass; one was combined with CNS relapse. Only one patient had an isolated medullary relapse. At the time of this analysis, 6 patients were dead and one is still alive in complete remission 2 years after relapse.

Conclusions: Despite achieving good outcome among our patients, relapse carried dismal prognosis. The unique pattern of relapse hints to the need to identify residual mediastinal/thymic tissue that may benefit from local therapy or novel targeted approaches, which should be delivered early during therapy.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0103 / #1573 | Overall and Event-Free Survival of Childhood Lymphoma in Greece: Results of the Nationwide Registry of Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST) Harmonized-Clinical Data (1996-2019)

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Background and Aims: Combined chemotherapy/radiotherapy for pediatric Hodgkin (HL) and non-Hodgkin (NHL) lymphoma have led to significant improvements. We aimed to assess overall (OS) and event-free (EFS) survival of children 0-14 years with HL and NHL over a 24-year follow-up period in Greece using harmonized clinical data and to explore the effect of sociodemographic, histopathological and clinical characteristics on disease outcomes.

Methods: Incident cases of HL (n=287) and NHL (n=389) were actively registered in the Nationwide Registry for Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST; www.narechem.gr). Kaplan-Meier curves were fitted, and EFS/OS were derived from Cox proportional models.

Results: Over the period 1996-2019, the OS_{5-year} was 96% for HL (range, OS₁₉₉₆₋₂₀₀₁: 93%; OS₂₀₀₂₋₂₀₁₉: 98%) and 85% for NHL (range, OS₁₉₉₆₋₂₀₀₁: 81%; OS₂₀₁₂₋₂₀₁₉: 89%). EFS_{5-year} was 86% for HL (range, EFS₁₉₉₆₋₂₀₀₁: 82%; EFS₂₀₀₂₋₂₀₁₉: 89%) and 81% for NHL (range, EFS₁₉₉₆₋₂₀₀₁: 76%; EFS₂₀₁₂₋₂₀₁₉: 86%). Older age at diagnosis, high maternal education and close proximity to treatment centers were linked to remarkably favorable HL outcomes. By contrast, stage IV patients showed a significantly increased risk of death (hazard ratio [HR]_{OS}: 14.11, 95% confidence intervals [CI]: 1.08-183.75) and of any unfavorable event (HR_{EFS}: 3.03, 95%CI: 1.00-9.21). Low hemoglobin levels also conferred higher risk for any unfavorable event (HR_{EFS}: 2.81, 95%CI: 1.09-7.22). Regarding NHL, OS (76%) and EFS (73%) were poor among high-risk patients. Moreover, NHL children with increased LDH levels and those treated with anti-CD20 antibody had higher risk of death (HR_{OS}, LDH>1000: 2.73, 95%CI: 1.32-5.67; HR_{OS}, antiCD20: 2.41, 95%CI: 1.26-4.59), and of any unfavorable event (HR_{EFS}, LDH>1000: 1.85, 95%CI: 0.98-3.47; HR_{EFS}, antiCD20: 1.66, 95%CI: 0.91-3.01).

Conclusions: Despite the significant outcome improvements during the recent 24-year period, especially for HL, we identified predictors of poor disease outcome pointing to further need for intensification of individualized treatments. Ongoing cancer registration entailing clinical components could contribute to use of state-of-the-art treatments.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0104 / #1624 | Pediatric Hodgkin Lymphoma in Morocco: A Multicenter Research

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Background and Aims: Hodgkin lymphoma (HL) is one of the most curable pediatric and adult cancers, with long-term survival rates now exceeding 90% after treatment with chemotherapy alone or combined with radiotherapy. The purpose of this study was to describe the clinico pathological features and therapeutic outcomes on Moroccan children presenting with HL.

Methods: We conducted a 5-year retrospective descriptive study of histologically proven HL under the age of 15 in all Moroccan University Hospitals from 2011 to 2015. Patients were classified according to the Ann Arbor and treated with the MDH protocols 04 and 11 according to the period. Assessment of therapeutic response was made using CT imaging +/- PET scan

Results: One hundred and sixty-six patients were included in our study. The median age was 10 years old. Sex ratio M/F was 2,6. The median delay of diagnostic was 4 months. Lymphadenopathy was the most common presentation (85,5%), sixty-five percent of the patients reported B symptoms and bulky disease was present in 46% of the patients. Nodular sclerosis was the predominant histological presentation of pediatric HL (59%). Ninety percent of the patients were unfavorable group. Survival was rated in patients treated (162 patients fulfilled the inclusion criteria). Seventy-nine percent of patients were in complete remission, 16% did not achieve it, and 7,4% relapsed. Six patients benefited from an autograft of hematopoietic stem cells. The 3-years OS was 87%. Seventy-eight patients are in remission maintained with a median recall of 35 months.

Conclusions: Treatment outcome remains satisfactory. The frequency of unfavorable forms and the long delay of diagnosis is a challenge for the developing countries.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PV0105 / #1674 | Profile of Children with Non-Hodgkin's Lymphoma – Single Centre Experience from South India – A Retrospective Descriptive Study

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Background and Aims: Non-Hodgkin's lymphoma (NHL) is an aggressive malignancy in children, however, is highly curable. There is a dearth of published literature from developing countries like India. We proposed to study the profile of NHL at our centre.

Methods: We performed a retrospective analysis from medical records of children (<18 years) diagnosed with NHL at our centre between January -2015 and December-2019. Diagnosis was based on histopathology/immunohistochemistry of tissue or flow cytometry from pleural fluid. Risk stratification with appropriate imaging, bone marrow and CSF examination was done. B-NHL was treated with FAB-LMB 96 (with Rituximab for group-C), Lymphoblastic lymphoma on high risk ALL protocol and anaplastic large cell lymphoma with ALCL-99 protocol. Clinical presentation, complications, type of NHL, risk stratification, treatment details and outcome were studied and analysed.

Results: Of the 67 children who had lymphoma, 47.7% had NHL with a median age of 7.5 years and 3:1 male: female ratio. Gastrointestinal involvement with abdomen mass was the most common presentation (56.3%). Rare clinical presentations included one with isolated lesion in base of tongue, another in thyroid and one more in paranasal sinus with cranial nerve palsy. Distribution of NHL included 40.6% Burkitt's lymphoma, 25% lymphoblastic lymphoma, diffuse large B cell lymphoma and anaplastic large T-cell lymphoma (12.5% each) and one each with primary mediastinal large B-cell lymphoma, subcutaneous panniculitis like T-cell lymphoma and primary CNS-lymphoma. Of the 28 children who initiated treatment, two succumbed to sepsis, one abandoned treatment and four had progressive disease / relapse. Three year EFS was 84%. Financial assistance including stay/food provided at Home away from Home services to carry on with the uniform treatment including chemoport and standard protocol with supportive care was required in 87.5% cases.

Conclusions: Timely diagnosis, appropriate risk stratification, tailor-made protocols, supportive care with dedicated social/financial support can lead to survival rates on-par with developed nations in most curable diseases like paediatric NHL.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PV0106 / #1681 | Clinical, Epidemiological and Evolutionary Profile of Burkitt's Lymphoma in Children in a Middle-Income Country

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Background and Aims: Burkitt's lymphoma (BL) is a very aggressive non-Hodgkin B lymphoma. Therapeutic advances in recent decades have made this lymphoma a chemocurable tumor with a good prognosis. Tunisian publications about BL are rare, especially in children. We aimed to describe the epidemiological, clinical, biological, radiological characteristics of BL in children and to analyze the therapeutic, evolutive and prognostic aspects of BL.

Methods: A descriptive analytical study carried out at the Pediatric Oncology Unit of the department A of the Children's Hospital of Tunis between January 1st 2001 and June 30th 2019. LMB01 and modified LMB01 Protocols were applied for the treatment.

Results: Our study included 52 children. The median age was 5 years. The sex ratio was 3.33. BL was revealed by abdominal pain in 69.2% and altered general condition in 40.3% of the cases. Abdominal location was predominant (92.3%). It was complicated by acute ileal invagination in 15 cases. Abdominal ultrasound made the diagnosis of BL in 73% of cases. BL has been confirmed by cytological exams (17 cases), histological exams (33 cases) or by bone marrow puncture (2 cases). A predominance of Murphy's stage III (71.2%) and therapeutic group B (82.7%) was noted. 43.1% of patients presented tumor lysis syndrome and 96.2% had febrile neutropenia. The CR rate was 65.3%. OS and EFS were respectively 80.3% and 73% at 2 years, 75.5% and 67.3% at 5 years. 42.8% of patients have a satisfactory response to induction chemotherapy COP. This factor was the only factor which influences significantly the OS ($p=0$).

Conclusions: The data in our study are consistent with the literature. In our analysis, the epidemiological and clinical features of BL match the occidental sporadic form. Our survival results are encouraging but have to be improved further, thanks to medical training of primary care physicians to early diagnosis and improvement of supportive care management.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PV0107 / #421 | Characteristics and Outcomes of Burkitt Lymphoma with CNS Involvement from Single Cancer Centre Institute in India

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Background and Aims: CNS involvement occurs in approximately 9–13% of cases of Burkitt Lymphoma (BL). While survival has improved up to 90% in developed countries using intensified chemoimmunotherapy for CNS involvement, survival in resource-limited countries is still inferior. There is only limited published data on the clinical characteristics and treatment outcomes of CNS involvement in BL from low middle-income countries (LMIC) including India

Methods: Children aged ≤ 14 years, pathologically diagnosed as BL at our institute between January 2007 and December 2017 were included for retrospective analysis. CNS involvement was diagnosed as ≥ 1 of following criteria; cranial nerve palsy, blast in CSF, isolated intracerebral mass, clinical spinal cord compression, and cranial or spinal parameningeal extension. All children received Modified LMB96 Protocol. Survival analysis was done using the Kaplan-Meier method.

Results: During 10 year study period, a total of 85 children with BL got treated at our institute. The percentage of CNS-positive BL was 11.7% ($n = 10$). The commonest presentation was symptoms of cranial nerve palsy ($n = 4$) and lower limb weakness ($n = 4$). CSF positivity was present in 2 children. Concomitant Bone marrow involvement occurred in 8 patients. Out of ten, 5 children had progressive disease, on chemotherapy. The median time to progression was 4.6 months. The 3 year EFS was 50% for 10 CNS positive children and 77.2% for 75 CNS Negative children. Seven of them had e/o culture positive sepsis and 3 had septic shock, but none of them succumb to infective complications.

Conclusions: CNS involvement was an adverse prognostic factor. For BL with CNS involvement, we could achieve a 3 year OS of 50% only with the limited resources available in LMIC settings. The careful addition of immunotherapy may improve outcomes.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0108 / #450 | Practice Patterns and Collaborative Research Interest in Nodular Lymphocyte-Predominant Hodgkin Lymphoma (NLPHL)

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Background and Aims: Global NLPHL One Working Group (GLOW) was formed to improve outcomes for nodular lymphocyte-predominant Hodgkin lymphoma (NLPHL). NLPHL is a rare subtype of Hodgkin lymphoma, and there is no consensus on the optimal first line treatment. This survey sought to assess practice patterns, resources, and collaborative research interest in NLPHL.

Methods: A prospective survey was designed by GLOW members that contained questions on the provider's scope of practice, practice patterns, institutional resources, and interest in collaborative research. The survey questionnaire was emailed to cooperative groups and individuals who manage NLPHL.

Results: 171 (27%) of the 630 invited providers from 47 countries participated in the survey, which consisted of 148 hematologists/oncologists, 16 radiation oncologists, 8 pathologists, 1 radiologist, 1 nuclear medicine physician, and 1 oncology nurse practitioner. 74 (42%) providers treated only pediatric patients up to age 18, 50 (28%) treated only adults ≥ 19 years, and 53 (30%) treated both age groups. The proportion who "always" or "usually" incorporated radiotherapy as upfront treatment of early-stage NLPHL was 7% in pediatric-only providers, 62% in adult-only providers, and 34% in providers of both children and adults ($P < 0.0005$). For advanced-stage disease, rituximab was incorporated by 57% of pediatric providers, 84% of adult providers, and 60% of providers who treat both groups ($P = 0.005$). The proportion of participants who were "very likely" to participate in international NLPHL research collaborations was 71% for retrospective database, 65% for prospective registry, and 61% for a prospective trial.

Conclusions: There is significant variability in the treatment of NLPHL. Compared to pediatric providers, adult providers are more likely to incorporate radiotherapy in early-stage NLPHL and rituximab in advanced-stage NLPHL. Given the high overall survival for NLPHL, evidence-based treatment regimens maximizing efficacy and minimizing toxicity are needed and can only be achieved through international collaboration.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0109 / #635 | Whole-Body MRI versus an FDG-PET/CT-Based Reference Standard for Early Response Assessment and Restaging of Paediatric Hodgkin's Lymphoma: A Prospective Multicentre Study

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Background and Aims: To compare WB-MRI with an FDG-PET/CT-based reference for early response assessment and restaging in children with Hodgkin's lymphoma (HL).

Methods: 51 children (ages 10-17) with HL were included in this prospective, multicentre study. All participants underwent WB-MRI and FDG-PET/CT at early response assessment. 13 of the 51 patients also underwent both WB-MRI and FDG-PET/CT at restaging. Two radiologists independently evaluated all WB-MR images in two separate readings: without and with DWI. The FDG-PET/CT examinations were evaluated by a nuclear medicine physician. An expert panel assessed all discrepancies between WB-MRI and FDG-PET/CT to derive the FDG-PET/CT-based reference standard. Inter-observer agreement for WB-MRI was calculated using kappa statistics. Concordance, PPV, NPV, sensitivity and specificity for correct assessment of the response between WB-MRI and the reference standard were calculated for both nodal and extra-nodal disease presence and total response evaluation.

Results: Inter-observer agreement of WB-MRI including DWI between both readers was moderate (κ 0.46-0.60). For early response assessment, WB-MRI DWI agreed with the reference standard in 33/51 patients (65%, 95%CI 51-77%) versus 15/51 (29%, 95%CI 19-43%) for WB-MRI without DWI. For restaging, WB-MRI including DWI agreed with the reference standard in 9/13 patients (69%, 95%CI 42-87%) versus 5/13 patients (38%, 95%CI 18-64%) for WB-MRI without DWI.

Conclusions: The addition of DWI to the WB-MRI protocol in early response assessment and restaging of paediatric HL improved agreement with the FDG-PET/CT-based reference standard. However, WB-MRI remained discordant in 30% of the patients compared to standard imaging for assessing residual disease presence.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PV0110 / #649 | Pembrolizumab in Pediatric Patients with Relapsed or Refractory Classical Hodgkin Lymphoma in Keynote-051: Response by 2007 Revised Response Criteria for Lymphoma and 2014 Lugano Classification

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Background and Aims: Pembrolizumab demonstrated antitumor activity and tolerable safety in pediatric patients with relapsed or refractory classical Hodgkin lymphoma (rrcHL) in the phase 1/2 open-label KEYNOTE-051 (NCT02332668) study. We evaluated clinical outcomes using the 2007 International Working Group (IWG) revised response criteria and the 2014 Lugano Classification by blinded independent central radiology (BICR) assessment with an additional follow-up of 15 months.

Methods: For initial eligibility, patients had to be aged 6 months to 17 years and have confirmed cHL that was relapsed or refractory to first-line or second-line therapy and PD-L1-positive (CPS ≥ 1). After a protocol amendment, an expansion cohort of patients aged 3 to ≤ 18 years with confirmed rrcHL was enrolled regardless of PD-L1 expression status. Patients received pembrolizumab 2 mg/kg Q3W. Primary endpoint was ORR; secondary endpoints included response

duration, PFS by BICR, and OS. Database cutoff date was January 10, 2020.

Results: Among a total of 22 patients from both protocol versions, median age was 15 years (range, 11-17); 8 patients had stage I/II disease and 14 had stage III/IV disease. At database cutoff, median study follow-up was 24 months (range, 4-43); 9 patients (41%) were on treatment for ≥ 12 months; 1 patient (5%) died and 17 (77%) discontinued study treatment; 21 (96%) were still on study. ORR was 55% (12/22; 95% CI, 32-76; 1CR, 11PR) per 2007 IWG and 64% (14/22; 95% CI, 41-83; 4CR, 10PR) per 2014 Lugano; median (range) response duration was 17 months (0+ to 29) and 9 months (0+ to 29), respectively. Median PFS was 8 months (95% CI, 4-19) per 2007 IWG and 8 months (95% CI, 4-12) per 2014 Lugano. Median OS was not reached.

Conclusions: On longer follow-up, pembrolizumab monotherapy showed comparable efficacy outcomes in patients with pediatric rrcHL as assessed by 2007 IWG criteria and 2014 Lugano Classification.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PV0111 / #952 | Bone Mineral Density After Treatment of Malignant Lymphoma in Childhood and Adolescence

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Background and Aims: The survival rate of patients with childhood cancer has improved to more than 80% over the past several decades. Reduced bone mineral density and increased fracture risk have been reported during and after treatment of children with cancer. The aim of the study is to evaluate the skeletal profile of paediatric patients with Hodgkin and non Hodgkin lymphoma the first year after diagnosis.

Methods: Children with Hodgkin and non Hodgkin Lymphoma diagnosed in our Oncology Centre were recruited during a two-year period and underwent dual-energy X-ray absorptiometry (DXA) for bone mineral density (BMD) assessment. A group of 22 sex- and age-matched healthy volunteers were served as controls.

Results: Twenty-two patients were evaluated (17 boys-5 girls) with mean age 12.2 years (3.5-15.8). Of those, 11 were diagnosed with Hodgkin and 11 with non Hodgkin Lymphoma. One girl with Hodgkin Lymphoma sustained a vertebral fracture one year post -diagnosis. Low levels of 25(OH)vitD (<20 ng/ml) were found in 18.2% (3/18) of

the patients (range: 12-20 ng/ml). These patients were subsequently supplemented with vitamin D. All patients underwent DXA scan. As a whole, they had significantly reduced BMD Z-scores of the lumbar spine (LS) and total body less head (TBLH), compared to controls. Regarding Hodgkin lymphoma patients, nearly half (5/11) had low-normal LS BMD (i.e. Z-score between -1 and -2) while one patient had low BMD Z-score (i.e. <-2) at the same site and further imaging revealed a vertebral fracture, for which the patient was started on zoledronic acid. Moreover, one patient with Hodgkin Lymphoma had low TBLH BMD. On the other hand, non-Hodgkin lymphoma patients had normal BMD Z-scores, with the exception of one patient, who had low-normal LS BMD.

Conclusions: Bone mineral density is adversely affected in paediatric patients with lymphomas the first year following diagnosis.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0112 / #1142 | Infections Post Haploidentical Transplant using Post-Transplant Cyclophosphamide Single Center Experience

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Background and Aims: The use of T-cell repleted haplo-identical transplant (haplo-HSCT) with post-transplant cyclophosphamide has increased the pool of available donors and the number of transplanted patients. However, it is associated with higher risk for infections due to more immunosuppression.

Methods: Retrospective analysis of all pediatric patients who underwent T-cell repleted haplo-HSCT using post-transplant cyclophosphamide at King Hussein Cancer Center from Jan-2013 to Dec-2019.

Results: Sixty-five patients underwent haplo-HSC, with median follow-up time 664 days (17-2619), the median age was 4.9 years (1-19.6). Twenty-nine patients had malignant disorders (3 years overall-survival, 59%) and 36 patients underwent transplant for non-malignant disorders (3 years overall-survival 83%). The median time for neutrophil and platelets engraftment was 13 days (11-43) and 18.5 days (10-93), respectively. Nine patients (19%) developed primary graft failure, thirty-eight patients (60%) developed acute graft vs host disease (GvHD) and 25 (40%) developed chronic GvHD. Fifty-seven patients (87%) experienced at least one infection after transplant, the median time for first infection was 20 days post-transplant, twenty-six patients had infections before transplant. The majority of these infections were viral (N=76); CMV viremia was detected in 42 patients (65%) using regular surveillance, none developed CMV disease; BK cystitis in 21 (30%), and EBV reactivation in 13 (20%). Gram positive bacterial infections were detected in 26 patients (40%). Twenty-seven (42%) patients developed gram negative bacterial infection. Only one patient devel-

oped invasive pulmonary aspergillosis diagnosed by lung imaging. Sixteen patients died: four due to primary graft failure, two of regimen related toxicity, nine of disease relapse, and one patient died due to respiratory failure after RSV infection.

Conclusions: Haplo-HSCT using post-transplant cyclophosphamide is feasible with similar incidence of infections as to the reported literature. There was no significant difference in overall survival between patients who developed infections and the infection-free patients. More efforts need to be done to minimize the risk of resistant infections in this population.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation

PV0113 / #1156 | Factors Affecting Peripheral Stem Cell Harvest for Autologous Transplant in Paediatric Solid Tumours

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Background and Aims: Peripheral blood stem cells (PBSC) are the preferred option for use in autologous HSCT resulting in improved outcomes. This study aimed to assess factors which influenced success of PBSC harvest (PBSCH).

Methods: Data were collected retrospectively for an 80-month period (January 2014 to October 2020) in a national tertiary paediatric oncology unit. Data included age, gender, weight, diagnosis, bone marrow (BM) involvement, conditioning regimen, peripheral CD34+ count, LDH rise and collection yield.

Results: 114 children underwent PBSCH over the time-period. Two children were included (primary diagnosis and relapse). In total, 223 collections were carried out. 72%(160/223) collections mobilized with chemotherapy. The most common diagnoses included neuroblastoma 36%(41/114), embryonal CNS tumours (eCNS) 25%(28/114), and Ewing sarcoma (ES) 24%(27/114). Unsuccessful PBSCH occurred in 9.5% (11/116) cases, all of whom had neuroblastoma. In children with primary neuroblastoma 28/39 children had BM involvement. 82%(23/28) had successful PBSCH. In children with no BM involvement, 72%(8/11) had successful PBSCH. Plerixafor was used as mobilization agent in 7/39 cases and resulted in successful PBSCH harvest in one child. Four children with relapsed neuroblastoma had previous chemotherapy. Two had BM involvement and both had unsuccessful PBSCH. Median peripheral blood predictive markers were highest in ES (peripheral CD34+ $\times 10^6$ /L 109, LDH rise 467u/L) and eCNS (peripheral CD34+ $\times 10^6$ /L count 60, LDH rise 582u/L) compared with neuroblastoma (peripheral CD34+ $\times 10^6$ /L 31, LDH rise 279u/L). This correlated with median CD34+ $\times 10^6$ /kg daily yields highest in ES(4.9) and eCNS(3.25) compared with neuroblastoma(1.95).

Conclusions: PBSCH was successful in more than 90% of cases. Lowest PBSCH success rate in children with neuroblastoma. A large pro-

portion of children with neuroblastoma had BM involvement and significant pre-treatment with chemotherapy, however BM involvement alone was not a predictor for unsuccessful PBSCH. Plerixafor did not demonstrate efficacy in our cohort. Peripheral markers of LDH rise and CD34+ count correlated with greater PBSCH success.

E-Poster Topic: Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation

PV0114 / #1197 | Survey of 'Clean Diet' Practices

Recommended for Paediatric Patients Receiving a Stem Cell Transplant (SCT)

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Background and Aims: 'Clean diet' is a non-specific term referring to dietary restrictions recommended to SCT patients. The key recommendations are that patients avoid foods considered at high risk for causing food-borne infection and adhere to food safety advice. It is apparent from an extensive literature search that there is no consensus of what constitutes a 'clean diet'. This gap in practice results in significant variations in the dietary restrictions recommended to SCT patients which may compromise nutritional status and quality of life. AIM: To investigate current international 'clean diet' practices being recommended in paediatric SCT centres.

Methods: A quantitative online questionnaire was developed using SurveyMonkey and circulated amongst healthcare professionals working in the area of paediatric SCT. The questionnaire explored differences in the duration of dietary restrictions, the foods and drinks considered high-risk, and the food safety advice recommendations. Data were exported into Microsoft Excel where descriptive statistics were undertaken.

Results: Thirty six respondents completed the online survey, of which 34 were eligible for analysis. Ten (34%) respondents recommend that allogeneic SCT patients follow a more rigid version of the 'clean diet' for a longer duration compared with autologous patients. Fourteen respondents (41%) permit eating out in restaurants and eating takeaway meals. Eight (24%) respondents permit probiotics. Three (9%) centres recommended cooled boiled water as the only source of permitted drinking water. The majority (>80%) of respondents gave general food safety advice to SCT.

Conclusions: The survey revealed significant differences in the 'clean diet' recommendations for SCT patients. The lack of a universally accepted definition on what constitutes a 'clean diet' hinders research into the effectiveness/need of such recommendations. There is an urgent need to develop evidence-based dietary guidelines for this cohort so as to optimise their nutritional intake at a time when they are already at significant nutritional risk.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0115 / #1223 | Dance and Movement Therapy in the Rehabilitation of Children with Acute Leukemia after HSCT

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Background and Aims: The methods of adaptive physical culture for children with acute leukemia is not sufficiently studied. The study was conducted to develop and implement in the process of adaptive physical education for children after HSCT with acute leukemia a set of exercises using elements of dance and movement therapy, aimed at increasing the level of development of coordination abilities.

Methods: The study was conducted from May 2019 to April 2021 of 15 children from 7 to 9 ages. We used a set of exercises consisting of 3 blocks, including exercises for developing dynamic and static balance and maintaining a sense of rhythm. Classes were held for 40 minutes 3 times a week for 3 months. The main methods were: heart rate monitor, selection of the optimal music, the load in accordance with the well-being, control of the fatigue process. As criteria for the effectiveness of the developed set of exercises using elements of dance and movement therapy: "Walking in a straight line", "Romberg Test", "Rhythm Preservation" (Meinel K., Schnabel G.).

Results: Improvements in dynamic balance were increased in all patients by an average of 5 cm from the baseline values according to the "Walking in a straight line" test. According to the Romberg Test, the static balance time improved in all patients by an average of 4 seconds. In the test for the preservation of the sense of rhythm "Rhythm Preservation" each patient improved their result by an average of 3 seconds.

Conclusions: Dance and movement therapy has a positive effect on the development of static and dynamic balance, a sense of rhythm. Musical accompaniment, as well as dance exercises, had a positive impact on the mental state of not only children, but also parents. Dance and movement therapy can be used in the rehabilitation program for children with acute leukemia after HSCT.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0116 / #1243 | Factors Affecting Outcomes in Children with Acute Leukemia Post Hematopoietic Stem Cell Transplantation – A Retrospective Study from a Tertiary Referral Centre in India

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Background and Aims: Pre-transplant measurable residual disease(MRD) and post-transplant graft versus Leukemia(GVL) effect are important determinants of successful outcome following hematopoietic stem cell transplantation(HSCT) in children with acute leukemia. We analyzed the impact of graft versus host disease(GVHD) on relapse free survival(RFS) in our cohort

Methods: We performed a retrospective analysis of children with acute leukemia who underwent HSCT in our unit between 2002 and 2018. Data analysed included pre-transplant MRD status by flow cytometry(only for children included after 2009), conditioning regimen, graft source, incidence of GVHD and RFS

Results: A total of 135 children aged between 6 months to 18 years with leukemia underwent HSCT. HSCT was performed in first complete remission(CR1) in 65(48%)[ALL (n=35), AML (n=18), MPAL(n=9) and CML in blast crisis(n=3)] and following relapse in 70 children(52%)[ALL(n=45) and AML(n=25)]. The myeloablative conditioning regimen in ALL was 12Gy total body radiotherapy and busulfan or melphalan based in AML. Stem cell source was peripheral blood in 112 (83%), cord in 15(11%) and bone marrow in 8(6%) children. We documented engraftment in 132/135 and GVHD in 106 (79 %) children. In children without GVHD, pre-emptive strategies employed to achieve a GVL effect were early withdrawal of immunosuppression, pre-emptive donor lymphocyte infusion (DLI)[n=12], with additional lenalidomide(n=7) and 13/19 children developed GVHD which translated into RFS in 11(58%) children. The mortality in the entire cohort was 61(45%) [relapse of the disease-26(41%), sepsis-17(29%), severe GVHD-14(23%) and regimen related toxicity-4(7%)]. After a median follow up of 32 months the RFS was 55%. Those who underwent HSCT in CR1 had better survival(63%) compared to HSCT after relapse(47%). GVHD of any grade was associated with a lower risk of relapse(p=0.008).

Conclusions: Our study reinforces the importance of GVHD or strategies aimed at inducing GVL effect in reducing the incidence of relapse in children undergoing HSCT for acute leukemia even in resource constrained settings.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0117 / #1435 | Creation of Program of Hematopoietic Stem Cell Transplantation in Children of the First Year of Life with Malignant Neoplasms in N.N. Blokhin NMRC of Oncology

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Background and Aims: Hematopoietic stem cell transplantation (HSCT) – is a key treatment option for several malignant neoplasms (MNO) in children. Risk-adapted HSCT required for children of the first year of life.

Methods: Risk-adapted program of HSCT in children of the first of life with MNO established in our institution. The following options of optimization of therapy included to our program: nutritional and infectious status estimation, conditioning regimen adaptation, special care for sing and mucosa. Five HSCT were made during 2020. All patients were girls with age median 11 (10-12) months. Diagnoses: neuroblastoma (n=4), medulloblastoma (n=1). Neuroblastoma conditioning included a combination of Treosulfan and Melphalan, in a child with medulloblastoma, Thiophosphamide and Carboplatin were used.

Results: Four patients survived. One patient died early on due to post-transplant complications (infection). Following complications were found: skin toxicity 1-2 gr. (n=4), 3-4 gr. (n=1), oropharyngeal mucositis 1-2 gr. (n=5), neutropenic enterocolitis gr. 1 (n=2). These complications were treated with systemic and topical therapy. Nutritional deficiency was treated with use of a special nutritional support. No significant infections were recorded (excluding died patient).

Conclusions: HSCT in children of the first year of life is unsolved problem. Children of first year of life require a special care due to higher toxicity. Intense HSCT lead to serious complications such as infections and malnutrition. Multidisciplinary team needs to be involved for care of patients of the first year of life during HSCT.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0118 / #1594 | Adoptive Transfusion of Cytomegalovirus-Specific Cytotoxic T Lymphocytes (CMV-CTLs) for Pediatric Patients with CMV Diseases after Allogeneic Hematopoietic Stem Cell Transplantation

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Background and Aims: Cytomegalovirus-specific cytotoxic T lymphocytes (CMV-CTLs) therapy provides an alternative approach for patients with CMV diseases after hematopoietic stem cell transplantation (HSCT). However, the safety and efficacy of CTLs in pediatric patients remain unclear.

Methods: Ten pediatric patients with CMV diseases after allogeneic HSCT were given CMV-CTLs therapy in addition to conventional antiviral treatment between May 2018 and September 2020, and their clinical data were analyzed.

Results: This study included five boys and five girls. The median age was five years (range 1 to 7 years). Primary diseases were thalassemia

major and severe aplastic anemia in 8 and 2 patients, respectively. Nine patients received haploidentical HSCT, and one received HSCT from a matched sibling donor. CMV serostatuses were D+/R+ and D+/R- in 7 and 3 patients, respectively. Severe CMV pneumonia was observed in all patients, among which one patient also developed CMV retinitis. Two patients developed grade III to IV acute graft-versus-host disease (GVHD). Another two patients developed moderate chronic GVHD. Co-infections or reactivations with other viruses, including hepatitis B virus, parvovirus B19, Epstein-Barr virus, and adenovirus, were observed in 7 patients. Three patients had bacterial enteritis. Three patients developed invasive fungal infection. Patients were given conventional anti-CMV treatment consisted of ganciclovir and foscarnet. CMV-CTLs were administered at a median of 19 days (range 12-29 days) after the onset of CMV infection. All except one patient were given two transfusions of CMV-specific CTLs at an interval of 2 weeks. CMV viral load turned negative at a median of 23 days (range 15-33 days) after the last infusion of CMV-CTLs. All patients had resolutions of CMV pneumonia approximately two months after the second infusion of CMV-CTLs. However, one patient had blindness caused by CMV retinitis.

Conclusions: CMV-CTLs transfusion is safe and effective for pediatric HSCT recipients and should be considered in those with CMV diseases.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0119 / #169 | The Distribution of GSTA1 Promoter Variants and Functional Diversity of GSTA1 in Human Populations

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Background and Aims: GSTA1 is a member of a family of enzymes that function to add glutathione to electrophilic compounds, including therapeutic drugs. GSTA1 holds much importance in the field of oncology, as it is involved in the metabolism of many important chemotherapeutic agents such as busulfan, thiotepa, doxorubicin, cyclophosphamide, and chlorambucil. GSTA1 has several functional SNPs within its promoter region that are responsible for a change in its expression by altering promoter function. We investigated the distributions of GSTA1 promoter haplotypes and diplotypes across different human populations and assess their functional impact on the GSTA1 expression.

Methods: PHASE 2.1.1 was used to infer haplotypes and diplotypes of six GSTA1 promoter SNPs on 2501 individuals from 26 populations classified by the 1000 Genomes Project into five super-populations: African, American, East Asian, European, and South Asian. Pairwise FST analysis was used to compare sub-populations. Luciferase reporter assay (LRA) was used to evaluate the impact of SNPs and SNP-SNP interactions on GSTA1 expression. HOMER (v4.11) was used to search for transcription factors at six SNP sites.

Results: The distributions of GSTA1 promoter haplotypes and diplotypes were significantly different among the different human populations. Three new promoter haplotypes and six new promoter diplotypes were found. LRA demonstrated that SNPs at -52 and -69 has the most impact on GSTA1 expression. Other SNPs have a significant impact on transcriptional activity and interact with SNPs at -52/-69. Results of LRA and in silico transcription factors prediction were used to propose a new model of SNP interaction.

Conclusions: Due to the significant differences in GSTA1 diplotype distribution among human populations, future pharmacogenomics or disease-related studies would benefit from the inclusion of the complete GSTA1 promoter haplotype that is based on the newly proposed metabolic grouping derived from the LRA results.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0120 / #344 | Reexamining Donor Source of Hematopoietic Cell Transplant as a Risk Factor for GRAFT-Versus-Host Disease

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Background and Aims: Graft-versus-host disease (GvHD) remains a significant source of morbidity and mortality following allogeneic hematopoietic cell transplant (HCT). Donor source has been suggested as a risk factor for GvHD; patients who receive grafts from unrelated and mismatched related donors may be at increased risk of developing this complication as compared to those with matched related donors. Therefore, many centers utilize higher intensity GvHD prophylaxis regimens for these higher risk patients. However, emerging data is challenging this principle of relative protection among recipients of matched donor grafts. Here we describe the incidence of acute and chronic GvHD among matched related transplant recipients as compared with unrelated and mismatched related transplant recipients. Additionally, we compare mortality between these groups.

Methods: Data from all patients aged 0-25 years who underwent allogeneic HCT at our center from 2000-2020 were retrospectively collected. GvHD and mortality rates across groups were described using contingency table analyses.

Results: Data from 222 transplants were available. Transplant sources include 112 matched related donors (50.4%), 95 unrelated donors

(42.8%), 5 related haploidentical donors (2.3%), and 10 unrelated cord blood donors (4.5%). Incidences of acute GvHD were similar between patients who received grafts from matched related donors as compared to all alternative donors (27.68% vs. 33.64%, $p = 0.3831$, OR 0.76, 95% CI 0.43 to 1.31). Chronic GvHD occurred at similar rates between these groups as well (16.1% vs. 12.7%, $p = 0.5676$, OR 1.31, 95% CI 0.63 to 2.70). Finally, the all-cause mortality rate had no differences between the groups (36.6% vs. 28.18%, $p = 0.1987$, OR 1.47, 95% CI 0.83 to 2.56).

Conclusions: Rates of GvHD do not significantly differ based on donor source when comparing matched related versus alternative donors. Further studies examining the potential benefit of additional GvHD prophylaxis for matched related donor graft recipients may help optimize GvHD prevention in this group.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0121 / #732 | Autoimmune Cytopenia after Hematopoietic Stem Cell Transplant in a Pediatric and Young Adult Population

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Background and Aims: Autoimmune cytopenias are an increasingly recognized complication of allogeneic hematopoietic cell transplantation (allo-HCT). There are few existing studies that document the incidence of autoimmune cytopenia post allo-HCT in pediatric populations. This study aims to estimate the incidence of autoimmune cytopenias in pediatric populations following allo-HSCT. Furthermore, we sought to describe key variables about patients who develop this complication.

Methods: We used In Text Explorer (ITE), an electronic medical record searching software, to search for following keywords in patients ages 0-25 years who have received allo-HCT between January 2000 and December 2020: "hemolytic anemia", "autoimmune hemolytic anemia", "ITP", "immune thrombocytopenia", and "autoimmune neutropenia". Patients who developed autoimmune cytopenias following allo-HCT were included for further review.

Results: The incidence of autoimmune cytopenia in our pediatric population was 5.4% ($n=12/222$). The majority of cases were autoimmune hemolytic anemia (AIHA) (66.67%, $n=8/12$). Fewer patients developed concomitant AIHA and immune thrombocytopenia (ITP) (16.66%, $n=2/12$), both autoimmune neutropenia and ITP (8.3%, $n=1/12$), or ITP alone (8.3%, $n=1/12$). The average age at bone marrow treatment was 9.14 years (1.58-19.66 years). AIHA was diagnosed on average of 236 days (45-1084 days) after transplant. One-third of patients who developed autoimmune cytopenia ($n=4/12$) were treated for a non-malignant disease (e.g. aplastic anemia). Alemtuzumab was used in the conditioning regimens for 58.3% of patients ($n=7/12$).

Acute graft versus host disease preceded autoimmune cytopenia in 41.7% of patients (n=5/12). Cytomegalovirus was reactivated prior to cytopenia development in 3 patients (25%, n=3/12).

Conclusions: This study found an 5.4% incidence of autoimmune cytopenia following allo-HCT in pediatric and young adult patients. Future studies should focus on an evidence-based approach to management of this HCT complication.

E-Poster Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PV0122 / #974 | Treatment of Two Jordanian Siblings with Congenital Dyserythropoietic Anemia - Type 1B- using Interferon-Alpha-2A

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Background and Aims: Congenital dyserythropoietic anemias (CDAs) are a group of rare hereditary anemias characterized by marked ineffective erythropoiesis and characteristic morphologic abnormalities of the erythroblasts. Three major types have been described based on characteristic morphologic abnormalities of the erythroblasts, and the molecular approach for the causative genes. CDA type 1 is inherited as an autosomal recessive disorder caused mainly by a mutation in the CDAN1 gene encoding codanin-1 protein. Therapy includes red-cell transfusion, iron chelation therapy, interferon-alpha (INF alpha), and allogeneic stem cell transplantation.

Methods: We are reporting two transfusion-dependent siblings diagnosed with CDA type 1, including their treatment with IFN and outcome.

Results: Two boy siblings with CDA type 1 were referred for stem cell transplant at our center at the ages of 11 and 12 years. They have first-degree consanguineous parents and both born with low birth weight (1677 gm and 1500 gram, respectively) and have been transfusion dependent since birth. Both were diagnosed using genetic testing and found to be homozygous at C15ORF41, consistent with a diagnosis of CDAN1B. Both are short, have mild splenomegaly but no dysmorphism. Their ferritin levels were 1500 ng/ml and 1220 ng/ml respectively. Bone marrow evaluation showed quantitatively increased erythroid precursors. The erythroid maturation is megaloblastic and shows binucleation in 10% to 15% of the erythroid cells. The skeletal survey was normal. They were started on interferon-Alfa-2a as one million units weekly, with excellent response, no blood transfusion requirement since starting therapy, iron chelation has stopped. INF-alpha dose was decreased to every 2 weeks after one year of therapy, with stable hemoglobin; both patients have been receiving INF-alpha for more than 12 months, without toxicity. The yearly cost of weekly interferon-alpha is 2,400 USD vs 100,000 USD for allogeneic-hematopoietic transplant, in addition to the cost of post-transplant complications.

Conclusions: We report two Jordanian siblings with transfusion-dependent CDA-1 caused a homozygous mutation of the C15ORF41 gene. Both showed dramatic response to IFN-alpha-2a which was maintained for at least one year with no toxicity.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PV0123 / #1098 | A Novel Surgical Approach to Cervico-Thoracic Neuroblastoma with Excellent Efficacy and Minimal Morbidity

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Background and Aims: The trap door and clam shell approaches with modifications have been the standard surgical approaches to cervico-thoracic neuroblastomas (CTNB). These tumors pose unique challenges as they often involve major blood vessels as well as the brachial plexus. We describe a novel surgical approach to CTNB that enables safe resection.

Methods: Our prospective study from January 2018 till January 2021 included children up to 18 years of age diagnosed through tissue biopsy to have CTNB. The surgical approach included a small cervical incision and a median sternotomy. The incision did not include a thoracotomy but provided adequate exposure in all patients with minimal morbidity. All structures from the neck to the descending aorta all the way to the posterior mediastinum could be visualised.

Results: Patient A is one-year-old boy who had presented with stridor, severe airway obstruction and superior mediastinal syndrome. He underwent surgery following four cycles of neo-adjuvant chemotherapy (NACT) which significantly shrunk the tumor and now involved the right cervicothoracic region, and four cycles of adjuvant chemotherapy. He remains in remission one-year post-treatment. Patient B is two-year-old girl with neuroblastoma with a paraspinal extension from C6-D7 vertebrae and encasing C6-T1 roots of the brachial plexus. She underwent

surgery after eight cycles of NACT and remains in remission 3 years post-treatment. Patient C is a five-year-old boy presented with a recurrent cervicothoracic ganglioneuroblastoma on the left side who underwent a complete tumor excision. All three children were operated through a uniform cervico-thoracic approach and tolerated the procedure well. They were all discharged on the fifth day post-surgery with minimal pain, no shoulder joint instability or displacement of clavicle. All patients had Horner's syndrome.

Conclusions: We would recommend this novel surgical approach using a cervical incision and a sternotomy for all children with CTNB as it paves the way for improved disease control with optimal post-surgery outcomes.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PV0124 / #1107 | Increased Sensitivity to Chemotherapy by Targeting Endogenous P21CIP1/WAF1 Expression in High-Risk Neuroblastoma Cell Lines

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Background and Aims: Neuroblastoma (NB) is the most common solid extracranial tumour in early childhood. Relapse remains a major therapeutic challenge and it is today acknowledged that both genetic and non-genetic mechanisms contribute to development of resistance. The cyclin-dependent kinase inhibitor p21^{Cip1/waf1} can act either as a tumour suppressor or as an oncogene depending largely on the cellular context. Here, we aim to examine the sensitivity of high-risk NB cell lines to the p21^{Cip1/waf1} inhibitor UC2288 (p21i) and explored mechanisms to increase the efficacy of chemotherapy.

Methods: Here, a panel of NB cell lines, BE(2)-C, IMR32, SK-N-AS, SHSY-5Y, Kelly, SK-N-SH, SK-N-FI, and SK-N-DZ, differing in TP53 status, were investigated for endogenous expression of p21^{Cip1/waf1} using immunocytochemistry and qRT-PCR. Cell cycle progression of p21^{Cip1/waf1} expressing cells were investigated using immunocytochemistry. Cell viability and regrowth capacity were evaluated following treatment with the p21i either as a single agent or in combination with chemotherapy (cisplatin or doxorubicin).

Results: All tested cell lines showed a subpopulation of cells with endogenous p21^{Cip1/waf1} mRNA and protein expression at baseline i.e. without prior treatment. Moreover, we provide novel evidence that the endogenous expression of p21^{Cip1/waf1} was not a result of cellular stress and/or DNA damage, but rather a subpopulation of p21^{Cip1/waf1} expressing cells was identified with active proliferation (Ki67 positive), suggesting an oncogenic role of p21^{Cip1/waf1}. Treatment with the p21i resulted in increased sensitivity for cisplatin in the SHSY-5Y, Kelly and BE(2)-C cell lines. Furthermore, delay in growth was observed for Kelly and BE(2)-C following combination treatment with p21i and doxorubicin.

Conclusions: Taken together, our results suggest a dual role of p21^{Cip1/waf1} in high-risk NB. Furthermore, we demonstrate a potential synthetic lethality between the p21i and chemotherapy which could be a novel potential therapeutic strategy for NB.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PV0125 / #1145 | MTHFD2 Expression is a Novel Prognostic Indicator in Neuroblastoma

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Background and Aims: MTHFD1 and MTHFD2 are two of key enzymes in the folate cycle, which is an important metabolic pathway central to nucleotides generation, DNA methylation and redox homeostasis. MTHFD1 or MTHFD2 has been found to be a potential therapeutic target in several cancers, but its role in neuroblastoma (NB) has not been fully elucidated.

Methods: MTHFD1 and MTHFD2 expression were detected using immunohistochemistry in 62 patients with newly diagnosed NB, and the data were used to evaluate its relationship with clinical features, MYCN status and prognosis.

Results: Both MTHFD1 and MTHFD2 were expressed in all of 62 NB patients. 16 patients (25.8%) had strong expression of MTHFD1, and 37 patients (40.3%) had strong expression of MTHFD2. 15/16 patients with strong expression of MTHFD1 had strong expression of MTHFD2 simultaneously, so we only analyzed the prognostic significance of MTHFD2 expression status. At a median follow-up time of 36 months, the 3-year progression-free survival (PFS) rates were 43.5%, and the 3-year overall survival (OS) rates were 67.2%. 3-year PFS and 3-year OS rates in the MTHFD2-strong expression group were both significantly lower than those in the MTHFD2-weak expression group (24.3% vs. 72.0%, $P < 0.001$, for 3 year-PFS; 56.8% vs. 86.0%, $P < 0.033$, for 3 year-OS). Multivariate survival analysis showed that strong expression of MTHFD2, stage 4 and MYCN amplification were independent adverse prognostic factors for PFS ($p = 0.002$, $p = 0.006$ and 0.007 , respectively). **Conclusions:** In conclusion, we found that the majority of NB cases were MTHFD1 and MTHFD2 positive, and strong expression of MTHFD2 was significantly correlated with poor outcomes, indicating the potential role of MTHFD enzymes as therapy targets for NB.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PV0126 / #1148 | The Efficacy and Safety of Irinotecan Plus Temozolomide and Vincristine in Neuroblastoma with Brain Recurrence: A Retrospective Study

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Background and Aims: Brain metastasis is common in relapsed neuroblastoma patients with poor prognosis, this study was to evaluated

the efficacy and safety of the combined regimen with irinotecan and temozolomide and vincristine (VIT) for neuroblastoma patients with brain recurrence.

Methods: The retrospective, single-institutional, observed study was conducted in Sun Yat-sen University Cancer Center in China. Pediatric neuroblastoma with brain recurrence treated with VIT regimen (irinotecan: 50 mg/m²/day intravenously for 90 mins on day 1-5; temozolomide, 100 mg/m²/day orally 1 h before irinotecan on day 1-5; vincristine, 1.5 mg/m²/day intravenously on day one, repeated every 3 weeks) between 2010 to 2019 were evaluated.

Results: Sixteen were enrolled in the study, including 5 female and 11 male, with median age of 36.5 months (8.2 to 66.1 months) at diagnosis and median time of 12.9 months (8.4 to 40.0 months) from initial diagnosis to brain relapse. Nine patients with brain metastases only and 7 patients with both brain and other site recurrence. Three patients received emergency surgery for the symptoms of intracranial hypertension. Thirteen patients were evaluable for response: 1 CR, 8 PR, 1 SD, 5 PD. The overall response rate was 69.2%. 11 patients received whole brain radiation with the median dose of 30Gy (25.4-34.5Gy). Seven patients alive without tumor and 9 patients died of disease at the last follow-up. The median overall survival time and progression free survival time after brain relapse was 18.2 months (5.76-59.9 months) and 8.7 months (0.7-59.9 months), respectively. Anemia (68.5%), leucopenia (43.0%), diarrhea (35.0%), neutropenia (31%), abdominal pain (27.6%) and vomiting (11%) were the most common adverse events. No fatal adverse events have been observed.

Conclusions: VIT regimen was an effective salvage treatment for high-risk NB patients with brain recurrence. Further prospective randomized controlled clinical study is warranted.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma*

PV0127 / #1153 | Pathogenic Variants in The NF1 Gene in Neuroblastoma

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Background and Aims: Neurofibromatosis type 1 (NF-1) is a neuroblastoma predisposition syndrome, but limited cases have been described. Somatic pathogenic variants in the NF1 gene are involved in neuroblastoma pathogenesis; however, they are considered as a rare event. The aim was to study the frequency of pathogenic variants in the NF1 gene in patients with neuroblastoma.

Methods: For the period from April 2019 to April 2021, next-generation sequencing was performed on DNA of 74 neuroblastoma

patients. The germinal status of the variants identified in the tumor was confirmed by Sanger sequencing on the patient's normal tissue.

Results: Eight (10.8%) patients with pathogenic variants in the NF1 gene were identified. 7/8 patients were assigned to the high risk group due to stage 4 (n = 6) and/or the MYCN gene amplification (n = 4), 1/8 - intermediate risk group. In 4/8 of the cases, germinal genetic variants were found in the NF1 gene, in all cases there were clinical manifestations of NF-1: café-au-lait spots (n = 4), optic nerve glioma (n = 2). In 4/8, pathogenic variants in the NF1 gene were somatic with the alternative allele frequency 25.5-65%. In 1 case, analysis of paired samples revealed a pathogenic variant only at relapse. The following aberration were detected: 2/8 - missense, 3/8-nonsense, 2/8 - mutations at the splice site, 1/8 - frameshift deletion. All germinal variants were loss-of-function (2 nonsense, 1 deletion and 1 splice site). Clinically, all high risk patients had an unfavorable disease course: 6/7 - relapse/progression, 1/7 - refractory disease.

Conclusions: Pathogenic variants in the NF1 gene were detected in 10.8% of patients with neuroblastoma who underwent molecular genetic analysis. High-risk neuroblastoma patients with NF-1, as well as with clinically significant somatic pathogenic variants in the NF1 gene, had a poor prognosis, however, studies on a larger nonselected population of patients are needed.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma*

PV0128 / #1200 | Autologous Stem Cell Transplantation in Paediatric Oncology; Poor Mobilizers, Plerixafor and Pegfilgrastim

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Background and Aims: High-dose (HD) chemotherapy followed by autologous stem cell transplantation (ASCT) is an important part of the standard treatment for high-risk (HR) solid malignancies in paediatric oncology. Collection of a sufficient amount of Hematopoietic Stem and Progenitor Cells (HSPCs; CD34+ cells) from granulocyte-colony stimulating factor (G-CSF)-mobilized peripheral blood remains a challenge. We aimed to identify characteristics of patients that are poor mobiliz-

ers and compare success of mobilization and hematopoietic recovery between different mobilization regimens in paediatric cancer patients.

Methods: In this multi-centre retrospective study we analysed medical records of 281 patients who underwent mobilization from 2000 to 2019. Mobilization regimens consisted of daily subcutaneous Filgrastim or a single injection of Pegfilgrastim +/- Plerixafor (CXCR4 inhibitor) after chemotherapy. Patients were classified as poor mobilizer when harvest counts were $<2 \times 10^6$ CD34+cells/kg or in case Plerixafor was used.

Results: Thirty-eight out of 281 patients were poor mobilizer (13.5%). Associated with poor mobilization were: bone marrow involvement at diagnosis and start of mobilization, prior chemotherapy, diagnosis of neuroblastoma, and lower leukocyte and platelet count at the start of mobilization. The use of Plerixafor resulted in successful harvest in 75% of poor mobilizers, showing higher harvest yield per session and less apheresis sessions to achieve target harvest yields (median 1.30 vs. 0.63×10^6 cells/kg; $p=0.005$ and 2.5 vs. 4.0 ; $p=0.041$). Patients mobilized with Pegfilgrastim showed superior harvest per session compared to the Filgrastim-only group (median 9.10 vs. 3.45×10^6 cells/kg; $p<0.001$).

Conclusions: The use of Plerixafor was safe, tolerable and effective, resulting in successful harvest in 75% of poor mobilizers. We recommend the use of Pegfilgrastim over Filgrastim because of superior harvest yields and less patient discomfort. An in depth analysis of HSPC phenotype is ongoing and will reveal whether the different mobilization regimens have an effect on HSPC characteristics.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0129 / #1226 | Dinutuximab Beta Plus Conventional Chemotherapy for Relapsed/Refractory High-Risk Neuroblastoma: A Single Centre Experience

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Background and Aims: Outcomes for children with relapsed and refractory high risk (HR) neuroblastoma (NB) are dismal. Anti-GD2 mediated chemoimmunotherapy has notable anti-tumor activity in patients with relapsed or refractory neuroblastoma. The aim of the study was an evaluation of safety and effectiveness of the combination of immunotherapy with dinutuximab beta (DB) and chemotherapy (CT) in patients with relapsed or refractory HR-NB.

Methods: All patients had received TPOG NB 2009 national protocol for high-risk NB treatment at diagnosis (dose-intensive induction chemotherapy $\pm$ surgery $\pm$ radiotherapy or high-dose chemother-

apy followed by autologous stem cell transplantation). Salvage treatments have been administered after progression or relapse including ICE (ifosfamide+carboplatin+etoposide), TVD (topotecan+vincristine+doxorubicin) or TVC (topotecan+vincristine+cyclophosphamide) $\pm$ temsirolimus $\pm$ bevacizumab, RIST (rapamycin, irinotecan, sunitinib, temozolomide). Patients could not achieve remission in primary or metastatic sites were included in the study.

Results: Since January 2020, eight ($n=8$) patients received a total of 42 cycles of DB+CT. The most common CT scheme was intravenous irinotecan (50 mg/m^2 per dose, on days 0-4) and oral temozolomide (100 mg/m^2 per dose, on days 0-4). Dinutuximab beta was administered intravenous 10 days continuous infusion with 10 mg/m^2 per day (on days 1-10). Patients received 2 to 10 successive cycles with a duration of 35 days each. Disease evaluations were performed after cycles 2, 4, and 6 and every 2 to 3 cycles thereafter. Objective (complete or partial) responses were observed in 6/8 (75%) including complete remission (CR) in 3/8 patients and 3/8 partial response (PR). Stable disease was also observed in 1 patient. The remaining 1 patient developed bone and bone marrow progression after 4 cycles of treatment. Common grade ≥ 3 toxicities were leukopenia (62%), thrombocytopenia (27%), hypertransaminasemia (24%), fever (14%) and rash/itching (12%).

Conclusions: Our study suggests that dinutuximab beta-based chemoimmunotherapy is effective and safe in patients with relapsed/refractory HR-NB.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0130 / #1237 | Treatment of Relapse/Refractory Neuroblastoma: Dokuz Eylul University Experience

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Background and Aims: To evaluate clinical characteristics and treatment responses of patients with relapse&refractory neuroblastoma.

Methods: Between 2000-2020, there were 40 patients with diagnosis of relapse&refractory neuroblastoma in our center. They had been treated accordingly the Turkish Pediatric Oncology Group Neuroblastoma (TPOG-NB) protocols. Medical records were analyzed retrospectively. The clinical characteristics, treatment responses, relapses, response to salvage treatments and survival rates were analyzed.

Results: Median age at diagnosis was 3.3yrs, M/F ratio:1.5. There was one patient in LRG(Stage4S), one patient in IRG(Stage2A). There were 38 patients in HRG(stage3 ($n=4$), stage4 ($n=34$)). Tis-

sue diagnosis couldn't be taken in patient with Stage4S-NB, this case died with progression despite chemo and radiotherapy. In HRG(n:38), MYCN was analyzed in 31 cases and found amplified in 29% of cases. Del 11q was analyzed in 5 cases with nonamplified MYCN. Tumor localization was surrenal(n:31), paraspinal(n:4), midline(n:5). Metastatic sites were bone marrow(73%), bone(68%), lymph node(60%), lung(10%), CNS (7.5%). At diagnosis, treatment response to induction was 58%PR,25%NR/PD. Delayed surgery had been performed in 65% patients (complete resection in 14, partial in 12). Radiotherapy was given to 26 cases: primary tumor(n:14), metastasis(n:4), primary+met(n:8). Six patients died before RT, family refused RT in one. There were 11 patients with refractory disease, 7 progressive disease, 20 with systemic relapse, two patients with primary relapse. Median relapse time was 13mos(3-64mos). Median progression time for responsive patients was 17mos. Salvage therapies were average 3 courses(1-8 courses)TCV in 18; 3 courses ICE in 9 patients. Salvage chemo(containing irinotecan, temozolamide, topotecan, etoposide) and dinutuximab beta immunotherapy was used in nine patients. HDCT-ASCT was performed in 16 patients. TCV, ICE, RIST and temozolamide were given to 9 NR cases despite HDCT-ASCT. Also RIST protocol, irinotecan-temozolamide-temsirolimus and metronomic oral cyclophosphamide/etoposide were given as 3rd line treatment to NR patients. Median follow-up time for patients was 27mos(5mos-9.2yrs); one and two years OS were 77% and OS63% respectively. Twelve patients were followed more than 4 years.

Conclusions: Relapse/refractory neuroblastoma group consists of advanced stage, unfavourable histology, 29% positive MYCN amplification. Progression-free survival rates prolonged up to 6 years and disease became chronic with different treatment strategies. CR can be achieved by chemo and immunotherapy in 2 cases and it was the best response.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0131 / #1268 | Survival of Neuroblastoma Patients in Countries with Limited Access to ANTI-GD2 Immunotherapy

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Background and Aims: The use of anti-GD2 immunotherapy significantly improved the survival rates of high-risk neuroblastoma patients. However, anti-GD2 antibodies are still not available in many resource-limited countries including China.

Methods: A total of 76 pediatric patients that were diagnosed with neuroblastoma (64 patients, 84%) or ganglioneuroblastoma (12 patients, 16%) between January 2016 and October 2020 were reviewed for their clinical data, retrospectively.

Results: This study included 34 (45%) girls and 42 (55%) boys, with a mean age of 3 years (range 0.5 to 13 years). Thirty-four (45%) patients

had metastasis of the bone marrow at the initial diagnosis. Eight (11%) patients were detected to have MYCN gene amplifications. Patients were categorized into stages of L1 (22%), L2 (18%), M (55%), and MS (5%), according to the International Neuroblastoma Risk Group (INRG) staging system, respectively. Patients were treated according to Children's Oncology Group ANBL0531 or ANBL0532 protocols, or a domestic clinical trial. Five patients (7%) received autologous stem cell transplantation, among which 3 of them (5%) received a single transplant and the other 2 patients (2%) received tandem transplants. Three (5%) patients died due to relapses or progressions of diseases. The estimated overall survival rates at 5 years were 67.74% for total patients and 59.38% for stage M patients. Patients in other stage groups were all alive. The estimated relapse rates at 5 years were 46.95% for total patients, and 29.41% and 64.67% for stage L2 and M patients, respectively. Patients in other risk groups did not have relapses of diseases.

Conclusions: The survival of neuroblastoma patients in higher risk groups are still concerning in countries that do not have anti-GD2 antibodies. There is an urge to use novel immune or target therapies to improve clinical outcomes of neuroblastoma patients in these countries.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0132 / #131 | Fatty Acid Analysis by Oral Intake of Ω 3/ Ω 6 Unsaturated Fatty Acids in Neuroblastoma Mouse

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Background and Aims: In TH-MYCN Tg mouse, the group fed high levels of ω 6 unsaturated fatty acids showed superior survival compared to the ω 3 group. (2020 SIOP Takeuchi et al.). On the other hand, there was no significant difference in immunological background between the two groups. In the present study, we investigated the differences by fatty acid analysis of mouse fed a diet rich in ω 3/ ω 6 unsaturated fatty acids.

Methods: Female hemizygote TH-MYCN Tg mouse were started on a diet rich in ω 3/ ω 6 unsaturated fatty acids at 6 weeks, sacrificed at 12 weeks, and blood samples were taken. The groups were also divided according to the presence or absence of tumors, and fatty acid analysis was performed in each group.

Results: Mouse fed a diet rich in ω 3 unsaturated fatty acids for 12 weeks showed that increased levels of eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA) and docosapentaenoic acid (DPA), and decreased levels of oleic acid and linoleic acid. On the other hand, oleic acid, arachidonic acid, linoleic acid, and docosatetraenoic acid increased in mouse fed a diet rich in omega-6 unsaturated fatty acids. In addition, stearic acid and saturated fatty acids increased in mice without tumors compared to mice with tumors, regardless of the diet.

Conclusions: Oral nutritional immunotherapy had an effect on tumor-bearing mouse, with $\omega 6$ having a greater effect than $\omega 3$. In addition, stearic acid was increased in the presence of tumors, suggesting that it may be involved in the early development of tumors. We need to increase the number of analyses in the future.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0133 / #1320 | Clinical and Behavioral Impact of Serial Digital Education on Anti-Gd2 Therapy for Relapsed/Refractory High-Risk Pediatric Neuroblastoma

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Background and Aims: Advances in anti-GD2 treatment for pediatric neuroblastoma have the potential to improve outcomes for patients with relapsed/refractory high-risk neuroblastoma. Clinicians are challenged to integrate newer agents, demonstrating gaps in treatment administration and side-effect management. Supportive care for patients and their families is often-overlooked yet critical for coping with treatment and side effects. Education in these areas can improve team-based care for pediatric patients with neuroblastoma.

Methods: Two 1-hour CME activities were broadcast live-online to US and international learners in December 2020 and January 2021 and remains on-demand through January 2022 at OMedLive.com. The educational initiative was designed for a multidisciplinary team and included expert panelists discussing integration of anti-GD2 monoclonal antibodies, team-based administration/management, and insights from families. Data analysis to determine practice patterns and clinical impact were based on assessments of knowledge/competence at three timepoints, behavioral impact, and practice preference questions.

Results: To date, 724 clinicians participated in the activities. Across all eight CME test-questions, improvements in knowledge/competence were observed in understanding novel therapies, differentiating outpatient vs inpatient considerations, side effect management, and shared decision-making. Baseline comparisons of US and international learners demonstrated knowledge differences regarding mechanisms of action and similarities in competence managing side effects. At 2-months follow-up, 65% reported impact on clinical practice, and 53% on patient experience/outcomes. Practice pattern comparisons of US and international learners illustrated similarities in treatment approach and perceptions of challenging side-effects; differences

were observed in clinician attitudes toward utilization of anti-GD2 therapies.

Conclusions: The educational initiative revealed clinician perceptions, baseline knowledge and competence at a turning point in the treatment landscape of pediatric neuroblastoma. The initiative was successful in improving clinical and behavioral impact across the care team related to newer therapies, their administration/management, treatment adherence, and family communication.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0134 / #1366 | High-Risk Neuroblastoma Treatment: Demethylating Agents didn't Improve the Outcomes

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Background and Aims: Survival of patients with high-risk neuroblastoma remains unsatisfactory. We aimed to represent our experience in improvement of treatment of patients with high-risk neuroblastoma.

Methods: Twenty two patients with high-risk neuroblastoma treatment in 2018 – 2020: 13 (59.1%) were boys and 9 (40.1%) – girls. Age median – 54.2 (19 – 100) months. All patients had metastatic lesions of bones and bone marrow. Patients were divided into 2 groups: 1 COG-oriented polychemotherapy (PCT) with demethylating agent (5-Azacytidine) in the first and second induction block (n = 14; 63.6%); 2 Standard COG-oriented PCT (n = 8; 36.4%). Surgical treatment, high-dose chemotherapy, and radiation therapy were performed in both groups. Consolidation was carried out in the standard mode with treosulfan and melphalan with the inclusion of a demethylating agent (5-Azacytidine). 10 patients from the groups (45.5%) who did not achieve the full effect of induction chemotherapy received systemic radiotherapy with ¹³¹I-meta-iodobenzylguanidine (¹³¹I-MIBG).

Results: In the median follow-up of 31 months the direct effectiveness (ratio of the number of total and partial effects obtained) of induction chemotherapy in group 1 was 71.1%, in group 2 - 75.1%. Bone marrow sanitation (BMS) during the induction chemotherapy in group 1: after first PCT – 58.5%, rest of pts. achieved BMS after 2-5 blocks. Group 2: BMS was observed in 62.8% cases. For one patient in this group wasn't registered before consolidation. Two-year OS of patients with stage 4 in group 1 was $62.6 \pm 14\%$, in group 2 - $57.1 \pm 14.7\%$, EFS - 37.4 ± 14.5 and $33.1 \pm 11.7\%$ respectively. Use of systemic radiotherapy correlates with the absence of progression or recurrence of the disease.

Conclusions: Demethylating agents didn't show efficacy in high-risk neuroblastoma. Patients with residual tumor showed effectiveness of ¹³¹I-MIBG therapy.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0135 / #1425 | Neuroblastoma in Adolescents and Young Adults, a Single Center Experience

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Background and Aims: While neuroblastoma (NB) is relatively common younger children, its much less often seen in older ones and less than 5% cases are diagnosed in adolescents and adults (AYAs). Although this subgroup is usually characterized by unfavourable prognosis due to indolent clinical course it is usually treated according to common paediatric guidelines. This is a retrospective analysis of a single institution experience in treatment of AYAs diagnosed with NB.

Methods: A total of 13 AYA patients (median age 14, range 10-28 years at diagnosis) with undifferentiated NB in 8(62%) or ganglioneuroblastoma in 5 (38%) cases were included. The primary tumor had mediastinal localization in 7 cases, also, there 3 abdominal and 3 pelvic tumors. In most cases there were metastases to lymph nodes (n=7), bones (n=6) or bone marrow (n=4). All patients were tested for MYCN amplification and segmental chromosomal abnormalities; in 4 cases targeted sequencing was performed. Apart from local control all patients received several lines of systemic treatment with high-dose consolidation used in 8/13 (62%) cases.

Results: The MYCN amplification and del 1p36 was found only in one patient with most aggressive clinical course. The targeted sequencing revealed clinically relevant mutations in all 4 cases, in 2 cases they were found in ATRX gene. Although 7/13 (54%) patients are alive with a median follow-up of 48 (8-104) months, only one of them still maintains long-term response with 5-year EFS of just 8%. Nine of 13 cases showed signs of primary resistance with subsequent chemotherapy lines also being ineffective. There were, however, some dramatic responses to targeted therapy (n=3) or immunotherapy with immune checkpoint inhibitors or their combination with other immunotherapy modalities.

Conclusions: AYAs diagnosed with NB comprise a cohort characterized by distinct tumor biology and treatment course. Although the usual treatment modalities are mostly ineffective, additional treatment options may prove useful.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0136 / #1427 | Clinical Validation of the Compass System for Two Matched Elekta Versa HD Linear Accelerators

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Background and Aims: Back ground Quality Assurance (QA) in radiotherapy is an important studying issue in order to test the performance of medical Linac machine and verify the delivering accuracy of patient treatment plan. Aim Clinical validation of using compass system in VMAT QA for abdomen site pediatric patients

Methods: Method Ten VMAT neuroblastoma plans were created using Monaco TPS. CT images, RT Structures, RT Plans, and RT doses for all cases were exported to COMPASS system through DICOM network then COMPASS computations were performed using collapsed cone algorithm for all patients with a calculation grid resolution of 2 mm from the imported DICOM RT plans. Finally, a comparison was performed based on dose difference and all VMAT plans were measured at two matched linear accelerators.

Results: The mean dose of PTV and D99 %for the ten cases were measured. The comparison between the Monaco dose calculation, compass computed dose and compass reconstructed dose showed a good agreement within maximum deviation 4% for the PTVs and critical organs in point doses and volume doses.

Conclusions: Compass system is a reliable effective system to perform 3D dose verification of VMAT plans for neuroblastoma patients. So Compass QA system can be used as an effective tool before VMAT treatment. The variations of dose between TPS and compass are due to mode of calculation, compass algorithm uses dose to water for calculation while Monte-Carlo algorithm uses dose to medium for dose calculation.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0137 / #153 | Combination of anti-PD-1/anti-PD-L1 Antibodies Promotes Tumor Cell Phagocytosis By Bone Marrow Derived Dcs In A Murine Neuroblastoma Model

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Background and Aims: We previously showed that co-administration of anti-PD-1/anti-PD-L1 monoclonal antibodies (mAbs) significantly suppressed tumor growth, and demonstrated significant increase of CD11c⁺ phagocytes to the tumor microenvironment in a murine neuroblastoma model (Inoue et al. SIOP 2018 Kyoto). Herein we demonstrate the role of the co-administration of mAbs to the phagocytosis using bone marrow derived dendritic cells (BMDCs).

Methods: Green fluorescent protein (ZsGreen)-expressing murine Neuro2a neuroblastoma (ZsGreen-Neuro2a) cells were created and

incubated with both anti-PD-1 and anti-PD-L1 mAbs. Those mAb-treated cells were co-cultured with BMDCs that were generated from bone marrows of A/J syngeneic mice by GM-CFS and IL-4 stimuli together with CpG-ODN. ZsGreen⁺CD45⁺CD11c⁺ cell that phagocytosed mAb-opsonized ZsGreen-Neuro2a cells were evaluated using a flow cytometer on the day 2 and 5 following co-culture. Expression of PD-1, PD-L1 and MHC Class II on CD11c⁺ BMDCs was also examined simultaneously.

Results: Opsonized ZsGreen-Neuro2a cells with anti-PD-1/PD-L1 mAbs were significantly phagocytosed by CD11c⁺ BMDCs on the day5, compared to the isotype-match IgG-treated control group, but not observed on the day 2. CD11c⁺ BMDCs that expressed PD-L1 and MHC class II molecules were transiently decreased on the day 2 following the co-culture of BMDCs and mAb-treated ZsGreen-Neuro2a cells. However, there were no significant difference of the ratio of PD-L1⁺ CD11c⁺ and MHC class II⁺ CD11c⁺ cells between mAb-treated and control groups on the day 5.

Conclusions: Antibody-dependent tumor cell phagocytosis were promoted in vivo when CD11c⁺ BMDCs were co-cultured with neuroblastoma cells treated with anti-PD-1/PD-L1 mAbs. Taking together with significant increase of CD11c⁺ phagocytes in vivo from our previous results, PD-L1 and MHC II on CD11c⁺ BMDCs may play a critical role to induce the antibody-dependent tumor cell phagocytosis in an antibody-mediated neuroblastoma treatment.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0138 / #158 | ALK Protein Expression in Neuroblastoma and its Correlation to Morphology, MYCN Amplification, Outcome and Potential Value in Risk Stratification

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Background and Aims: Neuroblastoma, an enigmatic tumor has diverse outcomes and risk stratification is important for its management. The International Neuroblastoma Risk Group (INRG) Classification assigns risk as very low, low, intermediate and high-risk groups. Anaplastic Lymphoma Kinase (ALK) gene mutations and ALK protein expression in neuroblastoma are important cellular events. The aim of this pilot study was to evaluate ALK protein expression in prognostication of neuroblastoma.

Methods: A total of 54 cases of neuroblastoma were evaluated. A fine needle aspiration biopsy cytology with cell block histology was performed in all cases. Histological type (undifferentiated/poorly differentiated), MKI and MYCN status by FISH (n=46) were determined. Staging and risk assignment as per INRG was performed. ALK protein expression was evaluated by immunohistochemistry (D5F3

clone) and tumoral percent positivity and intensity (0,1,2,3) determined. H-score was calculated [(% negative)+(% 1+positive)+(% 2+positive)+(% 3+positive)] giving a range from 0-300.

Results: The mean age was 29 months (21days-9years) with a male:female ratio of 1.7:1; INSS stage-30(M), 20(L2), 2(L1) and 2(MS); 23 undifferentiated and 31 poorly differentiated; MYCN amplified in 22/46 cases; INRG-31 high, 13 intermediate and 10 low-risk category. At mean follow-up of 469 days (7-835 days); 22/50 were disease free and 28/50 showed relapse/died of disease. Alk protein showed granular cytoplasmic heterogeneous expression in 35 (65%) cases. Alk H-Score did not correlate independently with, MYCN amplification (p=0.35), INRG risk groups (p=0.52) and overall survival (p=0.2). Alk positive poorly differentiated tumours was identified as a subset with good prognosis (p=0.02). Cox proportional hazard model showed Alk negativity to be associated with poor outcome (hazard ratio=2.36).

Conclusions: Alk protein expression is potentially useful for prognosis along with traditional parameters and is worthy of further exploration. The moderate to high levels in 65% neuroblastomas makes it an attractive therapeutic target and an additional step towards precision medicine.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0139 / #1596 | 4Sc-205, First Oral KIF11 Inhibitor, Exhibits Antitumor Activity and Potentiates Standard and Targeted-Therapies in Primary and Metastatic Neuroblastoma Models

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Background and Aims: Neuroblastoma remains incurable for a majority of patients with high-risk disease. These tumors are highly proliferative, invasive and are or acquire resistance to current therapies. Mitotic-related proteins are consistently altered in high-risk neuroblastomas. Kinesin family member 11 (KIF11), is a motor protein which is essential for bipolar spindle formation and mitotic progression in human cells and a therapeutic target in multiple tumors. Here, we

have investigated the effects of the first oral KIF11 inhibitor, 4SC-205, in high-risk neuroblastoma using preclinical primary and metastatic models.

Methods: KIF11 prognostic value was analyzed by mining publicly-available mRNA expression datasets and by immunohistochemistry in neuroblastoma tissue microarrays. Genetic and 4SC-205 inhibition of KIF11 was analyzed in clinically-representative neuroblastoma cell lines. Effects on cell viability, target inhibition, and mechanism of action were assessed in vitro. Findings were validated in 3D spheroids, and in vivo primary and metastatic xenograft models. Combinations with chemotherapy and neuroblastoma targeted therapies were investigated in vitro to inform future clinical trials.

Results: KIF11 expression is associated with poor overall survival and genetic variables associated with aggressive neuroblastomas such as MYCN amplification, -1p36 and +17q23. 4SC-205 recapitulates KIF11 silencing phenotypic effects in vitro and in vivo. 4SC-205 cytotoxicity is mediated by cell cycle arrest in mitosis and subsequent induction of apoptosis. In vivo, 4SC-205 inhibited primary tumor growth in subcutaneous and orthotopic patient-derived xenografts, and reduced metastatic burden. Finally, 4SC-205 potentiates the therapeutic effect of standard chemotherapies (i.e. cisplatin, doxorubicin and topotecan) and targeted therapies such as ALK and RAS/MAPK inhibitors.

Conclusions: These findings suggest that KIF11 inhibition using 4SC-205 alone or in combination with conventional and/or targeted therapeutics, is a potential new therapeutic strategy to treat high-risk neuroblastomas and warrants further clinical development.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0140 / #1609 | Neuroblastoma Risk and Maternal Lifestyle Factors: A Systematic Review and Meta-Analysis Including the Greek Nationwide Registry of Childhood Hematological Malignancies and Solid Tumours (NARECHEM-ST) Data

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Background and Aims: The etiology of childhood neuroblastoma remains largely unknown. The aim of the present systematic review and meta-analysis was to summarize and quantitatively synthesize published evidence on the association of maternal modifiable lifestyle factors with neuroblastoma risk including the unpublished primary Greek data.

Methods: We searched MEDLINE up to December 31, 2020 for eligible studies assessing the association of maternal lifestyle factors (smoking, alcohol consumption and nutritional supplementation during pregnancy) with childhood (0-14 years) neuroblastoma risk. Random-effects models were run, and pooled odds ratios (OR) and 95% confidence intervals (95% CI) on the relevant associations were calculated, including estimates derived from the primary data of the Greek Nationwide Registry for Childhood Hematological Malignancies and Solid Tumours (NARECHEM-ST) case-control study (n=103 cases, n=103 controls; 2009-2017).

Results: Twenty-one eligible studies amounting 5,163 cases participating in case-control and cohort/linkage studies were included in the meta-analysis. Maternal smoking and alcohol consumption were not significantly associated with neuroblastoma risk (pooled OR_{smoking}: 1.08, 95% CI: 0.96-1.22, I²=12.0%, n=17 studies; pooled OR_{alcohol}: 1.01, 95% CI: 0.82-1.18, I²=0.0%, n=8 studies). By contrast, maternal vitamin intake during pregnancy was associated with significantly lower neuroblastoma risk (pooled OR: 0.57, 95% CI: 0.34-0.95, I²=58.9%, n=4 studies). Multivariable analysis of the NARECHEM data showed that maternal alcohol consumption before pregnancy was positively associated with neuroblastoma risk (OR: 3.74, 95% CI: 1.78-7.85) whereas supplementation with folic acid/vitamins during pregnancy, with lower neuroblastoma risk (OR: 0.34, 95% CI: 0.14-0.79).

Conclusions: The results of the largest to-date meta-analysis point to a significant inverse association between maternal vitamin intake during pregnancy and childhood neuroblastoma risk and a modest positive association with maternal smoking during pregnancy. Future longitudinal studies are warranted to confirm and further specify these associations and to allow the implementation of preventive efforts focused on modifiable maternal risk factors of childhood neuroblastoma.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0141 / #1648 | Management Limitation of Neuroblastoma in Iraq: A Limited-Resource Setting Experience

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Background and Aims: Neuroblastoma is the most common extra-cranial solid tumors of childhood and the most common in the first year of life, accounting for 8-10% of all childhood malignancies. the aim is To study the demographic features of neuroblastoma in children whom treated in Children Welfare Teaching Hospital in Iraq. And to assess the response to chemotherapy treatment as they only line of therapy with lack of other lines.

Methods: A descriptive retrospective study was done on newly diagnosed children with Neuroblastoma whom admitted to the oncology unit in the Children Welfare Teaching Hospital/ Medical City/ Baghdad over a 4-years period from the 1st of January 2010 to the 31st of December 2013.

Results: In total, 119 cases with neuroblastoma were identified from Children Welfare Teaching Hospital records between January 2010 and Dec. 2013. Male to female ratio is 1.9:1. The median age at the time of diagnosis was 32 months and the range was (2-155) months. Fever was reported in 79(66.4%) of the patients, followed by pallor 71(59.7%), and weight loss which was reported in 24(20.2%) patients. seventy-nine patients (66.4%) were diagnosed with stage IV neuroblastoma, while stage I were reported in two patients only. Abdomen was the main primary site of the tumor, (86%). Suprarenal mass was detected in 84 patients (70.6%). about 90% of the patients were considered as high risk group. Patients were followed up for three years with mean of 57 months. At end of first eight chemotherapy courses; five patients showed complete remission, twelve patients died, ten patients were lost, twenty-nine patients had signs of progressive disease, and eighty-four patients had signs of partial response. There was no significant association between 3 years' survival and risk group as well as bone marrow metastases, p value 0.2 and 0.9 respectively. Three years' overall survival was 41.4%.

Conclusions: Management was limited by the lack of resources and genetic studies

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0142 / #1703 | Socioeconomic Factors and Distance from Treating Centre do not Predict Survival in South African Children with Neuroblastoma

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Background and Aims: Optimal management of neuroblastoma depends on accurate risk stratification at time of diagnosis. Many low-and-middle-income countries (LMIC) lack access to the specific genetic tests used globally for this purpose. This study aimed to determine whether socioeconomic factors could be used to predict prognosis in neuroblastoma and so provide alternative measures for risk stratification in resource-constrained settings.

Methods: This retrospective record review included 145 patients with biopsy-proven neuroblastoma in Johannesburg, South Africa, between 1 January 2000 and 31 December 2018. Records were obtained from the three main paediatric oncology units (POU) in Johannesburg: Charlotte Maxeke Johannesburg Academic Hospital, Chris Hani Baragwanath Academic Hospital and Wits Donald Gordon Medical Centre. Kaplan-Meier survival analysis was performed in relation to various biological and socioeconomic factors which included serum ferritin and lactate dehydrogenase (LDH), age, stage at diagnosis, parental employment status, nationality and distance of residence from treating facility. Cox proportional hazards regression analysis was used to determine significance of prognostic factors in both univariate and multivariate models.

Results: Factors with a significant effect on survival were younger age ($p=0.030$), lower stage ($p<0.001$) and serum LDH level $<750\text{U/L}$ ($p=0.041$). None of the socioeconomic factors observed had a significant effect on survival (mother employed $p=0.215$, father employed $p=0.125$, South African nationality $p=0.563$). The association between distance of residence from treating facility and stage at diagnosis was not significant (Kendall tau-b coefficient 0.108, $p=0.060$).

Conclusions: Socioeconomic factors did not prove to be significantly associated with neuroblastoma survival in this study. Age, stage and LDH level did, however, suggesting that tumour biology exerts an over-riding influence on prognosis in this type of cancer.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0143 / #171 | Results of Mass Screening for Neuroblastoma in 18-Month-Old Infants in Osaka, Japan

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Background and Aims: Nationwide mass screening for neuroblastoma in infants aged 6 months has been performed in Japan since 1985. In 2004, the Japanese government halted the mass screening because it tended to overdiagnose localized tumors with a favorable prognosis but did not reduce neuroblastoma-related mortality. However, new

mass screening programs, which changed the targets to 18-month-old infants, were performed in the Osaka prefecture. The change in the age group was made according to the assumption that most favorable tumors must have regressed by 18 months and unfavorable tumors could still be detected. In this study, we assessed the efficacy of mass screening in detecting unfavorable neuroblastoma cases among 18-month-old infants.

Methods: Health institutions in Osaka provided test kits to the guardians of infants who received 18-month-old checkup, during 2004-2017. Patients whose standardized urinary levels of vanillylmandelic acid or homovanillic acid exceeded the threshold were further examined and treated in two specific regional hospitals. Screening-positive patients with and without neuroblastoma were retrospectively reviewed.

Results: Among 142,423 children who attended the new mass screening program, 85 had positive results and 14 were diagnosed with neuroblastoma. Twelve cases were classified as very low risk, while two were classified as high risk according to the International Neuroblastoma Risk Group. MYCN amplification was detected in one case. Total resection was performed in 12 cases, and open biopsy, followed by chemotherapy and stem cell transplantation, was performed in one case. One high-risk group patient died of recurrence at the age of 5. The proportion of patients with neuroblastoma confirmed by imaging among the positive screening cases was lower than that in the previous screening for 6 months.

Conclusions: The new mass screening for neuroblastoma in 18-month-old infants did not detect unfavorable cases of neuroblastoma efficiently, although few participants benefited from it.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0144 / #286 | Prostaglandin E2 Receptor Subtype EP2 as a Target for High-Risk Neuroblastoma

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Background and Aims: Neuroblastoma is the most common extracranial solid tumor in childhood, accounting for nearly 15% of all cancer-related deaths in young children. Despite remarkable advances in tumor diagnosis and management, the five-year survival rates for patients with high-risk neuroblastoma remain below 50%. Cyclooxygenase-2/Prostaglandin E2 (COX-2/PGE₂) cascade fosters inflammation-enriched microenvironment in neuroblastoma, promoting tumor proliferation and invasion. Our study aims to 1) elucidate the specific downstream PGE2 receptor (EP) subtype that is directly involved in promoting neuroblastoma growth; 2) evaluate the feasibility of inhibiting a PGE2 receptor subtype as a new strategy to treat high-risk neuroblastoma.

Methods: We analyzed COX-2/PGE₂ cascade gene expression profile from five major neuroblastoma patient datasets. A time resolved fluorescence resonance energy transfer (TR-FRET) method was adopted to reveal the key G α s-coupled receptor that mediates PGE2-initiated cAMP signaling in high-risk neuroblastoma cells. The anti-tumor effects of our recently developed novel EP2 antagonists were determined both in vitro and in vivo. The outcomes are validated using neuroblastoma cell lines in which the EP2 receptors is abated by CRISPR/Cas9-mediated genome editing.

Results: Based on the patient survival data from Versteeg-88, Kocak-649, SEQC-498, Primary NRC-283, and Oberthuer-251, we identified COX-2/PGE2/EP2 signaling axis is highly associated with the aggressiveness of human neuroblastoma. TR-FRET assay demonstrated EP2 receptor as the culprit G α s-coupled receptor that induces cAMP signaling in high-risk neuroblastoma. Our recently developed selective and bioavailable small-molecule EP2 antagonists have demonstrated a potent suppressive effect on the tumor growth and associated angiogenesis in human neuroblastoma xenografts mice models. The tumor suppressive effect originated by blocking EP2 receptor was further confirmed in EP2 knockout neuroblastoma cell line.

Conclusions: Collectively, our results suggest that the PGE2/EP2 pathway contributes to the growth and malignant potential of human neuroblastoma cells; pharmacological inhibition on EP2 receptor by our drug-like compounds might provide a novel therapeutic strategy for this deadly pediatric cancer.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0145 / #294 | The Role of BIT1 Protein in Promoting Malignancy of Neural Crest Cell-Derived Pediatric Tumors

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Background and Aims: Bit1 is a transcription regulator of anti-apoptotic molecule Bcl2. It maintains cell survival and induces cell death in response to integrin-dependent cell adhesion. Recently, the loss-of-function mutations of Bit1 were identified in patients presenting developmental impairments of skeletal muscle, endocrine, and nervous systems resulting in a syndrome called Infantile-onset Multisystem Nervous, Endocrine, and Pancreatic Disease (IMNEPD). Contrastingly, overexpression of Bit1 was reported to promote malignancy and metastasis in several cancers. We conducted immunohistochemistry (IHC) on E13.5 mouse embryo and found that Bit1 was expressed in the neural crest cell (NCC)-derived organs including dorsal root ganglions. Neuroblastoma (NB) and medulloblastoma (MB)

are the pediatric tumors arising from NCC. Besides the origin, NB and MB share several similarities such as MYCN amplification and 17q abnormality. Based on these facts, we hypothesized that Bit1 has a common role in NB and MB. We aim 1) to confirm the expression of Bit1 in NB and MB by IHC, 2) to clarify the association of NB, MB and Bit1 by database analysis, and 3) to identify the function and mechanism of Bit1 in NB and MB by in vitro experiments.

Methods: IHC were performed on formalin-fixed paraffin-embedded tumor sections. Database analysis for MB were performed using Toronto cohort (Cavalliet al., Cancer Cell. 2017). To start cellular experiments, TaqMan-probe qPCR was performed.

Results: IHC showed Bit1 expression was increased in the higher grade and advanced stage of NB and SHH subgroup of MB. Database search for MB showed Bit1 expression was upregulated in group 3 with poorer overall survival. qPCR showed Bit1 expression was different for each cell line.

Conclusions: Our IHC and database analysis results suggested Bit1 is associated with more severe disease conditions. Further investigations are scheduled to reveal the relevance of Bit1 to NB and MB.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0146 / #365 | Meta-Iodobenzylguanidine (MIBG) Induced Thrombocytopenia in Neuroblastoma Patients May Be Caused by Selective Uptake of MIBG via the Serotonin Transporter in Megakaryocytes

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Background and Aims: The therapeutic use of [¹³¹I]meta-iodobenzylguanidine ([¹³¹I]MIBG) is often accompanied by hematological toxicity, primarily thrombocytopenia. We hypothesize that this is caused by selective uptake of MIBG via the serotonin transporter (SERT) located on platelets and megakaryocytes. In this study, we have investigated whether cultured human megakaryocytes are capable of selective plasma membrane transport of MIBG and whether pharmacological intervention with selective serotonin reuptake inhibitors (SSRIs) may prevent this radiotoxic MIBG uptake.

Methods: Peripheral blood CD34⁺ cells were differentiated to human megakaryocytic cells using a standardized culture protocol. Prior to [³H]serotonin and [¹²⁵I]MIBG uptake experiments, the differentiation status of megakaryocyte cultures was assessed by flow cytometry. Real-time quantitative polymerase chain reaction (RT-qPCR) was used to assess SERT and NET (norepinephrine transporter) mRNA expression. On day 10 of differentiation, [³H]serotonin and [¹²⁵I]MIBG

uptake assays were conducted. Part of the samples were co-incubated with the SSRI citalopram to assess SERT-specific uptake. HEK293 cells transfected with SERT, NET, and empty vector served as controls.

Results: Cultured human megakaryocytes are capable of selective plasma membrane transport of MIBG. After 10 days of differentiation, megakaryocytic cell cultures from three different hematopoietic stem and progenitor cell (HSPC) donors showed on average 9.2 ± 2.4 nanomole of MIBG uptake per milligram protein per hour after incubation with 10^{-7} M MIBG (range: 6.6 ± 1.0 to 11.2 ± 1.0 nanomole/mg/h). Co-incubation with the SSRI citalopram led to a significant reduction (30.1% - 41.5%) of MIBG uptake, implying SERT-specific uptake of MIBG. A strong correlation between the number of mature megakaryocytes and SERT-specific MIBG uptake was observed.

Conclusions: Our study demonstrates that human megakaryocytes cultured in vitro are capable of MIBG uptake. Moreover, the SSRI citalopram selectively inhibits MIBG uptake via the serotonin transporter. The concomitant administration of citalopram to neuroblastoma patients during [¹³¹I]MIBG therapy might be a promising strategy to prevent the onset of thrombocytopenia.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0147 / #384 | Analysis of Clinical Features and Prognostic Factors of Neuroblastoma with Bone Metastasis in Infants

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Background and Aims: To summarize the clinical features of infant neuroblastoma (NB) with bone metastasis and analyze the prognostic factors.

Methods: The retrospective research was carried out to analyze the medical records of 32 patients of NB from January 2010 to December 2019 who were diagnosed and systematically treated in Beijing Children's Hospital(BCH) Affiliated to Capital Medical University. Children with age ≤ 12 months and imaging findings indicating signs of distant bone metastases were included in the study. Children without regular treatment or lost to follow-up were excluded.

Results: 32 NB infants with bone metastases accounted for 16.2%(32/198) of infants diagnosed with NB during the same period. There were 12 males(37.5%) and 20 females(62.5%). Among the 32 cases, 14 cases(43.8%) had simple bone metastasis, 19 cases (59.4%) had distant lymph nodes. Bone metastases mainly occurred in the skull, with 11 cases of single bone metastases and the rest with 2 or more bone metastases. Compared with 168 infants NB without

bone metastasis during the same period, the prognosis of children with bone metastasis was worse, 3-year OS was 97.6% vs 82.7%, $P=0.001$. Univariate analysis showed that the prognosis of children with bone marrow metastasis, meningeal and intracranial metastasis, MYCN gene amplification, and high-risk group was poor ($P<0.05$). A total of 30 children were included for efficacy evaluation and prognostic analysis. A total of 18 cases survived event-free, and 12 cases had an event. The time of the event was 7 months (1.5-32 months). Among them, 7 cases survived and 5 cases died. 3-year EFS and OS are 57.1% and 82.7%, respectively.

Conclusions: The median age of onset of NB in infants with bone metastasis was 9 months, and the skull was the most common site of bone metastasis. These infants had a poor prognosis, with an OS of 82.7% at 3 years, and infants with bone marrow metastasis had a worse prognosis than infants with bone metastasis alone.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma*

PV0148 / #394 | NKG2D CAR-T Cells: An Innovative Therapy for Neuroblastoma

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Background and Aims: Neuroblastoma (NB) represents 11% of pediatric cancer deaths in patients younger than 15 years. Natural Killer Group 2D (NKG2D) receptor are expressed in cytotoxic immune cells. Its ligands, NKG2DLs are upregulated in many types of tumor including neuroblastoma. However, neuroblastoma cells may develop mechanisms of resistance to NKG2D mediated cell-lysis such as NKG2DL release. NKG2D CAR-T lymphocytes may overcome immune evasion and become an effective therapy for treating neuroblastoma. **OBJECTIVES:** Determining NKG2D CAR-T cells efficacy and overcoming resistant mechanisms against NB cells.

Methods: NKG2DL expression was analyzed in 9 neuroblastoma cell lines by flow cytometry. NKG2DLs was evaluated in 8 paraffin-embedded neuroblastoma samples by immunohistochemistry. CD45RA- memory T cells were isolated from PBMCs. NKG2D CAR-T cells were generated by lentiviral transduction of CD45RA- T cells. NKG2D CAR-T cell-cytotoxicity against NB cells was evaluated in vitro by Europium-TDA assays. NB cell lines were pre-incubated with specific blocking NKG2DL antibodies. Soluble NKG2DLs (sNKG2DL) in cell lines supernatants were evaluated by ELISA assay. NB cell lines with MICA overexpression were generated by lentiviral transduction.

Results: All neuroblastoma cell lines analyzed showed heterogeneously expression of at least one type of NKG2DL ($MFI \geq 1.5$). Indeed, NKG2DLs were found in the neuroblastoma paraffined samples. NKG2D CAR-T cells were cytotoxic against 2/9 neuroblastoma cell lines, while the other cell lines showed resistance to NKG2D CAR-T cell. Soluble NKG2DLs were found in cell lines supernatants. MICA overexpression in resistant NB cell lines allowed NKG2D CAR-T mediated-lysis.

Conclusions: Analyzed neuroblastoma cell lines expressed NKG2DLs, but only 2/9 were susceptible to NKG2D CAR-T cells lysis. The concentration of sNKG2DLs found in the supernatants of neuroblastoma cells could partially explain the resistance observed in the unsusceptible cell lines. NB cell lines were become in susceptible to NKG2D CAR-T lysis after MICA overexpression.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma*

PV0149 / #461 | Single-Cell Transcriptomic Landscape of Transgenic Mouse Model of Neuroblastoma

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Background and Aims: Neuroblastoma (NB) is the most common and deadly malignancy of infancy. It is an embryonal neuroendocrine tumor arising from the neural crest lineage. Amplification of MYCN oncogene is found in approximately 20% of NB tumors, making up the most common genetic abnormality and represents a critical stratifying prognostic marker. Given that MYCN plays a key role in NB tumorigenesis and aggressiveness, TH-MYCN transgenic mouse model is widely used in NB research. This model is characterized by the overexpression of MYCN in neuroectodermal cells, driven by tyrosine hydroxylase (TH) promoter, giving rise to tumors exclusively in the sympathoadrenal system, reflecting human NB features. We aimed to explore the transcriptional landscape of TH-MYCN tumors by single-cell RNA sequencing (scRNA-seq).

Methods: Tumors from three homozygous and two hemizygous TH-MYCN mice were profiled by scRNA-seq (Chromium 3' v2 kit, 10x Genomics), sequenced on the NextSeq platform (Illumina), and analyzed using pagoda2 and conos packages.

Results: scRNA-seq of tumors from TH-MYCN mice revealed 17 major cell populations, spanning stromal, immune and tumor compartments. The substantial differences between homozygous and hemizygous TH-MYCN tumors were observed in the adrenergic (ADRN) population. The heterogenous ADRN population consisted of 7 distinct subclusters, predominantly resembling sympathoblasts (ISL1, STMN1, STMN2, NEFL, NEFM), notably proliferating sympathoblasts (MKI67,

CDK1, AURKB, AURKA, TPX2). However adrenal chromaffin cells were rare (TH, DBH, ADM, PNMT, CDKN1C). The ADRN cell composition observed in TH-MYCN tumors is highly reminiscent of human NB. Interestingly, we discovered a cluster of immature ADRN cells with distinct gene signatures associated with neural crest development (SOX11, ATRX, XIST, PTPRS, NEDD4, TCF4), which may contribute to chemoresistance and relapse.

Conclusions: Examining the cellular composition of TH-MYCN tumors at a transcriptional level gives a detailed view of tumor cell heterogeneity at an unprecedented resolution. This comprehensive atlas of TH-MYCN tumors provides a resource for unmasking novel therapeutic targets.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0150 / #470 | Treatment of Neuroblastoma Targeting Serine and Arginine Metabolism

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Background and Aims: We previously identified the super poor prognosis group within neuroblastomas with 11q loss of heterozygosity based on the DNA methylation profile. Combined with expression analysis, we extracted PHGDH as a candidate gene characterizing aggressive features. High PHGDH maintains upregulated serine biosynthesis, one of the patterns of cancer metabolism. We showed PHGDH inhibition suppresses the growth of neuroblastoma in vitro. Moreover, arginine depletion, known to affect serine metabolism, inhibited neuroblastoma cells' proliferation and enhanced PHGDH inhibitors' effect by altering cellular metabolism. Since these results were obtained in vitro using a few cell lines, validation using more cell lines and in vivo study are required.

Methods: We conducted RT-PCR to confirm the expression of PHGDH and ASS1 (a gene involved in arginine synthesis) in eight cell lines. We observed the in vitro toxicity to these cell lines of serine synthesis inhibition by PHGDH inhibitor and arginine depletion by arginine deiminase. We also assessed the effects of their simultaneous administration. For in vivo study, we subcutaneously injected either of two cell lines (IMR-32 with high PHGDH and normal ASS1 and SK-N-SH

with low PHGDH and deficient ASS1) into NOG mice. We administered PHGDH inhibitors and arginine deiminase either singly or simultaneously and assessed their effects.

Results: The expression of ASS1 was deficient in four of the eight cell lines. Arginine deiminase was more toxic to ASS1-deficient cells. In vitro, PHGDH inhibitor and arginine deiminase showed combined effects, especially in ASS1-deficient cells, and were synergistic (combination index < 1) at high concentrations. In the mouse model, PHGDH inhibitor and arginine deiminase both inhibited tumor growth as single agents, and the combination of the two agents almost stopped tumor growth. No adverse events such as weight loss were observed.

Conclusions: Treatments targeting serine and arginine metabolism are promising for aggressive neuroblastomas, and further clinical investigation for practical use is warranted.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0151 / #629 | Management of The Infant Adrenal Mass Lesion: Single Center Experience

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Background and Aims: According to the Turkish Pediatric Oncology Group- Infant Adrenal Mass Follow-Up (TPOG-ISKIP) Protocol, asymptomatic infants with localized adrenal mass lesions are followed up without surgical intervention. Herein we aimed to evaluate ≤ 12 months of age infants with localized adrenal mass lesion who diagnosed and followed up in our center.

Methods: The medical reports of patients were analyzed retrospectively. There were 12 infants with adrenal mass lesion who were eligible for the TPOG-ISKIP Protocol between 2007 and 2021. Clinical characteristics, radiologic, serum and urinary biochemical and histopathologic evaluations were analyzed.

Results: There were 15 patients were eligible. Median age at diagnosis was 60 days (3 days-10 months). M/F=1.5. Adrenal mass was detected by antenatal US in 6 patients. In remaining 9 patients complaints were constipation, respiratory distress, restlessness and prolonged jaundice during admission. There were 46% cystic, 33% solid and 13% cystic&solid components. The adrenal mass was unilateral in all patients; 53% were on the left. The longest diameter of the masses were 0.5cm-6cm. "Wait and see" strategy was performed. The adrenal mass lesions disappeared in 8 infants during follow-up. In four cases adrenal mass lesions were completely removed by surgery; the remaining three cases were lost to follow-up. The median regression time of the mass was 7months (2months-17months) in 8 patients. The indication for

surgery was an increase in the size of the adrenal mass in 3 cases and no change in size in one case. The surgery times were 8th, 30th, 40th and 44th months of follow-up. Histopathologic examination revealed adrenal tissue containing calcification, hyalinization in one, adrenal adenoma in one, and adrenal epithelial cyst in two cases.

Conclusions: With the TPOG-ISKIP infant adrenal mass monitoring protocol, spontaneous regression occurred and unnecessary surgery was avoided in 67% of our cases. Surgical intervention was required in 4 cases during follow-up and performed in safe conditions.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0152 / #698 | ALK Inhibitors for Treatment of Adult-Onset Neuroblastoma

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Background and Aims: Neuroblastoma (NB) is a tumor of childhood with >90% diagnosed before 5 years. Adult-onset NB (AON) is extremely rare and differs significantly from childhood disease. AON, while more indolent, is usually metastatic at diagnosis, generally chemotherapy-resistant, and almost invariably lethal. Standard therapies for NB are poorly-tolerated by adults. Prior studies have shown AON is enriched for genomic aberrations especially ALK. Although ALK inhibitors (ALKi) are effective against other ALK-enriched tumors, use in AON has not been reported.

Methods: Retrospective review of patients >18 years at diagnosis seen at Memorial Sloan Kettering Cancer Center. Response to ALKi evaluated using International Neuroblastoma Response Criteria; objective responses were noted. Progression-free (PFS) and overall survival (OS) were calculated using Kaplan-Meier methods.

Results: Since 1979, 52 patients with AON were seen. Of 23 patients evaluated, 14 (61%) harbored somatic ALK mutations. Seven were treated with FDA-approved ALKi. One, initially diagnosed at 12 years, was treated with ALKi following a late relapse. All ALKi were well-tolerated; adverse events included grade 1-2 nausea and vomiting (n=6), and neurologic symptoms including hallucinations, drowsiness, and dizziness (n=1 patient each) which resolved after stopping ALKi. Four patients received >1 ALKi either due to progressive disease or intolerance. Most responded to the first ALKi with objective though partial response (n=5). Median time to progression for initial ALKi was 15.5 (10-45) months. Of the patients treated with a second ALKi (n=4), 1 had CR and 2 had PR after not progressing previously, while the other had stable disease. Median OS from beginning treatment with first ALKi was 46.5 (17-74) months.

Conclusions: AON is a rare disease and a therapeutic challenge. Enrichment of ALK mutations permits consideration of targeted therapy. ALKi were well-tolerated, often associated with significant responses,

and treatment with serial ALKi yielded ongoing benefit. Given this, ALKi should be considered for future AON cases.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0153 / #720 | Association of BRAF V600E Mutations with Vasoactive Intestinal Peptide Syndrome in MYCN-Amplified Neuroblastoma

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Background and Aims: Rarely, vasoactive intestinal peptide-related diarrhea (VIP-D) is observed in patients with high-risk neuroblastoma (HR-NB). The associated fluid and electrolyte abnormalities pose a clinical challenge for administering multimodal cancer-directed treatment. Two patients with HR-NB developed VIP-D during induction and were found to have unique genomic aberrations on tumor sequencing, thereby leading to the discovery of a novel treatment modality for their VIP-D.

Methods: With IRB approval, clinical and genomic information on two patients with VIP-secreting HR-NB treated at MSKCC was retrospectively analyzed. Tumor sequencing was performed by CLIA-certified platforms: either MSK-IMPACT or Foundation One.

Results: Two females with MYCN-amplified HR-NB acutely developed massive VIP-D after two cycles of induction chemotherapy (as per ANBL0532). Both patients had a somatic BRAF V600E mutation. Patient #1 (19 months at diagnosis) had persistent VIP-D despite chemotherapy, octreotide, and tumor resection. Upon dabrafenib (BRAF inhibitor) initiation, diarrhea and serum VIP levels decreased within days. Four weeks later, the patient relapsed in brain with BRAF wild-type neuroblastoma. Dabrafenib was discontinued and diarrhea recurred. Upon trametinib (MEK inhibitor) initiation with dabrafenib resumption, diarrhea improved. However, progressive disease led to death 13 months post-diagnosis. Patient #2 (11 months at diagnosis) had prompt resolution of diarrhea and decrease in serum VIP levels after starting dabrafenib and trametinib with induction cycle five. These agents were discontinued after post-induction chem-immunotherapy without return of diarrhea or increased VIP levels. She remains progression-free 43 months post-diagnosis. Primary adrenal tumor was BRAF wild-type, but metastatic sites (liver/lymph nodes) demonstrated BRAF V600E mutation.

Conclusions: HR-NB patients who develop VIP-D during induction may have somatic BRAF V600E mutations. The demonstrated tumor heterogeneity suggests that combination therapy with BRAF and MEK inhibitors leads to rapid improvement in symptoms and favorable outcomes. Tumor sequencing of HR-NB patients with VIP-D can result

in timely therapy with appropriate TKIs and facilitate ongoing tumor-directed treatment.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0154 / #726 | Skeletal Muscle Metastases in Neuroblastoma: An Ostensible “Forme Fruste” Of Stage MS Disease

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Background and Aims: While subcutaneous metastases are often observed in patients with stage MS neuroblastoma, an entity that usually resolves spontaneously, skeletal muscle metastases (SMM) have been rarely described and their significance is uncertain.

Methods: After Memorial Sloan Kettering (MSK) IRB approval, we conducted a retrospective analysis of patients with SMM of neuroblastoma seen at MSK between 1995 and 2021. Staging and response were recorded by International Neuroblastoma Risk Group Staging System and International Neuroblastoma Response Criteria respectively. Progression-free and overall survival were measured from diagnosis.

Results: Sixteen patients with neuroblastoma SMM, all detected at diagnosis at a median age of 3.9 (0.1-15.6) months, were identified. All had metastases at other sites and 14/16 (88%) had ≥ 2 SMM in disparate muscle groups. Two, thirteen, and one patients had high, intermediate, and low-risk disease. Fourteen patients had tumors with favorable histology without MYCN-amplification, 2 tumors were MYCN-amplified. Of the 14 patients evaluated with MIBG scan the majority (78%) had MIBG-avid SMM. Only 1 patient (with MYCN-non-amplified neuroblastoma) had disease progression. All survive at a median follow up of 32.7 (3.9-305.5) months. Biological markers including histology and chromosomal aberrations were not prognostic, nor was initial management. SMM completely resolved in 10 patients by 12 months post-diagnosis. Of the four patients managed with watchful observation alone without any cytotoxic therapy, three achieved and have maintained continual complete remission with SMM resolving by 5, 13, and 21 months post-diagnosis. The fourth received intermediate-dose chemotherapy after progressive disease and remains in partial remission with substantially reduced SMM size.

Conclusions: Children with neuroblastoma SMM have an excellent prognosis, with a clinical course suggestive of stage MS. Based on these results a wait-and-watch approach for MYCN-non-amplified disease, especially if supported by genetic profiling, could eliminate or reduce exposure to cytotoxic therapy while preserving a favorable prognosis.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.a Neuroblastoma

PV0155 / #749 | Integrating Non Psychoactive Phytocannabinoids and their Cyclodextrin Inclusion Complexes into the Treatment of Pediatric Cancers

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Background and Aims: The non-psychotropic cannabidiol (CBD), deriving from Cannabis sativa is endowed with anticancer, antioxidant and genoprotective properties, which along with its safe profile suggest its usage in cancer therapies. The low solubility of CBD hampers its therapeutic potential. Cyclodextrins (CDs) are oligosaccharides used to incorporate apolar molecules inside their hydrophobic cavity, increasing their stability, solubility and bioavailability. CBD-inclusion complexes with CDs is a good nanomedicine-based formulation strategy. The current research aims to study the potential cytotoxic effect of CBD and CBD-CDs [CBD-RM β CD (randomly methylated β -cyclodextrin), CBD-HP β CD (hydroxypropyl- β -CD)] on neuroblastoma cells.

Methods: CBD is diluted in 10% DMSO and CBD/CDs solutions are prepared by mixing solid CBD, solid CDs and dH₂O. Phase solubility studies are conducted to determine improvement in CBD's solubility from the addition of CDs. For the biological assays SH-SY5Y and BE(2)-M17 neuroblastoma cells are incubated at a range of concentrations of CBD, CBD-RM β CD, CBD-HP β CD, RM β CD and HP β CD for 24, 48 and 72 hours. Analysis of cell viability is performed with Alamar Blue and MTT assays.

Results: CBD's solubility is enhanced in the presence of both CDs. CBD and CBD/CDs exert cytotoxicity in a dose and time-dependent manner; as their concentration and time of exposure increase, cell viability is reduced. RM β CD and HP β CD at the highest concentration of 4 mg/ml exert antitumor action per se. IC₅₀ values are calculated as 0.29, 0.21, 0.06 mg/ml for CBD, 0.046, 0.035, 0.03 mg/ml for CBD-RM β CD, and 0.029, 0.023, 0.021 mg/ml for CBD-HP β CD at 24, 48, and 72 hours, respectively for SH-SY5Y cells. For BE(2)-M17 cells IC₅₀ values are 0.3, 0.27, 0.19 mg/ml for CBD, 0.1655, 0.1644, 0.1534 mg/ml for CBD-RM β CD, and 0.1086, 0.1020, 0.071 for CBD-HP β CD at 24, 48, and 72 hours, respectively.

Conclusions: The results of our study could afford the basis of research regarding the use of natural products and their inclusion complexes as anticancer agents.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0156 / #781 | Morphology, MYCN Amplification and their Correlation to Survival in Neuroblastoma Patients in India

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Background and Aims: The International Neuroblastoma Pathology Classification (INPC) morphological types and Mitotic karyorrhectic index (MKI) have a bearing on prognosis of neuroblastoma. MYCN amplification is important in INRG risk stratification. We evaluated these parameters in a cohort of patients of neuroblastoma from India to ascertain their prognostic value in our population.

Methods: A total of 54 cases of neuroblastoma were evaluated. Fine needle aspiration biopsy cytology with cell block histology was performed and all cases categorized using the INPC. Mitotic karyorrhectic index (MKI) was determined by digital image visual assessment (DIVA) of smears and cell blocks. MYCN status was determined on cytology smears or liquid based preparations by FISH (n=46). Staging and risk assignment by INSS and INRG was performed. Follow-up was available in 50 cases.

Results: The mean age was 29 months (21days-9 years) with 22 children <18months and 32 children ≥18 months; male: female ratio of 1.7: 1; INSS stage-30(M), 20(L2), 2(L1) and 2(MS); INRG-31high, 13intermediate and 10low-risk categories. MKI was high in 39(72.2%), intermediate in 7, low in 4 cases. The morphological type was 21 undifferentiated, 31 poorly differentiated and 2 ganglioneuroblastoma respectively. MYCN was amplified in 48%(22/46) cases. At a mean follow up of 469 (7-835) days 22/50 were disease free and 28 had relapsed/died. The overall survival correlated with age (<18 vs ≥18 months, p=0.01), INRG group (0.0001) and morphology (UD vs PD, p=0.036). MYCN status independently did not correlate with age (p=0.5), INSS stage (p=0.2) and overall survival (p=0.4). MYCN amplification correlated with morphology with twice as many positive in the UD group (70%UD vs 35%PD, p=0.01).

Conclusions: Our cohort of neuroblastoma showed a higher frequency of cases ≥18 months age, having high MKI and MYCN amplification and consequently more cases in the high-risk category. This regional variation in the neuroblastoma spectrum must be taken into account when planning global studies.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0157 / #860 | Scoliosis in Children with Neuroblastoma and Spinal Canal Involvement

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Background and Aims: Spinal canal involvement (SCI) is observed in 5-15% of patients with neuroblastoma (NB) leading to neurological deficits and spinal deformity. The aim of the study was to analyze the frequency of scoliosis in patients with NB and SCI.

Methods: The analysis included 61 patients with NB and SCI treated at Dmitry Rogachev NMRC PHOI from 01.2012 to 12.2018 according to the German NB-2004 protocol. The presence of scoliosis and the degree of its severity were assessed initially and 2 years after the diagnosis according to imaging data using Cobb angle measurement.

Results: Scoliosis was initially detected in 7/61 (12%). The median age at diagnosis was 8.0 months (2.3-11.8). Male:female ratio was 2.5:1. The primary tumor located retroperitoneally in 4/7(57%) and in posterior mediastinum - 3/7(43%). Most patients had stage 2 and 3 according to INSS - 4/7(57%). 5/7(71%) were stratified to the low risk group. In 6/7(86%) tumor invaded spinal canal at the thoracic level. In 6/7(86%) total infiltration of the spinal canal was noted. Neurosurgical treatment was performed in 4/7 (57%). Scoliosis was assessed as mild in all patients. Among patients without initial scoliosis, anatomic images after 2 years from diagnosis was available in 38/54 (70%). Scoliosis was detected in 9/38 (24%). The median age at diagnosis - 8.2 months (0.8-42.3), the male:female ratio - 2:1. The primary tumor located in the mediastinum in 7/9(78%). 7/9 (78%) patients were stratified to the low-risk group. SCI at the thoracic level was noted in 8/9 (89%). Tumor infiltrated canal from 33% to 66% of the diameter in 4/9 (44%). Neurosurgical treatment was performed in 6/9 (67%). In this group, mild scoliosis was observed in 7/9(78%) and moderate in 2/9(22%).

Conclusions: Scoliosis in patients with NB and SCI is detected both at diagnosis and in the long-term period and requires a multidisciplinary approach including orthopedists.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0158 / #862 | Neuroblastoma in Children: Epidemiology, Clinicopathology and Therapeutic Outcomes of Tunisian Case Series about 33 Patients

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Background and Aims: Neuroblastoma (NB) is the most common extra-cranial pediatric solid tumour. We aimed to evaluate epidemiology, clinicopathology and therapeutic outcomes of tunisian children with NB.

Methods: We retrospectively reviewed the data about all children treated for NB at the medical oncology department of CHU Habib Bourguiba between 2005 and 2019.

Results: Thirty-three children were collected with median age of 2,3 years (78.7%<4years). The primary tumor site was mostly the adrenal gland (60.6%). The most common metastatic sites were bone(57.5%), lymph nodes(42.4%) and bone marrow(36.3%). Histological diagnosis was made in 29 cases (NB =89.6%, ganglioNB =10.3%). The N-myc status, performed in 11 children, showed an amplification in 45.4% of cases. The disease was at stage II, III and IV in 3%, 21.2% and 75.8% respectively. Induction chemotherapy was delivered in all children with stage IV tumors (1 to 13 cycles) using mostly the SFOP protocol (88%). Objective response was noted in 14 cases (56%). Surgery was then performed in 3 cases (12%). Intensified chemotherapy was administered in 2 cases (8%) followed by curative radiotherapy. In localised stages, all children (n=8) received neoadjuvant chemotherapy with an objective response rate of 75% (partial response = 6) and stability in one case. Surgery was then performed in 7 cases (87.5%) (R0=4, R1=1, R2=2). One patient with incomplete (R2) surgery showed a metastatic progression after 11 months. Better 3-year overall-survival was noted in localised stages (75% vs 19.3%(p=0.04)), stage IV diagnosed < 12 months of age (100% vs 9.9%(p=0.012) and children <18 months at diagnosis (77.8% vs 14.6%(p=0.009). No difference was noted according to N-myc status (p=0.453).

Conclusions: The prognosis of NB is still poor, especially in later age and advanced disease. Management in our country is limited by the lack of resources and genetic studies. Further research is needed to develop treatment protocols adapted to the limited resources.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0159 / #878 | Chromosome 10 Abnormality Predicts Prognosis of Neuroblastoma Patients with Bone Marrow Metastasis

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Background and Aims: Neuroblastoma (NB) is the most common extracranial solid tumor in children. It is known for high heterogeneity and concealed onset. In recent years, the mechanism of its occurrence and development has been gradually revealed. The purpose of this study is to summarize the clinical characteristics of children with NB and abnormal chromosome 10, and to investigate the relationship between the number and structure of chromosome 10 abnormalities and NB prognosis.

Methods: Chromosome G-banding was used at the time of diagnosis to evaluate the genetics of chromosomes in patients with NB and track their clinical characteristics and prognosis. All participants were diagnosed with NB in the Medical Oncology Department of the Beijing Children's Hospital from May 2015 to December 2018 and were followed up with for at least one year.

Results: Of all 150 patients with bone marrow metastases, 42 were clearly diagnosed with chromosomal abnormalities. 13 patients showed abnormalities in chromosome 10, and chromosome 10 was the most commonly missing chromosome. These 13 patients had higher LDH and lower OS and EFS than children with chromosomal abnormalities who did not have an abnormality in chromosome 10. Eight patients had both MYCN amplification and 1p36 deletion. Two patients had optic nerve damage and no vision, and one patient had left supraorbital metastases five months after treatment.

Conclusions: The results indicated that chromosome 10 might be a new prognostic marker for NB. MYCN amplification and 1p36 deletion may be related to chromosome 10 abnormalities in NB. Additionally, NB patients with abnormal chromosome 10 were prone to orbital metastases.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PV0160 / #902 | Development and External Validation of a Novel Target-Based Prognostic Nomogram for Children with Neuroblastoma

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Background and Aims: The accuracy of the current staging systems, INPC histology categories and COG risk group, for predicting the overall survival of patients with neuroblastoma are still unsatisfactory. We aimed at developing a nomogram to accurately predict individual survival. This study sought to externally validate this nomogram using the Chinese cohort.

Methods: The records of 499 patients of neuroblastoma using the TARGET database's newest database with the publication time ranging from 2007 to 2015 were retrospectively analyzed. Additionally, 155 patients of neuroblastoma, who were treated between 2011 and 2019 at our institute, were included as an external validation set. The nomogram was established by using prognostic factors determined by LASSO regression and the predictive accuracy was verified with the concordance index and the AUCs of ROC curves.

Results: Gender, age, MYCN stage, INSS stage, diagnostic category and primary site were significant variables used to develop a nomogram for OS, which had a concordance index of 0.745 (95% CI, 0.700-0.790). The AUCs of ROC curves of 3-, and 5-year OS were 0.730 and 0.806, respectively. The calibration curved for the probability of postoperative 3-, and 5-year OS showed good agreement between the predicted and observed outcomes. The external validation showed the higher AUCs of ROC curves of 3-, 5-year OS were 0.840 and 0.824, respectively. Regard to the Chinese validation set, C-index (0.816 (95% CI, 0.747-0.885)) was also enhanced compared with the TARGET training set. Furthermore, according to the nomogram-predicted scores, the patients were respectively stratified into 2 and 3 groups with different risks to compared with the INPC histology categories and the COG risk group. The nomogram showed a better accuracy in predicting OS than the two current staging systems.

Conclusions: The nomogram can predict the 3-, and 5-year OS of patients with NB and show satisfactory utility in clinical practice.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0161 / #1036 | Differential micro-RNA Expression in Histological Typing & Prognosis of Wilms Tumor

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Background and Aims: Wilms tumor (WT) outcome has significantly improved in the last decade, however still a proportion of cases either relapse or are resistant. Histological subtyping is used in risk stratifi-

cation with blastema predominant cases portending a poor outcome. Differential miRNA expression profile may correlate with histological subtyping and prognostication. In the present study, global miRNA expression analysis was compared with histological subtypes and outcome.

Methods: Microarray expression analysis of 1205 mature human miRNA was performed in 15 fresh resection samples of WT using Agilent Human miRNA expression microarray (Santa Clara, USA). miRNA showing differential expression between histological and outcome groups were identified by T test using T-Bioinformatics platform followed by principal component analysis & unsupervised hierarchical clustering based on top 10 significantly differentially expressed miRNA, with focus on more than 2 log fold change. Expression of selected apoptosis related miRNA were evaluated by real time PCR analysis using abmBrightGreen 2x qPCR MasterMix-i-Cycler in a separate set of 10 cases of WT.

Results: The signature miRNA of WT was found to be miR483-3p with consistent up-regulation in 93.3% of cases irrespective of the histological subtype. Three miR (miR410, miR495, miR4446) were found to be significantly differentially expressed among different histological subtypes of WT (p<0.05) by Mann Whitney test and a panel of 10 miRNA (miR-126-5p, miR-299-5p, miR-3196, miR-499a-5p, miR-1, miR-491-5p, miR-888-5p, miR-658, miR-133a, miR-509-5p) helped in the histological subtyping of WT as determined by hierarchical clustering and heat map analysis. miRNA 143, under-expression was related to good outcome and confirmed by RT-PCR (5/8 cases). Necrosis and rhabdomyoblastic differentiation led to change in the miRNA expression profile.

Conclusions: A set of useful microRNA was derived based on the statistical analysis of differential expression, useful in histological subtyping of WT, potentially useful for serum analysis to derive pre-chemotherapy information of histological and prognostic data profiles.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0162 / #1066 | Predictors of Outcome in Relapsed and Refractory Wilms Tumor in a Resource-Limited Setting

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Background and Aims: Wilms' tumor (WT) is the most common renal tumor in children. Overall survival for patients with relapsed WT is around 50% in developing countries. We conducted this study to review the predictors of outcome for relapsed/refractory WT in resource-limited setting.

Methods: Retrospective review of relapse/refractory WT treated at our institution from January 2010 to December 2019. Kaplan-Meier

analysis and log-rank test were used for survival analysis and comparison.

Results: Of 243 patients with WT, 43 (17%) patients had relapsed/refractory WT (male: female 1.4:1, mean age: 3.9 years (+/- 2.2 year). 58% (n=25) and 42% (n=18) had refractory and relapse respectively. Histology at initial diagnosis was favorable in 86%(n=37). Stage at initial presentation was I (9%), stage II (19%), stage III (42%), stage IV (23%) and stage V (7%). 67% (n=29) received Anthracyclines and 63% (n=27) radiation therapy during initial therapy. At relapse, 98% (n=42) were stage IV. 47% (n=20) received carboplatin-etoposide-cyclophosphamide-doxorubicin and 28% (n=12) ifosfamide-cyclophosphamide-etoposide (ICE). 16% (n=6) were treated with palliative intent. Overall survival (OS) was 50% at 5 years, with a mean survival duration of 41.7 months (+/-6.8). There were 11 deaths (4: disease progression, 7: sepsis/cardiac dysfunction). Factors associated with better outcome were relapse after achieving remission vs primary progression (OS 60%, 8%, p-value=0.8), male gender vs female (60%, 40%, p-value=0.7), initial low stage (I/II) disease vs high stage disease (75%, 28%, p-value=0.13), no Anthracycline exposure (75%, 30%, p-value=0.14) and no radiation (60%, 0%, p-value=0.12) during initial treatment. Patients treated with carboplatin-etoposide-cyclophosphamide-doxorubicin had OS of 85% compared to 0% at 5 years for ICE (p-value=0.014).

Conclusions: Factors like the high stage at initial presentation, primary progressive disease, and previous Anthracycline and radiation exposure predict the worse outcomes for relapsed/refractory WT patients that are not improved by the use of intensive regimens like ICE. In view of limited resources, the selection of good-risk patients for curative treatment should be considered.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0163 / #1183 | Outcome of Patients with Relapsed Low/Intermediate-Risk Wilms Tumor after Upfront Treatment with Vincristine and Actinomycin-D only, Based on The Siop-93-01 and 2001 Protocols

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Background and Aims: In the current SIOp-RTSG protocol, treatment recommendations for relapsed Wilms tumors (WT) are stratified by intensity of first-line treatment. To increase evidence for the recommended regimen for patients relapsing after upfront vincristine and actinomycin-D (VA) treatment, we retrospectively evaluated rescue treatment and survival of this group of patients.

Methods: Patients from the SIOp 93-01 and SIOp 2001 studies who relapsed after primary stage I/II, low/intermediate-risk WT initially treated with VA and without radiotherapy were included. Univariate Cox regression analysis was performed to study the effect of relapse treatment intensity on event-free survival (EFS) and overall survival (OS). Relapse treatment intensity was classified into two groups: vincristine, actinomycin-D and doxorubicin/epirubicin (VAD/E), and more intensive therapies (ifosfamide/carboplatin/etoposide (ICE)) / > 4 drugs/high dose chemotherapy with hematopoietic stem cell transplantation (HD HSCT)).

Results: A total of 109 patients were analysed. Relapse treatment regimens included either VAD/E, cyclophosphamide/carboplatin/etoposide/doxorubicin (CyCED), or ICE backbones. Radiotherapy was administered in 62 cases, and 15 patients were treated with HD HSCT in addition to conventional chemotherapy. Overall, 5-year EFS and OS after relapse were 72.3% (95% CI: 64.0-81.6%) and 79.3% (95% CI: 71.5-88.0%), respectively. Patients treated with VAD/E did not fare worse when compared to patients treated with more intensive therapies (Hazard Ratio (HR) EFS: 0.611 (95% CI: 0.228- 1.638) (p-value= 0.327) and HR OS: 0.438 (95% CI: 0.126- 1.700) (p-value=0.193)).

Conclusions: Patients with relapsed WT after upfront treatment with VA showed no inferior EFS and OS when treated with VAD/E regimens compared to more intensive rescue regimens. This suggests that a subset of patients relapsing after VA may benefit from less intensive rescue treatment than ICE/CyCED based regimens. These patients deserve to

be pinpointed by identifying additional (molecular) prognostic factors in future studies.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0164 / #1234 | Single Cell Derived Organoids from Infant Wilms Tumour Reveal Novel Insights into the Rest Tumour Suppressor Gene

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Background and Aims: Wilms tumour is the most common renal cancer of infancy, affecting approximately 1 in 10,000 infants worldwide. Several genes with non-redundant roles in tumorigenesis show a predilection for younger patients but our knowledge of this genetic landscape remains incomplete. We utilised patient-derived organoids to study the phylogenetics of infant Wilms development, through whole genome sequencing of tumour and surrounding normal tissue.

Methods: Tumour biopsies and normal tissue, including kidney and fat, were collected from a resected kidney specimen, alongside blood, obtained from The Princess Maxima Centre. Two bulk normal kidney and 3 bulk tumour organoids were established, as previously reported, from which 9 single cell derived organoids were cultured. DNA was extracted and whole genome sequenced according to Illumina HighSeq X protocols. After alignment to the reference genome and filtering, single nucleotide variants, indels, copy number variants and structural rearrangements were called. Germline and somatic variants were distinguished using a one-sided exact binomial test on the counts of variant reads and depth. Phylogenetic trees were determined using a maximum parsimony approach.

Results: Calling somatic variants identified a double hit in the REST gene. The first of which was a nonsense Q405* mutation occurring in early embryogenesis. This mosaic mutation was present in fat, blood but only one of the kidney samples. Amongst the variants underlying the tumour trunk, we found a second frameshift indel in REST R220fs*3. Reconstructing tumour phylogeny revealed extremely early divergence of the single cell derived clones, after fewer than 30 mutations. Whilst subclones from the superior aspect of the tumour were closely related, subclones from two spatially distinct areas sampled from the inferior tumour exhibited substantial heterogeneity.

Conclusions: We demonstrate that patient-derived single cell organoids provide a unique opportunity to study infant Wilms phylogenetics and identify the first case of an early embryonic mosaic REST mutation predisposing to tumour development.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0165 / #1249 | High Rate of (EPI) Genetic Predisposing Factors and an Important Role for DIS3L2 in A Nationwide Wilms Tumor Cohort

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Background and Aims: Wilms tumor (WT) is associated with (epi)genetic predisposing factors including Beckwith-Wiedemann Spectrum (BWSp) and WT1-related syndromes. The prevalence and distribution of predisposing factors have not been well characterized. We aimed to correlate germline findings to phenotypic and tumor characteristics, and identify novel WT predisposition genes.

Methods: All children diagnosed with WT in the Netherlands between 2015-2020 were referred to a clinical geneticist and included in this study. Phenotypic data and diagnostic test results were collected. If no germline mutation was identified by targeted diagnostic testing, germline (trio-) whole exome sequencing (WES) and BWSp testing on normal kidney tissue-derived DNA was offered.

Results: A total of 126 patients were identified, including one familial WT (0.8%). (Epi)genetic predisposing factors were present in 42/126 patients (33.3%). Heterozygous DIS3L2 variants were identified as a novel predisposing factor in five patients (4.0%), with a second somatic hit in 4/4 (100%) tumors tested. Twenty patients (15.9%) were diagnosed with BWSp, including patients with a molecular diagnosis in blood-derived DNA (N=8), normal kidney tissue-derived DNA with at least one additional feature of BWSp (N=8), or solely a clinical diagnosis of classical Beckwith-Wiedemann syndrome (N=4). Four patients without additional BWSp features harbored 11p15 methylation defects in normal kidney tissue. Remaining findings included WT1-related syndromes (N=10, 7.9%), Fanconi anemia (N=1), REST (N=1) and NF1 (N=1) mutations. (Likely) pathogenic variants in adult-onset cancer predisposition genes (BRCA2, PMS2, CHEK2, MUTYH) were identified in 5/56 (8.9%) patients with available WES data. Candidate WT predisposition genes were identified which require validation in larger cohorts.

Conclusions: (Epi)genetic WT predisposing factors, including mosaic 11p15 aberrations, were present in at least 33.3% of patients with WT in this national cohort, with an important role for constitutional heterozygous DIS3L2 variants. Based on these results, we encourage standard genetic testing after counseling by a clinical geneticist.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0166 / #1342 | Characteristics and Outcome of Children with Renal Tumors in the Netherlands: First Five Year Single National Center Experience

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Background and Aims: In the Netherlands, pediatric renal tumor care has been centralized in the Princess Máxima Center since 2015. To describe epidemiology, clinical characteristics and outcome of all dutch children that presented with a renal tumor in the first 5 years of centralized care.

Methods: All patients presenting with suspicion of renal cancer were included after consent for data collection and storage of biomaterials.

Results: From 2015-2020, 164 patients presented with renal tumours. One family refused consent, 163 remaining cases included Wilms tumour(WT)(n=118), nephroblastomatosis/nephrogenic rests(n=6), clear cell sarcoma of the kidney(CCSK)(n=2), malignant rhabdoid tumor of the kidney(MRTK)(n=7), renal cell carcinoma(RCC)(n=8), congenital mesoblastic nephroma(CMN)(n=5), cystic nephroma(CN)(n=5), cystic partially differentiated nephroblastoma(CPDN)(n=2), angiomyolipoma(n=5) and others(n=5). It appeared, that, all malignant renal tumor cases registered in the Dutch Cancer Registry in that period, were diagnosed in Princess Máxima Center. In 149/163 children, DWI-MRI was performed at presentation, and chest-CT was performed in 121/135 children with WT, CCSK, MRTK or RCC. Central radiology(before start of treatment) and pathology review was performed respectively in 118/163 patients and 151/152 (partial-)nephrectomized cases. Treatment stratification

according to SIOP-RTSG protocols was established after agreement in the multidisciplinary tumor board. 19 patients had upfront surgery and 133 patients underwent pre-operative chemotherapy. Altogether, 135 underwent unilateral tumor nephrectomy, (n=17 nephron-sparing-surgery, n=11 no surgery). Of 133/152 patients, biomaterial was stored. Three WT patients and five metastatic MRTK patients died.

Conclusions: Centralization of care for children with renal tumors in the Netherlands led to centralized care of all malignant renal tumor cases, in the Netherlands. Twenty-four percent of all presenting cases included non-WT. We established a multidisciplinary approach for all renal tumor patients, which allowed best available care based on standard review based diagnostics, and enabled development of molecular and other innovation based treatment for the future, that will further enhance survival.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0167 / #1457 | Quality Assessment of Pathological Reporting of Wilms Tumor and Violation of Histologic Reporting Standard in A Nigerian Teaching Hospital

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Background and Aims: A key contributor to better patient outcomes of renal tumours has been refinement of therapy based on clinical and biologic risk factors. Following chemotherapy based on SIOP protocol, pathologic assessment of the percentage of non-viable and viable tumour, and the percentage of different histological components of the viable tumour is important for risk stratification. The purpose of this study was to evaluate the characteristics of violations of standardized pathological reporting among children who underwent surgery for renal tumours in a Nigeria teaching Hospital

Methods: The pathological reports of all patient that had nephrectomy at our institution from 2011 to 2020 were retrospectively reviewed based on the Royal College of Physician Dataset for histopathological reporting of renal tumours in childhood(G057). Sixteen data set reviewed included clinical, macroscopic and microscopic information. The frequency, type, and number of pathological protocol violations (PPVs) were assessed.

Results: A total of 19 pathological reports were assessed including 13(68.4%) girls at a median age of 31months. There were violations in all the reports. The overall incidence of violation was 70.1% (213 of 304 data set). The mean number of PPV was 11.21 ± 1.08 . Protocol violation per patient ranged from 56.3% to 87.5%. The most common violated items were non-reporting of percentage of necrosis 19(100%), proportion of blastema cells 19(100%), sinus invasion 18(94.7%), perirenal fat invasion 18(94.7%), not stating status of margin 18(94.7%), and renal vein invasion 17(89.5%). The least violated

items were lack of tumour site reporting 0(0%) and tumour size reporting in 1(5.3%) patient. Risk stratification was not done in 17(89.5%) patients

Conclusions: Most of dataset for histopathological reporting of renal tumours was violated in this study. Risk classification was also not routinely done. There is need to adapt user-friendly pathological reporting guidelines and checklists for renal tumour in this setting to improve quality of care of Wilms tumour

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0168 / #1602 | Incidence of Childhood Non-Wilms Renal Tumors: A Comparison of the Nationwide Registry of Childhood Hematological Malignancies and Solid Tumors, Greece with Seer, USA Data

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Background and Aims: Non-Wilms tumors (non-WT) in childhood represent ~10% of renal neoplasms and <1% of all childhood malignancies. We estimated the incidence-time trends of non-WT and compared the Nationwide Registry for Childhood Hematological Malignancies and Solid Tumors (NARECHEM-ST), Greece data to those of the Surveillance, Epidemiology, and End Results Program (SEER), USA.

Methods: Forty-seven pediatric (0-14 years) renal tumor cases other than WT (ICD-O-3: 8960/3) were registered in NARECHEM-ST (2001-2020), whereas 412 cases were extracted from SEER

(1990-2017). Age-standardized incidence rates (AIR) were calculated, whereas temporal trends were evaluated through Poisson regressions.

Results: In NARECHEM-ST, non-WT accounted for 19.5% of all renal tumors, which was higher than that in SEER (12.2%). Thirty-two percent of non-WT were diagnosed in infancy (21% before 6 months) in NARECHEM-ST vs. 19% of infant cases in SEER. Most common non-WT were congenital mesoblastic nephroma (CMN; 30%), renal cell carcinomas (21%), clear cell sarcoma (CCS; 20%) and rhabdoid tumors (RT; 13%). The male-to-female (M:F) ratio was 0.9 in NARECHEM-ST vs. 1.2 in SEER. Boys outnumbered girls with CCS in both NARECHEM-ST (M:F=4.0) and SEER (M:F=2.0). Overall, the AIR was 1.6/10⁶ children in Greece, marginally higher than in SEER (1.1/10⁶). In NARECHEM-ST, the highest incidence was noted in infancy (7.7/10⁶) falling steeply in older children. A similar pattern was detected in SEER, with a significantly lower incidence among infants (2.9/10⁶, p<0.0001). This high incidence of infant non-WT was mainly attributed to CMN (NARECHEM-ST: 6.7/10⁶; SEER: 2.7/10⁶) followed by RT (NARECHEM-ST: 1.0/10⁶; SEER: 1.6/10⁶). A significantly increasing trend of +7.2% (95% confidence intervals[CI]: 5.7, 8.7) was noted for SEER and a similarly increasing trend in NARECHEM-ST (+3.9%, 95%CI: -1.2, 9.2).

Conclusions: Variations in incidence/time trends of non-WT by age, sex and subtype imply a potential effect of environmental and genetic factors on disease etiology, whereas cross-country comparisons point the differential healthcare delivery patterns.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0169 / #1695 | Renal Cell Carcinoma in Children: Single Center Experience

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Background and Aims: Renal cell carcinomas (RCC) is rare pediatric tumor diagnosed in 2-6% of all cases of malignant renal tumors. Most pediatric RCC are localized. Surgery is the primary treatment with survival more than 90%. Children with RCC with distant metastases have a poor prognosis. We aimed to demonstrate our experience of treatment of children with RCC.

Methods: We performed retrospective analysis of treatment of patients with RCC in N.N. Blokhin National Medical Research Center of Oncology during July 2009 – April 2021. Twenty three patients were enrolled to analysis. All patients underwent CT and MRI of abdomen and chest with contrasting.

Results: Median age at diagnosis was 128 (3–200) months. M:F = 9 (39.2%) : 14/23 (60.8%). Tumor symptoms were: pain (30.4%), macro-hematuria (39.2%) and abdomen enlargement (30.4%). Localization: left – n=9 (39.2%), right – n=14 (60.8%). One patient developed a metachromic tumor in the contralateral kidney (histology – RCC). Staging: n=15 (65.2%) – stage I, n=2 (8.7%) – stage II, n=5 (21.7%) – stage III (tumor thrombus) and in one case (4.4%) V stage (right and left I stage). Treatment: 1 pt. – AV neoadjuvate chemotherapy, 22 pts. – surgery. Surgery: 18/23 (81.3%) – midline laparotomy, nephroureterectomy; 5/23 (12.5%) – kidney resection. One patient in the postoperative period receives immunotherapy (up to the present time). Histological variants of RCC: n=1 (4.4%) translocation RCC (Xp 11.2), n=2 (8.7%) papillary variant, n=1 (4.4%) medullary variant, n=1 (4.4%) chromophobic variant, n=18 (75%) RCC 1 type. At the median follow-up 53 (1–179) months, 3 (13%) patients died, 1 (4.4%) patient had three relapses during immunotherapy, the rest of the patients survived without events.

Conclusions: Our experience demonstrates the importance of a comprehensive examination, in-time diagnostics and success of multidisciplinary approach. Radical surgery correlates with better prognosis.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours*

PV0170 / #1746 | Baseline Cardiac Dysfunction in Wilms Tumor Patients in Botswana

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Background and Aims: Cardiac dysfunction is a well-documented side effect of anthracycline treatment for patients with cancer. There is a paucity of data on cardiac dysfunction in children with cancer in African settings.

Methods: We conducted a retrospective review of the Botswana Pediatric Oncology Database (BPOD) to evaluate the prevalence of cardiac dysfunction among children diagnosed with selected malignancies for which anthracyclines are included in standard treatment protocols from January 2007 to December 2019 at Princess Marina Hospital. Demographic, diagnosis, cumulative doxorubicin dose, HIV status, nutritional status and cardiac function data were collected. Ejection or shortening fractions of ≤ 0.55 or < 0.28 , respectively, defined cardiac dysfunction

Results: A total of 36 patients had 41 echoes. The median age was 4.1 years (interquartile range (IQR) 2.6–8.2). Males comprised 69.4% and 16.7% of the patients were HIV positive. Wilms tumour was the most common diagnosis (64%) while 47.2% were malnourished. The median cumulative doxorubicin dose was 142.5 mg/m² (IQR 90–220 mg/m²). The mean ejection fraction (EF) in the initial echocar-

diograms was 0.62 (standard deviation, STD 0.1) and mean shortening fraction (SF) was 0.34 (STD 0.07). Four patients (21%) had cardiac dysfunction on initial echocardiogram. All these patients were males, were six-years old or younger and had Wilms tumor. Two of the Wilms patients remain alive after treatment with normal cardiac function, one demised and one was lost to follow-up. None of the post-treatment echocardiograms of the whole cohort met criteria for cardiac dysfunction with a median time from diagnosis of 2.4 years (IQR 1.2–4.4).

Conclusions: In settings where echocardiograms are unavailable, treatment with anthracyclines at less than 300 mg/m²/lifetime without routine cardiac surveillance should be considered. However, initial echocardiograms should be prioritized for children with Wilms tumor due to the higher risk for pre-treatment cardiac dysfunction demonstrated in this cohort.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours*

PV0171 / #222 | Locoregional Control Using Highly Conformal Flank Target Volumes and Volumetric-Modulated Arc Therapy in Pediatric Renal Tumors: Results from the Dutch National Cohort

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Background and Aims: In pediatric renal tumors, conventional two opposing photon beams have been used to cover the postoperative flank target volume for decades. This single center study describes the locoregional outcome using highly conformal flank target volumes adjusted for postoperative changes and intra-fraction motion combined with Volumetric-Modulated Arc Therapy (VMAT).

Methods: Between 01-2015 and 12-2019, 36/161 newly diagnosed patients with renal tumors underwent flank only irradiation (n=30) or flank+whole lung irradiation (n=6) using highly conformal target volumes in line with the SIOP-RTSG consensus statement. VMAT consisted of full-arc 10MV photon beams optimized for constraints of the organs at risk. In case of locoregional relapses, image co-registration and dose reconstruction was performed. Each relapse was classified as either 'infield' ($V95\%_{\text{relapse}} \geq 99.0\%$), 'marginal' ($V95\%_{\text{relapse}}: 20.0\% - 98.9\%$) or 'outfield' ($V95\%_{\text{relapse}}: 0\% - 19.9\%$).

Results: At a median follow-up from diagnosis of 3.1 years (range: 0.4–5.7), the estimated 2-year Locoregional Control Rate, Disease-Free Interval and Overall Survival were 94%, 91% and 94%, respectively.

Locoregional relapse was observed in two patients. One patient had a combined tumor bed and regional recurrence, classified as infield ($V95\%_{\text{relapse}}: 100\%$) and outfield ($V95\%_{\text{relapse}}: 1.2\%$). The second patient had a regional relapse in the inferior vena cava classified as marginal recurrence ($V95\%_{\text{relapse}}: 93\%$). Relapses would not have been adequately covered by conventional beams.

Conclusions: This single center analysis provides encouraging evidence that excellent locoregional control can be obtained by using highly conformal flank target volumes with VMAT in pediatric renal tumors. The safety of this approach will be validated in a prospective multicenter study.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours*

PV0172 / #278 | Optimizing Resources in Relapsed Wilms Tumour (WT) While Improving Outcomes

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Background and Aims: While overall outcomes in relapsed WT have improved, salvage in high(HR) and very high risk(VHR) relapses remains a challenge.

Methods: Children(<18 years) treated for relapsed WT at our paediatric oncology unit from June 2012-December 2019 were analyzed. Relapses were stratified into Standard risk (SR: favorable histology(FH) WT relapsed after only surgery and/or vincristine/dactinomycin), HR: FH WT initially treated with >three drugs, and VHR: recurrent anaplastic or blastemal-predominant WT. Chemotherapy included carboplatin/doxorubicin/cyclophosphamide/etoposide in SR, and carboplatin/etoposide cyclophosphamide/etoposide in HR and selected VHR. Local treatment with surgery/radiotherapy was per stage. Children with suboptimal staging, or not treated according to standard guidelines during firstline treatment were termed to have 'inadequate treatment'. Event-free survival(EFS) and overall survival(OS) was calculated by Kaplan-Meier Analysis and univariate/ multivariate analysis of prognostic factors by cox- proportional-Hazard model using IBM SPSSTM v 24.0.

Results: Of 67 children(42 boys) with relapsed WT-median age 49months(IQR 37-79), 49 presented to our centre at relapse. First-line treatment was inadequate in 16(24%) children. Median time from initial diagnosis to relapse(TTR) was 11 months(IQR 7-17), and 43(64%) were symptomatic. Locoregional -24 (36%) and lung -20(30%) were common sites of relapse;30(45%) relapses were SR, 29 (44%) HR, and 8(11%) VHR. Fifty children were treated with curative intent,15(including 7/8 VHR) palliated and 3 defaulted.

Of 42 children who achieved a second remission, 21 had a second relapse, but none could be salvaged. Two-year EFS and OS post curative treatment were 48.6%(95%CI 35-61.8) and 56.6%(95%CI 43.3-70.5), in SR relapses -52.6% and 64.1% and HR relapses-41.5% and 47.3%. Factors predictive of inferior EFS on multivariate analysis were: adequate firstline treatment(HR:2.6), TTR <2 years (HR:5), and of superior EFS: locoregional relapse(HR:0.4), male gender(HR:0.3), age >4 years at initial diagnosis(HR:0.3). Of the 21 long-term survivors,5 have renal and 2 cardiac late toxicities.

Conclusions: Careful selection of patients (late relapses, inadequate frontline treatment) can salvage half of children with relapsed WT. Salvage in VHR relapses remains a challenge. Long-term survivors need careful monitoring for cardiac and renal dysfunction.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours*

PV0173 / #324 | Pediatric Small Renal Masses: Can Tumor Size Predict Histology and The Potential for Nephron-Sparing Surgery?

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Background and Aims: The majority of pediatric renal masses suspicious for malignancy are treated with radical nephrectomy (RN). Nephron sparing surgery (NSS) is reserved for patients with bilateral Wilms tumor (WT), predisposition syndromes, solitary kidney, or select non-WT histologies. We hypothesized that small renal masses (SRM) are more likely to have benign or non-WT histology, and therefore candidates for NSS.

Methods: A single-institution, retrospective review was performed on patients aged 0-18yrs who underwent removal of a unilateral renal mass from 2005-2019. Neoadjuvant chemotherapy and unknown final histology were exclusion criteria. A 4cm cut-point was used to define a SRM. Analysis was performed to describe the predictive characteristics of preoperative tumor size (<4cm vs ≥4cm) on final histology. Histologies deemed appropriate for RN were WT, clear cell sarcoma of the kidney, and rhabdoid tumor of kidney, while those deemed appropriate for NSS were renal cell carcinoma or benign histology. We also analyzed whether preoperative tumor size on imaging correlated with tumor size on final pathology.

Results: 134 patients met inclusion criteria. 97 (72.3%) tumors were WT. There were 12 tumors <4 cm. Tumors ≥4cm compared to tumors <4cm had a significantly higher likelihood of being WT vs non-WT (OR 17.54, 95% CI 3.53, 87.22), of being malignant vs benign (OR 4.73, 95% CI 1.34, 16.72), and of having RN-appropriate histology vs NSS-appropriate histology (OR 30.42, 95%

CI 5.92, 156.3). There was a strong positive Spearman correlation of 0.87 ($p < 0.0001$) between preoperative and postoperative tumor size.

Conclusions: The probability that a pediatric renal mass is WT increases with tumor size. 4cm is a logical cut-point that can be used to define a pediatric SRM. This cut-point may be able to predict the probability of WT and determine patient eligibility for NSS with intraoperative FS.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0174 / #372 | Anaplastic Wilms' Tumour Stage I - How Much Treatment is Needed? A Report from the SIOP WT 2001 GPOH and UK-CCLG Wilms Tumour Studies

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Background and Aims: Anaplasia is an unfavourable histological feature in Wilms' tumour (WT) associated with poor outcomes. However, historically, patients with stage I anaplastic WT (AWT) had good outcomes, albeit with more treatment than for stage I non-anaplastic WT. Since the SIOP-WT-2001 study, patients with focal anaplasia WT (FAWT) have been classified as intermediate-risk and received less intense treatment than patients with diffuse anaplasia WT (DAWT). We present the results of SIOP-WT-2001 treatment of patients with stage I FAWT and DAWT.

Methods: Retrospective analysis of clinico-pathological features and outcomes of 59 patients with stage I AWT (19 FAWT, 40 DAWT) from the SIOP-WT-2001 GPOH and UK-CCLG groups. The FAWT subgroup were treated as intermediate-risk WT, with eight weeks of actinomycin and vincristine (AV)(four weeks preoperatively, and four weeks postoperatively). For comparison, we also assessed outcomes in 818 patients with stage I intermediate-risk non-anaplastic WT (IR-non-AWT). The DAWT subgroup were treated for 31 weeks (four weeks pre-operatively with AV, and 27 weeks postoperatively with AV + doxorubicin). No subgroup received radiotherapy.

Results: Median follow-up was 67.6 months. For 19 patients with FAWT, the 4-year EFS and OS estimates were 87% (95%CI 72%-100%) and 100%, respectively. For 40 patients with DAWT, the 4-year EFS and OS estimates were 85% (95%CI 74%-98%) and 93% (95%CI 85%-

100%), respectively. The 4-year EFS and OS in the IR-non-AWT group (818 patients) were 91% (95%CI 89%-93%) and 98% (95% CI 97%-99%), respectively.

Conclusions: We showed that the SIOP approach of classifying and treating FAWT as intermediate-risk tumour is justified since the outcomes for patients with stage I FAWT were comparable with those of other patients with stage I IR-non-AWT, who received the same treatment. Patients with stage I DAWT also showed good outcomes, although with more intensive chemotherapy than IR-non-AWT, but, importantly, without radiotherapy.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0175 / #388 | Outcomes of Children with Wilms Tumor Treated in a Large Referral Center in Western Kenya

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Background and Aims: Wilms tumor (WT) is a curable type of cancer with 5-year survival rates of over 90% in high-income countries, whereas this is less than 50% in resource-limited countries. We assessed outcomes of children with WT treated at a Kenyan teaching and referral hospital.

Methods: We conducted a retrospective medical record review of all children diagnosed with WT between 2013-2016. Treatment protocol consisted of six weeks of pre-operative chemotherapy, surgery, and four to eighteen weeks of post-operative chemotherapy depending on disease stage. Event-free survival (EFS) was estimated using Kaplan-Meier method. A competing risk analysis with cumulative incidence functions was performed. Differences between groups were analyzed with Gray's test and Fine-Gray analysis was performed to calculate subdistribution hazards. Two-sided p-values of ≤ 0.05 were considered statistically significant.

Results: Ninety-two patients were treated between 2013-2016, of which 69% presented with high stage disease. Two-year probability of EFS (pEFS) was 44%. Twenty-seven percent of children died, 19% abandoned treatment and 11% suffered from progressive or relapsed disease. Patients who were treated in 2015-2016 compared to 2013-2014 showed higher pEFS ($p < 0.001$). Furthermore, the subdistribution hazards of abandonment and progressive or relapsed disease were respectively 0.42 (95% confidence interval (CI) 0.15-1.18, $p = 0.10$) and 0.11 (95% CI 0.02-0.86, $p = 0.04$) when comparing 2015-2016 to 2013-2014. The cumulative probabilities of outcomes in 2013-2016 were

not dependent on disease stage, insurance status, distance to hospital, or duration of symptoms. Twenty-nine children received radiotherapy in addition to chemotherapy and surgery. Two-year pEFS in this group was 86%.

Conclusions: Outcomes of children with WT treated at a Kenyan teaching and referral hospital improved over the years despite advanced stage at presentation. Survival probabilities of patients who received complete therapy including radiation are approaching those of patients in high-income countries. Additional improvement could be achieved by decreasing abandonment rates and working on strategies for earlier diagnosis.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0176 / #426 | MRI-Characteristics of Pediatric Renal Tumors: A SIOP-RTSG Radiology Panel Delphi Study

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Background and Aims: The International Society of Pediatric Oncology-Renal Tumor Study Group (SIOP-RTSG) does not advocate invasive procedures to determine histology before start of therapy. This may induce misdiagnosis-based treatment initiation, but only for a relatively small percentage of non-Wilms tumors (non-WTs). The aim of the current study is to identify MRI-characteristics that may be used for discrimination of newly diagnosed pediatric renal tumors.

Methods: A two-round Delphi procedure, consisting of online questionnaires, was initiated by the SIOP-RTSG radiology panel. Participants had ≥ 5 years of experience in MRI of pediatric renal tumors and/or assessed ≥ 50 MRI-scans of pediatric renal tumors in the past 5 years. The cut-off value for consensus and agreement among the majority was $\geq 75\%$ and $\geq 60\%$, respectively.

Results: The first round was completed by 23 participants, the second round by 20 participants. Consensus or an agreement among the majority on specific characteristics mainly concerned the discrimination between WTs and non-WTs, and WTs and nephrogenic rest(s) (NR)/nephroblastomatosis. The presence of bilateral lesions (75.0%) and NR/nephroblastomatosis (65.0%) are MRI-characteristics indicated as specific for the diagnosis of a WT, and 91.3% of the participants agree MRI is useful to distinguish NR/nephroblastomatosis from WT. Furthermore, all participants agree that age influences their prediction in the discrimination of pediatric renal tumors.

Conclusions: Although the discrimination of pediatric renal tumors based on MRI remains challenging, this study identified some specific characteristics for tumor subtypes, based on the shared opinion of carefully selected experts. These results may guide future validation studies and innovative efforts.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0177 / #468 | NRP1 Knockdown Inhibits The Invasion of Rhabdoid Tumor of the Kidney Cells

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Background and Aims: Rhabdoid tumor of the kidney (RTK) is a rare, highly aggressive malignant tumor, accounting for 2% of all renal tumors in childhood. The survival of patients with RTK is generally poor. To improve their prognosis, novel therapeutic targets are

strongly demanded. We therefore performed an integrated analysis of mRNA-seq and miRNA-seq to detect candidate oncogenes of RTK and evaluated their roles in RTK in vitro.

Methods: An integrated analysis of mRNA-seq and miRNA-seq was performed to determine the expression profile of exosome-derived miRNAs and mRNAs in human RTK-derived cell lines (WT-CLS1, G401) as well as the human embryonic renal cell line HEK293T, which served as a control. A gene ontology (GO) enrichment analysis was performed to analyze the functional characteristics of differentially expressed mRNAs in RTK cells. Quantitative reverse transcription-polymerase chain reaction (qRT-PCR) was performed to validate the expression of candidate mRNAs. A Matrigel invasion assay was performed to evaluate the cell invasive ability.

Results: An integrated analysis of mRNA-seq and miRNA-seq revealed 40 mRNAs that were highly expressed in RTK cells targeted by exosomal miRNAs, the expression of which was lower in RTK cells than in HEK293T. A GO enrichment analysis showed that these highly expressed mRNAs were mainly related to cell adhesion. Of these mRNAs related to cell adhesion, we selected Neuropilin 1 (NRP1) as a candidate oncogene, since its elevated expression is associated with a poor prognosis in many types of tumor. qRT-PCR showed that the expression of NRP1 in RTK cells were significantly higher than that in controls. A Matrigel invasion assay showed that RTK cells in which NRP1 had been knocked down exhibited a decreased invasion ability.

Conclusions: The present study indicated that NRP1 acts as an oncogene by promoting the invasion of RTK and can serve as a therapeutic target.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours*

PV0178 / #617 | Impact of Patient Navigation on Treatment Delay Among Patients with Wilms Tumor in Rural Rwanda: Preliminary Findings at One Year of Intervention

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Background and Aims: Wilms tumor (WT) is highly curable. In Rwanda, each treatment modality (chemotherapy, surgery, and radiotherapy) is offered at a different health facility, contributing to various care delays. With support from Sanofi Espoir, a new patient navigation (PN) program for patients with WT was established at Butaro Hospital. We report on its impact on treatment delays and acceptability among caregivers.

Methods: Newly diagnosed patients with WT were enrolled prospectively in PN from March 2020. Clinical data were abstracted from med-

ical files. Demographic and disease characteristics and timeliness of treatments were compared between patients in PN and a historical control. We assessed program acceptability through surveys with caregivers at the end of treatment.

Results: A total of 39 and 34 patients were included in the PN and control cohorts, respectively. There were no differences in sociodemographic or baseline clinical characteristics between the two groups, except more patients presented with metastatic disease in the PN group than the control (73.7% vs 32.4%, $p=0.001$). Treatment was more prompt among patients in PN; specifically, the median time between end of preoperative and start of postoperative chemotherapy was reduced by 8 days (14 days (IQR:13-15) in PN group vs 22 days (IQR: 17.5-27.5) in the control, $p<0.001$). Of the eighteen caregivers surveyed, most found the navigator role helpful in understanding the disease and treatment, linking them to different providers, reminding them of appointments, and providing emotional support. All caregivers supported the continuation of the PN and suggested it be expanded to all pediatric patients.

Conclusions: Results from the PN program are encouraging. There was an increase in timeliness of treatment during the navigation period and caregivers appreciated the program for both logistical and emotional support. Further analysis will be done to evaluate longer-term impact of PN and how to expand services to include other cancers.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours*

PV0179 / #703 | Updated Risk Stratification in Children with Stage I Favorable Histology Wilms Tumors. A Report from The Children's Oncology Group AREN0532 and AREN03B2 Studies

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Background and Aims: Patient age (<2 vs 2+ years) and tumor-nephrectomy weight (TNW; < 550g vs 550g+) were used to risk stratify children with Stage I Favorable Histology Wilms Tumor (FHWT) in AREN0532. To consider risk stratification for future studies, children from AREN0532 were pooled with like children enrolled on biology study AREN03B2. Maximum tumor diameter (TD) was also assessed as a potential new prognostic factor.

Methods: Children with confirmed Stage 1 FHWT by central review and available data were selected. Univariate Cox models stratified by treatment received isolated the continuous effect of each factor on event-free survival (EFS). Overall significance tests and tests for non-linearity of each factor's association with EFS were performed. Cox models investigated the impact of treatment on the risk of EFS as a continuous function of each factor. Children without abdominal lymph node sampling were excluded.

Results: A total of 658 Stage I FHWT children were included (135 AREN0532, 523 AREN03B2 alone). Median age (range) was 23 months (0 to 16 years), TNW was 429g (32-1716g), and TD was 10cm (1-23cm). Treatment was 72% EE4A, 4% DD4A, and 24% observation. Risk of EFS did not vary by age ($p=0.96$) or TNW ($p=0.17$), but larger TD was associated with poorer EFS ($p = 0.04$). None of the factor effects were significantly non-linear. The outcomes for children with older age, heavier TNW, and larger TD treated with observation only vs. chemo experienced trended toward increased risk that were not significant.

Conclusions: Among Stage I FHWT children enrolled on AREN03B2/532, neither age nor TNW was associated with EFS. Larger TD may be associated with poorer EFS and warrants further study. These findings support broadening Stage I FHWT "observation only" enrollment to include older children or those with any TNW.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0180 / #786 | The Status of Active Vitamin D (1,25(OH)₂D₃) Production in Wilms Tumor Survivors

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Background and Aims: The active vitamin D (1,25(OH)₂D₃) is produced in the kidneys. It is known that 1,25(OH)₂D₃ conversion might be impaired in living kidney donors, and in early-stage renal failure patients with normal creatinine. Here, we aimed to investigate the

status of the 1,25(OH)₂D₃ production, and bone health parameters in Wilms tumor (WT) survivors.

Methods: Our single-center prospective study included WT survivors (time-off therapy ≥ 6 months), congenital solitary kidney (SK) patients, and healthy controls. Age, gender, body weight, height, body mass index (BMI), and arterial blood pressure measurement results were recorded. Urea, creatinine, calcium, phosphorus, alkaline phosphatase, glomerular filtration rate (GFR, calculated by Schwartz formula) 25(OH)D₃, parathormone, 1,25(OH)₂D₃, and urinalysis test results were evaluated. Vitamin D deficiency was defined as a serum 25(OH)D₃ level less than 20 ng/ml. 1,25(OH)₂D₃ levels were interpreted according to age-based reference values.

Results: Twenty-two WT patients, 27 SK patients, and 25 healthy children were included. The median age (42 boys, 32 girls) was 9.37 (range: 1.33-18.58) years. Age and gender distribution were comparable among groups. The mean GFR levels were found to be lower in both the WT group and SK group in comparison to that of the healthy controls. At a median follow-up of 7.91 years, none of the WT patients had GFR levels <90 ml/min/1.73 m². The WT group had the highest frequency of Vitamin D deficiency (77.3%) and had the highest 1,25(OH)₂D₃ level and 1,25(OH)₂D₃/25(OH)D₃ ratio. Hypocalcemia and elevated ALP levels were more common in Vitamin D deficient patients.

Conclusions: WT patients have the highest prevalence of Vitamin D deficiency, without a 1,25(OH)₂D₃ conversion defect. This result may be due to a relatively shorter follow-up duration and the absence of a significant decrease in GFR. Our study indicates the importance of follow-up in terms of vitamin D support as well as nephrological long-term side effects of WT patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0181 / #871 | Clinical and Molecular Characteristics of Cystic Nephroma in Children

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Background and Aims: Cystic nephroma (CN) is a rare kidney tumor arising during childhood and associated with hereditary DICER1 syndrome. The aim of the study was to analyze the clinical and molecular characteristics of CN in children.

Methods: The study included 8 patients with histologically verified CN. All patients were consulted or received treatment in Dmitry Rogachev NMRC PHOI for the period of 2012-2020 (9 years). We have analyzed clinical, radiological, molecular characteristics and results of CN treatment.

Results: The median age was 14.2 months (8.6-197.0). Male:female ratio was 7:1. Symptoms included an increase in abdominal circumference and palpable abdominal mass in 6 (75%) cases, in 1 (12.5%) - arterial hypertension and in 1 (12.5%) case a tumor was found by routine abdomen US. CT showed multicystic lesions with thin septa that accumulated a contrast. Tumor volume ranged from 59.7 ml to 1293.1 ml (median 626.3 ml). The left kidney was involved in 5 (62.5%) and the right kidney in 3 (37.5%). 5 (62.5%) patients received neoadjuvant chemotherapy per the decision of the local physicians but no tumor regression was observed in any case. 3 (37.5%) patients underwent an initial surgery. Nephrectomy was done in 6 (75%) cases and nephron-sparing surgery in 2 (25%). Mutation analysis was done in 6 patients and showed DICER1 somatic and germline mutations in 5/6 (83.3%) cases. One family had carriers with DICER1 germline mutation. The median follow-up was 17.6 months (1.7-58.9). All patients are alive with no evidence of disease.

Conclusions: CN must be included in a differential diagnosis of cystic renal masses. The presence of cystic kidney tumors is an indication for initial surgery. Patients with histologically confirmed CN are required consultation of medical genetics and molecular analysis of DICER1 status with subsequent surveillance to earlier detection of other metachronous DICER1-related tumors.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0182 / #894 | Renal Failure Associated with Wilms Tumor Treatment in Wagr Patients – An Analysis of The International WAGR Syndrome Association Registry (IWSA)

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Background and Aims: Children with WAGR Syndrome are predisposed to intrinsic renal disease and Wilms Tumor (WT). Detailed analyses of WT treatment on the development of renal disease is lacking. The purpose of this study is to compare renal function in children with WAGR who developed WT to those who did not.

Methods: Self-reported data from parents of children with WAGR was analyzed. Children with WAGR without WT were compared with children with WAGR and WT. Using standard definitions for Stage 1-5 renal failure, categorical comparisons were made with chi-squared test and continuous comparisons were made using t-test and ANOVA. A multivariable logistic regression was conducted to identify predictors of renal failure.

Results: Of 145 children with WAGR, 64 developed WT. Mean age at follow up was 7.5(IQR 3-15) years for children with WAGR and WT and 11(5-22) years for children without WT. Despite being a younger cohort ($p=0.017$), children with WAGR and WT had a greater association with renal failure (any stage) than WAGR children without WT 34/64(53.1%) compared with 5/81(6.17%, $p<0.0001$). Of the thirty-seven children who were on at least one hypertensive medicine, 33 had WT; 5 did not ($p<0.001$). Two children with WT required hemodialysis, one of them subsequently received a kidney transplantation. By univariate analysis renal failure at any stage requiring medical management was associated with total nephrectomy as initial treatment for the WT (30/37, 81.1%) compared with nephron sparing surgery (NSS) (3/23, 13.0%). Other significant factors were chemotherapy duration greater than 12 months 7/8 (87.5%) and greater than two-drug therapy. On multivariate analysis, nephrectomy was the only significant factor associated with renal failure ($p=0.0002$).

Conclusions: Although children with WAGR are predisposed to develop intrinsic renal disease, renal failure is higher in children with WAGR and WT than in those without WT. When possible, NSS surgery should be encouraged.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0183 / #907 | Identification of A HUB lncRNA AC016723.4 In Wilms Tumor –Based On Two lncRNA Signature Models

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Background and Aims: To identify the prognostic hub lncRNA AC016723.4 based on two lncRNA signature models in Wilms tumor (WT).

Methods: The data including RNA expression and clinical information of pediatric WT patients were downloaded from the Therapeutically Available Research to Generate Effective Treatments (TARGET) database. The differentially expressed genes by metastasis status or

pathological type were respectively identified and 2 prognostic signature of pediatric WT was generated according to univariate and multivariate Cox analysis. Receiver operating characteristic (ROC) curve and area under curve (AUC) was used to evaluate the accuracy. The common lncRNA in both models was identified.

Results: 38 significantly different lncRNAs were found in the comparison of stage 4 patients and non-stage 4 patients, consisting 33 up-regulated genes and 5 down-regulated genes. Then three key prognostic lncRNAs were identified by univariate and multivariate Cox regression (RP11-138H8.6, AC016723.4, AC007879.2). We divided the patients into high risk and low risk groups according to the expression of the three genes, there was significant difference in survival rate between the groups ($p=5.121 \times 10^{-4}$). With the model constructed by the 3 lncRNAs, AUCs were 0.729 in 5-year survival rate and 0.72 in 10-year survival rate. The same methods were used in grouping with pathological classification, a total of 285 different lncRNAs were found in the comparison of favor histology patients and diffuse anaplasia patients, consisting 215 up-regulated genes and 70 down-regulated genes. A total of 10 lncRNAs were found to construct risk groups after univariate and multivariate Cox regression (AF127577.8, RP11-515O17.2, AC009501.4, RP11-693J15.4, AC010907.5, RP13-436F16.1, AC016723.4, AC016723.4, RP11-242J7.1, RP11-242J7.1). AUCs of the model were 0.862 in 5-year survival rate and 0.835 in 10-year survival rate. Finally, the lncRNA AC016723.4 was identified as the hub prognostic lncRNA.

Conclusions: The lncRNA signature can predict the prognosis of patients with WT. AC016723.4 may play an important role in the carcinogenesis.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0184 / #911 | Stage II Intermediate Risk Wilms Tumors: Do They All Need The Same Treatment?

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Background and Aims: Wilms tumour (WT) histopathological type and stage are critical for patients' treatment. According to the current

SIOP protocol, stage II intermediate risk WT (IR-WT) are treated post-operatively with actinomycin D and vincristine for 27 weeks (AV2) with excellent outcomes. The aim of the study is to investigate the relative prevalence of reasons/criteria for stage II as a first step to assessing if some patients could be considered for reduced therapy, similar to stage I.

Methods: A retrospective study of patients with unilateral IR-WT stage II from the SIOP-WT-2001-UK and IMPORT cohorts (2001- 2020). Patients were treated with 4 weeks AV preoperative chemotherapy followed by surgery. Stage II tumours were divided into pattern groups: A. perirenal fat infiltration, B. soft tissues of the renal sinus infiltration, C. lymphatic/blood vessels of renal sinus infiltration, D. renal pelvis wall/ureter wall infiltration, E. infiltration of the vena cava/adjacent organs.

Results: 168 patients with IR-WT were included (median age 36.5 months, range 2 to 199). 15/168 (8.9%) patients relapsed, and 6/168 (3.6%) died. In 106 cases (63.1%) there was only one criterion for stage II, in 53 cases (31.5%) there were two, and in 9 cases (5.4%) three criteria present. Pattern A was observed in 43 (25.6%), pattern B in 121 (72%), pattern C in 65 (38.7%), pattern D in 8 (4.8%), and pattern E in 3 cases (1.8%). Because of high survival and few events in IR-WT, there were no significant survival differences between the five sub-groups of stage II.

Conclusions: Expansion to larger numbers of stage II anatomical sub-groups and inclusion of further features (molecular profiles, blastemal volume) with comparison with Stage I IR-WT group may allow better understanding of prognostic staging criteria in order to propose cohorts for reduced post-operative treatment.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PV0185 / #912 | Identification of A Three-mRNA Signature as a Novel Potential Prognostic Biomarker in Pediatric Wilms Tumor

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Background and Aims: The prognosis of Wilms tumor (WT) is significantly related to its clinical stage. The aim of this study was to generate a prognostic model based on clinical stage and to predict survival outcome in pediatric WT.

Methods: The data including mRNA expression and clinical information of pediatric WT patients were downloaded from the Therapeutically Available Research to Generate Effective Treatments (TARGET) database. The differentially expressed genes were identified between 4 and non-4 stage, a prognostic signature of pediatric WT was generated according to the results of univariate and multivariate Cox analysis. Receiver operating characteristic (ROC) curve

was used to evaluate the three-mRNA signature in pediatric WT patients.

Results: We identified 167 differentially expressed mRNAs between 4 and non-4 stage, including 25 downregulated mRNAs and 142 up-regulated mRNAs. Univariate and multivariate Cox analysis identified 3 mRNAs (BMP3, APOBEC3H, C1QL1) to establish a predictive model. The risk score formula is as follows: Risk score = $(-0.1053) \times \text{BMP3} + 0.107128 \times \text{APOBEC3H} + (-0.1446) \times \text{C1QL1}$. The pediatric WT patients were divided into low-risk group and high-risk group based on the median risk score. The receiver operating characteristic (ROC) curve analysis revealed good performance of the 3-mRNA prognostic model (the area under the curve [AUC] was 0.693 in 5-year survival and 0.721 in 10-year survival).

Conclusions: The three-mRNA signature can predict the prognosis of patients with pediatric WT. It makes sense in the understanding of targets for pediatric WT patients. However, Further clinical and molecular verification is needed.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0186 / #915 | Identification of A Six-MRNA Signature as a Novel Potential Prognostic Biomarker in Pediatric Wilms Tumor

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Background and Aims: The prognosis of Wilms tumor (WT) is significantly related to its pathology subtypes. The aim of this study was to generate a prognostic model based on pathology subtypes and to predict survival outcome in pediatric WT.

Methods: The data including mRNA expression and clinical information of pediatric WT patients were downloaded from the Therapeutically Available Research to Generate Effective Treatments (TARGET) database. The differentially expressed genes were identified between diffuse anaplasia patients and favor histology patients, a prognostic signature of pediatric WT was generated according to the results of univariate and multivariate Cox analysis. Receiver operating characteristic (ROC) curve was used to evaluate the six-mRNA signature in pediatric WT patients.

Results: We identified 843 differentially expressed mRNAs between diffuse anaplasia patients and favor histology patients, including 225 downregulated mRNAs and 618 up-regulated mRNAs. Univariate and multivariate Cox analysis identified 6 mRNAs (MAGEC2, RERGL, L1TD1, SPAG11A, C10orf71, C10orf71) to establish a predictive model. The risk score formula is as follows: Risk score = $(0.200472) \times \text{MAGEC2} + (-0.10455) \times \text{RERGL} + (0.117739) \times \text{L1TD1} + (-0.28229) \times \text{SPAG11A} + (-0.19131) \times \text{C10orf71} + (0.1322) \times \text{KRTAP17-1}$. The pediatric WT patients were divided into low-risk group and high-risk group

based on the median risk score. The receiver operating characteristic (ROC) curve analysis revealed good performance of the 6-mRNA prognostic model (the area under the curve [AUC] was 0.804 in 5-year survival and 0.824 in 10-year survival).

Conclusions: The six-mRNA signature can predict the prognosis of patients with pediatric WT. It makes sense in the understanding of targets for pediatric WT patients. However, further clinical and molecular verification is needed.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0187 / #943 | Prognostic Significance of Post-Chemotherapy Blastemal-Type Histology in Children with Wilms Tumor (WT)

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Background and Aims: Post-chemotherapy BT-WT is a high risk histological subgroup in WT which benefits from intensified treatment. We report the presentation and outcomes of children with BT-WT treated at our centre.

Methods: All children with BT-WT treated at our centre from January 2013 - December 2019 were analyzed. While resectable tumours were operated upfront, unresectable tumours received preoperative chemotherapy (Vincristine-Dactinomycin(VA)x4 weeks in non-metastatic and VA+Doxorubicin x 6 weeks in metastatic tumors) followed by delayed nephrectomy. Children with anaplastic histology and post-chemotherapy BT-WT were termed High-Risk (HR), and received Regimen I (Vincristine, doxorubicin, cyclophosphamide, etoposide x 24 weeks) with local radiation and local treatment of metastatic sites. Presentation and outcomes (Event free survival (EFS) and overall survival (OS)) were analyzed by histology.

Results: Of 203 children treated during this period, 27 (13.3%) had HR histology. 22 had BT-WT, 1 diffuse anaplasia (DA) and 4 a combination of BT-WT and DA, 18 were male. Median age at diagnosis was 48 months (range 12 months-18 years) in BT-WT and 40 months (3 months-16.5 years) in non-HR patients. The number of patients in stage I, II, III, IV and V were 3 (11.1%), 2 (7.4%), 9 (33.3%), 11 (40.7%) and 2 (7.4%), compared to 29 (13.6%), 35 (20.6%), 59 (33.5%), 46 (26%) and 10 (5.7%). At a median follow-up of 24 months, 12 are in remission, 14 relapsed (1 salvaged, 10 died). One child with Fanconi anaemia had a second malignancy-medulloblastoma and subsequently died. The 2-year EFS and OS of BT-WT were 45.5% (95% CI 31.9-66.7) and 63.8% (95% CI 38-72), lower than 75.6% (95% CI 72-118) and 82% (95% CI 133-154) in non-HR WT. By stage, 2-year EFS, OS in BT-WT were 75%, 75% (stage 1 and 2), 31%, 52% (stage 3) and 18.2%, 47% (stage 4). All 5 children with stage 1/2 BT-WT are in remission at median 46 months (32-70 months).

Conclusions: Children with BT-WT present at a higher median age and stage. While overall outcomes are far inferior to non-HR WT, lower stage BT-WT can be treated successfully with intensified regimens to achieve long-term survival.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.b Renal Tumours

PV0188 / #981 | Combination Chemotherapy and CIS-Retinoic Acid (cis-RA) for Treatment of Children with Bilateral Renal Lesions: Report of Feasibility of Multiple Chemotherapy Regimens and Prolonged cis-RA Exposure

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Background and Aims: Pediatric bilateral renal lesions present diagnostic and therapeutic challenges. Discrimination of Wilms tumor (WT) from nephrogenic rests (NR) radiographically is challenging. Response of NR to WT therapy is often minimal and renal-sparing surgery often not feasible. Previous reports describe response of nephroblastomatoses to cis-retinoic acid (cis-RA), and safety of prolonged cis-RA is established in neuroblastoma. We report feasibility of combining cis-RA with two chemotherapy regimens in children with no option for resection which would preserve functioning renal parenchyma.

Methods: Three children with extensive bilateral lesions and a fourth post-unilateral nephrectomy with extensive residual contralateral disease were treated at 3 institutions from 2016-2021. Surgical resection of renal lesions would have rendered them anephric. Hence, concurrent chemotherapy (21-day cycles) and cis-RA (days 1-14 of 28-day cycles, 160mg/m²/day) was utilized. Timing of chemotherapy and cis-RA cycles was independent. All consented to treatment. Demographic and treatment data were abstracted from clinical records.

Results: Age at initial presentation was 3-23 months. Cis-RA was added to vincristine/actinomycin (VA) prior to surgery for 2 children and to vincristine/irinotecan/temozolomide (VIT) for 2 children with persistent renal lesions after initial multimodal therapy. Children completed

3-36 cycles of cis-RA; 3-8 with chemotherapy, 3-28 as monotherapy. No unexpected adverse events or treatment delays occurred. All are alive 1-12 years after presentation; one heavily pre-treated child has grade 1 chronic kidney disease. Three remain on cis-RA monotherapy; one is off-therapy. One required partial nephrectomy after combined therapy (VA + cis-RA) for stage I favorable histology WT. The remaining 3 have not required further intervention for concern of malignant transformation following treatment with cis-RA; none developed metastases.

Conclusions: Combining cis-RA with VA and VIT chemotherapy for prolonged duration was feasible and well tolerated in this small series. Outcomes were excellent, but the additive benefit of cis-RA is unknown. A larger series is needed to establish efficacy.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.c Bone Tumours

PV0189 / #1285 | Ewing Sarcoma: Prognosis Value of the Histological Response to Chemotherapy in Real Life

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Background and Aims: Ewing sarcoma is an aggressive bone cancer. It represents 2% of pediatric cancers and the second most frequent bone sarcoma. The prognosis of this neoplasm has improved since the introduction of cytotoxic treatment regimens. The aim of this study is to assess the prognosis value of the histological response after induction chemotherapy and the predictive factors influencing this response in real life.

Methods: A mono-centric retrospective study was conducted over a period of 19 years between 2000 and 2018 including only operated patients after induction chemotherapy at the pediatric oncology unit of Salah Azaiz Institute. Patients were treated according to the EWING-99 protocol. Only metastatic patients and bad responders had radiation treatment. 10% is the threshold used to assess the histological response.

Results: Forty nine patients (37 male, 12 female) were included. The mean age was 12 years (3-18). There were 36 patients with peripherally localized tumors and 13 with axial tumors. The average size was 11cm. Twelve patients had metastatic disease at presentation (11 lung and 1 bone). Twenty three patients (46.9%) were classified as good responders (GR) to chemotherapy. Eighteen had localized disease (48.6%) and 19 with a peripherally localized tumor (52.7%). For non-metastatic patients, overall survival (OS) at 2 years was 81.7% for GR and 62.4% for bad responders (BR) (p=0.000). Disease free survival (DFS) at 2 years was 57.8% for GR and 38% for BR (p=0.004). Hematological toxicity and the location of the primary tumor were found to be prognosis factors influencing the response to chemotherapy (respectively p=0.005, p=0.039). The multivariate analysis showed that size

and metastatic status are prognosis factors for histological response ($p=0.048$, $p=0.007$).

Conclusions: A good histological response to induction chemotherapy is a prognostic factor influencing disease free survival and overall survival. These results highlight the importance of optimizing the induction chemotherapy to attain a good response.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0190 / #1379 | Survival Outcomes in Paediatric Patients with Osteosarcoma and Ewing Sarcoma of the Bone: A West of Scotland Review

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Background and Aims: Osteosarcoma and Ewing sarcoma are the commonest malignant bone tumours occurring in children. Overall survival is dependent on various factors, including presence of metastatic disease. We sought to determine survival outcomes for our paediatric patients treated for osteosarcoma or Ewing sarcoma of bone.

Methods: This is a single centre, retrospective study (2001-2016) of paediatric patients diagnosed with osteosarcoma or Ewing sarcoma of the bone. Electronic records and databases were used to identify patients and collate information.

Results: Fifty-four paediatric patients were included (34 Ewing sarcoma and 20 osteosarcoma), 29 male and 25 female. Median age of diagnosis was 13.3 years (range 0.87 years to 16.1 years). 32/54 patients had localised disease. Twenty two (41%) of the patients had metastatic disease at presentation (sixteen (47%) of the patients with Ewing sarcoma and 6 (30%) of the patients with osteosarcoma). The commonest site of metastatic disease in both groups was lung. Seventeen (30%) of paediatric patients with bone tumours relapsed, with a median time to relapse of 1.1 years (range 0.2 – 8.3 years). 32% of patients with Ewing sarcoma relapsed and 30% of patients with osteosarcoma relapsed. Overall survival for all patients was 72% (71% in patients with Ewing sarcoma and 75% in patients with osteosarcoma). Overall survival in patients with metastatic disease was 32% (6/16 – 38% in patients with Ewing sarcoma and 1/6 – 17% in patients with osteosarcoma).

Conclusions: We have reported the results of a single centre retrospective review of paediatric patients with bone tumours. Relapsed rates were similar for patients with Ewing sarcoma and osteosarcoma. The commonest site of metastatic disease was lung. Overall survival exceeds 70%. As expected, overall survival was much lower in patients with metastatic disease, particularly with osteosarcoma.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0191 / #1409 | The Role and Optimal Timing of Local Control Measures in High-Risk Ewing Sarcoma Patients Receiving High-Dose Chemotherapy with Autologous Hemopoietic Stem Cell Transplantation

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Background and Aims: While outcome in patients with Ewing sarcoma depends on both local and systemic control efficiency, the former may be inferior in patients with inoperable or metastatic disease. This is a retrospective single-center study aimed to determine an optimal local control option in patients treated in a transplant clinic.

Methods: Sixty four of 80 autologous hemopoietic stem cell transplantation (auto-HSCT) recipients also receiving local control measures before and/or after transplant with median age of 12 (1.3-19) years were included in the study. In most cases effective local control was hindered by large (>200 ml) primary tumor volume ($n=49$, 77%) and/or its axial localization ($n=55$, 86%). Surgery, radiation therapy (RT) or both were used prior to auto-HSCT in 12(15%), 18(23%), and 15(19%) cases, accordingly. In 9(11%) cases surgery was used before and RT after auto-HSCT, in 10(12%) cases all options were used afterwards. Among 44 patients with axial localization radical surgery, RT or both were applied prior to auto-HSCT in 7 (16%), 10 (23%), and 11 (25%) cases, accordingly. In 7 (16%) cases RT was performed after auto-HSCT and surgery before it. Past auto-HSCT surgery and RT were used in 3 (6%) and 6 (14%) cases, accordingly.

Results: The 5-year OS in patients receiving only surgery, RT or combination of surgery and RT prior to auto-HSCT was 66%, 42% and 64% accordingly, while in patients with delayed local control it was just 18% ($p=0.04$); the 5-year EFS was 62%, 32%, 55% and 12% accordingly ($p=0.04$). In patients with axial tumor receiving combination of surgery and RT before or after auto-HSCT the 5-year OS and EFS were 61% and 17% ($p=0.008$), 61% and 11% ($p=0.002$), accordingly.

Conclusions: The surgery or its combination with RT provides optimal local control prior to auto-HSCT. After auto-HSCT the local control effectiveness is much worse, especially in patients with axial tumor.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0192 / #1441 | Personalized Treatment of Ewing Sarcoma: Intensification of Therapy for Outcomes Improvement

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Background and Aims: Ewing sarcoma (ES) relapses are still a problem. Most of the patients have relapses in distant sites. New schemes of therapy required for improvement of outcomes. We aimed to analyze our experience in treatment of relapses of ES and propose personalized program of intensified treatment.

Methods: Two hundred and seventy four patients treated during 2008-2020, 262 included to analysis. Age median 11.6 (6 months – 18 years) years. Age of the most of patients was between 11 and 17 years old. Forty eight patients among 262 (18,3%) with relapses included to our study. Most of the patients (70.8%) were with isolated, 39.6% – with combined and 14.5% - with local relapses. Metastatic lung disease was found in 66.6%. Second-line therapy was used in case of early relapses with VIT/VCT schemes. Intensified scheme of primary treatment (shorten intervals) was used in case of late relapses.

Results: In the median of follow-up 39,3 months OS of the group with relapses was 53 ± 8,1%. After the completion of treatment second relapse diagnosed only in 9 patients. VIT response rate was 60% and VCT – 39%. In this regard, to intensify therapy we proposed a new program with autologous hematopoietic stem cell transplantation (aHSCT) with treosulfan and melphalan conditioning. Ten patients treated with such scheme. With a median follow-up of 4.2 (2-9) months all patients are alive without progression. Depending on individual features we've adapting course intervals. Study continuous.

Conclusions: ES relapses are still a problem. There is no golden standard. Possible successful approach for treatment is intensification and personalization of treatment with the help of aHSCT.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0193 / #1506 | Molecular Changes in Progressive/Relapsed or Refractory Osteosarcoma

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Background and Aims: Each year 30%-40% of pediatric osteosarcoma (OS) patients present with relapsed or refractory disease with a five-year overall survival below 30% despite intensive multimodal therapy. The high degree of intra- and intertumoral heterogeneity in OS complicates novel treatment approaches. Molecular characterization of druggable driver genes or therapeutically approachable genomic alterations like BRCAness, a defect in the HR pathway, are the most promising basis for new therapy strategies.

Methods: We retrospectively analysed clinical data and molecular alterations evaluated based on a pre-defined target prioritization from 62 OS patients enrolled in the INFORM registry during 2015–2019. Alterations were compared to those described in literature. In addition, data from whole exome sequencing and low coverage whole genome sequencing from BRCAness positive tumours, as defined by our methylation classifier was reviewed.

Results: The most frequently altered genes in our cohort of relapsed OS included TP53 (21%), PTEN (15%), CDKN2A/B (15%), RB1 (13%), MYC (13%), EPHA2 (10%) and IGF1R (8%). Overall, no significant difference in tumour biology could be observed when comparing data to gene alterations described in literature. Potentially druggable targets were identified in 69% of patients. Therapy in concordance with the INFORM report was administered in 21%. However, only few targets were labelled with a “very high” or “high” priority score (5%). BRCAness was detected in 2/3 of cases.

Conclusions: In our data we observed several canonical tumour genes and showed that relapsed OS shows a high degree of heterogeneity with little recurrent genomic alterations. Furthermore, a high frequency of tumors with BRCAness was identified, similar to other tumours bearing BRCAness features. How these molecular alterations can be translated into new therapeutic strategies remains to date an unresolved therapeutic challenge in OS.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0194 / #1535 | Therapeutic Results of EWING99 in a Tunisian Population. About 60 Cases

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Background and Aims: Ewing sarcoma family of tumors (ESFT) arise in bony sites. With current multimodal programs, including combination chemotherapy, surgery and radiotherapy in the Ewing 99 protocol, the 5-year survival rate is respectively 65%, 28% for localized and metastatic forms. The aim of our study is to evaluate EWING 99 results in our population.

Methods: Retrospective study of 60 cases of ESFT treated with EWING 99 protocol at FarhatHached University Hospital Sousse Tunisia from 2003-2016. Statistical analysis was made with SPSS10.0 and survival rate with Kaplan Meyer method.

Results: Median Age at diagnoses was 20 years (3-53years) with a sex ratio of 0.93. Swelling was the most common symptom (58%) associated with pain in 54%. The median time to diagnosis was 4.6 month (1-30 months). The tumor was localized in the bone in 78.3% with predominance in the lower limb (33.7%). The median tumor size was 10.3 cm (1-22). Metastatic forms were observed in 45% of cases with pulmonary metastasis in 57%. Neoadjuvant Chemotherapy was performed for all cases complicated of hematological toxicity grade 3 and 4 in 46.7% of cases. Histological good responder rate was 27.6%. Surgical resection was performed for 65% of patients and was conservative in 94% of them. Primary resection was performed for 44% of the metastatic forms. Radiotherapy and adjuvant chemotherapy were performed respectively for 30% and 80% of cases. The VAI protocol was used in 91.2% of them. The five-year overall survival was 58% with respectively 68% and 42% for localized and metastatic forms. The univariate analysis demonstrates that metastasis, spine primary tumors are significantly worse prognostic with respectively $p=0.007$ and $p=0.004$.

Conclusions: Our results are slowly inferior comparing with EWING 99 protocol which is explained by the lack of access to scheduled therapeutic modalities and the high rate of metastatic forms at diagnosis with the delay of diagnosis. More challenging is needed with the new EuroEWing 2012 protocol.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0195 / #1592 | Site-Wise Radiotherapy Treatment Results in Ewings' Sarcoma Family of Tumors: Last 3 Years Experience

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Background and Aims: Management of Ewing's sarcoma requires multimodality approach, wherein chemotherapy is the mainstay of

treatment. Local treatment includes either surgery or radiation depending on the site or both of them. In this article we reviewed treatment results of radiotherapy in patients of Ewing's sarcoma treated in the last three years at our institute.

Methods: A retrospective detailed analysis of patients with biopsy proven Ewing's sarcoma from any site of the body that reported to our hospital was done from year 2017 to 2019. A total of 130 patients were registered out of which only 36 patients had received radiotherapy and they were analysed in a more stringent fashion with variables including site of primary, metastatic disease status, response to radiation. Patients were either treated by conventional or conformal radiotherapy planning.

Results: Out of total 36 patients, 25 were male and 11 were female. Site wise, Upper limb (UL) – 4, Lower limb (LL) – 11, Pelvis – 7, Spinal/Para spinal tumours – 7, Rib – 4, Head and Neck – 2, Retro peritoneum – 1 patient. 7 patients underwent surgery as local treatment after chemotherapy (UL – 1, LL – 3, Rib – 3). 20 patients received curative radiotherapy, 16 palliative (local progression or metastatic). 6 patients received local radiotherapy and lung bath. Curative or adjuvant radiotherapy dose was 54 Gy/30# 1.8 Gy per fraction, 5 days a week Palliative RT (30 Gy/10# or 20 Gy/5#), Lung Bath – 12 Gy/8#. At the end of 3 years, out of 20 patients (Curative RT) 9 patients had no evidence of local disease. (UL – 2, LL – 4, Pelvis – 0, Ribs – 2, Spinal/Paraspinal – 1).

Conclusions: Radiotherapy is an integral part of multi-modality management in Ewing's sarcoma irrespective of site of primary disease (only radical radiotherapy after chemotherapy or palliative), local treatment (either adjuvant or radical), metastatic disease (Lung Bath or Pall RT to metastatic site).

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0196 / #1652 | Graphene Oxide Nanoparticles Induce Apoptosis in Wild-Type and CRISPR/Cas9-IGF/IGFBP3 Knocked-Out Osteosarcoma Cells

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Background and Aims: Background Osteosarcoma (OS) affects both adolescents and adults, and some improvement in the survival rate for affected patients has been reached in the last decade. Still, non-specificity and systemic toxicity may limit traditional therapeutic approaches to some extent. The insulin growth factor 1 (IGF1) and its binding protein (IGFBP3) have been implicated in tumorigenesis. Nanoparticles, such as graphene oxide (GO), can effectively treat cancer as they can specifically target cancer cells while reducing undesired side effects. Aim This study aimed to evaluate the toxicity of GO on

osteosarcoma in vitro using tumor cell lines with and without knocking out the IGF and IGFBP3 genes.

Methods: Method Human osteosarcoma cell lines, U2OS and SAOS2, and the normal osteoblast cell line hFOB1.19 were used. The IGF1 and IGFBP3 genes were eliminated using CRISPR/Cas9. Tumor cells were cultured and treated with GO. Apoptosis and reactive oxygen species (ROS) were analyzed by Annexin V-FITC and ROS assays. The nuclear factor erythroid 2-related factor 2 (NRF2), a crucial regulator of cellular resistance to oxidants, was investigated by Western blotting.

Results: We found a significantly higher apoptosis in the OS than hFOB1.19, especially in U2OS cells in which IGF1 and IGFBP3 were knocked out. ROS increase due to GO exposure was remarkably time and concentration dependent. Based on the rate of apoptosis, ROS, Nrf-2 decrease, and cytomorphological changes, GO has a significant cytotoxic effect against OS.

Conclusions: Conclusion Targeting the IGF1 and IGFBP3 signaling pathways may strengthen GO-related cytotoxicity with the potential to increase the survival of patients affected by this tumor.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.c Bone Tumours

PV0197 / #276 | Long-Term Follow-Up of Survivors of Childhood Osteosarcoma (OS): A Report From The Childhood Cancer Survivor Study (CCSS)

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Background and Aims: Advancements in chemotherapy combination strategies and surgical techniques have improved survival for children with OS. Better characterization of long-term effects of these differing strategies should allow for improved survivorship.

Methods: Five-year survivors of childhood OS treated from 1970-1999 in CCSS were evaluated for late (death > 5 years from diagnosis) mortality (n=1,799 eligible), risks for subsequent malignant neoplasms (SMN) and chronic health conditions (CHC; n=1,257 eligible) through longitudinal surveys using piecewise exponential regression. Four chemotherapy strategies: Methotrexate+Doxorubicin [MA], Methotrexate+Doxorubicin+Bleomycin+Cyclophosphamide+Dactinomycin [MABCD], Methotrexate+Doxorubicin+Cisplatin [MAP], and Other along with three surgi-

cal approaches: amputation, limb-salvage, neither were evaluated adjusting for race, sex, and age.

Results: OS survivors had a median age of 14.3 years at diagnosis (range 1.8–21.0) and a median age of 37.9 years at time of last follow up (range 8.6–63.8). The 35-year cumulative all-cause late mortality and SMN incidence were 21.9% (95% Confidence Interval [CI]: 19.8–24.1) and 10.4% (CI 7.6–13.3), respectively, with no statistically significant difference among the chemotherapy groups. Adjusting for the chemotherapy groups and demographic variables, limb-salvage had a similar all-cause mortality rate (Relative Rate [RR] 0.9; CI 0.7–1.3) compared to amputation. Adjusted incidence rates of any CHC were elevated for MAP (RR 1.3; CI 1.0–1.6) relative to MA and for limb-salvage surgery (RR 1.4; CI 1.2–1.6) relative to amputation. Similar results were obtained when CHCs were restricted to severe/disabling, life threatening, or fatal conditions.

Conclusions: No appreciable differences were seen in all-cause mortality and SMN incidence across the chemotherapy groups. MAP, the current standard of care, is associated with elevated CHC risk, relative to MA, as is limb-salvage surgery relative to amputation. Given the limited changes in MAP therapy over time, this study provides important information about the risks current patients will likely face after treatment and can help guide their care as they age.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.c Bone Tumours

PV0198 / #383 | Outcome of Metastatic Ewing Sarcoma in Children Treated with Interval Compression

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Background and Aims: Interval compression(IC), defined as 2-week long cycles of alternating VDC (Vincristine-Doxorubicin-Cyclophosphamide) and IE (Ifosfamide-Etoposide), was proved to improve survival of localized Ewing Sarcoma. There are no data concerning the outcome of patients with metastatic disease treated using this approach. In this study, we reviewed the outcome of our metastatic patients who were treated using IC.

Methods: This is a retrospective chart review of pediatric patients with metastatic Ewing sarcoma treated with interval compression at out center in the time period between Jan-2013 to Mar-2020. Data was collected from our electronic medical records. We used descriptive analysis; with Kaplan-Meier method to estimate EFS and OS, and logrank -test to perform specific univariate comparisons.

Results: We identified 34 patients (median age,11.6years; range,2.7to17.1). Site of primary tumor was distributed into axial (N=20,59%), extremity (N=11,32%), and soft tissue (N=3, 9%). Twenty-four patients (71%) had pulmonary metastasis, and 14(41%) had extra-pulmonary metastasis. The site of extra-pulmonary

metastasis was bone in 11 patients, lymph nodes in 2, and intra-spinal in 1. All patients underwent local control; modality was surgery-alone in 7(21%) patients, radiotherapy-alone in 18(53%), or both in 9(26%). The estimated 5-year OS and EFS were $46\% \pm 11\%$ and $35\% \pm 9.3\%$, respectively. Patients with pulmonary only metastasis had EFS of $44\% \pm 13\%$ in comparison to those with extra-pulmonary metastasis of $21\% \pm 13\%$ ($P=0.18$). There was no statistically significant difference in survival according to age group (above and below 12 years) or site of primary. As for local control, EFS in surgery-only group was $83\% \pm 15\%$, in combined surgery and radiation $30\% \pm 18\%$, and in radiation-only $15\% \pm 10\%$ ($p=0.048$).

Conclusions: The outcome of our metastatic patients treated using IC is in line with recent published literature using other regimens. Our results also suggest the value of optimal surgical local treatment. Larger reports or randomized trials may better illustrate the value of IC in metastatic patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0199 / #402 | TNF-Mediates CD83, ICAM-1 and MHC Class I Upregulation and Enhances CHM1319 Specific TCR-transgenic T-cell Cytotoxicity Against Ewing Sarcoma Cell Lines

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Background and Aims: Ewing sarcoma (EwS) constitutes a pediatric solid tumor entity with low somatic mutational burden and a low rate of tumor infiltrating T-cells, indicating a low extent of immunogenicity. In EwS, immunogenicity may furthermore be significantly diminished by a predominantly M2 macrophage driven pro-tumorigenic tumor microenvironment. In the past, our research group demonstrated that CHM1³¹⁹ specific TCR-transgenic T-cells are able to control EwS growth in a preclinical mouse model as well as in a patient with metastatic disease. However, new adjuvant techniques to induce long lasting and curative CHM1³¹⁹ specific TCR-transgenic T-cell mediated anti-tumor responses needed to be identified.

Methods: In this work, we sought to identify a technique to improve the cytotoxic effect of transgenic T-cells by altering the immunogenic cell surface marker expression on EwS cell lines using different cytokines. We analysed the effect of the addition of certain cytokines on the surface expression on EwS cells lines as well as the gene expression on RNA level by flow cytometry and microarray assays, respectively. Next, we assessed the effect of altered cell surface and gene expression on T-cell mediated recognition and killing of EwS cells.

Results: Tumor necrosis factor (TNF) is upregulated during inflammatory processes. We demonstrate that TNF particularly causes pro-immunogenic CD83, MHC class I and II as well as ICAM-1

upregulation in EwS cell lines. This observation was associated with significantly improved recognition and killing of the tumor cells by EwS specific CHM1³¹⁹/HLA-A*02:01 restricted TCR-transgenic T-cells.

Conclusions: Conclusively, TNF mediated CD83 upregulation constitutes a new tool to improve T-cell mediated anti-tumor responses in EwS in vitro. There are several ways to induce TNF secretion in the clinical setting, e.g. irradiation and hyperthermia. In future clinical protocols, this finding may be important in order to identify appropriate conditioning regimens and appropriate time points for adoptive T-cell based immunotherapy in EwS patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0200 / #423 | The Failure of Hyperthermic Intrathoracic Chemotherapy (HITHOC) For Pleural Dissemination

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Background and Aims: Pleural dissemination (PD) is severe advanced stage of pediatric solid malignancy and is difficult to control even with multimodal therapy. To improve their prognosis, we have performed hyperthermic intrathoracic chemotherapy (HITHOC) following cytoreductive surgery (CRS). But some patients developed tumor recurrence even after HITHOC. Here we report our results to reveal the failure of HITHOC.

Methods: We conducted a retrospective study of 11 patients who underwent CRS+HITHOC from 2013 to 2020 in our institution. We performed HITHOC at the same time of CRS when PD was evident before surgery, meanwhile in cases we noted PD at first during CRS or by postoperative pathological findings, we performed it after CRS. The underlying diseases were following: 6 osteosarcoma, 1 germ cell tumor, 1 malignant rhabdoid tumor, 1 desmoplastic small round cell tumor, 1 Wilms tumor, and 1 hepatocellular carcinoma. The median age was 13.2 years and median follow-up period was 6 months(0-27), 7 cases were male. We divided cases into two groups; the group faced recurrence at HITHOC lesion (group R: n=6) and the other that had no recurrence(group NR: n=5).

Results: We performed HITHOC 14 times for 11patients and noted 6 recurrences at the pleura. Five cases died of tumor recurrence. The median number of tumors extirpated by CRS was 24 (1-105) in group R, 5 (1-49) in group NR. In group R, 33% of the patients underwent CRS and HITHOC concurrently, whereas 50% in group NR. Because of numerous impalpable disseminations or tumor invasion to major vessels, tumors pathologically or radiologically remained after 5 times of surgery we aimed CRS in group R, whereas none in group NR($p=0.003$).

Conclusions: Tumor recurrent cases after HITHOC showed high mortality and CRS tended to be insufficient in these cases. HITHOC may insignificant for the cases that is difficult to accomplish CRS.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0201 / #491 | Genomic Characterization of Pediatric Bone Sarcomas by Next Generation Sequencing, Utility for Clinical Management

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Background and Aims: Osteosarcoma (OS) and Ewing's sarcoma (ES) are the most frequent pediatric primary malignant bone tumors. Survival of patients with bone sarcomas has barely improved in recent decades. New tools are required to improve the clinical management of these patients. The aim of this study was to identify genetic alterations of potential clinical application using next-generation sequencing (NGS).

Methods: Genomic data were obtained from tumor tissue of 45 patients initially diagnosed with OS and ES and analyzed using the OncoPrint Childhood Cancer Research Assay (Thermo Fisher), a NGS-based tool.

Results: Patient cohort consisted of 45 pediatric and young adult patients 58% (n=26) initially diagnosed with OS and 42% (n=19) with ES. Sex distribution: 24 males/21 females. Mean age: 6.6 years (range 0.6-26.9). Eleven patients were metastatic at diagnosis (24.4%). Primary sites included lower extremities (48.9%), upper extremities (24.4%) and trunk (26.7%). The most frequent mutations in patients diagnosed with OS were in TP53 (28%) and RB1 (28%) genes and amplifications of MYC, KIT and PDGFRA. As expected, EWSR1-FLI1 rearrangement was the most frequent alteration found in ES and usually not accompanied by other alterations (TP53, 12.5%). In 4 patients (9%) there was a change in the initial diagnosis after genomic characterization. Three patients were initially diagnosed as ES and the final diagnosis were: myxoid liposarcoma (FUS|DDIT3), rhabdoid soft tissue tumor (SMARCB1) and angiomatoid fibrous histiocytoma (EWSR1|CREB1). In one patient initially diagnosed of OS the rearrangement of EWSR1-FLI1 was detected and thus classified as ES. Moreover, in those diagnosed as ES, the pathognomonic rearrangement EWSR1-FLI1 was found in 87.5% of the samples (14/16) by NGS while only in 75% of cases the fusion was detected by pathological analysis (12/16).

Conclusions: Genomic characterization should be part of the diagnostic process of pediatric sarcomas, complementing pathology due to relevant findings for the clinical management of these patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0202 / #535 | Targeting B-Catenin Enhances Immune Checkpoint Inhibition Efficacy in Osteosarcoma

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Background and Aims: Osteosarcoma (OS) is the most common bone malignancy in childhood and adolescence. Though the prognosis of OS patients has been improved by multimodal therapies, patients with metastatic disease still have a very poor prognosis. We previously reported a novel β -catenin inhibitor Tegavivint suppresses OS primary tumor and lung metastasis using Patient-derived cell lines. Wnt/ β -catenin signaling is reported to influence intratumoral T cell infiltration, and we have investigated the influence of Tegavivint on the OS tumor immune microenvironment using murine syngenic OS models.

Methods: We used murine osteosarcoma cell lines F331 and F420 for in vitro and in vivo studies. We performed cytotoxicity studies using the CCK-8 assay and western blotting to investigate downstream β -catenin signaling perturbations secondary to Tegavivint. Intratibial injections were performed into C57BL/6 mice using both cell lines. After tumors were found palpable, Tegavivint and anti-PD-1 antibody was administered as monotherapy or in combination. After treatment we compared the tumor volume in the control, monotherapy and combination therapy groups. We have assessed T cell tumor infiltration using immunohistochemistry.

Results: Tegavivint suppresses the growth of F331 and F420 cell lines with IC₅₀s of 9.52nM and 49.4nM, respectively. Tegavivint monotherapy and combination therapy significantly suppressed the growth of murine tumors (p<0.05). PD-1 monotherapy did not show any significant anti-tumor activity on OS tumors. IHC stainings showed that CD8-positive T cells increased in Tegavivint monotherapy and combination therapy mice compared with the control mice.

Conclusions: These results indicate that Tegavivint can alter the tumor immune microenvironment and enhance the activity of immune checkpoint blockade in osteosarcoma. Future directions include flow cytometry and imaging mass cytometry for more comprehensive tumor immune microenvironment profiling, including further characterization of T cell properties and assessing macrophages, NK and dendritic cells.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0203 / #619 | Machine Learning Predicts Survival in Ewing Sarcoma Based on Gene Expression Profiles in Small Cohorts

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Background and Aims: Risk stratification of patients is crucial and individualized therapy is warranted. This is especially needed for children, where drugs have severe and long-term side effects. Therefore, accurate prediction of prognosis is needed at diagnosis.

Methods: We built machine learning (ML) models on three Ewing Sarcoma (EwS) datasets to predict survival based on gene expression. We analyzed all datasets with annotated survival in GEO: GSE17618 (n=44), GSE63155 (n=46), and GSE63156 (n=39). Patients of each cohort were classified by long (overall survival longer than 5 years) vs short survival (dead within 5 years). To avoid overfitting and to determine performance in each data set, we used repeated crossvalidation.

Results: Depending on the cohort, the performance of the models varies, but is superior to random prediction for each dataset. The ROC AUC ranges from 0.67 (GSE63156), 0.75 (GSE63155), to 0.87 (GSE17618). Moreover, we show that a comparably low sample size is enough to obtain good results. This is advantageous as for rare entities like EwS it is often hard to acquire large data sets. As our models work well based on approximately 40 patients, it means that individual hospitals can also gather their own data sets, and create and apply their own models. For each model, the expression of 100s of genes is involved. This demonstrates the complex interplay in tumor cells, which can hardly be condensed to a few genes, even though single genes are published now and then to be survival markers. This emphasizes the importance of ML methods in order to generate models that are able to incorporate such a high number of variables and make use of this complementary information.

Conclusions: With gene signatures incorporating 100s or 1000s of genes, we can predict patient's overall survival very well. This provides an opportunity for improved individual risk assessment, which can also individualize therapy.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.c Bone Tumours

PV0204 / #751 | Ewing's Sarcoma of the Sphenoid and Ethmoid Bones in Infants and Toddlers

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Background and Aims: Primary Ewing's sarcoma (ES) of the skull accounts for 1% of all ES and occurs most commonly in the tem-

poral, parietal, and occipital bones. The sphenoid/ethmoid location is extremely rare and its outcome is uncertain.

Methods: We describe three cases with primary ES of sphenoid/ethmoid bones treated in our institution.

Results: All three cases presented with unilateral proptosis. Magnetic resonance imaging (MRI) showed a sphenoid/ethmoid lesion with a soft tissue component. Metastatic work up was negative. The first case was a 13-month-old boy previously diagnosed with cystic fibrosis. Sinus infection was initially suspected. Case two and three were a 5-year-old boy and girl respectively. Case one and two had molecular studies that showed EWSR1-FLI1 translocation. Treatment included chemotherapy and proton beam radiotherapy and complete remission (CR) was achieved without relevant issues. They remain alive and disease free 24 months after. Case number 3, presented 14 years ago. She was initially diagnosed with esthesioneuroblastoma and no molecular analysis was made. Patient was treated with chemotherapy, gross total resection and photon beam radiotherapy. Remain in CR for fourteen years, when she presented to the emergency department with acute respiratory distress and a mediastinal mass was found. Biopsy was made and histological and molecular studies were consistent with ES. Patient died due to disease progression one month later. Retrospective molecular test of the initial ethmoid bone tumor was positive for EWSR1-FLI1 translocation.

Conclusions: ES of the sphenoid/ethmoid bone is very rare. Surgery is usually not feasible and proton beam radiotherapy should be considered. Some cases could be mistaken with other entities. Definitive diagnosis should include the identification of signature chromosomal translocations (EWSR1-ETS). Prognosis does not seem to be worse than other locations.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.c Bone Tumours

PV0205 / #772 | Factors that Impact the Outcomes in Ewing's Sarcoma Family of Tumors in Low and Middle-Income Countries

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Background and Aims: Ewing's sarcoma family of tumors (ESFT) is common in second decade of life. Achieving good outcomes in ESFT requires a multimodality approach. We report the Clinico-pathological features, treatment, and survival outcomes of patients with ESFT treated at our center.

Methods: Patients diagnosed and treated for ESFT at our center from 1 st January 2009 to 31 st December 2017 were included in this study. Data was collected from the patient's case records. Vincristine, doxorubicin, and cyclophosphamide alternating with ifosfamide and etoposide

every 3 weeks was the chemotherapy given. Event-free survival (EFS) and overall survival (OS) were estimated using Kaplan–Meier method.

Results: The study included 173 patients among whom 44 (25%) patients were metastatic at diagnosis. Median age of patients was 16 years (range 6 months to 68 years). The most common site of the primary tumor was Pelvis (19%), followed by long bones. Extra-skeletal primary was observed in 46 (24%) patients. The median follow-up was 75 months and the five-year EFS and OS were 61% and 67% respectively for the overall cohort whereas for the localized disease was 72% and 78.5% respectively. Metastatic disease, tumor volume > 200 ml, tumor diameter > 8 cm, pelvic site, hemoglobin < 10 gms%, elevated lactate dehydrogenase, positive margin, and necrosis < 90% were significantly associated with inferior EFS and OS on univariate analysis. On multivariate analysis, metastasis disease, tumor diameter > 8 cm, and necrosis < 90% were significantly associated with inferior EFS and OS.

Conclusions: Outcomes in our study are comparable to reports in the literature. Whether the use of dose-dense chemotherapy and/or autologous stem cell transplant would improve outcomes without increased toxicity in resource-limited settings needs to be explored. Factors unique to LMICs like treatment default, bulky disease, more metastatic disease at presentation, and poor nutrition have a negative impact on the outcome.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0206 / #776 | Localized Osteosarcoma: Predictive and Prognostic Factors in Tunisians

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Background and Aims: Osteosarcoma is the most frequent bone tumor. Histological response to neoadjuvant chemotherapy is a major prognostic factor influencing survival. Predictive factors of response to neoadjuvant chemotherapy have been identified in western series but not identified in the Tunisian population, hence the interest of our work.

Methods: From 1980 to 2018, in the Salah Azaiz Institute, 103 patients diagnosed with a histologically proved localized osteosarcoma were included in a retrospective descriptive longitudinal study.

Results: Median age was 21 years (1-70) and sex ratio was 1.39 in favour of men. Fifty percent were located in the femur. Conventional osteosarcoma was the most frequent histological subtype. Seventy five percent received high-dose methotrexate neoadjuvant chemotherapy and the median dose was 12g/m. Within patients treated by neoadjuvant chemotherapy, 77% underwent surgery and 19.4% were good responders. Mean delay between neoadjuvant treatment and surgery was 62 days and 54 days between surgery and adjuvant chemotherapy. Median survival was 32 months and median disease-free survival was

21 months. In our study, a statistical trend was in favour of a good histological response in patients with methotrexate rates above average and lower delay between symptoms onset and treatment. Upon univariate analysis, tumor site, elevated Lactate dehydrogenase levels, low Methotrexate doses, Methotrexate at 24 hours lower than 4.4 $\mu\text{mol/l}$, no clinical response, incomplete surgery and recurrence before 2 years after treatment were associated with lower survival.

Conclusions: Early patient care and methotrexate rates above average would be a way of improving the prognosis of localized osteosarcoma.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0207 / #834 | Should Osteosarcoma Treatment Continue be Avoided in LMIC Settings Without GCSF? A Feasibility Study in Togo, West Africa

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Background and Aims: A pilot study was undertaken to establish feasibility of treatment for patients with pediatric osteosarcoma in Togo, West Africa, without availability of granulocyte colony-stimulating factor (GCSF). No established treatment for osteosarcoma currently exists in-country. We utilized the Memorial Sloan Kettering Cancer Center P6 regimen (MSKCC P6)—an aggressive chemotherapy regimen consisting of vincristine, doxorubicin, and cyclophosphamide, alternating with ifosfamide and etoposide given over six months.

Methods: Patients diagnosed with osteosarcoma per physical and radiological findings from June 2019 to December 2020 were offered treatment with MSKCC P6, opposed to the current standard of palliative care. Pathology was confirmed by Norton Children's Hospital (USA). If a patient experienced either one neutropenic episode resulting in chemotherapy delay, or two episodes of neutropenic fever (NF), or two grade IV mucositis episodes, dosing of Ifos/ Cyclophosphamide was reduced by 25%. NF protocol included empiric antibiotics of Cef-tazidime or Meropenem, with Fluconazole until blood cultures results dictated narrowing.

Results: Eleven patients were diagnosed with osteosarcoma during the designated time period. Nine patients consented to treatment, one patient was not offered treatment due to advanced bilateral disease, and one refused treatment. The median age was 16.5 years, and 64% were female. Six of nine patients who began the protocol were able to complete therapy. Twenty-three neutropenic fever (NF) episodes were documented, with bacteremia confirmed in 5/23 NF episodes. Two patients with initial extensive metastatic disease expired prior to cycle three secondary to sepsis. Five of 9 patients remain alive with a current survival range of 9-20 months.

Conclusions: We demonstrated the treatment of osteosarcoma is feasible in an LMIC setting despite the lack of GCSF, but may not be

beneficial for patients who present with metastatic disease. Further studies are needed to determine efficacy, survival, and proper eligibility of candidates for chemotherapy.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0208 / #840 | High-Dose Methotrexate-Induced Hepatotoxicity in Childhood Osteosarcoma: A Study from the Taiwan Pediatric Oncology Group

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Background and Aims: High-dose methotrexate (HD-MTX) is a standard chemotherapeutic drug for osteosarcoma. Hepatotoxicity caused by HD-MTX is commonly observed; however, reports are scarce on the frequency of this adverse event in Asians. This study explored the incidence of HD-MTX-induced toxicity in childhood osteosarcoma in Taiwan.

Methods: We retrospectively identified children who had newly diagnosed osteosarcoma of the extremities in 2009-2016 and whose treatment followed the TPOG-OS-2009 protocol, which was derived from the EURAMOS-1 trial and included 12 courses of HD-MTX (12 g/m²/course) as the backbone. The pharmacokinetics of HD-MTX and grades of toxicity and their effects on patient survival were analyzed.

Results: A total of 61 patients were recruited and 661 HD-MTX courses were analyzed. Grade III-IV hepatotoxicity was experienced by 60 patients (98%) and observed in 384 courses (58%). Younger age and MTX peak level at 4 h $\geq 1,000 \mu\text{mol/L}$ were risk factors for grade III-IV hepatotoxicity. Grade IV hepatotoxicity (observed in 54% of patients) was associated with a delay (6.3 days) in the interval between MTX to next chemotherapy. A positive correlation trend was observed between MTX dose intensity and survival, but statistic power is limited. The higher frequency of grade III-IV hepatotoxicity had no significant effect on survival. In terms of the incidence of grade III-IV hepatotoxicity caused by HD-MTX, studies from other countries have reported rates between 11% and 80% of childhood osteosarcoma patients, which were lower than the rate of 98% reported in this study. **Conclusions:** An extraordinarily high incidence of MTX-related hepatotoxicity in childhood osteosarcoma was found in Taiwan, which

frequently interfered with the schedule of chemotherapy. More pharmacogenomic studies of HD-MTX with regard to ethnic variation and risk prediction of toxicities are required in the new generation of precision medicine to improve treatment outcome and reduce toxicity for childhood osteosarcomas in Asian populations.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PV0209 / #856 | Feasibility and Safety of Outpatient Administration of Ifosfamide and Etoposide for Paediatric Patients with Ewing Sarcoma in a Resource-Limited Setting Amid the Covid-19 Pandemic

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Background and Aims: Paediatric Ewing sarcoma (ESW) chemotherapy, Ifosfamide/etoposide (IE), requires 24hrs MESNA (2-mercaptoethane sulphonate sodium) hydration to prevent haemorrhagic cystitis(HC) leading to increased admission burden and delay in treatment due to the unavailability of inpatient beds. We developed a pathway to administer IE as outpatient. This study was conducted to assess the feasibility and safety of outpatient administration of IE in a resource-limited setting

Methods: A retrospective descriptive study of paediatric patients with ESW who received IE as outpatient from January 2020 till January 2021, at a single institution. Protocol for safe outpatient administration of 1800mg/m² of ifosfamide for 5 days was developed with MESNA bolus before the start of infusion and at end of 9 hours of hydration. We selected patients older than 10 years who were able to drink at home, with no pre-existing renal dysfunction or history of haematuria. The patients were monitored for HC with a urine dipstick before, mid-way, and at end of infusion. A urine analysis (UA) was done 24 hours after the last dose of ifosfamide

Results: Total of 47 cycles were given as outpatient. Thirty-five (74%) cycles were given on time, two (4%) cycles were delayed due to mucositis, two (4%) due to delayed count recovery and 8(18%) due to lack of availability of outpatient appointments. All cycles were completed in the outpatient setting 47(100%), with no episode of HC. UA was not done at 24hrs for 27(58%) cycles so microscopic haematuria may have been missed. For our population, this saved PKR. 2,155,138(PKR 45,854 /cycle) (\$299/cycle) in cost and 282 days of hospitalization.

Conclusions: Outpatient administration of IE for paediatric patients with ESW is feasible and safe with a reduction in the cost of chemotherapy and days of hospitalization. This was particularly useful in continuing timely treatment of EWS during the COVID19 pandemic when inpatient beds were limited.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0210 / #1095 | The Long-Term Follow-Up Results of 62 Children with Thyroid Tumor

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Background and Aims: The incidence rate of thyroid neoplasms has been increasing year by year in children, which has attracted extensive attention from the oncology profession. Thyroid adenoma and thyroid cancer are the main thyroid tumors in children. However, due to the lack of attention to thyroid tumors in children in China, there are few clinical data. The purpose of this study is to improve the understanding of pediatric surgeons on thyroid tumors in children, so as to achieve early detection, early diagnosis, early treatment and improve the clinical prognosis.

Methods: 63 children with thyroid tumors (24 boys and 38 girls) with a median age of 11 years (1-17 years) were treated in the Department of oncology, Shanghai Children's medical center from October 1989 to October 2018,

Results: Among 62 cases of thyroid tumors, 1 case was medullary carcinoma, 1 case was thyroid mesenchymal tumor, 33 cases were thyroid adenoma, the rest were papillary carcinoma. All 62 children with thyroid cancer received corresponding thyroidectomy. 56 cases (56 / 62, 90.3%) were followed up by telephone. The follow-up time was 2-8 years. More than 80% cases were followed up for more than 5 years. There was no death in 56 cases. 45 cases (45 / 56, 80.3%) had long-term hypothyroidism, and the effect of thyroxine replacement therapy was satisfactory. 15 cases (15 / 56, 26.7%) had hypocalcemia after operation, which was gradually relieved after symptomatic treatment. 8 cases (8 / 56, 14.2%) had distant metastasis after operation, and received iodine 131 radiotherapy. Three cases (3 / 56, 5.3%) recurred, with an average recurrence time of 15.6 months.

Conclusions: The incidence rate of thyroid tumors in children is very low. Postoperative long-term complications are mainly hypothyroidism and hypocalcemia. Hypocalcemia can be relieved in most 6 months. The overall treatment effect is pretty satisfactory without death.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0211 / #1150 | Gene Expression Profiles Analysis Identifies a Novel Biomarker in Ewing's Sarcoma

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Background and Aims: This study integrated several datasets from gene expression omnibus (GEO) database to identify a new biomarker in Ewing's sarcoma (ES).

Methods: We analyzed 154 samples of ES patient tissues (GSE12102 and GSE34620) and 39 samples of normal controls (GSE12474, GSE48574, GSE38417, GSE39454, GSE38680, and GSE13608). Differentially expressed genes (DEGs) were calculated by DESeqR. The DEGs were then evaluated for GO and KEGG. The protein and protein interaction (PPI) network was constructed by Cytoscape software and the hub genes were identified by the cytoHubba plug-in. The GEPIA (gene expression profiling interactive Analysis) website was used for prognosis analysis. Furthermore, hub genes were evaluated by Q-PCR and western blotting (WB) in ES cell lines. CCK-8 assays, colony formation assay and flow apoptosis test were performed to explore the role of MAD2L1 in ES cell lines.

Results: A total of 1171 DEGs, including 555 up-regulated DEGs and 616 down-regulated DEGs were identified. GO analysis indicated that DEGs were particularly enriched in molecular interaction. KEGG analysis results demonstrated that upregulated DEGs were particularly enriched in cell cycle signaling pathway. The top 10 hub genes including CDK1, MAD2L1, CDC20, UBE2C, BUB1B, CCNB1, TOP2A, TPX2, NDC80, and KIF11. The high expression of the four genes including CDK1, MAD2L1, BUB1B, KIF11 indicated poor survival for ES patients. The results of Q-PCR demonstrated MAD2L1 was the highest expression one among these 4 genes, and the results of WB revealed that MAD2L1 was upregulated in 3 ES cell lines including A673, RDES and SKNMC. CCK-8 assays and colony formation assay revealed that the knockdown of MAD2L1 significantly impaired the growth of Ewing's sarcoma cell lines. si-MAD2L1 induced apoptosis of ES cells in vitro.

Conclusions: MAD2L1 may be a potential biomarker and therapeutic target for ES.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0212 / #1407 | Epigallocatechin Gallate Inhibits In-Vitro Cell Growth of Rhabdomyosarcoma

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Background and Aims: Epigallocatechin gallate (EGCG), a major polyphenol of green tea, has shown inhibitory potential on cell proliferation in various malignancies. In this study the effect of EGCG on two rhabdomyosarcoma (RMS) cell lines was investigated to provide insight

into molecular level understanding of the mechanisms involved in cell metabolism and apoptosis.

Methods: Two human RMS cell lines (RD, embryonal histology and RH30, alveolar histology) were used. Growth inhibition of EGCG was measured by MTT-assay, colony formation via colony forming assay (CFA), gene expression by real-time PCR, migration by transwell assay and intracellular signaling by Western blot.

Results: EGCG significantly reduced cell viability in RD and RH30 cells in a dose-dependent manner. In RH30 cells cell viability was significantly reduced at 25 μ M EGCG ($p=0.05$) and in RD cells at 50 μ M EGCG concentration ($p=0.0001$). Exposure of EGCG on RD and RH30 cells also resulted in a decrease of migration ($p=0.01$). Colony formation was reduced by 40% at 2.5 EGCG concentration in RD cells ($p=0.01$) and by 60% at 3 μ M EGCG concentration ($p=0.0001$). As one major molecular mechanism downregulation of the hedgehog pathway was identified, especially in Gli-I protein.

Conclusions: EGCG potently inhibits proliferation of RMS in vitro. It might therefore be a promising agent in the clinical setting, especially in chemo-resistant or advanced RMS. Further studies seem justified and are necessary to assess the potential of EGCG against RMS.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0213 / #1453 | Evaluation of the Immune Profile of Murine Embryonal Rhabdomyosarcoma after Hyperthermia Treatment with Magnetic Resonance Guided High Intensity Focused Ultrasound

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Background and Aims: Rhabdomyosarcoma (RMS) most commonly affects children and young adults. Current treatment involves surgery, chemotherapy, and/or ionizing radiation with overall five-year survival around 30% in the most aggressive form of the disease and thus, new treatments are needed. Immunotherapies are a burgeoning treatment group that have not been explored thoroughly in RMS. To study immunotherapies murine models are employed but understanding of the baseline immune infiltrate and corresponding microvasculature is lacking. To increase immune response hyperthermia (HT) has been used and one method to generate non-invasive localized HT is magnetic resonance guided high intensity focused ultrasound (MRgHIFU). In our project we are examining the tumor microenvironment and immune

response in an immunocompetent murine embryonal RMS (ERMS) model before treatment and in response to MRgHIFU HT.

Methods: We implemented targeted heating using MRgHIFU. A pre-heating cycle of 1.5 minutes was completed to raise the tumor temperature to the target point (42°C $\pm$ 1°C). HT was maintained for 20 minutes. The mice were recovered and underwent necropsy 4h, 24h, 48h, 72h or 7d after treatment. IHC staining of CD45 was performed on the tumor samples.

Results: Tumors were stained to determine the presence of leukocytes and were qualitatively analyzed by a sarcoma pathologist for density of immune infiltrate. 84% of tumors (16/19) had a moderate to high level of immune infiltration. Additional staining to confirm the presence of microvasculature was performed to ensure that there was no confounding due to lack blood flow. 95% of tumors showed a moderate-high level of microvasculature (18/19). Based on these results we aim to further characterize the immune infiltrate seen in these tumors with specific staining for T-cells, B-cells, and macrophages.

Conclusions: With the collaboration of a multidisciplinary translational team we have successfully adapted MRgHIFU treatment to examine the effect of HT on the immune microenvironment in our murine ERMS model.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0214 / #1658 | Pediatric Rhabdomyosarcomas in Algeria, National Survey

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Background and Aims: Rhabdomyosarcoma is the most common tumor of soft tissue tumors in children, and one of the typically pediatric tumors. There is very little data reported in the Maghreb and Africa.

Aim is to evaluate the epidemiological pattern, clinical features, histological and outcome in Algerian children with rhabdomyosarcoma, and to compare findings with high outcome countries.

Methods: This is a national, multi-institutional and retrospective study, from January 2013 to December 2017. A survey was conducted in 08 pediatric oncology departments in Algeria, including the following items: age, sex, location, tumor size, initial imaging and extension

assessment, surgery and histological type, treatment and outcome of patients.

Results: We collected 138 patients; the average age is 04 years (range from 02 months to 15 years). The sex ratio M / F: 1.26. The size of tumor was > 5 cm in 75% of cases. The most frequent locations that have been found are: head and neck 38%, genitor-urinary 30 %, and others 32%. The diagnosis was made most often after primary surgery (total excision 16%, partial excision 12%, and biopsy 72%). The most common pathological type was embryonal 77%. Disease was localized in 87% and metastatic in 13 % of cases. Chemotherapy was used in all patients according to the 2005 RMS protocol in 75% of cases, MMT95 / 98 in 25% of cases. Overall survival is 68% and recurrence-free survival 55%.

Conclusions: Despite the large size of the tumor the rate of metastases is low; this study should lead to national recommendations to standardize chemotherapy protocols, indications of local treatment to improve outcome.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0215 / #1735 | Infantile Fibrosarcoma (Ifs): Towards a Unifying Concept of NTRK-Rearranged Spindle Cell Neoplasms in Infants

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Background and Aims: Infantile Fibrosarcoma (IFS) is a congenital/infantile fibroblastic tumour, locally aggressive, rarely metastasizing. An ETV6-NTRK3 fusion has been identified in the majority of cases. Other alternative fusions involving MAP-kinase pathway have been reported, however there are no systematic studies investigating their frequency, prognostic impact and the differential criteria (apart age), from NTRK-rearranged spindle cell neoplasm (NTRK-SCN). The aim of this study is to analyse a consecutive series of cases diagnosed as IFS to better define their molecular and clinico-pathological features.

Methods: 44 cases with histologic diagnosis of IFS and available RT-PCR for ETV6-NTRK3 were collected as part of EpSSG studies or AIEOP central review. All patients were younger than 2 yrs. H&E stained slides were reviewed and translocation-negative cases were studied by FISH (1 case), next-generation sequencing (NGS and/or RNAseq) to identify potential fusions.

Results: Based on molecular features, 3 distinct groups emerged: Group 1- 33 IFS with ETV6-NTRK3 fusion and morphology showing oval-to-elongated cells, minimal/absent myxoid stroma, prominent vascular pattern. Group 2- included 3 NTRK1-positive tumours (2 TPM3-NTRK1 with dermatofibrosarcoma protuberans(DFSP)-like features, 1 LMNA-NTRK1 with lipofibromatosis-like neural tumour morphology); 3 RAF-rearranged tumours (one identified by FISH with unknown partner, 2 with CLIP1-RAF and GOLGA4-RAF1 fusion, respectively), all showing a very bland, fibromatosis-like morphology. One case showed FGFR1-CKAP5 fusion. Group 3- 4 fusion-negative tumours at RNAseq. All patients are alive and well, but one (TPM3-NTRK1) experienced multiple recurrences and lung metastases.

Conclusions: ETV6-NTRK3 fusion was found in 75%; while other fusions were identified in 15.8% (NTRK1 or RAF both in 6.8% and a new FGFR1-CKAP5 fusion in one case). Although subtle distinctive morphologic features were identified, all cases shared a continuum morphologic spectrum, common clinical features and prognosis supporting the unifying concept of IFS independently from the fusion identified. Further investigations are needed for NGS-negative tumours.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0216 / #1766 | Rhabdomyosarcoma in Infants and Children Under 36 Months of Age Treated on the European Paediatric Soft Tissue Sarcoma Group (EPSSG) RMS2005 Study

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Background and Aims: Infants (defined as children under 12 months of age) with rhabdomyosarcoma have historically been known to have poorer survival than the older age groups. Here we present outcomes for children with localised rhabdomyosarcoma less than 36 months with particular emphasis on infants.

Methods: This study includes children less than three years of age enrolled on the EpSSG-RMS2005 study for localised disease. Treatment comprised chemotherapy, local surgery and/or radiation therapy adapted to risk group and age in the very young. The main outcome measures were event free (EFS) and overall survival (OS).

Results: Outcome data were available for 485 out of 490 patients aged less than 3 year, 110 of whom were infants. Median follow up for alive patients (n=393) was 72.7 months (range 6.9-158.2). Radiotherapy was delivered to 32% of infants, with 41.7% receiving brachytherapy. Radical surgery was performed in 62% of infants. All received chemotherapy according to risk group and there were no toxic deaths. Five year OS for infants is 88.4% (95%CI 80.3-93.2) which is significantly better compared to that of 1-3 year olds OS of 78.0% (95%CI 73.2-82.0; p=0.0204). The OS for 3-18 year olds is 81.0% (95%CI 78.5-83.2). Five year EFS for infants is 72.5% (95%CI 62.8-80.0; p=0.3263), 66.1% (95%CI 61.0-70.7) for 1-3 year old and 72.4% (95%CI 69.7-74.9) for 3-18 years old.

Conclusions: Infants treated on RMS2005 study have excellent OS which compares favourably to series published so far. The EpSSG-RMS2005 chemotherapy regimen, combined with increase in application of adequate local therapy, and improvements made in imaging and supportive care may have contributed to these results.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0217 / #189 | Clinical Characteristics and Outcomes of Infants Compared with Children Diagnosed with Rhabdomyosarcoma: Analysis of Surveillance, Epidemiology, and End Results Data from 2000 to 2016

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Background and Aims: Rhabdomyosarcoma (RMS) is the most common soft-tissue sarcoma of childhood, but occurs infrequently in infants (< 1 year). Historically, infants with RMS have a worse survival compared to other pediatric age groups. We assessed the clinical features and treatment factors associated with survival comparing infants to children aged 1-9 years diagnosed with RMS.

Methods: Children aged < 10 years diagnosed with RMS between 2000-2016 were identified using the Surveillance, Epidemiology, and End Results (SEER) database. Descriptive statistics were used to assess demographic, clinical, and treatment characteristics of infants and children with RMS. Kaplan-Meier estimates and Cox proportional hazards regression were performed to assess for factors associated with survival.

Results: Age < 1 year was independently associated with an increased risk of mortality (hazard ratio for death [HR] 1.69; p=0.005). Compared to children aged 1-9 years, fewer infants received standard of care therapy, i.e. chemotherapy combined with local control (surgery and/or radiation) (86.8% vs 75.7%; p=0.009). Infants were less likely to receive radiation therapy (34.0% vs 66.4%; p<0.001) and more likely to receive surgery (68.9% vs 57.5%; p=0.02) than children aged 1-9 years. Across age groups, chemotherapy combined with local control was significantly associated with reduced mortality (HR 0.43; p=0.008). Alveolar histology (HR 1.61; p<0.001) metastatic disease (HR 3.20; p<0.001) and Hispanic ethnicity (HR 1.58; p<0.03) were negatively associated with survival.

Conclusions: Age of < 1 year is an independent risk factor for increased mortality from RMS. However, infants treated with chemotherapy combined with local control experience improved outcomes compared to those who receive less intensive therapy. Withholding standard of care therapy in infants out of concern for late effects must be considered with caution. Factors that may contribute to differences in survival including molecular differences, Intergroup Rhabdomyosarcoma Study group and stage, and specifics of therapy, should be further explored.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0218 / #352 | Can Response to Second Line Therapy in Relapsed/Refractory Rhabdomyosarcoma Predict Subsequent Treatment Response(s) in Children and Young People?

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Background and Aims: Rhabdomyosarcoma is the commonest soft tissue sarcoma in children and young people. Disease progression during or after multimodality treatment continues to be associated with poor outcomes. Whilst conventional cytotoxic chemotherapy is usually offered at first recurrence/progression, few data exist to evaluate the role of systemic therapy beyond second line treatment.

Methods: We undertook a retrospective institutional review of patients aged <25 years with relapsed rhabdomyosarcoma (<6 or ≥6 months after completion of treatment) or on-treatment progression (with or without initial response) diagnosed between January 2000 and December 2018, treated with ≥2 chemotherapy lines.

Results: 64 patients (33 alveolar, 39 metastatic) received ≥2 treatment lines including 110 (94%) with conventional chemotherapy and 7 (6%) with targeted agents. Recorded response rate to 2nd line therapy was 9/34 (27%) for on-treatment progressions and 17/30 (57%) for relapses. 31 patients (48%) received 3rd line, 15 (23%) 4th line, 3 (5%) 5th line, 2 (3%) 6th line and 1 (2%) 7th and 8th line therapy. Recorded objective response rates (complete or partial response) were: 3rd line, 7/30 (23%), 4th line, 2/13 (15%), 5th line 1/3. No patients responded to 6th or later lines of treatment. Among the responders to 3rd line therapy, 4/7 had previously shown response to 2nd line. Of 10 patients with first relapse occurring ≥6 months post-treatment, no patient responded to 3rd line treatment unless they had previously shown response to 2nd line treatment.

Conclusions: Our institutional data demonstrate that some patients with relapsed rhabdomyosarcoma can respond to multiple lines of treatment. Although validation in a larger cohort is required, in patients who relapse following complete remission or stable radiologic residual disease ≥6 months after frontline treatment, failure to respond to systemic therapy at first relapse may predict response failure to subsequent salvage regimens. For such patients, novel therapeutic approaches may be indicated.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas*

PV0219 / #374 | Kaposi Sarcoma in Childhood: A Report from Single Centre

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Background and Aims: Kaposi sarcoma (KS), a soft tissue tumor with a variety of presentations, is caused by HHV-8. Four clinical forms (classical, endemic, HIV- related, iatrogenic) are described. The objective of this study was to report our institutional experience with KS over a 30-year period.

Methods: From 1990 to 2020, 307 children with non-rhabdomyosarcoma soft tissue sarcomas (nRMS-STs) were admitted to the Pediatric Oncology Department at Hacettepe University. Of these, nine children with KS were evaluated retrospectively.

Results: There were 4 males, 5 females. Median age was 10 (3–16) years. Presentation symptoms were red to violaceous nodules and

plaques on the skin in 6 patients, and on gingiva in one. Skin lesions were solitary in two cases, limited in one and disseminated in three. Two patients were admitted with lymphadenopathy. One of these patients also had fever and hepatosplenomegaly. Posttransplantation iatrogenic KS was determined in four cases. In addition, one patient had primary immune deficiency. Three of those four presented with mucocutaneous involvement while one had systemic disease. Endemic form was determined in one patient who had a single cervical lymph node. Three patients had classical KS with skin involvement. After total excision, no more treatment was applied for two patients. In patients with posttransplant KS, doses of immunosuppressive drugs were reduced. Cyclophosphamide was given to two patients, interferon-α to two, corticosteroid to one, and immunoglobulin and corticosteroid to one. One patient died of renal failure. Eight patients (60%) were alive and disease free at a median of 42 (18 to 172) months. Five-year overall and event free survival rates were 83.5% and 35.6%.

Conclusions: Kaposi sarcoma is rare in childhood. In our study, it constituted 2.9% of nRMS-STs. The most common (44.4%) type was iatrogenic form. Although recurrent/progressive disease was seen in 55.5% of cases, none died of KS.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas*

PV0220 / #503 | Safety and Pharmacokinetics of Nivolumab for Japanese Pediatric Patients with Relapsed or Refractory Solid Tumors or Hodgkin Lymphoma (HL) (NCCH1606): A Single-Center, Phase-1 Trial

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Background and Aims: This phase-1 trial aimed to determine the safety, pharmacokinetics, and recommended phase-2 dose (RP2D) of nivolumab, a PD-1 blocking monoclonal antibody for Japanese pediatric patients with relapsed/refractory solid tumors and Hodgkin lymphoma (HL).

Methods: We conducted a single-center phase 1 study for patients aged 1–24 y with relapsed/refractory solid tumors or HL who had undergone ≥ 2 previous treatments. Patients received 3 mg/kg nivolumab over 60 min every 2 wk. The primary endpoint was the frequency of adverse events equivalent to the dose-limiting toxicity.

Results: Twenty-six patients with a median age of 13.5 y (range 2–20 y) were enrolled between May 29, 2017 and August 20, 2019. No adverse events equivalent to dose-limiting toxicity were observed, and 3 mg/kg nivolumab was determined as the RP2D. The most common grade-3–4 adverse events were lymphopenia (26.9%), anemia (19.2%), gamma-glutamyl transpeptidase increase (11.5%), neutropenia (7.7%), and maculo-papular rash (7.7%). Most immune-related adverse events

were low grade; the prevalence was as follows: hyperthyroidism (2 patients), hypothyroidism (2 patients), and allergic rhinitis (1 patient). The pharmacokinetic data of Japanese children were similar to those of Japanese adults. One patient with HL was enrolled and showed complete response. Among the 16 evaluable patients with solid tumors, 1 with neuroendocrine carcinoma achieved partial response, although this tumor was of maternal origin (N Engl J Med 2021; 384:42-50). No objective responses were observed in patients with neuroblastoma, soft tissue sarcoma, and osteosarcoma.

Conclusions: Nivolumab is safe and well-tolerated in Japanese children and adolescents and showed clinical activity in HL. The defined RP2D of nivolumab in Japanese pediatric patients was equal to the RP2D determined in the Children's Oncology Group (COG) phase-1 trial. Nivolumab showed no significant anti-tumor activity in common pediatric solid tumors. Funding: Ono pharmaceutical Co., Ltd. Clinical Trial Information: UMIN000026497

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0221 / #512 | Cell-Free DNA in Rhabdomyosarcoma: Unravelling The Potential

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Background and Aims: In children with rhabdomyosarcoma, 30% experience relapse, despite current treatment based on risk stratification. We aim to improve risk stratification and response monitoring by detection of tumor-derived, cell-free DNA (cfDNA). Here we investigated different approaches for detection of tumor-derived cfDNA in children with rhabdomyosarcoma.

Methods: cfDNA was isolated from plasma from Dutch patients with rhabdomyosarcoma, treated on the EpSSG RMS2005 protocol. Droplet digital PCR (ddPCR) for the hypermethylated tumor-suppressor gene RASSF1A and for the patient-specific breakpoint of the fusion gene in the alveolar subtype was performed. Shallow Whole Genome Sequencing (shWGS) for copy number aberration (CNA) detection, cell-free Reduced Representation Bisulphite Sequencing (cfRRBS) and Whole Exome Sequencing (WES) on cfDNA were also performed.

Results: Overall, tumor-derived cfDNA was detected in 35/54 (65%) at diagnosis by at least one technique. Hypermethylation of RASSF1A was detected in 14/25 patients at diagnosis. The patient-specific fusion breakpoint was detected in 3/3 patients at diagnosis and in 1/1 patient

at relapse. CNA were detected in 16/31 patient-samples by shWGS at diagnosis and in 1/2 at relapse. cfRRBS of 26 diagnostic samples from patients with embryonal subtype, accurately identified 21 as having embryonal rhabdomyosarcoma, 3 were classified as alveolar and in 2 no tumor-derived cfDNA was detected. WES of cfDNA was performed in 3 patients at diagnosis, which showed a well-recognized mutation in 1 patient (NRAS) and several non-recurrent mutations in the other two patients.

Conclusions: This pilot study demonstrated ability to detect tumor-derived cfDNA in 65% of patients with rhabdomyosarcoma at diagnosis, both embryonal and alveolar subtype. Furthermore, tumor-derived cfDNA can successfully be investigated by shWGS, WES, cfRRBS and ddPCR. Follow-up studies are initiated with prospective standardized sample collection in a uniformly treated cohort, to determine which techniques can be used to guide therapy at diagnosis and throughout treatment.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0222 / #533 | Targeting Bet Proteins in Combination with Salvage Therapy as a Novel Therapeutic Option for Aggressive Pediatric Rhabdomyosarcomas

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Background and Aims: Outcomes for patients with high-risk or relapsed rhabdomyosarcoma (RMS) remain poor with survival rates below 30%, and effective therapies are needed. Alveolar RMS (ARMS) is characterized by either of two transcriptionally active fusion proteins, PAX3/7-FOXO1. There is growing evidence that bromodomain-extraterminal domain (BET)-containing protein, specifically BRD4, can stabilize PAX3/7-FOXO1 leading to increased upregulation of downstream target genes such as IGF1R, MYCN, and ALK. With the development of potent and selective BET inhibitors, we hypothesize that treatment of ARMS cell lines with either monovalent BET inhibitor (birabresib [OTX-015]) or bivalent BET inhibitor (AZD5153) in combination with salvage cytotoxic therapy would serve as an efficacious and well-tolerated therapeutic option.

Methods: Topotecan was prioritized for combination as a well-established and safe component of ARMS salvage therapy. Pediatric ARMS cell lines (Rh30, Rh41, and SJCRH30) were treated with

different dose ratios of OTX-015 or AZD5153 in combination with topotecan for 5 days. Cell proliferation was determined by methylene blue staining. IC50 and combination index values were determined using the Chou and Talalay method and CalcuSyn v2 software. To set the stage for combination therapy approaches, dose-finding studies for AZD5153 and topotecan were conducted in patient-derived xenograft (PDX) ARMS models established at our institution.

Results: Combination index and Bliss independence analysis identified dose ratios of OTX-015 or AZD5153 plus topotecan that resulted in additive growth inhibition in ARMS cell lines. Dose-finding studies indicated inhibition of PDX growth at 1.25 mg/kg – 2.5 mg/kg AZD5153 twice daily and 2.5 mg/kg topotecan once daily.

Conclusions: In a panel of pediatric ARMS cell lines, BET inhibition in combination with topotecan produces additive cytotoxic effects in vitro. Studies are underway to investigate the antitumor effects of these drug combinations in ARMS PDX models, and to better understand the BET/BRD4-dependent mechanisms driving these effects.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0223 / #677 | Immunotherapy Based on Autologous Hematopoietic Cells in Pediatric Cancer

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Background and Aims: Cell immunotherapy through autologous dendritic cells (DC) presents a hopeful strategy for the treatment of adult tumors. An objective response is 30%. To estimate results of treatment of children with solid tumors using immunotherapy (dendritic cell vaccine).

Methods: Twenty-six patients aged 2–17 years with solid tumor received adjuvant immunotherapy with autologous DC vaccine between 2016 and 2021. The mean age was 8.6 years, the median age was 6.6 years. In the main soft-tissue sarcoma was diagnosed (13 children or 50%). Four patients had glioblastoma (15.4%), 4 children (15.4%) had osteogenic sarcoma, 3 (11.5%) – sarcoma Ewing, 1 (3.85% each) – neuroblastoma and atypical teratoid/rhabdoid tumor. Nine patients (34.6%) were treated with DC vaccine in combination with the third-line chemotherapy regimens, 10 children (38.5%) – with the second-line chemotherapy and 7 (26.9%) – with the first-line

chemotherapy. To estimate treatment effectiveness, the paired-comparison method was used. The control group consisted of 36 patients received chemotherapy in a maintenance regimen.

Results: The mean number of DC vaccine administrations was 26 (from 2 to 139). An objective clinical response with a maximum duration of 67 weeks was achieved in 7 patients (26.9%), a recurrence was revealed in one case (3.8%) and the progression of disease was detected in 16 cases (61.5%). Two patients refused to continue immunotherapy. No complications with immunotherapy have been recorded. The overall survival in the group with dendritic cell vaccine was 34.6%, and in the control group – 0%. The median time to progression in the experimental (DC vaccine) group was 5.1 months, in the control group – 1 month.

Conclusions: DC immunotherapy is feasible and safe, this treatment may generate tumor-antigen-specific immune responses in many patients, and has high potential in combination therapy of malignant tumors in children.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0224 / #735 | Chest Wall Reconstruction in Pediatric Oncology Patients: A Systematic Review

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Background and Aims: Resection of pediatric chest wall tumors may result in large defects requiring reconstruction for function and cosmesis. Multiple reconstructive methods with variable outcomes have been described. We performed a systematic review of the literature to describe commonly used approaches and outcomes.

Methods: A systematic literature search was performed of English-language publications describing chest wall tumor resection and reconstruction using implantable materials in patients <22 years old. Soft tissue resection only, sternal resection, and reconstruction by primary repair or muscle flaps alone were excluded. Data were collected on demographics, diagnosis, reconstructive method, and outcomes. Results were compared between patients who underwent rigid chest wall reconstruction and patients who had mesh reconstruction.

Results: There were 56 articles with 252 patients included. The median age was 12 years old. Most tumors were malignant (n=172, 68.3%). The most common diagnosis was Ewing's sarcoma (n=565, 25.8%), followed by Askin's tumor (n=16, 6.3%) and osteosarcoma (n=16, 6.3%). A median of 3 ribs were resected (range 1–12). Synthetic mesh was the most common reconstruction material (n=191, 75.8%), followed by biologic mesh (n=27, 10.7%). Rigid prostheses were used in 34 patients (13.5%). There were 19 post-operative complications reported, including eight seromas and surgical site infections, two titanium bar dislocations, and one subclavian artery puncture by a titanium patch. There were 28 cases of scoliosis reported. There were no significant

differences in complications (23.3% rigid vs. 11.2% mesh, $p=0.1$) or scoliosis (23.8% rigid vs. 21.9% mesh, $p=0.8$) by reconstruction method. The median duration of follow-up was 24 months.

Conclusions: Chest wall reconstruction following tumor resection in children poses many challenges, with a need to choose a durable material that restores chest wall support, while considering future growth. In this review of the literature, there were no differences in post-operative complications or scoliosis development by reconstruction method.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0225 / #752 | Impact of First-Line anti-GD2 Therapy in Patients with Newly Diagnosed Ewing Sarcoma

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Background and Aims: Despite multimodal therapy, the prognosis of patients with metastatic Ewing sarcoma (ES) remains poor, indicating the urgent need for new therapies. The disialoganglioside GD2, a well-established tumor-associated antigen, has been shown to be expressed by 40–90% of ES cells, making it a suitable therapeutic target in ES. Here we report 4 cases with newly diagnosed localized or metastatic, GD2-positive ES treated with the anti-GD2 antibody dinutuximab beta (DB).

Methods: Three patients presented with metastatic disease and 1 patient with localized, but progressive disease. GD2 expression was confirmed in all 4 patients by immunohistochemistry in paraffin-embedded tumor specimen. All patients received 6 cycles of VIDE (Vincristine, Ifosfamide, Doxorubicin, Etoposide) induction chemotherapy, followed by 8 cycles of VAI (Vincristine, Actinomycin-D, Ifosfamide) consolidation chemotherapy. After the first VAI cycle, the patients received 1 cycle of DB (10 mg/m²/day over 10 days), followed by surgery and irradiation of the primary tumor (54 Gy). The remaining 4 cycles of DB were given after VAI cycles 3, 5 and 7, and after completion of VAI therapy.

Results: One patient had no primary tumor after the first VAI/DB cycle, whereas 3 patients exhibited remaining primary tumors. All of these 3 patients presented with complete necrosis of their primary tumor upon histologic evaluation of the resected tumor tissue. Granulocyte recovery was faster in VAI cycles with DB than in those without DB. All 4 patients achieved complete remission, without evidence of relapse. The median duration of response from diagnosis was 13 months [range 12–21 months]. Treatment with DB was well tolerated, with no severe side effects reported.

Conclusions: First-line anti-GD2 immunotherapy in patients with GD2-positive ES appears to be feasible and promising, with no severe toxicity. A confirmatory clinical trial should be carried out.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0226 / #798 | Potential Benefit of New Target Inhibitors in Neurotrophic Tropomyosin Receptor Kinase (NTRK) Fusion Transcript Tumors: A Historical Cohort Analysis

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Background and Aims: NTRK fusion transcripts have been identified in numerous tumors associated with variable outcomes, some requiring aggressive therapy, and sometime new TRK inhibitors (TRKi). We aimed to assess the benefit of these promising drugs through a retrospective multicenter historical cohort analysis.

Methods: Fifty-eight patients with NTRK fusion transcript tumors (NTRK-FTT) were identified between 2001 and 2019, including 36 infantile fibrosarcomas (IFS) of soft tissue or kidney, 13 non-IFS conjunctive and 9 central nervous system (CNS) tumors. Treatment burden before any TRKi therapy was retrospectively compared with outcome, i.e., survival without aggressive local therapy (SWALT, defined by absence of mutilating surgery or radiotherapy), survival without aggressive medical therapy (SWAMT, survival without anthracyclin or alkylating agents); and survival without any aggressive therapy (SWAT). **Results:** Median ages were 0.2y [0.1-0.5], 3.8y [1-20] and 12y [1.2-14.5] respectively for IFS, non-IFS conjunctive and CNS groups. For

non-IFS conjunctive tumors, pathological diagnosis was spindle cell sarcoma (6), benign tumors (4), lipofibromatosis-like neural tumor (1), myxoid liposarcoma (1), and osteosarcoma (1) and for CNS group: high grade glioma (4), glioneuronal (2) or neuro-epithelial tumor (2), ATRT (1). Overall, 50 patients had surgery (3 mutilating), 36 chemotherapy (including 17 with alkylating agents/anthracyclin), 9 radiotherapy, 2 "wait and see strategy" and 9 had TRKi. After a median follow-up of 66.7 months [range, 1-226], patients status were (unknown n=1): complete remission (n=31), alive with stable residue (n=10), partial response (n=1), stable disease (n=3), progressive disease (n=2) and death (n=10). In the IFS group, 5y-SWALT was 88.2% [95%IC, 77.9-99.8], 5y-SWAMT and 5y-SWAT were 74.5% [61.4-90.5]. In non-IFS conjunctive tumors, 5y-SWALT was 64.3% [38.5-100], 5y-SWAMT and 5y-SWAT were 49% [24.0-99.8]. In CNS tumors, 5y-SWALT and 5y-SWAMT were 22.2% [6.5-75.4]; 5y-SWAT was 11.1% [1.8-70.5].

Conclusions: Survival of IFS-patients is significantly higher than other NTRK-FTT. This latter group and specific cases of IFS might therefore potentially benefit from TRKi therapy.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas*

PV0227 / #815 | Larotrectinib in NTRK Fusion-Positive Locally Advanced or Metastatic Soft Tissue Sarcomas in Children: A Case Series

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Background and Aims: NTRK (neurotrophic tropomyosin receptor kinase) gene fusions are oncogenic drivers in a variety of pediatric and adult tumors. Larotrectinib, is an orally administered highly selective TRK inhibitor, that is suggested to be effective in some cancers harboring NTRK fusions.

Methods: Larotrectinib [provided through compassionate use (Single Patient Access)] was used in five children with NTRK gene fusion-positive locally advanced or metastatic soft tissue sarcomas. Demographic characteristics and outcome were assessed.

Results: Five patients with a median age at diagnosis of 2.6 years (range, 0.5-7.5 years) used larotrectinib at a median age of 5.3 (1.2-10.5) years. The diagnosis were infantile fibrosarcoma (IFS) (2 patients), rhabdomyosarcoma (RMS) (1 patient), non-rhabdomyosarcoma soft tissue sarcoma (NRSTS) (2 patients). Three had NTRK1 and two NTRK3 fusions. All patients used larotrectinib after at least one first-line chemotherapy. In one of the locally progressive IFS patients, partial response was attained and the tumor could be surgically removed with negative margins. He is with alive no evidence of disease for 12 months. Larotrectinib was used as the 2nd (IFS), 7th (NRSTS), 8th line therapy (RMS) in three patients and stable

disease (SD) was attained for a median of 3 months (2-7 months), after which progression occurred. The patient with IFS is alive with disease (AWD). Two patients (RMS, NRSTS) died due to progressive disease (DOD). A patient with NRSTS has recently begun larotrectinib for local microscopic disease. In one of the four patients (25 %) with measurable disease objective response was attained. In three heavily pretreated patients SD for 2-7 months were attained. No major adverse effect due to larotrectinib was observed.

Conclusions: Larotrectinib, an oral TRK inhibitor, was well tolerated in this limited series. For children with locally advanced or metastatic NTRK fusion-positive STS, larotrectinib, an oral TRK inhibitor therapy may be considered.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas*

PV0228 / #838 | Rhabdomyosarcoma in Pediatrics: A Population-Based Study of Secondary Primary Neoplasms

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Background and Aims: To investigate, in a population-based study, the risk of developing second primary neoplasms (SPNs) after a diagnosis of primary rhabdomyosarcoma.

Methods: Data on a cohort of patients with a diagnosis of primary rhabdomyosarcoma as their first malignancy were extracted from the Surveillance, Epidemiology, and End Results US registry from 1975 to 2017. Observed-to-expected ratio of developing SPNs was calculated to estimate relative risk (RR) and associated 95% confidence interval (CI), which were compared to a reference population (RP) matched for age, sex, race, and calendar year.

Results: A total of 1,342 patients (801 males, 541 females) with primary rhabdomyosarcoma were identified, 3.5% of whom developed SPNs (49/1,381). Mean age at rhabdomyosarcoma diagnosis was 8 years old, while that at SPNs diagnosis was 23 years old. There was a significant risk of SPNs development in rhabdomyosarcoma patients compared to the baseline [RR: 5.9 (95% CI: 4.4-7.8); p < 0.05]. Embryonal rhabdomyosarcoma is associated with the highest risk of developing SPNs [RR: 5.6 (95% CI: 3.8-8.1); p < 0.05].

Conclusions: Patients with primary rhabdomyosarcoma experience a 3.5% increased risk of SPNs compared to the RP. Embryonal rhabdomyosarcoma is associated with the highest risk of developing SPNs. Patients who are diagnosed with primary rhabdomyosarcoma should be followed with these risks in mind.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas*

PV0229 / #853 | EBV Associated Smooth Muscle Tumors in Children with Primary Immunodeficiency Disorders

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Background and Aims: Epstein-Barr virus positive smooth muscle tumor (EBV-SMT) is a very rare tumor. It is usually seen HIV infection or immunosuppressive treatment after solid organ transplantation. In a small group of pediatric patients, EBV-SMT is associated with primary immunodeficiency disorders (PID). Up to date, 16 cases of EBV-SMT associated with PID have been reported. Here, we present institutional experience with 9 patients.

Methods: Nine patients with EBV-SMT associated with PID diagnosed and followed-up between the years of 2000 and 2021 were included in this study. The clinical features and treatment results of patients were recorded from patient files and hospital information system retrospectively.

Results: The median age was 10 years (4.3-18 years; M/F=6/3). The associated PIDs were; combined immunodeficiency(n=4), JAK3 deficiency(n=1), ITK deficiency(n=1), DOCK-8 deficiency(n=1), STAT-1 gain of function deficiency (n=1), Hyper IgE syndrome (n=1). Tumor was located in the adrenal (n=6), liver (n=4), gall bladder (n=2), vertebral intradural (n=2), spleen (n=1), lung (n=1), ileum (n=1), femur posterolateral soft tissue (n=1). Five patients presented with multifocal lesions while 3 patients presented with a solitary lesion. Complete surgical excision was performed on six patients. Three patients were treated with chemotherapy and one patient received radiotherapy. Median follow-up time was 22 months. Out of the 5 patients who died, 3 died with disease while other 2 died due to immunodeficiency or hematopoietic stem cell transplantation (HSCT) related reasons. Four patients are alive at 3, 52, 98, 162 months from the diagnosis.

Conclusions: Patients diagnosed with EBV-SMT should be evaluated for PIDs in the absence of HIV infection or solid organ transplantation. Complete resection of the primary tumor is the mainstay of the treatment however the long-term survival is poor without definitive immune reconstitution. Combined immunodeficiency was the most common diagnosis among patients with EBV-SMT and curative treatment is HSCT.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0230 / #870 | Children with Progressive and Relapsed Pleuropulmonary Blastoma: An European Collaborative Analysis

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Background and Aims: Background Children with progressive (PD) or relapsed (RD) disease of pleuropulmonary blastoma (PPB) type II/III are known to have a very poor outcome.

Methods: A retrospective review of children with type II/ III PPB and PD and RD registered in national, European databases and CWS trials was performed. Patients with diagnosis of primary disease between 2000-2018 were selected.

Results: Nine patients with initially IRS-group II (n=1), IRS-group III (n=6) and IRS-group IV (n=2) experienced PD despite conventional surgery/chemotherapy (CHT, n=9) and radiotherapy (RT, n=1) with a median time to progression of 0.58 years [range 0.02-1.27] from diagnosis. All died despite multimodal treatment. Twenty-six children experienced RD after primary disease with IRS group I (n=4), II (n=5), III (n=12) and IV (n=5). Chest relapse occurred in 12 patients and metastatic in 14: central nervous system (n=10), bone (n=1), not specified (n=3). Median age at relapse was 4.3 years [1.7- 15.1], time to relapse was 1.03 years [0.03- 2.95]. Patients were treated with salvage CHT (n=20), surgery (n=10), and/or RT (n=10). A second complete remission was achieved in 10/26 patients: Seven patients were alive in second CR at a median of 4.24 years (2.09-14.63), 2 in partial remission/ stable disease (under new agent treatment after 2nd relapse) and one patient was lost of follow up after CR. Five-year EFS and OS for patients with RD is 37% (±19, CI 95%). Convince local therapy (salvage surgery and salvage RT) had an impact on OS, (p=0.03 and 0.02 respectively). Local treatment of CNS metastases seems very important.

Conclusions: Until now, children with progressive PPB could not be salvaged. New therapy is warranted for these patients. Cure of patients with RD of PPB type II/III is possible, even if rare, with multimodal treatment including new resection and radiotherapy associated to second line chemotherapy.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0231 / #895 | Malignant Peripheral Nerve Sheath Tumor (MPNST) and MPNST-Like Entities Defined by DNA Methylation Profile in Pediatric and Juvenile Population

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Background and Aims: MPNST accounts for 3-10% of pediatric sarcomas, with 50% arising in type 1 neurofibromatosis (NF1). Sporadic MPNST diagnosis may be challenging, due to the absence of specific markers, apart from immunohistochemical H3K27me3 loss. DNA methylation profile (DM) is a useful tool for brain and mesenchymal neoplasms categorization. MPNSTs exhibit a specific DM. Recently, another group was recognized, defined as MPNST-like, including MPNSTs with retained H3K27me3 expression and clinical/histological features still not well explored in pediatric patients. This study aims to evaluate DM in pediatric/juvenile MPNSTs and its usefulness for diagnostic purposes.

Methods: The study includes 2 groups of pediatric/juvenile malignancies (age <25 years). Group 1: Neoplasms with MPNST diagnosis selected from the Bambino Gesù Children's Hospital (OPBG) Pathology database and consultation files. The DM for these cases was analyzed. Group 2: Neoplasms with MPNST and MPNST-like DM according to sarcoma classifier, with MPNST or other histological diagnoses, selected from the DM reports performed at OPBG. Clinical/histological and genetic/molecular information were collected and further immunohistochemical (Desmin/MyoD1/Myogenin/BCOR) and molecular (RT-PCR/NGS) tests were performed

Results: Group 1: DM analysis in ten MPNST (age range 0.2-24 years; 5 in NF1) confirmed the diagnosis in 4 (all in NF1). Two cases showed a MPNST-like (1 undifferentiated sarcoma and 1 Triton tumor), and one mixed (MPNST in NF1). The remaining 3 cases were classified as DICER1, BCOR and RMS-MYOD1mut. NGS confirmed these findings. Group 2: 8 out of 150 sarcomas with available DM, excluding those of group 1, showed a MPNST-like DM. The heterogeneous category included two groups: high-grade undifferentiated sarcomas/MPNST (n=2) and low-grade mesenchymal neoplasms (n=6) with tyrosine-kinase gene fusion (n=2).

Conclusions: DM is useful in diagnosing MPNST in NF1 and discriminating morphologic mimics classified as MPNST in pre-molecular era (BCOR-sarcoma, DICER1-associated-sarcomas). MPNST-like category is a 'hybrid' group in pediatrics and deserves further investigation.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0232 / #903 | BCOR-ITD Pediatric Tumors from Soft Tissue/Kidney Display Specific DNA Methylation Profiles

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Background and Aims: BCOR-ITD is a genetic alteration common to different pediatric malignancies including primitive myxoid mesenchymal tumors/undifferentiated sarcomas (PMMTI/UND), clear cell sarcomas of kidney (CCSK) and brain high-grade neuroepithelial tumors (HG-NET). Recently this genetic alteration has been reported in high-grade endometrial stromal sarcomas (HGESS). Despite common gene expression profile, HG-NET and PMMTI/UND show different immunophenotype. Our aims were to investigate DNA methylation (DM) and copy number variation (CNV) profiles in pediatric BCOR-ITD to better define the contribution to the diagnostic work-up and to explore epigenetic differences between brain- and soft-tissue-BCOR tumors.

Methods: We analyzed DM and CNV profiles by Illumina EPIC-beadChip of 35 pediatric primary tumors (18 PMMTI/UND, 11 CCSK, 3 HG-NET, 1 BCOR-CCNB3, 2 Spindle cell/NTRK fusion and 4 relapses (3 PMMTI/UND, 1 HG-NET) from 36 patients (age range 0-7 years). BCOR-ITD was tested for 32 samples and identified in 22 cases. YWHAE-NUTM2 fusion was identified in 2 cases. The methylation data were analyzed by the sarcoma classifier and by means of bump hunting strategy, to identify differentially methylated regions (DMRs) in soft-tissue- vs brain-BCOR tumors

Results: Unsupervised hierarchical clustering analysis of DM data revealed two distinct clusters, one formed by HG-NET-BCOR, the second including the most of PMMTI/UND and CCSK. Cases with YWHAE-NUTM2 fusion classified as HGESS. The BCOR-ITD negative cases clustered with other entities including rhabdoid and MPNST-like tumors. Functional enrichment analysis of DMRs showed significantly deregulated pathways, suggesting different developmental

programs. CNV showed an overall flat profile in 18 out of 32 evaluable cases. Eleven cases showed 1q-gain (n=7, with CDKN2A/B deletion in 2 cases) or 10q-loss (n=3).

Conclusions: BCOR-ITD pediatric tumors share distinctive methylation profile representing a useful diagnostic tool. The evidence of 1q gain, and losses at oncosuppressor genes loci deserve further investigation for their role in the clinical behavior of these malignancies.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0233 / #906 | Localized Rhabdomyosarcoma: The Role of Local Control in a Single Centre

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Background and Aims: Rhabdomyosarcoma (RMS) is a heterogeneous aggressive disease both in presentation and histology. Local control is a mainstone in the multimodal treatment. Our aim was to verify if the way local control was obtained and the type of remission had a role in overall-survival (OS) and in event-free-survival (EFS).

Methods: Italian patients affected by rhabdomyosarcoma, treated following the EPSSG RMS2005 protocol between October 2005 and December 2016 were included. Exclusion criteria were: patients aged >25 years at diagnosis, evidence of metastatic lesions, tumors previously treated, pre-existing illness that could prevent treatment, previous malignant tumors and an interval between the diagnostic surgery and treatment >8 weeks. Data including histology, type of treatment, type of remission and follow-up length were collected. Survival probabilities were estimated using the Kaplan-Meier method.

Results: We identified 389 patients. 56/389 were excluded. The remaining 333 were divided in 4 groups depending on how complete remission (when obtained) at the end of treatment was determined: histologically (free margins after surgery) (HCR=113), radiologically (with an imaging showing no residual disease) (RCR=133), clinically (treatment completed with good results, but possible disease residual) (CCR=52). 35 patients did not obtained a complete remission. Relating to tumor location, HCR were higher in patients with RMS localized in so-defined "surgical sites" (p=0.0001). The statistical analysis showed that surgical sites, Low-risk group, lymph-node negative at diagnosis, IRS 1, HCR were associate with the best prognosis with significant higher OS (p<0.0001). Interestingly, in our cohort the OS in patients with CCR tended to move towards patients with HCR Focusing on RMS

localized in surgical sites, only the complete resection was associated with higher OS.

Conclusions: Patients with localized RMS in surgical sites without lymph-nodal localization who achieved HCR had the best EFS and OS. RCR does not seem to give a prognostic advantage compared to a CCR.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PV0234 / #913 | Integrative Clinical and Biopathological Analyses in Malignant Peripheral Nerve Sheath Tumors (MPNST): Results of a French Expert Center

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Background and Aims: MPNST are very rare and aggressive diseases, affecting adolescents and young adults. Identification of these tumors is still difficult without precise diagnostic criteria in pathology. Molecular tools such as CGH array (aCGH) and RNA-sequencing (RNAseq), which allow to perform fusion gene detection together with clustering and expression signature analyses, could help to define a genetic profile specific to MPNST. The aim of this study is to correlate the genetic alterations and expression profiles to the clinical data of patients. We will focus on the specific histological subtype of MPNST presenting a rhabdomyoblastic differentiation (Malignant Triton Tumors: MTT).

Methods: Clinical and biological data of MPNST patients were collected in Leon Berard Center, and centralized histopathological review was performed for all cases. We analyzed aCGH and RNAseq available profiles and will complete the analyses for all the patients. Additional WES analyses will be performed for cases with residual available material.

Results: aCGH analyses of 36 tumors (including 9 (25%) from children) pointed recurrent chromosome alterations (CDKN2A homozygous deletion, NF1 heterozygous deletion, gain of 2, 7 and 17q, loss of 9p, 10 and 11, isochromosome 8). RNAseq analyses of 29 tumors (including 8 (27%) from children) found no specific fusion transcript but indicated a cluster of 24 MPNST (83%), including 11 MTT (84%). Five other MPNST (17%) were apart from others (near rhabdoid tumors for one of them) and will be the subject of more in-depth studies. A search for clinical correlation (including NF1 status) and diagnostic/prognostic signatures, as well as a study of the oncogenic signaling pathways involved in these tumors, will complete these analyzes.

Conclusions: The use of molecular biology techniques seems to be useful to achieve the histological diagnosis of MPNST, proving the need for

integrative analysis, and could help to find new therapeutic strategies for these patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0235 / #920 | Synovial Sarcoma in Pediatrics: A Population-Based Study of Secondary Primary Neoplasms

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Background and Aims: Synovial sarcoma is a very rare soft tissue tumor that is seen more often in patients younger than 30 years old. The purpose of this study is to investigate, in a population-based study, the risk of developing second primary neoplasms (SPNs) after a diagnosis of primary synovial sarcoma.

Methods: Data on a cohort of patients with a diagnosis of primary synovial sarcoma as their first malignancy were extracted from the Surveillance, Epidemiology, and End Results US registry from 1975 to 2017. Observed-to-expected ratio of developing SPNs was calculated to estimate relative risk (RR) and associated 95% confidence interval (CI), which were compared to a reference population (RP) matched for age, sex, race, and calendar year.

Results: A total of 221 patients (120 males, 101 females) with primary synovial sarcoma were identified, 5.4% of whom developed SPNs (12/221). Mean age at synovial sarcoma diagnosis was 14 years old, while that at SPNs diagnosis was 31 years old. There was a significant risk of SPNs development in synovial sarcoma patients compared to the baseline [RR: 5.2 (95% CI: 2.7–9.0); $p < 0.05$]. Stratification by primary site revealed that synovial sarcoma of the lower extremities and hip was associated with increased risk of SPNs [RR: 6.7 (95% CI: 2.3–13.8); $p < 0.05$].

Conclusions: Patients with primary synovial sarcoma experience a 5.4% increased risk of SPNs compared to the RP. Synovial sarcoma is associated with the highest risk of developing SPNs. Patients who are diagnosed with primary synovial sarcoma should be followed with these risks in mind.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PV0236 / #935 | Tailored Liquid Biopsy Analyses of Pediatric Rhabdomyosarcoma Patients

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Background and Aims: Liquid biopsies (LBs) hold great promise for non-invasive molecular classification and surveillance of neoplasms. In contrast to tissue biopsies, LBs harbor the potential to capture spatial and temporal tumor heterogeneity. We have applied different analytical technologies to a cohort of fusion-positive ($n=13$) and fusion-negative ($n=17$) rhabdomyosarcomas (RMS) aiming to identify the most suitable approach for tumor detection.

Methods: RMS ($n=30$) were subjected to DNA methylation profiling in order to obtain information on molecular tumor class and copy number variations (CNV). LBs ($n=39$) were collected at various time points during therapy and cell-free DNA and RNA (cfDNA/RNA) were co-isolated from plasma samples. Low-coverage whole-genome sequencing (lcWGS) was performed for all LB samples. To a subset of LBs ($n=20$) we additionally applied DNA methylation profiling.

Results: Higher chromosomal instability within fusion-negative compared to fusion-positive RMS, demonstrated by frequency of CNVs, could also be retrieved from LBs based on lcWGS. lcWGS was found to be more reliable for CNV detection from LBs compared to DNA methylation profiling. DNA methylation-based sarcoma classification from LBs is feasible but heavily depends on cfDNA input amounts and quality.

Conclusions: While lcWGS was found to be reliable for monitoring fusion-negative RMS, alternative technologies, e.g., for fusion transcript detection, may be more suitable for fusion-driven RMS with low CNV content. Our study highlights the importance of considering distinct subgroup-specific molecular features for successful clinical translation of LB-based diagnostic approaches.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.e Retinoblastoma

PV0237 / #1067 | Ocular Salvage in Retinoblastoma: Outcome of 325 Eyes from a Tertiary Care Eye Hospital in North India

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Background and Aims: Early diagnosis and optimal management of Retinoblastoma (RB) can lead to high cure rates even in resource

limited settings. The rates of ocular salvage vary greatly from center to center attributable to the extent of disease and facilities available. With a view to optimize treatment for RB, a multidisciplinary team was established in 2014 at a charitable hospital in North India that primarily caters to low socio-economic strata. EUA, systemic and intravitreal chemotherapy, topical treatments were available from the outset. IAC was introduced in 2019 and brachytherapy in 2020. We hereby report the epidemiological profile and rates of ocular salvage of this cohort.

Methods: All patients' records consecutively presented to the hospital in the time period March 2014-March 2021 were analyzed retrospectively. Demographic, clinical, radiological data as well as details of treatment given and outcomes with regards to ocular salvage were recorded. All patients underwent orbital imaging to rule out optic nerve or extraocular extension. Systemic chemotherapy comprised of 6–12 cycles of vincristine, etoposide and carboplatin.

Results: A total of 286 patients presented during the study period. Out of these, data of 230 patients (325 eyes) could be analyzed. Rest either abandoned/ opted for another centre/ were referrals for prosthesis or had incomplete data. The median age at diagnosis was 22 months for unilateral and 15 months for bilateral cases. Male: female ratio was 1.25:1. Unilateral and bilateral involvement was seen in 135 (58.7%) and 95 (41.3%) patients respectively. Intraocular and extraocular disease was seen in 187 (81.3%) and 43 (18.7%) cases respectively. Enucleation was done in 175 (53.8%) eyes. Overall globe salvage could be achieved in 150 (46.1%) eyes.

Conclusions: Even with access to expanded armamentarium of multimodal treatment options in developing countries, the challenges are real with advanced stage at presentation, significant burden of extraocular disease and high rates of abandonment.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.e Retinoblastoma

PV0238 / #1101 | Comparison of Eye-Survival after Two Different Systemic Chemotherapy Regimens in Children with Retinoblastoma

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Background and Aims: Eye-preserving therapy in retinoblastoma comprises systemic chemotherapy, but studies analyzing the efficacy of different chemotherapy regimens are scarce.

Methods: The efficacy of two different eye-preserving chemotherapy regimens (VEC vs. CyVEC) was compared in a prospective non-interventional observational study including children diagnosed with retinoblastoma between 2013 and 2019 in Germany and Austria. CyVEC was given from 2013-2016 and VEC from 2016-2019. Event-free eye survival (EFES) and overall eye survival (OES) of all 164 eyes (118 patients) treated with both regimens were calculated and clinical

characteristics were investigated as risk factors. Ototoxicity and incidence of second cancers were assessed as side effects.

Results: The EFES after VEC (2-year EFES: 74.3%) was higher than after CyVEC (2-year EFES: 51.8%), ($p_{\text{logrank}} < 0.001$). Advanced ICRB group was prognostic for a lower EFES ($p_{\text{logrank}} < 0.001$). The multivariate analysis showed that age at diagnosis older than 12 months and ICRB A/B/C were associated with better EFES. No second malignancy or ototoxicity was reported after a median 3.14 years of follow-up after diagnosis of retinoblastoma.

Conclusions: Despite omitting cyclophosphamide, the EFES was higher after VEC chemotherapy that contains higher doses carboplatin compared to CyVEC. The major risk factor for recurrence and loss of the eye was advanced ICRB tumor grouping. Randomized clinical trials on efficacy and side effects of eye-preserving systemic chemotherapy are required to tailor the adequate treatment protocol for retinoblastoma patients.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.e Retinoblastoma

PV0239 / #1195 | Epidemiologic Analysis of Death in Eye's Cancer and Annexes in the 0 – 19 Years Old Group in Brazil Between 2009 and 2019

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Background and Aims: The most common malignant neoplasms of the eye in pediatric patients are retinoblastoma and medulloepithelioma. These neoplasms have a rare incidence mainly because they are congenital, representing about 3% of tumors in children. This work aim is to trace the mortality profile in pediatric patients who have eye cancer and its attachments in Brazil during the year 2009-2019.

Methods: Epidemiological study with data from the Mortality Information System - SIM / National Cancer Institute - INCA.

Results: The total number of deaths registered was 355. Of these deaths, 149 of the cases (42%) were men between 0 and 4 years old and 109 of the cases (31%) were among women. When analyzing which region of the country has the most cases, the North region had an incidence rate of 0.29 cases in Brazil in the 0-4 years old group, and 0.02 in patients in the 15-19 years old group. On the other hand, the Central-west region leads in cases with patients between 5 and 9 years old, with 0.06. The highest number of cases were in the 0–1-year-old group.

Conclusions: The malignant neoplasms of the eye generate a greater number of deaths at the age when there is the highest incidence of the disease, at the beginning of life, with more cases in men patients than women. A fact that should be considered is the incidence in

specific regions of the country, and further studies are needed to understand this occurrence, given that this pathology has a genetic background.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.e Retinoblastoma

PV0240 / #1686 | The Outcome of Advance Stage Retinoblastoma in Indonesia: A Single Centre Experience

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Background and Aims: In Indonesia, retinoblastoma is the second most common malignancy in children, after leukemia. As the national referral hospital, our mortality rate is high because most patients come with advanced stages. The cause of late referral is multifactorial, including the lack of public awareness, long traveling distance to the health care, parent's fear, socio-economic problems, and lack of knowledge or poor health care facility. In this study, we describe the profile and outcome of retinoblastoma patients who came to our pediatric hematology-oncology division in 2019.

Methods: A retrospective data analysis of newly diagnoses retinoblastoma patients who come to our division and received chemotherapy in the pediatric hematology-oncology division from January to December 2019.

Results: Thirty-five new patients consulted initially with a pediatric oncologist in the outpatient clinic. There were 18 boys and 17 girls, all patients came with extraocular tumor, with a mean age of 27.7 months (range: 8-43 months). At initial presentation, there were 36 eyes affected, with 16 patients had the tumor on their left eye, 12 patients on their right eye, and 6 patients with bilateral tumor. We found 7/35 patients (20%) with marrow infiltration at the initial bone marrow aspiration. Thirteen patients died and fourteen patients were lost to follow up at the time we reported.

Conclusions: Same as other developing countries, our retinoblastoma mortality rate was high in our center due to the advanced stage at the initial diagnosis. The early detection and immediate referral is the main key to lower the mortality rate, especially in archipelago country. The data registry and health care facilities should be improved in all aspects. We need strong action and commitment from all stakeholders to improve our retinoblastoma care in Indonesia.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.e Retinoblastoma

PV0241 / #242 | Functional Vision and Eye-Related Quality of Life In Pediatric Retinoblastoma Patients

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Background and Aims: Little is known about the impact of retinoblastoma (RB) and its treatment on the quality of life (QoL) of childhood survivors.

Methods: This cross-sectional study included children ages 5-17 years who completed treatment for RB. After informed consent, parents and children completed the Pediatric Eye Questionnaire (PedEyeQ) to assess eye-related QoL. Results are reported on a scale from 0 to 100. Standardized normals for PedEyeQ score >90. Visual acuity (VA) was also assessed. Multivariate linear regression determined associations between domain scores and enucleation, chemotherapy, and bilateral VA at last follow-up appointment.

Results: Mean (SD) age for the 10 enrolled patients was 1.3 (1.6) years at diagnosis and 9.9 (3.9) at study enrollment. Male:female ratio was 1:1. Most RB patients had unilateral disease (60%), enucleation (60%), chemotherapy (80%), no motor (89%) or speech (100%) delay, and no hearing loss (90%). At least one domain score from the PedEyeQ was lower than normal (<90) for every child and for 9/10 parents. Children with worse bilateral VA had poorer scores in "functional vision" (β -coefficient -45.5; 95% CI: -72.3, -19.0; $p=0.007$); and were "bothered by eyes and vision" (-41.46; 95% CI: -75.2, -7.71; $p=0.02$). Parents reported worse "functional vision" for children with poorer VA (-50.8; 95% CI: -73.5, -28.6; $p=0.002$) and post-enucleation (18.3; 95% CI 1.7, 34.8; $p=0.04$). Parents of RB patients with poorer VA was also associated with greater "worry regarding child's visual function" (-51.7; 95% CI: -58.7, -44.8; $p<0.01$). Parents of patients post-enucleation (24.8; 95% CI: 19.7, 29.8; $p<0.001$) and those who received chemotherapy (-14.5; 95% CI: -19.9, -9.23; $p=0.001$) also demonstrated inferior scores. **Conclusions:** RB children had decreased QoL and functional vision as measured by the PedEyeQ. Enucleation, chemotherapy, and worse VA were associated with poorer functional vision, QoL, and parental worry.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.e Retinoblastoma

PV0242 / #712 | Clinical Features, Evolutionary Profile and Therapeutic Results of Retinoblastoma

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Background and Aims: Retinoblastoma is the most common eye tumor in children, but it is rare. Its incidence is 1/20 000 births. Current protocols based on chemotherapy and local treatments have improved outcomes. In this work we present the clinical, therapeutic and evolutionary characteristics of retinoblastoma

Methods: This is a retrospective study carried out at the clinical hematology and pediatric oncology department of Casablanca, over a period of 3 years from 2017 to 2019. The variables analysed were age at diagnosis, date of diagnosis, ocular affection (unilateral or bilateral), pathological staging, treatment modalities used (radiotherapy, surgery, and chemotherapy), and date of last visit.

Results: A total of 70 patients were included in this study. Male to female ratio was 1. The median age at presentation was 24 months and the median delay to diagnosis was 3 months. The disease was bilateral in 44% of cases. Leukocoria was the most common presenting feature (76%) followed by strabismus (38%). Treatment evaluation was possible in only 52 (75%) patients. The others were lost to follow up during or at the end of treatment. Initial chemotherapy was given to 76% of patients. Enucleation was performed in 48 (68%) eyes and 41% of cases have benefited from conservative treatment. Radiation therapy has not been administered to any patient. A successfully treated was achieved in 67% of patients, while deaths were reported in 7% of cases. At 3 and 5 years OS was 67% and 65% respectively.

Conclusions: Retinoblastoma is the most common intraocular malignancy in children. The diagnosis is essentially clinical. Education of doctors of first-line recognition of the telltale signs of retinoblastoma is essential for early diagnosis and satisfactory functional and aesthetic results.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.e Retinoblastoma

PV0243 / #797 | Survival Outcomes of Unilateral Retinoblastoma with Upfront Enucleation: A Decade of Experience from a Developing Country

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Background and Aims: Retinoblastoma is the most common childhood intraocular malignancy. In high income countries over the past decade upfront enucleation for unilateral retinoblastoma is least favored due to other alternatives that can help in globe preservation, but in low-middle income countries for advanced disease it is still the preferred option due to lack of resources and expertise.

Methods: This retrospective study was carried out at Shaukat Khanum Memorial Cancer Hospital and Research Centre, Pakistan. A total of 113 patients diagnosed with unilateral retinoblastoma who had upfront enucleation from July 2009 till January 2019 were included.

Results: Total 113 patients were referred to our facility diagnosed with unilateral retinoblastoma, with mean age at diagnosis 37.4 months ($\pm$ 24.5). The most common clinical presentation was leukocoria (74.3%). Patients who underwent enucleation had advanced disease; group D present in 62.8% followed by group E (32.7%). Histopathology revealed high risk features in 29 patients (25.7%) and intermediate risk in 54 patients (47.8%). Disease progression was seen in 8 patients and among them 7 patients had high risk pathology. The 3-year over-all survival was 75.2%, whereas event-free survival was 73.5%.

Conclusions: Primary enucleation for advanced unilateral disease remains the best treatment modality in low-middle income countries. Awareness about the early symptoms among the general population and health care personnel at a nationwide level is needed to facilitate early detection and lessen disease-related morbidity and mortality.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.f Liver Tumours

PV0244 / #1041 | Epidemiological Profile of Hospitalizations for Malignant Neoplasm of the Liver and Intrahepatic Biliary Ducts in Children and Adolescents in Brazil

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Background and Aims: Liver tumors are rare in the pediatric population and when they occur, are usually malignant. The most common malignant liver neoplasms are hepatoblastoma, followed by hepatocellular carcinoma. Likewise, tumors located in the intrahepatic bile ducts have a low pediatric incidence, with biliary rhabdomyosarcoma being the most common. This study aims to analyze the epidemiological profile of hospitalizations due to the liver and the intrahepatic biliary tract tumors in the pediatric population in Brazil.

Methods: Data were collected through the Hospital Information System (SIH). Between January 2008 and January 2021, data of children and adolescents diagnosed with liver and intrahepatic biliary tract tumors were analyzed. The studied variables were hospitalizations, deaths, length of stay, age group (less than 1 year, 1 to 4 years, 5 to 9 years, 10 to 14 years), sex and ethnicity (white, black, brown, yellow and indigenous).

Results: In the analyzed period, a total of 4,576 hospitalizations were reported among the studied population. The age group from 1 to 4 years had the highest number of hospitalizations (53.85%), days of stay (53.46%) and death (41.62%). The male gender (58.63%) had a higher number of hospitalizations, as well as the white ethnic group (42.09%).

Conclusions: Although rare in children, liver and intrahepatic biliary tract neoplasms are responsible for high numbers of hospitalizations, since the treatments commonly used to ensure patient survival are complete surgical resection of the tumor, chemotherapy and in some cases hepatectomy and organ transplantation. These tumors have a difficult initial diagnosis due to nonspecific symptoms, the absence of a current consensus classification system and low incidence, so there is a need for new prognostic, diagnosis and treatment tools to ensure an accurate initial diagnosis and develop less invasive treatments to provide the cure and well-being of patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0245 / #1058 | High Expression of 5-Hydroxymethylcytosine is Associated with Favorable Prognosis of Hepatoblastoma in China

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Background and Aims: This study aimed to analyze the expression of 5-hydroxymethylcytosine (5-hmC) with survival outcomes for Chinese patients of hepatoblastoma.

Methods: We collected and analyzed clinical data concerning patients aged <15 years with hepatoblastoma treated at the Sun Yat-Sen University Cancer Center or First Affiliated Hospital of Sun Yat-sen University from 2010 to 2018, based on the Children's Oncology Group protocol. Pathological specimens were collected through biopsy or surgical resection before initiation of chemotherapy. The immunohistochemical expression of 5-hmC were analyzed in tissue samples from 81 patients. The prognostic value of 5-hmC were evaluated by Cox regression and Kaplan-Meier analyses.

Results: We enrolled 81 patients with hepatoblastoma (median follow-up, 41.0 months; range, 21.1–131.6). The total recurrence rate was 25.9%. 3-year overall survival (OS) rates of strong and weak 5-hmC expression were 93.5% and 80.0%, respectively. 3-year event-free survival (EFS) of strong and weak 5-hmC expression were 86.1% and 55.9%, respectively. We discovered that strong 5-hmC expression were associated with higher OS and EFS. Multivariate analysis indicated that 5-hmC were independent prognostic indicators for OS and DFS.

Conclusions: Our findings showed that strong 5-hmC expression hepatoblastoma had lower recurrence and higher EFS and OS rates. 5-hmC expression may have prognostic relevance in hepatoblastoma.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0246 / #1104 | Clinical Analysis of 6 Cases of Neonatal Hepatoblastoma

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Background and Aims: Hepatoblastoma diagnosed one month after birth can be called neonatal hepatoblastoma. At present, surgery and chemotherapy in neonatal hepatoblastoma are challenging. We investigated the diagnosis and treatment of neonatal hepatoblastoma.

Methods: Clinical data of 6 children with hepatoblastoma diagnosed in Shanghai Children's medical center from April 2017 to December 2020 were retrospectively analyzed. All patients were clinically diagnosed as hepatoblastoma in the neonatal period. The clinical stage, risk group, vascular tumor thrombus, lung metastasis and follow-up information were retrospectively analyzed.

Results: Six patients included 2 boys and 4 girls. The gestational age was 41 weeks (38-48 weeks), birth weight was 3.9kg (3.2-4.4kg), first diagnosis age was 6 days (1-11 days). There were 1 case in stage I, 3 cases in stage II and 2 cases in stage III. There were 2 cases of epithelial mesenchymal mixed type, 2 cases of embryo fetal mixed type and 2 cases of undetermined type. There were 4 cases in high risk group, 1 case in medium risk group and 1 case in low risk group. One stage I and one stage II patients underwent tumor resection directly. Postoperative pathology confirmed that the tumor thrombus was found in the vascular, so they received postoperative adjuvant chemotherapy with an average of 5.5 courses. The remaining 4 cases were treated with preoperative adjuvant chemotherapy with an average of 2.75 courses, followed by radical hepatectomy and postoperative chemotherapy with an average of 5.5 courses. There were no postoperative complications. The average follow-up time was 39 months, and Event free survival(EFS) and Overall survival(OS) was 83.3% and 100% respectively.

Conclusions: Neonatal hepatoblastoma is very rare. It can be accurately diagnosed by clinical manifestations, abdominal enhanced CT and alpha fetoprotein level. After active surgery and adjuvant chemotherapy, the prognosis is pretty good, but it still needs further experience in diagnosis and treatment.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0247 / #1108 | Characteristics and Outcome of Children with Hepatoblastoma and Lymph Node Involvement

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Background and Aims: Hepatoblastoma is the most common liver malignancy in childhood. Risk-stratification based on the Pre-Treatment Extent of Tumor (PRETEXT) system has become the standard approach which in turn relies on defining the extent of liver involvement and a number of associated “annotation factors”. Lymphadenopathy is an uncommon occurrence in hepatoblastoma. Here we assembled a cohort of patients with lymph node positive hepatoblastoma and presented their clinical features.

Methods: We reviewed the population-based pediatric oncology database from Hong Kong (1996-2020, n=76) and the Chinese Children's Cancer Group hepatoblastoma (CCCG-HB) clinical registry (2016-2020, n=353) for children (<18 years) with hepatoblastoma. Patients with lymphadenopathy based on initial imaging were included. Baseline characteristics, treatment profile and outcome of these patients were summarized.

Results: Sixteen patients (4%) with lymphadenopathy were identified, all with intra-abdominal nodal involvement. Nine were male and the median age of diagnosis was 1.4 years (range: 0.5-5.2). PRETEXT staging was I in 1 patient, II in 4, III in 2, IV in 8 and unassigned in 1. Distant metastasis (lung) was present in 2. Additional annotation factors present included caudate involvement in 2, extra-hepatic involvement in 2, multifocality in 7, rupture in 1, portal vein involvement in 6, hepatic vein involvement in 4. Initial chemotherapy regimens included PLADO in 8, C5V in 6, C5VD in 1 and cisplatin in 1. Among patients where surgical details were available (n=14), 11 had complete tumor resection, 1 had liver transplantation, and two did not undergo surgery (1 died due to rupture). Among patients with data on outcome (n=14, median follow-up=1.2y, range: 0.1-21.6), 5 patients progressed (median time-to-progression=0.7y). Sites of progression included lung (2), transplanted liver (1), multiple lymph nodes (1), and was unknown in 1.

Conclusions: Lymphadenopathy is uncommon in children with hepatoblastoma and often associated with extensive disease involvement. Recurrence solely as lymphadenopathy may rarely occur.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0248 / #117 | Prognostic Factors and Nomogram to Predict Overall Survival for Children with Hepatoblastoma Based upon Seer Database

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Background and Aims: To investigate the prognostic factors of children with hepatoblastoma(HB) based upon the database of National Cancer Institute's Surveillance, Epidemiology, and End Results(SEER) and constructing and verifying nomogram for forecasting 3/5/10-year overall survival(OS).

Methods: From 2000 to 2016, clinical data of 717 HB patients aged under 18 years were acquired from SEER database. Both univariate and multivariate Cox's regression analyses were performed for determining independent prognostic factors for HB. Results of Cox analysis were displayed by forest plots. Then a nomogram for predicting 3/5/10-year OS rates was constructed based upon the known prognostic factors. The nomogram was validated internally and its reliability evaluated by concordance index(C-index) and calibration curve.

There were 441 boys and 276 girls. The value of OS was and the median OS was 53 months. OS rate was respectively. Cox regression multivariate analysis indicated that Age at diagnosis, tumor size, tumor stage, surgical approach and chemotherapy were independent prognostic factors for and they were further applied for constructing a prognostic nomogram. The value for internal validation of nomogram was and internal calibration curves also showed an excellent agreement between actual survival and nomogram prediction.

Conclusions: Age at diagnosis, tumor size, tumor stage, surgical approach and chemotherapy are correlated with survival and prognosis of HB children. A nomogram offers accurate and individualized prognosis and assists clinicians in customizing therapeutics.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0249 / #1447 | Outcome of Children with Hepatoblastoma: Analysis of a Single Center Experien in Tunisia About 21Cases

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Background and Aims: Hepatoblastoma (HBL) is the most common liver tumor of early childhood. According to the SIOPEL group; peri-operative chemotherapy followed by surgical resection has improved the outcome for more than 90% and the challenge is to desescalade treatment without losing efficacy. The aim of the study was to describe the clinical characteristics of our population and the survival rate.

Methods: This is a retrospective study of 21 cases of HBL treated in our center during the last 23 years (1995 - 2017). The data was collected retrospectively and analysed with SPSS19.0.

Results: The mean age at diagnosis was 2.2 years (2months-8years) with a sex ratio of 0.9. The most common presenting symptom was abdominal mass (66.6%) and the median Alpha Fetoprotein (AFP) level at the time of diagnosis was 296,500 ng/ml (extremes). All patients benefited from the couple ultrasound / TDM for the diagnosis. Liver biopsy was performed in 17 cases. Lung metastases at diagnosis were found in 3 cases. Tumors were unifocal in 17 patients and multifocal in 4 patients. Neoadjuvant chemotherapy was performed for 17patients(81%). Surgery was performed in 14 children (66.6%) followed by chemotherapy in 9 cases. The median follow-up was 53 (1 day-21 years). Evolution has been marked by a complete remission in 9 cases, local recurrence in 3 cases, and 5 cases lost of seen.

Conclusions: The prognosis of children with hepatoblastoma was improved substantially during the past few years through the efforts of all the cooperative study groups. In limited resource country, outcome need more care for multidisiplinary management.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0250 / #157 | Radiological Estimation of Liver Regeneration in Children With Hepatoblastoma Following Neoadjuvant Chemotherapy and Hepatic Resection

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Background and Aims: The liver is unique amongst solid organs, having potential for regeneration. However, platinum-based chemotherapy commonly used in treating hepatoblastoma may negatively impact this process. The aim of this study is to determine the extent of liver regeneration by the first radiological imaging following neoadjuvant chemotherapy and major hepatic resection for paediatric hepatoblastomas.

Methods: The records of all children who had liver resection for hepatoblastoma in our institution from 2012-2021 were reviewed. Expected liver volume for weight was calculated using the Noda formula, [liver volume in cm³ = 50.12(Wt^{0.78}), Wt=weight in kg]. Computed tomography (CT) imaging at diagnosis, and after completion of chemotherapy and surgery, were reviewed by a radiologist blinded to treatment regime. Liver volumetry was calculated using copyrighted software. Operative reports and chemotherapy regimens were reviewed. Patients received chemotherapy according to the International Childhood Liver Tumours Strategy Group (SIOPEL) protocol.

Results: There were 6 eligible patients (median age 2 years, range 9 months – 10 years) who received the following protocols: 1 SIOPEL 4 high risk, 2 SIOPEL 4 standard risk, and 3 SIOPEL 6 standard risk. Surgical resections done included 1 extended right hepatectomy, 1 extended left hepatectomy, two right hepatectomies, and two left hepatectomies with segment VIII segmentectomy. Histology included pure fetal type (n=1), mixed fetal/embryonal (n=2), and mixed epithelial/mesenchymal (n=3). The liver volumes of all patients measured on CT scans done at 6-8 weeks post-surgery regenerated to 95-149% of the expected volume for weight, irrespective of the duration of chemotherapy and type of hepatic resection performed. Highest volume for weight (149%) occurred in the patient post extended right hepatectomy.

Conclusions: The pediatric liver regenerates to the appropriate volume for age after neoadjuvant/adjuvant chemotherapy and hepatic resection for hepatoblastoma. Resection type and Cisplatin based chemotherapy does not seem to impede hepatic regeneration. However, larger prospective cohort studies are required.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0251 / #1668 | Comparing Outcomes of Patients with Multifocal Hepatoblastoma Treated Either with Liver Transplantation or Partial Hepatectomy

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Background and Aims: Multifocal HB has traditionally been associated with increased rates of local recurrence, unresectability after chemotherapy, and has historically been treated with transplantation (OLTx). We compared the outcomes of patients with multifocal HB treated at our institution with either OLTx or hepatic resection to determine outcomes and identify risk factors for recurrence.

Methods: We performed a retrospective chart review on all patients < 18 years of age with multifocal HB treated at our institution between 2002 and 2019. Patient demographics, operative procedure, post-operative course, pathological data, laboratory values, short- and long-term outcomes were analyzed

Results: A total of 46 patients treated for multifocal HB were identified. Of this cohort, twenty-five (54%) underwent OLTx and twenty-one (46%) underwent partial hepatectomy. Median length of

follow-up across all patients was 19.7 months (IQR 9.4-43.6 months). There was a higher number of patients with histopathology demonstrating pleomorphic features who underwent resection vs transplant (56% vs 23%). Cohorts differed in PRETEXT staging with a higher number of PRETEXT IV patients in patients undergoing OLTx compared to resection (68% vs 24% respectively, $p=.008$). Overall survival across all study patients was 76%. There was no difference in rates of recurrence (OLTx=28%, Resection=14%) or overall survival (OLTx=72%, Resection=81%) in patients who underwent either resection or OLTx ($p=0.55$ & $p=0.89$ respectively). Older patients (>72 months) experienced worse recurrence rates and survival ($p=0.005$ & $p=0.003$ respectively). Histopathology demonstrating pleomorphic features independently associated with worse rates of recurrence ($p=0.03$). The presence of associated tumor thrombus was associated with worse recurrence rates and overall survival.

Conclusions: Through proper patient selection, multifocal HB was adequately treated with either partial hepatectomy or OLTx with comparable outcome results. HB with pleomorphic features, increased patient age at diagnosis, and the presence of associated tumor thrombus for multifocal disease may be associated with worse outcomes regardless of the local control surgery offered.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0252 / #1670 | Class I Histone Deacetylase Inhibitor Therapy in the Treatment of Hepatoblastoma

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Background and Aims: Relapse/refractory hepatoblastoma (rHB) has poor survival with limited treatment options. Histone deacetylase inhibitors (HDACi) have shown promising results in treatment of various malignancies. We sought to characterize HDAC gene expression and determine the most effective HDACi strategy for HB

Methods: HDAC gene expression was screened in a cohort of 50 HB patients compared to uninvolved liver with microarray analyses and validated by quantitative polymerase chain reaction (qPCR) in several HB cell lines and pooled normal liver. Several HDACi agents including panobinostat (pan inhibitor), mocitinostat (HDAC1-3) entinostat (HDAC 1&3), and romidepsin (HDAC 2-3) were tested with cytotoxicity assays (MTT). Immunoblotting assays were used to assess changes in drug targets and induction of apoptosis (PARP cleavage) with cell lines.

Results: Patient samples showed clear differential expression of HDAC genes including strong upregulation of class I HDAC genes HDAC 1-4

with selective downregulation of HDAC 10-11 confirmed by RT-PCR. Overexpression of HDAC 2-3 was shown to be the most upregulated as compared to normal liver with statistically significant increases in all tested HB cell lines ($p<0.05$). Panobinostat (IC₅₀ of 0.01-0.05 mM) showed strong effects in in vitro assays for viability while immunoblotting with panobinostat-treated cells showed an increase in acetylated histone H3 and H4 and an increase in cleaved PARP. In vitro assays also demonstrated efficacy of HDACi mocitinostat (IC₅₀ of .556-4.31mM), entinostat (IC₅₀ of 3.06-10.91 mM), and romidepsin (IC₅₀ of .001-0.02 mM). HB cell lines treated with HDACi therapy demonstrated increase in p21 gene expression with downregulation of MDM2 & MDM4. HDACi therapy also demonstrated downregulation of VEGF & c-MYC.

Conclusions: Class I HDAC gene expression is significantly altered in HB may serve as a marker of HDACi responsiveness in HB. Panobinostat and romidepsin showed strong preclinical efficacy in HB suggesting that targeting of HDAC2&3 may be a novel strategy for treatment development of rHB.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0253 / #1753 | Our Experience of 88 Patients with Liver Masses; a Study from Pakistan

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Background and Aims: Liver is the third-most-common site for intra-abdominal malignancy in children after Neuroblastoma and Wilms tumor. Two thirds of liver tumors in children are malignant. Complete surgical excision remains the cornerstone of cure in liver cancer. The aim of this study is to share our experience of surgical outcome in the management of liver tumors.

Methods: This was a retrospective study conducted at The Children's hospital & The Institute of child health, Lahore over the period of 15 years. Malignant tumors were first dealt with neo-adjuvant chemotherapy. Hepatic resection was performed with harmonic scalpel. All the patients' undergone liver surgery for benign & malignant liver tumors were included. Demographic details, pre & post operative events were noted and long term follow-up was retrieved.

Results: Total 88 children with liver mass were included. The mean age of patients was 42.92 ± 41.53 months. Male to female ratio 1.58:1. Most common malignant tumor was Hepatoblastoma (HB) (57.95%) second being Mesenchymal hamartomas 14.8%. Other were hepatocellular carcinoma (7.95%), Metastatic and Biliary cystadenoma. Right-sided involvement was more common. Right lobectomy was most commonly performed procedure. The mean follow-up was 42.84 ± 48.40 months. Most important postoperative complication was recurrence, seen in 6 patients (6.8%), other complications included subphrenic abscess (3.40%) and wound infection (9.09%). There was no operative

mortality. Majority patient received both neoadjuvant and post-operative chemotherapy for malignant type of liver tumor (80.7%). For long-term results, 21 patients could not be traced. So follow up was available for 67 patients. Seven died within one year of surgery, and 14 at 3 years follow up, 53 patients are alive and healthy.

Conclusions: Hepatoblastoma is the most common tumor with male preponderance and right-sided involvement. Complete surgical resection markedly improved survival in these patients. Although Pakistan is a resource limited country, our outcome is still comparable with the most of world.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.f Liver Tumours*

PV0254 / #330 | Germline APC Testing in Apparently Sporadic Hepatoblastoma and Correlation of Germline APC Variants with Clinical Characteristics and Outcomes

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Background and Aims: Germline mutations in the adenomatous polyposis coli (APC) gene have been identified in patients with hepatoblastoma without a family history of familial adenomatous polyposis (FAP). Defining the prevalence of germline APC mutations in apparently sporadic hepatoblastoma is essential since identification of FAP in childhood facilitates initiation of cancer surveillance and detection. The primary objective of this study was to determine the prevalence of germline APC mutations in cases of apparently sporadic hepatoblastoma.

Methods: This study was a multicenter, retrospective cohort analysis. Patients were eligible for the analysis if: 1) 0-18 years at diagnosis 2) diagnosed with hepatoblastoma between January 1, 2005 and March 1, 2021 3) germline APC testing performed 4) lack of family history of a) FAP, b) polyposis, c) early onset (<50 years) colectomy or colorectal cancer in a first, second, or third degree relative.

Results: 108 patients from 7 centers were eligible. Median age at diagnosis was 18 months (IQR 11-31 months). 21 (19%) had a history of prematurity (<36 weeks). Median AFP at diagnosis was 225,110 (IQR 71,000 – 782,350). PRETEXT stage and annotation factors were avail-

able for 93%; patients were PRETEXT 1 (6%, n=6), PRETEXT 2 (30%, n=32), PRETEXT 3 (36%, n=39), and PRETEXT 4 (21%, n=23). Annotation factors were positive in 74% (n=80). Gross total resection was achieved in 99 (92%) patients. 99 (92%) patients were alive at time of last follow-up. 12 (11%, 95% exact confidence interval: 0.06-0.19) had germline APC mutations; of those 3 (25%) were pathogenic and 9 (75%) were variants of uncertain significance (VUS).

Conclusions: Germline APC mutations were identified in 11% of patients with apparently sporadic hepatoblastoma. We recommend germline APC testing for all hepatoblastoma patients. Functional studies are needed to characterize the VUS detected. Multi-institutional collaborations are indicated to elucidate the association between germline APC mutations, prognostic features, and outcomes.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.f Liver Tumours*

PV0255 / #620 | Children with Hepatoblastoma Treated on the SIOPEL Protocol at a Tertiary Hospital in India- Identifying Predictors of Poor Outcome

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Background and Aims: There is paucity of data on Hepatoblastoma(HB) outcomes from LMICs. We report our experience in children with HB treated by a multidisciplinary team using the SIOPEL-based strategy.

Methods: Retrospective analysis of children (<15 years) with HB, treated between July'2013 -Dec'2019 was performed. Baseline evaluation included serum AFP, CT/MRI imaging and biopsy (select patients). SIOPEL risk stratification schema was followed. Treatment included single agent Cisplatin for Standard Risk(SR-HB) and SUPERPLADO for High Risk(HR-HB). Response evaluation was by serial AFP monitoring and imaging. Local treatment included partial hepatectomy/OLT (from Jan'2019). Survival analysis was performed by Kaplan-Meier Analysis, and analysis of prognostic factors by cox proportional-Hazard model(IBM SPSS™v 24.0).

Results: 107/140 children registered during the time period, were eligible for final analysis. Median age was 20.4months(Range:2-156) and M:F was 2:1. Twenty-five(23%) and 82(77%) patients were SR-HB and HR-HB respectively. Median AFP was 4,10,857.5ng/ml(Range:35-5928110). 7 children underwent OLT. At a median follow-up of

29 months, there were 17 disease progression, 11 relapses and 3 toxic deaths. Three-year EFS/OS was $64.3 \pm 4.9\%$ / $71 \pm 4.8\%$ (entire cohort), $96 \pm 3.9\%$ / $96 \pm 3.9\%$ (for SR-HB) and $54.6 \pm 5.9\%$ / $63.2 \pm 5.9\%$ (HR-HB). 44(45%) patients with $>90\%$ decline in AFP after cycle-2 had a higher 3-year EFS/OS ($79.1 \pm 6.7\%$ / $87.5 \pm 6.1\%$) compared to those with $<90\%$ decline ($56.8 \pm 7.1\%$, $65.2 \pm 6.7\%$) respectively. AFP half-life >8.9 days predicted inferior survival (2-year OS 69% vs 89%, $p=0.0052$). On univariate analysis, PRETEXT-IV[HR:4.7(95%CI:1.8-12.04; $p=0.001$)], metastatic(M+) disease[HR:3.5(95%CI:1.7-7.3); $P=0.001$], multifocal(F+) disease[HR:2.3(95%CI:1.1-4.8; $p=0.027$)], tumor rupture(R+) [HR:4.6(95%CI:1.1-19.6), $p=0.0360$], HR-HB[HR:13.4(95%CI:1.8-98.1); $p=0.010$] and $>90\%$ reduction in AFP after Cycle 2[HR:0.37(95%CI: 0.16-0.83) $p=0.016$] had prognostic significance on EFS, while on multivariate analysis only PRETEXT-IV[HR:2.85(95%CI:1.1-7.9; $p=0.045$)] and $>90\%$ reduction in AFP after C2 [HR: 0.4(95%CI: 0.2-1.01); $p=0.05$] continued to retain significance.

Conclusions: Standard risk hepatoblastoma has excellent outcomes with Cisplatin Monotherapy. Delayed AFP clearance predicts poor prognosis and mandates treatment intensification or additional therapies.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PV0256 / #877 | EZH2 Supports Hepatoblastoma Development through the Regulation of Two Dual-Specificity Phosphatases

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Background and Aims: Hepatoblastoma (HB) is the most common form of liver cancer in children. The 5-year survival is about 70%, except for patients presenting high-risk tumor. For these patients the standard medical care includes the combination of surgery and chemotherapy using high-doses of cisplatin. Our team recently performed transcriptomic studies of 25 HB and paired normal livers by RNA-seq (Hooks KB et al, Hepatology 2018). The analysis highlighted the existence of 3 independent groups of HB referred to as C1, C2A and C2B. The highly proliferative and poor-prognosis C2A group was molecularly characterized by an increase in several components of the polycomb repressive complex-2 including the methyltransferase EZH2 and its cofactors SUZ12 and EED.

Methods: Through the complementary use of 2D and 3D cell-based approaches and of an in vivo model in chick embryo,

Results: We confirmed the oncogenic potential of EZH2 in C2A-deriving HB cells. In addition, we demonstrated the benefit of combining cisplatin and GSK126, a specific inhibitor of EZH2, to block HB cell proliferation and survival. By proteomics we identified new genes regulated by EZH2 in HB cells and focused our attention on 2 dual-specificity phosphatases known to inhibit MAPK/ERK pathway by dephosphorylating ERK1/2 kinases. Expression of these two DUSPs

strongly and significantly correlated with the expression of EZH2 in patient samples. Our analysis also showed that the deregulation of EZH2 and these DUSPs is associated to various poor-prognosis parameters including patient survival, tumor recurrence and lung metastasis.

Conclusions: We confirmed the oncogenic role of EZH2 and its relevance as a therapeutic target in C2A HB. As GSK126 potentializes the antitumor effect of cisplatin in HB cells, its use could limit the toxicity of cisplatin-based chemotherapy. Among the new genes regulated by EZH2 we identified 2 DUSPs. Their role in proliferative HB is currently under investigation.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.g Germ Cell Tumours

PV0257 / #1160 | A Multi-Institutional Pooled Analysis

Demonstrates that Circulating miR-371A-3P alone is Sufficient for Testicular Germ Cell Tumour Diagnosis

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Background and Aims: Circulating microRNAs have clear potential for improving malignant germ cell tumor (MGCT) diagnosis. Here, we address the central issue of whether measurement of a single microRNA is sufficient for detecting testicular MGCTs, or whether it is necessary to quantify other members of the four-microRNA panel previously identified (miR-371a-3p/miR-372-3p/miR-373-3p and miR-367-3p).

Methods: We performed a pooled analysis of available published data where all four panel miRNAs had been assessed using pre-amplification PCR technology (four studies; total 329 patients). Two studies using identical methodology (and identical normalization using endogenous miR-30b-5p) were used in the discovery phase (n=51 patients: 17 MGCT, 34 controls). The two other studies (n=278 patients: 140 MGCT, 138 controls), which assessed the same test panel but with different normalization approaches (endogenous miR-93-5p, exogenous cel-miR-39-3p), were used for the validation phase. We derived sensitivity, specificity, positive- and negative-predictive-values (PPV/NPV) for the detection thresholds that maximised the Youden Index (YI).

Results: In the discovery-phase, the YI was 0.97 for miR-371a-3p (sensitivity=1, specificity=0.97), 0.71 (miR-367-3p), 0.68 (miR-372-3p), and 0.50 (miR-373-3p). These findings were confirmed in the validation-phase, with YI of 0.75 for miR-371a-3p (sensitivity=0.90, specificity 0.85), 0.55 (miR-367-3p), 0.47 (miR-372-3p), and 0.51 (miR-373-3p). Importantly, no combination of markers added additional diagnostic benefit to miR-371a-3p alone, in either the discovery or the validation phase.

Conclusions: Quantifying circulating miR-371a-3p alone is sufficient for testicular MGCT diagnosis. PCR measurement of this single miRNA marker will be more cost-effective and easier to interpret, facilitating future incorporation into routine clinical practice.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.g Germ Cell Tumours

PV0258 / #1493 | Whole-Exome Sequencing of Germ Cell Tumors in Childhood

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Background and Aims: Germ cell tumors (GCTs) are a rare group of neoplasms that affect about 3.5% of pediatric patients. GCTs are benign or malignant neoplasms that occur in gonadal and extragonadal sites, comprising a heterogeneous set of tumors. In contrast to many other cancers, previous studies show that GCTs have a relatively low mutational burden which comprise embryonic epigenetic state, copy number alterations (CNAs) and single nucleotide mutation or somatic insertion/deletion (SNV/indel). According to studies performed in adult patients, molecular alterations in the KIT, KRAS and NRAS genes and chromosome 12p amplification are frequently found. However, there is a lack of information in children. Therefore, the aim of this study is to perform a genomic profile by whole-exome sequencing (WES) of pediatric patients diagnosed with GCTs at the Barretos Children's Cancer Hospital.

Methods: We reported the WES using Illumina paired-end sequencing strategy of 16 cases and respective matched normal, including 10 ovarian, 5 testicular and 1 mediastinal tumors. Data analysis was performed as follows: Mutect, Pindel and Mutsig for SNVs/Indels; HMM-copy, Nexus Copy Number and Gistic for CNAs; and Signal (Cosmic v3) for mutational signatures. The somatic alterations found were compared with data available on adult GCTs.

Results: The analyses of Single base substitution signatures (SBSS) showed SBSS39 and SBSS22 in 68.75% and 12.5% of samples, respectively. The CNAs were found on chromosomes 4, 7, 8, 10, 12, 21, and 22, with gain of genes KRAS, CCND2, ETV6, KDM5A, MYC, and OLIG2, and loss of genes KIT, FBXW7, PDGFRA, and PTEN. In addition, mutations were observed in

following genes MTOR (19%), KIT (12%), CTNNB1 (12%), and ATM (12%).

Conclusions: Therefore, our analyzes have identified the molecular characteristics of pediatric GCTs, which correspond to the alterations established in adult GCTs. Further studies with whole-exome sequencing are necessary to confirm gene mutations associated with pediatric.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.g Germ Cell Tumours

PV0259 / #645 | Use of Peripheral Intra-Venous (PIV) Access in Adolescent and Young Adult (AYA) Patients Undergoing Chemotherapy for Testicular Cancer

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Background and Aims: AYA patients requiring chemotherapy for testicular cancer (TC) may have treatment administered through pIV access or central venous access devices (CVAD) such as peripherally inserted central catheters (PICCs) or implanted vascular access devices (VAD). Despite differences in morbidity and cost, there is no standard best practice regarding IV access for this population. We evaluated AYA patients undergoing chemotherapy for TC and compared baseline characteristics and complications related to the use of pIVs versus CVADs.

Methods: We included patients ≤ 25 years old with TC who underwent first-line chemotherapy at the University of Colorado Hospitals and Children's Hospital of Colorado from 2000-2021. Data were collected on baseline demographics, pIV or CVAD use, and associated complication rates. Data analyses were performed to compare complication rates and risk factors for each group.

Results: Fifty-seven patients met inclusion criteria, of which 55 had IV administration details. Of these patients, 27 (49%) used VAD, 15 (27%) PICC, and 13 (24%) pIV for access. Twelve of 13 (92%) patients started on therapy with pIV access completed all cycles of therapy without transitioning to PICC/VAD. There was a difference in incidence of DVT by delivery method with the highest incidence in the PICC group (40%), VAD (15%), and lowest in pIV (0%), $p=0.019$. Access-related complication rates were higher in PICC (60%) and VAD (30%) groups when compared to pIV (15%), $p=0.035$. Length of chemotherapy and increased tumor stage were observed to be risk factors for access-related complications.

Conclusions: pIV use for first-line chemotherapy in AYA patients with TC appears to be well-tolerated with high rates of therapy completion and low rates of complications. These data provide rationale to utilize pIV in delivering chemotherapy in AYA patients with TC.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.g Germ Cell Tumours*

PV0260 / #738 | High Index of Suspicion for Mature B-Cell Non Hodgkin Lymphoma can Guide a Conservative Surgical Approach to Abdominal Mass Presentation

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Background and Aims: Abdominal mass is a frequent clinical presentation of mature B-cell Non-Hodgkin lymphoma (B-NHL). This is often accompanied by acute signs and symptoms of partial or complete bowel obstruction which, if present, pose an oncologic and surgical emergency. Definitive treatment of abdominal masses due to B-NHL is chemotherapy, thus surgical intervention at presentation has been limited to diagnosis and to management of abdominal catastrophe.

Methods: We reviewed medical records of pediatric cases with the diagnosis of abdominal B-NHL treated at our children's hospital from 2010-2020. We describe operative techniques used, postoperative course, and outcome.

Results: Eighteen patients (age 3-13 years, 3 females) were identified with abdominal mass presentation of B-NHL, four had signs of bowel obstruction. Surgical biopsy was performed in 10, while eight underwent resection (44%), with primary anastomosis in six and ileostomy in two. Two of the patients who underwent primary anastomoses suffered anastomotic leak. One leak was contained, while the other resulted in peritonitis, abscess, and required open abdomen. Both required takeback and ileostomy. One of the patients developed ileostomy prolapse, which was manually reduced under general anesthesia, but required PICU admission. All four patients who presented with bowel obstruction underwent resection of the abdominal mass, one with primary anastomosis and three with ileostomy. All patients were alive at last follow up (median 3.3 years, range 0.25 - 8.3 years).

Conclusions: Surgical resection of abdominal masses due to B-NHL at diagnosis is accompanied by a high risk of complications and attendant delay in the start of definitive chemotherapy. These findings support guidelines stating that in the absence of an abdominal catastrophe, only biopsy should be performed and systemic therapy initiated.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.g Germ Cell Tumours*

PV0261 / #889 | Clinical Management and Treatment Results of Germ Cell Tumors: Experience from a Pediatric Oncology Clinic

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Background and Aims: The heterogeneity of pediatric germ cell tumors (GCTs) in terms of age, sex, site, histology, stage and prognosis requires specialized evaluation and treatment. Here we report the clinical characteristics and treatment results of pediatric GCTs treated at our center.

Methods: Medical records of patients were analyzed retrospectively.

Results: One hundred and thirty-three patients (6.6%) of 2006 pediatric cancers were GCTs. The median age at diagnosis was 4years (0-18), M:F ratio was 0.7. Primary tumor was gonadal in 63%, extragonadal in 33% and intracranial in 4% of patients. Histopathology as follows: mature teratoma (36.8%), embryonal carcinoma (11.3%), mixed germ cell tumors (11.3%), yolk sac tumor (9.8%), immature teratoma (9%), disgerminoma (7.5%), endodermal sinus tumor (6%), sex-cord stromal tumor (3%), others (5.3%). Stage distribution of gonadal GCTs was stage-I (44%), stage-II (18%), stage-III (24%), stage-IV (14%); and extragonadal GCTs was stage-I (25%), stage-II (11%), stage-III (48%), stage-IV (16%). For malignant types, average four courses (2-10) of cisplatin based chemotherapy regimens were given. Radiotherapy was performed in 5 patients: intracranial (n:4), extragonadal (n:1). Gonadal GCTs : Primary surgery and delayed surgery were performed in 78 and four cases, respectively. Forty-two patients received chemotherapy. Four of them died with progression. Extragonadal GCTs: Primary surgery was performed in 35 (complete resection:29, partial:6), with only biopsy in nine patients. Seven of them had metastatic disease, all received chemotherapy, and 4 died with progression. Median follow-up-time was 4 years (1 mos- 32 yrs); 10 and 15-years-EFS were 67% and 65%; 10 and 15-years-OS were 84%.

Conclusions: The survival rates were acceptable. Prognosis in most patients with extracranial GCTs is excellent. High number of teratomas may have a positive effect on these results. Heterogeneity of this tumor group and the data from a single center have some limitations.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.g Germ Cell Tumours*

PV0262 / #900 | The Accuracy of Alpha Fetoprotein as Marker of Malignant Recurrence in Patients Who Underwent Excision of Sacrococcygeal Tumor: A Systematic Review

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Background and Aims: Sacrococcygeal tumors are the most common extragonadal tumors in children. Recurrences are associated with a higher risk of malignancy. Patients are monitored for recurrence by physical and rectal examination, and serial monitoring of alpha fetoprotein (AFP), a tumor marker elevated in yolk sac tumors. Increase in AFP may indicate recurrent or metastatic disease. The objective of the study was to determine the diagnostic accuracy of AFP elevation relative to histopathology as a marker of malignant recurrence in patients who underwent excision of sacrococcygeal tumors.

Methods: The study employed electronic databases including PubMed, Google Scholar, and Cochrane. Citations in pertinent articles were also reviewed. No study year limit was set. The index test was AFP which was compared to histopathology in patients with malignant recurrence after the initial excision. The quality of eligible studies was analyzed independently by two reviewers using JBI Critical Appraisal Checklist for Diagnostic Test Accuracy Studies. Standardized data extraction tables were used. A third author reviewed the quality assessment and accuracy of the extracted data. The data were analyzed with RevMan 5.3.

Results: Seventeen studies were included, most were retrospective studies of outcomes of patients who underwent excision of sacrococcygeal tumors, with a few studies on factors associated with malignant recurrence. The usual reason for exclusion was vagueness in reporting whether the index test was done. The sensitivity of AFP in detecting malignant recurrence ranged from 0.25 to 1, with a wide range of variability. The specificity of the test on the other hand was 0.91 to 1.

Conclusions: AFP of patients who underwent excision of sacrococcygeal tumors is highly specific but with limited sensitivity. It can be used in determining who among the patients may not undergo costlier diagnostic examinations to further investigate presence of malignant recurrence or metastatic disease. Future studies may look into combining physical and rectal examination findings with AFP for a more sensitive surveillance.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0263 / #1011 | A Rare Pediatric Case of Extranodal Natural Killer/T-Cell Lymphoma, Nasal Type and Its Successful Management

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Background and Aims: Extranodal natural killer/T-cell lymphoma, nasal type (ENKL) is a rare peripheral T-cell lymphoma in North America. Pediatric cases are extremely rare, with only few reported worldwide, and optimal therapy is not defined. Here we present a pediatric case of ENKL initially misdiagnosed as chronic rhinosinusitis and its successful management with sequential chemoradiotherapy.

Methods: Authorization for release of health information pursuant to HIPAA was obtained and chart review was done to obtain the relevant history, laboratory values, imaging, and pathology results.

Results: A thirteen-year-old male presented with swelling on his nasal bridge and difficulty breathing through his right nare. Initial nasal endoscopy showed ulcerated and friable mucosa with polyposis. One of the polyps was resected and sent for pathological evaluation. Initial impression was chronic rhinosinusitis but work-up showed fragments of necrosis and an atypical lymphoid proliferation of T/NK, cytotoxic cell phenotype, EBV+ positive for CD3, CD5, CD7, CD8 > CD4, CD45Ro, focal CD57, focal CD56, TIA, granzyme, perforin, and EBER1 ISH with MIB1/Ki67 proliferation up to 40-50%, supporting a diagnosis of ENKL. No evidence of metastatic disease was found. Patient was diagnosed with ENKL, Nasal type, Stage IE. The asparaginase-based SMILE (dexamethasone, methotrexate, ifosfamide, L-asparaginase, etoposide) regimen was chosen for its superior response rates and similarity to other well tolerated pediatric protocols. Patient was in complete remission after the initial two cycles and completed a total of four cycles, followed by proton beam radiation. Therapy was well tolerated with no serious adverse events. He remains in remission one year since his initial diagnosis.

Conclusions: This case report emphasizes the need for maintaining a broad differential for pediatric nasal obstruction. It also demonstrates the safe and effective use of the SMILE regimen along with radiation in the management of this rare tumor in a pediatric patient.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0264 / #1019 | Spontaneous Resolution of Metastatic Thymoma with Prednisone in a Patient with Juvenile Myasthenia Gravis

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Background and Aims: Thymomas are rare in children and associated myasthenia gravis (MG) often portends poor prognoses. We present a patient with an aggressive thymoma and juvenile MG who had spontaneous resolution of his thoracic tumor burden with prednisone.

Methods: A 12-year-old boy with newly diagnosed MG was found to have large, partially calcified anterior mediastinal and pleural masses on CT. Pleural biopsy revealed an invasive Type B2 thymoma, consistent with stage IVa disease given pleural metastases. He received three cycles of neoadjuvant cyclophosphamide, doxorubicin, and cisplatin with no reduction in tumor size. He then underwent a successful thymectomy, extended radical parietal and visceral pleurectomy and decortication, diaphragm resection, and mesh reconstruction. Unfortunately, he developed tumor recurrence within one year. Subsequent imaging showed disease progression with tumor encasement of mediastinal structures, prompting prednisone initiation and plan for palliative debulking. Unexpectedly, repeat CT two months later showed near complete resolution of the pleural disease. He remains on maintenance prednisone with stable tumor burden on surveillance imaging.

Results: Thymomas comprise less than one percent of mediastinal tumors in children. Type B2 thymomas are moderately aggressive and often associated with MG. The limited data on thymomas in children suggests younger patients more often have later stage disease at the time of diagnosis. However, there is no consensus on thymoma management in children, and published treatment strategies have been adapted from thymoma guidelines in adults. First-line therapy involves thymectomy, while the benefit of adjuvant chemotherapy remains unclear. Additionally, steroids may be beneficial in patients with MG and metastatic thymomas. While this patient's disease recurred following neoadjuvant chemotherapy and aggressive surgical intervention, he achieved stable disease on prednisone.

Conclusions: Given the rarity of thymomas in the pediatric population, continued characterization of the spectrum of clinical presentations and outcomes is crucial for developing treatment modalities for these patients.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0265 / #1103 | Proton Therapy for Uveal Melanoma in Adolescents/Young Adults (AYAs) and Adults: A Matched Cohort Analysis

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Background and Aims: To assess the clinical outcomes of proton therapy (PT) for uveal melanoma (UM) in adolescents and young adults (AYAs)

Methods: Between January 1997 and December 2007, a total of 270 AYAs with UM were treated at the Jules-Gonin Eye Hospital/Paul Scherrer Institute with PT, using a total dose of 60 Cobalt Gray equivalents (CGE) in 4 fractions. Patients with at least one follow-up visit (233/270 pts) were retained for analysis and compared with 233

matched adult patients over 40 years of age with similar tumor size and location.

Results: In the group of 233 AYA patients, median age was 34.2 years (range: 15-39) with a male/female ration of 0.66 (1.08 in the group of matched adult patients). Average tumor thickness was 6.0 mm (range: 1.0-13.1), average largest tumor diameter was 16.1 mm (range: 3.8-25.9). Median and mean follow-up times were 62 and 71 months, respectively (range: 6-238), in the AYA group (46 and 57 in the group of matched adults patients, range: 6-202). Local recurrences were observed in 7 AYA patients (3.0%) and in 5 patients in the matched group (2.1%). Distant metastases occurred in 32 patients (13.7%) in both groups. At the last control visit, Snellen visual acuity was at least 0.1 in 105 (45.1%) of AYA patients and in 98 (42.1%) of matched patients.

Conclusions: Following PT, AYAs have comparable local tumor control and metastasis rates as our older UM patients. Useful vision is maintained in a lot of cases.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0266 / #118 | Reevaluation of Metanephric Stromal Tumor Two Decades after it Was Named: A Narrative Review

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Background and Aims: Objective: The aim of this narrative review is to provide an overview and update of metanephric stromal tumor (MST).

Methods: Materials and methods: All English language studies published from January 1, 2000 to December 31, 2019 in PubMed, EBSCO, Elsevier ScienceDirect, Springer Link and Taylor & Francis databases were searched with the search terms "metanephric stromal tumor" for this review.

Results: Seventeen eligible case reports representing 47 patients according to inclusion and exclusion criteria were included in this study. The average age of the patients was under 4 years (range from 2d to 56y) and over half of the cases (52.1%, 25/47) were diagnosed as MST by accident or during examinations for other diseases. Morphologically, tumor specimens of almost all cases presented concentric "onion-skin cuffing" or characteristic collarettes around renal tubules under low power. There were 79.2% (18/25) of patients exhibited BRAF V600E mutations. Immunohistochemistry (IHC) is characterized by CD34 (+), Vimentin (+), Desmin (-), S-100 (-), SMA (-). Most patients underwent surgeries, and no metastasis or recurrence was found except for one case.

Conclusions: Conclusion: MST is a rare benign pediatric renal tumor with surgical treatment as the first choice. CT examinations and ultrasonography are two widely accepted techniques for the diagnosis of MST. Percutaneous renal biopsy (PRB) is an effective and accurate way of preoperative diagnosis, however, it is not recommended for children under 10 years or with a cystic mass in CT images. The relationship between BRAF V600E mutations and mild clinical manifestations of MST is in need of further verification by biological experiments and clinical studies.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PV0267 / #126 | Congenital Juvenile Xanthogranuloma with Large Intracranial Tumor and Successful Treatment for a Preterm Infant

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Background and Aims: Juvenile xanthogranuloma (JXG) is a rare histiocytic disorder that commonly presents as a solitary cutaneous lesion, comprising various phenotypes. However, the etiology and pathogenesis of JXG are unclear. We report the case of a preterm infant with a histiocytic tumor that presented as a large intracranial mass.

Methods: Hydramnios and intracranial mass were detected via routine ultrasonography in a female fetus at 30 weeks gestation. Magnetic resonance imaging (MRI) revealed the presence of a large tumor in the posterior fossa, similar in appearance to an embryonal brain tumor. Due to the worsening hydrocephalus and increasing tumor size, cesarean section delivery was performed at 33 weeks of gestation.

Results: There were no external deformities or bulging anterior fontanel (Apgar score = 6/9). However, the infant could not swallow, and breathing was weak. To control the hydrocephalus, an Ommaya reservoir was implanted on day 4. Due to the overall deterioration of the infant's condition, the tumor was partially excised on day 28. Histopathological examination from biopsy revealed diffused infiltration of histiocytes with several Touton-type giant cells (CD68-positive, CD163-positive, CD1a-negative), the final diagnosis being JXG. Chemotherapy (including prednisolone, vincristine, and cytarabine) was administered from day 34. Following treatment, MRI showed regression of the tumor and improvement in the compressive status of the brainstem and cerebellum.

Conclusions: We experienced a preterm infant with congenital JXG in her posterior cranial fossa. Interdisciplinary interventions were necessary to control the congenital brain tumor. This case study could be the first report of JXA outlining the form of onset with successful treatments.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PV0268 / #1374 | Intralesional Corticosteroid Injections as a Therapeutic Option in the Treatment of Giant Cell Lesions of the Jaws in Patients with Noonan Syndrome

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Background and Aims: RASopathies (RP) are one of the best known groups of malformation syndromes with an increased cancer risk caused by germline mutations in genes that encode components of the RAS/MAPK pathway. Giant cell lesions (GCL) are benign tumor-like lesions most frequently affecting the jaws. The pathogenesis of GCL formation is not completely understood, the occurrence of GCL lesions has been described in association with different RP, as Noonan Syndrome (NS). Therapy options are limited to a few case reports and there is no standard of care to date.

Methods: Case 1 (A). a girl with a complex phenotype of NS, presented with bilateral giant lesions in the maxilla extending to the mandible, in which we found a germline mutation in exon 4 of RIT1 (c.170C>G) and a heterozygote NF1 mutation (c.3038C>T). Intralesional corticosteroid injections were performed after surgical curettage for a total of 3 times. The patient was lost to follow up for 2 years when she presented with systemic disease. Multiple different therapeutic approaches were applied without durable response. Case 2.(B). a 16 years old boy presented with progressive indolent swelling of the lower jaw and mild syndromic features indicative for NS. Genetic testing was performed and a heterozygote SOS1 mutation in Exon 11 (c.1654A>G) detected. The patient underwent surgical curettage and weekly intralesional steroid injections over a period of 8 weeks. Nearly complete regression was shown by MRI-scan and further 4 injections were performed to achieve complete remission.

Results: Our NS-patients showed completely different response to the same therapeutic approach. The alteration of two genes encoding for component of the RAS-MAPK signaling cascade might be responsible for the more severe form in patient A.

Conclusions: The mode of action of corticosteroids in GCL is not fully understood. Still, it is likely that the more aggressive approach in case B induced the better response.

E-Poster Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PV0269 / #1376 | Results of Surgical Treatment of Localized and Locally Advanced Adrenal Cortical Carcinoma: Single Center Experience

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Background and Aims: Adrenal cortical carcinoma (ACC) is most common cancer of adrenal gland, but still rare in pediatric population. Mortality rates of ACC are 0.04 – 0.2% from all cases of pediatric tumors. Radical surgery is the golden standard for treatment of local disease. We aimed to determine risk factors of relapse based on our results of surgical treatment of localized and locally advanced ACC.

Methods: Twenty-eight patients analyzed. Age median – 47.8 (06-216) months. Stage I, II, and III revealed in 12 (45%), 7 (25%), and 9 (30%), respectively. In 19 (68%) cases the secretion of one or more hormone observed. Macroscopically and microscopically complete resection were performed in 26 (93%) and 23 (82%) patients, respectively. The median tumor volume was 183 (3.6-1608) cm³ and the median tumor weight was 207.9 (48-710) gr.

Results: Five-year OS and DFS were 71% and 69%, respectively. OS and DFS according to stage I, II, and III were 100% vs. 71% vs. 17% and 100% vs. 71% vs. 14%, respectively. The radical surgical resection and the level of Ki-67 expression influenced significantly the rates of OS and DFS (p<0.001).

Conclusions: Radical surgical resection is crucial for success of ACC treatment with stages I-III. This should be taken into account when planning postoperative therapy. Some of biological characteristics of the tumor could also significantly affect the results of treatment.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0270 / #1401 | Non-Langerhans Cell Histiocytoses – The German Collection and Consultation Service

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Background and Aims: Rare or Non-Langerhans cell histiocytoses (non-LCH) include many malignant and benign, localized or diffuse diseases. In pediatrics, the most frequent are juvenile xanthogranulomatosis (JXG) and Rosai-Dorfman disease (RDD) followed by Erdheim-Chester disease (ECD), and malignant histiocytoses. Frequently, the precise histological diagnosis is undefined. Recently, molecular genetic alterations, particularly in the ras-raf kinase pathway, were identified.

Methods: In 2012, the German registry and consultation study for non-LCH - part of the International Rare Histiocytic Disease Registry - was initiated.

Results: Ninety-two patients were reported (36/50 female/male, 6 unknown; aged 0-4 years:40; 5-9:14; 10-14:10; 15-19:12; 20-39:5; >39:10). About 60 patients are in observation (0-6 years), 6 patients

died (aged: 1, 10, 42, 42, 51, 54 years); the others lost to follow-up (lfu). Among 34 JXG patients (all survivors or lfu), 2 had localized, 4 generalized skin involvement, 2 bone, 5 CNS, 2 eye, 3 deep soft tissue, 2 skin/liver involvement, 2 mediastinal/tracheal compression, 2 intrathoracic extension. 5 patients received steroids/polychemotherapy. Two ECD patients were lfu. Among 16 RDD patients (all survivors or lfu), 9 only had cervical lymphadenopathy (1 received steroids), 6 had extended disease (2 meningeal, 1 mediastinal/tracheal compression, 2 bone, 1 liver/lungs; 5 received steroids/polychemotherapy). Five had histiocytic sarcoma (2 died despite polychemotherapy; one survived after polychemotherapy/BMT). One patient had Langerhans cell sarcoma (survivor after polychemotherapy), 1 had H syndrome. Non-LCH was undefined in 27 (4 deaths after different treatment attempts). Three patients with ALK positive undefined non-LCH received molecular therapies. Complete genetic workup was only performed in a subgroup of patients.

Conclusions: Rare histiocytoses represent a very heterogeneous disease spectrum. Appropriate consultation depends on international prospective data collection, which should be further propagated.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0271 / #1408 | Langerhans Cells Histiocytosis in A Child with Barth Syndrome

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Background and Aims: Barth syndrome is a rare X-linked mitochondrial disease characterized by cardiomyopathy, skeletal myopathy, neutropenia, growth delay and organic aciduria. We report a case of an 8-year-old boy with Barth syndrome who had Langerhans Cell Histiocytosis.

Methods: Review of the clinical notes and diagnostics of the patients.

Results: Tachypnoea with hyperlacticaemic acidosis and bi-ventricular hypertrophy on echocardiogram in the neonatal period led to the suspicion of Barth syndrome later confirmed by genetic testing. Treatment with bicarbonate was initiated with improvement and later arginine was added. He was also maintained under propranolol. Growth and development stayed normal, as well as heart function, but severe cyclic neutropenia persisted associated with marked monocytosis. He presented some infections and prophylactic filgrastim was introduced. At 5 years of age, a right submandibular hard lymph node was detected, which led to a biopsy, and Langerhans Cell Histiocytosis (LCH) was diagnosed. The molecular biology didn't show any BRAF mutation. The staging revealed a lytic lesion in the right

parietal region defining a multisystem form of LCH. After induction with prednisolone and six doses of vinblastine, there was a regression of the lesions; an induction scheme was repeated, without remission. Constitutional neutropenia implied delays in treatment, so a more aplasiant therapeutic regimen was considered contraindicated. We continued maintenance treatment with the same drugs, he achieved complete remission and received a total of 30 doses of vinblastine. He stays disease free 2 years after the end of treatment.

Conclusions: To our best knowledge no other case of LCH in patients with Barth syndrome has been described. The authors discuss the challenge of treating LCH in children with cyclic neutropenia.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis*

PV0272 / #1636 | Paediatric Nasopharyngeal Carcinoma a Rare Entity; Experience in Children Hospital Lahore

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Background and Aims: Nasopharyngeal carcinoma (NPC) is a rare pediatric tumor representing about 1% of childhood carcinomas. Its incidence is high in South East Asia, southern China, and Africa among 10 to 20 years of age. Epstein-Barr virus, family history, and HLA class I genotypes are considered important risk factors for NPC. It has long asymptomatic course, and at the time of diagnosis, it is in an advanced stage. These children present with a neck mass and non-specific nasal, auditory or neurological symptoms. Histopathology confirms NPC, and the extent of the disease is determined by CT scan and MRI. Radiation Therapy (RT) is given for stage I and II disease, and for advanced disease, Chemotherapy (CT) is given in addition to RT. Overall survival has increased with a combination of RT and CT.

Methods: In this Retrospective Observational Study, we aimed to review the diagnosis at presentation, treatment, and outcome in children presented to our hospital with NPC from Jan 2016 to Dec 2020. Thirty children of age from 1 to 20 years having histopathological confirmation of NPC on biopsy were included in this study by non-probability convenience sampling. Their data was collected from hospital records, and it was analyzed by SPSS version 25.0.

Results: Male children were 22 (73.3 %) and 8 (26.7%) were females. Mean age was 11.83±1.86 years (minimum 7years and maximum 15years). 24(80%) belonged to rural areas and 6(20%) to urban areas. 2(6.7%) had stage II, 26(86.7%) stage III, and 2(6.7%) stage IV disease. 14(46.7%) are well after treatment, 12(40%) expired, 2(6.7%) have the residual disease, and 2(6.7%) abandoned treatment. 19(63.3%) received combined RT and CT, 7(23.3%) CT alone, and 4(13.3%) CT and surgery.

Conclusions: Children who presented with NPC belonged to a younger age group compared to the global trend. They had advanced disease at presentation, which resulted in decreased survival.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis*

PV0273 / #1639 | Morphologic Spectrum of INI1 [Integrase Interactor 1] Deficient Tumours in Children

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Background and Aims: Structural or functional deletion of INI1/hSNF5 gene located on chromosome 22q11.2 is a predisposing factor to the development of variety of tumours. Although INI1 is a unifying factor, tumours with INI1 loss are diverse. We undertook this retrospective study with the aim of studying the spectrum of all INI1 deficient tumours in paediatric patients as compared to adults.

Methods: We reviewed histopathology and complete immunohistochemistry profile of tumours from 137 patients including adults which showed loss of INI1.

Results: 52% of tumours showing loss of INI1 were seen in children. A strong association of loss of INI1 expression and 'rhabdoid' phenotype across different tumour types was demonstrated. Commonest tumours seen in children were extra-renal rhabdoid tumour (62%), rhabdoid tumours of kidney (21%) which constituted pure 'rhabdoid' morphology. Epithelioid sarcoma (8%), poorly differentiated chordomas (5%) and epithelioid malignant peripheral nerve sheath tumours (4%) were other tumours showing INI1 loss. Non paediatric patient population had wider morphologic and diagnostic spectrum associated with loss of INI1 and included epithelioid sarcoma, myoepithelial carcinomas, renal medullary carcinomas and INI1-deficient Carcinoma of the Sinonasal Tract. Only 6% of all epithelioid sarcomas were seen in children, the rest being in young adults.

Conclusions: Tumours with INI1 loss are diverse with respect to their clinical presentation, prognosis, morphology and also immunohistochemistry profile. Rhabdoid phenotype associated with loss of INI1 is a strong indicator of specific entities in children such as renal or extra renal rhabdoid tumours; while in adults, the morphologic and diagnostic spectrum associated with INI1 loss was found to be much wider. Presence of pure rhabdoid phenotype was less often seen in adults. Epithelioid sarcoma showed the widest morphologic spectrum of all tumours showing loss of INI1. All tumours with loss of INI1 and rhabdoid phenotype were highly aggressive in children.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0274 / #1646 | Solid Tumors in Infants, Ten-Year Study in a Single Institute in Iraq

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Background and Aims: Solid tumors represent about 50% of all malignant neoplasms in children aged less than 15 years. The incidence and histological type of solid tumors is age-dependent. They make up more than 80% of all malignancies in infants. The solid tumors to leukemia's ratio is 2:1 in children aged 1-14 but 5:1 in infants less than 1 year.

Methods: A retrospective study of one hundred & thirty-five infants' patient done in the department of pediatric hemato-oncology / Children Welfare Teaching Hospital / Medical City Complex in Baghdad, during a total period of 10 years from January 2007 to December 2016. Demographic data (date of birth, age, gender, family history), clinical signs and symptoms, tumor site, histopathology, mode of diagnosis, treatment modalities, date of last follow up and cause of death (if possible) all were reviewed from previously collected unit based data, inpatients and out patients' clinic files.

Results: From total 135 cases analyzed; males were predominating over females 75 (55.6%). Local (swelling- distention- mass) was the most common presentation in 109 (81%). One hundred twenty-two (90.4%) cases were malignant and seven cases (5.2%) were benign. Diagnosis of cases was based mainly on excision biopsy in 78 (58%). Neuroblastoma and Wilms tumor were the most common histological subtype in 38 (28%) and 36 (27%) respectively. Stage III and II made the higher percentage in 47 (35%) and 34 (25.8 %) respectively. Chemotherapy was received in 124 /130 (95.4%) of cases, surgery alone as primary surgical excision in 6/130 (4.6%) of cases. As an outcome; 67 (51.5%) alive, 35 (27%) died, 8 (6.2 %) lost to follow up and 20 (15.3%) abandoned treatment. Events free survival rate and overall survival were 49.2% & 51.2% respectively.

Conclusions: The principal neoplasms were neuroblastoma, followed by wilms tumor. More than half of cases were diagnosed as advanced stages at presentation

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0275 / #1663 | A Child with Hemophagocytic Lymphohistiocytosis and Suspected X-Linked Lymphoproliferative Disease Associated with C.131G>A Variant of Unknown Significance Involving The SH2D1A Gene

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Background and Aims: Type 1 X-linked lymphoproliferative disease (XLP1) results from pathogenic variants in the SH2D1A gene. XLP1 causes hemophagocytic lymphohistiocytosis (HLH). We report a child diagnosed with HLH and a c.131G>A variant of unknown significance (VUS) involving the SH2D1A gene.

Methods: A 12-month-old-male of mixed ancestry presented with prolonged fever, diarrhea, rash, and hepatomegaly. Laboratory exam revealed pancytopenia (WBC 13,900/ μ L, ANC 280/ μ L, hemoglobin 10.6 g/dL, platelet count 120,000/ μ L), and transaminitis (AST 140 U/L, ALT 129 U/L). LDH (800 U/L), ferritin (438 ng/mL, peak 5,490 ng/mL), and triglycerides (250 mg/dL) were elevated, with decreased fibrinogen (150 mg/dL). Bone marrow aspirate/biopsy and CSF were negative for leukemia and hemophagocytosis. Qualitative EBV plasma PCR was positive, and marrow biopsy was positive for HHV6. Interleukin 2 levels were elevated (14,100 pg/mL) with normal natural killer cell function. Two maternal uncles died in early childhood with unknown febrile illnesses. Genetics was consulted given concern for primary HLH.

Results: Hereditary HLH panel revealed a c.131G>A VUS in SH2D1A. This variant segregated with his mother. HLH treatment included high-dose steroids and etoposide, then ecilizumab. He developed neurologic changes/seizures, and brain MRI revealed punctate pontine infarcts, diffuse leptomeningeal enhancement, and cerebral edema. Brain biopsy revealed EBV positive T-cell lymphoproliferative disease. An Ommaya reservoir was placed and intraventricular rituximab was initiated. Haplo-identical bone marrow transplant was performed, complicated by acute graft-versus-host disease. The patient died at 31 days post-transplant from respiratory failure secondary to suspected pneumonia.

Conclusions: Pathogenic variants in the SH2D1A gene cause XLP1. The c.131G>A VUS results in an amino acid substitution of cysteine to tyrosine, where there is a large physiochemical difference in this highly conserved residue. Given the patient's significant clinical course and family history, it is likely to be disease-causing.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0276 / #1667 | Unusual Sites of Rosai Dorfman Disease in Children: Case Series from a Tertiary Care Pediatric Hemato-Oncology Centre

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Background and Aims: Rosai-Dorfman disease (RDD) is a rare histiocytic disorder of unknown etiology. Painless, bilateral cervical lymphadenopathy is the most common presentation. Involvement of unusual nodal sites like mediastinum and involvement of extra nodal sites are rare in children. We present series of children with RDD involving rare sites.

Methods: We analysed three children with unusual presentation of RDD.

Results: G, 14-year girl presented with bleeding fleshy mass protruding from right ear for six months and snoring. Whole body PET-CT revealed FDG avid lesion in nasopharynx, and cervical/ submandibular/mediastinal lymph nodes. Biopsy from mass was suggestive of RDD. As she was steroid refractory, she required salvage therapy with Cladribine/cytarabine and radiation therapy. She is in remission for past 2years. Master. A, 6-year boy child presented with bilateral neck swelling for 3-months with noisy breathing. Lymph node biopsy was suggestive of RDD. PET scan finding revealed generalised lymphadenopathy and was treated with methylprednisolone. Due to features of airway compression was followed by maintenance therapy (6-mercaptopurine+methotrexate). PET-CT showed complete metabolic response at the end of one year. 11year old boy presented with multiple neck swelling without organomegaly. PET-CT showed homogenous soft tissue in right lacrimal sac, maxillary ostium, bilateral nasal cavity, with lytic lesions in left distal humerus/ bilateral talus/ right navicular bone/ left calcaneum/left metatarsal bone/ bilateral femur. Lymph node biopsy was suggestive of RDD. As he relapsed after steroid treatment, he was treated with Lenalidomide and Dexamethasone, to which he responded well.

Conclusions: RDD is a pathological diagnosis and often mimics tuberculosis/ lymphoma in terms of clinical presentation. As these sites (sinus cavities, bone and mediastinum) are rarely involved, high index of clinical suspicion and coordination with pathologists are mandatory to arrive at timely diagnosis.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis*

PV0277 / #1692 | The Tumor Microenvironment May Trigger Lymphoproliferation in Cardiac Myxoma

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Background and Aims: Cardiac myxomas (CM) and primary cardiac lymphoproliferative disorders are rare primary cardiac neoplasms, and both children and adolescents may be affected. The composite occurrence of LPD in CM has been occasionally reported, and chronic inflammation in response to viral infection has been suggested to be at the basis of oncogenesis. Cancers can upregulate autophagy to endure microenvironmental stress and to increase local growth and aggressiveness. CM exhibit a dichotomous separation in low and high inflammatory grades (LIG vs. HIG).

Methods: We studied 23 CMs in youth and adults using autophagy-related proteins and NanoString technology for gene expression. Formalin-fixed and paraffin-embedded human cardiac myxomas were retrieved from University of Alberta Hospital files over 6 years, 2010-2016. A histological and immunohistochemical study was performed using antibodies against Beclin-1, LAMP-1, LC3, p62, LMP1, and Epstein-Barr virus in situ hybridization (EBER). We examined 3-4 micrometers-thick sections for the presence of Beclin-1, LAMP-1, LC3, LMP1, p62, and EBER positive cells.

Results: Autophagy-related proteins (Beclin-1, LAMP-1, LC3, and p62) were demonstrated in both tumor and stromal cells. ATG genes showed a progression of involvement in CM using an 8-gene signature. They were associated with Epstein-Barr virus (EBV) encoded latent membrane protein 1 (EBV LMP1) activation.

Conclusions: We suggest that CM can upregulate autophagy, creating a favorable environment for EBV-driven oncogenesis. To the best of our knowledge, the present study is the first to report on the TME using the expression of autophagy-related genes and proteins in CM. The microenvironment of CM is dynamic, with a variety of cell types and different molecular pathways at play, and this study may warrant further investigation.

E-Poster Topic: *Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis*

PV0278 / #1778 | Microrna Expression Profile Identifies Subgroups of Pediatric Adrenocortical Tumors with Different Clinical Outcomes

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Background and Aims: Pediatric adrenocortical tumors (ACTs) are rare aggressive neoplasms of the adrenal glands. Patients' overall five-year survival (OS) ranges from 30 to 70%, which reflects the heterogeneity of ACTs. Unlike adult cases, reliable prognostic factors or predictive biomarkers to discriminate between benign and malignant tumors are difficult to establish for pediatric patients. MicroRNA (miRNA) expression has been associated with development, progression, diagnosis, treatment response, and prognosis of several types of cancer. However, little is known about miRNA expression in childhood ACTs. The aim of this study was to evaluate the expression patterns and prognostic implications of miRNAs in ACTs.

Methods: Here, we performed miRNA microarray profiling on a cohort of 37 pediatric ACTs and nine non-neoplastic adrenal (NNA) samples, and evaluated their impact on patient's clinical features using the Mann-Whitney test. Survival analyses were carried out based on Kaplan-Meier curves and the log-rank test.

Results: We identified a total of 98 abnormally expressed miRNAs, which expression profile discriminated ACTs from NNAs. Among the 98 deregulated miRNAs, 30 were significantly associated with the presence of ACT unfavorable events (recurrence or death). Of the 30 miRNAs, higher expression levels of hsa-miR-630, -139-3p, -125a-3p, -574-5p, -596, -564, -1321, and -423-5p and lower expression levels of hsa-miR-377-3p, -126-3p, -410, -136-3p, -29b-3p, -29a-3p, -337-5p, -143-3p, and 140-5p were concomitantly associated with poor survival, tumor relapse, and/or death. Moreover, expression profile of these 17 miRNAs stratified patients into two groups of ACTs with different clinical outcomes. Although some individual miRNAs exhibit potential prognostic values in ACTs, only the 17 miRNA-based expression clustering was considered an independent prognostic factor.

Conclusions: Here we present the first study showing associations between miRNA profiles and pediatric ACT prognosis, highlighting miRNAs with promising values as predictors of favorable and unfavorable clinical outcomes. (FAPESP, Grants Numbers: 2010/07020-9 and 2014/20341-0, and CNPq, Grant Number: 471885/2013-4).

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0279 / #385 | Childhood Colorectal Cancer: Experience from a Tertiary Care Cancer Hospital in Bangladesh

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Background and Aims: Background : Colorectal cancer (CRC) is a rare childhood malignancy. Sign symptoms are nonspecific and overlap with common childhood gastrointestinal symptoms. Thus it is frequently overlooked as a consequence diagnosis is delayed. Aims: To determine the clinical profile of childhood colorectal cancer, site of involvement, histopathological pattern and surgical treatment received.

Methods: : Retrospective review of hospital records from January 2009 to December 2019

Results: Fifty two children aged between 5-19 years were reported with a diagnosis of colorectal cancer during the study period. There were male preponderance including 37 male and 15 female, M:F ratio was 2.47:1. Mean age was 13.55 years and most commonly affected age group was 11-15 years 37(71.15%). Most common presentation was abdominal pain 26(49.1%), per rectal bleeding 22(41.5%), constipation 7(13.2%), vomiting 7(13.2%), intestinal obstruction 6(11.3%). Other presenting symptoms were fever, difficulty in defecation, pain around the anus, abdominal distention, altered bowel habit, pain during defecation, weight loss, jaundice, anorexia. Subset of 31 patients could be analyzed for location of the tumor, the most common site of involvement were the rectum 17(54.83%), sigmoid colon 4(12.9%), rectosigmoid junction 3(9.67%). Histopathological diagnosis was available in 39 cases, among them adenocarcinoma was the most common 33(84.62%) followed by mucinous adenocarcinoma 5(12.82%), signet ring adenocarcinoma 1(2.56%). In 22 patients type of surgical treatment was noted including right hemicolectomy in 6, colostomy in 9, ileostomy in 4, resection sigmoid colon 2 cases respectively and 1 case was inoperable.

Conclusions: Conclusion: Common presenting feature of CRC was abdominal pain and per rectal bleeding. Rectum was the most commonly affected part in this study. Adenocarcinoma is the major histopathological type in children.

E-Poster Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PV0280 / #452 | Clinical Profile and Outcome of Pediatric Nasopharyngeal Carcinoma with Combined Modality Therapy – 5-Year Experience from a Tertiary Cancer Center, Bangladesh

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Background and Aims: Nasopharyngeal carcinoma (NPC) in children mostly present with advanced stage and poor to undifferentiated histology – treated with combined modality therapy like neoadjuvant chemotherapy(NAC) followed by concurrent chemo radiotherapy (CCRT). National Institute of Cancer Research & Hospital(NICRH), Bangladesh is unique for such combined-modality therapy. In pediatric oncology department up to 18yrs of children with malignancy

are treated. Increased incidence of NPC is usually observed at age 15-19 years. The study found opportunity to see clinical profile and outcome of children with nasopharyngeal carcinoma with combined-modality therapy.

Methods: The prospective observational study enrolled children with nasopharyngeal carcinoma consecutively from January 2016 to December 2020 at NICRH. All clinical Information regarding age, sex, presenting features and staging of tumor was recorded in a data sheet. Majority children were assigned to receive three cycles of NAC followed by CCRT. Some received NAC or CCRT only. Follow-up given 3-6 monthly till December 2020. Data were edited and analyzed statistically with SPSS version.23.

Results: The study enrolled 46 children with NPC; age ranges 7.5 to 17yrs with mean 12.6 ± 2.85 SD. Majority (69.6%) cases were below 15yrs age. Male: Female =2.28:1. Commonly presented with Neck swelling (87%), then epistaxis (39.1%), headache (26.1%), nasal blockage (24%), fever (19.6%), ear ache & otitis media (15.2%). Majority (87%) cases were in advanced stage with mostly (97.7%) poor to undifferentiated histology. Among total 46 children, 67.4% was treated with NAC + CCRT and 26.1% with either NAC or CCRT or 6.5% palliative only. The study found complete remission 50%, relapse 24% and death 24%. Remission achieved mostly (95.7%) from combined modality treated group. From non-combined treated group, majority (91.7%) children developed relapse and death.

Conclusions: The study found majority children with NPC from less than 15 years age group and remission mostly (95.7%) from combined-modality treated children.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0281 / #668 | Pleuropulmonary Blastoma in Children and Adolescents: The Expert/Partner Diagnostic and Therapeutic Recommendations

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Background and Aims: Pleuropulmonary blastoma (PPB) is a rare lung cancer occurring in early childhood and is often associated with germline DICER1 mutations. Three forms exist: Type I (cystic) with a favorable prognosis, Type II (solid/cystic), and III (solid) with unsatisfactory prognosis. Due to its rarity, a standard treatment has not been established.

Methods: The Consensus Conference Standard Operating Procedure methodology (which grades the levels of evidence and the strength of the recommendations) has been adopted by the European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT) to elaborate diagnostic and therapeutic recommendations for children with PPB, as part of the PARTNER project.

Results: In Type I PPB, a conservative complete tumor resection should be planned [Grade IV, Level A]. Thoracotomy is the recommended surgical approach [IV, B]. After an R0 resection, adjuvant chemotherapy might be discussed [V,C]. For types II-III PPB, tumor resection should be considered if a complete resection (R0/R1) is achievable and there is no evidence of nodal or metastatic disease [IV,B]. Core-needle (16-18 Gauge) or open biopsies are possible options [IV,B]. In all other cases, resection should be considered after neoadjuvant chemotherapy [IV,A], even there is no evidence of residual lesion, to confirm the absence of viable residual cells [V,B]. Some data support the use of a doxorubicin-containing regimen. We recommend radiotherapy in type II-III PPB with residual viable tumor after chemotherapy and surgery. All PPB cases should be referred for genetic consultation and DICER1 mutation screening [IV,A].

Conclusions: As for many rare tumors, we are aware that some of the statements presented are more expert opinions than evidence-based but we hope this document represents a useful guide for clinicians in

the management of PPB and will serve as a basis for further international collaboration. EU Funding: PARTNER GA n.777336 - 3rdHP (2014-2020).

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0282 / #695 | Evaluation of Acute and Chronic Toxicity in Tunisian Children Treated for Nasopharyngeal Cancer

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Background and Aims: Juvenile nasopharyngeal cancer is relatively common in Tunisia. Therapeutic modalities follow adult standards based primarily on radiotherapy and chemotherapy. However, the use of such treatment in young people could be the cause of sequelae that are often poorly tolerated. The objective of this work was to describe the acute and chronic toxicity in children treated for this cancer.

Methods: In this retrospective study we reported a series of 68 children treated for nasopharyngeal cancer at the Salah Azaiez Institute between January 2004 and December 2014.

Results: The median age of our patients was 14.7 years (8-18 years) with a median age of 15 years and a sex ratio of 2. Toxic side effects of chemotherapy were observed in 66% of cases. In our series, 62% of patients had digestive toxicity after chemotherapy. No patient experienced renal toxicity. We noted only one toxic death following febrile aplasia after initial chemotherapy. The therapeutic sequelae were mainly secondary to radiotherapy (95% of cases). For these patients, radiotherapy was of the 2D or 3D type. The average number of consultations during radiotherapy was 7. In our series, 88.6% of the patients had radiomucitis, 85.9% of the patients presented with radioepithelitis. Late radiation-induced sequelae were dominated by permanent xerostomia (83.8%) and cutaneous dystrophy (92%). We also observed a trismus rate of 33.8% and a rate of dental complications of 27.9% such as dental caries and necrosis. Hypothyroidism, a consequence of irradiation of the cervical region and the hypothalamic-pituitary axis, has been noted in 20% of cases. Ten patients had secondary amenorrhea. Hearing loss was noted in 21% of cases. The development of a second cancer was not noted in our series.

Conclusions: Innovative radiotherapy techniques, in particular conformational radiotherapy with or without intensity modulation, seem promising and could overcome toxicity problems while maintaining an excellent rate of local control.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0283 / #777 | Melanotic Neuroectodermal Tumor of Infancy. A French Multicenter Retrospective Study

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Background and Aims: Melanotic neuroectodermal tumors of infancy (MNTI) are rare tumors derived from neural crest. They mainly develop in infants in the first year of life in the bones of the craniofacial region. There is no recommendation for treatment and the management varies from one center to another. No french study has attempted to collect all patients with MNTI in France. Our work is to bring all the cases of MNTI and to describe their presentations, their treatments and their evolutions over the last years in France.

Methods: We carried out a retrospective, multicenter, national study by interviewing French pediatric cancer centers and the national register of rare tumors. Patients treated between 1980 and 2021 were included.

Results: Thirty cases of Melanotic neuroectodermal tumors of infancy were identified. The sex ratio (male/female) was 1.9 and the mean age at diagnosis 5.8 months. Head, neck and cranial vault locations were the most frequent. One patient was metastatic. Therapeutic management consisted mainly of surgical excision. 34.5% of patients received chemotherapy and 10.3% radiotherapy. All the patients with cerebral or metastatic form died of their disease. A total of 8 patients relapsed and 4 patients died. The average duration of patient follow-up in this study is 85 months.

Conclusions: We have identified 30 cases of MNTI over the past 40 years. The treatments vary according to the centers, the locations and the possible relapses. Due to the rarity of this tumor, it would be interesting to conduct international multicenter studies and promote the inclusion of patients in the registry dedicated to this disease. It seems necessary to increase biological and genetic data. Surgery with healthy margins seems to be the first choice to adopt when it's possible. If there are negative factors, chemotherapy could be offered, similar to neuroblastoma protocols.

E-Poster Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.h Rare Tumours and Histiocytosis

PV0284 / #787 | BCOR-ITD Tumors: Comprehensive Clinical, Histopathological and Molecular Analysis of a French National Serie

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Background and Aims: BCOR-ITD tumours form an emerging family of aggressive entities with an internal tandem duplication (ITD) in the last exon of the BCOR gene. The family gathers cerebral localisations, termed High-Grade Neuroepithelial Tumours with BCOR alteration (HGNET-BCOR), and sarcomatous types described in the kidney as Clear-Cell Sarcoma of the Kidney (CCSK), in the endometrium as High-Grade Endometrial Stromal Sarcomas (HG-ESS), in soft tissues as Undifferentiated Round Cell Sarcomas (URCS). Based on a series of 33 retrospective cases, we interrogated the homogeneity of the entity regarding clinical, radiological, histopathological findings, and molecular signatures.

Methods: Thirty-three cases of BCOR-ITD tumours were selected as being diagnosed in France between 2010 and 2020 with available RNA-seq data. These included 10 HGNET-BCOR and 23 BCOR-ITD sarcomas (7 CCSK, 4 HG-ESS, 12 URCS). All cases were centrally reviewed for clinical presentation, therapeutic strategies, histopathological and radiological data. Whole transcriptomic sequencing (28 cases) and DNA methylation profiling (24 cases) were used for unsupervised clustering.

Results: BCOR-ITD tumours affect mostly young children with a median age at diagnosis of 2.1 years [range 0;62.4]. Median overall survival is 3.9 years and progression-free survival is 1.4 years. This dismal prognosis is shared among all localisations except CCSK cases. Histopathological and radiological reviewing reveal marked differences between HGNET-BCOR and BCOR-ITD sarcomas. Consistently, unsupervised clustering defines two distinct groups. The HGNET-BCOR cases cluster together and do not exhibit preferential proximity with glial versus embryonal brain tumours. BCOR-ITD sarcomas cluster independently, beside other BCOR-rearranged sarcomas.

Conclusions: RNA-seq and DNA methylation data consistently identify HGNET-BCOR and BCOR-ITD sarcomas as two separated groups of tumours within a single coherent family. Clinics, histopathology and radiology further ascertain this result. Proximity among the two groups may result from common somatic changes within key pathways directly related to the novel activity of the BCOR-ITD protein.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0285 / #1004 | Usefulness of Next Generation Sequencing in the Diagnosis and Therapeutic Management of Pediatric Central Nervous System Tumors

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Background and Aims: Pediatric central nervous system (CNS) tumors are the second most common childhood malignancy. Little is known regarding the pathogenesis of most brain tumors in children. The implementation of next-generation sequencing (NGS) in pediatric neuro-oncology may impact diagnosis, prognosis and therapeutic strategies.

Methods: We prospectively analyzed 10 neuro-oncology patients (4 boys and 6 girls, average age 10.8 years) who underwent NGS tumor profiling using the OncoPrint™ Childhood Cancer Research Assay (Thermo Fisher) on frozen tissue obtained at diagnosis (3 ependymomas, 2 pineoblastomas, 2 low-grade gliomas (LGGs) and 3 high-grade gliomas (HGGs) from January 2019 to December 2020. This tool analyzes the mutational state of 203 genes including 82 mutation hotspots, 24 CNVs (copy number variants) targets, 44 genes with full exome coverage (specifically tumor suppressor genes) and a RNA panel for 97 genes (with >1700 fusion isoform variants).

Results: Fifty percent of patients (5 of 10) had some type of alteration including rearrangements, SNV/Indels and amplifications. In one patient with ependymoma two different rearrangements were found namely AKAP13-NTRK3 and RUNX1-TRPS1. This patient was treated with Larotrectinib, a specific NTRK inhibitor. One diagnosis of ependymoma was changed to HGG after NGS analysis (H3F3A p.(Gly35Arg) variant was identified in this tumor). Patients with pineoblastoma did not present any type of alterations. By contrast, in all patients with HGG different alterations were found, especially pathogenic or likely pathogenic variants. Dabrafenib was used in one of those patients presenting a pathogenic variant in BRAF gene. In LGGs no significant alterations were found.

Conclusions: NGS is a very useful tool for the diagnosis of tumor alterations that can determine the therapeutic strategy and can complement and redirect anatomical pathology diagnosis. Moreover, the results obtained seem to be tumor-type specific. Studies with more patients are needed and will allow progress in the molecular characterization of pediatric CNS tumors.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0286 / #1014 | Functional Analysis of a Novel CTNNB1 Mutation in Pediatric Medulloblastoma

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Background and Aims: Over the past 10 years novel insights into the biology and genomic complexity of medulloblastoma (MB) has allowed the definition of specific variants and provided the basis for the identification of target genes involved in the initiation and progression of tumors. We present a novel mutation in CTNNB1 gene not previously reported in pediatric MB.

Methods: After surgical resection histology, immunohistochemistry and FISH were analyzed in MB samples. Tumor DNA isolation and mutational analyses of the CTNNB1 gene were performed. For functional analyses, we transiently transfected U2OS human osteosarcoma cells with two constructs of two β -catenin variants (one our patient's). Cellular protein extracts were analyzed by Western blot. Luciferase reporter assays were performed. Structure of proteins was performed by Phyre2.

Results: A case of 9-year old previously healthy with holocraneal cephalic and MRI cerebellar tumor centered on the IV ventricle. Cerebrospinal fluid suggested metastatic spread. Histology was classic MB. Immunohistochemistry analysis showed positive staining for β -catenin, YAP1 and OTX2 expression and negative for GAB1 expression. FISH analysis revealed monosomy 6, and no c-myc nor n-myc amplifications. A heterozygous c.109-111del (p.Ser37del) was detected in the CTNNB1 gene, resulting in the in frame deletion of a TCT codon and loss of Ser37 residue. Δ S37 β -catenin variant displayed in vitro a significant higher protein expression and significantly increased the TCF/LEF transcriptional activity at higher levels than wild type β -catenin. This mutation targets specifically the GSK-3 β phosphorylation domain, resulting in its constitutive stabilization, nuclear accumulation and contribution to tumorigenesis by transcriptional activation of gene targets in Wnt/ β -catenin pathway.

Conclusions: We report a novel CTNNB1 mutation c.109-111del (p.Ser37del) in a classic MB. The structure of the mutated protein differs from wildtype one. Functional analysis discloses gain-of-function properties for the novel Δ S37 β -catenin variant contributing to a more active Wnt pathway and tumorigenesis.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0287 / #1017 | Clinical Characteristics, Treatment, and Survival Outcome of Childhood Atypical Teratoid/Rhabdoid Tumors (ATRT): A Retrospective 20-Year Follow-Up Study

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Background and Aims: ATRT is a rare and aggressive central nervous system tumor which prognosis has improved over the years due to multimodal therapy, including radiotherapy. We wanted to evaluate clinical features, therapeutic approaches, and outcome in children with ATRT in a single institution.

Methods: We retrospectively collected clinical information on 34 pediatric patients with ATRTs treated from 2000 to 2020 and analyzed the data for this series.

Results: The median age at diagnosis was 1 year (0-14 years), and male/female ratio was 1.1:1. Vomiting was the most common symptom at diagnosis (59%). Intracranial hypertension was present in 61.7%. Two patients had other synchronous tumors. The tumor's localization was supratentorial in 45.5%, infratentorial in 54.5% and spinal cord in 2.9%. Gross total resection of the primary tumor was achieved in 47.1% of patients, and 23.5% of patients received Radiotherapy (RT) (median age 4.5 years). The 1-year event-free survival (EFS) and 1-year overall survival were 40.6% and 41.9%, the 5y-EFS and 5y-OS were 12% and 18.6%, respectively. High-dose chemotherapy (HDCT) followed by autologous stem cell transplantation (ASCT) was performed in 11 (32.3%) and the 5y-EFS and 5y-OS were 15.2% (p=0.012) and 30.3% (p=0.009) respectively. Patients who received upfront RT had a 5y-OS of 56.3%, while children who did not receive radiotherapy in first treatment had a 3y-OS of 5% (p=0.002). Chemotherapy was performed in 33 of 34 patients and the most commonly used regimen was Head-Start III. Relapse/progression occurred in 20 patients (58.8%), and just two of these patients are alive. The first underwent complete resection, chemotherapy (ICE regimen) and craniospinal RT3D, the second is receiving MEMMAT protocol.

Conclusions: HDCT followed by ASCT did not improve EFS and OS in our analysis whereas upfront radiotherapy improved survival rates in our study.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0288 / #1018 | Hypermethylation of miR-124A-1, miR-124A-2, and miR-124A-3 in Pediatric Gliomas

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Background and Aims: MicroRNA (miR) hypermethylation has been implicated in various tumors. In adult glioma, miR-124a hypermethylation status is largely studied, but it is not investigated in pediatric glioma. Herein, we analyzed miR-124a methylation status and DNA methyltransferase enzyme (DNMTs) expression in pediatric glioma.

Methods: Methylation status of three loci encoding for miR-124a (miR-124a-1, miR-124a-2, and miR-124a-3) was investigated using methylation-specific PCR in 74 pediatric glioma. DNMTs expression was assessed by immunohistochemistry. Results were correlated with clinico-pathological parameters.

Results: MiR124a-1, miR124a-2, and miR124a-3 were detected hypermethylated in 17.6%, 21.6%, and 10.8% of pediatric gliomas, respectively. miR124a hypermethylation was correlated with glioma grade IV and glioblastoma IDH wild type histopathologic subtype. MiR-124a-2 hypermethylation was associated with male patients. Overexpression of DNMT1, DNMT3a and DNMT3b was detected in 100%, 21.6% and 9.5%, respectively. No significant association was detected between DNMTs overexpression and clinicopathological features. DNMT3a was correlated with hypermethylation of the three loci, while DNMT3b was correlated with miR-124a-1 and miR-124a-2 hypermethylation.

Conclusions: Our results suggest that miR-124a hypermethylation could be associated with pediatric glioma development and may be an interesting prognostic factor. DNMTs overexpression could be implicated in miR-124a hypermethylation. Further studies are needed to confirm these findings.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0289 / #1024 | Investigation of miR9-1, miR9-2 and miR9-3 Methylation in Pediatric Gliomas

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Background and Aims: MiR9 is an important tumor suppressor microRNA regulated by DNA methylation in various types of cancers. In gliomas, many studies have found that miR9 expression was repressed, but there is no evidence that low expression level of miR9 was due to its promoter hypermethylation. In this study, we investigated the methylation status of miR9 as well as DNA methyltransferase enzymes (DNMTs) expression in pediatric gliomas.

Methods: We analyzed the methylation status of the 3 members of the miR9 family in 74 pediatric gliomas. We also assessed DNMTs expression by immunohistochemistry. The relationship between miR9 methylation, DNMTs expression and clinicopathological parameters was analyzed using SPSS.

Results: We found that 17.6% of pediatric glioma cases had a methylation in at least 1 of the 3 loci of miR9. The highest rate of methylation was found in miR9-2 (14.9%), followed by miR9-3 (8.1%) and miR9-1 (8.1%). The promoter hypermethylation of miR9 family

was associated with glioma grade IV, glioblastoma IDH wild type histopathologic subtype and supratentorial tumor location. MiR9 hypermethylation was also associated with DNMT3a overexpression. However, no significant correlation was found between miR9 methylation and DNMT3b overexpression.

Conclusions: These results suggest that miR9 hypermethylation is a frequent epigenetic event in pediatric gliomas and may be involved in gliomagenesis.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0290 / #1029 | Investigation of Human Papillomavirus Infection in Pediatric Gliomas

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Background and Aims: Gliomas pathogenesis remains not completely understood and their etiological factors remain indefinable. Human papillomavirus (HPV) were supposed to be involved in gliomagenesis. The aim of this study was to investigate the occurrence of HPV in a series of 74 pediatric gliomas with different histological subtypes.

Methods: PCR assays were performed to amplify regions within the L1 and E7 genes of HPV genome. In addition, we performed an in situ hybridization analysis using HPV biotinylated DNA probes for the detection of high-risk HPV types (HPV16 and 18). The association of HPV with tumor features and patient outcome was evaluated using SPSS.

Results: HPV DNA was identified in seven samples (9.5%) by PCR, using the broad spectrum consensus primer GP5+/GP6+. Five cases were GBM IDHwt, one case was diffuse midline glioma and one pilocytic astrocytoma. PCR HPV typing identified HPV16 and HPV18 in two and five cases, respectively. In situ hybridization analysis confirmed PCR results. Significant association was identified between HPV presence and glioma grade IV ($p=0.009$) and patient age >10 years ($p=0.012$).

Conclusions: Our study supports the HPV occurrence in pediatric glioma, suggesting their probable causative role in gliomagenesis. Further analyses are required to more explore the implication of HPV in the glioma pathogenesis.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0291 / #1033 | Investigation of Human Cytomegalovirus in Pediatric Gliomas

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Background and Aims: Over the last decades, chronic viral infections have been associated with a number of human malignancies, including primary central nervous system tumors. However, evidence of association between infection with human cytomegalovirus (HCMV) and certain brain tumors such as gliomas has been an area of considerable controversy. In this study, we investigated the presence of HCMV among 74 pediatric gliomas.

Methods: HCMV infection was examined by PCR technique for the detection of a 240pb sequence in IE2 gene, by in situ hybridization for RNA transcript presence and by immunohistochemistry to detect immediate early antigen (IE), early antigen (E) and late antigen. The association of HCMV with tumor features and patient outcome was evaluated using SPSS.

Results: HCMV DNA was identified in 8 (10.8%) gliomas, including 3 (13%) glioblastoma IDH wild type, 3 (7.6%) pilocytic astrocytomas and 2 (28.5%) diffuse astrocytomas IDH wild type. However, neither in situ hybridization nor immunohistochemistry have detected HCMV in all tested samples.

Conclusions: Our results suggest that HCMV may play a causative role in the development of a subset of pediatric glial tumors. Further investigations are required to more elucidate the promising involvement of HCMV in these aggressive tumors.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0292 / #1035 | Investigation of Novel Human Polyomaviruses in Pediatric Gliomas

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Background and Aims: Polyomaviruses were detected and involved in several human malignancies. Their implication in gliomas pathogenesis has been studied, but has never been proven. Previous studies have reported the detection of SV40, JCV and BKV in glial tumors, but the involvement of novel polyomaviruses, such as MCV, KIV and WUV has been not yet investigated. Herein, we investigated the prevalence of MCV, KIV and WUV in pediatric gliomas.

Methods: The DNA sequences of polyomaviruses were examined by PCR on 74 pediatric glioma samples using archived tissues. Tumor characteristics and patient outcome were collected. The association of polyomavirus infection with tumor features and patient outcome was evaluated using SPSS.

Results: MCV DNA was identified in 8 gliomas (10.8%). However, neither KIV nor WUV were detected in all samples. No association was identified with the tumor features ($p > 0.05$) and patient survival (Log Rank test, $p > 0.05$).

Conclusions: Our study supports the occurrence of MCV infection in pediatric gliomas. Further investigations are required to more reveal the potential implication of this DNA polyomavirus in gliomagenesis.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0293 / #1096 | Bevacizumab: Good for the Eyes and the MRI's! A National UK Assessment of Visual and Radiological Outcomes of Bevacizumab Therapy in Paediatric Low-Grade Glioma

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Background and Aims: Whilst Bevacizumab is a proven effective third-line treatment for recurrent/refractory Paediatric Low-Grade Glioma (PLGG), its potential to rescue optic pathway glioma (OPG)-associated visual-loss alongside front-line therapy remains unexplored. We evaluated Bevacizumab's potential future role as front-line therapy; assessing its efficacy & tolerability.

Methods: We performed a multi-centre retrospective analysis of UK PLGG patients treated with Bevacizumab. Medical records, visual acuity (LOGMAR) & radiological (RANO) assessments were reviewed. Clinical, radiological & visual outcomes were compared.

Results: 83 patients from 10 UK PTCs (63% female) with PLGG were included: 65% pilocytic astrocytomas, 17% non-pilocytic astrocytomas, 18% unbiopsied. 13% tumours exhibited BRAF V600E mutation, 35% BRAF-KIAA1549 fusion, 22% BRAF wildtype. Median age 24 months (1-162) at diagnosis; 71 months (7-201) commencing Bevacizumab. Median duration from PLGG diagnosis to Bevacizumab 38 months. 24% patients had NF1 & 30% diencephalic syndrome.

88% tumours involved optic nerve (DODGE stage 1), 39% chiasmatic (DODGE stage 2), 48% optic chiasm and optic tracts (DODGE stage 3). 5% PLGGs were not OPGs, & 20% patients had metastatic disease. Bevacizumab was used: 1st-line in 10%, 2nd-line 4%, 3rd-line 73% & 4th line+ 13%. Average number Bevacizumab doses 22 (2-49). Bevacizumab was combined with irinotecan in 88% & was generally well-tolerated, except proteinuria (10%), vasculopathy (5%), hypertension (2%) & tubulopathy (1%). 92% patients survived. 86% demonstrated either stable disease or disease response on radiological assessment: 1% CR, 39% PR, 46% SD. 14% demonstrated on-treatment tumour progression. Visual acuity (VA) was assessed from 122 evaluable eyes; 104 excluding 18 blind eyes. 89 eyes (86%) demonstrated stable or improved VA (>0.2 LOGMAR) an average of 6 months into treatment. 15 eyes (14%) deteriorated VA >0.2 LOGMAR. 18 eyes (17%) progressed after Bevacizumab cessation (average off-Bevacizumab follow-up 14 months). 37% radiological and visual outcomes were concordant.

Conclusions: Bevacizumab was generally safe and effective for both PLGG tumour control and rescue/preservation of vision. Most patients demonstrated stable/improved radiological responses (86%), & stable/improved vision (86%). Bevacizumab's addition to front-line therapy should therefore be considered.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0294 / #114 | Ophthalmological Evaluation in Children Presenting with a Primary Brain Tumor

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Background and Aims: Children with a brain tumor are prone to develop visual impairment, which to date is often underestimated and unrecognized. Emphasis on this subject is necessary for its severe consequences in daily life and long-term sequelae. We assess the prevalence of ophthalmological evaluation and type of abnormal ophthalmological findings and their association with demographic and tumor-related characteristics in children presenting with a primary brain tumor.

Methods: Medical records of all 90 children diagnosed with a primary brain tumor between June 2018 and May 2019 and treated at the Princess Máxima Center for Pediatric Oncology, a tertiary referral center in The Netherlands, were retrospectively reviewed. Univariate logistic regression analysis was used to investigate associations between demographic and tumor-related characteristics and abnormal ophthalmological findings.

Results: 60 children (34 male [56.7%]; median [range] age, 9.3 [0-16.9] years; 32 infratentorial [53.3%], 11 supratentorial [18.3%], 17 suprasellar [28.3%]) were seen for ophthalmological evaluation within

a timeframe of 6 weeks around diagnosis of a brain tumor. The most common abnormal ophthalmological findings were eye movement disorders (66.0%), visual field defects (58.1%) (5 hemianopia [16.1%], 3 concentric defect [9.7%], 1 central scotoma [3.2%], 5 inferior/temporal/nasal defect [16.2%], 6 blind spot enlargement [19.4%]), and papilledema (44.1%). Eye movement disorders occurred more frequently in patients with an infratentorial tumor (OR, 4.71; 95% CI, 1.03 – 21.65). The risk of papilledema was associated with older age (OR, 1.19; 95% CI, 1.05-1.34), hydrocephalus (OR, 9.63; 95% CI, 2.68-34.61), infratentorial (OR, 9.11; 95% CI, 1.77-46.78) and supratentorial (OR, 13.13; 95% CI, 1.92-89.52) tumors.

Conclusions: The majority of children with a primary brain tumor underwent ophthalmological evaluation around diagnosis, 33% of the children were not timely evaluated. The high prevalence of abnormal ophthalmological findings stresses the importance of early standardized ophthalmological evaluation to detect visual impairment and provide timely treatment and follow-up to potentially prevent permanent visual loss.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0295 / #1151 | Dextro-Amphetamine Treatment for Hypothalamic Obesity in Children Surviving Brain Tumors or Other Causes

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Background and Aims: Hypothalamic obesity (HO), due to e.g. suprasellar tumors or genetic syndromes, may have devastating consequences for the child and its family. Currently, no effective treatment is yet available for HO. Amphetamines, commonly used in children with attention deficit/hyperactivity disorder, are known for their stimulant side effect on REE and suppressing appetite, resulting in weight loss. We here present our experiences of dextro-amphetamine treatment in children with HO.

Methods: A retrospective cohort evaluation was performed of patients treated with dextro-amphetamine at two endocrine pediatric clinics. Off-label use of dextro-amphetamine was initiated in patients with

progressive therapy-resistant HO with an acquired (brain tumor survivors), congenital, or genetic diagnosis. BMI SDS, REE (% of predicted value), (hyperphagic) behavior, and side effects were assessed at start and follow-up.

Results: Nineteen patients (of which 11 brain tumor survivors) were treated with dextro-amphetamine, with a mean age of 12.3 years $\pm$ 4.0 and mean treatment duration of 19.5 months $\pm$ 12.9. Fourteen children (10 acquired, 3 genetic, and 1 congenital) responded to therapy by weight loss or stabilization of BMI (mean Δ BMI SDS of -0.6 ± 0.8). Mean REE difference was $+9.7\% \pm 10.6$ (measured in 11 responders). Three children gained weight (mean Δ BMI SDS of 0.5 ± 0.1). Of two children BMI could not be evaluated, one stopped after 1 month due to hypertension and one had intensive lifestyle intervention simultaneously with dextro-amphetamine treatment. Thirteen children (10 brain tumor survivors) reported improvement of hyperphagia, energy level, and/or behavior. Reported side effects were hypertension, difficulty falling asleep, and negative behavior changes. Ten acquired and two genetic children with HO continued treatment at last moment of follow-up.

Conclusions: In children with HO, due to e.g. suprasellar tumors, treatment with dextro-amphetamine may significantly lower BMI, reduce reported hyperphagia, and improve reported activity level. Future studies are needed to support these results.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0296 / #1152 | Generation, Characterization and Functional Validation of Novel Pre-Clinical Models for Human Pediatric Glioma

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Background and Aims: Tumors of the central nervous system are the most common form of childhood malignancy, and remain the leading cause of cancer-related morbidity and mortality in children. Recently it has become apparent that pediatric gliomas differ from their adult counterparts, explaining in part why the therapies extrapolated from adult gliomas have failed, and further supporting the need to understand the biology of the childhood disease. Deep sequencing studies on brain tumor specimens within various cancer genome consortia (e.g. ICGC PedBrain, TCGA) has yielded a wealth of information about molecular aberrations that had accumulated in tumors. This has unravelled a considerable degree of heterogeneity and thus resulted in the delineation of different molecular subgroups. Nevertheless, validation of oncogenically relevant genes and therapeutically actionable targets

in an in vivo context still remains a major challenge in the process of developing novel treatment approaches.

Methods: In order to assess the role of these mutations we cloned the mutated genes into Cre/loxP-inducible lentiviral vectors. The lentiviruses were intracranially injected to Cre transgenic post-natal 1 (PN1) pups.

Results: We have generated state-of-the-art in vivo methodologies based on somatic gene transfer for generating animal models faithfully recapitulating the molecular and clinical characteristics of the respective human counterpart. In this study, we focused on NTRK2 fusions and FGFR1 mutations and characterized the tumors obtained by either in utero electroporation or lentiviral injection into postnatal (P0-P1) mice. Tumor derived cell lines grown in stem cell media were tested for conventional therapies (e.g. TMZ) and screening of available small inhibitor compounds. In addition, we have characterized the tumor microenvironment of these lesions, mainly the infiltrating immune cell populations.

Conclusions: This study represents an important contribution to the understanding of pediatric glioma biology as well as to translational efforts in developing new treatment approaches for these malignant tumors.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0297 / #1162 | Low Resting Energy Expenditure Correlates with Clinical and Radiological Hypothalamic Damage in Children Surviving A Brain Tumor

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Background and Aims: Children with (supra)sellar tumors are at risk for hypothalamic dysfunction (HD). HD may lead to decreased resting energy expenditure (REE) and/or hyperphagia contributing to the development of hypothalamic obesity. Our aim was to assess whether outcomes of REE measurements are associated with clinical severity of HD or radiological posterior hypothalamic damage.

Methods: A retrospective study was performed evaluating all children with weight gain or obesity at risk for HD in whom REE measurement was performed between 2014 and 2020. Measured REE (mREE) was

compared to predicted REE (pREE) using the Schofield equation. Radiologic hypothalamic damage was scored using Muller grading score plus additional neuroradiological characteristics on MRI. The mREE/pREE quotient was correlated to a clinical score for HD and to radiological hypothalamic damage.

Results: Sixty-seven children, with a mean BMI SDS of $+2.3 \pm 1.0$, were included. Of all children, 94.0% had been diagnosed with a brain tumor (craniopharyngioma, low grade glioma, germ cell tumor, pineoblastoma), while in four other types of brain injuries were present. Children with severe HD had a significant lower mean mREE/pREE quotient (0.65 ± 0.17) compared to children with no HD: 0.89 ± 0.12 , mild HD: 0.86 ± 0.19 , and moderate HD: 0.88 ± 0.15 . Mean mREE/pREE quotient of children with posterior hypothalamic damage on MRI at time of REE measurement (0.74 ± 0.20) was significantly lower compared to children with no damage (0.90 ± 0.12), or anterior damage (0.87 ± 0.13). In the multivariate linear regression, severe clinical HD and tumor progression/recurrence were significant risk factors for mREE/pREE quotient.

Conclusions: Low REE correlates to clinical and radiological scores for hypothalamic damage. REE measurements in children surviving a brain tumor may be useful to distinguish between those who may benefit from obesity treatment that increases REE from those who would be better helped using other obesity interventions.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0298 / #1188 | Molecular Characterization of Long-Term Survivors of Metastatic Medulloblastoma Treated with The 24 Gray of Craniospinal Irradiation

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Background and Aims: Current standard treatment for metastatic medulloblastoma consists of 36 Gray (Gy) of craniospinal irradiation (CSI) supplemented with local irradiation and adjuvant chemotherapy after surgery. Although contemporary protocols have been designed to limit a radiation dose using risk-adapted CSI dosing to reduce neurocognitive morbidity, high-dose CSI remains the standard of care for patients with the disease. Recently, the molecular classification of medulloblastoma has been emerging its clinical significance but has not been established particularly in patients with metastatic medulloblastoma treated with lower dose of CSI.

Methods: We molecularly analyzed three cases of metastatic medulloblastoma treated with 24.0 Gy of CSI by DNA methylation analysis using the Illumina EPIC array.

Results: All three patients had spinal metastases at the time of diagnosis and received treatment which consists of 24.0 Gy/16 Fr of CSI, chemotherapy, and high-dose chemotherapy. There was no disease progression observed during the 7.2-, 5.9-, and 4.0-years post-diagnosis follow-up. The molecular diagnosis was Group 3/4 in all three cases. Cases 1 and 2 belonged to Subtypes 7 and 4, both of which were reported to be good prognostic subtypes among the group. Case 3 belonged to Subtype 5 with MYC amplification.

Conclusions: The present cases suggest that the novel subtype classification in Group 3/4 medulloblastoma may be useful for risk stratification of patients with metastatic medulloblastoma who received lower dose of CSI than standard treatment.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0299 / #1258 | Genetic Predisposition to Brain Tumors in Pediatric Patients

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Background and Aims: A noticeable subset of mutations in pediatric brain tumors derive from the germline as part of cancer predisposition syndromes. The searching of specific mutations in cancer-related genes is important for risk prediction of second malignancies and possible application of targeted therapy. Our aim was to evaluate the impact of germline mutations in patients with different types of brain tumors.

Methods: Forty-one patients aged 3-22 years (22 boys) were enrolled in the study. The most common tumor was astrocytoma (n=18), followed by medulloblastoma (n=10), low-grade glioma (n=4), ependymoma (n=3), and also craniopharyngioma, ganglioglioma, pineoblastoma, teratoid tumor, meningioma and undifferentiated embryonal tumor. Genomic DNA was isolated from peripheral blood leukocytes and next-generation sequencing (NGS) on the NextSeq500 Illumina platform was performed. A multigene panel included coding sequences of 415 cancer-associated genes. For the analysis of NF1 large rearrangements MLPA assay was performed.

Results: In total, 13 patients (32%) carried different mutations in cancer-related genes. The most common were mutations in the NF1 gene found in 8 patients (19%). Patients with NF1 mutations had optic pathway glioma (n=3), pilocytic astrocytoma (n=4) and brain stem tumor (n=1). Three patients with astrocytoma had clinical features of neurofibromatosis type 1, but no NF1 point mutations or large

deletions were revealed. Other findings included pathogenic heterozygous variants in APC and MUTYH genes; mutations in NF2, SMARCB1 and PMS2 genes. A patient with NF2 mutation had multiple brain tumors, a patient with compound heterozygous PMS2 mutation (constitutional mismatch repair deficiency syndrome) has developed astrocytoma and two second malignancies (colorectal and ovarian cancer). The SMARCB1 mutation was associated with atypical teratoid tumor.

Conclusions: Significant proportion of pediatric tumors is due to rare mutations in cancer-associated genes. Second malignancies were developed in 4.7% of patients. Mutations in the NF1 gene had a significant pleiotropic effect and were associated with different brain tumor subtypes.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0300 / #1283 | Congenital Dicer1 Mutant CNS Sarcoma Evolving from Neural Crest Cells with Double Germline Alteration in The Context of DICER1 Syndrome and NRAS-Mutant Neurocutaneous Melanosis

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Background and Aims: DICER1 mutant sarcoma is a newly defined entity that can arise at different body sites and shows various morphologies. Tumors occur sporadically or in the context of DICER1 syndrome and most cases present with additional mutations in RAS-pathway genes. Although the mutational landscape has been investi-

gated, not much is known about the cellular origin of this entity. We present a male neonate with hereditary DICER1 syndrome, paraplegia and multiple melanocytic nevi. Magnetic resonance imaging showed a large arachnoid cyst with cerebellar hypoplasia, brain melanosis and a thoracic intra- and paraspinal tumor.

Methods: DNA methylation profiling and genome-wide high-throughput sequencing was applied to different tumor areas and biospecimens to dissect the complex genetic background involved in tumor formation.

Results: The tumor demonstrated variable degree of maturation. Low-grade mesenchymal areas containing numerous pseudo-meissnerian corpuscles predominated over undifferentiated and sarcomatous areas. Both components variably expressed neural markers (e.g. S100, SOX10, CD56). Methylation profiling revealed a primary CNS sarcoma, DICER1 mutant, with biallelic inactivation of DICER1 being confined to the malignant tumor component. A clonal NRAS p.Q61K mutation was detected in low- and high-grade tumor areas as well as in several skin biopsies resulting in the diagnosis of postzygotic NRAS mosaic neurocutaneous melanosis. The combination of a developmental disorder affecting melanoblasts and a tumor with variable lineage differentiation argues for a neural crest cell with both constitutional DICER1 and NRAS mutation as origin. The patient was treated according to the EU-RHAB protocol combined with MEK-inhibition and is in remission two years after diagnosis.

Conclusions: DICER1 mutant CNS sarcoma may arise in the context of two co-occurring cancer predisposing conditions. The analysis of an unusual spinal tumor in a neonate with hereditary DICER1 syndrome and NRAS-mutant neurocutaneous melanosis provides evidence of a neural crest cell origin of DICER1 mutant sarcoma.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0301 / #1307 | Function and Dependency of NF-kB Activity in ZFTA-Rela Fusion Ependymoma

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Background and Aims: Brain tumors are the most common cause of cancer related death in children. Ependymoma is an aggressive type of pediatric brain tumor resistant to chemotherapy, with treatment to date limited to surgical resection and radiation. Thus, identification and validation of molecular targets that can translate into clinical trials in ependymoma is desperately needed to improve patient outcomes. Over 70% of supratentorial ependymoma are driven by an oncogenic fusion between ZFTA and Rela (denoted ZR^{FUS}). ZR^{FUS} is sufficient to initiate ependymoma development in mice by acting as an oncogenic transcription factor and disrupting gene expression programs. It

is hypothesized that specific ZR^{FUS} interacting proteins are required for tumor formation and represent lead therapeutic targets.

Methods: To study ZR^{FUS} ependymoma, a natively-forming tumor model of ZR^{FUS} generated by in utero electroporation of the developing mouse brain was utilized. Tumor cells were isolated and then subjected to nuclear Rapid Immunoprecipitation and Mass Spectrometry Analysis of Endogenous Proteins (RIME) of HA-tagged ZR^{FUS} protein. Immunoprecipitation and Western Blot (IP-WB) were utilized to probe for leading protein interactions.

Results: We identified both canonical and non-canonical NF-κB proteins consistent with RelA mediated transcription. In addition, we identified a large series of novel chromatin-binding proteins as candidates potentially required for ZR^{FUS} mediated tumorigenesis.

Conclusions: Further study is ongoing in order to validate key ZR^{FUS} protein interaction dependency and elucidate functionality. In addition, future studies will employ ChIP-Seq (chromatin immunoprecipitation with massively parallel DNA sequencing) and CUT&RUN (cleavage under target and release using nuclease) to further analyze DNA binding sites and genomic interactions of these target proteins with the ZFTA-RelA fusion. By interrogating these mechanisms, novel therapeutic targets may be identified that can be brought to clinical trials in order to improve outcomes in pediatric ependymoma.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0302 / #1351 | **Setting Up A Nationwide Central Nervous System Tumor Awareness Campaign in Denmark**

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Background and Aims: Tumors of the central nervous system (CNS) are the second most common tumors among children with cancer. Diverse initial presenting symptoms complicates the diagnostic process, which might result in a diagnostic delay leading to a higher mortality and a high degree of long-term morbidity. Our aim is to launch a nationwide CNS tumor awareness campaign with the goal of reducing time from first symptom to diagnosis by increasing the awareness of symptoms of CNS tumors among the public and healthcare professionals.

Methods: In the United Kingdom, a CNS tumor awareness campaign 'HeadSmart' was launched in 2011. As a result, time from symptom onset to diagnosis was reduced from a median of 9.1 weeks to 6.7 weeks. Adapted to a Danish context, we will issue a similar national CNS tumor awareness campaign 'Hjernetegn'. The primary target group is healthcare professionals in the primary and secondary healthcare system including general practitioners, pediatricians and specialists in contact with children to increase the awareness of symptoms with a high suspicion of a CNS tumor in children. A website will

be set up providing evidence-based information on symptoms including a diagnostic tool to guide healthcare professionals. Information in layman's terms will be available, promoting timely contact to healthcare professionals. Prior to launching the campaign, we will investigate the diagnostic pathway in patients aged 0-17 years with CNS tumors, and continuously monitor the effect on lag time to diagnosis. The campaign is nested within the Danish Collaborative Comprehensive Childhood CNS tumor Consortium and will be initiated in a collaboration between the Danish Cancer Society and the Danish Childhood Cancer Foundation.

Results: The Danish CNS tumor awareness campaign will be launched spring 2022.

Conclusions: We will present our experiences, planning and setting up a CNS tumor awareness campaign, providing an overview of our campaign with insight into obstacles and limitations.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0303 / #1353 | **Survival and Risk Factors Including Molecular Subtypes in Children with Medulloblastoma**

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Background and Aims: Medulloblastoma (MB) is the most common malignant brain tumor in children, and although the treatment results have significantly improved, there is a patients group with unsuccessful treatment outcome. The aim of the study was to assess survival with determination of risk factors including molecular factors in children with medulloblastoma treated in our center in 2004-2015

Methods: The clinical picture of 49 patients with MB was analyzed, taking into account the tumor location, the extent of resection, histopathological and molecular type, advancement, and response to treatment.

Results: The localized form of MB was found in 32 (65.3%) children, and the disseminated form in 17 (34.69%) children. After completing the first-line therapy, CR was found in 37 (75.5%) patients, further disease progression was observed in 5 (10.2%) cases, and 7 children (14.3%) died before completing the first-line therapy, these are children with disseminated form and dominant anaplastic type. During the 5-year follow-up, overall-survival (OS) was 70%, while EFS (event-free-survival) was achieved in 64% of patients. In 43 cases, the molecular type of the tumor fragment was assessed, among them 8 patients had WNT type (18.6%), 2-SHH type (4.7%), 7 (16.3%) patients - Group 3 type, while 12 patients were classified as Group 4 (27.9%) In 14 cases (32.5%) the molecular subtype was not established. The highest survival rates were obtained in the WNT group (EFF and OS 87.5%), the lowest in Group 3 (EFF 80%, OS 58%).

Conclusions: The initially disseminated form, anaplastic type and belonging to group 3 in the molecular classification are the most important risk factors for medulloblastoma treatment failure.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours**PV0304 / #1362 | Pediatric Brain Tumors – Experience from A Tertiary Care Hospital in Pakistan**S. Riaz¹, M. Saeed², K. Afridi², A. Goraya², L.-U. Rehman¹, S. Zaman¹¹Children Hospital and Institute of Child Health, Pediatric Hematology Oncology And Bmt, Lahore, United States of America, ²Children Hospital and Institute of Child Health, Pediatric Hematology Oncology And Bmt, Lahore, Pakistan

Background and Aims: Brain tumors are the most common pediatric solid tumor under age of 15 years and also very frequent cause of cancer related mortality and morbidity. Advancement in radiology, and molecular biology has led to improved diagnosis and management of brain tumors in developed countries but pediatric neuro oncology is still at a nascent stage in the developing countries and morbidity associated with CNS tumors is significant in terms of physical deficits as well as neuropsychological and endocrine sequelae. There are only a few reports on the multidisciplinary approach and outcomes of pediatric brain tumors in developing countries.

The aim is to identify the clinicopathological profile and outcome of Pediatric CNS tumors in a tertiary care center.

Methods: A prospective study of clinical, radiological & pathological profile, and outcomes of children <18 years diagnosed with brain tumors from October 2018 to March 2021 at our institution. Histopathological categorization was done per WHO classification 2007. Statistical Analysis was done by SPSS version 22.

Results: 200 patients were diagnosed with 59% male. Highest incidence was in age group of 5-10 years. Most common diagnosis was medulloblastoma (27%) followed by pilocytic astrocytoma (19%), craniopharyngioma (10%), ependymoma (10%), GBM (10%), brain stem glioma (10%), other high grade gliomas (5%), other low grade gliomas (4%), multiple brain SOL (1.25%), choroid plexus tumors (1.25%), meningioma (1%), and CNS Lymphoma (2%). Amongst them, 54% received treatment (resection, radiotherapy and chemotherapy), 20% are on current waiting list, 4% were offered palliation due to inoperable tumors, and 22% abandoned/died. Overall 38% are on follow up/active treatment, 35% lost to follow up, while 27 % died.

Conclusions: Brain tumors are still the leading cause of cancer related morbidity and mortality in children. Multidisciplinary team is essential to improve outcome. Higher rates of treatment abandonment are documented in view of complexity and cost of the treatment.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours**PV0305 / #1364 | Germinoma as A Cause of Pituitary Stalk Thickening: Symptoms, Treatment and Long Term Results. The Children's Memorial Health Institute Experience**

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Background and Aims: Germinomas are type of germ cell tumor predominantly seen in pediatric populations. Most of them are localized in a pineal gland but germinoma may also affect other structures of central nervous system (CNS). Our study presents cases of patients with pituitary stalk thickening.

Methods: Study included 21 patients (9 girls) with pituitary stalk thickening (defined as pituitary stalk dimension as 3.25 +/- 0.56 mm in transverse diameter at the optic chiasm and 1.91 +/- 0.40 mm) visualized in MRI imaging. To assess pituitary function laboratory tests are collected: fT4, TSH, cortisol, ACTH, LH, FSH, prolactin. Also AFP and beta-HCG as tumor markers were collected

Results: The most common reason for starting the diagnostics was diabetes insipidus (14 patients). However analysing patients previous history in fact, the first symptom was decreased growth velocity, preceding the onset of diabetes insipidus by up to 2 years. 2 patients (1F, 1M) presents precocious puberty connected with high serum level of beta-HCG. 16 patients were diagnosed with germinoma, 3 with histiocytosis, one case of teratoma and germinoma mixed tumor and atypical teratoid rhabdoid tumor. Mean pituitary stalk dimensions in the first MRI were: AP 5.85 +/- 2.63 mm, transverse 5.80 +/- 3.21 mm. Velocity of tumor growth varied from 1.5 to 3.7 mm within first 6 months of observation. 2 patients underwent pituitary stalk biopsy, 1 had bone marrow biopsy. In 10 cases diagnosis was based on postoperative material histopathology. 7 patients were treated without histopathological confirmation, diagnosis was based on laboratory tests and typical MRI image.

Conclusions: Patients with diabetes insipidus and decreased growth velocity should undergo MRI imaging of the CNS. The diagnosis of germinoma should be taken into account in any case of pituitary stalk thickening. Chemoradiotherapy should be the first-line treatment in patients with strong suspicion of germinoma, even without histopathological confirmation.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours**PV0306 / #1381 | Constitutional Mismatch Repair Defect; A Case Series**

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Background and Aims: Constitutional mismatch repair deficiency is a distinct childhood cancer predisposition syndrome showing autosomal recessive pattern of inheritance. This rarely seen disorder has been reported to manifest in infants or young adulthood with an incidence of one per million patients. DNA Mismatch Repair is a system for detection and subsequent repair of errors in newly synthesized DNA during DNA replication. These errors may result from erroneous deletions

or faulty incorporation of bases. MMR system comprises of four genes named as MLH1, MSH2, MSH6 and PMS2.6. Heterozygous or monoallelic germline mutations of these genes lead to autosomal dominant syndrome Lynch Syndrome while homozygous or biallelic mutations of any of these aforementioned genes may result in a cancer predisposition syndrome called CMMRD.

Methods: Case series of CMMRD.

Results: Case no.1: 11years old male diagnosed as Glioblastoma Multiforme (GBM), IDH- wild type WHO grade IV. Immune-staining retained for MLH1 and PMS2. He underwent gross total resection followed by radiotherapy. Meanwhile he developed shortness of breath and further work up revealed T cell Leukemia. His elder sibling was diagnosed as T cell lymphoma two years back but succumbed to death before receiving any treatment. The paternal uncle has lost 5 children owing to malignancies. Case no.2: 6 years old female was diagnosed as T cell lymphoblastic lymphoma. Her brother died of GBM 2years back. Histopathology of mediastinal mass revealed retained immune-staining for MLH1, MSH2 and MSH6. Case no.3: 8 years old boy with suspicion of CMMRD and GBM with two diseased siblings diagnosed as GBM. He and his elder sister retained expression for MSH2, PMS2, MLH1 with P53 positive while his elder brother showed retained expression for PMS2 and MLH1 with p53 positive.

Conclusions: CMMRD is a highly malignant syndrome with poor prognosis. Early diagnosis for abnormal expression of genes can guide to start early surveillance in these children.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0307 / #1473 | Re Irradiation For Relapse Medulloblastoma: Safe Approach For A Palliative Disease

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Background and Aims: Many studies have demonstrated in the last years that once medulloblastoma has recurred, the probability of regaining tumor control is poor despite salvage therapy. The optimal approach to treating relapsed medulloblastoma remains in doubt. Although reirradiation has an emerging role in others relapsed CNS tumours, there is a lack of strong data on re-irradiation for medulloblastoma

Methods: This is a retrospective cohort study of patients aged 18 years or under, treated at least for a second course of external beam radiotherapy for recurrence medulloblastoma at Garrahan Hospital between 2009 and 2020.

Results: Twenty-four patients met eligibility criteria for inclusion. All patients received upfront radiotherapy as part of the curative-intent first radiotherapy (RT1), either craniospinal radiotherapy followed by posterior fossa boost in 20 patients (12 SR and 8 HR) or focal posterior fossa radiation in 4 patients. The second course of radiation con-

sisted of CSI in 15 (infant=4, SR=7, HR=4) and focal in 9 (SR=5, HR=4). The median OS from initial failure for patients who received 2 courses of CSI was 3 years, while for patients who received focal radiation as RT2 field was 2 years. The median OS from RT2 for patients who received 2 courses of CSI was 1.18 years while patients who received a focal RT2 was 0.81. No symptomatic intratumoral haemorrhagic events or symptomatic radionecrosis were observed in our cohort. All the survivors fell within mild to moderate intellectual disability range in Wechsler Intelligence Scale, with a median IQ at last assessment of 63 (range 43-69).

Conclusions: Although re irradiation could be considered a safe approach for a palliative care disease, the high cognitive cost is a concerning limitant factor.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0308 / #1481 | Timing of Radiotherapy and Effects on Disease Outcomes in Childhood Medulloblastoma Tumors: A Report from the Pediatric Proton/Photon Consortium Registry (PPCR)

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Background and Aims: Current practice for treatment of non-infant medulloblastoma is a course of minimally interrupted radiotherapy (RT) started within 4-5 weeks of surgery. Here we analyze the effect of RT timing on event-free (EFS) and overall survival (OS), reported by known risk factors for medulloblastoma.

Methods: Patients treated for medulloblastoma tumors between 2012-2020 enrolled on the PPCR were included if they had no pre-RT chemotherapy, no prior cancer, standard RT dosing, and available follow-up information. Time interval between surgery to RT and duration of RT delivery were evaluated for effects on EFS and OS using Kaplan-Meier method.

Results: 215 patients met inclusion criteria. Median follow-up and age at RT were 3.4 years (<1-19.6) and 9.1 years (3.0-21.8). Cohort characteristics include: standard-risk n=148 (68.6%), intermediate/high-risk n=67 (31.2%), and classic/desmoplastic histology n=169 (78.6%). Median time from surgery to RT was 33 days (7-98), 27 patients (12.6%) had >40 days interval. Median RT duration was 42 days (30-51), 15 patients (6.9%) had prolonged RT (>45 days). For the entire cohort, SR, and IR/HR patients, 5-year EFS were 83.1%, 88.1%, and 71.8%, and 5-year OS were 89.9%, 94.4%, and 79.0%, respectively. For HR/IR patients only, delayed start of RT >40 days trended towards worse outcomes (5-yr OS 41.7 vs 82.5%, p=0.13). Patients treated <28 days (vs 29+ days) had worse 5-yr EFS and OS (75.3 vs 85.5%, p=0.04 and 78.3 vs 93.3%, p=0.04). Prolonging RT >45 days hinted towards worse 5-yr EFS (71.4 vs 83.9%, p=0.22), but OS was no different.

Conclusions: Outcomes in this proton-treated cohort show excellent disease control. Patients were treated expeditiously after surgery, without much RT prolongation. There was a trend towards improved outcomes in HR/IR patients who started RT <40 days from surgery. Unexpectedly, we found starting RT <28 days of surgery was correlated with worse EFS/OS.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0309 / #1521 | CNS Sarcomas in Children and Adolescents: Experience of an Institution

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Background and Aims: Intracranial sarcoma (IS) is a rare disease. Soft tissue sarcomas are 6% of childhood cancers, and less than 0.2% of CNS tumors are primary intracranial sarcomas. Their origin are pluripotent primitive mesenchymal cells. Objective: to evaluate and report experience on the incidence and characteristics of a rare CNS tumor.

Methods: Reviewed of CNS tumors database at IOP/GRAACC/UNIFESP for the period 2008-2021, searching for all types of sarcomas that involved the CNS.

Results: The case series included 5 male and 2 female patients, average age was 11.3 years. Supratentorial region (71.4%) was the most common site, and also posterior fossa and intramedullary (1 each) were found. The PF sarcoma presented as a second neoplasm after 2 years finished treatment for medulloblastoma. Histology and immunohistochemistry studies revealed: spinocellular sarcoma, fusocellular and myxoid sarcoma, gliosarcoma, meningeal sarcoma, Ewing sarcoma, condrosarcoma and synovial sarcoma. Work-up excluded other primary tumor origin outside CNS. The presenting symptoms varied with tumor location, specially hemiparesis, seizure and headache. All

patients underwent multimodal treatment. Gross total resection was achieved in 6 (85.7%) cases, 1 after second look surgery. Chemotherapy involved mainly ICE Protocol (57.1%), also Head Start Protocol and Ewing Sarcoma Protocol. Molecular investigation was performed in 3 (42.8%) patients, 1 with alteration (NRAS c. 181C > A; p. Gln61Lys), but by now with no significant meaning for the disease. Follow up ranged from 6 months to 13 years, all patients are alive, 28.5% dealing with endocrinal issues after radiotherapy. Three patients are still in treatment, and tolerating it well. The mean overall survival (OS) was 36.4 months.

Conclusions: CNS primary sarcomas were mostly observed in the supratentorial region and in older children, with a male predilection. Multimodal treatment seems to benefit patients outcomes. More patients and longer follow-up periods are needed to further elucidate the biological features of this tumors.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0310 / #1531 | Central Nervous System Tumors in Infants - Disease Spectrum, Outcome and Morbidities Based on a Territory-Wide Cohort

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Background and Aims: Central nervous system (CNS) tumors presenting during the first year of life carry specific challenges in management due to their rarity as well as vulnerability of the developing nervous system. Here we report the disease spectrum, clinical characteristics and long-term outcome of infant with CNS tumors in Hong Kong.

Methods: Retrospective review of the population-based Hong Kong Paediatric Haematology/Oncology Study Group database for CNS tumors diagnosed during first year of life from 1999-2020. Clinical status as of February 28 2021 was used. Incidence rate was calculated based on population-size from the government census data.

Results: Forty-two patients were identified (crude incidence=4.1/100,000 infants/year). Male:female ratio was 17:25. Four patients were diagnosed antenatally, while the median age of diagnosis for the remaining patients was 5.5 months (range: 0.1-11.7). Histologic diagnoses (2 based on post-mortem examination) included glioma=13 (low-grade=5, high-grade=7, undefined=1), ATRT=4, choroid plexus papilloma=3, craniopharyngioma=1, ependymoma=3, ganglioglioma=1, germ cell tumor=4, medulloblastoma=6, and "PNET"=5; 2 patients were diagnosed with high-grade glioma and

SEGA respectively based on imaging. Seven patients had metastasis. Cancer predisposition syndromes were confirmed in two patients (tuberous sclerosis, Gorlin syndrome). Treatment included surgery in 38 (gross-total resection=22), systemic therapy in 21, and focal irradiation in 4. Twenty patients died (disease=18, pneumonia=1, cerebral edema=1); while 22 were survivors with median follow-up of 7.9 years (range: 0.9-21.2), including 2 patients with hemispheric high-grade glioma who received respectively baby-POG and no chemotherapy (follow-up: 4.3 years and 7.8 years). Morbidities in survivors include neurodevelopmental disorders in 10/22, epilepsy in 3, visual impairment in 1 and growth disturbances in 4 (overweight=2, short stature=1, precocious puberty=1).

Conclusions: Infantile CNS tumors are heterogeneous in disease biology and behavior with half of our cohort being survivors. Multidisciplinary intervention is essential to mitigate chronic health deficits.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0311 / #1585 | SHH Desmoplastic Medulloblastoma and Down Syndrome: An Extremely Rare Association

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Background and Aims: Medulloblastoma (MB), the most common malignant brain tumour of childhood, has a very low incidence in Down syndrome (DS). This protective characteristic has not been clarified yet. This case report represents the fourth case described in the literature.

Methods: We describe the only case in Garrahan Hospital, of a patient with Down Syndrome and Medulloblastoma. To our knowledge the only case described in Argentina and the fourth in the world.

Results: Thirteen-year-old female with DS and one month history of vomiting and march instability. The physical examination showed an increase in the support base, normal cranial nerves exam, without dysmetria. Brain MRI showed an expansive lesion involving the right cerebellar hemisphere with associated ventriculomegaly of the lateral and third ventricles. A complete surgery was performed as initial management and the pathology disclosed a desmoplastic medulloblastoma SHH subgroup. Lumbar cerebrospinal fluid was collected and showed no signs of malignant spread. It was staged as non-metastatic. She received a dose reduced craniospinal irradiation plus posterior fossa boost followed by adjuvant chemotherapy based on CCG9961B. No evidence of tumor recurrence was observed. The overall survival was 8 years. No tumour family history was described.

Conclusions: Future collaborative studies might contribute to define better standards of care and help improve quality of life for this particular subgroup. A reduced radiotherapy dose may be considered to avoid further neurocognitive damage.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0312 / #163 | Unexpected Binding of Tozuleristide “Tumor Paint” to Cerebral Vascular Malformations: A Potentially Novel Application of Fluorescence-Guided Surgery

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Background and Aims: Background Fluorescence-Guided Surgery (FGS) is under investigation as a means to improve the extent of resection for primary central nervous system (CNS) tumors. Tozuleristide, known also as “Tumor Paint”, is an investigational tumor-targeting agent covalently conjugated to a derivative of the fluorescent dye indocyanine green. We report the finding of avid intraoperative fluorescence of tozuleristide on cerebral vascular malformations thought to be primary intracranial neoplasms and discuss the wider role this marker may serve in the future for both neoplastic and non-neoplastic lesions.

Methods: Our institution is participating in a Phase 2/3 study of intraoperative near-infrared fluorescence detection of pediatric primary CNS tumors in patients receiving intravenous tozuleristide and imaged with the Canvas system. Our site enrolled 2 patients with intracranial lesions, suspected preoperatively of possibly being gliomas that proved to be vascular malformations after resection.

Results: Each lesion had a dark blue mulberry appearance and each fluoresced avidly with tozuleristide. Each was completely resected and the patients recovered without deficit. Pathologic assessment showed cavernous angioma for both cases. Tozuleristide fluorescence is postulated to result from binding to matrix metalloproteinase-2 and annexin A2, and literature review demonstrates expression of both these ligands on multiple oncological and cerebrovascular lesions, including cavernous malformations

Conclusions: This finding deserves further investigation to determine if tozuleristide “Tumor Paint” may have a wider role in the identification of neoplastic and non-neoplastic intracranial pathologies, as well as the benefits this agent may have over other fluorescent markers of tumor and vascular tissue.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0313 / #1637 | Capacity Building for Pediatric Neuro-Oncology in Pakistan- A Project by My Child Matters Programme

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Background and Aims: The landscape of Pediatric Neuro Oncology (PNO) has changed in Pakistan since 2014, when a PNO twinning program was initiated between the Aga Khan University Hospital (AKUH) and the Hospital for Sick Children (SickKids), Toronto. This twinning initiative identified the dire need of broadening the PNO services in other centers of Pakistan.

Methods: The project for the Capacity building of PNO services was initiated in 2019 with the support of a 'My Child Matters' grant by the Sanofi Espoir Foundation. The grant's main purpose was to enhance the capacity for PNO services in 13 public and private sector hospitals across Pakistan. Activities included establishment of National PNO tumor boards, educational workshops and fellowship to train and raise awareness of health care professional (HCP), patients and families.

Results: Monthly tumor boards were started in 2019 covering all provinces of Pakistan. To date 223 patients have been discussed. Three PNO workshops attended by 154 HCP, 25 lectures as part of a lecture series were delivered by 13 international faculties with > 1800 participants from 48 countries. Most activities were virtual amongst the mayhem of the pandemic. This project also led to the establishment of Pakistan's first Pediatric neuro-oncology fellowship program at AKUH. In addition, Pakistan's National Guidelines for the management of Medulloblastoma were developed endorsed by the Pakistan Society of Pediatric Oncology (PSPO) and the Pakistan Society of Neuro-oncology (PASNO). Lastly AKUH was able to host the Pakistan's First PNO Virtual Symposium in November 2020 which brought together 29 national and international luminaries in the field of PNO, it was attended by 1126 participants from 58 countries across the globe.

Conclusions: This program has shown its capacity to transform the management of pediatric brain tumors in a low middle-income country with a very large pediatric population. Further follow-up is needed to assess the impact on survival.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0314 / #1649 | Endoscopic Extended Transsphenoidal Surgery for Newly Diagnosed Paediatric Craniopharyngiomas

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Background and Aims: Advances in endonasal surgery has resulted in increased popularity of endoscopic extended transsphenoidal surgery (EETSS) for treatment of craniopharyngiomas. The aim of this study is to assess the outcome of EETSS in paediatric patients with newly diagnosed craniopharyngiomas.

Methods: Patient details were obtained from a prospective database of all endoscopic transnasal operations performed by a single surgeon. Outcomes including visual function, pituitary function, body mass index (BMI), postoperative neurological deficit, extent of resection and recurrence on follow-up were obtained.

Results: Between January 2011 and March 2021, 16 of 17 children (2-18 years old) with newly diagnosed craniopharyngiomas underwent EETSS. Gross total (GTR), near total (NTR) and subtotal resection (STR) was achieved in 4, 6 and 6 patients respectively. Postoperative radiotherapy was used in 6 patients. Normalisation of visual fields (VF) occurred in 9/13 patients with preoperative VF defects. With a median follow-up (FU) period of 75 (1-111) months, 2 patients required further surgery for tumour progression, both originally had STR for preservation of the pituitary stalk (PS). Only one patient out of 6 in whom the PS was anatomically preserved retained normal pituitary function at last FU. At the time of last FU, 15/16 children had anterior panhypopituitarism, 14/16 had DI and 1 patient developed new onset obesity. 2 patients who were obese preoperatively, were no longer obese at last FU. There were no episodes of postoperative CSF leak, stroke or mortality.

Conclusions: EETSS can be performed as the first option in the majority of children with newly diagnosed craniopharyngioma, despite factors such as small nose, non-pneumatised sphenoid sinus, small sella or purely suprasellar tumour location. Panhypopituitarism is very common at long term follow-up. Preservation of the pituitary stalk at the expense of leaving residual tumour may not be in the best interests of the patient.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0315 / #1676 | Clinical and Histological Characterization of Parenchymal Pineal Tumors in Adolescent and Young Adults: A National Retrospective Cohort

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Background and Aims: Parenchymal tumors of the pineal region (PPTs) are very rare and comprise several subgroups with distinct histological and clinical features. While pineocytomas (PC) are benign tumors arising in adults only, pineoblastomas (PB) and pineal parenchymal tumors of intermediate differentiation (PPT-ID) may occur both in adolescence and young adulthood. In this study, we aim to report a retrospective national cohort of adolescent and young adults (AYA) PPTs and describe the clinical and histological specificities for each subgroup.

Methods: Clinical and histological data were retrospectively collected.

Results: Over 13 years, 31 patients aged 12 to 35 years old were reported. Clinical and follow-up data was available for 14 patients: 10 patients with PPT-ID and 4 patients with PB. Molecular analysis were retrospectively performed leading to clarification of the initial histological diagnosis in light of the latest updates in PPTs molecular understanding. KBTBD4 was found to be recurrently altered in PPT-IDs. PBs in this adolescent population were associated with microRNA processing pathway deficiency. PBs were mainly treated according to high-risk primitive neuro-ectodermal tumors protocol with favorable outcome in 3 over the 4 patients. Treatments received for PPT-IDs were heterogeneous, especially regarding the modalities of adjuvant irradiation; only 1 patient died.

Conclusions: This cohort of AYA patients reports the specificities of PPTs arising at that age and the heterogeneity of their management, underlining the necessity for strategies evaluation and harmonization through dedicated clinical trials.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0316 / #1716 | Childhood CNS Tumor Survivors: Morbidity, Surveillance Needs, and Progressing Health Changes during Mid-Adulthood

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Background and Aims: New life-saving treatments contribute to higher survival rates followed by yet unknown late morbidity risks and aftercare needs of childhood central nervous system (CNS) tumor survivors. As part of a longitudinal project, this study investigated self-experienced persistent late effects (SPLE) and health changes occurring decades after initial diagnosis.

Methods: The study addressed 647 Swedish mid-adulthood very long-term survivors diagnosed 1982-2001 (mean age=32). For a fair picture of severity of late occurring problems, and to avoid data loss due to some survivors being unable to handle study questionnaires independently because of morbidity, parent-proxies were approached with questionnaires to assist with answering for survivors from whom questionnaires were not received. Assessment of SPLE included open-end questions for survivors to respond in their own words. Outcomes were analyzed quantitatively and qualitatively. Health changes were examined with prior follow-up six years earlier as the reference. Needs of clinical surveillance was examined.

Results: After 22 verified deaths (3.2%) since prior follow-up and non-responder attrition, data was provided for 507 survivors, 333 (65.7%) of whom experienced SPLE within a range of neurological, cognitive, sensory, endocrinological, and psychological origin. SPLE were experienced in the considerable to devastating span by 58%, varying with illness treatment and demographic factors. Twenty-one percent of severe-SPLE survivors experienced an unmet need of clinical surveillance. Over half of survivors with SPLE experienced changes in health status since prior follow-up, implying a deterioration for 76%. Changes included tumor growths, and recurrences or new cancers, or both. Health status six years earlier predicted current late morbidity.

Conclusions: A majority of mid-adulthood survivors experience disabilities intruding upon health, functioning, and quality of survival. Range of problems tend to expand over time, with significantly aggravated morbidity and complete general disability for some. A subgroup with severe sequelae witness unmet needs of specialized clinical surveillance, revealing necessity of improved life-time aftercare.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0317 / #1718 | Effects of Polyguanidine on the Cytoskeleton of Medulloblastoma Cells

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Background and Aims: Background. Medulloblastoma is the most common CNS malignancy in the pediatric population. Treatment frequently causes permanent and severe morbidity. Brain tumors have aberrant glycosylation resulting in hyper-sialylation, a potential treatment target. Purpose. The present in vitro study investigates the treatment efficacy of a polyguanidine construct (GuaDex) with high affinity to sialic acid.

Methods: DAOY (ATCC HTB-186) desmoplastic cerebellar medulloblastoma cells and MB-LU-181 neurospheres, and primary medulloblastoma cells were used. The cells were incubated with GuaDex at 0.05-10 μ M concentrations, during 4, 24, and 72 hours. Cytotoxicity was measured using a fluorometric cytotoxicity assay (FMCA). Cell

cycle analysis by DNA content determination was performed after 24 hours incubation. Fluorescence-labeled GyaDex was used to study the cell interactions. Sialic acid (Sia) expression was confirmed with a fluorescence labeled Sia binding lectin. FITC labelled phalloidin was used to detect actin and Alexa Fluor®488 Anti- β -Tubulin antibody as microtubule marker. The cell interactions were studied by confocal microscopy.

Results: GyaDex IC₅₀ at 72 hours incubation with medulloblastoma cells was 281 nM for DAOY cells and 223 nM for MB-LU-181 cells. FMCA assay showed growth inhibition on DAOY and MB-LU-181 cells incubated with 5 μ M GyaDex after 4 hours incubation. A significant decrease at G1 cell cycle phase was observed with an increase of G2 (two-way ANOVA, followed by Tukey post-hoc analysis). *S. nigra* FITC labelled lectin showed high expression of Sia on DAOY medulloblastoma cell lines. FITC-phalloidin and Alexa Fluor®488-Anti- β -Tubulin antibody results showed that GyaDex incubation caused polymerization of actin filaments and microtubule.

Conclusions: GyaDex showed potent growth inhibitory efficacy on medulloblastoma cells at nanomolar concentrations. DAOY cells showed disruption of the actin filaments indicating disruption of the microtubule. The results are very promising and warrant further studies.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0318 / #1759 | The Impact of Molecular Subgroup Classification of Medulloblastoma in Greece through an International Collaborative Program

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Background and Aims: Medulloblastoma is the most common central nervous system (CNS) tumor in children, accounting for 20% of all childhood primary CNS tumors. Whereas it was previously considered one disease, advances in genome and epigenome research have identified a compendium of at least four different diseases, defined by unique molecular profiles, clinical behavior, and outcomes. Thus, the 2016 World Health Organization classification of CNS tumors

proposed apart from the histopathological, a molecular classification for medulloblastoma. In many countries, including Greece, molecular classification of medulloblastoma has not been available until recently, due to financial and infrastructural reasons. Since 2018, a collaborative program between ACCC/DKFZ (Athens Comprehensive Cancer Center/German Cancer Research Center) supports the molecular analysis of tumor samples aiming to identify patient-specific alterations and molecularly classify tumors within a clinically relevant timeframe.

Methods: To evaluate the importance of integrated diagnostic procedures in patients with medulloblastoma, a retrospective analysis of clinical, histopathological, and molecular data was performed.

Results: For the period 2012-2020, among 30 children histologically diagnosed with medulloblastoma (26 classic, 2 desmoplastic, 2 anaplastic), 14 patients (7 males/7 females) with mean age 8.5 years (9 months-16 years) were further analyzed with immunohistochemistry, targeted next-generation sequencing, and genome-wide methylation arrays at the DKFZ and the Department of Neuropathology at University Hospital in Heidelberg, through PTT2.0-study and INFORM-registry (13 primary diagnoses/1 relapse). Based on DNA methylation, they were subclassified to WNT-activated Medulloblastoma (2 patients), SHH-activated with TP53-wildtype (1 infant patient) or TP53-mutant (2 patients), Group 3 (2 patients) and Group 4 (7 patients). Altered therapeutic actions have been taken in 8/14 patients (57%). Germline alterations predisposing to tumors (NBN, APC, TP53) were detected in 3/14 patients and further consultation was provided to their families.

Conclusions: Molecular subgroup classification offered clinically significant information to our patients and their families, improved diagnoses, and treatment options, highlighting the importance of integrated diagnostic procedures for children with CNS tumors through international collaborative programs.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0319 / #220 | HeadSmart Jordan: The Impact of an Hour of Zoom Lecture about Pediatric Brain Tumors Symptomatology on the Knowledge of the Health Care Providers

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Background and Aims: Brain tumors in children are commonly diagnosed late, particularly in Low-middle-income countries. Inspired by HeadSmart and funded by MyChildMatters program and King Hussein Cancer Center, we launched the HeadSmart Jordan project to help raise awareness about symptoms of pediatric brain tumors among health care providers in Jordan. We aim to evaluate the knowledge gained after one-hour Zoom lecture given to health care providers.

Methods: Thirty-six lectures were held to elaborate on the symptoms of pediatric brain tumors and the importance of early diagnosis. Questions were arranged randomly and distributed as a pre- and post-lecture test to assess attendees' knowledge improvement. They assessed the proper management of a child with symptoms that may suggest a brain tumor. T-test was used to assess the baseline knowledge (pre-test) and any improvement (difference between pre and post test scores). Data of the medical and nursing groups were analyzed separately.

Results: A total of 495 medical students, 176 new medical graduates, and 118 residents (pediatrics, family medicine and emergency), in addition to 288 nursing students attended the lectures. Male: female ratio was 1:1.25 (medical), and 1:3.5 (nursing). The residents' baseline knowledge was the highest; pre-test mean score was 69.7 (SD±13.2) compared to 57.0 (SD± 13.6) for medical students, 54.3 (SD±12.9) for graduates and 41.8 (SD ±14.1) for nursing. All groups showed significant improvement in the post-test mean scores ($p \leq 0.001$) regardless of gender (p -value > 0.05).

Conclusions: An hour of education on the importance of early diagnosis of pediatric brain tumors and their symptoms, helped to improve health care providers' knowledge. Integration of a similar education in the universities' curricula would strengthen this knowledge. It is important to demonstrate in future if this leads to a shorter time to diagnose children with brain tumors in Jordan.

E-Poster Topic: Disease Orientated / ASO3 Brain Tumours

PV0320 / #269 | Molecular Characterization of Serial Atypical Teratoid Rhabdoid Tumors Occuring in a Young Man 15 Years Apart

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Background and Aims: Central nervous system (CNS) Atypical Teratoid Rhabdoid Tumors (ATRT) are aggressive malignancies usually occurring in young children. While intense multimodal therapy improves cure, most children die of progressive disease within a few years. Biallelic SMARCB1 or SMARCA4 inactivation are the defin-

ing ATRT genetic alterations, while germline mutations define Rhabdoid Tumor Predisposition Syndrome (RTPS). We describe a young man with RTPS treated for a posterior fossa ATRT at 3.5-years-old and a spinal cord ATRT at 19-years-old.

Methods: Medical record review; germline and tumor molecular analyses

Results: Case: A 3.5-year-old boy developed nausea, emesis, and headache. CNS magnetic resonance imaging (MRI) demonstrated a localized cerebellar mass. Pathology following gross total resection (GTR) showed classic rhabdoid and undifferentiated tumor cells with loss of INI1 expression, confirming ATRT. He received craniospinal radiation and multiagent chemotherapy (CCG99701). At 19-years-old he developed intrascapular back pain. MRI showed a C7-T1 intraspinal mass. Pathology following GTR confirmed ATRT. He completed re-irradiation and is undergoing multi-agent chemotherapy. Peripheral blood mutational analysis demonstrated a SMARCB1 splice-site pathogenic variant c.986+1G>C in the patient, 2 siblings and their father, confirming RTPS. Genomic evaluation of the initial ATRT showed the SMARCB1 variant (91% variant allele frequency) with chromosome 22q double inactivation. Genomic evaluation of the subsequent ATRT showed the SMARCB1 pathogenic variant (81% variant allele frequency), loss of chromosome 22q, two somatic NF1 pathogenic variants, and multiple copy number changes. Methylation analyses matched both tumor specimens to ATRT. Both siblings developed ATRT-one at 18-months-old, currently alive without disease following treatment, and one at 46-months-old, deceased from treatment-related sepsis. The father died at 48-years-old from lung adenocarcinoma with biallelic SMARCB1 pathogenic variants.

Conclusions: This patient's spinal cord ATRT may represent a new primary with potentially targetable NF1 mutations. As more children survive childhood ATRT, surveillance strategies to identify subsequent malignancies are crucial, especially in individuals with RTPS.

E-Poster Topic: Disease Orientated / ASO3 Brain Tumours

PV0321 / #274 | Childhood Central Nervous System Tumors Characteristics and Survival in Ten Colombian Cities

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Background and Aims: Data on overall survival (OS) for central nervous system (CNS) tumors in children in low-and-middle income countries (LMIC) is scarce. We describe clinical characteristics and OS of children with CNS tumors in ten Colombian cities.

Methods: We analyzed prospectively collected data from 2009 to 2020 of children (<15 years) with primary CNS tumors included in VIGICANCER (Childhood Cancer Outcomes Surveillance System established in 2009 with the Sanofi-Esper-Foundation support) currently operating in ten Colombian cities. VIGICANCER does not include craniopharyngiomas, and germ cell tumors were excluded from this analysis. We used the Kaplan-Meier method for survival estimates.

Results: During the study period, 764 CNS tumors were diagnosed in 5063 children with cancer. Cohort median age was 7.5 years, 55% were males, and 10% afro-descendants. The cerebrum was involved in 48%, cerebellar in 27%, brain stem (BST) in 14%, and other brain structures/spinal cord in 11%. Pathology diagnoses were comprised of 29% supratentorial gliomas, 26% embryonal CNS tumors (medulloblastoma in 80%), 20% gliomas in other locations, 11% ependymomas, 4% neuroglial, 2% pineal, 1% neuroepithelial, and 7% other. Cohort 5y-OS was 44% (95%CI: 39, 48), and in less than 3 years of age 5y-OS was 36% (95%CI: 26, 47). OS improved from 36% (2009-2013) to 46% (2014-2020); $p=0.001$. Five-year OS were 85% (95%CI: 72, 92), 59% (95%CI: 41, 73) and 14% (95%CI: 6, 26) for grade I, grade II and grade III/IV supratentorial gliomas, respectively. Five-year OS for medulloblastoma and brain stem tumors were 46% (95%CI: 35, 56) and 8% (95%CI: 2, 18), respectively.

Conclusions: Our results contribute to the limited data on childhood CNS survival in LMIC. We found a significant increase in OS from 2014-2020 compared to 2009-2013. This survival improvement is modest compared to advances in high-income countries, achieving 5y-OS of 70 to 75% in malignant CNS tumors.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0322 / #297 | End-Of-Life Care for Pediatric Patients with Brain Tumors: Does Management by a Pediatric Palliative Care Unit Make a Difference?

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Background and Aims: Pediatric brain tumors constitute a major cause of morbidity and mortality. Pediatric palliative care (PPC) aims to improve the patient's quality of life and overall care. We aimed to compare the characteristics and last-month-of-life care of patients with CNS tumors based on if they were attended by a PPC Unit (PPCU) or not in a tertiary hospital.

Methods: Descriptive retrospective analysis of clinical records of deceased patients with CNS tumors between 1/1/2010-12/31/2020 in

a university tertiary hospital. Variables included epidemiological, cancer diagnosis and treatment, outcome, support measures, and hospital admission days in the last month of life and death location. Statistical analysis (non-parametrical or parametrical when applicable) was performed (established significance <0.05).

Results: 48 patients with CNS tumors died in the period of study, 75% attended by the PPCU. No differences were found regarding sex, age at diagnosis or at death, number of treatment, metastatic disease at diagnosis or during therapy or ICU admissions. Regarding the last month of care, no differences were found in chemotherapy or radiotherapy administration, red blood or platelet transfusions or invasive interventions. Patients attended by the PPCU stayed less days in hospital in their last month (median of 2 vs 9.5; $p=0.02$) and last week (median of 0.5 vs 7; $p<0.01$) and were more prone to die at home (50% vs 0%; $p<0.01$).

Conclusions: The PPCU attended 75% of the deceased patients with a CNS tumor. Whether the patients were attended or not by the PPCU did not seem to be influenced by their epidemiological or oncological features. Patients attended by the PPCU spent less days in hospital in their last month and were more prone to die at home. Further data on other cancers and specific interventions of PPC Units are needed.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0323 / #345 | Simultaneous Analysis of Medulloblastoma 4-Molecular Subgroups and MYCN Amplification by nCounter Assay

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Background and Aims: Medulloblastoma (MB) is the most frequent pediatric malignant brain tumor. It is a heterogeneous disease divided into 4 molecular subgroups: WNT, SHH, Group3, and Group4. In the last three MB subgroups, the amplification of MYCN oncogene is associated with a more aggressive phenotype and therefore may be used for patient risk stratification. The detection of MYCN amplification can be achieved by different methodologies, yet most of them require high amount/quality material, are laborious, and have a high cost. Herein we propose to perform the simultaneous analysis of the medulloblastoma molecular subgroups and MYCN amplification using an RNA-based technology (nCounter).

Methods: FFPE of 50 MB samples from Barretos Cancer Hospital were analyzed. Following RNA isolation, the MB molecular classification was done using the 22-target panel along with the MYCN. The nSolver software was used for the evaluation of mRNA counts (normalized by housekeeping genes) and the IBM SPSS software for the

statistical analysis. All 50 MB were also evaluated for MYCN amplification by FISH (ZytoLight® SPEC MYCN/2q11 Dual Color Probe).

Results: Of the 50 cases, 54% were SHH (n=27), 12% group 3 (n=6) and 34% group 4 (n=17). We found MYCN amplification by FISH in 3/50, being all SHH. These 3 amplified cases exhibited a median MYCN mRNA normalized counts of 14.901, a value significantly higher (Mann-Whitney test, $p<0.0001$) than the 47 non-amplified cases (median of 1.864 MYCN mRNA normalized counts).

Conclusions: The present findings suggest it is possible to simultaneously analyze the medulloblastoma 4-molecular subgroups and MYCN amplification in a single nCounter assay with no additional cost, neither additional biological material in a routine setting.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0324 / #44 | Cardiac Remodeling in Patients with Childhood-Onset Craniopharyngioma – Results of Hit-Endo and Kraniopharyngeom 2000/2007

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Background and Aims: Hypothalamic obesity caused by childhood-onset craniopharyngioma results in long-term cardiovascular morbidity. Knowledge about clinical markers and risk factors for cardiovascular morbidity is scarce.

Methods: A cross-sectional study on transthoracic echocardiographic parameters was performed to determine the associations with clinical and anthropometric parameters in 36 craniopharyngioma patients.

Results: BMI correlated with the thickness of interventricular septum in diastole (IVSd) ($r=0.604$, $p<0.001$) and left ventricular posterior wall thickness in diastole (LVPWd) ($r=0.460$, $p=0.011$). In multivariate analyses on risk factors for cardiac remodeling, sex hormone replacement therapy, BMI and male gender were positively correlated with increased left ventricular internal diameter in diastole (LVIDd), $R^2=0.596$, $F=10.323$, $p<0.001$. BMI and insulin resistance were selected as significant independent determinants of IVSd, produced $R^2=0.655$, $F=29.441$, $p<0.001$. Due to wide range of disease duration, 17 pediatric and 19 adult patients were analyzed separately. In the adult subgroup (age at study ≥ 18 years), BMI correlated with IVSd ($r=0.707$, $p=0.003$), LVPWd ($r=0.592$, $p=0.020$) and LVIDd ($r=0.571$, $p=0.026$). In the pediatric subgroup (age at study <18 years), no correlation between TTE parameters and BMI was observed. Only LVIDd correlated with disease duration ($r=0.645$, $p<0.001$). All cardiac functions were within the normal range, indicating no association with functional impairments.

Conclusions: Cardiac remodeling in patients with craniopharyngioma correlated with the degree of hypothalamic obesity, disease duration, sex hormone replacement therapy, male gender and insulin resistance.

As echocardiography has limited sensitivity in patients with obesity, further research on more sensitive techniques for cardiac diagnostics in craniopharyngioma patients is warranted.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0325 / #45 | Body Composition and Nuchal Skinfold Thickness in Pediatric Brain Tumor Patients

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Background and Aims: Obesity, cardiovascular disease (CVD), and relapse/progression have impact on prognosis in pediatric brain tumor (BT) patients.

Methods: In a cross-sectional study, we analyzed nuchal skinfold thickness (NST) on MRI follow-up monitoring as a parameter for body composition (BC) and CVD in 177 BT patients (40 WHO grade 1–2 BT; 31 grade 3–4 BT; 106 craniopharyngioma (CP)), and 53 healthy controls (HC). Furthermore, BMI, waist-to-height ratio (WHtR), caliper-measured skinfold thickness (cSFT), and blood-pressure (BP) were analysed.

Results: CP patients showed higher BMI, WHtR, NST and cSFT when compared with BT and HC. WHO grade 1–2 BT patients were observed with higher BMI, waist circumference and triceps cSFT when compared to WHO grade 3–4 BT patients. NST correlated with BMI, WHtR, and cSFT. NST, BMI and WHtR had predictive value for CVD in terms of increased BP. In multivariate analysis, only BMI was selected for the final model resulting in an odds ratio of 1.25 (1.14–1.379). In CP patients with hypothalamic involvement/lesion or gross-total resection, rate and degree of obesity were increased.

Conclusions: NST could serve as a novel useful parameter for assessment of BC and CVD risk in BT patients.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0326 / #454 | Incidence and Survival of Childhood Central Nervous System Tumors in Denmark Between 1997 and 2019

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Background and Aims: Central nervous system (CNS) tumors are the second most common childhood malignancy and have a high mortality. Incidence rates in Denmark remain among the highest in the world. Survival rates, however, have improved in the past decades in high-income countries. Our aim was to analyze incidence and survival rates and depict treatment modalities for children with CNS tumors over a 23-year period in Denmark.

Methods: We analyzed incidence and survival of childhood CNS tumors in Denmark diagnosed between 1997 and 2019 based on data from Danish Childhood Cancer Registry (DCCR) and information on histological types, tumor localization and treatment information abstracted from medical records.

Results: In 1997-2019, 949 children < 15 years were diagnosed with a CNS tumor. Age-standardized incidence from 1997-2019 was 42.1 per million person-years (95% CI, 39.4-44.6) and stable over the period. Age-specific incidence was highest for children aged 0-4 years (47.7 per million person-years). More than one third (n=374, 39.4%) were solely treated with surgery. Survival probability 5 and 10 years after diagnosis were 77.6% (95% CI, 74.7-80.2) and 74.7% (95% CI, 71.7-77.5), respectively. Overall, the 5-year survival improved from 73.0% (95% CI, 68.9-76.7) in 1997-2008 to 83.2% (95% CI, 79.2-86.4) in 2009-2019 (p-value <0.0001) in both children aged 0-4 years (p=0.0006) and children aged 5-14 years (p=0.16). Survival, however, differed considerably between histological subtypes. The histological type with the poorest 5-year survival was embryonal tumors (59.3%, 95% CI 50.9;66.7) and the highest was intracranial and intraspinal germ cell tumors (93.2%, 95% CI 75.1;98.3).

Conclusions: Incidence rates of CNS tumors in Denmark were stable in the period 1997-2019 but still remain among the highest in the world. Despite an improved survival in both younger and older children, some subtypes still have a poor prognosis.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0327 / #455 | **A Multi-Institutional Polish Study of Paediatric High Grade Gliomas**

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Background and Aims: High grade gliomas (HGG) constitute of less than 5% of pediatric brain tumors. Due to the rarity of them data on this topic are scarce. Our study aimed to investigate the long-term results of treatment of children with HGG and to identify factors related to better survival.

Methods: We performed a retrospective analysis of patients treated for HGG who had the main tumor location outside the brainstem. The analysis of children diagnosed with and treated due to HGG in 7 major polish institutions was carried out. The evaluation of factors which correlated with better survival was performed with the Cox proportional-hazard model. Survival was estimated with the Kaplan-Meier method.

Results: Study group consisted of 82 consecutive patients. All of them underwent surgery as primary treatment. Chemotherapy was applied in 93% of children with one third treated with temozolomide. After or during the systemic treatment, 79% of them received radiotherapy with median dose of 54 Gy. Median follow-up was 122 months and during that time 59 patients died. One, 2-, 5-, and 10-year overall survival was 78%, 48%, 30% and 17%, respectively. Patients with radical (R0) resection and temozolomide-based chemotherapy had better overall survival. Progression-free survival was better in patients after MR-confirmed R0 resection and radical radiotherapy.

Conclusions: The best outcome in HGG patients is observed in patients after R0 resection with immediate postoperative temozolomide-based chemotherapy and radical radiotherapy.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0328 / #46 | **Cerebral Infarction in Childhood-Onset Craniopharyngioma Patients – Results of Kraniopharyngeom 2007**

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Background and Aims: Cerebral infarction (CI) is a known vascular complication following treatment of suprasellar tumors. Risk factors for CI, incidence rate, and long-term prognosis are unknown for patients with childhood-onset craniopharyngioma (CP).

Methods: MRI of 244 CP patients, recruited between 2007 and 2019 in KRANIOPHARYNGEOM 2007, were reviewed for CI. Risk factors for CI and outcome after CI were analyzed.

Results: Twenty-eight of 244 patients (11%) presented with CI based on reference assessment of MRI. One CI occurred before initial surgery and one case of CI after release of intracystic pressure by a cyst catheter. 26 of 28 CI were detected after surgical tumor resection at a median postoperative interval of one day (range: 0.5–53 days). Vascular lesions during surgical procedures were documented in 7 cases with CI. There was a trend ($p=0.094$) towards higher initial presurgical tumor volume in CI patients compared with non-CI patients. No relevant differences with regard to surgical approaches were found. In all 12 irradiated patients, CI occurred before irradiation. Multivariable analyses showed that hydrocephalus and gross-total resection at the time of primary diagnosis/surgery both were risk factors for CI. PFS was lower after CI (median survival=1.16 years) when compared with the subgroup of patients without CI (median survival>5.62 years). After CI, quality of life (PEDQOL) and functional capacity (FMH) were impaired.

Conclusions: CI occurs in 11% of CP cases. Degree of resection and increased intracranial pressure are risk factors, which should be considered in the planning of surgical procedures for prevention of CI.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0329 / #49 | Pregnancies after Childhood

Craniopharyngioma – Results of Kraniopharyngeom 2000/2007

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Background and Aims: Data on female fertility, pregnancy, and outcome of offspring after childhood-onset craniopharyngioma (CP) are rare.

Methods: Observational study on pregnancy rate and outcome of offspring after CP in postpubertal, female patients recruited in KRANIOPHARYNGEOM 2000/2007 since 2000.

Results: 451 CP patients (223 female) have been recruited. 269 CP patients (133 female) were postpubertal at study. Six of 133 female CP patients (4.5%) with a median age of 14.9 years at CP diagnosis had 9 pregnancies, giving birth to 10 newborns. Three patients achieved complete surgical resections. No patient underwent postoperative irradiation. Five natural pregnancies occurred in 3 CP patients without pituitary deficiencies. Four pregnancies were achieved in 3 CP with hypopituitarism under assisted reproductive techniques (ART) (after median 4.5 cycles, range: 3–6 cycles). Median maternal age at pregnancy was 30 years (range: 22–41 years). Six babies (60%) were delivered by caesarean section. Median gestational age at delivery was 38 weeks (range: 34–43 weeks); median birth weight was 2,920

grams (range: 2,270–3,520 grams), the rate of preterm delivery was 33%. Enlargements of CP cysts occurred in 2 women during pregnancy. Other severe complications during pregnancy, delivery and postnatal period were not observed.

Conclusions: Pregnancies after CP are rare and achieved in 45% after ART. Close monitoring by an experienced reproductive physician is necessary. Due to a potentially increased risk for cystic enlargement, MRI of cystic CP is recommended during pregnancy. Severe perinatal complications, birth defects, and postnatal morbidity of mothers and offspring were not observed.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0330 / #495 | Challenges in Determining The Global Burden of Low-Grade Pediatric Brain Tumors

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Background and Aims: Low-grade brain tumors cause significant morbidity and mortality in children, but clarity on their global burden is challenging in part due to inconsistent inclusion in population-based cancer registries. We performed a systematic review of the cancer registry sources used by the Global Burden of Disease (GBD) study to determine availability of data reporting low-grade brain tumors.

Methods: Cancer registry data sources catalogued for use in the GBD study by August 8, 2020 were included. Two blind reviewers categorized datasets based on whether they included, excluded, or did not specify inclusion of low-grade brain tumors. Tumors of uncertain behavior were included as low-grade for this analysis. We report the number and location of datasets by low-grade brain tumor inclusion and by World Bank country classification.

Results: In total, 67 (17.3%) datasets across 23 countries reported low-grade brain tumors in a category distinct from malignant tumors and 71 (18.3%, 86 countries) reported low-grade brain tumors within their malignant brain tumor totals. Pediatric datasets that were reported using International Classification of Childhood Cancer (ICCC) coding could not be distinguished by behavior and were categorized as including both malignant and low-grade brain tumors unless the data were

specified to contain only malignancies of the brain. Of the 23 countries reporting low-grade brain tumors separately, seven were low- or middle-income countries (LMIC). Only 20 of the datasets across ten countries (one LMIC) additionally reported deaths separately by brain tumor behavior.

Conclusions: While global cancer burden estimation frameworks like the GBD study may be useful for estimating the burden of low-grade brain tumors where data is not present, broader reporting of these tumors separately from malignant brain tumors will be needed to fully capture differences in their epidemiology around the world.

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PV0331 / #50 | Craniopharyngiomas Diagnosed as Incidentalomas - Results of Kraniopharyngeom 2007

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Background and Aims: Childhood-onset craniopharyngiomas (CP) are diagnosed due to clinical symptoms (symCP) or incidentally (incCP). We investigated clinical manifestations and outcome in incCPs and symCPs.

Methods: IncCP were discovered in 4 (3m/1f) and symCP in 214 (101m/113f) CP recruited 2007–2014 in KRANIOPHARYNGEOM 2007. Age, sex, height, body mass index (BMI), tumor volume, degree of resection, pre- and postsurgical hypothalamic involvement/lesions, pituitary function and outcome were compared between both subgroups.

Results: Reasons for imaging in incCP were cerebral palsy, head trauma, nasal obstruction, and tethered-cord syndrome, whereas headache (44%), visual impairment (25%), and growth retardation (17%) lead to imaging in symCP. Tumor volume at diagnosis was smaller in incCP (median 2.39 cm³; range: 0.14–4.10 cm³) when compared with symCP (15.86 cm³; 0.002–286.34 cm³). Age, gender, BMI, height, hydrocephalus, tumor location, and hypothalamic involvement at diagnosis of incCP were within the range of these parameters in symCP. Complete resections were achieved more frequently (3/4 patients) in incCP when compared with symCP (20%). Surgical hypothalamic lesions were distributed similar in incCP and symCP. Irradiation was performed only in symCP (33%). No noticeable differences were observed concerning survival rates, endocrine deficiencies, BMI,

height, functional capacity and quality of life of the 4 incCP cases when compared with the symCP cohort.

Conclusions: IncCP are rare (1.8%) and characterized by lack of endocrine deficiencies, resulting in normal height and BMI, no hydrocephalus, and smaller tumor volume at diagnosis when compared with symCPs. Outcome of the observed incCP is similar with symCP.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0332 / #569 | Patterns of Failure and Long-Term Outcomes for Pediatric and Young Adult Patients with Myxopapillary Ependymoma after Radiotherapy

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Background and Aims: Myxopapillary ependymoma is a rare, typically slow-growing subtype of spinal ependymomas. There are no standard guidelines for radiotherapy and long-term outcomes for pediatric and young adult patients with myxopapillary ependymoma remain under-characterized.

Methods: This is an IRB-approved retrospective cohort study of 19 patients (age <27 years) with myxopapillary ependymoma treated 2000–2019 at Massachusetts General and Brigham Hospitals. Clinical and treatment characteristics, and long-term outcomes were recorded.

Results: Of the 19 patients, five (26.3%) were female. Median age at radiation was 16.4 years (range: 9.3–26.5). Median follow-up was 7.4 years (range: 1.4–18.1). At radiotherapy, fifteen patients (78.9%) had gross residual disease (GRD), five (26.3%) had metastatic disease above the primary site at the conus, and none had intracranial involvement. Sixteen patients (84.2%) received proton radiotherapy, and all radiation fields included the lumbosacral spine, primary tumor near the conus at diagnosis, and metastatic sites above the primary tumor. One (5.2%) and three (15.8%) received whole spine radiotherapy and craniospinal irradiation, respectively. Median dose was 50.4 Gy/Gy(RBE) (range: 46.8–54.0). While all patients were alive at last follow-up, five- and seven-year disease-free survival was 53%. Nine recurred at a median of 2.2 years (range: 0.1–12.7) after radiotherapy. One recurrence was in-field and at sites of GRD. This patient later developed intracranial progression. Five patients had intracranial recurrences at first progression with three having disease confined below T11 and two having higher spinal metastases at radiotherapy. Three patients recurred in the spine above the previous radiation field. Salvage therapy included radiation (77.8%), surgery (55.6%), and systemic therapy (44.4%).

Conclusions: Pediatric and young adult patients with myxopapillary ependymoma remain at risk for recurrences in the spine above the

radiation field and intracranially after radiotherapy. Further study is needed to determine the appropriate radiation field based on the extent of metastases in the spine.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0333 / #584 | Outcomes of Intraventricular 131-I-8H9 and External Beam Radiotherapy in Patients with Medulloblastoma, Ependymoma, and Pineoblastoma

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Background and Aims: Intraventricular compartmental radioimmunotherapy (cRIT) with radio-iodinated murine IgG1 monoclonal antibody 131-I-8H9 offers a therapeutic strategy for primary brain tumors. Most patients also receive external beam radiotherapy (EBRT) to a portion or full craniospinal axis (CSI), many with protons. Patterns of failure after 131-I-8H9 cRIT after combination with EBRT are unclear.

Methods: Patients with medulloblastoma, ependymoma, or pineoblastoma who received 131-I-8H9 cRIT were included. Extent of disease at cRIT initiation was assessed (no evidence of disease [NED] v evidence of disease [EOD]) as a predictor of response and progression-free survival (PFS) by Kaplan-Meier analysis with log-rank tests.

Results: Thirty-two patients (21 medulloblastoma, 8 ependymoma, 3 pineoblastoma) were included with median follow-up of 1.5 years from cRIT (range, 0.3-12.8). Median age at cRIT initiation was 12.3 years (range, 3.2-40.1). Median number of cRIT injections was 2.5 (range, 1-4) with median total dose of 72 millicurie (range, 2-192). Nearly all medulloblastoma patients had EBRT prior to cRIT (n=20, 95%) with median time to cRIT of 0.6 years (range, 0.1-4.9). Median CSI dose was 2340cGy (range, 2300-3600) with median boost to 5400cGy (5400-6000). Ten received CSI with protons and 15 also had focal EBRT. All ependymoma patients had prior EBRT (4 received protons); 2 pineoblastoma patients had prior EBRT. At cRIT initiation, 10 medulloblastoma (48%), 6 ependymoma (75%), 1 pineoblastoma (33%) patients had EOD. Post-cRIT for medulloblastoma, 1/11 initially NED patients progressed vs 6/10 EOD patients (p=0.02). For ependymoma, the 2 NED patients did not progress, 5/6 EOD patients progressed (p=0.11). Medulloblastoma patients NED at cRIT initiation had significantly improved PFS (p=0.024). All progressions involved the brain, 3 involved the spine.

Conclusions: Extent of disease at time of cRIT was associated with improved PFS in medulloblastoma patients. These data suggest EBRT for local control prior to cRIT may play a role in optimizing treatment outcomes.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0334 / #605 | BRAF V600E Inhibitors for Pediatric Low and High-Grade Glioma: J.P. Garrahan Hospital Experience

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Background and Aims: The BRAF V600E point mutation plays a key role in the tumorigenesis of many low- and high-grade gliomas. Its inhibition is part of the innovative and promising therapies emerging in recent years. Knowing the role and tolerance of these treatments is essential in the molecular era. The aim of this publication is to describe the oncologic response and adverse effects of targeted BRAF V600E inhibition in paediatric population with gliomas harbouring said mutation.

Methods: Descriptive, retrospective and single-center study, collecting information from paediatric patients with BRAF V600E gliomas who received Vemurafenib, Dabrafenib or both in the Neuro-Oncology service of Hospital Garrahan, a tertiary care center in a developing country.

Results: Thirteen patients were treated with BRAF inhibitions (Vemurafenib=10, Dabrafenib=1, both=2) in the last 7 years: nine were low grade gliomas (3 disseminated) and 4 high grade gliomas (1 disseminated). The median age at diagnosis was 8.63 years (0.89-14.04), and at start of target therapy was 11.62 years (3.64-15.42). All patients had a previous surgical procedure, and 92% had received another therapy prior BRAF inhibition: eighty three percent conventional chemotherapy (in one case, up to 4 different protocols) and 33% radiotherapy. Under target therapy, tumour response was obtained in 76% of patients (with size reduction greater than 25%), and the best response was observed in the first 6 months of treatment in 70%. Four patients progressed under treatment (all high-grade gliomas) and 2 progressed shortly after stopping the inhibitor (both low grade gliomas). Tolerance was acceptable: thirty eight percent had grade 3-4 toxicity, with complete recovery after dose reduction or treatment suspension.

Conclusions: A good response of this therapy in paediatric population with BRAF V600E gliomas is described, with an acceptable tolerance.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0335 / #691 | Systematic Review of Survival Benefit from Different Radiotherapy Modalities for Recurrent Ependymoma

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Background and Aims: Re-irradiation (re-RT) plays an important role in the management of recurrent ependymoma. Historically, re-RT has been performed with more focal techniques, but contemporary studies suggest benefits with craniospinal irradiation (CSI), with acceptable toxicities. The initial purpose of this study was to perform a meta-analysis comparing survival outcomes of CSI versus tumor bed irradiation (TB-RT) for re-RT for pediatric patients with recurrent ependymoma.

Methods: A systematic review of PubMed, Embase, and Lilacs was performed in July 2020, searching for studies reporting on survival outcomes of patients with recurrent ependymoma and submitted to re-RT as part of their treatment. Two authors independently selected and assessed the studies regarding eligibility criteria and risk of bias, and a third author revised it. The fixed-effect model was performed for median overall survival (OS) and progression-free survival (PFS). Non-heterogeneity was verified by the I² and Q-Cochran test.

Results: Six retrospective studies met final inclusion criteria for all purposes, including survival from re-RT, with a total of 216 patients. No randomized studies were found, one study encompassed 101 patients. After an appraisal of the quality of the studies, it was not possible to compare TBRT to CSI re-RT, due to the small number of studies. Median OS and PFS from the date of first recurrence were 30.8 months (95% CI 23.55 - 38.00) and 21.6 months (95% CI 16.13 - 26.97), respectively. Two of these studies reported surgery attempt, histology, CSI if distant failure, interval to first recurrence, and female gender as prognostic factors associated, on multivariable analysis, to OS.

Conclusions: Recurrent ependymoma remains a disease with dismal outcomes. For the time being, there is insufficient data to compare TBRT to CSI re-RT. A large contemporary study suggests good results with CSI.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0336 / #701 | Differentially Expressed-Aberrantly Methylated HUB Genes in Medulloblastoma SHH

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Background and Aims: Medulloblastoma (MB) is the most common malignant brain tumor in childhood. Currently, MB is classified in

according to molecular characteristics in four subgroups: WNT (Wingless), SHH (Hedgehog), Group 3 and Group 4. Dysregulation of DNA methylation is important for MB tumorigenesis and diagnosis; since genes methylation status could be included to MB classification. In this context, we aimed to identify differentially expressed-aberrantly methylated hub genes in MB SHH.

Methods: Gene expression data (GSE85212 and Pfister dataset); and gene methylation data (GSE85212 and GSE93646) were selected for further analysis. R2, Genomics Analysis and Visualization Platform, was used to obtain differentially expressed-aberrantly methylated genes between SHH and other MB subgroups. Pathway enrichment was performed by Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway. STRING platform was used to elucidated Protein-protein interaction (PPI) network; while that hub genes were find out through cytoHubba app in Cytoscape software. Association between survival analysis and hub genes expression were also performed using R2 Genomics Analysis and Visualization Platform.

Results: Overlapping between databases evidenced 135 downregulated and hypermethylated genes in SHH subgroup compared with to MB subgroups. These genes were mainly associated with neuron projection maintenance, signal transduction by p53 and apoptosis pathway. In PPI network, we found 8 top hub genes: AIPL1; UBR4; ABCA4; IMPG2; TBR1; OTX2; GNB3 and CHRM2. IMPG2 and TBR1 were associated with overall survival (OS). While that 250 genes were upregulated and hypomethylated in SHH MB; they were mostly involved in regulation of cell migration, central nervous system development, and regulation of apoptotic process. In this context, top 8 hub genes included EGFR; PAX6; PTCH1; MCM2; SDC2; ACTN1; BCAR1 and ACDY8. MCM2 and BCAR1 showed associated with OS in MB.

Conclusions: IMPG2, TBR1, MCM2 and BCAR1 are hub genes differentially expressed-aberrantly methylated in medulloblastoma SHH, associated with MB patient's survival.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0337 / #736 | Investigation of Molecular Sub-Groups by Immunohistochemical Methods and its Prognostic Importance in Medulloblastoma

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Background and Aims: Medulloblastoma is the most frequent brain tumor in children. Recently molecular and genetic studies revealed subtypes that lead to contribute the prognosis. This study evaluates with new comprehensive classification and discloses the relationship between prognosis and the subtypes.

Methods: Pathology blocks of patients diagnosed with medulloblastoma and whose treatment was completed were retrospectively evaluated. Molecular subgroups were defined using immunohistochemical methods, and the clinical data of the patients and their relationship with prognosis were analyzed.

Results: Thirty patients were assessed (F: 15; M: 15). The median age was 7,8 years (9 months- 16,5 years). Thirteen patients (43,3%) were classified in the low-risk group, while seventeen (56,7%) were in the high-risk group. The median follow-up time was 31,5 months (7-99 months). The five-year EFS rate was %57,6, and the OS rate was 68,4%. Any significant differences were noted among risk groups ($p>0,05$). According to molecular subtypes, 5 patients (16,7%) WNT (+), SHH p53 1 patient (3,3%), active SHH p53 wild 11 patients (36,7%), non WNT/nonSHH 13 patients (43,3%). 5 years EFS and OS rates respectively were 75,100% for WNT, 42,8;66,7% for SHH, 58,3;53,8% for non WNT/non-SHH. Among all patients, mutant p53 was detected in 5 (16,7%) patients, whereas wild p53 was in 25 (83,3%) patients. For mutant p53 individuals, five years EFS and OS were 40,80%, respectively. For p53 wild type, five years EFS and OS were 63,2;66,1% respectively. No significant differences were noticed between these two groups ($p>0,05$).

Conclusions: In this comparative study, no significant differences in the outcome were noted between clinical risk groups. The distribution of molecular subtypes and each subtype's prognosis were compatible with the literature. WNT tumors have a much better prognosis than all other subtypes. Molecular features were thought to be more crucial than clinical risk groups to determine survival.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0338 / #770 | NTRK2 Gene Fusions are Uncommon in Pilocytic Astrocytoma

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Background and Aims: Pilocytic astrocytoma (PA) is the most frequent pediatric primary brain tumor. Despite its overall good prognosis, the complete surgical resection is sometimes unfeasible, especially for patients with deep-seated tumors. For these patients, the identification

of targetable genetic alterations such as NTRK fusions raised as a new hope for therapy. The presence of gene fusions involving NTRK2 has been rarely reported in PA. The aim of the present study was to investigate the frequency of two described NTRK2 gene fusions (QKI:NTRK2 and NACC2:NTRK2) in a Brazilian PA cohort.

Methods: It was included 69 patients with PA, aged from 0.3 to 53.4 years old (median age=9.1 years old) from Barretos Cancer Hospital, Brazil. The analysis of NTRK2 fusions was performed using a dual-color break-apart fluorescence in situ hybridization (FISH) assay.

Results: NTRK2 fusions were successfully evaluated by FISH in 62 PA. No evidence of NTRK2 gene rearrangements neither NTRK2 copy number gains were found in our PA cohort.

Conclusions: NTRK2 fusions are uncommon genetic events in pilocytic astrocytomas, regardless of patients' clinicopathological and molecular features.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0339 / #805 | Adaptive Functioning Associated with Posterior Fossa Syndrome in Survivors Treated for Medulloblastoma

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Background and Aims: Pediatric patients who experience posterior fossa syndrome (PFS) following surgical intervention for medulloblastoma have long-term neurocognitive deficits; however, consequences for adaptive functioning are unknown. The goals of the present study were to 1) compare adaptive, behavioral, and emotional functioning between survivors with and without a history of PFS; and 2) examine the association between adaptive and intellectual functioning in this sample.

Methods: Comprehensive neuropsychological evaluations were completed for survivors treated for medulloblastoma (N = 45) with and without a history of PFS, including performance-based measures of intellectual functioning, and caregiver reports of adaptive, behavioral, and emotional functioning. We evaluated 1) differences in adaptive, behavioral, and emotional functioning; 2) rates of impairment; and 3) correlations between neurocognitive and adaptive functioning.

Results: Following surgical resection for medulloblastoma, 15 of 45 cases (33%) met criteria for PFS. Participants were primarily male (67%) and ranged in age from 7-18 years at the time of evaluation, with evaluations conducted 3.84 years post-radiation on average. The PFS group had a greater frequency of hydrocephalus requiring a shunt ($p=0.003$) and lower total radiotherapy dose to the tumor bed ($p=0.021$). Adaptive functioning was significantly worse in the PFS group for Practical ($d=0.81$) and Global Adaptive ($d=0.71$) skills compared to those without PFS. The PFS group also exhibited higher

rates of clinical impairment than expected normatively ($p < 0.05$). Despite having lower intellectual functioning scores, Full Scale IQ was uncorrelated with adaptive outcomes in the PFS group ($r < 0.2$). No significant group differences or increased rates of impairment were observed for behavioral and emotional outcomes ($p > 0.10$).

Conclusions: Survivors with PFS compared to those without are rated as having significant deficits in global adaptive functioning, with specific weakness in Practical skills. Findings suggest an increased risk for ongoing functional deficits despite acute recovery from symptoms of PFS, highlighting the need for early intervention to mitigate risk.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0340 / #836 | Health-Related Quality of Life Among Survivors of Childhood Central Nervous System Tumour in Malaysia

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Background and Aims: Survivors of childhood central nervous system (CNS) tumour experience significant late effects affecting their health-related quality of life (HRQOL) despite improvement in their survival rate. The current study aimed to evaluate HRQOL among Malaysian childhood CNS tumour survivors.

Methods: A comparative cross-sectional study was conducted in a tertiary centre in Malaysia. Forty-six survivors of childhood CNS tumour and ninety (age and gender-matched) survivors of childhood acute lymphoblastic leukaemia (ALL) had been off-treatment for at least one year were recruited. HRQOL was assessed using the Pediatric Quality of Life Inventory (PedsQL V4.0 and PedsQL Cancer Module V3.0) and child behaviour checklist (CBCL).

Results: Most participants were male: thirty in the CNS tumour survivors group (65.2%) with a male:female ratio of 1.9:1.0 and fifty-two in ALL survivors group (57.8%) with a male:female ratio of 1.4:1.0. At the time of recruitment, the median age were 13.7 years old in CNS tumour survivors and 11.84 years old in ALL survivors group. The mean PedsQL V4.0 total scale were significantly lower in all domains ($p < 0.05$) among CNS tumour survivors especially teenagers compared with healthy children and ALL survivors. Special education need and physical impairment were associated with lower total PedsQL V4.0 score ($p < 0.05$). The median PedsQL Cancer Module V3.0 score of CNS tumour survivors was significantly lower than ALL survivors in total scale, cognitive problem and communication ($p < 0.05$). Abnormal internalising CBCL score was associated with lower psychosocial health summary ($p = 0.007$) indicating poor HRQOL in CNS tumour survivors.

Conclusions: CNS tumour survivors reported lower HRQOL than ALL survivors and healthy children. Clinicians need to be vigilant of HRQOL needs among CNS tumour survivors especially those with special education need, physical impairment and internalising behavioural

difficulties. Adequate physical and psychosocial interventions tailored to individual patient are essential to improve the HRQOL.

E-Poster Topic: Disease Orientated / AS03 Brain Tumours

PV0341 / #979 | Radiotherapy after High Dose Chemotherapy and Stem Cell Rescue May Improve Survival in Patients with CNS Retinoblastoma – A Single Center Experience

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Background and Aims: Central nervous system (CNS) retinoblastoma (RB) (trilateral or stage 4b metastatic RB) is associated with a poor prognosis. More effective therapy is urgently required. However, little data exists on the clinical and treatment characteristics which can serve as the basis for planning future collaborative multi-institutional trials.

Methods: We performed an IRB approved retrospective chart review of all patients with CNS RB who underwent induction chemotherapy with an intent to proceed to myeloablative consolidation chemotherapy with stem cell rescue (ASCT) at MSKCC between 1993-2020. Descriptive statistics were used for analysis.

Results: 14 Pts (10 f, 4 m) with a median age at metastatic RB diagnosis and ASCT ($n=12$) of 1.4 (0-3.5) and 3 (0.8-6.9) years, respectively, were identified. Seven had disease metastatic to the CNS, 7 had trilateral disease. 4 had localized CNS disease and 10 had leptomeningeal dissemination. Two progressed early on during induction chemotherapy and did not proceed to ASCT. Disease status prior to ASCT was complete response ($n=6$), very good partial response ($n=4$), stable disease but persistent CSF cytology positive ($n=2$). Seven relapsed post-ASCT, all in the CNS (median 6 mos; range 3.4 -15.5 mos). Three of 5 long term RB free survivors (survived at least 5 yrs post diagnosis) received irradiation post-ASCT prior to relapse: 1 received 2340 cGy craniospinal irradiation (CSI), 1 received 2340 cGy CSI and radio-immunotherapy (RIT) with intra-ventricular I-131 radiolabeled 3F8 (murine monoclonal antibody directed against a cell surface protein GD2), and 1 received RIT with 3F8 alone.

Conclusions: While the majority of Pts with CNS RB achieve a good response to induction chemotherapy, most relapsed within the first year post-ASCT. Radiotherapy via RIT and/or low dose proton CSI post consolidation chemotherapy may have clinical utility in maintaining remission in patients with CNS RB.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0342 / #985 | Prognostic Factors for Patients with Relapsed Central Nervous System Non-Germinomatous Germ Cell Tumors

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Background and Aims: We aim to identify prognostic factors that may help better understand the behavior of relapsed central nervous system (CNS) non-germinomatous germ cell tumors (NGGCT).

Methods: We performed a literature review of the PubMed database for manuscripts that included patients with relapsed CNS NGGCT.

Results: We identified nine studies, including 101 patients.; 33 patients (33%) were alive 12 months post-initial relapse. Five patients had pure CNS germinoma at relapse; 12 patients lacked histopathological evaluation or tumor marker levels at relapse. Sixty percent of patients with serum/cerebrospinal fluid (CSF) alpha-fetoprotein (AFP) level ≤ 25 ng/ml at initial diagnosis were survivors compared with 28% among patients with serum/CSF AFP level > 25 ng/ml ($p = 0.01$). Seventy-one percent of patients who achieved complete response/continued complete response (CR/CCR) by the end of therapy at relapse were survivors compared with 7% among patients who had less than CR/CCR ($p < 0.0001$). Forty-eight percent of patients who received marrow-ablative chemotherapy followed by autologous hematopoietic cell rescue (HDCx/AuHCR) following relapse were survivors compared with 12% among patients who did not receive HDCx/AuHCR at relapse ($p = 0.0001$). Association between survivorship at relapse and site of relapse, relapse timing, the extent of surgical resection, and radiotherapy at relapse were not statically significant.

Conclusions: Serum/CSF AFP below 25 ng/ml at initial diagnosis, achieving CR/CCR by the end of therapy following relapse and HDCx/AuHCR were favorable prognostic factors even after excluding patients who did not receive irradiation at initial diagnosis or relapsed with pure CNS germinoma, or unknown histopathology and tumor marker levels at relapse.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0343 / #1076 | Clinical Characteristics and Therapeutic Effect of Cervical Neuroblastoma in Children

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Background and Aims: To summarize the clinical characteristics and therapeutic effect of cervical neuroblastoma in children.

Methods: From January 2014 to May 2019, the clinicopathological data of children diagnosed with primary cervical neuroblastoma and treated in the department of surgical oncology of Children's Hospital Zhejiang University School of Medicine were analyzed retrospectively. The recorded data include: age at the initial diagnosis, symptoms, tumor size, pathological type, treatment process and prognosis.

Results: A total of 10 children were included in the present study, with an average age of 33.7 months; the main primary complaint was the unintentional discovery of cervical masses (9 cases), in which 1 case was paroxysmal shortness of breath with hoarseness caused by tumor blockage of laryngeal cavity after 9 days of birth; the average size of the tumor at the initial diagnosis was 39.8cm^3 ; the primary diagnosis was by biopsy (5 cases were by part excision biopsy, 3 cases were by complete excision, 2 cases were by needle biopsy). In the primary diagnosis, 9 cases were diagnosed with neuroblastoma and 1 case was ganglioneuroblastoma; 7 cases were treated with chemotherapy after biopsy, and then operated with complete resection, 2 cases were treated with direct complete resection (1 case was respiratory obstruction at admission), 1 case was treated with complete resection after needle biopsy; 1 case was treated with no chemotherapy after operation, and the rest were treated with chemotherapy after operation. 1 case recurred with multiple metastasis, 1 case recurred locally, and the rest did not recur.

Conclusions: Primary cervical neuroblastoma in children is rare, and mainly discovered with mass at initial diagnosis. The combined treatment based on surgery is recommended as the main treatment, while surgical excision as the main treatment for Ganglioneuroblastoma. Further, during the diagnosis and treatment process, attention should be paid to airway patency, cervical vascular and nerve involvement.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0344 / #1202 | A Systematic Review of Presentation, Management, and Outcome of Thymolipoma in Children: A Rare Cause of Pediatric Mediastinal Mass

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Background and Aims: Thymolipoma is an uncommon cause of anterior mediastinal mass in children. Although a benign tumor, it is known to attain large size and cause compression symptoms, thus necessitating removal. This systematic review aims to review presentation, management, and outcome in the pediatric population.

Methods: A 11-year-old boy presented with a history of progressively increasing respiratory distress with increased dyspnea and cough for 2 weeks. Chest radiograph and CT revealed a large ill-defined soft tissue mass with a mediastinal shift to the left. FNAC was inconclusive. A possibility of teratoma or lymphoma was being considered. Biopsy was suggestive of thymolipoma. Complete resection of the mass by thoracotomy was done, despite encasement of the superior vena cava. The postoperative course was unremarkable. The histopathology confirmed the diagnosis. MEDLINE/PubMed, and Google Scholar databases were searched for studies in English regarding Thymolipoma in patients < 14 years old. Twenty-seven patients were included including the index case.

Results: The median age of presentation was 7 years (Range: 6 months-14 years), with male predominance (M: F= 1.7:1). Incidental detection on routine Chest radiograph was the most common presentation in 40.7 % of patients with the next common presentation being cough (33%) and respiratory distress (22%). Chest radiograph invariably shows diffuse opacification in hemithorax or could simulate cardiomegaly. CT is helpful in differentiation in most cases, with biopsy being diagnostic. The mean diameter was 16.2 cm (Range: 6.5cm -33cm) with a mean weight of 806.5 gm (154 gm- 2235gm). Association with Lymphocytosis, asthma, and erythroid hypoplasia has been occasionally described. Thoracotomy has been the choice of exposure in 62.5 % of cases, with sternotomy and thoracoscopy as viable alternatives. No malignant transformation has been reported.

Conclusions: Thymolipoma is a rare mediastinal mass in children. Despite attaining frighteningly massive size, complete surgical resection generally suffices and has an excellent long-term prognosis.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PV0345 / #154 | **Metastatic Medullary Thyroid Carcinoma (MTC) as a Source of Ectopic Adrenocorticotrophic Hormone (ACTH) Leading to Cushing's Syndrome in a 10-Year-Old Child**

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Background and Aims: Ectopic Adrenocorticotrophic hormone (ACTH) syndrome (EAS) as a cause of Cushing's Syndrome (CS) is not only difficult to diagnose but also difficult to manage. Medullary thyroid carcinoma (MTC) causing EAS is rarer, especially in the pediatric population. The management of fulminant CS in such a patient poses a complex dilemma.

Methods: A 10-year-old male presented with features of CS with increased weight gain, hyperpigmentation, facial puffiness, easy bruising, and decreased height gain. The diagnosis was confirmed by raised Serum Cortisol levels, 24 hours free urinary cortisol, and dexamethasone overnight test. A raised serum ACTH level and normal MRI brain led to suspicion of EAS. A 68- Gallium DOTANOC scan revealed somatostatin receptor-expressing nodules in both lobes of the thyroid with cervical nodal and bilateral pulmonary metastasis. A Fine needle aspiration cytology of the thyroid nodule confirmed the diagnosis of MTC, corroborated by raised serum calcitonin (>2,000pg/ml) levels. There were no features of multiple endocrine neoplasia (MEN). CS-induced hypertension and dyselectrolytemia was barely manageable by pharmacotherapy. Due to deteriorating clinical condition and unpredictable response to inhibitors of adrenal steroidogenesis, an upfront bilateral adrenalectomy was performed to control the symptoms. However, the ultimate outcome would remain poor due to metastatic medullary carcinoma thyroid.

Results: The use of DOTANOC in EAS is a useful tool in the identification of the primary site causing EAS. Bilateral adrenalectomy offers effective and expeditious management of uncontrollable Cushing syndrome, albeit requiring hormone replacement later on.

Conclusions: CS is a life-threatening disease in children which could rarely be caused by ectopic ACTH production by MTC. The significant morbidity of CS associated with the poor prognosis of metastatic MTC poses a daunting therapeutic challenge. Tackling fatal CS first by adrenalectomy followed up by resection of the primary tumor, later on, is a viable treatment option.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PV0346 / #1768 | **Primary Small Blue Round Cell Tumors of The Breast in Adolescents**

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Background and Aims: Primary malignant tumors of the breast in adolescents are very rare and of unknown etiology. The purpose of the study is to evaluate the result of the diagnosis and treatment of these small blue round cell tumor (SBRC) in our hospital.

Methods: This is a retrospective analysis of consecutive patients from 2017 to 2019 treated with (SBRC) primary mammary glands

Results: Two patients were treated with SRBCT breast tumors from 2017 to 2019. Both girls were 14 years old with no family's medical history. Symptoms presented were a painless lump with no nipple discharge from 3 to 5 months of evolution. An ultrasound was performed and MR in both. Primary open biopsy was performed in one, and a core needle biopsy in the other. The result was Embryonic Rhabdomyosarcoma T3N0M0. Received 6 cycles of neoadjuvant chemotherapy Vincristine, actinomycin and cyclophosphamide. A right mastectomy with axillary lymph node dissection was performed. Alveolar rhabdomyosarcoma pathology result with positive T3N1M0 margins. She received 2 cycles of adjuvant chemotherapy, but she developed progressive disease, with invasion of the thoracic wall and pleural effusion. She died at 10 months after diagnosis. The diagnosis in the second patient was non-Hodgkin's lymphoma (B cell). The tumor size measured 2 cm in the MR review. She underwent an initial resection of the breast mass it was resected of first intention since it was diagnostic and therapeutic. When the pathology report came back, she was started on chemotherapy with Etoposide, Ifosfamide, Cytarabine, after 63 days of treatment, she developed neutropenia and sepsis that was fatal.

Conclusions: 1. Primary malignant breast tumors of SRBC are very rare in adolescents, nevertheless, they should be considered in the differential diagnosis. 2. Molecular study should complement the histopathology to offer a most precise diagnosis, and to give a more accurate prognosis.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PV0347 / #235 | Use of ICG for Intraoperative Identification and Resection of Small Cell Undifferentiated Hepatoblastoma

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Background and Aims: Indocyanine green (ICG) is a water-soluble fluorescent dye with variant tissue uptake, cleared by the liver and excreted into the biliary system. Fluorescence guided surgery (FGS) capitalizes on abnormal ICG retention by malignant tissue for intraoperative tumor detection. ICG has been demonstrated to be differentially retained by hepatoblastoma and resection of primary and metastatic lesions by FGS is being increasingly utilized. There is currently no published data on ICG uptake by the rare and aggressive small cell undifferentiated (SCUD) form of hepatoblastoma.

Methods: A retrospective chart review was performed with full consent, within institutional ethical guidelines.

Results: The patient presented at 18 months with a large exophytic left sided PRETEXT II hepatoblastoma. Alpha-fetoprotein (AFP) was > 700,000 ng/mL. He received six rounds of cisplatin monotherapy and a margin negative left hepatectomy. Surgical pathology was consistent with mixed epithelial hepatoblastoma and the AFP normalized. Approximately 18 months post-resection, despite normal AFP levels, a recurrence in the tumor bed containing SCUD components was diagnosed. After two cycles of carboplatin plus doxorubicin and one cycle of cisplatin he underwent tumor resection. Pathology confirmed a SCUD hepatoblastoma with minimal necrosis and positive microscopic margin on the diaphragm surface. He received external beam radiation and four cycles of vincristine plus irinotecan. Following adjuvant treatment, an enlarging hepatic cystic lesion in segment VII, posterior to the right hepatic vein and abutting the vena cava was observed. It was stable on imaging for over six months prior. FGS excision was undertaken. Definitive tumor retention of ICG (0.5mg/kg, 72 hours prior) assisted in complete resection with negative margins. He is now two months post-resection with no evidence of tumor recurrence.

Conclusions: This case highlights FGS with ICG for the management of SCUD hepatoblastoma metastases. Despite poor differentiation, SCUD hepatoblastoma strongly retained ICG, enhancing tumor resection and preservation of normal parenchyma.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PV0348 / #289 | A Decade of Experience with Implanted Port for Pediatric Oncology Patients in a Tertiary Care Center

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Background and Aims: Background: Total Implantable Venous Access Devices for pediatric oncology patients have been available for more than three decades. These devices facilitate the administration of chemotherapy, other medications and blood products. We started implanting these devices for our pediatric Oncology patients in 2009. We present our one-decade experience from a tertiary care center in the Kingdom of Bahrain.

Methods: Design: The research design was a retrospective study of a prospective database. Setting: Salmaniya Medical Complex which is a tertiary care center in the Kingdom of Bahrain. Population: Pediatric Age Group Patients between the ages of 0 to 14 years from January 2009 to December 2018 were studied. Methods: A total of 186 Pediatric Oncology Patients requiring Total Implantable Venous Access Devices were included in the study. Aim was to review our experience and analyze the outcome. A record of all complications and final outcome were analyzed.

Results: Data collected included; age, sex, nationality, diagnosis, insertion site and complications. Over a ten-year period, 186 Total Implantable Venous Access Devices were placed. Ages ranged between 4 months and 14 years with a mean age recorded at 5.2 years. Majority were male (N=119, 66%), Leukemia was the main oncological diagnosis (N=101, 54.3%) followed by Solid Tumors (N=58, 31.2%) & Lymphomas (N=27, 14.5%) The Right vein was used in almost 60% of the patients. No reported complications in 85% of the implanted devices.

Conclusions: Although complications demanded the removal of the implanted port and possible reinsertion at some scenarios, their benefits, including the cost effectiveness argue for continued use in the pediatric oncology patients. It is safe and has many advantages compared to traditional central lines. Meticulous implantation technique, expertise and training to facilitate adequate handling are mandatory.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0349 / #583 | Is Biopsy Necessary for Residual Mass in Children with PTLT in the 18F-FDG PET/CT Era?

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Background and Aims: For children with post-transplant lymphoproliferative disease (PTLD), tissue biopsy is often used to confirm remission in the setting of residual radiographic mass after completion of therapy. Some have speculated that ¹⁸F-Fluorodeoxy Glucose Positron Emission Computed Tomography (FDG-PET/CT) might make invasive biopsy unnecessary. However, there is limited data addressing the sensitivity and specificity of ¹⁸F-FDG-PET/CT to accurately assess residual viable PTLT.

Methods: We present two pediatric cases illustrating the decision-making process with regard to obtaining a tissue biopsy to confirm remission versus residual active PTLT, in light of ¹⁸F-FDG- PET/CT findings.

Results: Case-1: 12-year-old with history of cardiac transplantation in infancy presented with a large abdominal mass. Tumor biopsy revealed monomorphic PTLT/diffuse large B-cell lymphoma (PTLT/DLBCL). After completing 4 cycles of LMB-based immune-chemotherapy, ¹⁸F-FDG-PET/CT showed avid uptake (Deauville 4) in a segment of thickened small bowel. Post-treatment biopsy did not show evidence of PTLT/DLBCL. Case-2: An 8-year-old post-cardiac transplantation presented with extensive abdominal and mediastinal masses. Biopsy confirmed Burkitt lymphoma. After completing LMB-based immune-chemotherapy, imaging showed residual mediastinal mass which was ¹⁸F-FDG PET/CT negative (Deauville 2). No biopsy

was performed and no further therapy was administered. Both patients remain in remission 8, 9 months post completion of therapy, respectively.

Conclusions: As a screening tool for identifying persistent disease post therapy, ¹⁸F-FDG-PET/CT may have limited value in identifying true positive residual disease and has not replaced tissue biopsy in cases where the residual mass shows increased metabolic activity. However, ¹⁸F-FDG-PET/CT might eliminate the need for tissue biopsy when the residual mass shows no metabolic activity on ¹⁸F-FDG-PET.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0350 / #847 | An Unusual Left Atrial Recurrence of Wilms Tumor

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Background and Aims: Majority of relapses in Wilms tumor occur in the lung, tumor bed, or liver and within 2 years. Described here is an unusual recurrence site that occurred 2.5 years after the primary diagnosis. The patient presented with shortness of breath secondary to atrial mass.

Methods: Description of the case and the investigations and management.

Results: A four-and-a-half year old boy had been previously completely treated for non-metastatic favorable histology right Wilms tumor with chemotherapy, surgery and local flank radiation. Thirty months from initial presentation and nine months after the completion of therapy patient presented to our centre with shortness of breath and features of cardiac failure. On evaluation was found to have left mediastinal mass infiltrating the left pulmonary vein and extending into the left atrium and intermittently obstructing the left atrial outflow. Child underwent an urgent medial sternotomy with excision of left atrial mass and biopsy of the mediastinal mass on cardio-pulmonary bypass. The histopathology confirmed recurrent WT and so he received 6 cycles of ICE (Ifosfamide-Carboplatin-Etoposide) chemotherapy followed by re-excision of the residual mediastinal mass and followed by local post-operative radiation. He remained tumor free for 1 year when follow up CT again showed re-recurrence of left mediastinal mass. The child was started on third line salvage chemotherapy.

Conclusions: Mediastinal recurrence with left atrial extension without parenchymal lung involvement has not been described in WT. Alternate salvage chemotherapy with surgical intervention on cardio-pulmonary bypass is effective and can be offered.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0351 / #940 | The Global Pediatric Oncology Facility Integrated Local Evaluation Surgical Tool (ProFILE-ST) to Evaluate the Impact of Non-Biological Factors in Pediatric Cancer Outcomes

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Background and Aims: Background: Non-biologic factors' influence on Pediatric cancer outcomes are more prevalent and actionable than biologic factors. There is no tool to perform a systematic assessment of non-biological factors related to pediatric surgical oncology. Objective: To develop the global Pediatric Oncology Facility Integrated Local Evaluation Surgical Tool (ProFILE-ST).

Methods: A partnership between IPSO and St Jude Research Hospital was established to develop ProFILE-ST. The development process includes five phases: construction, content validation, pilot testing, regional validation, and dissemination. During the seven-month construction phase, an expert team selected questions from the existing ProFILE tool that could be incorporated or adapted to ProFILE-ST and expert surgeons proposed new questions. Consensus about final elements to include was reached via bi-weekly online meetings. Eleven IPSO international experts were invited to complete a four-week content validation process.

Results: The resulting ProFILE-ST tool has 12 modules and 257 questions, which comprehensively capture the pediatric surgical oncology patient care continuum. Multiple choice, quantitative, and qualitative questions were included. Scores were assigned to most of the questions to facilitate analysis and illustration of results. Of the 257 questions in the ProFILE-ST tool, 159 (62%) stem from ProFILE, 70 (27%) were adapted and 28 (11%) new surgical questions were made. Validation of the tool is being performed by 11 multi-regional international IPSO experts on the field. Six experts have completed the validation process and agree that the tool would be helpful for situational analysis and assessment of improvement over-time. Suggestions were analyzed by the core team to incorporate them.

Conclusions: ProFILE-ST is a tool to systematic assess pediatric surgical oncology care. It is the first of its kind and will assist pediatric oncology surgeons, PHO teams and institutions to identify and monitor improvement strategies for optimal surgical care delivery. ProFILE-ST will begin end-user pilot testing (beta-testing) soon in 4 countries.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)*

PV0352 / #990 | Excision of Paediatric Limb Soft Tissue Sarcomas – Functional, Oncological and Cosmetic Outcomes

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Background and Aims: Soft tissue sarcomas (STS) are relatively uncommon, accounting for 4-7%¹ of malignancy in children. The extremity site is a peculiar location for STS of children and adolescents. Surgical resection is the main modality of management in STS – to prevent recurrence, wide excisions are performed, aiming for clear margins. This can have aesthetic and functional consequences. In this study, we investigate functional, oncological, cosmetic outcome and growth impact of immediate functional reconstruction performed in conjunction with limb-sparing resection of limb STS in children.

Methods: Retrospective evaluation of fifteen patients with STS in the extremities, treated over a period of ten years. The age of patients during surgery ranged from three months to thirteen years. The average length of patient follow-up was three years. Collected data included tumour histological subtype, surgical excision and reconstructive techniques, resection margin status, peri-operative complications, adjuvant therapy, tumour recurrence, and subsequent management.

Results: Three patients received neoadjuvant therapy and one received adjuvant therapy. There was no tumour recurrence and no mortality. Clear margins were achieved in 60%; two patients required re-excision which then resulted in clear margins. Two patients experienced peri-operative complications – both were post-operative infections. Eleven patients had a free flap reconstruction, and one had a functioning pedicled Latissimus Dorsi flap. Free tissue transfer was accomplished by groin flaps (n=7), functional gracilis (n=3) and anterolateral thigh flap (n=1). Eleven patients had no reduction in range of movement or sensation at the end of the follow-up period. There was no growth disturbance or fixed flexion deformity seen in thirteen patients.

Conclusions: Clear surgical excision is a good mainstay treatment of STS and has good functional and growth outcomes in children, despite wide excisions being required. The soft tissue reconstruction in the setting of sarcoma resection in children is safe, efficacious and helps to minimize aesthetic and psychological morbidity.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.b Radiation Oncology (PROS)*

PV0353 / #1556 | Radiotherapy of Pancreatic Fistula: A Case Report

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Background and Aims: Pancreatic fistula remains a significant problem in pancreatic disease, trauma and surgery. The treatment of established fistulas remains difficult and challenging despite improved diagnostic and management techniques. Radiotherapy remains an important aspect of treatment but has been relatively poorly documented.

Methods: We report a case of external pancreatic fistula treated with external radiotherapy at our institution in 2013.

Results: This is the case of a male patient, aged 16, with no previous history. The story goes back to 2013 marked by a fall from a bicycle with an abdominal impact point causing a complete section of the pancreatic isthmus with a peripancreatic effusion. Initially placed under absolute diet, parenteral diet and Sandostatin. On post-traumatic D10, the patient presented feverish peaks with epigastric pain with the CT scan showing the presence of 2 pseudocysts next to the fracture site with a high abundance of septate intraperitoneal effusion. Drainage was done bringing back 450cc of purulent liquid from which the culture isolated *Klebsiella pneumoniae*. He was operated on urgently for generalized peritonitis. Cobalt radiotherapy on the pancreatic compartment has been scheduled. Before the treatment, the fluid brought back on average was 200cc per day. At 12Gy, we notice a decrease in the liquid reduced to 100cc per day. The decision was to boost radiotherapy up to 20Gy in conventional fractionation spreading. The evolution was marked by the complete and final drying up of the fistula

Conclusions: Conclusion : Radiotherapy is a therapeutic modality that deserves a special place in the treatment of external pancreatic fistulas.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.b Radiation Oncology (PROS)

PV0354 / #371 | Incentives for Children Receiving Radiotherapy – Sharing Experience from East Malaysia

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Background and Aims: Cancer treatment including radiotherapy can be challenging and stressful for children, especially younger ones. Some of them may require sedation or general anaesthesia for simulation and daily treatment. Incentives are one of many ways to alleviate anxiety and fear among children receiving radiotherapy. We describe our experience providing radiotherapy services to paediatric patients at Sabah Women and Children's Hospital, which is the regional centre providing paediatric oncology and radiotherapy services in the state of Sabah in East Malaysia.

Methods: The Radiotherapy and Oncology Department has started a paediatric-friendly environment for paediatric patients. A toy box is placed at the Radiotherapy Unit for children to choose toy(s) they wish to receive upon completion of their radiotherapy treatment along with certificate of completion. For children with head and neck and brain tumours requiring beam direction shell (BDS) mask, superhero or cartoon character of their choice can be painted on the mask. They can also bring the mask home at the end of treatment. The first certificate of completion was given to a 4-year-old child with medulloblastoma completing craniospinal irradiation on 31st December 2016, along with a toy he had chosen prior to start of radiotherapy. Statistic is collected retrospectively as part of yearly data collection for submission to the Ministry of Health.

Results: A total of 72 children aged 3 to 12 years old underwent radiotherapy treatment between 2016 to 2020. Their diagnoses included brain tumours including medulloblastoma, leukemia, sarcoma and nasopharyngeal carcinoma. Two of them had total body irradiation in 2019 and 2020 respectively, prior to bone marrow transplant. The children looked forward to their daily treatment. All of them completed radiotherapy and received certificates of completion as well as toy(s) of their own choice.

Conclusions: This paediatric-friendly environment is a continuous effort to help children undergoing radiotherapy to reduce fear and anxiety.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PV0355 / #1424 | Securing Access to Innovative Anticancer Therapies for Children, Adolescents and Young Adults Outside Clinical Trials: Sacha Study of The French Society of Pediatric Oncology (SFCE)

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Background and Aims: Access to innovative anticancer therapies for children, adolescents and young adults within early phase trials in France is steadily increasing in the last years. However, clinical trials portfolio is still insufficient to address patients' needs. Access to innovative medicines outside their marketing authorization and clinical trials (through compassionate and off-label use) is a fact.

Methods: The SACHA study is a French prospective observational developed by SFCE to prospectively collect toxicity and efficacy data of innovative therapies administered to patients ≤ 25 years-old with relapsed or refractory pediatric cancers provided through the French temporary authorization use (compassionate use) program or through off-label use of anti-cancer medicines that have been first approved in adults after 2007. Prior to SACHA inclusion, multicenter tumor board discussion defining alternative available therapeutic options and clinical trials for each patient is mandatory. Data are collected by a validated pharmacovigilance tool (VIGINOM).

Results: SACHA was launched in March 2020, with an initial estimation of 150 patients/year. Up to 16th April 2021, 177 patients have been enrolled from 27 SFCE centers (89%). Main innovative therapies prescribed were MEK inhibitors as single agents (n=44) or in combination with BRAF inhibitors (n=25), followed by anti-angiogenic tyrosine kinase inhibitors (n=29). Main cancer types were central nervous system (CNS) tumors (49%), followed by non-CNS tumors (33%) and leukemias (18%).

Conclusions: SACHA is the first comprehensive national prospective observational study to provide high quality real world data on anti-cancer innovative therapies used outside trials in children and adolescents. The overarching goal is to warrant efficient pharmacovigilance, to trigger pediatric development when an efficacy signal is observed and to stop prescribing ineffective or unsafe medicines, while not jeopardizing enrollment in trials. First year enrollment was higher than anticipated and toxicity/efficacy analysis of the main cohorts is currently ongoing.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics*

PV0356 / #1586 | Patients Included in Phase I, II and III Clinical Trials Between 2015 and 2020 from the Pediatric Hematology and Oncology Division Vall D'Hebron Hospital, Spain

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Background and Aims: Despite good survival figures for most childhood cancers, disease such as high-risk neuroblastoma, CNS tumours or metastatic sarcomas have poor outcomes. While novel anticancer drugs are increasing in adult oncology, providing access to innovative therapies for children with cancer remains a challenge. In recent years,

novel therapeutic approaches and targeted therapies have emerged. We aim to study the access for patients to participate in Phase I and II clinical trials of new drugs, and enrolment on frontline Phase III trials; as well as analyzing trends over time.

Methods: Retrospective analysis of patients included in clinical trials from the Division of Pediatric Hematology and Oncology Vall d'Hebron Hospital (VH), Barcelona, Spain. Data collected from 2015 to 2020 have been analysed. Characteristics of trials and patients recruited were descriptively analysed for two periods: 2015-2017 and 2018-2020.

Results: A total of 138 pediatric patients participated in a clinical trial in our hospital between 2015 and 2020, 37 were phase 1, 27 phase 2 and 74 phase 3. Median age was 7.81. Of these, 48 (34.8%) patients were recruited from 2015-2017 and 90 (65.2%) from 2018-2020. For phase 1-2 trials, 20 (30.3%) patients were recruited from 2015-2017 and 44 (66.7%) from 2018-2020. In this period, early phase clinical trials included a novel targeted agent in 9 of 11 (81.8%) of cases during 2015-2017 vs. 20 of 22 (90.9%) during 2018-2020. For phase 3 trials, only 6 of 17 (35.3%) of them included targeted agents.

Conclusions: The inclusion of patients in clinical trials has seen a large increase in recent years. It is estimated that this increase will continue, facilitating access to innovation for a greater number of pediatric cancer patients; leading to improvements in treatment quality, efficacy of new drugs and achieving increased survival rates and improved quality of life.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics*

PV0357 / #353 | Market Access to New Medicines for Children with Cancer in Europe

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Background and Aims: There is an alarming delay between anticancer medicines becoming accessible to children in comparison to adults. After medicines have been approved by the EMA (European Medicines Authorisation), Health Technology Assessment (HTA) agencies will evaluate (comparative) effectiveness, safety data, and total costs. As part of the SIOPE Access to Medicines project, this study aimed to evaluate how HTA evaluations take place for oncology medicines newly indicated for paediatric use in Europe and to explore where the delays for market access originate.

Methods: We obtained HTA reports available in the public domain for nine European countries regarding a basket of three new oncology medicines (i.e. dinutuximab, blinatumomab and tisagenlecleucel). We evaluated the time elapsed between the European license for a paediatric indication and final national HTA decisions, reported the nature of the HTA decision and flagged when no HTA report was available.

Results: Out of 23 HTA decisions (four countries had no report for blinatumomab), 16 were fully positive, four with restrictions, three decisions were negative. For blinatumomab, tisagenlecleucel and dinutuximab, the median time to an HTA recommendation after regulatory approval of paediatric use was 353 days (range 193-751), 137 days (range 73-365) and 512 days (range -3-777), respectively. Individual countries showed large variability on time to decision. For blinatumomab, market authorization followed 1008 days after the indication in adults. HTA applications were submitted a median of 167 days (range 141-450) later, with a recommendation after 145 days (range 49-587).

Conclusions: This study shows for three oncology medicines, with a European license to be used in children, ample variability in HTA decision making in nine European countries. Collaboration between clinicians, academia, pharmaceutical companies, EMA and HTA organisations is needed to facilitate consensual and robust scientific HTA assessments, also considering the specific methodological challenges of collecting evidence in paediatric oncology.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PV0358 / #431 | Metro-Pd1: A Phase I Feasibility Study Evaluating Anti-Pd1 (Nivolumab) in Combination with Metronomic Chemotherapy in Children and Teenagers with Refractory/Relapsing Solid Tumors or Lymphoma

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Background and Aims: A multicenter Phase I study was designed to evaluate Nivolumab in combination with 3 different metronomic chemotherapy (MC) regimens in children with refractory /relapsing solid tumors or lymphoma. Objectives were to identify the MC regimen deemed feasible when given in combination with Nivolumab, to evaluate the safety profile, Overall Survival (OS), and Progression-Free Survival (PFS).

Methods: Patients aged 6 -18 years were enrolled. Patients > 18 years could be included if diagnosed before 18. Patients received intravenous

(IV) Nivolumab combined with oral cyclophosphamide and IV vinblastine (Arm A), oral capecitabine (Arm B), or cyclophosphamide, vinblastine and capecitabine (Arm C). Arm A and B were allocated sequentially. Arm C opened only if A and B were deemed safe. Dose-limiting toxicities (DLTs) were evaluated over the first two 28-day cycles. Patients were evaluable if they received > 2 cycles and > 70% of the planned dose.

Results: Sixteen patients were enrolled between March 2019 and September 2020, 3 in arm A, 6 in arm B, and 7 in arm C. Median age was 11.5 years (5-19). Median time between diagnosis and study entry was 32 months (6.5-117.5). Patients previously received a median of 3.5 (1-4) lines of systemic treatment, surgery (87.5%) and/or radiotherapy (69%). Median number of cycles was 2 (1-11). Median treatment duration was 56.5 days (28-342). Three patients were not evaluable (early tumor progression). No DLT was observed whatever the treatment arm. Grade 3 adverse events (AE) and Serious AE were observed in 43.8% and 12.5% respectively over the first 2 cycles. No grade 4 occurred. With a median follow-up of 7.3 months, 3-months PFS and OS were 35% and 74% respectively, in the whole population.

Conclusions: Combining Nivolumab with cyclophosphamide, vinblastine and capecitabine (Arm C) is safe. Randomized phase II trial comparing Arm C +/- Nivolumab is ongoing.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PV0359 / #433 | Risk and Benefit for Targeted Therapy Agents in Pediatric Phase II Trials in Oncology: A Systematic Review with Meta-Analysis

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Background and Aims: For research with human participants to be ethical, risk must be in favorable balance with potential benefits. Little is known about the risk/benefit ratio for pediatric cancer Phase II trials testing targeted therapies. Our aim was to conduct a systematic review of preliminary efficacy and safety profiles of Phase II targeted therapy clinical trials in pediatric oncology.

Methods: Our protocol was prospectively registered in PROSPERO (CRD42020146491). We searched Embase and PubMed for cancer Phase II pediatric trials testing targeted agents. We included solid and hematological malignancy studies published between 1 January 2015 and 27 February 2020. We measured risk using drug-related grade 3 or higher adverse events (AEs), and benefit by response rates. When possible, data were meta-analyzed. All statistical tests were 2-sided.

Results: We identified 34 clinical trials (1,202 patients) that met our eligibility criteria. The pooled overall response rate was 24.4% (95% confidence interval [CI] 14.5% to 34.2%) and was lower in solid tumors, 6.4% (95% CI 3.2% to 9.6%), compared with hematological malignancies, 55.1% (95% CI 35.9% to 74.3%); $p < 0.001$. The overall fatal (grade 5) AE rate was 1.6% (95% CI 0.6% to 2.5%), and the average grade 3/4 AE rate per person was 0.66 (95% CI 0.55 to 0.78).

Conclusions: Despite the expectation that targeted therapy pose a more favorable ratio of risk and benefit for trials than standard chemotherapy, our findings for Phase II in pediatric oncology trials do not confirm this tendency.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PV0360 / #440 | In Vitro Analysis of The Chemotherapy Drugs Azacitidine and Carboplatin in Cisplatin-Resistant Germ Cell Tumours

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Background and Aims: Germ cell tumors (GCTs) are benign or malignant neoplasms derived from primordial germ cells that occur in gonadal and extragonadal sites, presenting distinct histological patterns. The diagnosis is established with the aid of clinical data, histopathological analyzes, radiological exams and biological markers. Therapeutic planning takes place after histological evaluation and tumor staging and, for some patients, chemotherapy with cisplatin is the treatment of choice. However, a significant number of patients relapse to cancer due to acquiring resistance to this compound, by means of multifactorial mechanisms that have not yet been elucidated. Thus, the aim of this work is to analyze the in vitro effect of carboplatin and azacitidine drugs in cisplatin-resistant germ cell tumor lines.

Methods: Cisplatin-resistant cells (NTERA-2R) were derived by incubation with stepwise increasing cisplatin concentrations for 8 months and colony formation and cell migration were evaluated. It was observed that NTERA-2R had an increase in the colony formation and migration compared to parental cells. Then, NTERA-2R was treated with the chemotherapy drugs carboplatin and azacitidine and the cell viability was analyzed.

Results: The treatment with carboplatin showed IC_{50} 30.141 μ M for NTERA-2R and 1.734 μ M for parental cell, suggesting that carboplatin is less active in resistant embryonal carcinoma cells likely due to cross-resistance with cisplatin, a drug that shares the same mechanism of action, and so the treatment of the parent cells was more effective. Azacitidine treatment in both parental and resistant NTERA-2 cells has been shown to be very effective (IC_{50} 7.680 μ M and 6.461 μ M, respectively), most likely due to its differentiated pharmacological mechanism of action.

Conclusions: Therefore, it is necessary new studies to elucidate the effect of treatment with different chemotherapeutics agents on cisplatin-resistant cells, resulting in the development of new therapeutic approaches.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PV0361 / #444 | Oral Selpercatinib in Pediatric Patients with Advanced Ret-Altered Solid or Primary CNS Tumors: Preliminary Results from The Phase 1/2 LIBRETTO-121 Trial

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Background and Aims: Activating RET-alterations are oncogenic drivers of select pediatric/adult cancers. Selpercatinib, a first-in-class highly selective and potent, CNS-active RET-kinase inhibitor, is approved in multiple countries for adults/adolescents with thyroid cancer and adults with NSCLC.

Methods: LIBRETTO-121 (multicenter phase 1/2 trial in patients 0.5-21 years old with advanced, RET-altered solid or CNS tumors) began in June-2019 and enrollment is ongoing (NCT03899792). Selpercatinib is administered orally as capsule/liquid-suspension BID-continuously. Primary phase 1 objective: evaluate safety and dose limiting toxicities (DLTs); primary phase 2 objective: determine overall response rate.

Results: As of 2-Oct-2020, 11-patients (6 male) aged 2-20 years (medullary thyroid cancer, n=8/papillary thyroid cancer, n=2/osteosarcoma, n=1) had been treated (phase 1, n=4/phase 2, n=7). RET-alterations included fusions (n=2), activating mutations (n=8) and mutation with unknown clinical significance (mut^U, n=1). Prior therapies included surgery (n=8)/chemotherapy (n=1)/vandetanib (n=1)/radiotherapy (n=3), while 3-patients were previously-untreated. Time on selpercatinib ranged from 0.9-13.4 months; 9-patients remain on treatment. One patient experienced a dose reduction; 2-patients experienced dose interruptions due to treatment-emergent adverse events (TEAEs) (elevated alanine

aminotransferase [ALT] and bilirubin). There were no DLTs and no TEAEs that led to discontinuation of selpercatinib. TEAEs in >20% patients included elevated alkaline phosphatase (ALP), constipation, and headache. No drug-related TEAEs occurred in >20% patients. One patient reported TEAEs $\geq$ grade 3 (elevated ALT and AST, grade 3) related to selpercatinib. Best response was unconfirmed partial responses in 4-patients, stable disease in 6-patients (two lasting $\geq$ 16 weeks) and progressive disease in 1-patient with mut^U.

Conclusions: These findings appear consistent with the adult trial results, showing preliminary evidence of safety and efficacy of selpercatinib in pediatric patients with RET-altered solid tumors. Phase 1 portion in patients $\leq$ 2 years and phase 2 portion for patients >2 years are ongoing. ©2021 ASCO, Inc. Reused with permission. This abstract was accepted and previously presented at the 2021 ASCO Annual Meeting. All rights reserved.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics*

PV0362 / #606 | The Efficacy and Toxicity of Dexrazoxane use in Children with Cancer: A Population-Based Study from Maritimes, Canada

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Background and Aims: Anthracycline induced cardiotoxicity is a well-recognized complication in the pediatric population. The use of the cardio-protective drug dexrazoxane has gained traction despite its unclear efficacy and toxicity.

Methods: A retrospective, population-based cohort study was completed using chart and database information children treated with anthracycline at the IWK Health Centre from 2009-2015 (n=178). The efficacy of dexrazoxane was defined as a lack of undesirable deviations in identified cardiac parameters on echocardiogram. Toxicity of dexrazoxane was defined as chemotherapy delays (by 3 days or more) from any of: decreased absolute neutrophil count (ANC), decreased platelets, increase in viral/bacterial episodes and febrile neutropenia (FN) episodes. Patients were stratified into groups based on the total amount of anthracycline (<150mg/m², 150-250mg/m² or >250mg/m²) received and whether they receive dexrazoxane.

Results: Mean ages at diagnosis were 7.9 years for males and 7.6 years for females with a gender ratio of 46% male and 54% female. The most common diagnosis was high-risk acute lymphocytic leukemia (HR-ALL) (n=48), followed by low-risk acute lymphocytic leukemia (LR-ALL) (n=48). Regardless of anthracycline dose, we found no significant relationships regarding cardiac function in the untreated and dexrazoxane treated groups. However, we found that patients who were treated with >250mg/m² of anthracycline and dexrazoxane experienced significantly more platelet delays but no cardiac benefit (n_{dex}=31; n_{nodex}=11, p=0.007). When separated by diagnosis, we also found that dexra-

zoxane treated patients diagnosed with LR-ALL were likely to experience a delay in treatment due to both low ANC (n=12, p=0.0001) and the development of FN (n=12, p=0.02) whereas HR-ALL patients were likely to experience treatment delays due to platelet counts (n=25, p=0.03), low ANC (n=25, p=0.0001) and FN (n=25, p=0.0001).

Conclusions: Despite finding no significant differences regarding the efficacy of dexrazoxane as a cardio-protectant, we have shown that its use induces non-cardiac toxicities in children with cancer that contribute to treatment delays.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics*

PV0363 / #724 | Pediatric Precision medicine at National Cancer Center Japan (NCCJ): "Top-Gear" Project

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Background and Aims: Each year, approximately 2,100 children and 900 adolescents are diagnosed with cancer in Japan. National Cancer Center Japan has been taking the lead in laying the foundation for precision medicine through the Trial of Onco-Panel for Gene profiling to Estimate both Adverse events and Response by cancer treatment (TOP-GEAR) project, which was commenced in 2013. Here, we report the results of TOP-GEAR and TOP-GEAR Ped multigene panel sequencing of pediatric cancers.

Methods: We performed genomic analysis of matched tumors and normal tissue specimens obtained from pediatric and young adult patients with refractory or recurrent disease enrolled from August 2016 to March 2021.

Results: Among the 132 enrolled patients aged 1 to 28 years, 118 (89%) presented results evaluable for genetic analysis. Sixty-nine (52%) patients harbored at least one reportable germline or somatic alteration. The most commonly affected molecular processes were DNA maintenance (TP53), cell cycle regulation (CDKN2A, CDK4), transcriptional regulation (MYCN), RAS/MAPK signaling (NF1, KRAS, HRAS, NRAS), epigenetic modulation (ARID1A, SMARCB1), and receptor tyrosine kinases (ALK, FGFR, MET, and PDGFR). Consistent with previous findings, the overall tumor mutation burden was low, with a median 0.8 mutation/Mb. Nine (7%) patients carried pathogenic variants in cancer-predisposing genes. Potentially actionable findings were identified in 36 (27%) patients, and, to date, 13 (10%) patients have

received the recommended therapy based on their genetic profiles. Four patients had access to targeted therapy through clinical trials, whereas in nine patients, the therapeutic agents were used as an off-label.

Conclusions: The implementation of genomic medicine has been advancing our understanding of underlining tumor biology, which guides our mechanism-based approaches and planning for the development of new therapeutic strategies. Nevertheless, the paucity of proposed agents has been limiting the actionability of genetic findings, underscoring the importance of facilitating access to the targeted cancer therapies.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0364 / #1054 | "Effect of Prophylaxis with Alternate Days of Levofloxacin on Morbidity of Children with Lymphoblastic Leukemia Induction Therapy in a Developing Country. Pilot Study"

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Background and Aims: In our country, early mortality in acute lymphoblastic leukemia (ALL) is very high and severe infections are the most common cause, as well as the high tumor load and malnutrition due to diagnosis and late reference. Antimicrobial prophylaxis is intended to reduce mortality associated with infection, however, cost is the main limit in a country of limited resources. Objective: To know the effect of antimicrobial prophylaxis with Levofloxacin in alternate days in children with ALL induction to remission at the Hospital Materno Infantil ISSEMYM, Toluca, Mexico.

Methods: We designed a cases-controls study in children with high-risk ALL in induction to remission, without infection. They received prophylaxis with oral Levofloxacin 10 mg/kg/day in alternate days (every 48 hrs) and fluconazole 5mg/kg/day until end of induction or presence of infection. The controls were patients without prophylaxis paired for age, weight and gender. We evaluated morbidity, infection, type of antibiotics used, chemotherapy scheme, days of hospital stay. Central trend, chi2 and student T measurements were used.

Results: 11 patients were evaluable and 11 controls. No statistically significant differences were found between the overall characteristics of both groups. There were differences in the chemotherapy scheme at week 5 and 6, some patients receiving, cyclophosphamide, mercaptopurine and cytarabine vs etoposide and cytarabine. We observed that 3/11 (27.3%) patients receiving prophylaxis had some infection vs. 8/11 without prophylaxis (72.7%). This difference was significant (p. 0.043). Two deaths were reported in the group without prophylaxis vs 0 in the prophylaxis group. There were only in the type and number of antimicrobial schemes used.

Conclusions: Our pilot study, suggests that prophylaxis with Levofloxacin in alternate days may be useful in reducing the frequency and severity of infections, decreasing high mortality during the induction in our environment. Multicenter pharmaco-economic studies with more patients are indispensable, we continue to work on this line of research.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0365 / #1055 | Patient Profile and Outcome of Oncology Patients Admitted in PICU of a Tertiary Care Hospital in a Developing Country – A Retro – Prospective Study

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Background and Aims: Cancer remains a major cause of death in children, but recent advances in therapeutics and supportive care have considerably improved the prognosis. These patients need PICU (Pediatric Intensive Care Unit) management for various indications. Developing countries face issues like late presentation, antibiotic misuse and limited resources. Here, we aim to report our experience in managing oncology patients admitted to the PICU in a tertiary hospital in Mumbai.

Methods: It is a retrospective & prospective observational study done in a level III Pediatric Intensive care unit of a tertiary care hospital in Mumbai. Demographic data, admitting diagnosis, indication for ICU admission, PRISM (Pediatric Risk of Mortality) III score, SOFA (Sequential Organ Function Assessment), treatment needed, and outcome of all oncology patients from 1 month to 18 years of age, who required PICU care between January 2016 & December 2019 were studied.

Results: Out of 709 Oncology patients at our centre in mentioned period, 160 children (22.5%) required PICU care. 82 of them had a haemato-lymphoid malignancy & 78 had solid organ malignancy. The common indications of PICU admission were post surgical care (30%), febrile neutropenia (26%) with septic shock (13%) or respiratory distress (12.5%) and tumor lysis syndrome (7.5%). Post PICU care 106 children (66%) survived whereas 54 children died (33.75%). The average PRISM III score of patients at the time of admission in ICU was 12 for those who expired & 3.6 for those who survived.

Conclusions: There is increasing evidence that better supportive care during chemotherapy results in lesser morbidity & mortality. An efficient Pediatric ICU is an indispensable part of managing Pediatric oncology patients. Timely transfer to the ICU (when PRISM III score is not high), translates in better survival during ICU care. Hence, stringent watch must be kept on these patients during therapy & ICU care should be considered early.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0366 / #1069 | Complications of Central Venous Access Devices in Paediatric Patients with Newly Diagnosed Acute Leukaemia During Induction Therapy

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Background and Aims: Background: Central Venous Access Devices (CVADs) are essential for management of children with leukaemia. Due to associated complications, the optimal insertion timing and CVAD choice in newly diagnosed children with leukaemia is undefined, with variation in institutional practices. Objective: To define CVAD use and complications during induction therapy in paediatric patients newly diagnosed with leukaemia in a tertiary oncology centre.

Methods: Method: A retrospective analysis of patients undergoing CVAD insertion for newly diagnosed leukaemia at the Queensland Children's Hospital, Brisbane from January 2018 – June 2020.

Results: Findings: A total of 126 patients were identified, with a median age of 4.9 years (range 0.25-17). Most common diagnoses were acute lymphoblastic (n= 102, 81%) and acute myeloid (n=21, 17%) leukaemias. Eighty-two patients (65%) were febrile peri-insertion, with 49 PICCs (59.7%), 19 Portacaths (23.1%) and 14 tunneled central venous catheters (TCVC, 17%) inserted. Of these, 19 (24.4%) had a central line-associated bloodstream infection (CLABSI) in induction. Of these, 8 (38%) had a PICC, 7 (33%) had an implanted Portacath and 6 (29%) had a TCVC placed. Although none required removal during induction, one (5.3%) necessitated removal beyond induction for repeated CLABSI. One patient (5.2%) had Portacath dehiscence with local infection requiring removal in induction. Four patients had thrombotic complications, all PICC-associated (6.3% of PICCs).

Conclusions: Being febrile at the time of CVAD insertion may result in an increased incidence of CLABSI during induction therapy. Although PICCs may offer benefit in patients febrile at time of insertion, they are associated with complications, particularly thrombosis. Larger cohort studies would be helpful to clarify the best CVAD for newly diagnosed patients febrile at time of CVAD insertion.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0367 / #1136 | Results of a Randomized Controlled Trial: Effects of Specific Strength Training on Activities of Daily Living for Pediatric Cancer Patients with Leukemia and Non-Hodgkin Lymphoma

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Background and Aims: Pediatric patients undergoing cancer treatment show muscular deficits and reduced physical activity (PA) that limit autonomous accomplishment of Activities of Daily Living (ADL). Increased muscle strength and regular PA are associated with better abilities to cope with ADL and thus with the greatest possible autonomy and mobility during therapy. General sports interventions confirm beneficial impact on mental and physical well-being. However, effects of strengthening exercises have not yet been examined. Therefore, this bicentric RCT investigated effects of a specific strength training accompanying treatment on coping with ADL in pediatric oncological patients.

Methods: Participants (4-18 years) were recruited post-diagnosis with ALL, AML, NHL and randomized. Intervention group (IG) received a specific strength training combined with a general sports program (standard care), control group (CG) was provided standard care without any targeted muscle strengthening. Both groups completed 2-3 weekly training sessions within in-/out-patient appointments until cessation of acute therapy. ADL-specific aspects were assessed 4 times using self-reported questionnaire Activities Scale for Kids© (ASK) and a test track with 7 everyday tasks. MOON-test was conducted to scale motor abilities.

Results: From 09/2017-06/2020, 41 patients (10.0±4.0 years; 34% female; IG n=21) were recruited 15.1±10.2 days post-diagnosis (recruitment rate: 98%). Over an intervention period of 7.4±2.4 months, participants completed 37±21 training sessions (range: 10-80), with 29.7±4.7min on average. Both groups scored cross-sectionally similar and longitudinally increased ASK© outcomes (IG: 64.6±19.2-85.4±24.3; CG: 68.1±20.8-86±12.2). Motor performance was reduced in nearly all abilities compared to reference values. At final assessment, IG showed slightly improved motor abilities compared to CG.

Conclusions: Regular instructed training interventions during treatment support patient's autonomy and promote abilities to cope with ADL. Participants show remarkable compliance regarding motivation and implementation of strength training. In this sample, training objective and content tend to play a subordinate role. For differentiated analysis of training interventions, a larger cohort is needed.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0368 / #1146 | Methodology of a Remote Qualitative Evaluation for a Pediatric Oncology Programme in Low- or Middle-Income Countries During a Global Pandemic

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Background and Aims: To determine impact, experiences and expectations of stakeholders involved in 15 years of Sanofi Espoir Foundation's (SEF) "My Child Matters" (MCM) childhood cancer support programme (60+ projects in 80 countries for 30,000 health care professionals trained and reaching 100,000 patients); to assess best practices, limitations, award scheme logistics and adjust strategy, budget, methods and communication as needed.

Methods: This rapid evaluation (2.5 months) had three parts. First, SEF created and analysed a 20-item survey (available in 3 languages) distributed to all 29 MCM project leaders in the current funding cycle (implemented 2019-2021). Second, at project level, SEF contracted experienced healthcare evaluators to conduct a literature and document review and key informant interviews for 11 ongoing projects. Projects were purposively selected by project theme (e.g. cancer registries, early detection, treatment abandonment) and budget, in Asia, Africa, Europe, Latin America and ranged from low-to upper-middle-income (LMIC) World Bank ranking by country income classification. Third, at programme level, SEF simultaneously contracted additional evaluators who examined programme impact according to major healthcare stakeholders in paediatric oncology (e.g. international and non-governmental organizations). Renowned and dedicated paediatric oncology mentors facilitated connections with highly ranked country-level authorities. Interviews were conducted in French, English and Spanish as appropriate.

Results: Interview guides created by the consultants were used in one-hour interviews (with respondent consent). Forty-one requests for interviews sent by SEF were all accepted. Additional interviews were conducted with respondents recommended to the evaluators by initial interviewees, using a 360° approach. Evaluators were responsible for data management and analysis. A final report described constructive feedback to inform future planning (anonymized transcripts and respondent categories included in annexes).

Conclusions: A fully remote, in-depth, qualitative evaluation of an international childhood cancer support programme can be successfully conducted by multiple contracted evaluators in 3 languages in LMIC worldwide despite a tight deadline and pandemic circumstances.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0369 / #1159 | Prenatal Diagnosis and The Pediatric Oncologist. A Ten Year Survey at Assistance Publique Hôpitaux De Marseille (APHM)

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Background and Aims: Solid tumor or predisposition syndromes are increasingly suspected before birth due to improved prenatal imaging and biological diagnosis techniques; however optimal management during pregnancy and long-term outcomes of the children remain unclear. We therefore have performed a ten year (2009-2019) retrospective study of all indications of prenatal diagnosis related to pediatric oncology in public hospitals in Marseilles.

Methods: Patients were identified and data obtained from the prenatal diagnosis center database, hospital imaging and general database and pediatric oncology department files.

Results: Fifty-one cases were identified, 40 cases presented with tumor or mass : adrenal : 17, sacrococcygeal 9, cardiac 7, abdominal 4 (including 2 lymphatic malformations), ovarian 1, cervical 2; developmental abnormalities suggestive of cancer predisposition syndrome in 8 cases, prenatal diagnosis in familial predisposition syndrome 3 cases (rhabdoid tumor SMARCB1 mutation 2, Li-Fraumeni 1); median detection time was 30 weeks of amenorrhea (13-39 weeks) Among the adrenal lesions 4 completely regressed before birth, 2 neuroblastoma underwent surgery at birth due to complications (both progressed and required secondary chemotherapy), 5 spontaneous complete regression occurred before 18 months, 6 cases underwent surgery for incomplete regression (2 neuroblastoma, 2 bronchogenic cysts, 2 masses without malignant criteria). Six sacrococcygeal lesions underwent surgery at birth (5 teratoma, 1 myelomeningocele). All children with tumor are alive disease free (3 lost for follow-up). Prenatal abnormalities included 4 omphalocele and 4 macroglossia, Beckwith-Wiedemann syndrome (BWS) was confirmed in 6 cases. Medical termination of pregnancy was decided for 4 cardiac lesions (suspected tuberous sclerosis) and 2 large sacrococcygeal tumors, 1 BWS, 2 SMARCB1 mutations. Eleven newborns required intensive care including 8 for mechanical complications.

Conclusions: Various solid tumors and cancer predisposition syndromes can be found before birth. A multidisciplinary collaboration is mandatory for optimal management before and after birth.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0370 / #1180 | Prevalence, Risk Factors and Predictors of Adverse Outcomes of Febrile Neutropaenia in Oncology Patients on Chemotherapy at Red Cross Children's Hospital: A Three Year Retrospective Study

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Background and Aims: Background: Febrile neutropaenia (FN) is the commonest fatal acute complication of cancer treatment in children. The need for regional clinical decision rules making for home-based care in those at low risk for adverse outcomes have been identified. Aims: To evaluate the prevalence and potential risk factors for FN, identify adverse outcomes as well as validate the Swiss Paediatric Oncology Group (SPOG) FN risk index in a cohort of children treated for cancer at Red Cross War Memorial Children's Hospital, Cape-Town, South Africa. **Methods:** Methodology: A retrospective cohort study from 1st January 2017 to 31st December 2019.

Results: In all, 179 patients had chemotherapy and 267 FN episodes occurred. Independent predictors of FN were AML ($p=0.039$), ALL ($p=0.020$) and intensive chemotherapy ($p<0.001$). Mucositis ($p=0.001$), CVAD ($p=0.004$), haematologic malignancies ($p=0.040$), blood transfusion during FN episode ($p<0.001$) and severe neutropaenia (white cell counts $<0.3 \times 10^9$ cells/L) ($p<0.001$) were risk factors for adverse outcomes. Mortality rate from FN was 3.57%. Independent predictors of adverse outcomes were AML ($p=0.001$), CVAD in-situ ($p=0.019$) and severe neutropaenia ($p=0.005$). Out of 121 subjects with various adverse outcomes, 77 (63.6%) scored at least 9 using SPOG FN risk index ($p=0.010$). Using SPOG with the established cut-off value of 9 yielded a sensitivity of 52.3% and specificity of 62.0%. Positive and negative predictive values were 56.3% and 58.2% with overall accuracy of 57.4%. The area under the curve was 0.612 which translates to 61.2% accuracy ($p=0.002$). In our cohort, coordinates of the curve with the best predictive value yielded a cut-off of 7.5 with a sensitivity and specificity of 54% and 62%.

Conclusions: A lower cut-off of 7.5 using the SPOG FN risk index best predicts adverse outcomes in our cohort.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0371 / #1196 | Longitudinal Documentation of Symptoms by Children and Proxy Using Pediatric PRO-CTCAE

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Background and Aims: Pediatric oncology patients experience significant treatment-related adverse events (AEs). The Pediatric

Patient-Reported Outcome version of the Common Criteria for Adverse Events (Pediatric PRO-CTCAE) allows patients and proxies to contribute to the reporting of AEs. Core PRO-CTCAE asks about frequency, interference and severity of 15 common symptomatic AEs. This study aimed to use core PRO-CTCAE to describe patients' AE experience longitudinally through cancer treatment to build a phenotype database as part of a larger study.

Methods: English-speaking patients 7-18y and proxies of patients 5-18y completed the Pediatric/ proxy PRO-CTCAE at up to 6 set points throughout multiple cycles of chemotherapy using a REDCAP version on iPads either at the hospital or at home via an email link.

Results: Thirty-nine patients out of 103 in the larger study were eligible (barriers included age and language). Twenty-six of 39 completed all time points; 18 had both patient and proxy for all. The most common diagnosis was B-ALL (26%), 33% were White, 21% African American, 15% Hispanic. The most frequently reported AE across all data points was anorexia, followed by anxiety, pain and trouble sleeping. In general, worse scores were at T1 and T2 with frequency declining for most of the AEs from T3 to T6. Patients and proxies reported similar rates core AEs across time.

Conclusions: Caregivers and patient completion of the Pediatric PRO-CTCAE was feasible in the hospital at multiple time points using iPADS and REDCAP; data collection by email was less reliable. Most children report experiencing 1 to 12 AEs at each data point with similar reports from proxies. Anorexia and anxiousness as well as trouble sleeping and pain appear to be most frequent. Efforts to identify patients with high AE burden early in treatment could guide supportive care efforts to minimize the experience of subjective AEs.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0372 / #1222 | Decrease in Mortality is Observed after Implementing a Pediatric Early Warning System in a Pediatric Oncology Unit of the General Hospital of Celaya, Mexico

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Background and Aims: Clinical deterioration events (CDE), defined as unplanned ICU transfers or ICU-level floor interventions in pediatric oncology patients, result in high mortality as a consequence of antineoplastic therapy toxicity. Our objective was to evaluate the impact on the quality of care and assess mortality rates in our service and compare the effects of the implementation of a Pediatric Early Warning System (PEWS/EVAT).

Methods: Since April 2018, the EVAT project began, and data was collected as a baseline for 11 months. Data collections for post implementation occurred from October 2019 to December 2020 (15 months total). The EVAT scale and algorithm was successfully implemented with the help of St. Jude Global Critical Care. Data collection was carried out in conjunction with St. Jude by comparing the database of our hospital with the database of St. Jude Children's Research Hospital.

Results: Before implementation, 21 CDEs with 3006 in-patient days were recorded, and in the after period 14 CDEs with 3759 in-patient days were observed. When comparing percent mortality in the period before (42.8%) with the one after (35.7%), a 16.67% reduction was observed upon EVAT implementation. Additionally, the rate of deterioration events decreased from 6.98 (before) to a 3.72 (after). A decrease in mortality rates from 2.9 to 1.3 was also reflected, as well as the use of vasopressors decreasing from 5.9 to 2.3. Lastly, transfers to ICUs also decreased from 3.6 to 1.8.

Conclusions: The implementation of PEWS in a pediatric oncology unit in the General Hospital of Celaya Mexico, resulted in early identification of clinical deterioration, allowing timely interventions and improving the results of hospitalized patients, which were ultimately reflected with the decrease in deterioration events, unplanned transfers, decrease in use of vasopressors. Altogether, these changes led to a decrease in the mortality of patients with a significant reduction in mortality in 16.67%.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0373 / #1232 | Strengthening Nutritional & Hygiene Needs for Children with Cancer Amidst Covid

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Background and Aims: CanKids KidsCan is a national society working on providing holistic support to a child with cancer along with the family. The treatment support programme (TSP) as one of the major programs of the organization encompasses supportive care services through nutrition support, blood support, hygiene and infection control services, and holistic accommodation. In LMIC malnutrition has always been a cause of concern, especially among children undergoing cancer treatment. The pandemic worsens the nutritional and hygiene services leading to a situation requiring high concern. Thus we launch Cansurvive as Cankids programs to educate and to create scope to uplift the health status of the patient for timely and successful treatment completion.

Methods: 1. Identification of stakeholders with the help of SST 2. Inter-department discussions and brainstorming on structuring the project 3. Feasibility assessments 4. Reintegration model and parents group involvement 5. SMT meetings and discussions 6. Identifying ample

funding 7. Vendor chain management through internal tender committee for logistics management

Results: Till 31st March 2021, the program status is as under: a) It is running successfully at 26 Cankids Hospital Support Unit (CHSUs) across 7 cities. b) 979 unique beneficiaries received COVID kits and 1781 received Canourish kits (for kids and family). c) 3376 beneficiaries were counseled on nutritional management and 3443 beneficiaries were educated on hygiene. d) 72 parents benefitted under the reintegration model. e) 7 City based vendor for Nutritional as well as Hygiene items has helped facilitated a timely supply chain.

Conclusions: The program has been successful in reaching the target beneficiary and hereon measures are been taken on its scalability and sustainability to promote better treatment tolerance amongst children with cancer with an improved health status. The program believes in reducing the vulnerability due to various obvious factors pertaining to LMIC and moving a step towards Right to Health for Children with Cancer.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0374 / #128 | Cognitive Fatigue and Cognitive Function in a Computer-Based Exercise Study in Pediatric Oncology: Results of a Questionnaire Survey and a Continuous Auditory Response Test

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Background and Aims: Cognitive fatigue (CF) is frequently overlooked as a side effect in pediatric oncology and includes a spectrum of deficits affecting motivation, emotion and cognition. The effect of CF on physical or cognitive performance is not fully understood and it is still under investigation. The objective was to compare CF results of a questionnaire survey and cognitive performance results of a continuous auditory reaction test in a computer-based exercise study in pediatric oncology.

Methods: Thirteen pediatric patients with mixed cancer (3 female; mean age=13.31±2.39 years) were asked to complete the PedsQL Multidimensional Fatigue Scale questionnaire and a continuous auditory response test (ART) was performed at two time points [T1, T2] during cancer therapy. ART has a duration of eight minutes with the task of physically responding to a rare stimulus (target) randomly embedded in a background of standard stimuli; reaction time (RT) from hearing the target to responding by pressing a button was measured, mean value of approximately 50 targets was calculated. All children had the opportunity to participate in a computer-based exercise program twice a week. **Results:** Mean activity level was 89.62±100.13 minutes/week (min:0; max:240 minutes/week). Mean time between start of cancer therapy and T1 was 14.08±11.91 and T2 was 82.54±15.77 days. Mean CF Score was 72.75±15.17 [T1] and 70.18±19.23 [T2]; mean RT was

0.580±0.083 [T1] and 0.579±0.085 sec [T2]. Between CF and RT there was a moderate negative relationship in T1 ($r=-0.541$, $P=0.056$) and a weak negative relationship in T2 ($r=-0.409$, $P=0.165$).

Conclusions: While the present study showed no change in CF and RT over the course of therapy, there was a negative relationship between CF and cognitive performance. To prevent cognitive decline in pediatric oncology more attention should be paid to side effects such as CF. To date pediatric oncology lacks valid assessments and sufficient therapy programs for CF.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0375 / #1284 | Use of a Positioning Inflatable Cushion for Children During Lumbar Puncture

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Background and Aims: Lumbar punctures (LP) are commonly and repeatedly performed for diagnosis or treatment in children with cancer or hematological diseases. Body posture, muscle release, and quietness of child have a direct impact on the success of the LP.

Methods: This pilot prospective monocentric observational study aimed to evaluate the feasibility of using a positioning inflatable cushion designed for improving position of children during LP. At the end of each procedure, patients and caregivers answered a survey assessing pain (visual analogical scale (VAS)) and satisfaction of patients, parents and caregivers. The premedication used, the number of attempts, the presence of hemorrhagic cerebrospinal fluid (CSF) and the success of the procedure were also recorded.

Results: Fifty patients (2-18 years) undergoing diagnostic or therapeutic LP between October 2020 and February 2021 were evaluated. All procedures were realized after local analgesia with Lidocaine patch, 88% with nitrous oxide inhalation, 64% with hydroxyzine premedication, and 6% with adjunction of an analgesic. 100% of procedures were successful allowing either LCR collection or chemotherapy injection. Thirty-four (68%) LP were achieved after 1 attempt. Among them 15% (5/34) were macroscopically hemorrhagic. The median pain score (VAS) evaluated by children was 10/100. Patients were mostly satisfied (58%) or very satisfied (24%), as for the parents (37 and 34% respectively), doctors (37% and 42% respectively) and other caregivers (28 and 46%). Post-LP syndrome was observed in 4% of patients.

Conclusions: This pilot study confirms the feasibility of evaluating the use of a positioning inflatable cushion for LP, with a high level of satisfaction of all the actors, and a low level of procedural pain. The next step will consist in a randomized prospective study comparing LP per-

formed with or without cushion in order to confirm the effectiveness and the interest of this device.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0376 / #1296 | Safety, Feasibility and Acceptability of Peripherally Inserted Central Catheter (PICC) Line in Paediatric Oncology Patients: Experience from Tertiary Oncology Centre in Central India

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Background and Aims: A long term venous access is essential in children when treating malignant disease. The peripherally inserted central catheter (PICC) is commonly used central venous access in children and adolescents. It's use has been increased over years, mainly in cancer patients, for the administration of intravenous chemotherapy and frequent transfusions. The purpose of study to determine safety, feasibility and acceptability of PICC use and to determine PICC related complication and reason of removal in Tertiary Care Pediatric Oncology Unit of Central India.

Methods: All Pediatric patients (age below 16 years) with hematolymphoid malignancy or solid tumor who underwent PICC line insertion at National Cancer Institute, Nagpur from February 2018 to December 2020 were enrolled in the study. Demographic features, primary diagnosis, catheter days, complications, and reasons for removal of device were recorded during this period.

Results: A total of 61 PICCs were inserted during February 2018 to December 2020. Insertion success rate was 100%. Catheter maintenance success rate was 57.3% and median catheter life span was 107 days per device. PICC line infection seen in 27.86%, catheter blockage in 9.83%, accidental removal in 4.91%. Infections and catheter occlusion were the common complications.

Conclusions: PICC line use is safe, reliable and well acceptable in pediatric patient for long term use. Though infection control is main challenge which require repeated training and counselling of nursing staff.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0377 / #1301 | Outcome of Covid-19 Among Pediatric Cancer Patients of a Tertiary Care Hospital in Pakistan

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Background and Aims: Children with cancer are frequently immunocompromised. While children are generally thought to be at less risk of severe SARS-CoV-2 infection than adults, comprehensive population-based evidence for the risk in children with cancer is unavailable. We aimed to find out the outcomes from SARS-CoV-2 infection in children with cancer.

Methods: We conducted a retrospective study at Indus hospital & health care network on children with cancer already on treatment as well as newly registered patients for workup of cancer seen from May 2020 to August 2020.

Results: Total 538 pediatric patients were tested for SARS-CoV-2 by PCR, out of which 72 (13%) were positive. For age <5 were 23(32%), 5-10 were 20 (28%) and >10 were 29(40%). Males were 41(57%). Leukemia/lymphomas were 48(67%) while 24(33%) were solid tumors. Asymptomatic were 19(26%) while fever as the most common clinical symptom was seen in 42(58%) followed by cough, sore throat and body pain 11(15%), 4(6%) and 3(4%) respectively. Total of 16(22%) were detected during admission screening, 24(33%) had come for workup and 12(17%) were screened prior to chemotherapy. Mean duration of stay in hospital was 6.1 ± 6.5 days. Neutropenia was observed in 27(37%) while <1.0 ALC was observed in 17(24%). Abnormal chest x-ray was found in 15(21%). Chemotherapy was delayed in 14(57%), surgery and radiation was delayed in 2(3%), left against medical advice (LAMA) 5(7%) and referred 4(5%) while 12(17%) were given treatment as planned. Total 5(7%) patients expired, 2 were on palliation, 1 was disease related, 1 treatment related and 1 died of COVID-19.

Conclusions: Children with cancer with SARS-CoV-2 infection do not appear at increased risk of severe infection. This is reassuring and supports the continued delivery of standard treatment to pediatric cancer patients.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0378 / #1305 | Assessment and Development of Safe Chemotherapy Preparation in Pediatric Oncology Units in Lower Middle Income Country (LMIC)

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Background and Aims: Manual preparation of cytotoxic drugs, although prevalent in LMIC's, is a high-risk activity given the prolonged exposure to carcinogens with significantly higher potential for errors. Chemotherapeutic drugs often have a low therapeutic index, making precision key in successful treatment and efficacy. The Oncology Nursing Society highly emphasises the need for nurses and chemotherapy

handlers to be trained in risks and handling of these agents. In Pakistan, an LMIC, arrangement and handling of chemotherapeutic drugs in public hospital is quite often done by non-skilled staff without adequate precautions within the ward.

Methods: With the help of the My Child Matters Grant awarded to the Indus Hospital & Health Network by the Sanofi Espoir Foundation, six paediatric oncology units were partnered with and assessed for chemotherapy handling and preparation standards. The person responsible for chemotherapy preparation was evaluated in terms of knowledge and training in handling chemotherapeutic agents and automation of prescriptions, checks and labelling.

Results: Out of 6, 4 centres installed biological safety cabinets in dedicated rooms within the Paediatric Oncology Department. Out of the 4 installed, 3 were double layered for extra safety measures. All units imposed mandatory personal protective equipment usage and electronic prescriptions within 1 year. Additionally, 2 centres appointed dedicated clinical pharmacists and the nursing staff responsible for chemotherapy preparation in the remaining centres were mandatorily enrolled in nursing and pharmacy courses due to a lack of technical knowledge of risks and automated practices. In the units without biological safety cabinets, trainings and fundraising is ongoing.

Conclusions: Considering the lack of resources and expertise in clinical pharmacy, training and quality check of available staff and mandatory availability of biological safety cabinets as a prerequisite to an oncology setup are recommended. Ensuring effective handling and administration of chemotherapy drugs is an essential step in successful cancer treatment.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0379 / #1311 | Impact of Covid-19 on The Children with Cancer in 6 Pediatric Oncology Units of Pakistan-A Multi-Center Study

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Background and Aims: The coronavirus disease 2019 (COVID-19) pandemic has rapidly escalated into a global crisis and has especially effected the health care sector of lower middle income countries including Pakistan. Our aim is to evaluate the impact of COVID-19

pandemic on children with cancer in the different Pediatric Oncology Units (POUs) of Pakistan.

Methods: A retrospective multicenter study; Data was collected from 1st April'19 to 30th June'19 for the first cohort and from 1st April'20 to 30 June'20 for the second cohort in order to compare the patient registrations and abandonment rate in six Pediatric Oncology Units (POU's) of Pakistan, namely The Indus Hospital (TIH), Children Hospital and Institute of Child Health, Multan (CH&ICH), Lahore General Hospital (LGH), Mayo Hospital, Children Hospital & Institute of Child Health Faisalabad and Pakistan Institute of Medical Sciences (PIMS).

Results: In this study we have observed male (60%) preponderance and mean age was 7.98 ± 4.2 , overall registered cases were 634, from which 379 (59.7%) were registered in first cohort and 255 (40.3%) were registered in second cohort in our all 6 POU's. Overall abandoned cases were 77, from which 45 (58.4%) were abandoned in first cohort and 32 (45.6%) were abandoned in second cohort. We have found significant difference in registered cases p -value = 0.001, while there is no significant difference in abandonment rate. Total 59 COVID-19 positive cases were reported in our 3 POU's, out of which 24 (41%), 4 (6.7%), 27 (45.7%) and 4 (6.7%) were admitted, referred, advised home isolation and refused treatment respectively

Conclusions: This study, although providing a small picture of COVID-19, as we have observed the significant reduction in new cancer registrations in POU's and few patient families refused treatment due to COVID-19. More research is needed to better understand the COVID-19 impact on pediatric cancer.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0380 / #1325 | Effect of The Covid-19 Pandemic on the Clinical Deterioration Events in Childhood Cancer Patients in a Tertiary Hospital in Mexico

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Background and Aims: Background and Objectives: Pediatric cancer is one of the leading causes of death in children in Mexico. Its mortality rate was 5.09/100000 inhabitants in 2018. Among the main causes of death associated with this pathology are Clinical Deterioration Events (CDE). The frequency and severity of CDEs at our hospital has decreased with the implementation of the Pediatric Early Warning System (PEWS). However, the COVID-19 pandemic generated various changes in hospital processes, including our application of PEWS. For this reason, we decided to monitor the effects of the pandemic on pediatric cancer care and the characteristics of CDEs in the

department of pediatric oncology at our hospital during the COVID-19 pandemic.

Methods: Cross-sectional study of all hospitalized patients experiencing CDEs in the department of pediatric oncology between January 2019 and December 2020. For research purposes, 2 periods of time were compared: Pre-pandemic (Year 2019 –Feb 2020) and Pandemic (March 2020-Dec 2021). Clinical deterioration data was obtained using a standardized case report form. Descriptive statistics were used to record variations between the different study periods.

Results: During the study period, 1001 pediatric oncology hospital admissions were recorded, of which 50 experienced at least one CDE (3.67 in pre-pandemic and 4.26 in pandemic for 1000 inpatient day). The percentage of infections increased by 13.67% in pandemic and septic shock increased by 29% in the same period. In pre-pandemic, CDE mortality rate was 0.86/1000-inpatient days and increased to 2.13/1000 inpatient days during the pandemic.

Conclusions: This study demonstrated the effects of the pandemic on CDEs at our hospital, which can be observed in the increase of number of CDEs for 1000 inpatient days, infections, septic shocks and mortality rate. We plan to monitor CDEs from January 2021 onward to analyze whether the measures implemented during the pandemic have been adequate for recovery.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0381 / #1380 | Colonization-Guided Protocol for Empirical Antibiotic Therapy in Children with Solid Tumors After HSCT: First Results of a Single Center Prospective Study

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Background and Aims: The rapid emergence of antibiotic-resistant bacteria and lack of new antibiotics necessitates optimization of antibacterial therapy after hematopoietic stem cell transplantation (HSCT). Preexisting colonization with antibiotic-resistant microorganisms is an important factor for the development of severe infectious complications. The aim of the study was to assess the effectiveness of innovative protocol for the empirical antibacterial treatment (EAT) according to gut colonization in children with solid tumors and HSCT.

Methods: During 8 months period (July 2020 – March 2021) 32 patients with solid tumors were enrolled in the study. Bacterial culture ($n=28$) and CHROMagar technology ($n=30$) were used for antibiotic-resistant microorganisms detection. The present study demonstrates preliminary results of EAT protocol in children after HSCT according to preexistent colonization with antibiotic-resistant bacteria. Antibacterial therapy was initiated from first day of febrile neutropenia (FN)

and was based on sensitivity to antibiotics of potentially pathogenic gut microorganisms. Patients without preexistent colonization with resistant microorganisms received escalation strategy of EAT and patients with colonization received de-escalation strategy.

Results: Colonization with antibiotic-resistant microorganisms is present prior to FN in 64% children with solid tumors and HSCT. Bacterial culture demonstrated colonization with antibiotic-resistant microorganisms in 2 children (6.7%) and CHROMagar technology in 18 (64%) ($p=0.001$). Extended-spectrum beta-lactamase bacteria were diagnosed in 12 (43%) children, carbapenem-resistant in 4 (14%) and vancomycin-resistant enterococcus – 2 (7%). Escalation strategy (mostly cefoperazone-sulbactam) was used in 17 (53%) patients. De-escalation strategy (mostly meropenem) was used in 15 children (47%). Second line EAT was administered in 19 patients (56%). Response to second line EAT was observed in 13 cases (68%).

Conclusions: EAT protocol for children with solid tumors and HSCT was started in RM Gorbacheva Research Institute. The usage of CHROMagar technology for gut colonization analysis improved detectability of antibiotic-resistant bacteria and optimized treatment of FN.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0382 / #1415 | Patient Initiated Discussions of Advance Care Planning: The Perspective of the Family Member, Friend or Health Care Provider

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Background and Aims: It is difficult for young adults (YA), their families and providers to talk about end-of-life (EoL) decision-making. Yet, more than 3,000 YA die annually from chronic illnesses in the United States. Voicing My CHOICES™ (VMC) is an advance care planning (ACP) guide designed for YA. Researchers aimed to evaluate the experience of those the YA shared their completed VMC with.

Methods: Participants included the family members, friends or HCP of a YA (aged 18-39) enrolled in an 8-center study examining psychosocial outcomes after completing VMC. During a structured phone interview, participants were asked about their conversation with the YA, including what the conversation was like for them, any changes in their relationship and if care changes had been made.

Results: Interviews were conducted with 40 family members, 6 friends, and 5 HCP. Fifty percent of family members and 33% of friends reported a positive change in their relationship with the YA following the discussion (improved communication, learned more about the YA). Fifty one percent found the conversation to be positive, 14% stated they felt relief and 30% stated having VMC as a guide made the conversation easier. Two participants felt left with more questions about the YAs preferences. While 51% indicated the conversation was emotionally difficult, 68% reported benefit despite the difficulty. Only 10% of HCP noted a treatment change following the discussion. Seventy six percent of family members, 67% friends and 50% HCP did not think the YA would have talked to them about their EoL preferences without the study.

Conclusions: The use of an ACP tool appears to enhance communication about EoL preferences between the YA patient and their family members, friends and HCP. Interventions to help facilitate ACP conversations outside of the research setting are still needed.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0383 / #1432 | Patients with Covid-19 Infection and Hematological Cancer - A Nationwide Survey in Greece

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Background and Aims: Background and aims. Covid-19 infection caused by Sars-Cov-2 virus has emerged as a global pandemic threatening public health. The data to date show that the infected children represent 8% of reported cases with a mild course and very few deaths. Children with cancer during chemotherapy are in danger of severe infections due to immunodeficiency and myelosuppression. This study aimed to present Covid-19 and hematological cancer cases found in a nationwide survey in six pediatric hematology and oncology centers.

Methods: Out of the 41 oncology patients with Covid-19 infection we studied the 24 with hematological malignancies, 16 ALL, 1 AML, 3 NHL, 4 HL. Twelve children were on treatment (6 in maintenance), while 12 were on follow-up.

Results: All the patients had contact with a Covid-19 positive person. Most of them suffered mild symptoms lasting from one to four days before diagnosis, including fever, nasal discharge, cough, congestion, anosmia. Six patients needed hospitalization and thorough laboratory examinations due to Covid-19. One patient with aleukemic leukemia cutis received antibiotics due to fever. One patient had a pulmonary embolism diagnosed after tachypnea, not requiring intubation. She was under induction treatment according to ALLIC 2009 protocol and was treated with remdesivir and heparin of low molecular weight for three months. One patient was five months after ALLO-BMT. Two patients received gamma-globulin. The hospitalized patients underwent examinations with ECG and laboratory tests for DDimers that were into normal values. One patient suffered polyarthralgia 4 weeks after a negative test Covid-19. Patients on maintenance chemotherapy discontinued it for two weeks or until they had two negative PCR test results and none needed hospitalization.

Conclusions: We conclude that Covid-19 infection usually has a mild course in most pediatric patients with hematological malignancies. More research is needed to understand the impact of the infection in patients with cancer.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0384 / #1446 | Implementation of Pediatric Early Warning Systems (PEWS) Improves Provider-Family Communication in Pediatric Cancer Patients During Deterioration Events

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Background and Aims: The quality of communication between providers and patient families is an integral part of clinical care. Family concern is an important and validated component of Pediatric Early Warning Systems (PEWS), however little is known about the relationship between PEWS and provider-family communication.

Methods: Semi-structured interviews were conducted with 83 nurses, pediatricians, oncologists, and intensive care (ICU) clinicians who

recently cared for a patient experiencing clinical deterioration at two pediatric oncology hospitals of different resource-levels: St. Jude Children's Research Hospital (n=42 participants) in Memphis, Tennessee or Unidad Nacional de Oncología Pediátrica (UNOP, n=41 participants) in Guatemala City, Guatemala. Interviews were conducted in the participants' native language (English or Spanish), transcribed, and translated into English. Each transcript was coded by two researchers and analyzed for thematic content surrounding family communication and concern.

Results: PEWS facilitated more interaction between providers and families, especially ICU providers, allowing for relationship-building, anticipatory guidance, and destigmatization of the ICU before transfer. PEWS assessments also provided a way for families to contribute to the identification of deterioration. Time spent interacting with families was an important factor in the perceived quality of communication across all disciplines at both centers. At UNOP, language barriers occasionally prevented proper communication, affecting providers' ability to assess family concern for PEWS.

Conclusions: Providers across all disciplines at both institutions recognize patient families as a valuable part of the care team, particularly during events requiring escalation of care. PEWS can improve quality of communication between providers and families by providing more opportunities for interaction, building relationships and trust. These interactions allow increased involvement of families in the care of their children during clinical deterioration. By facilitating care during deterioration, increased provider-family communication resulting from PEWS further demonstrates the importance of PEWS implementation in both high-resource and resource-limited pediatric oncology settings.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0385 / #1470 | Oncological Risk Factors for Mortality in Critical Illness in a Dedicated Pediatric Cancer Center in Latin America

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Background and Aims: Children with cancer have frequent critical illness requiring pediatric intensive care unit (PICU) admission. In this study, we evaluate risk factors for mortality among pediatric oncology patients with critical illness.

Methods: We conducted a retrospective cohort study of all PICU admissions in a dedicated children's cancer hospital in Mexico December 2013–December 2019. Patients without cancer and those transferred to outside facilities were excluded. Clinical data were extracted from the medical records. Outcome was categorized as recovery or adverse outcome. T-tests, chi-square, fisher-exact, and ANOVA tests were used to assess risk factors for mortality.

Results: A total of 460 pediatric cancer PICU admissions meet inclusion criteria. Among these, the PICU mortality was 5.7% (26 patients) and adverse outcome (PICU death or palliative discharge) rate was 7.7% (35 patients). Hematological malignancy, use of mechanical ventilation (MV) and severity of illness on admission (Pediatric Index of Mortality 2, PIM2) were significantly associated with mortality ($P=0.0209$, $P<0.0001$, $P=0.006$ respectively). Status of oncological disease (new diagnosis, relapse, remission), treatment prior to ICU admission and presence of mucosal barrier injury (MBI) were not associated with mortality. Solid tumor patients had more ICU-free-days and mechanical-ventilation-free-days ($p=0.009$ & $p=0.007$ respectively). Among emergency PICU admissions (excluding patients admitted after elective surgery) ($n=306$), mortality was 7.2% ($n=22$). Emergency patients with hematologic malignancy had higher severity of illness (PIM2, $p<0.001$) but not higher mortality and patients with solid tumors did not have higher ICU and MV free days.

Conclusions: Overall mortality for this cohort was lower than previously reported, even after excluding post-operative scheduled patients. Of note, oncological disease status, treatment intensity before PICU admission and MBI wasn't related to mortality in this series. This suggest high quality critical care (even in resource limited settings) may be important to attenuate the impact of these risk factors in onco-critical illness outcomes.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0386 / #1472 | **Differences in Palliative Opportunities Across Diagnosis Groups in Children with Cancer**

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Background and Aims: Childhood cancer causes significant morbidity resulting in physical and emotional stress. During these times, patients and families could benefit from palliative care (PC) to improve

symptom burden, family-centered goals of care, and quality of life. We defined and evaluated "palliative opportunities" and compared them across three main disease categories in pediatric oncology: leukemia/lymphoma (LL), solid tumors (ST), and central nervous system (CNS) tumors.

Methods: A priori, nine palliative opportunities were defined: disease progression/relapse, bone marrow transplant, phase 1 trial enrollment, admission for severe symptoms, social concerns or end-of-life (EOL) care, intensive care admission, do-not-resuscitate (DNR) status, and hospice enrollment. A single-center retrospective review was completed on patients with cancer aged 0–18 years who died from 1/1/12–11/30/17. Demographic, disease, and treatment data were collected. Descriptive statistics were performed. Opportunities were evaluated from diagnosis to death and compared across disease groups.

Results: Included patients ($n=296$) had diagnoses of LL ($n=87$), ST ($n=114$), and CNS tumors ($n=95$). Palliative opportunities were more frequent in patients with ST (median 8, range 1–21) and CNS tumors (median 7, range 0–31) versus LL (median 5, range 0–21, $p=0.001$). While patients with ST had more progression/relapse opportunities ($p<0.0001$), patients with CNS tumors had more EOL opportunities ($P<0.0001$). Palliative opportunities increased towards the EOL in all diseases. PC was consulted in 108 (36%) patients: LL (48%), ST (34%), and CNS (32%, $p=0.02$). Prior to death, patients with CNS tumors had earlier PC consultation, DNR status, and hospice enrollment.

Conclusions: Children with cancer incur many events that warrant PC support, regardless of diagnosis. Patients with ST and CNS tumors had more palliative opportunities than patients with LL yet received subspecialty PC less often. Understanding the type and timing of palliative opportunities within each disease can guide utilization of PC consultation to ease patient and family stress and symptoms during their disease course.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0387 / #1478 | **Pediatric Early Warning Score (PEWS): The Experience from a Tertiary Care Center in México**

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Background and Aims: Pediatric cancer patients are susceptible to developing life-threatening clinical deterioration and tools such as the PEWS have been implemented to initiate timely measures in the event of clinical deterioration. The objective of this study was to describe the experience in the use of PEWS.

Methods: A retrospective study of the events during the years 2019 and 2020 in a pediatric oncology service.

Results: The hospital admissions per year in 2019 and 2020 were respectively 1298 and 1302. In 2019 there were 29 events, the most

frequent was the use of vasopressors in 72.4% of patients (21/29), followed by the use of mechanical ventilation (MV) in 10.3%. In 75.8% of the events, the patients were transferred to an intensive care unit (PICU), who had an average stay of 12.9 days. 55.1% (16/29) of the patients died. In 2020 there were 24 deterioration events, the most frequent was the use of vasopressors in hospitalization in 50% of the cases (12/24), followed by MV with 8.3%. 45% of the patients died. 50% of the patients had an average stay of 9.4 days per patient. Considering that the average cost per day in the ICU was \$472.5, the cost for the ICU stay in 2020 was \$53,298. If the 12 patients in 2020 had had an average stay of 12.9 days (the one obtained in 2019), the cost of their care would have been \$73,143, thereby saving \$19,845 US-dls.

Conclusions: In our institution, despite the increase in the number of hospital admissions per year, the events of clinical deterioration have decreased and proportionally the number of deceased patients, the average days of stay in the PICU, and the use of vasopressors and MV, which shows better results in training in the use of PEWS and this has represented a decrease in annual expenses by attention in PICU.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0388 / #1505 | International Study on the Impact of Delayed High Dose Methotrexate (HDMTX) Elimination in Pediatric, Adolescent and Young Adult (AYA) Patients with Cancer

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Background and Aims: Methotrexate (MTX)-related toxicities can delay or interrupt cancer treatments. Monitoring MTX levels allows early detection of delayed elimination and enhanced supportive care measures to avoid MTX-related serious adverse events.

Methods: We analyzed 115 pediatric and AYA patients (0 to 49 years old) treated with 305 courses of HDMTX (defined as a dose above 500 mg/m²) from various data sources: the Guardian Research Network (101 US community oncology programs), the Cincinnati Children's Hospital and the Global Quality Improvement Program for HDMTX. Severely delayed elimination of methotrexate was defined by the Ramsey Criteria and an "elevated level at hour 72" was defined as

a level greater than 0.2 microM at 72 hours. Acute kidney injury (AKI) was graded using the AKI Network scoring system.

Results: Grade 2-3 AKI occurred in 42% of courses with persistently elevated MTX levels at hour 72 versus 8.2% of courses without elevated levels at hour 72 ($p < 0.001$). Of 305 HDMTX courses, 17 (5.6%) had delayed elimination severe enough to meet criteria to receive glucarpidase. None were managed with glucarpidase, and 10 (59%) had elevated levels at 72 hours. Courses with delayed elimination ($n=17$) had a mean length of stay (LOS) of 6.5 days compared to 2.8 days for courses without delayed elimination ($n=288$), ($p < 0.001$). Courses with both delayed elimination and elevated levels at hour 72 ($n=10$) had a mean LOS of 8.2 days compared to 2.9 days for courses without delayed elimination ($n=288$), ($p=0.003$) and compared to 4.6 days in those courses without elevated levels at hour 72 but with delayed elimination ($n=7$), ($p=0.02$).

Conclusions: Careful monitoring of MTX levels can help guide supportive care and reduce LOS. Effective access to glucarpidase, which reliably clears MTX by hour 72, would be expected to reduce LOS by 3.6 days, in addition to reducing pediatric kidney injury and delays in subsequent treatments.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0389 / #1545 | Experience of Longitudinal Assessment of Nutrition in Children Undergoing Cancer Treatment

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Background and Aims: Timely assessment of nutritional status plays a vital role in managing childhood cancer. A detailed nutrition assessment at diagnosis and timely interventions may help in improving supportive care of patients on cancer treatment.

Objective: To assess the changes in nutritional status of children over a period of 1 year while undergoing cancer treatment

Methods: The study was done in the children's cancer ward of Super Speciality Paediatric Hospital and Post Graduate Institute at Noida. Newly diagnosed patients <18 years between April 2019 to March 2020 were included. Nutritional assessment of patients included anthropometric measurements of height, weight, MUAC, BMI, through which the z-score of patients was calculated using WHO standard growth charts. The intervention provided included nutritional counseling along with a customized diet plan. High-calorie high-protein snacks, commercial complementary feeding and ready-to-use therapeutic food (RUTF) were used.

Results: Out of 56 children included, M:F ratio was 1.8:1. 40 patients had ALL, 5 had AML. 50% children were diagnosed to be malnourished

at presentation. On longitudinal follow-up, with nutritional intervention, improvement in height and weight were noted in 25% each and increase in MUAC in 44.6% at 3 months. BMI improved in 34% children. At 6 months, an increase in height was noted in 44%, weight in 42%, MUAC in 33%, and BMI in 34%. Mild malnutrition reduced from 18 to 11.6% over the period and SAM and MAM reduced from 23% to 9.3% and 6.9% to 2.3% respectively.

Conclusions: The trends of anthropometric assessment disclose that early diagnosis and intervention in MAM and SAM patients on cancer treatment may lead to improvement in nutritional status. While patients who are normally nourished at diagnosis may maintain the same status when provided regular counseling and intervention.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0390 / #1552 | The Cumulative Incidence of Cisplatin-Induced Hearing Loss in Young Children is Higher and Develops at an Early Stage during Therapy Compared to Older Children

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Background and Aims: Ototoxicity is a common adverse event of cisplatin treatment. We investigated the development of cisplatin-induced hearing loss (CIHL) over time in children with cancer by age, and examined the influence of other clinical characteristics on the course of CIHL.

Methods: Data from Canadian childhood cancer patients were retrospectively reviewed. Hearing loss was graded according to International Society of Pediatric Oncology criteria. The Kaplan-Meier method was applied to estimate the cumulative incidence of CIHL for the total cohort and per age. Cox-regression models were employed to explore the effects of independent variables on CIHL development up to three years after start of therapy.

Results: In total, 368 patients with 2,052 audiological assessments were included. Three years after initiating therapy, the cumulative incidence of CIHL was highest in patients ≤ 5 years (75%, 95% CI 66-84%) with a rapid increase observed to 27% (95% CI 21-35%) at three months and 61% (95% CI 53-69%) at one year, compared to

patients >5 years (48%, 95% CI 37-62%) ($P < 0.001$). Total cumulative dose (TCD) cisplatin at 3 months (per 100 mg/m² increase: HR 1.20, 95% CI 1.01-1.41), vincristine (HR 2.87, 95% CI 1.89-4.36) and total duration of concomitantly administered antibiotics (>30 days: HR 1.85, 95% CI 1.17-2.95) further influenced CIHL development over time.

Conclusions: In young children, the cumulative incidence of CIHL is higher compared to older children, and develops early during therapy. The course of CIHL is further influenced by TCD cisplatin and other ototoxic (co-)medication. These results highlight the need for audiological monitoring at each cisplatin cycle.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0391 / #1696 | Retrospective Analysis of Antifungal Prophylaxis with Itraconazole in Children with High Risk Leukemia from a Tertiary Care Centre in India

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Background and Aims: Invasive fungal diseases (IFDs) are an important cause of morbidity and mortality (10-57%) in patients with haematological malignancies, especially during prolonged neutropenia due to chemotherapy, particularly for high risk (HR) leukemias.

Methods: Retrospective analysis of children with HR leukemia (Acute myeloid leukemia (AML), Infant Acute lymphoblastic leukemia (ALL), Relapsed ALL, ALL defined high risk as per BFM protocol) who received Itraconazole prophylaxis during intensive chemotherapy was done from January 2019 to December 2020. All were initiated on Itraconazole at start of induction therapy except HR ALL, at beginning of HR blocks.

Results: Of the 44 children (43% were chemotherapy naïve), 14 were (32%) AML, 13 (30%) HR ALL, 12 (27%) Relapsed ALL and 5 (11.5%) Infant ALL. Itraconazole was well tolerated. The mean duration of Itraconazole was 90 days (range 9- 330 days), 54.5 % did not get any fungal infections, completed prophylaxis and 45.5% switched to empirical or curative antifungal therapy. Overall, 14/44 (32%) were diagnosed with proven or probable IFDs on basis of radiology and serology and was most commonly seen in Relapsed ALL (58%), followed by HR ALL (31%), AML (21%) and none with Infant ALL. The most common fungal infection was Aspergillus in the chest 64.3% (9/14) followed by Candida 35.7 % (5/14) (Candidemia 2; hepatosplenic candidiasis (HSC) 2; urinary tract infection 1) and Mucormycosis in 7% (1/14). Dual fungal infection (aspergillus in chest and HSC) was seen in 2 patients. During follow-up, 80% cleared the fungal infection and no deaths were related to IFDs.

Conclusions: Administration of Itraconazole did not seem to alter incidence of IFDs in HR leukemia to a significant extent. But this group did not have a life threatening IFD. Larger studies are required to explore

this benefit of Itraconazole in this subset of patients who otherwise see significant mortality due to fungal infections.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0392 / #1793 | Attachment to the Golden Hour Protocol in the Administration of Antibiotics on Patients with Febrile Neutropenia after an Implementation Science in Action (ISA) Course

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Background and Aims: Background: Febrile neutropenia is an emergency that leads to a rapid deterioration to sepsis, shock, or death if not attended on time. The early administration of antibiotics helps to reduce the morbidity and mortality associated with disseminated infection. Aim: Increase the percentage of adherence to the golden hour within 60 minutes of the administration of the antibiotic, in febrile cancer patients, from 88% to 95%, and decrease the unnecessary use of broad-spectrum antibiotics from 40% to 10% at HITO through an Implementation Science in Action (ISA) Course.

Methods: As part of an ISA Course in an 18-month learning system, a multidisciplinary team from every department involved in the system (pediatric emergencies, laboratory, nursing, infectious disease department, pharmacy, and quality department) sought improvement in antibiotic adherence protocol. During the development of the course, the team test and implement changes in the golden hour protocol to improve the system, collect data to measure the impact of the changes and analyze monthly reports with the results.

Results: During the course period, we had 87 patients with febrile neutropenia. The percentage of patients attached to the golden hour protocol increased to 95.4%, and the use of empiric broad-spectrum antibiotics was needed only in 10% of the cases.

Conclusions: Using the ISA model, we were able to optimize the hospital protocol for the golden hour in pediatric neutropenic patients. The multidisciplinary analysis and teamwork on a quality improvement protocol are determinants for success.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0393 / #212 | Patient/Carer Experience of Dietetic Care Following Move to Remote Working During Pandemic

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Background and Aims: Background/aims: Patient reported outcome measures (PROMs) are essential in gaining valuable feedback. Due to outbreak of COVID-19, dietitians at Great Ormond Street Hospital (GOSH) were moved off site to work remotely resulting in change of practice, where patients who were previously seen face-to-face, were reviewed by telephone. The blood, cells, cancer (BCC) dietitians undertook a PROMs questionnaire to understand patient/carers experience of dietetic service and was compared to experience pre-pandemic.

Methods: Parents/carers with a child under care of BCC dietitians from mid-June to September 2020 were eligible to complete online Nutrition and Dietetic Patient Outcomes Questionnaires (validated PROM tool). Children >8 years were eligible to complete young-person version of questionnaire. After review with dietitian, parents/carers were given website URL to allow optional and anonymous completion of questionnaire on own electronic devices. Responses allocated numerical values and analysed using Microsoft Excel.

Results: Thirty-five parent/carers and two young persons completed questionnaire. No difference in dietetic experience observed overall (score=26.1) since pre-pandemic (score=26.1). Those who had met a dietitian once scored more favourably (score=23.7) than those who had seen dietitian twice or more (score =27.5, 26.7 respectively); in 2019, those who had seen dietitian >5 times had more positive experience (score=23.6). Reported improvement in parents/carers feeling a dietitian helped them better understand how to manage child's condition (+7%), but worse experience around managing anxiety (-9%) and increasing socialisation/interaction (-22%).

Conclusions: Change to remote working during pandemic well-received by parents/carers under BCC dietitians at GOSH. Data collected during national lockdown likely impacted results around anxiety and socialisation; however, impact of face-to-face reviews in reducing anxiety must be considered. A combination of face-to-face reviews and remote working is likely to become new model of care. Repeat questionnaire is needed once new ways of working implemented. Methods of gathering data from young children needs further consideration.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0394 / #249 | Improving the Early Detection and Management of Constipation in Children and Young People Living with Cancer in the North East of England

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Background and Aims: Cancer-related constipation in children and young people is a major complication associated with adverse health outcomes and poor quality of life. There is currently limited consensus agreement on its management. We aimed to address this gap in childhood cancer supportive care by establishing a local clinical practice guideline for constipation that informs treatment decision-making processes.

Methods: A 3-step modified Delphi method was used to establish consensus between May–June 2020. A 17-member panel representing oncology patients, their carers, occupational therapy, dietetics, pharmacy, nursing, and paediatric oncology specialists were invited to participate. Step 1 utilised relevant patient data obtained between October 2019–April 2020, evidence from the critical appraisal of existing literature, and panel-member interviews using open-ended questions to develop comprehensive best practice statements. Steps 2 and 3 applied a 5-point Likert scale and a face-to-face departmental meeting respectively to identify consensus.

Results: Severe constipation-related complications among patients admitted to the paediatric oncology unit were reduced from 22% to 0% within 2-months of initiating the following key consensus points: 1) Iterative use of a “traffic light” risk assessment pathway and nurse-checklist for early identification of high-risk patients, 2) Providing all patients with a bespoke constipation information leaflet specifically developed for a paediatric oncology setting, 3) Initiating laxative treatment using a standardised treatment algorithm.

Conclusions: Our patient-centred consensus guideline for cancer-related constipation has improved the quality of patient care by using a novel risk assessment and management pathway. Avenues should be explored to trial this quality improvement process across other principal treatment centres.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0395 / #258 | Gastrointestinal Tracts Symptom Cluster among Children being Treated for Cancer: A Scoping Review

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Background and Aims: Co-occurring symptoms (“symptom clusters”) catalyze each other leading to higher levels of distress, greater symptom burden and poorer quality of life in children (ages 0-17.99 years) with cancer. Gastrointestinal tract-related (GI) symptoms are the most commonly experienced symptoms during active treatment. Further

investigation of symptom clusters that include GI symptoms is needed to improve their management. This review aimed to identify the scope of symptom clusters that include GI symptoms in children with cancer.

Methods: The scoping review was guided by the PRISMA extension for scoping reviews (PRISMA-ScR). Peer-reviewed, empirical studies of symptom clusters that included GI symptoms in children with cancer were identified through computerized databases searches of PubMed, EMBASE, and CINAHL through January 2021. Study characteristics, symptom cluster analysis procedure and findings were extracted using a standardized data extraction tool, and the results were described using frequencies and narration. Symptom clusters from each study that included GI symptoms were synthesized to determine the most commonly included GI symptoms across studies. GI symptoms reported in more than 50% of included studies were summarized as the comprehensive GI symptom cluster.

Results: Nine studies met eligibility criteria. Most used cross-sectional designs. Participants were mostly male, aged 2 to 29 years, and diagnosed with various cancers. To determine symptom clusters, most studies used the de novo approach informed by symptom prevalence ratings from broad symptom assessment tools. The comprehensive GI symptom cluster included nausea, vomiting, fatigue, lack of appetite, changes in taste, and weight loss. Most studies also identified either risk factors for or patterns of GI symptom clusters.

Conclusions: QOL for children could be improved by comprehensive assessment of GI symptoms and development of effective ways to relieve these co-occurring symptoms.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0396 / #268 | Analysis of Central Venous Catheter Care in Children with Cancer in Western Ukraine: A Prospective Study

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Background and Aims: Central venous catheters (CVC) guarantee a reliable vascular access, and are essential for cancer management in children. The aim of the Project and study was to evaluate the safety of CVC care and form a routine practice standard by dedicated team in specialized pediatric hospital providing chemotherapy in low income settings. The Project of improving CVC care in Western Ukraine was supported by My Child Matters Program.

Methods: The analysis was conducted as the prospective cohort study since July 2019 through April 2021. The data on removed CVC were captured by trained staff on the basis of pre-developed case report forms and hospital onsite source medical data. For each CVC, the total number of days were calculated from insertion to removal on 3 types of pre-defined complications: infection, thrombosis and dysfunction. The general complication rate per 1000 days was calculated. Analysis was performed to calculate frequency of different type of complications among the three different CVC types and diagnoses.

Results: Analysis included 184 removed CVCs in 133 children during study period. Three types of CVC were used in the study: tunneled Broviac-Hickman (51%); non-tunneled (39%) and port-catheters (10%). The total number of CVC days was 28,886 (mean: 157 days, range: 3–680). The rate of complications was 1.7 per 1000 CVC days. End of treatment was the removal indication for 151 of 184 CVC (82%). Infection was the most frequent complication out of CVC removal causes - 10.8%. At least one CVC-related complication was observed in 21.7% of patients.

Conclusions: The implementation of profound care for CVC by a dedicated team with adequate supply and prospective analysis showed decent results in reducing the CVC-related morbidity. The study was the first in the region to provide such data. The Project activities are replicated as a multicentre St Jude Global/EURADo Research Group study for 2021-2022.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0397 / #316 | Cisplatin-Induced Nephrotoxicity in Childhood Cancer: Comparison between Two Countries

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Background and Aims: Various definitions have been used to describe cisplatin-induced nephrotoxicity which potentially led to differences in the determination of risk factors. The objective of this study was to evaluate the incidence of kidney injury according to commonly used and alternative definitions in two different cohorts of children who received cisplatin.

Methods: This retrospective cohort study included children at three pediatric academic health centers in Vancouver, Canada (one centre) and Mexico City, Mexico (two centres), who were treated with cisplatin-based chemotherapy for various solid tumours. Serum creatinine-based definitions (KDIGO and pRIFLE), electrolyte abnormalities consisted of hypokalemia, hypophosphatemia and hypomagnesemia (based on National Cancer Institute Common Terminology Criteria for Adverse Events v5), and an alternative definition (Alt-AKI) were used to describe nephrotoxicity. Incidence with the different definitions, definitional overlap and inter-definition reliability were further analyzed.

Results: A total of 173 children (100 from Vancouver and 73 from Mexico) were included. Depending on the definition used, cisplatin-induced nephrotoxicity occurred in 38%–92% of the children. Nephrotoxicity and all electrolyte abnormalities were significantly more common in the Vancouver cohort than in the Mexico City cohort except when using the KDIGO definition. The most common form of electrolyte abnormalities was hypomagnesemia (88.9%, in Vancouver cohort) and hypophosphatemia (24.2%, in Mexico City cohort). The KDIGO definition provided the highest overlap of cases in Vancouver (100%), Mexico (98.6%) and the combined cohort (99.4%). A moderate overall agreement was found among Alt-AKI, KDIGO and pRIFLE definitions ($\kappa = 0.18$, 95% CI 0.1–0.27) in which KDIGO and pRIFLE showed moderate agreement ($\kappa = 0.48$, 95% CI 0.36–0.60).

Conclusions: Compared to pRIFLE and KDIGO criteria, Alt-AKI criteria detected more patients with cisplatin-induced nephrotoxicity. pRIFLE is more sensitive to detect not only the actual kidney injury but also the patients at risk of cisplatin-induced nephrotoxicity, while KDIGO seems to be more useful to detect clinically significant kidney injury.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0398 / #319 | Intrinsic: Intention to Limit the Therapeutic Effort of Physicians Caring for Children <12 Years with Advanced Cancer

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Background and Aims: Every year, 2080 children between 5 -15 years old are diagnosed with cancer in Colombia, approximately 40% die due to cancer. Cancer is currently the second leading cause of death in this age group. Despite the limited benefits, medical care with curative intent is a frequent practice in children with advanced cancer, generating more suffering for them and their family. Aim: To develop a scale to measure intentionality to limit the therapeutic effort (LTE) of physicians caring for children <12 years with cancer without curative option.

Methods: Design and construction of measurement scales. The development of this scale was based on the theory of planned behavior, which states intention as direct predecessor of behavior. Factors relevant to decision-making and medical behavior were identified through literature review and expert interviews. An iterative process of construction and reduction of items, psychometric properties (internal consistency analysis, discrimination and difficulty of items and content by expert judgment), and a pilot test of the first version of the scale with 36 participants was carried out.

Results: We propose a scale with four domains and 63 items: attitudes, subjective norm, perceived control and beliefs, using visual analogue scale for rating, ranking and paired-comparison (VAS-RRP) type scaling. The internal consistency of the domains fluctuated between 0.92 and 0.83. The discrimination and difficulty indices of the retained items were acceptable.

Conclusions: This is the first scale developed to measure the intention to LTE in children with cancer. The analysis of the psychometric properties of the scale shows a high internal consistency. It is necessary to continue evaluating the validity of the scale. In the future, the scale may have applications for research and clinical practice. It can also serve as a tool to measure interventions aimed to reduce end-of-life suffering.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0399 / #328 | Evaluating The Need for Integrated Pediatric Palliative Care Services in a Paediatric Oncology Setting: A Retrospective Audit

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Background and Aims: During the past decade, Pediatric palliative care has become an established area of medical expertise. Scant information exists regarding the triggers for referral and referral practice hence the need for the audit.

Methods: A retrospective audit of medical case records of pediatric oncology patients over a period of 1 year from September 30th, 2019 to September 30th, 2020 was conducted. Demographic details, diagnosis, staging, clinical parameters, the reason for referral, and palliative care plan were captured in a predesigned proforma.

Results: Among 126 children with cancer, 27 (21.4%) patients were referred to palliative care. The majority, 21 (77%) referrals were inpatient consults. Symptom management 17 (44.7%) was the most common trigger for referral followed by referrals for psychosocial support 12 (14.4%). Children with solid tumors 16 (59%) were more often

referred than hematological malignancies. Among those needing end-of-life care, 8 (88.8%) out of 9 families preferred home than a hospital.

Conclusions: Low incidence of palliative care referral and presence of symptoms as a trigger for palliative care referral suggests gaps in the integrated approach. The study findings prompt a review of palliative care referral criteria and referral practice in a pediatric oncology setting.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0400 / #337 | Documentation of Advance Directive and Advance Care Planning Discussions in Adolescent and Young Adult Recipients of Hematopoietic Stem Cell Transplant: A 5-Year Retrospective Review

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Background and Aims: Hematopoietic stem cell transplant (HSCT) carries significant risk of morbidity and mortality. Consequently, documentation of advance directives (ADs) and advance care planning (ACP) is crucial to promote patient-centered care. Historically, adolescent and young adult (AYA) patients, aged 15 to 39 years, have had less access to these conversations. We conducted a single center, retrospective review of the electronic medical records (EMR) of AYA patients who received HSCT in the past 5 years to characterize the frequency and context of documented ADs and ACP.

Methods: EMR documents for AYAs who received HSCT from 2015 through 2020 were screened. AD documents included text indicating completion of a signed paper directive, expressions of preferred code status, and identification of a healthcare proxy. ACP documents included discussions of goals of care, hopes and fears related to treatment decisions, and patient legacy. Documents with pertinent text directly copied from a previous note were excluded. Frequency and type of documentation were considered at the population and patient level. A Fisher Exact test was used to compare the proportion of documented ACP in surviving versus deceased patients.

Results: A total of 67,000 documents associated with 236 patients (202 surviving and 34 deceased) were reviewed. A total of 1009 pertinent documents were identified, screened for duplicated text, and reduced to 664 unique documents. More patients had a documented AD (78%) than ACP (38%). Deceased patients were significantly more likely to have documented ACP (85% versus 29%, $p < 0.0001$). Of all documented ACP conversations in deceased patients, 53% occurred in the last week of life.

Conclusions: The majority of AYA patients receiving HSCT did not have documented ACP and a significant majority of these conversations occurred with patients in their last week of life. These results support efforts at standardizing ACP implementation for AYA patients to promote more frequent and timely access.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0401 / #341 | A Pilot Exercise Program Utilising Limited Equipment for Children & Adolescents Undergoing Cancer Treatment within an Inpatient Setting

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Background and Aims: The positive impact of exercise on individuals with cancer is well established; however there is little evidence of the benefit of exercise with children and adolescents undergoing cancer treatment. We aimed to pilot a reproducible exercise program for children undergoing cancer treatment within an inpatient setting. The goal of the pilot was to determine the safety and feasibility of a standard exercise program utilising limited equipment. Additionally, we aimed to understand patient/family perceived benefits and challenges of participating in the program.

Methods: A standard exercise program was developed for children and adolescents undergoing cancer treatment in an inpatient setting. The program was designed to be suitable for individuals of varying levels of physical function. We measured heart rate to determine an appropriate exercise intensity and ensure patients exercised within appropriate aerobic capacity. We used the Borg scale to monitor perceived exertion. Exclusion criteria were agreed following consultation with the multi-disciplinary team. A questionnaire was given to patients and families to ascertain perceived benefits and challenges of the exercise programme.

Results: We piloted this programme with 20 patients. We found that a standard exercise program can be safely and feasibly delivered to children and adolescents undergoing chemotherapy within an inpatient environment. The results of the patient/family questionnaire demonstrate that a standard exercise program can help maintain engagement, be enjoyable and have a positive impact on their admission.

Conclusions: A standard exercise program can be delivered safely and feasibly in a paediatric oncology inpatient setting and has a positive impact on patient experience. Due to the limited equipment required, similar programs could be utilised in all income settings, including resource limited ones, with similar benefits anticipated. Future research with higher numbers of participants is required to measure the impact of similar programmes on children with cancer's self-efficacy, self-esteem and participation in activities of daily living.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0402 / #397 | Challenges for Pediatric Hematology Oncology during Coronavirus Pandemic in Pakistan

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Background and Aims: As the prime care is directed towards the management of infected cases during the COVID-19 pandemic, patients with non-communicable diseases are subject to lack of attention. Pediatric Hematology Oncology patients demand holistic care therefore might get affected tremendously during this pandemic. The objective of this study was to assess the challenges for Pediatric Hematology/Oncology in a developing country during the coronavirus pandemic.

Methods: An online cross-sectional survey was carried out among the Pediatric Hematologists-Oncologists from all over Pakistan. The questionnaire contained questions pertaining to assess the pandemic effects. Responses were received online, and data was analyzed in SPSS 16.0 in terms of descriptive statistics.

Results: The survey was sent to 96 Pediatric Hematologists-Oncologists with the response rate of 50%. Responders were working at 14 different Pediatric Hematology/Oncology centers in Pakistan. Majority (62%) were working in public sector hospitals, 14% reported to have stopped registering new patients, 68% reported reduced overall number of patients and 64% reported reduction of patients for scheduled chemotherapies; 78% reported affected supportive care services while 70% reported closure of various units of the department. Variable effect on hospital supplies, specialized investigations, tumor boards, oncological surgeries, radiation therapy sessions, stem cell transplant and academic activities was also reported. Availability of PPE was satisfactory, with 86% reported availability of surgical mask and 30% N95 mask. Sixty two percent reported unavailability of isolated febrile neutropenia rooms, 58% reported inability to exercise social distancing in ward and 90% reported application of short staff rosters. Significant level of personal and family health concern ($M > 2.5$) and stress (> 3.0) were reported by the pediatric hematologists/oncologists working during the pandemic.

Conclusions: Apart from overwhelming issues of COVID-19 patients and the doctors, patients with non-communicable diseases and the treating specialists are also tremendously affected in various aspects during the pandemic. Reported interrupted management of pediatric hematology oncology patients may lead to devastating long term effects.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0403 / #400 | Vicca: Vida Con Calidad, a Pediatric Palliative Care Educational Program for Health Professionals in Perú

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Background and Aims: Pediatric Palliative Care (PPC) is necessary for the health system; university education does not include studies in the PPC. The aim of this study was to evaluate the results of our educational program. We report herewith the results of the educational program experience.

Methods: The educational program was conceived into 4 modules addressing: the principles, symptom management, communication, and end-of life notions. Based on a multi-professional and intersectorial (from primary to tertiary sectors) approach, no eligibility criteria on the professional profile were required. To strengthen the impact of the programme and to favour care close to families living places, we aimed to identify PPC ambassadors in peripheral centres and located in the 4 peruvian regions.

Results: Thanks to My Child Matters grant, in 2019, 3 educational sessions were held at the institutional level and 4 during 2020. Due to COVID, 1 virtual course was held divided into 5 sessions, each session had a duration of 2 hours and counted 140 professionals from health. Also, 4 virtual workshops were held. As a complement to the training, educational material was specifically designed and distributed. To date, 1,370 health professionals have been trained: 414 physicians, 527 nurses, 108 psychologists, 25 social workers, and 296 including other health professionals. Thanks to this educational initiative and an alliance with the Ministry of Health, a PPC network was consolidated that unites 4 regions of the country. In addition, specific multi-professional PPC teams were established within other institutions throughout the country.

Conclusions: Integrative management of children with cancer should include palliative care by applying total family-centered care, regardless of whether children are receiving treatment directed at the disease. To this purpose training health professionals, especially at the primary level of care, is crucial to decentralize and optimize care and thus improve the quality of life of children and their families.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0404 / #401 | Complications in Patients with Fully Implantable Long-Term Vascular Devices

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Background and Aims: Long-term vascular devices (DV) offer patients a better quality of life, which is why the My Child Matters Program for the Quality Care in Pediatric Oncology project seeks to improve the quality of patient care through standardized management of long-term DVs providing safe vascular access with minimal complications. The aim to this work is to describe complications in fully implantable long-term DV patients.

Methods: Descriptive, retrospective study, in the period from July 2019 to March 2021. Complications were reported and recorded in the patient's medical record.

Results: Of a total of 58 fully implantable long-term VDs, 40% presented complications: perioperative 9% (2/23) and in relation to use 91% (21/23). 38% of the latter (8/21) corresponded to non-infectious complications: mechanical dysfunction 87.5% (7/8) and obstructive 12.5% (1/8). Catheter-related bacteremia-type (BAC) infectious complications corresponded to 62% (13/21). The infection rate was 4.4 / 1000 catheter days with a total of 2892 catheter days registered. In 22% (5/23) of complications, early removal of the DV was indicated, related to BAC 40% (2/5), mechanical dysfunction 40% (2/5) and obstructive 20% (1/5)

Conclusions: The infection rate in patients with long-term DV is high compared to figures reported in other catheter care and management reviews. Not so in terms of complications of the mechanical type that are within the reported ranges. Perioperative complications were mild and easily resolved. The creation of a standardized system for early recognition, registration, and targeted treatment improves the quality of patient care.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0405 / #424 | The Impact of 25(OH)Vitamin D Level on Outcome in Pediatric Patients with Chemotherapy-Induced Febrile Neutropenia

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Background and Aims: Febrile neutropenia is one of the most common and undesirable side effects of cancer therapy. It significantly contributes to cancer mortality, leads to treatment delays, and creates an economic burden. Different biomarkers have been investigated as prognostic factors with limited benefit. The aim of this study was to determine the impact of serum vitamin D level on the outcome of febrile neutropenia in children treated for cancer.

Methods: We prospectively evaluated the serum 25-hydroxy-vitamin D (25OHD) levels in children treated for chemotherapy-induced febrile neutropenia in our center. Vitamin D deficiency was defined as a serum 25OHD level less than 20 ng/ml. Death and/or antibiotic modification were accepted as treatment failures. The demographic data, microbiological and clinical outcomes, and the intensity of received treatment protocol were compared between groups depending on Vitamin D status.

Results: Seventy-four febrile neutropenia episodes in 34 pediatric cancer patients (median age: 5.6 years, 24 boys, 10 girls) were included. Sixteen, four and 14 patients had hematologic malignancy, central nervous system tumor, and extra-cranial solid tumors, respectively. In 51 episodes of febrile neutropenia, Vitamin D levels were below 20 ng/ml. Median Vitamin D levels were significantly lower in adolescents, and in patients who received higher intensity treatment protocols. Febrile neutropenia outcomes in terms of treatment failure, duration of neutropenia and antibiotherapy, were not influenced by Vitamin D levels.

Conclusions: The majority of our patients had low Vitamin D levels during febrile neutropenia episodes. Age and treatment with higher intensity protocols significantly correlated with Vitamin D deficiency. However, Vitamin D was not found as a prognostic variable for febrile neutropenia. The results of our study indicate to the necessity of follow-up of 25(OH)D levels in adolescents and heavily-treated patients. The impact of vitamin D supplementation on febrile neutropenia in vitamin D-deficient children is a subject of future research.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0406 / #448 | Paediatric Oncology Rehabilitation in a Low-Resource Context: SIOP Global Mapping Africa & the Formation of a Global Paediatric Oncology Rehabilitation Network

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Background and Aims: Children with cancer receive intense treatment that often has significant negative sequelae requiring rehabilitation. Appropriate rehabilitation vastly improves quality of life during treatment, at end of life, and survivorship including physical, mental and emotional health. There is no global paediatric oncology network

for rehabilitation professionals and limited understanding of rehabilitation infrastructures, particularly in low-and-middle income countries (LMIC). Aim: determine access to such services in Africa through the SIOP Global Mapping Programme and national rehabilitation organisations.

Methods: The SIOP Global Mapping Programme African baseline data identified physiotherapy services at a hospital/facility level, including location, services, and multidisciplinary team meetings. We also performed an extensive search for physiotherapy services in Africa through the World Physiotherapy global body.

Results: Of 100 complete responses from 46 African countries, 78 facilities (78%) representing 35 countries, reported physiotherapy services. Forty-two percent had dedicated paediatric oncology units/wards (34) or subspecialised wards (8) and 73 facilities in 35 countries had multidisciplinary team meetings. We identified 25 African national physiotherapy organisations. A working group, (including 13 physiotherapists from 8 African countries) formed a global network of paediatric oncology rehabilitation professionals to improve access to childhood cancer rehabilitation services in all settings and facilitate professional development.

Conclusions: A high number of paediatric oncology facilities reported physiotherapy services in Africa. Further investigation is required to understand the extent and accessibility of available services and the impact on children's quality of life. Specialised paediatric oncology training and units may result in increased access to physiotherapy, particularly if these professionals are included in multidisciplinary team meetings. As survival rates improve in LMIC, it is vital to increase awareness of paediatric oncology rehabilitation needs and attract/retain rehabilitation professionals. The new SIOP global rehabilitation network will advocate for appropriate paediatric rehabilitation and collaborate with WHO GICC and other key stakeholders to support local training and improve access worldwide.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0407 / #464 | Glucarpidase and Time of Administration in Childhood Cancer

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Background and Aims: Methotrexate is one of the drugs that can cause nephrotoxicity in some childhood cancers, mainly when it is used in high doses, which can lead to systematic toxicity if not cleared from the body. Supportive care, hydration, leucovorin and alkalisation can be used to aid the clearance. In cases of delayed clearance or excessive toxicity, Glucarpidase can be administered, to rapidly deplete systemic methotrexate. Glucarpidase is an expensive drug that has varied recommendations for use predominantly based on adult data. No standardised protocol exists internationally for the use of Glucarpidase.

To inform future recommendations on Glucarpidase use in paediatric patients, we conducted a retrospective study of 16 Paediatric centres in the UK.

Methods: A retrospective survey study included all the tertiary cancer children's hospitals in the UK, involving all the patients who had received Glucarpidase in the past 10 years. Proformas were sent to the clinical pharmacist leads in each hospital. Data on the patients' diagnosis, age, and changes in renal function, methotrexate concentration, and known toxicity profiles were collected, alongside the timing of Glucarpidase use.

Results: Out of sixteen paediatric centres, nine centres had used Glucarpidase in the past 10 years. We identified 29 patients treated with Glucarpidase, all at different time points and outcomes varied according to strategies used. The mean age was 16 (2-17) years. The average dose was 2080 (837.5 - 4000) units, which was given on an average of 72.4 (28.5 - 196) hours from Methotrexate infusion. The mean Methotrexate levels and serum creatinine prior to Glucarpidase administration were 45.97 $\mu\text{mol/L}$ and 115.3 $\mu\text{mol/L}$ respectively.

Conclusions: Our findings suggest that there is a wide range in the awareness, indication, and timing of Glucarpidase use amongst paediatric haematology-oncology centres in the UK. We propose to develop an international guideline for the use of Glucarpidase in a paediatric population.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0408 / #467 | Development and Pilot Testing of Proactive: A Pediatric Onco-Critical Care Capacity and Quality Assessment Tool

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Background and Aims: Hospitalized pediatric oncology patients frequently require critical care. Over 80% of children with cancer reside in low- and middle-income countries which face multiple challenges delivering high-quality pediatric onco-critical care (POCC). We recently identified POCC quality and capacity indicators for PROACTIVE (Pediatric Oncology cApaCity Assessment Tool for Intensive Care); a tool that evaluates strengths and limitations in POCC services in resource-limited settings. This study describes pilot testing of the PROACTIVE tool

Methods: The original consensus-derived PROACTIVE tool included 119 quality and capacity indicators. Indicators were converted to 182 questions with dichotomous, numerical, and Likert scale answers divided between 2 electronic surveys for pediatric intensivists and oncologists managing critically ill pediatric hematology-oncology (PHO) patients. Alpha-testing was completed by 4 pediatric intensivists with extensive experience in POCC to evaluate the content and format of the tool. We then recruited 11 centers from different countries, economic levels and array of pediatric critical care services for beta-testing, with the goal to assess the clarity, flow and applicability of PROACTIVE.

Results: Electronic surveys were administered in English using RED-Cap. Questions were modified following alpha-testing (November 2020). The 11 beta-testing sites ranged from general hospitals with no critical care units to specialized pediatric cancer centers, located primarily in low-middle and upper-middle income countries. Site leads recruited additional collaborators resulting in 38 multidisciplinary stakeholders with diverse experience in POCC. Beta-testing (January-May 2021) facilitated additional survey adjustments and the development of score-base facility reports highlighting areas of opportunity for improvement.

Conclusions: PROACTIVE identifies gaps in POCC services with the goal to improve the care and outcomes of PHO patients. Pilot testing of PROACTIVE evaluated the feasibility of implementing this tool in resource-limited hospitals. Survey data highlights limitations and barriers to POCC in resource-limited settings and will help guide implementation of PROACTIVE globally.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0409 / #476 | The Influence of Lymphocyte Subpopulation Counts and Immunoglobulin Levels on Prognosis of Febrile Neutropenia in Children Treated for Cancer

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Background and Aims: Febrile neutropenia (FN) is one most common oncological emergencies that necessitate prompt and

appropriate intervention. Various clinical and laboratory parameters have been investigated to prevent morbidity and mortality. In this study, we aimed to analyze the impact of lymphocyte subset counts and immunoglobulin levels on the prognosis of FN in children with cancer.

Methods: The prospective study included all children (0-18-years) with FN during cancer treatment. Age, gender, bodyweight and height, tumor type, and the intensity of treatment (according to Intensity of Treatment Rating Scale (ITR-3.0)) were recorded. Lymphocyte subset and immunoglobulin analyses were performed within 48 hours of FN. Any change in initial empiric antibiotic treatment and/or death was accepted as treatment failure.

Results: Fifty-two febrile neutropenia episodes in 34 pediatric cancer patients (median age: 6.86 years, 19 boys, 15 girls) were included. Twenty-one, two, and 11 patients had hematologic malignancy, central nervous system tumor, and extra-cranial solid tumors, respectively. Treatment failure was observed in 23 (44.2%) of the FN attacks. Absolute lymphocyte, T lymphocyte, and CD4 lymphocyte counts were significantly lower in the treatment failure group ($p < 0.05$). The IgG and IgM results were not different among the groups while the IgA level was found to be significantly higher in the group with treatment failure. According to ITR-3 scales, treatment failure was significantly more common in patients treated with higher intensity treatments. There was no statistically significant relationship between age, gender, body mass index, absolute neutrophil count, duration of neutropenia, and treatment success.

Conclusions: The absolute lymphocyte counts, particularly the CD4 counts, were found to be lower in patients who failed to respond to first-line therapy in FN. Diagnoses necessitating treatment with higher intensity regimens seemed to contribute as well. The impact of FN treatment modification on treatment success in the presence of these factors needs to be clarified in future research.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0410 / #497 | Relationship between Allergic Transfusion Reaction and Allergic Predisposition among Pediatric Patients with Hematological/Oncological Disease

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Background and Aims: Although blood transfusion remains one of the important supportive therapies in pediatric hematological/oncological disease, the occurrence of transfusion-related adverse reactions has continued to be a concern. Allergic transfusion reactions (ATRs) can occur particularly frequently in transfusion-related adverse reactions, and the onset of ATR may be related to a patient's atopic predisposition. Moreover, although reports suggest that pediatric patients with hematological/oncological disease are more likely to develop ATR, the relationship between allergic predisposition and ATR is yet to be fully clarified.

Methods: Patients who were diagnosed with pediatric hematological/oncological disease and received transfusion at two institutions were included. We determined patient background information related to a history of allergies, measured allergen-specific immunoglobulin E (IgE) using sera obtained upon diagnosis and analyzed their association with ATR onset. Allergen-specific IgE was measured using View Allergy 39®, with values ≥ 0.50 indicating positive items.

Results: Among the 363 patients analyzed, 141 developed ATR. Multivariate analysis identified age < 5 years, a history for food allergies, positivity for Japanese cypress, and Dermatophagoides pteronyssinus-specific IgE as risk factor for developing ATR. Moreover, our findings showed patients aged 5– < 10 years who had a history of asthma, allergic rhinitis, or pollinosis and those aged ≥ 10 years who were positive for dog dander-specific IgE were at risk for developing ATRs.

Conclusions: The current study showed that the allergic constitution of patients plays a role in the onset of ATR even in pediatric hematological/oncological diseases. Therefore, advance confirmation of a patient's allergic constitution may, to some extent, predict the onset of ATR. However, given that multiple allergic predispositions within complex mechanisms may be involved in the onset of ATR, further verification is required.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0411 / #499 | Evaluation of Aprepitant as an Add-On Therapy for Prevention of Chemotherapy Induced Nausea and Vomiting in Children

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Background and Aims: Chemotherapy-induced nausea and vomiting (CINV) is the most common and distressing adverse effects associated with antineoplastic treatment in children. Aprepitant is a selective neurokinin-1 receptor antagonist that has shown to be effective in preventing CINV in children when added to the standard antiemetic

regimen (combination of corticosteroid and 5-hydroxytryptamine receptor antagonist). The study aimed to evaluate the Aprepitant as an add-on therapy to the standard antiemetic regimen for the prevention of CINV in children receiving moderately (MEC) and highly (HEC) emetogenic chemotherapy.

Methods: This randomized control trial was carried out in Department of Paediatric Hematology and Oncology, BSMMU, Bangladesh on 46 patients aged 5-18 years with documented malignancy who scheduled to receive either MEC or HEC. After recruitment patients were randomly divided into Aprepitant arm (received Granisetron, Dexamethasone, and Aprepitant) and Control arm (received Granisetron and Dexamethasone). All the patients were followed up to 120 hours after completion of chemotherapy.

Results: The complete response rates for the Aprepitant versus control arm during the acute and overall phase were 82% vs. 40% ($p=0.003$) and 65% vs. 26% ($p=0.008$), respectively. However, a higher percentage of patients achieve complete response in the delayed phase was also observed, though statistically not significant (65% vs 40%, $p=0.077$). In the acute phase, there was a significant reduction in mild to moderate vomiting in the Aprepitant arm as compared to the control arm ($p=0.01$). In the overall phase, 35% of patients in the Aprepitant arm had mild to moderate vomiting as compared to 74% in the control arm ($p=0.027$). No major adverse effects were reported by patients or caregivers.

Conclusions: The addition of Aprepitant as an add-on therapy with standard antiemetic regimen was found to be effective and safe in the prevention of CINV in all phases specially in the acute phase in paediatric patients receiving MEC and HEC.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0412 / #553 | Kusankha Pamodzi: Healthcare Decision-Making Preferences among Patients with Cancer

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Background and Aims: Oncology teams are encouraged to include patients' preferences and goals of care in determining appropriate treatment courses. This is especially important in Malawi where many patients present with advanced stages of cancer resulting in challenging and complex treatment decisions. There is no existing data from Malawi exploring decision-making preferences among cancer patients to guide providers.

Methods: In the oncology clinic of Kamuzu Central Hospital (KCH) in Lilongwe, Malawi, 100 patients were surveyed related to healthcare decision-making. Survey data were summarized using simple descriptive statistics.

Results: A majority of cancer patients preferred to engage in shared decision-making regarding their cancer treatment: 68% (68/100) of patients preferred the medical team to make decisions regarding their care together with their input and 29% (29/100) of patients preferred for the medical team to make decisions without their input. About half (55%, 53/96) of cancer patients did not feel that their medical team involved them in decision-making regarding their treatment and 53% (53/100) felt they were never or only sometimes listened to by the medical team. Nearly all patients (91%, 91/100) preferred to have their medical team inform them how likely treatments are to lead to cure; and 62% (62/100) felt the medical team explained this to them in a way that was easy to understand. Most patients (63.6%, 63/99) felt the medical team adequately discussed emotional problems, such as anxiety or depression, related to their cancer treatment.

Conclusions: Shared decision-making is the preferred mode of treatment decision-making by the majority (68%) of surveyed cancer patients in Malawi. There is currently a gap, with about half of patients feeling the medical team did not involve them in their treatment planning. Nearly all patients (91%) wanted to know their risk of mortality. Oncology providers are encouraged to ask their patients' preference in decision-making and involve them in treatment decisions.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PV0413 / #562 | An Approach to Understanding Clinician Treatment Decision-Making at Diagnosis in Pediatric Cancer: Exploring Challenges Faced by Clinicians in Zimbabwe

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Background and Aims: Physicians treating children diagnosed with cancer in low- and middle-income countries (LMICs) face unique challenges when compared to physicians in high-income countries. These children often present with advanced disease and experience inferior outcomes which may lead to difficult treatment decisions, however, little is known on the factors that influence them. To gain an understanding of how physicians approach childhood cancer treatment decisions at the time of diagnosis in Zimbabwe, a modified Delphi approach was used.

Methods: Participants included physicians involved in treatment decisions for children with cancer in Zimbabwe as well as collaborating physicians from the United States. Childhood cancers were structured based on the International Classification of Childhood Cancer system

and further stratified by prognosis. In each round, a 'stoplight/robot' ranking of treatment intent for each cancer type-prognostic group was selected by each participant (green – curative; yellow – non-curative or palliative; red – comfort focused). After two Delphi rounds, each cancer type-prognostic group was discussed to understand decision-making rationale and ascertain consensus.

Results: Following the discussion phase, consensus was achieved for all cancer type-prognostic groups, with 27 cancer type-prognostic groups categorized as green, nine as yellow, and one as red. Recurrent themes of discussion included the potential for suffering without achieving cure and decreasing quantity of life from toxic death. While there was variability by cancer type, most participants would offer curative intent upfront and recommend further decisions based on treatment response.

Conclusions: Determining treatment plans for pediatric patients diagnosed with cancer in LMICs can be challenging. An initial approach to generating discussion and contextualizing some of these challenges is described. More research is needed to explore the factors that contribute to treatment decision-making in-depth as well as the circumstances when non-curative approach may be offered upfront, with the goal of improving outcomes and quality of life for children with cancer worldwide.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0414 / #570 | The Use of Palliative Oral Etoposide in Paediatric Solid Cancers - A Ten-Year Single-Centre Retrospective Series

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Background and Aims: Oral etoposide is often prescribed as palliative chemotherapy in paediatric cancer. However, the aims of therapy are not always clear and there is limited evidence to guide its use. We performed a single-centre 10-year retrospective audit of the use of etoposide in solid tumours to examine four outcomes of the use of oral etoposide as a palliative therapy: symptomatic relief, tolerability of side effects, effect on tumour progression and prolongation of survival.

Methods: All paediatric patients (< 16 years at initial diagnosis) with solid tumours who received palliative oral etoposide between January 2008 and December 2018 at Addenbrooke's Hospital, Cambridge, UK were identified and their medical records reviewed.

Results: Forty-one patients (20 with brain tumours) were identified. Symptomatic improvement was reported in 10 (24.4%) patients including stabilisation or reduction in neurological symptoms. Side effects were reported in 18 (43.9%) patients, sufficient to reduce quality of life in 9 (22%). One patient developed secondary leukaemia. Possible control of tumour progression was reported in 21 (51.2%) patients with an

apparent imaging response in 13 (31.7%). Sustained responses lasting ≥ 1 year were seen in 9 (22%) patients overall: 7 (35%) patients with brain tumours, 2 (9.5%) patients with extracranial solid tumours. The median duration of survival was 5.5 months (range 0.9 – 109+). There are 4 long-term survivors.

Conclusions: Oral etoposide provided symptom relief (24.4%) and control of tumour progression (51.2%) in a proportion of our patients, but this should be balanced with the incidence of side effects (43.9%). Sustained responses in tumour progression were seen in a higher proportion of patients with brain tumours (35%) than extracranial tumours (9.5%). Consensus guidelines on the use of palliative oral etoposide are needed and we recommend a multi-centre prospective study of oral etoposide as palliation in paediatric oncology to inform such a guideline.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0415 / #581 | Micronutrient Evaluation at Time of a Paediatric Cancer Diagnosis

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Background and Aims: Many children with cancer undergoing chemotherapy experience adverse events that could be exacerbated due to micro-nutrient abnormalities. A prospective pilot study is being performed to evaluate micro-nutrient and body mass (BMI) at diagnosis, then at 3 and 6 months while on chemotherapy. We are reporting the initial results taken at the time of diagnosis.

Methods: Participants were evaluated within 4 weeks of diagnosis. Serum was analyzed for vitamin A, D, E and trace elements copper (Cu), selenium (Se) and zinc (Zn). Weight, height, body mass index (BMI) and corresponding Z-scores were recorded. Blood was only collected at the time of routine blood work after informed consent was obtained.

Results: Twenty participants, ages 3-17 years, are enrolled to-date with the following diagnoses: acute lymphoblastic leukemia (n=6), Burkitts lymphoma (n=4), neuroblastoma (n=2), Ewings sarcoma (n=2), astrocytoma (n=1), rhabdomyosarcoma (n=1), medullablastoma (n=1), Hodgkins lymphoma (n=1), epithelioid sarcoma (n=1), Langerhans cell histiocytosis (n=1). Micronutrient abnormalities: low vitamin A in 4/20 (20%), low vitamin D in 10/20 (50%), high vitamin E in 8/20 (40%), low Cu in 3/20 (15%) and high Cu in 6/20 (30%), low Se in 4/20 (20%), high Se in 7/20 (35%) and low Zn in 12/20 (60%). Out of the 11 participants tested for albumin, 5/11 (45%) had low albumin. BMI values at diagnosis range between 14-26.2 kg/m² (average 18.7kg/m²). Fifty percent of participants have BMI Z-scores of -1 or lower or +1 or greater.

Conclusions: These preliminary results suggest that approximately 50% of participants had one or more micro-nutrient abnormalities at the start of chemotherapy. The study continues to evaluate micro-nutrient and BMI status at 3 and 6 months and will evaluate

specific insufficiencies or other abnormalities with adverse chemotherapy events, eg vitamin A insufficiency with mucositis, Zn insufficiency with fever and neutropenia.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0416 / #603 | Implementation of a Catheter Clinic in a Pediatric Oncological Service: Improvements in the Management of Prolonged Intravenous Therapy

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Background and Aims: The management of the pediatric cancer patient requires a safe vascular access that guarantees the administration of intravenous therapy without interruptions and with the minimum of complications. The creation of a specialized service called "Catheter Clinic", made up of a multidisciplinary team seeks to raise the standards of care for cancer patients. The aim to this work is to describe the activities carried out by the Catheter Clinic of the Pediatric Hemato - Oncology Department of the Hospital de Clínicas, FCM - UNA and analyze the results of their implementation.

Methods: Prospective study with intervention from July 2019 to March 2021. The habilitation of the Catheter Clinic in June 2020 was carried out within the framework of the Quality Care in Pediatric Oncology Project for its My Child Matters Program. The activity records, event reports and control of health personnel are under the supervision of the Catheter Clinic.

Results: Event reports showed a decrease in the infection rate from 9.9 / 1000 catheter days in 2019, to 5.1 in 2020 and 4.4 in the first quarter of 2021. The number of patients with vascular devices (VD) in place increased from 14 in 2019 to 36 in 2020 and 8 are registered in 2021, adding a total of 58 DV placed. 281 registered ratings (32% in 2020 and 78% in 2021) and 14 DV maintenance (43% in 2020 and 57% in 2021). They currently attend an average of 183 consultations / month. 7 training sessions were coordinated (5 national and 2 international) adapting to the virtual modality for the period of the pandemic.

Conclusions: The implementation of a Catheter Clinic improves the quality of care for pediatric cancer patients, leading and coordinating what is inherent to prolonged intravenous therapy through long-term DV, promoting educational, control and supervision activities.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0417 / #641 | Modular Exercise Program (MEP) in Pediatric Oncology for Intensity-Driven Exercise Control - Feasibility in Hematopoietic Stem Cell Transplantation

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Background and Aims: Exercise programs for children undergoing hematopoietic stem cell transplantation (HSCT) seem to be safe and feasible. However, standardized training programs with different intensities and evaluation of the effects on health are lacking. For the currently ongoing ANIMAL-trial (DRKS00019865) that evaluates the effects of two exercise-training intensities on immune recovery, the modular exercise program (MEP) was developed. This analysis presents the feasibility of and adherence to the different exercise arms of MEP.

Methods: MEP includes a moderate-high (HI) and a low-intensity module (LI). While the content and intensity differ, the amount and frequency of training is the same in both modules. Exercise heart rate (HR) was targeted at 60-80% (HI) and 50% (LI) of HR maximum (%HRmax). Five children undergoing HSCT (mean 9.6±3.5 yr) were randomized into one of the two modules. Outcomes were adherence, peak and mean HR during training, median duration of training interventions (DUR), and total duration per week.

Results: There were no significant differences between HI and LI in DUR (MED=26.7 vs. 29.1 min, p=.1, r=0) and DUR/week (MED=97.5 vs. 49.9 min, p=.564, r=.25). For training peak HR (MED=70.7 vs. 63.7 %HRmax, p=.083, r=.77) and mean HR (MED=56.8 vs. 50.9 %HRmax, p=.083, r=.77) the effect size Cohens' d indicates a strong effect. That was also found for adherence (MED=73.3 HI vs. 33.3 LI %, p=.083, r=.77). Adherence varied during HSCT phases with highest participation rates during HSCT week.

Conclusions: According to effect sizes and trends towards significance of training HR, MEP seems to be able to differentiate between high and low intensity training. Therefore, MEP might be a useful tool for training control in children undergoing a HSCT. However, the children seem to like the HI training more, which is reflected in the slightly better adherence. Further studies with a larger sample size should be conducted to support these results.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0418 / #673 | Participation in a Health Promotion Program in Pediatric Oncology: Families' Perceived Barriers and Facilitators

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Background and Aims: Health promotion programs represent a promising avenue to address the prevention or mitigation of childhood cancer's adverse effects. However, we don't know the most effective way to promote healthy behaviours in pediatric oncology or how families experience health promotion interventions during the very challenging time that is going through childhood cancer. The purpose of this study was to explore the perspectives of families affected by childhood cancer regarding their participation in a health promotion program. The program had a two to four years duration and consisted of psychosocial support, nutrition counseling and physical activity interventions.

Methods: We undertook a cross-sectional qualitative study using semi-structured interviews. Individual interviews (median 14 min; range 8 to 38 min) were conducted with 14 participants (nine parents, five teenagers affected by cancer), representing 12 families, purposively sampled within a larger study. Data was analyzed using thematic content analysis.

Results: Participants reported many barriers to participation including organizational barriers, physical or mental health issues and lack of interest or need. Key facilitators included flexibility in the delivery of interventions, tailoring of the interventions to individuals' needs and social support. About half of participants reported that the program had a positive impact on their health behaviour practices during cancer treatment. Participants also reported that the program improved their overall health and brightened up their days at the hospital. Even though all participants reported having a positive attitude towards the adoption of healthy behaviours, only about half of them mentioned having the intention to modify their health behaviour practices in the long-term. For future implementation, participants suggested adding more educational activities, increasing the frequency of physical activity interventions and improving coordination of research and clinical activities.

Conclusions: Participants' perspectives on barriers to participation and suggestions for future health promotion initiatives will guide future research to develop sustainable behaviour change interventions.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0419 / #750 | Veno-Occlusive Disease Following Non Transplant-Associated Chemotherapy in Spain: A Registry of the Pediatric Haematology and Oncology Society

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Background and Aims: Venocclusive Disease (VOD) or Sinusoidal Obstruction Syndrome (SOS) of the liver is a potentially life-threatening complication following hematopoietic stem cell transplantation (HSCT) and less often conventional chemotherapy. The actual incidence and mortality rate of SOS in non-transplant population remains unknown. We present a Registry of the Spanish Society of Pediatric Haematology and Oncology of SOS and describe clinical characteristics, treatment modalities and outcomes of this population.

Methods: Retrospective national survey to identify patients receipts of conventional chemotherapy who were diagnosed with VOD/SOS in a 15-year period. We reviewed for gender, age, oncology diagnosis, time to onset of VOD/SOS, previous chemotherapy received, receipt of abdominal radiation, specific and supportive VOD treatment and outcome. We retrospectively used the EBMT diagnostic criteria for grading the severity of VOD/SOS in our cohort.

Results: 23 events of SOS were identified in 22 children following conventional chemotherapy. Median age:5 years (0.1-14)16 patients had a solid tumor with a majority of nephroblastoma. Median time to develop VOD:4 months (0.4-12).Ascites and painful hepatomegaly were the most common symptoms. Relevant coagulopathy or respiratory distress were unusual. Thrombocytopenia was identified as an early diagnostic event .5 patients had received prior abdominal radiation. Actinomycin alone or with Vincristine were the leading chemotherapy agents identified as trigger of VOD. Using EBMT score, VOD was very severe in 10 cases, severe in 6, moderate in 2 and mild in 5. 14 patients required ICU admission and 3 died of VOD. Defibrotide was safely administered in 15 patients.

Conclusions: VOD in non-transplant pediatric population is a rare event, possibly under-diagnosed due to its predominant mild course. We identified a substantial proportion of life-threatening VOD. This should alert the oncologist to routinely screen patients of risk using validated scores as EBMT classification.Defibrotide resulted as a safe agent in our cohort with no major side effects.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0420 / #785 | Non-Infectious Morbidity During Pediatric Acute Myeloid Leukemia Chemotherapy in the Netherlands: A 15-Year Overview

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Background and Aims: While treatment intensification has improved pediatric acute myeloid leukemia (AML) outcomes, treatment-related toxicity consequently has profound implications for morbidity and mortality. The objective of this study was to evaluate non-infectious toxicity during AML treatment in the last three protocols of the Dutch Childhood Oncology Group, given that only prior protocols have been evaluated in this manner as of yet.

Methods: A retrospective chart review was performed on 245 Dutch patients diagnosed with de novo AML and treated according to ANLL-97/AML-12 (1998-2005) n=118, AML-15 (2005-2010) n=60, or DB-AML01 (2010-2013) n=67. Grade 3-4 toxicities (hematological toxicity, cardiotoxicity, respiratory toxicity, mucositis, typhilitis, nephrotoxicity, hepatotoxicity, neurotoxicity, pain, anaphylactic reaction, elevation of ALT or bilirubin) were defined according to Common Terminology Criteria for Adverse Event version 4.0, excluding infectious toxicity. Grade 5 toxicity (treatment-related mortality) was beyond the scope of this study. ICU admission data was additionally assessed. Toxicity frequencies were determined using descriptive statistics. Per-protocol comparisons were conducted via Chi-square test.

Results: Median age at diagnosis was 6.0 years [interquartile range (IQR) 1.0-12.0], 58% were male. The most common toxicities were 1) mucositis (n=80, 32.7%), 2) hepatotoxicity (n=67, 27.3%), 3) respiratory toxicity (n=58, 23.7%) and ALT >5x ULN (n=58, 23.7%). Eighty-three patients (33.9%) were admitted to the ICU at least once, for a median of 6.0 days [IQR 2.0-15.0]. More blood transfusion (erythrocyte and thrombocyte) administrations occurred during AML-12 compared to AML-15 (p=0.01) and DB-AML01 (p=0.02). There was a greater frequency of mucositis during AML-12 compared to DB-AML01 (p=0.01), and a greater frequency of bilirubin >3x ULN in AML-15 compared to AML-12 (p=0.01). There were no differences between AML-15 and DB-AML01.

Conclusions: Across all protocols, the most common non-infectious toxicity was mucositis. Toxicities were more prevalent during AML-12. High treatment-related morbidity rates stress the urge to improve supportive care and develop less toxic treatment, while maintaining efficacy.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0421 / #819 | Measuring Clinical Capacity to Sustain Pediatric Early Warning Systems (PEWS) in Resource-Limited Pediatric Cancer Hospitals: A Spanish-Language Tool

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Background and Aims: Pediatric Early Warning Systems (PEWS) improve identification of deterioration in hospitalized children with cancer, however, their sustainability has not been studied. Sustainability is critical to maximize impact of interventions like PEWS, particularly in low-resource settings. This study establishes the reliability and validity of a Spanish-language Clinical Sustainability Assessment Tool (CSAT) to assess clinical capacity to sustain interventions in resource-limited hospitals.

Methods: Participants included PEWS implementation leadership teams of 29 pediatric cancer centers in Latin America involved in a collaborative to implement PEWS. The CSAT, a sustainability assessment tool validated in high-resource settings, was translated into Spanish and distributed to participants as an anonymous electronic survey. Psychometric, confirmatory factor analysis (CFA), and multivariate analyses were preformed to assess reliability, structure, and initial validity. Focus groups were conducted after participants reviewed CSAT reports to assess their interpretation and utility.

Results: The CSAT survey achieved an 80% response rate (n=169) with a mean score of 4.4 (of 5; 3.8-4.8 among centers). The CSAT had good reliability with an average internal consistency of 0.77 (0.71 to 0.81) and CFA analyses supported the seven-domain structure (comparative fit index 0.825, RMSEA score 0.067, and SRMR score 0.067). CSAT results correlated with respondent's perceptions of the evidence for PEWS, its implementation and use in their center, and their assessment of the hospital culture and implementation climate. The mean CSAT result was higher among respondents at centers with longer time using PEWS ($p < 0.001$). Focus group participants noted the CSAT report helped assess their center's clinical capacity to sustain PEWS and provided constructive feedback for improvement.

Conclusions: We present validation of the CSAT tool, the first Spanish-language instrument to assess clinical capacity to sustain evidence-based interventions in hospitals of variable resource-levels. This assessment demonstrates a high capacity to sustain PEWS in these resource-limited centers with improvement over time from PEWS implementation.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0422 / #823 | Assessment of Physical Function in Children with Cancer: A Systematic Review

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Background and Aims: Appropriate selection of robust assessment tools is essential to monitor physical function in children with cancer during and after treatment. This systematic review aims to identify and critically appraise the measurement properties of physical function measures currently used for children with cancer.

Methods: Two systematic searches of seven electronic databases (CINHAL, Medline, PubMed, PsychINFO, Sportsdiscus, EMBASE, AMED) were completed to identify physical function measures used in children with cancer and to evaluate their measurement properties. Methodological quality and the adequacy of measurement properties specific to populations of children with cancer were critically appraised using the COSMIN framework to ascertain which measures have evidence to support their use in children with cancer.

Results: One hundred and one physical function measures were identified across 154 studies. Measurement property data were available for 12 measures. The measurement properties of only two outcome measures were assessed in more than one study. Despite some positive measurement property data, there was no assessment tool that had consistent and adequate evidence overall to recommend its use

in childhood cancer populations. Poor methodological quality of the included studies was the main limiting factor.

Conclusions: There is very limited population specific evidence to guide the selection of physical function measures in children with cancer. Further research into the reliability, validity and responsiveness of physical function measures in children with cancer is needed to provide a basis for more effective clinical assessment and management.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0423 / #825 | Physical Activity for Children Undergoing Acute Cancer Treatment: A Qualitative Study of Parental Perspectives

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Background and Aims: Little is known about how to facilitate participation in physical activity among children receiving acute cancer treatment. The aim of this study is to understand the parental perspectives on physical activity for children during acute cancer treatment and explore strategies to overcome physical inactivity.

Methods: A qualitative study was completed. Data were collected via semistructured interviews with parents of children (aged 4-18 years) who were in their first nine months of cancer treatment. Data were analyzed thematically.

Results: Twenty parents were interviewed. A childhood cancer diagnosis and subsequent treatment were described as setting in motion a spiral of physical inactivity. Parents identified movement restrictions as a result of commencing treatment and the hospital environment as factors initiating this decline. Parents described the subsequent impact of movement restrictions on their child over time including loss of independence, isolation, and low motivation. These three consequences further contributed to an inability and unwillingness to be physically active. Parents responded in a variety of ways to their child's inactivity, and many were motivated to overcome the barriers to physical activity yet exhibited a reduced capacity to do so. Suggested intervention strategies highlighted the need for comprehensive support from the organization providing treatment.

Conclusions: Reasons for reduced physical activity in children receiving acute treatment for cancer are complex and multifactorial. Inactivity cannot be addressed by children and parents alone but requires support from the oncology team through changes to the environment, services, and policies to promote physical activity. These findings may be used to inform targeted, effective, and feasible physical activity interventions.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0424 / #827 | Promoting Positive Physical Activity Behaviours for Children and Adolescents with Cancer: Development of the Canmove Intervention using the Behaviour Change Wheel

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Background and Aims: Increasing participation in physical activity has the potential to improve outcomes for children and adolescents with cancer. The objective of this study is to outline the theoretical process behind development of CanMOVE, a behaviour change intervention designed to increase participation in physical activity for children and adolescents with cancer.

Methods: The Behaviour Change Wheel was used to guide design. The three stages involved (1) understanding physical activity behaviour within the paediatric cancer setting, (2) identifying potential intervention functions, and (3) identifying appropriate behaviour change and implementation strategies. Qualitative and behaviour change literature relevant to the cancer treatment setting were used to inform results.

Results: An individualised and flexible approach to physical activity promotion, that considers both child and environmental factors, is required. Fifteen behavioural change strategies were identified to inform intervention components. Implementation strategies were identified in their ability to build motivation, opportunity and capacity towards participation in physical activity. Key intervention components include: standardised assessment and monitoring (physical activity, physical function and Health Related Quality of Life), education, provision of an activity monitor (both child and parent), active supervised participation in physical activity, and one-on-one capacity building sessions with a physiotherapist.

Conclusions: The CanMOVE intervention is a novel approach to physical activity promotion in the cancer treatment setting. The use of a comprehensive design process will aid evaluation and replication of CanMOVE as it is piloted for feasibility in a clinical setting.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0425 / #829 | Regional Adaptation of Education in Pediatric Palliative Care: EPEC-Pediatrics Eurasia

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Background and Aims: Pediatric palliative care was recently identified as a priority to improve pediatric hematology-oncology (PHO) care in Eurasia, encompassing countries in Central Asia and Eastern Europe. However, there are limited regional opportunities for palliative care education. We describe the adaptation and implementation of a bilingual end-user Education in Palliative and End-of-Life Care (EPEC)-Pediatrics course for PHO clinicians in Eurasia.

Methods: Priorities for palliative care education for PHO providers were identified by regional collaborators and recent studies assessing physician palliative care knowledge in Eurasia. Of the EPEC-Pediatrics curriculum, 10 modules were prioritized, translated into Russian, and regionally adapted by Eurasian bilingual pediatric palliative care experts. Due to COVID-19, this course was delivered virtually, consisting of 10 pre-recorded, asynchronous lectures (English and Russian) and a bilingual workshop (5-days, 10-hours total) with interactive lectures and small group sessions.

Results: Seventy clinicians managing children with cancer (74.3% physicians, 15.7% nurses, 10% other; 24.3% English- and 75.7% Russian-speaking) from 14 countries participated in the virtual Eurasian EPEC-Pediatrics course. At the time of abstract submission, 60% (n=42) participants had completed all components of the course, including the asynchronous pre-recorded lectures, 5-day virtual workshop, and pre- and post-tests. Of the participants who completed the

post-assessment, the majority felt more confident in their ability to prescribe opioids and treat pain for children with cancer (90%), to control child suffering at the end of life (92%), and understand how to hold difficult conversations (96%). Ultimately, 96% indicated that the EPEC-Pediatrics course provided them with knowledge and skills that will change their clinical practice.

Conclusions: We present a successful regional adaptation of the EPEC-Pediatrics curriculum, including novel delivery of course content via a virtual bilingual format. We plan to repeat this course annually to improve access to high-quality palliative care education for PHO clinicians in Eurasia.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0426 / #851 | Pre-Therapy Fasting Reduces Acute and Late-Life Cisplatin Toxicities

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Background and Aims: Both cancer patients and survivors experience a lower quality of life during and following cancer treatment. Especially for pediatric cancer survivors, who still have a whole life before them, lifestyle adaptations are often needed, due to late-effects caused by chemo- or radiotherapy. Many treatment-induced side-effects resemble features of ageing, thought to be the consequence of DNA damage inflicted by the therapy. We previously showed that short-term fasting triggers a 'survival response', delaying ageing, diminishing ageing-related pathologies and boosting defence mechanisms. Here we investigated in a preclinical setting whether fasting prior to cisplatin treatment prevents therapy-induced side-effects.

Methods: Wildtype mice of pediatric age were preconditioned with ad libitum (AL) diet or a two-day fasting period followed by treatment with cisplatin (10 or 20 mg/kg). Body weight, food intake and activity were monitored until moment of sacrifice (3 days, 8 days or 6 months after treatment). Subsequently, blood and several organs were collected for further analysis.

Results: Treatment with cisplatin led to decreased appetite in AL fed animals. Adverse effects on body weight, activity, food- and water-intake were less severe in fasted mice compared to AL fed controls. High-dose cisplatin treatment appeared to be lethal for AL fed animals, and death was prevented in 40% of fasted mice. Multiple signs of nephrotoxicity, including tubule deformation, tubule necrosis and kary-omegaly, were largely diminished by fasting in both low dose and high dose treated animals.

Conclusions: Toxicity caused by cisplatin observed in the presented preclinical model resembled side-effects seen in patients. Multiple features of nephrotoxicity were absent or less severe in fasted animals compared to AL fed controls. Survival after high dose cisplatin treatment was significantly improved in fasted animals. Translation of current findings into treatment regimens for paediatric oncology patients are in preparation and may diminish multiple short- and long-term health conditions and improve overall quality of life for cancer survivors.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0427 / #854 | Gut Dysbiosis in Pathogenesis of Neutropenic Enterocolitis

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Background and Aims: Neutropenic enterocolitis (NEC) is a dreaded complication of chemotherapy. There is scant literature regarding incidence, clinical features, and determinants. The understanding of gut dysbiosis in NEC and pediatric cancer is still evolving.

Methods: Pediatric cancer patients with neutropenia and gastrointestinal symptoms were evaluated for NEC with radiological evaluation. Clinical, imaging, and laboratory features were analysed. Fecal samples were analysed for fecal calprotectin by sandwich ELISA and gut microbiota by conventional culture and compared with healthy controls and children without NEC.

Results: Of 590 children receiving chemotherapy during study period (January 2019- December 2020), 72 were evaluated for NEC based on neutropenia and gastrointestinal symptoms. Diagnosis was established in 44 children with incidence of 7.4% (Four had recurrent episodes). Acute myeloid leukemia was the most common underlying malignancy (31%) followed by acute lymphoblastic leukemia (29%). Common manifestations included fever (98%), pain abdomen (88%), and diarrhoea (83%). Hypoalbuminemia was observed in 78% patients. Large bowel involvement (94%) with diffuse bowel involvement (63%) and pancolitis (64%) were common. Fecal calprotectin was significantly elevated in NEC group than non-NEC group and healthy controls (median 87, 53, and 42 $\mu\text{g/g}$ respectively). Higher degree of gut dysbiosis was observed in children with NEC with higher isolation of Bacteroides and infrequent isolation of Lactobacilli. Mortality rate of 23% was observed. Median duration of hospitalisation and intravenous alimentation were 10 and 6 days respectively. Only presence of free fluid predicted higher mortality. Though levels of fecal calprotectin

and gut dysbiosis were higher in NEC, they didn't increase mortality. Isolation of *Bacteroides* and absence of *Lactobacilli* predicted longer duration of intravenous alimentation.

Conclusions: NEC caused significant morbidity and mortality in pediatric cancer patients. Fecal calprotectin was significantly elevated in NEC group. Gut dysbiosis was also significantly higher in NEC group suggesting a key role in pathogenesis and influencing outcome. This highlights role of targeted interventions towards gut dysbiosis like prebiotics and probiotics.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0428 / #882 | Age-Appropriate Advance Care Planning in Children Diagnosed with A Life-Limiting Condition: A Systematic Review

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Background and Aims: Pediatric advance care planning (pACP) is an important strategy in pediatrics, including oncology, to support patient-centered care before the end of life. Involving the child in pACP adjusted to age and development is known to be difficult, yet paramount. It is unclear how pACP interventions suitable for pediatric oncology integrate the child's perspective. Objectives: To assess characteristics, content and evidence on pACP interventions and to evaluate their age appropriateness.

Methods: A systematic review was performed. CINAHL, Embase and MEDLINE were searched from January 1, 1998 to August 31, 2020, to identify peer-reviewed articles containing strategies and tools to facilitate pACP offered to children (0-18 years), with a life-limiting condition, and their families. A content analysis (characteristics, intervention content) of the tools was performed. Assessment of risk of bias and quality of reporting was performed using Cochrane tools and COREQ, respectively.

Results: Thirty articles were included, describing 17 unique pACP tools, including nine tools specifically used in oncology. The majority of tools was developed for adolescents and young adults. A focus on AYAs was motivated by their cognitive capacity and willingness to participate in ACP. Interventions mainly tried to disclose child's and family's preferences concerning (future) hopes, wishes and goals of care; and enhancing communication between children, parents and health care providers. The majority of tools mentioned the relevance of an age-appropriate approach, but none of the interventions described specific intervention elements related to age-appropriateness.

Conclusions: Tools to support ACP in pediatrics, including oncology, integrate age-appropriate elements to a very limited extent, mainly

focusing on adolescents. Involvement of children of all ages may need a more comprehensive approach.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0429 / #914 | Yellow Priority Assignment Code as a Predictor of Unplanned Intensive Care Unit Transfers

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Background and Aims: Pediatric oncohematologic (POH) patients often exhibit alert signs 24 hours prior to an event leading to an unplanned transfer to the Pediatric Intensive Care Unit (PICU). The study aim was to assess the ability of the Yellow Priority Assignment Code (YPAC) in identifying patients at risk for clinical deterioration and transference to PICU.

Methods: Case-control study including POH patients unexpectedly transferred to PICU between January, 2019 and December, 2020 compared to POH patients admitted to the ward during the same period. The primary endpoint was YPAC score. In case group, patients' vital sign measurements for calculating YPAC score were retrieved from their medical charts 24 hours prior to transference to PICU; in control group, these measurements were drawn 24 hours after hospital admission. Clinical outcomes after PICU admission were assessed as secondary endpoint.

Results: During the study period, 208 POH patients were admitted to PICU, comprising 57 unplanned transfers (27% of PICU admissions). Among unplanned PICU transfers, 86% had high (≥ 3) YPAC score 24 hours prior to transfer, with a statistically significant association between YPAC score and PICU transfer rate ($p < 0.00001$). YPAC risk factors associated with PICU transfer were analyzed using univariate logistic regression. All YPAC components, including arterial blood pressure, had statistically significant odds ratio (OR) > 1 , reflecting greater chance of being transferred to PICU. Staff concern was a major factor (OR=127). Among clinical outcomes, significant association was found between YPAC score and length of PICU stay ($p=0.0043$). Considering YPAC cut-off point ≥ 3 , its accuracy was 89%, sensitivity 86%, and specificity 93% for predicting PICU transfer.

Conclusions: The study suggests YPAC score may serve as a potentially useful tool for identifying POH patients who are at higher risk for clinical deterioration. It may identify such patients and drive clinical decision-making toward early intervention to improve clinical outcomes and decrease unplanned PICU transfers.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0430 / #923 | Prospective Evaluation of Nephrotoxicity during Platinum Chemotherapy among Paediatric Patients

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Background and Aims: The platinum chemotherapy agents, cisplatin and carboplatin are widely used in the treatment of paediatric cancers. The kidneys are particularly susceptible to platinum toxicity. Tubulointerstitial injury is accepted as the most predominant form of nephrotoxicity. The study aims to identify the risk of nephrotoxicity associated with platinum compounds and to assess the risk factors associated with them.

Methods: Prospective hospital based observational study of 42 patients who received platinum based chemotherapy for newly diagnosed malignancies between 01.03.2020 and 31.03.2021. Nephrotoxicity was assessed using serum creatinine, serum sodium, serum potassium, serum calcium, serum magnesium and serum phosphorous values done before starting each cycle and end of chemotherapy. The toxicities were graded in severity according to CTCAE 4.0.

Results: Forty two children were initiated on platinum based chemotherapy (30 received cisplatin, 10 received carboplatin and 2 received both). Median cumulative cisplatin and carboplatin dose was 360 mg/m² and 1400 mg/m² respectively. Median number of cycles in cisplatin and carboplatin groups are 4 and 6 respectively. In cisplatin group, 23 children(76%) had hyponatremia, 6(20%) had hypokalemia, 6(20%) had hypomagnesemia, 5(16.6%) had hypophosphatemia, 4 (13.3%) had grade 1 Acute Kidney injury(AKI) and 4(13.3%) had hypocalcemia. In the carboplatin group there were no acute kidney injury or hypokalemia or hypophosphatemia. But hyponatremia was noted in 5(50%), hypomagnesemia in 3 (30%) and hypocalcaemia in 2(20%). Pearson correlation showed significant inverse correlation with cumulative cisplatin dose and S. Magnesium levels(p= 0.003), cumulative carboplatin dose and S. Calcium(p=0.02), S. Magnesium(p= 0.003) and S. Potassium levels(p=0.039). Multinomial regression analysis did not show any significant association between cumulative dose of platinum compounds, age and sex with nephrotoxicity.

Conclusions: Grade I AKI was observed in 16.6% of children on cisplatin while no one had AKI in carboplatin group. Cumulative doses of platinum compounds had significant inverse correlation with serum electrolyte levels.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0431 / #942 | Clinical Course of Sars-Cov-2 in Children with Cancer in the South URAL Region

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Background and Aims: A new coronavirus infection is a potentially dangerous condition for cancer patients. There is an active accumulation of data both on the severity of the infection itself in children with cancer, and on the risks for this category of patients associated with the anti-epidemic measures carried out.

The aim of our study was to analyze the cases of MERS-CoV-2 infection in children with cancer in the region during two increases in the incidence rate (May-November 2020).

Methods: At the risk of SARS-CoV-2 infection for study period was 111 children in region. We registered prospective all PCR-positive cases of infection in pediatric patients with cancer. In total, during this period, the infection was detected in 16 patients. Of these, 9 patients were with leukemia (7 - ALL, 1 - AML, 1 - CML), 2 - with lymphomas (1 NHL, 1 HL), 3 - with soft tissue sarcomas, and 2 patients - with osteosarcomas.

Results: In 9 patients, the disease was asymptomatic, in 6 patients - in a mild form (lesion of the upper respiratory tract), in one child (relapse of AML, FLAI chemotherapy) in a moderate form, with specific pneumonia. Three children were treated in a hospital (one needed oxygen support), the rest were treated outpatient. The delay of chemotherapy ranged from 9 to 38 days. In two patients, chemotherapy was not canceled. On average, therapy was resumed on day 21. No progression of the main diseases was recorded. No patient died from SARS-CoV-2 infection.

Conclusions: It seems that the course of SARS-CoV-2 infection in pediatric patients with cancer did not differ significantly from the common pediatric cohort, and was characterized by minimal clinical manifestations. The exception was a patient with long-term severe agranulocytosis. We need further follow-up to assess the long-term consequences of both the infection itself and the associated delays in therapy.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0432 / #947 | Outpatient Management of Non-Neutropenic Fever in Pediatric Oncology Patients: A Single Center Experience

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Background and Aims: Fever is a common complication in pediatric oncology patients, though no standard practice for managing non-neutropenic fever exists: some institutions administer antibiotics

empirically, while others observe. A 3-10% incidence of bacteremia has been reported in patients with non-neutropenic fever, and patients with infections generally present with objective signs of illness. We characterized the presentation, management, outcomes, and rate of bacteremia in pediatric oncology patients with central venous catheters presenting with non-neutropenic fever at a major oncology center.

Methods: Retrospective review of patients treated at MSK KIDS receiving cancer-directed therapy with central venous catheters presenting to the Pediatric Ambulatory Care Center with non-neutropenic fever (reported/recorded temperature $\geq 38.0^{\circ}\text{C}$ with ANC $>0.5\text{K}/\text{mCL}$) between 1/1/2019–6/30/2019.

Results: We included 222 episodes. Median age was 5.85 years (0.5–34.1). Underlying diagnoses included neuroblastoma (51%, $n=114$), hematologic malignancy (21%, $n=47$), and sarcoma (18%, $n=40$); 47.3% ($n=105$) had relapsed disease. The majority had most recently received conventional chemotherapy (42%, $n=93$) or the anti-GD2 antibody naxitamab (31%, $n=69$). Based on hospital practice and clinician judgment, 97% ($n=216$) received parenteral antibiotics, most commonly ceftriaxone (90% of antibiotics administered, $n=201$). Blood cultures were positive in 2.3% of episodes ($n=5$), primarily with gram positive organisms (80%, $n=4$); antibiotics were empirically administered in 4 of these visits (80%). Three positive cultures occurred in patients with sarcoma and 2 in patients with neuroblastoma. One patient was admitted at initial presentation due to clinical concern; 4 were admitted after blood culture resulted positive, and none required the intensive care unit. No deaths from infection occurred.

Conclusions: Our experience reveals a low rate of bacteremia, suggesting that even in heavily pre-treated pediatric oncology patients, well-appearing patients with non-neutropenic fever may be managed with observation alone. Based on these findings, a prospective quality improvement project aimed at reducing administration of empiric antibiotics in well-appearing patients with non-neutropenic fever is ongoing.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0433 / #963 | Critical Care Basics: A Blended Learning Curriculum to Support the Care of Hospitalized Children with Cancer in Resource Constrained Settings

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Background and Aims: Despite program growth in treating pediatric cancer in sub-Saharan Africa, improvement in survival is modest. Survival during initiation of treatment is dependent on performance status, regardless of cancer type or stage, suggesting that improve-

ments in the quality of supportive care may be impactful. Available training has limited applicability for acutely ill hospitalized children and is costly to administer. Our aim was to develop and implement a blended learning curriculum of multidisciplinary modules (Critical Care Basics) linked to practice guidelines.

Methods: We iteratively developed sequential acute care courses for three levels of health care workers. Texas Children's Hospital Global HOPE acute care experts drafted the content which was reviewed and finalized with Global HOPE colleagues in Malawi. The curriculum includes online lessons which are completed independently followed by in person sessions with review of online content, skill demonstration and practice and simulation. There are pre and post tests and course evaluations for both online and in person sessions.

Results: The curriculum is symptom-based and expands upon the WHO Emergency Triage Assessment and Treatment (ETAT) program. The first course focuses on triage and communication. Intended learners are entry level health care workers with limited pediatric experience. The second level course covers respiratory, circulatory and shock evaluation and management. Intended learners are nurses, clinical and medical officers. The third level course includes advanced respiratory and shock topics. The intended learners are more experienced health care providers. Due to travel restrictions, implementation was limited to a pilot with learners at Texas Children's Hospital.

Conclusions: We used a collaborative, iterative process to develop a multidisciplinary curriculum for the care of acutely ill hospitalized children and piloted a virtual implementation process. Next steps include implementation at Global HOPE-affiliated hospitals in Africa and expanding the curriculum with lessons on neurologic symptoms, dehydration and fluid management.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PV0434 / #984 | Integrating Romiplostim into the Supportive Care of Ewing Sarcoma Therapy

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Background and Aims: Chemotherapy administered every two weeks is the standard care for patients with Ewing sarcoma. Maintenance of this dose density improves survival outcomes but is limited by myelosuppression. Romiplostim is a thrombopoietin receptor agonist that may be used to treat prolonged chemotherapy-induced thrombocytopenia (CIT). Variables associated with CIT and our institutional experience with romiplostim in Ewing sarcoma are described.

Methods: Patients aged between 3 and 33 years with Ewing sarcoma from 2010- 2020 from a single institution were reviewed. CIT was defined as a failure to achieve 75,000 platelets per microliter by day 21 after the start of any chemotherapy cycle. Associations with age, sex,

stage, disease site and local control were analyzed; patients treated with romiplostim were described.

Results: Twenty-seven out of 42 patients (64%) developed CIT, delaying one to four chemotherapy cycles per patient. Absolute neutrophil count was not a factor accounting for the delay. CIT occurred during consolidation therapy with ifosfamide/etoposide cycles in 21/27 (77.8%). Eleven of 21 (52%) had radiation, a significant CIT risk factor in a univariate analysis. Eight patients received romiplostim. The median starting dose was 3 (range 1-5) mcg/kg. The median dose used to achieve > 75,000 platelets per microliter was 4.5 (range 3-5) mcg/kg. A median of 11 days was needed for platelet recovery. Doses were escalated weekly by 1-2 mcg/kg to 4-10 mcg/kg and continued throughout chemotherapy. The correlation between the max dose of romiplostim and change in platelet count after romiplostim initiation is 0.76 ($p=0.03$). No romiplostim-related adverse events were identified aside from mild headache.

Conclusions: CIT is the primary factor associated with failure to maintain treatment intensity, particularly in patients receiving radiation. Romiplostim was safe and feasible in our cohort, and efficacy was associated with a higher romiplostim dose. A prospective study is needed to assess romiplostim efficacy.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0435 / #1059 | Standards for the Psychosocial Care of Children and Adolescents with Cancer: A Pilot Study in Peru

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Background and Aims: The diagnosis and treatment of pediatric cancer can deliver in psychosocial repercussions for the patients and their families that can affect their adjustment to illness, treatment's adherence, and quality of life. Improve psychosocial care in childhood cancer treatment centers is relevant to reduce psychosocial risks and promote coping strategies in patients and their families. Aim: Establish the scope of practice of the multidisciplinary team working in childhood cancer in Peru related to psychosocial care based on international standards, adapt them to serve as a model for other low and middle-income countries.

Methods: Within the framework of the WHO Global Initiative for Childhood Cancer in Peru, a situational analysis was conducted on December 2020 and January 2021 including a comprehensive needs assessment survey for parents/caregivers of children with cancer, an interview and a survey based of international standards for the psychosocial care in childhood cancer for the multidisciplinary team. The surveys were answered online by 25 health professionals from 9 institutions (Lima, Cusco, Arequipa, La Libertad) and 104 caregivers of children with cancer. A strategic plan was created including the

adaptation of new standards according to the needs detected in previous phase, the creation of educational material and the establish of multicenter working groups to design activities and tools to implement the standards. In this study we present the preliminary results.

Results: Most participants indicated that psychosocial care consistent with the Standards was poorly provided at their center, specially because of limited human resources. They also referred some important approaches to include in psychosocial care. After the analysis of the data, six standards were proposed: 1) Psychosocial assessment; 2) Psychoeducation and support; 3) Mental health attention; 4) Reintegration; 5) Psychosocial competences; 6) Transversal's approaches.

Conclusions: The Standards proposed in this study provides a starting point to improve psychosocial care in childhood cancer in Peru.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0436 / #1085 | Anxiety among Adolescent Cancer Patients and Caregivers Receiving Treatment Care and Support During Covid-19 Pandemic in India

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Background and Aims: BACKGROUND: Anxiety which is prevalent among cancer patients and families while undergoing treatment can impair their quality of life. The COVID-19 outbreak is likely to cause anxiety among children with cancer and this could have a negative impact on their treatment and mental outcome. Cankids Kidscan, a national society for change in childhood cancer that was functional throughout COVID-19 lockdown and has facilitated cancer treatment, finances and travel. The study was designed to understand the anxiety level among children and caregivers who continued to receive treatment during the lockdown. We hypothesize that the current (in-state) anxiety levels would be higher than usual (in trait) anxiety during the pandemic. AIM: To evaluate anxiety among adolescent cancer patients and families receiving treatment during COVID 19.

Methods: Cross sectional study, 50 adolescents (11-18 years) with cancer and treated across 7 centres in India and 100 parents. State-Trait Anxiety Inventory for Children and adults was used.

Results: Among children, the mean state anxiety score (mean=38.26; range 23-51; SD $\pm$ 7.3) was similar to trait anxiety score (mean =37.96; range 23-55; SD $\pm$ 6.2) p value= 0.74. Among parents also, the state anxiety score (mean 46.66; range 30-65; SD $\pm$ 7.14) was similar to trait anxiety score (mean 46.72; range 27-63; SD $\pm$ 6.93) ($p=0.921$).

Conclusions: Although we expected the anxiety level of adolescents and children to be higher during the lockdown period compared to their basal level, we were unable to demonstrate it. This might be due to the ongoing support provided by our organization. Women reported higher

state and trait anxiety levels when compared with children and men. The study recommends continued support by cancer support organizations to help alleviate anxiety among patients and parents during this pandemic.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0437 / #1112 | Pilot Test of Translation of the Italian Version of the Pediatric Quality of Life Inventory™ (PedsQL™) 3.0 Healthcare Satisfaction Hematology/Oncology Module: Reliability in Radation Oncology

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Background and Aims: Purpose of this study was to investigate reliability and linguistic validation of the PedsQL™ healthcare satisfaction Hematology/Oncology module from its original English version to Italian language.

Methods: Three phases standard procedure of cross-culture adaptation was used to create Italian version of PedsQL™ healthcare satisfaction Hematology/Oncology module. Forward translations and backward translations was performed. Finally, a pilot-testing for understandability of the 'pre-final' version was conducted with parents of children attending our Radiotherapy Center using two methodology of Cognitive Interviewing ("Thinkaloud Interviews" and "Respondent Debriefing"), in order to obtain the final Italian version of the PedsQL™ healthcare satisfaction Hematology/Oncology module.

Results: Five parents (1 father, 4 mothers) were recruited during their children's radiotherapy treatment and the grammatically and conceptually acceptable pre-final version of the PedsQL™ Healthcare Satisfaction Hematology/Oncology Module was administered. The questionnaire was well understood reflecting its linguistic adaptation. Compliance with questionnaire administration was optimal. All subjects stated that all the questions of each sections were clearly comprehensible and easy to understand, suggesting changes in a few words which were then corrected and rechecked with back translation. Furthermore, all the parents spontaneously asked to complete the questionnaire in order to judge the assistance received during radiotherapy.

Conclusions: Our Italian version of the PedsQL™ 3.0 Healthcare Satisfaction Hematology/Oncology Module seems to be a valid and func-

tional instrument to indagate Health-related quality of life. Moreover the instrument demonstrated appropriate language reliability to measure parental healthcare satisfaction in our group of pediatric patients. The desire of parents to express their response and opinion on the topic might suggest the usefulness of measuring HRQOL in clinical practice with parents of pediatric cancer patients. Our purpose is to continue to investigate parent opinion during radiotherapy in pediatric patients and a prospective study could be used to evaluate the utility of HRQOL measurement in clinical practice.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0438 / #1132 | Health-Workers Perception and Emotions of Teleconsultation in Pediatric Cancer Care

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Background and Aims: Teleconsultations in oncology are no innovation. Around the world several centers have implemented a telehealth system, mainly to provide care to populations living in rural areas. During the first lockdown of the novel Coronavirus pandemic (COVID-19), we developed a teleconsultations model to replace in-person visits. We conducted an online questionnaire-based study with the aim to assess the feasibility of teleconsultations, evaluate technical issues, the satisfaction, perception and emotions of health-workers.

Methods: This is a prospective monocentric study and data was collected from the 23/03/2020 until the 11/05/2020 in the Institute of Pediatric Hematology and Oncology (Lyon, France). Using the Auvergne-Rhone-Alpes health cooperation group platform, we realized synchronous, video-based consultations between health-workers and pediatric patients under 18 years, treated for blood or solid malignancies or blood benign diseases. Every health-worker fulfilled an online, pseudo-anonymized questionnaire including technical aspects and data concerning satisfaction, perception and emotions (items from QLQC30 and PMH scale) following the consultation.

Results: We collected 140 questionnaires. With a median length of 18 minutes (range:4-65), only 33% percent of consultations was perceived as urgent according to the health-worker. Reasons for consultation were medical announcements (9%), follow-up cares (67%), delivery chemotherapy at home (19%) and psychiatric interview (5%). We only included already diagnosed patients and 90 % had a teleconsultation for the first time. Health-workers same that delivered less consultations faced technical issues (p-value=0.03). The need of physical examination was expressed by 42% of the health-workers. A considerable pattern of emotions is expressed after the teleconsultation: the health-worker feels calm (88%), serene (81%), worried (37%), angry

(50%) or nervous (50%). Sixty percent of the physicians was globally satisfied of the video-consultation.

Conclusions: Teleconsultations are a healthy legacy of the pandemic and could find a complementary place to in-person visits for selected circumstances.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0439 / #1167 | Cognitive Impairment and Family Functioning of Survivors of Pediatric Cancer: A Systematic Review

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Background and Aims: The relationship between treatment-induced cognitive deficits and family function is not well understood among survivors of childhood cancer. The purpose of this systematic review was to: 1) describe the characteristics of studies, study populations, domains assessed, and study quality; and 2) evaluate the associations between family factors and cognitive impairment in survivors of pediatric cancer.

Methods: Studies evaluating associations of cognitive impairment and family factors among survivors of pediatric cancer were identified by searching Pubmed, MEDLINE, Embase, and PsychINFO. Independent raters identified eligible studies, rated study quality, and extracted data on study characteristics and results which were then summarized based on a qualitative synthesis.

Results: The search yielded 1617 unique publications, of which 32 studies met our inclusion criteria. The majority of studies (76%) focused on family sociodemographic characteristics, such as socioeconomic status (SES), in relation to cognitive outcomes, including indicators of global cognitive function, specific cognitive impairments, or academic performance. A substantial proportion of studies (55%) evaluated at least one aspect of individual, dyadic, or global family functioning in relation to cognitive impairment. Most, but not all, studies demonstrated a link between lower SES or worse family functioning and cognitive impairment in survivors. Few family-based interventions to improve cognitive outcomes were identified.

Conclusions: Even though studies investigating the links between cognitive impairment and family functioning are increasing, critical gaps remain and should be addressed by future research. There is a need for longitudinal studies, investigations integrating more than one aspect of cognitive and/or family functioning, and more nuanced conceptualizations of socio-demographic risk factors in subsequent research to

delineate how these differences influence the cognitive impact of cancer treatment.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0440 / #1212 | The Development of CAR-T Road Trip: An Educational Board Game for Children Undergoing CAR T-Cell Therapy

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Background and Aims: Providing anticipatory guidance and psychoeducation is a psychosocial standard of care in pediatric oncology. Chimeric antigen receptor (CAR) T-cell therapy is a promising treatment option for pediatric cancer, providing hope for patients, but may be associated with severe toxicities. Psychoeducation can increase knowledge, address expectations, and prepare children and families for the treatment process and uncertain outcomes. Developmentally appropriate resources are limited for children undergoing CAR T-cell therapy. An educational board game (CAR-T Road Trip) was developed to improve knowledge, preparation, and coping in an interactive, non-threatening way.

Methods: An interdisciplinary group of medical and psychosocial providers working with youth receiving CAR T-cell therapy collaborated on the game design and essential components using an iterative process. Considerations included elements of cellular therapy, reading level, ability to sanitize materials, and fit on a bedside table.

Results: Players choose a toy car game piece and use a spinner to travel through six sections of the "road", representing different treatment stages: Preparation, Collection, Infusion, Cytokine Storm, Home Stretch, Post CAR Evaluations. Question cards corresponding to each section allow the provider to assess the child's understanding of treatment and address knowledge gaps. Feeling cards allow children to express thoughts and concerns, and stop signs create set points where the child is prompted to ask questions. The first player to navigate their car through each section and cross the finish line "wins". The game is designed to be played with the child, a provider, and family member prior to leukapheresis.

Conclusions: Initial testing supports CAR-T Road Trip as an enjoyable, informative psychoeducational intervention to help prepare children for CAR T-therapy and allow expression of feelings and worries. Next steps include feasibility testing with additional children and soliciting

provider feedback. Adaptation of the game for other treatment centers will follow.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0441 / #1293 | Implementation of Evidence-Based Psychosocial Care in Six Pediatric Oncology Units Across Pakistan

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Background and Aims: Diagnosis and treatment of childhood cancer can be a vastly traumatic experience for patients and their families, particularly cases facing debilitating procedures. The scope for psychosocial care in patients and parents is based on a variety of ethical, social, and individual needs and not limited to psychotherapy. Social support and focus on mental health is a concept not commonly integrated with healthcare in Pakistan, a dismal situation affecting prognosis negatively even in cases where there is access to treatment. With these needs in mind, the Indus Hospital & Health Network initiated a psycho-social department for pediatrics offering disease-related counselling, preparation for procedures and interventions, end of life care services and psychological assessment for patients and parents.

Methods: Through the My Child Matters grant awarded to IHHN, development of psychosocial services was encouraged at 6 different pediatric oncology units through collaboration and training sessions. Of these, 3 Pediatric Oncology Units hired psychologists to offer standardized psychosocial care to patients and their families. Online training courses were then initiated for all healthcare workers in contact with pediatric oncology patients at these centers with focus on holistic care.

Results: A total of 18 healthcare workers were certified through the online course, with passing marks in pretests and posttests, exhibiting enhanced understanding of grief counselling and breaking bad news post course. Active participation of doctors and nurses was noted from centers where psychologists were not yet inducted. In centers where dedicated psychologists were present the rate of patients who Left During Treatment, significantly decreased from 2018 to 2020.

Conclusions: Psychosocial services are an essential stakeholder in the treatment journey of childhood cancer. Along with establishment of counselling services for patients and families, focused training target-

ing understanding and management of supportive care and mental health needs to be implemented in all healthcare workers involved in the patient experience.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0442 / #1344 | Enhancing Impact and Efficiency of Financial Support for Cancer Through Use of AI: A Nascent Initiative of Indian Cancer Society

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Background and Aims: Since 2011, Indian Cancer Society-Cancer Cure Fund(ICS-CCF), established through donations of mutual fund dividends, has funded \$23 million in treatment costs for 9800 under-privileged patients. For governance, a Due Diligence Team(DDT) of cancer experts from academic tertiary centers reviewed every planned treatment and authorized covering the treatment costs for evidence-based care. Since expert time is scarce, in 2019, ICS-CCF introduced NAVYA AI(AI) system to scale the prior authorization operation. AI is a clinically validated system that matches clinical data with evidence and expert recommended treatment options. The system can adapt to institutional guidelines such as ICS-CCF authorization criteria on published survival rates. In this prospective study, we explore application of AI in prior authorization, by assessing concordance with DDT expert decisions and potential time savings.

Methods: Clinical data of beneficiaries were input in AI and if the output treatment options matched the planned treatment, the application was authorized, otherwise rejected. If AI did not output a decision, the application was deferred to DDT. Concordance between AI and DDT decisions, and DDT time spent on deferred applications, were measured.

Results: From April 22nd, 2020 to February 3rd, 2021, 411 beneficiary applications with age ≤ 18 yrs were simultaneously reviewed by AI and DDT. AI "authorized" (334/411) or "rejected" (0/411), 81.3% (334/411) of applications; and "deferred" 18.7% (77/411) of applications. AI and DDT were 99.7% (333/334) concordant to authorize a beneficiary application. DDT spent an average of 3.6 minutes/deferred application. On average, in 40 weekly batched reviews, AI authorized or rejected 8 applications/week, potentially saving 28.8 minutes/week (8 applications * 3.6 minutes/application in DDT review) of experts' time.

Conclusions: A clinically validated AI system can assess majority of beneficiary applications with authorize or reject decisions, 99% concordant with cancer experts, promising a five-fold experts' time-save.

Subsequently, ICS-CCF is awarding financial support directly based on AI decisions of prior authorization.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0443 / #1456 | Development and Pilot Implementation of an Activity-Based Emotional Support Intervention for Caregivers of Children with Cancer

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Background and Aims: This study aimed to pilot PAX ("Play, Act & Interact"), an activity-based emotional support intervention for caregivers of pediatric cancer patients, which focuses on addressing their psychological distress and post-traumatic stress symptoms (PTSSs).

Methods: Sixteen mothers whose children were pediatric cancer patients participated in this 4-week intervention. Their children (n = 16; 14 males; median age at diagnosis = 124 months; median amount of time from diagnosis = 9 months) were at different treatment stages for a range of different diagnoses. Caregivers completed self-report instruments assessing their psychological distress including PTSSs and family functioning before and after the intervention and a brief open-response exit survey. Paired sample t-tests were computed to compare the pre and post-intervention scores.

Results: The Post-traumatic Stress Disorder Checklist scores significantly decreased from pre- (M=37.00, SD=14.75) to post-intervention (M=32.56, SD=15.52), $t(15)=4.25$, $p<.001$. There was also a significant difference between pre- (M=33.5, SD=3.18) and post-intervention (M=35.7, SD=3.14) scores on the Family Adherence subscale of the Family Adaptability and Cohesion Evaluation Scales III, $t(15)=-2.58$, $p=.02$.

Conclusions: PAX was a promising intervention for supporting caregivers' PTSSs and family adaptability. Future studies investigating the long-term effects and replicating the current study with more participants and a control group are needed.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0444 / #1485 | Concordance of Caregiver Proxy and Patient Self-Reports of Health-Related Quality of Life at End of Cancer Treatment Psychosocial Screening

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Background and Aims: Recent pediatric oncology studies suggest that caregiver and patient reports of health-related quality of life (HRQL) during treatment may not be concordant, especially for psychological stress (Mack et al., 2020). Using data from an end of cancer treatment psychosocial screening program, we evaluated (1) concordance between caregiver proxy- and patient self-reports of HRQL, and (2) association of caregiver characteristics with concordance.

Methods: Following end of treatment, families were approached in outpatient clinic to review an educational handbook and complete psychosocial screening surveys on an iPad. Assessment included the PRISM cognitive screen, BRIEF-2, PROMIS25, and the PedsQL Family Impact Module. For each measure the clinical cutoff was used to classify results as resilient (no needs identified) or at-risk (needs identified). This analysis focuses on a subset of 57 patient-caregiver dyads with complete data. Concordance was calculated as percent of pairs whose results were in agreement for resilient or at-risk. Chi-square analyses were used to evaluate the association of caregiver characteristics with concordance.

Results: Patients were 51% leukemia/lymphoma diagnosis, 51% male, and average age of 15 years. By domain, overall concordance and resilient-specific concordance were: pain=86% (of which 86% resilient), depression=82% (75% resilient), anxiety=77% (68% resilient), mobility=75% (58% resilient), fatigue=74% (65% resilient), social=72% (65% resilient), overall HRQL=75% (35% resilient), and cognitive concerns=47% (33% resilient). Chi-square analyses revealed associations between caregiver HRQL and concordance on anxiety ($p=.019$), depression ($p=.001$), and pain ($p=.019$) as well as family adjustment to cancer and concordance on anxiety ($p=.049$), depression ($p=.000$), fatigue ($p=.035$), and pain ($p=.042$).

Conclusions: Two of the most concordant reports were psychological symptoms of anxiety and depression, particularly for resilient caregivers. This is inverse of Mack et al. and other findings that generally report psychological symptoms as the least concordant. Varied concordance across measures highlights the importance of collecting patient self-reports whenever possible.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0445 / #1574 | The Effectiveness of a Medical Clowning Program on Improving Emotional Status Among Hospitalized Children Undergoing Cancer Treatment: A Quasi-Experimental Study

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Background and Aims: Children with cancer spend a long time in hospitals while receiving cancer treatment, which can last from months to years. The substantial burden and severity of the health conditions causes them to have a higher level of emotional distress compared with their peers. Specific strategies are needed to help these children cope

with their emotional distress. Our research aimed to examine the effectiveness of medical clowning on improving positive emotions among hospitalized children aged 4 to 18 undergoing cancer treatment, and to analyze whether age moderates this effect.

Methods: This quasi-experimental research study with nonequivalent control group design was conducted at pediatric oncology wards in Taiwan. Participants were recruited from June 2018 through April 2020. To evaluate the average efficacy of medical clowning, we adapted generalized estimating equation models with the binary coded variable Positive Emotion as the primary outcome.

Results: Changes in the probabilities of positive emotion were significantly different across groups (51.84% for the experimental group, 15.76% for the control group; $\Delta = 36.08$, $P = .001$), and the change was more than 2 times larger for the experimental group (effect ratio = 3.28, $P < .05$) than for the control group. When evaluating the moderating effect of age on the intervention, none of the coefficients reached the significant ($P < .05$) level, suggesting that age may not moderate the intervention effect.

Conclusions: Our findings clearly support the benefit of medical clowning on enhancing positive emotion among hospitalized children undergoing cancer treatment. Younger and older children benefited equally. This both reflects the importance of medical clowning to enhancing emotional well-being and breaks the stereotype that clowning works only for younger children. A well-established long-term relationship along with routinely clowning can substantially enhance positive emotion, demonstrating the value of medical clowns in child-friendly health care.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0446 / #1603 | Benefit Finding in Children with Advanced Cancer and their Families

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Background and Aims: Although childhood cancer causes significant stress, most survivors are resilient and do not exhibit severe or lasting psychopathology. Research is limited, but some survivors may even report benefit finding or positive outcomes following this stressful life event. However, considerably less research has included families of children who are unlikely to survive their illness. Thus, this study investigated benefit finding among parents and their children with advanced cancer, as well as associated demographic and medical factors.

Methods: Families ($n = 57$) of children with advanced cancer (ages 5-25) were recruited from a large pediatric hospital. Advanced cancer was defined as relapsed or refractory disease, a prognosis of $<60\%$, or

referral to end of life care. Participants completed a demographic survey and the Benefit Finding Scale at enrollment.

Results: Children, mothers, and fathers reported moderate to high benefit finding scores. Correlations between family members were weak and non-significant. Children reported significantly higher benefit finding than fathers. Demographic and medical factors were not associated with benefit finding in children, mothers, or fathers.

Conclusions: Families of children with advanced cancer reported moderate to high benefit finding regardless of background or medical factors. Children also identified benefits of their cancer experience independent of the experiences of their mothers and fathers. Larger studies should continue to examine factors associated with both positive and negative outcomes in the context of childhood cancer to inform more effective interventions.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0447 / #1715 | Why Do Patients Discontinue Treatment? An Analysis of Factors Leading to Treatment Abandonment

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Background and Aims: Childhood cancer is still considered a dreaded disease with many myths and stigma surrounding it. Cancer treatment is difficult and costly. Not all patients are able to complete the treatment and many discontinue the treatment for various reasons. The aim of the study was to analyze the reasons leading to discontinuation of treatment at a referral center in North India

Methods: Retrospective study on the profile of pediatric cancer patients at a tertiary care center who discontinued the treatment and took discharge against medical advice during the period of 48 months (01 April 2017 to 31 March 2021). The institute is a public sector hospital with funding from both state and central government for cancer treatment

Results: Out of 154 new pediatric cancer patients registered and advised treatment, 23 discontinued treatment during this period. Male: Female 2.8:1 Age range 0-5: 16, 5-10: 3 and 10-18:4 patients. Causes for discontinuation: poverty 11 (even though treatment was financially supported 100%, parents had difficulty in staying and continuing treatment in a different city with loss of daily income), lack of family support: 7, alternative treatment: 4. 3 were terminally ill and wanted end of life care at home. 3 patients did not cite any cause for the discharge

Conclusions: Malignancy is associated with catastrophic consequences for the patient and their family. Social and financial factors play major role as the cause for discontinuation of the treatment

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0448 / #204 | Psychological Distress in Parents of Children and Adolescents with Cancer During Sars-Cov-2 Pandemic: A Single-Center Experience

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Background and Aims: According to the Global Registry of COVID-19 in childhood cancer from SIOP, more than 1,600 patients with childhood cancer have experienced SARS-CoV-2 infection. There is scarce data about psychological distress in parents of children and adolescents with cancer or survivors. We aimed to assess disruptions and worries in parents during COVID-19 pandemic.

Methods: We performed an online 18-question survey about experiences, attitudes and behaviors of parents during the first two outbreaks of COVID-19 in Spain. The survey took place between 5th December 2020 and 4th January 2021.

Results: Of the 177 parents that we contacted, only 52 completed the survey (65% female). Children diagnosis were hematologic malignancy (63.5%), CNS tumor (21.2%), and solid tumor (15.4%). 32.7% of children were receiving anticancer treatment, while 67.3% had already finished their treatment (15.4% were long-term survivors). Only 9.6% of children had been infected by SARS-CoV-2. 92.3% of parents were worried about their children being infected with SARS-CoV-2, and 76.9% thought their children might develop severe disease because of immunosuppression. 82.8% of parents showed concerns about the risk of their children becoming infected in a hospital visit, but only 48.1% thought staff might be the source of contagion. 86.6% of parents said they actively monitored COVID-19 symptoms in their children and siblings. 55.9% of parents thought they had enough information about COVID-19 in children with cancer. Strikingly, only 44% had received this information from health staff. During COVID-19 pandemic, 48.1% had seen their appointments in hospital rescheduled, and 57% of parents were worried about delay in diagnosis of relapse as a consequence of the pandemic. 55.7% of parents were worried about the psychological impact of isolation in their children and siblings. 19.2% of parents had quitted their jobs and did not take siblings to school in order to avoid contagion to their children.

Conclusions: Parents of children with cancer showed concerns and life modifications during COVID-19 pandemic.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0449 / #283 | The Benefits of A Narrative, Strengths-Based Approach to Distress Screening in the Community for Parents of Paediatric Oncology Patients

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Background and Aims: Distress is an expected and understandable part of the paediatric cancer journey, through diagnosis, treatment and survivorship. Despite initially experiencing high levels of distress, many families are competent and will adjust and cope long-term (Kazak et al, 2007). However, unresolved emotional distress (low mood, anxiety, trauma and social isolation) may lead to worse psychosocial outcomes for young people and family members, and thus needs to be screened for regularly, throughout the cancer trajectory. The therapeutic relationships families build with community family support coordinators (FSCs) working with the Child Cancer Foundation of NZ can facilitate screening, knowledge and information sharing, as well as access to the appropriate services and people that young people and their families may need to support them through their cancer journey.

Methods: Through a collaboration between consumers, health professionals, FSCs and researcher clinicians, we designed a measure to capture wellbeing of parents in the community. Level of distress, current stressors, as well as current strengths and coping strategies, were incorporated into the 'Wellbeing Check-In'. This tool provides a way to have conversations that normalises distress, whilst identifying those in need of additional services. We aimed to explore the reported benefits for FSCs in using this check-in tool and screening measure, through the use of focus groups interviews.

Results: Using a collaborative research approach enabled the measure to be founded on principles important to the FSCs: strengths and enhancing and respecting resilience; having a narrative approach that is culturally appropriate and ensures a holistic model of wellbeing was considered when screening for distress. It was also important to provide useful information about common emotional responses to their child being diagnosed with cancer, as well psychosocial strategies that parents could utilise themselves, to support their wellbeing.

Conclusions: Narrative distress screening, using community FSCs, is an innovative and novel evolution in distress screening.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0450 / #362 | Ethical Challenges of Managing Suicidality of Bereaved Parents within Pediatric Institutions

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Background and Aims: Despite the increase in long-term survival rates for childhood cancer, some cancer diagnoses are associated with very poor prognoses, and roughly 15% of children die from their disease. Bereaved parents are at risk for suicidal ideation and completion. Although professional ethics and national standards recommend care for parents, there are ethical challenges associated with managing parent suicidality within a pediatric institution.

Methods: We describe two cases illustrating ethical challenges of responding to suicide risk in parents experiencing bereavement within pediatric institutions. The first case describes the mother of a toddler with leukemia hospitalized for end of life care who expressed a plan to kill herself when her child died. The second case is the father of a teen with progressive osteosarcoma who died at home on hospice. Father had amassed firearms and indicated that their family would be imminently reunited. Lack of systematic psychosocial support in both cases resulted in several ethical concerns.

Results: We reference the American Psychological Association's Ethical Principles of Psychologists and Code of Conduct and apply an ethical framework for providing psychological services to parents in pediatric settings (Andrews, 2021) to these two cases. We describe the ethical challenges of: Conflicting interests of parent and child, multiple relationships, documentation and record keeping, and informed consent. Other ethical concerns include provider competence, concern for financial burden to family, and challenges involving agencies outside of a pediatric institution when managing suicide risk of grieving parents.

Conclusions: We provide best practice recommendations regarding ethical management of suicide risk in parents experiencing bereavement within a pediatric institution. We discuss implications for clinical practice, research, and organizational changes/policies, as well as potential solutions to improve assessment and management of parental suicidality.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0451 / #489 | Assessing Barriers and Facilitators to Implementation of the Klik Pain Monitor App

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Background and Aims: The KLIK Pain Monitor app was developed to improve pain management of children with cancer in the home setting, by providing educational information about pain (management) and real-time feedback on reported pain scores (0-10) from healthcare professionals (HCP's). This paper describes the results of a project to study barriers and facilitators (B&F) for future implementation of the app, guided by the Knowledge-to-Action (KTA) Cycle.

Methods: The KTA Cycle is a planned action framework, designed to enhance implementation efforts. One phase of the cycle calls for systematic assessment of B&F to use of an innovation. As part of a feasibility study of the app, B&F for implementation were assessed by conducting standardized interviews and questionnaires with families and HCP's who used the app.

Results: Twenty-seven families (M age child = 7.3 years, 51.8% male) and 6 HCP's who participated in the feasibility study were interviewed and completed evaluative questionnaires. Barriers most often mentioned by families related to specific functionalities of the app (i.e. daily reminders not received consistently, the wish for an overview of previously reported pain scores) and unclear instructions on when app could be used (not during hospitalization). Family app use was facilitated by user friendliness of the app and perceptions it improved care for patients. HCP's most often mentioned time consumption and increased workload as barriers, and improved care for patients as facilitators. Based on the identified B&F, we were able to optimize the app to the specific context and wishes of end-users.

Conclusions: This patient-centered care approach should contribute to successful implementation in the future. Consecutive studies will closely monitor effects of the specific adaptations to implementation and use of the app in children at the Princess Máxima Center for Pediatric Oncology, and hopefully in a broader range of children in the future.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0452 / #518 | Psychological Adjustment of French Adolescents and Young Adults at the Beginning of Cancer Treatment: Initial Stage of a Longitudinal Study

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Background and Aims: The recognition of the unique psycho-social, developmental and medical needs of adolescents and young adults (AYA) with cancer has led to the creation of dedicated units and/or teams in some French hospitals. This study aims to follow the evolution over time of the psychological adjustment and adaptation to cancer in AYA patients hospitalized in two comprehensive cancer centers, Institut Curie, Paris and IHope, Lyon.

Methods: Every 3 months for 18 months, participants will complete online questionnaires investigating anxiety (STAI-Y), depression (BDI), social support (SSQ6) and coping strategies (Brief COPE).

Results: Forty-three patients ($N_{\text{men}}=21$; $N_{\text{women}}=22$) between 15 and 27 years old ($M_{\text{age}}=19.9$) and diagnosed with all types of cancer or cancer recurrence, have been recruited. Questionnaires of the initial stage of the study were completed within 2 months after diagnosis. At the beginning of treatment, descriptive statistics demonstrate a slight depression score ($M=10.7$, $SD=7.0$), a mean score of 45.7 ($SD=5.2$) for trait anxiety and the main use of positive thinking ($M=5.81$, $SD=1.86$) and acceptance ($M=6.84$, $SD=1.34$) as coping strategy. Spearman's correlations significantly demonstrate that depression score correlates with being a woman ($p<0.01$) and coping mechanism. Score of depression is positively correlate with venting ($p<0.05$), denial ($p<0.01$), and behavioral disengagement ($p<0.01$), and is negatively correlate with acceptance ($p<0.01$). Student tests demonstrate that AYA with slight depression score or over, tend to use behavioral disengagement ($t=-2.10$, $p<0.01$) compared to AYA without depression. On the other hand, AYA without depression are more likely to use acceptance ($t=4.45$, $p<0.05$). About trait anxiety and social support, no significant results were found.

Conclusions: Those preliminary results at the beginning of cancer treatment will be deepened by conducting a longitudinal methodology. This research will improve the understanding of psychological adjustment over time and thus allow family and health professionals to better adjust to AYA cancer patients' needs.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0453 / #542 | Online Cognitive-Behavioral Group Intervention for Young Adult Survivors of Childhood Cancer: A Pilot Study

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Background and Aims: Young adult childhood cancer survivors (YACCS) are a vulnerable group in need of psychosocial support, but tailored interventions are lacking. This study examined feasibility and preliminary effectiveness of an online group intervention (On Track Online for YACCS) aimed at teaching active coping skills and providing peer-contact, thereby reducing and preventing psychosocial problems in YACCS. The intervention is based on psycho-education, cognitive behavioral therapy, and aspects of acceptance and commitment therapy.

Methods: YACCS completed questionnaires pre- and post-intervention. Feasibility was based on attendance, drop-out, and an evaluation questionnaire (satisfaction). Preliminary effectiveness was evaluated with the Distress Thermometer, Mastery Scale (self-efficacy), Illness Cognition Questionnaire, Impact of Cancer - Childhood Survivors, and Pediatric Quality of Life. Preliminary effectiveness was investigated by testing differences on the psychosocial outcomes between T1 and T0 within respondents, using paired samples t tests and Cohen's d.

Results: 10 YACCS participated in the intervention and completed all questionnaires. There was no drop-out; 90% of participants attended at least five out of six sessions. Overall satisfaction with the intervention was high; 7.6 on a 0-10 scale. Participants' distress (Distress Thermometer, $\text{Mean}_{T0}=5.1$, $\text{Mean}_{T1}=3.7$, $p=.030$, Cohen's $d = -.6$) and feelings of helplessness (ICQ, $\text{Mean}_{T0}=10.4$, $\text{Mean}_{T1}=8.7$, $p=.001$, Cohen's $d = -.8$) reduced significantly from T0 to T1, while their self-efficacy (Mastery Scale, $\text{Mean}_{T0}=20.1$, $\text{Mean}_{T1}=22.8$, $p=.013$, Cohen's $d = .8$) improved.

Conclusions: On Track Online for YACCS is a feasible intervention, filling a gap in psychosocial services for YACCS. This pilot indicated preliminary effectiveness in reducing distress and feelings of helplessness, while improving self-efficacy, which is in line with previous modules for other populations with the On Track program. This intervention fills a gap in psychosocial services for YACCS during survivorship care. Points of improvement for the intervention include reducing the number of participants per group, and adding one or more sessions.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)

PV0454 / #676 | Satisfaction with Fertility Preservation Decisions Among Adolescent Males with Cancer: A Mixed Methods Study

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Background and Aims: Half of male childhood cancer survivors experience treatment-related fertility impairment, which can lead to distress. Access to fertility preservation (FP) has increased, yet sperm banking remains underutilized before treatment. Decisional dissatisfaction is linked to worse mental and physical health in long-term survivorship, and cancer survivors often regret missed opportunities for FP. However, little is known about short-term FP decisional satisfaction. This mixed-methods study examined FP decisional satisfaction among families of male adolescents one month after cancer diagnosis.

Methods: Data were part of a larger study examining FP decision-making among families of adolescent males newly diagnosed with cancer. Thirty-nine families completed the Brief Subjective Decision Quality measure one month after diagnosis. Decisional satisfaction was compared for participants (mothers, fathers, adolescents) who did and did not attempt to bank. Semi-structured interviews included the following question: How do you/your family feel about the banking decision now/in the future? Transcripts were analyzed for thematic content.

Results: Decisional quality scores were moderate-high ($M=5.74-6.33$ out of 7) with no significant differences between non-banking ($n=15$) and banking ($n=24$) families (adolescents: $p=0.83$, $d=0.08$; mothers: $p=0.18$, $d=0.45$; fathers: $p=0.32$, $d=0.44$). Three qualitative themes emerged among non-banking families: (1) Satisfaction with decision (50% of participants), (2) Acceptance of decision (60%), (3) Potential for future regret (40%). The only theme noted among banking families was satisfaction with the decision (93%).

Conclusions: Quantitatively, short-term decisional satisfaction was high regardless of banking decision. However, qualitative findings suggest experiences of families who did not bank may be more nuanced, as several participants discussed a potential for future regret. Tools are needed to better capture these variations in decisional satisfaction. Further, longitudinal research and clinical assessments at different ages/developmental stages are important to identify changes in satisfaction, acceptance, and regret, and manage any fertility-related distress that may occur across the treatment continuum.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0455 / #766 | **Assessment of Psychological Distress Among Parents of Children with Cancer**

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Background and Aims: The diagnosis of pediatric cancer is a potentially traumatic event and a major source of psychological distress for parents. The aim of our study was to determine the prevalence and the risk factors of psychological distress among parents of children diagnosed with cancer.

Methods: This is a descriptive study carried out on parents of children diagnosed with cancer and treated in the department of medical oncology at the Habib Bourguiba Hospital in Tunisia. Over a period of four months, From the beginning of December 2020 until the end of March 2021, forty three parents (34 mothers and 9 fathers) were assessed, using psychometric scales of anxiety, depression and post-traumatic stress according respectively to the Hamilton anxiety rating scale (HAM), the Beck Depression Inventory (BDI) and the Impact of Event Scale (IES).

Results: Sixteen percent of the parents had scores indicating acute stress and 21% had scores indicating post-traumatic stress. Eighty six percent of the parents had mild to severe depression and 95 % had minor to major anxiety. Post-traumatic stress and anxiety were significantly correlated with the female sex of parents. Significant correlations were also found between post-traumatic stress scores and symptoms of depression and anxiety. However, our study did not show a significant correlation between the other factors studied and the parents' psychological distress.

Conclusions: The prevalence of anxiety and depressive disorders was high among parents of children diagnosed with cancer and a significant correlation has been found between stress and anxiety with the female sex of parents. Our results were similar to several studies although other risk factors for psychological distress in parents of children with cancer have been described in the literature.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0456 / #799 | **Joy or Dread? Factors that Continue to Impact A Mother's Emotional Wellbeing Once Her Child's Treatment for Cancer has Completed: Implications for Emotional Support**

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Background and Aims: This research explores factors that may continue to impact a mother's emotional well-being once her child's

treatment for cancer has completed. Research indicates that, contrary to a general expectation of experiencing joy as treatment ends, some families experience very mixed emotions, with fear playing a large part, both leading up to treatment completion and, for some, continuing post-treatment. However, there is little literature that explores a mother's emotional well-being after a number of years' post-treatment. This research is a contribution toward addressing that deficit.

Methods: Five mothers were interviewed using semi-structured questions to gather data relating to their specific lived experiences relating to their feelings towards their child's treatment completion and their feelings post-treatment. The interviews were transcribed verbatim and analysed using Interpretative Phenomenological Analysis. This method is concerned with personal perception rather than with the formation of objective statements and seeks an insider perspective. Superordinate and subordinate themes are presented.

Results: Indicate that although end of treatment was longed for, there continues to be much uncertainty and fear post-treatment, and this can continue years after treatment has ended. Mothers described changes within themselves (e.g. new attitudes to living) and a need to adapt to a new normal. There were elements of grief for the loss of family life with which they were once so familiar. There was also a strong sense of wanting to support others, so that their own experiences were not wasted. All participants recognized that further emotional support for themselves would be beneficial.

Conclusions: Although there is a longing for treatment to finish, life after childhood cancer is not guaranteed to be joyful. There are clearly mixed feelings that include elements of dread, with the whole family often being affected years after treatment has ended. Emotional support offered post-treatment is suggested to be helpful.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0457 / #965 | Analysis of Drawings of the Human Figure in Children with Leukemia During Treatment: A Comparison with a Control Group of Healthy Peers

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Background and Aims: The drawing was identified as a good communication tool to bring out the experiences of the cancer patient and of the child in general (Rollins, 2005, Bianchi Di Castelbianco et al., 2007). The aims of this study are to assess the IQ and emotional difficulties of pediatric patients with leukemia during the Maintenance phase of treatment adopting drawings tests.

Methods: Participants were 25 children aged from 6 to 10 years old (Mean age in years = 7.62; SD = 1.45), of which 13 females and 12 males. In particular, 4 were diagnosed with Acute Myeloid Leukemia (AML), and 21 with Acute Lymphoblastic Leukemia (ALL). This study also included the presence of a control group made up of 25 healthy children (Mean age in years = 7.64; SD = 1.51), matched by age and gender to the patients in the experimental group. After the informed consent signature of parents, Human Drawing Test (Polacek & Carli, 1977) and DAP: SPED (Naglieri, McNeish & Bardos, 1991) were adopted to assess respectively IQ and emotional difficulties. A very good inter-rater agreements were obtained from two independent judges that scored the drawings (Spearman's rho attested between 0.95 to 0.99).

Results: Paired samples t tests were run and the results showed that children with leukemia scored significantly higher (Mean = 54, SD = 14.1) than healthy peers (Mean = 47.6; SD = 7.1) on the emotional assessment test ($t = 2.0$; $p = 0.04$), while in the IQ assessment the results didn't show relevant differences ($t = -1.20$; $p = 0.23$). No age or gender differences were reported in the clinic group. Days of hospitalizations didn't be associated with the emotional difficulties.

Conclusions: These empirical data could suggest the emotional vulnerability of the pediatric patients and psychological monitoring and interventions should be implemented.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0458 / #967 | Screening and Strengthening Activities of Fine Motor Skills in Pediatric Patients Under Treatment for Oncohematological Diseases

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Background and Aims: Difficulty in fine motor dexterity and visual-motor skills could be possible in children with oncohematological diseases. This study aims to: 1. Assess the fine-motor and visuo-spatial skills in pediatric patients; 2. Evaluate the possible efficacy of a strengthening intervention to promote the acquisition of manual dexterity, strength in the hands and to enhance grapho-motor skills.

Methods: Participants were 53 children with leukemia, 26 girls and 27 boys, with an average age of 5.86 years (SD = 2.04) recruited at the Pediatric Hematology, Oncology and Stem Cell Transplant Center (University of Padua). They are Caucasian (75.5%) and all non-Caucasian speaking Italian. Children were assessed with the manual dexterity scale from the MABC-2 and with VMI pre and post-intervention and

also healthy peers were assessed adopting the same instruments. A part of them was been enhanced (N=9), while another part no (N=9).

Results: Paired samples t-tests showed significant differences comparing clinical and control groups in the VMI standardized scores ($t=-4.58$, $p=0.0001$), with the clinic group with a worse performance ($M=108.73$; $SD=16.01$) than the healthy one ($M=123.75$, $SD=16.62$). Hierarchical regression model was run to identify the possible predictors on the standard Movement scores ($R^2 = 0.38$, $F(39) = 7.55$, $p = 0.01$) with gender ($\beta = 0.38$, $p = 0.022$), ethnicity ($\beta = -41$, $p = 0.004$), and the presence / absence of siblings ($\beta = 0.37$, $p = 0.01$) that influenced standard Movement scores. A repeated measures ANOVA test was performed to investigate the efficacy of the strengthening activities on Motor skills. Results showed that the effect of time with the presence of strengthening was significant for Movement ($F=6.31$, $p=0.02$, $np^2=0.33$; $b=0.64$) and VMI scores ($F=22.83$, $p<0.001$, $np^2=0.65$; $b=0.99$) unlike the time effect alone didn't impact significantly.

Conclusions: Specific motor psycho-educative programs should be implemented for the pediatric patients more at risk.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PV0459 / #970 | Efficacy of Strengthening Activities of Emotional Skills in Pediatric Patients Under Treatment for Oncohematological Diseases

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Background and Aims: Illness and hospitalization could compromise the acquisition of emotional competence that could induce coping problems. This study aims to: 1. Assess children's emotional competence during the acute phase of treatment and the possible factors impacting on it and 2. Evaluate the possible efficacy on child's emotional abilities adopting an empowerment training.

Methods: Participants were 44 children with leukemia (20 girls and 24 boys), with an average age of 6.37 years ($SD = 2.2$), recruited at the Pediatric Hematology, Oncology and Stem Cell Transplant Center (University of Padua). They are Caucasian (72.7%) and non-Caucasian Italian speaking (27.3%). Children were assessed with the NEPSY-II theory of mind (TOM) and the Affective Recognition (AR) tests. These pediatric patients were matched by age and gender with healthy peers evaluated with the same tests.

Results: Paired samples t-tests showed significant differences comparing clinical and control groups in the standardized scores ($t=-4.58$,

$p=0.0001$), with the clinic group with a worse performance ($M = 108.73$; $SD=16.01$) than the healthy one ($M=123.75$, $SD=16.62$). Comparing with standardized norms, 13.6% of these patients attested at a sub-clinical level for TOM and 13.6% for AR, while 18.2% attested at a clinical level for TOM and 29.5% for AR. The non-Caucasian group had a worse emotional competence than the Caucasian group in AR ($t= 2.34$; $p=0.024$) and in TOM ($t=3.75$; $p=0.001$). Pre-school children obtained best scores than the older ones both in TOM ($t=2.64$; $p=0.012$) and in AR ($t=3.15$; $p=0.003$) tasks. A repeated measures ANOVA test was performed to investigate the efficacy of the strengthening activities on Emotive skills. Results showed a significance for the effect of time with the presence of training for improving AR ($F=9.53$, $p=0.01$, $np^2=0.51$) unlike the time effect alone didn't show significance.

Conclusions: Specific emotive training should be implemented for more at risk children for their emotional competence.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fa Nursing - Education*

PV0460 / #1140 | Experience of a Nurse from a Lower-Middle-Income Country Tackling Advanced Education: An International Masters Program

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Background and Aims: Nurses in Pakistan have limited options for graduate education (masters or PhD). This is one country where there are fewer nurses than physicians according to World Bank statistics. Nurses are not encouraged to conduct nursing research or even join physician research project resulting in a dearth of evidence-based nursing practice. An experienced Pakistani pediatric oncology nurse is the first nurse admitted to Masters in Global Child Health offered by St. Jude Children's Research Hospital Graduate School of Biomedical Sciences fall 2020.

Methods: Learned about the program through an email from St. Jude Children's Research Hospital, which is an attractive program for candidates who wish to pursue higher education in the field of Pediatrics with focus on Pediatric Oncology. Applied for the program and got selected after critical review of application, personal statement, recommendation letters and interview.

Results: Getting in this program proved that Nurses can be accepted alongside Physicians for higher studies. Online Masters provide new experiences and challenges with a great opportunity to learn from best international faculties. Core curriculum includes important modules on epidemiology, Biostatistics, Foundation of Global child health, Research, health systems, health economics and leadership. Minimum six hours are needed weekly to complete online lectures, reading material, discussions and assignments. Each cohort has students from various healthcare disciplines from LMIC with passion for Pediatrics and Pediatric Oncology care. Diversity in the group help us learn and understand the similar challenges faced in health care systems and devise strategies to overcome them.

Conclusions: My participation in this program demonstrates that nurses are up to the task of true multidisciplinary collaboration in advanced education, clinical practice and research addressing nursing challenges, accomplishments and potential in health care. Nurses should step up in every aspect of health care as a partner highlighting the important role they play in improving the outcomes.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.f.a Nursing - Education*

PV0461 / #1786 | Progress Report: Implementation of a Multi-Site Pediatric Oncology Nursing Train-The-Trainer Course in Eurasian Hospitals

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Background and Aims: In many Eurasian countries, pediatric oncology nursing is not yet a recognized subspecialty and education opportunities are limited. In 2018, pediatric oncology nursing education was prioritized by Eurasian Alliance in Pediatric Oncology (EurADO) members. An adaptable regional nursing train-the-trainer course was designed based on a needs assessment, including modules: 1) pediatric cancer/treatment overview, 2) specific pediatric cancers, 3) chemotherapy, 4) oncologic emergencies and 5) vascular access

management. Ten sites implemented Module 1 in November 2019-December 2020. The course was placed on hold due to COVID-19 and adaptations were made before reinitiating in February 2021. Course adaptations, lessons learned, and progress to date will be presented.

Methods: Utilizing a train-the-trainer approach, the course consists of three components: 1) instructional theory and teach-back; 2) provision/review of course content and teaching instructors; and 3) instructors adapting, translating (as needed), and teaching each module to nurses in Eurasian hospitals. COVID-19 adaptations included adding a virtual educational delivery option for instructors and transitioning to Qualtrics electronic pre-/post-test and evaluation forms. Languages included English, Russian, Romanian, Ukrainian and Uzbek.

Results: Module 1 findings were presented at SIOP 2020. Five of ten sites have implemented Module 2 with additional site engagement planned. Of those, 194 nurses achieved a 79% mean pre-test score; post-test scores will be presented. Despite the pandemic, most instructors opted to teach in-person as opposed to virtually. Data quality has improved with electronic forms, although some instructors preferred using paper forms and manually entering electronic data. Barriers include competing clinical priorities and time. Course progress and findings to date will be presented.

Conclusions: Although COVID-19 caused a midway delay in course implementation, Eurasian instructors resumed participation one year into the pandemic, emphasizing the importance of this course despite a global pandemic and competing clinical priorities. Ongoing course adaptation, implementation, and assessment is planned.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.f.a Nursing - Education*

PV0462 / #198 | Nursing Intervention Protocol to Improve Chemotherapy Induced Neutropenic Complications among Children with Cancer: A Cluster RCT

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Background and Aims: Chemotherapy induced neutropenia (CIN), is a significant problem among cancer patients receiving chemotherapy, it may lead to life-threatening infections that may quickly lead to sepsis, and death. Severe neutropenia and febrile neutropenia are major causes of morbidity, treatment interruptions and dose reductions in patients undergoing chemotherapy. There is need to develop evidence based nursing protocols to provide safe and effective care for patients receiving chemotherapy. **Aims:** To assess the effectiveness of "Nursing Intervention Protocol" regarding nurse's and caregiver's knowledge and practices for management of CIN complications patients receiving chemotherapy.

Methods: A randomized controlled research design was utilized in the present study. A total enumeration sample of 51 nurses and 105 patients and their caregivers were enrolled in the present study. Out of them 27 nurses and 51 each patients and their caregivers were enrolled in control group from Radiotherapy Department of PGIMER, Chandigarh and 24 nurses and 54 each patients and their caregivers were enrolled in experimental group from pediatric hematology oncology. Data collection instruments included; demographic and work profile of nurses and caregivers, nurses and caregivers CIN knowledge questionnaires; clinical profile of patients; neutropenia guideline observation checklist for evaluating nurses and caregivers practice compliance with designed protocols related to hand washing, oral care, IV line care and sitz bath.

Results: Mean scores for nurses and caregivers post protocol knowledge as well as practice level were significantly higher than the pre protocol scores.

Conclusions: Designing and implementing evidence based nursing intervention protocol is important for improvement of nurse's and caregiver's knowledge and practice so that patient care is improved.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fb Nursing - Research*

PV0463 / #1086 | The Experience of Handling Oral Chemotherapy at Home for Caregivers of Pediatric Patients with Acute Leukemia at Children's Hospital 2 in Vietnam

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Background and Aims: Caregivers at Children's Hospital 2 in Ho Chi Minh City, Vietnam who have a child with acute leukemia must give oral chemotherapy at home. Caregivers are instructed how to handle drug at home by nurses and physicians. Patients who take chemotherapy at home may use doses of ¼ or ½ pill depending on the patient's weight. There are no studies or surveys in Vietnam and no policies or procedures at Children's Hospital 2 about the safety recommendations for using oral chemotherapy at home. Based on the toxicity guidelines of chemotherapy and manufacture's cautions, the nurses at our hospital were concerned about the safety of caregivers. **Aims:** The purpose of this paper is to describe the experience of handing oral chemotherapy at home for caregivers of pediatric patients with acute leukemia from June, 01st to July 31st, 2020.

Methods: Caregiver survey

Results: Eighty-seven caregivers were surveyed. 100% caregivers received instruction on safe handling of oral chemotherapy at home with nurses providing instruction 25% of the time. Mothers of 84% patients prepared medications of which 15% were pregnant. Medication was prepared in the family room and the bedroom.

Approximately three-quarters of the children needed medication divided into smaller doses with 87% using tablet crushers. Safeguards when handling chemotherapy were not used by 80% of the caregivers. 69% disposed of chemotherapy in the regular garbage; however, 77% of caregivers saved extra medication for future use. Of note, 60% of caregivers always washed their hands after handling chemotherapy.

Conclusions: Although 100% of caregivers received instruction on safe handling of oral chemotherapy; however, according to APHON guidelines, it is not enough and some practices are not safe for caregivers and community. Counseling should be improved to ensure safety for caregivers, pediatric patients and community. Currently, we are developing educational leaflets for caregivers

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fb Nursing - Research*

PV0464 / #1706 | Effect of Mentorship on New Graduate Nurses' Assertive Communication: An Evidence Review

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Background and Aims: • 10% of admitted patients are exposed to a safety event • Nurses have a "duty to act to prevent harm." • Inexperienced nurses require a supportive framework in which to develop professional confidence In new graduate nurses, how does participating in a mentorship program compared to no mentorship affect assertive communication during the first year of practice?

Methods: • Cumulative Index to Nursing and Allied Health Literature, MEDLINE, PubMed, Nursing and Allied Health, and Google Scholar databases were searched using the key words newly qualified nurses, newly graduated nurses, new nurses, novice nurses, mentor, mentorship, mentoring, assertive communication, assertiveness, speaking up, confidence, empowerment, work empowerment, structural empowerment, informal power, organizational empowerment, and moral courage. • Inclusion criteria were peer-reviewed articles written in English published between 2015 and 2020. • Exclusion criteria were books; dissertations; magazines; trade journals; articles in a language other than English; and studies pertaining to tool development, nurse retention only, or populations other than new graduate nurses. • John Hopkins Evidence Based Appraisal Tools were used to guide the appraisal of evidence.

Results: • The strength of the evidence is low • Despite the limited and weak nature of the evidence, the evidence is consistent, and the proposed question can be answered: → More than on strategy, including a supportive environment, should be used. → New graduate nurses must have physical and emotional access to a trusted mentor or preceptor in order for communication skills to develop.

Conclusions: • Leadership should create supportive environments. → Ensure new graduate nurses' voices are met with action and respectful responses. → Provide time for mentors and new graduate nurses to

meet to allow the growth of these relationships. • A mentorship program should be used in conjunction with another strategy: → A nurse residency program → Peer support → Opportunities to gain experience

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fb Nursing - Research*

PV0465 / #737 | Experiences of Parents Caring for Children with Cancer during the Covid-19 Pandemic

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Background and Aims: The coronavirus disease 2019 (COVID-19) pandemic caused the collapse of medical services in Japan and other parts of the world. Several people with non-COVID-19 related health problems could not get routine treatment during the pandemic, including children with cancer was no exception in that time. Here, we report on illustrative cases of children receiving cancer treatment and their families' recovery from adversities during this period.

Methods: Data were collected through semi-structured interviews involving the children's parents and analyzed using qualitative inductive methods after the children were discharged from the hospital. Individual statements and expressions were recorded in analyzing the data, and significant content and categories were compared. The local ethics committee approved this study.

Results: At the time of diagnosis, parents reported feeling overwhelmed and remorse for their past consultation behavior. Parents relied on both health care workers and their peers for emotional support. Throughout the pandemic, parents thought that children were secure in the hospital and that they could get rest being away from their child. But the social distance caused muscle weakness in the children, and the parents felt isolated. Social distancing significantly affected heavily both parents and children emotionally.

Conclusions: In Japan, cancer treatment for children is commonly done in the hospital, not outpatient wards, except for maintenance therapy. Hospital visits were limited to one hour during the pandemic. The parents could not spend time with their children sufficiently, which was eliciting a positive response from them. However, physical distancing had a negative psychological impact on both children and parents. It is also necessary for medical professionals to assess the parents' health simultaneously in order to provide care for both the child and the family. This work was supported by JSPS KAKENHI Grant Number 17K12374.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fc Nursing - Quality Improvement/Practice Project*

PV0466 / #1219 | Socially Distant Nursing Care: Creating A Remote Role for The Office Practice Nurse

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Background and Aims: The office practice nurse is a well-defined role at Memorial Sloan Kettering Cancer Center (MSKCC). The office practice nurse works collaboratively with the physician(s), advanced practice nurse(s), and office staff. The office practice nurse practices at the expert nurse's level because her role must include knowledge of the disease process, treatment, refined assessment skills, and knowledge of health care systems. In the setting of a global pandemic, this role became vital to keep patients connected and enabled them to continue to receive care at MSKCC.

Methods: In March 2020, COVID-19 was declared a global pandemic by the World Health Organization. New York City was an epicenter, and immediate plans went into effect to ensure patient safety and staff members at MSKCC. Through creativity and flexibility, the pediatric neuroblastoma office practice nurse's role moved to a remote role. Technology provided maximum communication between the interdisciplinary team taking care of medically complex patients in the clinic.

Results: With resiliency and flexibility, the role has been successful in the home setting. An office practice nurse's presence has allowed for better social distancing in the ambulatory care center while still enhancing patient care. Remotely, the office practice nurse has been able to triage patient phone calls, coordinate care with outside institutions, and provide patient education on a virtual platform. With a myriad of technological advances, including video conferencing, and e-mails, the office practice nurse has been able to remain connected to both the interdisciplinary team and patients all over the country and world.

Conclusions: Although nursing is a hands-on role, in the event of a global pandemic, with creativity, the OPN role relocated off-site without jeopardizing its integrity. The lessons learned during the pandemic can be used to use in the future to continue to provide high-quality patient care in a virtual setting.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy*

PV0467 / #1110 | Memory-Like NK Cells: A Novel Therapy for Pediatric Sarcoma

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Background and Aims: Sarcomas are heterogeneous and clinically challenging cancers that represent 12-15% of pediatric tumors. Current treatments are resulting in modest outcomes for patients with sarcoma. Therefore, innovative therapies are crucial to increase survival and improve the quality of life of these children and young adults. Natural Killer (NK) cells recognize and eliminate malignant cells without the need of prior sensitization. However, their short life span, limited persistence and the immunosuppressive tumor microenvironment (TME) dampen NK-based immunotherapy in solid tumors, making a requirement for ground-breaking approaches to ameliorate NK cell anti-tumor functionality.

Cytokine-induced memory-like (CIML) NK cells have been recently described as briefly pre-activated NK cells that exhibit potent anti-cancer responses. Moreover, several studies reported that ML-NK cells display long-term survival and in vivo persistence, and resistance to the immunosuppressive TME. The aims of this work were characterizing in depth CIML-NK cells and exploring their potential cytotoxic activity against sarcoma cells.

Methods: To generate CIML-NK cells, highly purified NK cells were obtained from healthy donors and pre-activated with cytokine cocktail. Cells were maintained 6 days for differentiation. Their cytotoxicity, degranulation capacity and cytokine production against sarcoma cells were evaluated by flow cytometry. Also, differential expression of CIML-NK receptors was characterized.

Results: CIML-NK cells exhibited a phenotype characterized by low CD16 expression and high CD25, CD57, NKG2D and DNAM-1 among other markers. Co-cultured with sarcoma cells, CIML-NK showed an increased IFN γ production but non-significant differences were found in CD107a expression as compared to controls. Four-hours cytotoxicity assays did not unveil significant differences when comparing CIML- and control-NK cells.

Conclusions: Our results strongly suggest that CIML-NK cells acquire potent anti-tumor capabilities against sarcoma cells, manifested by their enhanced IFN γ production. Otherwise, long-term cytotoxicity experiments with additional biological samples need to be considered. Altogether, our findings will raise CIML-NK cells as a promising step forward in sarcoma treatment.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0468 / #1290 | Clinical Utility of Targeted Panel Sequencing in Comparison to Whole-Exome Sequencing by A Computational System in Pediatric Tumors

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Background and Aims: The utility of extensive molecular profiling like whole-exome sequencing (WES) over smaller targeted next-generation sequencing (NGS) gene panels of pediatric tumors in routine clinical practice is a matter of debate. The aim of this study was to evaluate the performance of different sequencing panels by virtual gene panel analysis of WES results with a computational system using the digital drug-assignment (DDA) algorithm.

Methods: Between 2017 and 2020, 103 patients with high-risk pediatric cancer (<21 years) were involved in the precision oncology program of our clinical practice. 88 tumor samples were fit for WES analysis. Molecular profiles were processed by the DDA system, ranking molecular alterations by their importance. WES results were filtered by a 995-gene virtual panel selected based on literature search. Sequencing results were also virtually filtered for smaller, commercially available NGS gene panels. Results were evaluated by a molecular tumor board (MTB) to provide therapy recommendations.

Results: In comparison to the results of the virtual 995-gene WES analysis, DDA identified the most important actionable driver also in >90% of cases if the results were filtered for the commercially available 400-800-gene panels. DDA detected the strongest drivers also in >80% cases after filtering for virtual panels of 200-400 genes. But gene panels below 200 genes contained the most relevant driver only in less than 50% of cases, which would also have affected therapy selection by the MTB.

Conclusions: According to these results, DDA can identify the most important driver alterations in most commercially available diagnostic panels above 200 genes. Dynamic, real-time update of the virtual gene panel used for WES analysis based on growing scientific literature and increasing number of registered therapies will most likely continuously widen the advantage of WES for precision oncology with DDA in pediatric tumors.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0469 / #1315 | Evidence of Tumor Immunoediting in Pediatric Cancer Subtypes Through In-Silico Prediction of MHC-I Neoantigen Presentation

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Background and Aims: MHC-I genotype influences immunoediting of adult cancers with well-presented mutations preferentially lost during tumor development and poorly presentable oncogenic mutations retained across multiple tumor types (Marty et al., 2017, Cell). Somatic loss of B2M/HLA has been implicated as a mechanism of immunoediting. Immune evasion has also been associated with poorer outcomes in both adult and pediatric tumors. Here we sought to determine whether MHC-I genotype also affects the mutational landscape of pediatric cancer.

Methods: Pediatric oncology patients ($N_{\text{total}} = 149$; 32 acute leukemia, 46 glioma/non-glioma of CNS, 21 soft-tissue sarcomas, 12 bone sarcoma, 43 other) from a single center Next-Generation Sequencing (NGS)-based tumor profile with FoundationOne arrays between August 2011 and May 2020 were retrospectively analysed. Somatic mutations were evaluated for effective patient MHC-I presentation using patient harmonic-mean best-rank (PHBR) scores. Overall survival, relapse-free time, and tumor type specific analyses were conducted, comparing mutations with high vs. low PHBR scores.

Results: Clonal mutations were better presented than subclonal mutations ($p = 0.006$), largely driven by acute leukemia patients. Patients with B2M alterations accumulated mutations that should otherwise be presented by patient-specific HLA alleles ($p = 0.0004$). Patients with poorer outcomes (died within 1 year of relapse) had poorer presentation of mutations compared to patients with more favorable outcomes, though this did not reach statistical significance ($p = 0.06$).

Conclusions: Analysis of presentable mutations in B2M-mutated patients and presentable clonal mutations in this cohort point to evidence of immune surveillance in pediatric tumors, with immunoediting resulting in more invisible mutations overrepresented in patient tumors, acting as a potential barrier to immunotherapy. However we did not find that patient overall survival and relapse-free time was significantly associated with neoantigen mutation burden.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0470 / #1317 | Natural Killer Cells Dually Targeting Soft-Tissue Sarcoma and Its Inhibitory Microenvironment Exhibit Enhanced Anti-Tumor Activity

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Background and Aims: Outcomes for children with refractory or relapsed soft-tissue sarcoma are poor. Immunotherapy with chimeric antigen receptor (CAR)-expressing lymphocytes has shown promise. However, efficacy has been limited by lack of expansion within the tumor microenvironment (TME) and by off-tumor toxicity due to sarcoma antigens shared by normal tissues. We have developed natural killer (NK) cells that simultaneously target inhibitory cells of the TME and a sarcoma-associated antigen, MUC18. By using a "co-stimulation only" CAR to target MUC18, we hypothesized that dual-targeted NK cells would be activated only within the TME co-expressing MUC18 and TME-associated ligands, but not in MUC18+ normal tissues.

Methods: We generated a MUC18-directed CAR and confirmed specificity and functionality using MUC18 overexpressing and knockout targets. To define an optimal co-stimulatory(cs) endodomain in NK cells within the TME, MUC18 csCARs with 4-1BB, OX40, 2B4, and DNAM-1 endodomains were screened for expansion and cytotoxicity in a TME culture system comprised of alveolar rhabdomyosarcoma, Rh4, and inhibitory macrophages (M2s). Killing and proliferation of NK cells co-expressing MUC18.csCAR and an NKG2D-based cytotoxic CAR targeting Rh4 and M2s were assessed in TME cultures and against MUC18+ normal tissue.

Results: MUC18.OX40.csCAR NK exhibited increased proliferation in the TME (25-fold vs. unmodified NK cells) but not in MUC18+ normal tissue (1.1-fold vs. unmodified NK) without additional cytotoxicity, confirming OX40 as an optimal co-stimulatory domain. Dual-targeted NK cells demonstrated enhanced tumor killing in the TME (7.5-fold tumor decrease vs. 2.1-fold decrease by unmodified NK vs. 1.4 and 2.1-fold decrease by NK expressing either CAR alone).

Conclusions: We present a novel modular approach to combinatorial antigen recognition in pediatric sarcoma that integrates signals from two different sources (tumor cells and TME) to activate NK cells only in the correct environment. This NK cell approach has potential to improve the efficacy and safety of cell therapy for pediatric soft-tissue sarcomas.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0471 / #1318 | Ewing Sarcoma-Derived Extracellular Vesicles Impair Dendritic Cell Maturation and Function

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Background and Aims: Ewing Sarcoma (EwS) is characterized by an immunosuppressive tumor microenvironment and a scant T cell infiltrate. However, the mechanisms of immune evasion remain undetermined. Extracellular vesicles (EV) play a major role in the crosstalk between tumor and microenvironment. Therefore, we investigated whether EwS-derived EV interfere with the interface between the innate and adaptive immune system by impairing the maturation and function of dendritic cells.

Methods: EV were isolated by differential centrifugation from supernatants of EwS and MRC5 fibroblast cell lines. EV were quantified by nanoparticle tracking analysis, and characterized by western blot and flow cytometry (FCM). The effect of $0.75 - 3.0 \times 10^9$ EV particles on the maturation of 1×10^6 CD14+ monocytic-derived dendritic cells (moDC)/ml in vitro was determined by FCM and RNA sequencing. The stimulatory function of EV-treated moDC was assessed by T cell proliferation and IFN γ release. Peripheral blood mononuclear cells (PBMCs) of patients with EwS were characterized by FCM and RNA sequencing, and compared to healthy, age-matched children.

Results: EV preparations exhibited small EV characteristics, including size (100-170nm) and protein markers CD63, CD81 and tsg101. EwS EV, but not MRC5 EV induced a dose dependent pro-inflammatory cytokine release (Pearson's r IL-6 20,02; TNF α 6,24), and inhibited the upregulation of maturation markers CD86 ($p < 0.05$) and HLA DR ($p < 0.01$) on moDC. EwS EV-treated moDC significantly suppressed the proliferation of CD8+ ($p < 0.01$) and CD4+ ($p < 0.001$) T cells and reduced the IFN γ release (MRC5 EVs 28.2; TC32 EVs 22.3 ng/ml; $p < 0.05$). Concurrent with pronounced inhibition of CD4+ vs. CD8+ T cell proliferation, the frequency of CD4+ T cells among PBMCs was reduced in patients at diagnosis ($p < 0.05$) while CD8+ T cells were normal.

Conclusions: EwS cell line-derived EV induce a pro-inflammatory response and inhibit the maturation and stimulatory function of moDC. Thereby, EwS EV may contribute to the immunosuppressive microenvironment in EwS.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0472 / #1361 | Characteristics of Infections Identified from Implementation of the Golden Hour Protocol during the Covid-19 Pandemic in a Third Level Hospital in Mexico

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Background and Aims: Cancer is one of the leading causes of death in children worldwide and treatment-related infections play an important role in the mortality rate. In countries with middle and low-income, deaths secondary to infection are reported approximately 10 times more frequently compared to higher-income countries. The etiology for infections varies from region to region, even within the same country. For this reason, it is necessary to identify and develop hospital strategies for timely detection and treatment of infections in pediatric cancer patients. The Golden Hour (GH) is a strategy for timely administration of antibiotics in pediatric oncology patients with fever and proper management of the infections identified that our hospital has been implementing for more than two years with the support from St. Jude Children's Research Hospital.

Methods: Cross-sectional study of patients with infections identified through the GH protocol (GHP) from April 2019 to March 2021. For research purposes, 2 periods were compared: April 2019- March 2020, April 2020-March 2021 pandemic period. Descriptive statistics were used to determine differences between study periods and the most common pathogenic microorganism.

Results: Of 1,017 admissions to pediatric oncology, 8.36% activated the GHP with neutropenia febrile. 60% had bacteremia and the most frequently isolated bacterium was E. coli. Hospital stay days decreased by 23.4% from the first period to the pandemic period. Septic shock decreased by 50% from first to second period. No deaths from infections associated to febrile neutropenia in GHP were reported.

Conclusions: Implementing the GHP at our institution led to a reduction in hospital days, septic shock, and deaths due to infection, despite the COVID-19 pandemic. The most frequent bacterium isolated was E. coli in GHP.

E-Poster Topic: Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy

PV0473 / #1372 | Repurposing "Old" Blood Tests to Replace Costly Interim Analysis Imaging of Hodgkin Lymphoma in Low and Middle-Income Settings

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Background and Aims: PET-CT interim assessment is considered gold standard in the risk- and response-adapted management of Hodgkin lymphoma (HL), but this modality is often inaccessible in the low- and middle-income (LMIC) setting. The objective of this study was to investigate the correlation of low-cost, readily available blood test results with changes in PET-CT findings after two cycles of ABVD in children and adolescents with HL.

Methods: Levels of ferritin, lactate dehydrogenase (LDH), erythrocyte sedimentation rate (ESR), albumin, white cell count (WCC), absolute lymphocyte count (ALC), absolute total eosinophil count (AEC) were compared with Deauville scores on PET-CT at interim assessment of pediatric patients with HL. Deauville 1-3 denoted rapid early response (RER) while Deauville 4-5 denoted slow early response (SER). Missing values were imputed using the k-nearest neighbour algorithm. The baseline and follow-up blood test values were combined into a single difference variable. Data were split into training and testing sets for analysis using Python scikit-learn 0.23.1 with logistic regression, random forests, neural net, naïve Bayes, and support vector machine classifiers.

Results: Random forest analysis gave a test validated accuracy of 76.6% when predicting RER or SER from blood samples. The predictive accuracy of the optimal model applied to the full dataset was 85.7% with ROC AUC 82.1%. Differences in ferritin and LDH were the most predictive variables, together contributing more than half of the discriminative performance; differences in ESR, albumin, WCC and AEC each contributed 7–11%; differences in haemoglobin and ALC had little value. Other classifiers performed less well than random forests.

Conclusions: Ferritin and LDH may be used to substitute for PET-CT if unavailable in resource-constrained settings. These exciting preliminary results continue to be validated to predict RER versus SER, validity of this model in low, medium and high-risk patients, and the need for consolidation RT in resource-constraint settings.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy*

PV0474 / #1373 | **Analysis of Cell-Free DNA in the Plasma of Patients with Non-Resectable, Progressive or Relapsed Pediatric Malignancies**

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Background and Aims: Cell-free DNA (cfDNA) released by tumor cells provides the opportunity to improve diagnosis and treatment using minimally invasive approaches to molecularly survey clinically informative biomarkers and 'actionable' targets. The 'Individualized Therapy for Relapsed Malignancies in Childhood' (INFORM) registry aims to identify actionable therapeutic targets on a personalized basis. Relapsed or refractory high-risk tumors as well as selected rare entities, for which no further standard of care therapy is available, are characterized by state-of-the-art next-generation sequencing (NGS) technologies. Our study aims at integrating liquid biopsies into the diagnostic INFORM pipeline and pave the way for clinical translation of these technologies.

Methods: cfDNA for various downstream analyses is derived from plasma samples and isolated according to optimized protocols. Depending on the yield of cfDNA, data availability for the primary tumor, and tumor entity, we have performed low-coverage whole-genome sequencing (lcWGS), whole exome sequencing (WES) and targeted gene panel sequencing. Results are being compared with data obtained from standard tumor biopsies and normal control tissues following adjustment of established bioinformatics pipelines to suit liquid biopsy-derived data.

Results: Sensitivity and specificity of our approach is presented as a proof-of-concept study to compare the performance of different NGS platforms and their utility for analyzing cfDNA samples. We demonstrate robust tumor detection by lcWGS (n=115), mutation tracking by targeted approaches (n=74) and the advantages of the combinatorial use of various approaches (n=64).

Conclusions: We have demonstrated the feasibility of liquid biopsy analyses in advanced pediatric malignancies. We highlight the informative values of the liquid biopsy methods applied in terms of tumor detection, assessment of tumor heterogeneity, and identification of actionable targets. Our study is expected to guide and accelerate the implementation of liquid biopsies into prospective pediatric clinical trials.

E-Poster Topic: *Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy*

PV0475 / #1578 | A Current Review of Completed and Ongoing Car-T Cell Trials in Paediatric Solid Tumours

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Background and Aims: Despite treatment advances, cancer remains the leading cause of death in children. The major contributing factor being patients with relapsed/refractory solid tumours who continue to have poor outcomes. An urgent need for novel therapies in these patients exists. Although the use of chimeric antigen receptor (CAR)-T cell therapies is an emerging treatment for solid tumour malignancies, pooled data specific to treatment of paediatric patients in this setting is limited.

Methods: We performed a literature review to identify recently completed and currently ongoing clinical trials administering CAR-T cell therapy to patients <18 years of age for intra and extra-cranial solid tumour malignancies. Reports from October 2010 to October 2020 were identified using MEDLINE and Embase on the Ovid platform, and additional trials were identified using the Cochrane Central Register of Controlled Trials (CENTRAL), ClinicalTrials.gov, and the European Union Drug Regulating Authorities Clinical Trials (EUDRACT) database in addition to meeting abstract repositories.

Results: 30 trials were identified, 4 completed, 26 ongoing (active and recruiting or active but no longer recruiting). 1 trial tested first, 13 second, 4 third and 6 fourth generation CARs. Disease groups included: neuroblastoma (13), brain tumours (9), solid tumours (8), sarcomas (4) and liver tumours (1). Among completed studies 2 each targeted HER2 and GD2. Among ongoing studies, 12 targeted GD2, 4 B7H3, 3 HER2, 2 each IL13Rα2, EGFR806, PSMA, and CD19, and 1 each targeted CD171, GPC3, and CD44v6. Cumulative patient recruitment to completed and ongoing trials was as follows: neuroblastoma 74, sarcoma 14, brain tumours 7. Objective response or stable disease was seen in 32 patients (34%).

Conclusions: CAR-T cells have shown progress in haematologic malignancies however there remain practical difficulties with their use in solid tumours. Maturing data from ongoing studies will provide objective evidence to support the use of CAR-T cells in paediatric solid tumour patients.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0476 / #1001 | Development of a Dose-Adjusted Polygenic Risk-Model for Anthracycline-Induced Cardiotoxicity

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Background and Aims: Anthracyclines, while effective chemotherapeutic-agents, cause cardiac dysfunction and congestive heart failure in 57% and 16% of patients, respectively. Cumulative anthracycline dose is a crucial predictor of cardiotoxicity; however, dose alone cannot explain all cardiotoxic events. Strongly associated pharmacogenetic variants in SLC28A3, UGT1A6, and RARG have previously been shown to contribute to anthracycline-induced cardiotoxicity in pediatric patients and may help identify those most susceptible. The aim of this study is to develop a prediction model for anthracycline-induced cardiotoxicity incorporating pharmacogenetic variants and cumulative dose, and to determine if genetic information can improve risk-predictions of this dose-dependent toxicity.

Methods: 1,075 anthracycline-treated children were genotyped, and cardiotoxicity cases were identified. SLC28A3, UGT1A6, and RARG variants were used to model dose-adjusted cardiotoxicity-risk by logistic regression. Model performance was assessed in a separate validation cohort (n=358). Additionally, a dose-stratified analysis was performed using the same modelling techniques and grouping patients by cumulative anthracycline exposure (150mg/m², 151 – 250mg/m², and >250mg/m²).

Results: After adjusting for cumulative anthracycline dose, carriers of SLC28A3 were less likely to develop cardiotoxicity (OR 0.65 [0.41, 1.0]), and carriers of UGT1A6*4 or RARG were more likely to develop toxicity (OR 2.94 [1.41, 5.79] and 1.83 [1.21, 2.72]). The risk-model incorporating both cumulative dose and genetic information was able to better discriminate cases of cardiotoxicity (AUC 0.763), compared with models including cumulative dose only (AUC 0.723), or genetics only (AUC 0.638). In the dose-stratified analysis, genetics markedly improved cardiotoxicity predictions only in high-dose patients.

Conclusions: Combining pharmacogenomics with cumulative dose results in a risk-model that better differentiates those most susceptible to anthracycline-induced cardiotoxicity, especially at high-doses.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0477 / #1008 | Paediatric Oncology in a Shared-Care Centre in Ghana – A Review of Cases Seen in The First Year of Operation

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Background and Aims: Ghana, a lower middle-income African country (estimated population: 31million), has only two comprehensive childhood cancer treatment centres located in Accra and Kumasi. To improve geographic access to childhood cancer care, a Shared-Care Centre (SCC) was established in November 2019 at Cape Coast Teaching Hospital (CCTH) linked to the treatment hub in Accra, for basic cancer diagnosis, supportive care and administration of selected chemotherapy. CCTH is located 143km from Accra, and has a catchment population of about 7million. The aim of the study was to describe the cases seen at CCTH in its first year as a SCC.

Methods: A retrospective review of electronic medical records at the CCTH SCC from November 2019 to October 2020.

Results: Forty-eight children diagnosed with cancer were seen during the period of which 26 (54%) were female. The median age was 4 years (IQR: 2.75-9.5years). The commonest diagnoses were Non-Hodgkin lymphoma (23%), acute leukaemia (21%), nephroblastoma (21%) and neuroblastoma (13%). Median duration of symptoms prior to presentation was 1 month (IQR: 0.75-3months); however, patients with solid tumours reported later than those with haemato-lymphoid malignancies (3.7months vs 1.7months; p-value 0.05). One patient absconded after confirmation of diagnosis while 9 (19%) were referred upfront to the comprehensive treatment centre in Accra. Nineteen children (40%) started chemotherapy at the SCC of which 10/19 (53%) - comprising 9 with nephroblastoma and 1 with Burkitt lymphoma - successfully completed treatment at the SCC, 4/19 (21%) abandoned treatment, 2/19 (11%) relapsed and 3/19 (16%) died. A significant number of children (19/48; 40%) had advanced disease at presentation and were not started on curative therapy, with over half of them (11/19; 58%) dying within a week of admission to the SCC.

Conclusions: SCCs improve access to paediatric oncology services. Increased community education and family support may prevent delayed presentation and abandonment.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0478 / #1027 | Building and Sustaining an Infection Prevention and Control (IPC) Program at a Pediatric Cancer Unit (PCU) in a Hospital in Ecuador through a Twinning Partnership

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Background and Aims: Hospitals IPC programs are often insufficient to meet the needs of PCUs in low resource settings. We assessed the

effectiveness of a twinning partnership to build and sustain a dedicated IPC team for a PCU in Hospital SOLCA, Quito in Ecuador.

Methods: We created a partnership between St Jude Global Infectious Diseases (Global ID) Program and Hospital SOLCA Quito to build a local IPC team expertise and skills. This team focused on surveillance of healthcare associated infections (HAI), staff training and practice auditing in IPC essential practices. Twinning had three periods. Decreasing rates of HAI were indicators for quality of care in POU. We used descriptive statistics.

Results: Each period had multiple actions. 1st period (2007 – 2009), Planning and Preparing: local expert identification, training, financial and technical support, local leadership buy-in of the twinning. 2nd period (2010 – 2012), Building and Nurturing the IPC Team: team members identification, training and job performance description, IPC of PCU needs assessment, creation and approval of policies and procedures, auditing and feedback of IPC practices (surveillance of HAI, hand hygiene, and safe vascular access, surgical site infection and related to urinary catheter). 3rd period (2013 – 2019), Sustaining IPC Team: IPC team members, policies and procedures, and functions of the IPC became institutionalized. Since creation of IPC team at Hospital SOLCA Quito, 16284 children with cancer had a total of 80789 hospital day and 333 HAI events. At baseline (2010), there was 9 HAI / 1000 patient days. At the end of 2019, last measure before the start of the COVID-19 pandemic, was 2.6 HAI / 1000 patient days.

Conclusions: We successfully decreased HAI in a POU by a twinning partnership. We sustained this model at our institution, and we hope to demonstrate its utility and fidelity by replicating such model in other PCUs in Ecuador.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0479 / #1032 | Development of a Quality Improvement Educational Intervention in Febrile Neutropenia in Pediatric Cancer Patients in Peru: Preliminary Results of the Dott Project

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Background and Aims: Severe infections are the most important causes of morbidity/mortality in children with cancer. In Peru, a limitation for optimal treatment of children with fever and neutropenia due to chemotherapy is the delay in the administration of antibiotics. We performed an intervention to decrease the time to antibiotics (TTA) in febrile and neutropenic patients presenting to the emergency room (ER), increasing the perception risk of neutropenic fever in the ER staff and the importance of a timely administration of antibiotics. This study forms part of the DoTT project, an educational intervention for health

providers, aligned to the WHO Global Initiative of Childhood Cancer in Peru.

Methods: This study was performed at a tertiary care national hospital, Edgardo Rebagliati. We included hemato-oncological patients younger than 14 years, who arrived at the Pediatric ER. We defined TTA by measuring the time between patient's arrival to the ER and the administration of the first dose of an antibiotic. We compared the TTA between thirteen patients admitted from 7/2020-12/2020 with fever and neutropenia (after intervention), and historical data from 7/2016-2/2017 (23 patients).

Results: Median age was 7 years. Nine patients had leukemia and 4 patients had solid tumors, receiving oncological treatment. Mean pre-hospital delay was 176 minutes (range, 14-906) and TTA was 133 minutes (range, 34-400). One patient died of sepsis. Age, sex, source and timing of antibiotics did not significantly affect hospital stay, antifungal use and/or antibiotic turnover. Based on our historical data, mean TTA was 206 minutes (range, 137-390). Early results indicate a decrease in the TTA, although not statistically significant, likely due to the sample size

Conclusions: The TTA exceeds the recommended time at an international level, causing the evitable morbimortality. It is necessary to perform a multidisciplinary intervention to improve antibiotic start time. Ongoing educational intervention refinement and testing of the instruments are planned.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0480 / #1043 | Impacto: A Novel Mobile Application to Reduce Treatment Abandonment in Pediatric Cancer Patients: A Pilot Usability Study

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Background and Aims: In Peru, of a total of 1800 new cases of cancer in children and adolescents are estimated annually, approximately 20% abandon treatment, representing the first cause of treatment failure. Based on the successful experience of other countries, a mobile application was designed aiming to follow-up cases to reduce treatment abandonment. This pilot study aims to examine the app's basic design and functions and to establish its acceptability and usability.

Methods: Thanks to the Association of Pediatric Hemato-Oncology of Peru (HOPPE) and the WHO Global Initiative for Childhood Cancer, a mobile application named IMPACTo (Integrated Model for the Prevention of Abandonment of Cancer Treatment) was designed, based on an action algorithm where 3 alert levels are activated (green, yellow and red) according to the time elapsed since the patient misses an appointment in consultation or chemotherapy until contact is reestablished with the institution in order to implement early interventions. Userst' experience was evaluated through a mixed-method approach.

Results: For the use of the mobile application, treatment abandonment working committees were formed in 2 institutions in the country that serve pediatric cancer patients (National Institute of Neoplastic Diseases and the National Institute of Child Health San Borja). Committees were constituted of a multidisciplinary team (pediatric onco-hematologist, nurse educator, psychologist and social worker) to daily monitoring cases and activating alerts through the app. Professionalst' feedback showed that most of them considered the app's operativity as satisfactory and easy to use.

Conclusions: This pilot study suggest that the app can easily be used and is acceptable by the pediatric onco-hematology multidisciplinary team at public hospitals in Peru. Given the professionals' feedback, the app could help to monitor the cases of risk of cancer treatment abandonment in children, as a potential key tool to implement early interventions.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0481 / #1053 | Understanding Adolescent and Young Adult Oncology Care in Latin America: Interviews with Pediatric and Adult Hematologists and Oncologists in Central America/Caribbean and Mexico

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Background and Aims: Most adolescents and young adults (AYA: ages 15-39) with cancer live in low- and middle- income countries. However, little is known about the barriers to care and resources needed for AYAs with cancer in these settings. We aimed to address this gap through interviews with pediatric and adult hematologists/oncologists in Central America/Caribbean and Mexico.

Methods: We enrolled a convenience sample of pediatric and adult hematologists/oncologists in Central America/Caribbean and Mexico who treat AYAs with cancer. Semi-structured interviews were conducted to better understand the barriers to care and opportunities to improve care for this population. Interviews were conducted in Spanish, recorded, translated and transcribed into English. A thematic analysis of the interviews was performed.

Results: The nineteen respondents were from 7 countries, provided >20 hours of interviews, and were primarily pediatric hematologist/oncologists (64%). Barriers to care included: (a) lack of a multidisciplinary team with focus on AYA patients, (b) discontinuity of care, (c) missing collaboration between adult and pediatric hematologists/oncologists, (d) lack of psychosocial resources, including mental health and fertility support, and dedicated facilities, (e) competing for resources with adult oncology, (f) abandonment of therapy, (g) delay in diagnosis, and (h) treatment costs. Potential opportunities for improvement included: (a) multidisciplinary teams with expertise in AYA oncology, (b) standardized treatment protocols, (c) AYA cancer registry, (d) dedicated space (both clinical and for AYA patients), (e) mental health support (e.g., psychologists, support groups), (f) educational and vocation assistance, (g) financial support, and (h) AYA cancer awareness campaign (for physicians and general public).

Conclusions: This study identifies barriers to care and potential opportunities to improve care for AYA patients in Central America/Caribbean and Mexico. There are clinical and psychosocial barriers that limit AYA focused care, as well as strategies to improve care, which can be examined in future studies.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0482 / #1060 | Adolescent and Young Adult Oncology Care in Mexico: A Survey of Pediatric Hematologists and Oncologists

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Background and Aims: Little is known about the current landscape of adolescent and young adult (AYA) cancer care in low-and middle-income countries, including in Latin America. To address this knowledge gap we studied where AYA patients are treated and barriers and facilitators to providing care for this distinct population in Mexico.

Methods: This cross-sectional study involved an online self-administered survey, conducted among pediatric hematologists/oncologists in Mexico. Descriptive statistics characterized the respondents and evaluated survey responses.

Results: We received 100 survey responses (29% response rate). Respondents primarily worked in public hospitals (86%), for >10 years (49%), with >50 new diagnoses per year (51%). Thirty-eight percent of physicians reported the maximum age treated at their institution was 18 years; with as high as 22 years (2%) and as low as 14 years (1%) reported. These age cut-offs were mostly dictated by hospital (45%), government (23%) and department/division (15%) policies. However, 75% of physicians reported that a patient could stay at their institution until treatment completion even if they surpassed the maximum age. Only 24% of physicians stated they would feel comfortable treating patients older than they currently do, with another 45% stating it would depend on the type of cancer. In providing care for AYA patients, most physicians (72%) reported collaborating with adult hematology/oncology colleagues. Of services needed to provide care for AYA patients, physicians reported that mental health, transportation, financial and social services would be the most helpful.

Conclusions: Exposure to AYA patients among pediatric hematologists/oncologists is potentially restricted due to age limits, but collaboration with adult colleagues to address care of this patient population is possible and currently performed. Additional prioritization of services such as mental health and specific inclusion of this patient population in national cancer control plans could begin to address the challenges associated with caring for AYA patients with cancer.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0483 / #108 | The Incidence, and Outcome of Childhood Cancer in Fiji From 2002-2016: Pre and Post Twinning

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Background and Aims: The general health of children in Fiji has improved, cancer has emerged as an important unmet health care need but Fiji has limited health care resources for diagnosis and treatment of childhood cancer. To support the development of cancer care for children, teams in Fiji and New Zealand established a Paediatric Oncology Twinning Program in 2010; using national criteria with treatment based on SIOP PODC resource adapted protocols, this Twinning Program provides direct clinical support for diagnosis and treatment of selected curable cancers. To evaluate the impact of child cancer program we sought to describe incidence of child cancer and outcome of treatment before and after the introduction of Twinning Program.

Methods: A retrospective study of the clinical records of all patients registered on three hospital-based registries between 2002 and 2016 were reviewed. Data fields collected included age, gender, pathological diagnosis, stage, treatment received, survival and cause of death.

Results: Over 15-year period study; total of 356 patients under 15 years were diagnosed with cancer, giving an age adjusted incidence of childhood cancer in Fiji of 70-115 per million. Most common single diagnosis was Acute Lymphoblastic leukemia, followed by CNS tumours. Period before introduction of adapted treatment protocols and Twinning Program the overall survival was less than 10% and survival for those receiving treatment was 30%. Since introduction of the Twinning program overall survival improved to 52%. However, small number of cases and short study interval since commencing twinning program, it is too early to evaluate outcomes of specific diagnoses pre- and post-twinning.

Conclusions: Incidence and diagnoses are typical of other LIMC settings. Introduction of adapted treatment protocols for specific treatable cancers supported by Twinning Program has led to increase in treatment availability and overall improvement in the outcome for children in Fiji. Incomplete and possibly inaccurate hospital-based registration highlights the need to introduce a mandated child cancer registry.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0484 / #1102 | Interruption of Treatment during Covid Pandemic and its Effect on Treatment Outcomes of Childhood Cancer Patients – A Report from Children Cancer Centre -Nich, Karachi, Pakistan

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Background and Aims: Pediatric oncology department at NICH is a model of public- private partnership where free treatment is offered to children. The first case of COVID-19 was reported in March 2020 at Karachi. With strict SOPs in place, cancer care services were continued at our department. However, many patients abandoned treatment. This study was performed to assess the impact of Covid-19 pandemic on treatment outcomes of children with cancer who abandoned treatment.

Methods: Record of all the patients receiving standardized chemotherapy protocols, who dropped out for more than four weeks between March 23, 2020 and June 22, 2020 was reviewed. Demographic and clinical details of patients were recorded in especially designed forms. Phone calls were made by department's social worker to track lost to follow up patients. Duration of treatment abandonment was recorded for those patients who came back for continuation of treatment. They were also re-evaluated to confirm disease remission before resuming further treatment. Patients' record was further analyzed at six months follow up to assess their final outcomes. Patients who lost to follow up during diagnostic workup were excluded from the study.

Results: Out of the total 293 on chemotherapy patients, 62(21.1%) children lost to follow up during the study period. 49 children were

male and majority 75.8% were not local residents. About half of the children (n=32) had Acute lymphoblastic leukemia. Of them, 42(89.3%) were on Maintenance phase of chemotherapy. 29/ 62 children reported back; of them 16 had relapsed disease on arrival. The duration of treatment abandonment ranged from 29 - 88 days. Thirteen children; who were in remission, were continued on treatment. At six month f/up, another 3 (23%) children relapsed.

Conclusions: COVID-19 has posed substantial disruptions to pediatric cancer management. There is a marked rise in the number of patients abandoning treatment altogether, hence compromising their overall outcomes

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0485 / #1115 | Deserve - An Advocacy Campaign for Right to Childhood with Cancer

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Background and Aims: The lengthy and complicated process of treatment of cancer obliges the children estimated to be diagnosed in 3,00,000 number globally every year to lose their rights during their treatment and even after. The campaign, I Deserve is planned in the time frame of 2016 to 2020 with an aim of collecting 3, 00,000 supporting signatures backing 1 patient by 1 signature and to handover to the stakeholders of the Government of India appealing for the National Plan for Childhood Cancer in India.

Methods: Advocated through various car rallies across the country Reached out the various sectors of the societies in Hospital and Care Centers society for advocacy and request for consent signatures Approached various school and collages with the awareness and advocacy programs and request for their consent signatures Developed online page with a form requesting consent signature on the website of CanKids Reached out the families of the children with cancer supported by Cankids

Results: Collected 33, 216 signatures from 9 car rallies Collected 50, 926 signatures from the hospitals and care centers from PAN India, 34, 028 from schools and colleges Collected 12, 198 from the Cankids websites and 1, 21, 856 from various social media Collected 371 pledges from the key stake holders of the societies PAN India During lockdown due to COVID Crisis in 2020, we reached to 23, 000 families of the children with cancer supported by Cankids and collected 48, 678 signatures Ultimately in October 2020, we hit the set target with 3, 01, 219 signatures.

Conclusions: With the group of Survivors advocates and parents representatives, we expect of meeting Minister of Health & Family Welfare, Govt. of India and Prime Minister by 2021 and handover more than 3, 00,000 collected pledges with an strongly advocate appealing for the National Plan for Childhood Cancer in India.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0486 / #1117 | Two Decades of Childhood Cancer Care in Cameroon: 2000–2020

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Background and Aims: The WHO global initiative for childhood cancer aims at a survival rate of 60% by 2030. The aim of this study was to investigate progress made with pediatric oncology care in Cameroon over the last twenty years to identify gaps to address for the achievement of the WHO goal nationally.

Methods: This was a review of published literature related to childhood cancer care in Cameroon from January 2000 to October 2020.

Results: Forty articles were analyzed. Themes identified included awareness (2), epidemiology (10), management and outcomes (18), supportive care (4), palliative care (5), advocacy and partnerships (4), and traditional and complementary medicine (1). There was little government investment in childhood cancer care. International twinning was established between Stellenbosch University, South Africa and the Cameroon Baptist Mission Hospital complex in 2003, joined by Leeds children's Hospital, UK in 2015, as well as the inclusion of the pediatric oncology unit in Yaounde in the Francophone Africa Pediatric Oncology Group in 2003 with support by World Child Cancer, Sanofi My Child Matters and Beryl Thyer Memorial Africa Trust. Low community awareness was mitigated by education targeting health care professionals and the general public. Collaboration with imaging, surgery and pathology units were essential for diagnosis. A nutrition support program was established while specialist palliative care was limited. The Cameroon Pediatric Oncology Group was created in 2020 with formal support from the Ministry of Health leading to inclusion of pediatric specific priority actions in the national cancer control strategic plan for 2020 – 2024.

Conclusions: Childhood cancer care delivery has progressed constantly in Cameroon over the last two decades with the assistance of twinning programs and non-governmental organizations' support. With improved government prioritization and investment, Cameroon may be enabled to meet the WHO GICC objective of 60% survival by 2030.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0487 / #112 | Risk Factors of Childhood Cancer in Armenia: Case-Control Study

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Background and Aims: Childhood cancer is a leading cause of death among children aged 0-19 years worldwide. Each year, 300,000 new cases of childhood cancer are diagnosed around the world. Given the differences between pediatric cancer incidence rates, types and trends, it is important to elicit country-specific risk factors of this disease. The aim of this study was to identify possible risk factors for childhood cancer in Armenia.

Methods: We used a case-control study design enrolling participants from the only specialized pediatric hematology and oncology center in Armenia. Cases included patients ≤ 14 years old diagnosed and treated with a malignant disease between 2017-2020 in the centre. Controls included patients treated in the same center during the same period but diagnosed with non-malignant disease. We conducted telephone interviews with mothers of the participants asking questions on family sociodemographic characteristics, parental demographics, child health, pregnancy-related factors and family environmental exposures. Both univariable and multivariable logistic regression analyses were utilized.

Results: Overall, 234 participants (117 cases, 117 controls) were included in the study. Based on multivariable analysis, maternal usage of folic acid during pregnancy was protective against childhood cancer by almost 50% (OR=0.54; 95% CI: 0.31-0.94). On the contrary, experiencing horrifying/terrifying event(s) during pregnancy (OR=2.19; 95% CI: 1.18-4.07) and having induced abortions before getting pregnant with the given child (OR=2.94; 95% CI: 1.45-5.96) were associated with higher odds for a child to develop cancer.

Conclusions: This study identified three important modifiable risk factors for childhood cancer in Armenia, all related to the period of pregnancy. The findings indicate the need for improved pregnancy care, including education of reproductive age women and their family members on the importance of stress reduction during pregnancy, avoidance of induced abortions via practising safe birth control methods, and use of folic acid.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0488 / #1147 | Profile of Childhood Cancer Cases Seen at The Paediatric Oncology Unit of a Tertiary Hospital in Ghana – A 10 Year Review

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Background and Aims: Childhood cancers though curable have inequalities in outcomes between low- and middle-income countries and high income countries. The WHO Global initiative on Childhood cancer (GICC) has a goal of increasing survival rate at least by 60% by 2030 while reducing suffering and improving quality of life for children with cancer globally. The main aim for this study was to assess the trends in yearly proportions and numbers of cases seen and evaluate the survival patterns and also to serve as a baseline as interventions are implemented in tandem with the WHO GICC.

Methods: A retrospective study was conducted to review secondary data from the paediatric cancer registry of all patients diagnosed with cancer from 2011-2020. Demographic and clinical data stored in excel was exported to Stata and analysed. Outcome data included death or alive.

Results: Total number of patients diagnosed were 1,094 with a Male: Female ratio of 1.4:1. The commonest cancers seen was lymphomas forming about 38.1% of the new diagnosis made. The incidence of lymphomas reduced from 62.8% in 2011 to 23.0% in 2020. Proportions for Acute leukaemias also increased from 11.3% in 2011 to 23.8% in 2020. Number of patients alive in reference to the year of diagnosis was lowest in 2016 with 29% of patients being alive. Subsequent years have higher survival with 53% of patients in 2019 still alive. Leukaemia treatment is expensive and unaffordable and translated to poor outcomes as patients tend to abandon treatment.

Conclusions: This baseline preliminary data showed a reducing trend in the incidence of lymphomas with an increase in the number of leukaemia patients seen but poor outcomes in patients with leukaemia. This calls for concerted effort with interventions to improve on the outcome and be able to reach the 60% survival goal for the WHO GICC.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0489 / #1171 | Assessment of Hematological Parameters at the Time of Diagnosis in Childhood Cancers

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Background and Aims: In our study, we aimed to evaluate hematological parameters such as leukocyte count, neutrophil count, lymphocyte count, monocyte count, systemic immune inflammatory index, neutrophil-lymphocyte ratio (NLR), platelet-lymphocyte ratio (PLR),

lymphocyte-monocyte ratio (LMR), C-reactive protein (CRP), platelet-platelet ratio (PCT), erythrocyte sedimentation rate (ESR), procalcitonin level, lactate dehydrogenase (LDH), albumin, CRP / albumin ratio and HALP score that were looked at during the diagnosis of childhood cancers.

Methods: The oncology charts of children diagnosed with childhood cancers between August 2018 and December 2020 were retrospectively analyzed. In January August 2018 and January 2021, 125 patients diagnosed with childhood cancer, excluding leukemia and retinoblastoma, were included in the study. Demographic characteristics of patients, laboratory findings at the time of diagnosis, primary tumor diagnosis and stage, whether they received chemotherapy, whether they received radiation therapy, relapse status and their final status were recorded.

Results: More than half of the 125 patients included in the study had solid tumors with 81 (64.8%) patients, 21 (16.8%) had lymphomas and 23 (18.4%) had CNS tumors. There was a significant difference between lymphocyte counts, NLR, PLR, LMR and HALP scores between lymphoma and solid tumors (p values respectively 0.003; 0.003; 0.042; 0.003; <0.05). There was a significant difference between CRP and LDH levels between lymphoma and SSS tumors (p values respectively 0.027; 0.015). We found a significant difference between the progression of the tumor and lymphocyte count, platelet count, PCT, hemoglobin, albumin level, CRP concentration, ESR, LDH, procalcitonin, NLR, PLR, LMR, CRP-albumin ratio and HALP score in comparative statistical analysis.

Conclusions: There is a relationship between the progression of the tumor at the time of diagnosis and hematological parameters. Due to the lack of a study on hematological parameters in childhood malignancies, we thought that our study would be enlightening for subsequent studies.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0490 / #1179 | Complications of Central Venous Catheter in Children with Cancer: Retrospective Audit from a Tertiary Cancer Care Centre in India

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Background and Aims: Central Venous Catheters (CVC) in pediatric oncology patients has significantly reduced the trauma of pricks and chemotherapy extravasation. Complications always remain a concern with the use of CVC's especially secondary to central line-associated bloodstream infection (CLABSI) in LMIC like ours hence the audit

Methods: A retrospective audit for complications was done in a newly established division of pediatric oncology in a tertiary cancer care

centre having a dedicated nursing pool, standard operating procedure (SOP) for line management and stringent infection control practices for patients with CVC's from September 2019 to March 2021.

Results: Forty-nine CVC's were inserted in 39 children with cancer aged < 18 years. Majority 31(63.3%) were subcutaneous tunnelled CVC followed by triple lumen percutaneous central catheters 18 (36.7%). Median duration of CVC for tunnelled and percutaneous CVC was 105 (± 84.8) days and 13.5(± 9.53) days respectively. Majority of the patients 29 (59.18%) were aged less than 10 years with most children having the diagnosis of Acute Leukemia in 34(69.38%). Total of 18 complications were noticed of which CLABSI was observed in 13 (72%). The incidence of CLABSI was 3.26 per 1000 CVC days. Gram negative bacilli was the most commonly found organism 9(69.2%). Three (16.6%) triple lumen percutaneous central catheters and 6(20%) subcutaneous tunnelled CVC were infected, of which all 3 percutaneous catheters and two subcutaneous tunnelled CVC required emergency removal. Non-infectious complications were noted in 5(10%) cases (2- Wound gaping, 1-right atrial rupture requiring thoracotomy and atrial wall repair, 1- Pneumothorax, 1- Tunnel infection).

Conclusions: This study showed that even with stringent infection control practices CLABSI is still a risk factor mandating removal of the CVC. Gram negative bacilli pose more risk in our patients compared to reported literature from HIC probably mandating changes in antibiotic stewardship in patients with CVC's and fever in LMIC like ours.

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0491 / #1199 | The Distribution, Characteristics and Outcome of Childhood Cancer in Baptist Hospital Mutengene, Cameroon

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Background and Aims: Paediatric oncology registries are essential to estimate the burden of childhood cancers, plan care delivery and advocate for its recognition as a public health priority. This study aimed to provide information on the distribution, trends and outcomes of paediatric malignancies at Baptist Hospital Mutengene (BHM).

Methods: This was a retrospective cross-sectional study. Data from a hospital-based paediatric oncology networked database (POND) registry was analysed for children 0-14 years diagnosed from 2007 to 2016. Data were analyzed on SPSS version 25.0. Descriptive analyses were used to describe the distribution of diagnoses and clinical characteristics while Kaplan Meier Curves were used to estimate survival.

Results: Two hundred and eighty five patients were recorded from 2007 to 2016, predominantly from the Southwest Region (76.8%) of Cameroon. The male to female ratio was 1.4:1; median age at diagnosis was 8 (IQR 5.5 – 9.5) years. The abdominopelvic, maxillofacial and

eye regions were the most common tumour sites. The most common diagnoses were lymphomas (242, 84.9%), solid tumours (26, 9.1%), sarcoma (11, 3.9%) and leukaemia (4, 1.4%). The stage of the disease at diagnosis was Stage IV (22.6%), Stage III (65.8%), Stage II (4.6%) and Stage I (7.0%). The overall survival is 51.3% and was higher in children who completed treatment (60%).

Conclusions: Lymphoma, retinoblastoma, soft tissue sarcoma and leukemia were the most common paediatric cancers in BHM. There was a high rate of late diagnosis and survival was lower amongst patients who did not complete treatment. Early diagnosis initiatives and financial and psychosocial support to enable families to complete treatment are required for better survival rates. Acknowledgement: BTMAT, WCC

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0492 / #1239 | Establishing Profiles of Paediatric Oncology Units for Regular Assessment of Childhood Cancer Care in an LMIC

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Background and Aims: Treatment of childhood cancer in LMIC's faces barriers at each level. From lack of awareness and education, insufficient access and expertise, delays in diagnosis and treatment, therapy abandonment and an overarching lack of resources in hospitals and among patients. The My Child Matters Grant by the Sanofi Espoir Foundation aimed at improving outcomes of childhood cancer in LMIC's was awarded to the Pediatric Oncology Department of Indus Hospital & Health Network in 2019 for training, teaching and holistic improvement of childhood cancer treatment in Pakistan. As a tertiary care multi-disciplinary facility, the IHHN grant recipient team partnered with eight public sector pediatric units across different cities, motivating and empowering stakeholders to develop their pediatric oncology functionality via a multi-faceted strategy.

Methods: Team representatives visited all 8 POU's collaborating under the grant. A detailed performa was filled via interview regarding the facility, patient care, patient data management, human resources, baseline nursing standards, infection control standards and hand hygiene protocols. Chemotherapy preparation was observed and feedback taken from focal persons regarding local limitations and challenges.

Results: Out of the 8 POU's assessed, 5 units had drastically improved functionality in terms of patient capacity, standard of care, human resources and patient data management. Along with encouraging specialized training among local residents, regular tumor board participation, inter-hospital referral pathways and protocol discussions

were consistently ongoing. Dedicated room for bone marrow and intrathecal procedures was allotted in the ward. Chemotherapy handling and preparation was assigned to trained personnel with mandatory safety precautions and electronic prescription generation. Psychosocial aspect of care was integrated with treatment. Unfortunately, baseline nursing standards and infection control standards were not achieved in most centers

Conclusions: Continuation of collaboration and trainings is required to further improve care at existing partner POU's. The developed model should be mirrored in other local setups to improve care and offer widespread access to underserved childhood cancer victims.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0493 / #1251 | The Sanofi Mcm Award Supports The Cameroon Paediatric Oncology Group to Strengthen Childhood Cancer Services Nationally

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Background and Aims: The CPOG was founded in May 2018 at a national workshop of stakeholders, led by paediatric oncologists. A SWOT analysis was conducted and a five-year action plan was elaborated. World Child Cancer was identified as a mentor organization and Funding was obtained through the Sanofi Espoir Foundation through the My Child Matters Initiative to finance the activities of CPOG for three years.

Methods: The CPOG was founded in May 2018 at a national workshop of stakeholders, led by paediatric oncologists. A SWOT analysis was conducted and a five-year action plan was elaborated. World Child Cancer was identified as a mentor organization and Funding was obtained through the Sanofi Espoir Foundation through the My Child Matters Initiative to finance the activities of CPOG for three years.

Results: From September 2019 to April 2020, 923 children were impacted. Formal collaboration was obtained with the Ministry of health and the National Committee for the Fight Against Cancer. National protocols were elaborated for ALL and retinoblastoma, harmonized education leaflets were produced. Two palliative care workshops were organized and Palliative care outreach established at two centres. One a parent accommodation was acquired. Three annual national conferences were held. International Childhood Cancer Days were celebrated. Twinning was established with Leeds Children's hospital for capacity building. Consumable supplies were provided for COVID-19 prevention. Paediatric oncology specific priority actions were included in the National Cancer Control Plan 2020 – 2024.

Conclusions: Collaboration between paediatric oncology professionals and other stakeholders, with support from twinning partners and international NGO's led to significant service development. This project has laid the ground work for Cameroon to forge towards achieving the GICC goal of 60% survival by 2030.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0494 / #1254 | Effect of Covid-19 Pandemic on Teaching and Training Activities in Pediatric Oncology Across Pakistan

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Background and Aims: A multi-disciplinary approach is essential for success in pediatric oncology treatment. Updated protocols, quality nursing care, psycho social support, safe and standardised preparation of chemotherapy, infection control and effective data management are key shareholders for effective management of childhood cancer. The Department of Pediatric Oncology at Indus Hospital & Health Network (IHHN) initiated consistent teaching and trainings with the help of the My Child Matters Grant from Sanofi Espoir Foundation. These courses were conducted in person starting in 2019 and had to be postponed and restructured due to COVID-19 pandemic in early 2020.

Methods: Six courses (physicians, nursing, infection control, pharmacy, psychosocial care and cancer registry) were conducted virtually starting December 2020, open to all and free of cost. Lectures with Q&A sessions were conducted via zoom and recordings shared through email. Each course was conducted by the relevant department at IHHN with pre and post assessment conducted through google forms. Session feedback was taken through zoom polls and a comprehensive course feedback conducted after completion, e-certificates were awarded to successful participants according to a predetermined criteria.

Results: A total of 434 physician's registrations were done from around Pakistan and countries like Saudi Arabia, Malaysia, Jordan and Canada for the online physician's course, of which 110 received certifications after completing post-test and attendance criteria of 55%. Pharmacy, infection control, psychosocial care and cancer registry courses saw participation and certification of 51, 41, 24 and 14 participants respectively. Online sessions received positive feedback in terms of instructors, course content, convenience and access from over 90% participants.

Conclusions: Due to the ease in coordinating hectic schedules and cost effectiveness of online lectures, this virtual teaching model will persist despite the trajectory of the COVID-19 pandemic. Similar ventures

aimed at subspecialty trainings and refresher courses are needed in a widespread manner.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0495 / #1257 | Using Profile Abbreviated Version to Guide Pediatric Hematology-Oncology (PHO) Opportunities Prioritization and Gap Analysis in Chile

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Background and Aims: Childhood cancer services in Chile have been provided by the Programa Infantil para Drogas Antineoplásicas (PINDA) since 1988. The program includes 20 pediatric hematology-oncology (PHO) centers. The St. Jude Pediatric Oncology Facility Integrated Local Evaluation (ProFILE) tool was used to diagnose the PINDA's strengths and opportunities. We aim to describe the implementation of ProFILE in Chile.

Methods: During the preparation phase, the tool was translated into Spanish. All translated materials were validated by seven native Spanish speakers, including 4 PHO physicians. Sixteen PHO facilities were invited to participate. During the assessment phase, local PHO providers served as liaisons collecting and entering data into Epi Info 7 platform. Facility-specific descriptive and score-based reports were provided to each participating center. Finally, four 2-hour virtual sessions were conducted in March 2021 as a part of the interpretation and action phase.

Results: After a 7-month implementation process, heterogeneous strengths and opportunities were identified across the 16 PINDA centers. Aggregated results showed strengths in chemotherapy (87%), diagnostics (76%), service capacity (75%), supportive care (75%), surgery (75%), and patients and outcomes (75%). Areas of opportunity included: facility and local context (56%), service integration (60%), personnel (61%), national context (71%), and finances and resources (72%). Over sixty participants and stakeholders participated in four design-thinking exercises performed during the ProFILE virtual workshops. The priorities identified for the next three years target to improve: nursing specialization, strengthen core and extended PHO team, coverage for the diagnosis and treatment of childhood cancer in adolescent patients, delivery of palliative care services, interdisciplinary care meetings, access to molecular studies for brain tumors, and strengthening of the referral system.

Conclusions: The ProFILE dynamic 360o evaluation of health services conducted in Chile helped PINDA centers and stakeholders define priorities and design an improvement strategy for the next three years.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0496 / #1274 | Novel Approach and Outcomes of an Online National Palestinian Territories Childhood Cancer Workshop Utilizing The St. Jude Profile Tool

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Background and Aims: Due to political and geographic fragmentation, the Palestinian Territories (PT) had not previously participated in a national meeting to prioritize plans for capacity building in pediatric hematology and oncology (PHO). We conducted a national stakeholder workshop using a novel virtual platform to establish key national priorities and initial action plans.

Methods: In March of 2020, the abbreviated version of the St. Jude Pediatric Oncology Facility Integrated Local Evaluation (ProFILE) tool was implemented to assess infrastructure, capacity, and delivery of healthcare services across the PT at the four centers that provide pediatric oncology care. Both single-facility and national aggregated data reports were generated to inform a four-day design-thinking workshop in September 2020, including key stakeholders, which took place online using Zoom videoconferencing platform and MURAL visual collaboration software.

Results: The workshop included 22 participants from the PT representing the four facilities, the Ministry of Health, and the World Health Organization, to first review the data reports from ProFILE. Participants were then divided into four breakout rooms to identify the challenges during the patient journey for different diagnoses, and identify opportunities for improvement and key collaborators to involve. Four initiatives were prioritized: nursing specialization; implementation of interdisciplinary care meetings; increase in national and international collaborations; and patient data collection and management. Working groups were formed to develop and action plan with deliverables for each initiative.

Conclusions: The ProFILE tool can be successfully implemented using a virtual format. This novel approach can assist other global regions to define national priorities while allowing for multi-site and stakeholder collaboration.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0497 / #1279 | The St. Jude Global Childhood Cancer Analytics Resource and Epidemiologic Surveillance System (SJCARES) Hospital-Based Cancer Registry (HBCR) Network: An Interim Analysis of Implementation Outcomes**

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Background and Aims: Traditional population-based cancer registries (PBCRs) do not systematically collect the treatment and outcomes data required to inform childhood cancer control practices. HBCRs offer a complementary solution. The objective of this study was to report the first interim analysis of implementation outcomes from the SJCARES HBCR network.

Methods: Data captured in the SJCARES HBCR network between 9/20/2020 and 4/1/2021 are included in this report. Interim implementation outcomes were measured based on the SJCARES HBCR network onboarding process (described in detail at <https://www.stjude.org/global/sjcares/registry.html>). As a first step to HBCR access, SJCARES-specific data use agreements are signed. Sites next complete position-specific training. Training is divided into two parts: (1) asynchronous Cure4Kids modules and (2) dummy test case entry with individualized feedback on performance. Upon completion, teams gain access to a secure cloud-based registry. Case registration is rolling starting with basic demographics (presentation date), diagnosis/staging (3-months later), frontline treatment (6-months later) and follow-up/outcomes (annually).

Results: 135 institutions requested access to the SJCARES HBCR, 79 signed data use agreements and 33 identified teams. 101 individuals started asynchronous training, of whom 69% (69/101) completed training. Six hospitals are entering data into their secure registry instance. Essential information for 638 patients across all sites is included. Complete demographic information is available for 78% (500/638) of patients and diagnosis data for 83% (529/638), with the rest pending data entry. 95% (504/529) of registered patients had a confirmed cancer diagnosis (median age 6 years [interquartile range 3-11 years]). 56% were male (280/500). Lymphoid leukemias were the most common malignancy and comprised 25% of registered cases (126/504).

Conclusions: Global interest in the freely available SJCARES HBCR is high and data collection is ongoing. These data are critical for programmatic surveillance and implementation of quality improvement interventions. Next steps involve evaluation of data quality and supporting continued growth.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0498 / #1289 | Using The My Child Matters Grant for Training and Capacity Building in Pediatric Oncology Units Across Pakistan**

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Background and Aims: The My Child Matters Grant, awarded for teaching and capacity building in Pediatric Oncology Units by the Sanofi Espoir Foundation was given to The Indus Hospital in 2019 for a three year period. Pakistan, falling in the LMIC category, records 8000 documented cases of childhood cancer with an unfortunate survival rate of only 35%. A lack of pediatric oncology facilities and primary health care systems creates a difficult gap in diagnosis and timely treatment, a dismal situation particularly for the underprivileged portion of the population. A consequent lack of childhood cancer related data prevents development of evidence based protocols and calculation of true population-based incidence

Methods: The Pediatric Oncology Department at Indus Hospital collaborated with 10 small-scale POU's across Sindh and Punjab, guiding and motivating them to employ standardized practice meeting the quality indicators set for each stakeholder. Regular courses in 6 disciplines, hands on trainings and large scale public awareness campaigns through community health workers were conducted regularly. A detailed and basic Childhood Cancer Registry was also developed for appropriate documentation of malignant cases across these centers

Results: From 2019 to early 2021, a total of 3,463 children were inducted for cancer treatment in Indus Hospital and its associated MCM centers, with a greater number screened for malignancy in outpatient settings. Approximately 1,800 individuals of the general public were educated by each childhood cancer awareness campaign bi-annually. Online courses were able to certify and train: 257 physicians, 123 nurses, 18 psychosocial workers and 54 other healthcare workers (pharmacy, infection control).

Conclusions: Successful implementation of quality care in pediatric oncology requires consistent training, focused funding and community outreach encouraging early presentations. Grants and private partnerships such as the My Child Matters program is essential for LMIC's to reach the WHO Global Initiative to increase childhood cancer survival by 2030.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0499 / #1294 | Childhood Cancer Surveillance in Southern Africa

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Background and Aims: In Africa, population-level data on childhood cancers is sparse and incomplete, particularly information on staging and coding. Where population-based cancer registries (PBCRs) exist, they mainly focus on adult cancers. To address this issue, the Childhood Cancer Surveillance project aimed to establish national childhood cancer programme within five Southern African countries (Eswatini, Mozambique, South Africa (SA), Zambia and Zimbabwe).

Methods: All participating centres have functioning PBCRs with either national (SA, Eswatini and Zimbabwe) or regional (Mozambique and Zambia) coverage. Registrars were trained on the specifics of childhood cancer registration which included using the Tier 1 Toronto Childhood Cancer staging guidelines stage and to improve the accuracy of coding. All childhood cancers for the years 2018-2019 were retrospectively captured, staged, and information on treatment and vital status at end of 31 December 2021 were recorded where possible.

Results: Cancer cases recorded ranged from 150 in Eswatini to over 800 in Mozambique and SA. The most common cancers varied by country; retinoblastoma in Zambia (17%), leukaemias in Eswatini (13%) and SA (21%), Wilms tumour in Zimbabwe (16%) and Burkitts lymphoma in Mozambique (11%). Staging was assigned for <10% of cases captured before the training. After the training, staging improved to ≈45-50% across the region. Treatment and outcome information was rarely recorded before the project started.

Conclusions: National childhood cancer registration is feasible and cost-effective when incorporated within a PBCR. Tailored training on staging can effectively enable cancer registrars to code stage on childhood cancers. Information on treatment (type and initial date) is hard to find where medical records are on paper form and stored in different systems. Active follow-up requires additional resources. Clinicians need training to include sufficient information on staging, treatment and survival in order to strengthen surveillance of childhood cancers.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0500 / #1322 | Beliefs, Brokers, and Networks: Health System Determinants of Childhood Cancer Drug Access in Five East African Countries

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Background and Aims: Access to essential childhood cancer drugs (CCDs) is a core determinant of childhood cancer outcomes. The limited available evidence suggests that CCD access is highly variable across countries, particularly in low- and middle-income countries (LMICs) where the burden of childhood cancer looms largest. To support evidence-informed national and regional policies for improved childhood cancer outcomes, we investigated the barriers to and facilitators of CCD access in five East African countries – Ethiopia, Kenya, Rwanda, Tanzania and Uganda.

Methods: We conducted a qualitative analysis to: (i) investigate contextual determinants of CCDs within and across study jurisdictions; and (ii) the alignment between CCD inclusion on the National Essential Medicines List (NEML) of each country and the 2019 WHO Essential Medicines List for Children (EMLC). We undertook thematic analysis of academic literature, policy documents, and semi-structured interviews from a purposive sample of stakeholders involved in childhood cancer care, drug procurement and supply, and health policy and program development (n=72). The Advocacy Coalition Framework was applied to examine policy development and implementation dynamics.

Results: A review of each NEML against the WHO EMLC revealed variability in drug prioritization for key cytotoxic drugs across sites, with the least drug alignment noted in Rwanda, Ethiopia, and Tanzania. Principal barriers to CCD access were comparable among jurisdictions, including national budgetary limitations, weak cancer surveillance and drug monitoring systems, and inadequate global drug stock predictability. CCD access facilitators revealed the presence of strong political

will, institutional policy brokers, and core system beliefs that centralize child security as drivers for the development of policy and establishment of stakeholder networks between private, non-profit/civil and public sectors to advance implementation.

Conclusions: CCD access across East Africa remains vulnerable to global market perturbations for consistent drug supply. The presence of cohesive policy networks and prominent institutional policy brokers are key features buffering global and national supply chain constraints.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0501 / #1333 | Mortality Audit of Children with Cancer at a Tertiary Care Facility in Sub-Saharan Africa

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Background and Aims: Although the outcomes in childhood cancer have improved significantly in the recent past, same cannot be said of the disease in low middle-income countries. There is also paucity of data in this region. This audit sought to describe the mortality profile of cancer patients younger than 15 years at the paediatric oncology unit of Komfo Anokye Teaching Hospital.

Methods: A retrospective descriptive data on the demographic and clinical characteristics of children (0-14years) who died in the paediatric oncology unit from January 2020 to December 2020 were retrieved from their medical records and analysed.

Results: During the study period, 37 patients out of 115 patients seen within the study duration died placing the mortality rate at 32.17%. There was a female predominance noted (56.76%) among those that died with a median age of 7 years. The mean duration from first consultation to the start of treatment was 30days. Leukemias and lymphomas were the leading causes of mortality contributing about 54.05% to these deaths. Majority (56.76%) of the deaths occurred during treatment and the disease was uncontrolled at the time of death in 78.38%. A significant proportion (54.05%) of the deceased patients were on palliative care and 10.81% of patients that died had abandoned treatment.

Conclusions: Improvement in the mortality of childhood cancers is dependent on promoting early diagnosis and supportive care as well as enhancing access to quality care.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0502 / #1352 | Trends and Disparities in Cancer-Related ED Visits and Hospital Admissions: US National Estimates, 2012-2018

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Background and Aims: Cancer remains a major cause of morbidity and mortality affecting all age-groups. This study aims to examine trends and sociodemographic disparities in cancer-related ED visits and hospital admissions in the United States.

Methods: A multi-year secondary analysis of the National Hospital Ambulatory Medical Care Survey (NHAMCS) (2012-2018) was conducted to generate nationally weighted estimates of cancer-related ED visits and hospital admissions with corresponding 95% confidence intervals (CI). Estimates were stratified by race/ethnicity, sex, age-groups, and health insurance status.

Results: Cancer diagnosis accounted for 3.1% (95% CI, 2.9 – 3.3) and 9.3% (95% CI, 8.6 – 10.1) of the estimated 954,128,804 ED visits and 93,437,676 hospital admissions, respectively, for period 2012-2018. Annual cancer-related ED visit and admission rates remained stable during the period, ranging from 2.9% (95% CI, 2.6 – 3.4) in 2012 to 3.8% (95% CI, 3.3 – 4.3) in 2018—for ED visits and 8.5% (95% CI, 7.1 – 10.1) in 2012 to 9.9% (95% CI, 8.3 – 11.7) in 2018—for hospital admissions. Among cancer ED visits, annual hospital admission rates were stable, averaging 29.2% (95% CI, 26 – 32.7). Average annual cancer-related ED visit and admission rates for the study period were highest among individuals aged ≥65 years (10.7%, 95% CI, 10.0 – 11.5; and 14.1%, 95% CI, 12.8 – 15.4, respectively), non-Hispanic Whites (4.0%, 95% CI, 3.7 – 4.2, for ED visits), and Medicare beneficiaries (9.1%, 95% CI, 8.4 – 9.8; and 13.5%, 95% CI, 12.3 – 14.7), and lowest among individuals aged ≤24 years (0.3%, 95% CI, 0.2 – 0.4; and 1.1%, 95% CI, 0.7 – 1.8).

Conclusions: Annual rates of cancer-related ED visits and hospital admissions remained stable from 2012-2018, with rates particularly low (≤1%) in children and adolescents. The burden of cancer-related ED utilization and hospitalizations is largest among older adults (≥65 years), non-Hispanic Whites, and Medicare beneficiaries.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0503 / #1354 | Agreement between the 'End-User' Assessment and Scores Generated by the Profile Tool in Twelve Pediatric Hematology and Oncology (PHO) Cohorts

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Background and Aims: The Abbreviated Pediatric Oncology Facility Integrated Local Evaluation (ProFILE) tool is divided into 12 modules. Most questions have a score and a desirable threshold to facilitate

results visualization and allow comparison. At the end of each module, ProFILE enables the end-users to rate the status of PHO care delivery for the specific overarching theme. The purpose of this study was to determine the level of agreement between the end-user assessment and ProFILE scores among twelve cohorts who have completed the tool.

Methods: We retrospectively reviewed ProFILE aggregated scores and end-user scores for 12 cohorts that completed the assessment between January 2019 and March 2021. The cohorts were from Eastern Mediterranean, Sub-Saharan Africa, South America, and Eurasia, spanned 21 countries, and had a median of 7 facilities (range 3-17). We compared agreement between end-user and ProFILE score for each module using intraclass correlation (ICC).

Results: A total of 144 module entries were analyzed. Agreement between ProFILE scores and end-user scores was optimal for eight out of the twelve modules of the tool: National Context (ICC 0.80), Personnel (ICC 0.82), Service Capacity (ICC 0.91), Diagnostics (ICC 0.94), Chemotherapy (ICC 0.97), Supportive Care (ICC 0.89), Surgery (ICC 0.86), and Radiation Therapy (ICC 0.95). The ICC was < 0.65 (non-optimal) in four modules. End-user scores were higher than ProFILE scores for the Facility and Local Context and Service Integration modules. The reverse was observed for Finances and Resources, and Patients and Outcomes.

Conclusions: The observed good internal consistency between the ProFILE scoring system and the end-user impression in most modules speaks to the tools' validity and reliability. Non-optimal agreement may result from differences in end-user's awareness or perception of the relevance of specific or novel areas assessed in the tool. Results will inform qualitative inquiries to further improve the tool.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0504 / #1356 | Using the Re-Aim Evaluation Framework to Assess the Impact of the St. Jude Abbreviated Profile Tool: A 2-Year Experience

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Background and Aims: The Abbreviated Version of the Pediatric Oncology Facility Integrated Local Evaluation (ProFILE) tool was launched in April 2019. National and regional pediatric hematology and oncology (PHO) groups have used ProFILE to set priorities to improve PHO care. We aimed to operationalize the RE-AIM (Reach, Effectiveness, Adoption, Implementation, and Maintenance) framework to assess the tool's impact.

Methods: Each RE-AIM element has been defined as follows. Reach captures the number of facilities completing the assessment phase. Effectiveness refers to the outcomes obtained from the interpretation and action phase. Adoption reflects the number of St. Jude Global Regional Directors, local regional teams, and third parties willing to implement ProFILE and the additional assessment tools developed using ProFILE as the backbone ("sibling tools"). Implementation indicates the extent to which ProFILE was delivered as intended. Maintenance reflects the institutionalization of ProFILE as part of the routine organizational practices and policies of St. Jude Global.

Results: Reach: A total of 117 facilities providing PHO care in 23 countries have completed the assessment phase. The largest cohort had 17 PHO facilities. Effectiveness: During the interpretation and action phase, seven facility-based workshops and nine regional workshops were conducted. Only one cohort (8 facilities) was unable to complete the interpretation and action phase (due to political unrest). Adoption: Four St. Jude Global Regions have implemented ProFILE and it has been delivered in four languages (English, Russian, Spanish, and Portuguese). Five additional tools ("sibling tools") have utilized the ProFILE backbone. Implementation: Workshop adaptations were necessary during the COVID 19 pandemic, but the fidelity of the implementation process was conserved. Maintenance: Autonomous delivery of the interpretation phase has occurred in four regional cohorts.

Conclusions: The RE-AIM framework facilitates the evaluation of ProFILE outcomes. ProFILE meets essential framework elements and results suggest good chances of sustainable adoption and implementation in other global regions.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0505 / #1369 | Cancer Does Not Wait, and Neither Can We: An Experience Transferring Quality Improvement Knowledge and Skills in the Context of the Covid-19 Pandemic

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Background and Aims: In quality improvement collaboratives (QIC), knowledge transfer (KT) typically occurs between teams from different organizations or clinical areas collaborating towards a shared aim. In the context of the Mexico in Alliance with St. Jude Collaborative (MAS Collaborative), we assessed the feasibility of QI KT from the Golden Hour, a project aimed at improving timely administration of antibiotic

therapy for children with cancer and fever, to the “Cancer does not wait, and neither can we”, a project aiming to sustain the continuity of curative treatment for children with cancer during the COVID-19 pandemic.

Methods: From May 2019 to November 2020, 23 participating institutions adopted the Institute for Healthcare Improvement's Breakthrough Series Model (multidisciplinary teams collaborating around a common aim using a shared theory of change and measurement strategy) for the Golden Hour. An intermediate-level QI course was embedded into the MAS Collaborative to facilitate knowledge and skills development and transfer. In June 2020, a theory of change and measurement strategy for the “Cancer does not wait, and neither can we” project was introduced. Teams added new team members and virtual learning sessions and coaching were used to facilitate TK.

Results: Thirteen of the 23 participating teams actively participated in the “Cancer does not wait, and neither can we” project. Treatment continuity increased from a pre-COVID baseline of 77% (data from 466 visits) between July 2019 and March 2020 to 85% (data from 2,047 PHO visits) between August and November 2020. Teams reported this exercise as a valuable opportunity for KT and improvement.

Conclusions: KT to a second topic during the QIC demonstrated to be feasible. It not only helped teams sustain treatment continuity but also to improve it from baseline even in the context of a public health emergency. KT has the potential of maximizing the impact of QICs beyond their initial aim.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0506 / #1375 | Challenges and Obstacles to Optimum Delivery of Childhood Cancer Care

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Background and Aims: The epidemiology of Childhood cancer in Jordan shows geographic variation. Prior studies suggested under reporting and detection. A national initiative that aims to raise awareness on Non-Communicable Diseases (NCDs) has launched a project that sheds the light on the barriers and challenges of childhood cancer care through direct communication with the community.

Methods: This is an outreach project involving multidisciplinary stakeholders from all Jordanian health sectors targeting marginalized communities in underserved and rural areas in Jordan. The project organized community educational days to raise awareness on childhood cancer. There were pediatric screening clinics led by pediatricians, primary healthcare physicians, medical students and nurses. The team assessed the visitors and appropriate referrals were made. Post COVID-19 pandemic, the focus has shifted towards capacity building of primary healthcare centers personnel through virtual conferences and workshops.

Results: This is an ongoing project that has captured national media recognition with positive feedback. We describe the experience of

one of the projects focusing on childhood cancer awareness and early detection. The project has emphasized that many of the issues surrounding NCDs are women and child related. It showed the geographic variation in the awareness of childhood cancer and knowledge gap amongst primary healthcare providers. It outlined the social challenges that children with cancers and their families face.

Conclusions: A Childhood cancer awareness project as part of a novel national healthcare initiative that focuses on NCDs. The project revealed the wide knowledge gap in the field of NCDs and childhood cancer in particular; and uncovered regional variation in the care provided.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0507 / #1389 | The Challenge of Spreading Knowledge about Childhood Cancer during the Covid-19 Pandemic – The Experience of Online Training

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Background and Aims: A NON-GOVERNMENTAL ORGANIZATION DEVELOPED A PROGRAM TO PROMOTE TRAINING TO PRIMARY HEALTH CARE PROFESSIONALS. OVER 12 YEARS, MORE THAN 25,000 HEALTH PROFESSIONALS WERE TRAINED THROUGH IN-PERSON CLASSES. WITH THE APPEARANCE OF COVID-19 PANDEMIC AND THE NEED FOR SOCIAL DISTANCING, THE METHODOLOGY WAS TOTALLY ADAPTED TO A DIGITAL FORMAT AND THE TARGET AUDIENCE WAS CHANGED TO HEALTH STUDENTS,

Methods: The training was adapted to the digital format, enabling the classes to be given through the Zoom platform, with the participation of pediatric oncologists all the time. The training will be carried out in a virtual environment according to a previously scheduled calendar. The classes take place live, where the pediatric oncologist will teach the content and will display slides with didactic material aiming at a better explanation on the theme. In addition, during the presentation, quizzes are included, in order to make the class more dynamic and interactive. After the training, students receive complementary didactic material by e-mail. In addition, they receive complementary material on pediatric oncology on the website of the non-governmental organization to students who reach 75% or more of participation in the total workload of the training. The training has a total of 6 hours with the following topics: Epidemiology; Signs and symptoms; Caring for children with cancer; Referral of suspected case; Communication of diagnostic suspicion.

Results: After 3 months of a project in partnership with 10 organizations, 742 students (medicine, nursing and pediatric residents) were trained by the Program.

Conclusions: The topic of pediatric oncology is not part of the curriculum of most universities in the country. This fact makes the Program even more necessary because it knows that one of the measures to increase the cure rate of the disease is to prioritize early diagnosis to all health professionals and future professionals in the area.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0508 / #1393 | Fewer Pediatric Cancer Diagnoses during the First Year of the Covid-19 Pandemic: "Truth or Consequence?"**

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Background and Aims: The COVID-19 pandemic has a negative impact on Pediatric Oncology functioning. Countries notice a decrease in new cases of childhood cancer and adolescents, and delay in diagnosis. The objective is to analyze the epidemiology of new cases seen in our center in 2020 compared to 2014-2019 period

Methods: Review of new cancer cases registered in a Pediatric Oncology-Hematology Unit of a public tertiary hospital in Northern Spain. We also treat a large part of solid tumors from another neighboring region. All cases diagnosed in 2020 were analyzed and compared with six years before. Data were compared with other studies at national and international level. Possible causes are hypothesized

Results: The mean number of new cases 2014-19 was 63 (range 81-50; in frequency: CNS tumors> Leukemias> Soft Tissue Sarcomas> Lymphomas> Neuroblastomas). In 2020, 44 new cases were diagnosed and we observed a decrease in leukemias (59,5%), especially B-precursor acute lymphoblastic leukemia (BCP-ALL), soft tissue sarcomas (STS; 50%) and CNS tumors (31,5%). However, 68.8% more Lymphomas were diagnosed; mainly Hodgkin lymphomas in adolescents. No differences for other neoplasms. The findings of fewer diagnoses, especially in the first months of pandemic, have been referred too by other series. We didn't observe significant delays in diagnosis or increase in mortality. No patient had COVID-19 at diagnosis. Since 2020 March, a marked reductions of admissions due to other viral respiratory infections, has been observed

Conclusions: In the first months of the pandemic, 30% fewer childhood cancers were diagnosed, mainly leukemias, CNS tumors and STS. More information is needed on the epidemiological consequences of the COVID-19 pandemic in relation on lower pediatric cancer incidence. The possible effect of changes of viral respiratory infections should be studied deeply, mainly in BCP-ALL

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0509 / #1403 | Helicobacter Pylori Chronic Gastritis and its Mucosa Lesions Among Moroccan Population**

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Background and Aims: Infection with *Helicobacter pylori* is one of the most widespread bacterial infections implicated in chronic gastritis in the world which the progression leads to gastric carcinoma. The aim of this research is to study the mucosa histological aspects of *Helicobacter Pylori* chronic gastritis among Moroccan patients suffering from different disorders. The evaluated parameters are the lymphoplasmocytic infiltration, the glandular Atrophy, intestinal Metaplasia, dysplasia, lymphoid follicles, presence of Biermer's auto-immune disease and the potential adenocarcinoma.

Methods: 248 gastric biopsies samples were analyzed at the Anatomic Pathology Laboratory of Al Idrissi Regional Hospital of Kenitra, Morocco. Standard histopathological techniques were performed on 10 % formalin fixed paraffin embedded, 5µm-thick and hematoxylin & eosin stained sections. The Sydney System for the classification and grading histological parameters are used.

Results: All our samples showed chronic gastritis (chronical inflammation) where *H. pylori* was very common with a prevalence of 69%. The prevalence of glandular atrophy is 16%. It is associated with *H. pylori* infection in 75% of cases, and accompanied by Biermer's auto-immune disease in 8% of cases. Gastric metaplasia is observed in 16.53% of cases and 40% of which are associated with *H. pylori*. Moreover, one case of dysplasia with presence of *H. pylori* and a metaplasia was registered. one case of adenocarcinoma with presence of *H. pylori* was reported. Lymphoid follicles represented 32% of cases and they are associated with *H. pylori* chronic gastritis in 82 % of cases.

Conclusions: According to the present study, a close relationship could exist between *H. pylori* colonization and some pre-cancerous lesions. The lack of an early eradication and the progression of lesions could lead to gastric carcinogenesis. More investigations are needed to support these relation.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0510 / #1406 | Achieving and Sustaining Results in the Context of the Covid-19 Pandemic: Experience from the Mexico in Alliance with St. Jude Golden Hour Quality Improvement Collaborative**

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Background and Aims: We previously reported early results of the Mexico in Alliance with St. Jude Golden Hour Collaborative (MAS Collaborative), which aimed to increase the proportion of febrile pediatric hematology-oncology patients (fPHO) who receive the first dose of antibiotics in ≤ 60 min from 39% to 80% across 23 institutions in Mexico between May 2019 and November 2020. We now report on the results and sustainability achieved during the COVID-19 pandemic.

Methods: This 18-month quality improvement collaborative (QIC) followed the Institute for Healthcare Improvement's Breakthrough Series Model (including multidisciplinary teams collaborating around a common aim using a shared theory of change and measurement strategy) coupled with structural support (leadership engagement, infrastructure and continuous coaching) and a QI capability-building program. The NHS Sustainability Model was used to assess strengths and weaknesses around 10 established processes, staff, and organization factors that contribute to the sustainability of improvement initiatives. Each institution conducted a self-assessment based on the team consensus at three points in time (March, August, and November 2020).

Results: By March 2020, 1,073 fPHO visits had been reported, of which 781 (73%) received antibiotics in ≤ 60 min; 21 centers had showed modest, moderate, or significant improvements (QIC progress scale score ≥ 3.5), and 17 centers had met their local aim. By November 2020, centers had reported 1,621 fPHO visits, of which 1,183 (73%) received antibiotics in ≤ 60 min. Nineteen centers completed the NHS Sustainability self-assessment with an aggregated score of 3.43 (range 2.8-4; max=4) in March, 14 centers completed it in August (3.5, 2.1-4) and 11 centers in November (3.5, 2.3-4). Sustainability scores decreased for five centers and increased for nine, while results were sustained for 17 centers despite of COVID-19.

Conclusions: This first international pediatric cancer QIC demonstrates how QICs with proper structural supports can achieve and sustain results, even in the context of unprecedented health system disruptions.

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0511 / #1469 | Precision Medicine - Combining Information and Technology to Educate Patients and Health Professionals

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Background and Aims: Precision Medicine combines the data already used for diagnosis and treatment (signs, symptoms, personal and family history, as well as tests) with the person's genetic profile, currently being applied to pediatric oncology. Despite being widely discussed in the oncological context, it benefits few patients and its information is poorly accessible. In order to humanize health based on education, raise awareness and demystify the approach, as also to clarify and facilitate access to information on this new technology, an online mini-course was created to address this topic.

Methods: Beaba Institute, a non-profit entity for demystifying cancer and providing information about the disease and treatment, has developed the first Instagram mini-course in the health area with the theme of Precision Medicine, divided into 5 chapters: Precision Medicine, Precision Oncology, Genetics, Medicines and Glossary. Through co-creation, interdisciplinarity and Information Technology, the development was made with multi-professionals: oncologists working in genetics, psychologists, illustrators, art directors and technology developers.

Results: In the first 30 days, between February and March 2021, 561 users were reached with 1,097 visits and 387 followers with organic engagement. The profile of users according to the age group: 38% are between 25 and 34 years old, 31% are between 35 and 44 years old, 15% are between 18 and 24 years old and 16% are over 45 years old. According to gender, 89% of users are female.

Conclusions: The mini-course is an empowerment tool for patients and family members to make the decision shared with their health team, as well as it can be a way to clarify the possibilities of treatment in a given case. In order to democratize access to information, the mini-course is available for free on the Instagram profile @beabadamedicinadepre-cisao and does not have a deadline to be closed.

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0512 / #1486 | All Hands in to Improve Children's Lives: Innovative Approaches to Inspire Actions to Cureall

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Background and Aims: Since the World Health Organization (WHO) Global Initiative for Childhood Cancer launched in 2018, a common approach, CureAll, has been in development with global partners. Using four pillars (Centres of Excellence and Care Networks, Universal Health Coverage, Regimens for Management, Evaluation and Monitoring) and three enablers (Advocacy, Leveraged Financing, and Linked Policies/Governance), CureAll guides stakeholders to achieve the initiative's 2030 targets, increasing survival rates for six common childhood cancers to 60% globally, and saving one million additional lives while improving quality of life.

Methods: Supporting CureAll framework's launch in December 2020, St. Jude Global hosted 3 creative stakeholder engagement activities: a global photography storytelling campaign to answer "what does All Hands In mean to you?"; virtual sessions where participants co-created mind maps clarifying actions to CureAll using health systems thinking; and video vignettes with 12 WHO Headquarters/Regional Offices staff and country leaders in focus countries applying the CureAll approach.

Results: The photography campaign generated a collection of 20 stories from clinicians, foundations, hospital directors and managers from 17 countries (several featured in WHO events/materials). Two mind map sessions engaged 62 participants, generating 3 sub-maps across CureAll enablers that highlighted needed actions: advocacy for children's rights to health; stakeholder coordination; and national policy development and implementation for cancer control. Compiled vignettes were live screened to a global audience of over 105 participants as two video shorts, including a call for All Hands In expressed in 12 languages.

Conclusions: Photography storytelling, mind maps and videos can be innovative engagement and collaboration tools to cohesively narrate and personalize the CureAll approach of the Global Initiative for Childhood Cancer. Connecting goals with stakeholder emotions and inspiration can identify intersecting opportunities for impactful and sustainable actions to address childhood cancer as a global health priority.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0513 / #1504 | Cancer Diagnosis in Children and Adolescents in a World in Pandemic by Sars-Cov-2: Impact on the "Delayed Diagnosis" Experience at A Single Institution

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Background and Aims: The COVID-19 pandemic has disrupted childhood cancer care. The true impact of COVID-19 infection on children undergoing cancer treatment is at present unknown. The objective was to determine the impact on the time of diagnosis of childhood cancer during the pandemic.

Methods: A cross-sectional, retrospective cohort study in <18 years newly diagnosed with Cancer between February 2019 to January 2020(Pre-pandemic period) vs February 2020 to January 2021(Pandemic period) at Pediatric oncology department of Centro Medico Nacional "20 de Noviembre", department national referral and COVID-19 designated hospital. We compared the time to diagnosis in both periods. Time to diagnosis (TTD) was defined as time from initial symptoms to the start of treatment for cancer. Descriptive statistics was performed.

Results: During the pandemic period, 50 cases of childhood cancer were diagnosed compared to the pre-pandemic period in which 66 were diagnosed, reduction of 24% of cases, however, TTD was shorter during the pandemic period with an interquartile range of 61 against the interquartile range of 121 in the pre-pandemic period, observing a less dispersion in the TTD in the pandemic period, we found a decrease of 57% in the number of patients with brain tumors. The age group with the greatest decrease in the number of cases was 6 to 10 years. Cancer in advanced stages of the pre-pandemic period, 37 patients vs 26 in the pandemic period, not statistically significant.

Conclusions: In recent years we have worked on a TTD program, with a zero rejection policy, establishing direct contact with first contact doctors, during the pandemic we have established protocols in conjunction with hospital authorities to continue with the medical care of the pediatric patient with Cancer. There are reports that children with brain tumors represent a fragile population exposed to the indirect effects of the COVID-19 pandemic.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0514 / #1534 | Experience of using C5 Tool for Policy Change on Pediatric Oncology in Mongolia

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Background and Aims: The Mongolia pediatric hematology/oncology (PHO) has historically had little attention in policy and political arenas, with no approved PHO policy documents. Mongolia was one of the pilot countries to conduct collaborative priority-setting for PHO using C5's (Country Collaboration for Childhood Cancer Control) 5 modules. The in-country team included representatives from the ministry of health, clinicians, and civil society, who worked with St. Jude to propose national PHO priorities for planning and policy change.

Methods: The team worked through written and live group exercises and discussions to enrich data initially retrieved via desk reviews on Mongolia's health system (c1). Stakeholder mapping was conducted on service delivery institutions and governmental/non-governmental, professional, and training bodies (c2). These steps supported analyses of strengths, weaknesses, opportunities, and threats (SWOT) across the Mongolia health system (c3). Data from these first sections helped identify projects to address needs in module c4, with projects

prioritized based on the local team's ratings of political will, values, and timelines (c5).

Results: Collaborative efforts to identify PHO improvement priorities using C5 led to policy change and improved data. The team updated data across 35 dimensions of national health systems and PHO indicators. Maps captured treating facilities in the capital as well as 20 referring hospitals across Mongolia and 12 stakeholder organizations. 26 national projects were initially proposed to respond to 63 SWOT items. C5 results identified key areas for advancing the care of children with cancer in Mongolia across 6 health system dimensions, which were important for the development of the National Plan of Action on PHO for 2021-2023. This is the first comprehensive PHO policy in Mongolia, approved by Minister of Health Order.

Conclusions: C5's structured approach enabled engagement of stakeholders and was useful in capturing the national context. It facilitated comparisons with other countries and laid a foundation for developing PHO policy.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0515 / #1540 | **Collateral Damage of Covid-19; Challenges Faced during the First Wave of the Pandemic in India**

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Background and Aims: Pediatric cancers are highly curable when brought to hospital in time. The cure rate of pediatric cancers in low- and middle-income countries is much lower than expected. COVID-19 has brought many challenges into this system, thus widening the gap. We discuss challenges faced during the first wave of COVID-19 in India

Methods: The center caters to 75-85 new cancer registrations each year and has >400 children on follow up and >150 patients on active treatment at any point of time. Data recorded between March 2020 to February 2021 in this study. Patients were followed up in hospital and through teleconsultation during this period

Results: 82 new patients were registered, and 152 children were on active treatment during the study period. Telemedicine was offered to 58 children on ALL and APML maintenance and CML treatment. CBC monitoring was done locally, and medicines provided to them through CanKids NGO. 5/58 children discontinued treatment, of which one child resumed treatment after a break of 4 months. Two children were not traceable, and 2 others died presumably of ALL relapse. All new patients were treated as per standard protocol. Out of the 82 new registrations, 60 were ALL, 6 AML, 2 CML and rest solid tumours. 5 families refused treatment. 2 children died during induction because of late presentation due to strict lockdown. Chemotherapy was modified for few children with solid tumours. Interim maintenance added to ALL chemotherapy to reduce hospital visits. Surgery delayed for 3 children and radiotherapy delayed for 3 children

Conclusions: Its heartening that we were able to provide treatment to all patients despite hospitals being closed down due to COVID-19. Limited bed strength and healthcare staff, scarcity of blood products, delay in funding were crucial challenges for optimal management. NGOs played a vital role in bridging the gap

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0516 / #1548 | **Co-Designing Collaborative Health Systems Strengthening Priorities for Children With Cancer in Ukraine, Sri Lanka, and Mongolia using C5 Tool**

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Background and Aims: Engaging stakeholders to define priorities considering the national health system is vital to sustainably improve childhood cancer outcomes in low- and middle-income countries (LMICs). Based on a pediatric hematology/oncology (PHO) adapted Health Systems framework, C5 (Country Collaboration for Childhood Cancer Control) was developed with collaborators from 6 LMICs with self-administered modules. To support collaborative priority-setting, C5's 5 modules were adapted for sequential, team-based implementation in

a three-country pilot. In-country teams from Ukraine, Sri Lanka, and Mongolia representing ministries, clinicians, civil society, and public health professionals and administrators across 3 World Health Organization regions connected with St. Jude over 9 or more months to co-design national PHO priorities for strategic planning and refine C5.

Methods: Applying systems thinking, data was summarized and reconciled via triangulation, consensus discussions and online voting tools and activities to co-design summaries, including maps and matrices. Groups progressed through defining national context indicators (c1) and system-wide stakeholder mapping (c2), generating multi-level country summary maps. This supported national PHO strengths, weaknesses, opportunities, and threats (SWOT) analyses (c3), then used to create matrices of projects (c4), which in turn were refined based on political will, values, and timelines as Priority Collaborative Portfolios of key projects (c5).

Results: Iterative exercises enriched pre-filled/template data/graphics, including honing data sources, expanding sites/stakeholders, and defining indicators for strategic planning and advocacy. National maps inclusive of 20+ identified referring community-based sites were designed with each team, alongside depicting PHO burden and services' accessibility. Virtual platforms (Mural and Forms) facilitated real-time voting, visualizations, and discussions to refine SWOT analyses (200+ items) and projects (60+) into 5 or fewer prioritized elements per country.

Conclusions: C5 demonstrates a systematic participatory approach in identifying the collaborative landscape for PHO improvements. In-country teams with broad stakeholders are key to capturing national contexts, ensuring ownership, and defining priorities considering local values and political will.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0517 / #1607 | Impact of Covid-19 Lock-Down on the Emotional Health of Mothers of Children with Cancer in Pakistan

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Background and Aims: According to whom recommendation to reduce the viral transmission and implement better physical and social distancing, a strict lockdown started in Pakistan in Late March 2020. These measures adopted to contain the epidemic could have a significant psychosocial impact on the families with children having cancer. Not much has been explored about the impact of this epidemic on these mothers' emotional and mental health. Objective: To assess the extent and risk factors of psychosocial impact on these mothers during the phase of COVID-19 travel lockdown in Pakistan.

Methods: After 100 days of lockdown, interviews were conducted with these mothers using a survey questionnaire in Paediatric Haematology/Oncology at the Children's Hospital Lahore to assess the risk factors and depression among these mothers having children with cancer on active treatment in the department. CES-D (The Center

for Epidemiological Studies Depression Scale) score with a score of 15 used as the cut-off indicates a risk for depression. All data were analyzed using SPSS16.

Results: 36 mothers participated in this study. Of their children, 28% were on active treatment for Leukaemia, and 72% for solid malignancies. These families resided >100Km in 67% and <100KM in 33% of families, making their travel to the Primary Treatment center a significant challenge during a total lockdown. 34/36 mothers have scores above 15, and only two mothers have scores <15. 25/36 families did not have transportation to reach the PTC for chemotherapy sessions, and 50% have had severe HHM Household material hardships, 31% moderate, and 19% mild to none indicating their social distress. There was partial to a total shutdown of oncological surgeries ranging from 0-20% compared with April-June 2019.

Conclusions: Lack of transport, poor socioeconomic status, the anxiety of getting COVID-19, shutdown of diagnostic and curative services in the PTC could negatively impact the emotional health of mothers in this ordeal.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0518 / #1617 | Stress, Resilience and Moral Distress in Cancer Care Providers in the Time of Covid-19: The Children's Hospital Lahore Experience

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Background and Aims: The COVID-19 epidemic has a significant impact on the health system globally. During the national lockdown in the first wave in Pakistan from March 2020 onwards, the pediatric oncology health care providers underwent profound challenges to cope with the dynamics of health care service provision to these children with travel lockdowns, partial shutdown of outpatient, and surgical facilities. This study evaluates the early effects of the COVID-19 pandemic on the emotional health of oncology physicians and nurses and explores factors associated with stress resilience and moral distress.

Methods: This study was done in July to evaluate the impact of national lockdown from March-June 2020 on the HCP's working in the department of the Paediatric/Haematology/Oncology the Children's Hospital Lahore Pakistan assessing their stress, resilience, and moral distress using different scores during this unprecedented time of the global pandemic. The data were evaluated using SPSS.

Results: 85 HCPs were included in this study 74/85(87%) females with 1:6.7 M: F ratio and 78/85 (92%) < 40 years old and 19/85 (22%) doctors and rest of nursing staff 56/85 (67%) and paramedical and supportive staff 10/85 (11%). 54% single and 46% were married, and 37/85 (44%) having children <18 years old. Among them, 89% working inpatients, 5% in ICU, and 6% in ambulatory clinics. The majority of work is related to cancer programs in 93% HCPs. 72% showed high resilience and adaptability score. 62/85(73%) had higher stress with a

cutoff level of 12, and only 27% had a lower stress score. 20/85(23%) had moral distress above 2 (uncomfortable to worst possible).

Conclusions: The pandemic has brought significant emotional challenges for pediatric oncology HCPs. There is a need to provide continued psychological support to enhance their resilience and adaptability and reduce stress.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0519 / #1675 | Pediatric Cancer Patients with Covid 19: One Year Single Experience of West Java – Indonesia

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Background and Aims: Children with cancer who are ongoing chemotherapy are seemed to be at risk for exposure to SARS-CoV-2 infection and COVID-19. Moreover COVID-19 pandemic also has disrupted cancer care. This study aims to report the characteristics of COVID-19 in this population and also its impact on pediatric cancer care in West Java Indonesia.

Methods: This retrospective study in Hasan Sadikin General Hospital Bandung was conducted to report pediatric cancer patients undergoing active chemotherapy during April 2020 to March 2021. The subject were screened with purpose prior to hospital admission for chemotherapy or invasive procedures requiring sedation tested by nasopharyngeal swab PCR. Data was obtained from the Indonesia Pediatric Cancer Registry (Bandung branch) and Hospital Pediatric COVID-19 Registry. Data consisted of characteristics of demographic, clinical, and laboratory parameters. All confirmed positive of SARS-CoV-2 patients categorized into asymptomatic and symptomatic.

Results: Of 82 pediatric cancer patients tested for COVID-19, 26 were positive. Most cases were 5-10 years of age group (9/26), higher in males (19/26), and mostly with malnutrition (15/26). It was predominately acute leukemia (11/26). The main presentation was asymptomatic (19/26), whereas mild (2/26), moderate (3/26), critical infection (1/26), and MIS-C (1/26), respectively. The vast majority of cases with normal lymphocyte, while lymphopenia only exists in 2 cases. Persistent lymphopenia was noted in severe cases. The median NLR in symptomatic was 6.94 (0.42 - 7.37) while in asymptomatic patients 13.40 (0.60-14). Patients were treated by favipiravir in 3, antibiotics in 8, and oxygen support in 2 children. Of 6 deaths, four were not solely attributable to COVID-19, while two died during COVID-19 infection. All of the positive patients have been suspended and delayed in chemotherapy.

Conclusions: In this study we observed that most pediatric cancer patient with COVID-19 were mild. The impact of this pandemic caused suspended and delayed in treatment.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0520 / #1679 | Impact of Financial Support and Focussed Group Counselling on Treatment Abandonment in Children with Acute Lymphoblastic Leukaemia. Experience Over 22 Years from North India

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Background and Aims: Background: Efforts are being made worldwide to prevent treatment abandonment in cancer; however, no substantial data exists regarding the impact of these interventions from India.

Aims: To determine changes in treatment refusal, abandonment rates and its reasons in response to financial support and focussed group-counselling

Methods: A retrospective cohort-study was conducted at paediatric haematology-oncology unit, King George's Medical University, Lucknow, India among children<18years admitted with acute lymphoblastic leukaemia from 1995 to 2017. Based on interventions, study was divided into three phases; Phase-1 (1995-March 2003): Basic support, Phase-2 (April 2003-June 2009): Financial support (direct-costs) and Phase-3 (July 2009-2017): Financial (direct and indirect costs) and social support along with group-counselling. Phase-3 was further subdivided into phase-3a: group counselling and phase-3b: intensified group counselling. Baseline parameters, treatment refusal and abandonment proportions were compared over three phases. Parameters with $p < 0.1$ on univariate analysis were entered in logistic regression model with first phase as reference.

Results: Number of children registered for treatment during phase 1, 2, 3a, 3b was 176, 200, 360 and 305 respectively. Socio-demographic parameters were identical over phases except for a rise in rural population ($p=0.003$). Treatment refusal decreased significantly over time: 21% vs. 14.5% vs. 12.5% vs. 5.9% ($p < 0.001$), especially during phase-3b. Though no change was found in treatment abandonment during phase-2, abandonment significantly reduced in phase-3a (20.3%) as compared to phase-1 (30.2%), with the proportion of children abandoning, due to financial constraints, declining. Abandonment further reduced in phase-3b vs. phase-3a (11.1% vs. 20.3%) ($p=0.001$). After adjusting for variables, abandonment was found to decrease independently in phase-3 (a, b) as compared to phase-1 ($p_1=0.017$, $p_2=0.007$).

Conclusions: Though helpful, financial assistance unaccompanied by counselling may be insufficient to bring a radical change. Hence, parental counselling, emphasising on treatment adherence and the aftermaths of abandonment, is indispensable for preventing abandonment in semi-literate populations.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0521 / #1690 | Priority Setting in Childhood Cancer using a Novel Health Systems Framework in a Lower-Middle-Income Country**

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Background and Aims: In low- and middle-income countries like Sri Lanka where many health system gaps exist, prioritization of needs is essential to ensure efficient use of limited resources. C5 (Country Collaboration for Childhood Cancer Control) is a tool developed by St. Jude Global to help identify national priorities in childhood cancer through multi-stakeholder inputs.

Methods: C5 consists of 5 modules. For the first two modules, an online survey and live discussions facilitated multi-stakeholder review and updates of national health system indicators and identification of stakeholders. In module 3, a SWOT (Strengths, Weaknesses, Opportunities and Threats) analysis regarding childhood cancer was done through a visual workspace software. The fourth module enabled shortlisting of needs according to a framework adapted from WHO health system building blocks, identifying opportunities for local, regional and global collaborations. In module 5, priority projects were selected through rankings, visual matrices and discussions of time, local value and political will.

Results: Stakeholders participating in the project included representatives from paediatric oncology, paediatrics, pathology, surgery, cancer control, Ministry of Health, international agencies, charity organizations, and civil society. 38 hospitals (7 treating, 31 referring) and 13 other stakeholder organizations were identified and mapped to reflect their roles serving more than 700 children with cancer diagnosed each year. The SWOT analysis of the Sri Lankan system revealed twenty-seven strengths, seventeen weaknesses, ten opportunities, and eight threats. From 12 proposed projects, three were voted as having highest priority: 1) establishing a paediatric oncology nurses training

program, 2) establishing a hospital-based cancer registry, and 3) addressing medical products and technology shortages and stockouts. **Conclusions:** C5 is an effective way of identifying national priorities in childhood cancers. The success of the tool depends on effective stakeholder participation. C5 demonstrates how collaborative priority-setting is feasible using web-based surveys, virtual meeting platforms, and visual workspace solutions.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy***PV0522 / #1694 | Chronic Kidney Disease Risk with and Without Wilms Tumor in WAGR Syndrome**

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Background and Aims: WAGR syndrome is a disorder characterized by Wilms tumor (WT), Aniridia, Genitourinary anomalies, and Range of developmental delays. Approximately 50% of patients develop WT and renal failure has been estimated to occur in 36%-90% of those with WT and WAGR. Specific aims of this study were to evaluate the rate of chronic kidney disease (CKD) and clinical characteristics within a WAGR patient registry database.

Methods: We utilized self-reported data from a WAGR syndrome patient registry. CKD features included: proteinuria, focal segmental glomerulosclerosis, and kidney failure. Clinical characteristics included history of WT or nephrogenic rests (WT/NR) and the presence of cardiometabolic feature. The rate of features in the CKD group and no-CKD group were compared to determine the risk ratio (RR) for CKD by each clinical factor. The RRs for CKD were then evaluated within subgroups: those with history of WT/NR (+) and those without history of WT/NR (-).

Results: The cohort consisted of 73 participants (39 +WT/NR and 31 -WT/NR). Kidney failure was reported by 25% of all participants and 38.4% had at least one feature of CKD. The +WT/NR subgroup had more than two-fold CKD rate (51.3%) compared to the -WT/NR subgroup (22.6%). Cardiometabolic characteristics were associated with increased RRs for CKD and the highest RRs were observed in the -WT/NR group. Hypertension was associated with the CKD highest risk (RR 2.6-8.2) and approximately 3-fold risk was found for obesity.

Conclusions: Kidney failure and other features of CKD are common in all patients with WAGR syndrome. There is a two-fold overall incidence of CKD in those with a history of WT or NR. Cardiometabolic factors such as hypertension and obesity appear to increase the risk for CKD development. Patients affected by WAGR syndrome should be routinely monitored for kidney and cardiometabolic health.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0523 / #1733 | SARS-CoV-2 Persistence in Immunocompromised Children

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Background and Aims: The temporal dynamics of SARS-CoV-2 infectivity in immunocompromised children are unknown but have important infection control implications. We evaluated the length of time immunocompromised patients (ICP) remain positive for SARS-CoV-2 and identified factors associated with viral persistence.

Methods: We conducted a retrospective cohort study of SARS-CoV-2 positive ICP at a large quaternary pediatric hospital from March 2020-December 2020. Immunocompromised status was defined as primary immunodeficiencies, secondary or acquired immunodeficiencies due to medical comorbidities or immunosuppressive treatment. The primary outcome was time to first-of-two negative SARS-CoV-2 PCR tests at least 24 hours apart. Chi-square, Fisher exact, and Wilcoxon tests were used to compare demographic and clinical characteristics. Kaplan-Meier curve median event times and log-rank-sum tests were used to compare outcomes between groups. Subjects without 2 consecutive negative tests censored at the last test.

Results: Seventy-nine children met inclusion criteria. Median age was 15.5 years (IQR 7-18 yrs), 68% were male, 57% of children were white, and 48% were Hispanic/Latinx. Most (67%) were tested in outpatient settings, and 58% of children were asymptomatic. There were no significant differences in demographics, hospitalization, or testing indication between children with oncologic diagnoses and other immunocompromising conditions/medications. The median time to two negative tests was 42 days (IQR 25.0,55.0). Longer median duration of positivity (days) was observed with leukemia and lymphoma vs solid tumor diagnoses (49 [IQR 25-59] vs 31 [IQR 21-44], $p=0.06$), moderate/severe lymphopenia (absolute lymphocyte count (ALC) ≤ 1000) vs normal ALC (44 [IQR 25-59] vs 31 [IQR 26-45], $p=0.05$), with no statistically significant difference in symptomatic vs asymptomatic illness (44 [IQR 31-55] vs 33 [IQR 22-53] $p=0.3575$).

Conclusions: Most ICP with SARS-CoV-2 infection took >3 weeks to become negative. Longer time to negativity was observed in children with an oncologic diagnosis receiving chemotherapy and with a lower ALC.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0524 / #1750 | Delayed Diagnosis of Childhood Cancers and Health-Seeking Behaviors of Caregivers in Botswana

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Background and Aims: Previous research in Botswana showed that the average time to childhood cancer diagnosis was over 10 weeks. Understanding factors contributing to delays, such as caregiver health-seeking behaviors, can inform interventions.

Methods: Between June and December 2020, a structured questionnaire was administered to caregivers of children actively or previously under treatment for cancer in Botswana, supplemented by medical record review. Patient delay was defined as the time between recognition of symptoms by the caregiver and presentation to the health system, while healthcare delay was defined as the time from the first medical visit until diagnosis of cancer. Delays in care were analyzed according to demographics, health-seeking behaviors, cancer knowledge, and health information sources.

Results: Most caregivers interviewed were mothers (38/50, 76%), unemployed (36/50, 72%), with a minimum of high school-level education (44, 88%). The majority (37, 74%) reported visiting a healthcare facility within 6 days of symptom onset. Median healthcare delay was 75 days (inter-quartile range, IQR 30-154). Longer healthcare delays were associated with brain tumors (median 200 days) and children > 10 years of age (median 133 days), but not with maternal education. Health information sources used most often were healthcare facilities/workers (24, 48%), the Internet (13, 26%), and TV/radio (11, 22%). Caregivers believed that, or were unsure if, cancer was incurable (16, 32%), caused by eating the wrong foods (30, 60%), or caused by a curse or witchcraft (21, 42%).

Conclusions: Healthcare delay was longer than patient delay in this cohort. The COVID-19 pandemic could have influenced the time to diagnosis but was not evaluated in this study. Many caregivers endorsed misconceptions about cancer, despite high levels of education and previous counseling. Interventions aimed at the healthcare system, such as training of healthcare workers and improving the referral process, should be prioritized.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0525 / #1756 | Characterization of Solid Tumors in a Chilean Children Hospital. 2014-2018

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Background and Aims: In Chile, childhood cancer is the second cause of death in children 5 to 14 years old. Solid tumors represent 60% of cases, being this a very heterogeneous group. An early recognition and opportune referral to a specialized center is crucial so we can diagnose and treat the disease. We need to characterize solid tumors and train health professionals to promptly recognize them because a late referral contributes to morbidity and mortality. Objectives: To describe the distribution and clinical presentation at diagnosis, calculate incidence, survival curves, time between the start of symptoms and diagnosis, and evaluate differences in survival between metastatic and non metastatic tumors.

Methods: We evaluated patients with solid tumors treated at Roberto del Río Hospital, between years 2014-2018. We recorded clinical, laboratory and imagenological presentation at diagnosis. We calculated incidence, survival curves and time between start of symptoms and diagnosis. We created a data base that was analyzed with software Stata 13 SE.

Results: We evaluated 157 patients. We had an incidence of 78.2 cases per million people under the age of 15. The most common solid tumors were central nervous system tumors, extracranial germ cell tumors and lymphomas. Median age at presentation was 6 years of age. 15.2% had metastasis at diagnosis. Event free survival and overall survival were 71.8% and 79.6% respectively at 24 months from diagnosis for the whole group. There was no difference in time to diagnosis between metastatic and non metastatic tumors. There was statically significant difference in both event free ($p < 0.001$) and overall survival ($p < 0.01$) between metastatic and non metastatic patients.

Conclusions: Trough this research we were able to succesfully characterize solid tumors in our setting. There was no difference in time to diagnosis between non metastatic and metastatic patients. We demonstrate that metastatic tumors at diagnosis have worse prognosis than non metastatic tumors.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0526 / #1760 | **Socio-Economic Challenges of Afghan**

Parents of Children with Cancer: The Children's Hospital Lahore Pakistan Experience

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Background and Aims: Experience Background and Objective: There is no established pediatric comprehensive cancer care center in Afghanistan, forcing these families to travel to neighboring countries, including Pakistan, to seek treatment for their children, aggravating their battle against cancer. The research related to distress associated with living with cancer and psychosocial care needs is lacking in low-middle-income LMIC countries.

Methods: This is a mixed-method study with quantitative and qualitative collected through face-to-face interviews with Afghani parents

and caregivers of exploring their socio-economic challenges, financial and communication barriers, and the impact of this illness on their family and their hopes for a cure for these children.

Results: Families of 20 children included in this study with a mean age of 7.56 years, and 85% were males. 50% had Pushto and 50% Dari as their first language. 30% did not understand Urdu, while 55% having little and 30% no understanding of URDU, resulting in 85% unable to converse in Urdu. They have to travel 800 to 2300KM distance to reach the Children's hospital Lahore requiring 12 to 72 hours journey and 6000 to 30000 Afghan Rs for each trip resulting in 70% able to afford only one trip a year back home paid by relatives, neighbor or charity in 95%. These children were having ALL in 65% and solid tumors in 35%. Their family income ranged from none to 27000/month. 70% were accompanied by parents and the rest by grandparents. The expenses of remaining family members at home were sponsored by others 85%. Loans were taken by 95%, and siblings' schooling and emotional health were also affected by parents leaving homes with the sick child for indefinite times. 85% declared financial burden the most worrisome than communication and logistic barriers

Conclusions: These families need immense psychosocial support, and the need for parental support groups cannot be overemphasized

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0527 / #1770 | **Strategies to Support Treatment Completion in Children with Cancer in Low- and Middle-Income Countries: A Systematic Review of Cancer Treatment Abandonment and Adherence Interventions**

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Background and Aims: The disparity in pediatric cancer treatment outcomes between high-income countries and low- and middle-income countries (LMICs) is profound. Treatment abandonment (TxA), defined as an absence from curative treatment for more than four consecutive weeks, may be associated with other manifestations of poor adherence, and frequently leads to poor outcomes in LMICs. Different interventions have aimed to minimize TxA, recognizing its varied underlying contributors, including anticipatory interventions aimed at adherence. This systematic review characterizes the purpose, scope, feasibility, and outcomes of TxA and treatment adherence interventions in children with cancer in LMICs across the five WHO dimensions of adherence.

Methods: Two reviewers independently screened 4135 papers and extracted data on interventions to improve TxA and adherence to curative treatment for cancer in patients less than 20 years old in LMICs, from systematic searching of PubMed January 1, 2000 to December

31, 2020. Each intervention was categorized according to the WHO-defined five domains of adherence (education, psychosocial, care delivery, health systems, social/financial protection). Studies were assessed for heterogeneity and risk of bias.

Results: 40 studies from 21 countries met inclusion criteria, with most (85%) published after 2010; the lowest economy where interventions were conducted involved low-income settings in 11 studies, lower-middle-income settings in 23 studies, and upper-middle-income settings in 6 studies. Most studies (70%) depicted interventions addressing 2 or more domains, with 30% focused on education, 30% focused on psychosocial, 48% focused on care delivery, 50% focused on health systems, and 73% focused on social/financial protection.

Conclusions: Multi-dimensional interventions have been deployed in LMICs to improve adherence and minimize TxA in LMICs. LMIC interventions aimed at social/financial protection and health system strengthening are most commonly described in the published literature. While these interventions appear feasible, further work is needed to determine their efficacy and effectiveness in reducing TxA and improving survival outcomes.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0528 / #1795 | Quality of Life in Childhood Cancer Families Compared to Benign Hematological Illnesses

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Background and Aims: The quality of life of families with children having chronic illnesses is expected to be poor. The relationship between various childhood illnesses and quality of life is sparse in literature. The current study examined the associations disease, parental psychological distress and quality of life in children with blood diseases and cancers.

Methods: Cross sectional study of three groups families with diagnosis of pediatric cancer, thalassemia and hemophilia were selected for the study. ePROVIDE Clinical outcome assessment Parent report Ped-sQL 2.0 Family impact Module was used for assessment. It included 36 items comprising 8 dimensions. Emotional, social, cognitive functions of the family pertaining the illness were assessed.

Results: 50 families each of all 3 cohorts were enrolled in the study. Only one parent per family was interviewed. Age ranged from 1-18 years. Time from diagnosis ranged from 1 month to many years in case of thalassemia and hemophilia. Only those on active cancer treatment were included. Total score (62.8, 73.2, 68.7), parent health related quality of life summary of score (60.625, 68.75, 67.5) and family functioning summary score (68.75, 81.25, 85.935) were all significantly lesser in childhood cancer compared to thalassemia and hemophilia.

Conclusions: In spite of limited years of treatment, financial and psychosocial support provided to families, the QOL continues to be lesser in childhood cancer. Thalassemia and hemophilia, although are life-long illnesses are included in national policy and patients avail standard treatment without much difficulty. Development of a similar policy may aid families of children with cancer.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0529 / #236 | 60 Years of Childhood Cancer in the West Midlands Region

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Background and Aims: The West Midlands Regional Children's Tumour Registry is a specialist paediatric registry recording childhood cancer cases in the West Midlands region of the UK (representing 9% of the national population). Our aim was to produce an historical, regional overview of childhood cancer incidence and survival from 1957 onwards.

Methods: 7,913 cases were included; 4,411 males and 3,502 females, aged 0-14 years, resident in the West Midlands with a malignancy / benign CNS tumour diagnosed 01/01/57 - 31/12/16. Detailed statistics for 45 diagnostic groups were compiled. Survival by decade of diagnosis was calculated using Kaplan-Meier survival analysis. Age and sex-specific incidence rates per million and directly age-standardised incidence rates were calculated.

Results: Most diagnoses showed steady survival gains. Acute lymphoblastic leukaemia five-year survival was 8.9% [CI 4.6-15.1] during 1957-1966 but 91.8% [CI 88.1-94.3] by 2007-2016. Although not statistically significant, other diagnoses (notably NHL) showed decreased five-year survival between recent decades with survival during 2007-2016 at 77.4% [CI 66.4-85.9] but 85.9% [CI 74.7-92.4] during 1987-1996. Patients with CNS tumours showed improved survival overall. By 2007-2016 five-year survival for ependymoma, astrocytoma and medulloblastoma was 95.7% [CI 72.9-99.4], 86.9% [CI 80.3-91.3] and 68.0% [CI 52.0-79.7] respectively. Deaths within 1 year of diagnosis fell from 57.41% to 7.19% although 23.68% and 19.35% of deaths in patients with bilateral retinoblastoma and Hodgkin's disease occurred 20+ years from diagnosis, mainly from second malignancy.

Conclusions: Statistics were comparable to national data with a small number of diagnoses requiring further case-mix evaluation (notably NHL, neuroblastomas > 1 year and renal tumours). Patients with hepatoblastoma, ependymoma, astrocytoma and medulloblastoma experienced higher survival rates than those nationally, although small numbers and wide confidence intervals must be considered. Overall, steady, sustained survival improvements were seen, attributable

to advances in tumour biology, intensified multi-agent chemotherapy and new treatments. The importance of lifelong follow-up was clearly demonstrated.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0530 / #261 | Assessment of Early Death and Treatment Related Mortality in Sub-Saharan Africa: Report from Cancare Africa – Collaborative African Network for Childhood Cancer Care and Research

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Background and Aims: Estimated survival of childhood cancer is less than 20% in many countries in sub-Saharan Africa. Early deaths during treatment are common. A distinction between treatment related and disease related causes would guide interventions and requires reliable information about the absence of progressive disease at the time of death. The aim was to estimate the proportion of patients who died of treatment-related causes within three months of diagnosis.

Methods: We conducted a multi-centre prospective, observational cohort study in sub-Saharan Africa. Data were collected from September 2019 to March 2020. Children below 16 years with newly diagnosed cancer treated with curative intent were included and deaths within three months of diagnosis were evaluated. Data were abstracted in real time by trained personnel using standardised case report forms and confirmed by treating clinicians. The treating clinician's assessment of the cause of death and signs, symptoms and additional investigations of patients were documented and used to determine the cause of death.

Results: We included 252 patients (median age 6.0, range 0.2–15.0 years, 54% male). All data were complete. The most common cancer was Burkitt lymphoma (63/252, 25%). Fifteen percent of patients (37/252) died during the first three months of treatment. Of these 37 patients, at least 30 (81%) died of a treatment related cause.

Conclusions: Fifteen percent of patients in sub-Saharan African died within three months and 81% of deaths were considered treatment related. To improve survival, supportive care needs to be improved to be in balance with the intensity of treatment.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0531 / #265 | Microna Childhood Cancer Catalog (M3Cs) A Resource for Translational Bioinformatics toward Health Informatics in Pediatric Cancer

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Background and Aims: MicroRNAs (miRNAs) are small RNA molecule (≈ 22 nucleotide) that belong to class of non-coding RNA. They play an important role in gene regulation. Research has accumulated a large body of evidence implicating many miRNAs in the diagnosis, prognosis and risk of hematological and solid tumors of childhood cancer. Using bioinformatics, many online miRNA databases are freely available. The aim of these databases is to report predicted and validated miRNA-binding sites. No currently available miRNA database is designed to report the clinical utility of miRNA research in pediatric cancer to utilize miRNA research findings into the clinical side. That is why we have launched MicroRNA Childhood Cancer Catalog (M3Cs). This is the first specialized miRNA platform in childhood cancer. The principle of this platform is, based on the translational bioinformatics spectrum, to bring miRNA research into clinical significance in both pediatric cancer patient care and drug discovery and toward health informatics in childhood cancer. To achieve this goal, we have utilized two approaches; data-driven clinical significance and data-driven human pediatric cell line models.

Methods: M3Cs development passed with three phases: Phase 1: Literature Mining: It includes external database search and screening. Phase 2: Data processing that includes 3 steps: 1- Data Extraction, 2- Data Curation & annotation, 3- Web Development. Phase 3: Publishing: Shinyapps.io by Rstudio was used as web interface for deployment of M3Cs. M3Cs became online and can be accessed through <https://m3cs.shinyapps.io/M3Cs/>.

Results: For Data-driven clinical significance Approach, 538 miRNAs from 268 publications were reported in clinical domain while 7 miRNAs from 5 publications were reported in clinical & drug domain. For data-driven human pediatric cell line models approach, 538 miRNAs from 1268 publications were reported in cell line domain while 211 miRNAs from 177 publications in cell line & drug domain.

Conclusions: M3Cs is translational bioinformatics resource toward health informatics Pediatric Cancer

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0532 / #298 | Consequences of Maternal Preconceptional Radiation Exposure: The Risk of Hemoblastosis in Offspring

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Background and Aims: Background: Maternal preconceptional radiation exposure is considered as one of the potential risk factors for hemoblastosis in offspring. Mayak Production Association (PA) is the first nuclear production facility in Russia. Mayak workers cohort includes about 25% of female who is one of the few world's resources to estimate such risks directly. Aim: to assess the risk of hemoblastosis in the offspring of mothers who were exposed to preconceptive occupational radiation exposure.

Methods: We conducted retrospective study using nested case-control approach. Source cohort included all people who were residents of Ozersk located near Mayak PA during 1949-2009. Cases were defined as incident of hemoblastosis occurred in person under age 25 in 1949-2009 (n=81), controls were matched on sex, year of birth, parent's age at birth (n=324). The odds ratio (OR) and excess relative risk (ERR) per unit dose of external gamma exposure were calculated with a 95% confidence interval using the EPICURE program.

Results: Acute lymphoblastic leukemia was the most frequent childhood cancer. Average age at diagnose was 6 without sex difference. Dose range of female workers in both groups before conception was quite wide: 2.83-2797.8 mGy. The odds ratio OR 0.89 (0.54-1.46) indicated no relation between the level of maternal preconceptional radiation exposure and hemoblastoses in offspring. OR depends neither on the fact of maternal exposure nor on the accumulated dose of maternal preconceptional exposure. The ERR/Gy coefficients were insignificant in the analysis of the accumulated maternal dose. Non-parametrical analysis by dose categories did not show any elevated risk as well.

Conclusions: The dose dependence between maternal prolonged preconceptional external gamma exposure and the risk of hemoblastosis in offspring was not revealed. Cancer predisposition in offspring and the manifestation of the effect in the form of solid cancer at older age cannot be excluded, which requires continued monitoring.

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0533 / #303 | Creation of a Pharmacy Intervention Tool for use in a Pediatric Hematology/Oncology Program in Sub-Saharan Africa

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Background and Aims: Data from resource rich countries show that pharmacists have positively impacted cancer care through direct

patient care activities. Limited studies describe the role and impact of pharmacists in oncology care in low- and middle-income countries. The lack of data may be due to the lack of convenient data capture tools or software. The purpose of this project was to create a user-friendly tool for Global HOPE (Hematology/Oncology Pediatric Excellence) pharmacists in Botswana, Malawi, and Uganda to document clinical pharmacy interventions.

Methods: The tool was designed based on results of a comprehensive literature review and assessment of existing validated pharmacy practice tools. REDCap (Research Electronic Data Capture) hosted at Texas Children's Hospital was selected as a customizable, secure, web-based platform for capturing interventions. The tool was initially created as a document, translated into REDCap, tested by Global HOPE pharmacists and Pharmacy Advisory Council members, then finalized by the study team. End-users were trained via live webinar presentation with step-by-step instructions. Additional optimizations were made following a 1-month pilot testing period with final go-live in September 2020.

Results: The finalized tool captures key patient demographics, intervention specifics, duration of time spent on the intervention, and receptiveness of the medical team. It also serves as a pharmacovigilance tool to capture adverse event and toxicity data, with real-time notification of Global HOPE pharmacy leadership. Preliminary qualitative data indicates high user acceptance of the tool.

Conclusions: End users are receptive of the tool based on its simplicity and ease of use. This tool will allow us to describe the interventions made by Global HOPE Pharmacists and highlight the clinical role of the oncology pharmacist in Sub-Saharan Africa. In the future, we will survey the pharmacy team regarding their perception of the database to ensure it is a time-efficient, effective tool in daily practice.

E-Poster Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PV0534 / #360 | National Childhood Cancer Control Planning in Ukraine: Collaborative Implementation and Outputs of St. Jude Global's C5 Tool

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Background and Aims: St. Jude Global developed C5 (Country Collaboration for Childhood Cancer Control) as a structured tool to help in-country decision-makers identify priorities that strengthen health systems for children with cancer. After contributing to a preliminary self-administered C5, Ukraine was one of three countries selected to implement C5 as part of a team-based pilot. The objective was two-fold: to help the Ukrainian team develop the strategic childhood cancer control plan, and to test C5's usefulness and design.

Methods: From July 2020 through April 2021, the Ukraine and St. Jude teams completed C5's five modules sequentially. Through offline and online data collation, analysis and design exercises and discussions, the group generated: national context analyses; stakeholders maps; national Strengths, Weaknesses, Opportunities and Threats (SWOT) analysis; proposed projects; and prioritized pediatric hematology/oncology (PHO) needs and respective action plans to address them.

Results: The group mapped 35 hospitals and 15 stakeholder organizations impacting more than 1000 children with cancer annually, and generated a 75-item SWOT analysis spanning Ukraine's health system. Distilling 30 proposed projects via matrices considering political will, value and timelines, four priorities emerged: developing national PHO treatment regimens and guidelines; adapting international standards for unified PHO training programs; ensuring integrity and sustainability of the Population-Based Cancer Registry; and defining referral pathways and network sites for multidisciplinary care guided by capacity/resources. C5 outputs will be used for strategic planning and healthcare policy making by the Ministry of Health.

Conclusions: C5 is a useful and effective instrument to develop the national planning and advocacy strategy for childhood cancer in low- and middle-income countries. This process brings together multiple stakeholders and accumulates data that facilitates the prioritization and strategic vision of the needs to be addressed. C5 should be scaled up and offered to countries that want to improve the care and cure of children with cancer.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0535 / #366 | **Out-Of-Pocket Expenditure in Childhood Cancer during the Covid-19 Pandemic in an LMIC**

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Background and Aims: The covid-19 pandemic has put us in a situation of vulnerability not only in health but also economically. Pediatric cancer services have been significantly disrupted during the COVID-19 pandemic in Latin America, which has been more severe in countries with low healthcare expenditure. The aim of this study is to find out the out-of-pocket expenses assumed by parents of children and adoles-

cents with cancer in Peru during the COVID-19 pandemic and compare this to the pre-COVID era.

Methods: We conducted an anonymous cross-sectional survey of 235 parents and caregivers of all health services in Peru (MINSA, EsSalud, armed forces and police, private). The survey was distributed electronically through contacts obtained from hospital records and printed formats were also delivered to hospitals both in hospitalization and in consultation external for a period of one month. We also take care that the distribution of the survey is carried out to the different institutions that provide health services, both public and private. Foundations and parent committees help us with diffusion in Lima and in the regions.

Results: The average income per family prior to the pandemic was USD 334.40 (1,200 soles) and the average expenditure was USD 278.67 (1,000 soles). the average income during the pandemic was USD 278.67 (1000 soles) and the average expenditure was (USD 278.67) 1000 soles. The cost of having a child with cancer corresponds to 86% of the household's ability to pay before diagnosis and 75% after diagnosis.

Conclusions: Out-of-pocket health spending is a catastrophic expense for families, and it can put health and life at risk. We consider that specific policies for the protection of this vulnerable population are needed.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0536 / #368 | **ONCoped: Using Telemedicine during the Covid-19 Pandemic for Early Diagnosis of Childhood Cancer**

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Background and Aims: ONCoped is a smartphone application that allows the exchange of images and data between general practitioners and primary care pediatricians with pediatric oncologists. It was created in 2017 by the Association of Pediatric Hemato-Oncology of Peru (HOPPE) to promote the early detection of childhood cancer. Its development had the support of the Ministry of Health and international partners. Before the pandemic, ONCoped was used to carry out inter-consultations between primary care professionals and specialists regarding suspected cases of cancer and facilitate the transfer to Lima of patients to make the diagnosis and continue treatment. Since the beginning of COVID 19 Pandemic, the application was used also to follow up patients, now it has a new platform for parents, educators and healthcare professionals.

Methods: Thanks to a grant by the My Child Matters programme of the Sanofi Espoir Foundation and within the framework of the WHO Global Initiative for Childhood Cancer, in which Peru is one of the index countries implementing the initiative in the Region of the Americas, the use of ONCoped have been expanded to parents, teachers and non-medical staff to receive important information during the COVID-19

pandemic about the early recognition of symptoms and signs of childhood cancer and remote consultations.

Results: ONCOpeds has received 113 consultations from parents of children who have cancer. Thirteen physicians (pediatric oncologists) in Lima attend the consultation and establish a communication with the family or primary care health worker. More than 500 users have downloaded the app, which provide essential information to parents, teacher and health care workers.

Conclusions: Oncopeds was designed to promote early diagnosis of childhood cancer and helped specialists stay in touch with parents and primary care physicians who continued caring for them and is a tool to keep the care and follow-up of children beyond the hospital walls

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0537 / #398 | Invasive Fungal Infections in Immunocompromised Children: Novel Insight Following a National Study

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Background and Aims: To obtain a national overview of the epidemiology and management of invasive fungal infections (IFIs) in severe

immunocompromised children treated for acute leukemia or who had undergone an allogeneic hematopoietic stem cell transplantation (a-HSCT).

Methods: We performed a french national multicenter retrospective study to collect epidemiological data for proven and probable IFI in children with acute leukemia in first line or relapse treatment or had undergone an a-HSCT. We also conducted a prospective practice survey to provide a national overview of IFI management in pediatric hematology units.

Results: From January 2014 to December 2017, 144 IFI were diagnosed (5.3%) among 2721 patients: 61 candidiasis, 60 aspergillosis and 23 "emergent" fungi including 10 mucormycosis and 6 fusariosis. For IFI diagnosis, all centers used ultrasonography, computerized tomography-scan, and blood cultures. A vast majority of centers performed either a broncho-alveolar lavage (95%) to document IFIs in patients with pulmonary symptoms, either a biopsy (77%) to have a histopathological evidence of fungal elements. In practice, BD-glucan, galactomannan, and polymerase chain reaction were the most used biomarkers. In this national cohort of immunocompromised children, the higher IFI incidence rate arose in patients suffering from an acute myeloid leukemia (12.9%), appearing as a risk factor (OR 3.24, 95% CI [2.15-4.81], $p < 0.0001$). Interestingly, patients undergoing allogeneic hematopoietic stem cell transplantation had an IFI incidence of only 4.3%. For 83% of these patients, the use of primary antifungal prophylaxis, principally fluconazole, seemed to be associated with a lower IFI incidence rate (OR 0.28, 95% CI [0.14-0.60], $p = 4.90.10^{-4}$). However, these children largely presented IFI with emergent pathogens (41%), such as mucormycosis and fusariosis, resistant to this prophylaxis.

Conclusions: Emerging fungi and new antifungal resistance profiles uncovered in this study should enrich future reflection on IFI management in immunocompromised children.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0538 / #407 | Covid-19 Infection in Children and Young Adults Undergoing Hematopoietic Stem Cell Transplantation or CAR-T Cell Therapy

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Background and Aims: Patients who have undergone hematopoietic stem cell transplantation (HSCT) or chimeric antigen receptor T-cell (CAR-T) therapy have severe lymphocyte immunodeficiencies (prolonged B-cell aplasia following CAR-T). However, there is a paucity of data regarding COVID-19 in the setting of HSCT or CAR-T.

Methods: We retrospectively reviewed data from pediatric oncology patients at Children's Hospital Los Angeles who received HSCT or CAR-T and had PCR confirmed COVID-19 between February 2020 and March 2021.

Results: We identified 18 patients (1-21 years) with COVID-19 (6 CAR-T, 4 both CAR-T and allogeneic HSCT, 5 allogeneic-HSCT, and 3 autologous HSCT). The majority of patients were of Hispanic ethnicity (n=11). Half the patients (n=9) were overweight/obese. Fifteen had hematological malignancies. COVID-19 occurred at a median of 280.5 days following HSCT or CAR-T. Four of 18 (22%) had critical/severe COVID-19 by WHO criteria. These patients developed hypoxia and/or respiratory failure, three needed intensive care, and two had multisystem inflammatory syndrome (MIS-C). There were no COVID-19 related deaths. COVID-19 treatments included remdesivir (n=4), steroids (n=3), convalescent plasma (n=3), monoclonal antibody (n=1), and intravenous immunoglobulin (n=1). One patient (CAR-T, B-cell aplasia) had 6 hospitalizations due to persistent COVID-19. Eight patients had PCR positivity for >60 days. Three of the 7 patients who were <100 days following HSCT or CAR-T had severe/critical infection while one (CAR-T, B-cell aplasia) of 11 who were >100 days had severe/critical infection.

Conclusions: Our findings suggests a substantial rate (4 of 18 patients) of severe/critical infection and hospitalization/intensive care among COVID-19 infected patients with HSCT or CAR-T. These patients are potentially at risk for MIS-C, recurrent hospitalizations, and prolonged PCR positivity. COVID-19 infected patients within the first 100 days of HSCT or CAR-T, or those with B-cell aplasia may be at risk for severe/critical disease. Further studies will be needed to confirm findings from this limited sized cohort.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0539 / #451 | The National Survey on Experiences with Childhood Cancer Patients and Caregivers in Japan

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Background and Aims: Improving treatment outcomes and providing appropriate social services, such as educational opportunities for patients, financial support, and mental care for caregivers and siblings, are imperative for patients with cancer. Thus, the Japanese government has focused on these issues in their Third Cancer Control Plan. To evaluate the effects of current Cancer Control Activities, the current study sought to elucidate the experience of patients with childhood cancer in Japan.

Methods: The National Database of Hospital-Based Cancer Registries was used to identify patients diagnosed with cancer in 2014 and 2016, among whom those aged between 0 and 18 years old were selected.

Thereafter, a questionnaire asking about access to information, communication with medical staff, family's economic concerns, and education was completed by their caregivers. The average overall ratings of care, as well as the percentage of respondents for each question, was calculated.

Results: A total of 2,511 caregivers across 97 hospitals were invited, among which 49% responded. The average overall rating for medical care was 8.4 on a scale of 0–10. Among the 1,029 caregivers, 76% were satisfied with hospital staff communication, while 86% faced financial burdens, such as transportation or accommodation costs, although 99% received financial support for the treatment. Upon diagnosis, 513 patients were attending school, with most of those in elementary and junior high school (82% and 60%, respectively) transferring to hospital schools. Moreover, 62% of those in high school had to take a leave of absence, while 8.8% dropped out. Only 40% of caregivers stated the availability of suitable services for the patient's family.

Conclusions: The current study revealed that support for education and economic issues are especially needed. Additionally, necessary interventions that improve the quality of cancer care for children should be provided.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0540 / #479 | More Boys are Diagnosed with Cancer than Girls Globally: What are the Reasons of Gender Imbalance in Pediatric Cancer Incidence?

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Background and Aims: Boys and girls should not have different genetic predisposition to cancer unless sex chromosomes involved; no impact of sex hormones at these ages; no differential exposure to tobacco, alcohol and occupational carcinogens yet; children share the same environment and consume the same food. We assume incidence of cancer should be similar among boys and girls, and M/F close to 1.0. Previously we showed that more boys are diagnosed with cancer than girls, and here we analyzed subgroups to further evaluate gender imbalance.

Methods: We analyzed the global cancer statistics of the IARC. We extracted data for age groups and analyzed the data for country-based global income levels and for most common cancer sites among children.

Results: M/F was 1.24 globally and different among age groups (0-4 1.29; 5-9 1.37; 10-14 1.24 and 15-19 1.09); income of the countries (HIC 1.11; U-MIC 1.22; L-MIC 1.33 and LIC 1.22) and cancer sites (hodgkin 1.61, CNS 1.3, leukemias 1.39, NHL 1.68, kidney 1.02, liver 1.59 and thyroid 0.22)

Conclusions: Since the biology, lifestyle and environment of the boys and girls are more less similar in the childhood, there is no apparent reason to explain this difference. The imbalance was more evident among small children but disappeared at older ages. In L-MIC M/F was

higher; however in HIC M/F was 1.11, which could be explained by socio-economic discrimination against girls; which we suppose medical care is not sought for girls and many with cancer left undiagnosed. M/F differs significantly between various cancer sites as well. Lymphomas was 60% higher among boys whereas thyroid carcinoma was 4.5 times higher among girls. The incidence of cancers between boys and girls differs significantly without any apparent reasons to explain. Genetic, biologic and socio-economic factors seems to contribute into this imbalance and these factors needs to be further explored.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0541 / #492 | Pediatric Cancer in Tunisia, the State of the Art: About Data in Salah Azaiez Institute

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Background and Aims: Childhood cancers are rare and very different in nature and locations from those in adults. Salah Azaiez Institute is the reference center in Tunisia for the monitoring, diagnosis and treatment of cancers for all ages and particularly for children. The aim of this study was to analyze the epidemiological aspects of children treated at our institute.

Methods: We collected the files of children treated during the period between 1994 and 2015.

Results: The total number of children that were looked after at the institute during this period was 2062. The average age was 9.34 years with an interval varying from less than one year to 17 years; 6.5% of patients were infants, 23.5% were in infancy, 39.8% were in late childhood and 30.2% were adolescents. The majority of the children were male (58.8%). The most frequent tumor locations were the central nervous system (18.6%), hematologic malignancies (18.5%), the head and neck region (14%), the abdomen (13.9%) and bone tumors. (11.5%). The most frequent histological types for all ages were lymphomas (13.6%), especially Hodgkin lymphomas (11.2%). The distribution of tumors by age group was characterized by the predominance of abdominal tumors (35.1%) mainly of neuroblastoma type, followed by ophthalmological tumors mainly of retinoblastoma type (17.2%) in infants. For infancy, the two most frequent tumors were abdominal tumors (25.6%), mainly nephroblastoma type, and central nervous system tumors (19.6%). During the second childhood, it was tumors of the central nervous system (24.5%) with a proportion of 3.7% for glioblastomas, and hematologic malignancies (22.1%) with a predominance of hodgkin's diseases (14.9%). In adolescents, we noted a predominance of hematologic malignancies (20.7%) and bone tumors (18.8%) with a proportion of 6.1% for Ewing's sarcomas.

Conclusions: This is a study reporting the different epidemiological aspects of pediatric cancer in Tunisia. Another study evaluating the therapeutic characteristics in children would be interesting.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0542 / #500 | Delays in Seeking Care for Fever in Neutropenic Patients on Chemotherapy: A Retrospective Descriptive Study

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Background and Aims: Fever is found frequently in neutropenic children undergoing chemotherapy and can be the only sign of a serious infection. Despite instructions to seek medical care immediately after the first fever for an infectious workup and rapid administration of antibiotic treatment, patients often delay and/or attempt to manage fevers at home. This retrospective descriptive study sought to investigate possible reasons for families to delay instead of seeking immediate emergency care for neutropenic fever.

Methods: Charts were reviewed of LAC+USC pediatric oncology patients undergoing chemotherapy who presented for fever in the setting of neutropenia to LAC+USC Medical Center ER or clinic from 2015-2020. Whether a patient presented immediately after the first fever was the primary outcome, in addition to the time from first fever to presentation. Possible etiologies considered included primary language, distance, number of siblings, single vs. dual-parent home, single vs. multi-family home, number of prior neutropenic fever admissions, longest prior admission, chemotherapy intensity, and if the patient was in relapse.

Results: 80 charts were reviewed, and 41 charts meeting inclusion/exclusion criteria were abstracted. Delayed care after the first fever appeared to be associated with a history of > 2 prior admissions for neutropenic fever (odds ratio 4.78, 95% confidence interval 1.04-24.78). Additionally, patients who presented with cough were found in our study to be more likely to present to care immediately, whereas those patients with abdominal pain were more likely to wait for more than one fever.

Conclusions: A possible predictor of delayed care for neutropenic fever is a history of multiple prior neutropenic fever admissions; future studies evaluating this possible association with an a priori hypothesis may be illuminating.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0543 / #604 | Analysis of Melanoma Recurrence in Pediatrics: A Population-Based Study 1975-2017

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Background and Aims: To investigate, in a population-based study, the risk of developing disease recurrence (DR) after a diagnosis of primary melanoma in pediatrics.

Methods: Data on a cohort of patients with a diagnosis of primary melanoma as their first malignancy were extracted from the Surveillance, Epidemiology, and End Results US registry from 1975 to 2017. Observed-to-expected ratio of developing DR was calculated to estimate relative risk (RR) and associated 95% confidence interval (CI), which were compared to a reference population (RP) matched for age, sex, race, and calendar year.

Results: A total of 1,594 patients (631 males, 963 females) with primary melanoma were identified, 4.1% of whom developed second primary neoplasms (66/1,594). Mean age at melanoma diagnosis was 16 years old, while that at DR was 30 years old. There was a significantly increased risk of DR in melanoma patients compared to the baseline [RR: 2.28 (95% CI: 1.8–2.9); $p < 0.05$]. Superficial spreading melanoma is associated with the highest risk of DR [RR: 2.5 (95% CI: 1.7–3.6); $p < 0.05$]. Stratification by site revealed an increased risk of DR in the scalp and neck area [RR: 5.84 (95% CI: 2.8–10.7); $p < 0.05$]. Patients aged 10–14 years old experienced the highest risk of developing DR compared to the baseline [RR: 4.24 (95% CI: 2.2–7.4); $p < 0.05$].

Conclusions: Pediatric patients with primary melanoma experience an increased risk of disease recurrence. Patients aged 10–14 tend to have the highest risk of recurrence. The scalp and neck area is associated with the highest risk of recurrence. Pediatric patients who are diagnosed with primary melanoma should be followed with these risks in mind.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0544 / #616 | The Impact of Covid-19 Pandemic on Access to Treatment for Children with Cancer in India and Treating Centre Practices: Results of the Inpog-Acc-20-04 Study

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Background and Aims: The COVID-19 pandemic had led the Indian government to announce a nationwide lockdown on the 23rd of March 2020. This study (InPOG-ACC-20-04) aimed to explore the impact of this on the accessibility of care of children with cancer in India and to see strategies adopted by hospitals for service delivery during the lockdown.

Methods: Weekly average childhood cancer (<18 years) patient registrations during pre-lockdown period (Jan 1st 2020 to March 23rd 2020) were compared with the post-lockdown period (Mar 24th 2020 to May 31st 2020). The effect on the scheduled treatment was investigated for the post-lockdown period. A survey of health care providers was conducted to determine centres' strategies to deal with the effect of COVID-19.

Results: Of the 30 centres participating in this study, 1146 childhood cancer patients were registered from Jan 1st 2020 to May 31st 2020. The weekly average patient registration was 67.3 pre-lockdown and 35.5 post-lockdown which was a decline of 47.3% with travel distance being a factor. While most centres have experienced this decline, there have been a few who have seen an increase in patient registrations. Of those patients scheduled for treatment during the post-lockdown period, 36.1% experience delays in one or more modalities. Centres adopted several strategies to include modifications to treatment protocols, increased use of growth factors, and increased support from social organisations.

Conclusions: Our multi-center study from India suggests that the COVID19 pandemic and the lockdown impacted two out of three children with cancer. The effect of this on survival remains to be established.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0545 / #704 | The Paradox of High Social Vulnerability and Good Survival Rates in Chile's Public Health System: The Social Context of Children with Cancer

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Background and Aims: In Chile, 440–590 cases of childhood cancer are diagnosed annually; 84.7% are treated in public hospitals. Children's survival rate is close to 75%, as in developed countries. Although the explicit health guarantees system covers the treatment costs, additional elements often cannot be surrounded by the family. Aim: To

explore risk factors based on families' socioeconomic situation diagnosed with cancer in four pediatric oncology units of public hospitals.

Methods: Observational and descriptive study; the socioeconomic data from families were recorded by a social questionnaire (SQ). The SQ included five dimensions: i) sociodemographic children information, ii) parent's sociodemographic information, iii) Housing/environment, iv) participation/networks, and v) Spiritual. In 8 months, 104 new cases were surveyed. Ethics Committee approved this project. Quantitative data were tabulated for each hospital separately and then jointly at the total level.

Results: 99% of the children belong to the public health system, and 58% belong to the lowest income (US\$5 -US\$142 per capita). During treatment, 33% lived in a Shelter House. The caregiver is mainly the mother (91%), followed by the father (6%). The mother's educational level is secondary (42%); only 25% have completed technical/higher education. The family financial supporter is the father (56%), and his educational level reached is mainly secondary education. 70% reported good quality of housing with low overcrowding levels. 65% of families said that children share a bedroom, and one in four children share a bed. 32% reported using public transport during treatments. More than 30% reported not having internet access.

Conclusions: Despite family poverty levels, survival rates are high. The health system mediates as a protection factor. However, shelter houses, transports and connectivity provided by civil organizations are essential to mediate treatment access. Women's role as carer appears as a challenge in gender inequalities because of physical, mental, and socioeconomic consequences in the long term.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0546 / #755 | Implementing the Who Global Initiative for Childhood Cancer in Morocco: A Survival Study

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Background and Aims: In 2018, WHO launched the Global Initiative for Childhood Cancer aiming to reach at least 60% survival rate for children with cancer by 2030. Morocco was selected as a focus country but only little information was available about the current survival rate. We

report a national survival study conducted for the six WHO focus diseases, by the Moroccan Society of Pediatric Hematology and Oncology (SMHOP) in collaboration with the Moroccan Ministry of Health. The SMHOP was supported by the Sanofi Espoir Foundation.

Methods: We included children aged less than 15 years diagnosed from 1st January 2017 to 31 December 2019, in the six Moroccan PO centers for one of the six WHO priority diseases: Acute Lymphoblastic Leukemia (ALL), Burkitt Lymphoma (BL), Hodgkin Lymphoma (HL), Retinoblastoma (RB), Wilms tumor (WT) and Low grade glioma (LGG). We estimated 3-year overall survival (OS) for 878 eligible patients using Kaplan-Meier method.

Results: The most frequent cancer type was ALL (43.6%), followed by WT (15.8%), BL (15%), RB (12%), HL (11.3%) and LGG (2%). The one year overall survival (OS) was 80%, 2 years survival was 73.6% and 3 years survival 68.2%. The 3 - years was Event Free Survival 62.3%. Death was the most frequent event (14%) and the treatment abandonment rate was 4%. The 3-years OS was 88.8% for RB, 85.4% for HL, 81.2% for BT, 73.5% for LGG, 61.5% for ALL and 53.5% for BL. The difference was statistically significant. Patients aged less than 10 years and living less than 100 Km from the cancer center, had a significantly better prognosis.

Conclusions: Even though the number of cases reported was less than expected, this survival rate was used to define the 2030 survival rate goals for the Moroccan Childhood Cancer Plan. Setting up a national pediatric cancer registry was identified as a priority.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0547 / #778 | Pediatric Oncology Training in the Time of Covid-19 Pandemic: Report from the Francophone African Group of Pediatric Oncology

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Background and Aims: As an initiative within the Francophone African Group of Pediatric Oncology project, the African School of Pediatric Oncology created in 2015 an e-learning platform for the pediatric oncology's university diploma (Sanofi Espoir Foundation - My Child Matters program). This e-learning platform allowed us to continue providing training despite the restrictions related to COVID-19 pandemic. We report on the implementation of the platform and the progress since the start of the pandemic

Methods: The e-learning committee, a panel of multidisciplinary experts, contributed to the creation and maintenance of the platform and its adaptation to the African context. This platform was set up as an LMS (Learning Management System) with SPOC (Small Private Online Courses) online courses. To increase the platform's accessibility,

an e-learning mobile application was set up in 2019. The participants' satisfaction was assessed using a short survey to evaluate technical and pedagogical aspects of the platform and the mobile application.

Results: Since the start of the Covid-19 pandemic, the number of online courses has doubled. We were able to upload 41 courses and are in the process of developing new features which allow programming and recording online courses. At the moment, the e-learning platform counts 220 users and 223 downloads from android and IOS. 67 trainees surveyed "agree" or "totally agree" with the points related to the technical and educational aspects of the platform. Written comments included a desire to see an expansion of courses available on the platform, to have a discussion forum, and to expand the library section to include the research results of former DUCP candidates.

Conclusions: The platform initially developed as secondary learning tool quickly became the only training resource available during the pandemic. The feedback from the participants is highly encouraging and the experience should be replicated by other training programs

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0548 / #796 | Covid-19 in Children with Cancer in the Province of Quebec: 1 Year After the Pandemic

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Background and Aims: The clinical course of patients with COVID-19 who are undergoing cancer treatment is unknown. This study aims to describe the epidemiology and clinical course of COVID-19 in children with cancer and/or hematopoietic stem cell transplant (HSCT) recipients in the province of Quebec, the epicenter of COVID-19 during the first wave in Canada.

Methods: Baseline demographic and clinical data of children with cancer and/or HSCT recipients with confirmed SARS-CoV-2 infection by polymerase chain reaction in the four pediatric oncology centers in the province of Quebec were collected from a web-based registry.

Results: As of March 31, 2021, our cohort included 23 patients, representing 0.04% of the 60,827 reported pediatric COVID-19 cases < 20 years old in the province of Quebec. The median age was 11 [5-15] years, including 14 boys (61%). By cancer type, 10 (43%) had acute lymphoblastic leukemia (ALL), 1 (5%) acute myeloid leukemia, 4 (17%) brain tumors, 4 (17%) non-Hodgkin lymphomas, 2 (9%) neuroblastomas and 2 (9%) rare tumors. Four patients (17%) tested positive for

COVID-19 at cancer diagnosis, 12 (52%) while on treatment, 5 (22%) post-treatment and 2 (9%) post-HSCT. Ten patients (43%) were asymptomatic. Symptomatic patients had fever (n=8), cough (n=5) and dyspnea (n=1). Chemotherapy administration was delayed in 9 (39%) cases due to COVID-19, this delay lasted >2 weeks in 37% of cases. Two (9%) patients (2 and 15 year old), both treated for ALL, required hospitalization due to COVID-19. One presented with febrile neutropenia and the other with respiratory distress requiring non-invasive ventilation in the intensive care unit. There was no thrombo-embolic event, delayed multisystem inflammatory syndrome or COVID-19-related death.

Conclusions: In the province of Quebec, children with cancer and/or HSCT recipients generally had a benign clinical course of SARS-CoV-2 infection. However, COVID-19 induced several treatment delays, of which the long-term impact will need to be determined.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0549 / #961 | Implementation of Recommendations for Return of Research Results in the Children's Oncology Group

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Background and Aims: Background and aims: In North America, most pediatric oncology patients are treated according to a clinical trial at some point. The Children's Oncology Group (COG) has implemented a process for returning research results in lay language to patients, families, health care providers and the public via summaries posted on the COG public website. Benefits of returning results include recognition of participants' contributions to the progress resulting from research and provision of aggregate information potentially useful for future decision making. Return of results (ROR) aligns with ethical principles such as respect and beneficence. COG recommendations for the return of research results were previously published. This report describes implementation of those recommendations.

Methods: The multidisciplinary COG Return of Results (ROR) Committee collaborates with clinical trial team members to prepare lay summaries for phase I-III trials using a standard template. Summaries can be accessed on the website, once primary aim results are published, by keyword-guided search and/or pre-registering for notification when a summary is available. Patients and families are notified about the plan for posting of lay summaries at three timepoints: at initial study consent, end of therapy/transfer to survivorship clinic, and re-consent at age of majority. The ROR Committee oversees lay summary development and posting, monitors key metrics, and is responsible for continuous quality improvement.

Results: More than 30 lay summaries have been posted to date. There have been more than 1200 individual registrations to receive

summaries and more than 1900 visits to the website to download summaries. We have disseminated information about the ROR initiative within COG and at national nursing conferences.

Conclusions: Most of the recommendations have been implemented. Next steps include expansion of the lay summary portfolio, development of brief video education modules, and conduct of a survey to assess registrants' experience with the website in obtaining results.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0550 / #977 | The Unique Characteristics of Childhood Cancer Treatment Abandonment at an NGO-based Pediatric Oncology Unit in Syria

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Background and Aims: Treatment abandonment is the most important factor contributing to childhood cancer therapy failure in developing countries. No previous studies have focused on the characteristics of treatment abandonment in developing countries when treatment and transportation costs are financially covered for children and families. The aim of this study is to analyse the incidence and report the causes of treatment abandonment at a major referral NGO-based pediatric oncology unit in Damascus, Syria. Treatment abandonment is the most important factor contributing to childhood cancer therapy failure in developing countries. No previous studies have focused on the characteristics of treatment abandonment in developing countries when treatment and transportation costs are financially covered for children and families. The aim of this study is to analyse the incidence and report the causes of treatment abandonment at a major referral NGO-based pediatric oncology unit in Damascus, Syria.

Methods: Between June 2018 and February 2021, all newly enrolled patients at BASMA Pediatric Oncology Unit were included. Charts were reviewed retrospectively to identify treatment abandonment (SIOP PODC definition) in addition to causes.

Results: Among 736 patients, treatment abandonment was identified in 18 patients (2.5%). The incidence was higher in malignant bone tumors cases (n=9/91, 9.8%) compared to leukemia/lymphoma (n=2/151, 1.3%) and other tumors (n=7/494, 1.4%). The main reason for treatment abandonment was refusal of multidisciplinary surgical local control decision for bone tumors (7/18 [38%] for total cases and 7/9 [77%] for malignant bone tumors cases). Three out of 7 local control decisions were limb-salvage surgeries while the remaining 4 decisions were amputation.

Conclusions: Despite low incidence of treatment abandonment in our setting compared to other developing countries, refusal of appropriate surgical local control for malignant bone tumors remains a major contributing factor. Interventions are needed to be implemented in our clinical practice to solve this issue.

E-Poster Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PV0551 / #995 | Impact of the Covid-19 Pandemic on New Admissions and Clinic Attendance at a Paediatric Oncology Unit in Ghana

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Background and Aims: The SARS CoV-2 virus infection was declared a pandemic by the World Health Organization in March 2020. As at 15th April, 2021, Ghana had recorded 91545 cases. Among other devastating effects, the COVID-19 pandemic has disrupted health services globally, including posing challenges with access to and delivery of childhood cancer care. The aim of the study was to determine the effect of the pandemic on the number of new diagnoses and outpatient visits at the Paediatric Oncology Unit (POU) of Korle Bu Teaching Hospital (KBTH), Accra, Ghana.

Methods: A hospital-based retrospective study comparing data from March 2019 to February 2020 (pre-pandemic) and March 2020 to February 2021 (pandemic period) was conducted. Data were obtained on number of admissions, number of patients who received treatment and out-patient clinic attendance and transferred onto Microsoft Excel for analysis.

Results: A total of 400 children (age <18 years) were newly diagnosed with cancers at the POU, KBTH over the combined 2-year period – 191 pre-pandemic and 209 in the pandemic period. This represented a 9.4% increase in the number of new patients during the pandemic. There was also a 7.1% increase in number of patients on active treatment in the pandemic period (166 vs. 155). However, the average monthly outpatient clinic attendance declined by 9.0% from 222 in the pre-pandemic period to 202 during the pandemic (p-value: 0.046).

Conclusions: Despite the challenges of providing clinical care due to COVID-19, the POU at KBTH recorded an increase in the number of new cases of childhood cancers during the pandemic period. The decline in outpatient attendance may have been due to measures put in place to reduce crowding at the outpatient clinic such as longer appointment time intervals as well as increased use of telephone follow-up.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0552 / #1003 | Parental Perspectives on Healthcare Transition in Survivors of Pediatric Cancer

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Background and Aims: The healthcare transition (HCT) from pediatric to adult care can be challenging for childhood cancer survivors (CCS) and their parents. Parental readiness to transition is a key predictor of HCT success. However, there is little research examining parental needs during HCT. To investigate the needs of parents of CCS during HCT, we conducted an integrative review of the literature using the social-ecological model (SEM) of HCT. The social-ecological model is a theory- and data-driven model that has been previously applied to CCS transition readiness, and examines modifiable and preexisting barriers and facilitators for parents, patients and providers.

Methods: Studies included focused on cancer, HCT, and described the parental and/or survivor perspective. Web of Science, PubMed, PsycInfo, and CINAHL were searched using keywords for "survivors of pediatric cancer", "parents", "HCT", and "needs." The searches yielded 454 articles, including three hand-searched articles. Five articles were included in the final analysis. Quality was assessed using critical appraisal tools by the Joanna Briggs Institute.

Results: Data were extracted into nine factors derived from the social-ecological model. Articles were published within the last decade, largely qualitative, and mainly examined parents and CCS together. Parents most frequently mentioned relationships with their practitioner and CCS as contributing to HCT readiness, while abstract factors, such as goal-setting and expectations around HCT, were not mentioned.

Conclusions: Our results are limited by the dearth of research on this topic, the homogeneity of samples, and joint presentation of CCS and parent data. Nonetheless, our results indicate that parents do not weigh all aspects of the social-ecological model equally, with macro-level barriers, such as socio-demographic factors being viewed as less salient for HCT readiness. Parents mostly focused on interpersonal factors, such as their relationships with practitioners and CCS, indicating that practitioners should emphasize these in preparing parents for HCT.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0553 / #1048 | Adult Survivors of Childhood Cancer: Views on Covid19 and Vaccination

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Background and Aims: Adult childhood cancer survivors (ACCS) are at increased risk of developing late effects (LEs) as a result of their cancer and treatment, including cognitive delay, diabetes, metabolic

syndrome, and organ damage. Consequently, many ACCS may be at increased risk for worse outcomes with COVID-19 infection. It is important to determine ACCS views on the COVID19 pandemic and vaccination.

Methods: We obtained informed consent and emailed an online survey to 235 ACCS followed through the BC Cancer Late Effects, Assessment and Follow-up clinic, receiving 89 responses (37.9% RR) which were analyzed.

Results: Sixty-seven% of respondents believed that ACCS should be prioritised for vaccination, with 87% indicating they would receive a COVID19 vaccination if available. Seventy-seven% and 89% believed that COVID19 was a serious health problem for themselves or others, respectively. There were significant differences in views based upon education level and complimentary and alternative medicines (CAM) use. Education: Seventy-three% of respondents had post-secondary education (PSE); more patients with PSE believed that masks (92.3% vs 80.2%) and social distancing (90.8% vs 83.3%) were effective COVID-19 prevention measures. Respondents with PSE were less likely to always follow Health Canada guidelines (80% vs 91.7%). Respondents with PSE were less likely to use social media as a major source of news (87.5% vs 80%). CAM use: Twenty-nine% of respondents used alternative therapies (CAM). Patients using CAM were less likely to believe COVID19 would pose harm to them (69.2% vs 80.9%), believed less in the effectiveness of masks (80.8% vs 92.2%) and social distancing (84.6% vs 90.5%), and fewer claimed to always follow Health Canada guidelines (76.9% vs 85.7%).

Conclusions: Many ACCS appear to underestimate their risk from COVID19; PSE and CAM use appear to be correlated with these views. Future efforts toward assessment of ACCs' perspectives and how their views are determined, may help to modify attitudes regarding COVID-19 prevention.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0554 / #1077 | Factors Associated with Successful School Reentry of Childhood Cancer Survivors in Japan

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Background and Aims: In Japan, academic continuity is guaranteed during hospitalization, but there are many problems in school reentry of pediatric cancer patients after discharge of hospitals. Our aim is to reveal the factors associated with successful school reentry of pediatric cancer survivors.

Methods: We conducted a multi-institutional survey using self-administered questionnaires among 109 cancer survivors of ten years and over in 2019.

Results: A total of 109 responses were received from 42 patients and 67 guardians. Their diseases were as follows: hematological malignancies (58), solid tumors (15), brain tumors (18), and others (18). School ages at the reentry were as follows: lower grades of primary school (24), higher grades of primary school (36), junior high school (24), high school (10), and university school (6). Twenty-nine of 30 (97%) patients and 54 of 56 (96%) guardians reported their success of the school reentry, whereas three reported difficulties in school reentry. Univariate analysis showed that ages of junior high school at admission ($p=0.026$) and brain tumors ($p=0.007$) were significantly higher-risk factors for failure in school reentry. Ninety-three percent of the survivors had one or more advisers for school reentry. The reliable advisers reported were as follows; parents(17), close friends(5), doctors(11), nurses(12), previous school teachers(36), in-hospital school teachers(24), child-life specialists(CLS)(6) and medical social workers(MSW)(9). Previous school teachers were the most reliable advisers for parents and one of the top categories among those who reported success of school reentry.

Conclusions: This survey revealed that a special program would be needed to facilitate school reentry after cancer therapy for patients with brain tumors or at the ages of junior high school. It also showed less involvement of nurses as advisers for school reentry, although the nurses were supposed to be the most reliable staff for total care of the patients.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0555 / #1093 | Family Matters: Impact on Quality of Peer Relationships Over Time in Pediatric Brain and Solid Tumor Survivors

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Background and Aims: Parental acceptance, family emotional expressiveness, and general family functioning have direct influences on domains of social competence including peer relationships. Pediatric survivors of brain and solid tumors often find navigating peer relationships challenging. The aim of this study was to determine whether family functioning predicted was associated over time with quality of peer relationships during a 2-year follow-up in survivors.

Methods: Participants were youth diagnosed with either a brain or solid tumor and assessed at three time points (i.e., baseline, 12 months, 24 months) with the Family Adjustment Scale (FAD), the Attributions and Coping Questionnaire (ACQ)-Peers, and the PROMIS Pediatric Peer Relationships-Short Form (PPR-SF). A linear mixed modeling analysis was used, with PPR-SF as the dependent variable and FAD, ACQ-Peers at baseline, and their interaction as predictors. A between-person measure of FAD was derived by sub-

tracting the FAD grand mean from the person-specific 2-year FAD average.

Results: The mean age of survivors ($N=85$) was 10.88 ± 2.28 years with either a brain ($n=51$) or solid ($n=34$) tumor diagnosis. Greater family conflict during the 2-year period was associated with lower quality of peer relationships ($b = -0.29$, $SE = .09$, $t(73) = -3.20$, $p = .002$) and a higher tendency to respond to ambiguous social situations with sadness at baseline exacerbated this effect ($b = -0.17$, $SE = .08$, $t(73) = -2.11$, $p = .039$). However, this interaction term only approached significance when diagnosis type was included as a covariate ($b = -0.15$, $SE = .08$, $t(72) = -1.88$, $p = .064$).

Conclusions: Results suggest that family conflict and encoding/interpreting social cues constitute promising intervention targets. Clinical pediatric oncology teams should consider incorporating peer- and mood-focused screening and family- and survivor-focused psychosocial interventions designed to strengthen family communication and challenge negative automatic thoughts in an effort to promote the development of strong peer relationships for survivors.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0556 / #1186 | The Barriers and Facilitators to Physical Activity in Childhood Cancer Survivors: A Qualitative Interview Study with Parents

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Background and Aims: Childhood cancer survivors (CCSs) have an increased risk of cardiovascular complications in later-life. Physical activity (PA) levels influence cardiovascular risk, and evidence indicates that CCSs participate in less PA than peers without a cancer history. Although parents are key agents in their children's PA behaviour, little is known about how parents of CCSs view their child's PA. There is also little evidence regarding the factors which may act as barriers or facilitators to the PA behaviours of CCSs. This study aimed to explore what parents of CCSs perceive as the key influences on their child's PA behaviours.

Methods: Parents of CCSs were invited via outpatient clinics at two hospitals in England. Semi-structured interviews were informed by the Theoretical Domains Framework, a theory-based framework which enables the identification of key cognitive, affective, social and

environmental factors influencing behaviour. Thematic analysis was used to analyse the data, with codes relating to the barriers and facilitators to their child's PA participation, being mapped onto the Theoretical Domains Framework.

Results: A total of 22 parents were interviewed. Parents discussed several cognitive, emotional, social and environmental factors which, in their view, either hindered, or helped, their child's participation in PA. Key influences included the child's physical ability, the child's feelings about PA, how active their friends were, and their access to PA opportunities at home, school and in the local area. Parents also reflected on the important role of parental support and encouragement, and how they dealt with the specific challenges to their child's PA brought about by the experience of cancer.

Conclusions: The results of this study, along with findings from interviews with the CCSs themselves, will help to inform the development of an evidence-based and theoretically-informed intervention to promote, encourage and support PA for CCSs in follow-up care.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0557 / #1297 | **Run for Your Life: A Systematic Review of Exercise to Reduce Post Treatment Fatigue and Increase Quality of Life in Childhood Cancer Survivors**

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Background and Aims: Fatigue and poor quality of life is increasingly recognised in survivors of childhood cancer. Exercise is encouraged to decrease symptoms and increase quality of life. However the literature regarding this approach is slim. We wanted to provide a robust, evidence based platform to guide health care professionals, childhood cancer survivors and their families, when trying to minimise distressing symptomatology and to generate suggestions for future studies

Methods: A systematic review of MedLine, EMBASE, PubMed and SPORTdiscus identifying prospective studies, that employed validated tools, measuring the effect of exercise interventions on fatigue and Quality of Life in survivors of childhood cancer, aged under eighteen at diagnosis.

Results: Five hundred and fifty five papers were identified, two hundred and fourteen of which were screened following duplicate removal. Forty nine full text articles generated twelve studies of sufficient quality. Five randomised controlled trials were identified, the remainder were prospective observational studies. The largest had two hundred and twenty two participants, the smallest sixty eight. There were a total of six hundred and sixteen subjects and two hundred and sixty eight controls. Age range for intervention: 5-24 years. Statistically significant outcomes for fatigue and/or QoL and an overall positive outcome was seen in five studies. Statistically significant results but overall equivocal outcomes were seen in six studies. A negative

outcome was seen in one study. No studies reported a cost/benefit analysis.

Conclusions: Poor study design, a lack of standardised interventions and poor measurement of outcomes hampers research and the ability to give clear messages to childhood cancer survivors. A mixed methodology approach, employing recognised biomarkers, would define what interventions work well for children at different stages of development. COVID 19 has spurred interest in scientific evaluation of exercise intervention (UK Recovery Trial) and expertise and infrastructure from similar interventions worldwide should be harnessed for childhood cancer survivors.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0558 / #1350 | **Correlates of Meeting the Strength-Based and Combined Physical Activity Guidelines in Adolescent and Young Adult Survivors of Childhood Cancer**

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Background and Aims: Muscle-strengthening physical activity (PA) alone and combined with aerobic PA has positive effects on cancer survivors' physical functioning, fatigue and quality of life. Prior survivorship studies have focused almost exclusively on aerobic-based PA and few have examined engagement in the recommended strength-based PA needed to optimize health outcomes. This study aimed to examine engagement and correlates of meeting the strength-only and combined strength and aerobic PA guidelines in adolescent and young adult (AYA) survivors of childhood cancer.

Methods: AYA [n=379, Mage=19.9(SD=2.5), 52.8% female, 33.0% black, indigenous, people of color (BIPOC)] were recruited as part of a larger longitudinal study. At baseline, days/week of muscle-strengthening PA and days/week and minutes/day of moderate-to-vigorous aerobic PA (MVPA) were self-reported. Binary logistic regression models examined demographic (age group, sex, race/ethnicity) and medical (diagnosis type, age at diagnosis, length of treatment, time off treatment, late effects) correlates of meeting the Center for Disease Control and Prevention's strength-based and combined PA guidelines.

Results: On average AYA engaged in 1.9(SD=2.0) days/week of strength-based PA and 70.2(SD=59.0) minutes/day of MVPA on 3.4(SD=2.1) days/week. Less than half of AYA met the strength-based guideline (n=168, 44.3%) and only 109 (28.8%) met the combined guideline. Males and YA (age 18-25) were more likely to meet

the strength-only and combined guidelines compared to females and adolescents (age 16-17) ($p < .01$). When controlling for age and sex, hematologic cancer type ($\text{Exp}(B) = 2.05$, $p < .01$) and treatment length ($\text{Exp}(B) = 0.99$, $p = .03$) were associated with meeting the strength guideline. No medical variables were associated with meeting the combined guideline.

Conclusions: Just under half of AYA survivors are meeting the strength-based PA guidelines and even fewer are meeting the optimal combined guideline. Female survivors, adolescents and AYA who were treated longer or for a solid or neurological tumor may need additional PA support and targeted interventions to promote engagement in important muscle-strengthening PA.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0559 / #1358 | Impairment of Lung Diffusion Capacity in Survivors of Childhood Cancer: A Single Centre Study from a Low and Middle Income Country

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Background and Aims: With increasing survival rates in childhood cancer, late effects have assumed a greater importance in amongst this population. Derangements in pulmonary function, especially those associated with diffusion capacity of lungs is a well-known, yet an underestimated and underreported toxicity amongst children from low and middle income countries (LMIC'S). Our study reports the incidence and association of various risk factors for pulmonary toxicity in our cohort of childhood cancer survivors (CCS).

Methods: This cross-sectional study included CCS of various malignancies with either of exposures (chemotherapy agents like bleomycin, carmustine, high dose cyclophosphamide and methotrexate, busulphan, melphalan; irradiation to lungs/partial lungs or surgery involving Lung or mediastinum. Medical case records were reviewed for demographic details, disease characteristics and treatment exposures. The primary outcome was determined by Pulmonary function test, and was interpreted as pulmonary dysfunction, defined by GOLD criteria, and Diffusion Capacity of Lung for Carbon Monoxide (DLCO corr $< 75\%$ of predicted [grade ≥ 2 per CTCAE 4.03])

Results: Of a total of 500 CCS, 200 received pulmonary toxic therapy and 152 underwent PFTs. Mean age at the time of PFT was 21 years (10-41 years) and the mean time from diagnosis was 12 years (2-24 years). Although no patients reported respiratory symptoms, 10% (15/152) demonstrated a restrictive ventilatory defect, 7% (11/152) had airflow obstruction, 17% (26/152) had reduction in diffusion capacity and 1 patient had a moderate mixed obstructive/restrictive abnormality. Interestingly initial diagnosis with Germ cell tumor (GCT) had highest incidence (60%) of impaired DLCO, followed by Hodgkins lymphoma (17%) and acute lymphoblastic leukemia (11%) ($p < 0.005$). Risk factors for impaired DLCO included,

male sex, age at follow-up (> 18 years) and those who received both exposures (bleomycin and thoracic surgery) ($p = 0.006$, $p = 0.05$, $p = 0.01$).

Conclusions: Patients with GCT and Hodgkins lymphoma, and those receiving bleomycin and thoracic surgery were important predisposing factors for impaired lung diffusion in CCS. Hence these patients will need to be monitored for pulmonary disease to permit early interventions.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0560 / #1459 | Assessing the Needs of Survivors of Childhood Cancer: An Integrated Clinical and Research Screening Tool

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Background and Aims: Survivors of childhood cancer are at risk for a variety of medical and psychosocial late effects from their treatment. Risk-based screening is an important part of their appropriate after-care. The current study describes compliance rates of an online screening tool developed for a childhood cancer survivorship program as well as prevalence rates of the most commonly cited health problems.

Methods: The Alberta Children's Hospital Long-Term Survivor Questionnaire has been developed to guide survivors' clinical follow-up appointments. Survivors or parent-proxies are asked to complete the form via the secure online survey manager Research Electronic Data Capture (REDCap) prior to every appointment. Questions include current health problems, health behaviours (i.e., exercise, alcohol use) and standardized measures psychosocial wellbeing (i.e., anxiety, depression). Following completion of the questionnaire, patients were asked if they would like their data included in a research database.

Results: A total of 701 survivors are enrolled in the survivorship clinic and attend appointments at least once annually. Of these, 9.1% were stratified as low-risk for late effects based on disease and exposure risks, 45.0% medium, and 44.8% high-risk. From January to April 2021, 65.6% of patients completed the online screening tool ($n = 92/147$). Of those, 63.7% ($n = 58$, mean age = 18.4 years $SD = 7.3$, 42.1% male) consented to being included in the research database. The most common diagnosis group was solid tumour (40.8%) followed by leukemia (28.6%). Frequently reported complaints include headaches (33.3%), sleep problems (21.1%), anxiety (21.1%), and depression (19.3%).

Conclusions: Compliance rates for questionnaire completion are acceptable. Future work should aim to enhance compliance. Survivors report physical as well as psychosocial health problems. Ultimately we will establish a comprehensive, longitudinal database of outcomes that will be linked with demographic, diagnosis and treatment exposure data. The database will allow us to capture the burden of childhood cancer among survivors in Alberta, Canada.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0561 / #1498 | Gender Discrepancies in Risk Factors Associated with Bone Mineral Density in Long Term Childhood Acute Lymphoblastic Leukemia Survivors

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Background and Aims: Cohorts of childhood acute lymphoblastic leukemia (cALL) survivors reaching adulthood are increasing. Approximately 30% of cALL survivors are known to meet criteria for low bone mineral density (BMD) (Z-score < -1) 10 years post-diagnosis. Our objectives were to establish risk factors for low BMD in order to identify high-risk ALL survivors who will benefit from long-term follow-up.

Methods: We recruited 245 (50,6% female) long-term cALL survivors from the PETALE Cohort, who were treated with the Dana Farber Cancer Institute protocols, and did not experience disease relapse nor received hematopoietic stem cell transplant.

Results: Median age at recruitment was 21.7y (range 8.5 – 41y) and median time since diagnosis was 15.1y (range 5.4 – 28.2y). All participants underwent DEXA scans at the lumbar spine (LS-BMD), femoral neck and total body. Prevalence of low BMD (Z-score < -1) ranged between 21 and 22% depending on the site. Males exhibited significantly lower BMDs than females at the lumbar spine and total body sites (LS-BMD: 29% vs 15%). For all participants, low BMI was associated with decreased BMD. Treatment-related factors such as cumulative glucocorticoid (GC) doses and cranial radiation therapy (CRT) correlated with lower BMDs in both the full cohort (GC: $p=0.03 - 0.002$, CRT: $p=0.01 - 0.001$) and in females (GC: $p=0.02-0.05$, CRT: $p=0.001$). Lower levels of physical activity correlated with low BMDs in the female group ($\beta=0.1-0.2$, $p=0.02-0.003$).

Conclusions: This study emphasizes on gender-associated risk factors for low BMD in cALL survivors at the age of reaching maximal peak bone mass. Males ALL survivors are more affected than females, with cumulative GC doses, CRT, low levels of physical activity and low BMI as risk factors for lower BMD. Longer follow-up of BMD through time in this population is needed to establish if they exhibit a steeper BMD loss curve than aging healthy adults.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0562 / #150 | Physical Impairments and Activity Limitations of Childhood Acute Lymphoblastic Leukemia Survivors: A Petale Cohort Study

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Background and Aims: Long-term musculoskeletal complications represent a growing burden for childhood acute lymphoblastic leukemia (cALL) survivors. The aim of this study was to describe the physical impairments and activity limitations in cALL survivors of the PETALE cohort.

Methods: In this cross-sectional observational study, participants were recruited from the PETALE cohort (cALL long-term survivors treated with Dana Farber Cancer Institute protocols 87-01 to 05-01 who did not relapse or receive hematopoietic stem cell transplant). This sample aimed to recruit survivors at high risk of multiple morbidities studied in the PETALE cohort. Criteria for experiencing bone morbidities such as osteonecrosis (ON) and fractures were high risk of disease relapse (HRR), age at diagnosis >10 years and vertebral fracture. Survivors underwent magnetic resonance imaging of the hips, and standardized assessment by a physiotherapist including isometric maximal strength, the Five Time Sit-to-Stand Test and the 6-Minute Walk Test. Descriptive statistics were used to characterize physical impairments and activity limitations.

Results: The mean age of survivors (SD) and time since diagnosis (SD) were 24.3 (6.7) and 16.4 (5.7) years, respectively ($n=100$, 50% male, 73.2% HRR). Thirteen (14.1%) of the 92 survivors who underwent magnetic resonance imaging of the hips had hip ON (53.9% male). Survivors had lower values in muscle torque (hip abduction, ankle dorsiflexion) and isometric maximal strength (knee extension), and lower performance in Five Time Sit-to-Stand Test and 6-Minute Walk Test compared to reference values. Female survivors with hip ON showed lower hip abduction torque than those without ON ($p<0.05$).

Conclusions: Long-term cALL survivors in our cohort had greater physical impairments and activity limitations compared to expected values referenced from a healthy population. Sex differences emerged. These data contribute to the frailty phenotype often described in childhood cancer survivors.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0563 / #1522 | Developing a Cohort for a Multisite Longitudinal Study of Self-Management of Adolescent and Young Adult Survivors of Childhood Cancer: Recruitment, Retention, and Covid-Related Adjustments

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Background and Aims: We are creating a multisite longitudinal cohort of adolescent and young adult childhood cancer survivors (AYA) to study patterns of and predictors of self-management. We describe recruitment and retention, noting challenges during COVID-19.

Methods: The cohort will be 600 AYA (16-25 yo, > two years off treatment), and a caregiver, recruited within 2 years from cancer centers in Philadelphia, Cincinnati, and Los Angeles. At baseline (T1) and annually for two years, AYA complete measures of self-management. Measures of associates of self-management are included at T1 as well as caregiver report of perspectives of patient and family. Those who transfer to adult follow-up care complete an additional survey. Measures of COVID-related behaviors and exposure and impact were added. Because of COVID, recruitment was temporarily on hold, with much recruitment now taking place via phone.

Results: Since 2020, of 491 AYA approached thus far, 89% consented and, of those, 86.7% completed baseline (Mage=19.92, 48.17% male, 66% Non-Hispanic White, 52.9% liquid cancer). Those who did not complete T1 are more likely male and BIPOC. With half of the sample recruited by phone due to COVID, there is currently an 86% consent rate and 85.7% completion rate versus rates of 92% and 81% for in person. Of the 64 eligible for T2 thus far, 75% have completed (with many pending). Only one of 29 whose care was transferred to adult care have not completed their transfer survey. Caregiver recruitment for the 2 sites recruiting mostly by phone is 40% versus 80% for in person.

Conclusions: The recruitment and retention of a sample of 600 diverse AYA within two years is proving feasible, despite COVID-19 setbacks. Recruitment via phone has not significantly impacted consent or completion of AYA, but has impacted the ability to approach caregivers. Retention challenges are likely at T2 and T3.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0564 / #1523 | Implementation of Universal Cancer Predisposition Screening in the Pediatric Oncology Population

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Background and Aims: Growing availability of gene sequencing has increased awareness of pediatric oncology cancer predisposition syndromes (CPS). About 8.5% of pediatric oncology patients have a CPS. This study investigates facilitators and barriers of pediatric oncology CPS testing while providing patients the opportunity to receive genetic counseling and testing.

Methods: This study of pediatric oncology patients uses surveys administered at study entry, after genetic counseling, and after CPS testing with the Invitae™ Multi-Cancer Panel, an 84 CPS gene panel. Parents and patients > 15 years of age take the surveys. The surveys assess demographics, health literacy, cancer diagnoses, and the knowledge, attitudes, facilitators and barriers regarding CPS screening. After genetic counseling, patients are offered the CPS genetic testing. If a pathogenic CPS mutation is found, first-degree relative testing is offered. After the completion of all testing, the final survey evaluates knowledge and attitude developments during and satisfaction with the process.

Results: The study has 110 patients enrolled. From the 87 individuals whose genetic test results have been obtained, 37% had negative results, 54% had variants of uncertain significance and 8% (7 patients) tested positive for likely pathogenic cancer predisposition mutations. This statistic is consistent with the NCI PDQ estimation of 8.5% of pediatric oncology patients having a genetic cancer predisposition. The surveys found that 95% of parents and 100% of surveyed patients wanted patient testing for CPS. There was a correlation between high self-reported CPS understanding and strong interest in CPS testing. Confidentiality and privacy concerns were present for 42% of parents and 41% of patients. Concerns about CPS testing lessened after genetic counseling.

Conclusions: There is high interest and knowledge about CPS among the surveyed pediatric oncology patients and their parents. Patient and parent opinions trended together. Exit surveys showed positive satisfaction and expectations regarding CPS screening.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0565 / #159 | 'A Different Person Entirely': Adolescent and Young Adults' Experiences Returning to Education after Cancer Treatment

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Background and Aims: Adolescent and Young Adults (AYAs) establish their independent, adult identities as part of their psychosocial development, a process that is largely informed by educational experiences. Not only is a cancer diagnosis disruptive to this process but AYA cancer survivors (AYACs) face barriers as they attempt to reintegrate into educational systems. This study explores the experiences of AYACs as they return to education, to identify these obstacles and the implications for care teams.

Methods: In-depth semi-structured interviews were conducted with AYACs (n=8), aged 16-19 at diagnosis and 18-27 at time of interview. Interviews were transcribed verbatim and analysed using the principles of Giorgi's phenomenological analysis.

Results: Four major themes were identified: AYACs suffer from debilitating late effects (theme 1) post-treatment as they adjust to a loss of normality and other fundamental losses (theme 2) associated with a cancer diagnosis, such as irrecoverable future plans. The educational systems (theme 3) to which they return can be both accommodating, capable of making allowances, and uncompromising, unable to adapt to AYAC survivors' needs. Appropriate mechanisms to facilitate resilience (theme 4) amongst AYACs are vital for successful return to education.

Conclusions: This study supports previous findings that late effects and systemic barriers can hinder return to education but further research focused on this age group is required. We believe that treating clinicians and specialist services can facilitate the return of AYACs to education by providing warning and comprehensive information about late effects, as early as possible prior to treatment completion, as well as effective information-sharing with educational institutions.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0566 / #1641 | The Risk of Radio-Induced Valvular Heart Disease after Childhood Cancer: Contribution of Dose-Volume Histogram Indicators

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Background and Aims: Childhood cancer survivors are at increased risk of developing valvulopathies. Despite the large size of voxelized dosimetric data currently provided by individual dose estimates in radiotherapy, most studies are limited to single variable approaches

such as mean dose to the heart to study its relationship with the risk of valvulopathy. Therefore, we data from 7670 five-year survivors of the French Childhood Cancer Survivor Study (FCCSS) cohort, to investigate the potential predictive capability of dose-volume histogram indicators in the risk of valvulopathy after childhood cancer.

Methods: Individual dose volume histograms for the whole heart were obtained, and mean heart dose was calculated, as well as the doses (Gy) delivered to the v% of the heart volume (D_v in Gy), and the volume percentages of heart receiving at least d Gy ($V_{\geq d}$). Their role in the occurrence of valvulopathy was investigated using the Cox proportional hazard regression model or penalized Cox regression.

Results: Eighty two patients had developed a severe valvulopathy (grade ≥ 3). Patients treated with radiotherapy had an approximately 2-fold (95% CI: 1.16, 3.42) risk increase. The risk for of valvulopathy increased 12-fold (95% CI: 7.02, 21.83) when mean dose to the heart was over 20 Gy. The risk increased 30-fold (95% CI: 16.07, 58.8) as the volume having received $\geq$ Gy increased. Exposure to chemotherapy seems to increase by almost 2-fold the risk of valvulopathy in most of the alternative adjustments. Multivariable approaches seem to provide better predictions than the binomial model, but overall the model that studies irradiation dose-effect relationship adjusted on the mean dose to the heart appears to be the closest to the true model according to the AIC.

Conclusions: Findings may be useful for patients and doctors both before treatment and during long-term follow-up for VHD in survivors of childhood cancer.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0567 / #1701 | Dispositions of Reproductive Behavior in Families with Children Completed Special Treatment of Brain Tumors and Leukemias

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Background and Aims: Heterogeneity of cancer in terms of tumor localization, treatment, and toxicity makes it necessary to develop effective targeted family rehabilitation protocols based on specific impact of nosology on various areas of family life, including reproductive behavior.

The aim was to study the dispositions of reproductive behavior in families with children who finished the treatment of brain tumor (BT) or leukemia (ALL).

Methods: 639 mothers of children with ALL and 80 with BT, aged 25-45 (median 36), completed special 98-items questionnaires.

Statistical processing was performed in the SPSS 17.0 program using nonparametric methods. The secondary analysis of the all-Russian sample study was used as a control.

Results: The number of born children was considerably higher in the studied cohort than in control - 1.75 vs 1.28, $p < 0.01$, however in BT group significantly lower than in ALL - 1.5 vs 1.78, $p < 0.01$. Average desired number of children in BT and ALL groups - 2.62 vs 2.64; expected - 2.03 vs 2.08, correspondingly. Cancer in a child seriously affected the family planning - 52.5% in BT vs 37.6% in ALL, $p < 0.01$; the most of families wanted a newborn - 82.5% vs 78.4% but postpone the childbirth - 55% vs 47%; gave birth to another child after anti-tumor treatment - 10% vs 16% in BT and ALL, correspondingly. The additional analyses of birth motives showed that leukemia in children causes fewer irrational fears connected with cancer and myths over it than BT.

Conclusions: The results reflect less positive prognoses of survival and quality of life in children with brain tumors. Specific factors associated with the type of malignant tumor in a child do not play a decisive role, but can contribute to the total number of children in the family. It is necessary to provide medical, social and psychological assistance to parents at all stages of antitumor treatment.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0568 / #1734 | Latent Profile Analysis to Describe Outcomes of Survivors of Pediatric Central Nervous System Tumors

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Background and Aims: Survivors of pediatric central nervous system (CNS) tumors often experience multiple persistent and clinically significant late-effects following treatment completion. New approaches are needed to identify those experiencing significant morbidity. The current cross-sectional study aimed to 1) identify subgroups of survivors of pediatric CNS tumors experiencing similar patterns of global health, cognitive functioning, and psychosocial outcomes and 2) determine risk factors related to subgroup membership.

Methods: Participants included 70 survivors of CNS tumors (52.9% female, 65.7% non-Hispanic White, median age 12.5 years). Parents/legal guardians (90% mothers) completed Patient-Reported Outcomes Measurement Information System (PROMIS) parent-proxy measures as part of a larger, ongoing longitudinal institutional cohort study of childhood cancer survivors. Latent profile analysis (LPA) identified subgroups of participants reporting similar patterns of functioning across five PROMIS measures. Logistic regression analyses examined characteristics associated with identified profiles.

Results: Two distinct profiles were identified. Profile 1 contained 55.7% (n=39) of participants reporting functioning similar to peers

with lower levels of depression. Profile 2 (n=31) included those reporting poor global health with high levels of anxiety, depression, and fatigue. Females and those with high grade tumors were more likely to have impaired functioning (Profile 2). Controlling for sex and treatment, those with high grade tumors who had sub-total resection only (9.32 (95% CI: 1.07, 81.06)) or gross total resection (GTR) with chemotherapy and/or radiation (7.17 (95% CI: 1.18, 43.55)) were more likely to have impaired functioning compared to those with low grade tumors undergoing GTR only.

Conclusions: Innovative methods, such as LPA, can summarize multiple outcomes and identify subgroups with similar profiles within the larger survivor population. Consistent with prior literature, survivors with higher grade tumors were more likely to fall into the impaired group. This approach may guide risk-prediction models for both prevention and intervention efforts in current and future research with pediatric cancer survivors.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0569 / #174 | Long Term Morbidity and Mortality in 2-Years Hepatoblastoma Survivors Treated with Siopel Risk-Adapted Strategies

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Background and Aims: Survival of hepatoblastoma (HB) patients treated with cisplatin-based chemotherapy has widely increased so that current therapeutics strategies are developed to improve quality of cure by reducing treatment-related late toxicities especially in low and standard-risk HB. However no previous report detailed health status of very long-term HB survivors.

Methods: We studied long-term mortality and morbidity of 151 2-years HB survivors treated in France and included in HEPATOBIO database (sex-ratio M/F=1.6, median age at diagnosis=2.6 years [range=0-17.7], median year at diagnosis=2008 [1994-2017]). Fifty-three patients had loco-regional risk factors VPEFR, 12 were PRETEXT-IV and 30 were metastatic. All received cisplatin and 84

anthracyclines. Twelve had liver transplant. To assess hearing, renal and cardiac functions, audiograms were collected in 116/151 patients (76.8%), glomerular filtration rate in 113/151 (74.8%) and cardiac ultrasound in 65/84 (77.4%) anthracyclines-exposed patients.

Results: With a median follow-up of 9.4 years (range=2.1-25.8), four late relapses (including two first relapses), one second malignancy (Acute Myeloid Leukemia AML-M5) and two deaths (one from HB, one from AML) occurred. The 10-years OS and EFS were 95.5% (95%CI=91.9-99.1) and 98.7% (95%CI=96.8-100), respectively. Seventy non-oncologic health-events were observed: 57 cases of hearing loss (25 Brock 3-4), three liver cirrhosis (including 2 biliary cirrhosis post partial hepatectomy), five portal cavernoma (3 pre-operative and 2 post-operative), two focal nodular hyperplasia, two grade-1 chronic kidney disease and one asymptomatic grade-1 cardiac dysfunction.

Conclusions: With current risk-adapted strategies, hepatoblastoma is a highly curable disease, with very rare relapses, and few late effects except hearing loss which remains a serious condition in these very young patients.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0570 / #184 | Tubular Dysfunction and Treatment-Related Risk Factors in Long-Term Childhood Cancer Survivors; DCCSS-LATER 2: RENA

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Background and Aims: The aim of this nationwide cross-sectional cohort study was to determine the prevalence of and risk factors for tubular dysfunction in childhood cancer survivors (CCS).

Methods: For the Dutch Childhood Cancer Survivor Study (DCCSS) LATER cohort (1963-2001) part 2; clinical visit & questionnaire study; RENA project, 1,024 CCS (≥5 years after diagnosis), aged ≥18 years at time of study, treated between 1963 and 2001 with potentially nephrotoxic therapy (i.e. nephrectomy, abdominal radiotherapy, total body irradiation, cisplatin, carboplatin, ifosfamide, high-

dose cyclophosphamide (≥1g/m² per course or ≥10 g/m² in total), or bone marrow transplantation participated. Five-hundred age- and sex matched controls were used from the Lifelines cohort study. Renal electrolyte loss was defined as low serum levels in combination with increased renal excretion or receiving electrolyte supplementation. α1-microglobulin:creatinine ratio >1.7 mg/mmol was considered as low-molecular weight proteinuria (LMWP). Chi-Square analysis was used to compare the prevalence in both groups. Multivariable logistic regression analyses were performed to assess risk factors and expressed as odds ratio (OR) with 95% confidence interval (95%CI).

Results: Median age at diagnosis was 4.7 years (interquartile range [IQR] 2.4-9.2), at study 32.5 years (IQR 27.7-38.0), and follow-up time 25.5 years (IQR 21.4-30.3). Risk factor for magnesium loss was cisplatin (OR 10.1, 95%CI 3.9-26.0), for potassium loss these were ifosfamide (OR 2.4, 95%CI 1.2-4.7) and cisplatin (OR 3.5, 95%CI 1.6-7.5) and for phosphate loss ifosfamide (OR 2.3, 95%CI 1.2-4.3). LMWP was associated with ifosfamide (OR 2.8, 95%CI 2.0-4.1). Overall prevalence of renal losses in CCS (magnesium 5.6%, potassium 4.5%, phosphate 5.5%) was not higher compared to controls, while LMWP was significantly more prevalent (CCS 20.1% versus controls 0.4%, p<0.001).

Conclusions: After a median follow-up of 25.5 years CCS had an increased prevalence of LMWP compared to controls, but not of renal electrolyte losses. Ifosfamide and cisplatin were associated with tubular toxicity over time.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0571 / #185 | Hypertension in Childhood Cancer Survivors after Treatment with Potentially Nephrotoxic Therapy and the Added Value of Ambulatory Blood Pressure Monitoring; DCCSS-LATER 2: RENA

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Background and Aims: We evaluated the prevalence of and risk factors for hypertension in childhood cancer survivors (CCS) compared to

controls. Furthermore, a pilot study using 24-hour ambulatory blood pressure monitoring (ABPM) was performed.

Methods: In the Dutch Childhood Cancer Survivor Study (DCCSS) LATER cohort (1963-2001) part 2; clinical visit & questionnaire study; RENA project, 1,024 CCS participated; ≥ 5 years after diagnosis, aged ≥ 18 years at study participation, treated between 1963-2001 with nephrectomy, abdominal radiotherapy (RT), total body irradiation (TBI), cisplatin, carboplatin, ifosfamide, high-dose cyclophosphamide ($\geq 1\text{g}/\text{m}^2$ per course or $\geq 10\text{g}/\text{m}^2$ total) or bone marrow transplantation. We used 500 age- and sex matched controls from Lifelines. Hypertension was defined as blood pressure (BP) (mmHg) systolic ≥ 140 and/or diastolic ≥ 90 or receiving antihypertensive medication for previously diagnosed hypertension. Multivariable regression analyses were used. The ABPM-pilot study was performed in 77 CCS. Hypertension was defined as BP daytime: systolic ≥ 135 and/or diastolic ≥ 85 , nighttime: systolic ≥ 120 and/or diastolic ≥ 70 , 24-hour: systolic ≥ 130 and/or diastolic ≥ 80 . Outcomes were masked hypertension (MH), white coat hypertension (WCH) and abnormal nocturnal dipping (aND).

Results: Median age at diagnosis was 4.7 years (interquartile range [IQR] 2.4-9.2), at study 32.5 years (IQR 27.7-38.0), and follow-up time 25.5 years (IQR 21.4-30.3). The prevalence of hypertension was comparable in CCS (16.3%) and controls (18.2%). A decrease in glomerular filtration rate (GFR) per $10\text{ml}/\text{min}/1.73\text{m}^2$ was associated with hypertension in CCS (OR 1.2, 95%CI 1.1-1.3), but not in controls. Treatment risk factors were abdominal RT (OR 2.2, 95%CI 1.3-4.0) and TBI (OR 3.0, 95%CI 1.6-5.8). The ABPM-pilot study showed 7.8% MH, 2.6% WCH, and 20.8% aND.

Conclusions: Prevalence of hypertension was comparable in CCS and controls, but an association with GFR was only found in CCS. Treatment risk factors were abdominal RT and TBI. ABPM has added value by identifying masked hypertension and abnormal nocturnal dipping.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0572 / #225 | Healthcare Providers', Administrators', and Staffs' Perspectives of Barriers and Enablers to Physical Activity during Treatment: A Qualitative Study

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Background and Aims: Physical activity (PA) is safe and beneficial for young people (i.e., children, adolescents) on-treatment for cancer, but is not part of routine care in Canada. Understanding healthcare providers' (HCPs'), administrators', and staffs' perceptions of PA during treatment is critical for developing and implementing PA programs for this cohort. This study explored barriers and enablers to PA during treatment to optimize PA implementation efforts for young people.

Methods: A convenience sample of 16 HCPs, administrators, and staff who provide care to young people during cancer treatment in Alberta were recruited and interviewed by two trained authors (KE, EM) using

a semi-structured, theory-informed guide. Questions and probes were designed to elicit insights into perceptions of barriers and enablers to PA during treatment and to understand how to implement PA during treatment. Interviews were audio-recorded ($M_{\text{length}}=20$; $SD=8$ minutes), transcribed, and analyzed thematically by three authors (AW, KE, EM) using a deductive-inductive approach.

Results: Participants' perceptions were organized into 4 main themes: (1) "balancing the importance of PA during treatment" reflected participants' understanding of the benefits of PA, and their focus on prioritizing life-preserving care; (2) "multilevel factors impacting PA during treatment" captured the complex hospital systems and the personnel, infrastructure, and culture shifts that need to be navigated to successfully implement PA, and recommendations to augment a previously generated workflow to optimize clinical efficiency; (3) "design considerations to make PA safe, fun, flexible, and family-centred" described participant perceptions of essential aspects for PA program success, and (4) "reconsidering PA during treatment: the COVID-19 catalyst" captured participants' newfound beliefs related to innovation and the feasibility of online PA delivery.

Conclusions: Participants in this study were supportive of PA during treatment but identified important barriers, enablers, and strategies to enhance PA implementation. This work has informed an Alberta-based online PA program, IMPACT, that will begin Spring 2021.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0573 / #241 | Defining Opportunities for Palliative Care in Pediatric Oncology – Clues for Referral?

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Background and Aims: Although outcomes for children with cancer have improved greatly, there remains significant suffering and unmet needs. Using a holistic approach, Pediatric Palliative Care (PPC) aims to lessen this burden. Early referral has been correlated with improved quality of life and longer survival. Yet, lack of well-defined referral criteria hinders reaching out to children and families. The aim of this study was to evaluate opportunities for PPC referral in children who died from progressive disease (PD) in a Pediatric Oncology Department.

Methods: After a literature review, a list of referral criteria for PPC (opportunities) was developed, divided in "automatic" and "recommended". Medical records of children who died from PD between January 2015 and December 2019 were reviewed.

Results: One-hundred-thirteen patients were included, 51% male. Age-at-death ranged from 3m-18y (<1y=3%; 1-5y=31%; 6-10y=28%; 11-14y=23%; 15-18y=14%). Most children had solid tumors (46%; 27% hematologic, 27% central nervous system). Overall, median time admission-first opportunity was 19 days (IQR 0-155) and admission-death 1.4 years (IQR 0.7-3.0). The median number of opportunities per

child was 3 (IQR 2-4). The most common occasions for “automatic” referral were PD (15.5%) and hematopoietic stem cell transplant referral (10.2%), whereas relapsed disease was the most frequent “recommended” opportunity (33.9%). Admission to ICU (“automatic”) was the most common cause for referral in infants (40%); in all other age groups it was relapsed disease (17-39%).

Conclusions: Our study shows that all children with cancer who died from PD in our department had opportunities for PPC referral, some identified very early in the disease course. Our biggest limitation was the retrospective review, but if anything, we may have underestimated opportunities for lack of proper chart documentation. Validation and widespread use of referral criteria for PPC in children with cancer are needed to foster early integration and meet the needs of children and families.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0574 / #243 | Meta-Analysis of Psychosocial Interventions for Pediatric, Adolescent and Young Adult Cancer Survivors

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Background and Aims: Adolescent and young adult (AYA) cancer survivors have the worst psychosocial improvement in outcomes when compared with their pediatric or adult counterparts. Despite significant advances in medical treatments for AYAs diagnosed with cancer, their psychosocial care remains behind. This study systematically reviewed and meta-analyzed all existing clinical trials evaluating psychosocial care and survivorship outcomes for AYA cancer survivors.

Methods: We searched 11 electronic databases, 4 professional websites, and did a manual search of reference lists in existing reviews. We selected randomized controlled trials and controlled trials without randomization focusing on pediatric and AYA cancer survivors across six survivorship outcome domains.

Results: We included 61 studies (4,402 participants) published between 1987 and 2020. Overall, the risk of bias across studies was low. We identified an overall moderate and statistically significant treatment effect size for PAYA cancer survivors across six outcome domains, $d = 0.470$, 95 % CI 0.360, 0.579. Age significantly moderated the effect size of survivorship outcomes, $b = -0.019$, 95 % CI -0.031, -0.007. Studies targeting AYA survivors on average reported significantly smaller treatment effects than studies targeting pediatric and AYA survivors, and only pediatric survivors, $b = -0.307$, 95 % CI -0.545, -0.068 and $b = -0.329$, 95 % CI -0.533, -0.124, respectively.

Conclusions: Psychosocial, behavioral, and supportive interventions were overall effective for pediatric and AYA cancer survivors. In-person therapy and distance-delivered interventions were equally as effective. However, interventions were not effective for certain survivorship out-

come domains and less effective among AYA versus pediatric cancer survivors.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0575 / #284 | Risk Factors of Obesity in the Childhood Cancer Survivors: A Retrospective Cohort Study

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Background and Aims: The aim of this study was to identify cancer treatment related risk factors for adulthood obesity in adult childhood cancer survivors (CSS).

Methods: Of the 7670 five-year survivors of solid childhood cancers from the French Childhood Cancer Survivor Study (FCCSS) cohort, who were treated before age 20, between 1945 and 2000 in 5 centers in France, 3216 sent back at least 1 self-questionnaire and were included in this analysis. We compared their obesity rate ($BMI \geq 30$) with the expected one from French population's gender and age specific rates.

Results: Patients from FCCSS were less frequently obese than expected from French general population rates (9.7%, 95%CI=8.7-10.8, $n=312$, vs. 12.5%), this difference being similar in women (10.2% vs 13.3%) and men (9.3% vs 11.6%). Nevertheless brain tumour survivors experienced an increased risk of obesity compared to patients with other tumours (22%, $RR=2.5$, 95%CI: 1.6-3.7). As compared to patients who did not received radiotherapy, those who received 1 to 5 Gy, 5 to 20 Gy, 20 to 40 Gy, and 40 Gy or more to the pituitary gland, had a higher risk of obesity of 1.0 (95%CI : 0.6-1.6), 2.1 (95% CI: 1.4-3.2), 2.6 (95% CI: 1.9-3.6) and 3.9 (95%CI: 2.6-5.8) respectively. Young age at diagnostic (<5 years) and a high social deprivation index were also risk factors for later obesity ($RR=0.72$, 95%CI: 0.53-1.0, and $RR=1.2$, 95%CI: 1.1-1.3, respectively).

Conclusions: Childhood cancer survivors are less frequently obese in adulthood than general population. Pituitary gland irradiation and a young age at diagnosis are associated with an increased risk of obesity. Nevertheless, because obese patients could be at higher risk of life pathology, CCS who received pituitary irradiation should continue to benefit a close monitoring during adulthood.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0576 / #302 | Ventricular Repolarization during Cardiopulmonary Exercise Test in Childhood Acute Lymphoblastic Leukemia Survivors

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Background and Aims: Acute lymphoblastic leukemia (ALL) is the most common hematological malignancy in children, with a survival rate up to 90%. However, anthracycline-related cardiotoxicity is a major cause of mortality and morbidity in this population, and early detection of cardiac dysfunction remains a challenge. This study aims to explore ventricular repolarization during a maximal CPET in childhood ALL survivors. We hypothesized that the use of CPET allows detecting electrophysiological abnormalities.

Methods: A total of 250 childhood ALL survivors underwent a maximal CPET on ergocycle and their oxygen uptake was measured. All survivors were monitored continuously during the test with a 12-lead electrocardiogram. Measurements of QT interval were completed at rest, at the end of each stage of the CPET, and at the recovery period. The QT interval was defined as the period from the onset of the Q-wave to the end of the T-wave, measured linearly and was corrected (QTc). Participants were divided in two groups according to the median of survivors' cardiorespiratory fitness (group A: $<32.0 \text{ mL.kg}^{-1}.\text{min}^{-1}$ and group B: $\geq 32.0 \text{ mL.kg}^{-1}.\text{min}^{-1}$).

Results: All survivors included in the analysis ($n=200$) performed a maximum CPET. At rest, the QTc interval was $379.1 \pm 32.2 \text{ ms}$. We observed that 5 survivors had a prolonged QTc interval at rest ($496.8 \pm 39.9 \text{ ms}$). During the CPET, the mean QTc interval was $371.6 \pm 16.1 \text{ ms}$. The mean QTc interval was not different between groups A and B ($372.0 \pm 14.4 \text{ ms}$ and $370.1 \pm 17.6 \text{ ms}$, $p=0.218$). A trend shows that group B has a longer QTc interval during low to moderate intensity exercise.

Conclusions: Cancer and anthracycline treatments have an impact on ventricular repolarization. Low cardiorespiratory fitness in childhood ALL survivors is associated with a longer ventricular repolarization during an exercise. It shows the importance of studying the response to exercise to improve early detection.

E-Poster Topic: Disciplines / AS06 Survivorship

PV0577 / #310 | Attention Performance and Behavioral Outcomes in Young Chinese Survivors of Pediatric Osteosarcoma and Soft-Tissue Sarcoma

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Background and Aims: There is limited evidence on functional outcomes in Chinese survivors of pediatric/adolescent sarcomas. This study examined attention performance and behavioral functioning in a cohort of young survivors with sarcoma in Hong Kong.

Methods: This cross-sectional study recruited 91 Chinese survivors of osteosarcoma or soft-tissue sarcoma (STS) diagnosed at <18 years old and were ≥ 3 years post-treatment at follow-up. They completed assessments of attention (Conners Continuous Performance Test [CPT-III]), body image perception (Body Image Scale), and behavior functioning (Adult Self-Report Behavioral Checklist). Impairment was defined as ≥ 1.0 standard deviation worse than reference norms. Independent t-test and Pearson's correlation test were conducted to identify clinical and treatment factors associated with functional outcomes.

Results: Participants included 47 survivors with osteosarcoma (51% male, age 27.3 [8.0] years) and 44 survivors with STS (45% male, age 25.0 [5.6] years), who were 15.4 [7.5] years post-treatment. The major STS diagnoses were Ewing sarcoma (18.2%), embryonal rhabdomyosarcoma (13.2%) and synovial sarcoma (11.4%). Compared with norms, a minority were impaired on CPT measures of impulsivity/inhibition (3%–17%), sustained attention (11%–13%) and inattentiveness (2%–9%). Female survivors showed more sustained attention deficits than male survivors ($P=0.002$). Older age at diagnosis correlated with poorer inattentiveness ($r=0.217$, $P=0.047$) and vigilance ($r=0.234$, $P=0.032$) scores. Survivors reported significant symptoms of depression (66.3%), anxiety (51.7%) and sluggish cognitive tempo (70.8%). Survivors of osteosarcoma (versus STS, $P=0.010$) and those diagnosed at adolescence (versus pre-adolescence, $P=0.045$) reported poorer body image perception. Treatment regimens were not significantly associated with functional outcomes.

Conclusions: Impaired attention, emotional distress and body image dissatisfaction were observed in a minority of Chinese survivors of sarcoma, especially in survivors diagnosed during late adolescence. Future work includes in-depth assessment of survivors' neurocognitive function and its association with treatment predictors (cumulative drug doses and types of surgical interventions) and identifying modifiable risk factors for poorer functional outcomes.

E-Poster Topic: Disciplines / AS06 Survivorship

PV0578 / #320 | Examining the Connection between Cerebellar CRUS I and Executive Functioning in Survivors of Pediatric Brain Tumor and Healthy Controls

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Background and Aims: Survivors of pediatric brain tumors face challenges in social cognition including theory of mind (ToM); the mechanisms for these difficulties are unknown. The left CRUS I (LCI) region of the cerebellum has been shown to be related to ToM in other patient populations (e.g., autism spectrum disorder). The primary aim of this study was to identify possible relationships between LCI volume and ToM in survivors compared to healthy controls. We hypothesized that LCI volume would correlate positively with ToM.

Methods: Nineteen survivors (n=5 glioma, n=4 medulloblastoma, n=10 other tumours; n=6 chemotherapy, n=18 surgery, n=10 radiation; mean[SD] diagnosis age 7.0[4.1] years, current age 15.1[3.0] years; 52.6% female) and 18 controls (current age 12.0[3.2] years; 55.6% female) completed T1 BRAVO MRI (3T General Electric Healthcare) that yielded measures of LCI derived using FMRIB's Software Library (FSL). Survivors and controls also completed a measure of ToM using the computerized Pediatric Evaluation of Emotions, Relationships and Socialization (PEERS) administered on an iPad. Higher scores reflected greater accuracy in ToM tasks.

Results: LCI volume in survivors (3407.2[277.5] mm³) and controls (3387.7[366.1] mm³) did not differ significantly (p=0.50), adjusting for sex and age; ToM was not significantly different between groups (p=0.54). Linear regression analysis with ToM as the dependent variable and sex, age, group membership (i.e., survivor vs. control), LCI volume and an interaction for LCI volume and group membership as predictors was significant (R²=0.46, p=0.05). The interaction of LCI volume and group membership was significant (p=0.01). Examination of this interaction revealed LCI volume increased as ToM scores increased in controls. The same was not found for survivors.

Conclusions: The relationship between LCI volume and ToM is different among survivors and controls. Future research should aim to examine other potential mechanisms for social cognitive difficulties in survivors of pediatric brain tumor.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0579 / #376 | Second Malignancies in Children with Head and Neck Primary Tumors

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Background and Aims: Children with primaries in the head and neck region, especially those who receive radiotherapy (RT) are at increased risk for second malignancies (SM). This study aims to assess the incidence and outcome of SM in children with head and neck primary tumors.

Methods: During 1990-2017, 3214 children with cancer were treated in the Istanbul University, Oncology Institute. The primary tumor was in the head and neck region in 1414 (651 brain tumors, 289 retinoblastoma, 474 other). Survivors followed up for at least 3 years from diagnosis were evaluated for second malignancies.

Results: Thirty SM were identified in 28 survivors at a median of 12 years (2-26) from diagnosis. The primary diagnosis was nasopharyngeal carcinoma (NPC) in eight, bilateral retinoblastoma (RBS) in seven, embryonal brain tumors (CNST) in eight, Hodgkin lymphoma (HD) in three, rhabdomyosarcoma (RMS) in two. Twenty-six had received RT, dose (20-70 Gy) differed according to diagnosis. Two developed MDS and lymphoma. The others developed solid tumors as SM: eight sarcomas, six thyroid cancer, five carcinomas, one basal cell carcinoma (BCC), three meningiomas, one Langerhans cell histiocytosis (LCH), two CNS tumors, one peripheral nerve sheath tumor, one PNET. Two patients developed two SMs each: one with medulloblastoma (GBM and NHL) and one with NPC (sarcoma and carcinoma). As SMs, five thyroid cancer, two meningiomas, three sarcomas, three carcinomas, one BCC, LCH, CNST each are alive. All patients survived for a median of 3 (0.5-16) years after SM; 16/28 (57%) are with NED at a median of 8 (1-16) years; 12 died at a median of 1.8 (0.5-4.75) years after SM. All SM were detected early during regular surveillance.

Conclusions: Children with head and neck primaries are at risk for SM which may occur many years later, especially in RT sites. Patients need regular surveillance lifelong for early detection of SM. SM should be treated with curative intent, to achieve long term survival.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0580 / #379 | Subsequent Malignant Neoplasms in the Pediatric Age in Retinoblastoma Survivors in Argentina

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Background and Aims: Retinoblastoma survivors have a higher risk of subsequent malignant neoplasms (SMNs). However, retinoblastoma survivors in low- and middle-income countries, are exposed to high intensity treatments, so they may present an increased risk of early SMN

Methods: We followed 714 (403 (56.4%) non-hereditary and 311 (43.5%) hereditary) retinoblastoma survivors diagnosed from August 1987 to December 2016, up to the age of 16 years. We quantified risk of SMNs with cumulative incidence (CI) and standardized incidence ratios (SIR) analysis. Multivariate regression Cox model was used to determine the association of treatments and risk of SMNs

Results: With a median follow-up of 11 years (range: 0.18 -16.9), 24 survivors (3.36%) developed 25 SMNs (n:22 hereditary, n:2 non-hereditary) [SIR:48 (CI 95%=31.3; 75.1) vs SIR:4.1 (CI 95%=1.04; 16.6)]. SMNs included sarcomas (n:12), acute leukemias (n:5) and central nervous system tumors (CNS) (n:3). All cases of acute myeloid leukemia (AML) and most of Ewing sarcomas (ES) occurred within 5 years of retinoblastoma diagnosis. The type of SMN was the main indicator of mortality (5/5 patients diagnosed with leukemias, 6/12 with sarcomas and 0/3 with CNS tumors died). The risk of developing ES among hereditary survivors treated with radiation was 700-fold higher compared to the general population [SIR 781.3 (95%CI=252; 2422.6). The risk of AML was 140-fold higher among those exposed to chemotherapy [SIR 140.6 (95%CI=45.3; 436)]. The CI of SMNs for hereditary survivors was 5.4% (95%CI= 3; 9.5) at 10 years and 13.7% (95%CI= 8.4; 22.1) at 15 year. Cox analysis showed an increased risk among survivors treated with radiation and chemotherapy [hazard ratio (HR):2.8, 95%CI=0.8; 10.3].

Conclusions: Retinoblastoma survivors from Argentina are at higher risk of developing SMNs early in life, especially those treated with radiation plus chemotherapy. AML and ES presented within 5 years of retinoblastoma diagnosis and are associated to chemotherapy and radiation exposure.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0581 / #387 | A Survey of Ovarian Reserve and Quality of Life in Female Adolescent and Young Adult Survivors of Pediatric Cancer

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Background and Aims: Gonadotoxicity and reduced fertility are well-known side effects of cancer treatment. Currently fertility surveillance recommendations for female childhood cancer survivors (CCS) vary

among existing long-term follow-up programs. Anti-Müllerian hormone (AMH) has been shown to correlate well with antral follicle count (AFC) in healthy women but is currently not recommended as the primary surveillance modality in follow-up programs for female CCS. Psychological parameters related to fertility could affect quality of life for these patients and should be included in long-term follow-up. The aim of the study is to present the baseline data from inclusion for a prospective follow-up study of fertility surveillance where both medical and psychological aspects are considered.

Methods: Adolescent and young adult female survivors of pediatric cancer in Western Sweden were included in 2016–2018, in total 54 participants. Median age at inclusion was 21 (15–29) years and median age at cancer diagnosis was 10 (1–17) years. Anti-Müllerian hormone (AMH) levels, antral follicle count (AFC), and data on fertility were measured at inclusion and will be prospectively followed up every 2–5 years. The study includes questionnaires and interviews concerning quality of life, administered at inclusion and at 20/25–30/40 years of age.

Results: Eighteen out of 54 participants (33%) had AMH levels below 1.0 µg/L, and were considered to have high or very high risk of infertility, including all eleven stem cell transplant patients, of whom six had needed induction of puberty. Median AMH value was 2.50 µg/L. Psychological assessment showed that more than one-third of participants (n=20) had elevated anxiety scores.

Conclusions: Systematic surveillance of ovarian reserve in female CCS, using AMH and AFC, might help to identify those needing fertility cryopreservation and may serve as a model for future screening programs. Psychological wellbeing related to fertility affects quality of life and anxiety for these patients and should be addressed in follow-up programs.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0582 / #396 | Create Childhood Cancer Rehabilitation Program Development: Increase Access to Improve Outcomes

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Background and Aims: Cancer and treatment results in functional limitations for children and adolescents. Often persisting into survivorship, impairments hinder development and decrease quality of life. At our institution, we developed a sub-specialty pediatric oncology rehabilitation program with the goal of delivering focused, evidence-based rehabilitation care integrated into cancer treatment. We facilitated growth in rehabilitation services by utilizing the CREATE™ (Collaboration, Rehabilitation/Research, Education, Assessment, Treatment, Evaluation) model. Here, we report the growth of rehabilitation access and highlight structural and process changes enabling growth.

Methods: We identified and evaluated new patients with oncologic diagnoses, from 2002 to 2019, within our program's database. Descriptive data was collected and evaluated regarding timing and type of physical therapy (PT), occupational therapy (OT), and speech-language pathology (SLP) services utilized within 5 years of a child's diagnosis.

Results: A total of 1974 children were assessed for their use of rehabilitation services; 1580(80.0%) received care from at least one rehabilitation service. Between 2002 and 2019, the proportion of children receiving outpatient services grew in all disciplines (PT: 15→81%, OT: 9%→43%, SLP: 6→13%). Inpatient services expanded as well (PT 38→84%, OT 33→65%, SLP 9→29%). The earliest diagnostic group (acute lymphoblastic leukemia (ALL)) to have a standardized clinical pathway for rehabilitation reached 100% access to PT. Integrating therapists in the clinic reduced, by half, the time to access to outpatient services and also increased the number of visits attended. Notable factors contributing to growth included: identifying rehabilitation and oncology leaders, using a prospective surveillance model, integrating therapists into cancer clinic, and creating a rehabilitation gym within the outpatient clinic.

Conclusions: Structural and process changes, such as the CREATE™ model, within a cancer clinic supported by evidence-based clinical practice improve access to rehabilitation services that promote recovery, ongoing development and quality of life.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0583 / #435 | **Personalized Massive Open Online Course for Childhood Cancer Survivors: Feasibility Study**

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Background and Aims: Many studies have shown Childhood Cancer Survivors (CCS) are ill-informed about long-term follow-up (LTFU). Massive open online courses (MOOCs) could easily allow a deployment at an international level and make reliable information available for all CCS, wherever they live. The aim of the present study was to evaluate the feasibility of using a personalized MOOC to inform French young adult CCS about the LTFU conditions adapted to their cancer treatment.

Methods: The MOOC team (specialists of LTFU care, communication professionals and CCS associations) designed a MOOC named "Childhood Cancer, Living Well, After" including three modules addressing transversal issues (lifestyle, psychological support, fertility problems) and eight modules covering organ-specific problems. Detailed data on childhood cancer treatments were used to allocate specific modules to each participant. CCS were selected through cluster sampling from the French Childhood Cancer Observation Platform. A psychologist proposed a personalized psychological support by e-mail or telephone, according to the patient's choice.

Results: Among the 574 contacted CCS, aged 18-35y in 2020, 98 (17.1%) participated in the MOOC. Women were 2.5 more likely to participate than men ($P < 0.0001$). More than 90% of participants would recommend the MOOC to other CCS and 25% declared they want to improve their LTFU. Participating in the MOOC did not cause psychological distress.

Conclusions: Using a personalized MOOC to inform CCS about LTFU is feasible. We would now quantify this MOOC impact on the CCS knowledge about LTFU adapted to their own medical history, but also measure how this MOOC can improve the LTFU care of each CCS.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0584 / #462 | **Effectiveness of Play Therapy in Enhancing Psychosocial Outcomes in Children with Chronic Illness: A Systematic Review**

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Background and Aims: Cancer is one of the most common childhood chronic health conditions (CHC) associated with impairments to daily functioning causing distress, adjustment issues and worry. Evidence indicates that Play Therapy (PT) can support positive adjustment and emotion regulation skills in children with CHC. However, there has been no systematic synthesis of available evidence in this area. To address this gap, the aim of this systematic review was to examine the evidence for effectiveness of PT intervention in children with cancer and other CHC.

Methods: Systematic search of the databases Medline complete, PsycINFO, CINAHL, Embase and Sport Discuss to identify peer reviewed papers reporting original studies published in English between 1990-2020. Studies testing any type of PT with children with CHC aged between 3-11 years, or their parents if Filial Therapy

(FT), where outcomes targeted included psychosocial measures were eligible.

Results: The title and abstract search identified 6742 papers and six studies were eligible (three each for PT and FT). Cancer was the focus of 2 studies, one pre-post study of PT and one RCT assessing FT. Other studies focused on diabetes (k=1), thalassemia (k=1) and mixed CHC. Mostly outcomes included anxiety, depression, and self-concept but the assessment measures differed. Sample size were generally small (range:4-58) and study designs varied from feasibility trial to RCT. Results relating to PT were mixed but the study involving cancer suggested wellbeing and self-efficacy improved although small sample size (n=4) limited conclusions. More consistent effects were found for FT in reducing children's depression and reduced parental acceptance and stress.

Conclusions: With varied study designs, and small sample sizes, evidence regarding the effectiveness of PT in improving psychosocial outcomes for children with CHC is inconclusive. The field should move from small underpowered studies to randomised trials with comprehensive protocols and larger sample sizes.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0585 / #475 | Implementing Long-Term Follow-Up after Pediatric Cancer: Ethical Tensions and Survivor Perspectives

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Background and Aims: Recent studies have highlighted the long-term effects of some treatments for pediatric cancers. Survivors are sometimes at risk of sequelae, but not all of these effects were known at the time of treatment or post-treatment follow-up. New information and continuation of follow-up are then important as an adult. Our qualitative study aims to identify the main ethical tensions that may be raised by the implementation and organization of long-term follow-up for people who are no longer ill with cancer but who are now considered to be at risk of developing sequelae.

Methods: We adopt an empirical ethical approach, based on the perspectives of survivors. First, questionnaires (n= 845) were sent to members of the SALTO cohort, in which a follow-up consultation had been offered to all persons treated for pediatric cancer diagnosed between 1987 and 1999 in the French Rhône-Alpes region (168 respondents). Then, semi-structured interviews (n=55) were conducted with volunteers.

Results: Long-term follow-up needs and expectations (autonomy /support, specialized/global or psychosocial approaches) vary according to the time of life (entry into adulthood, employment, pregnancy). People's representations of their cancer (e.g., cancer as an episode from the past / cancer as chronic disease) also have an impact their needs and expectations. However, this view evolves with the follow-up consultations and new informations on the long-term effects of treatments.

Conclusions: Long-term follow-up should be modulated over time according to changes in health status, but also evolutions in survivors' needs and expectations. This research is part of the study START: Interventional Project on Long-Term Follow-Up Care: Virtual Help, Research and Transversality, INCa-ARC-LIGUE_11902

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0586 / #521 | Unresolved Grief. a Silent On-Going Burden for Childhood Cancer Survivors

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Background and Aims: Childhood cancer affects 17,500 children/adolescents under 14 years old in Latin America and the Caribbean yearly; survival is 65%. Rightfully, treatment emphasis is survivorship; consequences are not always positive and mental health long-term repercussions need attention. In 2019, the International Society of Paediatric Oncology (SIOP) Pediatric Oncology in Developing Countries Committee (PODC), Patient Family and Stakeholder Engagement Task Force explored patient and caregiver treatment engagement. Ten low and middle-income countries participated in 5/6 WHO regions. We report on Peruvian, El Salvadorean, and Mexican survivors' mental health accounts.

Methods: Nine survivors in Mexico; 9 El Salvador; 12 Peru (30 total) participated in the hospital (Peru) and local foundations (freestanding Mexico; on hospital grounds El Salvador). Interviews, recorded with consent, lasted ~1 hour. Survivors aged 13-31 had diagnoses of leukemia/lymphoma and solid tumors. Short-term survivors finished treatment <2 years and long-term >5 years. Data analyzed for content; one prominent theme was unresolved grief.

Results: Survivors reported that peers/friends they made during treatment who died were a significant source of emotional suffering then and now, despite their rational understanding of death's meaning. These experiences left marks that influence their life goals and approaches. Some developed complex and long-term emotional and mental distress: self-harming, depression, survivor guilt, feelings of loss and meaningless life. One positive reaction was understanding death as a life cycle.

Conclusions: Survivors reported a lack of information or follow-up by mental health professionals. A strong existential component

appeared in their anguish about the purpose or meaning of their own survivorship. Several struggled continually and mostly in isolation, with the wish/ideal of having a purposeful life. Most survivors will eventually adapt and cope, but a follow-up program is essential for collecting data to inform interventions and sources of help across Latin America and provide mental health services for this population.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0587 / #545 | Psychosocial Developmental Trajectory of Young Adult Survivors of Childhood Cancer: Survivors of Central Nervous System Cancer at Risk

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Background and Aims: Growing up with (a history of) childhood cancer, especially central nervous system (CNS) cancer, may have psychosocial consequences. Knowledge about possible delays in psychosocial development enables healthcare providers to aim for the most favorable psychosocial functioning of patients and survivors. The study aimed to compare the psychosocial developmental trajectory of young adult survivors of childhood cancer with that of a norm group of young adults from the general population.

Methods: From 2017 to 2020, 558 young adults (18-30 years, 51% female, 10.9% CNS) from the Dutch Childhood Cancer Survivor Study (DCCSS) LATER cohort (diagnosed 1963-2001) part 2 completed the Course of Life Questionnaire (CoLQ) that assesses achievement of milestones in Autonomy, Psychosexual and Social development. Differences between survivors and the norm group were assessed with ANOVA and Cohen's d (CoLQ-scales) and with logistic regression analysis and Odds Ratio (OR) (CoLQ-items); for the total group and for CNS survivors.

Results: The total group of survivors did not score less favorable than the norm group on any scale or milestone. The subgroup of CNS survivors scored significantly lower than the norm group on the scales of Autonomy (d -0.36; p 0.000) and Psychosexual (d -0.46; p 0.000) development. Additionally, on half of the items of Autonomy (OR 0.25-0.34), Psychosexual (OR 0.30-0.48) and Social development (OR 0.23-0.47),

CNS survivors were less likely ($p < 0.01$) than the norm group to have achieved the milestones.

Conclusions: Overall, the psychosocial developmental trajectory of survivors was as favorable as in the norm group, but CNS survivors appeared to be at risk of an unfavorable trajectory in all three developmental domains. Monitoring of the achievement of psychosocial development should be included in the standards of psychosocial (after)care in order to be able to trace and minimize delay in the psychosocial development in an early stage.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0588 / #643 | Neurological and Orthopedic Sequelae in 59 Pediatric Patients Diagnosed with Spinal or Paraspinal Tumors

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Background and Aims: Spinal or paraspinal tumors have a higher risk of orthopedic and neurological sequelae due to their location and possible involvement of the spinal cord. The objective is to describe the characteristics and the sequelae reported in patients diagnosed with spinal or paraspinal tumor, in a tertiary hospital.

Methods: Patients diagnosed between January-2000 and December-2019 were included. Clinical, radiological and histological data were collected after reviewing medical records.

Results: Over 20 years, 59 patients were diagnosed with paraspinal/spinal tumors. The median age was 36 months, without sex differences. The distribution by histological type was: neuroblastomas (45.8%), Ewing's sarcoma (18.6%), low grade glioma (10%) and other minority types (23.6%). Nearly three quarters of tumors were located at thorax or lumbar level. Spinal canal involvement was observed in 35.6%, secondary to extension through the conjunction foramina. In 25.4% initial clinical signs of spinal cord compression (SCC) were observed. Long-term orthopedic sequelae were reported in 35%, mainly spinal kyphoscoliosis (28.5%). There was no association between spinal canal occupation and orthopedic sequelae ($p = 0.57$). Neurological sequelae were reported in 23.7% and they were associated with clinical SCC at diagnosis ($p = 0.0005$). The most frequent neurological sequelae were motor alterations, including paraparesis and paraplegia, (13.5%; 53.3% of those with initial SCC); and sphincter dysfunction (8.5%). Due to SCC, 20.3% of patients underwent urgent intervention (11.8% chemotherapy and 8.5% surgery; 2 laminectomy

and 3 laminotomy). Patients that received urgent decompressive surgery suffered more long-term neurological sequelae than those who did not ($p = 0.0005$) and also more than those with urgent chemotherapy ($p = 0.06$).

Conclusions: Orthopedic and neurological sequelae were common in paraspinal/spinal tumors. Neurological permanent consequences were more frequent in patients with initial SCC and after decompressive surgery. These sequelae may have an impact on the quality of life of survivors.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0589 / #655 | Posttraumatic Stress, Posttraumatic Growth and Coping in Childhood Cancer Survivors

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Background and Aims: Surviving treatment of malignancies in childhood is associated with positive as well as negative consequences. Posttraumatic stress (PTSS) and posttraumatic growth (PTG) following pediatric cancer are frequently studied topics in childhood cancer survivors. Coping strategies used to deal with pediatric cancer experience influence the psychosocial adaptation of survivors. These strategies cannot be seen as universally adaptive or maladaptive. Their efficacy depends on the match of a specific coping strategy with the demands of a specific situation. The aim of this study is to analyze and compare the pattern of relationships of PTSS and PTG with coping strategies used by survivors to deal with their cancer experience.

Methods: Ninety-two childhood cancer survivors aged 14-28 years completed measures of PTSS (UCLA PTSD index for DSM-IV), PTG (Benefit Finding Scale for Children) and coping strategies (Kidcope). Correlation analysis was used to examine the relationship of 11 coping strategies with PTSS and PTG.

Results: PTSS positively correlated with 7 coping strategies and mainly with the escape-oriented ones (e.g. distraction, social withdrawal and wishful thinking). Negative relationship was found for 2 approach-oriented strategies (cognitive restructuring and social support). PTG positively correlated with 3 approach-oriented coping strategies (cognitive restructuring, positive emotion regulation and social support) and 1 escape-oriented strategy (resignation). Neither PTSS nor PTG were related to the strategy focused on problem solving.

Conclusions: Overall, our results support the connection of PTSS with escape-oriented strategies and the connection of PTG with approach-oriented strategies. However, the efficacy of several strategies such as resignation and problem solving appears to have context-specific nature in pediatric oncology settings. Different pattern of relationships of PTSS and PTG with coping strategies further supports the distinction

of PTSS and PTG as two separate concepts rather than two ends of the same continuum.

E-Poster Topic: *Disciplines / ASO6 Survivorship*

PV0590 / #661 | Late Effects of a South African Childhood Cancer Cohort – A Linkage Analysis of Health Care Engagement Patterns

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Background and Aims: There is limited data regarding childhood cancer late effects in Africa. We investigated health care engagements and late effects of a South African cohort.

Methods: A linkage analysis between data documented in the Tygerberg hospital childhood tumour registry and the hospital patient administration (Clinicom™) and laboratory results systems (TrakCare) was done. Data collection included demographic information (age, sex), cancer diagnosis, treatment received and health care engagement information for late effects and other health care issues.

Results: The study included 281 childhood cancer survivors (CCSs) with M:F ratio of 1.2:1; mean age at last contact 21.9 years (median 5 years; range 4-49 years) with a mean follow-up period of 15.9 years (median 14; range 4-37 years). The diagnoses were leukaemia (75/281; 26.7%), CNS tumours (57/281; 20.3%), lymphoma (49/281; 17.4%), renal tumours (34/281; 12.1%), germ cell tumours (12/281; 4.3%) and other cancers (54/281; 19.2%). Treatment modalities included chemotherapy (218/281; 77.6%), surgery (143/281; 50.9%), radiotherapy (102/281; 36.3%) and other (1/281; 0.4%). The mean number of health care visits post cancer treatment was 12 (median 7). The majority of engagements occurred at tertiary hospitals (67.6%), followed by primary care clinics (23.3%) and regional hospitals (9.1%) and were predominately ambulatory clinic visits (89.3%). CNS tumour survivors had the highest mean total number of visits (21 visits/person) and mean tertiary hospital visits (16 visits per person), with a mean of 2 hospital admissions/person. Late effects were neurological sequelae (48/281; 17%), endocrinopathy (36/281; 13%), psychiatric disorders (33/281; 12%), physical deformities (27/281; 10%), and thirteen (5%) CCSs were substance abusers. Five second malignancies and one pre-malignant condition were diagnosed.

Conclusions: CCSs had mostly ambulatory health care engagements at tertiary hospitals. CNS tumour survivors had the most late effects and required more health care. A consistent survivorship care plan is required to manage late effects.

E-Poster Topic: Disciplines / AS06 Survivorship**PV0591 / #683 | Hypertension and Lipid Abnormalities among Survivors of Childhood Acute Lymphoblastic Leukemia**

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Background and Aims: With the increased survival rates of childhood cancer, it becomes equally important that the need for evidence based surveillance of long term effects of cancer therapy like hypertension, altered lipid profile is addressed which was attempted in the present study.

Methods: This is an observational cross section study conducted at tertiary Cancer Centre among survivors of childhood ALL who were ≤ 14 years of age at diagnosis and had completed therapy at least 2 years earlier. Survivor was considered hypertensive if SBP and or DBP ≥ 95 th percentile in children < 18 years and SBP ≥ 140 mmHg or DBP ≥ 90 mm Hg in survivors aged ≥ 18 years. A diagnosis of dyslipidemia is made with one of three factors; Triglycerides ≥ 100 mg/dL in children aged 0-9 years or ≥ 130 mg/dL in children 10- 19 years old and ≥ 160 mg/dl in ≥ 20 year or HDL < 40 mg/dL or LDL ≥ 130 mg/dL in children and TC ≥ 240 mg/dl for adults as per (NCEP) Expert Panel on Cholesterol Levels in Children and adults

Results: Two hundred and Seventy seven survivors of childhood ALL were enrolled in study with median follow-up of 4 years (2-11.3 years). Median age of study participants was 13 years (IQR, 10- 15.6). Sixteen (5.8%) were identified having unrecognized hypertensive blood pressure. In our study, 4% (n=11) of children had systolic hypertension, and 3 % (n=8) had Diastolic hypertension. Prevalence of hypertension in males and females was 8.1% and 2.6 % respectively (p-value: 0.05). There was high prevalence of hypertension in overweight/obese children (p-value- 0.002). Dyslipidemia was prevalent in 41.9% (n= 166) survivors. Among survivors, 18.4% (n=51) had elevated TG, 14% (n=38) disclosed high LDL-C, and 28.5% (n=79) displayed low HDL-C levels.

Conclusions: Males and overweight/ obese survivors were at increased risk of hypertension

E-Poster Topic: Disciplines / AS06 Survivorship**PV0592 / #706 | Adaptive Functioning in Adolescent/Young Adult Cancer Survivors: The Role of Cognitive and Behavioral Symptoms**

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Background and Aims: Adolescent and young adult (AYA) cancer survivors have increased risk for executive functioning (EF) and psychosocial problems. Previous studies suggest that EF impairments contribute to lower adaptive functioning and quality of life in this population. The present study aimed to evaluate how the relationship between EF and internalizing symptoms impacts adaptive functioning.

Methods: Neuropsychological data were abstracted from clinically-referred AYA cancer survivors (n=208, 43% Brain Tumor, 39% Leukemia, MAge=15.75years, Range=12-23years, 58.7% male). Data consisted of a battery of parent-report questionnaires including the Behavior Rating Inventory of Executive Function, the Child Behavior Checklist, and the Adaptive Behavior Assessment. Path analysis was used to examine relationships between EF, internalizing symptoms, and adaptive functioning.

Results: Path analysis was significant and predicted 35-39% of AYA adaptive functioning domains ($p < 0.001$). Parent-reported EF was significantly related to parent-reported internalizing problems and all adaptive functioning domains, with greater EF impairments related to more psychological distress and lower adaptive skills ($p < 0.001$). EF was indirectly related to social skills through internalizing symptoms ($p < 0.05$), but internalizing symptoms did not mediate the relationship between EF and conceptual or practical skill domains. Demographic covariates did not significantly contribute to adaptive outcomes in the overall model.

Conclusions: Underlying EF impairments contribute to problems in psychosocial and adaptive functioning among AYA survivors. Internalizing symptoms appear to contribute to social functioning, but may not impact other adaptive domains above and beyond EF impairments. Attention to neuropsychological and psychosocial functioning in AYA survivors is important for identifying, assessing, and intervening upon adaptive outcomes and improving long-term quality of survival. Further research on these relationships is needed among survivor populations that are not routinely referred for long-term neuropsychological follow-up, such as individuals with solid tumors.

E-Poster Topic: Disciplines / AS06 Survivorship**PV0593 / #718 | Clinical Informatics Tools to Extract Cumulative Anthracycline Exposure from the EHR, Measure Cardiovascular Outcomes, and Assess Adherence to Guideline Recommendations for Children with Cancer**

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Background and Aims: Background: Cardiovascular disease (CVD) is a significant cause of late morbidity and mortality for survivors of childhood cancer. Clinical informatics tools could enhance provider adherence to echocardiogram guidelines for early detection of late cardiac effects. The primary aim of this study was to employ a clinical informatics approach to automatically extract cumulative anthracycline exposure data from the EHR to construct cardiovascular risk-adapted cohorts of childhood cancer patients to assess early CVD as well as adherence to guideline-based recommendations for echocardiogram surveillance.

Methods: Procedure: Cancer registry data was linked to EHR data based on medical record number (MRN). Structured query language (SQL) facilitated the construction of anthracycline-exposed cohorts at a single institution. Primary outcomes included heart failure, captured by International Classification of Disease (ICD) coding and compared to manual chart review, and adherence to echocardiogram guideline recommendations.

Results: Between July 1, 2013 and December 31, 2018, there were 385 pediatric oncology patients included in the final analytic cohort, among whom 194 had no anthracycline exposure, 143 had low anthracycline exposure ($<250\text{mg/m}^2$), and 48 had high anthracycline exposure ($>250\text{mg/m}^2$). Among patients with no, low, and high anthracycline exposures, approximately 2.0% ($n=4$), 4.3% ($n=6$), and 6.7% ($n=3$) had ICD codes for heart failure, respectively ($p=0.212$). Manual review of echocardiogram parameters for heart failure yielded a sensitivity of 75%, specificity of 98%, positive predictive value of 68%, and a negative predictive value of 99%. For patients with anthracycline exposure, 78.5% ($n=62$) were adherent to guideline recommendations for echocardiogram surveillance and there was a significant association for primary language ($p=0.003$) as well as race/ethnicity ($p=0.047$).

Conclusions: Extraction of treatment exposures from the EHR through clinical informatics methods and integration with cancer registry data represents a feasible approach to assess CVD outcomes and adherence to guideline recommendations for survivors.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0594 / #725 | Challenges in Using Standardized Evaluation Tools in Pediatric Cancer Rehabilitation

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Background and Aims: Rehabilitation is an emerging field due to the increase in survivorship of pediatric cancer patients. Cancer treatments and cancer itself may have deleterious impact in the physical, cognitive, social and functional aspects of the child life, that can even follow them into their adulthood. It is imperative that we can assess these side effects early on so the rehabilitation process can prevent and diminish their impact. The use of standardized evaluation tools may be useful to assess this population.

Methods: A small survey was completed by three different rehabilitation professionals who work in the Oncology Rehabilitation Center of Fundacion Nuestros Hijos: an occupational therapist, a physiotherapist and a clinical neuropsychologist. The range of years they have worked in pediatric oncology goes from 3 to 6, with 180 patients referred annually. They were asked to identify the standardized evaluation tools they use, including questionnaires, and to categorize them on "no difficulty in applying" and "some difficulties in applying". If one of them classified in the second group, they had to further describe what is the main issue.

Results: Across the three disciplines, a total of 18 evaluation tools were identified: 15 standardized tools and 3 were questionnaire type tools. Of the 15, all of them present problems while using them in the pediatric cancer population. The main difficulties were related to motor impairments, such as impossibility to walk (2 tests), ataxic syndrome (3 tests) and impossibility to write (9 tests). Only one had time-related difficulties.

Conclusions: While the use of standardized evaluation tools is a vital factor in measuring functional impairments in the pediatric cancer population, many of them were designed to use on typical children. It is necessary to adapt these tools for their use in this population to correctly assess their functional levels and improve their quality of life.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0595 / #727 | Rise in Bacteremia Cases due to Streptococcus Mitis Caused by the Delay in Referral during the Covid-19 Pandemic

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Background and Aims: Streptococcus Mitis bacteremia has been on the rise for the last 10 years in the pediatric cancer population. Risk factors include having an hematologic cancer, undergoing chemotherapy (especially after 3 weeks), the use of methotrexate, having mucositis and severe neutropenia. This has a special interest in our pediatric cancer population who are more at risk to develop severe complications related to these bacteria. Because of the COVID-19 pandemic, dental attentions were interrupted, and referrals stopped during quarantines. In the last two trimesters of 2020, when dental attention resumed, a rise of cases and severity regarding S. Mitis bacteremia occurred.

Methods: We revised the clinical information regarding dental attention since 2017, and we registered all the notifications regarding streptococcus mitis bacteremia. The information regarding the severity of the clinical manifestation was also recorded.

Results: Since 2017, 60 patients have been referred to dental treatment. Until March 2020, when the national lockdowns started and dental attentions had to be suspended, there were 0 cases of streptococcus mitis bacteremia registered. From September 2020, when attentions resumed, until March 2021, three (3) cases of S. mitis have been reported. The 3 were in the intensive care unit (ICU), and one of them was there three times due to the bacteremia. After their

discharged they were immediately referred to dental treatment to prevent more complications regarding the s. mitis.

Conclusions: Dental care and surveillance is a key factor when we talk about pediatric cancer patients. The COVID-19 pandemic has proven to us that, when untreated, patients can develop severe complications that can even be life threatening. It's important to educate patients, families, physicians and nurses about the importance of early dental intervention.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0596 / #733 | Evaluation of Cellular Senescence Biomarkers during Premature Aging Following Acute Lymphoblastic Leukemia Therapy

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Background and Aims: Acute lymphoblastic leukemia (ALL) survivors suffer long-term effects of therapy suggestive of a syndrome of premature aging. Senescent cells accumulation underlies many age-related diseases and occurs following cancer therapies. Our hypothesis is that T-cell receptor excision circles (TREC) and the senescence-associated secretory phenotype (SASP) blood measurements as biomarkers of immunosenescence represent a quantitative non-invasive strategy to assess the level of premature aging in this population. In this study, we aim to evaluate quantitatively biomarkers of senescence and premature aging (TREC and SASP) in ALL survivors in order to determine their biological age.

Methods: Using genomic DNA and serum from ALL survivors (n=248) and healthy control participants (n=281) blood samples, TREC and SASP were measured respectively by qPCR and multiplex ELISA. The biological age of the ALL survivors was determined by extrapolating from the healthy control group.

Results: Immunological age extrapolated from TREC levels in ALL survivors shows decades of accelerated aging (average 35.79 yrs, S.D. ± 12.54). This accelerated aging is associated to metabolic syndrome and physical deconditioning in females. Also, TREC significantly correlate with several SASP factors (CRP, IL-6, VEGF, Flt-1, IL-15, IL-12/IL-23p40, VCAM-1). Interestingly, for many SASP factors, the correlation with TREC is better than the correlation with chronological age, suggesting that TREC levels are an accurate predictor of biological/immunological age.

Conclusions: These results suggest that, compared to chronological age, TREC and several SASP factors levels could be better measures of actual aging state and predictors of early-onset chronic diseases in

childhood cancer survivors. Further studies are needed to refine these tools, which could contribute to optimized follow-up strategies for ALL survivors.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0597 / #746 | Age-Expected Participation in Survivors of Pediatric Brain Tumor: Frequency and its Association with Fatigue and Cognitive and Physical Functioning

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Background and Aims: Pediatric brain tumor survivors (PBTs) experience several adverse disease and treatment related sequelae, which can have serious implications on their participation abilities in daily life. The aim of this study was to investigate the frequency of participation limitations, problems in functioning and fatigue, and to identify factors influencing participation levels and fatigue.

Methods: Children (4-18 years) after treatment for a pediatric brain tumor or with an expectant approach were eligible (n=147). Data were collected using the parent-reported Child and Family Follow-up Survey (CFFS), developed and validated to measure outcomes (i.e. age-expected participation (CASP), impaired cognitive, physical and sensory functioning) in youth with acquired brain injury. Fatigue was assessed using both parent and child versions of the Pediatric Quality of Life Inventory Multidimensional Fatigue Scale (PedsQL-MFS). Associations between disease/treatment related factors, fatigue and participation limitations were explored using univariable analyses.

Results: Ninety-one PBTs (median age: 11.3 years, interquartile range [IQR]: 9.5-14.1) participated in this study. Time since last treatment was 3.9 years (IQR: 2-6.2). Limited participation was reported in 53% (CFFS-CASP<97.5). Parents reported increased fatigue in their children on all scales ($p<0.01$) and children themselves reported increased general and cognitive fatigue, compared to normative values ($p<0.02$). Impaired cognitive and physical functioning ($p<0.01$), and parent- and self-reported fatigue ($p<0.03$) were associated with participation limitations. An expectant approach corresponded with age-expected participation ($p=0.02$). Sex, age at diagnosis, time since diagnosis/treatment, infra/supratentorial tumor, high/low-grade tumor, radio-/chemotherapy or radical/partial resection did not influence participation levels ($p>0.07$). Increased fatigue was reported in children who have had a third ventriculostomy or ventriculoperitoneal shunt ($p=0.04$). Younger age at diagnosis, longer time since diagnosis,

chemotherapy, and radical/partial resection showed associations with more cognitive fatigue ($p < 0.05$).

Conclusions: PBTS are at risk for participation limitations and fatigue at short and long term follow-up. Impaired physical/cognitive functioning and fatigue are negatively influencing participation abilities.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0598 / #789 | Co-Creation of the International Pediatric Oncology Exercise Guidelines (IPOEG) Toolkits and Dissemination Plans

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Background and Aims: The published international Pediatric Oncology Exercise Guidelines (iPOEG) provide information about physical activity (PA) for children and adolescents affected by cancer (e.g., how much PA to engage in). Resources to move this knowledge to practice, and equip end-users (i.e., individuals responsible for adopting and using the iPOEG) with tailored information, are critical to ensure impact. This study engaged end-users to co-create iPOEG Toolkits (i.e., resources) and build dissemination plans.

Methods: A mixed-methods, integrated knowledge translation study was conducted with end-users, who were 'professionals' (i.e., health-care providers, exercise professionals, community-based organization) and 'patients/families' (i.e., children and adolescents affected by cancer, families/caregivers). End-users were engaged through a series of iterative steps and data were collected via online worksheets (closed/open-ended questions) and focus groups (open-ended questions). Questions focused on understanding what resources were needed, what information was important, and how information should be presented. In an iterative cycle, data were analyzed (descriptive statistics, content analysis), integrated into mock-ups by a design specialist and the study team, and provided to end-users for feedback.

Results: Each end-user group ($n=5-21$ participants) completed >2 online worksheets and participated in 1-2 focus groups. Professionals co-created educational resources to distribute to patients/families. Patients/families co-created resources to provide practical tips/tools to engage in PA. The iPOEG Toolkits include posters, infographics, and videos. Dissemination plans to support outreach include email, website, and social media strategies. The iPOEG Toolkits and dissemination plans are amendable to tailoring to meet program and site-specific needs.

Conclusions: The iPOEG Toolkits are moving the iPOEG publication to practice by equipping end-users with the information they need to recommend, support, or engage in PA. Future work will track the dissemination efforts and the impact of the iPOEG Toolkits on end-user behaviour.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0599 / #803 | Pulmonary Involvement in Children with Langerhans Cell Histiocytosis

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Background and Aims: Pediatric pulmonary Langerhans cell histiocytosis (pPLCH) is a rare disease, mostly a component of multisystemic LCH. Pulmonary involvement has been shown in 7-16% of all LCH. We aimed to investigate the clinical features and treatment results in children with pulmonary LCH.

Methods: The clinical characteristics and treatment outcomes of 36 patients with lung involvement of the LCH diagnosed and treated between 1974-2020 were analysed retrospectively.

Results: The lung involvement was detected in 36 (9.8%) out of 367 LCH patients. The median age was 1.7 years (min & max 0.4 & 17.7) with a M/F ratio of 2.3. Ten (27.7%) patients had respiratory symptoms on admission including coughing, wheezing, chest pain and dyspnea. Radiological investigations revealed variable appearance ranging from pneumonic infiltration, reticulogranular and nodular opacities to multiple pulmonary cysts depending on the stage of disease. Two patients were presented with pneumothorax, one patient had pulmonary mass. All patients had multisystem involvement except one with isolated lung involvement. Bone, skin, liver, BM, spleen & thyroid involvement were detected in 21, 20, 7, 5, 3 & 1 patients, respectively. Twenty-nine patients were treated with VBL containing regimens, the rest were treated with different regimens depending on the years of diagnosis at institution. Five-year event free (EFS) and overall survival (OS) were 54.5% and 63.1% in all patients. In children younger and older than two years of age, the 5-year EFS was 53.3% vs 56.3% and 5-year OS was 58.7% vs 68.8%. In children with and without risk organ involvement, five-year EFS was 54.3% vs 54.5% and five-year OS was 54.5% vs 66.8%.

Conclusions: The children with pPLCH have high survival rates. The young age and the presence of multisystem involvement are risk factors and requires careful management to prevent the excess mortality. New targeted therapies are promising in relapsed & refractory cases.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0600 / #835 | Teacher and Audiologist Perspectives of Needs of Children who Acquired Hearing Loss from Cancer Treatment

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Background and Aims: An unintended outcome of many childhood cancer treatment regimens is irreversible hearing loss. Typically, the children acquire bilateral sensorineural hearing loss primarily affecting the high frequencies which adversely affects their ability to understand speech. While they may be able to communicate reasonably well in quiet, one-to-one situations, the difficulty is greater when they need to listen in group situations, over distance and in the presence of background noise. Classrooms, by their nature, are unfavourable listening environments; children with impaired hearing require assistive listening equipment and special educational services from Teachers of the Deaf and Hard of Hearing. Barriers to provision and acceptance of these supports have not been examined specifically for children who acquired their hearing loss in the context of life-threatening cancer.

Methods: Audiologists working in Public Health clinics in British Columbia (N=36) and Teachers of the Deaf and Hard of Hearing in British Columbia (N=100) were invited to participate via online surveys consisting of both closed and open-ended questions; respondents could present up to four child cases. Questions focussed on the professionals' perception of child and family needs and the barriers to meeting those needs. Responses to the open-ended questions were rated using thematic analysis of narratives technique.

Results: Eight Audiologists responded presenting 26 child cases; 18 Teachers responded presenting 42 child cases. Teachers identified greatest child needs in academics, social-emotional adjustment and peer communication and family needs in support. Barriers to meeting these needs included the hearing loss, social-emotional issues and academic challenges. Audiologists identified the greatest need for both child and families as support; barriers included social-emotional, financial and medical needs and lack of access to services.

Conclusions: Teachers and Audiologists strive to help children who acquire hearing loss from their cancer treatment. They identified specific needs for the children and their families and barriers to meeting those needs.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0601 / #867 | Cisplatin-Induced Hearing Loss is Mediated by Toll-Like Receptor 4 in Childhood Cancer Therapy

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Background and Aims: Cisplatin is a highly versatile, platinum-based chemotherapeutic used in the treatment of various cancers and solid tumours in children. Unfortunately, it is associated with irreversible hearing loss, or ototoxicity, in over half the children treated. Not only does this impact the quality of life of childhood cancer patients, it also compromises the dosage of cisplatin that can be administered for cancer treatment. Cisplatin causes cell death and inflammation in the inner ear, but specific mechanisms are not well defined. Toll-like receptor 4 (TLR4) is an immune receptor that mediates contact hypersensitivity to nickel – but its interactions with platinum are unknown. It remains to be investigated if similar interactions between TLR4 and the platinum component of cisplatin are involved in mediating cisplatin-induced ototoxicity. In this study, we investigated the role of TLR4 in mediating cisplatin-induced ototoxicity by interacting with cisplatin and its platinum-based structure. Additionally, we evaluated the efficacy of small molecules in protecting against cisplatin-induced ototoxicity.

Methods: An inner ear cell line, a physiologically relevant model, was examined for inflammatory responses under cisplatin treatment following genetic inhibition of *tlr4*. Additionally, isogenic cell lines expressing or lacking TLR4 were treated with cisplatin/platinum or known ligands of TLR4 and measured for inflammation or cell death. Cisplatin-TLR4 toxicity responses were also evaluated in a zebrafish model of cisplatin-induced ototoxicity after genetic silencing of *tlr4*. Lastly, rescue effects of small molecule inhibitors of TLR4 were investigated in cells treated with cisplatin.

Results: The presence of TLR4 increased cell stress, inflammation, and death in response to cisplatin treatment both in-vitro and in-vivo and was involved in distinct interactions with platinum. Genetic inhibition and small molecule inhibitors of TLR4 protected against these effects.

Conclusions: TLR4 is involved in mediating cisplatin-induced ototoxicity, likely through interactions with its platinum component, and is a promising target for oto-protective therapies to prevent it in childhood cancer patients.

E-Poster Topic: *Disciplines / AS06 Survivorship*

PV0602 / #964 | Engaging Parents of Pediatric Cancer Survivors with Obesity in a Healthy Lifestyle Intervention during the Covid-19 Pandemic: Implications for Trial Recruitment and Data Collection

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Background and Aims: Nearly half of pediatric cancer survivors (PCS) are overweight/obese (OW/OB) and are at an increased risk for metabolic syndrome and other negative long-term physical health complications. We are implementing a cluster randomized, controlled, repeated measures trial (NOURISH-T+, NCT04656496) across four pediatric oncology clinics to engage parents of pediatric cancer survivors (age 5-12) with OW/OB. The onset of the COVID-19 pandemic in March 2020 required significant restructuring of in-person study protocols and had significant implications for participant recruitment and engagement approaches. The aim of this presentation is to describe the challenges, lessons learned, and adaptations to optimize participant recruitment, data collection, and overall engagement in this virtually-based study.

Methods: Since January 2021, a total of 14 parent-child dyads have been enrolled in the study. Parents and PCS are assessed on anthropometric measures, physical activity (PA) and dietary behaviors at baseline, 3-, 6-, and 12-months post-intervention. The coordinating team has rotating bi-weekly check-ins with clinic staff to discuss recruitment progress.

Results: Notable challenges impacting recruitment and engagement identified in discussions with clinic staff and the coordinating team include parental reports of zoom fatigue and feeling overwhelmed by such daily routines as overseeing online schooling of their children. The increased reliance on virtual communication compounds the existing challenge of engaging participants with low technological literacy. Changes made due to COVID-19 include shifting in-person data collection to occur remotely, diversifying recruitment techniques, and developing new procedures to accommodate this rapid shift. Strategies employed include offloading manualized tasks through REDCap automation, increasing flexibility in session schedule coordination, mailing resources when requested to reduce any virtual fatigue and support accessibility, and developing video tutorials and a web-based resource repository.

Conclusions: Despite challenges associated with COVID-19, thus far enrolled families appear engaged and report behavioral changes. Lessons learned are expected to augment recruitment, engagement, and retention in this study.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0603 / #1020 | **Forecasting Global Essential Childhood Cancer Drug Need and Cost: An Innovative Model-Based Approach**

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Background and Aims: Childhood cancer survival in low- and middle-income countries (LMICs) lags behind that in high income countries (HICs). Unreliable access to essential chemotherapy and supportive care medicines contribute to this gap. LMICs require information about total drug needs and associated costs to plan effectively. We developed FORxCAST, a customizable tool to predict needed current and future cancer medication quantity and cost.

Methods: The FORxCAST tool forecasts annual drug quantity and cost to treat 18 common pediatric cancers, customizable to region, regimen, stage distribution, and drug price. We used SIOP-adapted treatment regimens, supplemented with input from disease experts, to model actual treatment approaches. FORxCAST employs a global microsimulation model to estimate incidence for both diagnosed and undiagnosed (total) patients. As a baseline assumption, we have used the latest (2015) Management Health Sciences International Medical Products Price Guide (MSHIMPPG) to create estimates of drug cost to treat pediatric cancers. We quantified total cancer drug need both globally and for specific countries.

Results: The median projected annual costs of drugs to treat all children with 18 of the most prevalent cancers globally is USD \$209.7M (65.3-294.1) for currently diagnosed cases and USD \$355.5M (111.1-498.8) when also accounting for undiagnosed cases. Ranges reflect upper and lower price estimates depending on sources from MSHIMPPG. Comparative volume estimates by drug, region, and diagnosis are provided. The most expensive disease to treat is acute lymphoblastic leukemia, comprising 59% (USD \$124.4M) of estimated global pediatric chemotherapy cost. Drug volumes and costs varied substantially across included case studies comprising of four country specific predictions across three continents.

Conclusions: FORxCAST enables evidence-based, locally-relevant forecasting of childhood cancer drug need. Decision-makers could use FORxCAST to advance innovative procurement strategies at local, regional and international scales to drive global improvements in childhood chemotherapy access.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0604 / #1044 | **Implementation of Telephone Screening for Timely Diagnosis of SARS-Cov2 Infection in Children with Cancer in a Low-Middle Income Country**

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Background and Aims: Children with cancer are at high risk of infection due to immunosuppression, including from SARS-Cov2. A timely diagnosis reduces the risk of severe complications. Access to Covid-19 testing is limited in low-and-middle-income countries (LMIC). However, screening for signs and symptoms may detect Covid-19 infections. We implemented telephone screening for timely diagnosis of SARS-Cov2 infection in children with cancer in Tijuana, Mexico.

Methods: Telephone screening was conducted for children with cancer on active treatment (curative or palliative) or within 5 years off-therapy at Hospital General-Tijuana. Patients and immediate family were screened at least twice a week by a trained social worker. Screened signs and symptoms included cough, fever/chills, diarrhea/vomiting, myalgia/bone pain/fatigue, congestion/runny nose/sore throat, loss of smell/taste, difficulty breathing, pulse-oximetry information (if available) and travel/contact history. If the patient (or family member) screened positive for at least one sign/symptom, the patient was scheduled for a clinic/emergency visit and SARS-CoV-2 testing.

Results: From May 2020 to March 2021, 684 patients were screened via telephone. Sixty (8.7%) screened positive and were immediately scheduled for further evaluation and testing. Thirteen (21.6%) resulted positive for SARS-CoV-2, representing 38.4% of the total children with cancer and SARS-CoV-2 infection during the study period. Three (23%) of patients diagnosed via telephone screening presented moderate neutropenia (absolute neutrophil count 500-1000/mm³) and 80% were not neutropenic. None of the patients diagnosed by telephone screening developed severe symptoms nor required hospitalization or medication. Symptom monitoring continued after a confirmed diagnosis.

Conclusions: Over a third of patients with confirmed SARS-CoV-2 infection screened positive in the initial telephone screening. We demonstrated that in a LMIC, where access to COVID tests is limited due to strained resources, telephone screening may provide a feasible, low-cost and effective approach for COVID-19 timely diagnosis in children with cancer and may reduce unnecessary hospital admissions and complications due to SARS-CoV-2.

E-Poster Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PV0605 / #1063 | Trends in Pediatric Cancers from a Hospital Based Registry, Single Center Experience in LMIC over 10 Years

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Background and Aims: In the absence of population based cancer registries, hospital records can only highlight the importance of collection and analysis of data to address the burden of cancer among children. For this purpose, a hospital-based registry's pediatric cancer data (individuals aged 0-18 at diagnosis) from 2010 to 2019 was compiled and analyzed at The Aga Khan University Hospital (AKUH) in Karachi, Pakistan.

Methods: A paper-based system is used to record all interactions with patients at the hospital. A dedicated team of Cancer Registrars transcribe this data onto a US based registry software; CNExT Registry data was extracted through software generated reports and compiled for analysis.

Results: The total number of children presented at our center was 2,694, out of which 1,673 (62%) were males while 1,021 were females. Top 5 sites were bone marrow (n=887, 32.9%), central nervous system (n=481, 17.8%), lymphoid tumors (n=294, 10.9%), bone (n=255, 9.5%) and soft tissues (n=145, 5.4%). All other sites contributed to 632 cases, 23.5% of the total patient population. Furthermore, 806 patients were aged 0-4, 627 were aged 5-9, 684 were aged 10-14 while 966 patients were 15-19 years old. Geographically, 74.3% of the patient population was from the Sindh province, 25.8% of patients travelled from all over the country as well as Afghanistan covering distances as large as 1,200 km. A consistent increase of 210% was noted during the 10-year period.

Conclusions: Our database analysis shows that the top 5 cancer sites are consistent with other databases around the world. With a broad geographical representation, we can assume this reflects the population at large. Increase in numbers may reflect awareness and referral pattern emphasizing the need for capacity building at the national level to accommodate patients locally so families don't have to travel over provincial and international borders to receive treatment.

E-Poster Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PV0606 / #1111 | Development & Impact of Wish-Fulfillment Program for Pediatric Oncology Patients in a Not-For-Profit Tertiary Care Center in a Developing Country

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Background and Aims: In children, the diagnosis of a life-limiting illness such as cancer can have detrimental effects on their well-being and mental health. Wish granting is a positive intervention used mostly with children on supportive care to manage their distress, instill hope and create a normalizing experience. The present paper discusses the development and impact of wish fulfillment program for pediatric cancer patients at The Indus Hospital & Healthcare Network.

Methods: The program was initiated by the Social Services (SS) team of Pediatric Psychosocial Department of TIH&HN in 2019. The SS team operates the program through funding (financial and in-kind) received by private donors. Four separate categories of wishes were granted that included wishes for an experience, objects (toys etc.), to meet (with their favorite personalities) or to go. In this program children on supportive care were included that are referred by the Palliative Care (PC) team whereas children on curative treatment were also included but they are identified by the Child Life or SS team.

Results: Total 39 children diagnosed with cancer 27(69%) on supportive care, 12(31%) on curative care had their wishes fulfilled under this program. We have fulfilled the wish for an experience of 13(33%) such as being on television, objects 15(38%) for example getting a laptop, to meet 9(23%) such as with sports player Darren Sammy and to go 2(5%) such as for Muslim religious pilgrimage. Children looked forward to the activity, prepared in advance for it and considered it as the one of the best memory of their lives. Parents of these children also reported positive affect and improved coping overall.

Conclusions: The experiences narrated by children and their caregivers show that wish fulfillment can be considered as a positive supportive intervention that promotes well-being and hope in tough circumstances.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0607 / #1122 | Patient and Family Engagement in South African Paediatric Oncology: Knowledge, Communication, and Muzzles

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Background and Aims: The PODC task force assembled researchers from 9 countries in Africa, Asia, and South America for a global project to understand the scope, nature, and extent of patient and family engagement in paediatric cancer. The four facets of patient engagement (Higgins et al., 2018) are personalisation, access, commitment, and therapeutic alliance – this served as a framework to understand the experiences of patients and main caregivers during cancer care, and the role of power.

Methods: After obtaining ethics clearance and permission to conduct research from three hospitals in Gauteng, potential candidates were contacted and invited to volunteer for the interviews. Five women (two caregivers and three survivors) participated in audio-recorded interviews. These were transcribed and analysed using an adapted method of thematic content analysis (Braun & Clarke, 2006) and discourse analysis (Gee, 2011). Five themes were developed.

Results: The doctors were identified as specialist possessors of knowledge, and were obeyed. Doctors were also heavily relied on, generally without complaint. The patient/caregiver is compelled to be a diligent and attentive participant, with rights and responsibilities. Accessing information and asking questions of the professionals was regarded as important. With a strong emphasis on faith and trust in the doctors, relying on their expertise could, sometimes, be experienced as leading the participant to a particular set of actions. Communication was necessary to understand the treatment process though this was often experienced as cold and not reassuring. Emotions were frequently deliberately suppressed, among child patients and family. There was also an overarching imperative ‘to be strong’ even among those with

the same experiences. This was an internal/external invocation to master fear, and endure.

Conclusions: The influential nature of the relationships between doctors, patients and caregivers can negatively/positively affect the aims of the set, and understanding the nature and powers of the relationships will facilitate the process.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0608 / #1280 | Implementing a Pediatric Cancer Surveillance System in Mexico: A Successful Example of Replicability of a Project in Latin-America Supported by the “My Child Matters” Program

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Background and Aims: Clinical outcomes research provides key information to inform clinical decision-making and healthcare policies. In cancer care, survival as metric is complementary to incidence and mortality rates. Population-based surveillance systems for real-time and longitudinal monitoring of clinical outcomes are critical to improve cancer survival. In Mexico, cancer clinical outcomes surveillance systems are lacking. Herein, we describe the implementation of a pediatric cancer surveillance system (PACARSS) embedded in a Population-Based Cancer Registry in Tijuana, Mexico.

Methods: We used the successful VIGICANCER model established in Colombia in 2009 and supported by the “My Child Matters (MCM)” Program as a guide. In 2016, email/phone communication was initiated with the VIGICANCER team, facilitated by an MCM's mentor. In 2017, a needs-assessment visit and on-site training were conducted in Tijuana by the VIGICANCER team. Based on the needs-assessment, a collaborative project roadmap focused on long-term sustainability was developed, including personnel, processes, data collection/analysis methodology and auditing procedures. In 2018, the PACARSS team was assembled and comprised pediatric oncologists(1), project coordinators(1), data collectors(2), analysts(1), advisors(2: pediatric oncologist/epidemiologist from VIGICANCER & pediatric

oncologist/implementation scientist & MCM mentor). In 2019, a 3-day training in Cali was completed. In 2020, we completed VIGI-CANCER protocol adaptation for PACARSS; database development; bi-monthly web-based meetings for process follow-up/feedback (7); and scheduled audits/quality checks (1).

Results: In 2020, PACARSS was launched in Tijuana. After ethics committee approvals were obtained in participating institutions (6), data collection was started. Unfortunately, due to the COVID-19 pandemic, data collection was paused for 8 months. Nevertheless, we successfully collected data in 47 pediatric cases.

Conclusions: Comprehensive needs-assessments and adaptable roadmaps, coupled with intensive on-site and on-line training and VIGI-CANCER/MCM mentorship facilitated replicability of a pediatric cancer clinical outcomes surveillance system in a middle-income country. To ensure reliability, PACARSS data is currently being systematically validated by the VIGI-CANCER team.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0609 / #1298 | **Storytelling in Global Health Education: Teaching about Global Childhood Cancer Disparities and the Sustainable Development Goals Utilizing the How I Live Documentary**

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Background and Aims: The How I Live (HIL) Documentary was created in partnership between the Global Health Initiative at Dana-Farber/Boston Children's Cancer and Blood Disorders Center and Persistent Productions (P2). The film follows children with cancer, their families, and medical teams in Egypt, El Salvador, Ghana, and Myanmar, and documents the challenges of cancer treatment in low- and middle-income countries. As part of HIL's educational distribution and outreach, we developed an interactive virtual curriculum utilizing a 20-minute version of the documentary. The session aimed to help participants (1) understand the challenges of healthcare delivery in resource-limited settings using childhood cancer as a case study, (2) brainstorm interventions to address these challenges, and (3) use the United Nations (UN) Sustainable Development Goals (SDGs) as a framework to analyze key themes in the film.

Methods: Five learning sessions were conducted virtually at universities in the United States with 15-40 participants per site. Data collected included self-reported and real-time post-film reactions and key takeaways collected anonymously through Poll Everywhere. Data was analyzed in NVivo for common themes.

Results: Education sessions were either integrated into an existing class or offered as an extracurricular activity in partnership with a student interest group. Participants were a mix of undergraduate and

graduate (medical or public health) students. Most common immediate film reactions included both upsetting emotions including frustration and sadness as well as positive emotions including motivation and inspiration. Major themes from the key takeaways included: obstacles to treatment including infrastructure, access and patient retention, hope, equity, and the unique role of documentaries in global health.

Conclusions: Documentary filmmaking is an underutilized and compelling tool for global health education. Storytelling allows learners to gain a more nuanced understanding of a global health issue and accompanying facilitated and interactive discussion helps learners apply this understanding to practical solutions.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0610 / #1367 | **Feasibility of Implementing the Toronto Childhood Cancer Stage Guidelines in a Sub-Saharan Africa Hospital-Based Cancer Registry (GFAOP)**

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Background and Aims: Toronto Childhood Cancer Stage Guidelines target population-based registries, seeking to provide internationally comparable information to evaluate childhood cancer outcome. Two tiers are proposed to classify 16 tumor types, Tear-1 coding localized/ (regional)/metastatic disease and Tear-2 applying more detailed international codes. We conducted a pilot study to test feasibility of Toronto staging in a hospital-based registry in low-income and middle-income countries, in Sub-Saharan Africa.

Methods: Seven GFAOP centers with at least 40 new cases diagnosed per year from 2017 to 2019 were selected as pilot centers (Burkina Faso, Cameroun, Democratic Republic of Congo, Ivory Coast, Madagascar, Mali and Senegal). Staging was not possible for 4 tumor types, leukemias (CNS involvement was not available) and CNS tumors (histology was not available). Finally, we included 1766 children <18 years with one of the 12 remaining tumor types (Hodgkin's and non-Hodgkin's lymphomas, neuroblastomas, retinoblastomas, Wilms' tumors, hepatoblastomas, osteosarcomas, Ewing's sarcomas, rhabdomyosarcomas, non-rhabdomyosarcoma soft-tissue sarcomas, testicular and ovarian tumors).

Results: Tier-1 staging was performed for 1541 cases (87%). Tear-2 staging was considered possible for 6 tumor types (Hodgkin's and non-Hodgkin's lymphomas, Wilms' tumors, osteosarcomas, Ewing sarcomas and hepatoblastomas). Of the 1034 cases with these tumor types, 838 (81%) could be assigned the detailed Tear-2 stage.

Conclusions: Patient care is based on the capacity of the service to correctly diagnose and stage cancer, a major indicator for prognosis. This pilot study shows that Tier-1 staging can be implemented from the standard framework for most tumor types, and that Tier-2 level is possible for some of them. The staging for a few pathologies requiring a biopsy such as brain tumors is not feasible. The GFAOP hospital-based cancer registry plans to routinely implement Tier-1 and aims to improve completeness for the cancer groups already staged. Prospective staging of leukemia will be considered in these settings.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0611 / #1437 | Securing Continuation of Treatment for Children with Cancer in Times of Social Unrest and Pandemic

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Background and Aims: In Chile, Governmental programs ensure access to cancer treatment for all children and adolescents. Fundación Nuestros Hijos (FNH), a local NGO supporting childhood cancer, provides care for non-covered medical and psychosocial needs. Common financial difficulties in families of children and adolescents with cancer increased considerably when a wave of social unrest arose in Chile in October 2019 and the Covid-19 pandemic in March 2020 hit the country, leaving families of children with cancer facing greater challenges. Here we describe FNH's actions to support these families and prevent treatment delay.

Methods: A socio-economic survey was conducted to learn about the families' main needs. Concurrently, interventions were conducted to assist the families. The data from the survey had descriptive purposes. Four main interventions: Safe and private transportation, food, heating, and internet connectivity were implemented with the goal of preventing treatment discontinuation and reduce families' economic stress.

Results: The main results of the survey in which 525 (70%) of FNH's beneficiary families participated showed that 75% had only one breadwinner, and 52% had one unemployed family member. Almost 90% of job loss occurred during the months of social unrest and pandemic. To avoid COVID-19 contagion and treatment interruption, safe transportation services (1,062) were given. Food family baskets (420), meals (1,290) for parents of hospitalized children, and groceries gift cards (50) were provided. Gas purchase discount tickets (1,100) and stoves (312) were donated. Hospital school and rehabilitation services were re-organized to online format, and tablets (82), phones (95), and internet chips (50) were given.

Conclusions: The services provided helped the families to relieve some of their needs, facilitated the continuation of treatment during the pandemic, and made the caregivers feel supported and listened to. The online modalities for the school and rehabilitation programs will be kept as permanent-mixed services after the pandemic.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0612 / #1484 | The Tanzanian Pediatric Cancer Network: A Comprehensive Evaluation of the Incidence of Presenting Patients and Access to Care in Tanzania

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Background and Aims: In sub-Saharan Africa, cancer treatment facilities are often scarce and centered in urban areas, which means many patients must travel long distances to reach cancer care. Still, little is known about how this distribution of cancer centers impacts patients' ability to access care.

Methods: Records of all pediatric patients (<18yo) treated for cancer in Tanzania in 2019 through the Tanzanian Pediatric Cancer Network were reviewed. This network represents all 9 hospitals currently treating children with cancer in Tanzania. Demographic and diagnostic information was recorded. ArcGIS was used to estimate patients' travel times from home addresses to cancer care. AccessMod was used to calculate 4-hour access for the population.

Results: In 2019, a total of 821 pediatric patients were diagnosed with cancer with an incidence of 1.42/100,000 population. The most common diagnoses were Wilms tumor (17.2%, n=134), retinoblastoma (16.8%, n=131), and acute lymphoblastic leukemia (15.9%, n=124). There was regional variation by diagnosis, with the Lake Zone having the highest incidence of Burkitt lymphoma (0.29; average 0.15) and retinoblastoma (0.5; average 0.33). Reported regional incidence inversely correlated to distance to cancer treatment facility. Median travel time to cancer care for patients was 4.55 hours. Currently 74.5% of the population lives within 4 hours of existing cancer care facilities, and it would require adding cancer care capacity to 8 additional facilities to reach 95% coverage.

Conclusions: This is the first comprehensive evaluation of the incidence of presenting pediatric cancer patients in Tanzania. Only 18.8% of the estimated pediatric patients with cancer in Tanzania were treated in 2019. The reported incidence was lower in areas further away from treatment centers, suggesting the importance of increased

access to cancer care facilities as part of a future national scale-up strategy.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0613 / #1491 | Swahili Translation and Cultural Adaptation of the Pediatric Patient-Reported Outcome Version of the Common Terminology Criteria for Adverse Events

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Background and Aims: A pediatric version of the adult patient reported outcome adverse event survey (PRO-CTCAE) was developed and validated for use in pediatric cancer clinical trials to better capture the symptom experiences through direct self-report. The aim of this study was to develop a Swahili language version of the Pediatric PRO-CTCAE.

Methods: The measure was translated into Swahili using mixed qualitative and quantitative methods. The child and caregiver versions of fifteen core symptom AEs, and the corresponding questions, were selected from the Pediatric PRO-CTCAE library, then forward and back translated into Swahili. The translated items were further refined using concurrent cognitive interviewing. Each round of interviews included five children, ages 8-17 years-old, receiving cancer therapy at Bugando Medical Centre, the cancer referral hospital for Northwest Tanzania, and continued until >80% concordance reached.

Results: Three rounds of revisions were completed involving 13 patients. A total of 50% of questions (19/38) were fully comprehended after the initial evaluation. After three rounds of interviews, all questions obtained at least 80% comprehension with no further revisions required. Two AEs (peripheral neuropathy and anxiety) were the most difficult for patients to understand, correlating with education level and experience.

Conclusions: A Swahili Pediatric PRO-CTCAE was effective in eliciting patient reported outcomes related to cancer treatment, with good comprehension for children aged 9 to 17. This survey is important to incorporate patient self-reporting of symptomatic toxicities and is an effective tool to increase capacity for pediatric cancer clinical trials throughout East Africa, further reducing global disparities in cancer care.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0614 / #1551 | Needs of Parents of Children with Cancer for a Parent Supportive Meeting at Indonesia Academic Hospitals

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Background and Aims: Parent supportive meetings may assist families to reduce distress of having a child with cancer. However, little is known about what parents regard as their needs for support. Objectives of this study are: 1) to investigate needs and preferences of parents of children with cancer for parent supportive meetings; 2) to examine the relationship between these needs and various socio-demographic and clinical characteristics.

Methods: This cross-sectional study was conducted using a semi-structured questionnaire. All consecutive parents of childhood cancer patients who attended the inpatient and outpatient clinic of an Indonesian academic hospital were requested to join the study.

Results: Between March and July 2019, 200 parents participated in this study (response rate 95%). Parents were interested in group (90%) or individual (84%) support to help them deal with stress related to their child's illness. Best time to attend supportive meeting would be: during treatment (51%), shortly after diagnosis (43%), after treatment completion (6%). Best format: drop-in basis (83%), set number of sessions (18%). Best time schedule: monthly (58%), every 2 weeks (30%), weekly (10%), only once (2%). Most parents (77%) considered a supportive meeting with guest speakers in question-answer session very useful. Parents considered attendance very important to: find out about services and available support in community (70%), get advice about family or relationship issues (62%), learn about how other parents cope with stress and anxiety (59%). Significantly more mothers (93%) than fathers (84%) are interested in parent supportive meetings ($P=0.046$). More frequent meetings are preferred by more low-educated (48%) than high-educated parents (31%, $P=0.015$), and by more young (55%) than older parents (35%, $P=0.005$).

Conclusions: Parents of children with cancer at a large Indonesian academic hospital are interested in monthly drop-in based parent supportive meetings during the whole of treatment to help them better cope with stress related to their child's illness.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0615 / #1577 | The World of Picture Books for Children with Cancer: Content Analysis**S. Fazeli

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Background and Aims: Numerous studies have offered ways to help sick children cope with their illness and overcome their multiple challenges using arts-based approaches such as story-telling (Carter & Ford, 2013; Wilson, Hatson and Wyatt, 2015). The aim of this paper is to examine the actual themes that are being taken up in one genre of the story dealing with issues of childhood cancer, picture books, and their potential for supporting children with cancer.

Methods: I did a content analysis of 15 picturebooks that have been written for/by children with cancer. This approach is based on the methods suggested by Vandergrift (1999) and Hintz and Tribunella (2013) focusing on two main steps: analyzing each element in picture books in isolation and then analyzing all the elements together. Based on codes and thematic analysis I explore the themes, illustrations, messages, and interpersonal and textual meaning (Painter, Martin & Unsworth, 2013) of these books.

Results: In this study, I consider themes such as the coping mechanisms fictional characters use, treatment of emotions such as hope or anxiety, and the ways in which these books serve to entertain and educate children and their parents. Statistics show that every two minutes a child is diagnosed with cancer in the world (Unger, 2016). Physical changes in these children could affect their self-perception, confidence, and social interactions significantly (McCaffrey, 2006). Drawing on textual and visual representations of the books, I consider context-specific recommendations for children's authors and illustrators, along with educators and health care professionals who are working in this area with/for children with cancer.

Conclusions: Analysis suggests that these picturebooks have beneficial aspects in relation to acceptance, hope and dealing with challenges. However, I note that there are various gaps in relation to the age groups covered in these books, race, the tone of the texts and the educational messages they are delivering.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0616 / #1627 | Psychosocial Pediatric Oncology Traumas & Art Therapy Challenges**M. Elewah

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Background and Aims: A diagnosis of pediatric cancer marks a family journey full of emotional, psychological, physical and practical challenges. In the low and middle-income countries, childhood cancer treatment focuses on the treatment modalities, medications and basic

services. Based upon the increasing impact of childhood cancer cases over the last decades and the equally less advances available support services in the low and middle-income countries, the awareness and the ability to address the support services is extremely limited. The Art Therapy is the healing tool to re-address the psychological and psychosocial dimensions of childhood cancer aiming to improve entire family quality of life.

Methods: As part of the various interventions used in the art therapy workshops, different interventions shall be presented, explained in depth. In addition to MAC scale from IPOS used as assessment tool

Results: The prevalence of anxiety, depression and psychosocial problems during treatment received by childhood cancer patients have rarely been studied. The primary aim of this presentation is to determine the point of prevalence and cumulative prevalence of anxiety and depression in pediatric cancer patients and types of art therapy interventions that could ease the pain. The secondary aim is to evaluate the sociodemographic and clinical factors associated with anxiety and depression from diagnosis then remission and then end of treatment using MAC scale results.

Conclusions: However, emotional care is just as important as physical care throughout the cancer journey. Four recommendations are offered for advancing the support services using art therapy interventions: conduct research to better understand the psychological and psychosocial dimensions of pediatric cancer patients and their families in low, and middle income countries, expand the number of national pediatric cancer control plans that address pediatric cancer support services; develop and disseminate guidelines for pediatric cancer support services; and form regional partnerships to improve psychosocial better care, better cure, better life.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0617 / #1744 | Impact of Covid-19 on Treatment and Follow Up in Children with Pediatric Cancer, US and UK Comparison**R. Rohrer

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Background and Aims: To compare both reported experience in published literature and family interviews in the United States and the United Kingdom to determine similarities and differences in responses in the pediatric oncology communities in these counties as part of a global extensive study. The study attempts to determine if and how some stakeholders were able to mitigate the impact of COVID on treatment access, continued as opposed to interrupted care, and the use of support services to children and families.

Methods: The author surveyed articles and editorial comments in published articles as well as national newspapers and oral interviews where possible. Physician and hospital recorded interviews, and in the UK, statements, articles and interviews particularly from the NHS were critical sources.

Results: The COVID-19 crisis has brought to the forefront the greater challenges facing the US and UK healthcare systems. The results of this study point to the following overriding concerns by medical teams and families: interruptions in treatment protocols, delays in particularly follow up/ response to treatment CT or MRI imaging, reduced supportive services. Delays in all these areas were more significant in the NHS system; rural areas of the both the US and UK have experienced the most setbacks, beginning with the ability to travel to care and impacting all other services.

Conclusions: Roughly fourteen months into the COVID-19 pandemic, healthcare systems attempt to care for COVID and patients with other vital needs. Delayed preventative care, surgeries, and procedures have impacted both the pediatric and adult oncology communities. Will it even be possible for cancer treatment to “catch up” in many ways? How can we estimate how many have lost a chance for cure because of the COVID emergency? This study hopes to expand globally, particularly to include the experiences of Asian and African pandemic and healthcare responses.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0618 / #209 | Education and Childhood Cancer Patients ~ Interventions for Supporting Academic Success Throughout the Childhood Cancer Journey

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Background and Aims: Receiving a childhood cancer diagnosis and undergoing treatment can impact the academic success of any age patient. Supporting the educational endeavors of childhood cancer patients is a cornerstone in program delivery.

Methods: At diagnosis, iPads and/or eReaders provided at no cost to patients. Provide tools for patient students to connect with their school environment, engage in virtual learning, access to unlimited resources via the internet and various age appropriate educational software applications. Parents of on-treatment patients have access to specific financial assistance funds for things including 1:1 tutoring, assistive devices, special software or hardware. Post-Secondary School bursary available for both On Treatment and Off Treatment Survivors. The establishment of Education Support ~ Survivors acknowledges that education challenges can continue to exist years after the completion of treatment and at any age. Depending on the age of a child at the time of diagnosis, a childhood cancer survivor can be in elementary school or middle school.

Results: Including younger, school aged children in the scope of eligibility for this program and the provision of additional supports at a young age, the Education Support for Survivors program will hopefully lead to

earlier academic success, potentially decreasing the need for education support as the survivor gets older.

Conclusions: The desired outcome of providing education support throughout the childhood cancer journey is to make a positive impact on a survivors' long-term development, including the attainment of major societal goals that include education, employment and functional independence.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0619 / #245 | Study of the Differences between Cancer Patients with Fever According to the Place of Origin (Home Vs. Hospital)

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Background and Aims: Childhood cancer survival rates have improved in the last decades. Taking into account the great diversity among patients with febrile neutropenia, efforts continue to be made to differentiate between patients with low and high risk of serious complications and to characterise different subgroups. In this study, we analyzed the influence of patient location (home vs hospital) when fever was detected.

Methods: This is a prospective descriptive and analytical study. We evaluated febrile episodes in paediatric patients with cancer between January 2018 and December 2019 at an University Hospital. We collected data pertaining to admission date and diagnosis, medical history including underlying cancer characteristics, disease staging and comorbidities.

Results: Fever episodes were analyzed according to initial location (99 at home vs. 35 at hospital). Blood count values and number of hours of fever before receiving medical assistance were lower in hospitalized patients. Number of episodes with uncontrolled disease was higher in patients admitted. Original location of patients did not have any influence in the final diagnosis of bacterial infection. All episodes identified in patients already hospitalized, ended up also at hospital. Out of the 99 episodes initiated at home, 42.4% returned to home. Those who were finally admitted, had lower hemogram counts. The diagnosis of bacterial infection was more frequent in the group of admitted patients ($p=0.009$) while non bacterial infections was more common in those who were discharged home ($p=0.018$). All that finished at hospital received antibiotic and 54,7 % who were finally sent home. Final outcome was favorable in all patients.

Conclusions: The group of patients who are at home at the beginning of their fever episode, they are more likely to come back home if the following conditions are met: good general condition, controlled disease and normal counts of leucocytes. The probability of bacterial infection in this group will be lower.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0620 / #255 | Covid-19 Infection in Children with Cancer: Single-Center Experience from Turkey**N. Eker¹, A.G. Tokuç¹, B. Tufan², S. Aras¹¹Marmara University Medical Faculty, Pediatric Hematology And Oncology, Istanbul, Turkey, ²Marmara University Pendik Education and Research Hospital, Pediatric Hematology And Oncology, Turkey

Background and Aims: Covid-19 infection is thought to cause a milder clinical course in children. The general opinion is that Covid-19 infection may progress severely in children with cancer. We evaluated the outcomes of Covid-19 infection in children with cancer in a single center from Turkey.

Methods: We evaluated the patients from their files, retrospectively.

Results: In our center, between March 2020 and January 2021, the treatment of 163 patients (71 leukemia and 92 solid tumors) was ongoing. HSCT was performed in 6 patients with an urgent indication. Twenty-two patients were diagnosed with Covid-19 infection. Six (27.2%) patients received azithromycin and hydroxychloroquine, 5 (22.7%) patients received favipiravir and 11 (50%) patients did not receive any treatment for Covid-19 infection. Anticoagulant treatment was initiated in 6 (27.2%) patients with high D-dimer values. Considering the patients who received steroid treatment, one case was already taking steroids due to HLH. Two patients were diagnosed with PIMS, and one patient had received steroid therapy for septic shock. The median time of fever was 4 days. Only two of all patients needed oxygen and respiratory support. In the other twenty patients, six were treated at home, and 14 were inpatient and without serious complications. Considering the mean time to be negative for Covid PCR, it was 19 days in 18 cases, except for 3 cases that have not yet become negative and one case that has never become positive. During this period, chemotherapy of 14 (63.6%) patients was disrupted. The mean time was 20 days. Disease progression developed in only one of the patients whose treatment was disrupted.

Conclusions: The rate of severe complications in children with cancer was low in our cases. This result brings to mind the question of "Could being immunosuppressive prevent serious complications in children with Covid-19 infection?". Comprehensive studies are needed to answer this question.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0621 / #415 | Perceptions of Adolescents with Cancer Participating in Shared-Decision Making: A Mixed-Method Study**L.-M. Wu¹, S.-S. Chiou², P.-C. Lin², Y.-M. Liao³, H.-L. Su⁴¹Kaohsiung Medical University Hospital and Kaohsiung Medical University, Department Of Medical Research And School Of Nursing, Kaohsiung City, Taiwan, ²Kaohsiung Medical University Hospital and Kaohsiung Medical University, Division Of Hematology And Oncology Department Of PediatricsAnd College Of Medicine, Kaohsiung City, Taiwan, ³Kaohsiung Medical University Hospital, Division Of Hematology And Oncology Department Of Pediatrics, Kaohsiung City, Taiwan, ⁴Kaohsiung Medical University Hospital and Kaohsiung Medical University, Department Of Nursing And School Of Nursing, Kaohsiung City, Taiwan

Background and Aims: Adolescents have their rights to be involved their treatment decision making. However, health teams and parents desire to shield them from distressing information. This study was to describe the perception on levels of decision-making during cancer treatment for adolescents with cancer and examine the trajectory of their decisional conflicts over time.

Methods: An explanatory mixed-method was used, incorporating questionnaires and individual interviews. Participants were recruited from the medical center hospital in southern Taiwan. Inclusion criteria were: (1) at age between 13 and 20 years; (2) diagnosis with cancer and receive cancer treatment; and (3) able to communicate and provide a written consent. Each participant was assessed with decisional conflict and coping questionnaires and was evaluated at four time points (baseline, 1 month, 3 and 6 months). The interview data was analyzed qualitatively to gain an in-depth understanding of shared decision-making after receiving cancer treatment.

Results: Participants perceived significantly the highest decisional conflict, uncertainty, and values clarity at 2 months post cancer diagnosis (all $p < 0.05$). They were generally involved in minor decisions with the purpose of gaining their fully cooperation, making treatment more feasibility, felt sense of control and building trusting relationships. The level of shared decision-making in cancer treatment was: informed and followed, informed and consulted, active inquiry and sharing information, collaboration and control. Participants highlighted how their roles in health care discussions varied from direct participation to indirect involvement. Overall, doctors and parents decided their level of involvement communication and/or decision-making. Love and trust were key components during cancer treatments.

Conclusions: Adolescents with cancer perceived different levels of participating in shared-decision making during receiving cancer treatment. Participants generally valued their participation but some may choose to shy from distressing information. Future research needs to develop the best clinical care model based on the level of participants needs.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)***PV0622 / #486 | Preparing Children to Cancer Treatment: Multimodal Patients' Informing Approach in the Process of Care**

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Background and Aims: Proper organization of patients' preparing to the different types of treatment is an important challenge facing pediatric oncology. The quality of the information process affects children's psychological adjustment to treatment and further compliance.

Methods: Using interview method an analysis of patients and their families' needs regarding information for better preparing to treatment was conducted. The approach to patients' informing was analyzed through interviews with physicians. A comparison of the data obtained allowed identifying the effective approach to the informing.

Results: Analysis of interview data determined the need for tools facilitating the treatment' preparing for radiotherapy, HSCT, surgery from the perspective of vivid, clear and age-appropriate content – Teddy bear photo-tours stories. The photo-tours describes and shows the main procedures and conditions that children will face during cancer treatment. Using photo-tour story when informing helps physicians - to facilitate the process of identifying and negotiating family anxieties, prejudices; also families – to clarify expectations, discuss roles, and elaborate coping strategies if required. As a result, by feedback received there was a decrease in the level of family anxiety, increase in satisfaction with feeling of support and in understanding of treatment conditions and goals, as well as increase in compliance.

Conclusions: Applying a multimodal approach to informing provides the advantage in identifying prejudices, fears and timely provide the information or assistance required. Photo-tours as a visual support help to solve the issue associated with children age, families' sociocultural differences, and level of education more effectively.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0623 / #713 | Immigrant Disparities in the Lived Experience of the Covid Pandemic in Pediatric Oncology Patients in New York

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Background and Aims: Pediatric oncology patients receiving treatment at a large academic medical center in New York City (NYC) represented a microcosm of communities more severely affected by the COVID-19 pandemic. We describe parents' perspectives of the pandemic's impact on families.

Methods: A bilingual (Spanish/English) survey adapted for the pandemic and low literacy participants using established instruments was

administered by telephone in preferred language to parents/guardians of pediatric oncology/transplant patients aged 0-18 years, with data collection in REDCap and analysis in SPSS.

Results: We interviewed 30 families (ongoing); 18 preferred Spanish, 12 preferred English. Henceforth referred to as 'Spanish' and 'English'. In contrast to English, 50% of Spanish had <9 years of schooling and 41% needed help understanding written health materials. Average household crowding varied, with 4.8 (Spanish) v. 5.6 (English) members, though 50% of Spanish households had 3–5 persons sharing the patient's bedroom. Among all, up to 9 persons shared the patients' bathroom, and 23% of families shared their kitchen with another family; 58% of Spanish and 33% of English households included people working outside the home. Pandemic impact on finances resulted in 44% of Spanish (but no English) scoring low food security, with 88% (Spanish) and 42% (English) reporting more difficulty affording food. Independent of language preference, 41% reported family member job loss, 50% reduced spending on groceries, while 33% of Spanish reduced healthcare spending. If they lost current income, 6% of Spanish but 33% of English could survive one year. Overall, 27% lost at least one family member to COVID-19. Fear of SARS-CoV-2 exposure affected 87% when retrieving prescriptions, while 22% Spanish reported missing medical appointments due to fear of exposure in available means of transportation.

Conclusions: Pediatric oncology patients' families in NYC suffered multiple pandemic related challenges, with greater burden in those preferring Spanish, and far-ranging implications for care provision beyond the pandemic.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0624 / #760 | "Money was the Problem": Financial Difficulty is the Main Cause of Childhood Cancer Treatment Abandonment in South West Uganda

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Background and Aims: Treatment abandonment contributes significantly to poor survival of children with cancer in low-middle-income countries (LMICs). In order to inform an approach to this problem at our Cancer Unit, we investigated why caregivers withdraw their children from treatment.

Methods: In a qualitative study, in-depth interviews were conducted with caregivers of children who had abandoned cancer treatment at the Paediatric Cancer Unit (PCU) of Mbarara Regional Referral

Hospital (MRRH) in South Western Uganda, between May 2017 and September 2020. Recorded in-depth interviews with caregivers were transcribed and analyzed to identify themes of caregiver self-reported reasons for treatment abandonment. Ethical approval was granted by Research and Ethics Committee of Mbarara University of Science and Technology. Consent was sought from participants before each interview was conducted.

Results: Seventy-seven out of 343 (22.4%) children treated for cancer at MRRH abandoned treatment during the study period; 20 contactable and consenting caregivers participated in the study. The median age of children's caregivers was 37 years and most (65%) were mothers. At the time of this study, eight (40%) children were alive and 5 (62.5%) were males; with a median age of 6.5 years. Financial difficulties, other obligations, the child falsely appearing cured, preference for alternative treatments, belief that cancer was incurable, fear that the child's death was imminent and chemotherapy side-effects were the caregivers' reasons for treatment abandonment.

Conclusions: Treatment abandonment among children with cancer in Uganda is, most times, as a result of circumstances beyond the caregivers' control and interventions need to approach this issue with empathy and support, rather than judgement and condemnation.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0625 / #821 | Effects of Covid-19 on Pediatric Cancer Care, A Multicenter Study of 11 Middle Eastern Countries

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Background and Aims: During COVID-19 pandemic, major challenges are facing pediatric cancer centers regarding access to cancer centers, continuity of the anticancer therapy, hospital admission and infection protection precautions. This multicenter study discuss the effects of COVID19 on pediatric cancer care in different countries with different resources.

Methods: Pediatric oncologists actively treating children with cancer from 29 cancer centers at 11 countries were asked to answer a survey from May 2020 to August 2020 either directly or through the internet.

Results: COVID-19 pandemic affected access to pediatric cancer care in the form of difficulty to reach the center in 22 (75.9%) centers and affection of patients' flow in 21 (72.4%) centers. Health care professionals (HCP) were infected with COVID-19 in 20 (69%) surveyed centers. Eighteen centers (62%) modified treatment guidelines. Care of follow up patients was provided in-hospital in 8(27.6%) centers, through telemedicine in 10 (34.5%) centers and just delayed in 11 (38%) centers. Pediatric oncologists had different expectations about the future effects of COVID-19 on pediatric cancer care. Sixty-two pediatric oncologists think that children with cancer are more susceptible to COVID-19. Seventy-six percent pediatric oncologists think COVID-19 pandemic will increase the use of telemedicine. Fifty-five percent pediatric oncologists think if the COVID-19 pandemic persists, we will need to change chemotherapy protocols to less myelosuppressive ones

Conclusions: COVID-19 pandemic has affected pediatric cancer care. We recommend We recommend doing further collaborative studies to prioritize pediatric cancer management during COVID 19 era. (Clinical trial identifier: NCT04374838)

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0626 / #864 | Patients, Survivors and Parents Engagement in Childhood Cancer Treatment in Iran (2019): A Qualitative Social Study of Expectations and Experiences

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Background and Aims: Background and aims: Iran has 83 million inhabitants and 19,973 children were diagnosed with cancer over the last 10 years. MAHAK Charity hospital offers free treatment to all Iranian children with cancer. In 2019, we collaborated with the SIOP PODC Patient, Family and Stakeholder Engagement Task Force on field research to document recent and longer-term survivor and parents' perceptions, expectations and experiences of engagement during cancer treatment at various hospitals in Iran.

Methods: Fifteen in-depth interviews and three focus group discussions (FGDs) were conducted by an Iranian qualitative researcher (sociologist) with recent adolescent/young adult survivors (12-20 years), longer-term survivors (19-31 years) and parents (36-61 years; 6 mothers, 3 fathers) from diverse socio-economic and ethnic backgrounds. Data was recorded and transcribed in Farsi and translated into English for content analysis.

Results: The participants listed a wide range of factors affecting their relationship with health care professionals. Their expectations included inter alia improved communication and information flow among the key stakeholders including doctors, and nurses when speaking with the family members and the patients themselves. They emphasized the need for psychosocial support, child-specific psychological support, and the importance of peer support groups (for survivors and parent). Participants identified areas of satisfaction and dissatisfaction regarding their actual engagement and decision-making during the treatment journey.

Conclusions: Conclusion: A multidisciplinary approach needs to be developed and reinforced throughout the childhood cancer treatment procedures in Iran. The medical aspect of the cancer treatment is generally privileged in the health care system together with ignorance of the psychosocial aspects and services during treatment leading to poor attention to social and psychological needs of parents and patients. MAHAK Charity hospital was shown to be pioneering the introduction of such an interdisciplinary approach and found to represent much of the satisfactory aspects of social engagement for the family and patients.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0627 / #934 | Use of Mhealth Application to Improve Pediatric Cancer Treatment Protocol Compliance in Low Resource Settings

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Background and Aims: Low treatment compliance is a well-established barrier responsible for the survival gap between low- and high-income countries for pediatric cancer, including Burkitt Lymphoma (BL), one of the most common cancers diagnosed in East Africa. An mHealth application, mNavigator, was used to improve effective delivery of best practice care through algorithm-directed BL treatment. A pre-post study design was used to evaluate the impact of this application on provider protocol compliance.

Methods: A 21-section checklist was created based on the Tanzanian National Guidelines for BL treatment to systematically calculate protocol compliance. All patients <18 years of age diagnosed with BL at Bugando Medical Center in Mwanza, Tanzania from March 2016 to Feb 2021 were included in the evaluation. The patients were separated into a historic and prospective (intervention) cohort based on the introduction of mNavigator, in August 2019. The compliance scores were calculated and analyzed for each cohort using StataSE 16.

Results: A total of 82 patients were analyzed in this study, 49 in the historic cohort and 33 in the intervention cohort. There was a significant 9.2% ($p = 0.02$) increase in the total compliance mean after the implementation of mNavigator (from 47.0% in the historic cohort to 56.2% in the intervention cohort). Most notable increases in provider protocol compliance were seen at proper staging at diagnosis and interim evaluation with 39.4% and 45.2% increases, respectively. Significant increases in provider protocol compliance were observed for every cycle of chemotherapy.

Conclusions: The significant increases in provider protocol compliance following the implementation of mNavigator suggest the benefits of including mHealth applications to guide pediatric cancer treatment in countries with limited provider training capacity. Additional studies are required to evaluate the impact of mNavigator on treatment outcomes.

E-Poster Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PV0628 / #937 | Czech Childhood Cancer Information System

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Background and Aims: The knowledge of cancer burden in the population, its time trends and the possibility of international comparison is an important starting point for cancer programmes. A reliable interactive tool describing cancer epidemiology in children and adolescents has been non-existent in the Czech Republic. The goal of this study was to develop a new web portal entitled the Czech Childhood Cancer Information System, which would provide information on childhood cancer epidemiology in the Czech Republic.

Methods: Data on childhood cancers, which are recorded in the Czech National Cancer Registry, were validated using the clinical database of childhood cancer patients and combined with data from death certificates. These validated data were used to determine incidence and survival rates of childhood cancer patients aged 0–19 years who were diagnosed in the period 1994–2016 (N = 9,435). Data from death certificates were used to monitor long-term mortality trends.

Results: The web portal is available for anyone at <https://ccc-is.uzis.cz/index-en.php>, providing basic information for experts (i.e. analyses and publications) on individual diagnostic groups of childhood cancers. It involves an interactive tool for analytical reporting, which provides information on the following basic topics in the form of graphs or tables: incidence, mortality and overall survival.

Conclusions: The software is capable of processing data from multiple sources, is freely available to all users and makes it possible to carry out automated analyses even for users without mathematical background: a simple selection of a topic to be analysed is required from the user. This on-line available software tool makes it therefore possible for anyone to display long-term trends of childhood cancer incidence, mortality and survival, and to compare data from the Czech Republic to those from abroad.

E-Poster Topic: *Disease Orientated / AS03 Brain Tumours*

PV0629 / #1398 | **Mutant Braf Gene in Blood Plasma in Children with Brain Gliomas during Radiotherapy and Its Role in the Prognosis**

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Background and Aims: The search for genetic markers of prognosis seems promising for individualization of treatment programs for patients with brain tumors. We studied the dynamics of the concentration of mutant BRAF gene DNA in the peripheral blood plasma before, during and after radiotherapy (RT) and tried to find a correlation of those data with the effect of treatment.

Methods: Digital droplet PCR (ddPCR) was used to determine the concentration of mutant tumor BRAF gene DNA in 146 blood plasma samples (groups included 31 patients with pontine tumors and 29 patients with gliomas of other localization). The samples were obtained before the course of RT, during RT (at the 3-5th fraction, at the 14-18th fraction) and after RT. According to the results of RT, patients were divided into two groups: the group with no progression of the tumor for 6 months or more after the end of the course of RT, and the group with the tumor progression in less than 6 months after the end of the course of RT.

Results: A significant increase in the relative concentration of mutant (V600E) DNA of the BRAF gene in the peripheral blood at the beginning of the RT course and its decrease by the end of its course in the progression-free group was observed. Absence of changes or an increase in BRAFV600E levels by the end of the RT course was typical for patients with progression.

Conclusions: Methods of liquid biopsy allow to estimate changes in the amount of mutant (tumor) DNA in the peripheral blood plasma in children with brain gliomas during RT and to predict the effect of RT. Further study of the role of liquid biopsy in pediatric brain gliomas is planned.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PO0001 / #1061 | **Rate of Cardiotoxicity in Childhood Acute Lymphoblastic Leukemia Treated with Daunorubicin using Echocardiography and Troponin I**

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Background and Aims: Acute lymphoblastic leukaemia (ALL) is the commonest malignancy in childhood. Childhood ALL Survivors have a life long increased risk for cardiovascular morbidity and mortality compared to the general population, mainly caused by chemotherapy with daunorubicin. The aim of the study is to detect the rate of daunorubicin induced cardiotoxicity in children with acute lymphoblastic leukemia during induction phase chemotherapy.

Methods: This prospective observational study was conducted on 40 newly diagnosed patients of ALL aged 1 to 17.9 years who got daunorubicin in the department of Paediatric Hematology and Oncology, BSMMU. Complete blood count and echocardiography were done and troponin I was measured in all patients before and at the end of induction.

Results: Of the 40 patients, 8 patients (20%) had developed cardiotoxicity evidenced by reduction of left ventricular ejection fraction (LVEF) in echocardiography. Baseline LVEF was $68.80 \pm 5.98\%$ which was reduced to $65.32 \pm 7.07\%$ after induction phase chemotherapy ($p=0.023$). No significant alteration of troponin I was seen ($P=0.581$) between baseline and after induction. Male patients had a greater risk of developing cardiotoxicity than females but statistically not significant ($P=0.643$). There was no significant association between age of the patients and cardiotoxicity ($P=0.112$). Cardiotoxicity was higher in patient with initial high WBC count ($p=0.039$). Echocardiography also showed an increase tendency of mitral regurgitation and left ventricular hypertrophy after induction phase chemotherapy.

Conclusions: This study showed the rate of cardiotoxicity was 20% in ALL patients treated with daunorubicin at completion of induction. It also found that LVEF was decreased during induction therapy. Echocardiography can be used to detect early cardiotoxicity induced by daunorubicin.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0002 / #1113 | Interim Analysis of the Outcome of a Single Acute Lymphoblastic Leukaemia Cohort between 2008 and 2017 at Tygerberg Hospital

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Background and Aims: Overall survival (OS) for children with acute lymphoblastic leukaemia (ALL), treated in the Tygerberg hospital paediatric oncology unit (POU) was reported in 2006 as 66%. This was a subsequent analysis of OS for the cohort treated between 2008 and 2017.

Methods: This was a retrospective folder review study. Data collected were demographic data (age at diagnosis, sex, geographic region), initial white cell count, risk group, cytogenetics, treatment protocol, toxicity data, overall survival (OS) and event free survival (EFS). This is an interim analysis, but the final analysis will be presented.

Results: Seventy patients were included. The median age at diagnosis was 4.7 years (mean 5.6 years; range 0.6–14 years) with a male to female ratio of 1:0.7. The majority (56%), presented with a white cell count of less than $20 \times 10^9/L$ and 55 patients (76%) were B-cell ALL, but 41 patients (58%) had high risk features. The majority (77%) were

treated according to a Children's Oncology Group-based treatment protocol and the rest according to the 1995 Berlin Frankfurt Múnster protocol. Nearly 90% of children achieved morphological remission at the end of induction therapy. The 2-year OS was 79%. Relapses occurred in 16 patients (22%) and were dependent on the assigned risk group (3% for standard risk versus 36% for high risk). Thirteen of the sixteen patients with a relapse had an isolated bone marrow relapse.

Conclusions: Overall survival has improved from 66% previously to 79% in the subsequent time period and is comparable to OS in high income countries.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0003 / #1220 | Dynamics of Serum Vitamin B12 and Folate during Induction Chemotherapy of Childhood Acute Lymphoblastic Lymphoma

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Background and Aims: Acute lymphoblastic lymphoma[ALL] and the chemotherapy alters the nutrient absorption and nutritional status of children. This study aims to assess the serum levels of vitamin B12 and folate among children with newly diagnosed ALL during the induction phase of chemotherapy.

Methods: All children below 14 years of age with newly diagnosed ALL treated in pediatric oncology department over a period of 5 months from November 2020 to March 2021 were included in the study. Serum vitamin B12 and folate levels before and after the completion of induction chemotherapy were analysed. Enhanced chemiluminescence technique was used for estimation of serum vitamin B12 and serum folate levels.

Results: A total of twenty six children with male: female ratio of 1.9 : 1 were included. Mean age was 6y 6 months [range: 1.58 to 12.4 year]. 17 children had B- ALL and 9 had T-ALL. 16 children were high risk and 10 were standard risk. 1 child (< 5 year) had severe acute malnutrition and 1 child (> 5 years) was undernourished. Others had normal nutritional status. Mean serum vitamin B12 level before and after induction was 513.47 ± 212.27 pg/ml and 374.53 ± 189.52 respectively, the difference was statistically significant [$p=0.006$]. This difference was more evident in children with T - ALL [$p=0.02$] and high risk ALL [$p=0.003$]. Mean serum folate before and after induction was 7.64 ± 4.94 ng/ml and 5.15 ± 2.32 respectively, difference was significant [$p=0.02$]. Among the 2 children with severe acute malnutrition & undernourished status, neither had any vitamin B12 or folate deficiency before or after induction chemotherapy.

Conclusions: There was a significant decline in serum vitamin B12 and serum folate levels after induction chemotherapy among children with ALL. The risk factors for reduction in vitamin B12 levels post induction chemotherapy were T-ALL & high risk ALL. Hence it is important to monitor vitamin B12 and folate levels among ALL children on chemotherapy.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PO0004 / #1224 | Life-Threatening Complications at Diagnosis in Congenital Leukemia

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Background and Aims: Leukemia in newborns is extremely rare. Diagnosis and therapeutic decisions represent a challenge and initial supportive management is crucial for survival. Aim: Review of clinical, radiological and laboratory features of newborns diagnosed with leukemia
Methods: We reviewed medical reports of 65 children with leukemia in the last 5 years and selected 2 congenital leukemias. Analysis of clinical complications related to hyperleukocytosis and leukostasis and its initial management.

Results: We report 2 female newborns with loss of fetal well-being and severe symptoms due to leukemia after birth. Both presented skin lesions, hepatosplenomegaly, respiratory and neurological symptoms with massive extra-hematological organ involvement. CASE-1: Laboratory tests showed progressive increase of leukocytosis (up to $21.6 \times 10^9/l$), megakaryoblasts in blood smear and bone marrow that confirmed Acute Megakaryoblastic Leukemia (AML-M7) diagnosis. Due to the severity of CNS lesions in US/MR and poor neurological prognosis, palliative care was decided according to the family. Necropsy confirmed disseminated leukemic infiltration and severe brain involvement. CASE-2: Blood test revealed hyperleukocytosis ($900 \times 10^9/l$, 96% blasts), anemia and thrombocytopenia. Bilineal congenital leukemia was diagnosed with severe infiltration of the CNS, retina, skin, lungs, liver, spleen and pancreas. Due to clinical instability, severe respiratory distress and leukostasis, 2 exchange transfusions were performed on the 1st and 2nd day of life, achieving a decline in leukocytes counts and clinical improvement. She had prednisone poor response, morphological complete remission on day +33 and currently on-therapy.

Conclusions: The risk of life-threatening complications and death within the first days is high in congenital leukemia. Leukostasis can cause severe neurological, hepatic, pulmonary and multi-organ injuries.

Very high hyperleukocytosis is a medical emergency and cytoreductive procedures should be used urgently. The response to manual whole blood exchange transfusion was excellent in one patient and we outline its benefits for leukoreduction in newborns when umbilical vascular access is feasible

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia*

PO0005 / #1345 | QT Interval Prolongation in Patient with Acute T Cell Lymphoblastic Leukemia

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Background and Aims: The cause of QT interval prolongation in patients with hematologic malignancies continues to be uncertain.

Methods: Herein is reported a case of a 7-year-old male with high risk T-cell acute lymphoblastic leukemia.

Results: In 2020, the patient presented with a two-week history of fever, fatigue, weakness and loss of appetite. In physical evaluation general pallor was detected. There was no organomegaly. The bone marrow aspiration showed cellularity with 83% of malignant cells. The immunophenotyping revealed T-cell population and molecular cytogenetics detected no abnormalities. High risk therapy for acute lymphoblastic leukemia was initiated with ALL IC BFM 2009 protocol. MRD was performed only before the block HR-1' and detected 0% of malignant cells and 7% of B-precursor cells. Block HR-1' was initiated. The child received 3 doses of Dexamethasone, 1 dose of MTX, 2 doses of Cyclophosphamide. During the second day of the block, bradycardia (42-47) and long QT were detected. Echocardiogram and Holter Monitor showed no abnormalities. On the third day, heart rate self-regulated. Unfortunately, the block was interrupted. A month later the second attempt of block HR-1' was performed, but with dose reductions. On the first day, again bradycardia and QT elongation were detected. The block was interrupted, again. It should be reported that during the induction regimen and first block the patient received supportive drugs, such as Fluconazole, Ondansetron. As the main cause of cardiac complications was unknown, the regimen was changed and the drugs were given subsequently with dose reduction. The Dexamethasone, Fluconazole and Ondansetron were drawn. The heart rate remained normal during the modified block.

Conclusions: As the patient's heart rate was normal with modified blocks, the cause of cardiac complications still remains unclear. The association between the complication and Dexamethasone, Fluconazole, Ondansetron or combination of these drugs should be further investigated.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0006 / #1405 | Childhood Acute Lymphoblastic Leukemia Treatment in an Academic Hospital in Kenya: Treatment Outcomes and Health-Care Providers' Perspectives

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Background and Aims: Early deaths and treatment non-adherence are major reasons for low childhood acute lymphoblastic leukemia (ALL) survival in low and middle-income countries. This study assessed treatment outcomes of children presenting with ALL, and evaluated perspectives of health-care providers (HCP) on ALL treatment at a Kenyan academic hospital.

Methods: This was a combined retrospective medical records and cross-sectional questionnaire study. Treatment outcomes of 136 children diagnosed with ALL between 2010 and 2016 were collected. Questionnaires were completed by 245 HCP (response rate 86%) between September and October 2016.

Results: Childhood ALL treatment outcomes were: death (30%), progressive or relapsed disease (26%), abandonment (24%), and event-free survival (20%). Of all deaths, 80% were early deaths (prior or during induction), while 20% occurred in remission. Probability of event-free survival at 3 years was 18%. Only 57% of HCP believed childhood ALL can be cured with more doctors (96%) than other HCP (45%) believing in curability of ALL ($P < 0.001$). Majority of HCP (96%) thought that experienced doctors should put more time and effort into making parents understand diagnosis and necessity to complete treatment. According to HCP, reasons for protocol non-adherence included parental financial difficulties (94%), use of alternative treatment (79%) and lack of health-insurance (73%).

Conclusions: Event-free survival for ALL in Kenya is low. The primary reason for treatment failure is death from treatment-related complications. More efforts should be directed towards improving supportive care strategies. In the opinion of HCPs, improved communication with parents and supervision of junior staff will improve ALL treatment outcomes.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0007 / #1502 | Dietary Adequacy Evaluation of Pediatric Acute Lymphoblastic Leukemia Patients from 2013-2020 to Healthy Food Guidelines for Mexican Children and Adolescents

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Background and Aims: Acute Lymphoblastic Leukemia (ALL) is the most common type of cancer among children and adolescents. Malnutrition is frequent among ALL patients, affecting treatment effectiveness and overall survival. However, information about dietary intake among oncologic pediatric patients is scarce. We aimed to evaluate dietary adequacy of pediatric ALL patients from 2013-2020 and compare it to Mexican guidelines for children and adolescents.

Methods: After parental informed consent was signed, 142 children and adolescents aged 2-18 years old diagnosed with ALL were included. Subjects were divided by age into five groups (A= 2-3; B= 4-5; C= 6-8; D= 9-12; E= 13-15; F=16-18 years old), and dietary adequacy by food group was calculated using Mexican dietary guidelines. Wilcoxon test was made using NCSS; $P < 0.05$ was considered significant.

Results: Adequacy was significantly lower than the recommendation for fruits (B=72.09%; C= 79.95%; D= 58.13%; E= 38.74%; F= 45%), and vegetables (A= 32%; B= 30%; C=39%; D= 26%; E=14%; F= 29%). Additionally, sugar adequacy was above the recommendation (A= 110%; B= 140%; C=130%; D= 141%; E= 139%). Adequacy of protein, fat, dairy and legumes, was in line with recommended dietary intake.

Conclusions: Evaluation of dietary intake of cancer patients at the time of diagnosis provides information on nutritional risk and the opportunity for early nutritional intervention to prevent or reverse a deterioration of nutritional status during treatment the risk of developing malnutrition. LB-P received a fellowship from CONACyT. We thank "Hospital Civil de Guadalajara, Dr. Juan I. Menchaca" for their collaborative work.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0008 / #1566 | Patients with Down Syndrome and Acute Leukemia

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Background and Aims: Down syndrome (DS), or trisomy 21, affecting one in 1,000 live births, has been linked to an increased risk of developing acute leukemia. Transient abnormal myelopoiesis (TAM), occasionally may present with blasts in peripheral blood smear and spontaneous remission or may represent a preleukemic state in the neonatal period, preceding DS-Acute myeloid leukemia (AML). DS-AML is usually M7 acute megakaryoblastic leukemia. Patients with DS- Acute lymphoblastic leukemia (ALL) had historically inferior outcomes when they received risk-adapted therapy. This study aims to present DS-Acute leukemia cases from a nationwide survey in five pediatric oncology centers.

Methods: From 24 patients studied (14 male), 11 had ALL (8 male), 8 AML (4 male), and 6 TAM (3 male). The treatment regimens varied during the years. Median age at diagnosis was 4.9 years (1.75–12 years) for the ALL and 2 years (1.25–15 years) for the AML. Median WBC at diagnosis was $5.5 \times 10^9/L$ ($1.9\text{--}394 \times 10^9/L$). Median percentage of bone marrow blasts was 70% (21.6–90%).

Results: The patients with ALL had B-cell precursor ALL, most of them suffered severe infectious complications, and all achieved first complete remission. Two of 11 ALL relapsed, 18 and 34 months from diagnosis, but unfortunately, succumbed due to severe infections. Three of 8 patients with AML had M7 leukemia. All the patients with AML achieved a first CR. Of the 6 patients with TAM, 3 demanded treatment and all achieved CR. Only 1 progressed to AML 18 months later. Median follow-up was 12 and 11ys for ALL and AML respectively. Overall survival (OS) when treated after 2000 was 62.5% and 80% for ALL and AML, respectively.

Conclusions: The new protocols with less toxic therapies have led to improved survival rates the patients with DS-acute leukemia after 2000. Despite the increased cure rates, there still exist inferior outcomes due to infections.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0009 / #170 | Clinical and Epidemiological Profile of Patients with Acute Presentation Leukemia in the Pediatric Population Confirmed by Flow Cytometry from a Hospital in Medellin, Colombia, 2012-2017

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Background and Aims: The objective of this study was to characterize clinically and epidemiologically pediatric patients diagnosed with leukemia of acute presentation confirmed by flow cytometry from a hospital in Medellin, Colombia between 2012-2017.

Methods: A retrospective analytical study was carried out, which included pediatric patients diagnosed with acute leukemia confirmed by flow cytometry at a hospital in Medellín. A non-probability sampling of consecutive cases was carried out. The information was secondary, through medical records. The analysis was carried out through the IBM SPSS version 25 program. This work had the ethical approval of the institution.

Results: 54 pediatric patients were included; Regarding sociodemographic characteristics, it was found that 57.4% (31) were male, 48.1% (26) were <5 years old, 27.8% (15) between 5 and 9 years old and 24.1% (13) were ≥ 10 years old. Regarding the personal history, the main one was the presentation of acute infections with lymphocytosis with 18.5%; 9.3% of the patients had a family history of leukemia. 74.1% presented a diagnosis of lymphoma / lymphoblastic leukemia of B cells. 29.6% of the patients were admitted to the Intensive Care Unit, 59.3% had remission and 22.2% of the population died.

Conclusions: The pediatric population represents a high percentage of diagnoses of hematologic malignancies, although their diagnosis and management have state-of-the-art technological aids, it's still a disease with high mortality.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0010 / #1751 | Bilateral Kidney Leukaemia in a 2 Year Old African Male

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Background and Aims: Acute lymphoblastic leukaemia ALL is the commonest childhood cancer and manifest with fever, features of bone marrow and tissue infiltration resulting in pancytopenia and lymphadenopathy as well as hepatosplenomegaly. Usually kidney involvement at initial presentation is not very common and may manifest as renal enlargement which can mimic other pathologies. There are few reported cases of renal involvement in leukaemia as initial presentation and may often be associated with renal

impairment. Usually leukaemic cells may invade any part of the body and diagnosis can be made by identification of blast cells in the bone marrow and treatment often involves the use of chemotherapy Aim: Is to demonstrate bilateral nephromegaly as an early manifestation of ALL in an african boy and evaluate the outcome following treatment.

Methods: A case of a 2 year old male with presentation of abdominal pain and distension of 9days duration associated with facial puffiness, leg swelling as well as urinary frequency but no change in colour of urine or oliguria.No history of fever, weightloss, change in bowel output. Examination reveals submandibular lymphadenopathy 1-3cm, bilateral proptosis, non- pitting pedal odema and abdominal masses(the right and left lower quadrant measuring 10x8cm and 6x5cm respectively with both kidneys palpable). No hepatosplenomegaly. Other examination not remarkable. Fine needle aspiration is positive for malignancy and peripheral blood smear identifies 26%blast cells. Serum creatinine is 0.22mg/dl, urea-19.6mg/dl, uric acid 9.07mg/dl. Electrolytes, liver function test and urinalysis are not remarkable. Ultrasonography reveals enlarged kidneys bilaterally and child is managed for ALL

Results: Recieves treatment for leukaemic with standard risk protocols. Completed induction phase. Currently on consolidation phase and abdominal masses have reduced and bone marrow aspiration after induction reveals blast of 1%.

Conclusions: Bilateral renal enlargement can be an initial and early manifestation of acute lymphoblastic leukaemia in children and responds to standard chemotherapeutic agents.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0011 / #1758 | Primary Port Placement in New Leukemia Patients: An Observational Study

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Background and Aims: The practice of placing peripherally inserted central catheters (PICCs) during induction therapy for new diagnosis of leukemia was the standard practice at our institution secondary to concerns about neutropenia and healing. In early 2020, Jaffrey et al published a multicenter study raising concerns about thrombosis rates in the pediatric population for PICCs versus other central lines. In leukemia patients they showed an increased hazard ratio of 3.5 for PICCs relative to other central lines. Since April of 2020, we made an institutional change of placing ports at the time of diagnosis for leukemia patients and following them prospectively for wound complications and thrombosis rates. We hypothesized that wound complications and thrombosis rate would not be impacted by primary port placement.

Methods: All patients (n=20) presenting with a new or relapsed diagnosis of leukemia over a one-year period had a port placed and prospectively followed. Data collected included age, sex, type of leukemia, absolute neutrophil count (ANC), vessel cannulated, and the development of wound complications and thrombosis.

Results: The population studied included 10 males and 10 females with the average age of 8 years (range 1-16) at the time of diagnosis. The majority of patients presented with pre B ALL (n=15). All ports were placed in the internal jugular or subclavian vein. The median ANC at the time of placement was 525 cells/ μ L (range 0.05-2.89). There was a single minor wound dehiscence noted at the end of induction which healed without any intervention. There were 2 patients that developed non-occlusive thrombus on the catheter noted during ultrasound for other indications.

Conclusions: Patients with a new diagnosis of leukemia can successfully have a port placed at the time of diagnosis with minimal risk of wound complications and thrombosis. Plans are to continue collecting data at our institution and expand the study to include other centers.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0012 / #1772 | Early Mortality and Toxicity Associated with Intensified Therapy, Breaking the Treatment Reduction Paradigm for Lower-Income Countries

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Background and Aims: Progress in outcome for Acute Lymphoblastic Leukemia (ALL) has been made in the last years mainly through efforts led by high-income-countries and include optimized post-induction therapy with intensifications, costing greater treatment-related toxicity. ALL represents more than half of pediatric cancer in Mexico, and despite efforts to facilitate access to treatment, overall survival rates nationwide are lower than 50%. There is reluctance to adopt new treatment protocols, generally to avoid increasing the mortality associated with treatment, and attenuation is typically used to bypass these risks. Nevertheless since It has been demonstrated that treatment, its compliance and the correct evaluation of the response are necessary to improve the prognosis of patients with ALL, we decided fully reengineer the care strategy for our patients with ALL.

Methods: Since 2019 we began works to stablish a treatment strategy (chemotherapy, support, risk and response evaluation) based on the AEIOP-BFM-2017 protocol. Here we describe the results in terms of early effectiveness and safety for the first patients treated with this new approach. Toxicity was evaluated in standardized terms (NCI CTCAE Ver. 5.0). Since 2021 minimal residual disease was assessed by flow cytometry using the EUROFLOW methodology.

Results: We included 13 patients (52% women), aged 3 to 15 years (average: 8).15% diagnosed as T-ALL. Eight patients were assigned to early high risk. 77% of patients achieved undetectable minimal residual disease (MRD) levels at the end of induction, and 84% at the end of the consolidation. Treatment related mortality was 7% (one patient in Induction, due to aspergillosis), non-hematological grade IV toxicity was presented in 21% of the patients (neurological and hepatopancreatic) and 84% of the patients presented grade IV hematological toxicity, mainly in induction.

Conclusions: These early results show the importance of the early detection of infections, blood count surveillance, and present promising results that encourage the continuation of this treatment strategy.

Publication Only Topic: *Disease Orientated / ASO1 Haematology / ASO1.a Acute Lymphoblastic Leukaemia*

PO0013 / #1798 | Asparagine Dyshomeostasis May Reshape the Proteome in Leukemia and Other Significant Medical Conditions

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Background and Aims: Recent studies showing Asn-insufficiency is prognostic for diabetes (Refs.1a), are consistent with a hypothesis that therapeutic Asparaginase treatments modulate ribosome advancement at Asn-codons, contributing to remission of Acute Lymphoblastic Leukemia (ALL) and to its metabolic syndrome like side-effects (e.g., hyperglycemia and compromised vascular-integrity). Hyperglycemia might be caused by Asn-insufficiency inhibiting translation of IRS2 at the longest Asn(N)-Codon Repeat (N-CR) found in IRS2 of virtually every amniote. N-CR translation is Asn-susceptible, as is frameshift recoding at slippery-Asn-codons, demonstrated in SARS-CoV-2 and PEG10, and here postulated for (ALL-associated) Parvovirus B-19 (Refs.1b). PEG10 may be disproportionately susceptible to Asn, it encodes the second longest human N-CR and, with PNMA3, PNMA5, and ZCCHC5, constitutes a cohort-of-four-mammalian-domesticated-retrotransposons known to recode; all of which exploit slippery-Asn-codons. PEG10 is a diabetes associated regulator of vascularity that binds NSP-14 of SARS-CoV-2; it's dysregulation may cause vascular-restructuring, cerebrovascular-accidents, preeclampsia or modified disease-progression of COVID-19 (Refs2). Recoding of the cohort could plausibly contribute to side-effects of Asparaginase. We have demonstrated Asn-susceptibility of PEG10 recoding in-vitro. In the absence of Asparaginase, the Asn residue encoded at the frameshift site is variably incorporated into the frameshifted product of PEG10(sometimes) or SARS-CoV-2(almost always) in bacterial or cell-culture expression systems respectively. Asparaginase and Asn-insufficiency in-vitro appear to maximize frameshifting but the exact repositioning of the ribosomes on the mRNA may be different under conditions of Asn-sufficiency versus Asn-insufficiency, potentially

resulting in subtly different recoding outcomes. In-vivo, diabetes or obesity associated Asn-insufficiency might alter recoding outcomes of, e.g., SARS-CoV-2 or PEG10, resulting in the increased COVID-19 severity associated with diabetes or obesity. It is suggested that (Refs.3), increased awareness of variable outcomes at Asn-susceptible recoding sites at relevant viral and cellular genes could contribute to mitigating treatment complications associated with Asparaginase therapy for ALL. Refs.1a;b:PMID:29556673,29788285,26178806,5524673;22139573,32565969. Refs.2:PMID:27436286,Stukalov, A. in33464917, 24873829. Refs.3:PMID:26447468,15680415,31511883,32588943;28484491;33781551,32779861.[BJ1]

Methods: Intentionally leftBlank

Results: Intentionally leftBlank

Conclusions: Intentionally leftBlank

Publication Only Topic: *Disease Orientated / ASO1 Haematology / ASO1.a Acute Lymphoblastic Leukaemia*

PO0014 / #543 | Features of the Course of Herpes Encephalitis in a Child with Acute Lymphoblastic Leukemia in Remission

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Background and Aims: The introduction of chemotherapy group BFM protocols in Ukraine has significantly improved the treatment results of children with acute lymphoblastic leukemia (ALL). Up to 90.0% of children achieve complete remission. Some complications worsen treatment results, viral infection inflammatory brain damage in particular. Objective: to analyze the manifestations and course of herpetic encephalitis (HE) in a child with the remission of ALL during maintenance chemotherapy (MT).

Methods: Electroencephalography, head magnetic resonance imaging (MRI), liquor analysis.

Results: In July 2017 ALLIC BFM 2009 chemotherapy for the high-risk group began. In August 2018 preventive irradiation of the CNS performed with MT. In September 2019 at MT a gait disturbance and left leg lameness was observed. A month later, during of the MT, a gait disturbance with a lameness on the left leg was noted. Short-term episodes occurred of left leg twitching up to 5 minutes followed by left arm severe weakness, sagging, dysfunction and body drowsiness. The eye fundus was normal. MRI revealed focal changes in the right hemisphere cortical region, left side pyramid inflammatory changes. Third day symptoms were 20 minute convulsions, vomiting, forced body position, chanted speech, hyperthermia. Fourth day worsening symptoms: walk impairment, limited movements and weakness of left extremities, tendon reflexes from arms and legs D≤S. Liquor PCR revealed herpes type 1/2. Control brain MRI showed an increase of the right hemisphere parenchyma lesion area. Convulsions have being

resumed. Treatment included acyclovir for 3 months and anticonvulsant therapy for 12 months. Restored gait, strength in the limbs, normalized electroencephalography were observed. MRI revealed residual encephalomyelitis with cystic-gliotic changes in the affected area.

Conclusions: When treating children with ALL in remission, there is a risk of developing infectious complications, including viral ones. During the recovery period with HE, postnecrotic cysts are formed as the substitution of brain tissue.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0015 / #566 | Initial Results of Childhood Acute Lymphoblastic Leukemia Treatment on Protocols ALL-Mb-2008: A Retrospective Single-Center Study from Kyrgyz Republic

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Background and Aims: Вступление. Большой прогресс был достигнут в диагностике и лечении острого лимфобластного лейкоза (ОЛЛ) у детей за последние десятилетия. Подавляющее большинство излечиваются, однако существует потребность в улучшении, особенно в определенных подгруппах пациентов. Цели. Нашей целью было ретроспективно оценить заболевания и результаты лечения детей с ОЛЛ, включенных в единый центр в протокол лечения (ОЛЛ-Москва-Берлин-2008) в период с 2014 по 2018 год.

Methods: Материалы и методы. В общей сложности 78 пациентов в возрасте от 1 до 18 лет лечились от ОЛЛ в Национальном центре онкологии и гематологии в Кыргызской Республике, и наблюдались расчетная выживаемость без рецидивов (RFS), выживаемость без событий (EFS) и общая выживаемость (OS). в течение оцениваемого периода времени. Использовали метод выживаемости Каплана-Мейера.

Results: Полученные результаты. В группы стандартного, среднего и высокого риска было 39 (50,0%), 29 (42,2%) и 10 (12,8%) пациентов соответственно. Группа риска была установлена на 1-й и 36-й дни лечения, и в соответствии с иммунофенотипирования (100%) и цитогенетические (45%). Безрецидивная выживаемость, EFS и OS составили 87%, 78% и 76% соответственно. Смертность детей (n = 19) была связана с неблагоприятным иммунофенотипом, цитогенетическими аномалиями, синдромом острого лизиса опухоли, геморрагическими осложнениями, бактериальными и грибковыми инфекциями, нейрорлейкозом и рецидивом. Также в Кыргызстане нет возможности трансплантации костного мозга.

Conclusions: Заключение. Наш центр продемонстрировал показатели выживаемости, сопоставимые с некоторыми странами со средним уровнем дохода.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0016 / #672 | Acute B-Lymphoblastic Leukemia in Children and Adolescents: Descriptive Study and Treatment Results

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Background and Aims: Acute lymphoblastic leukemia is the most common malignant disease in children, phenotype B accounts for 85 % of ALL the aim of this study is to evaluate the epidemiological, clinical, biological, therapeutic and progressive characteristics of ALL B in children and adolescents.

Methods: This is a retrospective study carried out at the clinical hematology and pediatric oncology department of Casablanca, over a period of 5 years from 2014 to 2018.

Results: 75 patients were collected, the median age was 7 years (3 months - 20 years), the sex ratio was 1.2 (M / F), the most frequent reason for consultation was the infectious syndrome (60 %). 21 patients (28%) presented a major leukocytosis (> 50,000) at diagnosis. The initial neurological involvement was present in 2 cases. The karyotype was normal in 15 patients, shows the presence of the Philadelphia chromosome in 3 cases, hypoploidy in 3 cases, hyperploidy in 15 cases, a rearrangement involving MLL was found in 5 patients. Therapeutically, 72 patients were treated according to the national protocol (Marall protocol) (97%) and 3 patients with the Frall protocol. At the end of prephase (Marall) 20 patients responded well to corticosteroids (26.6%), The complete remission rate after induction was 62.6% (47 patients), partial remission was achieved in 9.3% (7 patients). 8% (6 patients) were in therapeutic failure. The evaluation after consolidation brings the number of patients in complete remissions to 54 patients (72%). Relapse occurred in 21 patients (28%) with 9 neurological relapses and 15 medullar relapses. At 3 years the overall survival was 59.1%, the event-free survival was 53.1%.

Conclusions: Considered to have a good prognosis, ALL B requires better stratification for therapeutic adaptation in order to avoid subsequent complications of chemotherapy, the search for residual disease has allowed early determination of the status allowing therapeutic adaptation.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0017 / #722 | Management and Outcome of Children with Relapse of Acute Lymphoblastic Leukemia

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Background and Aims: Leukemia is the most common malignant neoplasm in children. Considerable progress has been achieved through use of intensive chemotherapy protocols and risk stratification factors. A series of patients with acute lymphoblastic leukemia is reviewed to better define the clinical features of relapse and to evaluate the outcome.

Methods: Data were collected from patients admitted with leukemia between January 2005 and December 2020. All patients diagnosed with relapse are evaluated. The diagnosis was based on clinical features and cyto-hematological investigations. Flow cytometry was introduced since 2008th.

Results: During a mean follow-up of 34 months, thirty six (n=36) patients are identified with a sex-ratio of 1 and a median age at initial diagnosis of 5.5 years ranging from 0.6 to 15.5 years. Relapse delay varied from 3 months to 12 years from the first complete remission. Medullary relapse was reported in 79% of the patients (n=31), neuromeningeal in 30% (n=11), and gonadal in 10% of the patients (n=4). Two patients presented with isolated ocular involvement. Combined relapse was seen in 22% patients. Immunophenotype was available in 55% however, no patient benefit from minimal residual disease MRD assessment or cytogenetics study. All patients underwent second line chemotherapy with cranial and/or testicular radiation in 38% and stem cell transplantation was performed in 20% only. Overall survival is 20% with a mean follow up time of 30 months. Results seem better with gonadal relapse than other sites.

Conclusions: In the absence of minimal residual disease MRD assessment, patients staging is still based on classical criteria. Genetic studies are so recommended to identify patients at high risk of relapse and to adapt treatment strategies.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.a Acute Lymphoblastic Leukaemia

PO0018 / #833 | Leukemia Cutis, A Case Report

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Background and Aims: Leukemia cutis is a rare condition in which neoplastic cells infiltrate the skin in leukemia patients. The clinical manifestations result from the invasion of leukemic cells into the dermis, epidermis, or subcutaneous tissue. Occur in places with continuous inflammation such as the extremities, face and chest. Lesions have a wide range of characteristics, such as nodules, macules, petechiae, purpura, papules, and plaques. For the diagnosis, a skin biopsy with

immunophenotype with peripheral blood smear and bone marrow aspirate is required.

Methods: A 4-year-old male patient with a diagnosis of high-risk PRE T Lymphoblastic Leukemia 5 months ago, who in the consolidation phase begins with pruritic, nodular, granular and hypopigmented lesions. Treatment is given considering sarcoidosis, however it does not improve so a skin biopsy is taken.

Results: A biopsy result is obtained that reports monomorphic infiltrate, small mature lymphocytes with large blasts with little cytoplasm and a voluminous nucleus. The predominance of the infiltrate is perivascular and perianaxial, with a tendency to form germinative centers accompanied by vasculitis. The immunophenotype TdT (+) CD10 (+ focal) CD20 (-) and CD3 (-) with a diagnosis of LEUKEMIA SKIN INFILTRATION.

Conclusions: The clinical presentation of Leukemia Cutis is highly variable in appearance, symptoms and presentation. Currently the patient continues with chemotherapy in order to start radiotherapy. The use of whole-body electron-beam radiation seems to be a valuable alternative to conventional irradiation, avoiding systemic toxicity. The prognosis is poor, with many patients having other extramedullary disease and poor survival rates. Most patients die within months of diagnosis.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0019 / #124 | Therapy Related AML in a Boy with NHL Treatment Related All

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Background and Aims: Acute lymphoblastic leukemia (ALL) occurring in patients with a history of prior chemotherapy/radiotherapy exposure has been previously reported to be rare, accounting for <2.5% of ALL cases. Therapy-related leukemia (t-MDS/AML) is a well known complication of conventional chemoradiotherapy used to treat a variety of primary malignancies including Hodgkin lymphoma (HL) and NHL, ALL, sarcoma, and ovarian and testicular cancer. t-AML has a dismal prognosis and is one of the most frequent second malignant neoplasms which could be encountered by pediatric oncologists. In this case report, we would like to introduce a 9 year old boy with t-AML who had been diagnosed as NHL treatment related ALL.

Methods: Case report

Results: The patient was diagnosed as therapy-related acute myeloid leukemia

Conclusions: Chemo is known to be a greater risk factor than radiation therapy in causing leukemia. The risk gets higher with higher drug doses, longer treatment time, and higher dose-intensity (more drug given over a short period of time). Prospective evidence for

management of therapy-related acute myeloid leukemia (t-AML) is limited. Allogeneic HCT has generally been recommended because these cases respond poorly to traditional chemotherapy. The poor disease outcome may be explained by numerous potential factors, which include the prognosis of the primary malignancy and the toxic products of previous cytotoxic treatments that may compromise chemotherapy for AML. In addition to these facts, the frequent association between t-AML and poor cytogenetics appear to result in chemotherapy resistance. As the number of patients with t-AMLs is expected to rise, safety issues of cytotoxic therapies are becoming increasingly important in order to develop strategies to reduce the risk for therapy-related malignancies without compromising success rates for the respective primary disorders. The incidence of treatment related AML/MDS (t-AML/MDS) in children is extremely low. The majority of children with t-AML/MDS should have hematopoietic stem cell transplantation.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0020 / #1349 | Covid-19 Infection Presenting as Acute Monoblastic Leukemia

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Background and Aims: Acute monoblastic/monocytic leukemia (AMoL, M5a) is a subtype characterized by the proliferation of monoblasts.

Methods: We report a case of a 2-year-old boy with COVID-19 infection presenting as AMoL.

Results: In February, the child presented with a fever (38°C). The Covid-19 PCR test was negative. CBC showed Hb-10.3g/dL, PLT-47.0x10⁹/L, WBC-32.79x10⁹/L, MONO-17.0x10⁹/L (monocytosis-52.1%), the manual differential was consistent with poorly differentiated myeloblastic cells (38.0%), LDH was high (716.0U/L). Blood coagulation parameters were normal. Ultrasound showed lymphadenopathy and hepatosplenomegaly. CMV IgM, EBV IgM, EBV, CMV, HSV1-2, Covid-19 PCR tests were negative. Bone marrow aspirate showed normocellular BM with 78.0% blast cells, immunophenotyping by MFC were positive for CD33 (56.0%), CD11c (38.0%), CD38(71.0%), CD64(56.0%), CD38(71.0%), HLA-DR(90.0%), CD123(21.0%) and

negative for CD13 and CD117. Molecular biology by the FISH method using the diagnostic kits for t(8;21) (q21;q22)AML1/ETO, t(16;16)(p13;q22)CBFb/MYH11, t(15;17)(q24.1;q21.2)PML/RARA, MLL rearrangement, p53 gene, 5q, 7q, and 20q deletions were negative except for MLL rearrangement. The patient was monitored for two weeks, during that period developed pancytopenia without blast cells in peripheral blood, LDH value became normal (195 U/L). A second bone marrow aspirate film showed hypocellular BM with 5.0% blast cells. The US showed the normal size of lymph nodes, spleen, and liver. During this period the patient had a fever (up to 40.0°C), acute respiratory failure, and the chest X-ray showed bilateral pneumonia. The patient was transferred to the pediatric intensive care unit (PICU). Laboratory data showed elevated D-dimer (4260ng/ml), Anti SARS-CoV antibody (30.1 COI), CRP-(270.0mg/dl). Ten days after, the patient was stabilized and discharged with the following CBC data: Hb-10.1g/dL; WBC-7.99x10⁹/L; NEUT-2.95x10⁹/L, LYMPH-4.70 x10⁹/L, MONO-0.33 x10⁹/L, PLT-284x10⁹/L.

Conclusions: This clinical case demonstrates that COVID-19 infection can have also a leukemia-like presentation and further investigations are needed.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0021 / #1460 | Facial Myeloid Sarcoma: Unusual Manifestation of Acute Myeloid Leukemia

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Background and Aims: Myeloid (granulocytic) sarcoma is a rare tumor, occurring at any extramedullary anatomical site that can occur before or after the onset of acute myeloid leukemia. The Aim of this observation is to illustrate the clinical, biological and radiological appearance of unusual revealing manifestation of acute myeloid leukemia.

Methods: We report the case of a 22-month-old infant.

Results: The child was admitted for bilateral exophthalmia with periorbital ecchymosis. This is followed by a left cheek tumor. This child was referred to us from the maxillofacial surgery department for suspected histiocytosis. The patient subsequently underwent computed tomography and magnetic resonance imaging visualizing osteolysis of the face, associated with two retro-maxillozygomatic orbital masses. The diagnosis of myeloid sarcoma was obtained following an anatomicopathological study of a biopsy of the tumor. The bone marrow aspirate confirmed the coexistence of acute myeloid leukemia.

Conclusions: Making the diagnosis of a granulocytic sarcoma is a real challenge given its polymorphic clinical expression. It should be considered in front of facial tumor of uncertain origin.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0022 / #1514 | Fungal Infections Complicating Therapy of AML in Children with Sickle Cell Disease: A Report of Two Cases

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Background and Aims: Patients with sickle cell disease (SCD) have an increased incidence of leukemia and present a management challenge. Vasculopathy, increased thrombosis risk, and asplenia may contribute to complications during chemotherapy. We report two cases of acute myelogenous leukemia (AML) in children with Hgb SS and describe severe fungal infections complicating their treatment course.

Methods: This is a case series, compiled using retrospective chart review.

Results: Patient 1, a 13-year-old female with Hgb SS, presented with persistent neutropenia despite reduced hydroxyurea dosing. A bone marrow biopsy revealed AML with a normal karyotype (46 (XX)), and negative FISH and molecular work-up. CNS was negative. She achieved remission at the end of induction with cytarabine, etoposide, and daunorubicin per AAML1031, and completed five cycles of chemotherapy, including Intensification III per AAML0531. While awaiting count recovery, she developed colitis and liver lesions. Peripheral blood PCR showed high levels of Mucor. She then developed portal and hepatic vein thrombosis, acute liver failure, and multifocal stroke leading to herniation and cardiac arrest. Patient 2, a 9-year-old female with HgbSS, had myeloblasts noted on a routine CBC. Diagnosis of AML with a normal karyotype (46 (XX)) and negative FISH and molecular work-up was confirmed by bone marrow biopsy. CNS was negative. Treatment was initiated per AAML1031 as above, and she achieved remission with negative MRD at end of induction. Her course was complicated by trichosporon fungemia with persistent fungal elements seen on bone marrow biopsy until Intensification I, anakinra-resistant HLH, CMV viremia, and fungal pneumonia. Infections were treated with multiple antimicrobial therapies and HLH was treated with steroids and emapalumab. Due to risk of complications, Intensification III was not completed.

Conclusions: Two cases of children with sickle cell disease and leukemia highlight a population at increased risk of chemotherapy complications that may require a modified approach to therapy.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0023 / #1597 | Clinical and Hematologic Profile and Treatment Outcome of Pediatric Patients with Chronic Myelogenous Leukemia in a Government Tertiary Hospital

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Background and Aims: Chronic Myelogenous Leukemia accounts for 3 percent of childhood leukemia. It is characterized by proliferation of malignant cells due to the presence of BCR-ABL fusion gene. Tyrosine kinase inhibitors, such as Imatinib have improved the life expectancy of patients with CML. Objective: This study reviewed the clinical and hematologic profile and treatment outcome of patients diagnosed with CML in a tertiary hospital from 2014 to 2019.

Methods: This is a retrospective study which reviewed the medical records of CML patients in a tertiary hospital. Clinical and hematologic profile and treatment outcome of the patients were summarized.

Results: A total of fifteen patients with CML were included in this study. The most common presenting symptoms were weakness and fever. Majority of the patients presented with anemia, leukocytosis and thrombocytosis. Six patients were able to have a complete hematologic response, however, there were 3 mortalities reported.

Conclusions: Future research are needed to look into the association of the clinical and hematologic profile with the treatment outcome of CML patients which may help in developing clinical guidelines in its diagnosis and management.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0024 / #1629 | A Case Report with an Atypical Progression of a Secondary Thrombocytosis

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Background and Aims: Thrombocytosis is defined as the increased number of thrombocytes above 450.000 / mm³. It is quite in frequency in especially infants secondary to anemia or infectious episodes. Primary thrombocytosis is a myeloproliferative neoplasm that is rarely seen in children but more commonly seen in adults. We herein report a seven-years old girl who had thrombocytosis for nearly one and a half year and being diagnosed as secondary thrombocytosis.

Methods: A 7 yrs old girl was presented to the emergency room with subfebrile fever and chills. In physical examination there was no pathological finding with no enlarged lymphadenopathies or hepatosplenomegaly. Laboratory work up was performed.

Results: In complete blood count, White blood cell 10,700/mm³, hemoglobin 12,7 g/dl and thrombocytes 1.593.000/mm³ with no

atypical cells in the peripheric plate. Reticulocytes %1.65. Erythrocyte sedimentation rate was 15/1 hr. Blood chemistry was within normal limits including haptoglobin 44 mg/dl. For differential diagnosis of secondary thrombocytosis, vasculitis panel was negative. The thrombocytes of the mother and the father were within normal limits. Bone marrow aspiration and biopsy resulted in hyperactivity in megakaryocytes with no blastic infiltration and no fibrosis and probable mutations JAK-2, MPL and CALR, associated with primary thrombocytosis were negative. Despite no symptoms occurred, anti-aggregation treatment with low dose aspirin was initiated. Aspirin treatment was ended at the end of six months free of any symptomatic findings. Thrombocytosis resolved after one and a half year time. In the last complete blood count, thrombocyte was 643.000/mm³.

Conclusions: In the childhood, secondary thrombocytosis is frequently associated with iron deficiency anemia, major trauma, bacterial and viral infections, otoimmune diseases or surgery and rarely idiopathic. Our case is a secondary thrombocytosis with unknown etiology which achieved remission spontaneously.

Publication Only Topic: Disease Orientated / ASO1 Haematology / ASO1.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0025 / #178 | Myeloid Sarcoma in Mastoid Bone in a Child as the First Manifestation of Acute Promyelocytic Leukemia (APL) Relapsed: A Case Report

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Background and Aims: Recurrent extramedullary APL can affect many sites, but is most often in the central nervous system. Other less frequent locations are the skin and the external auditory canal. Extramedullary relapses often accompanies bone marrow involvement.

Methods: A descriptive case report.

Results: A 4-year-old boy is diagnosed with APL with high-risk t(15; 17), who received PETHEMA protocol without achieving morphological or molecular remission after induction and who achieved complete remission after 8 weeks of treatment with ATRA. Surveillance begins with complete hematological and molecular remission identified by PCR and FISH. At 2 months of clinical follow-up, he presented early relapse of his disease manifested by right earache and facial paralysis, a tomography was performed that revealed a mastoid mass which was biopsied and in which a granulocytic sarcoma was reported. Molecular relapse to bone marrow with t (15; 17) (q24; q21) bcr3 is documented by PCR (HemaVision) with normal FISH, negative immunophenotype by flow cytometry and CNS 3 status. The PETHEMA protocol is indicated due to the impossibility of giving treatment with ATRA / ATO, intrathecal chemotherapy is

started and it is proposed to consolidate treatment of relapse in the CNS with craniospinal radiotherapy and hematopoietic stem-cell transplantation.

Conclusions: Treatment of children with APL with protocols with ATRA and anthracyclines is associated with complete remission in more than 90% of patients, and relapses have become infrequent in the long-term follow-up of these patients. There is no international consensus on treatment for these patients, but arsenic trioxide is currently considered the most effective drug for the treatment of refractory or recurrent APL, it has good penetrance in the skin, soft tissues, and CNS, although there is no consensus that defines whether or not it should be combined with ATRA and whether consolidation therapy with hematopoietic stem cell transplantation should be used.

Publication Only Topic: Disease Orientated / ASO1 Haematology / ASO1.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes

PO0026 / #609 | Obserbation of Notable Therapeutic Response in a Patient with Systemic Juvenile Xanthogranuloma with Kif5B-Alk Fusion

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Background and Aims: Juvenile xanthogranuloma (JXG) is classified as histiocytosis that is generally diagnosed as a subcutaneous nodule in infancy. Some patients with involvement of the central nervous system (CNS) might have severe neurological sequelae.

Methods: The patient presented with multiple subcutaneous tumors of up to 6.0 cm in diameter when he was two weeks old. Enhanced computed tomography (CT) revealed multiple low-density areas of 1.0 cm in diameter in the liver and right kidney and nodular shadows in both lung fields. Magnetic resonance imaging (MRI) revealed enlargement of the left side of the medulla and massive tumor within the cervico-thoracic spinal cord. Biopsy of a cutaneous lesion was performed, and histopathological examination revealed JXG. Immunohistochemical examination revealed that the cells were positive for anaplastic lymphoma kinase (ALK) D5F3 and ALK1 active kinase. Developmental retardation has become increasingly accentuated in the first year. Spasticity, muscle weakness, and clonus of the lower extremities had progressed, and he needed tube feeding due to dysphagia. At 12 months of age, CT revealed a new lesion in the ischial bone. Brain MRI showed that enlargement of the CNS lesions.

Results: Then, next generation sequencing, FoundationOne

Conclusions: Histiocytosis should be also classified based on its actionable genes that directory link to targetable therapy. A high index of cognition could be maintained for such an actionable target in histiocytic neoplasms.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PO0027 / #728 | Acute Megakaryoblastic Leukemia (AMKL) with RBM15 MKL1- Fusion, Presenting Bone Lesions, Hepatosplenomegaly without Blasts in the Peripheral Blood: A Case Report

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Background and Aims: AMKL in infants usually presents bone marrow fibrosis and rapidly progressive disease. Translocation of t(1;22)(p13;q13), or RBM15-MLK1 fusion, is present in 1-2.4% of pediatric AML patients and restricted to AMKL.

Methods: Case report. Flow cytometry (FC), immunohistochemistry (IHC), and genomic profiling test in the bone marrow.

Results: A one-year-old female was admitted with fever, thrombocytopenia, and hepatosplenomegaly. Imaging showed bilateral lytic lesions in the mandibles, tibia, spine, and skull. Bloodwork: Hgb 45g/L, WBC 7.7 x10⁹/L without blasts, and platelets 4x10⁹/L. Bone marrow showed clusters of cells compatible with extramedullary origin. FC was not consistent with hematological disease (negative CD3, CD4, CD8, CD19, CD33, CD34, CD45, CD56, CD117, and HLA). Tibia biopsy showed undifferentiated cells marked by the proliferation of atypical round/spindle cells, with hyperchromatic/slightly pleomorphic nuclei, granular chromatin, and small nucleoli. IHC showed negative CD20, CD45, TDT, MPO, Actin, Myogenin, WNT, CD30, Cytokeratin, Tryptase, EMA, AE1, AE3, and CD1A and positive CD34, CD68, and CD43, and weakly positive for CD99. Genomic profiling was attempted twice, but the specimens did not yield sufficient RNA. The patient underwent adjuvant therapy for Ewing's Sarcoma: Vincristine, Doxorubicin, and Cyclophosphamide/Ifosfamide and Etoposide. Partial response was obtained, but no longer sustained and she presented disease progression after 12 courses. During re-evaluation, genomic profiling showed translocations of t(1;22) (p13;q13) or RBM15-MKL1 fusion. Bone marrow morphology, now marked by myeloid blast infiltration, confirmed AMKL phenotype without blasts in peripheral blood. Treatment according to AML protocol (NOPHO 93) was initiated.

Conclusions: AMKL with the t(1;22) usually presents in infants with extensive involvement of liver and spleen. This translocation finding may be useful as a diagnostic marker for atypical presentation.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PO0028 / #767 | Laboratory Monitoring of Transformation of Myelodysplastic Syndrome into Acute Erythroid Leukemia. Case Report

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Background and Aims: Acute myeloid leukemia, M6 type, or acute erythroid leukemia is an extremely rare variant of leukemia in both adults and children. Progression of the preceding myelodysplastic syndrome is one of the most common causes of this disease. The possibility to find two types of blast cells in the bone marrow of patients (erythroblasts and myeloblasts) is a special feature of acute erythroid leukemia.

Methods: A one-and-a-half-year-old child was referred for examination for a hemorrhagic rash on the skin of the trunk and limbs. A series of bone marrow and peripheral blood tests were required to diagnose a rare form of leukemia. Microscopic, flow cytometric, cytochemical studies of the samples were carried out in our laboratory for four months. This allowed us to observe the process of transformation of myelodysplastic syndrome into acute erythroid leukemia.

Results: Changes in the morphological features of the bone marrow and the immunophenotypical characteristics of blast cells were of particular interest. Our research has shown how the excessive percentage of myeloblasts was replaced by sharp proliferation of the erythroid lineage of hematopoiesis and an increase in the pathological erythroblasts count.

Conclusions: The described clinical case allowed us to follow the process of transformation of myelodysplastic syndrome into acute erythroid leukemia step by step. We hope that our research results will help laboratory specialists in the diagnosis of this extremely rare form of leukemia.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.b Myeloid Leukemias, Myelodysplastic and Myeloproliferative Syndromes*

PO0029 / #812 | Spontaneous Splenic Rupture in an Adolescent with Acute Myeloid Leukemia

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Background and Aims: Splenic rupture is extremely rare in children with hematologic malignancies associated with a high mortality rate. This study aims to describe clinico-biological characteristics and management of a patient with acute myeloid leukemia AML and splenic rupture.

Methods: We report a case of 13 year old girl with AML who developed a splenic rupture during induction chemotherapy course, successfully treated with conservative management.

Results: A 13year old girl was admitted with diagnosis of AML. On initial examination she was found to have pallor, without splenomegaly or hepatomegaly. She didn't have any lymphadenopathy Hematological investigations revealed major hyperleukocytosis at 174,400/ μ L, anemia at 6.7 g/dl without thrombocytopenia (platelet count at 182,000/ μ L) with increased plasmatic LDH level (6N). A blood smear showed 80 % of blasts associated with a massive bone marrow infiltration 90%. Flow cytometry was indicative of AML5. A chemotherapy Induction course was started, but at the 5 th day, the patient experienced acute abdominal pain with extreme pallor. Physical examination revealed shock manifestations with abdominal distension and splenomegaly extending 10 cm below the costal margin. Abdominal us revealed splenomegaly with a rupture at the junction of the inferior and medium third associated with a subcapsular Hematoma and a massive intraperitoneal effusion. Management consisted of transfusions, antalgics and antibiotics. No surgery was performed. She required repeated platelet and globular transfusions. On day 13th, she symptomatically improved and only a minimal subcapsular hematoma was visible on imaging on day 20th. Chemotherapy was continued and bone marrow transplantation was performed after complete remission. She is alive with a follow up time of 10 years without disease evidence.

Conclusions: Splenic rupture is a rare but also dramatic complication described in children with malignancies. Patients tend to exhibit very poor outcome. Progress in conservative management may improve survival. Key words: Adolescent, AML, splenic rupture

Publication Only Topic: Disease Orientated / ASO1 Haematology / ASO1.c Lymphomas

PO0030 / #1034 | Epidemiological Profile of Lymphomas Deaths in Brazil in the Pediatric Age Group

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Background and Aims: Lymphomas, both Hodgkin lymphoma and non-Hodgkin lymphoma, are a diverse group of diseases originated by a clonal proliferation of lymphocytes. Lymphoma represents one of the major groups of malignancies in the pediatric age group. This study aims to report the epidemiological profile of lymphomas deaths in the pediatric age group in Brazil.

Methods: Data were collected using the Mortality Information System. The parameters analyzed were the frequency of deaths by Hodgkin lymphoma, Follicular lymphoma, Non-follicular lymphoma and Mature T/NK-cell lymphomas. Additionally, were analyzed the child's color (white, brown, yellow, black and indigenous), the sex and the age (1-4 years old, 5-9 years old and 10-14 years old). The analyzed period was the interval between 2000 and 2019 in Brazil.

Results: Over the interval, 1,310 deaths were reported. The major number of deaths was represented by patients with Non-follicular lymphoma (72.2%) and Hodgkin lymphoma (21.9%). The most affected ethnic groups were white (46.6%) and brown (38.5%). Boys represented 70.2% of the cases. The distribution by age groups was 10-14 years old, 42.5%; 5-9 years old, 35.1%; and 1-4 years old, 22.4%. Over the years analyzed, the lowest number of deaths were reported in 2018 (53 deaths), and the highest was reported in 2006 (90 deaths), the average number of deaths was 65.5/year.

Conclusions: This study concludes that in Brazil, boys represented the major number of cases of death, and the most common lymphoma analyzed was Non-follicular lymphoma. Also, the ethnic group with the highest number of cases was white. It is possible to notice that the number of deaths increases with the age. Therefore, with the help of this epidemiological profile we can understand that the children with the highest chance of dying by lymphoma are white and older, thus the focus of the precocious diagnosis and treatment.

Publication Only Topic: Disease Orientated / ASO1 Haematology / ASO1.c Lymphomas

PO0031 / #1172 | Primary Bilateral Renal Non-Hodgkin's Lymphoma:Case Report and Literature Review

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Background and Aims: Bilateral renal involvement by Non-Hodgkin's lymphoma (NHL) is very rare. Extra-nodal lymphomas comprises about 25%–35% of NHLs. Primary urinary tract lymphoma comprised 0.1% to 0.7% of extra nodal NHLs. The differential diagnosis of renal non-Hodgkin's lymphoma from Wilm's tumor is difficult without renal biopsy barely.

Methods: We reported one patient with primary bilateral renal diffuse large B cell lymphoma. Radiological findings and clinical data were analyzed.

Results: A 3-year-old boy presented with hematuria for 2 days. He was admitted in Sun Yat-sen University Cancer Center, China. CT scan of abdomen showed multiple masses with lower signal intensity in T1W sequences and relatively hypointense on T2W sequences in kidney. In contrast, radiological findings of Wilm's tumor indicated pseudocapsule around tumor and hemorrhagic necrotic changes. According to literature review, bilateral Wilm's tumor accounted for 5% in Wilm's tumor and NHL presented in both kidneys at the same time (bilateral renal involvement) in 36.6% of primary renal lymphoma.

Conclusions: Bilateral kidney tumor is not contraindication for tumor biopsy. Tumor biopsy is recommended before therapy is initiated according to the characteristics of radiological findings.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0032 / #1248 | **Hodgkin Lymphoma Progression after the Covid-19 Infection**

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Background and Aims: Hodgkin lymphoma (HL) is one of the most common childhood malignancies, constituting 40% of all lymphomas.

Methods: We report a case of an 11-year-old boy with stage III nodular sclerosis classical Hodgkin lymphoma (NSCHL), with further progression after COVID-19.

Results: In August 2020, the patient was diagnosed with stage III NSCHL based on the following results: CT showed enlarged, multifocal lymph nodes of the left upper, middle, and lower cervical (8.2x3.6cm), left supraclavicular, axillary, mediastinal and hilar regions, with enlarged liver and spleen. The pathological evaluation concluded a diagnosis of NSCHL (Ki67=90-95%). The patient underwent 2 cycles of OEPA and 3 cycles of COPDAC chemotherapy regimen. During the 2nd cycle of OEPA, he presented with an allergic reaction to etoposide, which was replaced with etopophos. PET/CT scan after the 2nd cycle of chemotherapy revealed left sided cervical, supraclavicular, axillary and mediastinal lymph nodes with low metabolic activity (DS2) and upper jugular lymph nodes (1.2x0.8cm) with high metabolic activity (DS4). CT scan after the 4th cycle of chemotherapy showed multifocal lymph nodes of the left upper jugular (1.1cm), left supraclavicular and left prevascular areas. After the 5th cycle of chemotherapy, the patient tested positive for COVID-19 (February 2020). Ten days after,

he had persistent fever, up to 39°C, pericardial effusion, diarrhea, rash, lymphadenopathy. Lab tests were consistent with Pediatric Inflammatory Multisystem Syndrome associated with Covid-19 and the patient received IVIG. Afterwards, considering persistent fever, CT and biopsy of cervical lymph nodes were performed which confirmed disease progression. The patient is now scheduled to undergo 4 courses of chemotherapy according to ABVD and IEP schemes and autologous stem cell transplantation.

Conclusions: This clinical case demonstrates the necessity for a more thorough examination and careful monitoring of cancer patients with COVID-19. Larger studies are needed to evaluate the possible association between disease progression and COVID-19 infection.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0033 / #1383 | **Acute Hematologic Toxicity & Treatment Gaps during Radiotherapy in Children with Hodgkins Lymphoma:10-Year Experience of Radiation Therapy at A Tertiary Care University Hospital in Pakistan**

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Background and Aims: Radiation therapy (RT) is important component of curative treatment for pediatric Hodgkins Lymphoma (HL) for residual disease after initial chemotherapy and those having refractory/relapsed disease. Large volume of radiotherapy may pose a risk for acute hematologic toxicity which could result in radiation treatment gaps. We report our experience of RT for children with HL treated at tertiary care university hospital in Pakistan.

Methods: Radiotherapy was offered in multidisciplinary tumor board meeting to patients having less than complete response on early assessment with CT or PET-CT after initial chemotherapy. Dose of 19.8Gy/11 fractions was given for consolidation at the end of chemotherapy with 10Gy/5fractions boost to residual disease on early assessment. Patients with relapsed or residual disease were offered 25-30Gy. Treatment gaps and hematologic toxicity was observed in these patients.

Results: From January 2009-October 2020, 55 patients of HL were treated with RT in our hospital out of which 33 are included for analysis. mean age was 11(± 4.4) years, 26(79%) male and 7(21%) female patients. Neck was irradiated in 23(70%) patients, axilla in 8(24%) and mediastinum in 10(30%) patients. treatment volume included Spleen in 8(24%), abdominal nodes in 11(33%) and inguinal area in only 2 patients. Fourteen of thirty-three patients received supra & infra-diaphragmatic radiation out of whom 9(27%) receive them simultaneously and five(15%) received sequential radiation to these regions. Radiation was planned with 3D conformal technique in 23(70%) and

with IMRT in 10(30%) patients. RT field size or dose did not appear to impact hematologic toxicity. There were treatment interruption in 6(18.2%) patients but was not due to need of antibiotic therapy, hospital admission or cytopenia during the course of radiotherapy.

Conclusions: Radiotherapy can be given for large burden nodal disease without treatment gaps and acute hematologic toxicity. However, extended follow-up is to be reported for long-term hematologic and other consequences.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PO0034 / #1615 | Standard Dose Chemotherapy for Treating Children with Relapse Hodgkin Lymphoma -Experience from Resource Limited Center, Bangladesh

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Background and Aims: Majority children with Hodgkin lymphoma (HL) can be cured. But approximately 10 to 20% children develop relapse or refractoriness. High dose intensive therapy(HDCT) with autologous stem cell transplant (ASCT) is not feasible to treat relapse cases in resource limited settings. Pediatric practice adopts various risk stratified and response based non-transplant salvage approach with standard dose(SDCT) chemotherapy for relapse Hodgkin lymphoma. The aim of the study was to find out a standard dose second line chemotherapy for the children with relapse Hodgkin lymphoma in resource limited cancer center.

Methods: The prospective case series study was done at pediatric hematology and Oncology department of National Institute of Cancer Research & Hospital, Bangladesh from January to December, 2017. Total 5 children with relapse HL was included consecutively received ABVD protocol before. Age, sex, staging, time interval of relapse was recorded in data sheet. After written consent, children were assigned to treat with chemotherapy EPIC (Etoposide, Prednisolone, Ifosfamide and Cisplatin) protocol adopted from UK-trial. Radiotherapy was recommended to all relapsed sites after completion of 6 cycle chemotherapy. Outcome was measured and follow up to December, 2020. Statistical analysis done with SPSS version23.

Results: Age of the studied children were 7 to 15 years with mean 11.2 ± 3.63 , Male: Female=4:1. Relapse time was 12 to 30 months with mean $20.4 \text{ months} \pm 6.84 \text{ SD}$. Stage IA 20%, stage IIA 20%, IIIA 20% and stage IIIB 40%. Classical mixed cellularity 60% and nodular sclerosis 40%. The study found remission 40%, relapse 40% and Death 20% with EPIC protocol. All remission (40%) cases were from Stage IA and IIA with mixed cellularity type histology. Relapse (40%) from stage IIIA and IIIB with nodular sclerosis histology. Death ($n=1/5$; 20%) was from stage IIIB, mixed cellularity type cases.

Conclusions: The study found remission in children with low stage relapsed HL with non-transplant salvage chemotherapy.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PO0035 / #1631 | Pediatric Hodgkin Lymphoma: An Analysis of 5 Years Data from Pediatric Oncology and Hematology Unit, University Hospital Casablanca, Morocco

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Background and Aims: Hodgkin lymphoma (HL) is a rather rare malignancy in the pediatric population. However, it constitutes approximately 40% of all lymphomas that present during childhood and the cure rate for children with HL has steadily improved over the years. The purpose of this study is to describe the clinico-pathological features and therapeutic outcomes of pediatric HL in Casablanca.

Methods: We conducted a retrospective descriptive study of histologically proven HL, carried over a period of 5 years from January 2011 to December 2015 in a single Moroccan University Hospital, including all patients with HL aged fewer than 15. Patients were classified according to the Ann Arbor. Patients were treated with the MDH-04 protocol. Assessment of therapeutic response was made using CT imaging +/- PET scan.

Results: During the period study, Thirty-seven patients were diagnosed and treated for HL. The age was ranged between 3 and 15 years old, with a median age of 10. Sex-ratio M/F was 3,6. The median delay of diagnostic was 6 months (1-24). Lymphadenopathy was the most common presentation (90%). Sixty-five percent of the patients reported B symptoms. Bulky disease was present in 48,6% of the patients. Nodular sclerosis was the predominant histological presentation of pediatric HL (70%). Ninety percent of the patients were unfavorable group. Eighty-six percent of patients achieved complete remission, eleven percent did not, and eleven percent relapsed. Four patients benefited from an autograft of hematopoietic stem cells. No deaths were recorded in this series.

Conclusions: Despite the high prevalence in the unfavorable group, the therapeutic outcomes are satisfactory and the absence of death reassures the safety of the protocol. Furthermore, it is necessary to

reduce the duration and cumbersome treatment in the favorable group in order to avoid the long-term sequelae.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0036 / #1633 | Primary Refractory Hodgkin's Lymphoma in the Moroccan Pediatric Population: Epidemiological Profile and Therapeutic Results

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Background and Aims: Primary refractory Hodgkin's Lymphoma (PR REF HL) may generally be defined as progression of disease during induction treatment or a partial or transient response (< 60 days) to induction therapy. Improved outcomes after failure of primary induction therapy have been reported with myeloablative therapy and auto-grafting. The purpose of this study is to describe the clinic pathological features and therapeutic outcomes of children with PR REF HL in Morocco.

Methods: We conducted a retrospective descriptive study of children treated for HL, carried over a period of 5 years from 2011 to 2015 in all Moroccan University Hospital, including all patients with histologically proven PR REF HL aged fewer than 15. Patients were classified according to the Ann Arbor. Patients were treated with the MDH protocols 04 and 11 according to the period. Assessment of therapeutic response was made using CT imaging +/- PET scan. The 3-Years OS were calculated using the Kaplan-Meier method.

Results: During this period, 26 of the 162 patients treated for LH were considered in first line treatment failure (16%). The median of age was 12 years old (5-15). Sex ratio M/F was 2,7. The diagnostic delay was 5,5 months (1-24). Lymphadenopathy was the most common presentation (92%). Sixty-five percent of the patients reported B symptoms. Bulky disease was present in 38,5% of the patients. Nodular sclerosis was the predominant histological presentation of pediatric HL (65,4%). Eighty-eight point five percent of the patients were unfavorable group. Five patients are in complete remission after second line treatment. An autograft of hematopoietic stem cells supported three patients, ten of patients have died.

Conclusions: This current pattern for epidemiological features and outcomes of pediatric PR PEF HL enables delineate key populations in order to explore risk factors.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0037 / #1739 | An Abdominal Non-Hodgkin's Lymphoma Revealed by a Chylothorax in a Child: A Case Report

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Background and Aims: Chylothorax is described as a rare complication of lymphoma in adults. It's associated with a high mortality rate and recurrence. Paediatric cases of malignant lymphoma with chylothorax are rarely reported. We report on a case of a child with an abdominal non hodgkin's lymphoma and a chylothorax.

Methods: We report a case of a 5-year-old male presenting with a chylothorax revealing an abdominal malignant non-Hodgkin's lymphoma, successfully treated with conventional chemotherapy.

Results: A 5-year-old male was admitted for acute respiratory distress and general alteration, with a 4-week history of abdominal pain, fever and weight loss. Physical examination revealed a large right pleural effusion with hepatomegaly. Chest X-rays and thoracic CT scan confirmed the presence of the effusion, but no mediastinal lymph nodes were identified. Abdomen US showed homogeneous hepatomegaly and splenomegaly with multiple profound lymph nodes, however, there was no intraperitoneal tumor. Fine needle pleural ponction yielded 350 ml of a chylous liquid. Biochemical and cytological study confirmed the presence of chylomicrons with malignant Burkitt cells lymphoma. There was no bone marrow or neuromeningeal involvement. Chemotherapy according to LMB 2001 protocol was started with supportive care. The patient didn't require further aspirations. He achieved a complete remission. During a follow up of 11 years no recurrence occurred.

Conclusions: Chylothorax is uncommon in children. It is a redoubtable complication because of malnutrition, high recurrence rate and considerable mortality. Conservative management may represent an effective strategy.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0038 / #260 | Child Ovarian Mass - A Burkitt Lymphoma Find - Case Report

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Background and Aims: Burkitt's lymphoma (LB) falls under the classification of non-Hodgkin's lymphomas and is a rare condition that can also affect the ovaries, unilaterally or bilaterally. We bring the report of a case of an ovarian mass, diagnosed by imaging tests requested in the context of a 04- year-old child, complaining of diarrhea, vomiting and abdominal pain with 09 days of evolution. At the time, with imaging and staging exams, the possibility of germ cell tumor (GCT), lymphoma, rhabdomyosarcoma and neuroblastoma was raised. Right salpingo-oophorectomy and appendectomy were performed 05 days after hospital admission. Anatomopathological result showed that it was Burkitt's lymphoma (E III Group B LMB 89), and the child was submitted to adjuvant chemotherapy with good clinical evolution. Primary ovarian lymphoma is rare, corresponding to 0.5% of non-Hodgkin's lymphomas and constituting 1.5% of ovarian neoplasms. [1] Surgical treatment is not the treatment of choice for patients with lymphoma, but most patients with ovarian mass end up undergoing surgery - according to the literature, 60% of these patients undergo surgery [3]. Therefore, it is always necessary to make a differential diagnosis of ovarian masses in order to establish the best therapeutic scheme and prognostic definition, and lymphoma must be among the diagnostic hypotheses.

Methods: Retrospective study of the clinical case and review of medical records and literature on the subject.

Results: Call attention to the pediatric surgeon to consider all differential diagnoses in the evaluation of an ovarian tumor, and ovarian Burkitt's lymphoma, although rare, should be remembered in these diagnoses.

Conclusions: When a pediatric surgeon is faced with an infant ovarian mass, it is necessary to evaluate carefully the characteristics of the mass, in order to offer a treatment that does not compromise the patient's survival in case of malignant lesions and at the same time that does not endanger this child's fertility.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0039 / #524 | **Presentation and Outcomes of Childhood Hodgkin Lymphoma at a Tertiary Centre in Uganda**

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Background and Aims: Hodgkin Lymphoma (HL) is a highly curable form of childhood cancer. Limited data suggest that children with Hodgkin Lymphoma in sub-Saharan Africa have poor survival. We aimed to describe the presentation and treatment outcomes among children with Hodgkin Lymphoma at Mulago National and Referral Hospital, Uganda.

Methods: We retrospectively and also prospectively evaluated children with a histological diagnosis of Hodgkin Lymphoma from February 2019 to April 2021 at the Oncology unit at Mulago National and Referral Hospital in Kampala, Uganda. Kaplan-Meier methods were used to study 1-year survival.

Results: A total of 23 cases of Hodgkin lymphoma were analyzed. The median age at presentation was 9 years (IQR 5.3-12.8 years). There were 16 (70%) males. The mean duration of symptoms before diagnosis was 12 months. Lymphadenopathy was the most common presentation (95%). Morphology distribution by H and E showed that 22 cases (96%) were predominantly classical HL. The predominant histological subtype was mixed cellularity 48%, followed by Nodular sclerosing 22%, other histological subtypes were represented to a lesser extent. Immunohistochemistry findings were available for 15 samples. Staging classification the disease indicates a high frequency of advanced disease (Ann Arbor stage III and IV disease) in 71% of cases at diagnosis and almost all patients had bulky disease at 96%. One year survival rate was 80.9%; (95% confidence interval 60.9-100).

Conclusions: Survival among children with Hodgkin Lymphoma in Uganda is poor. Advanced stage disease may likely contribute to poor outcomes.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0040 / #600 | **Oepa / Copdac Protocol in Children with High-Risk Hodgkin's Lymphoma. Experience in an Oncology Service in Mexico**

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Background and Aims: Background and aims: Hodgkin lymphoma (HL) is the third most frequent cause of cancer in children, mainly between 15 and 19 years of age. Currently, more than 90% of patients diagnosed with HL survive by combined chemotherapy and radiotherapy treatments. The objective of this study is to describe the results of patients with high-risk HL who received treatment with the OEPA / COPDAC protocol. The high-risk criteria for these patients were: Hemoglobin less than 10.5 grams / deciliter, albumin less than 3.5 grams / deciliter, bulky mass greater than 8 cm, B symptoms, stages III and IV.

Methods: This study is retrospective and observational, it includes patients under 18 years old with a diagnosis of high-risk Hodgkin Lymphoma treated at the Hospital Civil Juan I Menchaca from January 2017 to March 2021.

Results: Children diagnosed with high risk HL were 17, an average of 3.4 cases per year, more frequent in males 13 cases, average age of presentation 11 years old. The incidence of cases increased in 2019 with 8 patients, doubling the number of cases presented in 2017. All patients with high-risk HL received OEPA protocol (Prednisone, Vincristine, Doxorubicin, Etoposide / COPDAC (Prednisone, Dacarbazine, Vincristine, Cyclophosphamide) achieving complete remission 16 of the 17 children, 3 required radiotherapy and 2 children apart from receiving 2 OEPA cycles and 4 COPDAC required Stanford V protocol. Side effects were febrile neutropenia 2 cases, anemia 2 cases, nausea and vomiting 3 cases, there were no relapses or deaths, currently 16 patients are under surveillance and adapted to daily life.

Conclusions: This group of patients with high-risk HL presented an excellent response to treatment with OEPA-COPDAC protocols, achieving complete remission in 94.1%, behavior similar to what is reported in the world literature, in addition to presenting minimal side effects and providing quality of life.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PO0041 / #669 | The Influence of «Bulky» Disease on Survival in Patients with Hodgkin Lymphoma

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Background and Aims: About 70% of patients with the Hodgkin lymphoma (HL) may be cured. The analysis of various predictive factors is applied to identification of the remained patients who need intensive therapy. «Bulky» disease is the most important parameter which directly has impact on survival of patients with HL. The assessment of tumor burden's components (a mediastinum/thoracic ratio – MTR, peripheral bulky disease, number of involved lymph nodes) in young adults with Hodgkin lymphoma and the analysis of their predictive value.

Methods: Data on 87 patients aged from 19 to 29 years (median 24±4) are submitted. The male-to-female ratio was 2.1:1. B symptoms occurred in 47 patients (54%). There were 47 patients (54%) with clinical stage I-II and 40 (46%) - with stage IV. Extranodal involves is revealed in 37 patients (42.5%).

Results: The optimum cut-off level according ROC method for MTR has been established at 0.33, for peripheral bulky disease has been established at 46 mm, and for number of involved lymph nodes has been established at 5. Use of the correlation and regression analysis allocated the best linear regression model for MTR with the maximum determination coefficient and the minimum residual dispersion ($\bar{y} = -0.928971 + 4.008252x$). There were statistically significant the equation by Fischer's criterion and determination coefficient ($F_{\text{fact}}(46.74) > F_{\text{theor}}(4.17)$). Linear and nonlinear regression models for “a peripheral lymphadenopathy”, “number of involved zones” didn't show significant influence on the outcome ($p = 0.9237$ and $p = 0.3385$). MTR significant influenced the late results. The disease-free survival in patient's group with MTR less than 0.33 was 100%, in group with MTR more than 0.34 was 41% (log-rank test 0.04649).

Conclusions: The study proved that mediastinum/thoracic ratio is the significant prognostic factor. Such factors as “a peripheral lymphadenopathy”, “number of involved zones” had smaller predictive value. These components are indicators of “bulky” disease and demand further studying.

Publication Only Topic: Disease Orientated / AS01 Haematology / AS01.c Lymphomas

PO0042 / #675 | Hodgkin's Lymphoma in Young Adults

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Background and Aims: Young adults represent a unique cohort of the population with high level of malignant tumors incidence. Hodgkin's lymphoma (HL) makes about 12% of all tumors at 19-29-year-old patients. Young adults isn't rather investigated since treatment it is carried out in various options according to different programs (pediatric and adult), and the remote consequences of therapy in such patients aren't studied. The purpose of our investigation was to assess of therapy for Hodgkin's lymphoma in young adults.

Methods: From 1995 to 2015 87 patients aged from 19 to 29 years old (median 24±4) with biopsy-proven HL were treated. There were 34 (39.1%) men and 53 (60.9%) women (2.1:1). B symptoms occurred in 47 patients (54%). There were 47 patients (54%) with clinical stage I-II and 40 (46%) - with stage IV. Extranodal involves is revealed in 37 patients (42.5%). All patients were given 2-11 cycles of chemotherapy (median

4) with ABVD (48 patients, 55.2%), BEACOPP (36, 41.4%), MOPP (2, 2.3%) and COPP (one patient). Chemotherapy regimen was followed by extend- or involved- or subtotal radiation with doses from 20 to 51 Gy in 57 patients (65.5%).

Results: All patients were assessed for the treatment results. With a median of 31 months of follow-up (3 - 156 months) the freedom from treatment failure of 3 years was 47.4% (SE 7.8%), event-free survival was 70.5% (SE 7.2%), overall survival was 82.7% (SE 5.2%). Of the 72 patients who achieved a complete remission, 19 (21.8%) relapsed at a median time of 31 months from initial diagnosis (7 - 80 months)

Conclusions: Low survival in young adults with HL confirm need of change of combined modality therapy role with transition on using principle of risk-adapted treatment that will allow to achieve good results, and to decrease side effects of follow-up.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0043 / #791 | Successful Treatment of EBV-Related Post-Transplant Lymphoproliferative Disorder with Low-Dose Nivolumab Therapy

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Background and Aims: Posttransplant lymphoproliferative disorders (PTLD) are among the major causes of both mortality and morbidity after allogeneic stem cell transplantation (allo-SCT). It mainly occurs due to EBV infections. The treatment includes reducing immunosuppressive therapy, rituximab, cytotoxic T-cell, and donor lymphocyte infusion. The immune checkpoint inhibitors can be a choice for patients unresponsive to treatment.

Methods: Here, we present a case diagnosed with EBV-related PTLD after allo-SCT and treated with nivolumab successfully.

Results: A 6,5-year-old girl diagnosed with EBER positive diffuse large B cell lymphoma (DLBCL). She was evaluated as stage III, and treatment according to NHL BFM 90 protocol R4 group has started. Through the treatment, rituximab was added because of progressive disease. However, the disease progressed again, and ifosfamide-carboplatin-etoposide (ICE) treatment with rituximab was started. She showed a good response in the early period, and then progression occurred again. Allo-SCT was performed from an HLA-matched unrelated donor with a conditioning regimen including melphalan, fludarabine, ATG.

The patient presented with generalized lymphadenopathy, graft versus host disease with complete chimerism, and positive EBV serology 17 months after SCT. Lymph node excision was performed, and the diagnosis was EBER positive DLBCL again. Despite the treatments such as reducing immunosuppressive therapy, rituximab, and R-CHOP regimen, the disease progressed. Ibrutinib was started. At

the end of the second month, fever, weight loss, arose levels of EBV DNA revealed disease progression. A higher level of PD-L1 was detected in tumor tissue. Nivolumab was started 0,5 mg/kg every two weeks. After the first dosage of nivolumab, fever, weight loss improved dramatically, and EBV DNA count decreased significantly. The dosage was increased by 1mg/kg every two weeks. PET-CT disclosed complete metabolic response in the second month of nivolumab. The patient has received this treatment for eight months every two weeks, and she has been in remission after initiation of nivolumab therapy.

Conclusions: Immune checkpoint inhibitors such as nivolumab can be beneficial and reliable treatment options for patients who developed PTLD after SCT.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.c Lymphomas*

PO0044 / #954 | Mortality in Children with Burkitt Lymphoma: Experience of Casablanca University Hospital

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Background and Aims: Burkitt lymphoma (BL) is one of the most aggressive cancer and yet one of the most curable, frequent causes of treatment failure in low income countries are treatment abandonment and death during treatment. The aim of our study is to report the main death causes of patient diagnosed with BL, from 2016 to 2019, in the university hospital of Casablanca, Morocco, in order to predict and prevent them.

Methods: All data concerning epidemiological, clinical, biological and evolutionary aspects were collected from medical files of children with BL from february 2016 to October 2019, treated according to GFAOP/LMB2005 protocol without rituximab.

Results: During the study period 24 patients were diagnosed with BL, of which 8 died, for a total of 33% deaths. A male predominance is observed, the median age is 8.5 years, 50% were undernourished. As for the location 75% had an abdominal location and among them 83% had a mass that exceeded 10 cm. 62% had a Murphy stage > 3, 72% belonged to the LMB group B and 28% to the group C. One patient died during prephase, he had both medullary and central nervous system involvement, and was severely undernourished, he didn't have tumor lysis syndrome and died of septic choc following an episode of febrile neutropenia. One patient died of septic choc during the management of grade IV neutropenia following consolidation therapy (CYVE). 1 patient died during hospitalization for salvage therapy by the progression of his lymphoma. The rest, 5, died at home, while taking metronomic therapy. The time between the cures was respected in 71% of the cases. The median survival from diagnosis to death is 6 months

Conclusions: 75%(6) of death are related to the progression of the disease, while the remaining 25%(2) could have been prevented. It contrasts with previous studies, and shows an improvement of supportive care in our context. This suggests the addition of rituximab to reduce deaths due to progression of the disease.

Publication Only Topic: *Disease Orientated / AS01 Haematology / AS01.d Stem Cell Transplantation*

PO0045 / #1119 | **Incidence and Clinico - Hematological Pattern of Diamond Blackfan Anemia: 10 Years Experience in a Single Center**

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Background and Aims: The study aimed to determine the incidence and clinico-hematological pattern of Diamond Blackfan Anemia (DBA), a congenital form of pure red cell aplasia (PRCA).

Methods: A retrospective observational descriptive cohort study was conducted in the Department of Pediatric Hematology and Transfusion Medicine, The Children's Hospital and The Institute of Child Health, Lahore, for a period of 10 years (January 2008 to December 2017). Data was analyzed using SPSS software.

Results: Among 19,475 cases reviewed, 74 patients (0.4%) were diagnosed as congenital PRCA with a male to female ratio of 4.2:1 and a mean age of 5 months. In this cohort, 23% of patients had a positive family history. Thirty-four percent of patients presented with congenital anomalies involving cleft lip and palate (23.5%), thumb and upper limb defects (20.5%), craniofacial anomalies (14.7%), and cardiac and urogenital malformations in 8.8 and 5.8% patients, respectively. Red blood cell (RBC) counts and hemoglobin (Hb) concentration levels were low with a range of $0.5-3.0 \times 10^{12}/\mu\text{l}$ and 2.5-6.5 g/dl, respectively. Fetal hemoglobin (HbF) levels remained within the range of 2-20%. Bone marrow aspirate showed <5% erythroblasts in all patients.

Conclusions: The specific clinico-hematological pattern as described in our study not only helps in early diagnosis and management of the disease and also paves way to establish a data registry.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PO0046 / #1185 | **The Results of Neuroblastoma's Treatment According to Nb-2004 Protocol**

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Background and Aims: Neuroblastoma is the most common extracranial tumor in children and takes third place among malignant children's tumors in Kazakhstan. The aim is to evaluate the results of neuroblastoma's treatment under the NB-2004 protocol in the Scientific center of pediatrics and children's surgery.

Methods: One hundred thirteen patients: 57 boys (50.5%) and 56 girls (49.5%) treated according to NB-2004 protocol between January 2015 and December 2020 were included in the survey. The median age of patients was 21.5 months (range 0.8-191.8). Tumor localization was retroperitoneally in 93 cases, in mediastinum - 12, both in the mediastinum and retroperitoneally -3, neck lesion-3, neuroblastoma without primary focus -2. Primary disseminated neuroblastoma was detected in 73.4%: metastases in bones, bone marrow, lymphatic nodes. N-MYC gene amplification was detected in 22.1%, 1p in 5.3%. According to risk group stratification «observation group» contained 26 patients (23%), intermediate-risk group (IRG) - 14 (12.4%), high risk (HGR) - 73 (64.6%).

Results: Local control was made in 96 cases: radiotherapy in 15 patients, surgery in 81. Autologous hematopoietic stem cell transplantation (auto-HSCT) was made in HGR in 37 patients (52%) with conditioning regimes CEM - 19 cases, melphalan +carboplatin - 1, TreoMel - 17. Engraftment was registered in all cases. Median follow-up in the whole group is 20,4 months (range 0.03-132.43), after auto-HSCT - 12.47 months (range 0.13-61.03). Five-year overall survival (OS) was - 61.9%. 2-year event-free survival - 35%. Disease progression was in 24 cases (21.2%). Relapses were in 18 patients (15.9%): in HGR - 83.3%, in IRG - 16.6%. OS in the HGR was 43.8%, in the IRG - 85.7%, in the low-risk group - 100%. OS after auto-HSCT is 54.1%.

Conclusions: The therapy of neuroblastoma according to the protocol is effective and the results of therapy in various therapeutic groups in our clinic are comparable to the GPOH group.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma*

PO0047 / #1337 | **Clinical Aspects, Diagnoses and Outcome of Patients with Neuroblastoma Treated in a Hospital Inside the Rio Grande do Sul**

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Background and Aims: Background and aims: Neuroblastoma (NB) is an embryonic tumor of the autonomic nervous system, typically developing in the adrenal gland or para-spinal ganglia. It is the most common cancer in the first year of life with an average age of 17 months at diagnosis. The clinical presentation is variable, having expansive mass characteristics in the region where it is housed. The objective of this study was to describe the clinical aspects, diagnosis and outcome of NB in children treated at the Pediatric Hematology-Oncology of the

University Hospital of Santa Maria in the state of Rio Grande do Sul in the very south of Brazil.

Methods: The information collected from the medical records between 01/01/1999 and 12/31/2015, and the data analyzed through averages and frequencies. After crossing the variables, the Pearson Correlation Coefficient used and, finally, the Kaplan Meier survival curves were constructed.

Results: We analyzed 28 patients and found 70.4% NB high risk, the mean age was 29 months, the most common symptom was fever (60.7%), abdominal mass (57.1%) and weight loss (46.4%). The MYCN gene tested in 82.15%, with 52.38% amplified. In 76% were treated with protocol Neuro-IX-2000 and Neuro-X-2010. Among the relapsed ($n = 15$), 20% were rescued. Overall survival was 53.6% in 5 years and event-free survival 5 years was 53.6%.

Conclusions: The results found are mostly consistent with the literature and that improvements needed in the treatment regimens offered, especially for patients with high-risk disease and those with relapses. Keywords: Neuroblastoma. Diagnosis. Survival. MIBG.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0048 / #1397 | The Impact of Radiation Therapy on the Treatment of Pediatric Neuroblastoma in the Tunisian Center

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Background and Aims: Purpose: To assess the benefit of radiotherapy in the treatment of neuroblastoma and to evaluate its impact in terms of toxicity.

Methods: A sample of thirteen patients with neuroblastoma treated in the Tunisian center between 1995 and 2018 was analyzed retrospectively using surgical and imaging reports.

Results: Nine patients were eligible for curative radiotherapy and 4 patients received palliative radiotherapy to metastatic sites. Patients irradiated with curative intent were classified as High risk in 7 cases and Non-High risk in 2 cases. The dose delivered was 36 Gy in most cases. Acute complications were dominated by radiodermatitis. No significant late toxicity was demonstrated. After a median follow-up of 20 months, 5/13 patients died from their disease. For the other eight patients, complete remission was achieved in 6 cases and partial remission was achieved in 2 cases.

Conclusions: In our study, radiotherapy was associated with improved local control and acceptable toxicity. Standardized medical care following international recommendations could improve our therapeutic results.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0049 / #1418 | Prognostic Factors of Neuroblastoma: A Retrospective Study about 56 Cases in the Center of Tunisia

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Background and Aims: Neuroblastoma has a heterogeneous clinical course, ranging from spontaneous regression to incurable high-risk disease. Besides classical clinic histological factors, molecular biological factors are challenging. The Aim of the study is to determine the prognostic factors in our population.

Methods: A retrospective study included 56NB treated at the medical oncology department of FarhatHached University Hospital from January 1996 to December2019 according to SFOP protocol initially than POG3941 since 2015. Statistical analysis was performed with SPSS 10.0 using the Kaplan Meyer method, X²test for univariate analysis ($p=0.05$)

Results: The average age was 3.5 years [2 months- 15 years] with 23% younger than 18 months. The adrenal glands were the primary site in 27% of cases. Metastatic forms were observed in 72% of cases. The amplification of the N myc gene was performed in 36 patients. 8 patients had 1p36 clonal deletion. The SFOP protocol was used in the neoadjuvant chemotherapy in 32% of cases with objective response in 40% of cases. Surgery was performed in 18 patients (26%). Adjuvant chemotherapy was administered to 19 patients. The 5 year overall survival was 53% with respectively 80% and 30% for localized, metastatic NB. Prognostic factors retained in the univariate analysis were age> 18 months and metastatic forms with respectively $p = 0.005$, $p = 0.0001$. However, the other prognostic factors described in the literature: N-myc oncogene amplification ($p=0.42$) and the LDH level were not significant ($p=0.37$) in our series.

Conclusions: Although consensus approaches to pretreatment risk stratification based on clinical⁴ and genomic tumor characteristics have been described. There is a need of developing molecular biology in our study especially with detecting mRNA in residual disease that may contribute to the development of a prognostic score for integration into clinical practice.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0050 / #1732 | Diagnosis and Treatment of Stage 4S Neuroblastoma with Bilateral Adrenal Involvement

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Background and Aims: The occurrence of bilateral adrenal neuroblastoma is extremely rare. According to the INRG classification, 4S neuroblastoma is defined by small primitive tumors with disseminated disease restricted to the liver, skin, and/or bone marrow and it affects children aged less than 1 year

Methods: Case-report

Results: We report a 2 month-old male infant with no previous history. He was hospitalized for abdominal distension. The clinical examination revealed an abdominal mass in the right flank and hepatomegaly. Abdominal ultrasonography showed two bilateral adrenal masses with heterogeneous tissue echostructure and calcification, associated with an enlarged heterogeneous multinodular liver with regular outlines. The CT scan confirmed the ultrasound findings, it allowed to study the loco-regional and distant extension of the tumor as well as to perform the biopsy of both masses. Urinary catecholamines and MIBG scintigraphy were not performed due to lack of means. The N-Myc gene was not amplified. The diagnosis of bilateral neuroblastoma, stage 4S with Pepper syndrome, according to the INRG classification, was established. The patient received two courses of neoadjuvant chemotherapy. The abdominopelvic CT scan concluded that there was partial regression in the size of the two retroperitoneal masses with a heterogeneous enhancement. The liver was enlarged, heterogeneous, and multinodular with regular outlines. The patient was operated on at the age of 5 months, we performed left tumor adrenalectomy and an adrenal-sparing tumorectomy in the right side. The histopathological examination revealed: Bilateral adrenal neuroblastoma with an apoplectic appearance of one of the two tumors. The follow-up was uneventful and complete remission was obtained with a good evolution in the medium term.

Conclusions: The prognosis of bilateral adrenal neuroblastoma with Pepper syndrome is favorable due to age less than one year, absence of bone metastases, and absence of N-MYC gene amplification in most cases. The treatment should be the less aggressive as possible

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0051 / #622 | Management of Dinutuximab Beta Extravasation in Two Patients with Neuroblastoma

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Background and Aims: We present the dinutuximab beta extravasation of two patients who received immunotherapy and irinotecan-temozolamide-containing chemotherapy due to relapse high-risk neuroblastoma.

Methods: Case 1-BGO A 10-year-old girl was diagnosed with a stage IV high-risk group neuroblastoma in September 2016. After first line treatment, complete remission was achieved in April 2017, autologous stem cell transplantation and then radiotherapy was performed. She had relapse with bone and liver involvement in April 2020. Chemotherapy and dinutuximab beta immunotherapy was started on 10.08.2020. On the 6th infusion day of fourth cycle, she described pain in the vascular access region and extravasation was detected with a swollen area of 2x3 cm. We stopped the infusion and removed the needle. Cold application and extremity elevation was performed. The patient was already receiving a broad spectrum antibiotic, meropenem, due to the high fever that developed on the 3rd day of infusion. Symptoms regressed and no complications occurred during follow-up. Case 2- EAK A 3-year-old girl was diagnosed with stage IV high-risk neuroblastoma in March 2019. After first line treatment, complete remission was achieved in September 2019, autologous stem cell transplantation was performed. In May 2020, she had relapse with the involvement of peritoneum and bone. Chemotherapy and immunotherapy was started in January 2021. On the 8th infusion day of first infusion, extravasation was seen in vascular access region. She had pain and a swollen area of 3x3 cm. Management and outcome were similar with the first case.

Results: Extravasation is a serious complication that can be seen in drug or fluid administration intravenously. In this report, we presented two cases in which monoclonal Anti-GD2 antibody extravasation was successfully managed with general extravasation principles and no sequelae was seen after long follow-up.

Conclusions: In the literature, monoclonal antibody extravasations are mostly defined as irritant. However, there are few reported publications on this subject. With increasing experience, understanding and managing complications of dinutuximab beta can be done with more standardized protocols.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0052 / #759 | A Single Center Clinical Analysis of Neuroblastoma in Children Younger Than 18 Months

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Background and Aims: Neuroblastoma (NB) accounts for 7–10% of malignant tumours among children. It is usually seen in early childhood. About a third of patients (36%) are diagnosed within the first year of life. Treatments stratified by risk classification have improved survival outcomes in patients with neuroblastoma. The purpose of this study is to evaluate the outcome of patients younger than 18 months with neuroblastoma.

Methods: Medical records and pathology reviews of 35 patients were evaluated, retrospectively.

Results: There were 35 patients with neuroblastoma who were younger than 18 months at diagnose between February 2007 and March 2021. Eleven patients (31.4%) were males, 24 (68.6%) were females. The mean age was 7 months old. There were 2 (5.7%) patients with Stage I disease, 9 (25.7%) with Stage III, and 16 (45.7%) with Stage IV and 8 (22.8%) Stage IVS. Thirteen (37.1%) patients had metastatic disease. The most common metastatic sites were liver (34.2%) and bone marrow (22.8%). MYCN amplification was detected in 6 (17.1%) patients. Total surgery was performed on 4 (11.4%) of the 35 patients at the time of diagnosis. Three patients relapsed during the follow up and one of them was diagnosed acute lymphoblastic leukemia during the maintenance therapy of neuroblastoma. Median surveillance time was 36 months. Three year overall survival for patients with neuroblastoma younger than 18 months (OS) 85.7%, event free survival (EFS) 82.9%, for patients younger than 12 months OS 87.1%, EFS 83.8% were found and OS was 68.8% was detected for patients with stage IV.

Conclusions: Satisfactory outcomes could be achieved in neuroblastoma in children aged within 18 months; the prognosis was similar in children who were younger than 12 months old compared with the patients were 12–18 months old. Intensive therapeutic approach could be avoided in these groups except for patients with stage IV disease in order to reduce the long-term adverse reactions.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0053 / #845 | Multiple Relapses of Neuroblastoma: Experience of Resource-Limited Setting

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Background and Aims: Neuroblastoma accounts for 15% of all pediatric cancer fatalities. Although advances in the treatment of neuroblastoma, relapses of high-risk neuroblastoma occur frequently.

Methods: Here we report cases of two patients with high-risk relapsed and refractory neuroblastoma with N-MYC gene amplification.

Results: In 2015 a four years old male was diagnosed with neuroblastoma, IIA stage. The treatment was started according to the VDC (vincristine, doxorubicin, cyclophosphamide) regimen followed by total surgical resection of the tumor. The patient underwent adjuvant radiotherapy (1.8Gy/21.6Gy) followed by chemotherapy by VDC-regimen. Six months after completion of treatment, the child had a local relapse and was treated according to the ICE-regimen. Three years later second local relapse was detected. The tumor was resected and chemotherapy started based on Topotecan/Cyclophosphamide-regimen, followed by autologous-HSCT. In 2020 third local relapse was detected. After surgery VIT (vincristine, irinotecan, temozolomide) regimen was chosen. During chemotherapy, health conditions got worsened and he died from DIC. In 2018 a three years old boy was diagnosed with neuroblastoma, IV stage. MIBG examination revealed metastatic lesions in the skeleton. The treatment was initiated by NB-2004-scheme followed by tumor resection and continuation of treatment with NB-2004-scheme. Afterwards, 131I-MIBG radiotherapy followed by autologous-HSCT was performed. In 2019 MIBG indicated residual multiple metastases without BM involvement and treatment was started by VIT regimen. In May 2020, again metastatic changes of brain were detected. Removal of metastatic lesion was followed by cranial radiation and monotherapy by Temozolamide. Six months later the child had pancytopenia, bone marrow aspiration was done and revealed disease relapse. Palliative chemotherapy with topotecan and cyclophosphamide was given. During chemotherapy, health conditions got worsened and DIC has developed, which lead to fatal outcomes.

Conclusions: In resource-limited setting, in patients with multiple relapsed or refractory neuroblastoma, chemotherapy may lead to prolonged disease stabilization.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.a Neuroblastoma

PO0054 / #897 | The Influence of Skeletal Muscle Mass on Pediatric Malignant Tumor

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Background and Aims: The decrease of skeletal muscle mass, quality and strength is defined as 'sarcopenia' and is known to be correlated

with a poor prognosis on adult patients. However, there have been few reports on the influence of sarcopenia on malignant tumors in pediatric patients. In the present study, we investigated whether or not the skeletal muscle mass affects the pediatric malignant tumor.

Methods: Twenty-eight patients with malignant tumor who were treated by chemotherapy and surgery at our hospital from 2007 to 2017 were divided into a progression-free (PFS) survival group (n=21) and a relapse/death (R/D) group (n=7). The patients who were treated by only surgery, or the patients that showed progress disease during the initial treatment were excluded. The rate of change in psoas muscle mass was calculated by comparing the psoas muscle volume (PMV) on computed tomography (CT) before and after treatment. The rate of change in the PMV, Kaup index, and serum albumin level were compared. Furthermore, we determined the cut-off rate of change in the PMV and compared the overall and progression-free survival.

Results: The kind of malignant tumor in this study was neuroblastoma (n=12), nephroblastoma (n=7), hepatoblastoma (n=4), rhabdomyosarcoma (n=4) and other tumor (n=1). The rates of change in the PMV were 1.45 and 1.07 in the PFS and R/D groups, respectively (P = 0.04). There were no significant differences in the rates of change in the Kaup index or the serum albumin level of the two groups. The patients whose rate of change in the PMV was >1.20 showed a prolonged overall (p=0.023) and progression-free survival (p=0.023).

Conclusions: A decrease in the skeletal muscle mass was suggested to be a significant prognostic factor for pediatric malignant tumor.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0055 / #1217 | Bilateral Wilms Tumour: Surgical Dilemmas in Management

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Background and Aims: Background and aims Synchronous BWT accounts for 4-7 % WT. As per SIOP protocol, neoadjuvant chemo followed by surgery is the standard practice. However, the surgical management is unlike the nephrectomy for unilateral Wilms tumor. The surgical dilemmas faced in each case are highlighted.

Methods: Methods Review of 2 cases of synchronous Bilateral Wilms Tumour during the covid pandemic (April 2020-April 2021)

Results: Case 1 Male infant presented at the age of 9 month with abdominal distension. Presence of multiple lesions replacing the renal parenchyma bilaterally on imaging was suggestive of Wilms tumour. As per SIOP protocol, after control of hypertension, he was taken up for sequential nephron sparing surgery. There were 3 nodules on the right and 4 nodules on the left, which were excised taking care to resect as well as preserve. At 6 months follow up after completion of chemotherapy, imaging revealed lesions bilaterally which have remained the same

size. Case 2 2 yr old girl child with abdominal mass, on imaging showed bilateral renal mass, right more than left. At 8 weeks of chemotherapy, she underwent sequential nephron sparing surgery. At surgery, the tumour lesions measured 20x15 cm on the right and 7x5 cm on the left, enabling partial nephrectomy. At 6 months follow up, persistent but stable lesion was noted on the left kidney without increase in the size.

Conclusions: Bilateral Wilms tumour management requires careful evaluation on a case to case basis to achieve maximal renal parenchymal preservation. Imaging by sonography can be an useful adjunct in the post operative follow up.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0056 / #1261 | Angiographic Occlusion and Intra-Arterial Chemotherapy in Wilms' Tumor

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Background and Aims: We had a positive experience of angiographic occlusion (AO) and intra-arterial chemotherapy since 1983 in 19 children with Wilms' tumor (WT). While this method has not been widely implemented in Europe and Russia, we decided to share our experience and offer a discussion of this problem.

Methods: Abdominal aortic angiography with selective renal artery angiography and embolization was performed in 19 children with stage 2-4 WT. 11 patients were 1-3 years of age, 8 were 4-8 years. Gianturco metal coils with a diameter of 0.3-0.7 cm were used as an occlusive material. The average time of AO was 40 minutes.

Results: Angiography was performed on the last day of preoperative polychemotherapy, while before the occlusion of the renal artery, doxorubicin or vincristine was injected into it. Echographic control after AO showed that the size of the kidney decreased by an average of 2-3 cm, foci of decay appeared in the tumor. Transperitoneal nephrectomy was performed on day 2-5 after AO in 15 patients, and on day 6-10 in 4 patients. Regardless of the timing of the operation after AO, there was a decrease in blood loss during surgery by 3-4 times, a decrease in the size of the tumor, which reduced the operation time by 30-40 minutes. In 18 of the 19 patients the integrity of the capsule was not damaged, in one patient the tumor sprouted through the main vessels and was not removed radically. The five-year survival rate of patients at stage 4 was 34 %, at stage 3 - 51%, at stage 2 - 100%.

Conclusions: Preoperative AO of the renal arteries for WT in children is an effective method that allows to reduce the risk of capsule damage during nephrectomy, increases its radicalism and ablastics and reduces blood loss and decreases the time of surgery.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0057 / #1359 | Non-Wilms' Renal Tumors in Children: An Analysis of 5 Years Data from Pediatric Oncology and Hematology Unit, University Hospital Casablanca, Morocco

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Background and Aims: Renal tumors other than Wilms' tumor are infrequent in childhood. Wilms' tumors account for 6% to 7% of childhood cancer, whereas the remaining renal tumors account for less than 1%. The most common non Wilms' tumors are clear cell sarcoma of the kidney, rhabdoid tumor of the kidney, renal cell carcinoma, mesoblastic nephroma, and multilocular cystic nephroma. Collectively, these tumors account for less than 10% of the primary renal neoplasms in childhood. The purpose of this study is to determine the actual incidence, age distribution, of patients with non-wilms tumors who were treated in a Public University Hospital in Casablanca Morocco from January 2015 to December 2019.

Methods: A retrospective study was carried out on all non-wilms tumor cases treated in our institution during this period. The variable analysed were age and date at diagnosis, pathological staging, treatment modalities used, and the last date of visit

Results: We identified 8 cases of non wilms tumor which represents 11% of renal tumors observed during our study period. The median age was 3 year with the extremes ranging from 5 month and 10 years old. These tumors occurred most commonly in patients less than one year old. We found a female predominance with a M/F ratio of 3:5. The most common histological entities were clear cell sarcoma (50%) and congenital mesoblastic nephroma (37%), followed by renal cell carcinoma (12%). In contrast to the wilms tumour, and despite the mesoblastic nephroma we found that the prognosis of these tumours is more poor, especially the clear cell sarcoma.

Conclusions: Non-Wilms' tumors may represent a significant proportion of renal tumors in children, especially in children aged less than 6 months or greater than 12 years. Most of our findings were generally similar to those reported in literature.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0058 / #1371 | Pathomorphological Features of The Non-Epithelial Component of Nephroblastoma or Wilms Tumor

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Background and Aims: According to the National Cancer Registry of Ukraine, malignant neoplasms are the second leading cause of pediatric mortality after accidents. Kidney tumors amount 7-8% of all tumors in the first 15 years of life. Tumors of the retroperitoneal space are the most common neoplasms, 85% of cases is nephroblastoma, or Wilms tumor. Classic nephroblastoma are triphasic tumor and composed of epithelial, blastemal, and stromal elements, but the diagnostic criterion for tumor is the presence in the kidney of undifferentiated tumor tissue or nephrogenic structures that are at different levels of development. The aim of this study was to establish the structural features of the non-epithelial component of nephroblastoma to optimize pathomorphological diagnosis and prognosis.

Methods: We studied postoperative material of 17 cases of Wilms tumor in children. The microscope slides were stained Hematoxylin and Eosin, van Gizon, Masson, PAS-reaction. Immunohistochemical reaction for cytokeratin, actin, vimentin and desmin was also analysed on all cases of Wilms' tumor.

Results: The non-epithelial component of the nephroblastoma is mosaic in microscopic pattern and is represented by tissues of mesenchymal origin of differing degrees of cataplasia. Moreover, myxoid, fibromatous, chondroid, osteoid, leiomyomatous and rhabdomyosarcomatoid structures were detected either separately or in different proportions. The morphological structure of the non-epithelial component of the nephroblastoma resembled a mono- or multicomponent mesenchyme. The stromal component of all nephroblastoma did not show differentiation into a specialized tissue type, and all cases had positive reactions to vimentin. Some of which had positive reactions to actin and only 5 of 17 cases with rhabdomyoblastic differentiation were positive for desmin.

Conclusions: In summary, the immunohistochemical stain for cytokeratin, actin, vimentin and desmin is helpful in confirming the presence of non-epithelial component and differing degrees of cataplasia of this component are significantly correlated with 5-year survival rate of patients.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0059 / #1436 | Treatment Outcomes of Nephroblastoma According to SIOP Protocols in the Center of Tunisia

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Background and Aims: Nephroblastoma (WT) accounts for approximately 5% of malignancies in children under age 15 years. SIOP pioneered therapeutic strategy based in upfront chemotherapy to identify therapeutic groups and adapt the adjuvant treatment. The aim of the study was to compare our results to those of SIOP.

Methods: Retrospective study of 84 WT patients treated between January 1994 and December 2019 according to SIOP93.01(Groupe1) and SIOP 2001(groupe2) since 2010. Statistical analysis and survival rate was calculated respectively with SPSS10.0 with Kaplan Meyer method.

Results: Median age at diagnosis was 3.5 years (3months-20years) with a sex ratio of 52%. Median time to consultation was 2month. The diagnosis was based on clinical and radiological presentation, biopsy was needed for 23patients(27.4%) and 22%of patients were metastatic. Neoadjuvant chemotherapy, surgery, adjuvant chemotherapy and radiotherapy were performed respectively for 98%, 94%, 94% and 21% patients according to SIOP93.01 and SIOP2001 respectively for 69% and 31 patients. OS and EFS were for all cohort was respectively 77% and 70%. For groupe1 OS was 63% and for groupe2 82%. The univariate analysis demonstrated significant prognostic factor in favor of Tumour volume<500ml, Therapeutic groupsI/II and localised tumours with $p<0.05$ for two groups. 21 children (25%) relapsed. Of these, 5 had local relapse, 12 metastatic relapse and 4 had local and metastatic relapse. Only 2 survived. We noted 5 toxic deaths and no late sequela

Conclusions: Despite our study showed lower rate survival comparing with SIOP results due to lack of resources, we note better results after using SIOP2001 protocol. However challenging treatment are needed for relapsed tumors

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0060 / #1477 | Clinical Case of VB Stage Bilateral Wilms Tumor, Treatment Efficacy According to the SIOP-2001 Protocol

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Background and Aims: Nephroblastoma (WT) is one of the most common malignant embryonic tumors in children and consists of derivatives of nephrogenic tissue with different degrees of differentiation. Bilateral nephroblastoma (BNB) is recorded in 4-10% of all kidney malignant tumors in children. BNB is a rare disease, that's why each new report about a child with BNB is an important scientific and practical experience.

Methods: This clinical case is about 4-year-old boy who was admitted to the oncology department as soon as a noticeable large formation in the left abdominal region was found and confirmed by CT scan, that

visualized a large formation with clear, straight contours in the left kidney (14.8x11.5x15.3 cm) and in the right kidney (2.0x1.8x1.7 cm).

Results: After neoadjuvant 6th week chemotherapy according to the Wilms-Tumor SIOP-2001 treatment protocol, the child underwent surgical intervention: laparotomy, tumor nephroureterectomy on the right (the size of the tumor of the right kidney was preserved with dimensions of in the posterior segment 1.8 x 1.9 x 2.0 cm). Histologically patient had mixed nephroblastoma, intermediate risk. Local tumor stage II. After 5 weeks in CT of the abdomen the tumor of the left kidney resolved and surgical treatment was excluded. After the end of chemotherapy total of 34 weeks, the following examinations were carried out: ultrasound of the abdomen, CT of the chest, MRI of the abdomen, scintigraphy, according to which, in the projection of the space left kidney, pathological formations are not determined, the right kidney with post-resection changes, the function of the left kidney is preserved.

Conclusions: This clinical case demonstrates the complexity of the treatment and management of a patient with BNB. The treatment of this patient according to the Wilms-Tumor SIOP-2001 protocol gave a positive result, both in the general condition of the patient and in his clinical, laboratory and instrumental picture.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0061 / #1720 | Pediatric Warning Signs Pointing to Adrenocortical Tumor

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Background and Aims: Corticosurrenalian adenomas are rare in children. They are usually revealed by Cushing syndrome including signs of virilization due to hypersecretion of androgens. our work aims to assess clinical presentation as well as the management of corticosurrenalian adenoma

Methods: We treat the case of two children who were hospitalized in our department for exploration of a surrenalian mass revealed by virilization signs.

Results: The first case is about of a 6-year-old boy who was admitted for gynecomastia evolving for 2 years. On examination, we have found bilateral gynecomastia, pubic and axillary hairiness, and an advanced weight and length for the age. In biology, we have found an elevated rate of estradiol. The bone age was 12 years. Abdominal US and CT scans have shown a left surrenalian calcified mass. Cerebral MRI was normal The second case is about a 5-year-old female who presented with a one-year history of virilization developing facial acne, coarse pubic hair, labia majora hypertrophy, and clitoral enlargement. A functional adrenal tumor was suspected. An abdominal ultrasound scan

identified unilateral left surrealian mass with calcifications. We completed both cases by a subsequent CT scan of the chest, abdomen, and pelvis for surgical planning and tumor staging. Treatment was surgical for both cases, thus, the infants underwent transperitoneal left adrenalectomy using a laparoscopic approach in one case, a laparotomy in the other one, and no complications were reported after surgery. The results of the histopathological examination have concluded to the diagnosis of corticosurrealian adenoma. After two years of follow-up, the virilization symptoms improved, we noted regression of the gynecomastia for the boy and the axillary and pubic hyperpigmentation has also clearly regressed for the girl.

Conclusions: Infantile virilization signs can be a mode of the revelation of corticosurrealian mass by elevation of estradiol's rate. In cases of adrenocortical adenoma, complete surgical excision can be curative

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0062 / #1726 | **Bilateral Pheochromocytoma in a Child**

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Background and Aims: Pheochromocytoma (PCC) is a catecholamine-secreting tumor, which is seen rarely in children. Early diagnosis and intervention are important for the successful management of this rare pediatric neoplasm. Routine genetic screening of these children is mandatory for the detection and timely management of familial PCC. Both open and laparoscopic approaches have been successfully employed

Methods: We reported the case of a child who was referred to us in order to treat his bilateral pheochromocytoma

Results: An 8-year-old male child referred to us for bilateral cortical adrenalectomy. He complained of pain abdomen. General physical examination and systemic examination findings were unremarkable except for a high blood pressure of more than 160/100 mmHg. Ultrasound examination (USG) of the abdomen showed multiple heterogeneous bilateral adrenal lesions that were confirmed on computed tomography (CT) scan. I131-Meta-Iodo-BenzyleGuanidine (MIBG) study was performed and showed specific bilateral fixations of MIBG by an adrenal mass on the right sparing the upper right mass and two on the left. Suspicion of bilateral pheochromocytoma was confirmed when 24 hr urinary normetanephrine was markedly elevated; The child was pre-operatively prepared with both alpha and Beta-blockers. After a bicostal incision, total right adrenalectomy was performed, and a left

cortical sparing adrenalectomy was done. The child remained hemodynamically stable postoperatively. In the follow up after two months, the child had normal blood pressure and serum cortisol level

Conclusions: A Cortical sparing adrenalectomy is advantageous in patients with bilateral disease to avoid long-term glucocorticoid deficiency, especially in children

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0063 / #173 | **Influence of Radiation Therapy Delay on Abdominal Tumor Recurrence in Pediatric Wilms Tumor. First Report of the Experience in Mexico (Hospital Infantil De Mexico)**

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Background and Aims: In the National Wilms Tumor Studies (NWTS) 1 and 2 it was established that the delay at the start of radiotherapy has been related to the lower control of the disease, however, these results are discordant in subsequent studies. We use the database of the Children's Hospital of Mexico (National Institute of Health).

Methods: Retrospective analysis of 50 patients under 18 years with resected unilateral non-metastatic WT treated adjuvant flank or whole abdominal radiotherapy from 2008 to 2018 at a single institution. The day of nephrectomy was considered day 0. RT delay after surgery was divided into 3 groups: 1 (day 0-9), 2 (day 10-14), and 3 (≥ 15 days).

Results: The median age at diagnosis was 4 years (1-15); 32 females (64%). Histology blastemal in 35 patients (70%). The most frequent stage was III in 36 patients. Twenty-five patients (50%) received neoadjuvant chemotherapy. The median delay was 21 days (6-145 days). The most frequent total dose was 10.8Gy (74%) and 37 patients (74%) received FRT. Thirty-seven patients (74%) received RT ≥ 15 days after surgery. The mean overall treatment time was 9 days. Recurrence occurred in 10 patients (20%) with a median of 10 months, 4 lung, 3 flank, 1 liver, 1 contralateral kidney, and 1 lung and liver. All recurrence were in group 3. RT delay of ≥ 15 days influences recurrence ($p < 0.0001$). Recurrence-free survival at 5 years after surgery was 68% for group 3 ($p = 0.058$). With a median follow-up of 64 months since diagnosis (9-130), 37 remained alive without treatment (74%), 1 alive with treatment, 5 die due to progression, and 7 lost follow-up.

Conclusions: Patients treated ≥ 15 days were the only ones who presented local and distant failures ($p < 0.0001$). Overall survival was not different between the groups analyzed. These results are consistent with what has been reported in the literature.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0064 / #1747 | Microscopic Structures of the Epithelial Component of Wilms Tumor

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Background and Aims: Child mortality from malignant neoplasms is high worldwide today, despite the achievements of pediatric oncology. Among tumors of the retroperitoneal space, nephroblastoma or Wilms tumor. Progress in the treatment of children with nephroblastoma are closely linked to the improvement of diagnostic methods, including morphological. The aim of this study was to examination the morphological of the structures of the epithelial component to ensure the reliability and objectivity of the histopathological diagnosis and prognosis of Wilms tumor.

Methods: Tissue morphological examination of 17 cases of postoperative nephroblastomas with different degree of differentiation and emphasis on the epithelial component was performed. The microscope slides were stained Hematoxylin and Eosin and immunohistochemical reaction for epithelial marker such as pancytokeratin was analysed in all cases of Wilms tumor.

Results: The epithelial component of nephroblastoma formed tubular (with or without lumen) and rosette; papillary or pseudobulb; solid (anaplastic and sarcomatoid) microscopic structures. The epithelial cells with a high degree of differentiation of the cubic shape were arranged in an orderly manner and formed approximately the same size tubular structures, mostly round with or without a lumen. Moderately differentiated epithelial cells of cylindrical and prismatic shape, located concentrically on the basement membrane, formed complexes resembling rosette-like structures found in neuroblastic tumors. Immunohistochemical typing of cells that formed rosette-like structures revealed their positive reaction with monoclonal antibodies to pancytokeratins, which confirms the epithelial nature of the cells. Therefore, the structures formed by these cells were regarded as pseudorosettes. In some fields of view, they very much resembled primitive glomerular structures. Low-differentiated and anaplastic nephrothelium formed solid alveolar structures that resembled metaplastic transitional cells, multilayer squamous non-keratinized epithelium, and in one observation resembled a picture of fibrosarcoma.

Conclusions: Morphological of the structures of the epithelial component and degree of differentiation are important of the histopathological diagnosis and prognosis of Wilms tumor.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0065 / #210 | Metastatic Anaplastic Nephroblastoma with Intracardiac Involvement

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Background and Aims: Background and Aims : Kidney cancer accounts for about 7% of all childhood cancer and the majority are nephroblastomas or Wilms tumor (WT). The involvement of the atria and the inferior vena cava in patients with WT is very low, almost 4%. The objective of this study is to report a 7-year-old boy with WT and intracardiac involvement.

Methods: A descriptive case report.

Results: A 7-year-old boy presented 1 week of evolution of asthenia, hyporexia, dyspnea and appearance of a mass in the right hemi-abdomen. Admission in poor condition, hypoventilated lung fields, globose, asymmetric abdomen, with a mass that involves the entire right hemiabdomen, collateral abdominal circulation, swollen lower extremities. A diagnosis of Anaplastic Stage IV Wilms Tumor with lung metastases is made, a patient with high oxygen requirements; An echocardiogram was made, finding a right atrium mass with 85% occupancy and global chamber dilation, tumor thrombus in the superior portion of the inferior vena cava and bilateral pleural effusion. He also presented disseminated intravascular coagulation with prothrombin time 22 seconds, international normalized index 1.92, and Dimero D of 4.233 nanograms / milliliter. He received first cycle of chemotherapy with Carboplatin, Ifosfamide and Etoposide. Control echocardiogram and lung scintigraphy one month after the start of chemotherapy: a healthy heart, without tumors; lung scintigraphy: apical subsegmental perfusion defect of the upper lobe of the right lung, suggestive of microembolism. Patient with noticeable clinical and imaging improvement, continuous management of WT.

Conclusions: The case of this patient with WT and intracardiac involvement is very interesting because after the first cycle of chemotherapy, he demonstrated an excellent response to the therapy used.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0066 / #390 | Single Center Study: Retrospective Study of 13 Cases of Synchronous Bilateral Wilms Tumors

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Background and Aims: Synchronous bilateral Wilms tumours (BWTs) make up 4% to 7% of all Wilms tumours (WTs) and does not necessarily carry a fatal prognosis, and it may be cured in a certain percentage of cases by radical surgery. The aim of this study is to describe clinical characteristics of BWTs of our population and to evaluate our therapeutic results.

Methods: From 1994 to 2020, 13 patients with synchronous BWT were treated with a preoperative chemotherapy with vincristine and actinomycin in the center of Tunisia. Chemotherapy (CT) was continued until safe nephron sparing surgery (NSS) could be performed for as long as radiological tumor response continued.

Results: Out of a total of 102 cases with WTs, there were 13 cases of BWTs (12.7%). the age at diagnosis ranged from 8 months to 5 years (mean 2.1 yrs) and the sex-ratio was 0.85. Two patients had lung metastases. 5 patients underwent a kidney biopsy to confirm diagnosis. The median duration of preoperative CT was 64.6 days (5-240 days). One patient had congenital dilated cardiomyopathy and died after one cycle of vincristine-actinomycin. 6 patients received the VAD regimen due to local advanced disease (4) and lung metastases (2). Chemotherapy had to be modified in five patients (30%) after 4 weeks of preoperative treatment. 3 patients progressed after neoadjuvant CT and died before being operated. 6 patients had a radical nephrectomy on the kidney with the larger tumor and NSS on the contralateral kidney, and 3 patients (33%) had bilateral NSS. Unfavourable histological type was revealed in 22%. One patient had a local relapse 5 weeks after surgery and another had local and metastatic (lung) relapse after 11 weeks. Currently, 7 children were alive without evidence of recurrence and disease progression. The overall survival was 6.8 years.

Conclusions: The management of bilateral nephroblastoma is binding and the therapeutic decision must be made after multidisciplinary consultation meeting.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0067 / #528 | Laparoscopic Nephron-Sparing Surgery for Unilateral Wilms Tumors: Initial Experience

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Background and Aims: Recent therapeutic focus for Wilms tumor has shifted to reduction in treatment options, especially chemo- and radiotherapy. Partial nephrectomy incorporated in multimodal treat-

ment for Wilms tumor patients with bilateral disease or predisposing syndromes. But laparoscopic nephron sparing surgery (LNSS), requiring careful selection of patients with Wilms' non-syndrome unilateral tumor remains more conservative.

Methods: Case 1. A 4-year-old boy was treated Wilms' tumor according to the SIOP-RTSG 2016 UMBRELLA protocol. The tumor volume was 12.2 cm³, with decreased by 82% after chemotherapy. LNSS was performed. Intraoperatively, laparoscopic ultrasound of the kidney was performed to determine the resection margins. Case 2. A 1-year-old girl was admitted for evaluation of Wilms' tumor. Computed tomography demonstrated 12.2 cm³ tumor mass which occupied the upper pole of the right kidney, and decreased by 93% after chemotherapy. LNSS was performed. During the operation with a thorough examination an intraparenchymal mass up to 1 cm in size was visualized. Resection was performed using cold scissors an indent with from surrounding renal tissue 3 mm indent without intraoperative ultrasound. No perioperative complications occurred in both cases.

Results: Случай 1. Гистологическая резекция R0 была завершена. Пациент оставался без опухоли после 12 месяцев наблюдения. Случай 2. Были обнаружены положительные края резекции R1. Была проведена операция второго вида с целью удаления почки. После 7-месячного наблюдения пациентка чувствовала себя хорошо, рецидивов не было.

Conclusions: The ideal surgical candidate for LNSS was characterized by (1) unifocal mass with a volume of less than 300 cm³; (2) favorable histology; (3) absent renal sinus or segmental vascular tumor invasion; (4) absence of lymph node metastasis, intraoperative tumor spill, or gross regional disease at surgery; and (5) a distinct interface between tumor and renal parenchyma or additional intraoperative imaging modalities like ultrasound.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours

PO0068 / #534 | Single Center Study on Bilateral Wilms' Tumor

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Background and Aims: Background: Management of bilateral Wilms' tumor is particularly challenging in decision making, considering nephron sparing surgery, the chances of recurrence and long-term renal function.

Aims: This study aimed to assess the outcome of patients with Bilateral Wilms tumor (BWT) treated according to SIOP protocol.

Methods: This was a retrospective and prospective randomized study. SIOP protocol was used to study patients of BWT in a single center and followed up over a period of 2 years. We had five children presenting with BWT from 2009 to 2020. All of them were appropriately staged and given neoadjuvant chemotherapy. Tumors were considered to have a 'good' response to chemotherapy if sufficient tumor shrinkage was observed. Nephroureterectomy/ Partial nephrectomy/Nephron-sparing surgery was considered in all and was performed when feasible, followed by completion adjuvant chemotherapy. All patients were followed up with serial ultrasound scans (3–6 monthly) and contrast enhanced computed tomography of abdomen (yearly). Blood urea and serum creatinine, hypertension, and proteinuria were assessed during follow-up visits.

Results: Of five patients, three were included in the study as two patients lost due to refusal and abandonment of treatment. Among the three patients, there were one girl and two boys with a median age of 1.5 years and mean regression in the size of tumor was 70%. The first child had poor response to chemotherapy and was considered for right radical and left partial nephrectomy. The second child had a 'good' response on left side and was planned for left partial and right radical nephroureterectomy based on pre-operative imaging analysis. The third child though, showed 'poor' response to pre-op chemotherapy and right partial and left radical nephroureterectomy was feasible. Recurrence was seen in patient with Wilms' tumor with teratoid differentiation.

Conclusions: Nephron-sparing surgery should be considered in all patients having BWT with favorable histology.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours*

PO0069 / #780 | Clear Cell Sarcoma of the Kidney in Children: Clinical Characteristics and Results of Treatment

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Background and Aims: Clear cell sarcoma of the kidney (CCSK) is a rare malignant renal tumor in children. The aim of the study was to analyze the results of therapy of patients with CCSK treated in Dmitry Rogachev NMRC PHOI.

Methods: Retrospective analysis of patients with a histologically confirmed diagnosis of CCSK treated for the period 01.2012-02.2020 (98 months) was done. Demographic characteristics, clinical symptoms, methods of diagnosis, and treatment modalities were analyzed. Patients were treated according to the protocols of the SIOP-RTSG. Overall and event-free survival was assessed by the Kaplan-Mayer method. The analysis of the results was carried out on 01.03.2021.

Results: The analysis included 10 patients with CCSK. The median age at diagnosis was 30.1 months (range 13.5-70.8). All patients were male. The diagnosis was established based on radiological data (n=9) and biopsy (n=1). Distant metastases were detected in 1 (10%) (lungs). The median tumor volume was 439 cm³ (range 256-996 cm³). Preoperative chemotherapy was performed in all patients (AV regimen in 7 (70%) patients). Tumor regression was noted only in 3 (in 1/7 after AV regimen), tumor progression in 5 and stable disease in 2 patients. Delayed nephrectomy was performed in all patients (10/10). In 1 (10%) case, intraoperative tumor rupture was documented. Distribution of patients by local stages: I - 4 (40%), II - 2 (20%), III - 4 (40%) (including 1 patient with distant metastases). Adjuvant chemotherapy included 4-drug regimen (n=7), AVD regimen (n=3). Radiation therapy was performed in 6 (60%) patients. The median follow-up was 35.7 months (range 12.0-70.7). Outcomes: 9 (90%) - alive, 1 (10%) patient died (non-tumor-related death). The progression of the disease was observed in 1 patient. 3-year EFS and OS were 78.8% (95% CI 52.5-100) and 90.0% (95% CI 71.4-100) respectively.

Conclusions: Intensive program therapy in patients with CCSK allows to achieve satisfactory results of treatment.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.b Renal Tumours*

PO0070 / #868 | Willm's Tumor: Experience of Hematology and Pediatric Oncology Department, IBN Rochd University Hospital Center

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Background and Aims: Wilms tumor is the most frequently occurring renal tumor in children and is one of the most treatment-responsive tumors. It is also the most common pediatric abdominal cancer, and the fourth most common pediatric cancer overall. The aim of this study is to determine the epidemiological and clinical profile of patients with willms tumors and to analyze the factors that significantly influence the survival of patients diagnosed with nephroblastoma.

Methods: We conducted a retrospective study of 64 children with over a 5-year period (2015-2019) at the Department of hematology and pediatric oncology hospital August 20, Casablanca. Tumors were divided according to histological type and histological risk group according to the SIOP 2001 classification. The patients were treated according to the GFA nephro 2005 protocol.

Results: The sex ratio M/F was 0.61 with a female predominance. Ninety two percent of Wilms tumors are diagnosed before six years of age with the median age of diagnosis being 3 years. The average time to diagnosis was thirty seven days and the abdominal mass was the most common presenting feature (72%). The disease was bilateral in 10% of cases and metastases were present in 45% of cases. Regressive histological type was the predominant type (35%). Tumors were divided into intermediate (69%) and high (22%) histological risk groups. Tumors were classified into stage I (36%), II (35%), III (27%), IV (2%), and V (12%). The treatment is based on a first chemotherapy associated with surgery, postoperative chemotherapy depending on the stage and radiotherapy in stages III and IV.

Conclusions: Wilm's tumor can be considered a model for successful multidisciplinary management of cancer, with improvement in survival from a mere 30% to more than 85% at present. However, a lot of efforts needs to be done in order to optimize the management of the nephroblastoma in our country.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours*

PO0071 / #1009 | Particularities of Pediatric Osteosarcoma in a Tunisian Case Series

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[Correction added on 3 December 2021: The co-author Sonda Masmoudi is added to the abstract “#1009 PARTICULARITIES OF PEDIATRIC OSTEOSARCOMA IN A TUNISIAN CASE SERIES”]

Background and Aims: Osteosarcoma is the most common childhood primary bone cancer. The purpose of this study was to report the epidemiological and clinicopathological particularities and treatment outcomes of tunisian case series of pediatric osteosarcoma with comparison to adults.

Methods: We retrospectively collected data about all patients treated for osteosarcoma at Medical Oncology department of Habib Bourguiba Hospital in Sfax between 2002 and 2017. They were then separated into two groups (≤ 18 years (n=33) and > 18 years (n=24)). Characteristics were then compared according to the age.

Results: The median age at diagnosis in children was 14 years old. Most children (54.5%) presented with pain and swelling in the primary tumor site. The tumor was localised at diagnosis in 78.8% of cases. Osteoblastic and chondroblastic osteosarcomas were the most common histological subtypes in children representing 39.4% and 21.2% of cases respectively. Sex, clinical presentation, metastatic status or histological subtype did not differ by age. Children had lower proportion of axial localisation of the primary tumor (3% vs 37.5%, $p=0.001$). All patients received neoadjuvant chemotherapy. Nineteen (57.6%) children showed a partial radiological response and 9 cases (27.3%) showed stability. Surgery of the primary site was performed in 30 cases of them with poor histological response to chemotherapy in 56.6% (n=17). No significant differences were noted in type of surgery (conservative vs radical) or histological response (poor vs good) in the two groups. In children, better 5-year overall survival was noted in localised than metastatic tumors (57.5% vs 28.6%, $p=0.012$) and in localised tumors with good histological response (94% vs 12.2%, $p=0.034$), but with no significant differences when compared to adults with same characteristics.

Conclusions: Characteristics of pediatric osteosarcoma were not significantly different when compared to adults except the primary tumor site. The prognosis remains poor regardless of age. Other factors seem to be more predictive of survival such as metastatic status and histological response.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours*

PO0072 / #1045 | Deaths of Children and Adolescents by Cancer Bone in Brazil, 1999-2018

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Background and Aims: Pediatric cancer represents 0.5-3% of all tumors in most populations. Within this group, bone tumors often occur in older children and adolescents. They represent about 3% of children's cancers. This project aim is to characterize the profile of deaths by pediatric bone tumors in Brazil between 1999 and 2018.

Methods: Epidemiological study with secondary data from the mortality information system - SIM / National Cancer Institute - INCA. The variables included were children's age group, sex and year.

Results: A total of 4,354 deaths were tabulated, of which 1,912 (43.91%) were women and 2,442 (56.08%) were men. Individuals from the age group 15-19 years old ($n = 2,395$; 55%) demonstrated higher death rates compared to groups 0-4 years old ($n = 191$; 4.38%), 5-9 years old ($n = 429$; 9.85%), 10-14 years old ($n = 1,339$; 30.75%). The highest number of men's deaths happened in the 15-19 years old group ($n = 1,514$; 62%), meanwhile, the highest number of women's deaths were registered in the 10-14 years old group ($n = 701$; 36.66%) and the 15-19 years old group ($n = 881$; 46.07%).

Conclusions: Therefore, from a clinical point of view, pediatric tumors have shorter periods of latency, in general they grow rapidly and are more invasive. However, the rarity and difficulty of the diagnosis make the treatment difficult. The data presented collaborated for a possible association between the final period of puberty, in both sexes, due to hormonal and anthropometric changes, and the intensification of the disease, causing death.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0073 / #1316 | Osteosarcoma Treated at the Red Cross War Memorial Children's Hospital (RCWMCH): 1995-2017

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Background and Aims: To review current practice and determine outcomes of patients with osteosarcoma.

Methods: We performed a retrospective analysis of data from records of 44 patients, ≤ 13 years with biopsy-proven osteosarcoma diagnosed at RCWMCH between January 1995 and December 2017.

Results: There were 44 patients (23 males and 21 females). Mean age was 9.9 years (range 4.8-13.8 years). The most common histological sub-type was conventional osteosarcoma ($n=37$, 84%). The most common primary site was the long bones around the knee ($n=36$, 82%). Thirty-two (73%) had localised disease and 12 (27%) had metastatic disease. Metastatic sites included pleuropulmonary ($n=8$, 18%), skeletal ($n=1$, 2%) and combined ($n=3$, 7%).

Patients with localised disease received a 2-drug neoadjuvant regimen (cisplatin/doxorubicin). Six with metastatic disease received an additional 3 drugs (ifosfamide/etoposide/carboplatin), 2 refused treatment and 3 died pre-chemotherapy. Twenty-four (55%) had limb-salvage, 13 (30%) had amputation and 7 (16%) did not undergo surgery.

Available histological results revealed 12 (39%) good responders (necrosis $\geq 95\%$) and 19 (61%) poor responders (necrosis $< 95\%$). Good responders continued the same regimens. Nine (47%) of the poor responders received high-dose methotrexate (HDMTX).

Twenty-five (57%) were alive and disease free, 18 (41%) died of disease and 1 (2%) died of treatment-related toxicity. The Kaplan-Meier

overall survival (OS) was 51%. The OS for localised disease was 66% and metastatic disease 17% ($p=0.0006$). Gender, age, tumour location, histologic subtype and necrosis were not significant predictors of survival. Considering localised disease, the patients who received HDMTX had a better survival (69%) than those who did not (53%) ($p=0.88$). Outcomes for limb-salvage surgery (81%) were superior to amputations (39%) ($p=0.00004$).

Conclusions: Outcomes for osteosarcoma at RCWMCH were reasonable for localised disease. Metastatic disease remained a key predictor of poor outcome. Toxic-related death was low, presenting scope to intensify treatment.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0074 / #1428 | Rare Localization of Ewing's Sarcoma Family Tumor a Single Center Experiments

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Background and Aims: In pediatric oncology, one of the mysterious neoplasms is an Ewing's Sarcoma (ES) family tumor. ES takes second place after osteosarcoma 10-25%, while in the structure of soft tissues it takes fifth place 0.09%. Ossal Ewing's sarcoma, extraosseous Ewing's sarcoma, Askin's tumor, a primitive neuroectodermal tumor are combined into one family due to their origin. The aim: to study rare localizations of tumors of the family of Ewing's sarcoma.

Methods: Pediatric oncological service exist in the Republic of Kazakhstan since 2013, in the Research Center for Children and Pediatric Surgery according to the protocol of EuroEwing2008 and CWS-2009 we analyzed 62 patients

Results: Twelve patients had rare localizations. The ratio of boys to girls was 7: 5. Localization: spine - 5, the cervical region - 1, thoracic - 2 and lumbar vertebrae - 2. Facial skeleton: zygomatic bone, upper jaw, lower jaw, orbit - a total of 4 patients. Bones of the skull: temporal bone -1, the clavicle -1, the second phalanx of the foot -1. By age category: under 5 years old - 2 (16.6%), under 12 years old - 5 (41.6%), under 18 years old - 5 (41.6%). The median age was 10 years. Distribution by stages: Stage I - 0 (0%), Stage II - 4 (33.3%), Stage III - 1 (8.3%), Stage IV - 7 (58.3%). Treatment outcomes: on therapy with a complete response - 2 (16.6%), complete remission achieved -4 (33.3%), lethal outcome - 6 (50%).

Conclusions: The implementation of the international protocols contributed to the accurate diagnosis, determination of the disease risk group and stratification of treatment. According to the results of the study in 50% of cases patients are admitted with stage IV of the disease, which is associated with an unfavorable outcome and contribute heightened awareness.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0075 / #1499 | Osteosarcoma in Children: The Rabat Children's Hospital Experience

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Background and Aims: Twenty years ago the prognosis of children with osteosarcoma was poor. The introduction of multimodal, risk adapted chemotherapy combined with limb-saving therapy is the treatment of choice in children with osteosarcoma and offered a chance to cure approximately 70% of patients. The aim of our study is to study the epidemiological, clinical, paraclinical, therapeutic and evolution of osteosarcoma and to identify the difficulties related to its management and the feasibility of the Os 2005 protocol.

Methods: This work is a retrospective study of 51 pediatric osteosarcoma files collected at the pediatric hematology and oncology center of Rabat

Results: Between 2011 and 2018, 51 children were treated for osteosarcoma at our institution. Their median age was 12.15 years (range, 6–15 years), with a sex ratio of 0.96. The mean time to diagnosis was 11.5 weeks. The lesion was located in 75% of cases in the region of the knee. 20% of patients were metastatic at the time of diagnosis. 94% of patients received preoperative chemotherapy. 51% of the cases were operated, 35% were conservative surgery and 65% were amputated, while adjuvant chemotherapy was received in only 49% of patients. The high rate of amputations can be explained by late diagnosis at a locally advanced or even metastatic stage, problems with adherence to the chemotherapy protocol related to the rupture of certain drugs leading to a prolongation of the inter-cure interval or even a change of protocol (os 2005 protocol was only respected in 69% of cases) as well as the failure to respect surgical deadlines exposing to tumor evolution and the administration of additional cures favoring the risk of chemotoxicity.

Conclusions: The prognosis of osteosarcoma has improved considerably with the introduction of multidrug therapy. However, results remain modest in developing countries; the overall survival rate in our study was 18% at 3 years.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0076 / #1677 | High-Dose Methotrexate Potential Side Effects during Osteosarcoma Treatment in the Limited Setting: A Single-Center Experience

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Background and Aims: High dose-methotrexate (HDMTX) is the mainstay of osteosarcoma treatment. Methotrexate serum level is not always available in the limited setting. Evaluation of the HDMTX potential side effects based on clinical manifestations is needed to improve our supportive measures.

Methods: We conducted a retrospective case series study of pediatric osteosarcoma in Dr. Sardjito General Hospital, Indonesia (April 2018-March 2021). Patients were grouped based on the high-dose methotrexate therapeutic dose: no methotrexate, 4 g/m² methotrexate, 8 g/m² methotrexate, and switched dose methotrexate (patients were previously received 4 g/m² then changed to 8 g/m²). We evaluate baseline characteristics and chemotherapy side effects based on the Common Terminology Criteria for Adverse Events version 5.0 2017 in each group.

Results: Twenty patients were enrolled in this study (50% males and 50% females). There were five patients in each no-methotrexate and 4 g/m² groups, seven patients in the 8 g/m² group, and three patients in the switched group. Seven patients were during chemotherapy, two did not finish chemotherapy, two had finished chemotherapy, and seven died. There were 12(60%), 5(25%), and 4(20%) patients classified as Enneking stage III, I/II, and IIB, respectively. In all treatment groups, the most common side effects were increased liver transaminases (100%), followed by neutropenia (65%), mucositis (65%), and increased serum creatinine (55%). Liver toxicity, kidney injury, and neutropenia were mainly found in the 8 g/m² group (12%) and 4 g/m² group (10%), respectively. Mucositis was found primarily on the 8 g/m² group (83%), but none was in the no-methotrexate group. The switched group appeared to have more side effects after the switch (7 cases) than before (5 cases).

Conclusions: The prevalence of liver and kidney injuries, neutropenia, and mucositis during the high-dose methotrexate treatment needs serious concern of supportive measures.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0077 / #200 | Ewing Sarcoma in Upper Egypt: Pediatric Versus Adult Experience from a Tertiary Care Cancer Center

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Background and Aims: There is a paucity of published research, particularly from developing countries that compared treatment outcomes in patients with Ewing sarcoma (EWS) treated in two different departments in the same institution. The aim is to assess the outcomes

of EWS in Upper Egypt, as an institutional experience in two different departments, with particular reference to factors of prognostic significance in patients treated in the pediatric oncology department compared with those patients treated in the medical oncology department.

Methods: We retrospectively reviewed medical records of 85 patients, including 59 patients treated in the pediatric oncology department, with a verified diagnosis of EWS at South Egypt Cancer Institute in the period from 2001 to 2015.

Results: With a median age of 11 years (Range: 1-18 y) for those patients treated in the pediatric oncology department with either VACD or VACD/IE protocols vs. 22 years (Range: 16-35 y) for those treated in the medical oncology department with VAIA protocol; the estimates of overall survival (OS) at 5 years were 54.5% and 31.7%; $p = 0.26$ for pediatric and adult patients, respectively. Using multi-variable Cox regression analysis, the effect of initially metastatic disease ($HR = 21.3$, 95%CI, 5.06-90.0; $P < 0.0001$) was more pronounced than surgery used as a local modality ($HR = 0.26$, 95%CI, 0.07-0.99; $P = 0.049$) in pediatric patients group, while the former effect was less pronounced ($HR = 4.61$, 95%CI, 1.11-19.2; $P = 0.036$) comparable to the effect of surgery ($HR = 0.048$, 95%CI, 0.004 -0.60; $P = 0.019$) in medical oncology patients. The anatomical tumor location, whether axial or appendicular, revealed to be marginally correlated to the OS ($HR = 3.71$, 95%CI, 0.87-15.7; $P = 0.076$) only in patients treated in the pediatric department.

Conclusions: Survival outcomes for patients treated in Upper Egypt at the department of Medical Oncology were inferior to that of their counterparts treated in the department of Pediatric Oncology, but with no statistically significant difference.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0078 / #359 | Osteosarcoma of Proximal Humerus with Limb-Salvage Surgery and Pulmonary Metastasectomy

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Background and Aims: Osteosarcoma is an aggressive malignant neoplasm of bone. Surgery with neo-adjuvant chemotherapy remains

the mainstay of treatment for metastatic osteosarcoma of the lungs. Metastatic disease to the lungs is often bilateral, and controversy exists as to the approach to these lesions (staged versus bilateral thoracotomies). Aim of the study to report a clinical case of osteosarcoma of proximal humerus with limb-salvage surgery and pulmonary metastasectomy of one lung.

Methods: A 16-year-old boy was admitted with a left shoulder pain. CT scan demonstrated an 8.4x8.7x6.8cm mass in the left humerus and multiple bilateral pulmonary metastasis ranging from 0.3mm to 0.7mm. Incisional biopsy was performed, revealing osteoblastic osteosarcoma. The treatment started according to OS-99 protocol. After 3rd and 7th cycles of chemotherapy interval imaging revealed partial response with reduction in size and number of pulmonary lesions. Nine months' post diagnosis limb-salvage surgery was performed with radical removal of the proximal humerus and autologous fibular grafting. The histology review of removed tissue detected fibroblastic osteosarcoma with chemotherapy regression Grade VI and clear margins. Following multidisciplinary tumor board discussions, the decision was made to continue treatment according to COSS regimen (carboplatin, etoposide). Three months' post resection of primary tumor, left thoracotomy with metastectomy was performed, with 9 lesions resected. Final pathology revealed no atypical cells within these lesions. Based on these findings, the decision was made to not proceed with contralateral metastectomy.

Results: After 12 months of follow-up of the patient, diagnostic workup revealed no sign of cancer recurrence.

Conclusions: Although limited by short follow up time, this case report demonstrates the rationale behind foregoing a contralateral thoracotomy when the initial metastectomy reveals no viable tumor cells during a staged approach for bilateral pulmonary metastases in the setting of osteosarcoma. Further studies are needed to confirm that this approach does not affect survival rates or other outcome parameters.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0079 / #568 | Clinicopathologic Characteristics and Treatment Outcomes of Children with Ewing Sarcoma: Experience in a Resource-Limited Setting

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Background and Aims: Ewing Sarcoma Family Tumors (ESFT) represent the second most frequent primary bone malignancy affecting children and adolescents. Owing to modern multidisciplinary treatment in the last three decades huge improvement in survival rates for patients with localized disease was observed and has reached about 70%. Despite improved outcomes in high-income countries, survival rates remain poor in the developing world. There are various diagnostic and therapeutic limitations contributing to survival disparity. This short case series study aimed to evaluate the outcomes of patients diagnosed with ESFT in Armenia and identify diagnostic and therapeutic limitations in the country.

Methods: A retrospective review of the medical records of 14 children (≤ 18 years old) with ESFT was conducted, who were diagnosed and treated at the Muratsan Hospital Complex of Yerevan State Medical University (2008 - 2019) and Pediatric Cancer and Blood Disorders Center of Armenia (2019 - 2021). Epidemiological, social, medical information was collected through the patient charts review.

Results: The total number of patients diagnosed with ESFT in our database during the mentioned period of time was 14. The median age was 12 years. Eight patients (67%) had metastatic disease at initial presentation. All patients received treatment according to VDC/IE chemotherapy regimen. The majority of patients (71.4%) had delays during the treatment due to chemotherapy complications. Four patients received part of diagnosis and/or treatment abroad; the most frequent reason for going abroad was PET-CT, otherwise none available in Armenia before 2020. Out of 14 patients 2 received radiotherapy for definitive local control, 4 patients underwent surgery and 8 patients received both. The majority of patients (71.4%) achieved complete remission. Four patients relapsed after initial treatment and received salvage therapy. Two patients had a fatal outcome.

Conclusions: Majority of patients with ESFT present at the clinic with advanced-stage disease, which dramatically affects the outcomes. Activities to increase the rate of early diagnosis, both among the public and healthcare professionals, are emerging.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0080 / #608 | Stem Cell Transplantation in Ewing Sarcoma: Results at Hospital Infantil De México

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Background and Aims: Ewing sarcoma (ES) is a highly aggressive cancer, with a survival of 70-80% for patients with standard-risk and localized disease and ~30% for those with metastatic disease. Objective: To analyze the global survival of children with ES treated with stem cell transplantation (SCT).

Methods: From January 2013 to June 2018, eight newly diagnosed patients with ES with high risk were enrolled in this study. Medical records were analyzed retrospectively. All patients were treated at Hospital Infantil de México Federico Gómez. Survival was determined using the Kaplan-Meier method.

Results: We included 4 females and 4 males, the median age at diagnosis was 11 years (range 2-16 years). Localized disease was diagnosed in 37.5% of the patients and 62.5% had distant metastases at time of primary diagnosis. Seven had ES osseo, 5 in axial bones. All patients were treated according to modified T9 protocol. The regimen used for consolidation with peripheral blood precursor cell rescue was Busulfan-melfalan in 7. The median of CD34+ cell counts was 2.9×10^6 (range 1.47-4.3). The median time to graft myeloid was 18.7 days (11-44). The most frequent complication were infections. The median follow up time was 42 months, with Overall survival in all patients was 62.5% and 50% at 2 and 5 years respectively.

Conclusions: In eligible cases the combination of chemotherapy with surgery/radiotherapy and consolidation with SCT with PBPC results benefit for children with ES

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0081 / #926 | Clinical outcome of Osteosarcoma in Children: A Single Center Experience

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Background and Aims: Osteosarcoma is the most common type of bone cancer in children. The treatment is multimodal. We aim to review our experience about clinical and survival of pediatric osteosarcoma in the center of Tunisia.

Methods: A retrospective study was conducted on 32 patients younger than 18 years with high grade osteosarcoma treated with Rosen T10, OS 94 or OS 2005 protocol at the medical Oncology Department of Farhat Hached Hospital from January 2003 to December 2020. The analysis of the data was carried out according to the Spss 20.0 software and survival curve according to the method of Kaplan Meyer.

Results: The study included 32 patients (68.8% males and 31.1% females). Median age was 13 years old [6-16]. One patient had a medical family history of osteosarcoma (3.1%). Osteosarcoma mainly affected the femur (40.6%), tibia (31.3%) and humerus 18.8%. Median size was 29mm. Osteoblastic osteosarcoma was the most frequent histological subtype (59.4%). Soft tissue invasion was found in 93.5% of cases. Metastatic disease was found in 9.7% of cases (3 patients). Lung metastasis represented 66.7% of cases when bone metastasis

represented 33.3% of cases. All patients received neoadjuvant chemotherapy. One patient had a complete response after chemotherapy (3.6%) when 8 patients had a partial response (28.6%), 8 patients had a stable disease (28.6%) and 11 patients were in progression (39.3%). Only 20% of patients had a good histological response. Patients had amputation in 62.5% of cases. The median overall survival was 26 months. The 3-year survival rate was 82% and the 5-year survival rate was 61%. Progression free survival rate was 62% at 3 years and 48% at 5 years.

Conclusions: Our experience indicates that There is a clear need for newer effective drugs for patients with osteosarcoma, especially for patients who present with metastatic disease or develop disease recurrence.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.c Bone Tumours

PO0082 / #980 | Clinical, Epidemiological Features and Prognostic Factors in Relation to Survival in Pediatric Patients with Diagnosis of Osteosarcoma at Guillermo Almenara Hospital – Peru

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Background and Aims: The aim of this study was to determine the association between clinical, epidemiological and other well-known prognostic factors for OS and outcome in children and adolescents with diagnosis of high-grade OS attended at Guillermo Almenara Hospital in Lima, Peru.

Methods: We retrospectively evaluated 22 patients under 18 years of age with diagnosis of high grade OS between January 2010 and December 2017. Demographic data (age, gender, place of residence), clinical characteristics, treatment features, overall (OAS) and event-free survival (EFS) data were analyzed.

Results: With a median time of follow-up of 24 months (range, 1.5-66), 5-year OAS and EFS for patients with localized disease were 54.2 ± 20 and $27 \pm 17\%$ and for patients with metastatic disease were 38.1 ± 23 and $22 \pm 14\%$, respectively. Median age was 13 years (range, 6-17 years), 14 of 22 patients (63.6%) were male. Six cases (27%) had metastatic disease at onset. Median time to diagnosis was 22.2 weeks (range, 4.5-108). Conservative surgery was performed in 12 of 20 patients (60%) and difference in OAS or EFS was not observed according to type of surgery. Place of residence (Andean vs Coast region), large tumor size (> 12 cm), primary axial tumor location and presence of pathological fracture were significantly associated with worse OAS and EFS rates ($p < 0.001$). Poor tumor necrosis ($< 90\%$) after

neoadjuvant chemotherapy was associated to inferior OAS ($p = 0.04$). Initial presence of metastasis, raised serum AP and LDH, histological subtype and surgical margins did not significantly affect prognosis, which may be related to sample size.

Conclusions: In our study, tumor size, necrosis response, axial location and pathological fracture predicted poor prognosis in pediatric patients with high grade OS. Patients living in Andean region of the country had inferior survival rates. Larger national multicenter studies are needed to clarify additional factors for improving survival.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0083 / #1021 | A Retrospective Analysis of 13 Pediatric Patients with Inflammatory Myofibroblastoma: A Single Center in South China

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Background and Aims: Inflammatory myofibroblastoma (IMT) is a rare soft tissue sarcoma. We aimed to explore clinicopathological characteristics, treatment and outcome of pediatric patients with IMT under the age of 18 years old at department of pediatric oncology in Sun Yat-sen University Cancer Center.

Methods: A retrospective analysis of clinical data was conducted with regards to symptoms, signs, diagnosis, treatment and outcome towards pediatric patients with confirmed diagnosis of IMT between August 2012 and June 2020. Descriptive method was adopted to analyze clinical data.

Results: Of 13 eligible patients, there were 9 boys and 4 girls with a median age of 7.2 years. Primary sites were abdomen and pelvis cavity in 6 patients, thigh in 2 patients, head in 3 patients, spleen in 1 patient, and lung in 1 patient, respectively. Resection was undergone in 12 patients. With a median follow-up of 5 years, 11 patients were alive and 2 patients died. Recurrence didn't occur in 9 patients after surgery. One developed recurrence at primary site 7 months after resection. One administered a combination of surgery and chemotherapy died after multiple recurrence. One relapsed at primary site one month after resection and died due to metastasis although crizotinib was administered. One patient was alive under the treatment of crizotinib. Anaplastic lymphoma kinase stained positive at 6 patients with being D5F3 positive in 3 patients.

Conclusions: IMT is a rare intermediate malignant neoplasm. Radical surgery is the mainstay treatment. ALK inhibitor is an effective treatment for patients with unresectable tumor and exerts its efficacy rapidly.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0084 / #1022 | Clinical Characteristics and Survival of Pediatric Patients with Rhabdomyosarcoma in a Hospital in Mexico, Experience of 15 Years

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Background and Aims: El rhabdomyosarcoma representa el 3,5% en niños de 0 a 14 años con cáncer, su tasa de supervivencia ha aumentado en base a los tratamientos actuales. El objetivo es describir las características y supervivencia de los pacientes con rhabdomyosarcoma en los últimos 15 años.

Methods: Retrospective observational study in children under 18 years of age with diagnosis of rhabdomyosarcoma at the Hospital Civil de Guadalajara, from 2005 to 2020.

Results: A total of 56 cases were reviewed, annual average was 3.7 cases, predominantly male in a 60%, medium age at diagnosis was 5 years. Genitourinary and trunk/retroperitoneum location were the most frequent found in 13 patients (23%) and 13 patients (23%) respectively, followed by head and neck location in 12 patients (21%), parameningeal in 9 patients (16%), orbit in 5 patients (9%), extremities in 3 patients (5%). At admission, Clinical Stage III constituted the majority of the cases with 22 patients (39%), followed by Stage IV seen in 16 patients (28%), Stage I with 13 patients (23%) and lastly Stage II with 5 patients (9%). Metastases to lung appeared in 9 patients (16%) and 8 patients had metastases to CNS (14%). Regarding histology, 48 patients (86%) had embryonic histology and 7 alveolar (12%). 28 patients (50%) had neoadjuvant therapy with the IRSG IV Protocol, radiotherapy was administered to 38 patients (67%). When conducting this investigation, 36 patients (64%) were alive without activity and 12 died (21%) due to tumor activity

Conclusions: Comparing in 2007, 12 patients (27%) were found alive without activity, 21 (27%) died due to activity. In 2011, 17 patients (30%) were alive and 25 (45%) died without activity. In our series, the patients presented an adequate response based on the IRSG IV protocol, with more than half of the patients (64%) being alive at the time of the study.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0085 / #1165 | Myoepithelial Lung Carcinoma in Child: A Case Report

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Background and Aims: Myoepithelial lung carcinoma is an extremely rare histological subgroup of primary salivary gland-type lung cancer. We herein report the case of a 13-year-old girl who presented myoepithelial lung carcinoma.

Methods: A 13-years-old girl with dyspnea, weakness, and hemoptysis was transferred to our clinic for further treatment. Examination revealed a central myoepithelial tumor of the right lung similar salivary gland epithelial neoplasms. Initially surgery was not possible due to "bulky" (128.8 cm³), muff-like transition to the right main bronchus, right pulmonary artery, compression of the vena cava superior and vena azygos, infiltration of the mediastinum, interlobular pleura, pericardium. Patient was treated according to the pediatrics sarcoma regimen: vincristine, dactinomycin, doxorubicin, ifosfamide. After 4 courses tumor has been reduced by 86%. The next stage the unique sparing surgical technique has been performed: right thoracotomy, pneumonectomy, marginal resection of the vena cava superior, wedge resection of the tracheal bifurcation.

Results: Histopathological examination confirmed radical resection. Chemotherapy is able to reduce tumor mass thus achieving operability in non-curatively operable cases of myoepithelial lung carcinoma. Patient achieve a complete remission after combined modality therapy (chemotherapy and surgery).

Conclusions: The described case is the only one experience of successful treatment for child with epithelial-myoepithelial lung carcinoma in the Russian Federation. The fact of an individual, interdisciplinary approach to the treatment of children with such a rare pathology is undeniable, but, unfortunately, today the experience of treating children with malignant myoepithelioma of the lung is limited. Standardized studies with large patient populations and long follow-up periods will shed light on the possibility of additional therapeutic options for children with myoepithelial tumors.

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PO0086 / #1190 | Malignant Skin Tumors Among Children and Teenagers in Brazil Underlying Conditions and its Impacts on the Health System

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Background and Aims: Malignant Skin Tumors (MST) are rare among children in the world. Despite that, because of the latitude and the sun exposure in Brazil, there are still a lot of cases among children in the country. They are aggressive tumors and can indicate an underlying genetic condition or another complicated medical process. The aim is to analyze the epidemiological profile and behavior of MST hospitalizations in Brazil to point out more targeted and efficient public policies.

Methods: Data were collected using the Hospital Information System (SIH). The population studied were children and teenagers from zero to 14 years old diagnosed with MST from January of 2008 to February of 2021 in Brazil. The variables analyzed were: number of hospitalizations, skin color, health care expenses, age group, mortality and hospitalization time.

Results: In the analyzed period, 3,537 hospitalizations due to MST were reported among the study's population, which were responsible for \$511,703.32 in health care expenses. The total hospitalization time for these cases among the years was 12,867 days, with an average of 5.1 days and an average mortality rate of 1.34. The highest number of hospitalizations (69.57%) were registered among brown and white children, group also responsible for 72.02% of the total hospitalization time.

Conclusions: Basal Cell Carcinoma (BCC) and malignant melanoma are the most common skin tumors among children related to sun-exposed areas. Besides that, there is a lack of information about MST among children and teenagers, and late diagnosis could be related to the different presentations in children from the same neoplasms in adults. That is problematic because children with MST probably have an underlying genetic or medical condition that predisposes the development of neoplasms, which should be identified early and need identification and intervention. This lack of specific knowledge causes long hospitalizations and preventable problems.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0087 / #1191 | NF (Neurofibromatosis) Associated Sarcomas in Pediatric and AYA Population from a Tertiary Care Centre- An Area with Unmet Need

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Background and Aims: NF associated sarcomas are rare tumors and not much is known about their biology and are considered to have poor prognosis. So far data from India is not there.

Methods: It was a retrospective study of a prospectively data base from January 2016 to 2021 from a dedicated Sarcoma medical/pediatric oncology clinic in tertiary centre in India. We conducted review of CAYA population ≤ 35 years with clinical diagnosis of NF and their outcomes were recorded.

Results: There were 26 patients, diagnosed with NF associated soft tissue tumors in during this interval. Family history of NF was there in 10/26 patients (38%). None was aware of NF1 before coming to our hospital. Of them 22(85%) patients were in CAYA population and this was our core group. Of 22 patients, median age was 24 years (10-35 years) and males (59%) were more than females. Of these, most common was MPNST (malignant peripheral nerve sheath tumors) 17(77%), plexiform neurofibromas 4(18%) and NF associated GIST (gastrointestinal stromal tumor) 1(5%). Of MPNST (n=17), Majority of them were metastatic (n=8,47%) and locally advanced (n=3,18%). In metastatic patients, most common site of metastasis was lymph nodes (n=4, 50%) followed by lung and pleura(n=3, 37%). Six patients received doxorubicin based chemotherapy in advanced setting and best response was Stable disease (SD) observed in 1 patient (16%). Pazopanib was used in 4 patients with best response SD in 1 patient (25%). NF with GIST patient didn't respond to Imatinib and sunitinib. Median overall survival in overall cohort was 13 months. The most important factor for outcome was presence of metastatic disease with median OS 7 months (p=0.001).

Conclusions: NF associated sarcomas are rare tumors more in CAYA population. Even with family history the awareness was nonexistent. In our setting, presentation at advanced stage was common and outcomes with metastatic disease was poor.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0088 / #1278 | Clinical Characteristics and Treatment Outcome of Undifferentiated/Spindle Cell Soft Tissue Sarcoma in Children -Experience from a Tertiary Care Cancer Center

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Background and Aims: Undifferentiated/ spindle cell soft tissue sarcoma (USTS) is rare among children and accounts for approximately 2% of soft tissue sarcoma in children. There is a scarcity of data regarding the outcome of undifferentiated sarcoma from developing countries.

Methods: Children, ≤ 14 years of age, diagnosed as USTS from 1st January 2007 to 31st December 2017, were included for retrospective analysis. Four-year event-free survival (EFS) and overall survival (OS) were noted and outcomes were correlated with different variables.

Results: Two-hundred twenty-nine children were diagnosed as soft tissue sarcoma, of which 13 (5.5%) had USTS. Median age was 120 months (range 7 – 168 months). Male: female ratio was 2.2:1. Nine (69.2%) and four (30.8%) children had group I and III disease, none of the children had group II or IV disease. Stage I, II, III, and IV disease were seen in three (23.1%), four (30.8%), six (46.1%), and zero children, respectively. One child was lost to follow up. All 12 children underwent surgery, six (50%) received chemotherapy (one vincristine, adriamycin, cyclophosphamide (VAdRC), one VAdRC + ifosfamide, etoposide, one VAdRC + vincristine, actinomycin D, cyclophosphamide and three ifosfamide, doxorubicin) and four (33.3%) received radiation. Three (25%) children relapsed and one (7.7%) had progressive disease and all died due to disease. At a median follow-up of 68 months (range 0-142), four-year EFS and OS were 75% and 75%, respectively. Four-year OS for group I and III disease was 100% and 25%, respectively ($P=0.004$). Four-year OS for stage I, II, and III disease was 100%, 100%, and 50%, respectively ($P=0.216$). Four-year OS of children with tumor size ≤ 5 cm and > 5 cm was 100% and 50%, respectively ($P=0.055$).

Conclusions: Indian children with USTS could achieve good survival with multimodality treatment. Gross residual disease (group III) and larger size of the tumor (> 5 cm) were poor prognostic factors.

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PO0089 / #1292 | Lipoblastoma in Children – A Single Centre Experience from South India

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Background and Aims: Lipoblastoma is a rare neoplasm that occurs mostly in infants and children. Although benign, it has a tendency for local recurrence. We here represent ten years of experience of children with lipoblastoma from a tertiary care cancer center.

Methods: Children, ≤ 14 years of age, pathologically diagnosed to have lipoblastoma between 1st January 2007 and 31st December 2017, were included for retrospective analysis. Presenting symptoms, post-surgery margin status, and recurrence was recorded.

Results: Fifty-nine children were diagnosed as non-rhabdomyosarcoma soft tissue sarcoma, of which 10 (16.9%) had lipoblastoma. Median age at the time of presentation was 16 months (range 8 – 72 months). Male: female ratio was 1.5:1. All children presented with progressively increasing painless swelling. One (10%) child had swelling over labia, one (10%) had inguinal swelling, two (20%)

children had swelling involving the upper extremity and six (60%) had involvement of lower extremity. Tumor size ranged from 3 to 9 cm. All children except one child had tumor size > 5 cm. Eight (80%) children had a pathological diagnosis of lipoblastoma and two (20%) had lipoblastomatosis. Nine (90%) children underwent wide excision with negative margins, while one child with swelling at the inguinal region had a microscopic positive margin due to nearby critical structure. At a median follow-up of 60 months (Range 27-149), four-year event-free survival and overall survival were 90% and 100%, respectively. One child with an initial positive margin had local recurrence requiring repeat surgery.

Conclusions: The prognosis of children with lipoblastoma is excellent after complete excision. Incomplete excision increases the risk of recurrence. Follow-up of children with lipoblastoma is necessary to monitor for recurrence.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0090 / #1400 | Clinical Outcome of Rhabdomyosarcoma in Children: Experience of Tunisian Children Hospital

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Background and Aims: Rhabdomyosarcoma (RMS) is a primary malignancy in children that arises from embryonic mesenchyme with the potential to differentiate into skeletal muscle. Most cases are diagnosed in children under the age of 6 years. Current treatment for RMS relies on surgery, with chemotherapy and radiation as adjunct therapies. The aim of this work was to study the clinical characteristics of our population and to evaluate our therapeutic results.

Methods: A total of 40 patients were collected at the department of oncology of Farhat Hached University Hospital of Sousse over a period of 18 years (2002-2020). Statistical analysis was done with SPSS23.0 software.

Results: The median age of our patients was 5 years (1-14 years). The sex ratio was 1.66. The head and neck region was the most common site of presentation and accounted for 35% of the patients followed by the limbs (15%) and the retroperitoneal (12.5% cases). The average consultation time was 2 months (1-12 months). The most common symptom was pain (24 cases), with an isolated lesion in 39 cases. The average tumor size was 6.7 cm (2-19 cm). 14 patients had a primary surgery that was of type R0 in 53.3%, R1 in 26.7% and R2 in 20% of cases. 11 received adjuvant radiotherapy and 10 received adjuvant chemotherapy. 23 patients received neoadjuvant chemotherapy with objective response in 79%. The secondary resection rate was 60%. Recurrence was noted in 27% of cases. Metastases were observed in 27% of cases. The 5 years overall survival was 41%. Incomplete

resection and metastatic disease are correlated with worse outcome with respectively $p=0.018$ and $p=0.02$.

Conclusions: RMS is the most common soft-tissue sarcoma of childhood. The management of RMS is multidisciplinary based on surgery, chemotherapy and radiotherapy. In our series, R0 surgery and adjuvant radiotherapy and / or chemotherapy were associated with better survival, consistent with literature data.

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PO0091 / #1429 | Organizing Pneumonia: A Clinical Challenge in a Child with Previous Rhabdomyosarcoma

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Background and Aims: Organizing pneumonia (OP) is very rare in children and most often cryptogenic. Secondary causes such as infections, immune diseases, chemotherapy, radiation or stem cell therapy have been described mainly in adults. We report a case of an 11- year-old boy who presented with OP eight years after treatment for a relapsed rhabdomyosarcoma.

Methods: Review of the clinical notes and diagnostics of the patient.

Results: At the age of 3 months, he presented with a pelvic non-alveolar rhabdomyosarcoma, and was treated with 7 IVA cycles (ifosfamide, vincristine, actinomycin D), 2 topotecan/cyclophosphamide cycles and surgery. Two years later, he had a prostatic relapse treated initially with cycles of topotecan carboplatin and later with doxorubicin, surgery and brachytherapy, being in remission since then. At the age of 11, he presented with fever, cough and posterior thoracic pain to his general practitioner and was treated with azithromycin, with mild improvement and then referred to the Oncology Unit. A chest x-ray showed nodular opacities in the left lung. Blood work-up revealed elevated C-reactive protein (68,6 mg/l), thrombocytopenia ($54,000 \times 10^9/L$) and mild leucocytosis ($14,343 \times 10^9/L$). Chest computed tomography (CT) showed two nodular lesions in the left upper lobe, one of them cavitated, and another nodular lesion in the left lower lobe. Abdominal ultrasound was normal. Diagnostic work-up, including bronchoalveolar lavage (BAL) cultures, was negative for infectious or immunological etiologies. A biopsy was performed and the histology showed an organizing pneumonia. In the meantime, there was a spontaneous improvement of the lesions and steroid therapy wasn't necessary.

Conclusions: This case presented a diagnostic challenge due to the rarity of this condition and its undetermined association with the patient's history of rhabdomyosarcoma. The authors stress the importance of this etiology in the differential diagnosis of pulmonary metastases.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0092 / #1468 | Parameningeal Rhabdomyosarcoma: The Experience of Pediatric Hematology and Oncology Center of Rabat (Morocco)

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Background and Aims: Approximately 30% of childhood rhabdomyosarcomas arise in the head and neck region; tumors that originate in parameningeal sites account for one half of all lesions in this location. Tumors arising in parameningeal sites are at risk for central nervous system (CNS) extension. The aim of our study is to describe the experience of our center in the management of parameningeal RMS in children and to evaluate the treatment results.

Methods: Our present study is a retrospective analysis of children with parameningeal rhabdomyosarcoma (RMS) treated in the Pediatric Hematology and Oncology Center of the children's hospital in Rabat (Morocco), from January 2014 to December 2018.

Results: Sixty three new cases of pediatric rhabdomyosarcoma attended the pediatric hematology and oncology center of Rabat (5% of childhood cancers). Thirteen cases of Parameningeal RMS were evaluated (21% of all RMS). The median age at diagnosis was 5 years (2- 11 years). Males predominated (9M: 4F). The average diagnostic time was 2.5 months. Parameningeal location and tumor size were confirmed by CT and / or MRI in all patients. 11 patients had large tumors > 5cm. Three patients had metastases at the time of diagnosis. The diagnosis is confirmed by histology in all cases. The most common histological type was the embryonal rhabdomyosarcoma. Chemotherapy was performed in all patients, according to two protocols: MMT95 and RMS2005. Radiotherapy was performed in 8 patients with a median dose of 50,4Gys. Surgical excision was performed in 2 patients. Five patients were alive in their last follow-up, 1 of them with disease, whereas 6 patients died due to the disease and 2 lost to follow-up.

Conclusions: Parameningeal rhabdomyosarcoma remains a diagnostic and therapeutic challenge with an unfavorable prognosis. Local failure is a major problem in children with parameningeal rhabdomyosarcoma and meningeal involvement.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0093 / #1482 | Childhood Head and Neck Rhabdomyosarcoma in Pediatric Hematology and Oncology Center of Rabat (Morocco)

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Background and Aims: Pediatric rhabdomyosarcoma (RMS) is the most common malignant mesenchymal tumor of the child. It represents 5% of all pediatric cancers. The head and neck location of the childhood RMS is the most frequent location. The aim of our study is to describe the clinicopathological characteristics and therapeutic results of a series of head and neck rhabdomyosarcoma.

Methods: Cases diagnosed as RMS affecting the head and neck region were retrospectively retrieved from the files of pediatric hematology and oncology center of Rabat (Morocco) from January 2014 to December 2018. Data on clinical features (sex, age and affected site), microscopic subtype, immunohistochemical results, treatment employed and follow-up status were obtained from the patient's medical charts.

Results: Twenty one patients were diagnosed with rhabdomyosarcoma of the head and neck (38% of RMS). The median age of patients was 5 years, the sex ratio was 2.5. The main diagnostic time is 2 months. The diagnosis of RMS was guided by imaging (CT and / or MRI) and confirmed by histology coupled with immunohistochemistry. The embryonal form was the most common. The location of the primitive tumor was parameningeal in 62%, orbital in 33% and non parameningeal in 5% of cases. The treatment consisted of chemotherapy in all cases, radiotherapy in 13 patients and surgery in 7 patients. Ten patients were alive in their last follow-up, 1 of them with disease, whereas 8 patients died due to the disease and 2 lost to follow-up and 1 discontinued treatment.

Conclusions: Head and neck RMS is an aggressive malignant neoplasm which demands especial concern to achieve early diagnosis and successful treatment. Its prognosis depends mainly on the initial location of the primitive tumor: orbital and non-parameningeal tumors have a better prognosis than parameningeal tumors.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0094 / #1525 | Long Term Follow Up of Bladder Neck Rhabdomyosarcoma in an Infant – Cure at A Cos

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Background and Aims: Rhabdomyosarcoma is the most common soft tissue sarcoma in children and is very rare in infants. Rhabdomyosarcoma in the bladder neck poses unique challenges as the cure comes at long term morbidity. We describe here one such journey treated at our centre with simple yet effective bladder surgery

Methods: A 3 month old male child presented to the pediatric urology department with complaints of dribbling of urine and crying during maturation. An ultrasound abdomen revealed a mass in the bladder neck for which he underwent a needle biopsy. A staging CT chest and bone marrow biopsy confirmed a non metastatic tumour. The histopathological examination revealed features of embryonal rhabdomyosarcoma and he was categorized as intermediate risk.

Results: He received neoadjuvant chemotherapy as per COG intermediate risk protocol and this was complicated by recurrent urinary tract infection, hyponatremia, seizures and febrile neutropenia. After 4 cycles of chemotherapy there was a significant reduction in the size of the mass and he underwent a radical cystoprostatectomy with ileal conduit and ileostomy. He had wound dehiscence and required re-suturing and a ureterostomy. He underwent ileostomy closure 6 months after the completion of chemotherapy. The closure of ureterostomy with bladder reconstruction would be staged after another 2 years. He has completed over 3 years follow up and is growing well with normal renal function and no long term effects of chemotherapy

Conclusions: The role of a paediatric urologist is central in the management of infants with bladder neck rhabdomyosarcoma whose outcomes are worse than older patients due to high rates of local failure. Concerns regarding morbidity and a reluctance to perform aggressive local control measures result in a higher rate of local failure. Families must be counseled regarding staged correction and bladder reconstruction over several years and the morbidity involved including loss of sexual function to ensure optimal outcomes

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0095 / #1580 | Nijmegen Breakage Syndrome and Pelvi-Perineal Rhabdomyosarcoma: A Rare but Potentially Dangerous Association

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Background and Aims: Nijmegen breakage syndrome (NBS) is a rare autosomal recessive disorder resulting from mutations in the NBS1 gene, which encodes for the DNA double strand break repair protein nibrin, and characterized by microcephaly, dysmorphic features,

immunodeficiency, and increased susceptibility to malignancy. Rhabdomyosarcoma (RMS) is the most common soft tissue sarcoma of childhood, but RMS has not typically been associated with this syndrome.

Methods: We present the case of a 10-month-old girl with NBS who presented with an introital lesion, initially diagnosed as a urethral prolapse. She was admitted to hospital for further investigations, examination under GA and excisional biopsy of the lesion. The pelvic ultrasound, MRI and histopathology confirmed the diagnosis of a stage III embryonal botryoid RMS arising from the bladder neck, without upper urinary tract dilatation. She completed 20 weeks of VAC, but follow-up imaging, cystovaginoscopy and a repeat biopsy demonstrated a residual mass in the bladder with extension to urethra, vagina and vulva. Due to her underlying condition, associated with a high risk of severe toxicity from chemotherapy and radiosensitivity, decision was made to proceed with surgical excision.

Results: At 18 months of age she underwent a laparoscopic radical cystectomy, hysterectomy and vaginectomy and en-block excision of the tumour through a perineal incision, with closure of the perineum and formation of bilateral terminal ureterostomies. She had a good initial postoperative course, but required revision of the ureterostomies due to obstruction at skin level. Subsequent follow-up MRIs show only inflammatory changes in the pelvis, without residual tumour.

Conclusions: RMS in patients with NBS is extremely uncommon but has been previously described. This association carries significant clinical implications in terms of potential need for therapy modification and follow up investigations. In addition, as previously suggested, there is a potential role for the NBS1 gene and the nibrin dependent pathway in the pathogenesis of RMS.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0096 / #1644 | Non-Rhabdomyomatous Soft-Tissue Sarcomas in Children; Experience from a Single Center in Baghdad-Iraq

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Background and Aims: Evaluation of demographic, clinical, and pathological patterns, in addition to treatment outcome of pediatric non rhabdomyomatous soft tissue sarcoma patients treated at the oncology unit – children welfare teaching hospital over a period of six years.

Methods: A retrospective analysis was undertaken on 33 patients with non- rhabdomyomatous soft tissue sarcomas who were admitted to children welfare teaching hospital through the years 2009-2014. In this study, information regarding gender, age, clinical signs and symptoms, pathological findings, treatment and overall survival rate were reviewed.

Results: From total 33 cases analyzed; males predominate over females 23 (69.7 %) with male: female ratio of 2.3:1. Most cases were observed in age group < 5 years, 20 (60.6 %). Painless swelling was the most common presentation in 26 (78.8%), followed by painful swelling in 7 (21.2%). As a primary site of involvement; abdomen and extremities shared the highest percentage in 11 (33.3%) for each respectively. Diagnosis of cases was based mainly on excision biopsy in 20 (60.6%). Chemotherapy was received in 24 /33 (72.7%) of cases. Three years' overall survival for those received chemotherapy with or without surgery was 50.6%.

Conclusions: Epidemiological pattern of the disease is compatible with the data from studies done in other countries. Adolescents more than 10 years tend to have long symptoms intervals than those less than 10 years. Complete surgical resection with or without chemotherapy as an adjuvant is the cornerstone therapy for localized non rhabdomyosarcoma soft tissue sarcoma.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0097 / #1647 | Non- Rhabdomyosarcoma Soft Tissue Sarcomas

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Background and Aims: Non- Rhabdomyosarcoma soft tissue sarcomas (NRSTS) are a heterogeneous group of rare sarcomas. The rarity and heterogeneity make their management challenging.

Methods: The data of the patients treated between 2010 and 2020 in a Pediatric Oncology Department were reviewed. Data for age, sex, histology, primary site, stage, chemotherapy, surgery, radiation therapy, and survival outcomes were analyzed.

Results: Nine patients (5 males, 55.5%) with NRSTS were identified. The predominant histological types were infantile inosarcoma (2), hemangioendothelioma, liver sarcoma, malignant rhabdoid tumor, desmoplastic sarcoma, myoepithelioma, undifferentiated sarcoma, malignant peripheral nerve sheath tumor. The diagnosis was confirmed by molecular studies. The median age at diagnosis was 2 years (0.3–17.4 years). The most common sites were the extremity (3), trunk (3), abdominal (2), head and neck (1). All of the patients had a diagnosis after biopsy, and 8/9 of the patients underwent a radical surgical excision. Upfront surgery was performed to 4 patients. Five patients received neoadjuvant chemotherapy according to CWS 2005 protocol. There were 3/9 IRS III cases. Major response to chemotherapy

was recorded in 3/5 and minor in 1/5 cases. Adjuvant chemotherapy received 2 patients. One of them received maintenance chemotherapy. Postsurgical radiotherapy (RT) was administered to 2 patients. One patient received chemotherapy, surgical excision and radiotherapy. There were 1 relapse and 1 progressive disease. These patients died due to non-responding disease. One of them with rhabdoid tumor progressed despite intensive chemotherapy and radiotherapy. The other with liver sarcoma obtained pulmonary metastases and died despite metastasectomy. At last follow-up (median 3 years, range: 0.5-6 years) 7 patients were alive tumor-free. The overall survival was 77.7%.

Conclusions: NRSTS is a group of rare tumors in pediatric population. Survival is influenced by the presence of metastases, high-risk histology, and surgically unresectable disease. Surgery is the cornerstone of treatment. Molecular studies will give a chance for target therapies.

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PO0098 / #1767 | Ovarian Masses in Pediatric Patients: Clinicopathological Characteristics and Surgical Management

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Background and Aims: The appropriate surgical treatment to pediatric patients with ovarian lesions are heterogeneous and ovarian preservation is desirable in children. The aim of this study was to describe the clinicopathological characteristics and management of surgically removed ovarian masses in our center.

Methods: A retrospective study carried out in 10 years on 27 patients under the age of 17. Medical records were reviewed retrospectively. Data regarding clinical findings, imaging and surgical management were evaluated.

Results: Median age at diagnosis was 7 years, with majority premenarchal. Abdominal mass/pain were the most common presenting symptoms. Median tumor size was 7.5 cm (range 3-13cm). Preoperative tumor markers were elevated in 15 % of cases. All patients had fertility-preserving surgery by cystectomy (ten patients) or unilateral salpingo-oophorectomy (17 patients) : nine via laparoscopy and eighteen via laparotomy. The most common pathologic finding was mature cystic teratoma (37%), followed by simple cysts of the ovary, and cystadenoma. No one received adjuvant chemotherapy. One patient had recurrence. All patients were alive at last follow up, one with disease.

Conclusions: Mature teratoma and functional ovarian pathologies are thought to be common in premenstrual ages and malign lesions are very rare in all age groups so we recommend ovarian fertility sparing surgery should be preferred.

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PO0099 / #1769 | Total Cystectomy and Transileal Urinary Diversion (Bricker Type) in a Paediatric Bladder and Prostate Rhabdomyosarcoma

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Background and Aims: Rhabdomyosarcoma (RMS) is the most common soft tissue tumor in children, and 15% to 20% a rise from the genitourinary tract. Contemporary therapy for bladder and prostatic RMS consist of risk-stratified multimodality treatment with chemotherapy and radiotherapy, with surgical intervention reserved for patients with a residual mass following upfront therapy.

Methods: A case report

Results: We report our experience of a case of rhabdomyosarcoma of the bladder in a 5-year-old boy who showed treatment failure and progression of disease despite chemotherapy and partial cystectomy. The decision to proceed to radical cystectomy was then made. A radical cystectomy was carried out with transileal urinary diversion (Bricker type) as described in adult patients.

Conclusions: Radical excision is reserved for residual viable tumour after chemotherapy and radiotherapy, when an organsparing approach is not possible. The choice of whether to create a continent or incontinent diversion after cystoprostatectomy is based on the extent of local disease and of organ resection, as well as the maturity of the child and capability of the family and caregivers.

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PO0100 / #291 | Pediatric Rhabdomyosarcoma in Oman: A 15 Year's Experience

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Background and Aims: Rhabdomyosarcoma (RMS) is the most common soft tissue sarcoma in children. There is limited data regarding pediatric RMS in Oman.

Methods: This is a retrospective descriptive study that includes all children, younger than 13 years old, diagnosed with RMS and followed up at the Royal Hospital from Jan 2006 to May 2020. Patients' data were retrieved from their electronic chart system (AlShifa).

Results: Twenty- eight children were included. The male-to-female ratio was 1:2. The mean age at presentation was 3-4 years. The two most common presenting complaints were body masses (64%) and

constitutional symptoms (25%). Noted that the commonest presenting symptoms were presented as a mass, mostly located in our study in the extremities 25%, then the orbits 14.3%, next in the neck 10.7%. The majority of patients had stage 1 or 2 (57%), while (25%) patients had stage 3, and (18%) patients had stage 4. In terms of risk stratifications, the majority of patients (53.5%) had intermediate-risk, while (28.5%) patients had low risk, and (18%) patients had high-risk disease. The five years' overall survival of all groups combined was 60% (p-value < 0.05); and the event-free survival was 58% (p-value < 0.05). The five years' overall survival for the high-risk, intermediate-risk, and low-risk was 40%, 50.8%, and 87.5% respectively. The five years' relapse-free survival for the high-risk, intermediate-risk, and low-risk was 75%, 53.3%, and 71.4% respectively.

Conclusions: This is the first study that describes the clinical characteristics and outcome of pediatric RMS in Oman. The survival outcomes were reasonably similar to published international data.

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PO0101 / #296 | Successful Treatment with Crizotinib in Refractory Congenital Embryonal Rhabdomyosarcoma with ALK Fusion Gene

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Background and Aims: Anaplastic lymphoma kinase (ALK) is a tyrosine kinase receptor that is aberrantly expressed in several malignancies, such as anaplastic large cell lymphoma and inflammatory myofibroblastic tumor, and the effectiveness of ALK inhibitors has been reported.

Methods: Case Report: A full-term newborn boy presented with a palpable mass on the right shoulder. Magnetic resonance imaging revealed a 4.0 × 4.5 × 5.0 cm mass under the right trapezius muscle. An excisional biopsy was performed and the pathological examination showed the features of both embryonal and spindle cell rhabdomyosarcoma (RMS). The tumor cells were positive for desmin, myogenin, myoD1, and muscle-specific actin (HHF35) by immunohistochemical analysis, and negative for PAX3/7-FOXO1 fusion by fluorescence in situ hybridization. After staging, he was diagnosed with embryonal RMS NOS stage2 (T2aN0M0), groupIII.

Results: He initially received conventional chemotherapy that consisted of vincristine, dactinomycin and cyclophosphamide (VAC). However, after two cycles of VAC and two cycles of vincristine and irinotecan, the mass had enlarged by 34%. The dose of cyclophosphamide was increased and the chemotherapy was continued in conjunction with proton beam radiation therapy (55.8 GyE). Even after the proton beam radiation therapy, the tumor size did not decrease and a needle biopsy revealed viable cells. From the 28th week, he received two cycles of chemotherapy with ifosfamide, carboplatin, and etoposide without effectiveness. Clinical sequencing detected the ATIC-ALK fusion gene. From the 36th week, he received crizotinib by off-label use. Crizotinib treatment for 33 weeks reduced the tumor volume by 48% without severe adverse effects. Finally, right upper limb-conserving surgery was performed.

Conclusions: The ALK inhibitor is a promising option for the treatment of refractory RMS with the ALK fusion gene.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0102 / #315 | Pazopanib Monotherapy Leads to Complete Remission of a Refractory Malignant Peripheral Nerve Sheath Tumor (MPNST), a Case Report

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Background and Aims: Malignant peripheral nerve sheath tumor (MPNST) are rare, aggressive sarcomatous tumors that arise from peripheral nerve sheaths. They commonly arise among patients with neurofibromatosis I (NF1). Definitive treatment is surgical excision. Post operative radiotherapy plays an important role in disease free survival rates and Chemotherapy is offered to people who have a systemic disease. Majority of MPNST are not sensitive to chemotherapy and progression after chemotherapy and radiation is common, currently there is no clear guidelines on what to offer patients with metastatic disease that progress after chemotherapy, radiation and surgical excision.

Methods: A 17 years old lady, with NF1, who complained of left foot mass. This was confirmed MPNST on biopsy, disease staging revealed bilateral multiple lung metastases. Patient underwent below knee amputation followed by 6 cycles of Ifosfamide/Doxorubicin chemotherapy, followed by radiation to the remnant lung metastases. 4 months off therapy evaluation showed multiple new lung nodules, one of the nodules was invading the right pulmonary artery, patient underwent surgical resection of the lung metastases followed by radiation. 2 months later, a new lung nodule ~ 1cm was detected on a follow up scan, patient was then started on pazopanib.

Results: Patient received pazopanib 800mg daily for 8 months before dose reduction by 50% due to diverticulitis, follow up Computerized Tomography scan of the Thorax, the lung nodule decreased by 25% in the first 3 months, and 50% by 5 months. Pazopanib was permanently stopped at 12 months after subsequent CT scans confirmed stable lung nodule ~ 3mm.

Conclusions: Patient is currently 12 months off therapy, alive and the lung nodule measured less than 2mm on the 9 months off therapy scan and thought to be fibrotic tissue. This is the first case report on pazopanib solo treatment which lead to complete remission of a refractory metastatic Malignant peripheral sheath nerve tumor.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas*

PO0103 / #548 | **The Influence of Prognostic Factors on Long-Term Survival in Children with Parameningeal Rhabdomyosarcoma**

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Background and Aims: Parameningeal localization of PMC is difficult to reach for early diagnosis, and the tumor is often detected at advanced stages of the disease. Purpose: to establish the influence of age, histological variant, stage of the disease on the prognosis of RMS in children with tumor localization in a single parameningeal region

Methods: The study group included 86 patients aged from 1 year to 17 years, with parameningeal RMS. The study included 44 (51.2%) male and 42 (48.8%) female. The median is 6 years old. The embryonic variant of RMS was detected in 61 (71.0%), alveolar - in 22 (25.6%) and pleomorphic - in 3 (3.5%) of cases. Predominance of patients in clinical group III - 74 (86.0%), in IV - 11 (12.7%) and in II - 1 (1.2%). Patients were treated according to international and local protocols.

Results: The overall 5-year survival rate for stage II was 72%, stage III - 62%, and stage IV - less than 39%. When the RMS was located in the infratemporal and pterygopalatine fossae, the overall 3- and 5-year survival rates did not practically differ (71.6% and 68.2%), but the 10-year survival was 63.9%. The overall 3-year survival (79.2%) in patients with the paranasal sinuses RMS does not change in the long-term follow-up. The overall 5-year survival rate in patients with intracranial component of the tumor was 52%. Poor 3-, 5- and 10-year overall survival (33.0%) rates were obtained in patients with damage to the middle ear. Patients with embryonic RMS had a 5-year overall survival (67%) compared with those with alveolar RMS confirmed (56%). The overall 5-year survival rate for patients with parameningeal RMS was 67.6%, and the 10-year survival rate was 65.3%.

Conclusions: Parameningeal RMS is rare in baby. The most unfavorable localization is in the middle ear and intracranial tumor spread.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas*

PO0104 / #694 | **Non-Rhabdomyosarcoma Soft Tissue Sarcomas in Children: A Single Institutional Experience**

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Background and Aims: Non-rhabdomyosarcoma soft-tissue sarcomas (NRSTS) are heterogeneous tumor groups consisting of many histological subtypes, clinical progress of which is very distinct from each other. This retrospective study evaluates the clinical and histological features and outcomes.

Methods: Retrospective analysis of clinical data was carried out in our center. According to WHO 2013, the patients were classified. GR1, middle grade rarely metastasis group, GR2, local aggressive group, GR3, malign tumor group. Clinical features, treatments, responses, and survival rates were analyzed.

Results: A total of 44 patients (male:23, female:21) were evaluated. The median age was 7,5 years, and the median follow-up time was 40 months (1-180 months). Five-year event-free survival(EFS) was 64,7%, whereas overall survival(OS) was 74,4%.

G1 (n:6): The most common diagnosis was dermatofibrosarcoma protuberans(n=3). Stage 1(n:5), and stage 2(n:1). The most common involvement site was the femur (n=4).

G2 (n:6): Aggressive fibromatosis was the most frequent diagnosis (n=4); all of these were in stage 3 and had > 5 cm tumor size.

G3 (n:32): Synovial sarcoma (n:5), and fibrosarcoma (n:4) in this group. The primer site of the tumor was extremities in 14 patients. At the time of diagnosis, stage I(n:9), stage II(n:2), stage III(n:10), and stage IV(n:11). Tumor size was > 5 cm in 21 patients. The most common metastasis site was the lungs(n=11). While biopsies were performed in 13 patients, subtotal resection was carried out in 3. Fifteen patients received neoadjuvant chemotherapy; sixteen patients received radiotherapy to the primer tumor site. 5-year EFS and OS were 60; 65,9%, respectively. Any difference was not detected between positive surgical margin, tumor size, and EFS rates. The OS rate was significantly low (p<0,05).

Conclusions: The most crucial factors in prognosis are tumor size and positive surgical margins. The outcome was poor in the patients, total excision, which can not be performed and received neoadjuvant chemotherapy.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas*

PO0105 / #705 | **Clinical Features & Treatment Results in Children with Head & Neck Rhabdomyosarcoma**

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Background and Aims: Rhabdomyosarcoma constitutes 3-4% of childhood cancers and nearly half of them are seen in the head and neck location. It is a challenge to improve the survival rates in this group of patients. Here, we aimed to investigate the clinical features and treatment outcomes in children with head and neck rhabdomyosarcomas

Methods: The clinical features, tumor location, histopathological diagnosis, stages, treatment and outcomes of 65 children with head & neck rhabdomyosarcoma diagnosed and treated between 2004-2018 were retrospectively analyzed.

Results: The median age was 5.8 years (2.8 mo-15.2 yrs; M/F=37/28). The primary location was parameningeal in 49.2%, orbital in 35.4%, other non-parameningeal 15.4% locations. Surgery was performed in 12.3% of the patients with total resection or microscopic residue; 87.7% with postoperative gross tumor mass. Partial resection was performed in only 18.8% of patients with the parameningeal location, while gross or partial resection was performed in 69.6% patients with the orbital location. Histopathologically, embryonal subtype was 73.8%; alveolar subtype 13.7%, spindle cell/sclerosing subtype 4.6%, unknown in five patients. The chemotherapy regimens of CDCV; VAC/VAdR; PIAV; VDC/IE were used depending to years of diagnosis. The five-year EFS was 41.2%, and the OS was 59.3%. Five-year overall survival rate in the orbital location was 85.2%, in the other non-parameningeal location 80%, and in the parameningeal location 34.2% ($p=0.01$). The five-year OS was 80.3% in the early stages 36.6% in the advanced stages ($p<0.001$).

Conclusions: Early diagnosis and local control of the disease are critical in patients with head and neck rhabdomyosarcoma. New innovative treatment approaches should be investigated to improve the outcomes in advanced stage, parameningeal site and refractory/relapsed diseases.

Publication Only Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PO0106 / #758 | Impact of Second-Line Anti-GD2 Therapy in a Patient with Relapsed, Metastatic Ewing Sarcoma

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Background and Aims: Relapsed Ewing sarcoma (ES) is an aggressive malignancy with poor prognosis, indicating the urgent need for new effective therapies. The disialoganglioside GD2 is a well-established tumor-associated antigen expressed in 40–90% of ES cells, making it a suitable therapeutic target in ES. We report a case with relapsed, metastatic, GD2-positive ES treated with the anti-GD2 antibody dinutuximab beta (DB).

Methods: A 17-year-old presented with ES of the distal left humerus. The patient received induction chemotherapy with VIDE (Vincristine, Ifosfamide, Doxorubicin, Etoposide), producing a good response. After 6 cycles, the residual tumor was removed, a MUTARS endoprosthesis implanted, and the elbow replaced. Tumor necrosis was 97%. Following consolidation chemotherapy with VAI (Vincristine, Actinomycin-D, Ifosfamide), the patient was in full remission. Three years later, the patient presented with pain and thickening near the MUTARS implant; 3D computed tomography (CT) and biopsy revealed local relapse. High-resolution CT detected two metastases on the left pulmonary lobe with pleural involvement. The patient received 15 cycles of irinotecan/temozolomide (I/T). After cycle 3, the local relapse was treated with radiotherapy (54 Gy). Following confirmation of GD2 expression, DB 10 mg/m²/day for 10 days (plus granulocyte colony-stimulating factor on demand) was given after I/T cycles 5, 7, 9, 11 and 13. Limb-sparing surgery was performed after cycle 7 and whole lung irradiation after cycle 8.

Results: Following irradiation and 2 cycles of DB, necrosis of the resected primary tumor was 100%. Lung metastases disappeared. As of April 2021, 18 months after relapse, the patient is still in complete remission, as demonstrated by CT. Treatment was well tolerated, with no severe side effects.

Conclusions: Second-line DB in a patient with relapsed, GD2-positive ES is feasible and appears promising, with no severe side effects. Confirmatory trials are warranted.

Publication Only Topic: Disease Orientated / ASO2 Solid Non Brain Tumours / ASO2.d Soft Tissue Sarcomas

PO0107 / #763 | The Prognostic Index and Risk Rank are Better Prognostic Markers for Children with Rhabdomyosarcoma

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Background and Aims: Currently, whether the inflammation-based prognostic scoring system has any prognostic significance for patients

with rhabdomyosarcoma (RMS) has not been previously investigated. To evaluate the prognostic potential of inflammation-based prognostic scoring systems.

Methods: We retrospectively collected clinical data on 83 eligible childhood patients diagnosed with RMS from May 2014 to January 2019. Potential prognostic markers included the Glasgow Prognostic Score (GPS), Neutrophil Lymphocyte Ratio (NLR), Platelets Lymphocyte Ratio (PLR), Prognostic Index (PI), Prognostic Nutritional Index (PNI) and C-reactive Protein (CRP) Albumin Ratio (CAR).

Results: In univariable analysis, PNI and PLR were each not significantly associated with overall survival (OS) and disease-free survival (DFS) in RMS patients, but GPS, NLR, PI and CAR in addition to risk classification were significantly related to OS and DFS in RMS patients. In the multivariable analysis, it was observed that RMS patients with higher PI and risk rank had reduced OS and DFS.

Conclusions: PI and the risk rank may serve as effective prognostic markers of OS or DFS for RMS patients and could potentially be used in routine clinical practice.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0108 / #849 | Synovial Sarcoma in Childhood and Adolescence: A Single Center Experience

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Background and Aims: Synovial sarcoma (SS) accounts for 7-8% of all soft tissue sarcomas (STS) and is the most common non-rhabdomyosarcoma soft tissue sarcoma (nRMS-STS) in childhood and adolescence. One third to half of the cases occur in patients younger than 20 years old. Here we present 30 years experience with patients with SS.

Methods: Among 307 children diagnosed with nRMS-STS between 1990-2020, 20 patients (6%) diagnosed and followed-up with SS were included in this study. The clinical features and treatment results of patients were recorded from patient files and hospital information system retrospectively.

Results: The median age was 10.8 years (5-15.8 yrs; M/F=15/5). The primary tumor was located in the extremities of 15 patients (75%), other tumor locations were abdomen (n=3), mandible (n=1) and orbit (n=1). Eight patients presented with pain while 12 patients presented with a mass. Total excision was performed in 13 patients. Histopathologically, 9 patients had monophasic spindle cell, 6 patients had biphasic, 1 patient had poorly differentiated subtype, and the subtype was not identified for the 4 patients. Five patients had a tumor size over

5 cm. Fifteen patients were treated with neoadjuvant (n=3) or adjuvant (n=12) chemotherapy, the chemotherapy regimens of IFO/DOXO; VAC/VAdRC; PIAV; CDCV; VDC/IE were used depending on the years of diagnosis. Five patients received radiotherapy. Five years EFS and OS rates were 63% and 92% respectively.

Conclusions: Surgery and chemotherapy are the mainstay of the treatment. Radiotherapy is needed in selected cases. The long term survival rates varies between 60% to 85%. The overall survival is comparable with the results from other centers. Further molecular work and innovative approaches should be investigated to improve the poor outcomes in refractory and relapsed cases.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.d Soft Tissue Sarcomas

PO0109 / #869 | Receptor Tyrosine Kinases as a Potential Therapeutic Target in Orbital Rhabdomyosarcoma

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Background and Aims: Orbital rhabdomyosarcoma (RMS) is associated with morbidity and mortality and there is a need to identify novel molecules for targeted therapy. The expression pattern of receptor tyrosine kinases (RTKs) and identification of molecular clusters which could be targeted for RTKs inhibitor treatment has been explored.

Methods: Eighteen patients with histopathologically confirmed orbital rhabdomyosarcoma and 5 skeletal muscles formed part of this study. qPCR gene expression profiles of 19 RTKs were generated. The patients were followed up for 28 months (18.6 ± 6). Clustering and statistical analyses was applied to identify the significant combination of receptor kinase genes associated with RMS.

Results: Mean age was 7.7 ± 6.9 years, male:female ratio was equal. Based on the AJCC staging 45% were in stage T1A and 55% in stage T1B. Follow-up was available in 72% and 23% patients died of disease. Overexpression of MET, AXL, EGFR was seen in 90-60% cases; EGFR3, IGFR2, FGFR1, RET, PDGFR1, VEGFR2, PDGFR2 in 60-30%; EGFR4, FGFR3, VEGFR3 and ROS, IGFR1, EGFR1, FGFR2, VEGFR1 in 30-10%. MET gene was overexpressed significantly and was associated with worst overall survival (P=0.019) in orbital RMS patients. There was a significant correlation of MET expression with RET, ROS, AXL, FGFR1, FGFR3, PDGFR1, IGFR1, VEGFR2 and EGFR3 genes. Association between MET gene expression and collective expression of RTKs was further evaluated by semi-supervised gene cluster analysis and Principal components analysis (PCA), which showed well-separated tumor clusters.

Conclusions: Overexpression of MET gene could be a useful biomarker for identifying high-risk patients of orbital rhabdomyosarcoma. Well-separated tumor clusters confirmed the association between MET gene and collective expression of RTK genes. Therefore, the effect of

multi-kinase inhibitors targeting MET, RET, ROS, AXL, FGFR1, FGFR3, PDGFR1, IGF1R1, VEGFR2 and EGFR3 needs to be evaluated in a larger cohort to explore the therapeutic potential of these receptors in orbital rhabdomyosarcoma. Acknowledgement: This study was supported by the University Grants Commission.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.f Liver Tumours

PO0110 / #973 | Rare Childhood Hepatic Tumor; Hepatoblastoma, Single Center Experience

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Background and Aims: The aim of this study is to evaluate the clinical, radiological and laboratory features, survival and prominent prognostic features of the patients with hepatoblastoma, who were diagnosed, treated and followed up in a single center.

Methods: Seven patients diagnosed with hepatoblastoma who were treated and followed up in our clinic between March-2013 and March-2021 were retrospectively analyzed. According to imaging findings, PRETEXT (PRE-Treatment EXTent of tumor) stage was determined. The appropriate surgical management strategy for hepatic tumors was planned at the initial radiological assessment and continued through the biopsy and tumor resection.

Results: The AFP median value of our patients at admission is 31000 µg / L. One patient (14.2%) in the PRETEX-2 group who had lung metastases at the time of diagnosis died due to intraabdominal bleeding after biopsy. Our other patient in the PRETEXT-2 group completed the neoadjuvant treatment, received the adjuvant treatment after mass resection without any problem. Our only patient in the PRETEXT-3 group came with local recurrence after chemotherapy and total mass resection, 6 months after treatment cessation, and died without responding to the second-choice treatments. Our first patient in the PRETEXT-4 group completed his treatment and has been followed for 5 years without disease; Another patient underwent liver transplantation after neoadjuvant therapy, but was lost due to post-transplant complications. Another patient underwent liver transplantation from his father after neoadjuvant treatment. Our last patient in this group has just been diagnosed and his neoadjuvant therapy is still ongoing.

Conclusions: In our case series, this rate is 57.1%. The fact that this rate is lower than in the literature is due to the late presentation of our cases and their being inoperable at the beginning, and the loss of 2 of our patients with complications after transplantation.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.g Germ Cell Tumours

PO0111 / #1471 | Clinical Presentation, Treatment and Outcomes Pediatric Ovarian Germ Cell Tumors: The Experience from a Tertiary Care Center in México

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Background and Aims: Germ cell tumors of the ovary (GOCT) are tumors that are most often seen in young women, they are fast-growing and are often unilateral and, if detected early, are curable. The objective of this study is to describe the clinical presentation of girls and adolescents with GCT.

Methods: A retrospective study of a series of clinical cases of all patients under 18 years of age was done from 2010 to march 2021 in a tertiary care center in Mexico.

Results: 34 patients were included. Mean age at the presentation was 11.7 years, the most frequent symptom was abdominal pain, which was present in 52% of the patients, followed by abdominal distention in 38%, only one case presented with primary amenorrhea. 59% of the patients had the tumor on the right side, 38% on the left side, 3% bilateral. Histology was reported: dysgerminoma 26.4%, immature teratoma 26.4%, mature teratoma 17.6%, endodermal sinuses 17.6%, mixed germ cell tumor 12%. The patients were classified as stage I in 41.1% of the cases, stage III in 55.9%, and stage IV in 3%. 29 patients (85.2%) underwent oophorectomy and 5 cases (14.7%) underwent salpingo-oophorectomy, the metastatic disease appeared at the pulmonary level in 4 patients and intra-abdominal metastases were present in 6 patients. Stage I patients only received surveillance and none presented progression, 20 patients were treated with PEB chemotherapy, 2 patients (10%) who had stage III at diagnosis required a second rescue protocol due to disease progression, all of whom died. With a follow-up time of 3.8 years, OS was 94%, EFS was 88%, and there were 2 relapses in patients with stage III at diagnosis. There was no abandonment.

Conclusions: This study confirms the good outcome for girls with ovarian germ cell tumors. Patients with advanced-stage relapsed frequently or presented progression of their disease.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.g Germ Cell Tumours

PO0112 / #1761 | Neurogenic Bladder After Resection of Sacrococcygeal Teratoma and its Management: About 2 Cases

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Background and Aims: Urologic complications are an underappreciated sequelae of the mass effect of sacrococcygealteratoma (SCT) and its resection. We present two cases of patients who were treated for SCT and presented after resection a neurogenic bladder.

Methods: We present two cases of patients who were treated for SCT and presented after resection a neurogenic bladder.

Results: Case presentation: The age at surgery of our patients was 2 and 4 months. It was Altman type 3 and they had complete surgical resection including removing the coccyx by combined abdominal and posterior sacral approach. Post-operatively, they presented bladder-voiding disorders and acute urine retention suggesting a neurogenic bladder confirmed by urodynamic findings which showed a poorly compliant high-pressure bladder. They were managed with clean intermittent catheterization and anticholinergic medications. Since they developed many urinary tract infection, a vesicostomy was performed in the two cases with good functional results.

Conclusions: Neurogenic bladder, when identified, should be aggressively treated in the patient with SCT to prevent the development of renal injury over time.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.g Germ Cell Tumours

PO0113 / #290 | Congenital Cervical Teratomas; Are They Really Benign?

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Background and Aims: Background: Teratoma is a term derived from a Greek word meaning monster. Classically it is composed of tissues from the three embryonic germ layers: ectoderm, mesoderm and endoderm. Most commonly found in the sacrococcygeal region, these tumors are rare in the neck region. Cervical teratomas are rare. They have an incidence of 1 in 20 000–40 000 live births and account for about 6% of all fetal teratomas. Airway obstruction in the newborn because of tracheal compression or occlusion has been reported as the cause of an 80–100% mortality rate in untreated cervical teratomas in the neonatal period.

Methods: Case Report

Results: Case: A 27-year-old mother of one, G3P1A2, came in premature labour at 28 weeks of gestation. Her prenatal ultrasound showed a Giant fetal neck mass, so she underwent an emergency Caesarean Section. A baby girl was born. At 1 minute she was found to have respiratory distress, and was immediately intubated with difficulty and this helped the baby to reach a normal APGAR at 10 minutes. Airway maintenance was extremely difficult due to the mass compression. The MRI study after stabilization confirmed a huge cystic teratoma in the neck, extending to the left side of the face and to the subcutaneous tissues of

the chest. It is compressing and displacing the neck structures. A high risk consent was taken from the parents and all possibilities explained thoroughly. The resection was complicated by desaturation episodes during anaesthesia and a quick Debulking was performed to bring the procedure to an end. She survived for 5 days and succumbed on the fifth post-operative day. The Histopathology's final report confirmed it to be an "Immature Cystic Teratoma Grade 3".

Conclusions: Conclusion: Malignant teratoma is exceptional in perinatal cervical cases. There have been reported malignant teratomas in the head and neck region in neonates.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0114 / #1184 | Two Cases of Adrenocortical Carcinoma Presented as Precocious Puberty, Which were Treated Effectively by Mitotane Therapy

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Background and Aims: Adrenocortical carcinoma (ACC) is secreting pure sex hormone is rarely reported, and its prognosis is unrevealed. We report two cases of ACC presented as precocious puberty showing relatively good prognoses after mitotane therapy.

Methods: Case 1. A 4.6yr-old girl presented with hirsutism and acne for 1 month. She presented clitomegaly with Tanner stage 3 hirsutism, stage 1 breast, and moderate acne. Bone age accelerated to 11 years. Evaluation for adrenocortical hyperplasia and malignancies showed non-specific. except high serum DHEA-S. Molecular genetic testing for CAH and other steroid hormone synthesis defects revealed as wildtype. Ultrasonogram showed normal pelvis. However, 18-months later, well-defined mass sized 8.5 cm was seen in left suprarenal area with pulmonary metastases. She had tumor removal surgery, and the biopsy revealed as ACC stage pT2, high mitoses, Ki-67 positive (30%). She received DEC chemotherapy with mitotane. After 3 cycles of chemotherapy, pulmonary metastases disappeared, She ended 10 cycles of chemotherapy, and maintained mitotane for another 2 years. She is relapse-free for 3 year and 6 months.

Results: Case 2. 4.5 year-old-girl presented with breast budding and acne for 3 months. She showed prepubertal response to GnRHST and high serum estradiol level, suggestive of peripheral precocious puberty. The serum estradiol, DHEA-S and 17-hydroxyprogesterone were elevated. BA was advanced at 11.0 yrs. Ultrasonography revealed no abnormal pelvic lesion. However, 6months later, 8.5cm sized mass in the left suprarenal area was found. After left adrenalectomy, biopsy revealed as ACC, 198g, Ki-67 10%. The patients was started mitotane and local radiation therapy. Three months after diagnosis, her breast regressed to prepubertal and DHEA-S level was normalized. Six month after diagnosis, the patient manifested mitotane-induced encephalopathy, and discontinued mitotane. After 10 months after discontinuation, she shows no evidence of tumor recurrence.

Conclusions: ACC can be revealed as various type of precocious puberty, and mitotane therapy would be the choice. However, the adverse effect hinders its long-term use.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PO0115 / #1275 | Refractory Multisystem Langerhans Cell Histiocytosis: Response to Braf Inhibitor Vemurafenib – A Case Report

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Background and Aims: Langerhans cell histiocytosis (LCH) is characterised by abnormal proliferation and accumulation of Langerhans cells. Clinical presentation is variable, with “single” or “multisystem” disease. Treatment depends on disease extent and ranges from observation alone to chemotherapy. The presence of a BRAF^{V600E} mutation, which is targetable, has offered another form of treatment in resistant cases. We report a case of a three-year-old with refractory multi-system LCH, who has had a dramatic clinical and radiological response to BRAF inhibition (vemurafenib).

Methods: Our patient presented when two months old with extensive seborrheic dermatitis, affecting his chest, trunk, scalp and flexures. Purpuric lesions were present within the rash and it failed to improve with topical treatment. A skin biopsy confirmed Langerhans cell histiocytosis (BRAF^{V600} sequence detected). Staging investigations were normal at presentation, confirming single system disease. Steroid therapy was commenced. He progressed on treatment to develop bony LCH with maxilla involvement. He was commenced on treatment as per LCH-IV protocol with vinblastine and further steroid therapy. His skin proved to be steroid responsive, but he had a persistent lesion in his maxilla. Cladribine therapy was then commenced but despite this, and subsequent indomethacin therapy, reassessment scans demonstrated new involvement of the pituitary gland and calvarium and progressive disease in the maxilla.

Results: Given his refractory disease, and presence of BRAF mutation, he was commenced on vemurafenib (BRAF^{V600} inhibitor). His treatment was complicated by diabetes insipidus secondary to pituitary involvement requiring desmopressin and recurrent ear infections. He has been on vemurafenib for 10 months. The calvarium and skin disease is in remission, and there has been a significant improvement in maxilla disease.

Conclusions: Our patient has had an excellent clinical and radiological response to vemurafenib. However, like many targeted therapies, we are now left with the challenge of if, when and how to discontinue therapy.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PO0116 / #1414 | Malignant Rhabdoid Tumors: A Single Center Experience

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Background and Aims: Malignant rhabdoid tumor is a rare and highly aggressive tumor that commonly occurs in infancy and young children. Those located in the central nervous system (CNS) are called atypical teratoid rhabdoid tumors (ATRT). As it is a rare tumor, there is no consensus on effective treatment yet. Despite multimodal therapy that combined with surgery, chemotherapy, and radiotherapy, the survival time is very short.

Methods: Our aim is to present our experience with this tumor by retrospectively evaluating malignant rhabdoid tumor cases diagnosed, followed up and treated in our clinic over a period of 10 years.

Results: The clinical features and prognosis of 14 malignant rhabdoid tumor cases diagnosed between 2010-2021 were retrospectively reviewed. During this period, 6 of 14 patients diagnosed with malignant rhabdoid tumor were female and 8 were male and the median age was 6 months (5.5 months-4.5 years). Cerebral (ATRT) in 8 of the tumor cases; 4 of them are renal; 2 of them were located paravertebral. Twelve of the patients were taken to a multimodal intensive treatment program, and 2 of them died before starting treatment. Progression of nine patients could not be prevented, and these 9 patients (9/14) died in a median of 4 months (11 days-15 months) from diagnosis. Two cerebral and two renal localized cases are in remission. One patient lost to follow-up with active disease. Alive patients have been followed up to 2, 3 and 4 years without any problems, respectively, starting from the end of treatment.

Conclusions: Diagnosis of rhabdoid tumor, which should be included in the differential diagnosis of renal and CNS tumors, carries poor prognosis especially in the first 3 years. Generally, total resection is not possible, despite the young age, radiotherapy is recommended and intensive chemotherapy is required. Studies on the treatment and stem cell transplantation with new treatment protocols suitable for the biological behavior of the tumor in the light of molecular studies are ongoing.

Publication Only Topic: *Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis*

PO0117 / #1475 | Childhood Giant Abdominal Lipoma: A Very Rare Case Report

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Background and Aims: Adipose tissue-derived tumors are rare in childhood. The majority of adipocytic tumors in children represent lipoblastoma and lipoma. Mesenteric lipoma is an extremely rare disease in children. We aimed to remind this rare tumor when diagnosing infantile and childhood tumoral masses.

Methods: We report a case of giant lipoma of the mesentery in a 5-year-old girl presenting with painless abdominal distension.

Results: A five-year-old boy presented with a painless progressing abdominal distension for last 2 months. On a physical examination, the child had a soft distended abdomen without tenderness. The alpha-fetoprotein and beta-hCG levels were normal. The patient subsequently underwent an abdominal and pelvic computed tomography (CT) examination that revealed a huge, well-defined, low density, fat attenuation, nonenhancing, intraperitoneal, mass lesion occupying majority of the abdominal cavity. The retroperitoneal structures and the bowel loops were displaced posterolaterally toward the abdominal walls. Multiple fibrous septations and few specks of calcifications were seen within this essentially fatty lesion. Based on these findings, the mass was suspected of being a lipoblastoma. An abdominal laparotomy was performed. A well-capsuled yellow mass was found attached to the mesentery of small intestine and the mesocolon. Although total surgical excision is recommended, our patient had subtotal tumor excision because the mass was not separable from sigmoid and rectum. Histopathological examination of specimen showed a tumor composed of mature white adipose tissue with no evidence of nuclear atypia or mitosis, confirming that it was a benign lipoma. The patient is followed with a small residue for one year under close monitoring with imaging.

Conclusions: Treatment of choice is total surgical excision with preservation of adjacent tissues. Total excision may not always be performed because of the risks of tissue and vessel damage. In these cases, close monitoring with imaging is recommended.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0118 / #1483 | When We Thought It's Phyllode Tumors and We Missed It! About Two Cases

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Background and Aims: Breast tumors are uncommon in children and adolescents. They are associated with a variety of benign, atypical, and malignant pathologic diagnoses.

Methods: We report two cases of 2 adolescents presented with breast tumors thought to be Phyllode tumors but the histological diagnosis was different.

Results: Case 1: A 12-year-old menarchal girl presented with a right breast lump increased in size over a 2-week period associated with a nipple discharge. She had a family history of colorectal cancer in the grand-father and stomach cancer in the aunt. Physical examination revealed a 6 cm, firm, well-defined and mobile mass in right breast sub areolar area. The axillary and supraclavicular lymph nodes were not palpable. On ultrasound, it was a huge heterogeneous tumor occupying almost the entire breast. A Phyllode tumor was suspected and the tumor was surgically resected. The histological diagnosis concluded to benign papilloma with clear margins of excision. The patient has been well without any recurrence for 18 months survey. Case 2: A 13-year-old girl visited our department since she had noticed, for 4 weeks, a painful palpable lump in her right breast gradually increased in size. She had no family history of breast cancer. Physical examination showed a 12 cm, firm, well limited, mass with local inflammatory signs occupying the right breast. Ultrasound showed an 8 cm heterogeneous mass with small cystic lesions. Total excision of the mass was done. The reconstruction was satisfactory. The histopathologic examination concluded to giant juvenile fibroadenoma without signs of malignancy.

Conclusions: Surgery for benign diseases of the mammary gland requires consideration both curability and functional preservation, but it's not always the treatment of choice. This is based on the nature of the tumor. This is why a careful decision-making regarding the treatment policy, including strict follow-up observation, is necessary.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0119 / #1495 | Outcomes for TMI in a Resource-Challenged Setting

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Background and Aims: Myofibroblastic tumors, represent an uncommon heterogeneous group of inflammatory tumors of unknown etiology, may affect children and young adults. The most frequent affected organ is the lung, up to 5%. The standard recommendation is complete resection surgery, which reduces the risk of recurrences, referred in up to 20-40% of cases. For unresectable disease, the use of NSAIDs, chemotherapy regimens, radiotherapy and ALK inhibitors have been proposed. We sought to describe: clinical presentation, treatment and results in a series of patients with myofibroblastic tumors, in the largest pediatric hospital in Honduras.

Methods: We retrospectively collected data, between January 2002 and July 2020, which included information on the clinical characteristics: age, sex, gender, location, histology, ALK status, treatment and current condition, with the ethics committee approval.

Results: A total of 9 patients were treated. The patient's age ranged from 3 years to 20 years (mean age 10.9 years), with a ratio F: 3.5: 1. The locations in order of frequency: Abdomen (3/9), lung (2/9), extremities (2 / 9) pelvis (1/9) face (1/9). ALK rearrangement was carried out in 2 patients, due to the lack of immuno-histochemistry in the country, resulting in one of them being positive. Complete resection was performed in 56% (5/9) of the patients and 44% (4/9) incomplete resection and all of them had local recurrence. Other therapies used was: chemotherapy 4/9, radiotherapy 2/9, ALK Inhibitors 1/9. During the median of 10 years of follow-up, the overall survival was 82%.

Conclusions: Complete resection continues to be the available option, as long as it is possible, to keep our patients in remission and reduce the possibility of recurrences. Considering the small sample, we described these findings and encourage further investigation, that can lead us to a better knowledge of the behavior and prognosis of these entities in those contexts.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0120 / #1613 | Infantile Myofibromatosis: A Case Report

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Background and Aims: Infantile myofibromatosis (IM) presents at birth or develops shortly thereafter, with 90% of cases debuting before the age of 2 years. IM is characterized by solitary or multiple firm nodules that are flesh-colored to purple ("myofibroma") and are generally painless. The reported incidence of IM is 1 in 150,000 live births. Disease-causing genes have been identified: PDGFRB and NOTCH3.

Methods: CASE REPORT

Results: We report a 17-year-old female patient who at 4 years, presented headache, which led to a left parietal biopsy in another institution with a diagnosis of myofibromatosis, with poor follow-up of the disease. Last year she consulted for mild oppressive headache and paresis of the right sixth pair, convergent strabismus in the left eye, without motor deficit. A PET CT scan showed increase uptake in lesions in various locations on the skull, spine, and left clavicle. We performed a biopsy of the lesion on C6, leading to the diagnosis of myofibromatosis. In Sequencing, it has the R561C variant in the PDGFR gene that is pathogenic. We planed treatment with methotrexate 30 mg/m² and vinblastine 5 mg/m² weekly x 26 weeks and then continue with the same biweekly schedule until completing 34 more weeks. She completed the first 12 weeks with good tolerance but the control PET CT scan showed an increase of metabolic activity in a left extra-axial intracranial nodular lesion that projects towards

the sphenoid sinus, not previously visible. We change to imatinib 800 mg/day. In the re-evaluation MRI, a marked decrease in the size of the nodular lesions was observed. She is currently on Imatinib treatment.

Conclusions: Considering how unusual the pathology is, we found it interesting to comment on the case, taking into account the atypical evolution with the first line of treatment

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0121 / #1665 | Gastric Teratoma—Unusual Cause for Bleeding of the Upper Gastro-Intestinal Tract in Children Gastric Teratoma—Unusual Cause for Bleeding of The Upper Gastro-Intestinal Tract in Children

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Background and Aims: Gastric teratoma is an extremely rare tumor, almost exclusively benign, accounting for < 1% of all teratomas in infants and children.

Aim: We aimed to study the clinical presentation, investigation modalities, intraoperative findings, histopathological types, and surgical outcome of gastric teratoma in children.

Methods: A retrospective study of pediatric patients with gastric teratoma was performed from 2009 to 2021 at Dhaka Medical College Hospital, Dhaka-1000, Bangladesh.

Results: There were two patients with gastric teratoma and all of them were male. One (50%) patients presented in infancy, while another (50%) patient was a neonate. The manifestations were as follows: a palpable abdominal mass (100%), abdominal distension (100%), anemia (100%), respiratory distress (100%), gastric outlet obstruction with recurrent vomiting (100%) in two, and melena in one (50%). Ultrasonography showed a solid cystic mass with mixed echogenicity in all the patients. Computed tomography scan of abdomen was performed in both (100%) patients, provided classical findings of teratomas. A growth was present on the posteroinferior wall near the greater curvature in two (100%) cases. Excision of the tumor was performed in all patients. Histopathology was mature type in both (100%). Complications were seen in one(50%) cases, and one (50%) mortality was observed in our series. Another patient was doing well in the postoperative follow up period.

Conclusions: Gastric teratoma has predilection for male sex and commonly presents as a palpable abdominal mass. Complete surgical excision is curative. The prognosis is excellent.

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PO0122 / #1698 | Intrapericardial Teratoma: Fetal Diagnosis and Neonatal Management

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Background and Aims: Intrapericardial teratoma is an extremely rare form of primary cardiac tumor. Till date, less than 100 cases have been reported in the literature. Fetuses diagnosed with pericardial teratomas (PT) almost die in utero from hydrops fetalis.

Methods: We report a male fetus with an early diagnosis of PT which was accomplished due to prenatal ultrasound and which permitted an optimal post-partum planning.

Results: A male fetus, coming from a spontaneous pregnancy followed by a primary care obstetrician, was diagnosed to have a mediastinal mass on ultrasound at 26 weeks' gestation. Fetal echocardiography showed a heterogeneous multi-lobed mass, measuring 28 x 22 mm and compressing the right atrium with moderate pericardial effusion. Fetal cardiac MRI showed a large mediastinal mass of heterogeneous signal with an intimate contact with the right heart cavities. No other structural abnormalities were detected. At 35 weeks' gestation, the mass increased in size (41x34mm) with appearance of a pericardial effusion of average abundance. Delivery was then anticipated after a course for fetal lung maturation. It was a caesarean birth of a boy, clinically term, eutrophic with good adaptation to the extra-uterine life. He had transient respiratory distress without signs of heart failure. Postnatal echocardiography showed a huge pericardial tumor in the right atrium (compressing its upper and lateral surface) and the left pulmonary artery with the presence of a circumferential pericardial effusion prevailing in the anterior. Its multi-lobed cystic appearance evokes a teratoma. Alpha foeto protein returned very high (111 335.50 IU / mL). The baby boy was clinically well and the search for another tumor localization came back negative. The parents were seized prenatally for a private pediatric cardiovascular surgery.

Conclusions: Intrapericardial teratoma is a rare primary cardiac benign tumor. Prenatal diagnosis, close monitoring for hydrops fetalis, antenatal pericardiocentesis for cardiac obstruction, and early neonatal tumor resection provide a good prognosis.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0123 / #1710 | Tumour Lysis Syndrome in an Infant with Langerhan's Cell Histiocytosis with Hemophagocytic Lymphohistiocytosis: Who is The Perpetrator?

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Background and Aims: Multisystem Langerhans cell histiocytosis (LCH) in infancy has overall poor prognosis. Association with Hemophagocytic Lymphohistiocytosis (HLH) is rare, but might increase the risk of mortality. Tumour lysis syndrome (TLS) in LCH and/or HLH is rare and at present not routinely monitored.

Methods: Case Study: An eight month old female presented with seborrheic eczema over scalp, scaly maculopapular lesions and hypopigmented lesions over trunk, palms and soles. She also had history of anemia requiring blood transfusion, hepatosplenomegaly but no lymphadenopathy. Complete hemogram revealed anemia and thrombocytopenia but normal leucocyte counts. Serum ferritin and triglycerides did not favor diagnosis of HLH at presentation. Serum B12 was low and replaced with intravenous doses without any improvement. Bone marrow examination revealed adequacy of all lineages with hemophagocytosis and few clusters or sheets of histiocytes negative for Cluster of differentiation (CD) 1a+ and MIB-1 ruling out LCH and malignant histiocytosis s/o HLH. Repeat HLH work-up met 5/8 established criteria for HLH. Skin biopsy confirmed diagnosis of LCH. Hence, final diagnosis was multisystem LCH with HLH. She was treated on LCH-III protocol with oral prednisolone and IV Vinblastine. She had decreased urine output after 12-18 hours of chemotherapy initiation. Investigations revealed TLS and Acute renal failure. Further doses of steroids were stopped. Supportive care was given for TLS along with peritoneal dialysis. She was persistently anuric for next 1 week and succumbed due to multi-organ failure.

Results: Discussion: Our literature review for TLS and Histiocytosis syndrome revealed 1 case of TLS with LCH and 4 cases of TLS with HLH in children. As of now, there is no recommendation for routine monitoring for TLS as it is considered a rare complication in Histiocytosis syndromes.

Conclusions: Conclusion: LCH along with HLH may have life threatening tumor lysis syndrome. Prophylaxis and aggressive monitoring after chemotherapy initiation should be done routinely to reduce risk of irreversible organ damage.

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PO0124 / #1752 | Renal Medullary Carcinoma in an Adolescent with Sickle Cell Trait

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Background and Aims: Renal Medullary Carcinoma (RMC) is a highly aggressive form of kidney cancer, primarily affecting adolescents and young adults, that is almost exclusively associated with sickle cell trait. We present a case of a 15-year-old female of Greek origin with metastatic RMC.

Methods: A 15-year-old female presented with a 2-week history of malaise, fever, dyspnoea, cough and back pain. Baseline imaging revealed pleural effusion, enlarged axillar and mediastinal lymph nodes and a nodule of the right pulmonary lobe. Abdominal MRI showed a cystic lesion in the right kidney, 4 cm in diameter, metastatic lesions in the liver and nodes in the right square lumbar muscle, right paraspinal muscles, bilateral pelvic sciatic bones and both femurs. Spinal MRI revealed metastatic infiltration of all the lumbar vertebrae and most of the thoracic ones. CA 125 was elevated in both serum (1,127 IU/ml) and pleural effusion. The patient underwent biopsy of the axillar lymph node and the diagnosis of RMC was established.

Results: Chemotherapy with weekly administration of Cisplatin, Adriamycin and Paclitaxel was initiated and transient clinical remission was observed. After 10 chemotherapy cycles, pleural effusion deteriorated and a second line chemotherapy, based on the MD Anderson protocol, with Gemcitabine, Ifosfamide and Cisplatin was administered. After 2 months, the patient had disease progression with metastatic mass in the right breast and underwent chemotherapy with alternating cycles of Cisplatin-Paclitaxel-Gemcitabine/ Bortezomib-Garboplatin-Gemcitabine (CPG/BCG). A right nephrectomy was performed after 6 months of alternating CPG/BCG cycles. Shortly after surgery, restaging examinations revealed progressive disease with metastatic mass in the left breast. Treatment with Bevacizumab and EGFR inhibitor Erlotinib is consequently planned.

Conclusions: RMC is a rare form of kidney cancer with extremely poor prognosis and limited therapeutic options associated with sickle cell trait. Hopefully, understanding the biologic mechanisms of RMC will lead to more efficacious therapeutic options.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0125 / #1757 | A Childhood Pheochromocytoma Case due to Von Hippel-Lindau Disease: A Very Rare Manifestation

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Background and Aims: Aim In childhood, pheochromocytomas are mostly due to genetic causes, of which von Hippel-Lindau (VHL) disease is the most frequent disorder. This tumor may be the only and/or initial manifestation of the disease, with delayed manifestations of the syndrome in other organs. We report a case of VHL disease who presented with pheochromocytoma.

Methods: A case report

Results: Case Description This patient was a twelve-year old boy, admitted with complaints of headache, hot flushes, palpitation and left back pain for the past one month. He was the first child of consanguineous parents. His birth history was unremarkable. Family history was positive for familial cancer syndromes; his mother was diagnosed with VHL disease and had underwent nephrectomy and surrenalectomy for clear cell renal carcinoma and pheochromocytoma. On physical examination, his blood pressure was high. General examination was otherwise normal. Laboratory tests showed an elevated 24 hour urinary VanillylMandelic Acid (VMA). Abdominal ultrasound and computed tomography revealed solid lesion 45*89 mm at the left adrenal gland compatible with pheochromocytoma. Genetic analysis confirmed the VHL disease. Follow-up according to the recommended pediatric screening protocol for children carrying a VHL mutation found out no additional VHL tumor. Total left adrenalectomy, including removal of the mass, was performed and the diagnosis of pheochromocytoma was confirmed histologically. Post-operative course was uneventful.

Conclusions: Conclusion Meticulous follow-up and early genetic testing may facilitate diagnosis and serve to prevent morbidity and mortality, as well as improving long term prognosis in VHL disease.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0126 / #586 | A Case of Successful Management of Kaposiform Hemangioendothelioma in a Resource Limited Setting

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Background and Aims: Kaposiform hemangioendothelioma (KHE) is a rare, locally aggressive vascular tumor, which typically occurs during early childhood. The treatment usually includes vincristine, steroids and rapamycin.

Methods: Here we report a case of a 10-months-old boy diagnosed with Kaposiform hemangioendothelioma.

Results: Ten-months-old boy was admitted to our clinic with complaints of general weakness and firm lesion in the sacrococcygeal region. From the age of three months, when the child's mother first noticed the lesion, it continuously increased in size. Laboratory tests revealed thrombocytopenia, anemia and leukocytosis. The level of D-dimer was highly elevated. Acute leukemia was suspected, but bone marrow aspiration didn't reveal any abnormalities. Transfusion therapy was effective for anemia, but thrombocytopenia was refractory. MRI was performed, which revealed a cutaneous vascular tumor with an enormous extent infiltrating and pushing aside bladder and rectum and infiltrating muscles from the dorsal thorax down to the right femur (4.7x0.9x5.0 cm). Considering clinical manifestations and results of laboratory tests Kaposiform hemangioendothelioma was suspected and skin punch biopsy was performed. Histological analysis confirmed initial diagnosis. Considering the fact, that determination of the typical GLUT1 marker was not available in Armenia, immunohistochemistry was not performed. The patient received dexamethasone monotherapy, which was ineffective. Afterwards, rapamycin was selected as the treatment option. During the first 2 weeks of treatment, platelet count was regulated and the level of D-dimer was decreased too, but was still abnormal. A month after the start of treatment, there was a significant decrease in the size of sacrococcygeal pathological formation. Now the condition of the child is stable and he is continuing to take rapamycin in a reduced dose.

Conclusions: Evaluation of clinical manifestations, laboratory tests and imaging studies can be indispensable for diagnosis of KHE in resource limited settings. It is proved that rapamycin is the most effective treatment option for KHE.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0127 / #702 | A Primary Adrenal Rhabdoid Tumor Mimicking a Stage 4 Neuroblastoma in a Neonate: An Exceptional Presentation to Report On

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Background and Aims: Background Rhabdoid tumor is a rare but also a distinct neoplasm characterized by a rapid and aggressive course. Most tumors are described in infants and affect classically the kidney. Aims This report describes a neonate exhibiting a primary adrenal rhabdoid tumor with diffuse metastases including liver, skin, skeleton and brain.

Methods: We report on a case of a neonate with an adrenal rhabdoid tumor. We describe clinical presentation and outcome.

Results: A 22 day-old girl presented with a 16 day history of progressive abdominal distension and subcutaneous nodules. On examination there were multiple blue subcutaneous nodules localized on the neck, back and limbs associated with an enlarged hepatomegaly. There

were no neurological signs. An abdominal computed tomography scan showed a large left adrenal mass (56 mm) containing calcifications with a nodular hepatomegaly and multiple osteolytic lesions on long bones x-ray suggesting a stage 4 neuroblastoma. Laboratory data were compatible with increased plasmatic LDH level (7N) with and bone marrow involvement but negative urinary catecholamine secretion. Within 5 days, she developed a general alteration with Focalized convulsions and rapid increasing abdominal distension. Laboratory findings revealed hypocalcemia, hyperkalemia and hypercreatinemia suggesting a tumor lysis syndrome. Cerebral CT scan showed multiple brain metastases. Pathological examination of cutaneous nodule biopsy was consistent with the diagnosis of rhabdoid tumor with loss expression of INI1. Patient underwent chemotherapy and symptomatic management but she expired rapidly as a consequence of progressive disease.

Conclusions: Rhabdoid tumor is rare and aggressive entity. Adrenal involving can mimic an aggressive neuroblastoma and cause a diagnosis delay. Behavior in neonates remains poor.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0128 / #801 | Retinoblastoma: Experience from a Single Center in Turkey

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Background and Aims: To analyze patients who diagnosed as retinoblastoma at our centre between 1987-2020.

Methods: The medical reports were reviewed retrospectively. Clinical characteristics, treatment details, treatment responses, and survival rates were evaluated.

Results: There were 71 eyes with retinoblastoma in 55 patients. Median age at diagnosis was 25mos (2-145 mos), M/F=1.5. The disease was unilateral in 71%, bilateral in 27% of patients, trilateral in one case. Four cases had familial retinoblastoma. The median time to diagnosis after the first symptoms was 3 mos(2weeks-28mos). The common findings at diagnosis were leukocoria (60%), strabismus (38%) and vision problems (21%). Reese-Ellsworth classification of 71 eyes were: RE-Group1 (1.4%), RE-Group2 (11.2%), RE-Group3 (14%), RE-Group4 (5.6%), RE-Group5 (56%); and unclassifiable 8 eyes. Tumor was unifocal in 42(59%), multifocal in 29(41%) eyes. Primary enucleation was performed in 30(54.5%) cases (5 of them bilateral) and 16 of them received adjuvant chemotherapy. Thirteen patients had extraocular disease with tumor positive optic nerve cut off. Chemotherapy response was 75%(CR+PR). Twenty-two patients (9 bilateral, 1 trilateral retinoblastoma) received neoadjuvant chemotherapy, and treatment response was PR in 77%. Surgery was performed after

chemotherapy in 12(21%) patients. Bilateral enucleation was required in 4 patients. Majority of patients received VEC chemotherapy 3 to 6 courses. External radiotherapy was performed in 14 patients(25%). Local therapies (thermotherapy, laser photocoagulation, cryotherapy), were given to 7 patients; intra-arterial melphalan/topotecan was applied to two patients. Median follow-up time was 3yrs(1mos–16yrs), 5 years EFS and OS were 31%, and 80% respectively. Ten patients died with progressive disease. One patient died because of progression in secondary leukemia.

Conclusions: More than half of our cases had advanced stage retinoblastoma at diagnosis affecting our results negatively. Retinoblastoma is one of the rare cancers of childhood. Early recognition of the complaints and advance diagnosis are important for saving lives as well as the eyes and vision. Multidisciplinary management is critical.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0129 / #816 | Mucoepidermoid Carcinomas of the Salivary Glands in the Pediatric Age Group at a Tertiary University Hospital of a Middle-Income-Country

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Background and Aims: Tumors in the pediatric age group have become epidemiologically more relevant, being the first cause of death in children (range one to nineteen years) in Brazil due to recent improvements in the control of infectious diseases. Given this, studies regarding not only the treatment but also the incidence of pediatric tumors became an important tool for health care planning. The present case study was carried out focusing on rare salivary gland tumors, more specifically the mucoepidermoid carcinoma - which was relevant in our data.

Methods: Review of the pediatric cancer charts registered over five years (2014-2019). The characteristics analyzed were age, sex, histology of the tumor, and clinical information available on patients' records.

Results: The review included 174 cases of cancer in children. Of these, twenty-two cases (12,6%) were deemed rare tumors, being four of them mucoepidermoid carcinomas (18,2% of all rare tumors). Three of these cancer cases were located in the parotid and one was located in the submandibular gland. The mean age at diagnosis was fourteen years (range eleven to sixteen years), and females were affected more frequently than males (ratio of 3:1). The main feature for its diagnosis was the tumor histology, along with relevant information

gathered at the physical exam, imaging, and lab studies. Surgical excision was the treatment of choice for all three cases treated at the hospital. Information on staging and disease status after treatment was lacking in all four patients' records.

Conclusions: Mucoepidermoid carcinoma is a rare tumor in children with a clinical presentation similar to that of adults.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0130 / #908 | Delayed Diagnosis of Childhood Thyroid Malignancy in South Africa: A Case Series and Review of Literature

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Background and Aims: Thyroid cancer is rare in children, accounting for only 1-3 %. Nodules are uncommon, but when found, 26% harbour malignancy. Differentiated Thyroid Cancer (DTC) in children has higher risk of extrathyroidal extension, locoregional and distant metastasis than in adults. Only a few cases have been documented in South Africa.

Methods: Retrospective review of children treated at our Institution for DTC. Review and comparison with current literature.

Results: Three patients were managed at our Institution for DTC in the last 7 years. A 12-year old female presented with a 2-year history of a progressively enlarging neck mass with dysphonia and lymphadenopathy. A lymph node biopsy demonstrated a papillary thyroid carcinoma (PTC). Total thyroidectomy and a modified neck dissection were performed. (T3N1bM0). A 7-year old male was initially seen 2 years prior with a 3-year history of an anterior neck mass which was progressive in size but was then lost to follow-up. He re-presented 2 years later. FNA demonstrated atypical follicular cells of undetermined significance. A left thyroid lobectomy and isthmectomy was performed. Histopathology showed papillary carcinoma with lymph nodes invasion. A completion thyroidectomy was subsequently performed (T3N1bM0). A 18-month old presented to another hospital with a diffusely enlarged nodular goitre. Ultrasound showed multiple solid nodes. FNA showed atypical follicle cells and atypia of uncertain origin. She was referred only six months later. Total thyroidectomy was performed. Histopathology showed a papillary carcinoma and capsular invasion. (T3N1bM0). All three patients were treated with post-operative Radioactive Iodine.

Conclusions: The awareness of DTC in children is low both in the population and general practitioners. In our series, the patients presented at a later stage than what is reported in the international literature. High suspicion is crucial to achieve prompt diagnosis and treatment and, therefore, better outcomes.

Publication Only Topic: Disease Orientated / AS02 Solid Non Brain Tumours / AS02.h Rare Tumours and Histiocytosis

PO0131 / #988 | Spontaneous Regression of an Orbital Langerhans Cell Histiocytosis After Biopsy: A Case Report

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Background and Aims: Langerhans cell Histiocytosis (LCH) is a rare benign pathology representing less than 1% of orbital tumors. Eosinophilic Granuloma, is the most benign form and the predominant clinical presentation of LCH. Clinical course and prognosis are unpredictable but some cases of spontaneously resolved Langerhans cell histiocytosis have been reported especially in localized forms

Methods: In this clinical case report, it is described a case of an 18-year-old male patient with eosinophilic granuloma of the left orbital roof that showed a spontaneous regression after biopsy without any additional treatment.

Results: An 18 year-old patient had been referred to our center after discovering a left orbital mass. The physical examination found a tumefaction of the supra-external left orbital angle associated to left exophthalmia. The cerebral CT and MRI showed a superolateral extraconal tissular mass extending through the superior orbital fissure and along the dura mater, measuring 36.6mm x 32.7mm. A biopsy was performed. The histopathological examination disclosed clusters of Langerhans cells admixed with numerous eosinophils and osteoclast-like giant cells. Initially, orbital radiotherapy was indicated. Computed tomography was taken to plan radiation therapy, and had shown no infiltrative lesion or abnormalities of the orbital bone to delineate. Inside of this CT-MRI discordance, we performed an Orbital MRI which revealed no abnormalities compared with the first one. So, we decided to follow the patient without any additional treatment. Nine months after biopsy, the patient showed no evidence of clinical or radiological recurrence. The lesion had completely disappeared in the later orbital MRI, the one at 9 months. And even after a follow-up of four years, there was no clinical or radiological evidence of disease.

Conclusions: The presented case supports an expectant attitude given the possibility of a spontaneous regression after the biopsy, especially in small lesions.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0132 / #1051 | Paediatric Meningioma in Unusual Intracranial Locations: A Report of Three Filipino Cases

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Background and Aims: Meningiomas rarely occur in the paediatric population where they have higher rates of aggressive histopathology. No cases of paediatric meningiomas in the Philippines have been published.

Methods: We retrospectively reviewed three cases of paediatric meningiomas excised in a tertiary hospital in Manila. All cases presented with signs of increased intracranial pressure secondary to significant tumour burden with preoperative diagnoses other than meningioma.

Results: A ten-year-old girl presented with generalized seizures and right sided weakness. MRI showed a lobulated, left insular tumour, with a preoperative diagnosis was glioma. She underwent gross total excision of the tumour and histopathology revealed an atypical meningioma. A two-year-old boy presented with ataxic gait and left ptosis. His MRI showed a large sellar and left parasellar tumour consistent with a chordoma. Subtotal excision of tumour was done with histopathologic findings of a clear cell meningioma. A nine-year-old boy presented with profound hearing loss of the left ear and facial asymmetry, for which a cerebellopontine angle tuberculoma with petrous bone osteomyelitis was considered. After a gross total excision, histopathology revealed an intraosseous meningioma. None of the children showed stigmata of neurofibromatosis 1 or 2. Intraoperative consideration of meningioma mandated maximal safe excision in an attempt to prolong progression-free survival and delay adjuvant therapy.

Conclusions: Paediatric meningiomas present more variably, are located in unusual locations, and have atypical imaging findings compared to those in adults. Filipino paediatric patients encounter multiple delays in diagnosis and intervention with larger tumour size during initial consultation. Owing to their rarity, meningiomas are seldom included in the preoperative list of differentials for paediatric tumours; therefore, all involved clinicians should always keep them in consideration.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0133 / #1175 | Radiotherapy for Pediatric Ependymoma

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Background and Aims: Ependymoma is the 3rd most common central nervous system tumor, and half of these are below 5 years of age. The prognosis of ependymoma remains poor. Factors associated with improved survival rate were essentially : total tumor removal and low histological grade. Surgical excision followed by radiation therapy was the primary mode of treatment. The place of chemotherapy remains to be defined. Purpose :To describe epidemiological, clinical characteristics and therapeutic results of pediatric ependymoma in a Tunisian population.

Methods: Between 1995 and 2018, 8 pediatric ependymomas treated in the Department of radiation oncology Farhat Hached Hospital, sousse, Tunisia were retrospectively reviewed.

Results: There was 5 males and 3 females with a ratio of 1.66. The average age at diagnosis was 6 years (range, 1-14 years). An infratentorial origin was identified in most cases (62.5%). Four patients had anaplastic tumor. Gross total resections were performed in all patients. 3 patients died before starting radiotherapy. 3 patients had post operative radiotherapy and 2 patients received adjuvant chemotherapy and post operative radiotherapy. Patients received focal radiotherapy on the bed tumor with doses between 54Gy and 59.4Gy using two-dimensional radiotherapy (2D-RT) for 2 patients and three-dimensional conformal radiotherapy (3D-CRT) for 3 patients. Spinal irradiation was performed according to various prognostic features in 2 cases. Acute complication was mostly transient and tolerable. No major late radiation effect was found. After a Median follow-up of 57.6 months, 4 patients were in complete remission and 1 patient died.

Conclusions: In our study, the characteristics of pediatric ependymoma are generally similar to those reported in the literature, with only minor differences. The prognosis remains poor with 3 – 7 years disease free survival, DFS has been reported to be ranging from 30% to 61%.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0134 / #1182 | The Costly Scenario of Brain Neoplasms Among Children and Teenagers in Brazil and its Impacts on the Health System

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Background and Aims: Brain neoplasms are prevalent among children and teenagers in Brazil. They are aggressive tumors and, even in non-metastatic cases, they are able to compress brain areas responsible for various functions, leaving sequels for life or even death. The aim is to analyze the epidemiological profile and behavior of brain neoplasm hospitalizations in Brazil and point out their expenditures for the health system.

Methods: Data were collected using the Hospital Information System (SIH). The population of the study were children and teenagers from zero to 14 years old diagnosed with brain cancer from February of 2008 to February of 2021 in Brazil. The variables analyzed were: number of hospitalizations, health care expenses, age group (<1; 1-4; 5-9 and 10-14 years old) and hospitalization time.

Results: In the analyzed period, 3,168 hospitalizations due to brain neoplasm were reported among the study's population, which were responsible for \$42,022,809.60 in health care expenses. Through-

out the years, a 118.5% increase in the number of notifications was observed between 2008 and 2018, with an 18.5% decrease between 2019 and 2020. The total hospitalization time for these cases among the years was 25,756 days. The highest number of hospitalizations (66%) were registered among children between 1 to 9 years old, age group also responsible for 64.93% of the total hospitalization time.

Conclusions: Brain neoplasms are very incident among children and teenagers in Brazil and, if not diagnosed early, can cause lifelong problems or even death. The expansion of access to non-invasive diagnostic techniques allowed the recognition of a greater number of cases; however, the technological inequality between the different regions of the country raises the possibility of underreporting cases. Besides that, the small number of specialized services in pediatric oncology and neuropediatrics limit these children's access to early diagnosis and effective treatment.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0135 / #1256 | Pediatric Brainstem Tumors: Clinical and Therapeutic Aspects About 15 Cases

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Background and Aims: Pediatric brainstem gliomas are relatively rare tumors of the central nervous system which have varying presentations and clinical outcome. This study aims to analyze the frequency, clinical and therapeutic aspects of pediatric brainstem glioma.

Methods: We retrospectively analyzed the data of patients, managed for pediatric brainstem glioma, aged under 20 years between January 2007 and July 2020 in the radiation and oncology department of the Ibn Rochd UHC.

Results: There were 15 patients (10 males and 5 females). The mean age of onset was 12 years (range 8 -14 years). The duration of symptoms varied from 1 month to 2 years. 9 of the patients had an intracranial hypertension due to a hydrocephalus, 6 had cranial nerve deficits at presentation, while 5 patients had cerebellar signs. The lesion was pontine in 12 cases. none of the patients had a tumoral resection, nine had a ventriculo-peritoneal shunt insertion for the hydrocephalus and 3 had a Stereotactic biopsy that revealed one astrocytoma grade 1, one low grade glioma and one glioblastoma. The radiotherapy was indicated in all the cases but only nine patients had a 3D radiotherapy with a total dose of 54 Gy. 3 patients received chemotherapy. 6 patients are still alive, 2 are lost to follow up and 7 patients are dead with a mean survival period of 6 months.

Conclusions: Brainstem glioma is a devastating disease with a bad prognosis. The clinical presentation is variable and the management is multidisciplinary. Our study illustrates the importance of treatment by radiation.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0136 / #1338 | Pediatric Primitive Central Nervous System Tumors: A Retrospective Study in the Center of Tunisia

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Background and Aims: Primitive Central nervous system tumors (PCNS) are the most common solid tumors in children and the second leading cause of cancer death. The aim of the study is to describe clinical and therapeutic characteristics of our population.

Methods: We retrospectively reviewed 25 patients diagnosed with PCNS and treated at the medical oncology department FarhatHached Hospital from January 1997 to June 2020. Statistical analysis was made with SPSS20.0

Results: Median age was 6 years ranging(0-14) with female predominance(52.2%). The average time to consultation was 2 months(1-6) in majority with an intracranial hypertension syndrome(65%). The tumor was infra-tentorial in 78.3%, localised in the fourth ventricle and vermis in 66.6%. Six histological subtypes were found: medullblastoma(n=20), anaplastic astrocytoma(n=1), PNET(n=1), ependymoma(n=1), pinealoblastoma(n=1), AT/RT(n=1). Twenty one patients (84%) underwent surgery, of which 47% had incomplete resection. Radiotherapy (RT) was delivered to 18 patients at a mean dose of 54 gray(30-54) with concomitant chemotherapy (vincristine) on 8 patients. The mean delay surgery-RT was 4.6 months. Twenty one patients received adjuvant chemotherapy based mainly on VP16-carboplatin-cyclophosphamide-vincristine(52.3%). Five patients received "8 drogues-in-one-day". Ten patients relapsed after a median delay of 6 months(1-26), two patients were operated, four patients had CT and none received RT. After a mean follow up of 37 months, twelve were in complete remission, nine were dead and two were lost of follow-up. The 5-year-OS and RFS were respectively 48% and 44%.

Conclusions: Multimodal treatment improved survival. In our study, the recurrence rate seems to be higher and the OS poorer which can be related to incomplete surgery, poor resources and RT-delay. A primary challenge in the years ahead will be how to integrate biologic study into the care of children in our population with the hope that molecular biologically based therapy will be more effective.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0137 / #1370 | Radiotherapy for Pediatric Glioblastoma

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Background and Aims: Glioblastoma in children accounts for no more than 3-5% of primary central nervous tumor in children. In the absence of concrete evidence for adjuvant chemotherapy, maximal surgical excision followed by adjuvant RT (in children >3 years of age) remains the current best treatment strategy for these tumors. The aims of this study is to describe clinical and therapeutic features of pediatric glioblastoma.

Methods: Between 1995 and 2018 we identified six paediatric glioblastoma confirmed by pathological examination in radiation oncology department, Farhat Hached Hospital Sousse Tunisia.

Results: The median age was 10 years (6-16 years) with ratio 2. The main reason for consultation was often intracranial high pressure. Right temporal locations were the most frequent. We describe 04 cases of common glioblastoma; 01 glioblastoma with oligodendroglial component and 01 gigantocellular glioblastoma. All patients underwent surgical intervention. Two patients had gross total resection (GTR) and four patients had subtotal resection. After surgery, all patients received radiation therapy associated to adjuvant chemotherapy in three cases. After a median follow-up time of 25 months, (5 days-120 months), 2 patients were in complete remission, 3 patients died and 1 patient was lost of follow up. The median overall survival was 14 months.

Conclusions: Paediatric glioblastoma is rare but distinctly different biological disease compared with adults; prognosis in the majority of children is better than the adults which in turn may be explained by a different biological make up of these tumors, the place of the genetic and molecular analysis is more than crucial to better understand this affection.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0138 / #1423 | Pediatric Brainstem Glioma, A Tunisian Single Center Experience

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Background and Aims: Brainstem gliomas are relatively rare tumors of the central nervous system. They have various presentations and clinical courses.

The aim of this study was to analyze the clinical profile and therapeutic outcomes of these tumors.

Methods: A retrospective study of 8 patients diagnosed with brainstem glioma in the department of radiotherapy of Farhat Hached hospital between January 2005 and December 2018.

Results: The median age was 6 years (4-13 years). There were 4 girls and 4 boys. The diagnosis was made urgently in 3 patients, after 1 month in 4 patients and 2 months in one patient. The tumor was pontine for all the patients. The most common manifestations were ataxia, half body paralysis and headaches. Symptoms related to cranial nerve dysfunction were present in all patients. Diagnosis was confirmed by characteristic imaging on Magnetic Resonance Imaging. The average tumor size was 43 mm (6-55 mm). Contrast enhancement was commonly not prominent. The lesion appeared as hypo-intense on T1 weighted images and hyper-intense on T2 weighted images. On spectroscopy, marked elevation of choline with reduced N-Acetylaspartic acid was

observed. Hydrocephalus was present in 4 patients. Three of them had ventriculoperitoneal shunt placement.

Only 5 patients were treated with exclusive conformal radiotherapy (RT) while the others died before treatment.

The median RT dose was 54 Grays (Gy) (32,4-54 Gy). One patient died under treatment. A partial response was observed in three patients. There was one case of clinical and radiological progression treated by salvage radiotherapy at dose of 18 Gy. After a median follow up of 10 months (4-42 months), 7 patients died and 1 patient was clinically and radiologically stable.

Conclusions: Brainstem tumors are aggressive with poor prognosis due to challenges in their management.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0139 / #1452 | Late Complications of Radiation Therapy in Children with Brain Ependymomas in Remission for More Than 3 Years

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Background and Aims: One of the important problems in the management of paediatric brain ependymomas (BE) is the development of late treatment complications. In this study we analyzed the late sequelae of BE treatment in patients who have been in remission for more than 3 years.

Methods: Of about 300 children treated in Russian Scientific Center of Roentgen Radiology for BE in 2000-2018, 50 had durable remission for a period of 3 to 12 years. All of those 50 patients previously, after surgical treatment, had received a single course of adjuvant local radiation therapy (RT) to a total focal dose of 54-55 Gy. The median age of the patients at the diagnosis of BE was 6.5 years. During RT, monotherapy with Vincristine was used. We studied the incidence of late radiation sequelae, according to the follow-up examinations of the patients.

Results: The following types of radiation-induced complications were identified: ototoxicity, endocrine disorders, growth disturbances, signs of encephalopathy (deterioration of learning performance, memory impairment). In 4 children (8 %) none of such complications were noted. In the remaining 46 (92%) there were various signs of radiation sequelae. Ototoxicity was observed in 3 patients (6%). 10 children from the study population (20%) do not attend school, although they have reached school age. The remaining 40 (80%) study at school, of which 23 (58%) have learning difficulties. 2 patients (4%) exhibit considerable memory problems, both children do not learn. 13 patients (26%) have such endocrine disorders as hypothyroidism, hypopituitarism, hyperprolactinemia, somatotrophic hormone insufficiency. Growth retardation was observed in 10 patients (20%).

Conclusions: The majority of patients in long-term stable remission after treatment for BE have various late radiation-induced sequelae, which reduce their quality of life. Further optimization of radiotherapy techniques should reduce the dose to the risk organs and thus decrease the risk of radiation damage.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0140 / #1501 | A Unique Case Report of Infantile Gliomasarcoma with PTR-NTRK1 Fusion Treated with Larotrectinib

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Background and Aims: Apresentamos aqui um caso raro de um bebê de nove meses com diagnóstico de gliossarcoma (OMS grau IV) no lobo frontal esquerdo.

Methods: Descrevemos o diagnóstico clínico, características histológicas, características radiológicas, aspectos moleculares e tratamento deste tumor raro. Uma análise analisada molecular foi realizada no tecido tumoral, que concluiu uma fusão do gene TPR-NTRK1

Results: O paciente foi tratado com um inibidor de TRK, larotrectinibe, e apresentou uma doença estável com lesão residual após três meses de terapia alvo.

Conclusions: O primeiro relato de um gliossarcoma infantil com rearranjo NTRK1 tratado com larotrectinibe.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0141 / #1557 | Epidermal Growth Factor Receptor (EGFR) C.787A>C Variation in a Child with H3K27M-Mutant Diffuse Midline Glioma

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Background and Aims: We present a H3.1 K27M c.83A>T (HIST1H3C p.Lys28Met), ATRX c.2169_2170del (p.Glu723AspfsTer9), TP53 c.338T>C (p.Phe113Ser), and EGFR c.2300_2308dup (p.Ala767_va1769dup) were detected in the genetic assessment of tumor tissue in a child with H3K27M-mutant diffuse midline glioma. Especially, detected EGFR c.2300_2308dup (p.Ala767_va1769dup) is extremely rare in children to our knowledge.

Methods: A ten-year-old girl presented with sudden onset left sided hemiparesis, headache, and ataxia. The patient's medical and family histories were unremarkable. Physical examination was normal except for left sided hemiparesis and ataxia. A hyperintense mass lesion involving the bilateral thalamus was detected in the axial T2-weighted and coronal FLAIR sequences on brain MRI. Due to the size and location of the tumor, the patient could not be operated on and was considered inoperable. Intended intensity modulated radiotherapy was intended for curative treatment to the patient because the radiological findings suggested a low grade glial tumor. The performed dose for primary radiotherapy was 50 Gy, 25 fractions, 5 fractions a week. After radiotherapy, curative surgery was planned, but only biopsy could be performed. As a result of the pathological examination, the high grade glial tumor was diagnosed. A chemotherapy protocol were instituted to the patient. After the third cycle chemotherapy, the mass was found to be progressed radiologically.

Results: H3.1 K27M c.83A>T (HIST1H3C p.Lys28Met), ATRX c.2169_2170del (p.Glu723AspfsTer9), TP53 c.338T>C (p.Phe113Ser), and EGFR c.2300_2308dup (p.Ala767_va1769dup) were detected in the genetic assessment of tumor tissue. The patient's treatment was re-arranged. Unfortunately, MRI showed progression after three cycles of chemotherapy. So, the patient's family refused treatment.

Conclusions: Molecular biologic studies should be performed in children with HGG tumors, particularly in bithalamic locations. In the light of these findings, it should be kept in mind that if there is a possibility

of targeted therapy, there may be a treatment option in this malignant disease with a poor prognosis.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0142 / #1604 | Intracranial Fibromyxoid Sarcoma: A Case Report

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Background and Aims: Low-grade fibromyxoid sarcoma (LGFMS) is a soft tissue tumor of mesenchymal origin. Primary sarcomas of the central nervous system (CNS) are rare, accounting for only 0.7% of all sarcomas, with an estimated incidence of 3 per 10 million people per year.

Methods: We present a case report of a 15-year-old female patient with an intracranial fibromyxoid sarcoma.

Results: We present a case of a 15-year-old female patient with a history of migraine medicated with topiramate, who presents on control MRI a space-occupying hyperintense T2 lesion in the vermian region with significant post-contrast enhancement and perilesional edema, with a spine MRI negative for metastases. Follow-up of the lesion was carried out due to its small initial size, showing lesion growth. Complete resection surgery was performed, which reported a predominantly myxoid proliferation including spindle cells, Ki67 of 10%, and positive immunohistochemistry for AML, desmin, vimentin, and MUC4, being compatible with low-grade fibromyxoid sarcoma. A negative PET-CT scan study was completed to rule out peripheral involvement. The patient continues to have migraine crisis eventually and remain under imaging control.

Conclusions: Less than 10 pediatric cases are described in the literature. The treatment of choice continues to be surgical excision, due to its rarity in the CNS, current recommendations for adjuvant therapy for intracranial lesions remain unclear.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0143 / #1640 | Papillary Tumor of Pineal Region Rare Pediatric CNS Tumor Case Series Treated in King Fahad Medical City KPMC

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Background and Aims: Papillary tumour of the pineal region (PTPR) is very rare neuroepithelial tumour of children first introduced in the World Health Organization classification of CNS in 2007 and classified as a WHO Grade II-III tumor the clinical behavior, prognosis, and appropriate treatment of PTPR in pediatrics not fully defined due to rarity of this tumors and limited numbers of reported pediatric cases in literatures with varieties of therapeutic modalities

Methods: Here we report 3 pediatric cases all were females treated in King Fahad Medical City the histological and immunohistochemical diagnosis was confirmed by analysis of genome-wide DNA methylation each case has its own clinical behavior which reflect tumor heterogeneity and lack of a consensus on the proper therapeutic and management plane,

Results: The first case presenting with spinal leptomeningeal dissemination with excellent tumor control with radiation therapy alone and continued further reduction of the tumor volume over time The second patient had aggressive course with recurrent local tumor growth despite radiation therapy and repeated surgical resections and she responded well to Pineoblastoma like intensive chemotherapy While The third patient remain stable without local recurrence after first total surgical resection indicating that surgery

Conclusions: This case series describes role of multimodal treatment in 3 pediatric patients with PTPR. The proper therapeutic strategy is still not clearer and the clinical behavior of PTPRs can be compared to that of ependymoma gross total resection is the mainstay of treatment and can be followed by Surveillance neuroimaging but we recommend that adjuvant therapies like radiation with or without or chemotherapy should be considered when complete surgical resection is not feasible.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0144 / #1657 | A Case of Polymorphous Low-Grade Neuroepithelial Tumor of the Young with Polysomy Of Chromosome 1 and 19 Lacking Braf Mutation

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Background and Aims: Polymorphous low-grade neuroepithelial tumor of the young (PLNTY) is a recently described entity comprising low-grade neuroepithelial tumor of the central nervous system, especially in children who exhibit epileptogenic tumors. PLNTY is characterized as oligodendroglial-like in appearance, heavy calcification, and CD34 expression. As described in literature, PLNTY often express BRAF-V600E mutations or fusions in the fibroblast growth factor receptors 2 and 3.

Methods: This is a descriptive case report of a 16-year-old previously healthy male who presented with a 6-month history of unresponsive, staring episodes associated with an aura. Imaging revealed 1.5 X 3.0 X 1.7 cm calcified, multiloculated cystic lesion involving the inferior right temporal lobe. Clinical features, characteristic microscopic findings, molecular profiling, and outcomes were reviewed.

Results: In this report, we described an adolescent male who underwent complete tumor resection for findings of a right temporal lobe brain neoplasm. Pathological examination revealed PLNTY. Histopathology was positive for CD34, glial fibrillary acidic protein (GFAP), and OLIG2 expression. The patient was negative for 1p or 19q deletion however exhibited polysomy of chromosome 1 and 19. On the contrary, fluorescence in situ hybridization (FISH) did not identify BRAF rearrangement associated with MAPK pathway mutations. The patient has not had further seizure-like activity following resection of the neoplasm.

Conclusions: This case demonstrates findings of PLNTY with expression of GFAP and OLIG2 favoring oligodendroglial-like characteristics. Our findings of CD34 expression supports the association between PLNTY and other tumors associated with epilepsy in children. Contrary to previous reported cases, our patient's findings did not reveal BRAF-V600E rearrangement. Rather, polysomy of chromosome 1 and 19 was expressed, which could suggest tumor hyperploidy. Next-generation sequencing with FoundationOne is currently pending.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0145 / #214 | Rapid Progression of Pediatric Pilocytic Astrocytoma after Discontinuing Molecularly-Targeted Therapy and Response to Restarting Targeted Therapy

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Background and Aims: Non-resectable pediatric low-grade gliomas (PLGG) pose significant therapeutic challenges. Progressive disease (PD) is common despite cytotoxic chemotherapy. Targeted therapies against BRAF mutations and fusions are promising, although drug selection, treatment duration, and response durability are not well-characterized. We describe 3 children with progressive PLGG who experienced sustained responses to extended BRAF/MEK inhibition but experienced PD following discontinuation of therapy

Methods: Comprehensive medical record review, panel-based tumor genomic evaluation, and literature review

Results: Patient-1 presented at 7-years-old with headaches and vision loss due to a hypothalamic-chiasmatic PLGG. He received carboplatin/vincristine (C/V) for 6 months, developed carboplatin allergy, then received temozolomide for 6 months. MRI showed stable disease. Biopsy following PD 5 years later revealed a KIAA1549-BRAF fusion-positive pilocytic astrocytoma (PA). Trametinib therapy yielded partial response (PR) on MRI. Trametinib was discontinued after 12 months. Twelve months later, MRI showed PD. Trametinib was restarted. Significant clinical-radiographic improvement continues 18 months later. Patient-2 presented at 3-years-old with vision loss due to a hypothalamic-chiasmatic PLGG. She received C/V and then bevacizumab-vinblastine therapy. MRI showed PD. Biopsy revealed a BRAFV600E mutation-positive PA. Twenty-three months of vemurafenib therapy resulted in a sustained PR. However, 4 weeks following drug discontinuation MRI showed PD. Dabrafenib/trametinib therapy was started and clinical-radiographic improvement continues 12 months later. Patient-3 experienced progressive, severe central apnea from 2 to 5-years-old. Biopsy of a large medullary mass revealed a BRAFV600E mutation-positive PA. She experienced life-threatening PD despite C/V and radiation. Vemurafenib therapy resulted in sustained PR for 36 months. However, 6 weeks following drug discontinuation, she experienced PD. Dabrafenib/trametinib therapy was started with rapid clinical-radiographic improvement that continues 12 months later.

Conclusions: Targeted BRAF/MEK inhibition can yield robust durable responses, but rapid symptomatic PD may occur following drug discontinuation. Re-initiation of BRAF/MEK inhibition can be effective. Optimal treatment duration remains undetermined. Continued surveillance is imperative.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0146 / #256 | Optic Glioma in Children: Single Center Experience

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Background and Aims: Optic nerve glioma (or optic glioma), a form of glioma which affects the optic nerve. They are usually pilocytic tumors. Optic gliomas are associated with neurofibromatosis type 1 in 30% of patients. Optic pathway/hypothalamic gliomas account for 4-6% of all brain tumors in children and 30% of all pediatric gliomas. Optic nerve gliomas have low mortality but extremely high prevalence of vision loss and exophthalmos in children. Currently, the main goal of treating optic gliomas is to preserve the patient's visions for as long as possible. Our aim is retrospectively evaluate the main characteristics and prognosis of optic glioma patients diagnosed and treated in our clinic

Methods: Forty five patients diagnosed with optic glioma in our pediatric oncology clinic during last ten years (2010-2020) were retrospectively evaluated.

Results: There were 26 girls and 19 boys and the mean age was 5 years. Twenty three of the patients (49.5 %) had NF-1. Optic gliomas of the patients with NF were generally at the anterior localisation (85% were prechiasmatic or chiasmatic) in contrast to the 75 % cases without NF whose gliomas were located at posterior region. In ophthalmological examination there were diminished vision and/or visual field defect in 18 (40%) cases. Chemotherapy was used in 23 (51%), surgery in 8 (17.7%), and radiotherapy in 4 (8.8) cases. 7 (15.5 %) patients were relapsed, but they had complete response after the second line chemotherapy. One patient died because of infection and 2 patients lost of follow up. 42 patients are in follow up in complete response. 6 (13.3 %) of the cases have diminished vision or visual field.

Conclusions: The survival and visual prognosis is promising in optic glioma patients when they are followed up and treated in experienced centers.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0147 / #652 | Reirradiation for Recurrent Pediatric Pontine Glioma : A Case Report

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Background and Aims: Pontine glioma is a rare and fatal brain tumor that occurs predominately in the pediatric demographic. Initial radiation therapy is the only proven intervention that has been able to affect survival in these patients. Even so, median survival is limited to approximately 9–12 months, with 2 year survival estimates as low as 10 %. Progression is frequently the evolution of these tumors. The concept of reirradiation at first pontine glioma progression has in past years emerged as a novel intervention with potential to improve clinical course and survival in pediatric brain tumors.

Methods: We report our first case of reirradiation for recurrent pediatric pontine glioma in Salah-Azaiez Institute.

Results: We report a case of an 8-year-old female diagnosed with pontine glioma in January 2020. Conformational radiotherapy of the brainstem was performed at the dose of 50.4 Gy. At the control MRI, a favorable outcome was noted with a decrease in tumor volume of 95%. Eleven months later, a systematic control motivated by a cerebellar syndrome revealed a progression of the tumor process about 37% with infiltration of the basal trunk, left cerebral peduncle and small supra tentorial hydrocephaly. Our patient received reirradiation with the Intensity modulated radiotherapy technique at a dose of 24 Gy in 12 fractions. Throughout the treatment, we witnessed some progress in the patient's case : At a dose of 8 Gy, the patient showed an improved mobility of her upper right limb as well as an improved muscle tone. At a dose of 20 Gy, the strabismus condition has improved and the ptosis

disappeared. Most importantly, at this dose, the patient started to walk. We did not note any acute toxicity.

Conclusions: Reirradiation appears to be one of the approaches that represents a step in the right direction to improve the prognosis of this disease.

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PO0148 / #765 | Primary Intracranial Germ Cell Tumours

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Background and Aims: Intracranial germ cell tumours(GCT) are rare in children. They are a heterogeneous group of neoplasms that show different clinical manifestations despite having a common origin. The peak incidence is during the second decade of life, median age at diagnosis of 10 to 14 years. There is a male preponderance. In the World Health Organization classification system, intracranial GCTs are divided into germinomas and nongerminomatous GCTs(NGGCTs). NGGCTs include embryonal carcinoma, endodermal sinus tumor, choriocarcinoma, teratoma and mixed tumors. Germinomas comprise 60 to 65 percent of all pediatric intracranial GCTs. Germinomas are highly chemo-radio sensitive, and therefore generally have a good prognosis, in contrast to NGGCTs. Our aim is to retrospectively analyse the characteristics, prognosis and prognostic factors diagnosed with intracranial germ cell tumours.

Methods: A retrospective analysis was carried out on the epidemiological and histological characteristics, clinical manifestations, and outcomes of 14 patients diagnosed with intracranial germ cell tumours in Marmara University Hospital during 2010 to 2020.

Results: A total of 14 patients were identified: 10 boys and 4 girls(M/F= 2.5). The mean age was 15.5 years(range 6.5-17 years). Histological confirmation of the diagnosis was obtained in all of the patients: 8(57%) were pure germinoma, 6(42,8%) were mixed germ cell tumours. The locations were 7 pineal(41%), 6 suprasellar(35%) and one globus pallidus. Main clinical symptoms were headache(79%), vomiting(77.7%) in suprasellar tumours, polydipsia and polyuria(100%) in pineal tumours. At diagnosis, 2(14%) patients had high serum and CSF AFP, 4(28,5%) had serum beta-hCG levels. Six(42,8%) of the patients were diagnosed with biopsy, and 3 subtotal, 2 near total and 3 with total excision. All of the patients received radiotherapy, and chemotherapy. Twelve(85,7%) of the patients are alive and in complete remission. They have no morbidity.

Conclusions: Intracranial GCT are generally germinomas. They are chemo-radiosensitive tumours and have good prognosis when treated and followed at experienced oncology centers.

Publication Only Topic: *Disease Orientated / AS03 Brain Tumours*

PO0149 / #806 | High Grade Glioma of Central Nervous System: Single Center Treatment Experience

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Background and Aims: To evaluate characteristics and treatment responses of patients with high grade gliomas(HGG) in our center.

Methods: Medical files of patients with malignant CNS tumors between 1987-2020 were analyzed retrospectively. There were 44 patients with HGG.

Results: Diagnosis of patients as follows: 21 pons glioma, 2 anaplastic astrocytoma, 11 anaplastic ependimoma, 7 glioblastoma multiforme, 1 glioblastoma, 2 gliomatosis cerebri. The median age at diagnosis was 6,5 yrs(7-17yrs), M/F:25/19. Age distribution: <5 yrs 12 patients, 5-10 yrs 18 patients, 10-18 yrs 14 patients. Surgery was performed to extrinsic component of mass in 3 patients of pons gliomas. For other HGG: 7 subtotal resection and 16 gross total resection had performed. Seven patients died before RT, other 37 patients received radiotherapy. Seven patients were not received chemotherapy, 3 of them died before chemo, and others received only RT. Chemotherapy protocols were changed over the years: For pons gliomas, MOPP in 3 patients, NOPP in 2 patients, only ICE in five patients, cyclophosphamide in one patient, temozolamide in one patient and temo+bevacizumab in one patient. Last 3 patients received nimotuzumab-vinorelbine after ICE and temo. For other HGGs, platin based regimens used for the first line treatment. Temozolamide-bevacizumab, irinotecan-temsirolimus combinations were the other options as second and also third line treatments. Median progression free survival time was 6 mos(2 weeks -25 mos) for pons gliomas, for other gliomas median progression free survival time was 14 mos(0 -74 mos). For pons gliomas: EFS rate for 6 mos was 75%, for one year was 17%, one year, 18 mos, and two years overall survival rates were 84%, 52% and 10% respectively. For other HGGs: EFS rate for one year and two years were 57% and 17% respectively. One year and two years overall survival rates were 73% and 36% respectively.

Conclusions: High grade glioma is a group of tumor in which still the helplessness experienced in treatment. Despite radiotherapy and chemotherapy, prognosis is very poor. The progression free and overall survival rates of patients were similar to literature. With new developments in molecular pathology, as the use of molecular target therapies, the progression free survival rates newly will improve.

Publication Only Topic: Disease Orientated / AS03 Brain Tumours

PO0150 / #951 | Integrating Saffron Metabolomics into Medulloblastoma Treatment

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Background and Aims: Crocus sativus (saffron) is endowed with anticancer, differentiating, antioxidant and genoprotective properties. Its pharmacological properties are attributed to crocins (CRCs) which are hydrophilic carotenoids. They are glycosylated esters of the amphiphilic natural carotenoid crocetin (CRT). The hemisynthetic derivative dimethylcrocetin (DMCRT) is similar to CRT and is soluble in many organic solvents. Saffron and its carotenoids are practically non-toxic substances. The current research aims to study the in vitro cytotoxic effect of the natural carotenoids CRCs and DMCRT on medulloblastoma.

Methods: CRCs are diluted to water and DMCRT to 10% DMSO. For the biological assays TE-671 medulloblastoma cells are incubated at a range of concentrations of CRCs (0.18-22.85 mg/ml) and DMCRT (0.18-11.43 mg/ml) for 24, 48 and 72 hours. Analysis of cell viability is performed with Alamar Blue and MTT viability assays.

Results: CRCs manifest a cytotoxic effect in a dose and time-dependent manner; as their concentration increases, cell viability is reduced for all time points. For the same concentration, as the time of exposure increases, the inhibitory effect is increased. The antiproliferative effect of DMCRT is less pronounced, observed only at concentrations higher than 5.71 mg/ml at 24 and 48 hours and higher than 1.43 mg/ml at 72 hours, whereas for lower concentrations the effect is not statistically significant. IC50 values for each time point are calculated as 3.230, 2.14 and 1.72 mg/ml for CRCs and 5.9, 5.6 and 3.712 mg/ml for DMCRT at 24, 48, and 72 hours, respectively.

Conclusions: The results of our study could afford the basis of research, regarding the use of natural carotenoids as an alternative solution to the toxicity of retinoids, while retaining the anticancer properties of retinoids. Acknowledgements: The research is co-financed through the Program "Support for researchers with an emphasis on young researchers-cycle B" (MIS 5048464) AND The Foundation for Education and European Culture.

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PO0151 / #978 | Second Primer Tumor after Burkitt Lymphoma: Medulloblastoma

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Background and Aims: Survival rates of children with cancer are constantly improving. We present the first case has CHEK2 mutation, second primer brain tumor medulloblastoma that developed after burkitt lymphoma.

Methods: Patient was 5 years old, was admitted to the hospital for the first time with complaints of abdominal swelling, walking disorder, foot pain, weight loss and right renal mass was detected by USG. 16x14 cm hard, smooth-surfaced mass in the right renal lodge extending to the pubic region were detected. On MRI, the mass is observed in the right half of the abdomen. Biopsy result was interpreted Burkitt lymphoma. AA-BB-CC-AA-BB-CC protocol was given to the patient with diagnosis stage 3 burkitt lymphoma. PET-CT performed after the first AA and BB cure was complete remission. Six months after diagnosis, his treatment was completed. He did not have any complaints or findings during the controls. After 3 years later, cranial MRI was taken due to intermittent and non-severe headache and a solid enhancing mass lesion with diffusion restriction was observed in the fourth ventricle. For the mass lesion in the ceiling of the fourth ventricle, the patient was consulted to neurosurgery and surgery was performed. In the pathological examination of the material taken, medullablastoma was evaluated histologically as classical type, grade 4.

Results: We found in literature, medullablastoma following Hodgkin lymphoma in 1 case and following non-hodgkin lymphoma in 1 case (lymphoblastic lymphoma). This is the first case in the literature as a medullablastoma develops after burkitt lymphoma. It was found that CHEK2 gene mutation was associated with brain tumor in only 1 patient in pediatric age group. Our patient is second.

Conclusions: The risk of secondary malignancy has increased throughout life after treatment in childhood cancers. Because of the sensitivity of tissues to cytotoxic treatments, increased survival and cancer tendency, secondary malignancies can be observed frequently.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PO0152 / #1013 | Twisted Giant Epiploic Cystic Lymphangioma in 3 Years Old Boy: Case Report

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Background and Aims: Lymphangiomas are benign tumors of pediatric population. Twisted lymphangioma is a very rare entity.

Methods: We are reporting a case of three years old boy with a giant twisted epiploic lymphangioma.

Results: A three years old boy with no past medical history presented to our emergencies with acute abdominal pain, fever and vomiting evolving since few hours. Physical examination showed stable hemodynamic with diffuse abdominal tenderness. Ultrasonography and CT scan exhibited a huge intraperitoneal cystic mass repressing the intestines without invasion. Their content was echogenic, concluding to cystic lymphangioma complicated by hemorrhage. The child was operated. This revealed a huge 20cm epiploic multilocular cystic lymphangioma. This mass arises pedunculated from the omentum and occupies the lowest abdomen without adhesion or invasion of the nearby structures. We noted two loose twists on the pedunculated origin of the mass, probably responsible for the acute abdominal pain. A total resection of the mass was done easily by ligating the origin of the tumor from the omentum. No intestine resection was needed. The liquid contained in the stalls was seromematic. Post operative follow up was uneventful and the child was discharged within five days.

Conclusions: Twisted epiploic cystic lymphangiomas are extremely rare. They can be revealed by abdominal acute pain caused by torsion. Pre-operative diagnosis is not always easy due to the rarity of the disease. The optimal treatment is surgical resection of the tumor avoiding bowel resection as much as possible. A complete resection guarantees no recurrence in most cases.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.a Surgery (IPSO)

PO0153 / #1016 | Atypical Presentations of Wilm's Tumor in Children: About 3 Different Cases

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Background and Aims: Nephroblastoma is one of the most frequent neoplasms in children. It commonly occurs in early childhood and manifests with abdominal mass. Thus, clinical presentation may be misleading, delaying the diagnosis and the management of the disease.

Methods: We are reporting three cases about patients presenting with nephroblastoma, in which diagnosis was not clear due to atypical clinical presentations.

Results: The first child was a 5 years old male with no medical history. He presented with abdominal pain, flaccid paralysis of the lower limbs, dysuria and anal incontinence. Radiological exploration showed a giant retroperitoneal mass of 20 cm with epidural extension, spinal cord compression, pulmonary and vertebral metastasis. Neuroblastoma was first suspected. Core needle biopsy was performed and histological analysis concluded to nephroblastoma. The second was a 4 years old boy with no medical history who presented to emergencies with acute

abdominal pain and fever with vomiting. Abdominal tenderness was extreme and there was leukocytosis and high CRP level in laboratory studies. Appendicular Peritonitis was the suspected diagnosis. Thus, ultrasonography showed a huge retroperitoneal mass with peritoneal effusion. CT scan confirm the diagnosis of ruptured nephroblastoma. The third patient was a 16 years old girl with no medical history who consults for recurrent left flank pain. She was diagnosed with a huge left renal tumor of 17 cm. renal cell carcinoma was first suspected. CT-guided core needle biopsy was performed and histological analysis concluded to nephroblastoma.

Conclusions: Clinical presentation of Wilm's tumor varies in patients. It may present as neuroblastoma with neurological manifestations, or acute abdominal pain with fever in a young infant with no medical history, as it may occurs in older children. Diagnostic process should always be well planned to avoid diagnostic traps that may lead to wrong management.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.b Radiation Oncology (PROS)

PO0154 / #446 | Percutaneous Treatment of Chest Wall Chondroid Hamartomas: The Experience of Bambino Gesù Children's Hospital

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Background and Aims: Mesenchymal hamartomas of the chest wall are rare benign lesions. Rarely symptomatic, they can cause compression on pulmonary parenchyma, leading to respiratory distress. Although spontaneous regression has been documented, the more common outcome is progressive growth. The treatment of choice is "en-bloc" excision of the involved portion of the chest wall, frequently leading to significant deformity. The aim of our study is to describe percutaneous techniques (Cryoablation, Radiofrequency thermoablation - RFT - and MicroWave thermoablation - MWT) to treat chest wall chondroid hamartomas.

Methods: Data of patients treated from 2005 to 2020 at Bambino Gesù Children's Hospital were collected. Different percutaneous techniques were adopted: RFT (12-minutes cycles at 80°C), MWT (7-minutes-cycles at 45W alternated with 5-minutes-cycles at 75W) and Cryoablation (2 cycles of 10 minutes at -40°C and 5 minutes of thawing).

Results: Five patients (median age 11.5 months; range 1-168.7) were treated for chest wall hamartomas; one patient showed bilateral location of the mass. All of them underwent CT-guided biopsy to confirm the diagnosis. Two patients underwent RFT, one MWT and two

cryoablation; one of the latter was shifted to RFT due to mass volume increase after 1st procedure. The median reduction of the volume of the tumors was 76.9% (range 13.9-96.3%). The patient treated with MWT showed volumetric increase of the mass and underwent surgical removal of the tumor. No major complication was reported.

Conclusions: Percutaneous ablation is technically feasible for expert radiologists and may represent a valid and less invasive treatment for chest wall chondroid hamartoma, avoiding skeletal deformities.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.c New Drugs/Experimental Therapeutics

PO0155 / #739 | Analysis of Our Molecular Profiling Results in Patients with Relapsed / Refractory Tumors

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Background and Aims: Today, by determining specific genetic changes in tumors, it is possible to give individualized treatment specific to each patient. However, the effectiveness of these treatments in childhood cancers is still controversial. Here, we present the genetic changes detected in the tumor tissues of relapsed / refractory patients in our clinic and the treatments initiated due to these genetic changes.

Methods: DNA isolation was performed from paraffin-embedded tumor tissues of the cases. The appropriate samples were subjected to clonal amplification and sequencing, and next-generation sequencing was performed. The pathogenicity of the determined variant and copy number changes (CNV) was evaluated.

Results: A total of 25 patients evaluated. There were 13 females(52%) and 12 males(48%). Median age was 6,5 years(2,5-16,5 years). 16 of those were diagnosed as CNS tumor consisting of medulloblastoma(n=2), low grade astrocytoma(n=4), high grade astrocytoma(n=8), ependymoma(n=1), ATRT(n=1). The remaining 9 patients were diagnosed as rhabdomyosarcoma(n=5), Wilms tumor (n=1), hepatoblastoma(n=1), Ewing sarcoma(n=1), alveolar soft part sarcoma(n=1). Pathogenic and/or possible CNV was detected in seven patients with CNS tumors. Deletion of NF-1 was determined in 3 patients. Significant variants such as SMARCA4(n=3), ARID1A(n=3), KMT2D(n=2), MSH(n=2), EGFR(n=1), FGFR3(n=1), MTOR(n=1) mutations were identified. Pathogenic and/or possible pathogenic variants and/or CNV were defined in 6 patients without CNS tumor. NF deletion(n=3), and MET amplification(n=1) were specified. Significant variants such as PTCH1(n=2), PIK3CA(n=1), SMARCA4(n=1), KRAS(n=1), RAC(n=1), MAX(n=1) were determined. One of the patients diagnosed as relapse rhabdomyosarcoma with KRAS and NF-1 mutation received trametinib, whereas anaplastic astrocytoma

with EGFR mutation was treated afatinib. Crizotinib was given to the relapse rhabdomyosarcoma with MET amplification. The patient with Wilms tumor, including RAC and MAX mutation, was medicated with regorafenib.

Conclusions: In evaluating tumor specimens of patients, numerous tumor mutations were recognized. However, cause of the lack and limitations of the studies about targeted therapies in children, each patient could not receive a specific drug. The treatment's validity and benefit can only be discussed with further studies.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PO0156 / #1080 | A Case Report of Infantile Leukemia with Co-Infection of Covid -19 and Invasive Pulmonary Candidiasis: Challenging The Treatment Norm

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Background and Aims: To highlight treatment challenges in infantile leukemia with co-infection of COVID-19 and Invasive pulmonary Candidiasis

Methods: Retrospective review of case record.

Results: An infant with MLL rearrangement positive B ALL presented with febrile neutropenia on day 6 of induction chemotherapy. On examination she was sick with tachycardia and tachypnoea. She had basal crepitations and hepatosplenomegaly. Her initial laboratory investigations revealed hemoglobin of 7.5 WBC 710 and platelet count of 61,000 with a high CRP 70. In view of prolonged neutropenia prior to diagnosis a screening galactomannan and fungal cultures were also sent. Due to recent international travel and a positive contact, RTqPCR for COVID-19 was asked for which came in positive within 24 hours of fever. Despite broad spectrum antibiotics, worsening clinical parameters and a positive Galactomannan intravenous amphotericin was started. Day 10 of induction the child had worsening tachypnoea and required oxygen. Her Blood cultures grew Candida parapsilosis and CT Chest revealed left lower zone fungal cavity with bilateral patchy consolidation. The inflammatory parameters worsened, CRP 87, NTPro - BNP 582, ferritin 2723, and D-dimer 1.95. She was started on IV Remdesivir, Ivlg 400 mg per kg for 3 days and IV micafungin based on culture susceptibility reports. She continued her chemotherapy schedule as per BFM 2002 protocol, however did not receive Day 8 Daunomycin. Gradually clinical and hematologic parameters improved by day 18 of induction. She received six weeks of antifungals. Repeat CT was suggestive of resolving consolidation and child achieved morphological remission at the end of induction chemotherapy.

Conclusions: COVID-19 might be an independent risk factor for invasive pulmonary fungal infections. Early diagnosis of invasive fungal infections allows prompt specific treatment which optimizes the management of underlying leukemia.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PO0157 / #1253 | Foot Reflexology in Paediatric and Young Adult Oncology and Haematology

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Background and Aims: Reflexology is a type of massage derived from traditional Chinese medicine that considers the feet as a representation of the body, in which different amounts of pressure are applied to stimulate or ease the corresponding part of the body. It can be used as a complementary therapy in order to improve treatment side effects.

Methods: The aim of this retrospective unicentric study was to describe the management of foot reflexology in our paediatric oncology and haematology unit between 2018 and 2020. One session lasts 15-30minutes and is provided by a single reflexologist, trained by the French Federation of Reflexology.

Results: 893 sessions were realized in 240 patients (3.7 sessions/patient). Among them, 116 were less than 15 years old, and 124 were aged between 15 and 25 years old. 106 had a solid tumor, 127 had a haematological malignancy, and 7 a benign haematological disease. Reflexology sessions were indicated for anxiety in 62.9% of cases, nausea and vomiting in 15.0%, intestinal transit disorders in 9.7%, pain in 9.2% and sleep disorders in 3.2% of cases.

Conclusions: This study describes the important place of foot reflexology in children and young adults patients treated for oncological or haematological diseases, presenting various symptoms. A prospective study is in progress to evaluate the effectiveness of this complementary therapy on target symptoms.

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PO0158 / #1276 | Description of Sophrology Activity in a Paediatric Oncology Unit

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Background and Aims: Sophrology and complementary therapies are increasingly developed as supportive care in paediatric oncology, and several French unit use it in their practice. Sophrology is a self-development method involving body and mind, and can be prescribed in case of stress, anxiety, sleep disorders, pain and other symptoms, sometimes combined with hypnoanalgesia.

Methods: This retrospective monocentric study describes the management of sophrology in our paediatric oncology unit in 2020. Session are prescribed by doctors and are provided by a single sophrologist.

Results: 504 sessions were realised for 170 patients (3.0 sessions/patient) treated for leukemia or solid cancer. Indications were relaxation in 46.7%, anxiety in 24.0%, pain in 7.9%, stress in 6.2%, and other reasons in 15.2% of cases. 16.7% of sessions were combined with hypnoanalgesia, and 83.3% were sophrology alone, among them a third were realized as a pair with foot reflexology.

Conclusions: This study describes the important place of sophrology as a complementary therapy in a paediatric oncology unit. A prospective study is in progress to evaluate the effectiveness of sophrology on target symptoms.

Publication Only Topic: Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care

PO0159 / #1488 | Chemotherapy-Induced Nausea and Vomiting Treatment with Fosaprepitant 3 vs 5 MGKG in Children with Cancer. Hospital Materno Infantil Issemym Toluca Mexico

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Background and Aims: Chemotherapy-induced nausea and vomiting (CINV) has a negative impact on quality of life of children with cancer. Fosaprepitant has been shown to be efficacy in adult patients with CINV. Previous studies suggest effectiveness of single dose of 4mgkdo in day 1 or repeated doses of 2 mg/kg, the cost is a limiting for its use in our environment. Know the effect of fosaprepitant at 3mgkgdo vs 5mgkgdo on the treatment of NVIQ in children with cancer at HOSPITAL MATERNO INFANTIL ISSEMYM, Toluca, Mexico

Methods: Randomized open clinical trial approved by the Ethics Committee. Children in treatment with highly and moderately emetizing chemotherapy received dexamethasone and ondansetron 5mgm2do day 1-5, plus fosaprepitant single dose 3mgkg vs 5mgkg, maximum 150 mg. Efficacy was considering in children with nausea and vomiting ≤ 2 according to National Cancer Institute's Common Terminology Criteria for Adverse Events (CTCAE) and without rescue medication. We also record adverse effects. Descriptive statistics were performed as well as χ^2

Results: We include a total of 60 chemotherapy events given to 31 children, 30 events /treatment group, age 12 months-14 years old, weight 11 to 38 kg, no differences in gender or type of cancer in both groups, 70% with solid tumors and highly emetizing chemotherapy in both groups. Efficiency of 42.6%. vs 57.4% for 3 and 5 mg/kg do respectively with $p=0.02$, with greater control of nausea and vomiting on days 3-5 of chemotherapy in the 5 mgk group. No local or systemic adverse events were recorded in any group.

Conclusions: Fosaprepitant appears to be safe and effective at doses of 3 to 5 mg/kg daily, in a real-life study, especially in the control of late nausea. The use of 3 mg/kg single dose could contribute to resource optimization in patients with moderate and highly metizing chemotherapy, which is vital in our environment

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0160 / #1519 | Real World Observation Study of Medical Cannabis: Product Concentration, Dosing and Pharmacokinetic Studies

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Background and Aims: Use of cannabis oil in the pediatric oncology population has been on the rise. Although these products are legally available for adult use in Canada, there is very little known about the efficacy and safety of these products in children. Rigorous studies of cannabis oil in children have not been conducted, and therefore an understanding of the metabolism, dosing, and toxicity of these products is lacking. We report a pilot real-world observational experience of cannabis oil use in pediatric oncology patients by parental instigation.

Methods: Patients actively using medical cannabis were consented into the study. Participants provided a sample of their own medical cannabis product during a single day study visit. Participants were given a typical dose of their CBD or THC oil with the dose determined and supplied by their parent as per their home routine. Serum samples and cardiovascular measurements (pulse, blood pressure) were collected prior to oil administration, and at 6 intervals within the subsequent 7 hours. A sample of each patients' cannabis oil was also obtained for comparison and compared to what was on the label. Samples were analyzed by HPLC/mass spectrometry for determination of cannabidiol (CBD), delta-9-tetrahydrocannabinoid (THC) and 9 other cannabinoid metabolites. Parents were given a short survey to investigate attitudes and beliefs around the use of CBD oil in the treatment of their child's cancer.

Results: Seven pediatric oncology patients, ages 5-18 years, with known active use of CBD or THC oil participated. Oil samples showed a wide range of metabolite composition. 5 products were obtained from licenced dispensaries or licenced producers, and 2 products were obtained from other sources. Results from oil and blood sampling showed a wide range of products and serum concentrations.

Conclusions: There is wide variability of products, dosing, and serum pharmacokinetics in children treated with medical cannabis.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0161 / #1550 | Case Series of Difficult Fungal Infections in Patients of Pediatric Cancer

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Background and Aims: Treatment of pediatric cancer patients poses many challenges and the most important of it is infections. The children on chemotherapy are at greater risk of infections and the most notorious and challenging of them being the fungal infection. Children with prolonged neutropenia are prone for opportunistic fungal infections. Prompt diagnosis and management is the key to successful outcome

Methods: 3 difficult fungal cases: One case was a 7 year old girl, a case of B cell ALL presented with CNS symptoms with mucormycosis brain abscess. Another child was a 9 years old boy with combined medullary and teticular relapse who had gluteal abscess and was diagnosed again with mucormycosis. Third case was a 11 years old boy with T cell ALL who developed multiple necrotic superficial skin lesions which was diagnosed with Fusarium. All the three cases had unusual presentations and were difficult not only to diagnose and manage.

Results: Fungal sepsis is a not as common as bacterial sepsis, hence they are not easy to diagnose and manage. Opportunistic fungal infections not only add to the burden of the disease, but also hamper the ongoing treatment of the cancer. Granulocytopenia is the most common risk factor associated with opportunistic fungal infections. Persistent fever in a neutropenic child should prompt the clinician to consider the invasive fungal infection and thus mandating the appropriate sets of investigations and use of early appropriate antifungal agents.

Conclusions: The invasive fungal infections can virtually affect any organ/ organ system namely, skin, eyes, sinuses, lungs, gastrointestinal tracts, CNS structures, genitourinary system, and cardiovascular system. It can also affect multi-organ system as a consequence of dissemination of the infection. Unless treated promptly and aggressively, they can result in irreparable damage and at times can even be fatal.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0162 / #1655 | Therapeutic Response and Functional Capacity with the Use of Metronomic Chemotherapy in Pediatric Oncology Patients

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Background and Aims: Metronomic chemotherapy is an opposite concept to conventional cytotoxic therapy in cancer treatment. The therapeutic target is tumor angiogenesis. This therapy was developed as an option for patients in whom cure is not a viable option. The objective was to describe the clinical characteristics of patients receiving metronomic chemotherapy. To assess clinical therapeutic response and functional status in middle-income countries.

Methods: Descriptive study, which was conducted in the paediatric oncology service of the Social Security Hospital in Mexico during February 2017 to March 2021. Oncology patients under 16 years of age were included, who received metronomic chemotherapy, in 42-day cycles, with oral intake of celecoxib, cyclophosphamide and methotrexate for up to 24 weeks. Living and deceased patients who received first- and second-line chemotherapy treatment and who were relapsed or refractory to treatment, living patients with terminal disease, who had at least an estimated life expectancy of more than 2 months and completion of the Karnofsky/Lansky performance status questionnaire were included.

Results: 25 patients were included, with a female predominance, 56%. The mean age was 8.3 years. CNS tumors predominated with 11 patients, 6 with osteosarcomas, 3 neuroblastomas, 2 germ cell tumors, 1 rhabdomyosarcoma, 1 Wilms tumor and 1 pleuropulmonary blastoma. According to RESIST criteria, 1 patient had complete disease response, 4 partial responses, 3 with stable disease and 17 with progression. The median survival observed in all patients was 6.7 months. In terms of functionality, 92% presented significant improvement, according to the Karnofsky/Lansky scale. Little toxicity was reported and there were no treatment-related deaths.

Conclusions: The use of metronomic chemotherapy is a viable therapeutic option for cancer patients with relapse or refractory to treatment. One third of the patients presented stabilization or decrease of tumor activity, which positively influenced in the increase of their functional capacity and survival.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0163 / #1656 | Designing A Scale from the Primary-Health-Level to Search The Best Quality of Life for the Terminally ILL Children

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Background and Aims: The population of Norther Panama City lives under vulnerable life conditions. During the pandemic, specifically during the curfew, the families ask for help, using cellphone. Those requests reach the basic palliative care team based of the community health center, where nurses, general physician, a pediatrician. The aim of the study is to design a score to predict if the problem could have a solution at the level of primary care attention or we would need to refer at a National pediatric hospital.

Methods: Descriptive retrospective study. The period August 15, 2020 to April 15 2021. The score items were: Is the problem a physical symptom management? Is this an oncological diagnosis? How much time a clinical palliative care team supported this case? When was the last home visit by health personnel? Does the end-of-life is addressed? Being the maximum punctuation 5. We collect the data in Excel and use the StatCalc of EpiInfo.

Results: We attended seventy-one request for help of ten families of children in palliative care. The oncologic diagnosis was rhabdomyosarcoma, osteosarcoma, T cell lymphoma, brain stem tumor, and acute lymphocytic leukemia. The other five patients were non-oncologic. 57.7% of the request were physical symptoms. In 87.3% of the request, the family was able to solve the problem at home with. 14.1% of the request, the basic palliative care team made a videoconference to the referral pediatric hospital to the oncologic on palliative care unit. We found a statistically significant difference if then punctuation of the score was less than 2 with the domiciliary solution of the problem $p=0.00013$

Conclusions: This study teaches us, as basic level health professionals, to search for clues in the story of the health attention of this family, that may need a referral to the tertiary level of attention.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0164 / #1794 | Using the Educational Method B-Learning Among Primary Caregivers to Reduce Sars-Cov-2 Infections in Children with Cancer in Limited Resource Setting

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Background and Aims: SARS-CoV-2 infection could be serious disease in children with cancer. In fact, in limited resources country as Mexico can lead to severe complications and death. We aimed to reduce the SARS-CoV-2 infection in children with cancer using the educational method B-learning (online and face-to-face) for the training of the primary caregivers in a public hospital in Mexico.

Methods: We used a B-learning educational method for training of the primary caregivers from March 1, 2020 to January 15, 2021 at the Hospital Civil de Guadalajara. A total of 120 caregivers of children with cancer were included in the study. 99 caregivers received b-learning training and were evaluated with activities on the Kahoot platform, scores >80% were included in the B-learning group ($n=75$), and scores <80% ($n=24$) and caregivers without training ($n=21$) in the control group ($n=45$). Positive cases of SARS-CoV-2 infections were identified.

Results: The primary caregivers trained with children with cancer negatives for SARS-CoV-2 infection obtained the score >80% ($n=93$) compared to the patients SARS-CoV-2 positives group ($n=6$) whom obtained scores <80%. The caregivers with B-learning training had higher degree of schooling ($p=0.001$), higher monthly income

($p=0.028$), and internet availability ($p=0.040$). The incidence of the SARS-CoV-2 infection was lower primary caregiver trained group (2.7%) versus the control group (15.6%)($p=0.001$). Of the 120 children with cancer, 101 did not presented SARS-CoV-2 infection, and 9 cases (7.5%) were SARS-CoV-2 diagnosed. Of these infected patients 6/9 were carriers, 2/9 presented mild respiratory symptoms, 1/9 presented dyspnea and hypoxemia, none had severe complications or death.

Conclusions: Educational interventions by using the B-learning training in the primary caregivers of the children with cancer can have positive impact in prevent the SARS-CoV-2 infections and could reduce serious complications resource limited settings.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0165 / #340 | Paediatric Oncology Rehabilitation during the Covid-19 Pandemic: The Experience of 4 Centres Globally

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Background and Aims: Paediatric oncology services in many settings were disrupted because of the COVID-19 pandemic. Rehabilitation services experienced many challenges due to the pressure on health infrastructure and the physical activity levels of children were adversely affected. Rehabilitation professionals were required to adapt and innovate. We present the subjective experience of four paediatric oncology physiotherapists from high- and low-income settings globally.

Methods: A survey-based methodology was used to identify both the challenges and opportunities faced by paediatric oncology physiotherapy professionals in low- and high-income settings during the pandemic. Subjective reports were used to qualify the effects of the pandemic on physical activity levels of children with cancer.

Results: Challenges to paediatric oncology rehabilitation services during the pandemic included significant disruption to services (100%), redeployment of staff (40%), staff sickness (20%) and distancing measures resulting in children's play/group activity being significantly restricted in 100% of services. All paediatric oncology rehabilitation professionals adapted and innovated to continue providing essential services to children e.g., via telemedicine (60%). Virtual opportunities enabled 60% of professionals new ways of sharing best practice and learning. Children's physical activity levels were adversely affected during the pandemic due to restrictions and lockdowns. Reduced opportunities for physical activity resulted in increased inactivity related symptoms. The experience of paediatric rehabilitation professionals globally shared many similarities in respect to challenges and opportunities.

Conclusions: Rehabilitation is essential to optimise physical and psychological recovery and ensure quality of life in children with cancer. Despite many significant challenges during the COVID-19 pandemic, paediatric oncology rehabilitation professionals adapted and innovated to continue delivering these essential services. Children's physical activity levels were negatively affected due to many factors, meaning an increased importance on the continuation of rehabilitation. Significant learning occurred around telemedicine and rehabilitation. Along with virtual learning opportunities, these lessons and innovations will be continued after the global pandemic ends.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0166 / #42 | Effective Planned CVAD (Central Venous Access Device) Removal Practice in Paediatric Haematology Oncology

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Background and Aims: Background and Aim A CVAD (central venous access device) is a catheter that is inserted into the central venous system. The rates for central venous catheter related bacteraemia (also referred to as catheter – related blood stream infection, or CRBSI) are 2.1 – 30.2 cases per 1000 catheter – days. Our aims were To review the time and practice of removal Central Venous Access Device (CVAD) in oncology and haematology patients, To establish a recommendation on time of removal central line for the End Of Therapy (EOT) patients, To ensure adequate documentation is available for CVAD removal.

Methods: Retrospectively review of the notes, the theatre entrances and the electronic system. The data collected through the proforma use the Microsoft Excel and kept anonymous.

Results: Result We found that during the 6x months period we had 12 patient who was at the EOT. Majority patient were female. the youngest patient was less than 1 year old, who been treated for neuroblastoma. During the review we find the patient who was treated for solids tumours were in majority. We found the majority patient had their line removed after 4 weeks. The latest line removal was after 100 days. We did not identify any patient who developed infection. However, one patient required the emergency procedure as the line was snapped.

Conclusions: Conclusion There is no clear-cut recommendation / guidelines with respect to elective removal of CVAD not in our Trust neither any other trust. However, we advise to remove the CVAD in 2 weeks' time. To keep CVAD longer can cause the bacteraemia, effect on community oncology nursing team and have psychological effect on patient who already in EOD. Currently we systematise the process of removal CVAD from patient in EOT, as elective procedure in 2 weeks and improving the communication between oncology and surgical team.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0167 / #430 | Cognitive Function in Children with Cancer Depending on Activity Level Compared to Healthy Children

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Background and Aims: High cognitive performance (CP) is essential in children and appears to be related to physical activity. Given the background that cancer, cancer therapy but also side effects like Fatigue can affect CP, aim of our research was to compare CP in children with cancer depending on activity level to healthy children.

Methods: Thirteen children each with [cwc] and without cancer [cwoc] matched for physical activity were included (cwc: mixed cancer, age 13.31 ± 2.39 ; cwoc: age 11.85 ± 2.44). Based on activity level before disease cwc was divided into a more active and a more inactive group; activity level was queried with questionnaire about type and duration of activity and was summarized in hours/month. For CP a continuous auditory response test (ART) was performed during cancer therapy and in cwoc. ART has a duration of eight minutes with the task of physically responding to a rare stimulus (target) randomly embedded in a background of standard stimuli; reaction time (RT) from hearing the target to responding by pressing a button was measured; mean value of approximately 50 targets was calculated. In addition, the drop in RT from the beginning to the end of the test was calculated (RT-diff).

Results: In cwc, mean time between start of cancer therapy and ART was 82.54 ± 15.77 days. Mean activity level before disease in all cwc was 35.96 ± 23.79 , in cwoc 40.74 ± 22.38 hours/month. Between more active [50.07 ± 24.22 hours/month] and more inactive cwc [19.50 ± 7.12 hours/month] there was no statistical difference in RT (0.550 ± 0.078 vs 0.611 ± 0.087 sec; $P=0.207$) and RT-diff (0.062 ± 0.067 vs 0.065 ± 0.076 sec; $P=0.941$). However, while mean RT in all cwc was 0.578 ± 0.085 and in cwoc 0.581 ± 0.090 sec ($P=0.934$) mean RT-diff in all cwc was 0.063 ± 0.068 , in cwoc 0.021 ± 0.025 sec ($P=0.048$).

Conclusions: CP appears to be decreased in oncology children during acute therapy. Activity level before disease had no effect on CP; however, general activity levels were rapidly reduced during therapy.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0168 / #550 | An Evidence-Based Guideline for Erythrocyte Transfusions in Children with Cancer

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Background and Aims: Erythrocyte transfusions play an important role in supportive care in children with cancer and those who are undergoing hematopoietic stem cell transplants. Nevertheless, evidence-based recommendations developed specifically for this population are lacking. This guideline aims to systematically review the available evidence and formulate recommendations on erythrocyte transfusions in children with cancer.

Methods: A multidisciplinary guideline panel of 20 Dutch experts and patient representatives developed evidence-based recommendations and - in the absence of evidence - expert opinion statements for erythrocyte transfusions in children (including neonates) with cancer. The guideline panel focused on the following eight topics for children with cancer: 1) in general; 2) during septicemia; 3) with cardiac and pulmonary comorbidity; 4) undergoing radiotherapy; 5) with hyperleukocytosis; 6) low or high-volume transfusion; 7) irradiated erythrocyte products; and 8) infusion rates. A systematic literature review was conducted in multiple electronic databases. Citations were independently appraised by two authors and subsequently assessed using the GRADE methodology. When evidence in children with cancer was limited, additional evidence was extracted from children and adult guidelines and appraised according to the AGREE II method.

Results: Five studies including 268 children with cancer formed the evidence base of this guideline. Recommendations were formulated for a total of eight different topics regarding children and neonates with cancer.

Conclusions: In this clinical practice guideline, we developed tailor-made evidence-based recommendations and best-practice statements regarding erythrocyte transfusion management in children with cancer.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0169 / #557 | Chemotherapy Administration Through Midline Catheters in a Newly Established Pediatric Oncology Setting: An Acceptable Option to Start With

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Background and Aims: Use of central venous access device needs specialized care to reduce risk of complications in pediatric oncology patients. In resource restricted setting, peripheral IV cannula is commonly used to administer chemotherapy. Midline catheters might serve as alternative to central venous access devices in this setting at lesser

cost and serious complications. To evaluate the utility and safety of midline catheters in pediatric oncology patients.

Methods: Single centre prospective observational study conducted at a tertiary care hospital in central India over a period of 18 months. 21 catheter placements were done. 4 types of catheters were used – 6 Central Venous Catheters(CVC), 5 Peripherally Inserted Central Catheters(PICC), 9 Midline Catheters(MC), 1 Chemoport. The primary end point was the occurrence of any complication.

Results: All the catheters were used in inpatient setting for Intravenous fluids, antibiotics and chemotherapy agents and were followed up for the development of complications. There were 18.9 complications per 1000 catheter days in CVC, 61.7 complications per 1000 catheter days in PICC, 27.3 complications per 1000 catheter days in MC. Median time for the development of complications was 52.6 days in CVC, 16.2 days in PICC, 36.5 in MC. Less serious complications like thrombophlebitis and leakage were observed in MC whereas more serious complications like Deep Venous Thrombosis (DVT) and Central Line Associated Blood Stream Infections (CLABSI) were seen with CVC and PICC.

Conclusions: Midline Catheters are an acceptable alternate type of catheter to PICC use in terms of reducing CLABSI, DVT even though nonserious complications like thrombophlebitis and leakage are more common.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0170 / #588 | A Case of Critically ILL Child Diagnosed with Ewing Sarcoma with Severe Clostridium Difficile Infection

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Background and Aims: Clostridium difficile (CD) is gram-positive, anaerobic, and spore-forming bacillus. The risk of CD is increased in critically ill children, who receive antibiotic therapy and have immunodeficiency. Hence, it can cause life-threatening complications in children with cancer. We report a case of severe CD in a young child diagnosed with Ewing sarcoma and treated at the Pediatric Cancer and Blood Disorders Center of Armenia (PCBDCA).

Methods: Diagnosis of CD in PCBDCA is made on detection of glutamate dehydrogenase (GDH) and CD toxins in the feces.

Results: A two-year-old female who was diagnosed with localized Ewing sarcoma was treated at PCBDCA. After the 5th cycle of chemotherapy according to VDC/IE treatment regimen, the child had diarrhea and the feces were sent for detection of CD. Consequently,

the diagnosis of CD was confirmed and treatment with oral metronidazole was started. Three days later the child had febrile neutropenia, pancytopenia (WBC $0.03 \times 10^3/\mu\text{L}$, PLT $6 \times 10^3/\mu\text{L}$, HGB-70 g/L, RBC $2.0 \times 10^6/\mu\text{L}$). Meropenem and G-CSF were started along with blood product transfusions. Inasmuch as symptoms were persistent, oral vancomycin was added and metronidazole was switched to intravenous therapy. One week later the general condition worsened and the child had severe gastrointestinal bleeding. The child had also coagulopathy (aPPT 190 sec, PT 82 sec, INR 11.5, TT 1.7 sec.), hypoalbuminemia, severe hypokalemia (1.4 mmol/L), and ascites. Fresh frozen plasma, tranexamic acid, albumin, and potassium supplements were added to the main treatment. After two weeks of intensive supportive care, the patient recovered.

Conclusions: Supportive care is an essential part of childhood cancer treatment. Considering the fact that children with cancer are at higher risk of CD infection, early detection should be performed in order to decrease complications.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0171 / #994 | Barriers to Effective Delivery of Palliative Care in a Teaching Hospital in Ghana

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Background and Aims: BARRIERS TO EFFECTIVE DELIVERY OF PALLIATIVE CARE IN A TEACHING HOSPITAL IN GHANA YIRIYALE J. B, PAINTSIL V, OBENG P, ASOOGO C PEDIATRIC HAEMATOLOGY ONCOLOGY UNIT, DIRECTORATE OF CHILD HEALTH, KOMFO ANOKYE TEACHING HOSPITAL, KUMASI, GHANA.

Inadequate palliative care delivery for seriously ill children is a global problem. Nurses caring for seriously ill children are normally faced with challenges in an attempt to effectively deliver palliative care. This causes children to have prolonged hospital stay and suffer unnecessarily.

The goal of this study was to explore barriers to effective delivery of palliative care in a tertiary hospital in Ghana by pediatric oncology nurses caring for children with cancers.

Methods: A survey was conducted amongst 14 nurses at the pediatric oncology unit at Komfo Anokye Teaching Hospital, Ghana from January to April 2021. Nurses perceptions were explored on : Uncertain prognosis, family not ready to acknowledge incurable disease, language barrier, cultural and social barriers about death and dying, misconception about opioid use, Insufficient education on pain and time constraints.

Results: Females. Out of these, 4(28.5%) were nursing officers (1 male, 3 females), 2 (14.2%) males were senior staff nurses, 6 (42.8%) were staff nurses (1 male, 5 females) and 1(7.1%) was a female enrolled nurse. All respondents reported 4 out of the 10 barriers listed in

the study questionnaire as frequently or almost always occurring. These barriers were uncertain prognosis (42.8%), family not ready to acknowledge incurable condition (35.7%), language barriers (14.2%), and time constraints (14.2%). They felt competent in effective delivery of pain management.

Conclusions: Uncertain prognosis and when the families are not ready to accept that a child has an incurable disease were the most common barriers to effective delivery of palliative care in our setting. It is hoped that by addressing these barriers, there would be an effective palliative care delivery.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.d Supportive Care and Palliative Care*

PO0172 / #999 | Pediatric Palliative Screening Scale (Papaz Scale) in Pediatric Cancer Patients: A Qualitative Study Approach

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Background and Aims: The initiation of pediatric palliative care (PPC) and referral to specialized teams still occurs near the end of life for children with life-limiting diseases, especially in a resource-limited setting. The aim of this study was to determine the need for PPC in children with cancer using the PaPaS Scale in a hospital in Peru.

Methods: Prospective, cross-sectional survey of parents and physicians, as well as a medical records review of pediatric cancer patients hospitalized at the National Institute of Neoplastic Diseases, Lima-Peru.

Results: Of 50 hospitalized patients, 46 parents completed the survey. Twenty-four patients (52.2%) were male and 22 (50.8%) females. The median age was 5 years (range, 8 month - 14 years). Four patients (8.7%) had the need to start PPC (≥ 25 points) and 33 patients (71.7%) had to be explained what PPC is and prepare for the start of it (≥ 10 points). Ninety-one percent of the parents did not know the meaning of PPC. The increase in hospitalizations by more than 50% in the last 3 months ($p=0.017$), the burden of treatment ($p=0.005$), the intensity of symptoms ($p=0.02$), the patient's conduct disorder ($p=0.009$), the physician's belief that the patient should start PPC ($p=0.005$) and life expectancy of less than 6 months ($p=0.05$) are strongly associated with the need for PPC

Conclusions: This tool allows us to evaluate the need to start CPP in children with cancer and thus provide multidisciplinary management earlier and not only at the end of life. Special thanks to Dr. Eva Bergstraesser for her help with the PaPaS Scale and the HOPPE Association for supporting the statistical analysis.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0173 / #1666 | Hospital-Based Schooling Program in Pediatric Oncology Units of Pakistan

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Background and Aims: Children with cancer cannot attend school regularly; most of them can also not go to school due to their compromised immunity. A lot of children's hospitals in resource-rich settings are running hospital-based schooling projects for such children so they do not lag behind their peers when they return to school. The aim of this paper was to identify availability of hospital schooling and bed-side education services in Pediatric oncology treatment centers of Pakistan.

Methods: The study sample includes pediatric oncology treatment centers in Pakistan which are providing pediatric oncology treatment. Data was collected on a standardized questionnaire from pediatric head nurses of the hospitals through a telephonic interview and other information was gathered from the web pages of the respective hospitals. Pediatric oncology Hospitals which are only referring patients to other tertiary care hospitals are excluded from this study because of unavailability of cancer treatment.

Results: It was found that there is bedside education system in 2 pediatric oncology units and schooling system in 1 hospital out of 19 pediatric oncology treatment centers.

Conclusions: Conclusion: Currently in Pakistan the educational needs of children with cancer are neglected and they have barely any opportunity to attend a hospital base school. Health care providers are not aware of importance of formal education inside the hospital.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0174 / #187 | Benefits of Using Psychosocial Resources in Carers of Paediatric Oncology Patients during the Treatment

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Background and Aims: The main objective of this paper is to evidence the importance of psychosocial resources used in carers of paediatric oncology patients during the treatment. Further objectives are described below: · To evidence the benefits of psychosocial resources used in carers and to highlight the most used. · To analyse if those resources could contribute to reduce the emotional impact of carers during the treatment. · To describe the role of the paediatric oncology psychologist so as to optimize psychosocial resources. To make a

meaningful contribution in order to give more tools to help families that are in treatment.

Methods: The method used for this study was systematization, based on an updated theoretical framework and the observation to groups of parents, during treatment workshops. The period selected was between 2016 and 2019

Results: During the period of the study, sixteen parent's devolutions were selected to be analysed. The results are: · Parents usually use a lot of resources that are classified as socio-organizational, psychological and personal resources. · Within psychological resources, there are some therapeutical skills used in some specific situations during the treatment. · Those resources are part of an integral learning during the paediatric oncology treatment that could help to balance the impact of the illness.

Conclusions: To sum up, families use and develop different psychosocial resources during the treatment. As professionals of psycho-oncology, we could contribute to improve those resources so as to help families to be with their kids during the treatment. Furthermore, it could help to reduce their emotional impact.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0175 / #370 | Effectiveness of A Comprehensive Rehabilitation Program for Children with Cancer and their Families

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Background and Aims: In the World Health System, children's oncology takes one of the first places in terms of the number of lost years quality of life. After basic therapy nearly all children and their families need rehabilitation treatment.

Methods: The assessment of the psychological state of children by scales of Spilberher-Hanin, psychometric research - questionnaire of the Center of injury and stress in pediatrics (Philadelphia), physical condition - test with a 6-minute walk. This work carried out within the program HUSKROUA/1702/8.2/0127

Results: A multidisciplinary team was created and a comprehensive program of rehabilitation of children with cancer was developed in a stable remission and their families based on a sanatorium "Maly-atko". The trial group included 20 children after the main treatment for leukemia and solid tumors aged 11±4.4 who received renewable therapy for 3 weeks. A separate group amounted to 20 members of their families who were around the clock with their children. After rehabilitation course there was an increase in the functional capabilities of cardiac and respiratory activity in 82%, reduction of manifestations of personal anxiety in 74% (32±3.2 score) and situational anxiety in 79% (30±4.1 score) with a tendency to form a positive psycho-emotional mood. A significant decrease anxiety in the family members

(92%, 36±2.8 score) was noted, which was due to the lack of information about child's disease, the ignorance of the main behavioral types of child's reaction, the absence of psychosocial support, a limited circle of communication of like-minded people.

Conclusions: High efficiency of the new direction of sanatorium rehabilitation in Ukraine was determined. Physical rehabilitation increases the functional reserves of the body of a child. Psychological correction contributes to improving social and psycho-emotional adaptation. Psychological assistance to parents reduces personal and reactive anxiety, which is important for stabilizing intra-family communication and improve the quality of life of all family members.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0176 / #411 | Social Impact of Covid-19 on Patients with Childhood Cancer in a Military Hospital in Colombia

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Background and Aims: Cancer cure and survival rates are low, between 20 - 40% in developing countries. Largely social factors are responsible for these outcomes as they affect adherence and compliance to treatment. Since March 2020, a state of public health emergency associated to SARS-CoV-2 / Covid 19 was declared in Colombia. This has generated deep socioeconomic changes that may impact cancer survival rates in the future. The objective of this study is to evaluate the social risk of patients who received cancer treatment between March 2020 to March 2021.

Methods: Design: Cross-sectional descriptive observational study
Methods: Evaluate the social risk of patients who received cancer treatment during the public health emergency associated to Covid 19 in Colombia. A scale was developed evaluating the following aspects of family dysfunction, access to the hospital, financial problems, psycho-family problems, desire to abandon, legal problems, health problems of caregivers and cultural problems. Population: Families of children with a diagnosis of cancer affiliated with the Colombian Military Forces healthcare system receiving treatment at the Hospital Militar Central in Bogotá, Colombia

Results: 53 patients were evaluated. Our finding included Family dysfunction 20/53 (38%), access to the hospital 37/53 (70%) due to not availability of public transportation, financial problems 15/53 (28%), psycho-family problems 13/53 (24%), desire to abandon 3/56 (6%), legal problems 2/53 (4%), health problems of caregivers 6/53 (12%) and cultural problems 1/53 (2%).

Conclusions: In the pandemic, new social factors that affect adherence and compliance to cancer treatment have emerged in Colombia. These

factors are beyond the health care system, will affect access to health care services and are likely to ultimately lead to late cancer diagnosis, treatment abandonment and low cancer survival and cure rates.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0177 / #471 | Wellness Program; Psychological Support for Children with Cancer

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Background and Aims: On April 12, 2002, the Children's Cancer Center of Lebanon (CCCL) opened its doors in Beirut as a regional reference center pioneering in the treatment of children with cancer, with an amazing average cure rate of 80%. Till date, CCCL has treated over 2,040 children with cancer from all over Lebanon and the region. We are currently supporting more than 300 patients yearly with successful survivors' stories. Since part of CCCL's mission is to provide excellence in psychosocial services and since cancer is not only a physical illness but also a mental one, the CCCL tackled this aspect by introducing the wellness program from the beginning of 2019. This program includes expressive yoga, music and art sessions that aid the children with cancer to improve their self-esteem, cope with their negative emotions and reduce their level of stress and pain. In addition, these sessions tackle an educational aspect where children learn new musical and artistic skills.

Methods: To be able to implement this program, we created a detailed procedure that includes a monthly schedule of the activities happening on a daily basis. In addition, a log sheet is used to log data about the sessions and about every patient, which includes the feedback of the coordinator and the teachers about the overall session and about each patient.

Results: By implementing these activities, the children were able to learn how to reduce stress and pain caused by the treatment and apply it at home with their families too. They were able to express their feelings and have a positive view about cancer.

Conclusions: This program helps the children express their feelings and share their pain and stress. In addition, it gives them the opportunity to learn new skills. CCCL is working on introducing these activities to patients at other partner hospitals all over Lebanon.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0178 / #953 | The "Twenty Thousand Leagues Under the Sea" Paradigm. Travel Literature in Hospital as an Aid to Therapy

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Background and Aims: Our idea stems from the need to provide readings that are not overly demanding (as often are the biographies of former patients, etc.) but which are still meaningful for our children and adolescent.

Methods: Taking up the tradition of phenomenological psychopathology (Binswanger, Minkowski), we recommend our guests those books that best fit the "world of life" (lebenswelt) of hospitalized patients for a serious illness: if the time of hospitalization is characterized by specific existential coordinates (sense of passivity, constrained space, etc.), then the book must also present a story that has the same vital coordinates within it, a fiction in which the reader can kindly recognize himself, without this causing any excessive activation that could lead to rejection of the book itself (and of his own experience). Recommended texts allow the young patient to identify himself - with no excessive fear - in the character of the book that deals with situations similar to his own in terms of time, space, alterity and lived body.

Results: In this sense, and in our experience ("Reading care in hospital" project, promoted by the ACP - Paediatric Cultural Association), "Twenty Thousand Leagues Under the Sea", by Jules Verne, represents the paradigm of inpatient literature, as it contains in its pages a series of lived variables that resonate with the healing process of our patients: the strict isolation that nonetheless allows the journey, the presence of dangers and the courage to face them, the curiosity, the surprise and anger.

Conclusions: Therefore, it should be easy for young readers (even the youngest ones, as there are also picture books) to identify with the characters of the novel, to live their own adventures in the gentle light of those narrated in the novel, if it somehow appears similar in terms of "lived world", similar as only paradigmatic stories can be.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0179 / #968 | "Being in the Landscape". Sharing The Truth as a Remedy for Loneliness in the Critical Stages of Treatment

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Background and Aims: Truth is an essential ethical tool when caring for the other, and especially when caring for patients. In critical stages of treatment, and when active cares become "active presence" during the last period of life, the truth takes a peculiar importance because of its sense of respect for dignity, as well as an antidote to the patient's loneliness due to the unawareness about his clinical situation. The danger in these cases, however, is the presence of a misalignment between

the positions of the patient and that of the clinicians, a gap that often does not allow the necessary sharing of the truth about his medical condition.

Methods: To understand the respective positions, in an effort to approach the life of the other, it is useful to recall the comparison between “landscape” and “geography” described by the psychologist Erwin Straus (1935). According to Straus, clinicians often do not understand the difference between the patient’s position (landscape), and their own position (geography): the clinicians believe they know something about the patient’s panorama, when they only know something regarding their own map.

Results: The clinician has to make an effort to descend into the patient “landscape” from his “geography” in order not to leave him in a concrete condition of loneliness, and should delicately make sure that the patient is completely aware of the situation, leading him in his choices following the principles of freedom and opportunity.

Conclusions: We clinicians can find the truth in that crossing point, at the limit between first (landscape) and third person (geography), the only place where sharing of experience and emotions is possible. Only after this rendezvous, we and our patients will be able to return to our new position, having gained the freedom that can only arise from the encounter between the level of experience and that of truth.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.e Psychosocial (PPO)*

PO0180 / #969 | The “Gift” as a Category of Lived-Space in a Group of Post-Therapy Adolescents

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Background and Aims: The concept of “gift” certainly has a prominent place in paediatric oncohematology. Aware or not, our children and their families are confronted since the first day of hospitalization with the gift of life, the gift of clinical and human care, the gift of biological substances, bone marrow, stem cells, blood, platelets, and with operators who donate themselves as they can for them.

Methods: Starting from this concept, in a weekly support group for post-therapy adolescents, who have successfully completed the cures and are in the first months of follow-up, the theme of the “gift” has been presented and discussed as their own offer of the illness’ experience and of their idea of present and future.

Results: The concept of “gift”, the group points out, reaches its full sense where there is “a space to fill”, an “empty place”, a “lack”. Thus, the gift of the bone marrow replaces the diseased bone marrow, the new organ replaces the defective one, blood and platelets take the place of dead cells. The gift thus appears as a concept that needs the dimension of a lived space and at the same time of a concrete space, in order to be understood. If there is no space available, it actually is an invasion.

During the sessions based on the concept of “gift”, the children began to discuss the gift of life, and the space that potential future parents should have in their minds to welcome a baby’s birth. Everybody needs space in their life so that some idea of the future could be installed in, a lived and concrete space that survived adolescents have conquered with endurance and bravery.

Conclusions: The concept of “gift” should be discussed in homogeneous post-therapy adolescent group sessions, in which the common experience of disease can be reorganized from different and transformative perspectives.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fc Nursing - Quality Improvement/Practice Project*

PO0181 / #1439 | Paediatric Cancer Infusion - Stopping Bubble Trouble for Both Patients and Practitioners

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Background and Aims: Vascular Air Embolism is a risk of every IV infusion, particularly affecting smaller children due to Patent Foramen Ovale and limited vasculature pathways. As such, additional efforts are made to remove air bubbles from IV fluids. However, common techniques such as flick and syringe, or inline filters, do not offer effective protection from occupational exposure to Hazardous Drug (HD) vapours, a leading cause of harm among healthcare workers. More than 20 million healthcare workers are affected each year by HD exposure, leading to increased infertility, miscarriage and cancers. Aspiration of bubbles usually requires the closed system to be opened, exposing the nurse to drugs that vaporise at room temperature. In-line filter devices are highlighted in EU policy documents as not suitable CSTD, given that they vent to atmosphere and do not filter the toxic drug vapours. New policies in USA and EU are seeking to mitigate the risk by promoting use of Closed System Transfer Devices (CSTD) to prevent environmental contamination.

Methods: A novel closed system aspirator device has been developed to automatically capture Air-in-Line without impeding the drug delivery. The novel device can be placed before the IV pump to minimise alarms that result from degassing of drugs that are poorly water soluble, such as Taxols, which use Cremaphor-CL as a carrier. This drug is among many known to degas readily, causing alarms and stoppages that require attention. By removing Air-in-Line as a Closed System Aspirator the device can improve IV delivery.

Results: A multi-centre clinical study is planned to validate the efficiency and economic benefit of implementing a closed system aspirator device in the Paediatric setting, to protect younger patients from micro-bubbles and nursing staff from occupational exposure.

Conclusions: The proposed study will investigate whether stopping bubbles in Hazardous Drugs can reduce IV interruptions and deliver healthcare quality improvements.

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.f Nursing / AS04.fc Nursing - Quality Improvement/Practice Project*

PO0182 / #166 | **A Global Challenge Demands Global Efforts**

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Background and Aims: The Princess Máxima Center is renowned for its integrated approach towards children with cancer. Since 2018, we cluster innovative care, research and education in one medical center. We are passionate and aim to be groundbreaking, not only in our treatments, but also in improving the quality of life of both patients and parents. In all our work, innovation is key. In the Netherlands, we form the spill of pediatric oncology, working closely together with 17 shared care centers. In the past years, we have already shown improvements in care due the centralization. For example the Solid Tumors department started a new protocol to deliver immunotherapy on an outpatient clinic basis. This method is not only safer and more efficient, but also much less toxic for children. Efficient immunotherapy is the result of the concentration of all cancer patients and the pooling of all expertise inside the center, for the benefit of patients.

However, curing children is not a nation-bound challenge. We propose that strengthening international collaboration in innovative pediatric oncology is crucial to get closer towards our mission of curing all children diagnosed with cancer and provide them optimum quality of life. Currently, we already work with researcher from 28 nationalities and have trained 400 healthcare workers from Low- and Middle Income Countries in our outreach program. To extend our international collaborations, we want to initiate four working groups to share experiences and exchange knowledge for innovative nursing.

The working groups are aimed at integrated research into nursing care, chemotherapy at home, professional leadership and outreach. The initiation of the working groups will take place before the SIOP 2021 congress. In our presentation, we will present the first results of the working groups and discuss the next steps in our international collaboration efforts.

Methods: Not applicable

Results: Not applicable

Conclusions: Not applicable

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy*

PO0183 / #1082 | **Prolonged ILEUS Secondary to Anti-Gd2 Immunotherapy for Neuroblastoma**

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Background and Aims: Immunotherapy has lately been incorporated into standard treatment of high-risk neuroblastoma. Important side effects of its use are emerging

Methods: We report the case of a 5.5-year-old Lebanese boy, diagnosed the previous year with high-risk neuroblastoma, with bone, bone marrow and lymph nodes metastases. He received treatment in Jordan, including chemotherapy, surgery of the primary tumor, ablative high-dose chemotherapy and radiotherapy of the primary tumor and bone metastases, achieving complete remission. He came to our Institution (Spain) for immunotherapy treatment. He received a course of high-dose isotretinoin followed by a course of dinutuximab beta 10 mg/m²/cycle IV continuous infusion for ten days. He required morphine for 4 days. Given subsequent excellent tolerance, he was discharged on day +6 and completed treatment on an outpatient basis. He was readmitted on day +9 because of vomiting, associated with no bowel movement on the last two days. He was dehydrated and without bowel sounds. Dinutuximab was discontinued. He received intravenous fluids and a nasogastric tube. An abdominal CT scan ruled out mechanical ileus and showed a sigmoid colitis. He stopped vomiting on the first day. He received parenteral nutrition and broad-spectrum antimicrobials (vancomycin, meropenem, liposomal amphotericin B) for 15 days. Enemas and prokinetic drugs (erythromycin and metoclopramide) were ineffective. An abdominal TC on day +17 showed pancolitis. On day +21, trophic feeding was started as an attempt to stimulate intestinal hormonal function, without success. On day +24, cisapride was started on a compassionate use. Finally on day +28, bowel sounds reappeared and progressively tolerated oral intake.

Results: According to the drug data sheet, ileus appears in the 1,5% cases. We found no literature describing its pathophysiology, duration or proper management. We report our experience.

Conclusions: Immunotherapy in neuroblastoma is an effective treatment but we should be aware of potentially severe, prolonged and difficult-to-manage side effects

Publication Only Topic: *Disciplines / AS04 Treatment and Care / AS04.g Tumor Biology, Immunology and Immunotherapy*

PO0184 / #232 | **Quality and Outcome of Bone Marrow Aspirates in Pediatric Hematology/Oncology: The National Oncology Centre Experience, Oman**

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Background and Aims: Bone marrow Aaspirate(BMA) is a fundamental diagnostic procedure in pediatric hematology/oncology (PHO). However, specimen quality is an important determinant of BMA

outcome. We aim to retrospectively evaluate BMA quality and outcomes in PHO department at the National Oncology Centre, Oman.

Methods: All BMAs done between January 2017 to December 2018 at PHO department at the National Oncology Centre were retrospectively reviewed to determine their quality and outcome. A BMA is defined optimal if it has a good number of bone spicules with minimal or no dilution, and a clear pathologic diagnosis can be easily made. A BMA is defined sub-optimal if it has some bone spicules with some dilution, but a clear pathologic diagnosis still can be made. A BMA is defined poor if it has no or low number of bone spicules with severe dilution, and no clear pathologic diagnosis can be made.

Results: A total of 428 BMAs were reviewed. Of which, 307 (72%) samples were done for diagnostic or reassessment of malignant hematological disorders, 50 (12%) samples for benign hematological disorders, 66 (15%) samples for metastatic work up for solid tumors, and 5 (1%) samples for other diagnosis. 198 (46%) samples were done for initial diagnosis, and 230 (54%) samples were done for disease reassessment. In terms of BMA quality, 268 (63%) samples were classified as optimal, 68 (16%) samples were classified as sub-optimal, and 92 (21%) samples were classified as poor. In terms of BMA outcome, 336 (79%) samples achieved a pathologic diagnosis, and in 92 (21%) samples there was no clear pathologic diagnosis was made. The main reasons reported as an explanation for poor or sub-optimal samples were haemodiluted or aparticle in 138 (32%) samples, anticoagulant EDTA effect in 16 (4%) samples, and combined effects in 5 (1%) samples.

Conclusions: The rate of poor BMAs in our study is within the reported rate in literature. The majority of BMAs achieved a clear pathologic diagnosis.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0185 / #1026 | Satisfaction of The Outpatient User Before and during the Covid-19 Emergency State in the Pediatric Oncology Unit of a Peruvian Tertiary Care Hospital

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Background and Aims: Providing quality care is the goal of every health establishment, therefore, knowing the satisfaction of the users is essential. Pediatric cancer patients have repeated consultations, during treatment, and after it has ended, they continue to attend the outpatient clinic for their check-ups for several more years, so it is important that they are satisfied with the care. During the COVID-19 pandemic, some changes were necessary for the outpatient clinic that could affect user satisfaction, this study sought to compare the level of satisfaction of outpatient users, before and during the state of emergency due to COVID-19, in the Pediatric Oncology Unit of the Edgardo Rebagliati Martins Hospital.

Methods: A quantitative, observational, cross-sectional, comparative study. 60 patients were included, who had at least one consulta-

tion between January 1 and March 15, 2020 (pre-pandemic) and also another consultation between March 16 and June 30, 2020 (during the pandemic). The SERVQUAL survey was applied to the companions of the patients.

Results: The mean age was 7 years, predominantly male (55%). The brain tumor was the most frequent with 33.3%. 81.7% came with the mother, 65% of the companions had a higher education level. In the pre-pandemic period, global satisfaction was 54.7%, the Reliability dimension obtained 49.7%, Responsiveness, 32.9%, Security, 78.3%, Empathy, 70.3% and Tangible Elements, 42.1%. During the pandemic, global satisfaction raised to 77.5%, the Reliability dimension reached 62.7%, Responsiveness, 77.5%, Security, 88.75%, Empathy, 92.3%, and Tangible Elements, 66.2%. The difference between satisfaction before and during the pandemic was statistically significant ($p = 0.003$).

Conclusions: Global satisfaction and the dimensions evaluated by the SERVQUAL survey reached a higher level of satisfaction during the pandemic than before the pandemic. It is recommended to permanently evaluate user satisfaction and implement strategies to improve and sustain satisfaction in the different dimensions.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0186 / #1479 | Challenges of the Early Diagnosis Program for Children's Cancer During The Covid-19 Pandemic in the West of Paraná- Brazil

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Background and Aims: Introduction: In 2020, Brazil and the world faced an outbreak of coronavirus 2 of the severe acute respiratory syndrome (SARS-CoV-2) in which it became a pandemic. The early diagnosis for children cancer program has been developed since 2008 in our region. The objective of program is assist health professionals in the recognition and early referral of suspected cases. In view of this new scenario, it was not possible to train health professionals on the topic of childhood cancer. The health system was overburdened, and the fear itself was directly involved in attending to this Pandemic. There was a need to seek other means to maintain the early diagnosis program through undergraduate students in the medicine, nursing and pediatric residents. Objective- To train medical and nursing students, as well as pediatric residents, to provide information on juvenile cancer.

Methods: Methodology - The course was administered online by the zoom platform, from October to December / 2020. The topics covered: epidemiology, signs and symptoms, discussion of clinical cases, communication of the diagnosis, necessary care for children undergoing cancer treatment, flowchart for referral of a suspected case.

Results: 192 medical and nursing students and 8 pediatric residents participated in the training. It trains students from 5 public and 4 private universities. The course provided an opportunity for reflection

and updating knowledge. Two students after the course, had the opportunity to participate directly in the early diagnosis of two children. This demonstrates the interest and learning of these students.

Conclusions: Comments: The ON LINE course for the early diagnosis of juvenile cancer was a great teaching experience. It is extremely important for medical and nursing students, as well as pediatric residents, to learn to suspect, diagnose and specially, to refer a suspected case of cancer to referral centers early.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0187 / #1520 | Epidemiology Characterization of Acute Leukemia in Pediatrics at a Referent Center of Paraguay

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Background and Aims: The acute leukemia (AL) is the most frequent cancer in childhood, presenting in two phenotypes: acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML). Is not recognized a causal agent, even if related to a genetic syndromes. The objective is to describe the epidemiology of cases of AL in children, in a reference center in the period between January 2019 to December 2020.

Methods: Data was collected from clinical files of pediatric patients diagnosed with AL. This is a retrospective and descriptive study.

Results: Of the total number of patients, 94.4% correspond to a new diagnosis of AL and 5.6% to patients diagnosed with a relapse of the disease. Of all these, 83.0% were due to the ALL phenotype, with 93.2% of these being of B line and 6.8% of T line. The remaining 17.0% correspond to patients with the AML phenotype. There were 5.7% of events with a diagnosis of acute promyelocytic leukemia. At the time of diagnosis, the age distribution was: patients with <1 year 1.9%, with between 1-5 years 56.6%, those between 5-10 years 28.3% and those with 10 to 15 years 13.2%. Male patients correspond to 52.8% and female patients to 47.2%. Down syndrome was found in 5.7% of the cases of AL, with this association occurring in 2.3% with the ALL and 22.2% concomitantly with the MLA.

Conclusions: Acute leukemia is the most common childhood cancer, most frequently found in children age 1 to younger than 10 years, similar data described in the literature. There are not meaningful differences between sex. ALL was found predominant over a AML. In patients with ALL the subtype B is most frequent. Acute promyelocytic leukemia represent a third of AML. Patients with Down syndrome present most frequently an association with myeloid lineage leukemia.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0188 / #1561 | The Clinical Utility of Total Oxidative Stress and Total Antioxidant Capacity in Childhood Febrile Neutropenia

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Background and Aims: The aim is to determine the oxidative status in childhood febrile neutropenia (FEN).

Methods: Demographic and clinical findings of pediatric patients with FEN were recorded. Blood samples were collected for total antioxidant capacity (TAC) and total oxidative status (TOS) in children with FEN at the baseline, 48th and 96th hours and in healthy children (once). The oxidative stress index (OSI) was calculated TOS/TAC ratio. A total of 25 FEN episodes of 18 patients were enrolled in this study

Results: The baseline TAC levels of patients with control group were 59.64 ± 25.2 mmol Trolox equiv/L and 15.4 ± 6.67 mmol Trolox equiv/L, respectively. The difference was statistically significant, ($p < .0001$). The TAC levels of FEN patients with and without antibiotic modification were higher than the control group (adjusted p values: .002 and .02, respectively). Also, the TAC levels of FEN patients with antibiotic modification was lower than without antibiotic modification ($p = .0224$). There was no statistical difference between the TOS values in children FEN and control group. The OSI value of children with FEN was lower than control group ($p < .0001$). The OSI values of FEN patients with and without antibiotic modification were higher than the control group (adjusted p values: .001 and $< .0001$, respectively). Also, the TAC values of the patients with antibiotic modification were higher than those of the patients without antibiotic modification ($p = .02$).

Conclusions: In conclusion, the oxidative status is affected in children with FEN, and it can give information about the follow-up of FEN.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0189 / #1691 | Large Mediastinum in Children: What if It's Not the Thymus?

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Background and Aims: Abnormalities of the mediastinum in children include mainly pathologies of the thymus and mediastinal masses. The discovery of an abnormality is most often made during the performance of a standard chest radiograph, especially in the context of respiratory symptoms. Detailed interpretation of the chest radiograph often allows differentiation of normal images, variants of normal and true pathology including tumor. So, we aimed to describe the clinical, radiological and etiological features of tumors revealed by a large mediastinum.

Methods: We conducted a retrospective study that collected the large mediastinum in the department of pediatric oncology of the Children's Hospital of Tunis between 2014 and 2020.

Results: We collated 20 children whose mean age was 5 years 3 months. Dyspnea (40%) and prolonged fever (35%) were the most frequent reasons for consultation. The clinical examination revealed an altered general condition (25%), peripheral adenopathies (40%), superior vena cava syndrome (20%) and thoracic deformation (15%). We collected 4 tumors of the anterior mediastinum (1 Hodgkin's lymphoma, 2 T-lymphoblastic lymphoma, 1 thymic hyperplasia), 7 tumors of the middle mediastinum (4 Hodgkin's lymphomas, 1 B-lymphoblastic lymphoma, 1 T-lymphoblastic lymphoma, 1 hemangioma) and 9 tumors of the posterior mediastinum (5 neuroblastomas, 2 ganglioneuromas, 2 malignant germ cell tumors). Treatment was based on chemotherapy alone (20%), chemotherapy combined with surgery (35%) and radio-chemotherapy (30%).

Conclusions: In front of a large mediastinum, an analysis of the thorax radiography and, if necessary, the ultrasound allows to establish the assessment in the majority of the cases. Recourse to CT+/- MRI and biopsy is often required.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0190 / #802 | Priorities for Improving Pediatric Hematology-Oncology Care in Eurasia; A Regional Experience Using the Profile Tool

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Background and Aims: Pediatric cancer care in Eurasia, or countries in Central Asia and Eastern Europe, faces multiple challenges. To help pediatric hematology-oncology (PHO) centers prioritize among potential improvement strategies, St. Jude Global developed Pediatric Oncology Facility Integrated Local Evaluation (ProFILE); a facility assessment of PHO context, workforce, diagnostics, therapy and patient outcomes. This study describes the implementation of ProFILE in Eurasia.

Methods: The abbreviated ProFILE, instructions, and training components were translated to Russian. Seventeen PHO centers from 11 countries part of the Eurasian Alliance in Pediatric Oncology (EurADO) participated. Centers identified ProFILE leads and implementation team members who completed ProFILE between March and October of 2020. Bilingual facility-specific reports and an aggregated regional report were generated and shared with local teams. All collaborators participated in a virtual 4-week asynchronous training to interpret their center's report, followed by a virtual ProFILE workshop in December 2020 to identify regional priorities for future work.

Results: The ProFILE assessment demonstrated heterogeneity in strengths and weaknesses of PHO services across the 17 participating centers. As a region, Eurasia performed best in the domains of surgery (85%), chemotherapy (82%), diagnostics (79%), and financing and resources (79%). Areas of opportunity included the national context (63%), service integration (65%), and supportive care (67%). Forty-four participants from 11 countries joined the synchronous virtual ProFILE workshop. Using design-thinking methodology and interactive engagement tools, 4 working groups identified regional priorities for

further collaboration: improving parental education in PHO, improving PHO training among core consultants, instituting tracking of quality measures, and strengthening chemotherapy safety practices.

Conclusions: The abbreviated PrOFiLE is an effective tool to guide countries and regions to prioritize among potential opportunities to improve PHO care. Priorities identified through implementation of PrOFiLE in Eurasia were presented to EurADO, which now plans to implement collaborative projects to address identified regional needs.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0191 / #804 | Exploring The Reasons for Late Report of Childhood Cancer Cases for Treatment at Komfo Anokye Teaching Hospital

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Background and Aims: Childhood cancer when reported early has a good prognosis and increases the survival rate. Normally, parents report these childhood cancer cases at stage 3 and 4 when it is more difficult and expensive to treat. The aim of this article seeks to explore the reasons of this late reporting of childhood cancers to the KomfoAnokyeTeaching Hospital (KATH) and how they can be mitigated

Methods: Parents of 70 patients of children diagnosed with cancer from January to December 2020 were interviewed on when they detected their children were showing early signs of cancer, when it was reported to our facility and why they reported at our hospital at that time.

Results: After careful analysis of the data and information, the study reveals that 55(78.6%) of these parents were female and 15(21.4%) were male. 64.3% of them were less than 5years old and the remaining 35.7% were between the age of 5 and 14years.The study also showed that common type of cancer presented was Acute Lymphoblastic Leukemia (18,25.7%) followed by Nephroblastoma,(14,20%), then Retinoblastoma (13,18.6%), Burkitts lymphoma,(12,17.1%), Non Hodgkins lymphoma(10,14.2%)and Rhabdomyosarcoma(3,4.2%). The main reasons for the late report of these childhood cancer case by parents are due to limited knowledge of childhood cancers and their treatments (60, 85.7%), superstition believes(7, 10%) and lack for funds for transportation and investigations (3, 4.3%).

Conclusions: Although, there were many reasons sighted, lack of education and limited knowledge on childhood cancer and its treatment has been the major cause of the delay in reporting childhood cancers. Parents were unable to detect early warning signs of the cancer and why it must be reported early. Stakeholder must step up childhood cancer education as even health personnel deployed into the communities have limited knowledge about childhood cancers.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0192 / #809 | Primary Paediatric Cancer Services in Rural Ghana: Bridging Gaps in Geographical Access to Specialized Care

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Background and Aims: An overwhelming majority of paediatric cancer specialists practice in tertiary care facilities within the two largest cities of Ghana. Most children with cancer and their families travelled long distances for at least 2-hours, at high out-of-pocket costs to access daycare. Although shared care services exist in some regional/secondary level facilities, there are no reports of specialized paediatric cancer services at the primary care level. Financial and geographical access to care significantly contribute to late presentation and poor survival. We explored the utility of a Paediatric Oncologist, PO to the paediatric cancer care cascade in primary care in rural Ghana, and identified some facilitators and barriers.

Methods: A single centre qualitative retrospective assessment of a PO-led paediatric cancer services (PCS) at a primary care facility, over a 3-year period using an in-depth interview guide.

Results: The motivated PO through expert leadership was the critical link to the care cascade. Administrative commitment to support staff training, hospital bill waivers to limit financial toxicity was essential for sustainability. A core multidisciplinary team comprising unspecialized nursing and pharmacy staff trained in a comprehensive centre provided the needed supportive care. However, all paediatric ward practitioners received training and in-person and electronic (WhatsApp) supportive supervision. We adapted chemotherapy ordering, preparation, administration, and monitoring standards that guided the family/patient-centered care. Chemotherapy was safely prepared on-site research laboratory which had a safety cabinet and provided laboratory support occasionally. Some barriers experienced include staff attrition; limited infrastructure and competing needs from common non-malignant conditions; bureaucratic constraints including non-reimbursement by the National Health Insurance Scheme for PCS at a primary care level; financial toxicity and inadequate supportive care services.

Conclusions: Indirectly, the PO's presence "averted the referral of several benign haematological conditions" to tertiary facilities, thus potentially limiting high patient volumes at these centers while enhancing geographical access to paediatric cancer care.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0193 / #813 | Patients' Charts Completeness at a General Tertiary University Hospital in a Middle-Income-Country

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Background and Aims: Hospital-Based Cancer Registries (HBCRs) are an important tool to evaluate the implementation of the guidelines and also study treatment outcomes for selected protocols and provide evidence for changes and updates of the treatment protocols. Unfortunately, the patients' charts that are the main source of data, usually exhibit missing information, compromising the quality of data. This study aimed to evaluate the completeness of pediatric cancer patients' charts at a tertiary university hospital in a middle-income country.

Methods: Review of the pediatric cancer patients' charts over five years (2014-2019) with evaluation of the percentage of missing data necessary for the HBCR' registration form.

Results: Eleven variables were analyzed by percentage of missing data. Results: Considering a review of 174 charts, the variable Classification of Malignant Tumors (TNM) shown the highest percentage of incompleteness (70.1%), followed by "State of the disease at the end of the first treatment in the hospital" with 68.96% of incompleteness and stage of pediatric cancer (different from TNM and age up to 18 years) with 66.66%.

Conclusions: The high rate of missing data demonstrates the importance of interventions to help health professionals be aware of the importance of correct filling of patients' charts and also the implementation of a standardized pediatric staging system properly designed for this age group such as the Toronto Consensus.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0194 / #892 | Relapse and Disease Progression Among Children With Cancer in Armenia Diagnosed during 2019-2020

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Background and Aims: There is scarce statistical data on recurrence and disease progression among children with cancer in Armenia. This study aims to analyze cases of cancer relapse and disease progression to have baseline statistics for further research and to develop respective actions.

Methods: We present descriptive statistics for all cancer cases with recurrence and disease progression in children aged 0-18 years who

received treatment at the Pediatric Cancer and Blood Disorders Center of Armenia during 2019-2020.

Results: We identified 23 cases of cancer relapse and disease progression. The median age at relapse/disease progression was 8.4 years, male/female ratio was 1.56. Median time from diagnosis to relapse/disease progression was 18.8 months. Overall, 8 patients had the following events prior to cancer relapse: two neuroblastoma cases were in the high risk group at diagnosis; one metastatic Ewing sarcoma had missed radiotherapy during primary cancer treatment; one child with ALL had treatment interruption due to heart attack, and toxic cardiomyopathy; there were two cases of AML, one of which had three relapses, the first one following the withdrawal of transplantation. A child with Hodgkin lymphoma having a relapse after 5 years, initially received the whole treatment as it was planned. Among 15 cases with disease progression, HSCT was delayed for technical reasons for the case of medulloblastoma; three patients with neuroblastoma were at high-risk and had subtotal resections. Patients with synovial sarcoma and embryonal rhabdomyosarcomas (orbital and testicular) with subtotal resection had large veins invasions. One patient had alveolar rhabdomyosarcoma with metastasis. Cases of metastatic neuroblastoma, metastatic osteosarcoma, ALL, Burkitt lymphoma were treatment-resistant. From these patients, one is alive, one is disease-free, nine are on treatment, twelve died.

Conclusions: Cancer type, unresectable tumor locations and incomplete local control, treatment resistance, missed or delayed therapy are potential triggers of relapse and disease progression in children with cancer in Armenia.

Publication Only Topic: *Disciplines / AS05 Epidemiology, Policy and Advocacy*

PO0195 / #896 | Central Venous Access Device Related Infections in Pediatric Oncology Patients: A Single Center Experience

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Background and Aims: Long-term intravenous catheters are required for the administration of antineoplastic chemotherapy, parenteral nutrition, and blood products in pediatric oncology units. Central venous access devices(CVAD) may contribute as an additional risk factor for infections in these patients. In this study port-a-cath(PAC) and central venous catheter(CVC) are considered as CVAD. This study describes the epidemiology of CVAD associated bacteremia and the profile of microorganisms responsible for CVAD associated bloodstream infections in pediatric patients with cancer treated in a single center.

Methods: The retrospective analysis included patients with cancer who had an implanted PAC or CVC, that were hospitalized between January 2016-March 2021 in our center. The medical records of patients were reviewed for demographic characteristics, diagnosis, CVAD related complications, and their management.

Results: In the study period, in total 84 PACs and CVCs were inserted to 71 patients. The number of the patients, received chemotherapy during that time was 511. Male to female ratio was 38:33. Mean catheter indwelling time was 8.5 months for all patients. The most common pathogens responsible for these catheter related infections were methicillin resistant-coagulase-negative staphylococcus. In 30 patients (42%), we detected an infectious agent in the venous access. All catheters were immediately removed after detecting an infectious agent, and they immediately received antibiotic lock-in procedures in order to prevent intraluminal occlusions and/or catheter colonization. As hematogenous complications, 9 patients (12%) developed infective endocarditis. Sepsis occurred in 24 patients (33%). All patients were successfully treated. No patient was lost due to infectious causes. No organisms were identified in 41 patients (57%) with CVADs.

Conclusions: Pediatric hematology and oncology patients have a long-term use of CVAD. CVAD is the most common source of the bacteremia or sepsis in neutropenic patients. The most important step to prevent sepsis in these patients should start with early detection of CVAD associated bloodstream infections, proceeds with antibiotic-lock-in process, appropriate antimicrobial therapy choice and immediately removal of the catheter.

Publication Only Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PO0196 / #929 | Results of Peripherally Inserted Central Catheter (PICC) Program and its Association with Infections in Pediatric Patients in a Third-Level Oncology Hospital

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Background and Aims: Central venous catheters in cancer patients are necessary and PICCs are an effective alternative for their safety and ease of management. The objective of this study is to describe more than 10 years of experience in using PICC and report the complications found during this time period in our hospital.

Methods: This is a prospective descriptive study of pediatric patients treated via PICCs at the "SOLCA Núcleo Quito" Hospital since 2009. The SOLCA nucleo cancer hospital in Quito has 40 beds for pediatric cancer patients.

Results: During the study period, 78 PICCs were placed. The majority of patients (77%) were diagnosed with leukemia or lymphoma. Ninety

four percent (n=47) of the patients were receiving the St. Jude TOTAL XV protocol and they were in the induction stage when PICC were placed. There were 13.9 complications per 1000 catheter days and 5.4 infections per 1000 catheter days. The median of catheter days were 43.5 days, IQR 25th percentile were 23.75 days, 75th percentile were 87.25 days. As of 2011 all PICCs were placed by a surgeon. The 75.7% of PICCs were placed in the basilic vein and 60.3% (n=47) patients had no complications. Of 31 complicated PICCs (39.7%), 12 infectious complications and 19 were non-infectious. The infectious complications were PICC-related bloodstream infections and the pathogens were 7/12 Gram-positive bacteria (Staphylococcus epidermidis 6, Staphylococcus hominis 1), 4/12 Gram negative bacilli (Klebsiella aerogenes formerly Enterobacter aerogenes, Pantoea spp, Stenotrophomonas maltophilia, Escherichia coli), 1/12 fungi (Candida tropicalis). One patient died from K. aerogenes secondary to a PICC-related bloodstream infection. The most frequent non-infectious complication was occlusion (n=13).

Conclusions: Our findings indicate the need to continue staff education and training in the insertion, care and maintenance of PICCs to decrease complications. Best practices guidelines adherence are also critical to reducing complications.

Publication Only Topic: Disciplines / AS05 Epidemiology, Policy and Advocacy

PO0197 / #957 | Capacity Building Efforts at an NGO-Based Pediatric Oncology Unit After End of War in Syria

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Background and Aims: War in Syria (2011-2018) has significantly affected the health care system of the country. Major challenges, including shortage of well-trained staff and economic crisis, have faced the first established NGO-based pediatric oncology unit in the country during war, which started two years after its establishment in 2009. Immediately after the end of war in the southern parts of the country, capacity-building efforts started.

Methods: This abstract will describe all capacity building efforts at BASMA Pediatric Oncology Unit in Damascus, Syria between June 2018 and March 2021. The unit provides free treatment for all newly diagnosed children with cancer between the age of 0-18 years. The description will include staff recruiting, infrastructure expansion, and the resulted change in new yearly-enrolled patients in the unit. The collaboration with international pediatric oncology centers will be described.

Results: An internationally trained pediatric oncologist was recruited in June 2018 to lead the capacity building efforts. Another consultant pediatric oncologist was recruited in September 2019. The total number of physicians has increased from three in May 2018 to six in March 2021. The total number of pediatric oncology nurses has increased

from (n=20) in May 2018 to (n=35) in March 2021. The first expansion of the unit was completed in March 2020, which increased the number of inpatient beds from 18 to 36 and outpatient chairs from 6 to 12. A specialized pediatric oncology critical care unit was established in December 2020. The unit has officially joined the St Jude Global Alliance in September 2020. The average total number of yearly enrolled patients has increased significantly from (n=92) between 2011 and 2017 to (n=352) in 2020.

Conclusions: Capacity building efforts has successfully lead to triple the number of newly enrolled patients in an NGO-based pediatric oncology unit in Syria after end of war.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0198 / #105 | The Reality of Pediatric Oncology Physiotherapists from Three Continents

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Background and Aims: Pediatric cancer survival has been on the rise for the last 20 years worldwide. While there are still differences in rates of diagnosis and survival, there is also a lack of information regarding pediatric cancer rehabilitation programs. This translates to a lack of consensus on how to treat those patients.

Methods: A survey was sent via email with a set of 10 question regarding their experience in oncopediatrics. It was answered within a period of two weeks and was returned for analysis of the answers. The physiotherapists(PT) were from 3 different continents: NorthAmerica, Europe and SouthAmerica

Results: Workplace: Two work in combined settings of outpatients and inpatients while one works in an inpatient hospital setting only. Only one of the three PT works exclusively with pediatric cancer patients and they have worked an average of 9 years in this setting. Evaluation tools:one doesn't use any evaluation tool, while the other two use standardized tools such as the 6-minute walking test(6MWT), pediatric balance test(PBS) and gross motor test(GMFM). Rehabilitation:average of 2 sessions per week, and a time in rehabilitation of between 5 months to 2 years.The most common diagnosis was brain tumors, leukemia and solid tumors. What does working in pediatric oncology mean?:"It means everything! Best feeling to be able to help a young patient and his/her family to attain a significant functional goal in this difficult period!" "Optimizing quality of life, maximizing function, helping children live life to the fullest, making happy memories" "It means giving hope while also creating better chances to the patients and their families. With physiotherapy interventions we can improve the quality of life of our patients and fulfilling dreams"

Conclusions: This survey of three completely different environments, countries and languages is the first step to understand what is currently being done in pediatric oncology rehabilitation.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0199 / #1109 | Cancer Survivors in the Time of Covid-19 by CCCL's Champion Circle

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Background and Aims: The Children's Cancer Center of Lebanon (CCCL) acknowledged the need to establish a club for the fighters, survivors and triumphant of the war against this horrible disease, and we called it the Champions' Circle! In the previous years, we discovered that our children were distinguished with a spirit of trust, hope and strength among themselves, and provided moral support to each other when they were facing the most challenging times in their lives. Accordingly, CCCL's Champions Circle saw the light during a special event held on April 3, 2018 grouping inspirational cancer fighters from all over Lebanon and the region. We are currently over 230 cancer survivors.

Methods: The recent COVID-19 pandemic has affected the world and has disproportionately affected and disrupted the lives of cancer survivors: those who have completed treatment, and those who are now living cancer-free. Fulfilling its objectives, and to avoid the impairment of the club ambitions, the committee organized online sessions to replace the physical meetings and events. During these sessions, survivors were encouraged to participate through online platforms with specified and valuable topics to assist them in dealing with this unusual situation.

Results: As a result, 10 online events were planned such as peer-to-peer support, eating healthy, positive thinking, COVID effects and vaccination awareness, the benefits of HPV vaccination, engaging them in the international childhood cancer day activation, sophrology and breathing sessions, in addition to future plans such as IT skills and leadership training and career orientations workshop.

Conclusions: Despite all physical restrictions forced during this worldwide pandemic situation, this new norm adapted by the club will allow survivors to virtually come together to ensure their self-development and share their experiences during the assigned sessions, to provide them with the needed support to cope with the post-treatment effects.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0200 / #1114 | Well-Being of Survivors of Childhood Cancers in India

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Background and Aims: With newer modalities of childhood cancer treatment, survival rates have greatly improved, leading to increased life expectancy. This study aimed at examining the well-being of survivors coming from lower socio-economic groups in India, thereby

understanding the unique psycho-social challenges they may face and their coping mechanisms.

Methods: A qualitative study using in-depth interviews was conducted with 30 survivors of childhood cancer between the ages 18 and 35 years. The interview guide consisted of 33 open-ended questions exploring the challenges faced by them in areas of relationships, education, work and career, sense of self, physical and mental health, as well as their support systems, coping mechanisms and overall well-being.

Results: A thematic analysis of the interviews showed that most participants showed moderate to high levels of satisfaction with their lives. However, all survivors faced issues of social stigma to varying extents. All participants mentioned concerns related to interpersonal relationships and work. Gender differences were seen with more men expressing feelings of anxiety and guilt due to the long-term financial burden their treatment inflicted on their families, whereas more women mentioned facing stigma and concerns in areas of relationships and marriage, especially in those survivors where the treatment had resulted in physically visible, lasting side effects. Several participants mentioned that while fear of relapse was high during the first five years after completion of treatment, it was almost non-existent post this period. No significant differences in spiritual beliefs were observed. Individual differences were seen in the importance of survivorship to one's sense of identity. The theme of family support was consistent in all narratives and for most, friends and schools were seen as secondary support systems.

Conclusions: Although childhood cancer survivors have high levels of well-being and general satisfaction with their lives, societal and economic burdens were noted by them as after-effects of cancer.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0201 / #1417 | Clinical Characteristics and Outcomes of Paediatric Cancer Survivors in a Tertiary Hospital in Uganda

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Background and Aims: Present-day regimens for the treatment of paediatric cancer have achieved long term survival in up to 75% of children treated globally. With the advent of multimodal therapies consisting of chemotherapy, surgery and radiotherapy, survival has been advanced even further but the clinical course of paediatric cancer survivors in Uganda remains largely unknown. This study aimed to describe clinical characteristics and outcomes of children who completed cancer treatment at Mulago National Referral Hospital.

Methods: We reviewed medical records of children diagnosed with cancer who completed treatment between February 2019 and April 2021 at the paediatric oncology unit. Diagnosis was established by

histology and immunohistochemistry where applicable and treatment was based on institutional risk adapted treatment protocols.

Results: Of the 96 patients who completed treatment aged 0-18 years, 52/96 (54.2%) were male. The median age at diagnosis was 5 years (IQR 2.1-10.0). Majority of patients had been treated for Acute Myeloid Leukaemia 19/96 (19.8%) and Wilms tumour 19/96 (19.8%). Most were alive 88/96 (91.7%) and 84/96 (87.5%) in clinical remission following treatment. Relapse occurred in 12/96 (12.5%) patients and included one case of relapse with a new cancer. Of those who relapsed, 9/12 (75%) died. The median duration of survival was 19.05 months.

Conclusions: While the overall survival of children who complete cancer treatment in Uganda is promising, survival of relapsed patients is dismal. This emphasizes the importance of initial remission and calls for the integration of further long-term survivorship care in the overall scale up of paediatric cancer care in Uganda.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0202 / #1558 | Infant Malignancies – 5 Year Experience in a Tertiary Care Oncology Center

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Background and Aims: Infantile cancers constitute less than 0.05% of all malignancies. Clinical, biological pattern and outcomes are different from the cancers in older children. We aim to analyse epidemiology, treatment related complications and outcome of infants with malignancies.

Methods: Retrospective analysis of data of infants diagnosed with malignancy between 2015-2020.

Results: Out of 865 children registered for treatment of malignancies, 118(13.7%) were infants. The most common malignancy was acute lymphocytic leukemia (22.8%). Sixty-three children (53.3%) had solid organ, fifty (42.3%) had haemato-lymphoid and five (4.2%) CNS tumours. Solid tumours included neuroblastoma(16.1%), Wilms tumour(8.4%), LCH(6.7%), hepatoblastoma(5.9%), GCT(5.9%), RMS (2.5%) and 1 case each of Ewings sarcoma and infantile myofibroma. Hemato-lymphoid malignancies included-ALL, AML, TAM, JMML, Burkitt and DLBL. Neuroblastoma was the most common malignancy in the first three months and ALL between three months to one year. Of the 27 patients with infant ALL, 22(81.4%) opted for treatment. Out of which, six expired in induction or before starting chemotherapy, 4 relapsed, 5 died during the course of treatment and 2 abandoned treatment. Four are doing well for 3 years now post treatment completion. Complications during infant ALL treatment were infections (bacterial (6/27), viral (2/27), tuberculosis(1/27) and HLH(1/27). Amongst 19 with neuroblastoma, four were stage 4s and 11 infants required chemotherapy. One of these patients died on treatment, one relapsed post treatment completion and three were loss to follow up. In renal

tumours, 10 had Wilms tumour, 2 had mesoblastic nephroma, 2 had clear cell sarcoma and 1 had malignant rhabdoid tumour of kidney.

Conclusions: Infant malignancies have different biology than malignancies in other age groups. Malignancies in infants range from self-remitting tumors to the ones with dismal prognosis. Immature organ function & immune system makes administering therapy challenging. Right assessment & optimised therapy decides outcome in this cohort.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0203 / #1628 | Love for Children with Cancer: Supporting Hospitalized Children with Film Brochures by Japanese Survivor

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Background and Aims: Sustained support from outside the hospital is vital for children and their families who focuses on treatment the closed environment of the hospital. In order to realize sustainable support for them, it is necessary to increase the number of supporters, but a versatile support method that anyone can participate in has not been studied. Therefore, in this study, a universal support model for sustainable support is examined. And a case study of its application in a hospital in Japan is described.

Methods: The general public is assumed to be the supporter of this model. The flow of the model is as follows. (1) The supporter buys a pamphlet at a movie theater in his/her residence, and mails it to the hospital staff. (2) The hospital staff will post the brochure in the pediatric ward. The feature of this model is the use of a movie pamphlet. This achieves the following two points: (1) Since new films are released every day, the supporters can concentrate only on selecting movies. (2) Children and their families can enjoy the movie without waiting for DVDs or online distribution.

Results: The study model was adapted to a hospital in Japan. The hospital's child life specialist (CLS) provided support as a stakeholder in the hospital. As a result, support has continued from ICCD 2019 to the present. Besides, even during the epidemic of the new coronavirus infection, the quality of support has been the same as before the epidemic.

Conclusions: The epidemic of the new coronavirus infection has required a transformation in the way support is provided outside the hospital. The proposed model is a model that can be applied to any area where there are movie theaters. It has high potential as a new way of providing support under the new coronavirus epidemic.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0204 / #1755 | Rehabilitation Camp for Children with Cancer as an Effective Technology of Social Reintegration

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Background and Aims: A malignant tumor and its treatment negatively affects not only the somatic state of the child, but also his psyche, development and formation of personality, complicates socialization, which actualizes the need to develop programs of psychological and social rehabilitation, including the camp program for children with cancer and siblings. The goal is to create conditions for children with cancer and siblings to overcome social deprivation and successfully integrate them into society.

Methods: The camp was held 2 times a year on the basis of a children's sanatorium for children aged 7 to 17 years. 1275 children received rehabilitation, 25% of them were healthy siblings. The rehabilitation program is conducted by a multidisciplinary team – a pediatrician, a clinical psychologist, a psychologist, teachers of additional education, counselors.

Results: The key aspect of effective rehabilitation is the formation of an adequate psychological, pedagogical and recreational program, taking into account the individual characteristics of children. The camp session can be divided into three stages, the first one is related to the adaptation of the child and takes about 3-5 days. The main task of this stage is to create an emotionally safe environment, to form effective patterns of interaction in the team. The second stage lasts 7-10 days and is aimed at individual achievements of the child in various types of creativity, physical recreation and intellectual activities. The third stage is marked by an emotional uplift and ends with the culmination of the program-a festival, carnival, etc.

Conclusions: The effectiveness of the camp program is connected with its daily adjusting with the established periods of activity, individual orientation, and providing the dynamics of group processes. The annual monitoring of the assessment of the qualitative and quantitative results of the psychological and social rehabilitation of disabled children in the camp has proved the effectiveness of this rehabilitation technology.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0205 / #217 | Experiences of Childhood Cancer Survivors Who Received Hematopoietic Stem Cell Transplantation

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Background and Aims: While hematopoietic stem cell transplantation is an effective treatment for hematologic malignancies, it is associated with physical and psychologic risks and may threaten the quality of life of patients. However, experiences of childhood cancer survivors

after undergoing hematopoietic stem cell transplantation are poorly studied. This study aims to explore the significance of the lived experiences of childhood cancer survivors after undergoing hematopoietic stem cell transplantation.

Methods: The fourteen survivors of childhood cancer who received hematopoietic stem cell transplantation participated in depth-interview, conducted face-to-face or online interview. Data collection period was October 2020 to January 2021. The analysis was based on Giorgi's phenomenological research method.

Results: The childhood cancer survivors thought that the transplant process itself was difficult, but they believed that receiving hematopoietic stem cell transplantation from their family or other people have given them a new life. Four themes, namely building a healthy body by regaining physical strength, a desire to express the overwhelming burden, cultivating positive thinking through reflection, and living a self-directed life, emerged in the lived experiences of childhood cancer survivors who underwent hematopoietic stem cell transplantation.

Conclusions: The findings of this study contribute to an understanding the experiences of childhood cancer survivors after undergoing hematopoietic stem cell transplantation. To improve their quality of life after transplantation, interventions that enable them to express the burden on their heart and help them recover their physical fitness are needed. Acknowledgements: This work was supported by the National Research Foundation of Korea grant funded by the Korea government (2019R1F1A1058944).

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0206 / #231 | Transdisciplinary Rehabilitation of Posterior Fossa Medulloblastoma

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Background and Aims: Posterior fossa medulloblastomas are one of the most common central nervous cancer in childhood. The incidence is 20% of all brain tumors and, due to the delicate zone they are located, sequels can go from low to severe impairments. These can even affect their independence, motor development, daily living activities and cognitive function for the long term. Because of the ample range of sequels, a transdisciplinary rehabilitation treatment program should be implemented in this population.

Methods: We gathered information of a typical clinical case of posterior fossa medulloblastoma who was referred to a rehabilitation service early on their diagnosis. The treatment consisted of a chemotherapy and radiotherapy combination for 18 months. We used standardized evaluation tools by occupational therapy, clinical neuropsychology, physiotherapy and pediatric dentist, when first referred in 2019 and 2021 for the final evaluation of the rehabilitation program.

Results: The length of time spent in rehabilitation was around 2 years. The balance impairments, which limited the ability to walk, diminished over time and have now receded to a functional level. All areas of occupation improved over time and independence was achieved. The cognitive

function has improved in the areas of speed processing, attention, working memory, and more importantly, school reinsertion was accomplished. The initial motor function impaired the ability to independently brush the teeth. This, added with bad oral habits, increased the risk of having infections related to lack of hygiene. With the work of the dentist combined with motor improvement over time, good oral health was attained.

Conclusions: Transdisciplinary interventions play a major role in posterior fossa medulloblastoma, and in pediatric cancer patients. Objectives that include more than one discipline help both the children and the therapists in achieving their best functional level, become independent and facilitate their inclusion and participation in society.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0207 / #633 | Daily Lives of Adolescent Survivors with Severe Chronic GVHD

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Background and Aims: Recent medical advances have increased long-term survival of patients after hematopoietic stem cell transplantation. However, the impacts of transplant-related complications are problematic. Patients with severe chronic GVHD require long-term treatment for complications, despite resolution of primary disease, and their quality of life is affected by symptoms that greatly limit their daily activities. Our study purpose is to clarify the perceptions of the current situation held by adolescents with severe chronic GVHD and their parents, and the impact of complications on their daily lives.

Methods: We interviewed two 18-year-old adolescent patients with severe chronic GVHD and their mothers about their perception of the current situation and daily life, and conducted qualitative descriptive analysis.

Results: Case A had muscle and joint contractures, vision impairment, and required full ADL assistance provided by her mother. Case B had bronchiolitis obliterans and planned to register for a lung transplant pending improvement in renal function. Both patients perceived their current physical condition as "physical pain due to current symptoms" and "inability to cope with daily symptoms", and their future physical condition as "uncertain what will happen in future" and "trying to improve my current condition". Their mothers also had the same perceptions. Case A felt that he had "no enjoyment in life" and "no social connections outside of the hospital" after graduating from high school. Case B, a high school student, felt that she had no friends to talk to and no relationships with similar-aged people with shared experiences.

Conclusions: In adolescents with severe chronic GVHD, physical symptoms made them feel uncertain about the future, but they had hope for improvement. Moreover, they had difficulty in acquiring ego identity due to limited social connections. Our findings suggest that physical care and continuous psychological and social care through a child's development is necessary.

Publication Only Topic: *Disciplines / AS06 Survivorship*

PO0208 / #972 | Adolescents and Young Adults Healed from Oncohematological Pathologies: Analysis of Life Experiences Through Efi-C

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Background and Aims: The literature on adolescent and young adult (AYA) cancer survivors was very huge, but a few studies had a qualitative approach. This study aims to: 1. Assess the general well-being and the personal growth of the participants; 2. Evaluate their possible associations with socio-demographic and illness variables.

Methods: Fifty AYA (25 males and 25 females) out of therapy for a mean of 8.2 years (SD=2.9), with a mean age of 19.3 (SD = 2.8) were recruited at the Pediatric Hematology, Oncology and Stem Cell Transplant Center (University of Padua) and, after the signature of informed consent, they were interviewed adopting the off therapy version of Ecocultural Family Interview-Cancer (EFI-C, Tremolada et al., 2018). A thematic analysis on EFI-C dimensions was run and the interviews were inserted in NVivo software, defining a network of nodes. A mixed-method approach was adopted using the word clouds and the relationship of the word frequencies with socio-demographic and illness variables.

Results: Healed patients showed problems related to physical well-being (54%), even if the influence of the disease on their lives (60%) and relationships (44%) were perceived as positive. The paired-sample T-test showed that females (Mean = 24.33; SD = 18.00) used more growth words than males (Mean = 16.92; SD = 14.41) ($t = -2.5$; $df = 48$, $p < 0.05$). The ANOVA test identified the presence of significant differences in the number of total words used in relation to the current age of the AYA ($F = 4.48$; $df = 2$; $p < 0.05$), with those aged 18-25 that used a higher number of words than those aged 15-18. A negative Pearson's correlation emerged between the number of words referring to the disease and the number of years that have elapsed since the end of the therapies ($r = -0.2$, $p < 0.05$).

Conclusions: Clinical suggestions for intervention could be derived from these results.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0209 / #1088 | Encouraging Outcomes of Pediatric High Risk and Relapsed Acute Leukemias-A Developing World Experience

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Background and Aims: Outcomes of high-risk and relapsed pediatric acute leukemias continue to be sub-optimal in the developing world. Allogeneic Hematopoietic stem cell transplantation (Allo-HSCT) is the only curative modality. However, less than 30% of patients have matched sibling donors (MSD) available. Hence, alternate donors [matched unrelated (MUD) and haploidentical] are being used to improve outcomes.

Methods: We retrospectively analyzed our data of all children with high risk/relapsed acute leukemias who underwent alternate donor HSCT at our center from April 2015 to July 2020.

Results: A total of 15 patients were included - three underwent MUD and 12 underwent Haploidentical HSCT (haplo-HSCT). Prior to HSCT, all patients were in Complete Remission (CR): CR1 - 1, CR2- 11 and CR3- 3. The median stem cell dose (CD34) was 6.5 million/kg for MUD HSCT and 5.6 million/kg for haplo-HSCT. All patients engrafted except one. Median time to neutrophil and platelet engraftment was 15 and 16 days, respectively. There were 3 transplant related mortalities (TRM). Two patients died of pneumonia - one before engraftment and the other one at Day +18 post-transplant. The third patient died due to Grade IV steroid refractory Graft vs Host Disease (GVHD). One patient was lost to follow-up. Remaining 11 patients remain in remission and are alive. The cumulative incidence of acute GvHD was 57.1% and of chronic GvHD was 21.4%. Overall Survival was 80% and the Event-free survival was 73.3%. The median follow-up of alive patients was 775 days (range: 333-2077).

Conclusions: Our experience shows encouraging outcomes for these high risk and relapsed pediatric patients of Acute leukemias using alternate donor HSCT from developing world.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0212 / #1260 | Leukoencephalopathy by Methotrexate in Patients with Acute Lymphoblastic Leukemia at National Pediatric Oncology Unit, Guatemala

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Background and Aims: Methotrexate (MTX) at high doses (HD) is used for many neoplasm treatments. However, it is associated with adverse events (AE), such as leukoencephalopathy, its prevalence reported varies 5-18%. Therefore, it is necessary to identify and evaluate its prevalence, and develop strategies to reduce complications.

Methods: Children diagnosed with acute lymphoblastic leukemia (ALL), aged 1-17.9 years at diagnosis received treatment according with protocol ALLG-0707 at UNOP (Pediatric Oncology National Unit), Guatemala. Treatment includes HD-MTX in 24-hour infusion (2 or 5 g/m²) in non-high risk (HR) or intensive polychemotherapy blocks in HR patients, and MTX age-adjusted intrathecal therapy (IT). We retrospectively reviewed 1473 consecutive cases (2010-2019), searching clinical or radiological findings suggestive of neurological AE.

Results: Only 42/1473 (2.8%) of 233 suggestive cases, were confirmed with leukoencephalopathy secondary to MTX, by imaging and neurologist evaluation. Forty were classified as subacute, median age at presentation was of 14.6 years [IQR 3.2-16.2], 62% were female, 90% corresponded to intermediate risk-group. The median accumulative MTX dose was of 6 g/m² [range 0-20] (3 patients had no previous HD-MTX). Over 75% occurred within 7 days after the 8th IT. All cases were admitted (average of 3.6 days), and treated with citicoline plus steroids (26), only citicoline (11), or only observation (5). Subsequent HD-MTX were limited and/or IT-MTX omitted. Only 3 cases continued without modification, no sequelae reported. Of note that 54% of the 42 cases relapsed, and at the time of the study 26% died after relapse.

Conclusions: UNOP have a lower prevalence leukoencephalopathy than reported. All cases were reversible, grade 1-2, without long-term motor or neurological sequels. No sequelae were reported after reintroducing methotrexate. Considering relapse rate, establish the treatment-guideline for neurological AE it is needed in order to a better use of steroids or avoid the modification of the ALL protocol.

Publication Only Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PO0213 / #1273 | Improving Access to Minimal Residual Disease Assessment: Lesson Learnt

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Background and Aims: Risk stratification and response-adapted therapy are the fundamental tenets for optimal management of childhood Acute Lymphoblastic Leukemia (ALL). Minimal Residual Disease (MRD) analysis at End of Induction (EOI) has emerged as the most important prognostic indicator and guides further therapy. All current treatment protocols incorporate MRD-based therapy. Access to accurate MRD assessment remains challenging for many centers in India. This project was initiated by a National Society for Childhood Cancer support to improve access to MRD.

Methods: Samples were collected at EOI for B-ALL and at end-of-consolidation for T-ALL. Patients' demographic profile and lab

parameters were documented. Flow cytometry-based MRD assay was performed as per the current best practice guidelines at one of the three designated laboratories. Concerted efforts were made to ensure quality control. Those with positive MRD were offered repeat testing after further therapy.

Results: Over a four-year period, 1850 samples from 1634 patients and 36 centers were processed. Distance traveled by MRD by sample ranged 3-2300 kilometers. Only 19 samples were unsuitable for analysis. Of the 1634 patients, 86.3% had B Lineage ALL, 12.9% had T ALL and 14 patients had relapsed ALL. Treatment protocols used were ICiCle in 52.3%, COG in 22.0%, BFM in 18.7%, and others 7%. Three seventy-five (26.6%) of patients with B lineage ALL and 22.6% had positive MRD at EOI and end of consolidation respectively. In over 75% of patients, more than 5 lakh events were analyzed. For patients with positive MRD at time point one, treatment was escalated in 235 patients.

Conclusions: With collaborative efforts, making MRD assay available to every child with ALL appears to be an achievable goal. MRD assessment allowed treatment modification for a proportion of patients. This may contribute towards improved outcomes. Ongoing follow-up will provide further insights regarding outcomes of MRD-based treatment adaptation in our setting.

Publication Only Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PO0214 / #1312 | Testing of a Nanostring Custom Gene Fusion Assay for the Rapid Diagnosis of Pediatric Cancers in Low-Middle Income Countries (LMICs)

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Background and Aims: Risk stratification and molecular targeting are key to increasing pediatric cancer cure rates in high-income countries. Precise diagnosis and successful treatment in low-resourced settings is often hindered by insufficient pathology infrastructure, including lack of molecular analysis. Given the high frequency of gene fusions in pediatric cancers, identifying such fusions would support cost-effective diagnosis, risk stratification, and precision medicine in LMICs.

Methods: The NanoString nCounter platform was chosen for implementation in LMICs due to minimal technical expertise required, ability to test samples with sub-optimal RNA quality, and rapid turnaround-time (2-3 days). Literature and public databases were used to obtain exact breakpoint sequence information for gene fusions associated with pediatric cancers. The hematologic malignancy panel was designed to detect 439 breakpoints for 223 non-IGH/TCR fusions

reported in leukemias and lymphomas. The solid tumor panel was designed to detect 436 breakpoints for 204 fusions.

Results: Assays (96 hematologic and 48 solid tumor) were performed using samples with known fusion status. In the initial assays, 90% had the expected fusion (or lack of any fusion) confirmed. Of those assays with unexpected results, 1 was unevaluable due to high background, 6 had too little tumor RNA based on concentration or tumor content, 5 were presumed false negatives, and 3 were presumed false positives.

Conclusions: The custom-designed fusion assay performed well with the most consistent results obtained with 100ng of RNA from a sample with >10% tumor content. Additional sequencing is needed to follow-up the unexpected results to determine final detection rate. Sequencing could confirm fusions that were not previously known to be present (false positives) or confirm novel breakpoints that are not in the assay design (false negatives). Implementation of this assay in LMICs, first planned for the Global HOPE site in Uganda, will be transformational in the care of these patients.

Publication Only Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PO0215 / #1490 | Microrna Expression Profiles in Pediatric Germ Cell Tumors: Implications in Distinct Histological Patterns

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Background and Aims: Germ cell tumors (GCTs) are benign or malignant neoplasms that presenting distinct histological patterns. GCTs are derived from primordial germ cells and can occur in gonadal and extragonadal sites and affects individuals of any age group. The main biological markers of GCTs are alpha-fetoprotein and the beta fraction of chorionic gonadotropin. However, there is no specific molecular marker for the diagnosis/prognosis of pediatric GCTs. Thus, the aim of this study was to evaluate the expression of miRNAs in pediatric ovarian and testicular GCTs and relate them to the clinical-pathological data.

Methods: A total of 42 GCTs pediatric patients were evaluated, including 11 testicles and 31 ovaries with distinct histological patterns. As a control group, we consider five samples of adjacent normal ovarian tissue and five testicular. The expression profile of miRNAs was analyzed by NanoString platform using the panel Human v3 miRNA Assay CSO. The data was analyzed with R/Bioconductor. The hierarchical clustering analysis showed that normal samples and teratomas clustered separately from the malignant tumors (endodermal sinus, embryonic carcinoma, dysgerminoma).

Results: Sixty-five miRNAs were differentially expressed between normal and malignant samples, which miR-302d-3p, miR-302b-3p,

miR-372-3p, miR-373-3p, miR-1246, miR-302a-3p, miR-302c-3p, miR-367-3p, miR-205-5p and miR-122-5p were overexpressed in malignant tumors, while miR-509-3p, miR-513b-5p, miR-508-3p, miR-513c-5p, miR-424-5p, miR-514a-3p, miR-507, miR-99a-5p, miR-506-3p and let-7c-5p were downregulated. Each histological pattern showed increased expression for specific miRNAs, including embryonic carcinoma (miR-1323, miR-520d-3p, miR-517b-3p), dysgerminoma (miR-142-3p, miR-150.5p, miR-29b-3p), endodermal sinus (miR-126-3p, miR-375, miR1247-5p). A Venn diagram was constructed to indicate the distribution of the expressed genes between testis and ovary. The members of the miR-371~373 and miR-302 clusters were the most differentially expressed in both testicular and ovarian tumors. The fold changes were most marked for miR-1915-3p in testicular tumors and miR-1246 in ovary tumors.

Conclusions: Our results indicates that microRNAs are promising candidate biomarkers in malignant GCTs.

Publication Only Topic: Disciplines / AS07 Childhood Cancer International (CCI)

PO0216 / #1497 | Pediatric Teratomas: A Review of the Brazilian Cohort

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Background and Aims: Teratomas are considered a subtype of germ cell tumors (GCT) that contain all types of embryonic germ cell layers and are classified as mature (MT), immature (IT) and malignant. This study evaluates the incidence and epidemiology of mature and immature teratomas in a Brazilian cohort demonstrating.

Methods: Two Brazilian pediatric clinical trials were selected (GCT-99 and GCT-2008) with a diagnosis of teratoma in any location were included. Patients were stratified by risk of disease and those classified as low and intermediate risk were treated with surgery only, while high risk patients received chemotherapy (CHEMO).

Results: 414 patients were included 290 MT and 124 IT. Most patients were female and the average age was 6.5 years MT and 5.5 IT. Ovary was the most frequent location for TMs (143) and ITs (52). Of these, the histology of 52 patients was Grade 1 and 28, Grade 2 and 42, Grade 3, and it is not possible to classify 2 patients. 31.5% of ITs had elevated AFP, 5.6% increased β HCG and 15% increased DHL. They had complete resection of the primary tumor 278 MTs and 99 ITs, macroscopic residual in only 1 MT and 7 ITs, 8 MTs and 9 microscopic residual ITs and 2 MTs and 5 ITs had isolated biopsy. There was no MT patient with distant disease and did not receive CHEMO. 12 IT patients (9.7%) had distant disease, the ganglion and lung being the most affected organs and these patients received CHEMO. 3 MT patients had recurrence, with the primary site being the affected site. 3 IT patients and 1 MT died as the first event.

Conclusions: There is no efficacy in the use of chemotherapy in MT and treatment with chemotherapy has not improved the results in IT patients.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0217 / #1562 | Epidemiological Aspects and Clinical Characteristics in Children with Nasopharyngeal Cancer in Tunisia

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Background and Aims: Introduction : In Tunisia, nasopharyngeal cancer in children accounts for about 18% of all nasopharyngeal cancers and 5 to 10% of pediatric tumors. The objective of this study was to describe the epidemiological profile and clinical characteristics of this disease in our patients.

Methods: Our study was retrospective. It included 68 patients aged less than or equal to 18 years followed for nasopharyngeal cancer and treated at the Salah Azaiez Institute between January 2004 and December 2014.

Results: The median age of our patients was 14.7 years with a sex ratio of 2. The average delay after the appearance of the first symptoms was 4 months. 63% of children consulted for cervical lymphadenopathy. 82% of patients presented rhinological signs. Otological signs were present in 50% of cases. 28% presented with neurological signs on initial examination. Nasofibroscopy was performed systematically in all of our patients. The tumor filled the entire cavum in 82% of patients. It happened to choanae in 12% of cases. The physical examination of our patients revealed lymph node involvement in 89.7% of patients. It was unilateral in 29.4% of cases and bilateral in 60.3% of cases. Cranial pair involvement was objectified in 11 patients. The trigeminal nerve (V) was most frequently affected (82%). Two patients in our series presented with digital hippocratism. All the patients were classified (or reclassified) according to the TNM 2010 classification: 51 patients were classified T3-T4 (79%) and 43 (63%) were classified N2-N3. The incidence of metastasis at the time of diagnosis of juvenile PNC ranges from 1.5% to 15%: one 17-year-old with bone metastasis to the pelvis and the other a 10-year-old girl with metastasis to a mediastinal lymph node

Conclusions: Juvenile nasopharyngeal carcinoma is relatively frequent in Tunisia. In most cases, it is a locally advanced disease with lymph node metastases. Treatment is usually multidisciplinary

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0218 / #1564 | Survival and Prognosis Factors in Nasopharyngeal Carcinoma in Tunisian Children

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Background and Aims: In Tunisia, pediatric cancers diagnosed before 18 years old represent about 5% of nasopharyngeal cancers and it is the

most common epithelial malignancy in pediatric oncology. The objective was to describe the course of the treated disease and to determine the prognostic factors involved in treatment failure and survival in juvenile nasopharyngeal cancer.

Methods: Our study was retrospective. It included 68 patients aged less than or equal to 18 years treated at the Salah Azaiez Institute between January 2004 and December 2014

Results: After a mean follow-up of 94 months, 78% of the patients were alive and in complete remission, 19% were in a situation of treatment failure with 16% of metachronous metastases. The occurrence of treatment failure was noted within 2 years following the end of treatment with an average delay of 16 months (6-24 months). The 5-year overall survival in the absence of treatment failure was 95% and that in the presence of treatment failure was 30%. The 10-year overall survival was 74%. The 10-year disease-free survival was 72%. Overall survival and 5-year disease-free survival in patients less than 14 years of age were better than in older patients. Overall survival and disease-free survival rates were better for females with no statistically significant difference. A statistical correlation with the occurrence of treatment failure was observed with the involvement of a cranial pair ($p=0.029$) and extension to the infra-temporal and / or intracranial fossa ($p=0.007$). A therapeutic protocol, including first chemotherapy, seems to decrease the risk of treatment failure in univariate and multivariate analyses

Conclusions: The combination of chemotherapy and radiotherapy appears to improve survival. However, metastatic treatment failure (the most common mode of failure) remains a major problem

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0219 / #1635 | Incidence of Childhood Cancer in Jordan, 2000-2017

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Background and Aims: The most recent population-based study of childhood cancer in Jordan dates back almost two decades. Current estimates are necessary to inform public health policy and improve cancer care. Therefore, we described the incidence of childhood cancer in Jordan using all available data from a population-based cancer registry.

Methods: We accessed data from the Jordan Cancer Registry (2000-2017) and retrieved all records of cancer diagnosed in residents of Jordan 0-14 years of age at diagnosis. We classified cases according to the International Classification of Childhood Cancer (ICCC) and calculated frequencies, age-adjusted incidence rates (AAIRs), incidence rate ratios (IRRs), and annual percentage changes

(APCs). We then followed up the main analyses with joinpoint trend analyses.

Results: Between 2000 and 2017, cancer was diagnosed in 3,933 children in Jordan (AAIR, 82.4 per million children). Boys outnumbered girls 2,265 (57.6%) to 1,668 (42.4%), and the corresponding AAIRs were 93.0 and 71.4, respectively (IRR, 1.30 [1.22–1.39]; $p < 0.001$). The five most common types of cancer were leukemia (29.5%; AAIR, 24.2), central nervous system tumors (17.5%; AAIR, 14.4), lymphoma (15.6%; AAIR, 13.1), soft tissue sarcoma (6.6%; AAIR, 5.4), and sympathetic nervous system tumors (6.1%; AAIR, 4.8). The incidence of leukemia, central nervous system tumors, lymphoma, and sympathetic nervous system tumors trended downward linearly between 2000 and 2017, while the incidence of soft tissue sarcoma was stable over the same period. The downtrend in the incidence of leukemia was the most pronounced (APC, -4.8% [-6.8% to -2.7%]; $p < 0.001$).

Conclusions: The distribution of childhood cancer cases in Jordan, according to the ICC, is consistent with international reports. The incidence of childhood cancer in Jordan is also consistent with regional estimates, which are generally lower than estimates from Europe and the Americas.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0220 / #1669 | Pre-Conception and Pregnancy Parental Exposure to Pesticides and Pediatric Neuroblastoma. A Meta-Analysis of Nine Studies

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Background and Aims: Background: The presentation of neuroblastoma in early childhood prompted researchers to investigate some relevant etiological agents, especially when maternal or parental exposure to harmful factors takes place earlier during conception, pregnancy, or exposure in early childhood. However, findings of existing case-control studies on the effects of pesticide exposure on childhood neuroblastoma have been often inconsistent. The objective of the study is to determine the potential association between parental exposure to pesticides and the incidence of neuroblastoma in children. **Methods:** Studies that investigated the link specifically between neuroblastoma and pesticides were searched from PUBMED, SCOPUS and Google Scholar from 1960 through August 2018. Only the case-control studies showing the link specifically between neuroblastoma and various pesticides were selected for a systematic review and meta-analysis.

Results: Our results demonstrated that parental pesticide exposure in preconception or pregnancy period is associated with increased risk of Neuroblastoma with an OR of 1.6 with Confidence interval (CI): 1.1, 2.3;

p -value = 0.013. While pesticide exposure after birth showed an OR of 1.0 with CI: 0.8, 1.3; p -value = 0.723]. Additionally, we also examined the relationship between parental occupational exposure to pesticides and neuroblastoma. Our result showed that this relationship is insignificant.

Conclusions: There appears to be an association between parental exposure to pesticides and neuroblastoma when the exposure takes place in the prenatal and pregnancy period. It is mention worthy that after SARS-CoV-2 pandemic, "stay at home" lifestyle is introduced by WHO, to prevent the transmission of virus. This policy, with all its advantages over controlling COVID-19, carries individual and environmental risks, thus people are more likely to exposed to increased disinfectant as well as pesticides used as home.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0221 / #1671 | Pesticides Exposure and Wilms Tumor in Childhood. A Prisma-Based Systematic Review with Meta-Analysis

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Background and Aims: Background: There exist hereditary forms of Wilms tumor associated with preconception exposures of the germ cells of either parent or somatic forms that may be associated with exposure to harmful environmental factors. Several case-control and cohort studies have examined pesticide exposure as risk factor for Wilms tumor. This review relies solely on case-control studies to determine the potential association between exposure to pesticides and the occurrence of Wilms' tumor in children.

Methods: PUBMED, SCOPUS and Google Scholar studies (1960-2018) were identified and systematically reviewed following PRISMA guidelines. Subgroup analyses were conducted after stratification for occupational vs. residential exposure and before birth (prenatal) vs. after birth (postnatal) exposure.

Results: Our results demonstrated that parental pesticide exposure in preconception or pregnancy period is associated with increased risk of Wilms tumor.

Conclusions: There is an association, only if exposure takes place before birth. Parental exposure to pesticides in household settings seems to show a strong association. Although this data is derived from six occupational exposure studies only, an OR > 2 should be considered highly relevant. These results are even more important in the recent SARS-CoV-2 pandemic, when "stay at home" lifestyle introduced by WHO, is encouraged and frequently adopted to prevent the transmission of virus. However, this policy, despite its advantages, also pose environmental risks as there is an increased likelihood of that people being exposed to disinfectant as well as pesticides used as home.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0222 / #1672 | Parental Exposure to Pesticides and Childhood Brain Cancer. A Systematic Review and Meta-Analysis

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Background and Aims: Background: Cancer is one of the most important causes of childhood death in developed countries, while malignant brain tumors are the second most common cancer in pediatric age. Pesticides may play an etiologic role, but the literature results are conflicting. This review provides a systematic overview and meta-analysis of published data on parental exposure to pesticides and childhood brain tumors.

Methods: We searched PubMed, SCOPUS, and Google Scholar for scientific literature published between Jan 1, 1966, and Aug 31, 2018, that assessed childhood brain tumors and parental exposure to pesticides. Then we undertook a meta-analysis addressing prenatal exposure, exposure after birth, occupational exposure, and residential exposure.

Results: We found that a prenatal exposure is associated with childhood brain tumor, odd ratio (OR) =1.32 (95% CI: 1.17 - 1.49; I^2 = 41.1%). The same was true for after birth exposure OR=1.22 (95% CI: 1.03-1.45, I^2 = 72.3%) and residential exposure to pesticides OR=1.31 (95% CI: 1.11-1.54, I^2 = 67.2%). Parental occupational exposure was only marginally associated with incidence of brain tumor (OR=1.17, 95% CI: 0.99-1.38, I^2 = 67.0%).

Conclusions: There was a statistically significant association between parental pesticide exposure before childbirth and childhood brain tumor. Similar associations were found for after-birth exposure and residential exposure, whereas a weak association was found for parental occupational exposure to pesticides. Awareness and education programs regarding the adverse effects of pesticide exposure on health may be beneficial to minimize the exposure. Notably, in the light of the recent SARS-CoV-2 pandemic, when stay-at-home has become a new norm, there is a risk of individuals being more exposed to pesticides in addition to disinfectants.

Publication Only Topic: *Disciplines / AS07 Childhood Cancer International (CCI)*

PO0226 / #1742 | Frequency of Hereditary Cancer Syndroms in Children- A Single Institution Experience

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Background and Aims: Although cancer in children is rare, it is the leading cause of death by disease past infancy among children. To improve patient outcome, an increased understanding of the underlying molecular changes, genetic mutations and their biological consequences is necessary. The extant clinically descriptive Cancer Predisposition Syndromes (CPSs) are rarely established for children with cancer. At our university center, we have investigated the frequency of a CPS between 2017 and 2020.

Methods: Suspicion of CPS was assessed twice: at the first presentation according to the Jongmans Pediatric Guidelines criteria, modified by the German Society for Pediatric Hematology and Oncology (GPOH), and supplemented by a center specific cancer associated dysplasia questionnaire. Secondly by validation of the cumulative toxicity at the end of therapy, defined as the total of all grades adverse events and severe adverse events. In patients with suspected CPS or relevant toxicity, whole-exome sequencing of germline DNA was performed, following genetic counseling and parental consent.

Results: In 324 of the 370 patients CPS was assessed (87,6%), of whom 85 (26,2%) were identified to be at risk for a childhood-onset CPS following the GPOH modified Jongmans criteria at first presentation (81,2%), or showed elevated toxicity (18,8%). 20 of these 85 patients (23,5%) harbored a confirmed CPS (n = 3 three patients with biallelic affection). Additionally 16 patients carried a pathogenic variant (PV) in a CPS gene. 12 of 85 (14,1%) had clinical features indicating CPS of those five (41,6%) were genetically verified.

Conclusions: These results indicate that a higher proportion of childhood cancer may be linked to predisposing germline variants than prior supposed. Therefore, a systematized investigation of underlying CPS, based on standardized guidelines, along with the analysis of the individual's cumulative toxicity at the end of treatment are necessary

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PO0228 / #252 | Systematic Review of Gastrostomy Complications and Outcomes in Pediatric Cancer and Bone Marrow Transplant

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Background and Aims: Nutrition support is essential in children with cancer, including those undergoing bone marrow transplant (BMT), to reduce the risk of malnutrition and associated deleterious outcomes.

Enteral nutrition is more commonly provided via nasogastric than gastrostomy tubes due to safety concerns with the latter in immunocompromised children. This systematic review investigated the incidence and type of complications and outcomes in pediatric cancer patients fed by gastrostomy.

Methods: Databases including CINAHL, Cochrane Central, EMBASE and MEDLINE were searched for randomized and observational studies investigating the use of any gastrostomy device in children aged <18 years with any cancer diagnosis, including those undergoing BMT. Five cohort and 11 case series studies were included. Risk of bias was assessed using the risk of bias in non-randomized studies of interventions and the National Institutes of Health quality assessment tool for case series. Due to clinical heterogeneity meta-analyses were not performed.

Results: Quality of evidence varied with five studies judged at serious risk of bias and poor quality, however the remaining 11 were considered at moderate risk of bias to good quality. Across studies, 54.6% of children developed ≥ 1 complication, of which 76.6% were classified as minor, 23.4% major. The most frequent complications included inflammation (52% of episodes), infection (42.1%), leakage (22.3%) and granuloma (21%). Evidence regarding infection rates in cancer/BMT patients compared with other disease states was inconclusive. Gastrostomy feeding was associated with improvement or stabilization of nutritional status in 77–92.7% of children. Length of admission was no different between children fed via gastrostomy versus nasogastric tube.

Conclusions: Gastrostomy feeding in this population is relatively safe and effective in stabilizing or improving nutritional status throughout treatment. Complications are frequent but mostly minor. Placement requires careful consideration of the complications, benefits, nutritional risk and status at diagnosis, treatment intensity and quality of life.

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PO0229 / #364 | Hematopoietic Stem Cell Mobilization and Collection for Children with Low Body Weight: Experience of N.N. Blokhin National Center of Oncology

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Background and Aims: Hematopoietic stem cell mobilization and collection is important stage of treatment for children with malignant disorders. Special group of patients is children with low body weight (under 15 kilograms). A special program for such patients required. This, we aimed to build such program and analyze results.

Methods: From January 2020 to April 2021 we performed 27 collections for children with low body weight. M:F=12:15. Age median: 39 months (12 months – 144 months). Weight median: 12.2 kg (7.7 –

15 kg). All patients were mobilized with G-CSF in a daily dose of 10–20 $\mu\text{g/kg}$. One poor mobilizer received preixafor in a dose of 120 $\mu\text{g/kg}$. CD34+ cells control: on 3rd day of mobilization on BD FACSCanto II flow cytometer. Collection: on Spectra Optia cell separator with MNC 6.9 and 11 software at a flow rate of at least 7.6–10 ml/min to ensure safe admission of the ACD-A. Collection was made with the help of central venous catheters (CVC): two-channel Certofix 18/20 catheter or single-channel Certofix 16G in femoral access for blood collection (in this case main CVC was used as return line). Extracorporeal circuit of the separator was filled with individually matched donor red blood cells irradiated with 25Gy. Preparation, mobilization and collection were made by multidisciplinary team. Special attention was paid for collection – all collections were made in ICU department.

Results: All patients received one collection. All the collections ($n=27$) were successful. No significant complications were registered. Only 3 patients have non-significant technical issues with CVC resolved successfully during the procedure. Average CD34+ cells collected in the final product was 15.25×10^6 per kg ($0.04 - 92.4 \times 10^6$ per kg).

Conclusions: Our experience showed the possibility of successful stem cell mobilization and collection for children with low body weight and program established in N.N. Blokhin National Center of Oncology.

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PO0231 / #459 | Novel Bioinformatics Approaches for Rna Analysis May Reveal Common Promising Molecular Markers Between Three Pediatric Tumors

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Background and Aims: Bioinformatics tools are essential for assisting and reinforcing scientific research searching genes with clinical impact or associated with the development of several cancer types. Common machine learning interpretability approaches produce logic rules to explain decisions through handle models with massive quantities of rules. The interpretability of these results may be facilitated by ExMatrix, considered a novel visualization method for classification model interpretability. Medulloblastoma (MB) and Ependymoma (EPN) are tumors of the Central Nervous System and Adrenocortical tumor (ACT) is an aggressive and rare neoplasm of the adrenal cortex. The aim of this study was to identify genes with similar expression patterns between MB, EPN and ACT pediatric tumors that could be useful as common potential biomarkers or therapeutic targets.

Methods: RNA-Seq was performed using tumor samples of 37 pediatric patients. Raw data were processed by TrimGalore, FastQC,

MultiQC, SeqMonk, STAR and DESeq2 R package's. Gene expression data was submitted to analysis based on logical rules, following a machine learning classification model and the results were visualized using the ExMatrix method. Enrichment analysis based on gene ontology (GO) and KEGG pathways were performed with <https://metascape.org>.

Results: We identified genes with similar expression patterns between MB and EPN, MB and ACT or EPN and ACT. Among pathways enriched by these genes, the Endocrine System Development showed SOX4 and POU3F2 up-regulated in MB and EPN, however to ACT both genes showed down-regulation; SOX3 up-regulated in EPN and down-regulated in MB and ACT; MSX1 gene up-regulated in EPN and ACT and down-regulated in MB. GO enrichment revealed the Process utilizing Autophagic Mechanism with the genes RALB, OPTN and GABARAPL1 up-regulated in ACT/EPN, but down-regulated in MB, and GFAP, SYT11 and RAB39B down-regulated in ACT and up-regulated in MB/EPN.

Conclusions: These genes might be investigated as common potential biomarkers or therapeutic targets for MB, EPN and/or ACT tumors.

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PO0232 / #501 | Human Papillomavirus (HPV) Vaccine Program

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Background and Aims: (CCCL) is a leading national organization dedicated to the treatment & support of kids and adolescents with cancer; A national advocate of childhood cancer control and leading NGO in Lebanon and the region. Inaugurated in 2002, the CCCL supports almost 50% of children with cancer up to 18 years in Lebanon (including non-Lebanese and in partnerships with 13 hospitals covering all treatment costs of patients without any discrimination and with average cure rate of 80%; Initiated The Champions Circle club of survivors to ensure continuous follow up and their protection from all future harm of cancers through various program including HPV.

Methods: As an experienced national advocate of childhood cancer control, the CCCL promotes awareness and education about childhood cancer; and has initiated the CCCL HPV Vaccine Program phase I for its network of the center's beneficiaries adolescents up to 15 years. The CCCL has expanded globally for development and committed to the United Nation Global Compact SDGs 3, 5, 8, 13, 17.

Results: On the World Health Day 2021, survivors of childhood cancer, Doctors and representative from the Country WHO office were invited to witness the launching of the CCCL HPV Vaccine Program. More than 40 attended the initial awareness sessions and resulted in over 100% registration for the program. The session has attracted interests from the international organizations and initiated a primary

discussion on the means, monitoring processes and methods to expand the program Nationally. A proposal was created and budget and needs is set for Phase II.

Conclusions: The CCCL aim in this abstract to demonstrate the success of the launch and to share the plan for reaching the WHO goals of 90, 70, 90. The CCCL commitment to public starts at birth and continue to protect all communities regardless of any backgrounds

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PO0233 / #530 | Biomarkers Kinetics during Febrile Episodes in Pediatric Patients with Cancer

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Background and Aims: In this study we investigated the kinetics of CRP, PCT, IL-6 and MR-proADM in a well-defined cohort of consecutive febrile patients with cancer in order to test the hypothesis that higher plasma concentrations and absence of rapid decrease in peak values would be associated with disease severity.

Methods: A prospective descriptive and analytical study of patients with cancer and fever (≤ 18 years of age) at an University Hospital was carried out between January 2018 and December 2019. It was collected: sex, age, diagnosis, date and symptoms at diagnosis and medical history (type of cancer, stage, treatment and comorbidities). The episodes were grouped into bacterial infection, no bacterial infection and SIRS.

Results: One hundred thirty-four episodes (14 males and 20 females) were included. The 55.2% of the episodes presented a solid tumor/Hodgkin's lymphoma, the 64.1% comorbidities and the 17.1% no control disease. Eighty-one episodes presented fever at diagnosis. Bacterial infection was met in 38 episodes. Biomarkers were measured at four different points: baseline, at 12-24 hours, at 25-48 hours and at 49-72 hours. All the biomarkers evaluated decreased progressively after peak level was reached. IL-6 and MR-proADM showed a trend towards higher levels in SIRS group although this elevation was significant only for IL-6 ($p < 0.005$). When SIRS episodes were excluded, IL-6 levels were higher in the bacterial infection group at baseline and at 12-24 hours ($p < 0.005$). Bacterial infections presented more frequently values of PCT above the cut-off point (> 0.5 ng/ml) at 12-24 hours.

Conclusions: In our experience, IL-6 kinetics is faster than PCT kinetics and both are faster than CRP in patients with fever and cancer who present a good outcome as it has been previously described in other groups of patients, including patients without cancer. Patients with good evolution shows a rapid increase and decrease of PCT and especially of IL-6 levels.

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PO0234 / #591 | Ruthenium and ⁹⁰Strontium Brachytherapy for Retinoblastoma

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Background and Aims: Retinoblastoma is the most frequent intraocular tumor of childhood. Over the years, brachytherapy proved to be an efficient treatment for retinoblastoma, offering an alternative to enucleation and external beam radiotherapy. Low energy gamma-ray emitting ¹²⁵I plaque is most used around the world. In contrast, the pure beta ray-emitting ¹⁰⁶Ruthenium (¹⁰⁶Ru) plaque is used mainly in Europe. However, only a few studies have been published on the use of ¹⁰⁶Ru and not a single study on the use of ⁹⁰Strontium (⁹⁰Sr) for retinoblastoma. To evaluate the efficacy of ¹⁰⁶Ru and ⁹⁰Sr plaque brachytherapy in the management of retinoblastoma.

Methods: In a retrospective study, we reviewed case series of 120 children with retinoblastoma (194 tumors in 137 eyes) treated with ¹⁰⁶Ru and ⁹⁰Sr plaques at the S. Fyodorov Eye Microsurgery Federal State Institution in Moscow between November 2007 and December 2020. Median follow-up was 52.6 months. The median patient age was 26 months. Tumors had a mean height and diameter of 2.9 ± 1.2 mm and 6.5 ± 2.5 mm, 1.9 ± 0.5 mm and 4.8 ± 1.7 mm, respectively. Median prescribed doses of ¹⁰⁶Ru and ⁹⁰Sr plaques at reference depth and outer surface of the sclera were 89 Gy and 330 Gy, 174 Gy and 700 Gy, respectively.

Results: Overall tumor control was achieved in 93%. Tumor recurrence was observed in 7% (14 tumors). Eye retention was achieved in 89% of cases (122 of 137 eyes). Radiation complications included retinal detachment (7%), non-proliferative retinopathy (14%), proliferative retinopathy (3%), optic neuropathy (25%), and subcapsular cataract (8%).

Conclusions: Brachytherapy using ¹⁰⁶Ru and ⁹⁰Sr plaques is an effective treatment for retinoblastoma with excellent local tumor control and few secondary complications.

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PO0236 / #773 | High Risk and Relapsed Pediatric Acute Myeloid Leukemia is No Longer Incurable in Developing World

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Background and Aims: Management of acute myeloid leukemia (AML) is a challenge, particularly in a developing country like ours. From dis-

mal survival for children with AML in last couple of decades, the results are gradually improving in our country. Limited data has been published in pediatric AML from India especially pertaining to hematopoietic stem cell transplant (HSCT).

Methods: We retrospectively analyzed our data of all children with high risk/relapsed AML who underwent HSCT at our center from April 2014 to April 2020.

Results: A total of 21 patients were included – 14 underwent Matched Sibling Donor (MSD) HSCT, 6 underwent Haploidentical HSCT (haplo-HSCT) and 1 underwent autologous HSCT. The patient who underwent auto-HSCT was a case of Acute Promyelocytic Leukemia and did not have any full-match donor. Prior to HSCT, all patients were in Complete Remission (CR) except one: CR1-3, CR2-15, CR3-2. The median cell dose (CD34) was 5.5 million/kg for MSD HSCT, 6.5 million/kg for haplo-HSCT and 5.2 million/kg for autologous HSCT. All patients engrafted. Median time to neutrophil and platelet engraftment was 14 and 15 days, respectively. There were two transplant related mortalities – one died of Grade IV acute gut graft vs host disease (GvHD) and the other one due to chronic GvHD and sepsis. Three patients relapsed and were lost to follow-up. Remaining 16 patients remain in remission and are alive. The cumulative incidence of acute GvHD was 23.8% and of chronic GvHD was 14.2%. Event-free survival was 76.2% and Overall Survival was 90.4%. The median follow-up of alive patients was 848 days (range: 381-2346).

Conclusions: HSCT is a reasonable option for children with high-risk or relapsed AML with good disease control even in resource-limited settings and our results are comparable with those reported from the west. Our experience offers ray of hope to all high risk AML patients in developing world.

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PO0237 / #89 | Clinical Analysis of Covid-19 Infections in Children with Cancer in a Tertiary Care Hospital in Bangladesh

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Background and Aims: Background and Aim : Children suffering from cancer are more vulnerable than others. This study performed to outline the clinical characteristics and outcome of children with cancer with severe acute respiratory syndrome corona virus 2 (SARS-CoV-2) infection in Dhaka Medical College Hospital, Dhaka.

Methods: Methods: Clinical data and epidemiologic history of 24 children with cancer with laboratory-confirmed COVID-19 infection by

real-time reverse transcription-PCR (RNA-PCR) were admitted in the Dhaka Medical College Hospital, Dhaka from May, 2020 to June, 2020. Clinical data and epidemiologic history of these patients were retrospectively collected and analyzed.

Results: Among the 24 cases, 16 (67%) were males and 8 (33%) were females. The median age was 5 years (range 0.11–12 years). The most common symptoms were fever (11 cases, 45%), runny nose (8 cases, 33%), cough (8 cases, 33%), asymptomatic (6 cases, 24%). Acute Lymphoblastic Leukemia were the most common (50%) and second most were Wilms tumor (17%). On admission, 17 (71%) had normal white blood cell counts, while only 2 (8%) had more than $10 \times 10^9/L$ and 5 (21%) had less than $4 \times 10^9/L$, respectively. 16 (67%) had normal neu-

trophil count and 7 (29%) had neutropenia. Lymphocyte counts were normal in 50% cases, lymphocytopenia in 10 (42%). C Reactive protein and serum ferritin were raised in 19 (79%); D-dimer, prothrombin time, activated partial thromboplastin time were raised in 8 (33%). Chest X-ray was abnormal in 6 (25%). 16 (66%) received intravenous antibiotics, 5 (21%) needed oxygen therapy and 4 (17%) needed corticosteroid. Nineteen (79%) were discharged from hospital, 3 (13%) were discharged on request and 2 (8%) were dead.

Conclusions: Children appeared susceptible to COVID-19 infections and there was significant gender difference. Clinical manifestations of COVID-19 infections in children were generally less severe than those of adults. Death rate is more in children with cancer than others.